A Two-Boundary Retrocausal Unification of Quantum Mechanics and General Relativity: Rigorous Core and a Conditional Master Theorem

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July 11, 2026

Abstract

We work in the setting of a fixed globally hyperbolic spacetime $(\mathcal{M}, g)$ carrying a finite-dimensional quantum matter system with two-boundary data $(\rho_{\rm i}, \{E_k\})$ in the sense of Aharonov–Bergmann–Lebowitz [5]. *Part I* proves, with complete proofs: (i) the final-state-conditioned density operator $\rho_2(\tau \mid k)$ evolves unitarily in τ with the background Hamiltonian (*unitary covariance*); (ii) the expectation of any Heisenberg observable satisfying a local covariant conservation law is itself covariantly conserved along the flow; (iii) a bulk no-signalling theorem for spacelike-separated local operations after marginalization over the terminal outcome; (iv) the posterior $\pi_\tau(k)$ is a closed bounded *discrete-time* martingale (indexed by the protocol sequence (s_n)) that converges a.s. to $\mathbf{1}_{\{K=k\}}$, under an explicit terminal record-completeness axiom; and (v) that taking the prior expectation over the terminal outcome K reduces the conditioned state to the ordinary forward state $\rho(\tau)$. These results constitute a self-contained operator-algebraic two-boundary framework on a curved spacetime background, rigorous modulo the explicitly stated axioms (in particular the causal-propagation axiom (M3) of Axiom 2.2 and the terminal record-completeness Axiom 7.1).

Part II states the paper's central claim: a *conditional unification* of quantum mechanics and general relativity. We exhibit a single universal constraint equation — the Two-Boundary Retrocausal Master Equation (TBRME) $\widehat{\mathfrak{C}}_k \Psi_k = 0$ on a universal Hilbert space $\mathcal{U}$ with two-boundary rigged data — and prove, as *Theorem 11.73 (Master Unification Theorem)*, that under an explicitly enumerated hypothesis bundle H (the Part I axioms, the bridge conjectures of Secs. 10 and 11.1, the structural/analytic hypotheses (T0)–(T13) of Conj. 11.56, and the regime data of Thm. 11.68) the TBRME is rigged-Hilbert well-posed and reduces *simultaneously* to (a) ordinary quantum mechanics (TDSE), (b) the rigorous Part I two-boundary conditioned quantum mechanics on curved spacetime, (c) the semiclassical Einstein field equation of general relativity, (d) the modified Wheeler–DeWitt constraint on pure geometry, and (e) the retrocausal modified TDSE (31) of App. C. The conditional implication $\mathsf{H} \Rightarrow$ unification is airtight (no unlisted hypothesis); the unconditional truth of the unification therefore reduces to the explicit open analytic and structural problems packaged in H.

Within Part II we adopt three structural postulates that absorb two of the seven classical obstructions to unification into the framework and explicitly scope the third to an effective-field-theory regime. First, the *thermal-time postulate* (Sec. 11.2, Pos. 11.11): the Page–Wootters clock Hamiltonian is identified with the modular Hamiltonian $-\log \Delta_\omega$ of the seed state ω, in the sense of Connes–Rovelli [33]. Time in the master equation is therefore *not* a manifold coordinate but a state-derived flow, and "temporal evolution" is Araki's monotone relative-entropy decay [52] of perturbations toward ω-equilibrium. Black-hole horizons enter the framework as the fixed subalgebras of thermal time; Wall's generalised second law [55] is read as monotonicity of exterior Araki relative entropy under modular flow. Second, the *entropic-gravity postulate* (Sec. 11.3, Pos. 11.18): the matter stress-energy $\widehat{T}^{\mu\nu}_{\rm m}$ is identified with the local density of the modular Hamiltonian K_ω via the first law of entanglement [58], and the operator-ordered linearised quantum Einstein tensor $\widehat{G}^{(Q)\,\mu\nu}$

is identified with the area-response of local modular Hamiltonians via Jacobson's operator-algebraic Clausius relation [56, 59]. The semiclassical Einstein equation is then an *equation of state* by construction rather than a dynamical conjecture — a postulate the paper diagnoses (Prop. 11.22) as emergent equation-of-state content rather than fundamental dynamics, under-determined in three named modes (trace/cosmological blindness, which-branch state selection, and the coupling G, itself an emergent response coefficient of the vacuum, $g = GE_\star^2/\hbar c^5 = O(1)$ fixed by the Standard-Model field content, Prop. 11.20); the diagnosis reframes but does not derive the postulate. Stationarity of stress-energy expectations along thermal time is Tomita KMS cyclicity (Prop. 11.24), while full semiclassical conservation $\nabla^\mu \langle \widehat{T}_{\mu\nu} \rangle_\omega = 0$ holds for every Hadamard state by the Wald–Hollands–Wald conservation axiom, independently of KMS (Lem. 11.25).

Third, the *Planck-cutoff postulate* (Sec. 11.4, Pos. 11.30): physical matter states are supported on the spectral range of the matter Hamiltonian below a fixed cutoff $E_\star \lesssim E_{\mathrm{P}}$, via the projection $P_{\leq E_\star}$. This is the standard Wilsonian effective-field-theory scope, and it makes Part I's finite-dimensional matter Hilbert space a structural consequence of the sub-Planckian regime rather than an ad hoc simplification.

Under these three postulates the hypothesis bundle H shrinks substantially: Conj. 11.10 becomes Prop. 11.15, (CP) of Conj. 11.1 and (T11) of Conj. 11.56 consolidate to the single commutation $[\widehat{H}_{\mathrm{m}}, \widehat{H}_{\mathrm{c}}] = 0$ (Hyp. 11.13), (T1), (T2), (T7) of Conj. 11.56 become Prop. 11.26, and Conj. 10.2's coupling-existence burden is absorbed into Pos. 11.18. Under Pos. 11.30, Prop. 11.33 further discharges (T5) spectral absolute continuity, (T6) band-limited slow-branch selection, (S3) spectral gap of ρ_{i}, the nuclearity of the test core Φ_k, and the Hadamard-free well-posedness of the entropic-gravity coupling, all as theorems of finite-dimensional linear algebra. This absorbs obstructions (1) (problem of time) and (5) (stress-energy on a dynamical metric) into structural identifications and explicitly scopes the framework to the sub-Planckian effective-field-theory regime, reducing the open content of H to the contraction estimate of Conj. 11.4, Conj. 11.5, Conj. 11.8, Conj. 11.1 (non-(CP) parts), and the residual rigged-Hilbert structural conditions (T0), (T3), (T4), (T7)–(T10), (T12), (T13) of Conj. 11.56.

We do not claim the residual open problems in H are solved; we claim, and prove conditionally, that if they are solved, quantum mechanics and general relativity are *both* conditional reductions of a single equation whose time parameter is the entropy-flow of a seed equilibrium state, whose gravity–matter coupling is the operator-algebraic Clausius relation, and whose matter content is restricted to sub-Planckian effective scope. Two concrete consistency checks strengthen the framework (Sec. 11.5): Prop. 11.39 shows that the record-adapted, branch-indexed semiclassical Einstein equation evades the Page–Geilker pathology [63] that falsifies naive forward semiclassical gravity, supplying a genuine empirical differentiator (with attribution to the final-boundary program of Kent [65, 66] and the decoherent-histories per-branch Einstein equations of Halliwell–Hartle [69, 70, 71] as precedents; Rem. 11.46 details what is distinctive in the present paper); Prop. 11.49 gives a closed-form lower bound on the Conj. 11.4 contraction constant in the finite-dim setting, converting the entropy-smallness conjecture to a constructive bound with an explicit numerical example ($C \geq 1/32$ nats in a two-level toy).

Section 12 sharpens the hypothesis bundle H to H': all structural conditions (T0)–(T13) of Conj. 11.56 *except (T11)* are promoted to theorems of this paper under the three postulates and the standard realisation Hyp. 12.9 on the gravity-sector Hilbert space (Props. 12.1, 12.3, 12.5, 12.7, 12.11, 12.13), and every bespoke conjecture of Part II discharges to either a theorem of the present paper or a single named open problem of the ambient operator-algebraic / QFT-on-curved-spacetime literature. Thm. 12.15 (Sharpened Master Unification Theorem) is the resulting statement. Four discharge appendices — App. D (Prop. D.2), App. E (Thm. E.7, Cor. E.8), App. F (Thm. F.2), and App. G (Thms. G.1, G.2) — settle the status of the four previously external open problems (E1)–(E4): (E3), (E4) are upgraded to theorems of the present paper and (E2) to a *conditional* theorem with explicit scope residuals (Rem. E.10). (E4) is closed unconditionally via Araki's 1973 relative-Hamiltonian perturbation theory and the Haagerup dual-weight theorem; (E2), (E3) are closed modulo two named ambient-literature inputs (Hyp. E.1, Hyp. F.1) labelled (E2a), (E3a) respectively — a uniform Sobolev Hadamard parametrix with mixed C^2 coincidence jets, and a Longo–Morsella quasi-local modular flow with τ-dependent generator on Ashtekar–Krishnan dynamical horizons. (E1) is restated as the explicit Wilsonian scope postulate H0 (Pos. 11.30): the CLPW Type II trace renders trace and generalised entropy UV-finite (Prop. D.2), which motivates but does not derive the matter cutoff. Eight further appendices close the remaining framework-level gaps: App. H (Thm. H.1) upgrades Hyp. 12.9 (existence of a gravity-sector Hilbert space) from existence assertion to theorem via the CLPW Type II crossed-product commitment, merging the realisation hypotheses into a single gravity-sector realisation and automatically discharging T-conditions (T0)–(T4), (T6), (T9) of Conj. 11.56; App. I scope-splits (E0) into (E0-lin), now a theorem (Thm. I.1) via Faulkner–Guica–Hartman–Myers–Van Raamsdonk [58] plus Oh–Park [243] plus the Casini–Huerta–Myers modular Hamiltonian [241], and (E0-nl), a single clean Fréchet-differentiability axiom (Ax. I.10) on the area functional whose sole generic-diamond residual is the QNEC-saturation equality (Hyp. I.5, Rem. I.6: an inequality the framework has versus the equality it needs); Sec. 13 records five concrete predictions — three empirical differentiators (the Page curve from two-boundary martingale convergence; coherent BMV-scale entanglement with the DSW horizon floor; and the universal quadratic phonon-cavity softening, the designated laboratory falsifier) and two structural consistency results (a parity-even CMB spectrum correction from the cosine clock-shift $\widehat{\Sigma}_\delta$, the earlier parity-odd claim retracted, reproducing the established quantum-gravitational term and unobservable by any projected survey; and a Type II-renormalised cosmological constant absent of the E_{P}^4 catastrophe, whose H^2 running coincides term-for-term with running-vacuum and holographic dark energy); App. J (Thm. J.1, Cor. J.3) accommodates the Standard Model (SU(3)×SU(2)×U(1), three generations, chirality, Higgs) without a new axiom via the locally covariant pAQFT machinery of [220, 264, 265, 266], with a state-independent anomaly correction to the Clausius identity and a Chamseddine–Connes spectral-triple consistency result showing the thermal-time flow is blind to generation count; App. K (Conj. K.1) organises UV behaviour via a Type II trace-stable inductive limit, with explicit honest scope flagging of big-bang physics as out of reach of the current Type II machinery; App. L (Conj. L.1) derives a cosmological-horizon selection principle for the terminal POVM $\{E_k\}$; App. M (Thms. M.1, M.2) recovers the Hawking radiation spectrum and the AdS/CFT reduction as consistency-check theorems; App. N (Prop. N.1) specifies a finite- dimensional FLRW toy model in which the five reductions (U1)–(U5) admit explicit numerical verification, released as a companion Python/QuTiP notebook. Under all commitments the remaining internal postulates of the framework are the nonlinear axiom Ax. I.10 (the nonlinear Clausius closure) and the realisation hypotheses (Hyp. D.1, through which the linearised sector of Pos. 11.18 and the Type II trace structure are theorems on de Sitter, stationary, and cosmological backgrounds; Hyp. I.3 for the de Sitter sector); the existence of $\mathcal{H}_{\mathrm{grav}}$ is a theorem. This is the strongest honest form the unification claim of the present paper takes: every conjectural ingredient internal to the framework is either discharged or carried as an explicitly named commitment; the complete residual commitment list — one nonlinear matter–geometry coupling axiom, the Planck-cutoff postulate H0 (with its ε-tail budget Hyp. 11.35 and the H0 co-scoping compatibility residual Rem. 11.72), the thermal-time postulate, the realisation hypotheses above, the record-adapted structural hypotheses ((vi-b), (x-a), (x-c), (x-d), (RP)) with the shell-summability residual (E5), the named ambient-literature inputs (E2a), (E3a), and the terminal-POVM selection Conj. L.1 — is consolidated in one place as Rem. 12.16, with (E4), Hyp. 12.9, the linearised AdS-anchored sector of (E0) (modulo Hyp. D.1), SM matter accommodation, Hawking recovery, and AdS/CFT reduction all closed and sub-Planckian UV behaviour organised (motivated, not derived) by the Type II trace. TBRME uniqueness (a separate companion paper, outlined in Rem. 12.18) is the only scoped deferral.

Contents

1 Scope and status of claims

This paper is divided into two parts.

At a glance. *Proven unconditionally* (Part I): a self-contained two-boundary conditioned quantum mechanics on a fixed globally hyperbolic background (unitary covariance, conservation, bulk no-signalling, posterior-martingale collapse, and a per-branch Margenau–Hill witness of the two-boundary link (Thm. 3.10) whose negativity marginalises away into the no-signalling identity), modulo two explicitly stated postulates — microcausality (M3, Axiom 2.2) and terminal record-completeness (Axiom 7.1). *Proven conditionally* (Parts II–III): that a single constraint equation reduces simultaneously to ordinary QM, the modified TDSE, semiclassical Einstein, and the modified Wheeler–DeWitt constraint (Thm. 11.68), the implication itself being an unconditional theorem (Thm. 12.15). *Assumed, not derived*: the entropic-gravity equation of state (E0), of which the linearised part (E0-lin) is here a theorem (Thm. I.1) and only a nonlinear Fréchet-differentiability axiom (E0-nl, Ax. I.10) remains primitive. *Remaining external residuals after sharpening*: besides (E0-nl), two named ambient-literature inputs — (E2a) a uniform Sobolev Hadamard parametrix (Hyp. E.1) and (E3a) quasi-local modular flow on dynamical horizons (Hyp. F.1); the pre-discharge Part II conjectures (E3)–(E4) are upgraded to theorems of this paper (Part III), (E4) unconditionally, and (E2) to a conditional theorem scoped to quasi-free seeds of the free massive scalar (Rem. E.10), while (E1) is restated as the explicit Wilsonian scope postulate H0 (Pos. 11.30), motivated but not derived by the Type II trace (Prop. D.2). *Out of scope*: the trans-Planckian / UV-complete regime and the big-bang singularity (App. K). We claim none of these residuals is solved here; we prove that if they hold, QM and GR are reductions of one equation.

Part I (Sections 2–9) is self-contained. Each numbered result is stated with precise hypotheses and proved here from those hypotheses together with standard background facts from finite-dimensional linear algebra, ODE theory for bounded Hamiltonians, Kolmogorov extension on countable products, and the upward theorem for conditional expectations. We emphasize that "self-contained" means "derived from the explicitly stated axioms of Part I"; in particular the causal-propagation axiom (M3) of Axiom 2.2 (see Rem. 2.3) and the terminal record-completeness Axiom 7.1 are postulates of the framework, not consequences of deeper principles. The setting throughout Part I is:

- $(\mathcal{M}, g)$ is a smooth, time-oriented, globally hyperbolic Lorentzian 4-manifold with a fixed smooth Cauchy temporal function τ [9]; the metric is a fixed background, not dynamical.
- the matter Hilbert space $\mathcal{H}$ is *finite-dimensional*, so all Hamiltonians are bounded self-adjoint operators and all operator identities are literal;
- the Hamiltonian $\tau \mapsto H(\tau) \in \mathcal{L}_{\mathrm{sa}}(\mathcal{H})$ is a continuous one-parameter family generating a two-parameter unitary propagator $U(\tau, \tau')$ via $i\hbar\partial_\tau U(\tau, \tau') = H(\tau)U(\tau, \tau')$, $U(\tau, \tau) = \mathbb{1}$;
- the boundary data $(\rho_{\mathrm{i}}, \{E_k\}_{k\in\mathcal{K}})$ satisfy Axiom 2.1;
- local observable algebras $\mathcal{A}(R) \subset \mathcal{L}(\mathcal{H})$ are specified as subsets of $\mathcal{L}(\mathcal{H})$ for each open region $R \subset \mathcal{M}$, with a microcausality hypothesis (Axiom 2.2) that is imposed where needed.

Part II (Sections 10–11.6) is conjectural, with a single conditional unification theorem. It contains: (a) the two gravity bridges — a semiclassical Einstein coupling (Conj. 10.2) and a modified Wheeler–DeWitt constraint (Conj. 10.13); (a′) the thermal-time postulate (Pos. 11.11, Sec. 11.2) identifying the Page–Wootters clock Hamiltonian with the modular Hamiltonian $-\log \Delta_\omega$ of the seed state, in the sense of Connes–Rovelli [33], which dissolves obstruction (1) (problem of time) and promotes Conj. 11.10 to Prop. 11.15; (a″) the entropic-gravity postulate (Pos. 11.18, Sec. 11.3) identifying the matter stress-energy with the density of K_ω via the first law of entanglement [58] and the operator-ordered linearised quantum Einstein tensor with the area-response of K_ω via Jacobson's Clausius relation [56, 59], which dissolves obstruction (5) (stress-energy on a dynamical metric) and promotes (T1), (T2), (T7) of Conj. 11.56 to Prop. 11.26; (a‴) the Planck-cutoff postulate (Pos. 11.30, Sec. 11.4) restricting all physical matter states to the spectral range of $P_{\leq E_\star}$ for a fixed cutoff $E_\star \lesssim E_{\mathrm{P}}$, which explicitly scopes the framework to the sub-Planckian effective regime, makes Part I's finite-dimensional matter setup a structural consequence rather than an ad hoc choice, and promotes (T5), (T6), (S3), nuclearity of Φ_k, and the Hadamard-free well-posedness of the entropic-gravity coupling to Prop. 11.33; (b) the remaining four speculative conjectures of Sec. 11.1 that close Part II's standing open problems; (c) the Two-Boundary Retrocausal Master Equation (Conj. 11.56), together with its conditional airtight well-posedness (Thm. 11.64); (d) the five-fold reduction theorem (Thm. 11.68) showing that the master equation reduces simultaneously to ordinary QM, Part I, the modified TDSE, semiclassical Einstein, and the modified Wheeler–DeWitt constraint; and (e) the consolidating *Master Unification Theorem* (Thm. 11.73), which combines Thm. 11.64 and Thm. 11.68 into a single statement: under the complete hypothesis bundle H of Part II, quantum mechanics and general relativity are both simultaneous conditional reductions of a single equation on a single state space. Each individual conjecture and each hypothesis in H is stated with the precise analytic conditions that would elevate it; the *implication* $\mathsf{H} \Rightarrow$ unification is proved unconditionally as a theorem, and is what we mean by "conditional unification of QM and GR" in the title and abstract. We use *unification* here in a precise structural sense: a single state space $\mathcal{U}$ and a single constraint equation of which QM and GR are simultaneous regime-reductions. We do *not* claim a first-principles derivation of gravitational dynamics — the gravitational equation of motion enters as the operator-algebraic Clausius equation-of-state postulate (E0) (Pos. 11.18, Sec. 12.5.1) — nor a UV-complete theory of quantum gravity (the trans-Planckian and big-bang regimes are explicitly out of scope, App. K); what is new is the architecture, that one equation carries both limits at once. We do *not* claim the individual conjectures of H are resolved; we do claim, and prove as a theorem, that if they are, QM and GR are reductions of the same equation. What is distinctive here, relative to string theory, loop quantum gravity, asymptotic safety and causal sets, is not a new ultraviolet completion of gravity — the framework assumes a sub-Planckian effective regime (Pos. 11.30) — but the conditional theorem (Thm. 11.73) that quantum mechanics and general relativity are simultaneous reductions of a single equation on a single state space, built from two-boundary retrocausal (ABL) conditioning and the identification of relational time with modular / thermal time; that distinctive content is largely orthogonal to the UV-completion question those programmes address.

Part III (Section 12) is the sharpened form of Part II. It consolidates the reductions of H

into the sharpened bundle H' and proves Thm. 12.15 (Sharpened Master Unification Theorem): every structural (T)-condition of Conj. 11.56 *except (T11)* is a theorem of this paper under the three postulates plus the standard realisation Hyp. 12.9 on the gravity-sector Hilbert space (with (T0)/(T8) holding on the open dense subset of generic boundary data of Prop. 12.7); every bespoke Part II conjecture is discharged to either a theorem of the present paper or to a single named open problem of the ambient literature. The residual, after all such discharge, is exactly one primitive matter–geometry coupling axiom (Pos. 11.18, the Jacobson–Bianchi–Myers Clausius identification, labelled (E0)) together with two surviving externally named residuals (E2a),(E3a) of Sec. 12.5 — the original external items (E1)–(E4) having been discharged down to these: (E1) is not an open problem but is retired into the Planck-cutoff postulate H0, (E4) is closed unconditionally (Araki relative entropy + Haagerup dual weight), and (E2),(E3) are reduced to the narrower residuals (E2a),(E3a). This is the strongest honest form the unification claim takes; the claim is conditional modulo the coupling axiom (E0) and the two external ambient-literature residuals (E2a),(E3a) — together with the standing structural assumptions enumerated in full in Rem. 12.16 (the Planck-cutoff postulate H0 and its co-scoping residual Rem. 11.72, thermal time, the record-adapted gravitational layer, and the cosmological CHT terminal-POVM selection) — and the conditional implication is a theorem of the paper.

Conventions. We use SI units; $\hbar, c, G$ are explicit. Planck units: $E_\mathrm{P} := \sqrt{\hbar c^5/G}$, $t_\mathrm{P} := \sqrt{\hbar G/c^5}$, $\ell_\mathrm{P} := \sqrt{\hbar G/c^3}$; $E_\mathrm{P} t_\mathrm{P} = \hbar$. The direction of these definitions is conventional SI bookkeeping; structurally (Prop. 11.20) the primitive of the gravitational anchor is the matter scale $E_\star$ and $G = g\,\hbar c^5/E_\star^2$ is the slaved response coefficient, with $g = O(1)$ fixed by the vacuum field content, so E_P and G carry the same single datum rather than two independent constants. The term "a.s." always refers to the probability measure $\mathbb{P}_\gamma$ of Section 5.

2 Setup

2.1 Spacetime and propagators

Let $(\mathcal{M}, g)$ and $\tau : \mathcal{M} \to \mathbb{R}$ be as above. By [9] $\mathcal{M} \cong \mathbb{R} \times \Sigma$ with smooth Cauchy slices $\Sigma_\tau := \tau^{-1}(\tau)$. Fix $\tau_i < \tau_f \in \mathbb{R}$. Let $\mathcal{M}_{[\tau_i,\tau_f]} := \tau^{-1}([\tau_i, \tau_f])$.

Let $\mathcal{H}$ be finite-dimensional with $\dim \mathcal{H} =: d < \infty$. Let $\tau \mapsto H(\tau)$ be continuous and bounded; by the fundamental theorem of ODE for linear equations in finite dimensions, the propagator $U(\tau, \tau')$ exists for all $\tau, \tau' \in [\tau_i, \tau_f]$, is jointly continuous, unitary, and satisfies $U(\tau_3, \tau_1) = U(\tau_3, \tau_2)U(\tau_2, \tau_1)$. In particular $U(\tau_f, \tau) = U(\tau_f, \tau')U(\tau', \tau)$ for all $\tau, \tau' \in [\tau_i, \tau_f]$.

2.2 Boundary data and local algebras

Axiom 2.1 (Two-boundary data)**.** $\rho_\mathrm{i} \in \mathcal{L}(\mathcal{H})$ satisfies $\rho_\mathrm{i} \succeq 0$, $\operatorname{Tr} \rho_\mathrm{i} = 1$; $\mathcal{K}$ is a finite set; $E_k \in \mathcal{L}(\mathcal{H})$ with $E_k \succeq 0$ for each $k \in \mathcal{K}$, and $\sum_{k\in\mathcal{K}} E_k = \mathbb{1}$.

Axiom 2.2 (Local algebras and microcausality)**.** To each open $R \subset \mathcal{M}_{[\tau_i,\tau_f]}$ we associate a $*$-subalgebra $\mathcal{A}(R) \subset \mathcal{L}(\mathcal{H})$ containing $\mathbb{1}$, and we impose the *outer-regularity compatibility*

$$\mathcal{A}(R) = \bigcap_{U \supset R,\, U \text{ open}} \mathcal{A}(U) \qquad \text{for every open } R \subset \mathcal{M}_{[\tau_i,\tau_f]}. \tag{1}$$

For an arbitrary subset $S \subset \mathcal{M}_{[\tau_i,\tau_f]}$ we then *define* $\mathcal{A}(S) := \bigcap_{U \supset S,\, U \text{ open}} \mathcal{A}(U)$. By (1) this agrees with the original assignment on open sets, and by construction the extended $\mathcal{A}$ satisfies isotony (M1) below for arbitrary subsets. We require:

(M1) *Isotony:* $R_1 \subset R_2 \Rightarrow \mathcal{A}(R_1) \subset \mathcal{A}(R_2)$.

(M2) *Microcausality:* if $R_1, R_2 \subset \mathcal{M}_{[\tau_i,\tau_f]}$ are spacelike-separated, then $[A, B] = 0$ for all $A \in \mathcal{A}(R_1)$, $B \in \mathcal{A}(R_2)$.

(M3) *Causal propagation:* for any open $R \subset \Sigma_\tau$ and any $\tau' \in [\tau_i, \tau_f]$, both

$$\text{(M3a)} \quad U(\tau',\tau)^\dagger\, \mathcal{A}(R)\, U(\tau',\tau) \;\subset\; \mathcal{A}\big(J(R) \cap \Sigma_{\tau'}\big), \tag{2}$$

$$\text{(M3b)} \quad U(\tau',\tau)\, \mathcal{A}(R)\, U(\tau',\tau)^\dagger \;\subset\; \mathcal{A}\big(J(R) \cap \Sigma_{\tau'}\big), \tag{3}$$

where $J(R) := J^+(R) \cup J^-(R)$. Both express that Heisenberg propagation of a localized operator between Cauchy slices keeps its support inside the causal domain of the original region. (M3a) and (M3b) are logically independent conditions; we impose both. In standard Haag–Kastler nets on a globally hyperbolic spacetime they are both satisfied. In algebraic-QFT terminology (M3) is the operator-algebra (Heisenberg-picture) analogue of the *primitive-causality / time-slice* property [81]; it is logically independent of microcausality (M2). Recent analyses of relativistic measurement establish that (M2) alone does *not* forbid superluminal signalling under spacelike-separated state updates, and that a time-slice / causal-factorisation condition — exactly the content of (M3) — is the necessary missing ingredient [82, 83, 84].

The terminal effects E_k are taken to be supported on Σ_{τ_f}, i.e. $E_k \in \mathcal{A}(\Sigma_{\tau_f})$; similarly ρ_{i} is associated with Σ_{τ_i}.

Remark 2.3 (Status of (M3) and derivation from no-signalling). The causal-propagation axiom (M3) is a genuine *postulate* on the pair $(\{\mathcal{A}(R)\}, H(\tau))$: in the present finite-dimensional setup $H(\tau)$ is an abstract bounded self-adjoint operator and (M3) does not follow from the other axioms. Operationally, (M3) is a *necessary* causal-cone constraint on the abstract propagator U that merely bounds the support of Heisenberg-evolved local operators; it is strictly weaker than the bulk no-signalling Thm. 6.1, which is recovered only when (M3) is combined with microcausality (M2) and the outer-regularity/normality argument used in its proof. Thus (M3) does not presuppose no-signalling — it is, in the necessary-but-not-sufficient sense of [84], the causal-propagation ingredient that microcausality alone does not supply. In a fundamentally local theory (e.g. Haag–Kastler nets obeying finite propagation speed) (M3) is a theorem; here it is an additional hypothesis used explicitly in the proof of Thm. 6.1. In finite-dim tensor-product settings where $\mathcal{A}(R)$ is the subsystem algebra of a tensor factor and $H(\tau)$ acts nontrivially only on factors in a local neighborhood, (M3) holds; beyond such settings it must be imposed.

Derivation from a weaker operational postulate. Under the Planck-cutoff Pos. 11.30 the matter algebra is finite-dimensional on any compact spatial region, and in this setting (M3) admits a derivation from an operationally weaker hypothesis. Soulas [79] proves that, in finite-dim quantum systems, operational *no-signalling* between spacelike regions — restricted to *ideal (projective) pointwise measurements* implemented in the projection-postulate sense, the statement that no choice of such local measurement in R_1 changes the marginal statistics of local observables in R_2 whenever R_1, R_2 are spacelike-separated — is logically equivalent to algebraic microcausality (M2) combined with tensor-factorization of the unitary propagator $U = U_1 \otimes U_2$ between spacelike-separated regions. (Soulas establishes the implication for ideal projective measurements; the extension to the general POVM/instrument no-signalling used in Thm. 6.1 is expected in finite dimensions by Naimark dilation of each local POVM to a projective measurement on an enlarged local factor, but we do not rely on it below.) Eggeling–Schlingemann–Werner [80] supply the Stinespring structure theorem showing that any semicausal (no-signalling) operation is semilocalizable. The Heisenberg propagation bound (M3) then follows from tensor-factorization plus isotony (M1): if $U(\tau',\tau)$ factorizes as $U_1(\tau',\tau) \otimes U_2(\tau',\tau)$ relative to any spacelike partition $R \sqcup R^c$ of Σ_τ, then $U(\tau',\tau)^\dagger \mathcal{A}(R) U(\tau',\tau) \subset \mathcal{A}(\text{causal evolution of } R)$ by construction, and the right-hand side lies in $\mathcal{A}(J(R) \cap \Sigma_{\tau'})$ because the causal evolution of a spacetime region is contained in its causal domain. We therefore have, under Pos. 11.30, the implication

(Operational no-signalling between spacelike Cauchy slices)

$\Rightarrow$ (Algebraic microcausality (M2) + tensor-factorization of U) $\Rightarrow$ (M3 causal propagation).

There is no circularity with Thm. 6.1: the hypothesis invoked there is operational no-signalling between spacelike Cauchy *slices* (a property of the propagator U), whereas the conclusion of that theorem is no-signalling between spacelike *bulk regions* after marginalisation over the terminal outcome — a distinct, strictly stronger derived statement. The former is an input constraint on U; the latter is a theorem about the full two-boundary Born law.

This lets the framework replace (M3) as a primitive postulate with the operationally weaker *no-signalling between spacelike Cauchy slices*; in finite dimensions (M3) is then a theorem. We retain (M3) in the statement of Ax. 2.2 for notational clarity and consistency with the Haag–Kastler literature. The reader who prefers to postulate no-signalling and derive (M3) via [79, 80] should note two scope conditions. (i) The Soulas equivalence is established for *ideal* (projective) measurements in the projection-postulate sense, whereas Thm. 6.1 is stated for general POVMs; the operational replacement is therefore rigorous as stated for the projective sub-class, and the POVM case requires in addition the Eggeling–Schlingemann–Werner semilocalizability step [80] (which the reader should verify upgrades projective no-signalling to the POVM statement in the case at hand). (ii) The tensor factorisation $U = U_1 \otimes U_2$ presupposes a type-I Hilbert-space split $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_{R^c}$, which holds in the finite-dimensional Pos. 11.30 scope but *not* for the type-III / Type II crossed-product algebras of Part II (Conj. 11.1); outside the Pos. 11.30 scope (M3) must be retained as a primitive. Within that scope the derivation loses no generality and gains one fewer primitive axiom. No stronger structure is used in Part I.

2.3 Forward state and backward effect

Definition 2.4 (Forward state and backward effect)**.** For $\tau \in [\tau_i, \tau_f]$ and $k \in \mathcal{K}$,

$$\rho(\tau) := U(\tau, \tau_i)\, \rho_{\mathrm{i}}\, U(\tau, \tau_i)^\dagger, \qquad \Lambda_k(\tau) := U(\tau_f, \tau)^\dagger E_k\, U(\tau_f, \tau).$$

Lemma 2.5 (Time-invariant Born law)**.** $p(k) := \mathrm{Tr}(\rho(\tau)\Lambda_k(\tau)) = \mathrm{Tr}(\rho_{\mathrm{i}} U(\tau_f, \tau_i)^\dagger E_k U(\tau_f, \tau_i))$ *is independent of* $\tau \in [\tau_i, \tau_f]$ *and satisfies* $\sum_{k\in\mathcal{K}} p(k) = 1$, $p(k) \geq 0$.

Proof. Using $U(\tau_f, \tau_i) = U(\tau_f, \tau)U(\tau, \tau_i)$,

$$\rho(\tau)\Lambda_k(\tau) = U(\tau, \tau_i)\rho_{\mathrm{i}} U(\tau, \tau_i)^\dagger U(\tau_f, \tau)^\dagger E_k U(\tau_f, \tau).$$

Taking the trace and using cyclicity,

$$\mathrm{Tr}(\rho(\tau)\Lambda_k(\tau)) = \mathrm{Tr}\big(\rho_{\mathrm{i}} U(\tau, \tau_i)^\dagger U(\tau_f, \tau)^\dagger E_k U(\tau_f, \tau) U(\tau, \tau_i)\big) = \mathrm{Tr}\big(\rho_{\mathrm{i}} U(\tau_f, \tau_i)^\dagger E_k U(\tau_f, \tau_i)\big),$$

which is independent of τ. Positivity follows because both $\rho(\tau)$ and $\Lambda_k(\tau)$ are positive semidefinite; more explicitly,

$$p(k) = \mathrm{Tr}\big(\rho(\tau)^{1/2}\Lambda_k(\tau)\rho(\tau)^{1/2}\big) \geq 0.$$

Finally,

$$\sum_k p(k) = \mathrm{Tr}\big(\rho(\tau) \sum_k \Lambda_k(\tau)\big) = \mathrm{Tr}\big(\rho(\tau) U(\tau_f, \tau)^\dagger (\sum_k E_k) U(\tau_f, \tau)\big) = \mathrm{Tr}\, \rho(\tau) = 1.$$

□

Remark 2.6 (Status of the Born weight). The weight $p(k)$ of Lem. 2.5 is an *input* of the framework: Lem. 2.5 establishes its τ-invariance, normalization, and positivity, but the trace form $p(k) = \mathrm{Tr}(\rho_{\mathrm{i}} U(\tau_f, \tau_i)^\dagger E_k U(\tau_f, \tau_i))$ is the standard quantum (ABL two-boundary) probability rule, taken as primitive. In the passive construction the marginal $\mathbb{P}_\gamma(K = k) = p(k)$ holds by construction (Sec. 5), not as a consequence of sequential Lüders reduction. The present paper therefore *assumes* the Born rule and derives record definiteness (Thm. 7.3) and conditional unitarity (Thm. 3.7) on top of it; it does not attempt a derivation of the Born weight itself, in the same status register as the microcausality (M3, Axiom 2.2) and terminal record-completeness (Axiom 7.1) postulates stated elsewhere in the paper. This is a deliberate scope choice consistent with the current state of the art: every known derivation program (Gleason's theorem and its POVM Gleason–Busch extension, the Deutsch–Wallace decision-theoretic argument, Zurek's environment-assisted invariance, and the Finkelstein–Hartle frequency argument) has been argued to rely on a non-trivial additivity assumption not reducible to non-contextuality and normalization [388]; for a current centennial assessment of these programs see [384].

3 Conditioned states and unitary covariance

Definition 3.1 (Conditioned state)**.** For $\tau \in [\tau_i, \tau_f]$ and $k \in \mathcal{K}$,

$$\rho_2(\tau \mid k) := \begin{cases} \dfrac{\rho(\tau)^{1/2}\,\Lambda_k(\tau)\,\rho(\tau)^{1/2}}{p(k)}, & p(k) > 0, \\ \mathbb{1}/d, & p(k) = 0, \end{cases}$$

with $\rho(\tau)^{1/2}$ the unique positive square root.

Remark 3.2 (Choice of branch). The ABL conditioning rule specifies probabilities for further measurements but does not single out a canonical density operator. Def. 3.1 adopts the symmetric (Hermitian, positive) "sandwich" branch $\rho^{1/2}\Lambda_k\rho^{1/2}/p(k)$, which is the choice made in the finite-dimensional core (App. A, Def. A.25); alternative (inequivalent) branches such as $\rho\Lambda_k/p(k)$ give the same Born probabilities but differ off-diagonally as operators. All results of Part I (Thms. 3.7, 4.1, 6.1, 7.3 and Cor. 5.4) are stated and proved relative to the symmetric branch. Downstream claims that treat $\rho_2(\tau \mid k)$ as an expectation-computing operator (notably the Bianchi-consistency remark under Conj. 10.2) also assume this branch.

Remark 3.3 (ρ_2 as an established two-time state). The symmetric branch is not an ad hoc choice: $\rho_2(\tau \mid k) = \rho^{1/2}\Lambda_k\rho^{1/2}/p(k)$, with mixed forward datum ρ and POVM backward effect Λ_k, is the positive symmetric representative of the *two-time (pre/post-selected) density operator* for mixed pre-selection and POVM post-selection [6]. The same object appears, under different conventions, as the superdensity / spacetime-density operator of [372] and as the pseudo-density "state over time" of [7]; the tomography and Kraus-isomorphism results of those works justify treating $\rho_2(\tau \mid k)$ as a legitimate (operationally reconstructible) state. With the paper's normalization $p(k) = \mathrm{Tr}(\Lambda_k\rho)$, the symmetric branch reproduces the generalized ABL probabilities $\mathrm{Tr}(\Lambda_k\rho)/\sum_{k'}\mathrm{Tr}(\Lambda_{k'}\rho)$ of that formalism for intermediate measurements. The operational content of this two-time structure—precisely *when* the forward state and backward effect are non-trivially linked—is made sharp by Thm. 3.10: the t_i-leg marginal of the associated Fullwood–Parzygnat state over time is the per-branch Margenau–Hill operator $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$ (not ρ_2, which is always positive), and for projective terminal effects $M_k \succeq 0 \iff [\rho(\tau), \Lambda_k(\tau)] = 0$, so its negativity is a dimension-free witness of temporal non-classicality on branch k that vanishes only under prior averaging (the no-signalling identity, Thm. 6.1).

Remark 3.4 (Response to the Barandes ABL critique). Barandes [125] recently argued that the ABL rule's apparent time symmetry is a re-ordering of measurements rather than genuine retrocausal dynamics, and flagged category errors in treating ABL probabilities as single-system quantities. This critique applies directly to interpretive frameworks that read ABL probabilities as retrodictive claims about a single realization's history. The present paper's use of ABL conditioning is *operator-theoretic*, not interpretive: we use $\rho_2(\tau \mid k) = \rho^{1/2}\Lambda_k\rho^{1/2}/p(k)$ as *a* self-adjoint positive symmetric representative of the two-time (pre/post-selected) state whose terminal-branch weight reproduces the generalised ABL branch probability $p(k) = \mathrm{Tr}(\Lambda_k\rho)/\sum_{k'}\mathrm{Tr}(\Lambda_{k'}\rho)$ (Def. 3.1; cf. Rem. 3.3), the symmetric-branch choice being non-canonical (Rem. 3.2) and the intermediate-measurement statistics being carried by the Margenau–Hill operator M_k, not ρ_2

(Thm. 3.10); and build the probability space $(\Omega_\gamma, \mathbb{P}_\gamma)$ in Sec. 5 as a Kolmogorov-extension construction on records, *not* as a per-realization history ascription. The martingale-collapse and prior-average results (Thm. 7.3, Cor. 5.4) are statements about the resulting probability space, not about retrocausal single-system dynamics. In this operator-theoretic reading the Barandes critique does not apply: we are not claiming ABL gives retrocausal dynamics; we are using the symmetric branch as a Hermitian positive operator consistent with ABL probabilities, on which Part I's theorems proceed. In particular, the marginal $\mathbb{P}_\gamma(K = k) = p(k)$ fixed in Sec. 5 is an *ensemble* statement on the Kolmogorov space $(\Omega_\gamma, \mathbb{P}_\gamma)$, and the collapse of Thm. 7.3 is an a.s./L^1 convergence over that ensemble, not the ascription of a pre-determined ABL value to an individual realization: K enters only as an ensemble random variable, never as a single-run retrodiction. The framework is therefore compatible with Barandes's position that ABL does not encode genuine retrocausality at the single-realization level.

Remark 3.5 (No weak values are used). The present construction nowhere invokes *weak values* or weak measurements: all intermediate measurements in Sec. 5 are described by ordinary (strong) Kraus/POVM operations, and the central object is the symmetric-branch density operator of Def. 3.1 rather than a weak-value or decoherence-functional construction (cf. the corresponding statement in the Part II overview). The companion critique of weak values [126] is therefore orthogonal to this framework, which makes no anomalous-weak-value claims. There is nonetheless a point of experimental contact worth recording: the per-branch quasiprobability $q_k(a) = \mathrm{Tr}(M_k P_a)$ built from the Margenau–Hill operator $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$ of Thm. 3.10 is exactly the real Kirkwood–Dirac/Margenau–Hill object that direct-measurement experiments reconstruct under *laboratory* postselection (weak-value tomography of pre- and postselected ensembles). What those experiments access is the lab-postselected instance; the *terminal-POVM* instance of q_k — postselection on the cosmological boundary data $\{E_k\}$ — is precisely the component that is not directly accessible, and the framework's two-time statistics reduce to the experimentally reconstructed ones when the terminal effect is replaced by a laboratory projector.

Lemma 3.6 (Basic properties of ρ_2). *For all $\tau \in [\tau_i, \tau_f]$ and $k \in \mathcal{K}$, $\rho_2(\tau \mid k) \succeq 0$ and $\mathrm{Tr}\,\rho_2(\tau \mid k) = 1$.*

Proof. If $p(k) = 0$, then $\rho_2(\tau \mid k) = \mathbb{1}/d$, which is positive and has trace 1. If $p(k) > 0$, then

$$\rho_2(\tau \mid k) = \frac{X^* \Lambda_k(\tau) X}{p(k)}, \qquad X := \rho(\tau)^{1/2},$$

so positivity is immediate because $\Lambda_k(\tau) \succeq 0$. For the trace,

$$\mathrm{Tr}\,\rho_2(\tau \mid k) = \frac{\mathrm{Tr}(\rho(\tau)^{1/2}\Lambda_k(\tau)\rho(\tau)^{1/2})}{p(k)} = \frac{p(k)}{p(k)} = 1.$$

□

The next theorem is the technical linchpin of Part I. It is not stated in the finite-dimensional core of App. A but is the natural dynamical upgrade of Definition 3.1.

Theorem 3.7 (Unitary covariance of ρ_2). *For all $\tau, \tau' \in [\tau_i, \tau_f]$ and all $k \in \mathcal{K}$,*

$$\rho_2(\tau \mid k) \;=\; U(\tau, \tau')\, \rho_2(\tau' \mid k)\, U(\tau, \tau')^\dagger. \tag{4}$$

Proof. Write $U := U(\tau, \tau')$, which is unitary. We consider the two cases separately.

Case $p(k) > 0$. By Def. 2.4, $\rho(\tau) = U\rho(\tau')U^\dagger$. Since the (unique) positive square root commutes with unitary conjugation,

$$\rho(\tau)^{1/2} = U\, \rho(\tau')^{1/2}\, U^\dagger. \tag{5}$$

(*Proof:* $(U\rho(\tau')^{1/2}U^\dagger)^2 = U\rho(\tau')U^\dagger = \rho(\tau)$ and $U\rho(\tau')^{1/2}U^\dagger \succeq 0$; uniqueness of the positive square root gives (5).) Using the composition law $U(\tau_f,\tau') = U(\tau_f,\tau)U$,

$$\Lambda_k(\tau') = U(\tau_f,\tau')^\dagger E_k U(\tau_f,\tau') = U^\dagger U(\tau_f,\tau)^\dagger E_k U(\tau_f,\tau)U = U^\dagger \Lambda_k(\tau)U,$$

so $\Lambda_k(\tau) = U\Lambda_k(\tau')U^\dagger$. Combining with (5) and $U^\dagger U = \mathbb{1}$,

$$\rho(\tau)^{1/2}\Lambda_k(\tau)\rho(\tau)^{1/2} = U\big[\rho(\tau')^{1/2}\Lambda_k(\tau')\rho(\tau')^{1/2}\big]U^\dagger.$$

Dividing by $p(k)$ (independent of τ by Lem. 2.5) gives (4).

Case $p(k) = 0$. $\rho_2(\tau \mid k) = \rho_2(\tau' \mid k) = \mathbb{1}/d$. $U(\mathbb{1}/d)U^\dagger = \mathbb{1}/d$, so (4) holds trivially. □

Corollary 3.8 (Von Neumann equation for ρ_2)**.** *In the Schrödinger picture, $i\hbar\,\partial_\tau\rho_2(\tau \mid k) = [H(\tau), \rho_2(\tau \mid k)]$ for all $k \in \mathcal{K}$. The initial condition is $\rho_2(\tau_i \mid k) = \rho_{\mathrm{i}}^{1/2}\Lambda_k(\tau_i)\rho_{\mathrm{i}}^{1/2}/p(k)$ when $p(k) > 0$, and $\mathbb{1}/d$ when $p(k) = 0$.*

Proof. Differentiate (4) at $\tau' = \tau$ using $i\hbar\partial_\tau U(\tau,\tau')|_{\tau'=\tau} = H(\tau)$. □

Remark 3.9. Theorem 3.7 says that, *conditional on the terminal outcome $K = k$*, the state evolves by ordinary unitary quantum mechanics with the same Hamiltonian $H(\tau)$ as the forward state. The two-boundary information enters only through the initial datum $\rho_2(\tau_i \mid k)$ on Σ_{τ_i}. The nontriviality of that two-boundary link is quantified by Thm. 3.10: it is operationally invisible exactly when $[\rho(\tau), \Lambda_k(\tau)] = 0$ (for projective terminal effects), in which case the symmetric branch ρ_2 coincides with the Margenau–Hill operator M_k and carries no quasiprobability negativity.

Theorem 3.10 (Per-branch Margenau–Hill witness of the two-boundary link)**.** *Let* $\dim H = d < \infty$. *Fix a slice $\tau \in [\tau_i, \tau_f]$ and a branch k with $p := p(k) = \mathrm{Tr}(\rho\Lambda) > 0$, where $\rho := \rho(\tau) \succeq 0$,* $\mathrm{Tr}\,\rho = 1$, *and $\Lambda := \Lambda_k(\tau) = U(\tau_f,\tau)^\dagger E_k\, U(\tau_f,\tau)$, $0 \le E_k \le \mathbb{1}$. Put*

$M_k := \frac{1}{2p}\{\rho,\Lambda\}$ (*Margenau–Hill / symmetric Jordan; Hermitian,* $\mathrm{Tr}\, M_k = 1$), $\rho_2(\tau \mid k) = \frac{1}{p}\,\rho^{1/2}\Lambda\rho^{1/2}$ (D

*and $f_k := \|M_k\|_1 - 1 = 2\sum_{\lambda_j(M_k)<0}|\lambda_j| \ge 0$. Let $\varrho_k = \frac{1}{2}\{\rho\otimes\mathbb{1},\ \mathscr{J}[\mathscr{E}_k]\}$ be the Fullwood–Parzygnat state over time [397, 398] built from the p-normalised Lüders branch channel $\mathscr{E}_k(X) = \frac{1}{p}E_k^{1/2}UXU^\dagger E_k^{1/2}$, with $\mathscr{J}$ the partial-transpose-on-A (Jamiołkowski) Choi operator (*not *the bare Choi C, which is PSD and carries no witness). Then:*

(i) Marginals. *$\varrho_k^\dagger = \varrho_k$,* $\mathrm{Tr}\,\varrho_k = 1$, *and*

$$\mathrm{Tr}_{t_f}\,\varrho_k = M_k, \qquad \mathrm{Tr}_{t_i}\,\varrho_k = \tfrac{1}{p}E_k^{1/2}\rho(\tau_f)E_k^{1/2}.$$

In particular M_k is the single-time (t_i-leg) reduction of ϱ_k, and $\mathrm{Tr}_{t_f}\,\varrho_k = \rho(\tau_i)$ is false (the branch channel is trace-decreasing, $\mathscr{E}_k^*(\mathbb{1}) = \Lambda/p \ne \mathbb{1}$).

(ii) Algebra. $\rho_2 \succeq 0$, $\mathrm{Tr}\,\rho_2 = 1$, *and*

$$M_k - \rho_2 = \tfrac{1}{2p}\big[\rho^{1/2}, [\rho^{1/2}, \Lambda]\big], \qquad M_k = \rho_2 \iff [\rho,\Lambda] = 0.$$

For every Hermitian A, $\mathrm{Tr}(M_k A) = \frac{1}{p}\Re\,\mathrm{Tr}(\Lambda A\rho)$; the same expectation is shared by ρ_2 iff $[A,\rho] = 0$. Since $\rho_2 = (\rho^{1/2})^\dagger\Lambda\,\rho^{1/2} \succeq 0$ always, the sign is carried by M_k alone.

(iii) Spectral witness. *For* projective *terminal effects $E_k = \Pi_k$ (so Λ is a projector):*

$$M_k \succeq 0 \iff [\rho,\Lambda] = 0, \qquad \textit{i.e.} \qquad f_k > 0 \iff [\rho(\tau),\Lambda_k(\tau)] \ne 0,$$

for all d, all ranks and degeneracies, with no support hypothesis. For general POVM effects only the clean direction $[\rho,\Lambda] = 0 \Rightarrow M_k = \rho\Lambda/p = \rho_2 \succeq 0$ is unconditional; the converse is generic-only *and fails on a set of positive measure (e.g.* $\rho = \mathrm{diag}(\frac{9}{10}, \frac{1}{10})$*,* $\Lambda = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right)$ *has* $M_k \succ 0$ *yet* $[\rho, \Lambda] \neq 0$*). The condition is unitarily covariant (Thm. 3.7).*

(iv) Certification (modulo NIM). *For an intermediate sharp* $A = \sum_a a\, P_a$ *the symmetrised two-time quasiprobability* $q_k(a) = \frac{1}{2p}\,\mathrm{Tr}(\{\Lambda, P_a\}\rho) = \langle a|M_k|a\rangle$ *satisfies* $\min_A \min_a q_k(a) = \lambda_{\min}(M_k)$*, so* $f_k > 0 \iff \exists\, A, a$ *with* $q_k(a) < 0$*. Hence* $f_k > 0$ *implies that* no joint Kolmogorov law *reproduces the symmetrised two-time statistics* $\{q_k\}$*, i.e. the full Halliwell NSIT+augmented-Leggett–Garg (necessary-and-sufficient) macrorealism criterion fails on branch* k*.* Under the explicit additional hypothesis of non-invasive measurability (NIM/adroitness) *this certifies that no non-invasive macrorealist model reproduces the within-branch statistics;* absent *NIM only "no joint Kolmogorov / no commuting model" is certified. Averaging restores classicality:* $\sum_k p(k)\, M_k = \frac{1}{2}\{\rho, \sum_k \Lambda_k\} = \rho \succeq 0$*, the licensing no-signalling identity (Thm. 6.1), so the negativity of the prior-averaged operator vanishes and the witness is irreducibly per-branch (note* $\sum_k p(k) f_k > 0$ *in general).*

Proof. Write $R := \rho^{1/2}$, $V := U(\tau_f, \tau)$.

(i) Marginals. For any linear $\mathscr{E}$ and its HS-adjoint $\mathscr{E}^*$, with $\mathscr{J}[\mathscr{E}] = \sum_{ij} E_{ji} \otimes \mathscr{E}(E_{ij})$ (PT-on-A of the column Choi),

$$\mathrm{Tr}_A \left[(\rho \otimes \mathbb{1})\mathscr{J}\right] = \sum_{ij} \rho_{ij} \mathscr{E}(E_{ij}) = \mathscr{E}(\rho), \quad \mathrm{Tr}_B \left[(\rho \otimes \mathbb{1})\mathscr{J}\right] = \sum_{ij} \rho E_{ji}\, \mathrm{Tr}[E_{ij} \mathscr{E}^*(\mathbb{1})] = \rho\, \mathscr{E}^*(\mathbb{1});$$

symmetrising in ϱ_k gives $\mathrm{Tr}_A\, \varrho_k = \mathscr{E}_k(\rho)$ and $\mathrm{Tr}_B\, \varrho_k = \frac{1}{2}\{\rho, \mathscr{E}_k^*(\mathbb{1})\}$. For the Lüders channel $\mathscr{E}_k^*(\mathbb{1}) = \frac{1}{p} V^\dagger E_k V = \Lambda/p$ and $\mathscr{E}_k(\rho) = \frac{1}{p} E_k^{1/2} V \rho V^\dagger E_k^{1/2}$, whence $\mathrm{Tr}_{t_f}\, \varrho_k = \frac{1}{2p}\{\rho, \Lambda\} = M_k$ and $\mathrm{Tr}_{t_i}\, \varrho_k = \frac{1}{p} E_k^{1/2} \rho(\tau_f) E_k^{1/2}$. Hermiticity is FP; $\mathrm{Tr}\, \varrho_k = \mathrm{Tr}\, M_k = \frac{1}{p}\mathrm{Tr}(\rho\Lambda) = 1$. Recovery $\mathrm{Tr}_B\, \varrho_k = \rho$ would need $\mathscr{E}_k^*(\mathbb{1}) = \mathbb{1}$, i.e. $\Lambda = \mathbb{1}$.

(ii) Algebra. $\rho_2 = \frac{1}{p} R^\dagger \Lambda R \succeq 0$ as $\Lambda \succeq 0$, and $\mathrm{Tr}\, \rho_2 = 1$. Expanding $[R, [R, \Lambda]] = R^2\Lambda - 2R\Lambda R + \Lambda R^2 = \{\rho, \Lambda\} - 2R\Lambda R = 2p(M_k - \rho_2)$. If $[\rho, \Lambda] = 0$ then $[R, \Lambda] = 0$ and $M_k = \rho_2 = \rho\Lambda/p$; conversely $M_k = \rho_2 \Rightarrow [R, [R, \Lambda]] = 0$, and pairing with Λ gives $0 = \mathrm{Tr}\big(\Lambda[R, [R, \Lambda]]\big) = 2\,\mathrm{Tr}(\Lambda^2 R^2) - 2\,\mathrm{Tr}(\Lambda R \Lambda R) = \|[R, \Lambda]\|_2^2$ (expand and use cyclicity; note that pairing with $K := [R, \Lambda]$ instead would give the identically vanishing $\mathrm{Tr}(K^\dagger[R, K])$ and no information), so $[R, \Lambda] = 0$ and $[\rho, \Lambda] = [R^2, \Lambda] = 0$. For all Hermitian A, $\mathrm{Tr}(M_k A) = \frac{1}{2p}(\mathrm{Tr}(\Lambda A \rho) + \overline{\mathrm{Tr}(\Lambda A \rho)}) = \frac{1}{p}\Re\,\mathrm{Tr}(\Lambda A \rho)$; for ρ_2, $\mathrm{Tr}(\rho_2 A) = \frac{1}{p}\mathrm{Tr}(\Lambda R A R)$ equals this iff $[A, R] = 0 \iff [A, \rho] = 0$.

(iii) Spectral witness. ($\Leftarrow$, all d, any effect) co-diagonalise: $M_k = \sum_i \frac{a_i \ell_i}{p} |i\rangle\langle i| \succeq 0$. ($\Rightarrow$, projective) In a basis with $\Lambda = \Pi = \mathrm{diag}(\mathbb{1}_r, 0)$ on $\mathrm{Ran}\,\Lambda \oplus \ker\Lambda$, write $\rho = \left(\begin{smallmatrix} A & B \\ B^\dagger & D \end{smallmatrix}\right)$; then $[\rho, \Pi] = \left(\begin{smallmatrix} 0 & B \\ -B^\dagger & 0 \end{smallmatrix}\right)$ and $2pM_k = \rho\Pi + \Pi\rho = \left(\begin{smallmatrix} 2A & B \\ B^\dagger & 0 \end{smallmatrix}\right)$, a Hermitian block with vanishing $(2,2)$-corner. *Zero-block lemma:* if $H = \left(\begin{smallmatrix} A & X \\ X^\dagger & 0 \end{smallmatrix}\right) \succeq 0$ then for any w in the second block $\langle (0, w)|H|(0, w)\rangle = 0$, so $(0, w)$ minimises the nonnegative form $v \mapsto \langle v|H|v\rangle$ and lies in $\ker H$, forcing $Xw = 0$ for all w, i.e. $X = 0$. Applied here, $M_k \succeq 0 \Rightarrow B = 0 \iff [\rho, \Lambda] = 0$. This is dimension/rank/degeneracy-free. The displayed $d = 2$ POVM datum gives $\det(2pM_k) > 0$ with positive diagonal, hence $M_k \succ 0$ while $[\rho, \Lambda] \neq 0$, refuting any unconditional POVM converse.

(iv) Certification. $q_k(a) = \frac{1}{2p}\mathrm{Tr}(\{\Lambda, P_a\}\rho) = \langle a|M_k|a\rangle$ for rank-one $P_a = |a\rangle\langle a|$; by Rayleigh–Ritz and Schur–Horn $\min_A \min_a q_k(a) = \lambda_{\min}(M_k)$ (higher-rank P_a only raise the value when $\lambda_{\min} < 0$). Thus $f_k > 0 \iff \lambda_{\min}(M_k) < 0 \iff \exists\, A, a$ with $q_k(a) < 0$. By Fine's theorem in Halliwell's form [395, 396], existence of a nonnegative joint law for the symmetrised two-time statistics is equivalent to $q_k \geq 0$ for every intermediate context, equivalently to the conjunction of the full NSIT set and the augmented LG inequalities, equivalently to macrorealism for the dichotomic variable; a context isolating $|a\rangle$ inherits the negative cell, so the criterion fails. The macrorealist-exclusion clause requires the NIM/adroitness hypothesis [401]: the paper's intermediate operations are strong Kraus/POVM maps and $p(k)$ is intervention-dependent (Thm. 6.1, Rem. 6.3), so the per-branch process is invasive by construction; absent

NIM only the joint-Kolmogorov exclusion holds. Finally $\sum_k \Lambda_k = V^\dagger(\sum_k E_k)V = \mathbb{1}$ gives $\sum_k p(k)M_k = \frac{1}{2}\{\rho, \mathbb{1}\} = \rho \succeq 0$. $\square$

4 Conservation of conditioned expectation values

Let $\widehat{O} : \mathcal{M}_{[\tau_i,\tau_f]} \to \mathcal{L}_{\mathrm{sa}}(\mathcal{H})$ be a smooth Heisenberg-picture operator-valued field, meaning that for every $x \in \mathcal{M}_{[\tau_i,\tau_f]}$, $\widehat{O}(x) \in \mathcal{L}_{\mathrm{sa}}(\mathcal{H})$ and for every $\tau \in [\tau_i, \tau_f]$, the restriction $\widehat{O}|_{\Sigma_\tau}$ is related to its value at τ_i by Heisenberg evolution:

$$\widehat{O}(x) = U(\tau(x), \tau_i)^\dagger\, \widehat{O}_{\mathrm{S}}(x)\, U(\tau(x), \tau_i), \tag{6}$$

where $\widehat{O}_{\mathrm{S}}(x)$ is the corresponding Schrödinger-picture operator on $\Sigma_{\tau(x)}$.

Theorem 4.1 (Outcome-wise conservation of conditioned expectations)**.** *Let $\widehat{O}(x)$ be a Heisenberg observable satisfying a local conservation law*

$$\nabla^\mu \widehat{O}_{\mu\nu\cdots}(x) = 0 \qquad \textit{in } \mathcal{L}_{\mathrm{sa}}(\mathcal{H}), \tag{7}$$

(for the case of a tensor observable; the scalar case is the statement $\nabla^\mu V_\mu(x) = 0$). Then for every $k \in \mathcal{K}$ with $p(k) > 0$,

$$\nabla^\mu \operatorname{Tr}\big(\rho_2(\tau(x) \mid k)\, \widehat{O}_{\mathrm{S},\mu\nu\cdots}(x)\big) = 0 \qquad \textit{on } \mathcal{M}_{[\tau_i,\tau_f]}. \tag{8}$$

In particular, on every realization $\omega \in \Omega_\gamma$ with $K(\omega) = k$, the conservation law (8) *holds with that value of k. We emphasize that the left-hand side of* (8) *depends on ω only through $K(\omega)$; no stronger record-wise statement is claimed, and indeed the analogous statement for $\rho^\gamma_{\mathrm{RC}}$ fails (cf. Rem. 10.4). We stress that* $\operatorname{Tr}(\rho_2(\tau(x) \mid k)\, \widehat{O}_{\mathrm{S}}(x))$ *is an* ensemble *expectation over the post-selected sub-ensemble $\{K = k\}$, not a quantity ascribed to an individual realization's history; "on every realization ω with $K(\omega) = k$" abbreviates "for the sub-ensemble conditioned on $K = k$" and carries no single-system ontological commitment. This is precisely the reading under which the Ensemble-Fallacy objection of Barandes [125] (Rem. 3.4) does not apply.*

Proof. By Thm. 3.7 and (6),

$$\operatorname{Tr}\big(\rho_2(\tau(x) \mid k)\, \widehat{O}_{\mathrm{S},\mu\nu\cdots}(x)\big) = \operatorname{Tr}\big(U(\tau(x), \tau_i)\rho_2(\tau_i \mid k)U(\tau(x), \tau_i)^\dagger \cdot \widehat{O}_{\mathrm{S},\mu\nu\cdots}(x)\big) = \operatorname{Tr}\big(\rho_2(\tau_i \mid k)\, \widehat{O}_{\mu\nu\cdots}(x)\big)$$

by cyclicity and (6). The state $\rho_2(\tau_i \mid k)$ is independent of x, so ∇^μ passes through the trace:

$$\nabla^\mu \operatorname{Tr}(\rho_2(\tau_i \mid k)\, \widehat{O}_{\mu\nu\cdots}(x)) = \operatorname{Tr}(\rho_2(\tau_i \mid k)\, \nabla^\mu \widehat{O}_{\mu\nu\cdots}(x)) = 0$$

by (7). $\square$

Remark 4.2 (Scope and interpretation). Theorem 4.1 is a statement about a fixed background g. If g is itself dynamical and is coupled to $\operatorname{Tr}(\rho_2(\tau \mid k)\widehat{O})$ via an Einstein-type equation, the Heisenberg propagator U, hence $\rho_2(\tau_i \mid k)$, depends on g; the conservation (8) then becomes a consistency condition that the coupled system must satisfy. This is the content of Conjecture 10.2 in Part II.

5 Filtration and posterior martingale

We construct a filtration on a chosen timelike worldline in $\mathcal{M}$ and derive the posterior process directly from that construction. The choice of worldline is explicit: different worldlines carry different filtrations.

5.1 Filtration along a worldline

Fix an inextendible, future-directed, timelike curve $\gamma : [\tau_i, \tau_f] \to \mathcal{M}$ with $\tau(\gamma(\tau)) = \tau$ (every Cauchy slice is crossed exactly once, by global hyperbolicity).

Fixed measurement protocol on γ. Fix, once and for all, a strictly increasing sequence $S_\gamma := (s_m)_{m\in\mathbb{N}} \subset \mathbb{Q} \cap [\tau_i, \tau_f)$ with $s_m \uparrow \tau_f$, and a *measurement protocol* $\mathcal{P}_\gamma = \big(\{M_j^{(m)}\}_{j\in J_m}\big)_{m\in\mathbb{N}}$ consisting of, for each m, a finite index set J_m and Kraus operators $M_j^{(m)} \in \mathcal{A}(\gamma(s_m))$ with $\sum_{j\in J_m} M_j^{(m)*} M_j^{(m)} = \mathbb{1}$. A *record* up to parameter τ is then the finite sequence $(j_1, \ldots, j_{n(\tau)})$ of outcomes at the times $s_m \le \tau$, where $n(\tau) := \max\{m : s_m \le \tau\}$ (set $n(\tau) := 0$ if $s_1 > \tau$). The POVM choices are encoded in $\mathcal{P}_\gamma$ and are not random. The restriction to a single countable *temporally ordered* sequence ensures that cylinder probabilities are defined by ordinary sequential composition of Kraus operators.

Passive (two-boundary) construction of $\mathbb{P}_\gamma$. We define the joint law of $(K, \text{records})$ by

$$\mathbb{P}_\gamma\big(K = k, \text{record} = (j_1, \ldots, j_n)\big) := p(k) \cdot \mathrm{Tr}\Big(\widetilde{M}_{j_n}^{(n)} \cdots \widetilde{M}_{j_1}^{(1)}\, \rho_2(\tau_i \mid k)\, \widetilde{M}_{j_1}^{(1)*} \cdots \widetilde{M}_{j_n}^{(n)*}\Big), \tag{9}$$

where each Kraus operator is transported to Σ_{τ_i} by the Heisenberg unitary of Thm. 3.7, $\widetilde{M}_j^{(m)} := U(s_m, \tau_i)^\dagger M_j^{(m)} U(s_m, \tau_i)$. The adjoint factors in (9) are therefore $\widetilde{M}_{j_1}^{(1)*}, \ldots, \widetilde{M}_{j_n}^{(n)*}$, and we use this Heisenberg-picture form throughout.

Kolmogorov consistency. Fix $k \in \mathcal{K}$ and a prefix $(j_1, \ldots, j_{n-1})$. Since $\sum_{j\in J_n} \widetilde{M}_j^{(n)*} \widetilde{M}_j^{(n)} = \mathbb{1}$, cyclicity of trace gives

$$\sum_{j_n\in J_n} \mathbb{P}_\gamma(K = k, (j_1, \ldots, j_n)) = p(k)\ \mathrm{Tr}\big(\widetilde{M}_{j_{n-1}}^{(n-1)} \cdots \widetilde{M}_{j_1}^{(1)}\, \rho_2(\tau_i \mid k)\, \widetilde{M}_{j_1}^{(1)*} \cdots \widetilde{M}_{j_{n-1}}^{(n-1)*}\big),$$

i.e. consistency under summing the latest outcome. Because the protocol S_γ is a single temporally ordered sequence, the cylinders form a refining chain and this latest-outcome consistency is the only condition required. Summing over k at $n = 0$ yields $\sum_k p(k) = 1$. The marginal $\mathbb{P}_\gamma(K = k) = p(k)$ holds *by construction* (not as a consequence of sequential Lüders reduction); this is the passive two-boundary probability space used throughout Part I.

Remark 5.1 (Why S_γ must be a sequence, not dense in $\mathbb{Q}$). If the protocol were instead indexed by a dense subset of $\mathbb{Q}\cap[\tau_i, \tau_f)$, Kolmogorov consistency would also demand that summing the cylinder probability at $\{s_1, \ldots, s_n\}$ over an *intermediate* extra outcome at some $s' \in (s_{m-1}, s_m)$ equal the original cylinder probability. That sum inserts a quantum channel $\Phi'(X) = \sum_{j'} \widetilde{M}'_{j'} X \widetilde{M}'^{*}_{j'}$ between $\widetilde{M}^{(m-1)}$ and $\widetilde{M}^{(m)}$, which generically changes the state, so consistency fails. Restricting to the ordered sequence S_γ eliminates intermediate indices and restores consistency.

By Kolmogorov's extension theorem applied to the $\mathbb{N}$-indexed cylindrical family (9), there is a unique probability measure $\mathbb{P}_\gamma$ on $\Omega_\gamma := \mathcal{K} \times \prod_{m\in\mathbb{N}} J_m$, equipped with the product of discrete σ-algebras (each J_m is finite and $\mathcal{K}$ is finite). We complete $(\Omega_\gamma, \mathbb{P}_\gamma)$; let $\mathcal{F}_\tau^\gamma$ be the sub-σ-algebra generated by records up to τ (i.e. by $(j_1, \ldots, j_{n(\tau)})$), and $\overline{\mathcal{F}}_\tau^\gamma$ its $\mathbb{P}_\gamma$-completion. By construction, $(\overline{\mathcal{F}}_\tau^\gamma)_{\tau\in[\tau_i,\tau_f)}$ is an increasing filtration.

Operational interpretation and relation to the active law. (9) is a retrodictive joint law: it is the distribution of $(K, \text{records})$ consistent with weighting each realization by $p(k)$ and propagating the conditioned state $\rho_2(\tau \mid k)$ under the unitary dynamics of Thm. 3.7. It is not produced by real-time sequential Lüders reduction; rather, it is the measure on Ω_γ under which the conditional dynamics is unitary and $\mathbb{P}_\gamma(K = k) = p(k)$ by construction. This is distinct from the "active" sequential-Born law used in Sec. 6, under which joint probabilities $P_M(j, l, k)$ are defined by nested Kraus actions on ρ_i and whose marginal over K need *not* equal $p(k)$ (cf. Remark after Thm. 6.1). The two laws live on the same outcome space Ω_γ but assign different probabilities in general; the martingale/collapse results of this section and of Sec. 7 are statements about $(\Omega_\gamma, \mathbb{P}_\gamma)$ in the passive sense (9), while Thm. 6.1 is a statement about the active law. We denote the active law by P_M (with subscript encoding the intervention) and reserve $\mathbb{P}_\gamma$ exclusively for the passive law.

5.2 Posterior and retrocausal state

Definition 5.2 (Posterior and retrocausal state)**.** For $\tau \in \mathbb{Q} \cap [\tau_i, \tau_f)$,

$$\pi_\tau^\gamma(k) := \mathbb{E}_{\mathbb{P}_\gamma}\big[\mathbf{1}_{\{K=k\}} \,\big|\, \overline{\mathcal{F}}_\tau^\gamma\big], \qquad \rho_{\mathrm{RC}}^\gamma(\tau) := \sum_{k \in \mathcal{K}} \pi_\tau^\gamma(k)\, \rho_2(\tau \mid k).$$

Lemma 5.3 (Properties of π_τ^γ and $\rho_{\mathrm{RC}}^\gamma$)**.** *For all rational $\tau < \tau_f$:*

(i) $0 \le \pi_\tau^\gamma(k) \le 1$, $\sum_k \pi_\tau^\gamma(k) = 1$ *a.s.* $\mathbb{P}_\gamma$*, and for rational* $s \le t < \tau_f$, $\mathbb{E}_{\mathbb{P}_\gamma}[\pi_t^\gamma(k) \mid \overline{\mathcal{F}}_s^\gamma] = \pi_s^\gamma(k)$ *a.s.*

(ii) $\rho_{\mathrm{RC}}^\gamma(\tau) \succeq 0$, $\operatorname{Tr} \rho_{\mathrm{RC}}^\gamma(\tau) = 1$ *a.s.*

(iii) $\mathbb{E}_{\mathbb{P}_\gamma}[\rho_{\mathrm{RC}}^\gamma(\tau)] = \rho(\tau)$.

Proof. For (i), conditional expectation preserves positivity and order, so $0 \le \pi_\tau^\gamma(k) \le 1$ a.s. Because conditional expectation is linear,

$$\sum_k \pi_\tau^\gamma(k) = \mathbb{E}\Big[\sum_k \mathbf{1}_{\{K=k\}} \;\Big|\; \overline{\mathcal{F}}_\tau^\gamma\Big] = 1$$

a.s. The martingale identity follows from the tower property:

$$\mathbb{E}\big[\pi_t^\gamma(k) \mid \overline{\mathcal{F}}_s^\gamma\big] = \mathbb{E}\big[\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_t^\gamma] \mid \overline{\mathcal{F}}_s^\gamma\big] = \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_s^\gamma] = \pi_s^\gamma(k).$$

For (ii), each coefficient $\pi_\tau^\gamma(k)$ is nonnegative and sums to 1, while each $\rho_2(\tau \mid k)$ is positive with trace 1 by Lemma 3.6. Hence $\rho_{\mathrm{RC}}^\gamma(\tau)$ is a convex combination of density operators, so it is positive semidefinite with unit trace. For (iii),

$$\mathbb{E}[\pi_\tau^\gamma(k)] = \mathbb{P}_\gamma(K = k) = p(k)$$

by the tower property, and therefore

$$\mathbb{E}[\rho_{\mathrm{RC}}^\gamma(\tau)] = \sum_k p(k)\rho_2(\tau \mid k) = \sum_k \rho(\tau)^{1/2}\Lambda_k(\tau)\rho(\tau)^{1/2} = \rho(\tau),$$

since $\sum_k \Lambda_k(\tau) = \mathbb{1}$. This prior-average is the matter-side no-signalling identity: the same marginalization $\sum_k p(k)\,(\cdot) = \rho(\tau)$ that returns the unconditional state here also sends the per-branch Margenau–Hill operator to $\sum_k p(k)\, M_k = \frac{1}{2}\{\rho, \sum_k \Lambda_k\} = \rho \succeq 0$, so the per-branch temporal non-classicality certified in Thm. 3.10 (negativity of M_k exactly when $[\rho(\tau), \Lambda_k(\tau)] \neq 0$) is washed out under the prior average; the positive representative $\rho_2(\tau \mid k)$ used here is the companion of that Margenau–Hill object. □

Corollary 5.4 (Reduction to standard QM in prior-average)**.** *For any Heisenberg observable* $\widehat{O}(x)$, $\mathbb{E}_{\mathbb{P}_\gamma}[\operatorname{Tr}(\rho_{\mathrm{RC}}^\gamma(\tau)\widehat{O}_{\mathrm{S}}(x))] = \operatorname{Tr}(\rho(\tau)\widehat{O}_{\mathrm{S}}(x))$*, which is the forward-evolution expectation of ordinary quantum mechanics with initial state* ρ_{i}.

Proof. Linearity of trace and expectation gives

$$\mathbb{E}\big[\operatorname{Tr}(\rho_{\mathrm{RC}}^\gamma(\tau)\widehat{O}_{\mathrm{S}}(x))\big] = \operatorname{Tr}\!\big(\mathbb{E}[\rho_{\mathrm{RC}}^\gamma(\tau)]\, \widehat{O}_{\mathrm{S}}(x)\big).$$

Apply Lem. 5.3(iii) to substitute $\mathbb{E}[\rho_{\mathrm{RC}}^\gamma(\tau)] = \rho(\tau)$. □

6 Bulk no-signalling

The following theorem is the correct no-signalling statement in the two-boundary setting. It says that *after marginalization over the terminal outcome*, no information can be sent through spacelike separation. It does *not* say that bulk interventions leave $p(k)$ invariant — indeed, $p(k)$ can change under intermediate operations, consistent with the retrocausal nature of the conditioning. The same marginalization that restores no-signalling here is the one that restores classicality of the two-time link: the per-branch Margenau–Hill operator M_k of Thm. 3.10 can be negative (temporal non-classicality / quasiprobability negativity on branch k), yet its prior-average $\sum_k p(k)\, M_k = \rho \succeq 0$ is exactly the no-signalling identity, so the negativity is irreducibly per-branch and vanishes under the marginalization performed below.

Theorem 6.1 (Bulk no-signalling)**.** *For $s \in \{1,2\}$, let $\tau_s \in (\tau_i, \tau_f)$ and let $O_s \subset \Sigma_{\tau_s}$ be relatively open in Σ_{τ_s} with $\overline{O_s}$ compact in Σ_{τ_s}; assume $\overline{O_1}$ and $\overline{O_2}$ are spacelike-separated as subsets of $\mathcal{M}$ (equivalently, $\overline{O_2} \cap J(\overline{O_1}) = \emptyset$). Let $\{M_j^{(1)}\}_{j\in J}$ be Kraus operators of a POVM with $M_j^{(1)} \in \mathcal{A}(O_1)$ and $\sum_j M_j^{(1)*} M_j^{(1)} = \mathbb{1}$, and let $\{F_l\}_{l\in L}$ be a POVM with $F_l \in \mathcal{A}(O_2)$. Here $\mathcal{A}(O_s)$ is defined by the subset convention of Ax. 2.2.*

For each choice of $\{M_j^{(1)}\}$, denote by $P_M(j,l)$ the joint probability of the bulk outcomes (j,l) under the active *sequential-Born law obtained by inserting the Kraus operators in chronological order determined by the time function τ and then marginalizing over the terminal outcome K. Explicitly $P_M(j,l) := \sum_k P_M(j,l,k)$, where $P_M(j,l,k)$ is the joint probability that the bulk POVMs in O_1, O_2 return (j,l) and a forward measurement of the terminal POVM $\{E_k\}$ returns k; since $\sum_k E_k = \mathbb{1}$ and each POVM is trace-preserving, $\sum_{j,l,k} P_M(j,l,k) = 1$, so P_M is a genuine probability distribution. It is the statistics seen by an experimenter who actually performs the terminal measurement and bins by k (cf. the active/passive discussion around* (9)*). (The explicit formulas for the two temporal orderings $\tau_1 \le \tau_2$ and $\tau_2 < \tau_1$ are written out in the proof. This active law is distinct from the passive law* (9) *of Sec. 5, which fixes $\mathbb{P}_\gamma(K = k) = p(k)$ by construction.) Then*

$$\sum_{j\in J} P_M(j,l) \;=\; P_\varnothing(l) \qquad \forall l \in L, \tag{10}$$

where $P_\varnothing(l)$ is the marginal probability of the O_2 outcome when no intervention is performed in O_1. In particular, the distribution of l is independent of the choice of POVM in O_1.

Proof. Pick any Kraus decomposition $F_l = \sum_\mu N_{l,\mu}^\dagger N_{l,\mu}$.

Case 1: $\tau_1 \le \tau_2$. Write $U_{21} := U(\tau_2,\tau_1)$, $U_{f2} := U(\tau_f,\tau_2)$, $U_{1i} := U(\tau_1,\tau_i)$. The active sequential-Born joint probability of (j,l,k) is

$$P_M(j,l,k) = \sum_\mu \mathrm{Tr}\big[E_k\, U_{f2} N_{l,\mu} U_{21} M_j^{(1)} U_{1i} \rho_{\mathrm{i}} U_{1i}^\dagger M_j^{(1)*} U_{21}^\dagger N_{l,\mu}^\dagger U_{f2}^\dagger\big].$$

Summing over k and using $\sum_k E_k = \mathbb{1}$ gives

$$P_M(j,l) := \sum_k P_M(j,l,k) = \mathrm{Tr}\big[F_l\, \sigma_j\big], \qquad \sigma_j := U_{21} M_j^{(1)} U_{1i} \rho_{\mathrm{i}} U_{1i}^\dagger M_j^{(1)*} U_{21}^\dagger,$$

independent of the Kraus decomposition of F_l.

Define $F_l^{(1)} := U_{21}^\dagger F_l\, U_{21}$. Since $U(\tau_1,\tau_2) = U(\tau_2,\tau_1)^\dagger = U_{21}^\dagger$, apply (3) of Ax. 2.2 with parameters $\tau = \tau_2$, $\tau' = \tau_1$, $R = O_2 \subset \Sigma_{\tau_2}$:

$$U(\tau_1,\tau_2)\, \mathcal{A}(O_2)\, U(\tau_1,\tau_2)^\dagger = U_{21}^\dagger\, \mathcal{A}(O_2)\, U_{21} \;\subset\; \mathcal{A}\big(J(O_2) \cap \Sigma_{\tau_1}\big).$$

Hence $F_l^{(1)} \in \mathcal{A}(J(O_2) \cap \Sigma_{\tau_1})$. Let $K_2 := J(\overline{O_2}) \cap \Sigma_{\tau_1}$. In a globally hyperbolic spacetime, $J^\pm$ of a compact set is closed [9], so K_2 is closed in Σ_{τ_1}. Spacelike separation of $\overline{O_1}, \overline{O_2}$ implies

$\overline{O_1} \cap K_2 = \emptyset$. The Cauchy slice Σ_{τ_1} is Hausdorff and $\overline{O_1}$ is compact, K_2 is closed, and they are disjoint, so by normality there exist disjoint open sets $V \supset \overline{O_1}$ and $W \supset K_2$ in Σ_{τ_1} with $\overline{V} \cap \overline{W} = \emptyset$. All points of Σ_{τ_1} are pairwise spacelike-separated in $\mathcal{M}$, so V and W are spacelike-separated open subsets of $\mathcal{M}_{[\tau_i,\tau_f]}$.

Because $J(O_2) \cap \Sigma_{\tau_1} \subset K_2 \subset W$ and $\mathcal{A}(S) := \bigcap_{U \supset S \text{ open}} \mathcal{A}(U)$, the algebra $\mathcal{A}(J(O_2) \cap \Sigma_{\tau_1})$ is contained in $\mathcal{A}(W)$ since W is one of the open sets appearing in the defining intersection. Thus $F_l^{(1)} \in \mathcal{A}(W)$. Also $M_j^{(1)} \in \mathcal{A}(O_1) \subset \mathcal{A}(V)$ by (M1). By (M2), $[F_l^{(1)}, M_j^{(1)}] = 0$ for every j.

Writing $\rho^{(1)} := U_{1i}\rho_{\mathrm{i}}U_{1i}^\dagger$ and using cyclicity with $[F_l^{(1)}, M_j^{(1)}] = 0$,

$$P_M(j,l) = \mathrm{Tr}\big[F_l^{(1)} M_j^{(1)} \rho^{(1)} M_j^{(1)*}\big] = \mathrm{Tr}\big[F_l^{(1)} M_j^{(1)*} M_j^{(1)} \rho^{(1)}\big].$$

Summing over j with $\sum_j M_j^{(1)*} M_j^{(1)} = \mathbb{1}$,

$$\sum_j P_M(j,l) = \mathrm{Tr}\big(F_l^{(1)} \rho^{(1)}\big) = \mathrm{Tr}\big(F_l\, U_{21} \rho^{(1)} U_{21}^\dagger\big) = \mathrm{Tr}\big(F_l\, \rho(\tau_2)\big) = P_\varnothing(l).$$

Case 2: $\tau_2 < \tau_1$. Write $U_{12} := U(\tau_1,\tau_2)$, $U_{f1} := U(\tau_f,\tau_1)$, $U_{2i} := U(\tau_2,\tau_i)$, and $\rho^{(2)} := U_{2i}\rho_{\mathrm{i}}U_{2i}^\dagger = \rho(\tau_2)$. The active sequential-Born joint probability of (j,l,k) is now obtained by chronological insertion of the Kraus operators in the order O_2 then O_1:

$$P_M(j,l,k) = \sum_\mu \mathrm{Tr}\big[E_k\, U_{f1} M_j^{(1)} U_{12} N_{l,\mu} U_{2i} \rho_{\mathrm{i}} U_{2i}^\dagger N_{l,\mu}^\dagger U_{12}^\dagger M_j^{(1)*} U_{f1}^\dagger\big].$$

Summing over k and using $\sum_k E_k = \mathbb{1}$ gives

$$P_M(j,l) = \sum_\mu \mathrm{Tr}\big[M_j^{(1)} U_{12} N_{l,\mu} \rho^{(2)} N_{l,\mu}^\dagger U_{12}^\dagger M_j^{(1)*}\big].$$

Now sum over j first and use cyclicity:

$$\sum_j P_M(j,l) = \sum_\mu \mathrm{Tr}\big[(\sum_j M_j^{(1)*} M_j^{(1)}) U_{12} N_{l,\mu} \rho^{(2)} N_{l,\mu}^\dagger U_{12}^\dagger\big] = \sum_\mu \mathrm{Tr}\big[U_{12} N_{l,\mu} \rho^{(2)} N_{l,\mu}^\dagger U_{12}^\dagger\big].$$

Unitarity of U_{12} and cyclicity then give

$$\sum_j P_M(j,l) = \sum_\mu \mathrm{Tr}\big[N_{l,\mu} \rho^{(2)} N_{l,\mu}^\dagger\big] = \mathrm{Tr}\big[(\sum_\mu N_{l,\mu}^\dagger N_{l,\mu})\rho^{(2)}\big] = \mathrm{Tr}(F_l \rho(\tau_2)) = P_\varnothing(l).$$

This proves (10) in both temporal orderings.

Note the two cases are structurally asymmetric: Case 1 (intervention O_1 to the past of O_2) is the only one that invokes causal propagation (M3b) and microcausality (M2); Case 2 (intervention to the future of O_2) uses neither, since summing over the *later* outcome j via $\sum_j M_j^{(1)*} M_j^{(1)} = \mathbb{1}$ removes the intervention by trace-preservation alone. Thus (M3) is load-bearing only for past-to-future no-signalling, consistent with its status as the principal locality postulate (Rem. 2.3). □

Remark 6.2 (Why $p(k)$ is *not* invariant). An analogous argument applied to $p(k) = \mathrm{Tr}(E_k U(\tau_f,\tau_i)\rho_{\mathrm{i}} U(\tau_f,\tau_i)^\dagger)$ would require $[U(\tau_f,\tau_1)^\dagger E_k U(\tau_f,\tau_1), M_j^{(1)}] = 0$. But $U(\tau_f,\tau_1)^\dagger E_k U(\tau_f,\tau_1) = \Lambda_k(\tau_1)$ is supported on Σ_{τ_1} in general, *not* spacelike- separated from the intervention region O_1. Consequently $\sum_j P_M(j,k) \neq p(k)$ in general. This is a genuine feature of the two-boundary framework and is not in conflict with Thm. 6.1: the distribution of the terminal outcome K is fixed by the boundary data $(\rho_{\mathrm{i}}, \{E_k\}, U(\tau_f,\tau_i))$, and if one changes the bulk dynamics (via an intervention), the propagator itself changes; $p(k)$ is defined with respect to a fixed dynamics. In the operational language of causal/localizable quantum operations [86], the change in $p(k)$ is a *causal influence* on a non-local quantity ($\Lambda_k(\tau_1)$ is not spacelike to O_1), *not* a signal: no marginal of an observable spacelike-separated from the intervention is altered (Thm. 6.1). The framework is thus retrocausal-yet-no-signalling in the precise sense that causal influence on the terminal record coexists with operational no-signalling.

Remark 6.3 (Relation to operational no-backwards-in-time signalling). Thm. 6.1 is the operator-algebraic, manifestly covariant counterpart of the operational no-backwards-in-time-signalling (NBTS) conditions of Guryanova, Silva, Short, Skrzypczyk, Brunner and Popescu [85]. In their model-independent two-time framework, NBTS for spacelike/parallel experiments is the marginal condition $p(a \mid x, y) = p(a)$; they prove that a two-time state avoids backward signalling for all intermediate operations *iff* the post-selection success probability is independent of the intermediate measurement (a "linear" two-time state). The present framework is deliberately *non-linear* in their sense — $p(k)$ depends on bulk interventions, cf. the preceding remark — so backward influence is permitted *within* the post-selected sub-ensemble, while Eq. (10) is exactly their marginal NBTS condition recovered after summing over the terminal outcome K. The role of the time-function τ and microcausality (M2)–(M3) is to render this marginal statement covariant and local, which the purely operational formulation leaves implicit.

Remark 6.4 (Information-theoretic shadow: Ji–Lloyd–Wilde retrocausal channel capacity). Ji–Lloyd–Wilde [369] prove a one-shot and asymptotic capacity theorem for a postselected closed-timelike-curve (P-CTC) channel in the Lloyd–Maccone framework [370]: the asymptotic retrocausal quantum and classical capacities equal, respectively, the average and the sum of the channel's max-information and regularised Doeblin information, achieved by amplified probabilistic teleportation on postselected favourable measurement outcomes. This is the *information-theoretic shadow* of the two-boundary machinery of Sec. 5: a P-CTC is mathematically a post-selection on a future boundary state, and the Ji–Lloyd–Wilde capacity quantifies information transfer under exactly the ABL conditioning structure of the present paper, specialised to the P-CTC sub-case. The result is *fully consistent with* Thm. 6.1: their nonzero retrocausal capacity is achieved only on postselected runs and vanishes operationally without a physical CTC. The capacity is a modular/postselected invariant of the channel, not a fine-tunable free parameter, supporting the Wood–Spekkens evasion of Thm. P.1. The recurrent journalistic overstatement of such results as "sending messages backward in time" is the version Thm. 6.1 forbids; the Ji–Lloyd–Wilde-style retrocausal influence *within a postselected sub-ensemble* is the version the framework predicts and which is, in fact, the operational content of the two-boundary structure.

Remark 6.5 (Relation to operational signal-locality and device independence). Eq. (10) is exactly the operational *signal-locality* condition — the "Local Agency" leg of the Local-Friendliness program [313] — underlying device-independent cryptography: no one-sided choice of local operation alters the marginal statistics of a spacelike-separated region. Hence device-independent randomness generation and key distribution, which require only operational no-signalling together with an observed Bell violation, are unaffected by the two-boundary structure. The residual frame-dependence resides not in the marginals (which are invariant by Thm. 6.1) but in the ordering of Kraus insertions by the time function τ in the active law (Sec. 5); the underlying two-boundary ontology is thus foliation-relative, consistent with the general lesson of the locally-mediated programme that augmenting quantum kinematics with a hidden ontology typically singles out a preferred ordering [320].

7 Posterior collapse along a worldline

Axiom 7.1 (Terminal record-completeness). K is $\overline{\mathcal{F}}^{\gamma}_{\tau_f-}$-measurable, where $\overline{\mathcal{F}}^{\gamma}_{\tau_f-} := \bigvee_{\tau<\tau_f,\tau\in\mathbb{Q}} \overline{\mathcal{F}}^{\gamma}_{\tau}$.

Remark 7.2 (Physical motivation from quantum Darwinism). Ax. 7.1 is an identifiability hypothesis: the terminal outcome K is recoverable from the cumulative pre-terminal record. It is motivated physically by quantum Darwinism [146, 147, 148]: when a quantum system interacts with a sufficiently large environment, pointer-state information becomes redundantly encoded across environment fragments, and any observer with access to multiple fragments recovers the pointer outcome. In the present framework the observer's worldline γ plays the role of an environment fragment accumulating records; when the record-encoding is redundant enough to determine the terminal pointer, Ax. 7.1 holds. Brandão–Piani–Horodecki [147] prove (in an information-theoretic sense) that for generic dynamics, near-maximal information in environment fragments implies each fragment's reduced state is close to a fixed pointer ensemble; the Spectrum Broadcast Structure framework of Korbicz [148] gives structural sufficient conditions. Neither paper formalizes the measure-theoretic filtration statement of Ax. 7.1 directly — that appears to be an open problem in its current form — but together they supply the physical reason to regard Ax. 7.1 as a mild hypothesis in settings with sufficient environmental coupling. In the language of repeated-measurement limit theorems [177, 175], Ax. 7.1 is the operator-algebraic form of the distinguishability/purification condition under which the posterior martingale collapses to a sharp pointer indicator; absent it, the martingale convergence theorem still yields convergence to the non-degenerate limit of Prop. 9.1. Prop. 9.1 gives the general collapse statement without this axiom; Cor. 9.2 quantifies the residual uncertainty when it fails. A 2026 operator-algebraic line makes this connection more native to Part II: Girard–Cheng–Cao [149] identify objectivity (observer agreement) with the *algebraic local recoverability* of a quantum code from an environment subalgebra, and recoverability in tracial von Neumann algebras is controlled by Petz-type recovery maps [170, 171]. This suggests recasting Ax. 7.1 as the statement that the terminal outcome K is recoverable from the subalgebra generated by the pre-terminal record $\overline{\mathcal{F}}^{\gamma}_{\tau_f-}$, with the exact filtration condition as the sharp ($\delta = 0$) limit of the approximate, redundancy-controlled recoverability of those works — a reformulation we flag as a natural direction rather than adopt here, since it would couple the identifiability hypothesis to the same von Neumann / modular machinery used in Part II rather than to mutual-information diagnostics alone.

Theorem 7.3 (Posterior collapse along γ)**.**

Remark 7.4 (Transience of redundancy and the regime of validity). The redundancy underlying Ax. 7.1 is generically *transient*: environmental mixing can destroy the Darwinian plateau at late times (the "rise and fall of redundancy" [172]). Since Ax. 7.1 is applied in the limit $s_n \uparrow \tau_f$, it is therefore not automatic from strong coupling alone. Persistence of redundancy is, however, possible precisely when the broadcasting interaction imprints distinct *conserved-charge or energy densities* on distinct terminal branches: Cao–Nussinov [173] show, via the large-deviation principle for subsystem thermalization, that redundancy survives thermalizing environmental dynamics in exactly this case. The operationally honest regime of validity for Ax. 7.1 is accordingly: either (i) the terminal records along γ are effectively frozen at some $t_{\text{freeze}} < \tau_f$ (no further γ-local re-mixing before τ_f), or (ii) the terminal pointer branches $\{k\}$ are energy-density-distinguishable along γ. When neither holds, identifiability degrades and the horizon-hidden residual entropy of Cor. 9.2 is positive even absent a horizon.

Assume Ax. 7.1. Then for all $k \in \mathcal{K}$,

$$\pi^{\gamma}_{s_n}(k) \xrightarrow[a.s.,\, L^1]{n\to\infty} \mathbf{1}_{\{K=k\}}, \qquad \left\|\rho^{\gamma}_{\mathrm{RC}}(s_n) - \rho_2(s_n \mid K)\right\|_1 \xrightarrow[a.s.]{n\to\infty} 0, \tag{11}$$

where $(s_n) = S_\gamma$ is the protocol sequence with $s_n \uparrow \tau_f$.

Proof. Set $\mathcal{F}_n := \overline{\mathcal{F}}^{\gamma}_{s_n}$. Because $(s_n) = S_\gamma$ is the protocol sequence with $s_n \uparrow \tau_f$, every $\tau < \tau_f$ satisfies $\tau \le s_n$ for some n, hence $\overline{\mathcal{F}}^{\gamma}_{\tau} \subset \mathcal{F}_n$. Therefore

$$\bigvee_{n\ge1} \mathcal{F}_n = \bigvee_{\tau<\tau_f,\, \tau\in\mathbb{Q}} \overline{\mathcal{F}}^{\gamma}_{\tau} = \overline{\mathcal{F}}^{\gamma}_{\tau_f-}.$$

Fix $k \in \mathcal{K}$ and set $X_k := \mathbf{1}_{\{K=k\}}$. By Ax. 7.1, X_k is measurable with respect to $\bigvee_n \mathcal{F}_n$. Since $X_k \in L^1(\Omega_\gamma, \mathbb{P}_\gamma)$ and $(\mathcal{F}_n)$ is increasing, Lévy's upward theorem — the a.s. and L^1 convergence of the closing martingale $\mathbb{E}[X_k \mid \mathcal{F}_n]$ generated by the fixed integrable variable X_k, which is therefore uniformly integrable — yields

$$\mathbb{E}[X_k \mid \mathcal{F}_n] \xrightarrow[\text{a.s., } L^1]{} X_k.$$

But $\mathbb{E}[X_k \mid \mathcal{F}_n] = \pi^\gamma_{s_n}(k)$ by Definition 5.2, proving the first convergence.

For the second convergence, write

$$\rho^\gamma_{\mathrm{RC}}(s_n) - \rho_2(s_n \mid K) = \sum_{k\in\mathcal{K}} \big(\pi^\gamma_{s_n}(k) - \mathbf{1}_{\{K=k\}}\big)\rho_2(s_n \mid k).$$

Using the triangle inequality and $\|\rho_2(s_n \mid k)\|_1 = \operatorname{Tr}\rho_2(s_n \mid k) = 1$ from Lemma 3.6,

$$\big\|\rho^\gamma_{\mathrm{RC}}(s_n) - \rho_2(s_n \mid K)\big\|_1 \le \sum_{k\in\mathcal{K}} \big|\pi^\gamma_{s_n}(k) - \mathbf{1}_{\{K=k\}}\big|.$$

The right-hand side converges to 0 a.s. because $\mathcal{K}$ is finite and each summand does. This proves the second convergence. $\square$

Remark 7.5 (Relation to QND martingale-collapse)*.* Thm. 7.3 is the two-boundary, operator-algebraic counterpart of the convergence theorem for repeated quantum non-demolition (QND) measurements of Bauer–Bernard [178] and Bauer–Benoist–Bernard [177]: the pointer-outcome posterior is a bounded martingale converging a.s. to $\mathbf{1}_{\{K=k\}}$. Those works additionally identify the *rate* of collapse with the per-step relative (KL) entropy of the indirect measurement; the same information-theoretic quantity is the natural candidate to set the timescale entering the Page-curve discussion (see Prop. 13.2). For general (non-QND) repeated measurements, the exponentially fast sector selection of Benoist–Greggio–Pellegrini [176] and the limit theorems of Benoist–Fatras–Pellegrini [175] apply under a purification/irreducibility condition, which is the quantitative counterpart of the identifiability Ax. 7.1. The collapse is per-branch: once $K = k$ is selected, the residual temporal non-classicality of the two-boundary link is carried by the symmetric Margenau–Hill marginal $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$, whose negativity $f_k > 0$ is equivalent (for projective terminal effects) to $[\rho(\tau), \Lambda_k(\tau)] \neq 0$ (Thm. 3.10); prior-averaging this witness restores positivity via the no-signalling identity $\sum_k p(k)M_k = \rho \succeq 0$ (Thm. 6.1), so the quasiprobability negativity is irreducibly intra-branch and does not survive the marginalisation that licenses the timescale of the Page-curve discussion.

Remark 7.6 (Worldline dependence)*.* The posterior π^γ_τ and its collapse depend on the worldline γ because different worldlines sample different local algebras. This is physical: an observer on γ sees the γ-filtration. The *terminal distribution* of K, however, is γ-independent by Lem. 2.5.

Remark 7.7 (Relation to retrocausal Born-rule derivations)*.* The collapse mechanism of Thm. 7.3 renders the terminal record definite but takes the weight $p(k)$ as input (Rem. 2.6). A complementary line of work seeks to *derive* the Born measure within time-symmetric, all-at-once retrocausal formulations of quantum theory: the fixed-point formulation of Ridley and Ridley–Adlam [385, 387] obtains the Born measure by summing a multiple-time wavefunction along fixed-point histories on the Keldysh contour, and Ridley [386] grounds it in self-location within time-extended Everettian worlds. The present construction is conceptually adjacent (two-boundary conditioning, a fixed-point terminal master equation) but differs in setting: it uses a fixed finite-dimensional matter factor and a per-worldline Kolmogorov measure $\mathbb{P}_\gamma$ on records rather than a global time-extended wavefunction over outcomes. Whether the passive marginal $\mathbb{P}_\gamma(K = k) = p(k)$ can be re-derived as a self-location measure over the γ-branches, rather than imposed by construction, is left open; importing such a derivation would upgrade Lem. 2.5 from a Born-input lemma toward a theorem.

8 Summary of Part I

1. Time-invariant Born law (Lem. 2.5).
2. Unitary covariance of $\rho_2(\tau \mid k)$ with generator $H(\tau)$ (Thm. 3.7).
3. Covariant conservation of $\mathrm{Tr}(\rho_2(\tau \mid k)\widehat{O}(x))$ on each realization, for $\widehat{O}$ satisfying a local covariant conservation law (Thm. 4.1).
4. Prior-average identity $\mathbb{E}[\rho^\gamma_{\mathrm{RC}}(\tau)] = \rho(\tau)$ (Lem. 5.3(iii)) and reduction to ordinary QM expectations (Cor. 5.4).
5. Bulk no-signalling under the active sequential-Born law (Thm. 6.1).
6. Posterior collapse under record-completeness (Thm. 7.3).
7. Generalized collapse without record-completeness (Prop. 9.1) and horizon-hidden residual entropy (Cor. 9.2).
8. Graded (fermionic) two-boundary framework under graded microcausality (Prop. 9.7).
9. Conditioned Ehrenfest theorem (Prop. 9.8) and semiclassical selection of classical initial data by the terminal outcome.

Each of these is proved in full from the stated axioms and standard background facts. They constitute a self-contained rigorous two-boundary framework on a fixed globally hyperbolic spacetime background carrying finite-dimensional matter.

9 Partial resolutions of open problems

In this section we upgrade three items from the open-problems list of Sec. 11 to theorems of Part I. All proofs are internal, using only the axioms and results already established. None of the conjectures of Sec. 10 is used.

9.1 Generalized collapse without terminal record-completeness

Axiom 7.1 is the strongest possible identifiability hypothesis: it says the terminal outcome K is exactly recoverable from the pre-terminal filtration $\overline{\mathcal{F}}^\gamma_{\tau_f-}$. In the presence of an event horizon intersecting Σ_{τ_f}, information about K may be hidden from any observer worldline γ that never crosses the horizon, and Ax. 7.1 can fail. The following proposition gives the collapse statement in *full generality*, without any identifiability assumption.

Proposition 9.1 (Generalized collapse)**.** *Without assuming Ax. 7.1, for every $k \in \mathcal{K}$,*

$$\pi^\gamma_{s_n}(k) \xrightarrow[a.s.,\ L^1]{n\to\infty} \mathbb{E}_{\mathbb{P}_\gamma}\big[\mathbf{1}_{\{K=k\}} \,\big|\, \overline{\mathcal{F}}^\gamma_{\tau_f-}\big], \tag{12}$$

and

$$\left\| \rho^\gamma_{\mathrm{RC}}(s_n) - \sum_{k\in\mathcal{K}} \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}^\gamma_{\tau_f-}]\, \rho_2(s_n \mid k) \right\|_1 \xrightarrow[a.s.]{n\to\infty} 0. \tag{13}$$

The limit in (12) *equals $\mathbf{1}_{\{K=k\}}$ a.s. for every $k \in \mathcal{K}$ if and only if Ax. 7.1 holds. Consequently Thm. 7.3 is the sharp special case of* (12)*.*

Proof. The proof of Thm. 7.3 used Ax. 7.1 only in the last step, to identify the limit $\mathbb{E}[X_k \mid \bigvee_n \mathcal{F}_n]$ with $X_k = \mathbf{1}_{\{K=k\}}$. Without that step, the upward theorem applied to $X_k \in L^1$ and the increasing filtration $(\mathcal{F}_n)$ with $\bigvee_n \mathcal{F}_n = \overline{\mathcal{F}}^\gamma_{\tau_f-}$ (established in the proof of Thm. 7.3) yields (12) a.s. and in L^1. Then (13) follows from the triangle inequality and $\|\rho_2(s_n \mid k)\|_1 = 1$ exactly as before. For the "if and only if" claim: a.s. limit equal to $\mathbf{1}_{\{K=k\}}$ for every k forces $\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}^\gamma_{\tau_f-}] = \mathbf{1}_{\{K=k\}}$ a.s., hence $\mathbf{1}_{\{K=k\}}$ is $\overline{\mathcal{F}}^\gamma_{\tau_f-}$-measurable, hence K is; the converse is Thm. 7.3. □

We stress what Prop. 9.1 does and does not relax. It weakens the *conclusion* of Thm. 7.3 — replacing the sharp indicator $\mathbf{1}_{\{K=k\}}$ by the conditional expectation $\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}^{\gamma}_{\tau_f-}]$ — but, by the "if and only if" above, it leaves the *all-or-nothing* character of Ax. 7.1 intact: the sharp collapse holds exactly when K is exactly $\overline{\mathcal{F}}^{\gamma}_{\tau_f-}$-measurable. Physical Darwinian arguments, by contrast, supply only *approximate* recoverability with a tolerance $\delta > 0$ (e.g. a δ-adequacy Holevo bound $\chi(\Pi_S : F) \geq (1-\delta)H_S$ in the functional-information measure of [174], which is explicitly not a certainty statement). A quantitative interpolation — a bound $\left\|\rho^{\gamma}_{\mathrm{RC}}(s_n) - \rho_2(s_n \mid K)\right\|_1 \leq g(\delta)$ with $g(\delta) \to 0$ as the functional-information deficit $\delta \to 0$ — remains open in that functional-information parametrization; an *entropy-parametrized* version is now a theorem: Thm. A.40 bounds the residual by $2\min\{H, \sqrt{H/2}\}$ with $H = H(K \mid \overline{\mathcal{F}}^{\gamma}_{\tau_f-})$ the residual conditional entropy, with a Fano converse showing the failure mode is exactly $H > 0$. Translating a Darwinian δ-adequacy bound into $H \leq f(\delta)$ would complete the δ-controlled collapse.

Corollary 9.2 (Horizon-hidden residual entropy)**.** *Let $M_{\mathrm{acc}} \subseteq \mathbb{N}$ index the protocol stages of S_γ whose records are accessible to a given observer class (e.g. the outcomes of instruments localized outside a black-hole horizon), and define the* accessible filtration *and its terminal join intrinsically,*

$$\mathcal{G}_\tau := \sigma\big(j_m : \ m \in M_{\mathrm{acc}}, \ s_m \leq \tau\big) \subseteq \overline{\mathcal{F}}^{\gamma}_{\tau}, \qquad \mathcal{G} := \bigvee_{\tau < \tau_f,\, \tau \in \mathbb{Q}} \mathcal{G}_\tau \subseteq \overline{\mathcal{F}}^{\gamma}_{\tau_f-}.$$

Then the accessible posterior $\pi^{\gamma,\mathcal{G}}_\tau(k) := \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \mathcal{G}_\tau]$ is a closed bounded martingale and converges a.s. and in L^1 to $\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \mathcal{G}]$. The quantity

$$H(K \mid \mathcal{G}) := -\sum_{k \in \mathcal{K}} \mathbb{E}\big[\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \mathcal{G}] \log \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \mathcal{G}]\big] \geq 0 \tag{14}$$

quantifies the terminal information inaccessible to γ; $H(K \mid \mathcal{G}) = 0$ iff K is $\mathcal{G}$-measurable (mod null). The residual is quantified by Thm. A.40 with $\overline{\mathcal{F}}_{\tau_f-}$ replaced by $\mathcal{G}$.

Proof. The family $(\mathcal{G}_{s_n})_{n\geq 1}$ is increasing with $\bigvee_n \mathcal{G}_{s_n} = \mathcal{G}$ *by construction*, so Lévy's upward theorem applies exactly as in Prop. 9.1; non-negativity and the zero-iff-measurable characterization are standard properties of conditional entropy. □

Remark 9.3 (Why the accessible information must be a filtration of records). For a general sub-σ-algebra $\mathcal{G} \subseteq \overline{\mathcal{F}}^{\gamma}_{\tau_f-}$, conditioning on the intersection filtration $\mathcal{G} \cap \overline{\mathcal{F}}^{\gamma}_{\tau}$ would converge only to $\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \bigvee_\tau(\mathcal{G} \cap \overline{\mathcal{F}}^{\gamma}_{\tau})]$, and the join of intersections can be *strictly* coarser than $\mathcal{G}$ — indeed trivial: with $X_1, X_2, \ldots$ i.i.d. fair bits, $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$, $U := \sum_i X_i 2^{-i}$ and $A := \{U \leq 1/3\}$, one has $A \in \bigvee_n \mathcal{F}_n$ but $A \notin \overline{\mathcal{F}}_n$ for every n (its conditional probability given $\mathcal{F}_n$ lies strictly in $(0,1)$ on the dyadic cell containing $1/3$), so $\sigma(A) \cap \overline{\mathcal{F}}_n$ is trivial mod null for every n and the martingale limit is $\mathbb{P}(A)$, not $\mathbf{1}_A$. Defining the accessible information intrinsically as the join of the accessible-record filtration, as in Cor. 9.2 (and as the horizon construction (44) in fact does), makes the limit σ-algebra equal to $\mathcal{G}$ by construction.

This resolves item (4) of Sec. 11: the collapse statement holds in full generality, and Ax. 7.1 is precisely the hypothesis that upgrades the general posterior limit to a sharp indicator. Horizon-hidden scenarios correspond to the regime $H(K \mid \mathcal{G}) > 0$.

9.2 Fermionic matter: the graded two-boundary framework

We now show that Part I transports verbatim to $\mathbb{Z}_2$-graded matter under a graded microcausality axiom, resolving item (5) of Sec. 11. A closely related spacetime / two-time treatment of fermionic systems, also bypassing Grassmann variables, appears in the superdensity-operator formalism [372] and its recent Hilbert-space, spacetime-symmetric extension [373]; the present construction differs in conditioning on a two-boundary (initial-state, terminal-effect) pair. Let $\mathcal{H} = \mathcal{H}_{\bar{0}} \oplus \mathcal{H}_{\bar{1}}$ be a $\mathbb{Z}_2$-graded finite-dimensional Hilbert space with grading operator $\Gamma := \mathbb{1}_{\mathcal{H}_{\bar{0}}} \oplus (-\mathbb{1}_{\mathcal{H}_{\bar{1}}})$, so $\Gamma = \Gamma^* = \Gamma^{-1}$. An operator $A \in \mathcal{L}(\mathcal{H})$ is *even* (degree 0) if $\Gamma A \Gamma = A$ and *odd* (degree 1) if $\Gamma A \Gamma = -A$. Write $\mathcal{L}(\mathcal{H}) = \mathcal{L}_+ \oplus \mathcal{L}_-$ for the corresponding decomposition, and for homogeneous A let $|A| \in \{0, 1\}$ be its degree.

Axiom 9.4 (Graded local algebras and graded microcausality)**.** Each $\mathcal{A}(R)$ is a graded $*$-subalgebra: $\mathcal{A}(R) = \mathcal{A}(R)_+ \oplus \mathcal{A}(R)_-$ with $\mathcal{A}(R)_\pm = \mathcal{A}(R) \cap \mathcal{L}_\pm$. Isotony (M1), outer regularity (1), the subset-extension convention, and the causal propagation (M3a,b) of Ax. 2.2 are imposed as before. Microcausality is *graded*:
(M2′) for spacelike-separated R_1, R_2 and homogeneous $A \in \mathcal{A}(R_1)$, $B \in \mathcal{A}(R_2)$,

$$AB - (-1)^{|A||B|} BA = 0. \tag{15}$$

Assume further that $H(\tau)$, ρ_{i}, the terminal effects $\{E_k\}$, and the protocol Kraus operators $\{M_j^{(m)}\}$ and intervention POVMs $\{M_j^{(1)}\}$, $\{F_l\}$ used in Sec. 6 are all *even*. For the state ρ_{i} and the effects E_k, evenness is precisely Γ-block-diagonality, i.e. the fermion-parity superselection rule (no coherent superposition of even- and odd-parity sectors); it is automatically consistent with positivity and normalization and imposes no constraint beyond grade-zero.

Remark 9.5 (Physical content of the evenness restriction)*.* In Haag–Kastler fermionic nets, odd (fermionic) field operators are gauge-variant and are not themselves physical observables; physical, gauge-invariant bulk observables (number densities, currents, stress-energy, bilinears) are even. Ax. 9.4 therefore formalizes the standard physical setting. Odd operators still enter the theory as building blocks but never as direct inputs to the probability axioms.

Remark 9.6 (Scope of the graded microcausality axiom)*.* We emphasize that (M2′) is imposed as an axiom that abstracts the output of the spin–statistics theorem and the CAR structure of free Fermi fields; we do not re-derive it. In the Haag–Kastler setting it is the graded-locality property of a fermionic field net, whose even part is the observable algebra. That even part satisfies *twisted* (rather than ordinary) Haag duality, implemented via the Klein transformation, and is known not to satisfy ordinary Haag duality in concrete models [375, 374]. Establishing (M2′) from first principles for a specific CAR net lies outside the present finite-dimensional scope. For the modular structure of such fermionic nets in the continuum see [271].

Proposition 9.7 (Graded two-boundary framework)**.** *Under Ax. 9.4 (replacing Ax. 2.2), all numbered results of Part I (Lem. 2.5, Lem. 3.6, Thm. 3.7, Cor. 3.8, Thm. 4.1, Lem. 5.3, Cor. 5.4, Thm. 6.1, Thm. 7.3, and Prop. 9.1) hold, with identical statements and proofs, provided the Heisenberg observables $\widehat{O}(x)$ in Thm. 4.1 are restricted to be even.*

Proof. Evenness of $H(\tau)$ and $\Gamma = \Gamma^*$ give $\Gamma H(\tau) \Gamma = H(\tau)$, so $i\hbar\partial_\tau(\Gamma U \Gamma) = H(\tau)(\Gamma U \Gamma)$ with $\Gamma U(\tau, \tau)\Gamma = \mathbb{1}$; uniqueness for linear ODEs yields $\Gamma U(\tau, \tau')\Gamma = U(\tau, \tau')$, i.e. $U(\tau, \tau') \in \mathcal{L}_+$ for all τ, τ'. Therefore Heisenberg transport $A \mapsto U^\dagger A U$ preserves grade, and (M3a,b) transports graded local operators to graded local operators of the same grade.

Lemma 2.5, Lemma 3.6, Thm. 3.7, Cor. 3.8: the proofs use only positivity of ρ_{i}, E_k, cyclicity of trace, unitarity of U, and uniqueness of the positive square root. None of these invokes the commutation structure of $\mathcal{A}(R)$; they transport verbatim.

Theorem 4.1: for even $\widehat{O}$, the trace $\mathrm{Tr}(\rho_2(\tau_i \mid k)\widehat{O}_{\mu\nu\cdots}(x))$ is a scalar function of x identical to the ungraded case; the proof is unchanged.

Theorem 6.1, Case 1: the only nontrivial step using the commutation axiom is $[F_l^{(1)}, M_j^{(1)}] = 0$. Both operators are even: $M_j^{(1)} \in \mathcal{A}(O_1)_+$ by hypothesis, and $F_l^{(1)} = U_{21}^\dagger F_l U_{21} \in \mathcal{L}_+$ because F_l is even and U_{21} preserves grade, hence $F_l^{(1)} \in \mathcal{A}(W)_+$ by the same (M3b) argument as before. With $|A||B| = 0$, the graded identity (15) reduces to $[F_l^{(1)}, M_j^{(1)}] = 0$. The rest of Case 1 proceeds unchanged. Case 2 uses only $\sum_j M_j^{(1)*} M_j^{(1)} = \mathbb{1}$ and unitarity of U_{12}, which are grade-independent, so it transports verbatim.

Filtration and passive law (9): all Kraus $M_{j_m}^{(m)}$ are even, so the cylinder expression is a trace of an even operator, non-negative by positivity of $\rho_2(\tau_i \mid k)$ and the product structure. Kolmogorov consistency uses only $\sum_j M_j^{(m)*} M_j^{(m)} = \mathbb{1}$ and cyclicity. Lem. 5.3, Cor. 5.4, Thm. 7.3, Prop. 9.1 are probability-theoretic statements about $(\Omega_\gamma, \mathbb{P}_\gamma)$ and the filtration $(\overline{\mathcal{F}}_\tau^\gamma)$, and do not depend on the grading at all. □

This resolves item (5) of Sec. 11 in full: the two-boundary framework works for $\mathbb{Z}_2$-graded (fermionic) matter under graded microcausality, the only modification being the standard restriction that intervention POVMs and Heisenberg observables be grade-zero (physically: gauge invariant).

9.3 Classical limit along realizations

The classical limit claim of item (6) of Sec. 11 cannot be stated as a genuine $\hbar \to 0$ Egorov theorem in a purely finite-dimensional setting, because without a Weyl-quantization scheme there is no classical phase space canonically attached to $\mathcal{H}$. However, the finite-dimensional analog — the Ehrenfest theorem for conditioned expectations — is immediate from Cor. 3.8 and deserves to be stated because it fixes the conditioned mean-value dynamics. We stress, however, that the Ehrenfest identity alone does not establish classicality: as emphasised by Ballentine, Yang and Zibin [376], mean-value motion is neither necessary nor sufficient for classical behaviour, and the semiclassical mean tracks a single classical orbit only up to the (logarithmically short, for chaotic H) Ehrenfest time. What selects a definite history on a realization $K = k$ and suppresses interference is decoherence/einselection [377] of the conditioned state $\rho_2(\tau \mid k)$ in a record/pointer basis — the same einselection machinery invoked in the quantum-Darwinism motivation (Rem. 7.2, [146, 147]); the Ehrenfest identity then merely fixes the conditioned mean once a quasi-classical branch has been selected.

Proposition 9.8 (Conditioned Ehrenfest theorem)**.** *For any $\widehat{O}_\mathrm{S} \in \mathcal{L}_\mathrm{sa}(\mathcal{H})$ (possibly time-dependent) and any $k \in \mathcal{K}$ with $p(k) > 0$,*

$$\frac{d}{d\tau} \operatorname{Tr}\big(\rho_2(\tau \mid k)\widehat{O}_\mathrm{S}(\tau)\big) = \frac{1}{i\hbar} \operatorname{Tr}\big(\rho_2(\tau \mid k)\, [\widehat{O}_\mathrm{S}(\tau), H(\tau)]\big) + \operatorname{Tr}\big(\rho_2(\tau \mid k)\partial_\tau \widehat{O}_\mathrm{S}(\tau)\big). \tag{16}$$

In particular, if $[\widehat{O}_\mathrm{S}, H(\tau)] = 0$ for all τ and $\widehat{O}_\mathrm{S}$ is time-independent, then $\operatorname{Tr}(\rho_2(\tau \mid k)\widehat{O}_\mathrm{S})$ is τ-independent on every realization with $K = k$ — even though $\rho_2(\tau \mid k)$ is determined by the retrodictive conditioning on k, not by ρ_i alone.

Proof. Differentiate using Cor. 3.8 and the product rule; cyclicity turns $[\cdot, H]$ into commutation under the trace. The time-independent commuting case is the direct specialization. □

Remark 9.9 (Semiclassical trajectory selection). In a semiclassical regime where $\mathcal{H}$ is replaced by an $\hbar$-parameterized family (e.g. Weyl quantization of a compact classical phase space, or coherent-state quantization), Egorov's theorem combined with (16) implies that the expectation $\operatorname{Tr}(\rho_2(\tau \mid k)\widehat{O}_\mathrm{S})$ follows, to leading order in $\hbar$, the classical Hamiltonian flow of the symbol of $\widehat{O}_\mathrm{S}$ evaluated along the classical trajectory through the semiclassical mean of $\rho_2(\tau_i \mid k)$. Because $\rho_2(\tau_i \mid k)$ depends explicitly on k through $\Lambda_k(\tau_i)$, different terminal outcomes select different classical initial data, and hence different classical trajectories. This is the sense in which, once decoherence has einselected a quasi-classical branch, the terminal outcome fixes which classical initial data — and hence which classical trajectory the conditioned mean follows — in the two-boundary framework. A rigorous $\hbar \to 0$ statement requires additional input beyond

Part I. For closed dynamics this input is a Weyl/coherent-state quantization scheme together with an Egorov theorem, but for chaotic H such closed-system correspondence holds only up to the logarithmically short Ehrenfest time. The physically appropriate — and now rigorous — route is the open-system one: for Markovian dynamics with phase-space diffusion strength D, quantum and classical evolutions agree on all observables for times exponentially beyond the Ehrenfest time once $D \gtrsim \hbar^{4/3}$, a threshold proven sharp [378, 380], with an Egorov-type correspondence for open systems with general Lindbladians now established [379]. Since the framework's master-equation regime (Part II) is itself an open-system description that supplies a decoherence/diffusion rate (cf. the horizon-decoherence floor of Sec. 13.2), the residual content of item (6) in that regime is to verify that the induced diffusion on $\rho_2(\tau \mid k)$ meets the $\hbar^{4/3}$ threshold; the finite-dimensional Ehrenfest identity above is the strongest dimension-independent statement available before such input is supplied.

This partially resolves item (6): the conditioned Ehrenfest identity is rigorously established here, and the remaining content of the open problem reduces to the well-defined technical question of supplying the external quantization/decoherence input — either a Weyl-quantization scheme with a closed-system Egorov theorem, or, in the open-system regime, a diffusion rate on $\rho_2(\tau \mid k)$ meeting the $\hbar^{4/3}$ threshold.

10 Outlook: conjectural bridges to gravity

Part II consists of two conjectures plus a Planck-suppression observation that fixes the low-energy phenomenology of the finite-difference clock perturbation. The conjectures are not theorems. Each is stated with the precise hypotheses that, if established, would elevate it. We present them in ascending order of ambition.

10.1 Conjecture 1: Final-state semiclassical gravity

Let the matter theory admit a stress-energy operator field $\widehat{T}_{\mu\nu}[g](x)$ (Heisenberg picture) that is a smooth symmetric tensor on $\mathcal{M}_{[\tau_i,\tau_f]}$ with $\nabla^\mu \widehat{T}_{\mu\nu}[g] = 0$ as an operator identity for every background g in the Sobolev ball below. (For finite-dim matter this is an additional structural assumption on how $\widehat{O}_\mu$ of Thm. 4.1 depends on g; for QFT it requires Hadamard renormalization [10, 16], which lies outside Part I's scope.)

Branch postulate (B). Throughout this subsection we adopt the symmetric branch of Def. 3.1 (Rem. 3.2) as the canonical density operator entering the gravitational source. The symmetric branch is selected by three jointly necessary properties that the asymmetric branch $\rho\Lambda_k/p(k)$ and the general $\rho^a\Lambda_k\rho^{1-a}/p(k)$ ($a \neq 1/2$) families fail: (B-i) *self-adjointness* of the conditioned density operator ($\rho_2^* = \rho_2$ requires $a = 1/2$, since $(\rho^a\Lambda_k\rho^{1-a})^* = \rho^{1-a}\Lambda_k\rho^a$ for self-adjoint ρ, Λ_k); (B-ii) *reality* of the resulting trace $\mathrm{Tr}(\rho_2[g]\widehat{T}_{\mathrm{S},\mu\nu}[g])$ as a real-valued symmetric tensor on g for self-adjoint $\widehat{T}_{\mathrm{S},\mu\nu}$ (the asymmetric ordering with $a \neq 1/2$ generically yields a complex-valued source on a real metric, since $\mathrm{Tr}(\rho^a\Lambda_k\rho^{1-a}\widehat{T})^* = \mathrm{Tr}(\rho^{1-a}\Lambda_k\rho^a\widehat{T}) \neq \mathrm{Tr}(\rho^a\Lambda_k\rho^{1-a}\widehat{T})$ unless $a = 1/2$); we make no positivity claim on the source tensor itself, since a positive state on a self-adjoint observable algebra only guarantees real-valuedness, not componentwise pointwise nonnegativity of the resulting symmetric tensor. (B-iii) *conservation per realization* via Thm. 4.1, which uses unitary covariance of ρ_2 on a fixed background. (B-iii) is satisfied by every $a \in [0, 1]$ branch (each conjugates unitarily as $\rho^a \to U\rho^aU^*$), but only the $a = 1/2$ branch satisfies (B-i) and (B-ii) simultaneously. Hence the symmetric branch is the unique branch for which the gravitational source is a self-adjoint, real-valued, covariantly conserved symmetric tensor. (The earlier wording, which attributed the selection to a Bianchi-identity failure of asymmetric branches, was inaccurate: the formal Bianchi calculation depends only on unitary covariance and goes through for every a; the genuine selection is by self-adjointness of ρ_2 together with reality of the resulting trace, not by conservation and not by any positivity statement about the stress tensor itself.) Other branches are excluded.

Record postulate (R): adapted sourcing, causal form. The branch postulate fixes *which* conditioned operator sources gravity on each branch; it does not by itself fix *when* the branch-conditioned source becomes gravitationally active. We adopt, as the second structural postulate of this subsection, *record-adapted (non-anticipating) sourcing* in its *causal (past-light-cone)* form: at the spacetime point x the gravitational source is conditioned on exactly the records whose events lie in the causal past $J^-(x)$, no more and no less. Conditioning instead on all records up to the exterior *time* $\tau(x)$ — the slicing-based form of an earlier draft of this subsection — is not causally safe: it updates the metric on an entire Cauchy slice at once, including points spacelike-separated from the record event, making the update foliation-dependent and, at the level of *distributions* of metric observables (not means), sensitive to the protocol choice outside its causal future. The causal form removes both defects by construction (Prop. 10.7(iv)), at the standard price of the feedback-gravity literature's causal-conditional prescription [134, 135]: posterior updates propagate as *null* shells on the future light cones of the record events, in the Barrabès–Israel thin-shell formalism [21].

Definition 10.1 (Adapted two-boundary source; causal form)**.** Fix the worldline γ, protocol $\mathcal{P}_\gamma$, filtration $(\overline{\mathcal{F}}^\gamma_\tau)$ and posterior $\pi^\gamma_\tau(k) = \mathbb{E}_{\mathbb{P}_\gamma}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}^\gamma_\tau]$ of Sec. 5, extended (i) to all real $\tau \in [\tau_i, \tau_f)$ by right-constancy, $\pi^\gamma_\tau := \pi^\gamma_{s_{n(\tau)}}$ with $n(\tau) = \max\{m : s_m \le \tau\}$ (records arrive only at the protocol times, so this is the unique right-continuous adapted version; $\pi^\gamma_\tau := p$ for $\tau < s_1$, with p the operative prior of (x-c) below), and (ii) through the terminal readout by $\overline{\mathcal{F}}^\gamma_\tau := \overline{\mathcal{F}}^\gamma_{\tau_f-} \vee \sigma(K)$ for $\tau \ge \tau_f$, so that $\pi^\gamma_\tau(k) = \mathbf{1}_{\{K=k\}}$ for $\tau \ge \tau_f$. For $x \in \mathcal{M}^{\text{ext}}$ define the *retarded record time*

$$t_\gamma(x) \;:=\; \begin{cases} \sup\{\tau : \gamma(\tau) \in J^-(x)\}, & x \notin J^+(\Sigma_{\tau_f}), \quad (\sup\emptyset := \tau_i) \\ \tau_f, & x \in J^+(\Sigma_{\tau_f}), \end{cases} \tag{17}$$

so that the records available at x are exactly those whose events lie in $J^-(x)$: $\gamma(s_m) \in J^-(x) \iff s_m \le t_\gamma(x)$ (the worldline enters $J^-(x)$ in temporal order). All causal relations here and in (23) below are those of the solution metric g_ω itself; the fixed-point map of (vii)/(x) evaluates them on its argument (g_0-cones at the zeroth iterate), and on $\mathcal{B}_{g_0}$ the $O(\kappa)$ cone perturbation shifts $t_\gamma(x)$ by $O(\kappa)$, within the contraction bookkeeping. The *adapted source* on a background g is the random tensor field

$$T^{\text{RC}}_{\mu\nu}[g](x;\omega) \;:=\; \operatorname{Tr}\!\big(\rho^\gamma_{\text{RC}}[g]\big(\tau(x); t_\gamma(x)\big)\, \widehat{T}_{\text{S},\mu\nu}[g](x)\big), \qquad \rho^\gamma_{\text{RC}}[g](\tau;t) = \sum_{k\in\mathcal{K}_+} \pi^\gamma_t(k)\, \rho_2[g](\tau \mid k), \tag{18}$$

i.e. the retrocausal state of Def. 5.2 with the weights evaluated at the *retarded* record time: branch states at the local time $\tau(x)$, posterior weights conditioned on the records in $J^-(x)$ only. For $x \in J^+(\Sigma_{\tau_f})$ this reduces identically to the per-branch source $\operatorname{Tr}(\rho_2[g](\tau \mid K)\widehat{T}_{\text{S},\mu\nu}[g])$ used in the post-terminal extension below; pre-terminally it is measurable with respect to $\sigma(j_m : \gamma(s_m) \in J^-(x)) \subseteq \overline{\mathcal{F}}^\gamma_{t_\gamma(x)}$ by construction. For the trivial protocol (no intermediate records) $\pi^\gamma_t(k) = p(k)$ pre-terminally — equal to the seed prior $p_0(k)$ exactly, since the trivial protocol is the zero-record sector where (60) holds under (T10) — and (18) reduces to the prior source $\operatorname{Tr}(\rho[g](\tau)\widehat{T}_{\text{S},\mu\nu}[g])$ by Lem. 5.3(iii).

Note the division of conditioning labor: the branch *states* $\rho_2(\tau \mid k)$ are two-time (future-conditioned) objects, but they enter the source only through the past-cone-measurable weights $\pi^\gamma_{t_\gamma(x)}(k)$; the total source (18) is causally adapted, and its prior mean is the forward source (Cor. 5.4). This is what will carry the no-signalling burden below. Two further specifications make (18) well-defined rather than implicitly circular.

Scaffolding continuation (no-further-records extension). The branch states $\rho_2[g](\tau \mid k)$ transport the terminal effects E_k backward through the metric on $J^+(\Sigma_\tau)$, which on a realization depends on records not yet available at x. The fixed-point construction of Conj. 10.2 therefore solves, for each finite record prefix r, for a *globally defined* scaffolding metric g_r under the convention that the posterior is frozen at its post-r value to the future of the last record cone of r (equivalently: g_r is the metric that would obtain if the protocol terminated after r, the terminal readout at Σ_{τ_f} retained). The physical metric is always the realized-record assembly of Conj. 10.2 — near x it is $g_{r(\omega,x)}$ with $r(\omega, x)$ the prefix realized in $J^-(x)$ — and the scaffolding enters only through the backward transport of Λ_k. Two admissible continuation conventions differ by $O(\kappa)$ in the transported effect and hence by $O(\kappa^2)$ in the assembled solution, within the contraction budget of (x-a); but it *is* a convention, which we fix once and for all as above.

Admissibility (RP): record permanence. Only *terminally readable* outcomes are eligible for gravitational conditioning: $\mathcal{P}_\gamma$ is admissible for Def. 10.1 iff each instrument outcome j_m is held in a pointer degree of freedom that remains recoverable from $\overline{\mathcal{F}}^{\gamma}_{\tau_f-}$ (no later instrument of $\mathcal{P}_\gamma$, and no dynamics on $[s_m, \tau_f)$, erases it). Quantum-erasable outcomes are excluded from the filtration used in (18): if an erasable outcome were gravitationally sourced and later coherently erased, the metric would retain a classical which-path memory and the framework would predict zero revival visibility in any delayed-choice-eraser protocol at arbitrarily weak gravitational coupling — an unintended falsifier. Under (RP) the erasable stage is never gravitationally conditioned and standard eraser phenomenology is untouched. (Consistency caveat: excising erasable outcomes from the gravitational filtration must respect the cylinder consistency of (9); marginalizing an erased outcome inserts the corresponding nonselective channel between the retained protocol steps, so (RP) requires the erasing stage to act as a single nonselective channel at the level of the retained records — Rem. 5.1 applied to the coarse-grained protocol. This is a genuine restriction on $\mathcal{P}_\gamma$, not bookkeeping.) Physically, (RP) is the gravitational pointer-basis principle articulated in Rem. 11.40: what gravitates is what has decohered irreversibly.

(Delayed-choice scope.) Because (RP) and the source Def. 10.1 are quantified over $\mathcal{P}_\gamma$ intrinsically, an agent making a spacelike deferred commit/erase decision is outside the protocol class $\mathcal{P}_\gamma$ by construction; that decision enters an (RP)-admissible protocol only once its record reaches J^- of the readout, so delayed-choice erasure is *causal, not anticipating* — the metric conditions on the erase/commit outcome strictly after it becomes a record in the worldline's causal past. (The reconciliation of *distinct* worldlines whose deferred choices are mutually spacelike remains the open multi-worldline corner of Rem. 10.9.)

Joint problem. Let $K : \Omega \to \mathcal{K}$ be the abstract terminal label, with prior law $p(k) = \mathrm{Tr}(\rho_i U(\tau_f, \tau_i)^\dagger E_k U(\tau_f, \tau_i))$ *evaluated on the dynamical background g*. We write $\rho_2[g](\tau \mid k)$, $\mathbb{P}^{[g]}_\gamma$, and $\Omega^{[g]}_\gamma$ for the Part I objects constructed with the propagator of $H[g]$, to make the dependence explicit.

Spacetime extension. For Part II we work on the maximal globally hyperbolic development $(\mathcal{M}^{\mathrm{ext}}, g_0)$ of which the Part I slab $\mathcal{M}_{[\tau_i,\tau_f]}$ is a sub-region; in particular $J^+(\Sigma_{\tau_f}) \cap \mathcal{M}^{\mathrm{ext}} \neq \emptyset$ and the Cauchy problem for the Einstein equation on $J^+(\Sigma_{\tau_f}) \cap \mathcal{M}^{\mathrm{ext}}$ with initial data on Σ_{τ_f} is well-posed in the standard sense. The matter source is extended past the terminal slice by unconditional Heisenberg propagation: for $\tau > \tau_f$ we set $\rho_2[g](\tau \mid k) := U_g(\tau, \tau_f)\, \rho_2[g](\tau_f \mid k)\, U_g(\tau, \tau_f)^\dagger$ (no further conditioning is applied past τ_f, since the terminal record has already been read out). Under this extension, $\widehat{T}_{\mu\nu}[g](x)$ and the source of (22) below are defined on the full Cauchy development of Σ_{τ_i} in $\mathcal{M}^{\mathrm{ext}}$, the Bianchi identity continues to hold per-realization on g_k by Thm. 4.1 applied with propagator U_g. The causal future of branch-distinguishing records is the region in which the realized metric g_ω genuinely deviates from g_0; under the causal form of Def. 10.1 this is a *structural* statement (support clause (23)(b) below: outside every record cone the metric is the prior-sourced background), not a gloss. On the post-terminal future $J^+(\Sigma_{\tau_f}) \cap \mathcal{M}^{\mathrm{ext}}$ the deviation is complete ($\pi^\gamma_\tau = \mathbf{1}_{\{K=\cdot\}}$ under Ax. 7.1) and the metric is the per-sector g_k.

Conjecture 10.2 (Final-state semiclassical Einstein equation; joint fixed-point form)**.** Fix a semiclassical background g_0 and a weighted Sobolev ball $\mathcal{B}_{g_0}$ around it. Under Pos. 11.30 the matter sector is finite-dimensional, so the positive-weight set $\mathcal{K}_+ := \{k \in \mathcal{K} : p_0(k) > 0\}$ is finite and $p_* := \min\{p_0(k) : k \in \mathcal{K}_+\} > 0$ is a well-defined seed-gap constant. Let $\varepsilon_{\mathrm{BFP}} \in (0, p_*)$ denote the Banach fixed-point threshold of (ii) and (vii) below; by Prop. 10.3 the set $\{k \in \mathcal{K} : 0 < p_0(k) \le \varepsilon_{\mathrm{BFP}}\}$ is empty, so the entire conjecture is asserted uniformly for every $k \in \mathcal{K}_+$ and no low-probability fallback is required.

Joint coupled system. Let $\mathcal{R}$ denote the countable tree of finite record prefixes $r = (j_1, \dots, j_m)$ of positive $\mathbb{P}_\gamma$-probability, augmented at depth ∞ by the terminal label $k \in \mathcal{K}_+$; write $r(\omega, x)$ for the prefix realized on ω in $J^-(x)$, i.e. up to the retarded record time $t_\gamma(x)$ of (17). The back-reacted metric is the record-indexed family $\{g_r\}_{r\in\mathcal{R}} \subset \mathcal{B}_{g_0}$ of scaffolding metrics (no-further-records extension, Def. 10.1), assembled into the random metric g_ω by the *causal-wedge assembly* $g_\omega(x) := g_{r(\omega,x)}(x)$, and defined as the unique Barrabès–Israel weak solution (modulo diffeomorphism) of the *distributional* field equation on $\mathcal{M}^{\mathrm{ext}}$

$$G_{\mu\nu}[g_\omega]+\Lambda g_{\omega,\mu\nu} \;=\; \frac{8\pi G}{c^4}\Big[T^{\mathrm{RC}}_{\mu\nu}[g_\omega](x;\omega) + \sum_{m:\, s_m \le \tau_f} \delta_{\mathcal{N}_m}\, \nu^{(\omega,m)}_{ab}[g_\omega]\, e^a{}_\mu e^b{}_\nu + \delta_{\Sigma_{\tau_f}}\, S^{(\omega,f)}_{ab}[g_\omega]\, e^a{}_\mu e^b{}_\nu\Big], \qquad g_\omega \in \mathcal{B}_{g_0} \tag{19}$$

where $\mathcal{N}_m := \partial J^+(\gamma(s_m)) \cap \mathcal{M}^{\mathrm{ext}}$ is the future *achronal boundary* (record cone) of the m-th record event. By the horizon-regularity theory of [22, 23], $\mathcal{N}_m$ is a *Lipschitz* hypersurface generated by null geodesics without future endpoints; it is therefore a rectifiable set carrying a well-defined $(n-1)$-dimensional surface (Hausdorff) measure — so the surface delta $\delta_{\mathcal{N}_m}$ (resp. $\delta_{\Sigma_{\tau_f}}$ on the terminal Cauchy slice) and its integration against test forms are unambiguous — and it is C^1, with a smooth generator congruence and well-defined transversal, off the *cut locus* $\mathcal{C}_m \subset \mathcal{N}_m$, the set of generator endpoints (where generators leave ∂J^+ or two generators meet). By [23], $\mathcal{C}_m$ has vanishing $(n-1)$-dimensional Hausdorff measure and is of codimension ≥ 2 in $\mathcal{N}_m$. We define the shell on the *regular record cone* $\mathcal{N}_m^{\mathrm{reg}} := \mathcal{N}_m \setminus \mathcal{C}_m$: the Barrabès–Israel transverse-curvature jump and the surface stress $\nu^{(\omega,m)}_{ab}$ are constructed on $\mathcal{N}_m^{\mathrm{reg}}$, where the generators and transversal N^a are C^1, and $\delta_{\mathcal{N}_m}$ in (19) is read as $\delta_{\mathcal{N}_m^{\mathrm{reg}}}$ (equal as measures, since $\mathcal{H}^{n-1}(\mathcal{C}_m) = 0$); $e^a{}_\mu$ is the inclusion pull-back into $T\mathcal{M}^{\mathrm{ext}}$ on $\mathcal{N}_m^{\mathrm{reg}}$. The caustic set $\mathcal{C}_m$ (codimension ≥ 2) carries no *thin-shell* (codim-1) layer in (19), so the shell source is *amputated* there. We stress that codimension alone does *not* exclude a codim-2 (conical/strut) concentration on $\mathcal{C}_m$ — such a defect *is* exactly codim-2; its absence is not automatic but an explicit assumption, the separate residual (vi-b′) below, and is not part of the null junction (20). On each null cone the surface stress-energy is of Barrabès–Israel form [21]: with generators ℓ^a, a transversal N^a ($N\cdot\ell = -1$), intrinsic coordinates (λ, θ^A), induced 2-metric σ_{AB}, and transverse extrinsic curvature $C_{ab} := \frac{1}{2}(\mathcal{L}_N g)_{ab}$ restricted to $\mathcal{N}_m$,

$$\nu^{ab} \;=\; \mu\, \ell^a \ell^b + j^A\big(\ell^a e_A^{\,b} + e_A^{\,a} \ell^b\big) + p\, \sigma^{AB} e_A^{\,a} e_B^{\,b}, \qquad \mu = -\frac{c^4}{8\pi G}\,\sigma^{AB}[C_{AB}], \quad j^A = \frac{c^4}{8\pi G}\,\sigma^{AB}[C_{\lambda B}], \quad p = -\frac{c^4}{8\pi G}\,[\tag{20}$$

where $[\,\cdot\,]$ is the jump across $\mathcal{N}_m$ (exterior minus interior of the cone). On the *spacelike* terminal slice the surviving shell is of Israel–Darmois form with the signature normalization made explicit,

$$S_{ab} \;:=\; -\,\epsilon\,\frac{c^4}{8\pi G}\Big([K_{ab}] - h_{ab}\, h^{cd}[K_{cd}]\Big), \qquad \epsilon := n^\mu n_\mu \;\; (= -1 \text{ for the timelike normal of } \Sigma_{\tau_f} \text{ in mostly-plus sign} \tag{21}$$

the ϵ-factor being the standard normalization of the Lanczos equation for non-timelike hypersurfaces [20] (an earlier draft quoted the timelike-shell form, $\epsilon = +1$, for these spacelike slices; the record shells, now null, are governed by (20) instead). The shell data are determined by the posterior jumps as follows. At $\mathcal{N}_m$ the jump in the bulk source across the cone is the posterior update $\pi^\gamma_{s_m} - \pi^\gamma_{s_m^-}$ applied to the branch sources; its normal-component mismatch (four functions on $\mathcal{N}_m$: the null Hamiltonian and momentum constraint sources) determines the four shell functions (μ, j^A, p) — the data count of the null junction matches the constraint mismatch exactly, which is the technical reason the null-shell formulation replaces the spacelike one: the spacelike Lanczos system (4 constraint mismatches for 6 components of $[K_{ab}]$) is underdetermined, whereas the null system is square once the transverse-traceless part of $[C_{AB}]$ (impulsive gravitational-wave data, not shell stress) is fixed; we impose the *zero-impulsive-wave condition* $[C_{AB}]^{\mathrm{TT}} = 0$. We stress that this is *not* a gauge choice: the transverse-traceless part of $[C_{AB}]$ is impulsive gravitational-wave data, physically distinct from the shell stress, and setting it to zero is the *physical no-radiation postulate* that a measurement record emits no gravitational radiation as it is laid down — the natural feedback-gravity statement that adapted measurement back-reaction, entering as a classical (record-sourced) channel, does not itself radiate (cf. [130, 132]). This is a substantive restriction on the source class, not a free selection, and it bears on *existence*, not merely uniqueness: a $[C_{AB}]^{\mathrm{TT}} = 0$ solution of the null junction exists *only if* the posterior-jump constraint mismatch lands in the pure-trace-plus-vector subspace (μ, j^A, p sector); a mismatch with a genuine transverse-traceless component cannot be absorbed by shell data alone, and the junction is then *obstructed* — the weak solution (19) fails to exist, not merely to be unique (absorbing it would require an impulsive gravitational wave, contradicting the postulate). We do *not* claim this holds for a generic record source: a generic stress-energy update across the cone has nonzero TT projection. What we assert, as a *named source-class residual* (an explicit clause of (vi-b), on the same footing as (vi-b′)/(vi-b″)), is a genuine restriction to a *quadrupole-free* source class — and we flag explicitly that this is *not* implied by the uniformly-informative class $\mathfrak{P}_{\mathrm{UI}}$. That class is defined *statistically* (QND-repeated protocols with step-uniform likelihood-separation), and its geometric signature is *parallel, caustic-free* cones ((x-b)); parallel-and-caustic-free is *not* spherical symmetry and does *not* suppress a time-varying mass quadrupole. A generic record back-reaction, *even in* $\mathfrak{P}_{\mathrm{UI}}$, carries a nonzero radiative (TT) multipole, for which the $[C_{AB}]^{\mathrm{TT}} = 0$ junction is *obstructed* (no solution). The certified sub-class is therefore the strictly narrower one of records whose back-reaction is *quadrupole-free* — exactly spherically symmetric sources (Birkhoff: no radiative multipoles), or sources engineered so that higher multipoles are non-radiative — for which the mismatch lands in the pure-trace+vector subspace by symmetry. Outside this quadrupole-free sub-class the no-radiation postulate fails and the null junction may have no solution; this is disclaimed, not asserted away. At Σ_{τ_f} the jump is between $\mathbf{1}_{\{K=\cdot\}}$ and $\pi^{\gamma}_{\tau_f-}$. Under Ax. 7.1 the terminal jump vanishes a.s. (Thm. 7.3), so $S^{(\omega,f)}_{ab} \equiv 0$ and the metric is continuous across Σ_{τ_f}; when Ax. 7.1 fails, $\left\|S^{(\omega,f)}\right\|_{L^\infty}$ is controlled by the horizon-hidden residual of Cor. 9.2 via Thm. A.40 (Rem. 10.5). The pair $(g_r, \nu^{(r,m)}_{ab})$ is determined *jointly* by (19)–(20), prefix by prefix along the record tree, as the fixed point of the null-junction weak-solution map $\mathcal{F}^{(r)}$ of (vii) and (x) below: at each iteration $\mathcal{F}^{(r)}$ emits the pair $(g, \nu[g])$ jointly, and the shells are components of the fixed-point output at every iterate, not post-hoc surface layers appended after a bulk equation has been solved. Consequently the *displayed* equation is (19), and the bulk form (22) below is derived from it by restriction; the per-branch equation of the post-terminal region is recovered verbatim on $J^+(\Sigma_{\tau_f})$, where $T^{\mathrm{RC}}_{\mu\nu} = \mathrm{Tr}(\rho_2(\tau \mid K)\widehat{T}_{\mathrm{S},\mu\nu})$ by Def. 10.1.

Bulk restriction and support. On the open complement of the shell surfaces (the record cones $\mathcal{N}_m$ and the terminal slice), where all surface delta-functions vanish, equation (19) reduces to the smooth bulk equation

$$G_{\mu\nu}[g](x) + \Lambda g_{\mu\nu}(x) = \frac{8\pi G}{c^4}\, T^{\mathrm{RC}}_{\mu\nu}[g](x;\omega), \qquad g \in \mathcal{B}_{g_0}, \tag{22}$$

which on $J^+(\Sigma_{\tau_f})$ is the classical semiclassical Einstein equation with the K-conditioned source, exactly as in previous formulations. Equation (22) is *derived from* (19) by restriction, not an independent equation; the shell terms (20) are what make the restriction globally consistent as a Barrabès–Israel weak solution on all of $\mathcal{M}^{\text{ext}}$. Under the Cauchy-problem formulation (initial data on Σ_{τ_i} inherited from g_0), the support restriction is now the *causal adaptedness restriction*

$$\begin{aligned} &\text{(a)} \quad g_\omega \restriction_O \ \text{ is } \sigma\big(j_m : \gamma(s_m) \in J^-(O)\big)\text{-measurable for every open } O \subset \mathcal{M}^{\text{ext}} \setminus J^+(\Sigma_{\tau_f}), \\ &\text{(b)} \quad g_\omega \ = \ g_0 \quad \text{on} \ \ \mathcal{M}^{\text{ext}} \setminus \Big(\textstyle\bigcup_m J^+(\gamma(s_m)) \ \cup \ J^+(\Sigma_{\tau_f})\Big) \qquad \mathbb{P}^{\text{univ}}_\gamma\text{-a.s.,} \end{aligned} \tag{23}$$

with $J^-(O) := \bigcup_{x \in O} J^-(x)$ and causal relations as in Def. 10.1: the metric near x depends only on records causally available at x, and outside every record cone it is the prior-sourced background — deterministic, identical for all realizations and all (RP)-admissible protocols. Clause (b) and the prior-mean identity $\mathbb{E}_{\mathbb{P}_\gamma}[\rho^\gamma_{\text{RC}}(\tau)] = \rho(\tau)$ (Lem. 5.3(iii), Cor. 5.4) generalize the support restriction of previous drafts: for the trivial protocol the union of record cones is empty and $\mathcal{M}^{\text{ext}} \setminus J^+(\Sigma_{\tau_f}) = J^-(\Sigma_{\tau_f}) \setminus \Sigma_{\tau_f}$ (the terminal slice is Cauchy), so (23) reduces *exactly* to the former statement $g_k \restriction_{J^-(\Sigma_{\tau_f})} = g_0 \restriction_{J^-(\Sigma_{\tau_f})}$ for every $k \in \mathcal{K}_+$, which is thus the zero-record sector of the present scheme. The physically correct retardation is built in: a test apparatus responds to the recorded branch source only after light-crossing from the record event (it must enter $J^+(\gamma(s_m))$); in a laboratory the crossing time is negligible and the Page–Geilker phenomenology of Prop. 11.39 is unaffected. Each continuation across a record cone is Barrabès–Israel admissible: the induced degenerate metric matches across $\mathcal{N}_m$, the transverse-curvature jump is encoded as the null shell (20), and each shell vanishes identically when the posterior does not jump at s_m (the smooth-junction sub-case; automatic at Σ_{τ_f} under Ax. 7.1). Generically g_ω is a classical solution only in the Barrabès–Israel weak sense (smooth off the shell surfaces, with the prescribed surface layers). Each shell is supported on a measure-zero hypersurface: the metric is C^0 across it while its first transverse derivative jumps, so shells are invisible to pointwise metric readings but *are* detectable in any open set straddling the surface (an impulsive tidal kick on crossing geodesics). Detectability is consistent with Prop. 10.7: the cone $\mathcal{N}_m$ bounds $J^+(\gamma(s_m))$, so any observation of its kink already lies in the causal future of the record it broadcasts. The k-sector metric deviates from g_0 only inside the record cones — structurally, by (23)(b) — and completely, i.e. with the full $\rho_2(\cdot \mid K)$-sourced branch geometry, wherever $\pi^\gamma(K) = 1$ on the available records, in particular on all of $J^+(\Sigma_{\tau_f})$ under Ax. 7.1, preserving Prop. 10.7 in the record-relative and causal form stated there.

Joint fixed point and universal sample space. The induced family $\{(g_k, \nu^{(k)}_{ab}[g_k], \rho_2[g_k](\cdot \mid k))\}_{k \in \mathcal{K}_+}$ is the joint fixed point of $\mathcal{F}^{(k)}$ in the sense of (vii). The realization-wise statement is phrased on the *universal sample space*

$$\Omega^{\text{univ}}_\gamma \ := \ \bigsqcup_{k \in \mathcal{K}_+} \Omega^{[g_k]}_\gamma \big|_{K=k}, \quad \mathbb{P}^{\text{univ}}_\gamma(A) \ := \ \sum_{k \in \mathcal{K}_+} p_0(k)\, \mathbb{P}^{[g_k]}_\gamma\big(A \cap \Omega^{[g_k]}_\gamma \mid K = k\big), \tag{24}$$

on which the random pair (K, g) is jointly defined by the causal-wedge assembly $g(\omega)(x) := g_{r(\omega,x)}(x)$ (equal to $g_{K(\omega)}$ on $J^+(\Sigma_{\tau_f})$ and, in the zero-record sector, throughout). Two glosses keep this construction aligned with the record-adapted dynamics. First, with a nontrivial protocol the superscript $[g_k]$ is shorthand for the *prefix-adapted* law: the cylinder probabilities of (9) are built recursively down the record tree with the transported Kraus operators $\widetilde{M}^{(m)}_j = U[g_{r_{m-1}}](s_m, \tau_i)^\dagger M^{(m)}_j U[g_{r_{m-1}}](s_m, \tau_i)$ carrying the scaffolding metric of the realized prefix r_{m-1}; Kolmogorov consistency survives prefix-dependent propagators, since $\sum_j \widetilde{M}^*_j \widetilde{M}_j = U^\dagger(\sum_j M^*_j M_j)U = \mathbb{1}$ telescopes prefix by prefix, and the construction closes at the (x-c) fixed point. Second, the weights displayed as $p_0(k)$ are the operative prior $p(k)$ of (x-c), which equals $p_0(k)$ exactly in the zero-record sector under (T10) and to $O(\kappa^0)$ in general ((60)). Because every $k \in \mathcal{K}_+$ has $p_0(k) \geq p_* > \varepsilon_{\text{BFP}}$ and hence is in scope of the fixed-point statement, $\sum_{k \in \mathcal{K}_+} p_0(k) = 1$ by Lem. 2.5 on g_0, so $\mathbb{P}^{\text{univ}}_\gamma$ is a true probability measure. No kinematic extension of the sample space and no sub-probability defect appears. Per-realization covariant conservation of the source on g_k holds by Thm. 4.1 applied to the background g_k.

Proposition 10.3 ($\varepsilon_{\mathrm{BFP}}$ strictly below the seed gap under Pos. 11.30)**.** *Under Pos. 11.30 (finite-dimensional matter under the spectral cutoff $P_{\leq E_\star}$) the positive-weight set $\mathcal{K}_+ = \{k \in \mathcal{K} : p_0(k) > 0\}$ is finite and $p_* := \min_{k\in\mathcal{K}_+} p_0(k) > 0$. Hypothesis (ii) of Conj. 10.2 produces a Fréchet-Lipschitz constant for $g \mapsto \rho_2[g](\tau \mid k)$ controlled by $p_0(k)^{-1}$ on the sub-ball $\mathcal{B}_{g_0}^{(k)}$; combined with the contraction requirement (vii), any threshold $\varepsilon_{\mathrm{BFP}} \in (0, p_*)$ places every $k \in \mathcal{K}_+$ in scope. Choose $\varepsilon_{\mathrm{BFP}} := p_*/2$ (any value strictly below p_* suffices; the factor $1/2$ supplies a margin of safety for Sobolev- ball perturbations of $p[g](k)$ around $p_0(k)$). Then*

$$\{k \in \mathcal{K} : 0 < p_0(k) \leq \varepsilon_{\mathrm{BFP}}\} \;=\; \{k \in \mathcal{K}_+ : p_0(k) \leq p_*/2\} \;=\; \emptyset,$$

since $p_0(k) \geq p_*$ for every $k \in \mathcal{K}_+$. The low-probability fallback of prior drafts is therefore vacuous under Pos. 11.30, and the universal sample space (24) *is a probability measure on $\mathcal{K}_+$ with no kinematic extension.*

Proof. Finiteness of $\mathcal{K}_+$ and positivity of p_* follow from finite- dimensional spectral cutoff (Pos. 11.30) applied to the Born-law probability simplex $p_0(k) = \mathrm{Tr}(\rho[g_0](\tau_f)E_k)$ with finite positive-operator partition $\{E_k\}$ on the finite-dimensional matter Hilbert space. The Fréchet-Lipschitz bound with constant $O(p_0(k)^{-1})$ is hypothesis (ii) combined with the operator-square-root Lipschitz bound on the gapped subspace $\rho_{\mathrm{i}} \succeq \eta I$ of Pos. 11.30 (cf. proof of Prop. 11.49). With $\varepsilon_{\mathrm{BFP}} = p_*/2 < p_*$, every $k \in \mathcal{K}_+$ satisfies $p_0(k) \geq p_* > \varepsilon_{\mathrm{BFP}}$ and the stated set equality is definitional. □

Hypotheses required for a proof. (i) a rigorous definition of $\widehat{T}_{\mu\nu}[g]$ as a Banach-space-valued function of g on $\mathcal{B}_{g_0}$, with $\nabla^\mu_g \widehat{T}_{\mu\nu}[g] = 0$ for every $g \in \mathcal{B}_{g_0}$; (ii) Fréchet differentiability of $g \mapsto \rho_2[g](\tau \mid k)$ on the sub-ball where $p[g](k) \geq \varepsilon/2$, with a Lipschitz constant controlled by ε^{-1} and the gap of $\rho[g](\tau)$ away from singularity; (iii) boundedness of the retarded Green operator $\mathcal{G}_{g_0}$ of the linearized Einstein tensor between the appropriate function spaces; (*iv*) a smallness condition $\sqrt{\kappa}\,T\,\sup_{k:p_0(k)>\varepsilon} \left\|\widehat{T}_{\mu\nu}[g_0]\right\|_{\rho_2[g_0](\tau|k),\,\mathrm{op}} < 1$ with $\kappa = 8\pi G/c^4$ the Einstein coupling (so that $\sqrt{\kappa}\,T\,\left\|\widehat{T}_{\mu\nu}\right\|$ is dimensionless when one consistently inserts c-factors via the Bianchi-conserved stress-energy on g_0), where $\|\cdot\|_{\sigma,\mathrm{op}}$ is the operator norm on the GNS space of σ; (v) Fréchet differentiability of the propagator $g \mapsto U[g](\tau_f, \tau_i)$ on $\mathcal{B}_{g_0}$ in the operator norm, with Lipschitz constant L_U depending only on g_0 and the standard parametric-ODE constants on $\mathcal{B}_{g_0}$. (*vi*) *Junction regularity.* The following ingredients are imposed separately — the source bound (vi-a) and solvability (vi-b), together with the caustic residual (vi-b′) and the non-crossing/composition scope (vi-b″) — since an L^∞ bound on the bulk source does not by itself bound the geometric junction data:

- (vi-a) (Source bound) the conditioned source $\mathrm{Tr}(\rho_2[g_0](\tau \mid k)\widehat{T}_{\mathrm{S},\mu\nu}[g_0])$ is in L^∞ on Σ_{τ_f} (for the induced metric $h[g_0]$) and on each record cone $\mathcal{N}_m$ (for the induced degenerate metric and a fixed transversal), uniformly in k with $p_0(k) > \varepsilon_{\mathrm{BFP}}$ and in m;
- (vi-b) (*Null-junction solvability; named hypothesis replacing the former spacelike junction map.*) For each record cone $\mathcal{N}_m$, the Barrabès–Israel null-junction system [21] for the posterior-jump data — the four null constraint mismatches on $\mathcal{N}_m$ prescribing the four shell functions (μ, j^A, p) of (20), under the zero-impulsive-wave selection $[C_{AB}]^{\mathrm{TT}} = 0$ (the physical no-radiation postulate of (20), not a gauge fixing; it requires the constraint mismatch to lie in the pure-trace+vector subspace, else the TT$=0$ junction is obstructed) — admits a unique solution, and the map $\mathcal{J}_{\mathcal{N}_m}$ from constraint mismatch to shell data is bounded with operator norm $\leq C_{\mathrm{J}}(g_0)$ depending only on the background geometry near $\mathcal{N}_m$ (in particular uniformly in m). The same is assumed at the spacelike Σ_{τ_f} for the ϵ-normalized Lanczos system (21) whenever the terminal jump is nonzero (i.e. only when Ax. 7.1 fails); there the four constraint mismatches *under*determine the six components of $[K_{ab}]$, and uniqueness is restored by the analogous transverse-traceless

gauge selection $[K_{ab}]^{\mathrm{TT}} = 0$. (This converts (vi-a) into an L^∞ bound on the shells: $\left\|(\mu, j, p)^{(m)}\right\|_{L^\infty} \le C_{\mathrm{J}} \,\|\text{constraint mismatch at } \mathcal{N}_m\|_{L^\infty}$. Unlike the elliptic spacelike case, null-junction solvability is a system of *transport equations along the generators* [24]: the shell data (μ, j^A, p) propagate along each generator of $\mathcal{N}_m^{\mathrm{reg}}$ from the vertex, and are well-defined *up to the generator's endpoint on the cut locus* $\mathcal{C}_m$ — past a focal point the generator has left ∂J^+ and the transported data is not defined. This is the true content of (vi-b): not an analytic existence postulate over an ill-defined surface, but the assertion that the generator-affine transport of (μ, j^A, p) (i) exists and is bounded, with operator norm $\le C_{\mathrm{J}}(g_0)$, on the non-focused portion $\mathcal{N}_m^{\mathrm{reg}}$, and (ii) terminates at the cut locus, where the record source is amputated (the $\mathcal{N}_m^{\mathrm{reg}}$ construction below (19)). We therefore state (vi-b) as a genuine hypothesis on $\mathcal{N}_m^{\mathrm{reg}}$, not a standard-estimate remark, and isolate the caustic contribution as the separate item (vi-b′) below. The Israel compatibility relations — the distributional conservation constraints coupling (μ, j^A, p) to the jump of the bulk source's normal components across $\mathcal{N}_m$ — are part of the system solved here, cf. Rem. 10.4.)

(vi-b′) (*Caustic residual; codimension-≥ 2.*) The cut locus $\mathcal{C}_m$ ($\mathcal{H}^{n-1}(\mathcal{C}_m) = 0$) carries no thin-shell layer. We assume the record source class is such that the amputation of $\nu_{ab}^{(\omega,m)}$ on $\mathcal{C}_m$ introduces no distributional codimension-2 (conical/strut) defect. A *sufficient* condition is that the transported data admit a limit with integrable transverse curvature as generators approach $\mathcal{C}_m$; note this is a genuine regularity restriction, *not* automatic, since generic focusing drives the congruence expansion (hence the transverse curvature) to $-\infty$ at the caustic, so integrability up to $\mathcal{C}_m$ constrains the focusing rate. For laboratory record cones the whole issue is void (the relevant $\mathcal{N}_m$ is caustic-free out to the readout, Rem. 10.9); in general it is a named residual, distinct from the junction solvability (vi-b) and from the summability residual (E5).

(vi-b″) (*Non-crossing scope; shell composition.*) The weak solution (19) is stated for record-cone configurations in which the $\mathcal{N}_m$ do not cross in the observation region. Where two record cones intersect on a 2-surface $\mathcal{S}$, the junction is not the sum of the two shells: crossing null shells satisfy the algebraic continuity (Dray–'t Hooft–Redmount) relation of [25, 26], under which a geometric shell tensor is continuous across $\mathcal{S}$ and the shell data are redistributed (an energy exchange). We *assume* (a named hypothesis, parallel to (vi-b′)) that for the transverse crossings arising in the certified class this redistribution produces no new codimension-2 concentration on $\mathcal{S}$; this is the generic-transverse outcome but is not forced in general, and *non*-transverse crossings — in particular crossings that occur *at or near a cut locus* $\mathcal{C}_m$, where the DTR transversality hypothesis fails — are outside the scope of both this clause and (vi-b′) and are not certified. In the uniformly informative class $\mathfrak{P}_{\mathrm{UI}}$ of (x-b), records are discrete and rare and the cones are effectively parallel and caustic-free out to the readout, so crossings (transverse or otherwise) do not occur in the certified region and the gap is void there; for configurations with (transverse, caustic-free) crossings we adopt the DTR composition rule at each crossing 2-surface as part of the junction system, and carry its uniform solvability as an additional clause of (vi-b). Outside $\mathfrak{P}_{\mathrm{UI}}$, where cone crossings proliferate, the thin-shell class is already disclaimed (x-b).

Combined, (vi-a) and (vi-b), together with the caustic residual (vi-b′) and the non-crossing/composition scope (vi-b″), make the null-shell data $(\mu, j^A, p)^{(m)}$ defining (20) well-defined elements of $L^\infty(\mathcal{N}_m^{\mathrm{reg}})$ and the resulting weak solution exists in the Barrabès–Israel sense on the certified (caustic-free, non-crossing) class. (*vii*) *Banach contraction estimate.* Let $\mathcal{B}_{g_0}^{(k)} := \{g \in \mathcal{B}_{g_0} : p[g](k) \geq \varepsilon_{\mathrm{BFP}}/2\}$ be the positivity sub-ball on which hypothesis (ii) is stated. The composite map $\mathcal{F}^{(k)} : \mathcal{B}_{g_0}^{(k)} \to \mathcal{B}_{g_0}^{(k)}$ sending $g \in \mathcal{B}_{g_0}^{(k)}$ to the unique Barrabès–Israel weak solution of (19) on $\mathcal{M}^{\mathrm{ext}}$ (bulk form (22) off the shell surfaces) with shells (20)–(21) is well-defined (i.e. $\mathcal{F}^{(k)}$ sends $\mathcal{B}_{g_0}^{(k)}$ into itself, which is automatic for a sufficiently small Sobolev ball by continuity of $g \mapsto p[g](k)$

around g_0 where $p_0(k) > \varepsilon_{\mathrm{BFP}}$) and is a contraction with Lipschitz constant $\mathrm{Lip}(\mathcal{F}^{(k)}) \le \kappa T\, L_T\, \|\mathcal{G}_{g_0}\|_{\mathrm{op}} < 1$ for every k with $p_0(k) > \varepsilon_{\mathrm{BFP}}$, where L_T is the operator-Lipschitz constant of $g \mapsto \mathrm{Tr}(\rho_2[g](\tau \mid k)\widehat{T}_{\mathrm{S},\mu\nu}[g])$ on $\mathcal{B}_{g_0}$ produced by combining (ii), (iv), (v), and the entropy-controlled bound of Conj. 11.4's "Non-vacuity" remark ($L_T \le L_T^{(0)} + O(\mathcal{S}^{1/2}/(p_*\eta))$ in the small- entropy regime), and $\|\mathcal{G}_{g_0}\|_{\mathrm{op}}$ is the operator norm of the linearised retarded Green operator from (iii). The smallness condition (iv) and the entropy bound of Conj. 11.4 ensure $\mathrm{Lip}(\mathcal{F}^{(k)}) < 1$ on $\mathcal{B}_{g_0}$, which is exactly the contraction required by Banach–Caccioppoli; without (vii) the relative-norm estimates (i)–(vi) are necessary but not sufficient. (*viii*) *Branchwise conditional-law existence.* For every k with $p_0(k) > \varepsilon_{\mathrm{BFP}}$, the back-reacted Born probability remains positive: $p[g_k](k) := \mathbb{P}_\gamma^{[g_k]}(K = k) > 0$, equivalently $\mathrm{Tr}(\rho[g_k](\tau_f)E_k) > 0$. (By the seed-prior identity (60) together with (T10) of Conj. 11.56, this holds automatically with $p[g_k](k) = p_0(k) > \varepsilon_{\mathrm{BFP}}$, so (viii) is implied by (T10) and not an independent assumption; we list it explicitly here so that the conditional law $\mathbb{P}_\gamma^{[g_k]}(\,\cdot \mid K = k)$ used in (24) is well-defined sector-by-sector.) (*ix*) *Observer stability across the metric ball.* The Part I worldline/filtration data $(\gamma, \tau, \mathcal{A}(\Sigma_\tau))$ persist for every $g \in \mathcal{B}_{g_0} \cup \{g_k : p_0(k) > 0\}$: τ is g-Cauchy temporal, γ is g-timelike with $\tau(\gamma(\tau)) = \tau$, and Ax. 2.2 (M1)–(M3) holds for every such g on the same algebra $\mathcal{A}$. By structural stability of Cauchy temporal functions [9] this is automatic on a sufficiently small ball $\mathcal{B}_{g_0}$; $\mathcal{B}_{g_0}$ is restricted to the open subset on which it holds. (This is exactly (T13) of Conj. 11.56.) (*x*) *Record-tree closure.* The fixed-point package (i)–(ix) is imposed with the branch index k replaced by the finite record prefix r throughout (the label k entering at the terminal stage), the scaffolding metrics g_r taken in the no-further-records extension of Def. 10.1, together with:

(x-a) *uniform contraction across the record tree*: the constants of (ii), (iv), (v), (vii) hold uniformly over prefixes r with $\mathbb{P}_\gamma(r) > 0$, so each $\mathcal{F}^{(r)}$ contracts with a common constant $q < 1$. The apparent obstruction — the record tree has no analogue of the seed gap p_* of Prop. 10.3, since $\inf_r \mathbb{P}_\gamma(r) = 0$ with depth while the terminal-stage Lipschitz constants scale as an inverse probability — is defused by convexity: the adapted source (18) is a π-weighted *convex combination* of the branch sources, so its g-Lipschitz constant is dominated by the worst terminal-sector constant, which the p_*-gap does bound. What (x-a) genuinely adds beyond the terminal-stage package is (1) that the scaffolding solutions g_r remain in the common ball $\mathcal{B}_{g_0}$ on which those constants were computed (fed by (x-b)) and (2) that the per-prefix conditional laws are well-defined (fed by (x-c)); (x-a) is asserted on the protocol class $\mathfrak{P}_{\mathrm{UI}}$ of (x-b);

(x-b) *summable record shells, on the uniformly informative protocol class*: let $\mathfrak{P}_{\mathrm{UI}}$ denote the class of (RP)-admissible protocols of QND-repeated type on the terminal pointer sectors whose per-step outcome likelihoods separate every pair of live sectors with a step-uniform gap $\delta > 0$ (the purification/irreducibility condition of the repeated-measurement limit theorems [177, 175, 176]). For $\mathcal{P}_\gamma \in \mathfrak{P}_{\mathrm{UI}}$ the posterior concentrates exponentially, $1 - \pi^\gamma_{s_n}(K) \le C_{\mathrm{UI}} e^{-\delta' n}$ a.s. for some $\delta' > 0$ (exponential sector selection [176]), hence the increments are a.s. geometrically summable, $\sum_m \left\|\pi^\gamma_{s_m} - \pi^\gamma_{s_m^-}\right\|_{\ell^1} < \infty$, and by the junction bound (vi-b) with the duality estimate $\left\|(\mu, j, p)^{(\omega,m)}\right\| \le C_{\mathrm{J}} M \left\|\pi^\gamma_{s_m} - \pi^\gamma_{s_m^-}\right\|_{\ell^1}$ the cumulative null-shell data $\sum_m \left\|(\mu, j, p)^{(\omega,m)}\right\|_{L^\infty(\mathcal{N}_m)}$ is finite a.s. and, for κT small, keeps $g_\omega \in \mathcal{B}_{g_0}$; the hypothesis content is the uniformity of $(C_{\mathrm{UI}}, \delta')$ over the metric ball. *Outside* $\mathfrak{P}_{\mathrm{UI}}$ the claim fails in general and we do not assert it: the martingale L^2-increment identity $\sum_m \mathbb{E}[(\pi^\gamma_{s_m}(k) - \pi^\gamma_{s_{m-1}}(k))^2] = \mathrm{Var}(\pi^\gamma_{\tau_f-}(k)) \le \frac{1}{4}$ controls only ℓ^2-summability in expectation, $\ell^2 \not\Rightarrow \ell^1$, and in the continuous-monitoring (diffusive) limit the posterior path has a.s. infinite total variation, so the cumulative shell data diverge and the thin-shell class is the wrong regularity class — that limit would require a white-noise-sourced formulation in the manner of [130], which we do not construct. (x-b) is therefore a genuine scope restriction to discrete, uniformly informative record protocols, not a technicality;

(x-c) *self-consistent Born weights and record law*: the Born map $w \mapsto \big(\mathrm{Tr}(\rho[g_\cdot(w)](\tau_f)E_k)\big)_k$, with $g.(w)$ the adapted solution built from prior w, has a unique fixed point p in the simplex neighborhood $\|p - p_0\| \leq O(\kappa T L_U M)$, and Part I is run with prior p. The fixed point is genuinely over the *pair* (prefix-metric family, record law): the transported Kraus operators of (9) inherit the scaffolding metric of the realized prefix (see the gloss after (24)), so the cylinder law is constructed recursively down the tree, and (x-c) asserts contraction of this joint update, of which the displayed simplex fixed point is the K-marginal. (Under the trivial-protocol support restriction the slab propagator was g_0-generated and $p = p_0$ exactly, Prop. 12.5; with adapted back-reaction that identity holds at zeroth order in κ and (x-c) is the fixed-point replacement anticipated in the footnote to (60). The martingale property of π^γ_τ, the collapse theorems, and Prop. 10.7 are insensitive to this replacement: they are statements about conditional expectations under *whichever* law the fixed point selects.);

(x-d) *terminal-POVM stability*: the terminal PVM $\{E_k\}$ is the CHT-selected pointer PVM of App. L (Conj. L.1), selected on the seed background g_0. It is hypothesized stable across the metric ball and the record tree: for every scaffolding metric $g_r \in \mathcal{B}_{g_0}$ the CHT selection on g_r exists and yields effects $E_k[g_r]$ with $\sup_k \|E_k[g_r] - E_k\|_{\mathrm{op}} \leq L_E \|g_r - g_0\|_{\mathcal{B}_{g_0}}$, the induced sector residual being absorbed into the (vii)/(x-a) contraction budget. Without (x-d) the conditioning target itself would be sector-relative (the $\delta\Lambda$-variance of App. U makes the horizon data g-dependent) and the fixed point would range over the POVM as well. (x-d) is the metric-ball rigidity input to the Cartan-masa selection of App. L: local rigidity of the Cartan selection under $\mathcal{B}_{g_0}$ perturbations is the stability half of the kill-test of Rem. L.3.

Under (i)–(x) the Banach fixed-point theorem applied prefix by prefix along the record tree, and separately on each k-sector with $p_0(k) > \varepsilon_{\mathrm{BFP}}$ at the terminal stage, yields the conclusion (the fixed-point map $\mathcal{F}^{(r)}$ of (vii) and (x) sends a $g \in \mathcal{B}_{g_0}$ to the unique Barrabès–Israel weak solution of (22) off the shell surfaces with the prescribed shells (20)–(21); (vi) and (x-b) ensure the shells are bounded and the weak solution is in $\mathcal{B}_{g_0}$, (vii) and (x-a) supply the actual contraction constant, (viii) and (x-c) make the per-sector conditional law well-defined, (ix) makes the Part I observer structure available on the back-reacted metrics, and (x-d) holds the conditioning target fixed across the iteration). Establishing (i)–(iii) rigorously in a QFT-on-curved-spacetime framework is open.

Remark 10.4 (Why the adapted ρ_{RC} source is conservation-consistent (and why naive ρ_{RC}-sourcing was not)). By Thm. 3.7, each branch $\rho_2(\tau \mid k)$ is a unitary flow, so each branch source is covariantly conserved per realization (Thm. 4.1). The weight field $x \mapsto \pi^\gamma_{t_\gamma(x)}(k)$ of (18) is piecewise constant: it is constant on each open region between consecutive record cones $\mathcal{N}_m, \mathcal{N}_{m+1}$ (there $t_\gamma(x) \in [s_m, s_{m+1})$ and records arrive only at the s_m), so on each such region the adapted source is a *fixed* convex combination of conserved tensors and $\nabla^\mu T^{\mathrm{RC}}_{\mu\nu} = \sum_k \pi^\gamma_{s_m}(k)\, \nabla^\mu \operatorname{Tr}(\rho_2(\cdot \mid k)\widehat{T}_{\mathrm{S},\mu\nu}) = 0$ by linearity. Across a record cone the posterior jumps, and the jump is absorbed as the Barrabès–Israel null shell $\nu^{(\omega,m)}_{ab}$ of (19). For a metric that is smooth off a shell surface with continuous induced metric, the contracted Bianchi identity holds distributionally as a geometric identity; imposing the field equation (19) then forces the total source to be distributionally conserved, which decomposes into exactly the junction system solved in (vi-b): the Lanczos-type relation (20) *plus* the compatibility relations coupling the shell data (μ, j^A, p) to the jump of the bulk source's normal components across $\mathcal{N}_m$ [20, 21]. We emphasize (correcting an earlier draft of this remark) that the Lanczos relation alone does not deliver conservation; it is the full junction system of (vi-b), whose solvability for arbitrary posterior-jump data is hypothesized there, that does. Granting (vi-b), conservation holds distributionally on all of $\mathcal{M}^{\mathrm{ext}}$ per realization. Coupling $\rho^\gamma_{\mathrm{RC}}$ to a *smooth* metric — the proposal rejected in earlier drafts of this remark — would indeed violate the Bianchi identity at every record update; the Israel-weak-solution formulation, already required at Σ_{τ_f}, is what renders the adapted source admissible. The genuinely inadmissible alternative is sourcing the observable past by the K-conditioned $\rho_2(\tau \mid k)$ itself: that source anticipates the terminal record, and future-conditioned (post-selected) sourcing is known to generate superluminal signalling in final-state models [137, 138]. The adapted scheme is the unique member of the family {prior, ρ_{RC}-adapted, $\rho_2(K)$} that is simultaneously conservation-consistent, non-anticipating, and Page–Geilker-consistent (Prop. 11.39).

Remark 10.5 (Terminal-shell information budget). Under Ax. 7.1 the terminal shell vanishes ($S_{ab}^{(\omega,f)} \equiv 0$ a.s.) and this remark is vacuous. When Ax. 7.1 fails, the terminal jump $\mathbf{1}_{\{K=\cdot\}} - \pi^{\gamma}_{\tau_f-}$ is nonzero with positive probability, and the shell at Σ_{τ_f} broadcasts into $J^+(\Sigma_{\tau_f})$ classical information about K that the pre-terminal records did not carry — at most $H(K \mid \overline{\mathcal{F}}^{\gamma}_{\tau_f-})$ bits (Thm. A.40(iii)). This is not a signalling pathology: the broadcast is confined to the causal future of the terminal slice. It is, however, a thermodynamic bookkeeping obligation: the generalized second law must charge the broadcast of $H(K \mid \overline{\mathcal{F}}^{\gamma}_{\tau_f-})$ bits against the horizon-entropy budget of Cor. 9.2, consistently with the GSL usage elsewhere in the paper (Rem. 10.12). Quantitatively the shell is bounded by the same residual, $\left\|S^{(\omega,f)}\right\|_{L^\infty} \leq C_{\mathrm{J}} M \left\|\mathbf{1}_{\{K=\cdot\}} - \pi^{\gamma}_{\tau_f-}\right\|_{\ell^1}$, whose expectation is $\leq 2C_{\mathrm{J}} M \min\{H, \sqrt{H/2}\}$ with $H = H(K \mid \overline{\mathcal{F}}^{\gamma}_{\tau_f-})$ (Thm. A.40(ii)). We record this as bookkeeping, not as a theorem: no quantitative GSL ledger for the terminal shell is constructed here.

Remark 10.6 (Coupling to post-selection; scope of the no-signalling defense). Because g_ω encodes the terminal record $k = K(\omega)$ via the dependence of $\rho_2(\tau \mid k)$ on $\Lambda_k(\tau_f)$, the semiclassical metric of (22) is *post-selection-dependent*: it is well-defined only relative to the two-boundary data $(\rho_{\mathrm{i}}, \{E_k\})$ specified at τ_i and τ_f. Thm. 6.1 addresses only the *matter-side* no-signalling problem: for any local POVM in $\mathcal{A}(\Sigma_\tau)$ at $\tau < \tau_f$, the marginal outcome distribution after summing over K is independent of any local intervention spacelike-separated from it, on a fixed background. (The same averaging-over-K collapse appears at the level of the two-time quasiprobability: by Thm. 3.10(iv) the prior-averaged Margenau–Hill marginal obeys $\sum_k p(k)\, M_k = \rho \succeq 0$, so any per-branch temporal non-classicality is invisible after marginalisation over the terminal record.) This does *not* prove that an observer who can probe the K-dependent classical metric g_ω at $\tau < \tau_f$ cannot recover information about K: the metric is not an element of the matter algebra $\mathcal{A}(\Sigma_\tau)$ and the theorem makes no claim about it. Operational consistency of Conj. 10.2 therefore requires the additional *geometric* no-signalling postulate that no pre-terminal classical observation of g_ω in any open set $O \subset J^-(\Sigma_\tau) \setminus J^+(\Sigma_{\tau_f})$ can statistically distinguish the sectors $\{g_k\}_{k:p_0(k)>\varepsilon_{\mathrm{BFP}}}$ at finer resolution than the realized record posterior π^{γ}_{τ}. Equivalently, the k-discrimination map $g \mapsto k$ from observed metric data to terminal label is a measurable function of the classical record already in hand, not an anticipation of the terminal readout. We record this as an explicit hypothesis on top of the matter no-signalling theorem; it cannot be derived from Thm. 6.1 alone. The conjecture's physical meaning is therefore: the metric is the *retrodictively reconstructed* background consistent with both the initial preparation and the terminal record, and pre-terminal observers see it only through the realized record posterior π^{γ}_{τ}; the prior p_0 governs the zero-record sector and the ensemble mean.

Proposition 10.7 (Geometric no-signalling, record-relative and causal form, automatic under (23))**.** *In the setting of Conj. 10.2 with the causal adaptedness restriction* (23)*, fix $\tau \in [\tau_i, \tau_f)$, any open $O \subset J^-(\Sigma_\tau) \setminus J^+(\Sigma_{\tau_f})$, and any classical (commutative) functional $\Phi : \mathrm{Met}(O) \to \mathbb{R}^N$ of the metric data on O. Let $\mathcal{M}_\tau$ denote the σ-algebra generated by all such $\Phi(g_\omega \restriction_O)$ on the universal sample space* (24)*. Then:*

(i) *(*No leakage beyond records.*) $\mathcal{M}_\tau \subseteq \overline{\mathcal{F}}^{\gamma}_{\tau}$ mod $\mathbb{P}^{\mathrm{univ}}_{\gamma}$-null sets, and*

$$\mathbb{P}^{\mathrm{univ}}_{\gamma}\big(K = k \bigm| \mathcal{M}_\tau \vee \overline{\mathcal{F}}^{\gamma}_{\tau}\big) \;=\; \pi^{\gamma}_{\tau}(k) \qquad \mathbb{P}^{\mathrm{univ}}_{\gamma}\text{-a.s.} \tag{25}$$

for every $k \in \mathcal{K}$ with $p_0(k) > \varepsilon_{\mathrm{BFP}}$: pre-terminal metric observations carry no information about K beyond the already-realized matter records.

(ii) *(*No anticipation.*) For $s \leq \tau$, conditioning on metric data in $J^-(\Sigma_s)$ alone gives $\mathbb{P}^{\mathrm{univ}}_{\gamma}(K = k \mid \mathcal{M}_s) = \mathbb{E}[\pi^{\gamma}_{s}(k) \mid \mathcal{M}_s]$, an $\overline{\mathcal{F}}^{\gamma}_{s}$-conditional quantity; in particular no functional of the* *metric on $J^-(\Sigma_s)$ is correlated with the record increments $(j_m)_{s_m>s}$ beyond their $\overline{\mathcal{F}}^\gamma_s$-conditional law.*

(iii) *(*Prior-mean form; zero-record sector.*) Unconditionally, $\mathbb{E}_{\mathbb{P}^{\text{univ}}_\gamma}[\mathbb{P}^{\text{univ}}_\gamma(K = k \mid \mathcal{M}_\tau)] = p(k)$ (the (x-c) fixed-point prior; $= p_0(k)$ in the zero-record sector and at $O(\kappa^0)$ in general); and on the zero-record region ($O \subset J^-(\Sigma_{s_1})$, or any O for the trivial protocol) $g_\omega\restriction_O = g_0\restriction_O$ is deterministic and $\mathbb{P}^{\text{univ}}_\gamma(K = k \mid \Phi(g_\omega\restriction_O)) = p_0(k)$ exactly — the statement of the previous version of this proposition.*

(iv) *(*Structural causal safety; protocol-counterfactual independence.*) If $O \cap J^+(\gamma(s_m)) = \emptyset$ for every m (in particular, whenever O is spacelike to all of γ's record events), then $g_\omega\restriction_O = g_0\restriction_O$ a.s., so every $\Phi(g_\omega\restriction_O)$ is a.s. constant. In particular its* law *— not merely its mean — is a point mass independent of both the realization and the protocol: any two (RP)-admissible protocols $\mathcal{P}_\gamma, \mathcal{P}'_\gamma$ induce the same (deterministic) metric statistics on every region spacelike to all of their record events. The distribution-level protocol sensitivity of the slicing-based scheme of earlier drafts is thereby closed by construction; inside the record cones, protocol dependence is confined to the causal future of the corresponding record events and is unobjectionable.*

Proof. (i) For $x \in O \subset J^-(\Sigma_\tau)$ and any record with $\gamma(s_m) \in J^-(x)$ we have $\gamma(s_m) \in J^-(\Sigma_\tau)$, hence $s_m \leq \tau$ (the worldline crosses each Cauchy slice once and $J^-(\Sigma_\tau) = \{\tau(\cdot) \leq \tau\}$); so $\sigma(j_m : \gamma(s_m) \in J^-(O)) \subseteq \overline{\mathcal{F}}^\gamma_\tau$, and by (23)(a) $g_\omega\restriction_O$ is $\overline{\mathcal{F}}^\gamma_\tau$-measurable; measurability of Φ gives $\mathcal{M}_\tau \subseteq \overline{\mathcal{F}}^\gamma_\tau$ mod null. Hence $\mathcal{M}_\tau \vee \overline{\mathcal{F}}^\gamma_\tau = \overline{\mathcal{F}}^\gamma_\tau$ and the conditional probability is $\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}^\gamma_\tau] = \pi^\gamma_\tau(k)$ by Def. 5.2. (ii) Tower property over $\mathcal{M}_s \subseteq \overline{\mathcal{F}}^\gamma_s$; the statement about record increments is the definition of conditional law given $\overline{\mathcal{F}}^\gamma_s$ plus $\mathcal{M}_s \subseteq \overline{\mathcal{F}}^\gamma_s$. (iii) $\mathbb{E}[\pi^\gamma_\tau(k)] = \mathbb{P}_\gamma(K = k) = p(k)$ is Lem. 5.3 under the (x-c) law; the zero-record statement repeats the proof of the previous version: such O is disjoint from all shell surfaces, (23)(b) gives $g_\omega\restriction_O = g_0\restriction_O$, so $\Phi(g_\omega\restriction_O) = \Phi(g_0\restriction_O)$ is a constant random variable, and conditioning on a constant gives $\mathbb{P}^{\text{univ}}_\gamma(K = k \mid \Phi(g_\omega\restriction_O)) = \mathbb{P}^{\text{univ}}_\gamma(K = k) = p_0(k)$ by (24). (iv) The hypothesis on O places it in $\mathcal{M}^{\text{ext}} \setminus (\bigcup_m J^+(\gamma(s_m)) \cup J^+(\Sigma_{\tau_f}))$, so (23)(b) gives $g_\omega\restriction_O = g_0\restriction_O$ a.s. under *each* admissible protocol's law separately; g_0 is the protocol-independent prior-sourced background (its construction uses only $(\rho_{\text{i}}, \{E_k\}, g_0)$-seed data), so $\Phi(g_0\restriction_O)$ is one and the same constant for $\mathcal{P}_\gamma$ and $\mathcal{P}'_\gamma$. □

Remark 10.8 (Status of Prop. 10.7). Prop. 10.7 is the gravitational counterpart of the matter no-signalling theorem (Thm. 6.1) and is logically independent of it: it follows from the structural adaptedness restriction (23) added to Conj. 10.2, which remains a postulate. Its content is now *record-relative*: the metric is exactly as informative about K as the classical matter records already in hand, and no more. This is the correct operational requirement — a metric that revealed *less* than the records would be inconsistent with the records' own gravitational footprint (cf. Prop. 11.39), while a metric revealing *more* would signal from the terminal boundary. It matches the causal-safety criterion of the measurement-feedback semiclassical-gravity literature: source adapted to the record filtration, unconditioned mean governed by a linear (here: the prior two-boundary) law [130, 132, 133, 136, 134]. The former first caution of this remark — that the slicing-based scheme needed a covariant refinement conditioning each point x only on records in $J^-(x)$, with null shells on $\partial J^+(\gamma(s_m))$ — has been discharged: that refinement *is* now the definition (Def. 10.1, (19)), and its causal-safety payoff is the structural clause Prop. 10.7(iv). Two residual cautions are recorded honestly. First, the scheme remains relative to the record structure $(\gamma, \mathcal{P}_\gamma)$, consistent with the observer-relative type assignments used elsewhere (Rem. 13.1); the reconciliation of *several* worldlines' adapted metrics is an open structural problem (Rem. 10.9). Second, protocol-independence of the *mean* pre-terminal metric holds exactly at linearized order (Cor. 5.4 plus linearity of the retarded Green operator $\mathcal{G}_{g_0}$); beyond linearized order the mean of the solutions need not solve the mean equation, and *no bound on this nonlinear protocol dependence of the mean is currently hypothesized* — an earlier draft attributed such an $O(\kappa^2)$ bound to hypothesis (x), which contains no such item; the same open problem affects the entire feedback-gravity literature [132]. What the causal form does close, by construction, is the *distribution-level* dependence outside the record cones (Prop. 10.7(iv)); inside the cones, protocol dependence is causal and unobjectionable. This is the precise sense in which the absence of gravitational signalling remains within Conjecture 10.2's conditional scope. We do *not* claim uniqueness of the adaptedness restriction from a geometry-side theorem; it is the minimal structural assumption extending the matter no-signalling content to the gravitational sector. Conjecture 10.2 is therefore a *record-conditioned reconstruction* of the metric consistent with the initial preparation, the realized records, and (in their causal future) the terminal record — not an anticipatory back-reaction of the terminal boundary on the observable past.

Remark 10.9 (Relational status of the pre-terminal metric). The assembled metric g_ω is defined relative to the record structure $(\gamma, \mathcal{P}_\gamma)$: it is the geometry *licensed for apparatuses whose records are included in* $\mathcal{P}_\gamma$ (in the laboratory reading of Rem. 11.40, γ carries the joint environmental record of the experiment, so every apparatus in the lab is covered). The framework currently has *no* theorem reconciling the adapted metrics of two worldlines γ, γ' with independent protocols: the assemblies g^γ_ω and $g^{\gamma'}_\omega$ agree outside both record-cone families (both equal g_0 there, by (23)(b)) and on $J^+(\Sigma_{\tau_f})$ (both equal g_K under Ax. 7.1), but may disagree inside one cone family and outside the other, unless each protocol's records lie in the other's causal past by the time they matter. A multi-worldline extension — a consistency condition on overlapping record structures, presumably a common refinement protocol whose filtration contains both — is an open structural problem of the same character as the QRF-relative type assignments of Rem. 13.1, and Prop. 11.39(ii) is scoped accordingly (test apparatuses whose own records are included in $\mathcal{P}_\gamma$). Until that extension exists, cross-observer statements about the pre-terminal metric are outside the framework's certified scope.

Proposition 10.10 (Adapted-metric residual: convergence to the branch metric)**.** *Assume hypotheses (i)–(x) of Conj. 10.2, and let $\widetilde{g}_K$ denote the per-branch fixed-point metric sourced by $\rho_2(\cdot \mid K)$ throughout $\mathcal{M}^{\text{ext}}$. Then for every τ:*

(i) *(*Source residual; finite-τ form of Thm. A.40.*) Almost surely*

$$\left\|\rho^\gamma_{\text{RC}}(\tau) - \rho_2(\tau \mid K)\right\|_1 \;\le\; 2\big(1 - \pi^\gamma_\tau(K)\big), \qquad \mathbb{E}\big[1 - \pi^\gamma_\tau(K)\big] \;\le\; \min\big\{H_\tau,\ \sqrt{H_\tau/2}\big\},$$

with $\pi^\gamma_\tau(K) := \sum_k \mathbf{1}_{\{K=k\}}\pi^\gamma_\tau(k)$ and $H_\tau := H(K \mid \overline{\mathcal{F}}^\gamma_\tau)$ the running conditional entropy of the terminal label given the records (the $\tau < \tau_f$ specialization of the algebra proving Thm. A.40(i)–(ii), via Cor. A.31).

(ii) *(*Metric residual.*) On $J^-(\Sigma_\tau)$, with $q := \text{Lip}(\mathcal{F}) < 1$ the common contraction constant of (vii) and (x-a) and M the stress-energy bound (S4) of Conj. 11.4,*

$$\|\, g_\omega - \widetilde{g}_K \,\|_{\mathcal{B}_{g_0}} \;\le\; \frac{2\,\kappa\, T\, M\ \|\mathcal{G}_{g_0}\|_{\text{op}}}{1-q}\ \sup_{\tau' \le \tau}\big(1 - \pi^\gamma_{\tau'}(K)\big) \;+\; \Delta_{\text{shell}}(\tau),$$

where $\Delta_{\text{shell}}(\tau)$ collects the record null-shell data, bounded under (x-b). The finiteness of $\Delta_{\text{shell}}(\tau)$ as $\tau \uparrow \tau_f$ under (x-b)/(E5) is what guarantees the pre-terminal assembly converges to a C^0/BV terminal-limit metric; its agreement with $\widetilde{g}_K$ across Σ_{τ_f} is the (x-c) fixed-point matching, not an automatic identity. In particular, for protocols in the uniformly informative class $\mathfrak{P}_{\text{UI}}$ of (x-b), on any realization and any τ with $\pi^\gamma_{\tau'}(K) \ge 1-\varepsilon$ for $\tau' \in [\tau_{\text{dec}}, \tau]$, the cumulative posterior variation on the window is bounded by $C_{\text{TV}}\,\varepsilon$ with $C_{\text{TV}} = C_{\text{TV}}(C_{\text{UI}}, \delta')$ (geometric summability of the increments from the first crossing of the $1-\varepsilon$ level), and every classical metric observation in $J^-(\Sigma_\tau) \cap J^+(\Sigma_{\tau_{\text{dec}}})$ agrees with the branch metric $\widetilde{g}_K$ up to $O\big(\kappa T M\, \|\mathcal{G}_{g_0}\|_{\text{op}}\, (1 + C_{\text{J}} C_{\text{TV}})\, \varepsilon/(1-q)\big)$. (Pointwise closeness $\pi(K) \ge 1 - \varepsilon$ alone would not *suffice: with infinitely many protocol epochs a sup bound on the posterior does not control its total variation; the window bound is supplied by the (x-b) class.)*

Proof. (i) $\rho^\gamma_{\mathrm{RC}}(\tau) - \rho_2(\tau \mid K) = \sum_{k\neq K} \pi^\gamma_\tau(k)\rho_2(\tau \mid k) - (1 - \pi^\gamma_\tau(K))\rho_2(\tau \mid K)$; the triangle inequality with $\|\rho_2\|_1 = 1$ gives the bound $2(1 - \pi^\gamma_\tau(K))$ (the algebra of Cor. A.31). For the expectation bound, condition on $\overline{\mathcal{F}}^\gamma_\tau$: $\mathbb{E}[1 - \pi^\gamma_\tau(K) \mid \overline{\mathcal{F}}^\gamma_\tau] = \sum_k \pi^\gamma_\tau(k)(1 - \pi^\gamma_\tau(k)) \leq -\sum_k \pi^\gamma_\tau(k) \log \pi^\gamma_\tau(k)$ (termwise $q(1 - q) \leq -q \log q$ on $[0, 1]$), and take expectations to get $\mathbb{E}[1 - \pi^\gamma_\tau(K)] \leq H_\tau$; the square-root bound is Pinsker plus Jensen exactly as in the proof of Thm. A.40(ii), using $\mathbb{E}[-\log \pi^\gamma_\tau(K)] = H_\tau$. (ii) g_ω and $\widetilde{g}_K$ are fixed points of the maps $\mathcal{F}^{\mathrm{RC}}$ and $\mathcal{F}^{(K)}$, which differ only through their sources; the standard Banach fixed-point perturbation estimate $\|g_1 - g_2\| \leq (1 - q)^{-1} \sup_g \|\mathcal{F}_1(g) - \mathcal{F}_2(g)\|$, the boundedness (iii) of $\mathcal{G}_{g_0}$, the duality bound $\left|\mathrm{Tr}((\rho^\gamma_{\mathrm{RC}} - \rho_2(\cdot \mid K))\widehat{T})\right| \leq M \left\|\rho^\gamma_{\mathrm{RC}} - \rho_2(\cdot \mid K)\right\|_1$ from (S4), and part (i) give the bulk term; the shell term is the (x-b) bound on the cumulative Barrabès–Israel data generated by the posterior jumps in $[\tau_{\mathrm{dec}}, \tau]$. For the window statement, on the event $\pi^\gamma_{\tau'}(K) \geq 1 - \varepsilon$ throughout the window, exponential concentration ((x-b), class $\mathfrak{P}_{\mathrm{UI}}$) makes the post-crossing increments geometrically decaying with total mass $\leq C_{\mathrm{TV}}\varepsilon$; the duality bound of (x-b) applied to the constraint components then bounds the window shell data by $C_{\mathrm{J}} M C_{\mathrm{TV}}\varepsilon$. Without the class restriction this step fails: a sup bound on $1 - \pi(K)$ does not control the total variation of the posterior over infinitely many epochs. □

Remark 10.11 (Positive status of Conj. 10.2). Within the record-adapted scope fixed by (23) and the matter+geometric no-signalling postulates of Rem. 10.6 and Prop. 10.7, Conj. 10.2 is a conservation- and no-signalling-consistent semiclassical Einstein equation sourced by the record-conditioned two-boundary state. (We do *not* claim uniqueness from a derivation; uniqueness up to the listed structural postulates is what is established. The support condition (23) is itself a postulate, as Rem. 10.6 makes explicit.) Its conditional content (existence, uniqueness modulo diffeomorphism, and contractive Banach fixed-point selection of the record-indexed metrics g_r and the per-sector metric g_k on $J^+(\Sigma_{\tau_f})$) is a Banach fixed-point theorem under the full hypothesis package (i)–(x); in particular (vi) (junction regularity, now in Barrabès–Israel null form, with (vi-b) a named solvability hypothesis rather than a standard-estimate remark) and (vii) (the actual contraction estimate) are load-bearing additions to the original (i)–(v), and (viii)–(x-d) ensure the per-sector conditional law, the Part I observer structure, the record-tree closure, and the stability of the conditioning POVM are available on the metric ball. The remaining open analytic problems are (i), (iii), the QFT-on-curved-spacetime inputs underlying (vii) (Hadamard renormalisation, linearised retarded Green operator bounds, and operator-Lipschitz stability of the propagator), the null-junction solvability of (vi-b) (generator-affine transport on the regular record cone $\mathcal{N}^{\mathrm{reg}}_m$, up to the cut locus) together with the codimension-≥ 2 caustic residual (vi-b′), and the constants of (x-b) — noting that (x-b) is now an explicit *scope restriction* to the uniformly informative protocol class, outside of which the thin-shell formulation is asserted to fail rather than conjectured to hold; (viii)–(ix) are structural and follow on a sufficiently small Sobolev ball by standard PDE structural-stability arguments. We therefore record the conjecture as a *conditional theorem on the post-terminal future given the full structural package*: its rigorous content is the existence and uniqueness assertion for $g_k \restriction_{J^+(\Sigma_{\tau_f})}$ for sectors with $p_0(k) > \varepsilon_{\mathrm{BFP}}$ (with the trivial $g_k = g_0$ for the low-probability sectors), and the pre-terminal sector dependence is exactly record-borne (Prop. 10.7) given the adaptedness postulate.

Remark 10.12 (Achronal-ANEC self-consistency of the conditioned source). We flag one global-consistency constraint that is *not* implied by the per-realization conservation of Thm. 4.1 nor by the state-independent spacetime-smeared quantum energy inequality recorded in App. E (whose conditioned-source floor (111) bounds only timelike-contracted smeared components): because the conditioned source $\langle \widehat{T}_{\mu\nu} \rangle_{\rho_2[g_k]}$ is a two-vector (ABL) functional of $(\rho_{\mathrm{i}}, \{E_k\})$ rather than a single Hadamard expectation, it is not *a priori* subject to the Fewster–Verch bound, and post-selected (weak-value) energy densities are known to admit anomalously large negative excursions that evade the single-Hadamard-state inequality. The rigorous per-branch form of this two-vector negativity is Thm. 3.10: the symmetric (Margenau–Hill) marginal $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$ that carries the sign of the two-time/ABL statistics fails to be positive precisely when $[\rho(\tau), \Lambda_k(\tau)] \neq 0$ (clean iff for projective E_k; generic-only for POVMs), whereas the operator $\rho_2[g_k]$ actually sourcing (22) is always positive ($M_k - \rho_2 = \frac{1}{2p}[\rho^{1/2}, [\rho^{1/2}, \Lambda_k]]$). Thus the anomalous weak-value negativity is a per-branch quasiprobability effect, not a property of the gravitational source itself, and it collapses under the prior average $\sum_k p(k) M_k = \rho \succeq 0$ (the no-signalling identity, Thm. 6.1); the achronal-ANEC admissibility condition below is the residual global constraint not captured by this per-slice positivity. We therefore impose, as an explicit admissibility condition on the two-boundary data $(\rho_{\rm i}, \{E_k\}, g_0)$, the *self-consistent achronal averaged null energy condition* of Graham–Olum [391]: every admissible per-sector solution g_k must satisfy $\int_\gamma \langle \widehat{T}_{\mu\nu} \rangle_{\rho_2[g_k]} \dot\gamma^\mu \dot\gamma^\nu \, d\lambda \geq 0$ on every complete achronal null geodesic γ of g_k. Under the record-adapted scheme the same admissibility is imposed on the *assembled* metric g_ω *including its shells*: a complete achronal null geodesic crossing (or generating a segment of) a record cone $\mathcal{N}_m$ picks up the impulsive contribution of the shell data (20) — a sign-indefinite δ-term weighted by $\nu_{ab}\dot\gamma^a\dot\gamma^b$ — and these terms are included in the ANEC integral realization by realization; per-sector ANEC of the smooth g_k alone does not suffice, and under (x-b) the cumulative impulsive contribution is finite so the condition is well-posed. In self-consistent semiclassical gravity this is precisely the condition that forbids the closed timelike curves, traversable wormholes and causality violations to which a retrocausal source is otherwise exposed; for minimally coupled fields it is the content of Wall's theorem deriving achronal ANEC from the generalised second law [55], the same GSL/Araki-monotonicity structure used elsewhere in this paper. Whether achronal ANEC of the conditioned source *follows* from seed-state ANEC plus the contraction estimate of Conj. 11.4, or must be retained as the independent admissibility postulate stated here, is left open; it is a residual external constraint of the same character as the items of Sec. 12.5.

10.2 Conjecture 2: Modified Wheeler–DeWitt constraint

Let $\mathcal{H}_{\rm clock}$ carry a self-adjoint operator $\widehat{H}_{\rm c}$ with absolutely continuous spectrum and generalized eigenbasis $\{|t\rangle\}_{t\in\mathbb{R}}$ with $\langle t|\widehat{H}_{\rm c} = -i\hbar\partial_t\langle t|$ in the sense of rigged Hilbert spaces [8]. Let $\widehat{S}_\delta := e^{-i\delta\widehat{H}_{\rm c}/\hbar}$ (a one-parameter unitary group), and let $\widehat{\Delta}_\delta := \widehat{S}_\delta - \mathbb{1}$ denote the (non-self-adjoint) forward-shift operator. Its self-adjoint symmetrisation is

$$\widehat{\Sigma}_\delta \;:=\; \tfrac{1}{2}\big(\widehat{\Delta}_\delta + \widehat{\Delta}_\delta^\dagger\big) \;=\; \cos\!\big(\delta\widehat{H}_{\rm c}/\hbar\big) - \mathbb{1}, \tag{26}$$

which is bounded self-adjoint with $\widehat{\Sigma}_\delta \preceq 0$, $\widehat{\Sigma}_0 = 0$, and $\widehat{\Sigma}_\delta = -\frac{1}{2}(\delta\widehat{H}_{\rm c}/\hbar)^2 + O(\delta^4\widehat{H}_{\rm c}^4)$ in the strong sense on analytic vectors of $\widehat{H}_{\rm c}$.

Conjecture 10.13 (Modified Wheeler–DeWitt constraint; rigged-Hilbert form)**.** Let $\Phi \subset \mathcal{H}_{\rm clock} \otimes \mathcal{H}_{\rm sys} \subset \Phi'$ be a Gel'fand triple with Φ a nuclear space of joint analytic vectors of $\widehat{H}_{\rm c}$ and $\widehat{H}_{\rm s}$, stable under $\widehat{\Sigma}_{\delta t}$, and let

$$\widehat{C}_{\alpha,\delta t} := \widehat{H}_{\rm c} \otimes \mathbb{1} + \mathbb{1} \otimes \widehat{H}_{\rm s} + \alpha\, \widehat{\Sigma}_{\delta t} \otimes \mathbb{1},$$

essentially self-adjoint on Φ (cf. Status paragraph). There exist constants $\alpha \in \mathbb{R}$ and $\delta t > 0$ such that the self-adjoint constraint

$$\widehat{C}_{\alpha,\delta t}\Psi \;=\; 0 \tag{27}$$

admits a non-trivial *physical state space*, namely the distributional kernel

$$\mathcal{H}_{\rm phys} \;:=\; \big\{\Psi \in \Phi' \;:\; \widehat{C}_{\alpha,\delta t}\Psi = 0 \text{ in } \Phi'\big\}, \tag{28}$$

equipped with the rigging (group-averaging) inner product

$$\langle \eta(\psi) \,|\, \eta(\varphi)\rangle_{\rm phys} \;:=\; \big\langle \psi \;\big|\; \delta(\widehat{C}_{\alpha,\delta t}) \;\big|\; \varphi\big\rangle, \qquad \eta(\psi) := \tfrac{1}{2\pi}\int_{\mathbb{R}} e^{is\widehat{C}_{\alpha,\delta t}/\hbar}\psi\, ds \;\in \Phi', \quad \psi \in \Phi, \tag{29}$$

defined via the spectral measure $E_C(\cdot)$ of $\widehat{C}_{\alpha,\delta t}$ at $\lambda = 0$ in the standard rigged-Hilbert sense [8]; concretely, for $\psi, \varphi \in \Phi$, $\langle\psi|\delta(\widehat{C}_{\alpha,\delta t})|\varphi\rangle := \lim_{\epsilon\downarrow 0}(2\epsilon)^{-1}\langle\psi|E_C([-\epsilon,\epsilon])|\varphi\rangle$, well defined whenever the spectral measure is absolutely continuous in a neighbourhood of 0 relative to Φ. The two-boundary data $(\Psi_i^{\rho_i}, \Psi_f^{\{E_k\}})$ define test elements of Φ via the *smeared* Page–Wootters embeddings $\iota_i, \iota_f : \mathcal{H}_{\mathrm{sys}} \to \Phi$ of Rem. 11.58 (the formal sharp embeddings $\langle\tau_i|, \langle\tau_f|$ live in Φ' rather than Φ, so a Gaussian time-window of width $\sigma_t \ll \delta t$ around $\tau_{i,f}$ is used to land the boundary data in the test space; the sharp limit is recovered as $\sigma_t \to 0$ in Φ'), and the two-boundary problem is non-degenerate in the per-sector compatibility sense

$$\left\langle \Psi_f^{(k)} \;\middle|\; \delta(\widehat{C}_{\alpha,\delta t}) \;\middle|\; \Psi_i^{\rho_i} \right\rangle \neq 0 \quad \text{for every } k \in \mathcal{K} \text{ with } p(k) > 0, \tag{30}$$

where $\{\Psi_f^{(k)}\}_{k\in\mathcal{K}}$ is the sector-indexed family of terminal-boundary test elements built from the POVM $\{E_k\}$ (the k-th vector encodes the projection of $\Psi_i^{\rho_i}$ onto the k-th effect via Tomita conjugation in the sense of Conj. 11.1, so $\Psi_f^{(k)} \neq \Psi_f^{(k')}$ for $k \neq k'$); the rigged pairing of Φ is taken with the boundary vectors understood as their smeared images $\iota_{i,f}(\cdot)$. The per-branch content of the non-degeneracy condition (30)—that each sector-indexed terminal effect E_k links non-trivially back to the initial datum ρ_i across the two boundaries—is closely related to the temporal (across-time) correlation diagnosed by the per-branch Margenau–Hill witness Thm. 3.10: for projective E_k the symmetric two-boundary marginal $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$ fails positivity exactly when $[\rho(\tau), \Lambda_k(\tau)] \neq 0$, while the prior-average $\sum_k p(k) M_k = \rho$ obeys the no-signalling identity (Thm. 6.1), so the across-boundary link is irreducibly per-sector. *Hilbert-space orthogonal projection onto* $\ker \widehat{C}_{\alpha,\delta t}$ *is not used and would generically be the zero operator*: the clock factor contributes absolutely continuous spectrum on $\mathbb{R}$, so $\ker \widehat{C}_{\alpha,\delta t} \cap (\mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{sys}}) = \{0\}$ in general, and physical solutions live as distributions in Φ', not as vectors in $\mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{sys}}$.

Status. (27) is not derived here and is not a theorem. The constraint operator on the left-hand side is *essentially self-adjoint on the algebraic tensor product* $\mathcal{D}_{\mathrm{c}}^{\mathrm{an}} \otimes_{\mathrm{alg}} \mathcal{D}_{\mathrm{s}}^{\mathrm{an}}$ *of analytic vectors of* $\widehat{H}_{\mathrm{c}}$ *and* $\widehat{H}_{\mathrm{s}}$ for every $\alpha \in \mathbb{R}$, $\delta t > 0$: the unperturbed sum $\widehat{H}_{\mathrm{c}} \otimes \mathbb{1} + \mathbb{1} \otimes \widehat{H}_{\mathrm{s}}$ is essentially self-adjoint on this domain because the factors commute and each is essentially self-adjoint on its own analytic vectors (Nelson's theorem); the perturbation $\alpha\, \widehat{\Sigma}_{\delta t} \otimes \mathbb{1}$ is bounded ($\left\|\widehat{\Sigma}_{\delta t}\right\| \leq 2$, so $\left\|\alpha\, \widehat{\Sigma}_{\delta t}\right\| \leq 2\,|\alpha|$), so by Kato–Rellich it is relatively bounded with relative bound 0 and preserves essential self-adjointness. (Note: $\widehat{H}_{\mathrm{c}}$ and $\widehat{H}_{\mathrm{s}}$ are unbounded, so the constraint operator is itself unbounded; the earlier statement that it is "bounded self-adjoint" was incorrect.) The physical-state condition is well-posed in the rigged-Hilbert-space sense of [8]. Essential self-adjointness here presupposes that $\widehat{H}_{\mathrm{c}}$ is itself essentially self-adjoint on the chosen analytic core; for generic relational clocks this can fail, with physical unitarity becoming clock-dependent and the rigging then contingent on a choice of self-adjoint extension (non-zero deficiency indices), as Gielen–Menéndez-Pidal [104] establish in minisuperspace with continuous spectrum at the relevant point. Under Pos. 11.11 the clock generator is the thermal-time/ modular Hamiltonian $\widehat{H}_{\mathrm{c}} = -\log \Delta_\omega$, which is self-adjoint by Tomita–Takesaki (and bounded within the cutoff scope), removing this extension ambiguity; the caveat therefore applies strictly to the out-of-cutoff, generic-clock regime. Projecting onto $\langle t| \otimes \mathbb{1}$ yields the symmetric finite-difference TDSE of Prop. 10.14, which reduces to the asymmetric form of the superseded companion note on the retarded (forward-causal) sector (see Rem. 10.16 and App. C). The *derivation* of (27) from a more fundamental principle (e.g. a path-integral definition of canonical quantum gravity in the Oppenheimer–Snyder collapse region of Sec. C.1) remains open. We record one orthogonality caveat so that no future reader overloads this open item: every ingredient such a derivation would supply — the constraint operator, the rigging, the constants $\alpha, \delta t$ — is homogeneous under the trace rescaling $\tau_R \mapsto \lambda \tau_R$, so even a complete derivation of (27) would not bear on the absolute count N of Rem. D.5: deriving the constraint and fixing N are independent problems.

Other recent routes to a genuine time in the Wheeler–DeWitt framework include the Hamilton–Jacobi–Einstein construction of [261], which inserts a local classical time before quantization, and the periodic-clock "trinity" of [262], which shows that the Page–Wootters, Dirac, and deparametrization pictures coincide and that the Page–Wootters conditional-probability rule must be refined for clocks with continuous spectrum; our rigged-Hilbert group-averaging prescription (29) addresses an analogous continuous-spectrum subtlety for our absolutely continuous $\widehat{H}_{\mathrm{c}}$ (cf. also the quantum-reference-frame equivalence theorem of [364]).

Proposition 10.14 (Conditional: symmetric-difference TDSE from Conj. 10.13)**.** *Assume Conj. 10.13. Let* $|\psi(t)\rangle := \langle t|\Psi \in \mathcal{H}_{\mathrm{sys}}$. *Then*

$$i\hbar\,\partial_t|\psi(t)\rangle \;=\; \widehat{H}_{\mathrm{s}}|\psi(t)\rangle + \frac{\alpha}{2}\big(|\psi(t+\delta t)\rangle + |\psi(t-\delta t)\rangle - 2|\psi(t)\rangle\big). \tag{31}$$

Equation (31) *is a delay–advance equation in* t*, not a first-order ODE: it does not generate a one-parameter unitary group on* $\mathcal{H}_{\mathrm{sys}}$ *from single-slice data, and the per-slice norm* $\|\psi(t)\|_{\mathcal{H}_{\mathrm{sys}}}$ *is not in general* t*-independent. The correct unitarity statement lives on the space of histories: the operator* $\widehat{H}_{\mathrm{s}}\otimes\mathbb{1}_t+\alpha\,\mathbb{1}_{\mathrm{sys}}\otimes\widehat{\Sigma}^{\mathrm{h}}_{\delta t}$ *is essentially self-adjoint on the algebraic tensor product of analytic vectors of* $\widehat{H}_{\mathrm{s}}$ *with* $C_c^\infty(\mathbb{R}) \subset L^2_t(\mathbb{R})$*, by Kato–Rellich applied to the bounded perturbation* $\alpha\,\widehat{\Sigma}^{\mathrm{h}}_{\delta t}$ *on top of the essentially self-adjoint commuting sum* $\widehat{H}_{\mathrm{s}}\otimes\mathbb{1}_t$ *on* $L^2_t(\mathbb{R};\mathcal{H}_{\mathrm{sys}})$*, where* $(\widehat{\Sigma}^{\mathrm{h}}_{\delta t}\psi)(t) := \frac{1}{2}(\psi(t+\delta t)+\psi(t-\delta t)) - \psi(t)$*, and the constraint* (27) *is the kernel of an essentially self-adjoint operator on a dense domain in* $\mathcal{H}_{\mathrm{clock}}\otimes\mathcal{H}_{\mathrm{sys}}$*. Per-slice unitarity is recovered only in the smooth/local limit, where* (31) *reduces to the Hermitian effective ODE of Prop. 10.27.*

Proof. Take the inner product of (27) with $\langle t|\otimes\mathbb{1}$; use $\langle t|\widehat{H}_{\mathrm{c}} = -i\hbar\partial_t\langle t|$ together with $\langle t|\widehat{S}_\delta = \langle t+\delta|$ and $\langle t|\widehat{S}_\delta^\dagger = \langle t-\delta|$, so that $\langle t|\widehat{\Sigma}_{\delta t} = \frac{1}{2}(\langle t+\delta t| + \langle t-\delta t|) - \langle t|$. Rearranging gives (31). Self-adjointness of $\widehat{H}_{\mathrm{s}}\otimes\mathbb{1}_t + \alpha\widehat{\Sigma}^{\mathrm{h}}_{\delta t}$ on $L^2_t(\mathbb{R};\mathcal{H}_{\mathrm{sys}})$ is immediate: $\widehat{H}_{\mathrm{s}}$ is self-adjoint by hypothesis, and $\widehat{\Sigma}^{\mathrm{h}}_{\delta t}$ is bounded with $(\widehat{\Sigma}^{\mathrm{h}}_{\delta t})^* = \widehat{\Sigma}^{\mathrm{h}}_{\delta t}$ by direct computation against the L^2_t pairing; the Kato–Rellich extension applies because $\big\|\widehat{\Sigma}^{\mathrm{h}}_{\delta t}\big\| \le 2$, hence $\big\|\alpha\,\widehat{\Sigma}^{\mathrm{h}}_{\delta t}\big\| \le 2\,|\alpha|$. For a rigorous treatment of the rigged-Hilbert-space projection see [8, Sec. II]. □

Proposition 10.15 (Sufficient condition for joint compatibility (30))**.** *In the setting of Conj. 10.13, suppose the spectral measure* $E_C(\cdot)$ *of* $\widehat{C}_{\alpha,\delta t}$ *admits, on a* Φ*-dense set of test elements* ψ,φ*, a Radon–Nikodym density* $\rho_{\psi,\varphi}(\lambda) := d\langle\psi|E_C(\lambda)|\varphi\rangle/d\lambda$ *which is jointly continuous in* (λ,ψ,φ) *on a neighbourhood of* $\lambda = 0$*, and suppose the* k-uniform rank-one rigged genericity hypothesis *holds: there exist a single common pair* $(\alpha,\delta t)$ *and a constant* $c_* > 0$ *such that*

$$\inf_{k\in\mathcal{K}:\,p(k)>0}\big|\rho_{\Psi_i^{\rho_{\mathrm{i}}},\Psi_f^{(k)}}(0)\big| \;\ge\; c_* \;>\; 0, \tag{32}$$

i.e. the spectral density at $\lambda = 0$ *paired against the sectorwise terminal family* $\{\Psi_f^{(k)}\}_{k\in\mathcal{K}}$ *of Conj. 10.13 is bounded below by a single positive constant uniformly in* k *(not* $|\mathcal{K}|$ *separate non-vanishing conditions). Then* (30) *holds for every such* k*, simultaneously, for the same* $(\alpha,\delta t)$*. The hypothesis* (32) *is generic: by Cauchy–Schwarz on the rigged pairing,*

$$|\rho_{\Psi_i^{\rho_{\mathrm{i}}},\Psi_f^{(k)}}(0)|^2 \;\le\; \rho_{\Psi_i^{\rho_{\mathrm{i}}},\Psi_i^{\rho_{\mathrm{i}}}}(0)\,\rho_{\Psi_f^{(k)},\Psi_f^{(k)}}(0),$$

so a uniform-in-k *lower bound follows from a single positive lower bound on the diagonal density* $\rho_{\Psi_i^{\rho_{\mathrm{i}}},\Psi_i^{\rho_{\mathrm{i}}}}(0) > 0$ *together with a uniform lower bound* $\inf_k \rho_{\Psi_f^{(k)},\Psi_f^{(k)}}(0) > 0$ *and a uniform lower bound on the cosine of the angle (in the rigged pairing) between* $\Psi_i^{\rho_{\mathrm{i}}}$ *and* $\Psi_f^{(k)}$ *at* $\lambda = 0$*; the third bound holds whenever* $\{E_k^{1/2}\Psi_f\}_{k\in\mathcal{K}}$ *has* $\widehat{C}_{\alpha,\delta t}$*-zero-mode component non-orthogonal to* $\Psi_i^{\rho_{\mathrm{i}}}$ *uniformly in* k*. In finite-dimensional clock truncations* $\mathcal{H}^{(N)}_{\mathrm{clock}} \to \mathcal{H}_{\mathrm{clock}}$ *in which the spectrum of*

$\widehat{C}_{\alpha,\delta t}$ *near* 0 *is discretised, the violating set has Lebesgue measure zero on the joint unit sphere; the* infinite-$\mathcal{K}$ *case is handled by the same single rank-one density inequality* (32)*, which is one condition on the spectral density and one uniform lower bound on the terminal family, not* $|\mathcal{K}|$ *independent conditions.*

Proof. By construction $\langle\Psi_f^{(k)}|\delta(\widehat{C}_{\alpha,\delta t})|\Psi_i^{\rho_{\rm i}}\rangle = \rho_{\Psi_i^{\rho_{\rm i}},\Psi_f^{(k)}}(0)$, which has absolute value $\geq c_* > 0$ by hypothesis, hence is non-zero for every k with $p(k) > 0$, simultaneously, with the same $(\alpha, \delta t)$. Joint continuity ensures the limit defining $\delta(\widehat{C}_{\alpha,\delta t})$ exists in the rigged pairing for each fixed k, and the genericity statement is the finite-dimensional Lebesgue-measure argument used identically in Rem. 11.67 below. □

Remark 10.16 (Recovery of the superseded asymmetric form). The asymmetric forward-shift TDSE $i\hbar\partial_t\psi = \widehat{H}_{\rm s}\psi + \alpha(\psi(t+\delta t) - \psi(t))$ of the superseded companion note (App. C, Rem. C.3) is *not* the projection of a self-adjoint constraint on $\mathcal{H}_{\rm clock}\otimes\mathcal{H}_{\rm sys}$, because $\widehat{\Delta}_{\delta t}$ is not self-adjoint. It coincides with (31) on the subspace of solutions that are retarded (causal): if $\psi(t)$ is supported on $t \geq t_i$ with $\psi(t) \equiv 0$ for $t < t_i$, then on the restricted Cauchy slice $t = t_i + \delta t$ the backward-shift term $\psi(t - \delta t)$ vanishes and (31) collapses to the asymmetric form (with $\alpha \to \alpha/2$). On general two-sided solutions of (27) the symmetric form (31) is the unitary representative of the same equivalence class.

10.3 The temporal link derived: from states over time to the cosine cocycle

This subsection replaces the heuristic "temporal bifurcation" picture behind the modified TDSE (cf. Rem. 10.16) by a derivation: the two-time object encoding "the particle linked to itself across δ" is constructed (not posited) as a Fullwood–Parzygnat state over time, and the clock-side correction $\alpha\widehat{\Sigma}_\delta$ of Conj. 10.13 is obtained as the *unique* effective one-leg reduction of that link, by reduction to Thm. B.5 (App. B). Throughout Definition 10.17–Remark 10.19, $\dim\mathcal{H} = d < \infty$ (the setting of Thm. 3.10 and App. A); Definition 10.20 and Lemma 10.22 are stated for an arbitrary self-adjoint clock on a separable Hilbert space. We write $\mathbb{F} := \sum_{ij} E_{ji}\otimes E_{ij}$ for the swap and, as in Thm. 3.10, $\mathscr{J}[\mathscr{E}] := \sum_{ij} E_{ji}\otimes\mathscr{E}(E_{ij})$ for the partial-transpose-on-A (Jamiołkowski) Choi operator of a channel $\mathscr{E}$.

Definition 10.17 (Two-time link state)**.** For a state ρ at time t and an interval span δ, the *temporal link state* on the doubled space $\mathcal{H}_t\otimes\mathcal{H}_{t+\delta}$ is the Fullwood–Parzygnat state over time [397] of the interval channel $\mathscr{S}_\delta := \mathrm{Ad}(\widehat{S}_\delta)$,

$$\varrho_\delta(\rho) \;:=\; \tfrac{1}{2}\big\{\rho\otimes\mathbb{1},\; \mathscr{J}[\mathscr{S}_\delta]\big\}. \tag{33}$$

Proposition 10.18 (Well-posedness and self-identification)**.** *The link state* (33) *satisfies:*

(i) ***(dressed swap / self-identification)*** $\mathscr{J}[\mathscr{S}_\delta] = (\mathbb{1}\otimes\widehat{S}_\delta)\,\mathbb{F}\,(\mathbb{1}\otimes\widehat{S}_\delta)^\dagger$*, and* $\mathscr{J}[\mathscr{S}_\delta]^\dagger = \mathscr{J}[\mathscr{S}_\delta]$*;*

(ii) ***(state axioms)*** $\varrho_\delta(\rho)^\dagger = \varrho_\delta(\rho)$*,* $\mathrm{Tr}\,\varrho_\delta(\rho) = 1$*, and* $\rho\mapsto\varrho_\delta(\rho)$ *is* linear*;*

(iii) ***(marginals)*** $\mathrm{Tr}_{t+\delta}\,\varrho_\delta(\rho) = \rho$ *and* $\mathrm{Tr}_t\,\varrho_\delta(\rho) = \widehat{S}_\delta\rho\widehat{S}_\delta^\dagger$*;*

(iv) ***(two-time statistics)*** *for Hermitian* A, B*,* $\mathrm{Tr}[\varrho_\delta(\rho)(A\otimes B)] = \Re\,\mathrm{Tr}\big(\rho\,A\,\widehat{S}_\delta^\dagger B\widehat{S}_\delta\big)$*, the symmetrised (Margenau–Hill / real Kirkwood–Dirac) two-time correlator of Thm. 3.10(iv);*

(v) ***(leg exchange = time reflection)*** $\mathbb{F}\,\mathscr{J}[\mathscr{S}_\delta]\,\mathbb{F} = \mathscr{J}[\mathscr{S}_{-\delta}]$.

Proof. (i) $(\mathbb{1}\otimes V)\mathbb{F}(\mathbb{1}\otimes V)^\dagger = \sum_{ij} E_{ji}\otimes VE_{ij}V^\dagger = \mathscr{J}[\mathrm{Ad}(V)]$ for any unitary V; Hermiticity holds because $(E_{ji}\otimes VE_{ij}V^\dagger)^\dagger$ is the $(i\leftrightarrow j)$ term of the same sum. (ii) Hermiticity of the Jordan product of Hermitian factors (using (i)); the trace is $\mathrm{Tr}[(\rho\otimes\mathbb{1})\mathscr{J}[\mathscr{S}_\delta]] = \mathrm{Tr}[\mathscr{S}_\delta(\rho)] = 1$ by the marginal identities in the proof of Thm. 3.10(i); linearity is manifest in (33). (iii) Specialise those marginal identities to the *unital* channel $\mathscr{S}_\delta$: $\mathscr{S}_\delta^*(\mathbb{1}) = \mathbb{1}$ gives $\mathrm{Tr}_{t+\delta}\,\varrho_\delta = \frac{1}{2}\{\rho,\mathbb{1}\} = \rho$, and $\mathrm{Tr}_t\,\varrho_\delta = \mathscr{S}_\delta(\rho)$. (iv) Direct computation, as in Thm. 3.10(iv) with Λ replaced by $\widehat{S}_\delta^\dagger B\widehat{S}_\delta$.

(v) $\mathbb{F}(E_{ji}\otimes E_{ij})\mathbb{F} = E_{ij}\otimes E_{ji}$; in an eigenbasis of $\widehat{H}_{\mathrm{c}}$ the dressing phase $e^{-i\delta(\xi_i-\xi_j)/\hbar}$ of the (i,j) term goes to its conjugate under the relabelling $i\leftrightarrow j$. □

Remark 10.19 (Why this is the honest form of the "temporal bifurcation"). Definition 10.17 repairs the three defects of the posited bifurcation ansatz $\psi(t,t+\delta t)=\psi(t)\otimes e^{i\phi\delta t}\psi(t+\delta t)$ (the ansatz behind the asymmetric form recovered in Rem. 10.16; for the corrected derivation see App. C, Sec. C.2): (a) that expression is *quadratic* in ψ, hence forbidden by linearity (Wootters–Zurek no-cloning); ϱ_δ is linear in ρ. (b) It clones the state onto two factors; ϱ_δ instead places *one* system on two *temporal* legs, with the self-identification carried by the swap in Prop. 10.18(i) — "the particle at $t+\delta$ is the same particle at t, shifted by $\widehat{S}_\delta$" is now an exact operator statement on $\mathcal{H}\otimes\mathcal{H}$, requiring no spacetime wormhole. (c) The ad hoc phase $e^{i\phi\delta t}$, $\phi\sim E$, becomes the fibrewise dressing $e^{-i\delta\xi/\hbar}$ on the spectral fibres ξ of $\widehat{H}_{\mathrm{c}}$. Moreover (33) is not one choice among many: the FP state over time is the essentially unique bipartite state over time satisfying the operational axioms of [398] (completeness, compositionality, classical conditionability, and time-reversal symmetry; those authors also show the original axioms of [397] do not by themselves single it out), and by [399] linearity in the initial state plus a quantum conditionability axiom already fix its Markovian multipartite extension uniquely, with the associated quasiprobability picture the temporal Kirkwood–Dirac distribution; for the unification of the temporal-state formalisms (pseudo-density matrices, superdensity operators, states over time) see [400, 7, 372]. Conditioning (33) on a terminal effect E_k (replace $\mathscr{S}_\delta$ by the p-normalised Lüders branch channel) reproduces the conditioned state over time ϱ_k of Thm. 3.10, whose t-leg marginal is the Margenau–Hill operator M_k; its negativity is the operational witness that the temporal link is non-trivially engaged by the two-boundary data.

Definition 10.20 (Effective one-leg reduction of the link)**.** Let $\widehat{H}_{\mathrm{c}}$ be self-adjoint on a separable Hilbert space $\mathcal{H}_{\mathrm{clock}}$. A family $D:\mathbb{R}\to\mathcal{L}(\mathcal{H}_{\mathrm{clock}})$, strongly continuous with $D(0)=0$, is a *link defect* if

(R1) **(clock covariance)** $D(\delta)\in\{\widehat{H}_{\mathrm{c}}\}''$ for all $\delta\in\mathbb{R}$;

(R2) **(translation cocycle)** $D(\delta_1+\delta_2)=\widehat{S}_{\delta_1}D(\delta_2)+D(\delta_1)$ for all $\delta_1,\delta_2\in\mathbb{R}$.

An *effective one-leg reduction* of the temporal link is $B(\delta):=\frac{1}{2}\big(D(\delta)+D(\delta)^\dagger\big)$ for a link defect D, subject to

(R3) **(reflection evenness)** $B(-\delta)=B(\delta)$ for all $\delta\in\mathbb{R}$;

(R4) **(normalisation)** for every $\psi\in\mathcal{D}(\widehat{H}_{\mathrm{c}}^2)$ the map $\delta\mapsto B(\delta)\psi$ is twice strongly differentiable at 0 with $B'(0)\psi=0$ and $-B''(0)\psi=\alpha\widehat{H}_{\mathrm{c}}^2\psi/\hbar^2$ for a fixed $\alpha\in\mathbb{R}$.

Remark 10.21 (Each clause is derived, not posited). (R1): $\mathscr{J}[\mathscr{S}_\delta]$ commutes with the diagonal clock translation $\widehat{S}_s\otimes\widehat{S}_s$ for all s (the dressing phases of Prop. 10.18(i) cancel fibrewise), so any reduction intertwining diagonal translations with one-leg translations lands in the commutant of $\{\widehat{S}_s\}_{s\in\mathbb{R}}$; for an ideal (multiplicity-free) clock — the Page–Wootters clock of Conj. 10.13 — this commutant is the abelian algebra $\{\widehat{H}_{\mathrm{c}}\}''$. (R2): the exact identity $\widehat{S}_{\delta_1+\delta_2}-\mathbb{1}=\widehat{S}_{\delta_1}(\widehat{S}_{\delta_2}-\mathbb{1})+(\widehat{S}_{\delta_1}-\mathbb{1})$ shows the canonical defect $\widehat{\Delta}_\delta=\widehat{S}_\delta-\mathbb{1}$ *is* a translation 1-cocycle; (R2) demands only that the reduction respect the group law of time translation on adjoining intervals — it replaces the bifurcation postulate and is the sole residual structural assumption (Rem. 10.26). The Hermitian-part prescription $B=\frac{1}{2}(D+D^\dagger)$: the constraint $\widehat{C}_{\alpha,\delta t}$ must be self-adjoint for (28) to be well-posed, and the Jordan symmetrisation $\frac{1}{2}\{\cdot,\cdot\}$ is already forced at the level of the state over time [398, 399]; taking the Hermitian part is the same symmetrisation applied on one leg (it is what produces M_k from $\rho\Lambda_k$ in Thm. 3.10(i)). (R3): by Prop. 10.18(v), exchanging the two legs of the link is $\delta\mapsto-\delta$; the two-boundary data assign no preferred leg, because the Born pairing $p(k)=\mathrm{Tr}(\rho_{\mathrm{i}}U^\dagger(t_f,t_i)E_kU(t_f,t_i))$ is invariant in t (Lem. A.5, App. A) and, by cyclicity, invariant under the ABL exchange $(\rho_{\mathrm{i}},E_k,U)\mapsto(E_k,\rho_{\mathrm{i}},U^\dagger)$; hence the effective correction cannot distinguish the forward from the backward reading of the interval. (R4): this is axiom (E) of Conj. 11.5 verbatim; it states only that the correction is second order in δ with a fixed scale α and no first-order clock drift (a first-order term would renormalise $\widehat{H}_c$, not link times).

Lemma 10.22 (Classification of effective one-leg reductions)**.** *Let $\widehat{H}_c$ be self-adjoint on a separable Hilbert space and let $B(\delta)$ be an effective one-leg reduction in the sense of Definition 10.20. Then*

$$B(\delta) \;=\; \alpha\,\widehat{\Sigma}_\delta \;=\; \alpha\big(\cos(\delta\widehat{H}_c/\hbar) - \mathbb{1}\big),$$

with α the constant of (R4)*. In particular the effective shift operator is $\mathbb{1} + \frac{1}{\alpha}B(\delta) = \frac{1}{2}(\widehat{S}_\delta + \widehat{S}_{-\delta}) = \cos(\delta\widehat{H}_c/\hbar)$: the symmetric superposition of the forward and backward clock shifts is* derived*, and the deviation from the identity is exactly the cosine cocycle* (26)*.*

Proof. We verify the hypotheses of Thm. B.5 (App. B) in the band-limit-free form of Rem. B.6: separability, boundedness and strong continuity of B with $B(0) = 0$, axioms (A)–(D) as listed in Prop. B.4, and the Peano normalisation (E′) of (87); the conclusion $B(\delta) = \alpha\widehat{\Sigma}_\delta$ is then that theorem's.

Step 0 (fibre calculus). By (R1) each $D(\delta)$ lies in the abelian von Neumann algebra $\{\widehat{H}_c\}''$. Fix a direct-integral diagonalisation $\mathcal{H}_{\mathrm{clock}} \cong \int^{\oplus}_{\sigma(\widehat{H}_c)} \mathcal{H}_\lambda\, d\mu(\lambda)$ in which $\{\widehat{H}_c\}''$ is the diagonalisable algebra ([30], Part II; the convention of Prop. B.4(iii), here taken over the full spectrum): every $T \in \{\widehat{H}_c\}''$ acts fibrewise as a *scalar* $t(\lambda)\mathbb{1}_{\mathcal{H}_\lambda}$ with $t \in L^\infty(\mu)$, $\widehat{S}_s$ acts as $e^{-is\lambda/\hbar}$, adjoints act as complex conjugates, and each operator identity among such elements holds fibrewise μ-a.e. (one null set per identity). Write $d(\lambda, \delta)$ for the fibre scalar of $D(\delta)$ and $b := \Re d$ for that of $B(\delta)$.

Step 1 ($B(0) = 0$; boundedness; (A); strong continuity). $B(\delta)$ is bounded self-adjoint by construction, and $B(0) = \frac{1}{2}(D(0) + D(0)^\dagger) = 0$. Strong continuity of $\delta \mapsto B(\delta)$ follows from that of D: for $T \in \{\widehat{H}_c\}''$, $\big\|T^\dagger\psi\big\|^2 = \int |\overline{t(\lambda)}|^2\, \|\psi_\lambda\|^2\, d\mu = \|T\psi\|^2$; applied to $T = D(\delta) - D(\delta_0)$ this gives $\big\|(D(\delta)^\dagger - D(\delta_0)^\dagger)\psi\big\| = \|(D(\delta) - D(\delta_0))\psi\| \to 0$. (The adjoint is not strongly continuous on $\mathcal{L}(\mathcal{H})$ in general; membership in the abelian algebra (R1) is what rescues it.)

Step 2 ((B): commutant membership). $B(\delta) \in \{\widehat{H}_c\}'' \subseteq \{\widehat{H}_c\}'$ since the abelian algebra is $*$-closed and contained in its own commutant; in particular $\widehat{S}_s B(\delta)\widehat{S}_{-s} = B(\delta)$. Only the commutant property is used by Thm. B.5 (Prop. B.4(b)).

Step 3 ((C): evenness). This is (R3) verbatim.

Step 4 ((D): the inhomogeneous cocycle (84)*).* This is the crux: (R1)–(R3) *imply* the d'Alembert cocycle for B. First, evaluating (R2) at $(\delta_1, \delta_2) = (-\delta, \delta)$ and using $D(0) = 0$,

$$D(-\delta) \;=\; -\,\widehat{S}_{-\delta}\, D(\delta). \tag{34}$$

Fix $(\delta_1, \delta_2) \in \mathbb{R}^2$ and intersect the μ-null sets of the finitely many operator identities (R2) at (δ_1, δ_2) and $(\delta_1, -\delta_2)$, (34) at δ_2, and (R3) at δ_2. Adding the two cocycle identities gives, fibrewise μ-a.e.,

$$d(\delta_1 + \delta_2) + d(\delta_1 - \delta_2) \;=\; e^{-i\lambda\delta_1/\hbar}\big[d(\delta_2) + d(-\delta_2)\big] + 2\, d(\delta_1), \tag{35}$$

suppressing the λ argument. *Sub-claim: for μ-a.e. λ with* $\cos(\lambda\delta_2/\hbar) \neq -1$*, the bracket in* (35) *is real, hence equal to* $2b(\delta_2)$. Write $d(\delta_2) = x + iy$, $c := \cos(\lambda\delta_2/\hbar)$, $s := \sin(\lambda\delta_2/\hbar)$. By (34), $d(-\delta_2) = -e^{i\lambda\delta_2/\hbar}(x+iy) = -(cx - sy) - i(sx + cy)$, so evenness $\Re d(-\delta_2) = x$ reads $sy = (1+c)x$, and

$$\Im\big[d(\delta_2) + d(-\delta_2)\big] = (1 - c)\, y - s\, x.$$

If $s \neq 0$ then $x = sy/(1+c)$ and $sx = s^2y/(1+c) = (1-c)y$, so the imaginary part vanishes; if $s = 0$ and $c \neq -1$ then $c = 1$, the evenness relation gives $x = 0$, and the imaginary part is again 0. This proves the sub-claim. Taking real parts of (35) on those fibres,

$$b(\delta_1 + \delta_2) + b(\delta_1 - \delta_2) = \Re\big\{e^{-i\lambda\delta_1/\hbar} \cdot 2b(\delta_2)\big\} + 2b(\delta_1) = 2\cos(\lambda\delta_1/\hbar)\, b(\delta_2) + 2\, b(\delta_1),$$

which is (84) fibrewise. Call δ_2 *good* if $\mu(\{\lambda : \cos(\lambda\delta_2/\hbar) = -1\}) = 0$. The exceptional set $(\pi\hbar/\delta_2)(2\mathbb{Z}+1)$ is countable, hence μ-non-null only if it contains an atom of μ; separability leaves at most countably many atoms $\{a_n\}$, so δ_2 fails to be good only if $\delta_2 = (2k+1)\pi\hbar/a_n$ for some n, k — a countable set. For good δ_2 (a co-countable, dense set) the fibrewise identity holds μ-a.e., so (84) holds as an operator identity for *every* $\delta_1 \in \mathbb{R}$. For arbitrary δ_2, pick good $\delta_2' \to \delta_2$: each of the four terms of (84) is strongly continuous in δ_2 at fixed δ_1 (Step 1; $\cos(\delta_1\widehat{H}_c/\hbar)$ is a fixed bounded operator), so the identity passes to the limit. Hence (D) holds for all $(\delta_1, \delta_2) \in \mathbb{R}^2$.

Step 5 ((E′)). (R4) is the twice-strong-differentiability form with $B(0) = 0$ and $B'(0) = 0$; by the vector-valued Taylor formula it implies the second-order Peano form (87), as noted in the statement of Thm. B.5.

Step 6 (conclusion). All hypotheses of Thm. B.5 hold, and by Rem. B.6 the band limit is inessential (only separability, self-adjointness of $\widehat{H}_c$, boundedness, strong continuity, and (E′) on $\mathcal{D}(\widehat{H}_c^2)$ are used). Therefore $B(\delta) = \alpha\widehat{\Sigma}_\delta$, and $\mathbb{1} + \frac{1}{\alpha}B(\delta) = \frac{1}{2}(\widehat{S}_\delta + \widehat{S}_{-\delta}) = \cos(\delta\widehat{H}_c/\hbar)$ by the spectral calculus. ☐

Proposition 10.23 (Existence, and provenance of axiom (D))**.** $D(\delta) = \alpha\,\widehat{\Delta}_\delta = \alpha(\widehat{S}_\delta - \mathbb{1})$ *is a link defect (*(R1)–(R2)*) whose reduction* $B(\delta) = \alpha\widehat{\Sigma}_\delta$ *satisfies* (R3)–(R4)*; effective one-leg reductions therefore exist and Lemma 10.22 is non-vacuous. In this canonical case the d'Alembert law is transparent: taking Hermitian parts of* (R2) *at* $(\delta_1, \pm\delta_2)$ *and summing, with* $\widehat{\Delta}_{\delta_2} + \widehat{\Delta}_{-\delta_2} = 2\widehat{\Sigma}_{\delta_2}$ *Hermitian and commuting with* $\widehat{S}_{\delta_1}$*,*

$$\widehat{\Sigma}_{\delta_1+\delta_2} + \widehat{\Sigma}_{\delta_1-\delta_2} \;=\; 2\cos(\delta_1\widehat{H}_c/\hbar)\,\widehat{\Sigma}_{\delta_2} \;+\; 2\,\widehat{\Sigma}_{\delta_1}.$$

Together with Step 4 of the proof of Lemma 10.22, the cocycle composition law (D) of Conj. 11.5 is thus a theorem *for Hermitian-part reductions of translation* 1*-cocycles, not an independent axiom; Prop. 11.6 and Thm. B.5 then consume it as input, and their existence directions are recovered.*

Proof. (R1), strong continuity, and $D(0) = 0$ are immediate; (R2) is the displayed identity of Rem. 10.21 scaled by α; (R3) is evenness of the cosine; (R4) holds strongly on $\mathcal{D}(\widehat{H}_c^2)$ by dominated convergence in the spectral calculus (as verified in Step 5 of the proof of Thm. B.5). For the display: $\Re[\widehat{S}_{\delta_1} \cdot 2\widehat{\Sigma}_{\delta_2}] = \frac{1}{2}(\widehat{S}_{\delta_1} + \widehat{S}_{-\delta_1}) \cdot 2\widehat{\Sigma}_{\delta_2} = 2\cos(\delta_1\widehat{H}_c/\hbar)\widehat{\Sigma}_{\delta_2}$, since the factors commute and $\widehat{\Sigma}_{\delta_2}$ is Hermitian. ☐

Remark 10.24 (Consistency: Born time-invariance, unitarity, no wormholes). (a) The modified constraint (27) replaces $\widehat{H}_c$ by $\widehat{H}_c + \alpha\widehat{\Sigma}_{\delta t}$ with $\alpha\widehat{\Sigma}_{\delta t}$ bounded, self-adjoint, and in $\{\widehat{H}_c\}''$; the modified clock group is therefore a strongly continuous one-parameter unitary group commuting with the original one, the propagator identities of Rem. A.1 hold verbatim for the modified propagator, and Lem. A.5 (App. A: time invariance of the Born distribution) holds *unchanged* for the modified dynamics, as does the conditional-unitarity Thm. 3.7. (b) Nothing in Definitions 10.17–10.20 refers to spacetime geometry: the "wormhole" of the informal picture is the dressed swap of Prop. 10.18(i), an operator on $\mathcal{H} \otimes \mathcal{H}$; the framework's exclusion of traversable wormholes/CTCs via the self-consistent achronal ANEC (Rem. 10.12) is untouched. (c) On $\mathcal{D}(\widehat{H}_c^2)$, $\alpha\widehat{\Sigma}_{\delta t} = -\frac{\alpha\,\delta t^2}{2\hbar^2}\widehat{H}_c^2 + O(\delta t^4)$, reproducing the modified TDSE (31) in the Page–Wootters reduction with $\widehat{H}_{\rm mod} = \widehat{H}_c + \alpha\widehat{\Sigma}_{\delta t}$.

Remark 10.25 (Dictionary: bifurcation language → two-time formalism). For the record, the informal vocabulary of the superseded temporal-bifurcation derivation (cf. Rem. 10.16; corrected derivation in Sec. C.2 of App. C) maps onto the present formalism as follows. *"Bifurcated wave function $\psi(t, t+\delta t) = \psi(t) \otimes e^{i\phi\delta t}\psi(t+\delta t)$"* → the two-time link state $\varrho_\delta(\rho)$ of Definition 10.17: linear in the state, defined on two temporal legs of one system, unique under the states-over-time axioms [398, 399]. *"Wormhole linking the particle to itself across δt"* → the dressed swap

$\mathscr{J}[\mathscr{S}_\delta] = (\mathbb{1} \otimes \widehat{S}_\delta)\mathbb{F}(\mathbb{1} \otimes \widehat{S}_\delta)^\dagger$ (Prop. 10.18(i)); its non-trivial engagement by two-boundary data is witnessed by the Margenau–Hill negativity of Thm. 3.10. *"Wormhole phase $\phi \sim E$"* $\to$ the fibrewise dressing $e^{-i\delta\xi/\hbar}$, $\xi \in \sigma(\widehat{H}_c)$. *"Temporal split to conserve energy and unitarity"* $\to$ the translation-cocycle property (R2) of the link defect plus the ABL leg-exchange symmetry (R3) of two-boundary conditioning. *"Modified term $\alpha\Delta t\,\psi$"* $\to \alpha\widehat{\Sigma}_{\delta t}\psi$, the unique effective one-leg generator correction (Lemma 10.22); the retrocausal content enters through the two-boundary conditioning of App. A, not through any nonlinearity.

Remark 10.26 (Honest gaps). (1) The states-over-time uniqueness theorems [397, 398, 399] are finite-dimensional; their extension to the rigged Page–Wootters clock of Conj. 10.13 is open (in infinite dimensions ϱ_δ is not trace-class, so Definition 10.17 is here a finite-dimensional motivation), though Lemma 10.22 itself holds for any self-adjoint clock on a separable Hilbert space. (2) The translation-cocycle clause (R2) is the residual physical postulate: it is exactly the group law of time translation imposed on the link defect, and is satisfied by the canonical defect $\widehat{\Delta}_\delta$, but a derivation of (R2) from the multipartite composition of states over time [399] (three-time consistency on adjoining intervals) has not been carried out. A derivation on this route in fact meets a structural obstruction, not merely an unfilled step: the link state ϱ_δ and its t-marginal channel $\mathrm{Ad}\,\widehat{S}_\delta$ are *phase-blind* — invariant under $\widehat{S}_\delta \mapsto e^{i\phi}\widehat{S}_\delta$ — whereas the canonical defect $D(\delta) = \alpha(\widehat{S}_\delta - \mathbb{1})$ is phase-*sensitive*, and no state-linear functional of a phase-blind object can return a phase-sensitive one. Recovering the finite $\widehat{S}_\delta$ from $\mathrm{Ad}\,\widehat{S}_\delta$ is a nonlinear Stone inversion, and the second-derivative anchor (R4) that would otherwise pin the phase presupposes the canonical generator $\widehat{H}_c$ as given datum: the rival lift $\widehat{H}_c + c\mathbb{1}$ yields the identical link state and satisfies (R4) against its own generator. Replacing the phase-blind contraction by a phase-*sensitive* degree-one object — the full complex Kirkwood–Dirac amplitude $K_{ab}(\delta) = \langle a|\widehat{S}_\delta|b\rangle\langle b|\rho|a\rangle$, linear in $\widehat{S}_\delta$ itself — does dissolve the no-go *at the defect level*: its adjoining-interval composition is the Chapman–Kolmogorov convolution $\langle a|\widehat{S}_{\delta_1+\delta_2}|c\rangle = \sum_b \langle a|\widehat{S}_{\delta_1}|b\rangle\langle b|\widehat{S}_{\delta_2}|c\rangle$, so the reconstructed defect obeys $D(\delta_1+\delta_2) = \widehat{S}_{\delta_1}D(\delta_2) + D(\delta_1)$ — (R2) *is* the group law, a theorem for that reconstruction. Two obstructions then *relocate* the postulate rather than remove it. First, the constraint $\widehat{C}_{\alpha,\delta t}$ must be self-adjoint for (27) to be well-posed, which forces the Hermitian part: the phase-sensitive D cannot enter (28), only $\mathrm{Herm}(D) = B = \alpha\widehat{\Sigma}_\delta$ does — the same phase-blind cosine object of Prop. 10.23, on which the Kirkwood–Dirac route pins exactly what the states-over-time route did. Second, the reconstruction r_{KD} divides by the coherences $\langle b|\rho|a\rangle$: it is basis- and state-dependent, not the state-linear universal reduction, and so inherits none of the axiomatic uniqueness of [397, 399] — it faithfully echoes any rival generator $\widehat{H}_c + c\mathbb{1}$, selecting no clock-energy zero. The residual therefore relocates from "the group law is a postulate" to "the clock-energy zero c is a supplied datum" — a datum the Hermitian construction of Prop. 10.23 already silently requires; the Kirkwood–Dirac route *exposes* it rather than creating it. That clock-energy zero is fixed canonically under the thermal-time identification (Pos. 11.11): there $\widehat{H}_c = -\log\Delta_\omega$ has $\Delta_\omega\Omega = \Omega$ by Tomita–Takesaki, so the modular generator annihilates the KMS vacuum and no additive-constant freedom survives — the *gauge* residual is thereby discharged, at the price of positing the modular seed ω. This does not, however, promote (R2) to a theorem of the cosmological arm, because the self-adjointness obstruction survives there as well. The modular flow enters that arm not as the physical evolution but as the reduced dynamics obtained by *conditioning* the self-adjoint constraint $\widehat{C}_{\alpha,\delta t}$ on the clock POVM (Thm. O.1(K1)), whose constraint operator is the phase-blind $\widehat{\Sigma}_\delta = \cos(\delta\widehat{H}_c/\hbar) - \mathbb{1}$; thermal time fixes only the self-adjoint *extension* of $\widehat{H}_c$, not the self-adjointness *requirement* that forces the Hermitian part. Thus the group law $\Delta_\omega^{i(\delta_1+\delta_2)} = \Delta_\omega^{i\delta_1}\Delta_\omega^{i\delta_2}$ is a Tomita–Takesaki theorem for the defect D, but the object that couples into (28) is $\mathrm{Herm}(D) = \alpha\widehat{\Sigma}_\delta$, whose cosine d'Alembert law is already a theorem without thermal time (Prop. 11.6). Thermal time therefore discharges the clock-energy-zero (gauge) residual but leaves (R2)'s phase-sensitive content physically inert; on both the generic-clock and the thermal-time constructions (R2) accordingly stands as a physical postulate at the level of the physical constraint. (3) The reduction is single-clock, inheriting the scope caveat stated after Conj. 11.5: uniqueness is "among single-clock perturbations," with refoliation covariance the open problem of Sec. 11. (4) The scale α and span δt remain undetermined inputs, as in Conj. 10.13. (5) Under terminal conditioning the two marginals of the conditioned link are asymmetric (Thm. 3.10(i)); the reduction above acts on the t-leg, matching the constraint's present-time action.

10.4 Parameter scaling and low-energy effective Hamiltonian

The constants $\alpha, \delta t$ in Conj. 10.13 are real parameters with $[\alpha]$ = energy and $[\delta t]$ = time. Dimensional analysis alone does not fix them. Because the symmetric shift (26) is even in δt, the leading low-energy effect is governed by the dimensionful combination

$$\mu \;:=\; \frac{\alpha\,\delta t^{\,2}}{\hbar^2}, \qquad [\mu] = \text{energy}^{-1},$$

together with the dimensionless ratio $\mu\,\widehat{H}_{\rm s}$ on the matter sector.

Proposition 10.27 (Hermitian low-energy effective Hamiltonian)**.** *Assume Conj. 10.13, so that (31) holds. If $|\psi(t)\rangle$ is sufficiently smooth in t that a Taylor expansion in δt converges in the relevant spectral subspace of $\widehat{H}_{\rm s}$, then*

$$i\hbar\,\partial_t|\psi(t)\rangle \;=\; \widehat{H}_{\rm s}|\psi(t)\rangle + \frac{\alpha\,\delta t^{\,2}}{2}\,\partial_t^2|\psi(t)\rangle + O\big(\alpha\,\delta t^{\,4}\,\partial_t^4\psi\big). \tag{36}$$

Substituting $\partial_t^2\psi = -\hbar^{-2}\widehat{H}_{\rm s}^{\,2}\psi + O(\mu)$ self-consistently yields

$$i\hbar\,\partial_t|\psi(t)\rangle \;=\; \widehat{H}_{\rm eff}|\psi(t)\rangle + O\big(\mu^2\widehat{H}_{\rm s}^{\,4}\psi\big), \qquad \widehat{H}_{\rm eff} \;:=\; \widehat{H}_{\rm s} - \frac{\mu}{2}\,\widehat{H}_{\rm s}^{\,2}. \tag{37}$$

The operator $\widehat{H}_{\rm eff}$ is self-adjoint for every real μ on the joint spectral domain of $\widehat{H}_{\rm s}^2$. In particular the leading low-energy correction is unitary; no first-order non-Hermitian distortion of $\widehat{H}_{\rm s}$ appears.

Proof. Expand

$$\tfrac{1}{2}(\psi(t+\delta t)+\psi(t-\delta t)) - \psi(t) \;=\; \tfrac{\delta t^2}{2}\,\partial_t^2\psi(t) + \tfrac{\delta t^4}{24}\,\partial_t^4\psi(t) + O(\delta t^6),$$

in which all odd-order terms cancel by the symmetry of (26) under $\delta t \mapsto -\delta t$. Insert into (31) to obtain (36). To obtain (37), write $i\hbar\partial_t\psi = \widehat{H}_{\rm s}\psi + R$ with remainder $R = O(\mu)$; then $\partial_t^2\psi = (i\hbar)^{-2}(\widehat{H}_{\rm s} + \partial_t R/(i\hbar))(\widehat{H}_{\rm s}\psi + R) = -\hbar^{-2}\widehat{H}_{\rm s}^{\,2}\psi + O(\mu)$, and substitution gives (37). Self-adjointness of $\widehat{H}_{\rm eff}$ follows because $\widehat{H}_{\rm s}^{\,2}$ is self-adjoint on the domain $\mathcal{D}(\widehat{H}_{\rm s}^{\,2})$ and $\mu \in \mathbb{R}$. $\square$

Remark 10.28 (Slow-branch axiom for the physical state space of Conj. 10.13). We henceforth augment Conj. 10.13 with the explicit *slow-branch axiom*: the physical state space $\mathcal{H}_{\rm phys}^{\rm slow} \subset \mathcal{H}_{\rm phys}$ of (28) is defined as the maximal $\widehat{C}_{\alpha,\delta t}$-invariant subspace whose elements are real-analytic in δ at $\delta = 0$ under the clock translation $e^{-i\delta\widehat{H}_{\rm c}/\hbar}$, equivalently whose joint spectral support of $\widehat{H}_{\rm c}$ lies in $[-\hbar/\delta_\star, \hbar/\delta_\star]$ for some $\delta_\star > \delta t$. This is the exact analog of (T6) of Conj. 11.56 and excludes the non-perturbative fast branch of Rem. 10.29 below at the level of admissible solutions. The TDSE projection of Prop. 10.14 and the low-energy Hermitian effective Hamiltonian of Prop. 10.27 are stated implicitly on $\mathcal{H}_{\rm phys}^{\rm slow}$.

Remark 10.29 (Slow vs fast branch). The substitution $\partial_t^2\psi = -\hbar^{-2}\widehat{H}_{\rm s}^{\,2}\psi + O(\mu)$ used to derive (37) keeps only the *slow branch* of solutions, i.e. those for which $\partial_t^2\psi/\psi = O(\widehat{H}_{\rm s}^{\,2}/\hbar^2)$. Equation (36), viewed as a second-order ODE in t, also admits a fast branch with $\partial_t^2\psi/\psi = O(2/(\alpha\delta t^2)) = O(2\hbar^2/(\mu\hbar^4))$, i.e. characteristic frequency $\omega_{\rm fast} = \sqrt{2/\mu}/\hbar$. The fast branch is non-perturbative in μ and lies outside the regime in which the Taylor expansion (36) converges; it is excluded by hand when one restricts to solutions analytic at $\mu = 0$. This is the standard effective-field-theory branch selection (cf. regularised finite-difference Schrödinger equations [19]); (37) and the Planck-scale phenomenology below refer exclusively to the slow branch.

Remark 10.30 (Planck-scale phenomenology). The leading correction in (37) has relative size $\mu\,\langle\widehat{H}_{\rm s}\rangle/2$. The natural "maximally aggressive" Planck identification $\alpha = E_{\rm Pl} = \hbar/t_{\rm Pl}$, $\delta t = t_{\rm Pl}$ gives $\mu = \alpha\delta t^2/\hbar^2 = t_{\rm Pl}/\hbar = 1/E_{\rm Pl}$, hence $\mu\langle\widehat{H}_{\rm s}\rangle = \langle\widehat{H}_{\rm s}\rangle/E_{\rm Pl} \sim 10^{-28}$ for laboratory energies $\sim 1\,\text{eV}$ — a Planck-suppressed Hermitian correction of the form expected from any effective-field-theory argument. This is consistent with current precision spectroscopy and optical-clock bounds on unitary deviations [19]: those experiments place much weaker constraints on Hermitian Planck-suppressed shifts than on order-unity non-Hermitian distortions. The asymmetric forward-shift form of Rem. 10.16, which would have produced an order-one non-Hermitian factor at $\lambda := \alpha\delta t/\hbar = 1$, is ruled out by these data; the symmetric form (27) adopted here is not.

11 Open problems

We list obstructions not resolved by this paper.

(1) **Type III algebras for interacting QFT.** Part I is finite-dimensional. Extension to interacting QFTs on curved spacetime, where local algebras are type III_1 and density operators do not exist, requires the crossed-product formalism [18]; the analog of Thm. 3.7 then becomes a statement about modular automorphisms rather than conjugation by unitaries.

(2) **Non-linear existence of solutions to** (22)**.** Conj. 10.2's hypotheses (i)–(iv) require estimates that are standard for bounded linear matter on compact time intervals but nontrivial for realistic matter on asymptotic backgrounds.

(3) **Derivation of Conj. 10.13.** The modified constraint is postulated. The superseded companion note gave a heuristic derivation, replaced here by the two-time link-state construction of Sec. 10.3 and the exact fiberwise reduction of Sec. C.2; a rigorous derivation of the constraint itself would require a non-perturbative definition of canonical quantum gravity in the collapse region. By Prop. 10.27 and Rem. 10.30, the symmetric form (27) produces only Planck-suppressed Hermitian corrections at low energy and is consistent with current precision-spectroscopy bounds; the asymmetric forward-shift form of Rem. 10.16 is excluded. The open problem is to derive (27) from a fundamental principle fixing $\alpha, \delta t$ rather than leaving them phenomenological.

(4) **Terminal identifiability in the presence of horizons.** *Partially resolved* in Sec. 9.1 (Prop. 9.1, Cor. 9.2): the general collapse statement holds without Ax. 7.1, and $H(K \mid \mathcal{G})$ (14) quantifies the horizon-hidden residual entropy. The remaining open problem is to identify $\mathcal{G}$ intrinsically as a horizon subalgebra in a QFT-on-curved-spacetime setting with dynamical horizons.

(5) **Fermionic matter.** *Resolved* in Sec. 9.2 (Prop. 9.7): Part I extends verbatim to $\mathbb{Z}_2$-graded matter under graded microcausality (15), with the restriction that intervention POVMs and Heisenberg observables be grade-zero (standard physical requirement of gauge invariance).

(6) **Classical ($\hbar \to 0$) limit.** *Partially resolved* in Sec. 9.3 (Prop. 9.8): the conditioned Ehrenfest theorem holds, showing that different terminal outcomes k select distinct classical initial data through $\rho_2(\tau_i \mid k)$. The Ehrenfest identity governs mean values only and does not by itself establish classicality [376]; a genuine $\hbar \to 0$ statement additionally requires either a Weyl-quantization scheme (closed-system Egorov, valid only up to the logarithmically short Ehrenfest time for chaotic H) or, in the open-system regime, a decoherence/diffusion input meeting the threshold $D \gtrsim \hbar^{4/3}$ [378]. Either is external input.

(7) **Cosmological clock.** The clock subsystem of Conj. 10.13 must be identified dynamically in a cosmological setting; this is the standard problem of time in quantum gravity.

11.1 Speculative conjectures aimed at the remaining open problems

The following conjectures are *speculative*. Each is proposed as a potential route to one of the outstanding items above and is formulated so that it could, in principle, be proved or falsified. They are distinct from the conjectures of Sec. 10 only in being less explored; we do not claim they are correct.

Conjecture 11.1 (Tomita two-boundary conditioning for type III algebras)**.** Let $\mathcal{N} \subset \mathcal{B}(\mathcal{H}_{\mathrm{QFT}})$ be a type III_1 local algebra on a globally hyperbolic spacetime with a cyclic and separating vector Ω representing the forward state ω, and let $\widetilde{\mathcal{N}} := \mathcal{N} \rtimes_{\sigma^\omega} \mathbb{R}$ be the modular crossed product of [18], equipped with its canonical semifinite trace $\mathrm{Tr}_{\mathrm{sf}}$ and the canonical (Haagerup) dual weight $\widehat{\omega}$ of ω on $\widetilde{\mathcal{N}}$ [39, Thm. X.1.17]. We record a bookkeeping correction to earlier versions: $\widehat{\omega}$ is a faithful normal *semifinite weight*, not a state — $\widehat{\omega}(\mathbb{1}) = \infty$, and $\widehat{\omega}$ does *not* restrict to ω on $\pi(\mathcal{N})$; indeed $\widehat{\omega}(\pi(a)) = \infty$ for every non-zero positive a, because the canonical map $T = \int \widehat{\theta}_p(\cdot)\, dp$ underlying the dual weight is Haagerup's *operator-valued weight* with $T(\mathbb{1}) = \infty$. In particular there is *no* normal conditional expectation $E_{\mathcal{N}} : \widetilde{\mathcal{N}} \to \mathcal{N}$ (a normal conditional expectation of the semifinite $\widetilde{\mathcal{N}}$ onto the type III $\mathcal{N}$ cannot exist), and earlier formulations that invoked one, or the identity "$\widehat{\omega} = \omega \circ E_{\mathcal{N}}$", are withdrawn. Consequently branch ensembles must be normalised on objects carrying *states*: on $\mathcal{N}$ itself with ω (branch (I) below), or on the observer-projected corner $\Pi\widetilde{\mathcal{N}}\Pi$ with a normal state such as the CLPW maximum-entropy tracial state $\tau_\Pi := \mathrm{Tr}_{\mathrm{sf}}(\Pi \cdot \Pi)/\,\mathrm{Tr}_{\mathrm{sf}}(\Pi)$ [18] (branch (II) below). On the unprojected crossed product any centralizer partition of unity has $\sum_k \widehat{\omega}(E_k) = \widehat{\omega}(\mathbb{1}) = \infty$, so at least one branch weight diverges and no normalised branch ensemble exists there; individual branches with $\widehat{\omega}(E_k) < \infty$ remain well-defined. Let $\sigma_t^{\widehat{\omega}}$ denote the modular flow of $\widehat{\omega}$ on $\widetilde{\mathcal{N}}$ ($\sigma_t^{\widehat{\omega}} = \mathrm{Ad}(\lambda_t)$; Lemma below). The conjecture is organised in two branches.

Branch (I) — Araki conditioning (primary; carries the two-boundary matter content). The terminal effects are an arbitrary POVM partition of unity $\{E_k\}_{k\in\mathcal{K}} \subset \mathcal{N}$ (or, dressed, $\subset \Pi\widetilde{\mathcal{N}}\Pi$), $0 \le E_k \le \mathbb{1}$, $\sum_k E_k = \mathbb{1}$ weakly, with *no* centralizer or modular-invariance hypothesis. The conditioned state is

$$\omega_k(x) \;:=\; p(k)^{-1}\, \omega\big(\Lambda_k^{1/2}\, x\, \Lambda_k^{1/2}\big), \qquad p(k) := \omega(\Lambda_k), \qquad \sum_{k\in\mathcal{K}} p(k) = 1, \tag{38}$$

a faithful normal state for every bounded positive $\Lambda_k^{1/2}$ (for dressed effects in $\Pi\widetilde{\mathcal{N}}\Pi$ the same formula is used with ω replaced by a faithful normal reference state on the corner, e.g. τ_Π or the forward matter datum, to which the Araki theory applies verbatim); its existence, perturbed-KMS structure, Connes-cocycle form, and crossed-product lift are *theorems* of App. G (Thm. G.1, Thm. G.2), via Araki perturbation theory — the entire-analyticity simplification is unavailable here, and is not needed. It is in this branch, and only in this branch, that the two-boundary link can be non-classical on the matter algebra: the Margenau–Hill witness of Thm. 3.10 is non-vacuous precisely when $[\rho(\tau), \Lambda_k(\tau)] \neq 0$, which requires effects with non-trivial modular spectrum or a forward matter datum distinct from the seed (Rem. 11.2).

Branch (II) — centralizer conditioning (computationally simple special case; clock/observer sector only). The terminal effects form a partition of unity *in the modular centralizer of the dual weight on the crossed product*,

$$E_k \;\in\; \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} \;\subset\; \widetilde{\mathcal{N}}, \qquad \sum_{k\in\mathcal{K}} E_k = \mathbb{1}, \tag{39}$$

equivalently $[E_k, \Delta_{\widehat{\omega}}^{it}] = 0$ for all $t \in \mathbb{R}$, where $\Delta_{\widehat{\omega}}$ is the modular operator of the dual weight. Each such E_k is a *gravitationally dressed observable of the observer's algebra*: the crossed product $\widetilde{\mathcal{N}}$ is not an auxiliary bookkeeping device but the algebra of operators dressed to the observer's worldline that survives the gravitational constraint [18, 96], and it is on this algebra that records are operationally read out (App. Z, Prop. Z.1; App. L). This branch enjoys the entire-analyticity simplification (the Petz element $L_k = \Lambda_k^{1/2}$ below is exactly computable), but it is *operationally classical on the matter algebra*: by Rem. 11.2, $\omega_k \restriction_{\pi(\mathcal{N})} = \omega$ for every k in the geometric-ergodic regime, so branch-(II) conditioning is Bayesian updating of the dressed clock/observer-energy distribution and carries no per-branch matter information. Its scope is nevertheless never empty: *Lemma (crossed-product centralizer).* With $\pi : \mathcal{N} \hookrightarrow \widetilde{\mathcal{N}}$ and $\{\lambda_s\}_{s\in\mathbb{R}}$ the canonical generators, the dual modular flow is unitarily implemented inside the crossed product, $\sigma_t^{\widehat{\omega}} = \mathrm{Ad}(\lambda_t)$, so that $\sigma_t^{\widehat{\omega}}(\pi(a)) = \pi(\sigma_t^{\omega}(a))$ and $\sigma_t^{\widehat{\omega}}(\lambda_s) = \lambda_s$ [39, Thm. X.1.17]. The fixed-point algebra $F := \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} = \widetilde{\mathcal{N}} \cap \{\lambda_s\}'$ is globally invariant under the dual action $\widehat{\theta}$ (which commutes with $\mathrm{Ad}(\lambda_t)$: the scalar phases in $\widehat{\theta}_p(\lambda_t) = e^{ipt}\lambda_t$ cancel under Ad), contains the covariant unitaries $\lambda(\mathbb{R})$ with $\widehat{\theta}_p(\lambda_s) = e^{ips}\lambda_s$, and has $\widehat{\theta}$-fixed subalgebra $F^{\widehat{\theta}} = F \cap \widetilde{\mathcal{N}}^{\widehat{\theta}} = \pi(\mathcal{N}) \cap \{\lambda_s\}' = \pi(\mathcal{N}^{\sigma^\omega})$ (using Takesaki duality $\widetilde{\mathcal{N}}^{\widehat{\theta}} = \pi(\mathcal{N})$ and $\mathrm{Ad}(\lambda_t)\pi(a) = \pi(\sigma_t^{\omega}(a))$). The triple $(F, \widehat{\theta}\restriction_F, \lambda)$ thus satisfies the hypotheses of W* Landstad–Takesaki duality [39, Ch. X.2], whence

$$\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} \;=\; \mathcal{N}^{\sigma^\omega} \rtimes_{\sigma^\omega} \mathbb{R} \;\cong\; \mathcal{N}^{\sigma^\omega} \,\bar{\otimes}\, \{\lambda_s\}'' \qquad (\{\lambda_s\}'' \cong L^\infty(\mathbb{R})),$$

the tensor form because σ^ω is trivial on $\mathcal{N}^{\sigma^\omega}$. (The one-point $\widehat{\theta}$-spectral subspaces of F are indeed $\pi(\mathcal{N}^{\sigma^\omega})\lambda_p$, but point spectral subspaces of an $\mathbb{R}$-action do not span σ-weakly in general; the duality argument, not a pointwise spectral-subspace decomposition, is what proves equality.) Two structural consequences. *First,* $\{\lambda_s\}''$ commutes with $\pi(\mathcal{N}^{\sigma^\omega})$ and with itself, so $\{\lambda_s\}'' \subseteq Z(\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}})$: the dressed centralizer has diffuse center and is *never a factor*; when $\mathcal{N}^{\sigma^\omega}$ is a II_1 factor the centralizer is a *finite* type II (non-factor) algebra, not a II_∞ subfactor. *Second,* $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ always contains the diffuse abelian algebra $\{\lambda_s\}''$ of bounded Borel functions of the *dressed generator* $\widehat{P}$ of the adjoined one-parameter group λ (the CLPW dressed boost/observer-energy generator). We caution that $\widehat{P} \neq -\log\Delta_{\widehat{\omega}}$ as operators: $\Delta_{\widehat{\omega}}^{it}$ and λ_t implement the same flow on $\widetilde{\mathcal{N}}$ but differ by a unitary in $\widetilde{\mathcal{N}}'$, so bounded Borel functions of $-\log\Delta_{\widehat{\omega}}$ need *not* lie in $\widetilde{\mathcal{N}}$; the admissible windows are those of $\widehat{P}$. Spectral windows of $\widehat{P}$ furnish non-scalar partitions of unity (39) *even when* $\mathcal{N}^{\sigma^\omega} = \mathbb{C}\mathbb{1}$; and when $\mathcal{N}^{\sigma^\omega}$ is itself nontrivial, undressed effects $E_k \in \mathcal{N}^{\sigma^\omega} \subset \mathcal{N}$ are a special case. Given (39), the analytic continuation

$$L_k \;:=\; \sigma_{-i/2}^{\widehat{\omega}}(\Lambda_k^{1/2}) \;\in\; \widetilde{\mathcal{N}}$$

is well-defined and equals $\Lambda_k^{1/2}$ on the centralizer ($\Lambda_k(\tau) :=$ Heisenberg propagate of E_k along the matter dynamics generated by $\widehat{H}_{\mathrm{m}}$, *provided* $\Lambda_k(\tau)$ remains in $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ for every $\tau \in [\tau_i, \tau_f]$; this is the content of the explicit **centralizer-preservation hypothesis** (CP) stated below, equivalent to (T11) of Conj. 11.56).

Hypothesis (CP) — centralizer-preserving matter dynamics. (Here (CP) abbreviates *centralizer-preserving*; it is unrelated to complete positivity, which holds for the map $x \mapsto L_k^* x L_k$ irrespective of (CP).) The matter Hamiltonian $\widehat{H}_{\mathrm{m}}$ commutes with the dual modular flow in the form $[\widehat{H}_{\mathrm{m}}, \Delta_{\widehat{\omega}}^{it}] = 0$ for all $t \in \mathbb{R}$ (under Pos. 11.11 this is the single commutation $[\widehat{H}_{\mathrm{m}}, \widehat{H}_{\mathrm{c}}] = 0$ of Hyp. 11.13). Equivalently, the canonical conditional expectation $E_{\mathrm{cent}} : \widetilde{\mathcal{N}} \to \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ commutes with Heisenberg propagation by $\widehat{H}_{\mathrm{m}}$, hence the propagated effect $\Lambda_k(\tau)$ stays in $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ for every $\tau \in [\tau_i, \tau_f]$, and the entire-analyticity simplification $\sigma_z^{\widehat{\omega}}(\Lambda_k(\tau)^{1/2}) = \Lambda_k(\tau)^{1/2}$ used to define L_k is preserved under matter dynamics. Without (CP) the propagated effect can leave the centralizer, in which case the full Petz transpose channel without the entire-analyticity simplification is required; this is precisely the branch-(I) Araki conditioning realised as a theorem in App. G, which needs no centralizer hypothesis. (CP) is therefore a hypothesis of the simple branch (II) only. (Scope.) The centralizer hypothesis on $\{E_k\}$ is a genuine structural restriction: the terminal effects must be $\sigma^{\widehat{\omega}}$-invariant in the sense (39), so that the conditioning element $L_k = \Lambda_k^{1/2}$ is a positive centralizer element of $\widetilde{\mathcal{N}}$ (not a unitary), acting on $\widetilde{\mathcal{N}}$ by the completely positive map $x \mapsto L_k^* x L_k$ (the Kraus form of the Petz transpose channel). Whenever $\widehat{\omega}(\Lambda_k) < \infty$ the functional $x \mapsto \widehat{\omega}(L_k^* x L_k)$ is a positive normal functional on $\widetilde{\mathcal{N}}$, and its normalisation ω_k restricts to a normal state $\omega_k \restriction_{\pi(\mathcal{N})}$ by plain restriction of a normal functional along the inclusion $\mathcal{N} \hookrightarrow \widetilde{\mathcal{N}}$ — no conditional expectation onto $\mathcal{N}$ exists or is used (cf. the bookkeeping correction above). By Rem. 11.2 this restriction equals ω itself in the geometric-ergodic regime: branch (II) conditions the dressed clock sector only. Physical realisability of the records is the observer-dressed one: $\{E_k\}$ are effects of the dressed algebra that a DEHK/CLPW observer measures (Prop. Z.1); after the CLPW positive-energy projection Π one may work in the type II_1 factor $\Pi\widetilde{\mathcal{N}}\Pi$, but a partition of unity there is a centralizer partition *of the tracial state* τ_Π, not of $\widehat{\omega}$; unless the E_k are functions of the dressed generator $\widehat{P}$, such effects fail (39) and belong to branch (I) (conditioning τ_Π by arbitrary $E_k \in \Pi\widetilde{\mathcal{N}}\Pi$ is again classical on the conditioned data, Rem. 11.2(ii)); the canonical selection of a terminal PVM there is Conj. L.1 (App. L), whose effects carry genuine matter support and are fed into branch (I). We do not require $E_k \in \mathcal{N}$ in branch (II): in the geometric backgrounds of interest that requirement is provably vacuous (see the standing hypothesis below), and in gravity it is the dressed algebra, not $\mathcal{N}$, that carries the gauge-invariant observables [18, 96]. General terminal effects with nontrivial $\sigma^{\widehat{\omega}}$-spectrum (in particular arbitrary undressed $E_k \in \mathcal{N}$) are handled by the branch-(I) Araki conditioning (38)/App. G — the *primary* mechanism of this conjecture, and a theorem. Define the branch-(II) *two-boundary conditioned state* ω_k on $\widetilde{\mathcal{N}}$ by

$$\omega_k(x) \;:=\; p(k)^{-1}\,\widehat{\omega}\big(L_k^*\,x\,L_k\big), \qquad p(k) := \widehat{\omega}(\Lambda_k) \;\in\; (0,\infty), \tag{40}$$

with the finiteness proviso $\widehat{\omega}(\Lambda_k) < \infty$ (automatic for bounded spectral windows of $\widehat{P}$; normalised branch *ensembles* live on $\Pi\widetilde{\mathcal{N}}\Pi$ or in branch (I), per the bookkeeping correction above). Then

(i) ω_k is a normal positive linear functional on $\widetilde{\mathcal{N}}$ with $\omega_k(\mathbb{1}) = 1$, hence a normal state, and is uniquely determined by (40) on the strong-operator-dense Tomita algebra of $\sigma^{\widehat{\omega}}$-entire elements;

(ii) when $\mathcal{N} = \mathcal{B}(\mathcal{H}_{\mathrm{sys}})$ is a type I factor with faithful normal state $\omega(\cdot) = \mathrm{Tr}(\rho\,\cdot)$, the crossed product $\widetilde{\mathcal{N}}$ is canonically isomorphic to $\mathcal{B}(\mathcal{H}_{\mathrm{sys}})\bar{\otimes}L^\infty(\mathbb{R})$ and the density operator of $\omega_k\restriction_{\mathcal{N}}$ is exactly $\rho_2(\tau \mid k) = \rho^{1/2}\Lambda_k\rho^{1/2}/p(k)$ of Definition 3.1;

(iii) the modular flow $\sigma_t^{\omega_k}$ on $\widetilde{\mathcal{N}}$ generates the analog of unitary covariance (Thm. 3.7) in the type III setting, and the prior-average identity (Lem. 5.3(iii)) holds in the normalisable settings: in branch (I), $\sum_k p(k)\,\omega_k = \omega$ on $\mathcal{N}$ is the theorem Thm. G.2(iv); in branch (II) it holds as the state identity $\sum_k p(k)\,\omega_k = \tau_\Pi$-normalised reference on the corner $\Pi\widetilde{\mathcal{N}}\Pi$, and on the unprojected $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ only as an identity of (infinite) weights. The finite-dimensional shadow of this prior-average is the no-signalling/averaging-collapse identity $\sum_k p(k)\,M_k = \frac{1}{2}\{\rho, \sum_k \Lambda_k\} = \rho \succeq 0$ of Thm. 3.10, under which the per-branch Margenau–Hill negativity (temporal non-classicality of the two-boundary link) collapses to a positive state; the type III lift conjectured here is the corresponding statement that the irreducibly per-branch conditioned states ω_k average back to $\widehat{\omega}$.

Motivation and verification of (ii). The map $x \mapsto p(k)^{-1}\widehat{\omega}(L_k^* x L_k)$ is the Petz transpose channel of $x \mapsto \Lambda_k^{1/2} x \Lambda_k^{1/2}$ with respect to $\widehat{\omega}$ [18]; it is completely positive (as $x \mapsto L_k^* x L_k$ is) and unital because $\omega_k(\mathbb{1}) = p(k)^{-1}\widehat{\omega}(L_k^* L_k) = p(k)^{-1}\widehat{\omega}(\Lambda_k) = 1$. The key simplification of the present formulation is that the centralizer hypothesis $\Lambda_k \in \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ *enforces* entire analyticity of $\Lambda_k^{1/2}$ for the modular flow (since $\sigma_z^{\widehat{\omega}}(\Lambda_k^{1/2}) = \Lambda_k^{1/2}$ for all $z \in \mathbb{C}$), turning what was a generic domain hypothesis in earlier formulations into a structural property of the boundary data. The restriction is physically natural: the terminal effects $\{E_k\}$ are record-bearing observables stable under the modular flow of the background state, which is the operator-algebraic analog of record-completeness (Ax. 7.1). To verify (ii) in finite dim, in $\mathcal{N} = \mathcal{B}(\mathcal{H})$ the modular automorphism is $\sigma_z^{\omega}(y) = \rho^{iz} y \rho^{-iz}$ on $\mathcal{N}$ and lifts to $\sigma_z^{\widehat{\omega}}$ on $\widetilde{\mathcal{N}}$ that acts trivially on the dual $\mathbb{R}$-factor; restricting to $\mathcal{N}$ gives $L_k\restriction_{\mathcal{N}} = \rho^{1/2}\Lambda_k^{1/2}\rho^{-1/2}$ and $L_k^*\restriction_{\mathcal{N}} = \rho^{-1/2}\Lambda_k^{1/2}\rho^{1/2}$; substituting,

$$\omega_k(x) = p(k)^{-1}\,\mathrm{Tr}\big(\rho\cdot\rho^{-1/2}\Lambda_k^{1/2}\rho^{1/2}\,x\,\rho^{1/2}\Lambda_k^{1/2}\rho^{-1/2}\big) = p(k)^{-1}\,\mathrm{Tr}\big(\rho^{1/2}\Lambda_k\rho^{1/2}\,x\big),$$

by cyclicity of the trace; this is exactly $\mathrm{Tr}(\rho_2(\tau\mid k)\,x)$. Note that this displayed computation, with $L_k\restriction_{\mathcal{N}} = \rho^{1/2}\Lambda_k^{1/2}\rho^{-1/2} \neq \Lambda_k^{1/2}$, is the *branch-(I)* case of a generic (non-centralizer) effect: it is consistent with a genuinely non-commuting ρ_2 precisely because Λ_k is *not* assumed to commute with ρ. Within the branch-(II) scope the type-I shadow of (39) forces $[\rho,\Lambda_k]=0$, $L_k=\Lambda_k^{1/2}$, and the display degenerates to the commuting $\rho_2=\rho\Lambda_k/p(k)$ (Rem. 11.2). The substantive content of Conj. 11.1 is therefore (a) the centralizer hypothesis on $\{E_k\}$ (a genuine structural condition; equivalent to $\sigma^{\widehat{\omega}}$-invariance of the spectral data of each E_k), (b) the identification of (38)/(40) as the canonical lift of ρ_2 to the crossed product, and (c) part (iii). The general type III_1 case in which $\{E_k\}$ has nontrivial $\sigma^{\widehat{\omega}}$-spectrum (so that $\Lambda_k^{1/2}$ is not entire under modular flow) is the branch-(I) Araki conditioning: it is *no longer open* — existence, normality, perturbed-KMS structure and crossed-product lift are Thms. G.1 and G.2 — and it is where the two-boundary matter content resides (Rem. 11.2). What remains conjectural is (b) and (c) above. The standing hypothesis of Conj. 11.1 is accordingly two-branched: *the terminal effects form a POVM partition of unity with* $|\mathcal{K}|\geq 2$ *and at least one non-scalar* E_k*, conditioned via the Araki branch* (38) *in general (branch (I), primary), and via the entire-analytic Petz form* (40) *when additionally* $\{E_k\}\subset\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ *(branch (II)).* The branch-(II) condition (39) is the **centralizer-rich scope** (every occurrence of "centralizer-rich scope" in this paper refers to this condition); it guarantees well-posedness of the *simple* case — and by the Lemma above it is non-empty for *every* faithful normal seed state, since $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}\supseteq\{\lambda_s\}''\cong L^\infty(\mathbb{R})$ — but the physical content of the conjecture lives in branch (I). Two regimes realise the centralizer-rich scope:

- *(a) Centralizer-rich seeds (undressed records possible).* Some faithful normal states on type III factors have large centralizers in $\mathcal{N}$ itself: on any III_λ factor ($0<\lambda<1$) with separable predual there exist periodic faithful normal states ($2\pi/|\log\lambda|\in T(\mathcal{N})$, Connes [47]), whose centralizers are nontrivial finite von Neumann algebras (for the Powers–Araki–Woods factors, the hyperfinite II_1 factor [47, 48]); and on the hyperfinite III_1 factor Haagerup's solution of the bicentralizer problem supplies a faithful normal state whose centralizer is an irreducible hyperfinite II_1 subfactor [40, Thm. 3.1]. For such seeds $E_k\in\mathcal{N}^{\sigma^\omega}\subset\mathcal{N}$ realises (39) with undressed effects. We caution that these are existence statements for *states*, not for KMS states of a prescribed dynamics; note also that a nontrivial centralizer is compatible with extremality of the KMS state — for fixed dynamics the normal β-KMS states form a simplex governed by the *center* of the centralizer $Z(\mathcal{N}^{\sigma^\omega})$ [195, §5.3], and Haagerup's state is a factor state with II_1 centralizer — so the operative condition is centralizer-richness, not non-extremality.
- *(b) Geometric KMS seeds (the physical case; dressed records only).* For the distinguished geometric states — the Minkowski vacuum on a wedge (Bisognano–Wichmann) and the Bunch–Davies/Hartle–Hawking state on the de Sitter static patch — the modular flow is the boost and acts *ergodically*: primary geodesic KMS states on de Sitter are weakly mixing [49], and the boost-invariant elements of the static-patch algebra are c-numbers [18]. Hence $\mathcal{N}^{\sigma^\omega}=\mathbb{C}\mathbb{1}$, no undressed centralizer partition exists, and — by the *equality* in the Lemma — (39) is realised *exclusively* by spectral windows of the dressed generator $\widehat{P}$ in $\{\lambda_s\}''\cong L^\infty(\mathbb{R})$: pure clock-window effects, whose conditioning is provably classical on the matter algebra (Rem. 11.2). Arbitrary partitions of unity in the observer-projected type II_1 factor $\Pi\widetilde{\mathcal{N}}\Pi$ [18] are centralizer partitions of the *tracial* state τ_Π, not of $\widehat{\omega}$; they generically fail (39) and are conditioned through branch (I), which is where App. L selects the canonical terminal PVM (with genuine matter support, Rem. L.2).

Remark 11.2 (Centralizer conditioning is trivial on the matter algebra). Branch (II) is exactly as classical as it looks; this remark proves it, and thereby locates the two-boundary matter content of the conjecture in branch (I).

(i) *(Matter marginal is the seed.)* Let $\Lambda = g(\widehat{P}) \in \{\lambda_s\}''$ be a positive clock-window effect with $0 < \widehat{\omega}(\Lambda) < \infty$ (by the Lemma, in the geometric-ergodic regime (b) these exhaust the admissible branch-(II) effects), and let $\omega_\Lambda(x) := \widehat{\omega}(\Lambda^{1/2} x \Lambda^{1/2})/\widehat{\omega}(\Lambda)$. Then

$$\omega_\Lambda \restriction_{\pi(\mathcal{N})} = \omega.$$

Proof. Write $h := g^{1/2}$ (real) and $h(\widehat{P}) = \int \widehat{h}(s)\, \lambda_s\, ds$ in the sense of spectral calculus, $\widehat{h}(-s) = \overline{\widehat{h}(s)}$. Using the covariance $\lambda_s \pi(a) \lambda_s^* = \pi(\sigma_s^\omega(a))$,

$$\Lambda^{1/2}\pi(a)\Lambda^{1/2} = \iint \widehat{h}(s)\widehat{h}(s')\, \pi\big(\sigma_s^\omega(a)\big)\lambda_{s+s'}\, ds\, ds',$$

and the dual weight evaluates the λ_0 (Plancherel) component: $\widehat{\omega}(\Lambda^{1/2}\pi(a)\Lambda^{1/2}) = 2\pi \int |\widehat{h}(s)|^2\, \omega\big(\sigma_s^\omega(a)\big)\, ds$ $\big(2\pi \int |\widehat{h}|^2\big)\, \omega(a) = \widehat{\omega}(\Lambda)\, \omega(a)$, the middle equality by modular invariance $\omega \circ \sigma_s^\omega = \omega$ and the last by Plancherel ($2\pi \int |\widehat{h}|^2 = \int h^2 = \int g$, and $\widehat{\omega}(g(\widehat{P})) = \int g\, dp$). In the type-I model $\widetilde{\mathcal{N}} \cong \mathcal{B}(\mathcal{H}) \bar{\otimes} L^\infty(\mathbb{R})$, $\widehat{\omega} \cong \omega \otimes m$, this is the elementary $\int \mathrm{Tr}(\rho a)\, g(p)\, m(dp) / \int g\, dm = \mathrm{Tr}(\rho a)$. □

Consequently the branch probabilities $p(k) = \widehat{\omega}(E_k) = \int g_k(p)\, dp$ are determined by the clock-energy distribution alone, independently of any matter datum: branch-(II) records carry *zero* matter information, and the conditioning is classical Bayesian updating of the dressed observer-energy variable.

(ii) *(The MH witness vanishes on the centralizer scope.)* In the type-I shadow the centralizer of $\widehat{\omega}$ is $\{\rho\}' \bar{\otimes} L^\infty(\mathbb{R})$, so (39) forces $[\rho, \Lambda_k] = 0$ when the forward datum is the seed; under (CP) with Pos. 11.11 the commutation $[\widehat{H}_{\mathrm{m}}, \widehat{H}_{\mathrm{c}}] = 0$ moreover makes the seed stationary under the matter dynamics, so $[\rho(\tau), \Lambda_k(\tau)] = 0$ at *every* intermediate time. Thm. 3.10 then gives $M_k = \rho\Lambda_k/p(k) \succeq 0$ and $f_k = 0$ on every branch: the two-boundary link is operationally invisible (cf. the gloss following Thm. 3.7). The same conclusion holds for conditioning the tracial state τ_Π by *arbitrary* $E_k \in \Pi\widetilde{\mathcal{N}}\Pi$, since the tracial reference density is proportional to Π and commutes with every effect.

(iii) *(Consequence: where the content lives.)* Non-trivial two-boundary *matter* physics requires either terminal effects outside the centralizer of the reference state — branch (I), Thm. G.1, where $[\rho(\tau), \Lambda_k(\tau)] \neq 0$ is possible and the Margenau–Hill negativity $f_k > 0$ of Thm. 3.10 is non-vacuous — or a forward matter datum distinct from the seed. Branch (II) is retained as the computationally simple sector (exact entire analyticity, exact (CP)) in which the records resolve the dressed clock/observer-energy distribution, and as the well-posedness anchor of the centralizer-rich scope; it is *not* the carrier of the retrodictive matter content. The witness model of Conj. 11.55 accordingly draws its terminal effects from branch (I); non-vacuity $f_k > 0$ is secured as a finite-dimensional theorem by Prop. 11.53 (the infinite-dimensional witness yields only POVM effects, for which Thm. 3.10(iii)'s clean converse is generic-only), while its clock-window sector is branch (II) and is labelled there as the classical sector.

Remark 11.3 (Correction: location of the record projections). Earlier versions of this hypothesis asserted, citing [18], that every type III_λ factor ($0 < \lambda \leq 1$) admits non-extremal KMS states whose $\mathcal{N}$-centralizer is a nontrivial type II_1 subfactor and that the de Sitter setting of [18] is of this form, with $\{E_k\} \subset \mathcal{N}$. That attribution was wrong on three counts: (i) [18] prove the *opposite* for their setting — the boost-invariant elements of the static-patch algebra are c-numbers, and the nontrivial invariant algebra appears only after adjoining the observer, in $\Pi(\mathcal{N} \rtimes \mathbb{R})\Pi$, whose projections are not elements of $\mathcal{N}$; (ii) type III_1 admits no discrete decomposition ($T(\mathcal{N}) = \{0\}$), so the Connes–Takesaki mechanism invoked does not exist there; (iii) for the geometric flow the physical KMS state is weakly mixing [49], hence has trivial $\mathcal{N}$-centralizer. The record projections are therefore relocated to the dressed algebra, (39), where the II_1/tracial centralizer statements are theorems [18, 40], and physical realisability is carried by observer dressing (App. Z) rather than by membership in $\mathcal{N}$. Undressed records remain available in the centralizer-rich regime (a) and, without any centralizer hypothesis, through the Araki-perturbation (non-(CP)) branch of App. G. Whether an Araki perturbation ω_V of the geometric KMS state by a *bounded local* V can have a nontrivial σ^{ω_V}-fixed algebra in $\mathcal{N}$ is open; an affirmative answer would restore undressed records in a physically motivated regime and is recorded as a refinement, not assumed. A second correction supersedes part of the first: the relocation alone was insufficient, because the relocated (centralizer) effects are provably matter-trivial (Rem. 11.2) — conditioning on them returns the seed on $\mathcal{N}$ and zero Margenau–Hill negativity. The present two-branch formulation therefore assigns the physical two-boundary content to the Araki branch (I) (App. G, a theorem, no centralizer hypothesis) and retains the dressed centralizer branch (II) as the exactly solvable clock sector.

Conjecture 11.4 (Relative-entropy smallness implies semiclassical Einstein well-posedness)**.** *Relation to existing well-posedness results.* Rigorous well-posedness of the semiclassical Einstein equation is known only in restricted settings: Pinamonti–Siemssen, Meda–Pinamonti–Siemssen [194] establish Banach-fixed-point contraction on FLRW with scalar matter via inversion of the retarded nonlocal operator; Juárez-Aubry (see the 2025 review [193]) extends to general globally hyperbolic spacetimes with a conformally coupled massless scalar via quasi-linear hyperbolic reduction. Neither result gives Lipschitz continuity of the stress-energy map $g \mapsto \langle \widehat{T}_{\mu\nu} \rangle_{[g]}$ on a weighted Sobolev ball of general metric perturbations with an *explicit* contraction constant, which is what Conj. 11.4 requires. The present conjecture therefore extends the MPS/Juárez-Aubry program by postulating that the required Lipschitz bound is controllable by the retrodictive relative entropy $\mathcal{S}(\rho_{\mathrm{i}}, \{E_k\})$; Prop. 11.49 verifies this explicitly in the finite-dim Planck-cutoff setting, giving the closed-form bound $C \geq 2(1-\mu L_T^{(0)})^2 p_*^2 \eta/(\mu M)^2$ of (55). Extending Prop. 11.49 to general metric balls (the non-finite-dim case of the conjecture) via the MPS inversion-of-nonlocal-operator trick is the natural technical path, and is the content that remains conjectural.

In the setting of Conj. 10.2, fix once and for all a background g_0, a time interval T, and *coercivity constants* $p_*, \eta, M > 0$ such that the standing hypotheses hold:

(S1) (i) and (iii) of Conj. 10.2, i.e. $\widehat{T}_{\mu\nu}[g]$ is Banach-valued on a weighted Sobolev ball $\mathcal{B}_{g_0}$ around g_0, and the linearised retarded Green operator $\mathcal{G}_{g_0}$ is bounded between the appropriate spaces;

(S2) (uniform terminal probability) $\min_{k \in \mathcal{K}:\, p_0(k)>0} p_0(k) \geq p_*$;

(S3) (spectral gap of ρ_{i}) the support projection of ρ_{i} has dimension equal to a fixed integer and the smallest nonzero eigenvalue of ρ_{i} is $\geq \eta$, so $\rho_{\mathrm{i}}^{1/2}$ is operator-Lipschitz with norm $\leq (2\sqrt{\eta})^{-1}$ in a neighbourhood of ρ_{i};

(S4) (uniform stress-energy operator-norm bound) $\sup_{k: p_0(k)>0} \left\| \widehat{T}_{\mu\nu}[g_0] \right\|_{\rho_{\mathrm{i}},\mathrm{op}} \leq M$ on the GNS Hilbert space of ρ_{i};

(S5) (background-Lipschitz propagator and stress-energy) the propagator $g \mapsto U[g](\tau_f, \tau_i)$ and the renormalised stress-energy $g \mapsto \widehat{T}_{\mu\nu}[g]$ are Lipschitz on $\mathcal{B}_{g_0}$ in operator norm with Lipschitz constants $L_U, L_T < \infty$; equivalently, the parametric ODE for $U[g]$ is Lipschitz-stable on $\mathcal{B}_{g_0}$ and the Hadamard-renormalisation prescription is locally uniform.

(S6) (junction-regularity inheritance) both (vi-a) and (vi-b) of Conj. 10.2 hold uniformly across $\mathcal{B}_{g_0}$: (S6-a) the conditioned source $\mathrm{Tr}(\rho_2[g](\tau \mid k)\widehat{T}_{\mathrm{S},\mu\nu}[g])$ is in L^∞ on Σ_{τ_f} (for the induced metric $h[g]$) and on each record cone $\mathcal{N}_m$ (for the induced degenerate metric and a fixed transversal), uniformly for $g \in \mathcal{B}_{g_0}$, in m, and for every k with $p_0(k) \geq p_*$; (S6-b) the Barrabès–Israel null-junction map of Conj. 10.2 (vi-b) — and, whenever the terminal jump is nonzero, the ϵ-normalized terminal Lanczos system (21) — has operator norm $C_{\mathrm{J}}(g)$ uniformly bounded on $\mathcal{B}_{g_0}$, so that the bulk source bound (S6-a) transports to a uniform shell bound. Together these are the analytic input needed to make the null-shell data well-defined and uniformly bounded across the metric ball; without (S6-b) a uniform L^∞ source bound does not yield a uniform shell bound.

(S7) (Lipschitz extension to the full interval) the operator-Lipschitz bound on $g \mapsto \rho_2[g](\tau \mid k)$ derived from (S5) and Powers–Størmer at $\tau = \tau_i$ extends to every $\tau \in [\tau_i, \tau_f]$ via Heisenberg propagation by $U[g](\tau, \tau_i)$, with the propagator-Lipschitz constant L_U of (S5) controlling the dependence on τ uniformly on the interval. (This is automatic from (S5) and propagator unitarity once (S3) gap and (S4) operator-norm bound hold uniformly along the interval, by Grönwall on the Heisenberg ODE; we list it explicitly so that the entropy bound below applies on the full interval, not only at $\tau = \tau_i$.)

(S8) (**seed-background contraction**) the bare contraction constant of Conj. 10.2 (vii) evaluated at the seed background is already strictly below 1: $\kappa\, T\, L_T^{(0)}\, \|\mathcal{G}_{g_0}\|_{\mathrm{op}} < 1$, where $L_T^{(0)}$ is the operator-Lipschitz constant of $g \mapsto \mathrm{Tr}(\rho_2[g](\tau \mid k)\widehat{T}_{\mathrm{S},\mu\nu}[g])$ on $\mathcal{B}_{g_0}$ in the zero-entropy limit $\mathcal{S} = 0$ (i.e. the Lipschitz constant coming from (S5) alone, before any $\mathcal{S}$-controlled correction of Rem. "Non-vacuity" below is added). Small retrodictive entropy only repairs *corrections* to the contraction constant; (S8) excludes the case of an already non-contractive seed for which $\mathcal{S} \to 0$ would not be enough.

Define the *retrodictive relative entropy on* g_0

$$\mathcal{S}(\rho_{\mathrm{i}}, \{E_k\}) := \sup_{k \in \mathcal{K}:\, p[g_0](k) \geq p_*} S\big(\rho_2[g_0](\tau_i \mid k) \,\big\|\, \rho_{\mathrm{i}}\big), \qquad S(\sigma \| \rho) := \mathrm{Tr}\, \sigma(\log \sigma - \log \rho). \tag{41}$$

(All g-dependence in $\mathcal{S}$ is at the seed background g_0; the Powers–Størmer / Pinsker chain below transports the bound to $g \in \mathcal{B}_{g_0}$ via L_U, L_T of (S5) and (S7).) There exists a constant $C = C(g_0, \kappa, T, p_*, \eta, M, L_U, L_T) > 0$, depending only on the standing data (S1)–(S8), such that whenever $\mathcal{S}(\rho_{\mathrm{i}}, \{E_k\}) < C$, the state-dependent hypotheses (ii) and (iv) of Conj. 10.2 hold for every k with $p_0(k) \geq p_*$, and the joint fixed-point system (19) of Conj. 10.2 admits a unique Barrabès–Israel weak solution — the scaffolding metrics g_r and the per-sector metrics $g_k \in \mathcal{B}_{g_0}$ for each such k — smooth off the record cones $\mathcal{N}_m$ and, only when Ax. 7.1 fails, off Σ_{τ_f}, with the prescribed null-shell data (20) on each $\mathcal{N}_m$ and the ϵ-normalized terminal Lanczos jump (21) on Σ_{τ_f} (the latter vanishing a.s. under Ax. 7.1, so the assembled metric is then continuous across Σ_{τ_f}); not a smooth classical solution in the presence of a non-vanishing shell.

Motivation. Hypotheses (S1) and (S3)–(S4) are properties of the matter operator and linearised gravity sector at the fixed background g_0 and of the prior data $(\rho_{\mathrm{i}}, \{E_k\})$ alone, while (S2) is a probabilistic restriction on observable records. (S3) is the gap condition required for operator-Lipschitz control of $\rho^{1/2}$ (without it the square root is only Hölder, and the trace-norm bound on $\rho_2[g] - \rho_2[g_0]$ degenerates near eigenvalue crossings). (S4) replaces a generic state-dependent operator norm by a uniform bound on the GNS space of ρ_{i} (*a priori* controllable from a Hadamard-renormalized stress-energy operator on the chosen background). Pinsker bounds $\|\rho_2(\tau_i \mid k) - \rho_{\mathrm{i}}\|_1 \leq \sqrt{2\,\mathcal{S}}$ uniformly in k with $p(k) \geq p_*$. (*Normalisation convention:* throughout this paper $\|\cdot\|_1$ is the *full* trace norm, $\|\rho\|_1 = \mathrm{Tr}\,\rho = 1$ for states, so quantum Pinsker reads $S(\sigma\|\rho) \geq \frac{1}{2}\|\sigma - \rho\|_1^2$ in nats, i.e. $\|\sigma - \rho\|_1 \leq \sqrt{2S}$; in the total-variation normalisation $\mathrm{TV} = \frac{1}{2}\|\cdot\|_1$ of App. A this is the familiar $\mathrm{TV}^2 \leq \frac{1}{2}S$. Earlier versions quoted $\sqrt{\mathcal{S}/2}$ here — the TV-form constant applied to the full trace norm — which overstated the bound by a factor of 2 and the derived contraction constants by a factor of 4; all displayed constants below carry the corrected factor.) Combining Pinsker with (S3) gives the Fréchet-derivative bound (ii) as an $\mathcal{S}$-controlled perturbation of the unconditioned case, with constants depending on p_*^{-1} and $\eta^{-1/2}$; combining Pinsker with (S4) and operator-norm continuity gives (iv) bounded by $M + O(p_*^{-1/2}\eta^{-1/2}\sqrt{\mathcal{S}})$. The Banach fixed-point contractivity constant on each k-sector is then controllable by $\mathcal{S}$ alone at fixed (p_*, η, M). $\mathcal{S} = 0$ iff $\rho_2(\tau_i \mid k) = \rho_{\mathrm{i}}$ for all k with $p(k) \geq p_*$, equivalently $\rho_{\mathrm{i}}^{1/2}\Lambda_k(\tau_i)\rho_{\mathrm{i}}^{1/2} = p(k)\,\rho_{\mathrm{i}}$ on the support of ρ_{i} for every such k. (Mere commutation $[\Lambda_k(\tau_i), \rho_{\mathrm{i}}] = 0$ is necessary but not sufficient: e.g. $\rho_{\mathrm{i}} = \frac{1}{2}\mathbb{1}_2$ and $\Lambda_k = \mathrm{diag}(1, 0)$ commute but yield $\rho_2 = \mathrm{diag}(1,0) \neq \rho_{\mathrm{i}}$.) When $\mathcal{S} = 0$, (22) reduces to ordinary semiclassical gravity with stress-energy $\mathrm{Tr}(\rho_{\mathrm{i}}\widehat{T}_{\mathrm{S}})$ in every sector.

Scope. The hypotheses (S3)–(S4) restrict Conj. 11.4 to gapped finite-dimensional matter (or to a gapped subsector controllable by spectral projection to an interval bounded away from 0 in ρ_{i}'s spectrum). Vacuum states and KMS states of interacting QFT, whose modular spectrum accumulates at 0, lie outside this regime and require a separate argument (cf. Conj. 11.1 for the type III lift, which does *not* use a gap hypothesis). Making the bound quantitative would resolve item (2) in the regime of weak retrodictive conditioning relative to a controlled background, with the cutoff p_* a physically natural restriction (vanishingly improbable records are unobservable) and (S3)–(S4) the standard coercivity assumptions required by any rigorous mean-field semiclassical Einstein theorem.

Non-vacuity of the entropy bound. The role of $\mathcal{S}$ in the gapped regime (where (S4) already gives an operator-norm control representation-invariantly) is *not* to bound $\left\|\widehat{T}_{\mu\nu}[g_0]\right\|_{\rho_2,\,\mathrm{op}}$, which is already controlled by M, but to bound the *operator-Lipschitz constant of the map* $g \mapsto \rho_2[g](\tau \mid k)$: by the Powers–Størmer inequality and the (S3) gap, $\|\rho_2[g](\tau \mid k) - \rho_2[g_0](\tau \mid k)\|_1$ is controlled by a constant times the trace-norm propagator perturbation $\|U[g] - U[g_0]\|_{\mathrm{op}}$, *but* the constant in this Lipschitz bound diverges as the support of $\rho_2[g_0](\tau \mid k)$ approaches the boundary of the support of ρ_{i}; small $\mathcal{S}$ is exactly the condition that $\rho_2(\tau_i \mid k) \succeq (1 - O(\sqrt{2\,\mathcal{S}}/\eta))\rho_{\mathrm{i}}$ on the support of ρ_{i}, hence that the Lipschitz constant $L_T(\mathcal{S})$ for hypothesis (ii) of Conj. 10.2 is finite and bounded by $L_T \leq L_T^{(0)} + O(\mathcal{S}^{1/2}/(p_*\eta))$ (with $L_T^{(0)}$ the bare Lipschitz constant on g_0 from (S5)). This bound is non-vacuous: $\mathcal{S}$ controls a quantity (L_T) that (S1)–(S5) alone do *not* bound, and the QFT regime in which (S3)–(S4) fail is exactly the regime in which both this Lipschitz bound and the conjecture's conclusion are out of scope; there is no regime in which $\mathcal{S}$ is doing no work.

Conjecture 11.5 (Composition-law selection of the modified Wheeler–DeWitt constraint)**.** Let $(\widehat{H}_{\mathrm{c}}, \{|t\rangle\})$ be a Page–Wootters clock and $\widehat{H}_{\mathrm{s}}$ a matter Hamiltonian on $\mathcal{H}_{\mathrm{sys}}$. Let $\widehat{T}_s := e^{-is\widehat{H}_{\mathrm{c}}/\hbar}$ denote the clock translation group. Consider the family of candidate clock+matter constraints

$$\widehat{C}_{B,\delta} := \widehat{H}_{\mathrm{c}} \otimes \mathbb{1} + \mathbb{1} \otimes \widehat{H}_{\mathrm{s}} + B(\delta) \otimes \mathbb{1}, \qquad B(\delta) \in \mathcal{B}(\mathcal{H}_{\mathrm{clock}})_{\mathrm{sa}}, \tag{42}$$

where $\delta \mapsto B(\delta)$ is a strongly-continuous family of bounded self-adjoint clock operators. Require:

(A) **Self-adjointness:** $B(\delta) = B(\delta)^*$ for all $\delta \in \mathbb{R}$ (so that $\widehat{C}_{B,\delta}$ is self-adjoint and the constraint surface is well-posed);

(B) **Clock-translation covariance:** $\widehat{T}_s B(\delta) \widehat{T}_{-s} = B(\delta)$ for all $s \in \mathbb{R}$ (equivalently $B(\delta) \in \{\widehat{H}_{\mathrm{c}}\}''$);

(C) **Time-reversal covariance:** $B(-\delta) = B(\delta)$ (the constraint should not distinguish past from future at the level of the clock-side perturbation);

(D) **Cocycle composition law (d'Alembert):**

$$B(\delta_1 + \delta_2) + B(\delta_1 - \delta_2) \;=\; 2\,\cos\!\big(\delta_1 \widehat{H}_{\mathrm{c}}/\hbar\big)\, B(\delta_2) \;+\; 2\,B(\delta_1) \tag{43}$$

for all $\delta_1, \delta_2 \in \mathbb{R}$. (This is the operator-valued analog of d'Alembert's functional equation; together with (B) it forces $B(\delta)$ to act on each spectral fibre of $\widehat{H}_{\mathrm{c}}$ as a constant multiple of $\cos(\delta\xi/\hbar) - 1$.)

(E) **TDSE limit:** $B(\delta) \to 0$ in strong-operator topology as $\delta \to 0$; for every $\psi \in \mathcal{D}(\widehat{H}_{\mathrm{c}}^2)$ the map $\delta \mapsto B(\delta)\psi$ is twice strongly differentiable at $\delta = 0$ with $B'(0)\psi = 0$ and $-B''(0)\psi = \alpha\,\widehat{H}_{\mathrm{c}}^2\psi/\hbar^2$ for some $\alpha \in \mathbb{R}$ (so that the leading low-energy correction has the fixed scale α). The identity for $-B''(0)$ holds in the strong sense on $\mathcal{D}(\widehat{H}_{\mathrm{c}}^2)$, not in operator norm: $B(\delta)$ is a bounded operator family while $\widehat{H}_{\mathrm{c}}^2$ is unbounded, so the second strong derivative is necessarily a quadratic-form identity on the domain of $\widehat{H}_{\mathrm{c}}^2$ (this is consistent with $\widehat{\Sigma}_\delta = \cos(\delta\widehat{H}_{\mathrm{c}}/\hbar) - \mathbb{1}$, whose strong second derivative at $\delta = 0$ is $-(\widehat{H}_{\mathrm{c}}/\hbar)^2$ on $\mathcal{D}(\widehat{H}_{\mathrm{c}}^2)$).

Then $B(\delta) = \alpha\,\widehat{\Sigma}_\delta = \alpha(\cos(\delta \widehat{H}_{\rm c}/\hbar) - \mathbb{1})$, and (27) is the unique element of (42) satisfying (A)–(E).

Motivation. Covariance (B) and the spectral theorem for $\widehat{H}_{\rm c}$ force $B(\delta) = f(\widehat{H}_{\rm c}, \delta)$ for a real Borel function $f(\xi, \delta)$. Time-reversal covariance (C) restricts $f(\xi, \delta)$ to be even in δ. The cocycle relation (43) of (D) specialises on each spectral fibre ξ of $\widehat{H}_{\rm c}$ to the scalar d'Alembert equation $f(\xi, \delta_1+\delta_2)+f(\xi, \delta_1-\delta_2) = 2\cos(\delta_1\xi/\hbar)f(\xi, \delta_2)+2f(\xi, \delta_1)$, whose strongly-continuous, even-in-δ, $f(\xi, 0) = 0$ solutions are exactly $f(\xi, \delta) = c(\xi)(\cos(\delta\xi/\hbar)-1)$ for some real-measurable c (this is d'Alembert's classical theorem on $\mathbb{R}$ applied fibrewise; the kernel of ξ giving zero frequency produces $f \equiv 0$, consistent with c free there). The normalisation (E) fixes $c(\xi) \equiv \alpha$ (state-independent of ξ), giving $B(\delta) = \alpha\widehat{\Sigma}_\delta$. We emphasise that axiom (B) is covariance under the *fixed* clock $\widehat{H}_{\rm c}$ only: every real Borel function of $\widehat{H}_{\rm c}$ satisfies it, so (B) selects within the class of single-clock perturbations and does not impose invariance under a change of clock or foliation. The uniqueness statement should therefore be read as "unique cosine deformation among single-clock perturbations," not as "the unique covariant modification"; refoliation/change-of-clock covariance of the selected δt is left as the open problem of Sec. 11.

Proposition 11.6 (Operator-sense d'Alembert reduction; rigorous step of Conj. 11.5)**.** *Assume (A), (B), (C), (E) of Conj. 11.5, plus the strengthening of (D) to the strong-operator equality* (43) *on the dense domain $\mathcal{D}(\widehat{H}_c^2)$, with $\delta \mapsto B(\delta)\psi$ strongly continuous on $\mathcal{D}(\widehat{H}_c^2)$ for every $\delta \in \mathbb{R}$. Then $B(\delta) = \alpha\,\widehat{\Sigma}_\delta$ for a unique $\alpha \in \mathbb{R}$ fixed by (E), and* (27) *is the unique element of* (42) *satisfying (A)–(E).*

Proof. By (B) and the spectral theorem, $B(\delta) \in \{\widehat{H}_{\rm c}\}''$, so there is a real Borel function $f(\cdot, \delta) : \mathbb{R} \to \mathbb{R}$ such that $B(\delta) = f(\widehat{H}_{\rm c}, \delta)$ in the spectral calculus of $\widehat{H}_{\rm c}$. Boundedness of $B(\delta)$ implies $f(\cdot, \delta) \in L^\infty(\mathbb{R}, d\mu_{\widehat{H}_{\rm c}})$. (C) gives $f(\xi, -\delta) = f(\xi, \delta)$ $\mu_{\widehat{H}_{\rm c}}$-a.e. Read on each spectral fibre ξ, the operator equation (43) reduces to the scalar functional equation

$$f(\xi, \delta_1 + \delta_2) + f(\xi, \delta_1 - \delta_2) \;=\; 2\cos(\delta_1\xi/\hbar)\, f(\xi, \delta_2) + 2\, f(\xi, \delta_1),$$

first for all rational pairs (δ_1, δ_2) off a single $\mu_{\widehat{H}_{\rm c}}$-null set (countably many pairs, one null set per pair). For each fixed $\xi \neq 0$ the symmetrization argument of App. B (Lem. B.2: swap $\delta_1 \leftrightarrow \delta_2$, use evenness, subtract) shows with *no* regularity input that $h := f(\xi, \cdot)$ is exactly $h(\delta) = c(\xi)(\cos(\delta\xi/\hbar) - 1)$ on $\mathbb{Q}$ for some real constant $c(\xi)$; no fibrewise continuity in δ need be extracted from strong continuity, which is used only in the final reassembly step below. (This replaces the classical continuity-based d'Alembert classification [29], §2.4.1, which is not needed here because the cosine kernel in (43) is known.) On the zero-eigenvalue block $\xi = 0$, i.e. on $E_{\widehat{H}_{\rm c}}(\{0\})\mathcal{H}$, the equation reduces at the operator level to the parallelogram equation, whose strongly continuous solutions are exactly $B(\delta)E_{\widehat{H}_{\rm c}}(\{0\}) = \delta^2 Q$ with Q bounded self-adjoint (Lem. B.2(iii)); hypothesis (E) forces $-B''(0)\psi = \alpha\widehat{H}_{\rm c}^2\psi/\hbar^2 = 0$ for $\psi \in \ker \widehat{H}_{\rm c}$, hence $Q = 0$ and $f(0, \cdot) \equiv 0$. (An earlier draft excluded $Q \neq 0$ by a boundedness argument that was a non-sequitur — boundedness of each $B(\delta)$ does not bound $\delta \mapsto h(\delta)$ on $\mathbb{R}$; the normalization (E) is what kills the kernel block.) This matches the $\xi \neq 0$ formula $h(\delta) = c(\xi)(\cos(\delta\xi/\hbar) - 1) \to 0$ as $\xi \to 0$. Measurability of $\xi \mapsto c(\xi)$ follows from measurability of $f(\cdot, \delta)$ for each rational δ. Hypothesis (E) requires $-B''(0) = \alpha\widehat{H}_{\rm c}^2/\hbar^2$ on $\mathcal{D}(\widehat{H}_c^2)$; evaluated along the rational sequence $\delta_n = 1/n$ and read fibrewise (norm convergence gives a.e. fibre convergence along a subsequence, uniformly over a countable dense set of vectors — the extraction is carried out in Step 2 of the proof of Thm. B.5 in App. B), this gives $c(\xi)\xi^2/\hbar^2 = \alpha\xi^2/\hbar^2$, hence $c(\xi) = \alpha$ for $\mu_{\widehat{H}_{\rm c}}$-a.e. $\xi \neq 0$. Reassembly under the spectral theorem gives $B(\delta) = \alpha(\cos(\delta\widehat{H}_{\rm c}/\hbar) - \mathbb{1}) = \alpha\widehat{\Sigma}_\delta$ for every rational δ, and strong continuity of both sides extends the identity to all $\delta \in \mathbb{R}$. Uniqueness is immediate from the chain of equalities. □

Remark 11.7 (Status of Prop. 11.6). Prop. 11.6 is rigorous; it converts the heuristic selection argument into a theorem under the explicit strong-operator strengthening of (D) on $\mathcal{D}(\widehat{H}_c^2)$. The remaining content of Conj. 11.5 is therefore the *physical postulate* that any candidate clock-side perturbation $B(\delta)$ entering a constraint of the form (42) should satisfy (A)–(E) on a domain at least as large as $\mathcal{D}(\widehat{H}_c^2)$. Under that postulate, the self-adjoint shift $\alpha\widehat{\Sigma}_\delta$ of Conj. 10.13 is no longer arbitrary: it is the unique operator satisfying the package, by Prop. 11.6. Establishing the operator-sense (D) for a candidate $B(\delta)$ remains an analytic check on the candidate, but does *not* require any additional functional-equation theorem. Establishing the package (A)–(E) on the domain $\mathcal{D}(\widehat{H}_c^2)$ would turn the postulated constraint of Conj. 10.13 into a *derived* object and resolve item (3). A substantial step in that direction is taken in Sec. 10.3: for effective one-leg reductions of translation 1-cocycles (the derived form of the two-time temporal link), the composition law (D) is a *theorem*, not an axiom (Prop. 10.23 and Step 4 of the proof of Lemma 10.22), so within that class the entire package reduces to the covariance, evenness, and normalisation clauses, each of which is itself derived from the two-boundary structure (Rem. 10.21).

Conjecture 11.8 (Horizon subalgebra as modular centralizer)**.** *Standing scope and relation to prior work.* We work on the crossed-product horizon algebra $\widetilde{\mathcal{A}}(\mathcal{H}^+) := \mathcal{A}(\mathcal{H}^+) \rtimes_{\sigma^{\omega_{\rm H}}} \mathbb{R}$ of [18], equipped with the dual weight $\widehat{\omega}_{\rm H}$. For *stationary / bifurcate Killing horizons* — Schwarzschild, Kerr, de Sitter static patch — the identification of horizon subalgebras with modular centralizers and the associated generalised-entropy formula have been established in the recent crossed-product programme: Jensen–Sorce–Speranza [87] give the Type II identification for general subregions on stationary backgrounds; Kudler-Flam–Leutheusser–Satishchandran [88, 89] extend to bifurcate Killing horizons and FLRW observational cosmology; Faulkner–Speranza [92] derive the generalised second law on arbitrary cuts of Killing horizons via half-sided modular translations; Kirklin [392] extends the algebraic GSL to all orders in perturbative gravity without a UV cutoff, via a boost-invariant Type II crossed product with null-translation invariance and dynamical-cut quantum reference frames, at the cost of a free-energy correction term for the cut degrees of freedom. These citations close the stationary case of the present conjecture. The genuinely open content of Conj. 11.8 is the extension to *dynamical horizons* (non-Killing, Ashtekar–Krishnan trapping/apparent horizons) on generic globally hyperbolic backgrounds, where no existing result supplies the modular centralizer identification. The equivariance hypothesis (KE) below is conceptually motivated by QRF-intertwining theorems of De Vuyst–Eccles–Höhn–Kirklin [96, 97]; making (KE) a theorem in the dynamical-horizon case is the content of this conjecture. The original BW/HH state $\omega_{\rm H}$ on $\mathcal{A}(\mathcal{H}^+)$ has trivial $\mathcal{A}$-centralizer for the geometric (weakly mixing) modular flow, in which case any $\mathcal{A}$-level protocol records only scalars and is operationally vacuous; the crossed product, by contrast, *always* has a non-trivial $\widehat{\omega}_{\rm H}$-centralizer, since by the Lemma of Conj. 11.1 $\widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\rm H}}} = \mathcal{A}(\mathcal{H}^+)^{\sigma^{\omega_{\rm H}}} \rtimes \mathbb{R} \supseteq \{\lambda_s\}'' \cong L^\infty(\mathbb{R}) \supsetneq \mathbb{C}\mathbb{1}$ (diffuse abelian in the geometric-ergodic case; when $\mathcal{A}(\mathcal{H}^+)^{\sigma^{\omega_{\rm H}}}$ is a II_1 factor — the centralizer-rich regime — it is a *finite* type II algebra with diffuse center $\supseteq \{\lambda_s\}''$, never a factor, by the Lemma; the observer-projected CLPW algebra with dressed generator bounded below is a II_1 factor [18]), so the construction below is never vacuous. Throughout this conjecture "centralizer" means the crossed-product centralizer $\widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\rm H}}}$; states and conditioned functionals descend to the original $\mathcal{A}(\mathcal{H}^+)$ by plain restriction of normal functionals along the inclusion $\mathcal{A}(\mathcal{H}^+) \hookrightarrow \widetilde{\mathcal{A}}$ (no normal conditional expectation $\widetilde{\mathcal{A}} \to \mathcal{A}(\mathcal{H}^+)$ exists — the canonical map is Haagerup's operator-valued weight; cf. the bookkeeping correction in Conj. 11.1). The resulting projections are *undressed* observables of the underlying QFT iff they additionally lie in $\mathcal{A}(\mathcal{H}^+)^{\sigma^{\omega_{\rm H}}}$, which for geometric flows forces them to be scalars; in general they are gravitationally dressed observables of the observer's algebra, physically realisable in the observer-relative sense of Rem. 11.3 and Prop. Z.1.

Let $\mathcal{H} \subset \mathcal{M}$ be a future event horizon generated by null geodesic congruence ℓ^a with associated modular automorphism group $\sigma_t^{\omega_{\rm H}}$ on the horizon algebra $\mathcal{A}(\mathcal{H}^+)$ of the Bisognano–Wichmann / Hartle–Hawking state $\omega_{\rm H}$, and write $\sigma_t^{\widehat{\omega}_{\rm H}}$ for the dual modular flow on $\widetilde{\mathcal{A}}(\mathcal{H}^+)$. Let γ be a timelike worldline that remains in the exterior $\mathcal{M} \setminus J^-(\mathcal{H})$. Let $\widetilde{\mathcal{A}}(\gamma(\tau))$ denote the crossed-product local algebra of a small causal-diamond neighbourhood of $\gamma(\tau)$, and $\widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\rm H}}}(\gamma(\tau)) := \widetilde{\mathcal{A}}(\gamma(\tau)) \cap \widetilde{\mathcal{A}}(\mathcal{H}^+)^{\sigma^{\widehat{\omega}_{\rm H}}}$ its intersection with the crossed-product centralizer of $\widehat{\omega}_{\rm H}$ (non-trivial by the Lemma of Conj. 11.1). For notational continuity with the rest of the conjecture body we write $\mathcal{A}^{\sigma^{\omega_{\rm H}}}(\gamma(\tau)) := \widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\rm H}}}(\gamma(\tau))$ throughout Conj. 11.8 and Prop. 11.9 below; by the Lemma of Conj. 11.1 this has the form $\mathcal{A}^{\sigma^{\omega_{\rm H}}} \rtimes \mathbb{R}$ localised at $\gamma(\tau)$ — diffuse abelian ($\cong L^\infty(\mathbb{R})$) in the geometric-ergodic case, a non-trivial finite type II algebra with diffuse center (never a factor, by the Lemma) in the centralizer-rich regime (a II_1 *factor* only after an observer-positivity projection, which the standing scope does not impose) — and in either case it carries a faithful normal semifinite trace and non-scalar partitions of unity, which is all the construction below uses. Fix once and for all, at each time s_m, a maximal abelian sub-von-Neumann-algebra $\mathcal{Z}(\gamma(s_m)) \subset \mathcal{A}^{\sigma^{\omega_{\rm H}}}(\gamma(s_m))$ (which exists by Zorn's lemma applied to the lattice of abelian subalgebras of $\mathcal{A}^{\sigma^{\omega_{\rm H}}}(\gamma(s_m))$), and a *Borel partition scheme* $(\mathcal{B}^{(m,n)})_{n\in\mathbb{N}}$ which is a strictly increasing sequence of *finite* Borel partitions of the joint spectrum of a generating self-adjoint element $q^{(m)}$ of $\mathcal{Z}(\gamma(s_m))$, generating $\mathcal{Z}(\gamma(s_m))$ in the limit $n \to \infty$ in the strong operator topology (when the generator has discrete spectrum the scheme is constant and the unique finite spectral resolution suffices; when $q^{(m)}$ has diffuse spectrum the scheme refines indefinitely). For each fixed $n \in \mathbb{N}$ the finite spectral projections $P_j^{(m,n)} := \mathbf{1}_{B_j^{(m,n)}}(q^{(m)}) \in \mathcal{Z}(\gamma(s_m)) \subset \widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\rm H}}}(\gamma(s_m))$ define a *finite POVM in the centralizer* summing to identity, hence a measurement protocol in the sense of Sec. 5; in the diffuse case the canonical record process is the Kolmogorov inverse limit of the protocol family $\{\mathcal{P}_\gamma^{\mathrm{cent},n}\}_{n\in\mathbb{N}}$ over the natural refinement maps, which exists by Kolmogorov's extension theorem applied to the consistent finite-stage record laws. Define the *canonical centralizer-resolving protocol family* $\mathcal{P}_\gamma^{\mathrm{cent}} = (\mathcal{P}_\gamma^{\mathrm{cent},n})_{n\in\mathbb{N}}$ on γ: for every n the stage-n Kraus collection $\{P_j^{(m,n)}\}_{j\in J_{m,n}}$ is finite (so each stage is a bona-fide protocol of Sec. 5), and the diffuse case is accessed only via the inverse-limit record process. The protocol always exists (a finite Borel partition of any spectrum exists) and depends on the choice of $(\mathcal{Z}, q, (\mathcal{B}^{(m,n)})_n)$: different choices yield protocols whose σ-algebras of records are *measure-theoretically isomorphic* as probability spaces (Prop. 11.9) ($\mathcal{G}_\gamma$ is therefore protocol-canonical *as a measure-theoretic record process modulo measure-preserving isomorphism*; its isomorphism class as a probability space, and in particular the residual entropy $H(K \mid \mathcal{G}_\gamma)$, is intrinsic, while a particular σ-algebra of records depends on the choice of ($\mathcal{Z}$, generator), and the underlying masas $\mathcal{Z}, \mathcal{Z}'$ need not be inner conjugate as subalgebras of $\mathcal{A}^{\sigma^{\omega_{\rm H}}}(\gamma(s_m))$ (cf. Prop. 11.9)). The records of $\mathcal{P}_\gamma^{\mathrm{cent}}$ generate the maximal centralizer-invariant sub-σ-algebra of $\overline{\mathcal{F}}_{\tau_f-}^{\gamma,\mathcal{P}_\gamma^{\mathrm{cent}}}$ relative to the chosen $\mathcal{Z}$. Define the *intrinsic accessible algebra* as the σ-algebra of *classical records* produced by $\mathcal{P}_\gamma^{\mathrm{cent}}$ before τ_f,

$$\mathcal{G}_\gamma \;:=\; \sigma\Big(\big\{\pi_\tau^{\gamma,\mathcal{P}_\gamma^{\mathrm{cent}}} \;:\; \tau < \tau_f,\, \tau \in \mathbb{Q}\big\}\Big) \;\subset\; \overline{\mathcal{F}}_{\tau_f-}^{\gamma,\mathcal{P}_\gamma^{\mathrm{cent}}}, \tag{44}$$

where $\pi_\tau^{\gamma,\mathcal{P}_\gamma^{\mathrm{cent}}} : \Omega_\gamma \to \prod_{s_m\le\tau} J_m$ is the τ-record map of $\mathcal{P}_\gamma^{\mathrm{cent}}$ in the sense of Sec. 5 (the classical Kolmogorov-extension projection of the centralizer Kraus schedule onto the realised outcome path). This is by construction a sub-σ-algebra of the record σ-algebra (a measure-theoretic object), *not* a sub-von Neumann algebra of $\widetilde{\mathcal{A}}$; the operator-algebraic lift back to $\widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\rm H}}}$ is the (commutative) joint spectral algebra of the chosen $\mathcal{Z}(\gamma(s_m))$, and the type-mismatch between an algebra of operators and a σ-algebra of records that the earlier formulation conflated is now made explicit. (*Quantifier choice.* We do not quantify over all admissible protocols: the trivial protocol $J_m = \{1\}$, $M_1^{(m)} = \mathbb{1}$ is admissible and contributes the trivial σ-algebra, so $\bigcap_{\mathcal{P}_\gamma}$ collapses $\mathcal{G}_\gamma$ to $\{\emptyset, \Omega_\gamma\}$; conversely $\bigvee_{\mathcal{P}_\gamma}$ over-estimates accessible information by counterfactually combining mutually incompatible POVM schedules. The canonical protocol $\mathcal{P}_\gamma^{\mathrm{cent}}$ is the operationally distinguished maximal informational schedule compatible with the modular structure of $\omega_{\rm H}$, and (44) is protocol-canonical in this sense.) Then, *subject to the K-equivariance hypothesis* (KE) of Prop. 11.9, $\mathcal{G}_\gamma$ is exactly the information about the terminal outcome K that is accessible to γ, and the residual entropy $H(K \mid \mathcal{G}_\gamma)$ of Cor. 9.2 equals $H_{\mathrm{cl}}(K \mid \mathcal{Z}_\gamma)$, where $\mathcal{Z}_\gamma$ is the classical (commutative) sub-σ-algebra of $\overline{\mathcal{F}}^{\gamma}_{\tau_f-}$ generated by the joint measurements of all mutually commuting elements of $\mathcal{A}(\mathcal{H}^+)^{\sigma^{\omega_{\mathrm{H}}}}$ (the modular-invariant subalgebra, i.e. the centralizer of ω_{H}) that are localisable on γ, and the conditional entropy is taken with respect to the classical probability measure induced by ω_{H} on the joint spectrum of these commuting observables. Equivalently, $\mathcal{Z}_\gamma$ is the spectrum of the maximal abelian sub-von Neumann-algebra of $\mathcal{A}(\mathcal{H}^+)^{\sigma^{\omega_{\mathrm{H}}}}$ that is contained in $\mathcal{A}(J^-(\gamma))$.

Motivation. In finite-dim toy models (random tensor networks with a horizon cut), the accessible exterior algebra is exactly the modular-invariant (centralizer) subalgebra of the boundary state restricted to the worldline's causal past. The centralizer $\mathcal{A}(\mathcal{H}^+)^{\sigma^{\omega_{\mathrm{H}}}}$ is in general a proper subalgebra of $\mathcal{A}(\mathcal{H}^+)$ (extremality of ω_{H} as a KMS state is equivalent to the centralizer being a *factor* — trivial *center*, not commutativity: Haagerup's extremal state on the hyperfinite III_1 factor has an irreducible II_1-factor centralizer [40, Thm. 3.1] — while for the geometric, weakly mixing flow the centralizer is $\mathbb{C}\mathbb{1}$ [49], and a non-scalar invariant algebra appears only after passing to the crossed-product "dressed" algebra of [18]), and $\mathcal{Z}_\gamma$ extracts its commutative, γ-localisable part, on which classical conditional entropy is well-defined. This formulation sidesteps the type-III pathology that the centre of $\mathcal{A}(\mathcal{H}^+)$ itself, or of the type II_∞ crossed product $\widetilde{\mathcal{A}} = \mathcal{A}(\mathcal{H}^+) \rtimes_{\sigma^{\omega_{\mathrm{H}}}} \mathbb{R}$, is trivial (a single factor); the operational invariant is instead the joint spectrum of the commuting modular-invariant exterior observables. The conjecture asserts that the same structural identification holds in QFT on a fixed spacetime with a horizon. If true, it gives a unitary, modular-invariant exterior algebra and a finite, classical conditional entropy $H_{\mathrm{cl}}(K \mid \mathcal{Z}_\gamma)$ along γ, recovering the operational "what the worldline can know about the behind-horizon record" interpretation while remaining covariant under modular flow.

Proposition 11.9 (Measure-theoretic invariance of the accessible algebra)**.** *In the setting of Conj. 11.8, let $(\mathcal{Z}, q, (\mathcal{B}^{(m,n)})_n)$ and $(\mathcal{Z}', q', (\mathcal{B}'^{(m,n)})_n)$ be any two choices of maximal abelian subalgebra of $\mathcal{A}^{\sigma^{\omega_{\mathrm{H}}}}(\gamma(s_m))$ together with generating self-adjoint elements and refining finite Borel partition schemes, with corresponding canonical protocol families $\mathcal{P}^{\mathrm{cent}}_\gamma$ and $\mathcal{P}^{\mathrm{cent}\,\prime}_\gamma$. By the Lemma of Conj. 11.1, $\mathcal{A}^{\sigma^{\omega_{\mathrm{H}}}}(\gamma(s_m))$ is a non-trivial von Neumann algebra of the form $\mathcal{A}(\mathcal{H}^+)^{\sigma^{\omega_{\mathrm{H}}}} \rtimes \mathbb{R}$ localised at $\gamma(s_m)$ (diffuse abelian in the geometric-ergodic case, a finite type II algebra with diffuse center — never a factor — in the centralizer-rich regime), carrying a faithful normal semifinite trace* $\mathrm{Tr}_{\mathrm{sf}}$ *inherited from the crossed-product trace [18]. Assume in addition the **K-equivariance hypothesis*** (KE)*: the conditioning channel $\Phi_K : x \mapsto p(K)^{-1}\widehat{\omega}(L_K^* x L_K)$ of Conj. 11.1 (with L_K the analytic-continuation operator of* (40)*, and with $\mathcal{N}$ of Conj. 11.1 identified with the joint local algebra $\mathcal{N} \supseteq \mathcal{A}(\mathcal{H}^+)$ of QFT on the spacetime, so that both Φ_K and the masas $\mathcal{Z}, \mathcal{Z}' \subset \widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\mathrm{H}}}} \subset \widetilde{\mathcal{N}}$ act on the common crossed product $\widetilde{\mathcal{N}}$) commutes with the conditional expectations onto $\mathcal{Z}$ and $\mathcal{Z}'$ in the form $E_{\mathcal{Z}} \circ \Phi_K = \Phi_K \circ E_{\mathcal{Z}}$ and likewise for $\mathcal{Z}'$, equivalently K commutes with the chosen masas in the modular sense. (KE) is automatic when $\{E_K\}$ generates a subalgebra of the common dressed centralizer $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} \cap \widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\mathrm{H}}}}$ (cf.* (39)*). Then:*

(1) *the abstract abelian von Neumann algebras $\mathcal{Z}$ and $\mathcal{Z}'$, equipped with the restrictions of* $\mathrm{Tr}_{\mathrm{sf}}$ normalised by the dual normal weight $\widehat{\omega}_{\mathrm{H}} \restriction_{\mathcal{Z}}$ to a probability state *(the trace is semifinite with* $\mathrm{Tr}_{\mathrm{sf}}(\mathbb{1}) = \infty$ *in the type II_∞ case; the physically relevant probability measure on the joint spectrum is $\widehat{\omega}_{\mathrm{H}}(\,\cdot\,)$, which restricts to a faithful normal state on $\mathcal{Z}$ and gives a probability measure $\mu_{\mathcal{Z}}$ on the joint spectrum), are* measure-theoretically isomorphic *as standard probability spaces $(X_{\mathcal{Z}}, \mu_{\mathcal{Z}}) \cong (X_{\mathcal{Z}'}, \mu_{\mathcal{Z}'})$ [30];*

(2) *under (KE) the isomorphism of (1) can be chosen K-equivariant: the joint laws of $(K, \pi^{\gamma,\mathcal{P}^{\mathrm{cent}}_\gamma})$ and $(K, \pi^{\gamma,\mathcal{P}^{\mathrm{cent}\,\prime}_\gamma})$ on $\mathcal{K} \times X_{\mathcal{Z}}$ and $\mathcal{K} \times X_{\mathcal{Z}'}$ respectively are isomorphic as standard probability spaces. In particular the conditional law $\pi^{\gamma,\mathcal{P}^{\mathrm{cent}}_\gamma} \restriction_{K=k}$ at each fixed k corresponds to $\pi^{\gamma,\mathcal{P}^{\mathrm{cent}\,\prime}_\gamma} \restriction_{K=k}$ under the same measure-preserving isomorphism, hence* the residual entropy $H(K \mid \mathcal{G}_\gamma)$ of (44) is independent of the choice $(\mathcal{Z}, q, (\mathcal{B}^{(m,n)})_n)$. *(Without (KE), part (1) still gives marginal isomorphism but the joint law with K may differ between protocol choices,*

and the conditional entropy is not intrinsic; the standing scope of Conj. 11.8 therefore includes (KE) as part of the data, automatically realised when $\{E_k\}$ is a centralizer partition of unity by (39)*.)*

We do not *claim that $\mathcal{Z}$ and $\mathcal{Z}'$ are inner conjugate as subalgebras of $\mathcal{A}^{\sigma^{\omega_{\mathrm{H}}}}(\gamma(s_m))$ (this is generically false in non-type-I factors: e.g. in type II_1 factors there exist non-conjugate Cartan and singular masas). The invariance asserted is at the weaker, measure-theoretic level of the record process, which is nevertheless sufficient for the entropy invariant.*

Proof. (1) Each $\mathcal{Z}$ is a separable abelian von Neumann subalgebra of a σ-finite (semifinite, after restriction by the conditional expectation onto a finite trace ideal) factor; equipped with the restriction of the dual normal weight $\widehat{\omega}_{\mathrm{H}}\restriction_{\mathcal{Z}}$ (which is a faithful normal state on $\mathcal{Z}$, normalising the semifinite ambient $\mathrm{Tr}_{\mathrm{sf}}$ to a probability state on the masa) it is a probability measure algebra, and the classification of separable abelian von Neumann algebras with faithful normal weight gives isomorphism to $L^\infty(X_{\mathcal{Z}}, \mu_{\mathcal{Z}})$ on a standard Borel space, unique up to measure-preserving Borel isomorphism [30, Ch. I, §7]. Two such algebras $\mathcal{Z}, \mathcal{Z}'$ on the same factor are then measure-theoretically isomorphic to one another (both being isomorphic to the same canonical model). (2) Under (KE) the conditioning channel Φ_K commutes with the conditional expectations $E_{\mathcal{Z}}, E_{\mathcal{Z}'}$, hence the K-conditional probability measures $\Phi_K^*\widehat{\omega}_{\mathrm{H}}\restriction_{\mathcal{Z}}$ and $\Phi_K^*\widehat{\omega}_{\mathrm{H}}\restriction_{\mathcal{Z}'}$ are intertwined by the same measure-preserving isomorphism that (1) provides on the unconditioned masa, giving a K-equivariant isomorphism of the joint probability spaces $(\mathcal{K}\times X_{\mathcal{Z}}, \mathbb{P}_K\otimes\mu_{\mathcal{Z}|K}) \cong (\mathcal{K}\times X_{\mathcal{Z}'}, \mathbb{P}_K\otimes\mu_{\mathcal{Z}'|K})$. The Shannon conditional entropy $H(K \mid \mathcal{G}_\gamma)$ of the joint law is then invariant under this K-equivariant measure-preserving isomorphism (Shannon entropy is an invariant of the joint distribution). $\square$

Conjecture 11.10 (Cosmological clock)**.** Fix a cosmological model whose total matter algebra $\mathcal{A}_{\mathrm{tot}}$ is a (type I, II_∞, or III) von Neumann factor with a distinguished cyclic-separating background state ω. Call a subfactor inclusion $\mathcal{A}_{\mathrm{clock}} \subset \mathcal{A}_{\mathrm{tot}}$ (equipped with a faithful normal conditional expectation $E_{\mathrm{clock}} : \mathcal{A}_{\mathrm{tot}} \to \mathcal{A}_{\mathrm{clock}}$) *admissible* if it satisfies:

(i) **Monotone coarse-graining:** the entropy $S_{\mathrm{clock}}(\tau) := S_{\mathcal{A}_{\mathrm{clock}}}(\omega_\tau\restriction_{\mathcal{A}_{\mathrm{clock}}})$ is monotone non-decreasing in τ on $[\tau_i, \tau_f]$, with $dS_{\mathrm{clock}}/d\tau > 0$ on a subset of full Lebesgue measure, where $S_{\mathcal{A}_{\mathrm{clock}}}$ is the canonical entropy on $\mathcal{A}_{\mathrm{clock}}$ for its type: (I) the von Neumann entropy of the reduced density operator (when $\mathcal{A}_{\mathrm{tot}} = \mathcal{B}(\mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{sys}})$ this reduces to $S(\mathrm{Tr}_{\mathrm{sys}}\,\rho(\tau))$), ($\mathrm{II}_\infty$) the Segal entropy with respect to the semifinite trace, or (III) the CLPW crossed-product entropy $S_{\widetilde{\mathcal{A}_{\mathrm{clock}}}}$ on the modular crossed product of [18], all of which agree on type I factors. In type I one then has the entropy bound $S_{\mathrm{clock}}(\tau_f) - S_{\mathrm{clock}}(\tau_i) \le \log\dim\mathcal{H}_{\mathrm{clock}}$; in the cosmologically relevant type II_∞/III setting this bound is replaced by the modular-flow entropy growth of [18], which is unbounded.

(ii) **Retrodictive compatibility (clock-coupled protocol):** there exist a designated clock-coupled measurement protocol $\mathcal{P}_\gamma^{\mathrm{clock}}$ (one whose Kraus operators all lie in $\mathcal{A}_{\mathrm{clock}}$ *and jointly generate a maximal abelian subalgebra $\mathcal{Z}_{\mathrm{clock}} \subset \mathcal{A}_{\mathrm{clock}}$* of full clock entropy *at the seed time τ_i*, $S_{\mathcal{A}_{\mathrm{clock}}}(\omega\restriction_{\mathcal{Z}_{\mathrm{clock}}}) = S_{\mathrm{clock}}(\tau_i)$; this is automatic on type I clock factors (where a spectral MASA realises the full von Neumann entropy of the seed state) and is a genuine non-trivial admissibility restriction on type II_∞/III clock factors, selecting clock subfactors that admit a MASA carrying the full Segal/CLPW modular entropy of the seed state; in the II_∞/III regime where no such MASA exists for the chosen ω, Conj. 11.10 asserts only the existence of the pair $(\mathcal{A}_{\mathrm{clock}}, \mathcal{P}_\gamma^{\mathrm{clock}})$ satisfying (i) and the weaker (ii)$'$ with $\varepsilon \in (0,1)$ strictly (no seed-time full-entropy MASA), and the uniqueness conclusion below is not claimed in this weakened sub-scope; this excludes the trivial protocol $\{M_1 = \mathbb{1}\}$, which records no clock information and would render (45) vacuous with $\varepsilon = 0$; we do *not* require full-entropy recovery at every $\tau \in [\tau_i, \tau_f]$ simultaneously — doing so would impose joint diagonalisability of the modular-evolved states across the interval, which is generically false off type I) and a constant $\varepsilon \in [0,1)$ such that, for the distinguished cosmological worldline $\gamma_{\cos}$ (the comoving observer of the background state ω, i.e. the integral curve of the modular flow σ^ω restricted to the centre of the cosmological symmetry algebra, which is unique modulo isometries fixing ω), for every $\tau \in [\tau_i, \tau_f]$, the Araki relative entropy of the clock-restricted retrocausal state, defined via the precomposition $\rho^{\gamma,\mathcal{P}^{\text{clock}}_\gamma}_{\text{RC,clock}}(\tau) := \varphi^{\gamma,\mathcal{P}^{\text{clock}}_\gamma}_{\text{RC}}(\tau) \circ \iota_{\text{clock}} \in (\mathcal{A}_{\text{clock}})_*$ (the state-level restriction to $\mathcal{A}_{\text{clock}}$ along the inclusion $\iota_{\text{clock}} : \mathcal{A}_{\text{clock}} \hookrightarrow \mathcal{A}_{\text{tot}}$, applied to the normal state $\varphi^{\gamma,\mathcal{P}^{\text{clock}}_\gamma}_{\text{RC}}(\tau) \in (\mathcal{A}_{\text{tot}})_*$; equivalently $\rho_{\text{RC,clock}} = \varphi_{\text{RC}} \circ \iota_{\text{clock}}$ uses only the algebra inclusion and is well-defined for type I, II_∞, and III without requiring a density-matrix representative, eliminating the type-confusion in writing $E_{\text{clock}}(\rho)$ for a state ρ; note the predual $(E_{\text{clock}})_* : (\mathcal{A}_{\text{clock}})_* \to (\mathcal{A}_{\text{tot}})_*$ of the conditional expectation goes in the *opposite* direction from restriction, so the earlier identification $\rho_{\text{RC,clock}} = (E_{\text{clock}})_*(\varphi_{\text{RC}})$ was a typographical category error and is here replaced by the correct restriction $\varphi_{\text{RC}} \circ \iota_{\text{clock}}$), relative to the clock-restricted forward state $\omega_\tau \restriction_{\mathcal{A}_{\text{clock}}}$ satisfies

$$S_{\text{Araki}}\Big(\rho^{\gamma,\mathcal{P}^{\text{clock}}_\gamma}_{\text{RC,clock}}(\tau) \;\Big\|\; \omega_\tau \restriction_{\mathcal{A}_{\text{clock}}}\Big) \;\leq\; \varepsilon\, S_{\text{clock}}(\tau) \qquad \text{for all } \tau \in [\tau_i, \tau_f] \tag{45}$$

(deterministically as a relation between the algebra-level states; the per-record almost-sure version is recovered, when (PW) below holds, on the explicit Part-I record probability space $(\Omega^{\text{rec}}_\gamma, \mathbb{P}^{\text{rec}}_\gamma)$ canonically associated with the clock-coupled protocol via Kolmogorov inverse limit of finite-stage refinements as in Conj. 11.8; in non-(PW) cases the relation is the deterministic inequality above, since no canonical record-event probability space is available beyond what the inverse-limit construction produces). Both sides are well-defined invariants of the inclusion $\mathcal{A}_{\text{clock}} \subset \mathcal{A}_{\text{tot}}$: the right-hand side is the Segal/CLPW entropy of the clock-restricted background state by (i), and the left-hand side is the Araki relative entropy [18] of two normal states on the same von Neumann algebra $\mathcal{A}_{\text{clock}}$, which depends only on the abstract inclusion and the modular structure, not on any chosen scalar coordinate on a spectrum. (At $\varepsilon = 0$ the clock-restricted retrocausal state coincides with the clock-restricted forward state, i.e. the protocol records no additional information beyond the prior; for $0 < \varepsilon < 1$ the protocol records up to a fraction ε of the available clock entropy. Quantifying over the clock-coupled protocol is appropriate because measurement schedules that ignore $\mathcal{A}_{\text{clock}}$ trivially fail to record clock information. The earlier formulation in terms of a "posterior variance" is replaced because variance requires a chosen scalar coordinate on the clock spectrum and is therefore not an invariant of $\mathcal{A}_{\text{clock}} \subset \mathcal{A}_{\text{tot}}$; relative entropy is.)

(iii) **Minimality:** no proper subfactor of $\mathcal{A}_{\text{clock}}$ also satisfies (i) and (ii) with the same ε.

(iv) **(PW) Page–Wootters split data (optional; required only for the clock-bridge conclusion below).** There exists a unitary tensor-factor split $U_{\text{split}} : \mathcal{H}_{\text{tot}} \xrightarrow{\sim} \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{sys}}$ identifying $\mathcal{A}_{\text{clock}}$ with $\mathcal{B}(\mathcal{H}_{\text{clock}}) \otimes \mathbb{1}_{\text{sys}}$ and $\mathcal{A}_{\text{sys}} := \mathcal{A}'_{\text{clock}} \cap \mathcal{A}_{\text{tot}}$ with $\mathbb{1}_{\text{clock}} \otimes \mathcal{A}^{(0)}_{\text{sys}}$, together with a self-adjoint clock generator $\widehat{H}_{\text{c}}$ on $\mathcal{H}_{\text{clock}}$ generating a covariant time observable $\widehat{T}_{\text{c}}$ (POVM or self-adjoint, as in Sec. 10) and a total Hamiltonian $\widehat{H} = \widehat{H}_{\text{c}} \otimes \mathbb{1} + \mathbb{1} \otimes \widehat{H}_{\text{m}} + \widehat{V}_{\text{int}}$ relative to which the Wheeler–DeWitt-style constraint of Conj. 10.13 can be formed. (PW) is automatic in type I factor cosmologies admitting a Page–Wootters tensor-product structure; it can fail in type II_∞/III where the *algebraic* clock $\mathcal{A}_{\text{clock}}$ exists but no tensor-product split does (e.g. when $\mathcal{A}_{\text{clock}}$ is the centre-of-modular-flow subalgebra of an irreducible type III factor and admits no commutant-tensor-factor decomposition).

Then the set of admissible inclusions is non-empty and forms a *single unitary-equivalence class within the (PW)-split class*: any two admissible subfactors $\mathcal{A}_{\text{clock}}, \mathcal{A}'_{\text{clock}} \subset \mathcal{A}_{\text{tot}}$ *both equipped with (PW) split data with the same* $(\mathcal{H}_{\text{clock}}, \widehat{H}_{\text{c}})$ *data type* are related by a unitary $U : \mathcal{H}_{\text{tot}} \to \mathcal{H}_{\text{tot}}$ implementing an automorphism of $\mathcal{A}_{\text{tot}}$ that maps $\mathcal{A}_{\text{clock}}$ into $\mathcal{A}'_{\text{clock}}$ and intertwines the modular flows of ω together with the (PW) split unitaries $U_{\text{split}}, U'_{\text{split}}$. (Without restricting to a fixed (PW)-split class the uniqueness fails by simple finite-dimensional counterexamples: on

$\mathcal{A}_{\rm tot} = M_2 \otimes M_3$ both $\mathcal{A}_{\rm clock} = M_2 \otimes \nVdash$ and $\mathcal{A}'_{\rm clock} = \nVdash \otimes M_3$ can be admissible for suitable ω and protocol choices, yet are not unitarily related as subfactors of $\mathcal{A}_{\rm tot}$; the (PW) data fixes a canonical clock factor type and excludes such sliding.) Under (PW), the Page–Wootters construction of Sec. 10 applied to $\mathcal{A}_{\rm clock}$ yields a τ-parameterization equivalent to the background time function τ of $(\mathcal{M}, g)$ up to reparameterization, and Conj. 10.13 follows with that clock identification. *Without (PW), the algebraic clock structure (i)–(iii) still holds, but the bridge to the Wheeler–DeWitt formulation of Conj. 10.13 requires the separate split data and is not derived as a theorem-level conclusion of the present conjecture.*

Motivation. Monotone non-decreasing entropy with positive growth on a full-measure set is the operational content of "having a good clock": the clock must coarse-grain information at least as fast as the system thermalizes, but not necessarily at a uniform positive rate (which would conflict with the entropy bound $S_{\rm clock} \le \log\dim\mathcal{H}_{\rm clock}$ in any finite-dimensional realization). The type II_∞/III generalisation accommodates unbounded modular-flow entropy growth in the cosmologically relevant limit. Retrodictive compatibility ensures the clock's records are the ones the observer uses to update the posterior; the minimality clause (iii) eliminates the trivial obstruction that one could always enlarge $\mathcal{H}_{\rm clock}$ by adjoining an extra factor with no dynamics. Uniqueness is asserted only *up to unitary equivalence between admissible decompositions*, the natural notion in this setting (an intertwining unitary need not preserve the explicit tensor product on the nose, only the modular structure). In toy spin-chain models this triple of conditions uniquely selects (up to such isomorphism) the subsystem playing the clock role. Extending this to continuum cosmology would address item (7). The "trinity theorem" of Höhn–Smith–Lock [102, 103] proves rigorous equivalence between three formulations of relational time — Dirac-quantized relational observables, Page–Wootters Schrödinger picture, and deparametrization/Heisenberg reduction — via quantum reduction maps using covariant POVM clocks; this is the strongest published rigorous backbone for the Page–Wootters identification used in Conj. 10.13 and the thermal-time postulate Pos. 11.11 below, and should be cited as the structural foundation on which both rest. In the relativistic case — which is the relevant one here, since the modular generator $\widehat{H}_{\rm c} = -\log\Delta_\omega$ has spectrum symmetric about 0 — this equivalence is established *per frequency superselection sector* [103], with a covariant POVM clock rather than a sharp self-adjoint time operator across sectors (Pauli's objection); the sector-wise statement is recorded in App. O (K4).

11.2 Entropic time: the thermal-time postulate

Obstruction (1) to unification — the problem of time — is the incompatibility between QM's preferred time parameter and GR's diffeomorphism invariance. We now show that the present framework contains the ingredients needed to *dissolve* this obstruction via the Connes–Rovelli thermal-time hypothesis [33]: the Page–Wootters clock generator of the master equation is identified with the modular Hamiltonian of the seed state, and the parameter "time" is derived from the state rather than postulated on the manifold; this is the time-emergence mechanism of the modern crossed-product program [18, 35]. Under this identification, the user's physical intuition — that time does not exist independently, that what is observed is monotone entropy flow, and that black holes are the loci at which that flow saturates — is realised literally as theorems of modular theory: Araki's monotonicity theorem [52], Bisognano–Wichmann modular flow at horizons, and the generalised second law [53, 55].

Postulate 11.11 (Thermal-time identification)**.** Let ω be the seed background state of Conj. 11.1 on the local matter algebra $\mathcal{N}$, with modular operator Δ_ω on the standard form of $\mathcal{N}$, and modular automorphism group $\sigma_t^\omega(A) := \Delta_\omega^{it} A \Delta_\omega^{-it}$. The Page–Wootters clock Hamiltonian $\widehat{H}_{\rm c}$ appearing in the kinetic generator $\widehat{K}_k$ of Conj. 11.56 is defined to be the (dressed crossed-product lift of the) modular Hamiltonian of ω:

$$\widehat{H}_{\rm c} \;:=\; -\log\Delta_\omega \qquad \text{(equivalently, } \sigma_t^\omega = e^{-it\widehat{H}_{\rm c}/\hbar}\cdot e^{it\widehat{H}_{\rm c}/\hbar} \text{ with } \hbar = 1 \text{ in modular units).} \quad (46)$$

In the crossed-product setting $\widetilde{\mathcal{N}} = \mathcal{N} \rtimes_{\sigma^\omega} \mathbb{R}$ of [18], $\widehat{H}_c$ is the self-adjoint generator of the $\mathbb{R}$-action adjoined in the crossed product, and the CLPW dressing makes its spectrum discrete on bounded band-limited subspaces, consistent with (T6) of Conj. 11.56.

(Bare vs. dressed — a standing distinction, not a notational nicety.) The symbol $\widehat{H}_c$ carries two inequivalent readings that must be kept apart. The *bare* $-\log \Delta_\omega$ on the uncompressed type III_1 algebra $\mathcal{N}$ has, by the Connes invariant $S(\mathcal{N}) = [0, \infty)$, *unbounded, full-line, absolutely continuous* spectrum — and it is exactly this unboundedness that supplies the thermal-time sector its content: the Pauli time-of-arrival evasion (Thm. Y.1), the de Sitter boost witness, and the KMS/half-sided-modular structure used for conservation and ANEC. The *dressed* generator — the CLPW crossed-product observer energy $\widehat{P}$ compressed to the band-limited corner $P_{\leq E_\star}$ — is *bounded* with discrete spectrum, and it is this compressed object, *not* the bare $-\log \Delta_\omega$, that the finite-dimensional rigour discharges of Prop. 11.33 (T5),(T6),(S3) operate on. The two are *distinct operators* ($\widehat{P} \neq -\log \Delta_{\widehat{\omega}}$, cf. the crossed-product centraliser Lemma following Conj. 11.1), related by the observer compression Π; that this compression preserves the modular/KMS data the clock and gravity sectors require — so that the bounded-generator rigour and the unbounded-generator thermal time can hold *simultaneously* on their respective legs — is the co-scoping assumption (H0-co-scoping), carried as a named residual (Rem. 11.72), not a proven identity. Wherever a bound $|\sigma(\widehat{H}_c)| \leq C(E_\star)$ is asserted it refers to the dressed compressed generator; the bare $-\log \Delta_\omega$ is not claimed bounded.

Remark 11.12 (Physical content of Pos. 11.11). The postulate has four consequences, each corresponding to a well-posed statement in existing mathematical physics:

(i) **Time is not fundamental.** There is no preferred time coordinate in the master equation; what is specified is a state ω, from which time is derived as its modular parameter. Different seed states give different time flows, related by the Connes cocycle $(D\psi : D\omega)_t \in \mathcal{U}(\mathcal{N})$. Because the cocycle is *inner*, the two modular flows coincide as *outer* automorphisms: there is a single state-independent flow on $\mathrm{Out}(\mathcal{N})$ (the Connes–Takesaki canonical flow of weights), while the state-dependent inner part is what distinguishes individual observers' clocks. The statement that observers in different thermodynamic states experience different times is therefore precise only at the inner level; modulo inner automorphisms the derived time is canonical, which is what makes the de Sitter boost (Bisognano–Wichmann) a state-independent witness. Diffeomorphism invariance of the underlying gravitational theory is therefore preserved at the level of the master equation: (66) specifies a state, not a coordinate.

(ii) **Time is entropy flow.** For any normal state ψ on $\mathcal{N}$ with $\psi \ll \omega$, the Araki relative entropy $S_{\mathrm{Ar}}(\psi \| \omega)$ is the unique quantity that is (a) non-negative, (b) zero iff $\psi = \omega$, and (c) monotone non-increasing under completely positive unital channels [52]. Under (46), modular evolution $\psi_t := \psi \circ \sigma^\omega_{-t}$ is the unique automorphism of $\mathcal{N}$ under which ω is KMS and under which any perturbation relaxes monotonically toward ω-equilibrium. The monotone modular-flow entropy growth axiom (i) of Conj. 11.10 is therefore not a hypothesis but a theorem (Prop. 11.15 below).

(iii) **Centralizer hypothesis is a commutation identity.** (CP) of Conj. 11.1 and (T11) of Conj. 11.56 both read $[\widehat{H}_m, \Delta^{it}_{\widehat{\omega}}] = 0$; under (46) this is the single commutation $[\widehat{H}_m, \widehat{H}_c] = 0$ on the dressed centralizer $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$. In Part I language this is the operator-algebraic content of covariant conservation: matter Hamiltonians that preserve the seed thermodynamic state commute with its derived time. (CP) and (T11) are thereby consolidated into a single hypothesis, Hyp. 11.13 below.

(iv) **Black holes are thermal attractors.** At a Bisognano–Wichmann horizon $\mathcal{H}^+$ the modular flow of $\omega \restriction_{\widetilde{\mathcal{A}}(\mathcal{H}^+)}$ is generated by the boost, with KMS temperature equal to the surface-gravity Hawking temperature $T_H = \kappa/2\pi$. The horizon modular centralizer $\widetilde{\mathcal{A}}(\mathcal{H}^+)^{\sigma^{\widehat{\omega}_{\mathrm{H}}}}$ of Conj. 11.8 is the fixed-point subalgebra of thermal time, and Wall's generalised second law [55],

$$\frac{dS_{\rm gen}}{dt} \;=\; \frac{dS_{\rm out}}{dt} + \frac{1}{4G\hbar}\frac{dA}{dt} \;\geq\; 0,$$

is, under (46), the monotonicity of modular relative entropy of the exterior state restricted to $\widetilde{\mathcal{A}}(\mathcal{H}^+)$ relative to $\omega|_{\widetilde{\mathcal{A}}(\mathcal{H}^+)}$. This is the precise sense in which black-hole horizons "push time forward": they are the loci at which thermal time has non-trivial fixed subalgebra, and exterior matter flowing toward them is matter being absorbed into that fixed subalgebra, driving $S_{\rm gen}$ monotonically upward.

Hypothesis 11.13 (Consolidated centralizer condition)**.** Under Pos. 11.11, the matter Hamiltonian $\widehat{H}^{(k)}_{\rm m}[\widehat{\mathfrak{g}}]$ appearing in the kinetic generator $\widehat{K}_k$ of Conj. 11.56 (T3) commutes with the thermal-time generator:

$$[\widehat{H}^{(k)}_{\rm m}[\widehat{\mathfrak{g}}], \widehat{H}_{\rm c}] \;=\; 0 \qquad \text{on } \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}.$$

This is (CP) of Conj. 11.1 and (T11) of Conj. 11.56 combined.

Remark 11.14 (Hyp. 11.13 is a reduction to seed-weight invariance of the matter flow). Hyp. 11.13 is a *reduction*, not an opaque postulate. Its consumers read it as the affiliation of the matter flow to the dressed centralizer — $e^{i\tau \widehat{H}_{\rm m}} \in \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$, equivalently $[\widehat{H}_{\rm m}, \Delta^{it}_{\widehat{\omega}}] = 0$ (Prop. 11.62). *Provided* the dressed $\widehat{H}^{(k)}_{\rm m}[\widehat{\mathfrak{g}}]$ is affiliated to the crossed product $\widetilde{\mathcal{N}}$ — a standing presupposition of the hypothesis, entangled with the bare–dressed co-scoping straddle carried as a named residual in Rem. 11.72 (bounded functions of $-\log \Delta_{\widehat{\omega}}$ need not lie in $\widetilde{\mathcal{N}}$) — the centralizer-of-a-weight theorem of modular theory gives, for the unitary $u = e^{i\tau \widehat{H}_{\rm m}} \in \widetilde{\mathcal{N}}$,

$$u \in \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} \qquad \Longleftrightarrow \qquad \widehat{\omega} \circ \mathrm{Ad}(u) = \widehat{\omega}.$$

Here $\widehat{\omega}$ is the dual *weight* ($\widehat{\omega}(\mathbb{1}) = \infty$; it does not restrict to ω), so the equivalence is the KMS/modular statement — $\widehat{\omega}$-preserving automorphisms commute with $\sigma^{\widehat{\omega}}$ [39, Ch. VIII], [195, Ch. 5.3] — and not a state substitution. Hyp. 11.13 is therefore equivalent to the named physical condition that the matter dynamics *preserve the seed thermodynamic weight* $\widehat{\omega}$: the operator-algebraic form of covariant conservation along thermal time, and the continuous-flow analogue of the Tomita–Takesaki mechanism already used for parity in Lem. 13.7. It upgrades the expectation-value stationarity of Prop. 11.24 to the operator/affiliation level that proposition leaves open, at the cost of the affiliation premise above.

This is a reduction, not a closure, and it is substantive exactly where the matter sector is *non-stationary.* Where matter time *is* thermal time — the stationary Bisognano–Wichmann corner, in which the matter generator is a multiple of the boost $\widehat{H}_{\rm c}$ itself — the condition holds *tautologically*, as already recorded for the de Sitter witness's classical sector, where (CP) holds because the static-patch dynamics is generated by the boost that generates $\sigma^{\widehat{\omega}}$; but that corner is matter-trivial ($\omega_k \restriction_{\mathcal{N}} = \omega$), and by the scope of Prop. 11.24 a genuinely local, partially smeared matter generator ($f \not\equiv 1$) does *not* commute with the full modular flow. Hence the invariance — and with it Hyp. 11.13 — genuinely *fails* for interacting matter (non-geometric modular flow; self-adjointness of $\widehat{H}_{\rm m}$ itself open in $d = 4$), for $O(\kappa)$ backreaction that breaks the seed's Killing symmetry, and on non-stationary backgrounds. That failure set, not a closure, is the precise physical content of the "except (T11)" exemption.

Proposition 11.15 (Clock admissibility is automatic)**.** *Assume Pos. 11.11 and the standing scope of Conj. 11.10 (cosmological total algebra $\mathcal{A}_{\rm tot}$, cyclic-separating background state ω). Take $\mathcal{A}_{\rm clock}$ to be the abelian von Neumann subalgebra generated by $\{\Delta^{it}_{\omega} : t \in \mathbb{R}\}''$ inside $\widetilde{\mathcal{A}}_{\rm tot}$, with $E_{\rm clock}$ the canonical conditional expectation of the crossed product onto this subalgebra. Then $\mathcal{A}_{\rm clock}$ is admissible in the sense of Conj. 11.10:*

(i) *Monotone modular-flow entropy growth (axiom (i)) is Araki's theorem [52]: for any $\psi \ll \omega$ the relative entropy $t \mapsto S_{\mathrm{Ar}}(\psi_t \| \omega)$ is non-increasing under the CP channel induced by E_{clock}, equivalently the clock entropy $S_{\mathrm{clock}}(\tau)$ is non-decreasing in the thermal-time parameter $\tau := t$.*

(ii) *Retrodictive Araki-compatibility (axiom (ii), eq. (45)) holds with $\varepsilon = 0$ for the clock-coupled protocol that reads out σ^{ω}-invariant spectral projections of $\widehat{H}_{\mathrm{c}}$ (guaranteed to exist on band-limited subspaces by (T6)).*

(iii) *Minimality (axiom (iii)) holds because $\{\Delta_{\omega}^{it}\}''$ is, by construction, the minimal abelian subalgebra intertwining the ω-modular flow.*

(iv) *(PW) split data exist in the CLPW crossed product with $\mathcal{H}_{\mathrm{clock}} = L^2(\mathbb{R})$, $\widehat{H}_{\mathrm{c}} = -i\partial_t$, and covariant time observable $\widehat{T}_{\mathrm{c}}$ equal to multiplication by t. This is the standard Naimark dilation of $-\log \Delta_{\omega}$'s spectral measure. The multiplication-operator realisation is the sharp projection-valued case on the unconstrained $L^2(\mathbb{R})$; in general, however, the time observable conjugate to the modular generator $-\log \Delta_{\omega}$ is only an* unsharp *observable—a covariant (Holevo-type) positive operator-valued measure rather than a self-adjoint operator canonically conjugate to $\widehat{H}_{\mathrm{c}}$—which is precisely the content of the Naimark dilation invoked here. This is the rigorous status of Connes–Rovelli thermal time established by van Neerven–Portal [101] (already for the harmonic oscillator and the free massless relativistic particle). Covariance $U(s)\,\widehat{T}_{\mathrm{c}}(B)\,U(s)^* = \widehat{T}_{\mathrm{c}}(B+s)$ is all that Page–Wootters conditioning requires, so admissibility is unaffected; the discreteness invoked via (T6) above is that of the band-limited compression, the ambient $\widehat{H}_{\mathrm{c}}$ retaining continuous spectrum.*

In particular, the algebraic *clock content of Conj. 11.10 — clauses (i)–(iii) — is a theorem under Pos. 11.11, and clause (iv) holds as the covariant (Holevo/Naimark) POVM clock on the $L^2(\mathbb{R})$ leg. We do* not *promote (iv) to the stronger claim of a tensor-product Page–Wootters split $U_{\mathrm{split}} : \mathcal{H}_{\mathrm{tot}} \to \mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{sys}}$ of the* physical *algebra in the type III regime: as Conj. 11.10(iv) itself cautions, that split can fail in type II_∞/III even where the algebraic clock exists, and what is constructed here is the split on the auxiliary crossed-product leg, which suffices for Page–Wootters conditioning but is not the physical tensor factorisation.*

Proof. (i) Araki's monotonicity of relative entropy under CP unital maps [52] applied to E_{clock}, combined with the KMS property of ω for σ^{ω}, gives the stated monotonicity; see [18, Thm. 3.3] for the crossed-product statement used here. (ii) Follows because the spectral projections of $\widehat{H}_{\mathrm{c}}$ lie in the dressed centralizer $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ by construction: with $\widehat{H}_{\mathrm{c}}$ realised as the dressed generator $\widehat{P}$ of the adjoined one-parameter group λ (Pos. 11.11), they are bounded Borel functions of $\widehat{P}$, hence elements of $\{\lambda_s\}''$ by the Lemma of Conj. 11.1 (*not* bounded functions of $-\log \Delta_{\widehat{\omega}}$, which need not lie in $\widetilde{\mathcal{N}}$; and *not* elements of $\mathcal{N}$), so the clock-restricted retrocausal state coincides with its modular-averaged counterpart and the Araki relative entropy (45) vanishes identically. (iii) $\{\Delta_{\omega}^{it}\}''$ is abelian and generates the ω-modular flow on $\mathcal{N}$; any smaller intertwining subalgebra would fail to contain a spectral resolution of $\widehat{H}_{\mathrm{c}}$, contradicting (46). (iv) Standard Naimark dilation; the CLPW crossed product [18] acts on the Hilbert space $\mathcal{H}_{\mathrm{QFT}} \otimes L^2(\mathbb{R})$ (*note* $\widetilde{\mathcal{N}} \ncong \mathcal{N} \bar{\otimes} \mathcal{B}(L^2(\mathbb{R}))$ as algebras — the crossed product of a type III_1 algebra by its modular flow is type II_∞, whereas the tensor product is type III; the split refers to the underlying Hilbert space and to the clock subalgebra $\{\lambda_s\}'' \cong L^\infty(\mathbb{R})$), with $\widehat{T}_{\mathrm{c}}$ multiplication and $\widehat{H}_{\mathrm{c}} = -i\partial_t$ on the $L^2(\mathbb{R})$ leg. □

Remark 11.16 (Effect on the hypothesis bundle of Thm. 11.73). Under Pos. 11.11, the hypothesis bundle H=(H0)–(H6) of the Master Unification Theorem shrinks: (a) Conj. 11.10 is no longer an independent hypothesis but a consequence of the postulate (Prop. 11.15); (b) (CP) of Conj. 11.1 and (T11) of Conj. 11.56 collapse to the single commutation condition Hyp. 11.13 (after the two-branch restructuring of Conj. 11.1, load-bearing only for the branch-(II) centralizer special case; branch (I) is closed by App. G without it). What remains of (H3) are Conj. 11.1 (less (CP)), Conj. 11.4, Conj. 11.5, Conj. 11.8, together with Pos. 11.11 itself as a new primitive. The net reduction is from five independent speculative conjectures plus two independent centralizer hypotheses to four speculative conjectures plus one consolidated commutation condition plus the postulate; the de Sitter witness of Conj. 11.55 satisfies the postulate canonically (the boost generator is $-\log \Delta_\omega$ by Bisognano–Wichmann on the wedge), so the witness is not weakened by the change of bundle. The obstruction of time is thereby absorbed into a structural identification rather than a separate conjecture.

Remark 11.17 (What Pos. 11.11 does and does not do). *Does:* dissolves obstruction (1) (problem of time) by making time a state-derived quantity rather than a manifold coordinate; promotes Conj. 11.10 to a theorem; consolidates (CP) and (T11) into a single commutation; makes the de Sitter witness canonical; identifies horizons as fixed subalgebras of thermal time with the generalised second law as Araki monotonicity. *Does not:* fix the state ω or the algebra $\mathcal{N}$ (these remain modelling inputs, exactly as in Connes–Rovelli); remove obstructions (2) UV completion of gravity, (4) measurement on dynamical metric, or (7) information at horizons; nor does it canonically fix the *rate* of thermal time relative to local proper time, which (for near-equilibrium seeds) is governed by the Tolman–Ehrenfest relation $T\sqrt{g_{00}} = \text{const}$—"temperature as the speed of time" [263, 34]—so that thermal time is sharply interpretable only in the (near-)equilibrium sector and acquires a position-dependent lapse otherwise. Generic two-boundary conditioned states (Conj. 11.56) are out of equilibrium; for them (46) furnishes a formal generator whose thermodynamic reading (the Araki arrow of time) is guaranteed only on the dressed centralizer $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$. Obstruction (5) (stress-energy on a dynamical metric) is addressed *next*, in Sec. 11.3 below, via a second structural identification that consolidates (T1)–(T2) of Conj. 11.56. The thermal-time postulate does not itself unify QM and GR; it removes obstruction (1) and prepares the ground for the entropic-gravity postulate that removes obstruction (5). The qualitative fusion of monotone entanglement-entropy growth, KMS/modular flow, and Page–Wootters relational time into a single emergent-time parameter has also been proposed, at the conceptual and parametrization level, by Weberszpil–Sotolongo-Costa [105]; the present contribution differs in supplying the operator-algebraic realisation (the CLPW/DEHK crossed product of App. H) and the two-boundary (ABL) closure that fixes the seed state, neither of which appears in that synthesis.

11.3 Entropic gravity: the stress-energy identification

Obstruction (5) — the definition of matter stress-energy and the quantum Einstein tensor on a dynamical (operator-valued) metric $\widehat{\mathsf{g}}$ — is the reason hypotheses (T1) and (T2) of Conj. 11.56 flagged those operators as "existence assumed." In standard QFT on curved spacetime, $\widehat{T}_{\mu\nu}[g]$ is constructed by Hadamard renormalisation [10, 16]: pick a fixed smooth g, build the Hadamard parametrix $H(x, y; g)$, and define $\widehat{T}_{\mu\nu}$ as the coincidence-limit remainder. Every step presupposes a classical background; on a quantum $\widehat{\mathsf{g}}$ neither $H(x, y; \widehat{\mathsf{g}})$ nor the inverse metric $\widehat{\mathsf{g}}^{-1}$ nor the Christoffel symbols $\widehat{\Gamma}[\widehat{\mathsf{g}}]$ have a canonical operator meaning. The same obstruction afflicts $\widehat{G}^{(Q)}_{\mu\nu}[\widehat{\mathsf{g}}]$, which is polynomial in $\widehat{\mathsf{g}}^{-1}$ and $\partial\widehat{\mathsf{g}}$.

The present framework, having adopted Pos. 11.11, now has access to three established *theorems* of modular theory and holography (for which recent rigorous support includes van Neerven–Portal [101] proving thermal time as a covariant POVM in harmonic-oscillator settings, Höhn–Smith–Lock [102, 103] proving equivalence of Dirac, Page–Wootters, and deparametrization formulations of relational time, and De Vuyst–Eccles–Höhn–Kirklin [96, 97] on QRF-intertwining theorems) that make a second structural identification available:

- **First law of entanglement [58].** For any state ω on a local algebra $\mathcal{N}$ with modular Hamiltonian $K_\omega := -\log \Delta_\omega$, and for any first-order perturbation ψ around ω,

$$\delta S(\psi\|\omega) \;=\; \delta\langle K_\omega\rangle_\psi,$$

with δS the first variation of the Araki relative entropy and $\langle\cdot\rangle_\psi$ the expectation in ψ. In integrated form with $K_\omega = \int_\Sigma \xi^\mu \widehat{T}_{\mu\nu}\, d\Sigma^\nu$ for the boost-Killing vector ξ of a Rindler-like wedge, this *identifies* matter stress-energy with the local density of the modular Hamiltonian.

- **Jacobson's equation of state [56].** For a small causal diamond at x with Unruh temperature $T = \hbar\kappa/2\pi$ and Bekenstein area-entropy $\delta S = \delta A/4G\hbar$, the Clausius relation $\delta Q = T\,\delta S$ applied to the horizon of the diamond yields $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ as an equation of state. In operator form with $\delta Q = \delta\langle K_\omega\rangle$, this identifies $G_{\mu\nu}$ with the area-response operator of the local modular Hamiltonian.
- **Vacuum-entanglement derivation of Newton's constant [59].** The area-law coefficient of the vacuum entanglement entropy on a local Rindler horizon is $1/4G\hbar$; no independent input fixes G.
- **JLMS relation (boundary modular Hamiltonian = bulk modular Hamiltonian + area operator).** Jafferis–Lewkowycz–Maldacena–Suh [245] prove, to leading order in the bulk gravitational coupling, $\delta K_\partial = \delta K_{\mathrm{bulk}} + \delta\widehat{A}/4G\hbar$, with $\widehat{A}$ the area operator of the extremal surface; this is the precise holographic operator-algebraic content of the area-response identification (48), and the semiclassical bridge to the crossed-product constraint of [238, 18].
- **Nonlinear entanglement-to-Einstein extensions.** Faulkner–Haehl–Hijano–Parrikar–Rabideau–Van Raamsdonk [106] extend the first-law derivation to *nonlinear* (second-order) Einstein equations in holographic CFTs via relative-entropy positivity; Paul–Roy [115] derive linearized Einstein around pure BTZ from the entanglement first law, establishing the result holds beyond the AdS vacuum. Jacobson [116] recasts the original Clausius derivation as an entanglement-equilibrium variational principle: the semiclassical Einstein equation is equivalent to stationarity of entanglement entropy on small geodesic balls at fixed volume, closing conceptual gaps in the 1995 argument.
- **Spacetime connectivity from entanglement.** Van Raamsdonk [117] shows that classical connectivity of spacetime emerges from entanglement between boundary region states in holographic setups, providing conceptual context in which the entropic-gravity identification below is natural.

Postulate 11.18 (Entropic-gravity identification)**.** Let ω be the seed background state of Pos. 11.11, with modular Hamiltonian $K_\omega = -\log\Delta_\omega$ on the matter algebra $\widetilde{\mathcal{N}}$. Fix a smearing kernel $f^{\mu\nu} \in C_c^\infty(\mathcal{M}^{\mathrm{ext}}; \mathrm{Sym}^2 T\mathcal{M})$ supported in a neighbourhood of a smooth local causal diamond D_x at $x \in \mathcal{M}^{\mathrm{ext}}$, with boost-Killing vector ξ on D_x. Define:

(EG1) **Matter stress-energy as modular density.** The smeared matter stress-energy $\widehat{T}^{\mu\nu}_{\mathrm{m}}[\widehat{\mathrm{g}}; f]$ of (T2) is the unique self-adjoint operator on $\mathcal{D}_k$ satisfying

$$\int f_{\mu\nu}(x)\,\xi^\mu(x)\,\widehat{T}^{\mu\nu}_{\mathrm{m}}[\widehat{\mathrm{g}}; f](x)\,d\mathrm{vol}_{\widehat{\mathrm{g}}}(x) \;=\; K_\omega[f] \qquad \text{(first-law integrand)}, \tag{47}$$

where $K_\omega[f]$ is the f-smeared modular Hamiltonian of $\omega\restriction_{D_x}$.

(EG2) **Einstein tensor as area-response.** The operator-ordered linearised quantum Einstein tensor $\widehat{G}^{(Q)\,\mu\nu}[\widehat{\mathrm{g}}; f]$ of (T2) is the unique self-adjoint operator on $\mathcal{D}_k$ satisfying the operator-algebraic Clausius relation

$$\int f_{\mu\nu}(x)\,\xi^\mu(x)\,\widehat{G}^{(Q)\,\mu\nu}[\widehat{\mathrm{g}}; f](x)\,d\mathrm{vol}_{\widehat{\mathrm{g}}}(x) \;=\; \frac{8\pi G}{c^4}\,K_\omega[f], \tag{48}$$

i.e. the area-response operator of the local modular Hamiltonian normalised by the Bekenstein coefficient $1/4G\hbar$ of [59]. Here the identification of the entanglement area-coefficient with $1/4G\hbar$ uses the standard induced-gravity assumption that the UV-divergent vacuum entanglement entropy is universally absorbed into the renormalisation of Newton's constant, independent of the matter-species content [56, 59, 62].

(EG3) **Consistency with classical limits.** On classical backgrounds $\widehat{\mathrm{g}} = g_0$ with Hadamard ω, (47)–(48) reduce to the standard Hadamard-renormalised $\widehat{T}_{\mu\nu}[g_0]$ and the classical linearised $G_{\mu\nu}[g_0]$, respectively, by [58] and [56].

Remark 11.19 (Physical content of Pos. 11.18). The postulate has four substantive consequences:

(i) **Stress-energy is not a Hadamard remainder; it is an entanglement response.** The matter stress-energy on a dynamical metric is no longer an output of a parametrix subtraction on a classical background. It is defined *intrinsically* from the modular Hamiltonian of ω via (47). Obstruction (5) is therefore dissolved the same way obstruction (1) was: not by building a complicated object on $\widehat{\mathfrak{g}}$, but by identifying the object with a pre-existing modular quantity of ω that requires no $\widehat{\mathfrak{g}}$-dependence in its definition.

(ii) **The Einstein tensor is an equation of state, not a fundamental operator.** (48) is Jacobson's Clausius relation in operator form: $\widehat{G}^{(Q)\,\mu\nu}$ is the area-response of local modular Hamiltonians, so "quantum gravity" is not a new quantisation of a classical Einstein equation but a statement about how modular structure responds to local deformations. This is the operator-algebraic version of the entropic-gravity programme of [56, 116, 61, 62, 117, 106, 115, 108].

(iii) **Stationarity along thermal time is Tomita cyclicity; conservation is Hadamard renormalisation.** K_ω generates σ^ω, which is ω-KMS, so $\omega(\sigma^\omega_t(A)B) = \omega(B\sigma^\omega_{t-i}(A))$ is a theorem (Takesaki [39]); under (47) its operator-algebraic content is *stationarity*: the ω-expectation of the boost current $\xi^\mu \widehat{T}_{\mu\nu}$ is constant along the modular flow (Prop. 11.24), with no Christoffel symbols of $\widehat{\mathfrak{g}}$ needed. The full covariant conservation $\nabla^\mu \langle \widehat{T}_{\mu\nu}\rangle_\omega = 0$ required by Thm. 4.1 of Part I is *not* a KMS statement; it holds for every Hadamard state by the Wald conservation axiom of the renormalised stress tensor (Lem. 11.25), on the classical backgrounds where (EG3) applies.

(iv) **The Einstein equation is tautological in the semiclassical limit.** Combining (47) and (48),

$$\int f_{\mu\nu}\xi^\mu\, \widehat{G}^{(Q)\,\mu\nu}\, d\mathrm{vol} \;=\; \frac{8\pi G}{c^4} \int f_{\mu\nu}\xi^\mu\, \widehat{T}^{\mu\nu}_{\mathrm{m}}\, d\mathrm{vol},$$

i.e. the smeared semiclassical Einstein equation holds *by definition of the two operators.* This is not a coincidence; it is the content of Jacobson's derivation. Conj. 10.2 then becomes the statement that the final-state-conditioned source also satisfies this identity, which under the branch postulate (B) and the thermal-time postulate is inherited from the unconditional identity by self-adjointness of ρ_2.

(v) **Averaged null energy is a modular-monotonicity statement, not a property of K_ω.** The boost modular Hamiltonian $K_\omega = -\log \Delta_\omega$ of (47) has full real-line spectrum (cf. the dressing discussion at Pos. 11.30), so (47) on its own places *no* lower bound on smeared energy and does not by itself give the averaged null energy condition. The object whose positivity yields ANEC/QNEC is instead the null-cut *shape derivative* of the Araki relative entropy: by Ceyhan–Faulkner [234], with the rigorous half-sided-modular-inclusion proof of Hollands–Longo [236], the averaged null energy along a cut equals $\partial_{\mathrm{null}} S_{\mathrm{rel}}(\omega\|\omega_0)$ and is nonnegative by monotonicity of relative entropy. Connecting the present seed sector to this result — i.e. establishing $\int_\gamma \langle \widehat{T}_{\mu\nu}\rangle_\omega \dot\gamma^\mu\dot\gamma^\nu\, d\lambda = \partial_{\mathrm{null}} S_{\mathrm{rel}}(\omega\|\omega_0) \geq 0$ — requires two nontrivial inputs not supplied here: that the dressed crossed-product algebra $\widetilde{\mathcal{A}}$ carries the half-sided modular inclusion of the CLPW construction, and that ω has finite averaged null energy. We record this as the correct route to seed-state ANEC within the present machinery — inherited from the same Araki monotonicity used in Pos. 11.11, rather than from the boost first law (47) — and leave its verification for the dressed algebra open. The half-sided modular inclusion is, however, *not* a structure of the dressed Type II crossed product in the standard sense: a Type II factor cannot carry a *standard* half-sided inclusion (one whose reference vector is the unique translation-invariant vacuum, by Wiesbrock's theorem), and the finite-dimensional band-limit destroys it outright — though a *bare* (non-standard) inclusion is not excluded, and Chandrasekaran–Flanagan [225] construct one on a Type II_∞ horizon crossed product precisely because standardness fails there. The inclusion proper lives on the uncompressed Type III_1 seed net — by Bisognano–Wichmann plus the spectrum condition; the null translation then lifts to a covariant automorphism of the dressed algebra, and what descends to the Type II crossed product is *not* the inclusion structure but the relative-entropy monotonicity (with the associated ANEC/QNEC inequality) — which is precisely the input the primitive-count reduction of Rem. I.6 draws on (the inequality the framework already has; the residual that reduction isolates is instead the saturation *equality* and the inversion), a theorem of the ambient literature [92, 95]. On Killing/bifurcate horizons the monotonicity (not the inclusion itself) descends, discharging the ANEC leg's dependence on a *dressed* HSMI; on a *generic* causal diamond the descent runs on the diamond's bifurcate null boundary (Chandrasekaran–Flanagan [225]), so what genuinely remains *of the ANEC/first-law leg* is (a) the H0 co-scoping straddle of Rem. 11.72 — that the finite-dimensional compression does not sever the ANEC leg from the Type III_1 modular structure it must run on — and (b) the finite averaged null energy of ω on complete cuts, a mild Hadamard-relative-to-vacuum integrability condition assumed on the seed, not a theorem. The saturation *equality* (Hyp. I.5) and the null-to-tensor inversion are the further, separate residuals carried by Rem. I.6/Ax. I.10, not discharged here.

Proposition 11.20 (No fundamental G: Newton's constant is an emergent response coefficient)**.** *Write* $E_{\mathrm{P}} = \sqrt{\hbar c^5/G}$ *for the Planck energy and define the dimensionless* gravitational anchor ratio

$$g \;:=\; \left(\tfrac{E_\star}{E_{\mathrm{P}}}\right)^2 \;=\; \frac{G\,E_\star^2}{\hbar c^5}. \tag{49}$$

Call a dimensionless parameter of the framework fundamental *if its value is a free input — the axioms admit a one-parameter family of internally consistent models labelled by it, with nothing in the axioms or the vacuum field content constraining it to a bounded interval — and* an emergent response coefficient *if instead it enters the axioms* only *through a fixed functional of other framework data (the vacuum field content and a single scale) that pins it to a bounded interval of definite sign. Assume, in addition to* (E0) (Pos. 11.18) *and the anchor clause* $E_\star \lesssim E_{\mathrm{P}}$ *of* (H0) (Pos. 11.30)*:*

($\mathrm{H}_\star 1$) (Unique anchoring scale.) *$E_\star$ is the sole dimensionful primitive entering the induced-gravity anchor g of* (49)*: the matter-side scale to which the Bekenstein coefficient $1/4G\hbar$ — and hence G/E_{P} — is tied is $E_\star$ alone, and no sharper dimensionful matter datum re-fixes g below $O(1)$. This scopes only the UV/gravitational-anchoring sector; it makes no claim that the sub-Planckian Standard-Model scales (the Higgs vacuum expectation value v, fermion masses, the modified-TDSE step δt) are functions of $E_\star$ — those enter $S = A/4G\hbar$ only through the species count, which cancels by L3.*

($\mathrm{H}_\star 2$) (Induced coefficient, imported.) *The vacuum entanglement / induced-gravity coefficient of* (EG2)*, evaluated on the actual Standard-Model field content, is finite, of the correct sign, and of order unity: a framework-internal Standard-Model heat-kernel supertrace evaluation gives* $\mathrm{str}[k_1] \approx -14.83$ *(cf. Visser's order-unity estimate* $\mathrm{str}[k_1] \approx -1$ *[404]), whence $g \approx 0.42$ (equivalently $E_\star \approx 0.65\,E_{\mathrm{P}}$), the entanglement area-coefficient and the induced-G coefficient being the* same *heat-kernel datum (Fursaev–Solodukhin; [403, 405]). We stress that the induced-G coefficient is the* signed, spin-weighted *supertrace* $\mathrm{str}[k_1]$ *(bosons minus fermions, with non-minimal-coupling weights),* $|\mathrm{str}[k_1]| \approx 14.83$*, and* not *the unsigned species multiplicity $N \sim O(100)$ that enters the Dvali/Caron–Huot–Li species-scale estimate (see clause* (ii) *of Rem. 11.21): these are distinct heat-kernel functionals of the same field content, so $g \approx 0.42$ is not in tension with that bound. Indeed the species bound of Caron–Huot–Li [189], applied to the actual Standard-Model content by its own spin-weighted dispersive form (their Eq. (4.15), a loop-suppressed* positive *count* $n_0 + 5.6\,n_{1/2} + 9.2\,n_1 + \cdots \lesssim O(1)/(GM^2)$*), places the field-theory cutoff at* $M \lesssim (2\text{–}4)\,E_{\mathrm{P}}$ — above*, not below, the Planck scale — consistent with those authors' own remark that for the Standard Model the graviton self-energy turns pathological only near*

$1.4\times10^{19}\,\mathrm{GeV} \approx E_\mathrm{P}$, "not much lower than the naive Planck scale, essentially due to loop factors." The naive $E_\mathrm{P}/\sqrt{N}$ reading (which would give $\sim 0.1\,E_\mathrm{P}$) omits precisely these one-loop factors and is a conservative over-estimate, not the sharp bound; $E_\star \approx 0.65\,E_\mathrm{P}$ satisfies the sharp bound.

Then G is not a fundamental parameter of the framework: its sole physical datum g is an emergent response coefficient of the vacuum — determined in sign and magnitude by the field content and $E_\star$, and un-sharpenable below $O(1)$.

Proof (structural). The argument is a chain of four steps. *(L0, only g is content.)* The dimensionful value of G in any unit system is a convention: rescaling the mass unit rescales G with no invariant effect, so no unit-independent statement singles out a value of G; only dimensionless combinations carry content, and the sole such combination in which G appears here is g of (49) [402]. *(L1, slaving.)* Within the framework's axioms G occurs only in (EG2), and there *only* through the Bekenstein coefficient $1/4G\hbar$; and by the anchor clause $E_\star$ is tied to $E_\mathrm{P} = \sqrt{\hbar c^5/G}$; hence G enters the framework only through the combination g. This alone gives non-independence, but it is symmetric in $(G, E_\star)$: it establishes only that G and $E_\star$ are not both free. *(L2, provenance breaks the symmetry.)* The symmetry of L1 is broken by ($\mathrm{H}_\star$1): $E_\star$ is posited as the sole dimensionful primitive anchoring g, and by the anchor clause of L1 the Planck energy $E_\mathrm{P} = \sqrt{\hbar c^5/G}$ — hence G — is tied to it, so $E_\star$ is the free scale and G its image, not conversely. (Knowing g alone cannot break this tie: g is symmetric in $(G, E_\star)$, so it is ($\mathrm{H}_\star$1), not the value of g, that designates the primitive.) ($\mathrm{H}_\star$2) then *values* the resulting anchor ratio: g is the induced-gravity heat-kernel functional of the vacuum field content evaluated at $E_\star$. Thus G's content g has internal provenance — a fixed functional of the field content and the single primitive scale — in the sense of the definition. *(L3, no residual dial.)* No sharper-than-$O(1)$ value of g can be extracted from the trace/counting data: in induced gravity the species multiplicity that would "count" the coefficient enters the vacuum entanglement entropy and the renormalised $1/G$ identically and cancels in $S = A/4G\hbar$ [406, 405]; and the single additive-constant freedom of the CLPW Type-II trace is already spent, in this framework, canonically fixing the generalised-entropy zero-point of the cosmological-constant accounting (Prop. 13.12), leaving no free dial to fix g below the $O(1)$ of ($\mathrm{H}_\star$2). The claim now follows positively, not by elimination: ($\mathrm{H}_\star$2) certifies that g *is* the value of a fixed functional of the vacuum field content evaluated at the single scale $E_\star$ (the induced-gravity heat-kernel sum) — the emergent-response-coefficient box by definition, bounded and of definite sign; L0–L1 certify that this g is the *sole* content in which G appears; and L3 certifies that no residual free dial re-promotes g (or $E_\star$) to a fundamental input. A content-free *sharp* determination of g — by a symmetry, an anomaly-cancellation, or a renormalisation-group fixed point, with no field-content input (the tertium quid conceded for asymptotic safety in (iii) of Rem. 11.21) — is excluded by ($\mathrm{H}_\star$2)'s provenance clause, which locates g's value in the field-content functional rather than in any content-independent condition. □

Remark 11.21 (Scope of Prop. 11.20: what it does and does not assert, and its robustness). (i) Guardrails. The proposition asserts that G's *status as an independent fundamental constant* is a category error within the framework: its content is the emergent coefficient g. It does *not* assert that G is empirically incorrect or ill-defined — G remains a perfectly good *measured* effective coupling, and every Newtonian and post-Newtonian result of the paper (Prop. 11.43) is untouched, with G there the observed coefficient. Nor does it assert that the framework *predicts* G: by L3 the coefficient is fixed only to $O(1)$ (a consistency check of sign and magnitude), never to a sharp number. The proposition trades two dimensionful constants $(G, E_\star)$ for one posited scale $E_\star$ plus a proof that G is slaved to it — a reduction of independent inputs, not their elimination; $E_\star$ itself is posited, not derived ($\mathrm{H}_\star$1). In particular, Prop. 11.20 operates *downstream* of (E0) (Pos. 11.18), which it assumes; it therefore does not bear on, and does not upgrade, the postulated status of the entropic-gravity identification or of the semiclassical Einstein equation.

It removes a category error about G's status as an independent constant; it does not convert the classical arm from postulated to derived. (The no-residual-dial step L3 is moreover conditional on Prop. 13.12, which spends the single Type-II additive constant canonically fixing the generalised-entropy zero-point of the Λ accounting — the Planckian vacuum energy itself being absorbed by the separate induced-gravity counterterm.) (ii) Species-dependence vs. divergence-universality. Hypothesis ($\mathrm{H}_\star 2$) uses the species content to fix the finite $O(1)$ coefficient; this is consistent with the species-universality invoked at (EG2), which is a statement about the leading *divergence* absorbed into $1/G$, not about the finite induced coefficient. (iii) Robustness (adversarial survey). The emergent status is not an artefact of the entropic arm. Replacing (E0) by a Chamseddine–Connes spectral action gives the Einstein coefficient $1/16\pi G_0 = N f_2 \Lambda^2/48\pi^2$ (here Λ is the spectral-action ultraviolet cutoff, not the cosmological constant of the surrounding discussion), i.e. $g^{-1} \propto f_2$ with f_2 a free moment of the cutoff profile that must be tuned to the observed G [267, 270]: the input is merely relocated, "posit G" $\to$ "posit f_2", and lands on the same $O(1)$, so the spectral arm *instantiates* Prop. 11.20 rather than evading it (and would in addition surrender the Type-II cosmological-constant mechanism of Prop. 13.12). More broadly, no approach in the literature predicts the dimensionless gravitational coupling as a sharp, field-content-independent number: asymptotic safety fixes only a scheme-dependent $O(1)$ fixed point $g_\star$, with G an experimentally matched relevant coupling; loop quantum gravity fixes the Barbero–Immirzi parameter by *demanding* $S = A/4G$, which already contains G (circular); causal sets, causal dynamical triangulations and group field theory carry G as a bare/tuned/emergent-from-free-parameters input; and the string and large-extra-dimension / warped constructions *relocate* the hierarchy to a modulus, a dilaton vacuum expectation value, a compactification volume, or an $O(10)$ warp exponent. The last of these genuinely *softens* the tuning — to the logarithm of the hierarchy — but still leaves a free input. In every case G is an input, a tuned parameter, a circular match to $A/4G$, or a relocated modulus, never a parameter-free prediction, exactly as Prop. 11.20 requires.

Proposition 11.22 (General relativity as an emergent equation of state, incomplete in three named modes)**.** *Call a tensor field equation* fundamental dynamics *if it is an independent law posited on its own degrees of freedom, and an* emergent equation of state *if it is instead the equilibrium first-law response of an underlying entropic/modular functional, inheriting its content from that functional and determined by it only up to that functional's own blind directions. Under the entropic-gravity identification* (E0) (Pos. 11.18) *— whose linear content is the entanglement-first-law theorem* Thm. I.1 *(modulo Hyp. D.1), whose nonlinear content is the Fréchet–Clausius closure* Ax. I.10 *with named generic-diamond residual* Hyp. I.5*, and whose source is the record-adapted* Def. 10.1 *— the semiclassical Einstein equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, \langle \widehat{T}_{\mu\nu} \rangle$ is* not *fundamental dynamics but an emergent equation of state, and is under-determined by the entanglement–modular substrate in exactly three modes:*

(1) ***Trace/cosmological blindness.*** *The substrate — the null-modular first-law data — fixes $G_{\mu\nu}$ only up to $+c\, g_{\mu\nu}$: the trace/conformal mode, hence the cosmological constant, carries no entanglement-first-law content and must be supplied separately (the fixed-volume equilibrium constraint; the single Type-II additive constant, allocated in Prop. 13.12). See Rem. I.6.*

(2) ***State-selection incompleteness.*** *The source $\langle \widehat{T}_{\mu\nu} \rangle$ is defined only relative to a which-branch rule the equation's own formalism does not contain; the framework supplies it as record-permanence conditioning — what gravitates is what has decohered irreversibly (Def. 10.1, Conj. 10.2, Rem. 11.40; the per-branch classical-metric idea is due to Kent and to Halliwell–Hartle).*

(3) ***Coupling as response coefficient.*** *The coupling G is not an independent fundamental constant but an emergent response coefficient of the vacuum (Prop. 11.20).*

The equilibrium content itself is a theorem at linear/AdS scope (Thm. I.1) *and a named-residual postulate* (Hyp. I.5) *at generic scope; it is in no regime an independent dynamical law.*

Proof (structural). (L0, content.) Within the framework the semiclassical content of gravity is exhausted by the smeared first-law identity of (EG1)–(EG2): the quantum Einstein tensor is *defined* as the area-response of the local modular Hamiltonian, so the field equation holds by construction of the two operators (Rem. 11.19(iv)). *(L1, demotion.)* That identity is the equilibrium first-law response of the area/relative-entropy functional (Thm. I.1, Ax. I.10); the Einstein equation is therefore an equation of state by construction, not a law posited on independent (graviton) degrees of freedom. *(L2, the two blind directions.)* An equation of state inherited from a functional is determined only where the functional carries content. The first-law/null-modular functional is (i) *conformally blind* — it constrains $G_{\mu\nu}$ contracted on null generators, which annihilates $+c\, g_{\mu\nu}$ (Rem. I.6) — leaving the trace/cosmological mode free (mode 1); and (ii) a functional *of a macrostate*, undefined until the state is selected, leaving the which-branch source free (mode 2). The framework names the fillers: the fixed-volume constraint and the Type-II additive constant of Prop. 13.12; the record-permanence rule of Def. 10.1. *(L3, the coupling.)* The one remaining scalar, G, is not a free dial but the emergent response coefficient of Prop. 11.20. Modes 1–3 exhaust the substrate's under-determinations of the equation of state. □

Remark 11.23 (Scope of Prop. 11.22: diagnosis not derivation, the asymmetry as content, and Weinberg–Witten selection). (i) Guardrails. The proposition asserts that the semiclassical Einstein equation is emergent equation-of-state content, under-determined in the three named modes. It does *not* derive the Einstein equation (the equilibrium content is a theorem only at linear/AdS scope, Thm. I.1; at generic scope it rests on Hyp. I.5 and the trace-inversion seam). It does *not* solve quantum gravity: it *reframes* it — one quantises the substrate, not the equation of state — and the intrinsic dynamical closure on a generic diamond remains open. It does *not* solve the cosmological-constant problem: Prop. 13.12 predicts only the absence of the E_{P}^4 catastrophe and an H_0^2 running, with Λ_0, c_Λ free. And it does *not* assert that general relativity is empirically wrong: the post-Newtonian sector is inherited unchanged ($\gamma = \beta = 1$; Prop. 11.43). The three modes are corollaries of one cause (the equation-of-state demotion), not three logically equivalent statements. (ii) The QM/GR asymmetry is the content, not a defect. That the quantum-mechanical arm (U2) is *derived* from the constraint structure while the gravitational arm (E0) is *postulated* is not a liability of the master theorem but its diagnostic content: the quantum dynamics is the algebra's own internal (modular) flow, which the framework produces; the Einstein equation is a consistency condition the algebra imposes on the background it lives on — a different logical type, inherited rather than derived. This is why the gravity arm need not, and cannot, be quantised as a fundamental field, and why G is a response coefficient rather than a coupling: an equation of state is not the same category of object as a fundamental law. (iii) Weinberg–Witten selection. The Weinberg–Witten theorem [57] forbids a theory with a Lorentz-covariant conserved stress tensor from supporting a massless spin-2 quantum. The framework is compatible *by construction*: the quantum Einstein tensor $\widehat{G}^{(Q)}$ of (EG2) is the area-response of the modular Hamiltonian, never a fundamental Lorentz-covariant spin-2 field, so no composite graviton of the forbidden kind is claimed. Weinberg–Witten does not merely fail to obstruct the arm; it positively *selects* the emergent equation-of-state reading over any fundamental-graviton alternative, which it excludes.

Proposition 11.24 (KMS stationarity along the modular flow)**.** *Under Pos. 11.11 and Pos. 11.18, let ω be the seed state with modular flow σ^ω on $\mathcal{N}$, and let $f^{\mu\nu}$ be a smearing kernel supported in a local causal diamond D_x with boost-Killing vector ξ. Then:*

(i) (Invariance.) $\omega \circ \sigma_t^\omega = \omega$ *for all $t \in \mathbb{R}$: a KMS state is invariant under the flow for which it is KMS [195, Ch. 5.3].*

(ii) (Stationarity/commutation along the flow.) *For every $A \in \mathcal{N}$ entire analytic for σ^ω,*

$$\frac{d}{dt}\Big|_{t=0} \omega\big(\sigma_t^\omega(A)\big) \;=\; \omega\big(\delta_\omega(A)\big) \;=\; 0, \qquad \delta_\omega(A) = -i\,[K_\omega, A],$$

with δ_ω the generator of σ^ω. In particular the ω-expectation of the smeared first-law integrand $\int f_{\mu\nu}\xi^\mu \widehat{T}^{\mu\nu}_{\mathrm{m}}\,\mathrm{dvol} = K_\omega[f]$ of (47) *is constant along thermal time, by* (i) *alone. (We do* not *claim $K_\omega[f]$ is affiliated to the dressed centralizer $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$: a partially smeared generator, $f \not\equiv 1$ on D_x, does not commute with the full modular flow in general; only the unsmeared generator does. Constancy of the expectation needs only state invariance.)*

(iii) (Geometric form.) *If σ^ω_t acts geometrically on D_x as the boost flow ϕ_t of ξ (the Bisognano–Wichmann reading underlying* (47)*; cf. Rem. 11.12 (iv)), then* (i) *applied to $\widehat{T}_{\mathrm{m}}[\phi^*_{-t}(f\,\xi)]$ and differentiated at $t = 0$ gives, distributionally on D_x,*

$$\big\langle \mathcal{L}_\xi\big(\xi^\mu \widehat{T}_{\mu\nu}\big)\big\rangle_\omega \;=\; 0,$$

i.e. the ω-expectation of the boost energy–momentum current $\xi^\mu \widehat{T}_{\mu\nu}$ is stationary along the modular flow ($\mathcal{L}_\xi$ the Lie derivative).

Scope. *This is expectation-value stationarity in the single state ω, along the single direction ξ. It is* not *the operator equation $\nabla^\mu \widehat{T}_{\mu\nu} = 0$; it constrains neither the directions transverse to ξ nor states other than ω, and it does not by itself discharge the covariant-conservation hypothesis* (7) *of Thm. 4.1. The covariant conservation used by the semiclassical (Clausius) machinery downstream is the state-independent Hadamard statement of Lem. 11.25 below, which holds independently of the KMS structure and consumes no Killing geometry — the boost form (iii) is therefore never required on non-Killing backgrounds.*

Proof. (i) Modular KMS of ω for σ^ω is Tomita–Takesaki's theorem [39]; invariance of a KMS state under its own flow is standard [195, Ch. 5.3]. (ii) Differentiate the constant function $t \mapsto \omega(\sigma^\omega_t(A))$ at $t = 0$ on the dense $*$-algebra of entire analytic elements, where $\delta_\omega(A) = -i[K_\omega, A]$ by (46); the constancy claim for $K_\omega[f]$ is the case $A = K_\omega[f]$ of the invariance (i), extended to the affiliated element by spectral truncation. (iii) Under the geometric action, $\sigma^\omega_t(\widehat{T}_{\mathrm{m}}[f\,\xi]) = \widehat{T}_{\mathrm{m}}[\phi^*_{-t}(f\,\xi)]$, so by (i) the function $t \mapsto \omega(\widehat{T}_{\mathrm{m}}[\phi^*_{-t}(f\,\xi)])$ is constant. Differentiating at $t = 0$ and using $\mathcal{L}_\xi \xi = 0$ and ϕ_t-invariance of $d$vol on D_x (ξ Killing there) yields $\int (\mathcal{L}_\xi f)_{\mu\nu}\, \xi^\mu \langle \widehat{T}^{\mu\nu}_{\mathrm{m}}\rangle_\omega\,\mathrm{dvol} = 0$ for every f; integration by parts gives the displayed identity in the sense of distributions. ☐

Lemma 11.25 (Semiclassical covariant conservation for Hadamard states)**.** *Let $(\mathcal{M}, g_0)$ be a smooth globally hyperbolic spacetime and let $\widehat{T}_{\mu\nu}$ be the point-splitting–renormalised stress-energy tensor of the free scalar matter field in the conservation-corrected prescription of Wald and Moretti (Hadamard parametrix subtraction plus the local counterterm proportional to $g_{\mu\nu}[v_1]$). Then:*

(a) (Conservation, all Hadamard states.) $\nabla^\mu \langle \widehat{T}_{\mu\nu}\rangle_\omega = 0$ *identically on $\mathcal{M}$ for* every *Hadamard state ω and every spacetime dimension $D \geq 2$ [13, Thm. 2.1(a)] (for a position-dependent external potential the divergence equals the classical source $-\frac{1}{2}\langle\widehat{\varphi}^2\rangle_\omega \nabla_\nu V'$, which vanishes for a constant mass term). The correction term is not optional: the bare Hadamard-parametrix subtraction has anomalous divergence equal to the gradient of a local curvature term, and the unique local counterterm removing it is forced [12]. For $D = 4$, conservation moreover holds at the* operator *level in the extended algebra of Wick products, $:(\nabla \cdot T)_X(f):\, = 0$ for all vector fields X and test functions f [13, Thm. 3.2(c)], which is the free-field form of hypothesis* (7)*. The Dirac-field analogue — a covariantly conserved Wick-polynomial stress tensor with its trace anomaly — is [15].*

(b) (Uniqueness and anomaly caveat.) *The prescription is unique up to addition of conserved local curvature tensors [11][10, Sec. 4.6]. The price of conservation is the trace anomaly: for the conformally coupled massless field $g^{\mu\nu}\langle \widehat{T}_{\mu\nu}\rangle_\omega = -v_1(x,x)/4\pi^2 \neq 0$ [12][13, Thm. 2.1(b)]. The anomaly affects the trace only; conservation is exact.*

(c) (Interacting fields.) *For any polynomial interaction Lagrangian in dimension > 2, the renormalisation freedom of time-ordered products can be consistently fixed by the Leibniz-rule ("perturbative agreement") conditions so that the interacting stress tensor is conserved to all orders of perturbation theory [14]; cf. the review [16].*

None of this uses the KMS property: conservation is a theorem of the Hadamard renormalisation prescription alone, valid for every Hadamard state and in every direction, on the classical backgrounds where (EG3) *of Pos. 11.18 identifies* (47) *with the Hadamard-renormalised stress tensor. It is this statement — not Prop. 11.24 — that supplies the conservation input of the semiclassical Einstein / Clausius machinery; on the dynamical background* $\widehat{\mathsf{g}}$*,* (7) *remains a hypothesis of Thm. 4.1.*

Proposition 11.26 (The gravity–matter coupling is well-defined)**.** *Under Pos. 11.18, the symmetric ordering* (62) *of the gravity–matter coupling* $\widehat{Q}^{(k)}[f]$ *is a well-defined symmetric operator on the common core* $\mathcal{D}_k$ *of (T1), and hypotheses (T1) and (T2) of Conj. 11.56 hold automatically.*

Proof. By (47), $\widehat{T}^{\mu\nu}_{\rm m}[\widehat{\mathsf{g}}; f]$ is the unique self-adjoint operator on $\mathcal{D}_k$ realising the smeared first-law integrand; its essential self-adjointness on $\mathcal{D}_k$ follows from essential self-adjointness of $K_\omega = -\log \Delta_\omega$ on the Tomita modular core, which is Tomita–Takesaki's theorem [39]. By (48), $\widehat{G}^{(Q)\,\mu\nu}[\widehat{\mathsf{g}}; f]$ is a scalar multiple of the same operator, hence also essentially self-adjoint on $\mathcal{D}_k$. The joint domain inclusion of (T7) is immediate because both operators are proportional to $K_\omega[f]$ and therefore commute, so the symmetric ordering (62) reduces to the single self-adjoint operator $\frac{8\pi G}{c^4} K_\omega[f]^2$ on $\mathcal{D}_k$, which is self-adjoint by spectral calculus. □

Remark 11.27 (Further shrinkage of the hypothesis bundle). Under Pos. 11.11 and Pos. 11.18, the hypothesis bundle H=(H0)–(H6) of the Master Unification Theorem shrinks further:

- (T1) and (T2) of Conj. 11.56 are no longer independent open analytic problems; they are consequences of the postulates (Prop. 11.26).
- (T7) (joint domain inclusion) becomes trivial because under (47)–(48) both operators are proportional to $K_\omega[f]$ and therefore commute on $\mathcal{D}_k$.
- Conj. 10.2, which asserts the well-posedness of the final-state-conditioned Einstein coupling, is reduced to the entropy-controlled fixed-point estimate of Conj. 11.4: the coupling identity is built into Pos. 11.18, and only the conditioning-perturbation contraction remains conjectural.
- The covariant-conservation assumption (7) of Thm. 4.1 is not discharged by KMS: Prop. 11.24 yields only stationarity along the modular flow. Its semiclassical form holds for all Hadamard states, and its free-field operator form holds in the extended Wick algebra, by Lem. 11.25 — independently of the postulates; on the dynamical background $\widehat{\mathsf{g}}$ it remains a hypothesis.

Net effect on H: (H4) loses (T1), (T2), (T7) as independent hypotheses; (H3) loses Conj. 10.2's coupling-existence burden (only the contraction estimate remains). The residual open content of H is: (i) Pos. 11.11 + Pos. 11.18 as primitives; (ii) Conj. 11.1 (less (CP)) on the two-branch scope of its standing hypothesis — branch (I), the Araki-conditioning branch carrying the physical two-boundary content, discharged by the App. G theorems (Thms. G.1, G.2); branch (II), the entire-analytic centralizer special case, retained as the exactly solvable clock sector — so that the consolidated commutation Hyp. 11.13 ((CP)+(T11)) is needed only for the branch-(II) convenience form, branch (I) bypassing it entirely; (iii) the entropy-smallness contraction Conj. 11.4; (iv) Conj. 11.5 on the clock shift; (v) Conj. 11.8 on horizon algebras; (vi) the remaining (T0), (T3)–(T6), (T8)–(T13) of Conj. 11.56 — structural conditions on rigged Hilbert data, save for the essential-self-adjointness content of (T3),(T4), which is conditional on the operator-existence Hyp. 12.9 (Rem. 12.17).

Remark 11.28 (What Pos. 11.18 does and does not do). *Does:* dissolves obstruction (5) by identifying the matter stress-energy and linearised quantum Einstein tensor with modular quantities of the seed state, not with Hadamard-renormalised operators on a dynamical metric; promotes (T1), (T2), and (T7) of Conj. 11.56 to theorems; promotes KMS stationarity of stress-energy expectations along thermal time to a theorem (Prop. 11.24), with semiclassical conservation supplied state-independently by Hadamard renormalisation (Lem. 11.25); makes the semiclassical Einstein equation a *definition* rather than a dynamical postulate, consistent with Jacobson's equation-of-state reading [56]; builds into the framework the operator-algebraic content of the first law of entanglement [58] and the vacuum-entanglement derivation of G [59]. *Does not:* fix ω (still a modelling input); remove obstructions (2) UV completion, (4) Cauchy slicing on dynamical metrics, or (7) information at horizons beyond what Pos. 11.11 already gave. The postulate is a structural identification, not a derivation of gravity from nothing; it says "if the first-law and Clausius relations of modular theory hold operationally, then the master-equation coupling is forced."

Remark 11.29 (Relation to rival unification programs). It is worth stating plainly where this places the framework. At its gravitational core the present construction is an *operator-algebraic* member of the entropic / equation-of-state lineage already named above: (E0) is the Jacobson–Bianchi–Myers Clausius identification [56, 59], lifted from a thermodynamic relation to an operator identity on the Type II crossed product by modular theory (Rem. 11.19). It therefore inherits both the strength and the limitation of that lineage. *Strength*: the semiclassical Einstein equation is an equation of state *by construction*, not a separately conjectured dynamics. *Limitation*: as in Jacobson's original derivation, the identification is controlled only at linear order in the metric perturbation (hence the scope of (E0-lin), Thm. I.1); recovering full nonperturbative gravitational dynamics is outside its purview and is exactly the residual (E0). We stress that the framework does *not* invoke the Verlinde entropic-*force* argument [61], cited above only as lineage and whose internal consistency has been disputed; the coupling here is the deterministic, time-symmetric Clausius identity, not a thermodynamic force law. Relative to the nonperturbative ultraviolet programmes — loop quantum gravity / spin foams, asymptotic safety [187], and causal sets [185] — the framework is *not* a candidate UV completion (cf. the precedents-and-caveats discussion of Sec. 11.4); it is an effective, sub-Planckian, background-fixed structure (Pos. 11.30) whose distinctive content — two-boundary retrocausal conditioning and the modular / thermal-time identification of relational time — is largely orthogonal to the UV-completion question those programmes address.

11.4 Planck-scale matter cutoff: the EFT scope postulate

The two preceding postulates identify abstract objects of the master equation with modular quantities of a seed state. A third postulate, orthogonal to both, commits the framework to an effective-field-theory scope: physical matter states are restricted to sub-Planckian energies. This is the standard Wilsonian cutoff of effective quantum field theory, the discrete-spectrum regime of the CLPW dressed crossed product (noting that the CLPW Type II reduction is driven by adjoining the modular/clock generator and, in the de Sitter case, a lower bound on the *clock* spectrum, rather than by the matter upper cutoff alone), and the operational content of the witness model's spectral truncation $\rho_{\mathrm{i}} := P_{\leq E_\star}\rho_\beta P_{\leq E_\star}/Z$ of Conj. 11.55. Promoting it from witness-local to structural shrinks the hypothesis bundle H of the Master Unification Theorem by converting several analytic assumptions into theorems of finite-dimensional linear algebra.

Postulate 11.30 (Induced-gravity anchor scale $E_\star$ (matter cutoff))**.** Fix once per physical setup an energy cutoff $E_\star > 0$ with $E_\star \lesssim E_{\mathrm{P}}$. Every physical matter state ρ on the matter algebra $\mathcal{N}$ of Conj. 11.1 is supported on the spectral range of the matter Hamiltonian below $E_\star$:

$$\rho \;=\; P_{\leq E_\star}\,\rho\,P_{\leq E_\star}, \qquad P_{\leq E_\star} \;:=\; E_{\widehat{H}_{\mathrm{m}}}\big([0, E_\star]\big), \tag{50}$$

where $E_{\widehat{H}_{\mathrm{m}}}$ is the projection-valued spectral measure of $\widehat{H}_{\mathrm{m}}$. Equivalently, the physical matter Hilbert space is $\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}} := \mathrm{Ran}(P_{\leq E_\star})$, finite-dimensional on any compact spatial region, and the physical matter algebra is $P_{\leq E_\star}\mathcal{N}\,P_{\leq E_\star}$. Similarly, the terminal effects $\{E_k\}$ and the initial state ρ_{i} of Axiom 2.1 are taken in $\mathrm{Ran}(P_{\leq E_\star})$, and all Heisenberg propagation is restricted to this subspace via $P_{\leq E_\star}U(\tau,\tau')P_{\leq E_\star}$.

Remark 11.31 (The anchor scale is fixed by the field content, not a free guess). The bound $E_\star \lesssim E_\mathrm{P}$ reads, on its face, as a postulated Wilsonian cutoff whose Planckian order is put in by hand — and that is how an earlier framing of this postulate ("Planck-scale matter cutoff") presented it. It is worth recording that the framework separately *fixes* this scale, so that its Planckian order is fixed by the field content rather than guessed. In the induced-gravity accounting of Prop. 11.20, Newton's constant is not a fundamental input but a response coefficient of the vacuum, $g := GE_\star^2/\hbar c^5 = (E_\star/E_\mathrm{Pl})^2$, and the Standard-Model field content fixes the dimensionless ratio through the heat-kernel supertrace $\mathrm{str}[k_1]$, giving $g \approx 0.42$, i.e. $E_\star \approx 0.65\,E_\mathrm{P}$. The anchoring scale is therefore not an adjustable ultraviolet knob: once the matter content is fixed, the scale at which the vacuum's response coefficient equals the observed G is fixed with it, and it lands Planckian because G is Planckian. The *same* supertrace sum services both gravitational poles at once — one str-sum, two allocations: its quadratically divergent part $\sim E_\star^2$ sets the inverse Newton constant $1/G$ (the ultraviolet pole), and its quartically divergent part $\sim E_\star^4$ is what the Sakharov cosmological counterterm absorbs (the infrared pole). We emphasise — and Rem. D.4 makes this precise — that these are two *logically independent* counterterms: fixing $1/G$ does not fix Λ, whose observed value remains an independent subtraction. The "two allocations" share a single heat-kernel expansion but not a single free constant, so this is bookkeeping, not a cross-check (see Prop. 13.12, Rem. 13.13). What Pos. 11.30 still posits is the *existence* of a single sharp band edge $E_\star$ and its use as a spectral projector; what it no longer needs to posit is the numerical *order* of that edge, which the field content fixes to be Planckian.

Remark 11.32 (Physical content of Pos. 11.30). Four consequences, each corresponding to a standard EFT statement:

(i) **Finite-dimensional matter on bounded regions.** On any compact spatial region R, the physical matter Hilbert space $\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}$ is finite-dimensional by microcanonical counting: Weyl asymptotics bounds the number of *single-particle* modes below $E_\star$ by $N(E_\star, R) \lesssim E_\star^3|R|$, and the total-energy constraint $\widehat{H}_\mathrm{m} \le E_\star$ then bounds the *Fock* dimension through the microcanonical entropy, $\log\dim\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}(R) \lesssim S_{\mathrm{micro}}(E_\star, R) \sim (E_\star^3|R|)^{1/4}$ up to $\mathcal{O}(1)$ species factors (radiation-type counting). An earlier version of this item quoted "$\dim \lesssim (E_\star|R|)^{3/2}$," conflating the single-particle mode count with the Fock-space dimension; the finiteness conclusion is unchanged. Part I's finite-dimensional assumption on $\mathcal{H}$ is therefore no longer ad hoc: it is the restriction of the full matter algebra to the physical sub-Planckian scope mandated by Pos. 11.30.

(ii) **Discrete matter spectrum.** $\widehat{H}_\mathrm{m}\restriction_{\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}}$ has discrete spectrum $\{E_0, E_1, \ldots, E_N\} \subset [0, E_\star]$. The modular Hamiltonian $K_\omega = -\log\Delta_\omega$ is a bounded self-adjoint operator on the finite-dim GNS space of ω. Tomita–Takesaki theory reduces to finite-dimensional linear algebra; the subtleties of Type III factors (Conj. 11.1) are bypassed in the Pos. 11.30 scope.

(iii) **Scope interpretation.** Gravitational dynamics at scales above $E_\star$ — topology change, Planckian curvature, non-perturbative quantum-geometric effects — is explicitly *out of scope*; above $E_\star$ the framework makes no claims. This is the same scope restriction carried by every effective field theory with a Wilsonian cutoff, every lattice regularisation, and every finite-dimensional algebraic quantum gravity proposal. It does not narrow the paper's content; it makes the content's domain of validity honest.

(iv) **Hawking robustness: the adiabatic-vacuum condition.** A hard cutoff at $E_\star$ raises the trans-Planckian question for any Hawking-type statement: the outgoing modes of a horizon trace back to exponentially blue-shifted, super-cutoff progenitors. Unruh and Schützhold [192] prove that the Hawking spectrum is reproduced to lowest order under modified (cutoff-scale) dispersion given three rather general *sufficient* conditions, chief among them that the field state at the cutoff scale is the local adiabatic (free-fall) vacuum near the horizon — and they exhibit explicit counter-examples with strong deviations when the conditions fail, so universality is conditional, not automatic. We therefore record the *adiabatic-vacuum condition as an explicit condition of the H0 scope*: the seed states to which Pos. 11.30 is applied near Killing horizons are assumed adiabatic (free-fall regular) at the band edge. The equilibrium KMS recovery of Thm. M.1 is insensitive to this (it is a benchmark about the Hartle–Hawking state, with no dynamical-flux claim), but any extension of the framework to a dynamical Hawking flux inherits the Unruh–Schützhold condition as a hypothesis, not a theorem.

Proposition 11.33 (Shrinkage of the hypothesis bundle under Pos. 11.30)**.** *Assume Pos. 11.30. Then the following are theorems:*

(i) ***(T5) Spectral absolute continuity at*** 0***.*** *The constraint operator* $\widehat{\mathfrak{C}}_k$ *restricted to the physical subspace* $\mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{\mathrm{phys,(k)}}_{\mathrm{matter}}$ *has, generically in the two-boundary data, non-degenerate zero eigenvalue with spectral measure absolutely continuous on a neighbourhood of* $\lambda = 0$ *in the rigged-Hilbert sense. (Proof: discrete spectrum of* $\widehat{H}_{\mathrm{m}} \restriction_{\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}}$ *plus the spectral theorem on finite-dim plus rigged-Hilbert extension to the clock and geometry factors.)*

(ii) ***(T6) Band-limited slow-branch selection.*** *The physical matter space* $\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}$ *is band-limited by construction:* $E_{\widehat{H}_{\mathrm{c}}}([-\hbar/\delta_\star, \hbar/\delta_\star])\Psi_k = \Psi_k$ *for every* $\delta_\star \le \hbar/E_\star$ *and every physical* Ψ_k*. The distinction between* Φ_k *and* Φ_k^{bl} *of Conj. 11.56 (T6) collapses, with* $\Phi_k^{\mathrm{bl}} = \Phi_k$ *the full physical nuclear core.*

(iii) ***(S3) Spectral gap of*** ρ_{i}***.*** *For generic initial data* ρ_{i} *on* $\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}$*, the smallest non-zero eigenvalue is bounded below by a constant* $\eta = \eta(E_\star, \dim \mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}) > 0$ *depending only on the cutoff and the physical Hilbert-space dimension. Operator-Lipschitz control of* $\rho_{\mathrm{i}}^{1/2}$ *follows from finite-dimensional spectral calculus without the Hölder-only regime that (S3) was designed to exclude.*

(iv) ***Nuclearity of*** Φ_k***.*** *The test space* $\Phi_k \subset \mathcal{D}_k$ *can be taken to be the full physical finite-dimensional space, which is nuclear trivially.*

(v) ***Well-posedness of Hadamard-free stress-energy.*** *The identification* (47)–(48) *of Pos. 11.18 is rigorous without Hadamard renormalisation: on the compressed corner the* dressed K_ω *is bounded self-adjoint on the finite-dim GNS space of* $\omega \restriction_{P_{\le E_\star}\mathcal{N}P_{\le E_\star}}$*, so Prop. 11.26 reads as a theorem of finite-dim linear algebra rather than a result of Type III Tomita–Takesaki theory. This is subject to the same caveat as (ii): the finite-dim reading uses the dressed compressed generator, whereas* (47)*'s* $K_\omega[f]$ *is* a priori *the bare type-III modular Hamiltonian smeared against a matter stress tensor; their agreement across the compression is the H0 co-scoping residual (Rem. 11.72), of which EG1 is the sharpest stress point, so the finite-dim discharge here holds* modulo co-scoping.

Proof. (i) On $\mathcal{H}^{\mathrm{phys,(k)}}_{\mathrm{matter}}$, $\widehat{H}_{\mathrm{m}}$ has finite spectrum $\{E_0, \dots, E_N\}$; combined with the discrete clock spectrum on the band-limited subspace (T6-derived by (ii) below), the constraint $\widehat{\mathfrak{C}}_k$ has discrete spectrum, and generic two-boundary data avoid exact resonance with 0. Spectral measures on discrete + continuous components are absolutely continuous on a neighbourhood of 0 by construction of the rigged extension on clock and geometry factors. (ii) $P_{\le E_\star}$ restricts $\widehat{H}_{\mathrm{m}}$ to bounded spectrum $[0, E_\star]$; the relevant clock generator here is the *dressed* band-limited generator (the CLPW observer energy $\widehat{P}$ compressed to $P_{\le E_\star}$), *not* the bare $-\log \Delta_\omega$ (which in type III_1 is unbounded, and whose unboundedness is retained for the thermal-time sector; see the bare/dressed distinction in Pos. 11.11 and Rem. 11.72). On the compressed corner the dressed Δ_ω has spectrum in a bounded interval $[a, b] \subset (0, \infty)$, so the spectral support satisfies $|\sigma(\widehat{H}_{\mathrm{c}})| \le \log(b/a) =: C(E_\star) < \infty$ by the operator-norm bound on the *compressed* algebra — this bound is on the dressed generator and holds *modulo* the H0 co-scoping assumption (Rem. 11.72) that the compression preserves the clock-sector modular data. (iii) Weyl's inequality for perturbations of Hermitian matrices: on a finite-dim space, the smallest non-zero eigenvalue of ρ_{i} is bounded below by a generic constant uniform in the weighted Sobolev ball $\mathcal{B}_{g_0}$ by continuity of the eigenvalue map. (iv)–(v) Standard finite-dim linear algebra. □

Remark 11.34 (The exact band limit is an idealization). Prop. 11.33 consumes Pos. 11.30 in its *exact* zero-tail form $\rho = P_{\le E_\star}\rho P_{\le E_\star}$. No physical state satisfies this exactly: thermal and Hadamard states carry super-exponentially suppressed but non-vanishing high-energy tails $\sim e^{-\beta E_\star}$, and the band limit is strictly stronger than analyticity of the orbit (Rem. 11.57). The discharge theorems above are therefore statements about an idealized scope representative, and their physical applicability rests on the following named hypothesis.

Hypothesis 11.35 (ε-tail robustness of the cutoff discharge)**.** Let ρ be a state with $\|(\mathbb{1} - P_{\le E_\star})\,\rho\,(\mathbb{1} - P_{\le E_\star})\|_1$ ε (and correspondingly tail-bounded terminal effects and propagator leakage). Then the conclusions of Prop. 11.33 hold with explicit $O(\varepsilon)$ error budgets: (T5) with the non-degenerate zero eigenvalue of $\widehat{\mathfrak{C}}_k$ perturbed by $O(\varepsilon)$ in eigenvalue norm (Weyl / Bauer–Fike), remaining isolated and non-degenerate provided the surrounding spectral gap of $\widehat{\mathfrak{C}}_k$ around $\lambda = 0$ exceeds this $O(\varepsilon)$ shift — this is the *constraint* zero-mode gap, set by the $\widehat{H}_{\rm m}$ spectral gaps and the generic non-resonance margin that (T5)'s own hypothesis supplies (a *different* operator from the initial-state eigenvalue gap η of (S3)), so (T5) inherits its robustness from that constraint non-resonance margin rather than from an absolutely-continuous density shift; (T6) with the band-limited projection error $\left\|(\mathbb{1} - E_{\widehat{H}_{\rm c}}([-\hbar/\delta_\star, \hbar/\delta_\star]))\Psi_k\right\| = O(\varepsilon^{1/2})$ propagated through the Taylor-remainder bound of Prop. 10.27; (S3) with the spectral gap η degraded to $\eta - O(\varepsilon)$; and nuclearity replaced by an $O(\varepsilon)$-approximate finite-rank truncation of Φ_k.

Remark 11.36 (Status of Hyp. 11.35). Hyp. 11.35 is stated, not proven: no ε-tail error budget currently exists for Prop. 11.33, and we flag the full ε-robust cutoff-shrinkage theorem as the *highest-value open strengthening* of the UV sector — it would simultaneously convert H0 from an exact idealization into a statement about the physical states that actually populate its scope, and immunize the discharge theorems against sub-Planckian new physics that perturbs the band edge. The template for the required tracking already exists inside the paper: App. S (Prop. S.1) tracks cutoff leakage through modular flow with explicit $(E/E_\star)^\alpha$ suppression factors, which is exactly the bookkeeping Hyp. 11.35 requires for the four discharge items. For thermal/Hadamard tails the natural budget is $\varepsilon \sim e^{-\beta E_\star}$, far below every other error term in the paper; the point of the hypothesis is not the size of ε but that the discharge theorems' dependence on it is currently unquantified.

A dedicated proof attempt partially closes this. The two *spectral* items admit unconditional explicit-remainder proofs: (S3) the gap degradation follows from Weyl's inequality for the Hermitian perturbation (Bauer–Fike condition number 1), and the nuclearity item from a trace-norm finite-rank truncation. The tail bookkeeping also sharpens: for a *general* above-cutoff tail the state-approximation rate is $\|\rho - P\rho P\|_1 = O(\varepsilon^{1/2})$ — the amplitude leakage $\|(\mathbb{1} - P_{\le E_\star})\Psi\| = \varepsilon^{1/2}$ (from $\|Q\rho Q\|_1 = \|Q\Psi\|^2$ for the pure case) dominates the trace-norm bound — improving to the stated $O(\varepsilon)$ only for tails that *commute* with the band projector, which includes precisely the thermal/Hadamard states $\varepsilon \sim e^{-\beta E_\star}$ that populate H0's scope. The residual is localised to (T5): the zero-eigenvalue shift of $\widehat{\mathfrak{C}}_k$ is controlled by the *operator*-norm leakage $\left\|(\mathbb{1} - P_{\le E_\star})\widehat{\mathfrak{C}}_k P_{\le E_\star}\right\|_{\rm op}$ — the "tail-bounded terminal effects and propagator leakage" clause already carried in the hypothesis above — not by the state trace-norm bound (a vector norm does not feed Weyl/Bauer–Fike); quantifying that operator budget in terms of ε is the one un-closed step.

Three further elementary sharpenings pin the rate structure down (write $P := P_{\le E_\star}$, $Q := \mathbb{1} - P$). *(1) The general-state rate $\varepsilon^{1/2}$ is sharp.* The coherent band-edge superposition $\Psi_\varepsilon = \sqrt{1-\varepsilon}\,\varphi + \sqrt{\varepsilon}\,\psi$ ($\varphi \in \operatorname{Ran} P$, $\psi \in \operatorname{Ran} Q$) has tail weight $\|Q\rho Q\|_1 = \varepsilon$ exactly, yet $\|\rho - P\rho P\|_1 \ge \|P\rho Q\|_1 = \sqrt{\varepsilon(1-\varepsilon)}$, matching the general upper bound $2\varepsilon^{1/2} + \varepsilon$; no better state-norm rate exists, so the $\varepsilon^{1/2}$ bookkeeping above is Θ, not merely O. *(2) The $\varepsilon^{1/2}$ is confined to state/vector norms; eigenvalue budgets are second order in the leakage.* For any state, positivity gives the Cauchy–Schwarz ceiling $\|P\rho Q\|_{\rm op} \le (\|P\rho P\|_{\rm op}\|Q\rho Q\|_{\rm op})^{1/2} \le \varepsilon^{1/2}$, and the leakage blocks

$P\rho Q + Q\rho P$ are purely off-diagonal in the (P, Q) grading, so first-order eigenvalue shifts vanish identically; a Schur-complement/Weyl bootstrap then bounds every eigenvalue shift of ρ against the block-diagonal part $P\rho P \oplus Q\rho Q$ by $2\,\|P\rho Q\|_{\mathrm{op}}^2/\delta \le 2\varepsilon/\delta$ once $\varepsilon^{1/2} \le \delta/2$, with δ the relevant unperturbed gap. Hence the (S3) in-band gap degrades only by $O(\varepsilon/\eta)$ for *arbitrary* — also non-band-commuting — tails (numerically tight, linear in ε), under the two-cluster reading $\mathrm{spec}(\rho) \subset [0, O(\varepsilon)] \cup [\eta - O(\varepsilon/\eta),\, 1]$: the tail block contributes a separate cloud of $O(\varepsilon)$-small eigenvalues, so "smallest non-zero eigenvalue" in (S3) refers to the in-band cluster, whose separation from the cloud is what the operator-Lipschitz control of $\rho_{\mathrm{i}}^{1/2}$ consumes. The band-commuting/thermal improvement is therefore needed only for the trace-norm items; the (T6) vector norm and the nuclearity truncation are $\varepsilon^{1/2}$-sharp regardless of tail shape, and nothing downstream distinguishes the two classes at $\varepsilon \sim e^{-\beta E_\star}$. *(3) The (T5) operator clause localises to the gravity–matter sector and enters quadratically.* Since $\widehat{H}_{\mathrm{c}}$, $\widehat{H}_{\mathrm{grav}}$ and $\alpha\widehat{\Sigma}_{\delta t}$ act on tensor legs distinct from the matter band while $\widehat{H}_{\mathrm{m}}[g_0]$ commutes with its own spectral projector,

$$Q\,\widehat{\mathfrak{C}}_k\,P \;=\; Q\left[\left(\widehat{H}_{\mathrm{m}}^{(k)}[\widehat{\mathfrak{g}}] - \widehat{H}_{\mathrm{m}}[g_0]\right) \;-\; \kappa^{-1}\widehat{Q}^{(k)}[f]\right] P,$$

i.e. the operator leakage is sourced solely by the geometry-dressing of the matter generator and the $T\!\cdot\!G$ coupling; and the zero-mode shift obeys the same quadratic bound $2\left\|Q\widehat{\mathfrak{C}}_kP\right\|_{\mathrm{op}}^2/\delta_{\mathfrak{C}}$, where $\delta_{\mathfrak{C}}$ is the (T5) non-resonance margin read on the block-diagonal part $P\widehat{\mathfrak{C}}_kP \oplus Q\widehat{\mathfrak{C}}_kQ$ (so the genericity clause must also keep $\mathrm{spec}(Q\widehat{\mathfrak{C}}_kQ)$ away from 0 — a band-edge clock-compensation margin). The un-closed step thus needs only $\left\|Q\widehat{Q}^{(k)}[f]P\right\|_{\mathrm{op}} \lesssim (\delta_{\mathfrak{C}}\,\varepsilon)^{1/2}$-type control — quadratically weaker than a linear budget — and the missing lemma is a buffered two-scale cross-band bound $\left\|(\nVdash - P_{\le E_\star})\widehat{Q}^{(k)}[f]P_{\le E_\star - \Delta}\right\|_{\mathrm{op}}$ from the Fourier decay of the smearing f: App. S is the template but bounds expectation values in $E \ll E_\star$ states, not operator norms, and a hard band edge gives only power-law suppression there. Whether the two-boundary thermal-time states lie in the exactly band-commuting class is itself co-scoping-conditional (their KMS property is with respect to modular flow, not $\widehat{H}_{\mathrm{m}}[g_0]$; Rem. 11.72), but by (2) no error budget now hinges on that membership: the sole substantive residual of Hyp. 11.35 is the (T5) cross-band coupling lemma.

Remark 11.37 (Effect on the modified TDSE parameters). Under Pos. 11.30, the two phenomenological parameters $\alpha, \delta t$ of the modified TDSE admit the normalization convention $\delta t \sim \hbar/E_\star$ and $\alpha \sim E_\star$, reducing the two-parameter phenomenology of Rem. 11.71 to a one-parameter family controlled by the single cutoff scale. This scaling is a convention on the $(\alpha, \delta t)$ split, not a derived tie: it is in tension with the Born–Oppenheimer-matched value of the physical combination $\mu = \alpha\,\delta t^2/\hbar^2$; see Rem. C.4, whose convention question is now settled empirically *against* the Planck-natural point by GRB photon timing (Rem. X.4). The Planck-natural identification $E_\star = E_{\mathrm{P}}$ recovers the Planck-suppressed regime of Prop. 10.27 exactly; any $E_\star < E_{\mathrm{P}}$ gives larger (but still sub-Planckian) deviations testable by precision spectroscopy. The cutoff therefore does not merely restrict scope; it also tightens the framework's falsifiable content.

Remark 11.38 (What Pos. 11.30 does and does not do). *Does:* commits the framework to sub-Planckian effective scope, making Part I's finite-dimensional setup structurally motivated rather than ad hoc; converts (T5), (T6), (S3), nuclearity of Φ_k, and the Type III subtleties of Prop. 11.26 into theorems of finite-dim linear algebra; ties the modified-TDSE phenomenology to a single physical parameter $E_\star$ via Rem. 11.37. *Does not:* address obstructions (2) UV completion of gravity, (4) Cauchy slicing on dynamical metrics with macroscopically distinct causal structure, or (7) horizon information beyond what Pos. 11.11 already supplies. In particular, Pos. 11.30 does not make the framework non-semiclassical; it makes its semiclassical scope honest and explicit. The superselection rule (61) between branches remains in force, and coherent metric superpositions with macroscopically distinct causal structure remain outside the framework's purview.

Precedents and caveats. The Planck-cutoff postulate is structurally analogous to choices made by several established sub-Planckian or discrete quantum-gravity programs: causal set theory (Surya [185]) provides an intrinsically discrete Lorentz-invariant UV regulator with a path-sum two-boundary formulation in toy 2d settings; loop quantum cosmology (Agullo et al. [186], Ashtekar–Lewandowski via the review in [186]) provides a Planck-density bounce mechanism with effective continuum equations matching the full quantum dynamics for sharply-peaked states; asymptotic safety (Bonanno et al. [187], Eichhorn–Schiffer [188]) provides momentum-dependent gravitational form factors below the fixed-point scale that can be interpreted as effective sub-Planckian corrections. None of these programs produces a rigorous continuum-limit or UV-completion theorem the present framework can rely on; they are cited as independent motivation that a sub-Planckian effective regime with finite-dim matter is a well-established setting in the quantum-gravity literature, not a peculiarity of the present paper.

One honesty caveat must accompany the asymptotic-safety entry in this list: *fundamental* asymptotic safety and a strict sub-Planckian band limit are mutually exclusive, not mutually supporting. The Reuter fixed point is generated precisely by the trans-Planckian fluctuations that Pos. 11.30 excises; and even *effective* asymptotic safety — the scenario in which the fixed-point scaling is an intermediate regime below a fundamental cutoff, as in the string-theory embedding of de Alwis et al. [190] — requires that new-physics scale to sit *above* the Planck scale ($\Lambda_{\mathrm{UV}} > M_{\mathrm{P}}$), i.e. it requires exactly the trans-Planckian scaling window that a band limit at $E_\star \lesssim E_{\mathrm{P}}$ forbids. H0 is therefore a *falsifiable structural bet against the Reuter fixed point*: evidence that gravitational RG flow continues through a genuine trans-Planckian fixed-point regime would falsify the band-limit postulate, not merely scope it. What survives as convergent motivation from that program is *operational* rather than fundamental: Percacci–Vacca [191] show that asymptotic safety itself implies a minimal length in the operational sense (no experiment resolves structure below the fixed-point scale), so both the AS scenario and H0 predict the same sub-Planckian phenomenological regime even though their trans-Planckian ontologies are incompatible.

Beyond these discrete programs, recent amplitude-bootstrap results give a first-principles upper bound on the EFT-breakdown scale that the postulate can adopt: Caron–Huot and Li [189] show that unitarity and analyticity of $2 \to 2$ graviton amplitudes imply a species-type bound $G\,\Lambda^{d-2}N \lesssim \mathcal{O}(1)$, where Λ is the scale above which local effective field theory must be modified and N counts parametrically light species. Two cautions on how this bound is to be read, both drawn from [189] itself, are needed before applying it here. First, the sharp bound is not the schematic $E_{\mathrm{P}}/\sqrt{N}$ but a *loop-suppressed, spin-weighted* dispersive inequality: their Eq. (4.15) reads $n_0 + 5.6\,n_{1/2} + 9.2\,n_1 + 157\,n_{3/2} + \cdots \lesssim \mathcal{O}(1)\,\log(M/m_{\mathrm{IR}})/(GM^2)$, an *unsigned* positive count of each spin's light modes (distinct from the *signed* induced-G supertrace $\mathrm{str}[k_1]$ of ($\mathrm{H}_\star 2$), which is a different heat-kernel functional of the same content). Second, the $E_{\mathrm{P}}/\sqrt{N}$ form omits precisely the one-loop factors that Eq. (4.15) retains; inserting the actual Standard-Model content ($n_0 \sim 4$, $n_{1/2} \sim 45$, $n_1 = 12$) gives a weighted count $\approx 3.7 \times 10^2$ and hence, up to the undetermined infrared logarithm of Eq. (4.15), a cutoff $M \lesssim (2\text{–}4)\,E_{\mathrm{P}}$ — *above* the Planck scale — consistent with Caron–Huot–Li's own numerical finding that for the Standard Model the graviton self-energy turns pathological only near $1.4 \times 10^{19}\,\mathrm{GeV} \approx E_{\mathrm{P}}$, “not much lower than the naive Planck scale, essentially due to loop factors.” The naive $E_{\mathrm{P}}/\sqrt{N} \sim 0.1\,E_{\mathrm{P}}$ estimate is therefore a conservative over-count, not a hard constraint that $E_\star$ must undercut; the framework's anchor $E_\star \approx 0.65\,E_{\mathrm{P}}$ of ($\mathrm{H}_\star 2$) sits comfortably below the sharp species-scale cutoff, and the conclusions are in any case robust to any $E_\star \lesssim E_{\mathrm{P}}$.

Bojowald [145] recently argued that in LQC with a more complete effective treatment the bounce can occur at *sub-Planckian* densities, with a new physical singularity possibly preceding it. This is a warning that identifying $E_\star = E_{\mathrm{P}}$ exactly is scheme-dependent; the framework's conclusions are robust to any $E_\star \lesssim E_{\mathrm{P}}$, and the precise scale at which the effective description breaks down is a separate open problem that Pos. 11.30 does not attempt to fix.

11.5 Consistency checks: Page–Geilker and an explicit entropy-smallness bound

The structural content of Part II is strengthened by two concrete calculations that test the framework on explicit problems. The first (Sec. 11.5.1) checks that the branch-indexed semi-classical Einstein equation of Conj. 10.2 evades the Page–Geilker pathology that falsifies naive forward semiclassical gravity. The second (Sec. 11.5.2) exhibits a closed-form lower bound on the entropy-smallness constant C of Conj. 11.4, converting that conjecture from existential to constructive in the finite-dim toy setting of Pos. 11.30.

11.5.1 Page–Geilker: the branch-indexed source evades the pathology

Page and Geilker [63] performed a Cavendish-type experiment to test whether the semiclassical Einstein equation $G_{\mu\nu} = 8\pi G\langle\widehat{T}_{\mu\nu}\rangle_\rho$ holds with the forward state ρ when matter is in a superposition of macroscopically-distinct configurations (e.g. a radioactive atom before read-out). The naive forward prediction is that the Cavendish balance responds to a weighted-average source $\sum_k p_0(k)\, T^{(k)}_{\mu\nu}$ regardless of whether the decay has been read out. Experimentally this is not observed: the balance responds as if the decay outcome is definite. Naive forward semiclassical gravity is thereby falsified. The two-boundary framework of the present paper predicts a different outcome, which we now verify passes the Page–Geilker test.

Proposition 11.39 (Record-adapted Einstein evades Page–Geilker)**.** *Assume Conj. 10.2 with the adapted sourcing of Def. 10.1 and the adaptedness restriction* (23)*, a terminal POVM $\{E_k\}$ resolving the decay record on Σ_{τ_f}, Pos. 11.30 establishing the sub-Planckian EFT scope, the superselection rule* (61) *on the universal sample space $\Omega^{\text{univ}}_\gamma$, and a protocol $\mathcal{P}_\gamma$ whose instruments include the environmental monitoring of the decay (the* record step*; in a laboratory this monitoring is performed automatically by decoherence, cf. Rem. 11.40). Then:*

- (i) *For each $k \in \mathcal{K}_+$ the post-terminal branch metric solves the per-branch Einstein equation $G_{\mu\nu}[g_k] = \kappa\,\langle\widehat{T}_{\mu\nu}\rangle_{\rho_2(\tau|k)}$ on $J^+(\Sigma_{\tau_f})$, with source determined by the branch-conditioned state of Def. 3.1; on the pre-terminal region the realized metric g_ω solves the adapted equation* (22)*.*
- (ii) *On each realization $\omega \in \Omega^{\text{univ}}_\gamma$ with $K(\omega) = k$, any test apparatus* whose own records are included in *$\mathcal{P}_\gamma$ (Rem. 10.9; in the laboratory reading this covers every apparatus in the lab) experiences, once inside the record cones — i.e. after light-crossing from the record events, a delay negligible in any laboratory (Def. 10.1,* (23)*) — the metric g_ω, which there agrees with the branch metric $\widetilde{g}_k$ —* not *the forward-averaged metric $\sum_k p_0(k)\, g_k$ — up to the residual of Prop. 10.10: $O\big(\kappa TM\, \|\mathcal{G}_{g_0}\|_{\text{op}}\,(1-\pi^\gamma_\tau(k))/(1-q)\big)$, with $\mathbb{E}[1-\pi^\gamma_\tau(K)] \le \min\{H_\tau, \sqrt{H_\tau/2}\}$ and $H_\tau = H(K \mid \overline{\mathcal{F}}^\gamma_\tau)$. In the decoherence-saturated Page–Geilker regime the branch-distinguishing records are in $\overline{\mathcal{F}}^\gamma_\tau$ before any gravitational apparatus can resolve the branch metrics (Rem. 11.40), so the residual is negligible and the apparatus responds to the realized branch.*
- (iii) *The ensemble-averaged test response over the universal sample space is a classical-probability mixture of per-branch responses with weights $p_0(k)$;* in the decoherence-saturated regime of Rem. 11.40 *(records precede any gravitationally resolvable branch difference) this is operationally indistinguishable from a classical random experiment in which the decay outcome is definite and the observer's uncertainty is epistemic. Outside that regime the pre-record response is prior-sourced and* does *differ from a definite-outcome model, precisely where no record yet exists (Rem. 11.48). This is the marginalization-restores-classicality face of Thm. 3.10: the per-branch temporal non-classicality witnessed by $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$ collapses under the prior average $\sum_k p_0(k)\, M_k = \rho \succeq 0$, the same no-signalling identity (Thm. 6.1) that makes the ensemble response indistinguishable from an epistemic mixture.*
- (iv) *The forward-averaged source of naive semiclassical gravity is recovered exactly as the* prior

expectation *of the adapted source,* $\mathbb{E}_{\mathbb{P}_\gamma}[\langle \widehat{T}_{\mu\nu}\rangle_{\rho^\gamma_{\mathrm{RC}}(\tau)}] = \langle \widehat{T}_{\mu\nu}\rangle_{\rho(\tau)}$ *by Cor. 5.4, never as a per-realization metric with nontrivial records.*

Consequently the Page–Geilker pathology does not arise: no realization's Cavendish balance responds to the averaged source once the decay is recorded anywhere in its environment; each responds to the recorded configuration of its own realization, switching at the recorded decay time with delay set by the decoherence/protocol timescale plus the light-crossing time from the record event *(the latter negligible in the laboratory); the forward-averaged source emerges only at the level of ensemble expectations, which is what the Page–Geilker experiment in fact observes. The switching statement sharpens the previous version of this proposition: the response is not merely binary per realization but tracks the realized record history in real time, consistent with the behavior reported in [63] (which compared time-averaged Cavendish responses against the definite-outcome and averaged-source predictions; it did not resolve switching in real time).*

Proof. (i) is Conj. 10.2 with Def. 10.1: on $J^+(\Sigma_{\tau_f})$, $\pi^\gamma_\tau(k) = \mathbf{1}_{\{K=k\}}$ and (22) is the per-branch equation; pre-terminally it is (22) as displayed. (ii) is Prop. 10.10(ii) evaluated on $\{K = k\}$, with the smallness of $1 - \pi^\gamma_\tau(k)$ supplied by the record step (Rem. 11.40 for the physical mechanism; the worked example below for an exact model). (iii) follows because for any test observable $\mathcal{O}$ depending only on the gravitational field,

$$\mathbb{E}_{\mathbb{P}^{\mathrm{univ}}_\gamma}[\mathcal{O}] \;=\; \sum_{k:p_0(k)>0} p_0(k)\, \mathbb{E}_{\mathbb{P}^{[g_k]}_\gamma|K=k}[\mathcal{O}]$$

by (24) and the tower property, which is the definition of a classical mixture with weights $p_0(k)$. (iv) is Cor. 5.4 applied to the stress-energy observable. □

Remark 11.40 (Records precede gravity in the Page–Geilker regime). Clause (ii) uses the following physical regularity, which we state explicitly as the *decoherence-first condition*: whenever two branch stress tensors differ sufficiently for any gravitational apparatus to resolve $\widetilde{g}_j$ from $\widetilde{g}_k$ within its response time, the interactions that make that difference macroscopic (including the apparatus's own gravitational coupling to the source mass) deposit branch-distinguishing records into the environment on a much shorter timescale, so $H(K \mid \overline{\mathcal{F}}^\gamma_\tau) \ll 1$ before the metric difference is resolvable. This is the standard amplification statement of decoherence theory (redundant records, [377, 172, 146]); in the gravitational case it is self-enforcing, since a superposition of mass configurations probed gravitationally decoheres through that very coupling [119, 133]. The condition is an empirical regularity of the decoherence-saturated regime, not a theorem of the framework; the regime where it could fail — macroscopically distinct sources held coherent — is precisely the mesoscopic BMV boundary already quarantined in Rem. 11.48, where the framework's metric assignment is governed by the prior source and its entanglement predictions are carried entirely by the quantum sector (Prop. 13.5; the adapted classical metric is a measurement-plus-feedback channel and cannot mediate entanglement [133]).

The decoherence-first condition also fixes *which basis gravitates*: this is the gravitational pointer-basis principle made operational by the record-permanence clause (RP) of Def. 10.1. Records eligible for gravitational conditioning are the terminally readable — redundantly broadcast, in-practice-irreversible — pointer outcomes, and it is the decoherence-first regularity that selects this class: the same environmental amplification that makes an outcome gravitationally resolvable makes it redundant, hence permanent. The "Wigner's friend with a gravimeter" fork — an outcome that gravitates and is later coherently reversed — is excluded by (RP): either the outcome is irreversibly broadcast (then it gravitates, and coherent reversal is thermodynamically excluded at the stated redundancy), or it remains coherently reversible (then it never entered the gravitational filtration and no metric which-path memory is created, so eraser revivals survive at full visibility). What gravitates is what has decohered irreversibly, and the selection is made by the admissible instruments of $\mathcal{P}_\gamma$ — the same decoherence structure that underwrites Ax. 7.1.

Remark 11.41 (Inflationary consistency is inherited, not re-derived). During inflation no classical records yet exist, so the trivial-protocol clause of (23)(b) applies: the classical metric on the whole pre-record region is the prior-sourced FLRW background g_0, and the inflaton/metric perturbations evolve entirely in the quantum sector on that background — which is exactly the standard QFT-on-FLRW mode calculation. The record-adapted layer first acts when decohered records form (horizon exit and subsequent classicalization), whereupon conditioning proceeds through the same posterior machinery as in the laboratory case. CMB phenomenology is therefore *inherited* from the standard calculation rather than re-derived, and the framework introduces no new free function into the primordial spectra beyond the $\widehat{\Sigma}_\delta$ correction already quantified in Sec. 13.

Remark 11.42 (PPN consistency for decohered matter). For fully decohered matter — the solar-system regime, where the branch-distinguishing records long precede any metric observation and $\pi^\gamma_\tau(K) = 1$ to within the residual of Prop. 10.10 — the record-adapted equation (22) reduces per realization to per-branch classical GR sourced by the realized configuration. The framework therefore inherits the parametrized post-Newtonian sector of GR unchanged ($\gamma = \beta = 1$), in agreement with the Cassini bound $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [28] and with lunar-laser-ranging and pulsar-timing constraints; no record-adapted correction survives at any post-Newtonian order once the source is decohered.

Worked example with the record step explicit. The bare two-level model of previous drafts is degenerate under the cosmological terminal POVM ($p_0(\text{undecayed at } \tau_f) = e^{-\Gamma\tau_f} \approx 0$) and, having no intermediate records, cannot say *when* the balance switches. Refine it minimally: let the monitoring epochs be $s_1 < s_2 < \cdots < s_N$ and $\mathcal{H} = \operatorname{span}\{|0\rangle, |1_1\rangle, \ldots, |1_N\rangle\}$, where $|0\rangle$ is the undecayed atom and $|1_m\rangle$ the decay products tagged by the (environment-recorded) decay epoch m; seed $\rho_{\mathrm{i}} = |0\rangle\langle 0|$. In the decoherent, epoch-atomic idealization the forward state is diagonal,

$$\rho(\tau) = e^{-\Gamma s_{n(\tau)}}|0\rangle\langle 0| + \sum_{m:\, s_m \le \tau} \big(e^{-\Gamma s_{m-1}} - e^{-\Gamma s_m}\big)|1_m\rangle\langle 1_m|, \qquad n(\tau) = \max\{m : s_m \le \tau\},\ s_0 := 0,$$

with the $|0\rangle$ weight frozen between epochs: in the epoch-atomic idealization the decay is attributed at the monitoring epochs, so the sum telescopes and $\operatorname{Tr}\rho(\tau) = 1$ for *all* τ (a previous draft wrote the continuously decaying weight $e^{-\Gamma\tau}$, which is subnormalized strictly between epochs), the terminal PVM is $E_\infty = |0\rangle\langle 0|$, $E_m = |1_m\rangle\langle 1_m|$ (so $K \in \{\infty, 1, \ldots, N\}$ is the decay epoch), and Def. 3.1 gives, for every τ,

$$\rho_2(\tau \mid m) = \begin{cases} |0\rangle\langle 0|, & \tau < s_m, \\ |1_m\rangle\langle 1_m|, & \tau \ge s_m, \end{cases} \qquad \rho_2(\tau \mid \infty) = |0\rangle\langle 0|.$$

Here $\rho(\tau)$ and every $\Lambda_k(\tau)$ are jointly diagonal, so $[\rho, \Lambda_k] = 0$ and by Thm. 3.10(iii) the per-branch Margenau–Hill operator coincides with the conditioned state, $M_k = \rho_2(\tau \mid k) \succeq 0$, with no temporal-quasiprobability negativity ($f_k = 0$): the Page–Geilker setup is decoherence-saturated precisely in the commuting (classical-record) regime where the witness of Thm. 3.10 is non-negative. The record step: at each s_m the protocol reads the pointer basis $\{|0\rangle, |1_1\rangle, \ldots, |1_m\rangle\}$ projectively (QND; physically performed by the environment). On a realization with $K = m_*$ the records are $j_m = 0$ for $m < m_*$ and $j_m = m_*$ for $m \ge m_*$, and the posterior of Def. 5.2 is exactly the classical hazard posterior: for $\tau < s_{m_*}$, $\pi_\tau(\infty) = e^{-\Gamma(\tau_f - s_{n(\tau)})}$, $\pi_\tau(m) = \big(e^{-\Gamma s_{m-1}} - e^{-\Gamma s_m}\big)e^{\Gamma s_{n(\tau)}}$ for $m > n(\tau)$ and 0 otherwise; for $\tau \ge s_{m_*}$, $\pi_\tau(m_*) = 1$. (N is finite for display only; under the standing protocol convention $s_m \uparrow \tau_f$ of Sec. 5 the unmonitored terminal window (s_N, τ_f) disappears and the posterior normalizes exactly, $\sum_k \pi_\tau(k) = 1$.) Substituting into (18), every live branch at $\tau < s_{m_*}$ has $\rho_2(\tau \mid k) = |0\rangle\langle 0|$, so

$$\rho^\gamma_{\mathrm{RC}}(\tau) = \begin{cases} |0\rangle\langle 0|, & \tau < s_{m_*} \quad \text{(recorded: no decay yet)}, \\ |1_{m_*}\rangle\langle 1_{m_*}|, & \tau \ge s_{m_*} \quad \text{(recorded: decayed at epoch } m_*\text{)}, \end{cases}$$

exactly, with zero residual: the adapted source is the definite mass configuration tracking the realized record history. Writing $\langle\widehat{T}_{\mu\nu}\rangle_{|0\rangle\langle 0|} = T^{(0)}_{\mu\nu}$ (undecayed atom) and $\langle\widehat{T}_{\mu\nu}\rangle_{|1_m\rangle\langle 1_m|} = T^{(1)}_{\mu\nu}$ (decay products):

- **Naive forward semiclassical gravity.** The source at τ is $\langle\widehat{T}_{\mu\nu}\rangle_{\rho(\tau)} = e^{-\Gamma s_{n(\tau)}}T^{(0)}_{\mu\nu} + (1 - e^{-\Gamma s_{n(\tau)}})T^{(1)}_{\mu\nu}$, a weighted sum over both configurations. The Cavendish balance in this theory experiences a metric interpolating between the two configurations on every realization, even after the decay record is read out. This is the Page–Geilker prediction that contradicts experiment.
- **Record-adapted semiclassical gravity.** On the realization $K = m_*$ the source is $T^{(0)}_{\mu\nu}$ for $\tau < s_{m_*}$ and $T^{(1)}_{\mu\nu}$ for $\tau \geq s_{m_*}$: the Cavendish balance experiences the undecayed-source metric until the recorded decay and the decayed-source metric thereafter, the transition being carried by the Barrabès–Israel null shell on the future light cone $\mathcal{N}_{m_*} = \partial J^+(\gamma(s_{m_*}))$ of the recorded decay event (an artifact of the epoch-atomic idealization; the microscopic decay dynamics smooths it), the balance switching after the light-crossing time from the record event — negligible in the laboratory. Over an ensemble of Page–Geilker trials the switching times are classically distributed with the hazard law, matching experiment. Ax. 7.1 holds (the epoch records determine K), so the terminal shell vanishes and g_ω continues smoothly across Σ_{τ_f} into the per-branch region.

For imperfect (POVM) monitoring with error probability ϵ per epoch, $1 - \pi_\tau(K) = O(\epsilon)$ after the first post-decay record and the metric residual is $O(\kappa TM \left\|\mathcal{G}_{g_0}\right\|_{\mathrm{op}} \epsilon)$ by Prop. 10.10 — far below any Cavendish sensitivity.

The distinguishing prediction is operational and concrete: the record-adapted framework gives per-realization responses that switch at the recorded decay time (classically distributed over the ensemble), while naive forward semiclassical gravity predicts an interpolated, non-switching response per realization. Prop. 11.39 therefore supplies a genuine physical advantage of the retrocausal framework over the forward theory in a regime where the forward theory fails empirically. This is the sharpest empirical differentiator the framework has from ordinary forward semiclassical gravity at present; modern optomechanical searches for the Schrödinger–Newton self-gravity deviation continue to return null results [394].

Correspondence limit: the classical two-body force. The consistency checks above are dynamical (*when* does the Cavendish balance respond, and *to which* source). We close the subsection with the complementary *static* correspondence check the framework has so far only asserted abstractly: that the record-adapted equation (19) reproduces, in its weak-field static limit, the Newtonian two-body force $F = Gm_1m_2/r^2$. This is the concrete counterpart of the PPN statement of Rem. 11.42: there we noted that fully decohered matter inherits the post-Newtonian sector of GR unchanged; here we exhibit the leading (Newtonian) term explicitly from the framework's own sourcing, tracking where the record/branch structure does load-bearing work and where it is inert. Throughout we take G as a *measured input*; nothing below derives G (the coupling-existence burden remains Conj. 10.2). What is derived is the two-body force at fixed G.

Proposition 11.43 (Newtonian two-body limit of the record-adapted Einstein equation)**.** *Work on $\mathcal{M}^{\mathrm{ext}}$ in the post-terminal region $J^+(\Sigma_{\tau_f})$, in mostly-plus signature* $(-,+,+,+)$, *with* $\Lambda = 0$ *at the Newtonian order under consideration. Consider two massive bodies of rest masses m_1, m_2, mutually at rest, with worldlines γ_1, γ_2 crossing a common static slice at spatial positions* $\mathbf{x}_1, \mathbf{x}_2$, $r := |\mathbf{x}_2 - \mathbf{x}_1|$. *Assume:*

(a) (Decohered configuration.) *Each body carries an (RP)-admissible terminally readable position record (Def. 10.1); on the realization ω the joint terminal label is* $K(\omega) = k = (\mathbf{x}_1, \mathbf{x}_2)$, *a definite two-body configuration, with* $\pi^\gamma_{\tau_f}(k) = \mathbf{1}_{\{K=k\}}$ *under Ax. 7.1 (Thm. 7.3).*

(b) (Weak, static, non-relativistic.) $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ *with* $|h_{\mu\nu}| \ll 1$, $\partial_t h_{\mu\nu} = 0$, *and the source rest-frame velocities are* $\ll c$, *so the branch stress tensor is rest-mass dominated,*

$$T^{\mathrm{RC}}_{00}[g](\mathbf{x};\omega)\big|_{K=k} \;=\; \rho(\mathbf{x})\,c^2, \qquad \rho(\mathbf{x}) = m_1\,\delta^3(\mathbf{x}-\mathbf{x}_1) + m_2\,\delta^3(\mathbf{x}-\mathbf{x}_2), \tag{51}$$

with $T^{\mathrm{RC}}_{0i} = O(v/c)\,\rho c^2$ *and* $T^{\mathrm{RC}}_{ij} = O(v^2/c^2)\,\rho c^2$ *negligible at Newtonian order.*

Then, on $J^+(\Sigma_{\tau_f})$ *the realized branch metric* $g_\omega = \widetilde{g}_k$ *solves* (19) *in the bulk form* (22), *and:*

(i) (Poisson reduction.) *Writing* $\Phi := -\frac{1}{2}c^2 h_{00}$ *(so* $g_{00} = -(1+2\Phi/c^2)$*), the* 00*-component of* (22) *reduces to the Poisson equation*

$$\nabla^2\Phi(\mathbf{x}) \;=\; 4\pi G\,\rho(\mathbf{x}) \;=\; 4\pi G\big(m_1\,\delta^3(\mathbf{x}-\mathbf{x}_1) + m_2\,\delta^3(\mathbf{x}-\mathbf{x}_2)\big). \tag{52}$$

(ii) (Potential.) *The decaying solution of* (52) *is*

$$\Phi(\mathbf{x}) \;=\; -\,G\left(\frac{m_1}{|\mathbf{x}-\mathbf{x}_1|} + \frac{m_2}{|\mathbf{x}-\mathbf{x}_2|}\right) \;=:\; \Phi_1(\mathbf{x}) + \Phi_2(\mathbf{x}). \tag{53}$$

(iii) (Force.) *A test body of mass* m_2 *following a timelike geodesic of* $\widetilde{g}_k$ *in the field* Φ_1 *of body* 1 *obeys, in the non-relativistic limit,* $\ddot{\mathbf{x}} = -\nabla\Phi_1$, *so the force on body* 2 *is*

$$\mathbf{F}_2 \;=\; -\,m_2\,\nabla\Phi_1(\mathbf{x}_2) \;=\; -\,G\,m_1 m_2\,\frac{\mathbf{x}_2-\mathbf{x}_1}{|\mathbf{x}_2-\mathbf{x}_1|^3}, \qquad |\mathbf{F}_2| \;=\; \frac{G\,m_1 m_2}{r^2}, \tag{54}$$

equal and opposite to $\mathbf{F}_1 = -m_1\nabla\Phi_2(\mathbf{x}_1)$. *Newton's law of universal gravitation is recovered.*

Proof. Why the source is the definite configuration (51), *not a superposition.* On $J^+(\Sigma_{\tau_f})$ the adapted source of Def. 10.1 reduces to the per-branch source $T^{\mathrm{RC}}_{\mu\nu} = \mathrm{Tr}(\rho_2[g](\tau \mid K)\,\widehat{T}_{\mathrm{S},\mu\nu}[g])$, because the posterior weight is the point mass $\pi^{\gamma}_{\tau_f}(k) = \mathbf{1}_{\{K=k\}}$ (Ax. 7.1, Thm. 7.3; the terminal shell (21) vanishes a.s.). The operator sourcing gravity is fixed by the branch postulate (B) of Sec. 10 (Def. 3.1, Rem. 3.2): the symmetric branch ρ_2 is the unique self-adjoint, real-trace, covariantly conserved conditioned density operator, and on the decohered configuration k it is the definite two-body state $|\mathbf{x}_1,\mathbf{x}_2\rangle\langle\mathbf{x}_1,\mathbf{x}_2|$. Hence the gravitating stress-energy is that of the *realized* configuration, not of a coherent superposition of configurations. This is the framework's structural resolution of the Page–Geilker problem (Prop. 11.39, Rem. 11.40): what gravitates is what has decohered irreversibly, so for two macroscopic bodies whose positions are continuously monitored by their environment ($H(K \mid \overline{\mathcal{F}}^{\gamma}_{\tau}) \ll 1$, decoherence-first) the source is (51) and the naïve $\sum_k p_0(k)\langle\widehat{T}_{\mu\nu}\rangle_k$ average is never the per-realization source. In particular the notorious ambiguity of semiclassical gravity—does the mean field or the collapsed field gravitate?—does not arise here: the record structure selects the collapsed field per branch, and the mean field re-emerges only as the ensemble expectation (Cor. 5.4, Prop. 11.39(iv)), which is not a physical metric. For decohered rest matter with $v \ll c$ the branch stress tensor is dominated by the 00 rest-mass component, giving (51): with the normalized static four-velocity $u^\mu = (c,\mathbf{0}) + O(h)$, $u_0 = -c + O(h)$, the dust form $\widehat{T}_{\mu\nu} = \rho\,u_\mu u_\nu$ yields $T^{\mathrm{RC}}_{00} = \rho c^2$ and the spatial components down by $O(v^2/c^2)$.

Step 1: linearised field equation. With $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, $|h| \ll 1$, the linearised Einstein tensor in the trace-reversed potential $\bar{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$ ($h := \eta^{\alpha\beta}h_{\alpha\beta}$) is, in Lorenz gauge $\partial^\mu\bar{h}_{\mu\nu} = 0$,

$$G_{\mu\nu}[g] \;=\; -\tfrac{1}{2}\,\Box\,\bar{h}_{\mu\nu} + O(h^2), \qquad \Box = \eta^{\alpha\beta}\partial_\alpha\partial_\beta = -c^{-2}\partial_t^2 + \nabla^2.$$

Static fields ($\partial_t = 0$) give $\Box \to \nabla^2$, so the bulk equation (22) (which holds off the record cones and on $J^+(\Sigma_{\tau_f})$ verbatim, by the bulk-restriction clause of Conj. 10.2) becomes $-\frac{1}{2}\nabla^2\bar{h}_{\mu\nu} = \kappa\,T^{\mathrm{RC}}_{\mu\nu}$ with $\kappa = 8\pi G/c^4$.

Step 2: extract h_{00}. At Newtonian order only $T^{\rm RC}_{00} = \rho c^2$ is non-negligible, so $\nabla^2 \bar h_{\mu\nu} = 0$ for $(\mu\nu) \neq (00)$ and (with decaying boundary conditions) $\bar h_{ij} = \bar h_{0i} = 0$, whence $\bar h_{00}$ is the only nonzero trace-reversed component. Then $h = \eta^{\mu\nu} h_{\mu\nu}$ and $\bar h = -h$ give $h_{00} = \bar h_{00} - \frac12 \eta_{00} \bar h = \bar h_{00} - \frac12 \bar h_{00} = \frac12 \bar h_{00}$ (using $\bar h = \eta^{00} \bar h_{00} = -\bar h_{00}$ and $\eta_{00} = -1$). The 00 equation $-\frac12 \nabla^2 \bar h_{00} = \kappa \rho c^2$ therefore reads

$$\nabla^2 h_{00} \;=\; \tfrac12 \nabla^2 \bar h_{00} \;=\; -\kappa\, \rho c^2 \;=\; -\frac{8\pi G}{c^4}\, \rho c^2 \;=\; -\frac{8\pi G}{c^2}\, \rho.$$

Step 3: identify the potential. The geodesic equation for a slow particle, $\ddot x^i = -\Gamma^i_{00} c^2 + O(v/c)$ with $\Gamma^i_{00} = -\frac12 \partial_i h_{00} + O(h^2)$ (static), gives $\ddot{\mathbf{x}} = \frac12 c^2 \nabla h_{00}$. Matching to Newtonian $\ddot{\mathbf{x}} = -\nabla\Phi$ fixes $\Phi = -\frac12 c^2 h_{00}$, equivalently $g_{00} = -(1 + 2\Phi/c^2)$. Substituting $h_{00} = -2\Phi/c^2$ into the Step-2 equation:

$$\nabla^2\Big(-\frac{2\Phi}{c^2}\Big) = -\frac{8\pi G}{c^2}\rho \implies \nabla^2\Phi = 4\pi G\, \rho,$$

which is (52). This is (i).

Step 4: solve and read off the force. The Green's function of ∇^2 with $\nabla^2(-1/4\pi|\mathbf{x}|) = \delta^3(\mathbf{x})$ gives, by linearity of (52) in ρ and the superposition of the two delta sources, (53). A test mass m_2 at $\mathbf{x}_2$ in the field Φ_1 obeys $m_2 \ddot{\mathbf{x}} = -m_2 \nabla \Phi_1$; since $\nabla_{\mathbf{x}}\big(-Gm_1/|\mathbf{x}-\mathbf{x}_1|\big) = Gm_1(\mathbf{x}-\mathbf{x}_1)/|\mathbf{x}-\mathbf{x}_1|^3$, evaluating at $\mathbf{x} = \mathbf{x}_2$ gives (54), of magnitude Gm_1m_2/r^2 and directed from body 2 toward body 1. By the symmetry $\mathbf{x}_1 \leftrightarrow \mathbf{x}_2$, $\mathbf{F}_1 = -\mathbf{F}_2$ (Newton's third law). This proves (ii)–(iii). $\square$

Remark 11.44 (What is framework-specific, and what is inert, at Newtonian order). Prop. 11.43 is textbook GR arithmetic once the source (51) is granted; the framework's content is entirely in *why* that source, and *where* it is supported. Four points.

(a) Record cones and support. Each body's records lie on its worldline γ_a, and by the causal adaptedness restriction (23) the metric deviation from g_0 is confined to $\bigcup_a J^+(\gamma_a) \cup J^+(\Sigma_{\tau_f})$: body 2 feels body 1's field only inside $J^+(\gamma_1)$, i.e. after light-crossing from body 1's record events (Prop. 10.7). In a static, long-established two-body configuration both bodies have been mutually inside each other's record cones for all relevant retarded times, so the retardation is invisible and Φ is the instantaneous (53); the framework nonetheless *predicts* the Newtonian potential is a retarded, causal quantity (its would-be action-at-a-distance is a cone-confined artifact of the static limit), consistent with Prop. 10.7(iv). This differs from textbook Newtonian gravity—which is instantaneous by fiat—but the difference is zero at Newtonian static order and first appears in the (standard GR) retardation/radiation sector, outside the present scope.

(b) Modified-TDSE correction to the two-body dynamics. The center-of-mass quantum evolution of each body is governed by the modified TDSE of Prop. 10.14, whose Hermitian low-energy reduction (Prop. 10.27) is $\widehat H_{\rm eff} = \widehat H_{\rm s} - \frac{\mu}{2} \widehat H_{\rm s}^2$ with $\mu = \alpha\, \delta t^2/\hbar^2$. Taking the two-body Hamiltonian $\widehat H_{\rm s} = \widehat p_1^2/2m_1 + \widehat p_2^2/2m_2 - Gm_1m_2/|\widehat{\mathbf{x}}_1 - \widehat{\mathbf{x}}_2|$ (the Newtonian potential of (ii) promoted to the relative coordinate), the framework predicts a fractional correction to the two-body dynamics of relative size $\mu\langle \widehat H_{\rm s}\rangle/2$. At the Planck-natural point $\mu = 1/E_{\rm P}$ (Rem. 10.30) this is $\langle \widehat H_{\rm s}\rangle/2E_{\rm P} \sim 10^{-28}$ for *per-quantum* laboratory energy scales ($\langle \widehat H_{\rm s}\rangle \sim 10\,\mathrm{eV}$) — utterly negligible for the force law, and consistent with the PPN bounds of Rem. 11.42. (For a macroscopic composite the naive centre-of-mass ratio is *not* small — $\langle \widehat H_{\rm s}\rangle/2E_{\rm P} \sim 10^{24}$ at solar-system orbital energies — so the per-quantum estimate does not extend to composite bodies as written; the composite-body ("soccer-ball") scoping of the $\widehat H_{\rm s}^2$ correction is recorded here as an explicit caveat, not resolved. The force-law conclusions below do not depend on it.) The correction is unitary (Hermitian) and does not modify the static potential (53); it perturbs only the *dynamical* propagation of a coherent non-stationary two-body state, which for a terminally decohered pair is the per-branch trivial case of (c).

(c) Two-boundary conditioning and no-signalling. Each body gravitates *per branch*: on the realization ω the source is the definite $k = (\mathbf{x}_1, \mathbf{x}_2)$. The two-boundary (future-conditioned) structure enters only through the past-cone measurable weights $\pi^\gamma_{t_\gamma(x)}(k)$ (Def. 10.1), and the

sum over branches restores the forward source, $\mathbb{E}_{\mathbb{P}_\gamma}[T^{\mathrm{RC}}_{\mu\nu}] = \langle \widehat{T}_{\mu\nu}\rangle_\rho$ (Cor. 5.4), which is the geometric no-signalling identity of Prop. 10.7: a Cavendish measurement of the two-body field learns exactly the already-realized positions and nothing about any terminal post-selection. For a genuinely decohered two-body system this is trivially satisfied ($\pi^\gamma_\tau(k) = \mathbf{1}_{\{K=k\}}$), which is why the two-body force is unambiguous and branch-independent in magnitude.

(d) Where the sourcing differs from textbook GR, and whether it matters. The record-adapted source (51) coincides pointwise with the classical dust source of GR on the decohered configuration; the shell terms of (19) vanish (no posterior jumps: the positions are already recorded and static, so $\nu^{(\omega,m)}_{ab} = 0$ and $S^{(\omega,f)}_{ab} = 0$), and the bulk equation (22) is the ordinary linearised Einstein equation. Hence at Newtonian order there is *no* deviation from textbook GR; the framework's distributional and two-boundary apparatus is entirely inert here, activating only when the source carries live (un-collapsed) posterior weight—the Page–Geilker/BMV mesoscopic regime (Rem. 11.48), not the classical two-body problem. This is the precise sense in which the framework passes the Newtonian correspondence test.

Remark 11.45 (Honest scope of Prop. 11.43). This is the *Newtonian correspondence limit*: weak field ($|h| \ll 1$), static ($\partial_t = 0$), non-relativistic ($v \ll c$), and with G taken as an empirical input. We do *not* claim: (i) a derivation of G—the existence of the gravitational coupling is Conj. 10.2, assumed here, and its numerical value is measured; (ii) higher post-Newtonian orders—the $O(v^2/c^2)$ corrections, though inherited unchanged from GR by Rem. 11.42 ($\gamma = \beta = 1$), are not computed here; (iii) the radiative sector—gravitational-wave emission from a time-varying two-body quadrupole lies outside the static limit and is moreover subject to the quadrupole-free source-class residual (vi-b) of Conj. 10.2 for the record shells, a genuinely open corner. The claim is exactly and only that the framework's record-adapted semiclassical Einstein equation reduces, in the stated limit, to the Poisson equation and yields $F = Gm_1m_2/r^2$. The conditional converse — that closing the linearised E0 tensor residual for SM matter would make this limit *derived* rather than postulate-based — is recorded as Cor. J.5.

Remark 11.46 (Relation to prior work: Kent and Halliwell). The idea that a final-boundary selection rescues semiclassical gravity from the Page–Geilker pathology is not original here. Kent [65, 67, 66, 68] has developed, since 2007, a program of final-boundary-selected semiclassical gravity in which a late-time mass-density record defines the preferred beable and a single global post-selection yields Born-weighted outcomes. Halliwell's decoherent-histories program [69, 70, 71] derives per-decoherent-branch effective classical equations, including semiclassical Einstein per branch in the minisuperspace/WKB regime. Both programs predate and anticipate the per-branch classical-metric picture used here.

What is distinctive in the present paper, relative to these precedents: *(a)* the branch-indexed source is the symmetric ABL two-time operator $\rho_2(\tau \mid k) = \rho^{1/2}\Lambda_k\rho^{1/2}/p(k)$ of Def. 3.1, not a forward decohered-branch state (Halliwell) nor a final global mass-density record (Kent)—it is a *both-boundaries* conditioning with a specific operator form selected by the branch postulate (B) of Conj. 10.2; *(b)* the ensemble-level probability structure is the rigorous finite-dim Part I probability space $(\Omega_\gamma, \mathbb{P}_\gamma)$ with the posterior-martingale and collapse theorems of Sec. 7, not a weak-value or decoherence-functional construction; and *(c)* the framework integrates with the modular-theoretic postulates Pos. 11.11 and Pos. 11.18 and the Planck-cutoff postulate Pos. 11.30, producing a single coherent conditional-unification structure rather than a measurement-only proposal. Kent's program does not branch-index the Einstein equation (uses one global post-selection); Halliwell's program does branch-index, but only via forward decoherence consistency without a terminal POVM. The synthesis is ours; the per-branch classical-metric idea is theirs and must be attributed.

Remark 11.47 (Relation to measurement-feedback semiclassical gravity). Record-conditioned sourcing of a classical gravitational field is an established programme: Tilloy–Diósi source Newtonian gravity from the realized record of a continuous-collapse process [130, 131], Kafri–Taylor–Milburn model the Newtonian interaction as a classical measurement-plus-feedback channel

[133], the causal-conditional prescription of Helou et al. and Liu et al. [134, 135] filters the Schrödinger–Newton source on the records in the *past light cone* — the Newtonian precursor of the null-cone conditioning now adopted in Def. 10.1 — and Oppenheim's classical–quantum gravity realises the metric as a stochastic classical field whose conditioned trajectories correlate with decohered matter configurations [129, 128]. In all of these, causal safety rests on two properties: the source is *adapted* (a functional of the past record only), and the record-average of the conditioned dynamics is *linear*, which blocks Gisin-type signalling; adapted *non-Markovian* feedback remains safe, losing only autonomy of the unconditioned equation [132], whereas *future-conditioned* (post-selected) sourcing makes the effective pre-selection dynamics nonlinear and signals, as demonstrated for final-state projection [137, 138]. Def. 10.1 sits exactly on the safe side of this classification, with two distinctive features. First, the source is not a filtered forward state but the two-boundary posterior mixture $\sum_k \pi_\tau^\gamma(k)\rho_2(\tau \mid k)$: the branch states are ABL two-time objects, the weights are past-adapted, and the prior mean collapses the two-time structure to the forward state (Cor. 5.4) — the linearity half of the criterion holds exactly and non-Markovianly. Second, the decoherence cost that the feedback literature identifies as mandatory [133, 128] is here supplied by the record instruments of $\mathcal{P}_\gamma$ themselves: the metric is deterministic given the record and stochastic unconditionally, so the decoherence–diffusion trade-off is met in the manner of a feedback theory rather than evaded, resolving the concern raised for deterministic-classical-metric hybrids. What none of the cited constructions contains is the terminal boundary: the TBRME source interpolates, along the record filtration, from the prior state to the terminal-conditioned branch state, with the interpolation error controlled by conditional entropy (Prop. 10.10, Thm. A.40).

Remark 11.48 (Scope of the Page–Geilker check). *Does:* show that Conj. 10.2, under the framework's classical-record structure, reproduces the empirically observed classical statistics of the Page–Geilker setup and is therefore consistent with the experimental result that falsifies naive forward semiclassical gravity. A genuine distinguishing prediction relative to the forward theory. *Does not:* verify the framework in regimes where macroscopic-metric interference (Version B of Rem. 11.38) might become observable. The Page–Geilker setup is a decoherence-saturated regime where the classical-record structure suffices; tests at the Bose–Marletto–Vedral mesoscopic scale [121, 124] probe the boundary with Version B and would constrain the framework's stance further. Christodoulou et al. [118] identify retarded-entanglement signatures as specifically indicative of Version B; Danielson–Satishchandran–Wald [119] show that any experiment near a horizon suffers unavoidable soft-graviton decoherence, which places fundamental bounds on the BMV program. Trillo and Navascués [127] caution that the Diósi–Penrose classical-gravity collapse model also predicts BMV-style gravitationally induced entanglement below a length scale, and more recently Tibau Vidal et al. [139] exhibit non-locally-tomographic mediators that generate BMV entanglement with no observable spacetime superposition at all; so a positive BMV signal does not by itself cleanly discriminate Version B from a decoherence-enhanced Version A. The complete discrimination requires combining entanglement signals with decoherence-rate and stochasticity bounds—the framework's distinguishing handle being the *coherent (non-diffusive)* generation of the entanglement (Prop. 13.5). The framework's stance is agnostic between these empirical possibilities, and its scope restriction to Pos. 11.30 is precisely the scope on which the agnosticism is internally consistent.

11.5.2 Explicit contraction bound for Conj. 11.4

Conj. 11.4 asserts the existence of a constant $C = C(g_0, \kappa, T, p_*, \eta, M, L_U, L_T) > 0$ such that $\mathcal{S}(\rho_\mathrm{i}, \{E_k\}) < C$ implies the Banach-fixed-point contraction (vii) of Conj. 10.2. We now exhibit C in closed form in the finite-dim toy model under Pos. 11.30, converting the existential conjecture to a constructive bound.

Proposition 11.49 (Explicit entropy-smallness bound). *Assume the standing scope (S1)–(S8) of Conj. 11.4 on a finite-dimensional matter Hilbert space under Pos. 11.30. Set* $\mu := \kappa T \|\mathcal{G}_{g_0}\|_{\mathrm{op}}$ *(dimensionless gravitational-coupling scale) and assume the seed-background contraction (S8):* $\mu L_T^{(0)} < 1$*. Then the contraction constant* C *of Conj. 11.4 satisfies the closed-form lower bound*

$$C \;\geq\; \frac{2\,(1-\mu L_T^{(0)})^2\, p_*^2\,\eta}{\mu^2\, M^2} \;=\; \frac{2\,(1-\kappa T L_T^{(0)}\,\|\mathcal{G}_{g_0}\|_{\mathrm{op}})^2\, p_*^2\,\eta}{(\kappa T M\,\|\mathcal{G}_{g_0}\|_{\mathrm{op}})^2}. \tag{55}$$

In particular $C > 0$*, so there exists a non-trivial regime* $\mathcal{S} < C$ *in which the fixed-point iteration contracts and Conj. 10.2's conclusions hold.*

Proof. The contraction estimate (vii) of Conj. 10.2 requires $\mu L_T < 1$, with the entropy-controlled operator-Lipschitz constant L_T of the source map. Pinsker's inequality in the full trace norm ($S \geq \frac{1}{2}\|\cdot\|_1^2$, see the normalisation convention after Conj. 11.4) gives $\|\rho_2(\tau_i \mid k) - \rho_{\mathrm{i}}\|_1 \leq \sqrt{2\,\mathcal{S}}$ uniformly in k with $p_0(k) \geq p_*$. On the gapped subspace where $\rho_{\mathrm{i}} \succeq \eta I$, the operator square root $\rho \mapsto \rho^{1/2}$ is Lipschitz with constant $(2\sqrt{\eta})^{-1}$ (Powers–Størmer-type bound [64] via functional calculus). Combined with (S4)'s bound $\sup_k \left\|\widehat{T}_{\mu\nu}[g_0]\right\|_{\rho_{\mathrm{i}},\mathrm{op}} \leq M$,

$$L_T \;\leq\; L_T^{(0)} \;+\; \frac{M\sqrt{2\,\mathcal{S}}}{2\sqrt{\eta}\,p_*} \;=\; L_T^{(0)} \;+\; \frac{M\sqrt{\mathcal{S}}}{\sqrt{2\eta}\,p_*}.$$

The contraction condition becomes

$$\mu L_T^{(0)} \;+\; \mu\,\frac{M\sqrt{\mathcal{S}}}{\sqrt{2\eta}\,p_*} \;<\; 1,$$

i.e.

$$\sqrt{\mathcal{S}} \;<\; \frac{\sqrt{2\eta}\,p_*\,(1-\mu L_T^{(0)})}{\mu M}.$$

Squaring both sides gives $\mathcal{S} < 2\eta p_*^2(1-\mu L_T^{(0)})^2/(\mu M)^2$, which is (55). Positivity of C follows from (S8): $\mu L_T^{(0)} < 1$ gives $(1-\mu L_T^{(0)})^2 > 0$, while $p_*, \eta, \mu, M > 0$ by (S2)–(S4) and (S1). □

Worked numerical example. Take the Page–Geilker toy of the previous subsection with $d = \dim \mathcal{H} = 2$, and for illustration set:

- $\eta = 1/4$ (spectral gap of ρ_{i}, consistent with e.g. $\rho_{\mathrm{i}} = \mathrm{diag}(3/4, 1/4)$);
- $p_* = 1/4$ (minimum terminal probability, consistent with a mildly-asymmetric $\{E_k\}$);
- $M = 1$ (stress-energy operator norm, in units where $\widehat{T}_{\mu\nu}$ is suitably normalized);
- $L_T^{(0)} = 1$ (zero-entropy Lipschitz constant of the source map);
- $\mu = \kappa T \|\mathcal{G}_{g_0}\|_{\mathrm{op}} = 1/2$ (dimensionless gravitational coupling over the slab duration).

Under (S8), $\mu L_T^{(0)} = 1/2 < 1$, so the seed-background contraction condition holds with margin $1 - \mu L_T^{(0)} = 1/2$. Substituting into (55):

$$C \;\geq\; \frac{2\cdot(1/2)^2\cdot(1/4)^2\cdot(1/4)}{(1/2)^2\cdot 1^2} \;=\; \frac{2\cdot(1/4)\cdot(1/16)\cdot(1/4)}{1/4} \;=\; \frac{1/128}{1/4} \;=\; \frac{1}{32}.$$

So in this toy, any two-boundary data satisfying $\mathcal{S}(\rho_{\mathrm{i}}, \{E_k\}) < 1/32$ nats yields a contracting fixed-point system; the universal sample space and the branch-indexed Einstein equation of Conj. 10.2 are rigorously defined, and Conj. 11.4 becomes an unconditional theorem in this regime. For comparison, $\log 2 \approx 0.693$ nats is the entropy of a maximally uninformative binary distribution; $1/32 \approx 0.031$ nats is the entropy of a distribution with probabilities roughly $(0.995, 0.005)$. The contraction therefore holds for any two-boundary data in which the retrodictive conditioning is not too far from the prior — the operationally natural regime where the terminal record only mildly disturbs the initial state.

Remark 11.50 (Scope of the explicit bound). *Does:* provide a closed-form lower bound on the abstract contraction constant C of Conj. 11.4 in the finite-dim toy model of Pos. 11.30, in terms of the standing data $(\kappa, T, \|\mathcal{G}_{g_0}\|_{\mathrm{op}}, p_*, \eta, M, L_T^{(0)})$; convert the conjecture to an unconditional theorem for two-boundary data satisfying $\mathcal{S} < C$ with C as in (55). *Does not:* fix C for general QFT matter sectors on Type III algebras without the finite-dim truncation; extend to regimes where (S8) fails (in which case $1 - \mu L_T^{(0)}$ is negative or zero and the contraction does not close, independently of $\mathcal{S}$); or determine the sharp value of C. The bound (55) is sufficient, not necessary.

Corollary 11.51 (Dimension-agnostic entropy-smallness bound under Pos. 11.30)**.** *Assume Pos. 11.30 with matter Hilbert space of dimension $d := \dim \mathcal{H}_{\mathrm{m}}^{[\le E_\star]} < \infty$ and terminal alphabet $|\mathcal{K}_+| \le d^2$ (the maximum rank of a POVM on $\mathcal{H}_{\mathrm{m}}^{[\le E_\star]}$). Let the uniform bounds be*

$$p_* \;\ge\; p_*^{(d)} \;>\; 0, \qquad \eta \;\ge\; \eta^{(d)} \;>\; 0, \qquad M \;\le\; E_\star, \qquad L_T^{(0)} \;\le\; c_L\, E_\star\, d, \tag{56}$$

where $p_*^{(d)} := \min_{k\in\mathcal{K}_+} p_0(k)$ and $\eta^{(d)} :=$ smallest non-zero eigenvalue of ρ_{i}, both strictly positive by Pos. 11.30, and $c_L > 0$ is the dimensionless Hadamard-stability constant from (S5) (independent of d). Set $\mu := \kappa T \|\mathcal{G}_{g_0}\|_{\mathrm{op}}$ and assume the seed-background contraction $\mu c_L E_\star d < 1$ (i.e. the slab duration T or the coupling window μ is chosen small enough to clear the worst-case $L_T^{(0)}$-scaling with d). Then the contraction constant of Conj. 11.4 satisfies the dimension-explicit lower bound

$$C \;\ge\; \frac{2\,(1-\mu c_L E_\star d)^2\,(p_*^{(d)})^2\,\eta^{(d)}}{(\mu E_\star)^2}, \tag{57}$$

which remains strictly positive for every finite d and every $(T, \|\mathcal{G}_{g_0}\|_{\mathrm{op}})$ satisfying the seed- background condition. Consequently Conj. 11.4 is an unconditional theorem on $\mathcal{H}_{\mathrm{m}}^{[\le E_\star]}$ whenever the two-boundary data satisfy $\mathcal{S} < C$ with C as in (57)*, in every finite dimension $d \ge 2$.*

Proof. The bound (57) is Prop. 11.49's (55) with the dimension-explicit substitutions (56). Positivity follows from $p_*^{(d)}, \eta^{(d)} > 0$ under Pos. 11.30 (the finite-dimensional spectral cutoff gives a strictly positive minimum Born weight and minimum non-zero eigenvalue of the projected ρ_{i}, both by finiteness of the probability simplex and the eigenvalue spectrum) and from the assumed seed-background contraction $\mu c_L E_\star d < 1$, which makes $(1-\mu c_L E_\star d)^2 > 0$. The upper bound $M \le E_\star$ is hypothesis (S4) combined with Pos. 11.30's spectral restriction. The linear scaling $L_T^{(0)} \le c_L E_\star d$ is the worst-case Hadamard-stability constant for the zero-entropy source Lipschitz bound on a d-dimensional matter factor under (S5): the bound on the Jacobian of $g \mapsto \mathrm{Tr}(\rho_{\mathrm{i}} \widehat{T}_{\mathrm{S},\mu\nu}[g])$ over the Sobolev ball scales linearly in $\mathrm{Tr}\,\rho_{\mathrm{i}} = 1$ times the operator-norm bound M times the dimension d of the effective matter factor, giving $c_L E_\star d$ uniformly. $\square$

Worked numerical examples in higher dimensions. $d = 4$ *(cosmological toy with finer terminal alphabet).* Take $\rho_{\mathrm{i}} = \mathrm{diag}(7/16, 5/16, 3/16, 1/16)$, so $\eta^{(4)} = 1/16$; $\{E_k\}$ a uniform rank-one POVM over $|\mathcal{K}_+| = 4$ outcomes with at least one weight $p_0(k) \ge p_*^{(4)} = 1/8$; $c_L = 1$ and $E_\star = 1$ (in suitable units) so $M \le 1$ and $L_T^{(0)} \le 4$. Choose the coupling window $\mu = \kappa T \|\mathcal{G}_{g_0}\|_{\mathrm{op}} = 1/8$, so $\mu L_T^{(0)} \le 1/2 < 1$ and the seed-background contraction holds with margin 1/2. Substituting into (57):

$$C \;\ge\; \frac{2\cdot(1/2)^2\cdot(1/8)^2\cdot(1/16)}{(1/8)^2} \;=\; \frac{2\cdot(1/4)\cdot(1/64)\cdot(1/16)}{1/64} \;=\; \frac{1/2048}{1/64} \;=\; \frac{1}{32}.$$

(The $(1/8)^2$ factors cancel, so the bound is $2\cdot(1/4)\cdot(1/16) = 1/32$; an earlier version misprinted this substitution.) So any two-boundary data on a $d = 4$ matter factor satisfying $\mathcal{S}(\rho_{\mathrm{i}}, \{E_k\}) <$

$1/32$ nats yields a contracting fixed-point system; the regime is non-empty for distributions whose retrodictive conditioning is very close to the prior.

$d = 2^n$ *(qubit cutoff).* For n qubits under the cutoff, $d = 2^n$, worst-case $p_*^{(d)} \geq 1/|\mathcal{K}_+| \geq 1/d^2 = 2^{-2n}$ and $\eta^{(d)} \geq 1/d = 2^{-n}$ (uniform ρ_{i}). With $\mu = 1/(4c_L E_\star d)$ (so the seed-background contraction holds with margin 3/4), (57) gives

$$C \;\geq\; \frac{2 \cdot (3/4)^2 \cdot 2^{-4n} \cdot 2^{-n}}{(1/(4c_L d))^2} \;=\; 18\, c_L^2\, d^2\, 2^{-5n} \;=\; 18\, c_L^2\, 2^{2n} \cdot 2^{-5n} \;=\; 18\, c_L^2 \cdot 2^{-3n},$$

which remains strictly positive but decays as 2^{-3n}, requiring exponentially tighter retrodictive conditioning as the qubit count grows. The scaling is the explicit price paid for the dimension-agnostic result: the finite-dim bound is genuinely d-dependent, not uniform, and the regime $\mathcal{S} < C \sim 2^{-3n}$ is the operationally natural sub- Planckian small-perturbation regime.

Remark 11.52 (What Cor. 11.51 achieves). *Does:* extend Prop. 11.49 from the single worked $d = 2$ example to every finite dimension d under Pos. 11.30, with an explicit dimension-dependent lower bound (57); make Conj. 11.4 an unconditional theorem for all finite-dim matter sectors under the sub-Planckian EFT scope; exhibit how C scales with d (polynomially in $p_*^{(d)}, \eta^{(d)}$, worst-case exponentially in qubit count through $p_*^{(d)} \sim d^{-2}, \eta^{(d)} \sim d^{-1}$). *Does not:* give a d-uniform lower bound (no finite-dim contraction constant is uniform in d without further structural hypotheses, and Cor. 11.51 does not claim one); extend to the genuine QFT / type III regime without the Planck cutoff; or sharpen the Powers–Størmer constant to match the optimal d-dependence of the Pinsker step. The corollary converts Conj. 11.4 from an unconditional theorem for $d = 2$ to an unconditional theorem for every finite d, at the explicit price of an $O(2^{-3n})$ regime shrinkage per qubit added.

Proposition 11.53 (Finite-dimensional non-vacuity of the two-boundary link)**.** *There exists a finite-dimensional two-boundary datum* $(\rho, \{E_k\})$ *with* ρ *a matter state distinct from the maximally mixed reference and* $\{E_k\}$ *a* projective *terminal measurement, such that the per-branch Margenau–Hill negativity is strictly positive,* $f_k > 0$*, on at least one branch, while the no-signalling identity* $\sum_k p(k) M_k = \rho$ *holds.*

Proof. Take $d = 2$, $\rho = \mathrm{diag}(\frac{3}{4}, \frac{1}{4})$, and the sharp terminal PVM $\{P_+, P_-\}$ onto the eigenvectors of σ_x, so $[\rho, P_\pm] \neq 0$. Then $p(\pm) = \frac{1}{2}$ and $M_\pm = \frac{1}{2}\{\rho, P_\pm\}/p(\pm)$ has eigenvalues $\{-0.059,\, 1.059\}$, whence $f_\pm = \|M_\pm\|_1 - 1 = 0.118 > 0$; by Thm. 3.10(iii) (projective effects, all d, no support hypothesis) $f_k > 0 \iff [\rho, \Lambda_k] \neq 0$, which holds here. Averaging, $\sum_k p(k) M_k = \frac{1}{2}\{\rho, P_+ + P_-\} = \rho \succeq 0$ (Thm. 6.1). Hence the two-boundary link is non-vacuous *in phenomenon* as a finite-dimensional theorem, independently of any type III, Cartan, or CHT structure. ☐

Remark 11.54 (What the finite-dim witness does and does not settle). Prop. 11.53 establishes that the operational signature $f_k > 0$ of the two-boundary link is realized by an explicit finite-dimensional projective datum, so the phenomenon the framework predicts is not empty. It uses Thm. 3.10(iii) in its *projective* form, where the clean equivalence $f_k > 0 \iff [\rho, \Lambda_k] \neq 0$ is a theorem; it does *not* rely on the type III/crossed-product machinery, on a Cartan masa, or on the cosmological selection Conj. L.1. It does *not* by itself show that the CHT-selected cosmological terminal POVM instantiates the phenomenon; that is the separate, open content of Conj. L.1 and Rem. L.7.

Conjecture 11.55 (Master conjecture)**.** The five conjectures Conj. 11.1, 11.4, 11.5, 11.8, 11.10, taken together within their respective scopes (the centralizer-rich (dressed) scope of Conj. 11.1, non-vacuous for every faithful normal seed by the Lemma there; the gapped or spectrally-projected matter sector of Conj. 11.4; the Page–Wootters clock regime of Conj. 11.5; the horizon centralizer regime of Conj. 11.8; the cosmological worldline regime of Conj. 11.10), upgrade the framework of this paper from a rigorous finite-dimensional result plus a cluster of gravitational extrapolations into a candidate rigorous formulation of two-boundary quantum gravity on Lorentzian manifolds in the regime where all five scopes overlap (a background state with non-trivial dressed (crossed-product) modular centralizer — automatic by the Lemma of Conj. 11.1 — a gapped matter sub-sector of bounded operator-norm stress-energy, a Page–Wootters-style clock, a horizon admitting a centralizer resolution, and a distinguished cosmological worldline), in which:

- **algebra** is type III crossed-product (Conj. 11.1),
- **dynamics** is semiclassical Einstein in the small retrodictive-entropy regime (Conj. 11.4),
- **constraint** is the uniquely selected modified Wheeler–DeWitt equation (Conj. 11.5),
- **information structure** at horizons is modular- centralizer (Conj. 11.8),
- **time** is the cosmologically emergent clock (Conj. 11.10).

None of these is proved here. The point of the master conjecture is to emphasize that a single coherent programme would resolve all remaining open items of Sec. 11 simultaneously, and to make explicit that the main content of any such programme lies in hard analytic estimates (Conj. 11.4, 11.5) rather than in additional physical postulates.

Non-vacuity of the common scope. We exhibit an explicit candidate witness model in which all five scopes overlap, ruling out the empty-intersection failure mode:

- **Background:** de Sitter spacetime $(\mathcal{M}, g_0)$ in the static patch of a generalised observer, with $\mathcal{N}$ the type III_1 local algebra of the patch and ω the Bunch–Davies (geodesic KMS) state at the de Sitter temperature [18]. Here $\mathcal{N}^{\sigma^\omega} = \mathbb{C}\mathbb{1}$ (the boost flow is weakly mixing [49]; the invariant elements are c-numbers [18]), so $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} = \mathbb{C} \rtimes \mathbb{R} \cong L^\infty(\mathbb{R})$ is diffuse abelian by the Lemma of Conj. 11.1. The unprojected crossed product $\widetilde{\mathcal{N}} = \mathcal{N} \rtimes_{\sigma^\omega} \mathbb{R}$ is a type II_∞ factor, and the CLPW *observer-projected* algebra $\Pi\widetilde{\mathcal{N}}\Pi$, with the dressed observer energy bounded below, is type II_1 with a maximum-entropy tracial state [18]. The witness's two-boundary data are drawn from *branch (I)* of Conj. 11.1 (Araki conditioning, Thm. G.1; no centralizer or (CP) hypothesis is needed there): the terminal effects $\{E_k\}$ are the spectral windows of a boost-smeared *local field* observable $A = \Pi\,\widehat{\varphi}(f)\,\Pi \in \Pi\widetilde{\mathcal{N}}\Pi$ with f *not* boost-invariant, i.e. the CHT-selected terminal PVM of App. L, which has genuine matter support (Rem. L.2) and is *not* a family of clock windows. The clock-window partitions of the dressed generator $\widehat{P}$ in $L^\infty(\mathbb{R})$ remain available as the branch-(II) *classical sector* of the witness: for them (CP) holds tautologically (the static-patch dynamics is generated by the boost $\widehat{K}$, which generates $\sigma^{\widehat{\omega}}$ itself), but by Rem. 11.2 they condition only the observer-energy distribution ($\omega_k\restriction_{\mathcal{N}} = \omega$, $f_k \equiv 0$) and are so labelled; they are not used as the witness's terminal data.
- **Matter sector (datum distinct from the seed):** a finite-dimensional spectral cut-off $\rho_{\mathrm{i}} := P_{\le E_*}\rho_{\beta'}P_{\le E_*}/Z$ of the thermal state $\rho_{\beta'}$ at inverse temperature $\beta' \neq \beta_{\mathrm{dS}}$ (a genuine excitation of the static patch: the forward matter datum differs from the Bunch–Davies seed restriction), with the gap $\eta > 0$ from the smallest non-zero eigenvalue of the truncation, satisfies (S1)–(S8) of Conj. 11.4 on a Sobolev ball $\mathcal{B}_{g_0}$ of de Sitter perturbations exactly as in the $\beta' = \beta_{\mathrm{dS}}$ case (the estimates use only the gap and the norm bounds, not the temperature). (The cut-off projection $P_{\le E_*}$ is taken inside the observer-projected CLPW algebra $\Pi\widetilde{\mathcal{N}}\Pi$, in which the dressed boost generator on the wedge is bounded below; on the original type III_1 algebra $\mathcal{N}$ alone the modular Hamiltonian has $\mathbb{R}$-spectrum, so the projection is well-defined only after the dressing. This is the operationally natural restriction to the dressed observer's energy budget.) *Non-vacuity of the two-boundary link ($f_k > 0$).* The terminal effects E_k are spectral windows of the boost-smeared local field $A = \Pi\,\widehat{\varphi}(f)\,\Pi$ compressed to the band-limited matter space $\mathcal{H}_{[E_*]}$. Two honest cautions apply. *(a)* $\Pi\widehat{\varphi}(f)\Pi$ is a priori only symmetric; we take f real and A to be (the closure of) a self-adjoint element of the II_1 corner $\Pi\widetilde{\mathcal{N}}\Pi$, so its spectral projections exist *in that corner.* *(b)* Compressing those projections further to $\mathcal{H}_{[E_*]}$ generically yields *POVM* effects, for which the clean $f_k > 0 \iff [\rho, \Lambda_k] \neq 0$ equivalence of Thm. 3.10(iii) holds only in the projective case. We therefore *do not* assert $f_k > 0$ from this

infinite-dimensional construction alone. Non-vacuity of the phenomenon is instead secured, as a finite-dimensional theorem, by Prop. 11.53 (matter datum $\rho_{\rm i} = P_{\le E_*}\rho_{\beta'}P_{\le E_*}/Z \neq$ seed; projective terminal PVM not commuting with $\rho_{\rm i}$), which exhibits $f_k > 0$ under Thm. 3.10(iii) with no POVM converse required. When the compressed windows happen to be projective (co-diagonalisation on $\mathcal{H}_{[E_*]}$), the same conclusion transfers verbatim to the witness — in sharp contrast to the clock-window sector, where $f_k \equiv 0$ identically (Rem. 11.2).

- **Clock:** the de Sitter time-translation Killing parameter inside the static patch defines an admissible Page–Wootters clock $\mathcal{A}_{\rm clock} \subset \mathcal{A}_{\rm tot}$ in the sense of Conj. 11.10 with monotone modular-flow entropy growth (i), Araki-relative-entropy compatibility (45) for any clock-coupled protocol that records the dressed observer's proper-time slice [18], minimality (iii) (the boost-generator C^*-algebra is minimal among Killing-time subalgebras realising the full seed-time MASA entropy), and (PW) split data (the wedge tensor-factorisation $\mathcal{H}_{\rm tot} = \mathcal{H}_{\rm clock}\otimes\mathcal{H}_{\rm sys}$ of [18] supplies $U_{\rm split}$ and the boost $\widehat{H}_{\rm c} = \widehat{K}$ with covariant time observable $\widehat{T}_{\rm c}$).
- **Horizon:** the de Sitter cosmological horizon $\mathcal{H}^+$ is a Bisognano–Wichmann horizon of ω, with non-trivial $\widetilde{\mathcal{A}}(\mathcal{H}^+)^{\sigma^{\widehat{\omega}_{\rm H}}}$ in the centralizer-rich scope of Conj. 11.8. The K-equivariance hypothesis (KE) of Prop. 11.9 is *automatic only for the branch-(II) clock-window sector*: a centralizer partition $\{E_k\} \subset \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ of (39) also lies in $\widetilde{\mathcal{A}}(\mathcal{H}^+)^{\sigma^{\widehat{\omega}_{\rm H}}}$ (both dual flows are generated by the boost $\widehat{K}$ up to CLPW dressing, so the two crossed-product centralizers coincide on the static patch), and then Φ_K commutes with any masa $\mathcal{Z} \subset \widetilde{\mathcal{A}}^{\sigma^{\widehat{\omega}_{\rm H}}}$. For the witness's actual (branch-(I), matter-supported) terminal PVM, (KE) is *imposed as part of the witness data* by choosing the masas $\mathcal{Z}(\gamma(s_m))$ inside the relative commutant of $\{E_k\}''$, exactly as the standing scope of Conj. 11.8 permits ((KE) is declared there to be part of the data when it is not automatic).
- **Composition law:** the static-patch clock satisfies (A)–(E) of Conj. 11.5 *via the spectral theorem applied to the boost generator*: let $\widehat{H}_{\rm c} := K$ be the boost generator on the de Sitter static patch (self-adjoint by Stone, with absolutely continuous spectrum on $\mathbb{R}$ on the original $\mathcal{N}$, discretised on bounded intervals after CLPW dressing). On the spectral subspace $\mathcal{H}_{[E_*]} := E_{\widehat{H}_{\rm c}}([-E_*, E_*])\mathcal{H}$ — which is *exactly* the $(\widehat{H}_{\rm c}, E_*)$-band-limited domain $\Phi_k^{\rm bl} \subset \Phi_k$ of (T6) of Conj. 11.56, so the witness lives precisely in the scope where (T6) supplies a uniform spectral gap (uniform spectral gap on the truncation: by CLPW dressing, the spectrum of $\widehat{H}_{\rm c}$ restricted to $\mathcal{H}_{[E_*]}$ is discrete with smallest non-zero eigenvalue $\Delta_* = \Delta_*(E_*) > 0$ uniform on $\mathcal{B}_{g_0}$ by the smallness of de Sitter perturbations [18], supplying the uniform spectral gap required by (S3) and (T6)) — consider the deformation

$$B(\delta) \;:=\; \alpha\big(\cos(\delta\widehat{H}_{\rm c}/\hbar) - \mathbb{1}\big) \;=\; \alpha\widehat{\Sigma}_\delta.$$

Verify the conj:wdw-selection axioms in their actual labelling: **(A) self-adjointness:** $B(\delta)$ is a real-valued bounded Borel function of the self-adjoint $\widehat{H}_{\rm c}$, hence self-adjoint on $\mathcal{H}_{[E_*]}$. **(B) clock-translation covariance:** $\widehat{T}_s B(\delta)\widehat{T}_{-s} = B(\delta)$ for all $s \in \mathbb{R}$ since $B(\delta) \in \{\widehat{H}_{\rm c}\}''$ by spectral calculus. **(C) time-reversal covariance:** $B(-\delta) = B(\delta)$ because cosine is even. **(D) cocycle/d'Alembert:** $B(\delta_1+\delta_2) + B(\delta_1-\delta_2) = 2\cos(\delta_1\widehat{H}_{\rm c}/\hbar)B(\delta_2) + 2B(\delta_1)$, by the scalar d'Alembert identity $\cos(x+y)+\cos(x-y) = 2\cos x\cos y$ on $\sigma(\widehat{H}_{\rm c}\restriction_{\mathcal{H}_{[E_*]}})$ via the spectral theorem, applied as a quadratic-form identity on the band-limited domain. **(E) TDSE limit:** $B(\delta) \to 0$ in strong-operator topology as $\delta \to 0$ on $\mathcal{H}_{[E_*]}$ (uniform on bounded spectral subspaces because $\|B(\delta)\| \le |\alpha|\cdot|\delta E_*/\hbar|^2/2 + O(\delta^4E_*^4)$); for every $\psi \in \mathcal{H}_{[E_*]} \subset \mathcal{D}(\widehat{H}_{\rm c}^2)$ (which is automatic on the band-limited subspace) the strong derivatives $B'(0)\psi = 0$ and $-B''(0)\psi = \alpha\widehat{H}_{\rm c}^2\psi/\hbar^2$ hold by direct computation on the spectral expansion. The verification of (E) requires the uniform spectral gap Δ_* to ensure that differentiation under the spectral integral is justified uniformly on $\mathcal{H}_{[E_*]}$ [31]; without this uniformity the strong second derivative would be only pointwise-spectral. Prop. 11.6 then applies on $\mathcal{H}_{[E_*]}$ and selects $B(\delta) = \alpha\widehat{\Sigma}_\delta$ uniquely (α fixed by (E); the family is even in δ, so $\delta \to -\delta$ at fixed

α is the only reparametrization freedom), with all axioms verified *on the (T6)-band-limited subspace* which is exactly the scope of Conj. 11.56, Conj. 11.5, and the semiclassical regime of Conj. 10.2.

This witness is finite-dimensional in the matter sector by spectral cut-off, and infinite-dimensional in the clock and gravity sectors by the dressed crossed-product construction; all five scope hypotheses are met by construction, so the common scope of Conj. 11.55 is non-empty. These two facts are carried by *different* objects and should not be read as one: the scope-non-emptiness is a property of this de Sitter construction, whereas non-vacuity *in phenomenon* ($f_k > 0$) is the separate finite-dimensional theorem Prop. 11.53, independent of the type III and Cartan machinery, and is *not* here shown to be realized inside the *same* de Sitter model that meets the five scopes (that coexistence is the open matter-supported $\wedge$ flow-invariant conjunction); had the terminal data been the branch-(II) clock windows the link would be provably invisible ($f_k \equiv 0$, Rem. 11.2). We do *not* claim the witness is unique or canonical, nor that the CHT-selected cosmological terminal POVM is simultaneously flow-(B)-invariant and matter-supported — that joint condition is open (Rem. L.7, Rem. L.3). We claim only that a working two-boundary datum with $f_k > 0$ exists, which refutes the empty-intersection failure mode.

11.6 A speculative unified master equation

We close by recording, in one place and in a single line, the speculative master equation that would unify every construction of this paper. The presentation is deliberately stylised: the equation is written in a condensed form that hides technical difficulties (domain issues, renormalisation, the meaning of the geometry Hilbert space) and is *not* proposed as a rigorous object. Rather, it is proposed as the natural target of the programme of Conj. 11.55: the single equation which, if the five conjectures of Sec. 11.1 are established, reduces simultaneously to Part I (finite-dim conditioned QM), to the semiclassical Einstein equation of Conj. 10.2, and to the modified TDSE of Prop. 10.14.

Caveat. We do not claim that the equation below is correct, well-defined, or unique. It is a concise summary of a proposal; its mathematical status is exactly that of the five conjectures on which it rests.

The universal state space

Let

$$\mathcal{U} := \bigoplus_{k\in\mathcal{K}_+} \mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}}, \tag{58}$$

where the direct sum is taken over $\mathcal{K}_+ := \{k \in \mathcal{K} : p_0(k) > 0\}$ (the only sectors that carry probability under the seed prior; the sectors with $p_0(k) = 0$ carry no physical state and are discarded from the universal sample space (24) as well as from the universal Hilbert space, eliminating the seminorm degeneracy that would arise from a $\sum_k w(k)\|\Psi_k\|^2$ weighted norm with $w(k) = 0$). $\mathcal{H}_{\mathrm{clock}}$ carries the Page–Wootters clock $(\widehat{H}_{\mathrm{c}}, |t\rangle)$ of Conj. 10.13; $\mathcal{H}_{\mathrm{geom}}$ is an (abstract, to be constructed) separable Hilbert space of spatial 3-geometries equipped with a self-adjoint constraint-generating family $\widehat{H}_{\mathrm{grav}}$; and $\mathcal{H}^{(k)}_{\mathrm{matter}}$ is the standard-form (GNS) Hilbert space of the two-boundary conditioned state ω_k on the matter algebra $\widetilde{\mathcal{N}}$ of Conj. 11.1 (in Part I, when $\widetilde{\mathcal{N}} = \mathcal{B}(\mathcal{H}_{\mathrm{sys}})$, this is just the GNS Hilbert space of $\rho_2(\tau \mid k)$, finite-dimensional). The direct sum over $\mathcal{K}_+$, rather than a tensor factor $\ell^2(\mathcal{K}_+)$, encodes the fact that $\mathcal{K}_+$ is a *classical superselection label* (the terminal record), *not* a quantum degree of freedom: no coherent superpositions across different k are physical, in accord with Rem. 3.2 and the branch postulate of Conj. 10.2. A universal state $\Psi \in \mathcal{U}$ is a tuple $\Psi = (\Psi_k)_{k\in\mathcal{K}_+}$ with $\Psi_k \in \mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}}$ and $\sum_{k\in\mathcal{K}_+} \|\Psi_k\|^2 < \infty$ *(plain Hilbert direct-sum norm, with every coefficient equal to one; the sector probabilities $w(k) = p_0(k)$ enter only through the universal sample space* (24) *and through* *the Born projector on each sector, never as Hilbert-space inner-product weights, ensuring $\langle\cdot,\cdot\rangle_{\mathcal{U}}$ is a genuine inner product and not a seminorm)*, where $w = (w(k))_{k\in\mathcal{K}_+}$ is the sector weight vector

$$w(k) \;:=\; p_0(k) \;=\; \mathrm{Tr}\big(\rho[g_0](\tau_f)\, E_k\big), \qquad w(k) > 0 \text{ on } \mathcal{K}_+, \quad \sum_{k\in\mathcal{K}_+} w(k) = 1, \tag{59}$$

with g_0 the seed background of Conj. 10.2; normalization $\sum_{k\in\mathcal{K}_+} w(k) = 1$ is then Lem. 2.5 applied to g_0. The earlier "self-consistent" identification $w(k) = \mathrm{Tr}(\rho_2[g_k](\tau_f \mid k)E_k)$ is *not* a normalized law on $\mathcal{K}$ (e.g. for projective E_k each summand equals 1); the seed prior p_0 is the unique conservation-consistent choice. *Branch-by-branch self-consistency (corollary of (T10)).* The apparent tension between branch-dependent dynamics and a branch-independent weight is resolved *as a corollary of* the shell-blind-matter hypothesis (T10) of Conj. 11.56 below, applied together with the adaptedness restriction (23) of Conj. 10.2. In the zero-record sector of (23) (trivial protocol, where $g_k \restriction_{J^-(\Sigma_{\tau_f})} = g_0 \restriction_{J^-(\Sigma_{\tau_f})}$ exactly), (T10) ensures matter-sector evolution on the slab $[\tau_i, \tau_f]$ is generated by a Hamiltonian that is *shell-blind* (i.e. depends on g_k only through $g_k \restriction_{J^-(\Sigma_{\tau_f})}$), the conditioned forward propagator $U[g_k](\tau_f, \tau_i)$ on the slab equals $U[g_0](\tau_f, \tau_i)$ for every $k \in \mathcal{K}_+$, hence by Lem. 2.5

$$w(k) \;=\; \mathrm{Tr}\big(\rho[g_k](\tau_f)\, E_k\big) \;=\; \mathrm{Tr}\big(\rho[g_0](\tau_f)\, E_k\big) \;=\; p_0(k) \qquad \text{for every } k \in \mathcal{K}_+, \text{ under (T10)}, \tag{60}$$

in the zero-record sector. That is, $w(k) = p_0(k)$ is *simultaneously* the seed prior and the Born law generated by the actual sector dynamics on g_k; the two coincide *by virtue of (T10)* because the back-reaction then lives in $J^+(\Sigma_{\tau_f})$ while the Born trace is computed at τ_f on the unconditioned slab and (T10) prevents the shell-supported back-reaction from leaking into matter dynamics on the slab. With a nontrivial record protocol the slab propagator is itself record-dependent (adapted back-reaction), the identity (60) holds at zeroth order in κ, and the exact statement is the Born-map fixed point (x-c) of Conj. 10.2: $w = p$ with $\|p - p_0\| \le O(\kappa T L_U M)$, which is precisely the branch-by-branch redefinition anticipated below. *Without (T10) even the zero-record identity* (60) *is not automatic and $w(k)$ would have to be redefined branch-by-branch as a fixed point of the Born map* $w \mapsto (\mathrm{Tr}(\rho[g_k(w)](\tau_f)E_k))_k$. The conjecture therefore couples branch-dependent dynamics to weights that, under (T10) and (x-c), are independently the unique self-consistent fixed point of the Born map, with no tension between the two. The fixed-point family $\{g_k\}$ of Conj. 10.2 acts on the sector wavefunctions Ψ_k but does *not* re-weight the *support* $\mathcal{K}_+$; the support is fixed by the seed two-boundary data $(\rho_{\mathrm{i}}, \{E_k\}, g_0)$ (this is consistent because the generic-compatibility hypothesis (T0) of Conj. 11.56 below ensures $\mathcal{K}_+ = \{k : p_0(k) > 0\}$, so no compatibility-induced rescaling occurs), while the operative weights on that support are $w = p_0$ exactly in the zero-record sector under (T10) and the (x-c) fixed point p (with $\|p - p_0\| \le O(\kappa T L_U M)$, $p = p_0$ at $O(\kappa^0)$) for nontrivial record protocols, as stated above. The components $\Psi_k(t, \mathsf{g}, \phi)$ are k-sector wavefunctions, and the physical observable algebra is the block-diagonal subalgebra

$$\mathcal{A}_{\mathrm{phys}} \;:=\; \bigoplus_{k\in\mathcal{K}} \mathcal{B}\big(\mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}}\big) \;\subsetneq\; \mathcal{B}(\mathcal{U}), \tag{61}$$

so that cross-sector matrix elements $\langle \Psi_k | \mathcal{A}_{\mathrm{phys}} | \Psi_{k'} \rangle = 0$ for $k \neq k'$ and relative phases between sectors are unobservable. This implements the classical superselection rule on $\mathcal{K}$ at the level of observables, not merely at the level of state vectors.

Let $|\Psi^{(k)}_{\mathrm{i}}\rangle \in \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}}$ purify ρ_{i} on Σ_{τ_i} in the standard form of ω_k (Tomita standard-form purification), let Λ_k be the Heisenberg-propagated terminal effect (pulled back to $\mathcal{H}^{(k)}_{\mathrm{matter}}$ along $\mathcal{H}_{\mathrm{geom}}$-valued Heisenberg transport), and let $\alpha \in \mathbb{R}$, $\delta t > 0$ be phenomenological parameters of the clock-shift term constrained by Rem. 10.30. Write $\widehat{\Sigma}_{\delta t} := \cos(\delta t \widehat{H}_{\mathrm{c}}/\hbar) - \mathbb{1}$ for the self-adjoint clock symmetrisation (26). Fix a real test-tensor coupling kernel $f^{\mu\nu} \in$

$C_c^\infty(\mathcal{M}_{[\tau_i,\tau_f]}; \mathrm{Sym}^2 T\mathcal{M})$ used below to smear all stress-tensor and Einstein-tensor operators into bounded or relatively-bounded self-adjoint operators (so that no pointwise unrenormalised operator products appear).

The master equation

Conjecture 11.56 (Two-Boundary Retrocausal Master Equation)**.** There exist, for each $k \in \mathcal{K}$ and the chosen kernel $f^{\mu\nu}$:

(T0) **Generic compatibility.** For every $k \in \mathcal{K}$ with $w(k) = p_0(k) > 0$ the rigged boundary-to-boundary pairing (70) below is non-zero. Equivalently, $\mathcal{K}_+ := \{k \in \mathcal{K} : p_0(k) > 0 \text{ and (70) holds}\} = \{k \in \mathcal{K} : p_0(k) > 0\}$, so the seed prior $\{p_0(k)\}_{k\in\mathcal{K}}$ already restricts to a probability law on $\mathcal{K}_+$ without re-normalisation. (This is a structural hypothesis on the two-boundary data $(\rho_{\mathrm{i}}, \{E_k\}, g_0)$: it rules out terminal effects whose Heisenberg propagate is orthogonal to the constraint surface in the rigged sense, which would otherwise force a sector-pruning rescaling incompatible with the seed-prior identity (59).)

(T1) a common dense form-core $\mathcal{D}_k \subset \mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}}$;

(T2) smeared self-adjoint operators $\widehat{T}^{\mu\nu}_{\mathrm{m}}[\widehat{\mathrm{g}}; f]$ (Hadamard-renormalised matter stress-energy contracted with $f_{\mu\nu}$) and $\widehat{G}^{(Q)\,\mu\nu}[\widehat{\mathrm{g}}; f]$ (operator-ordered linearised quantum Einstein tensor contracted with $f_{\mu\nu}$), each essentially self-adjoint on $\mathcal{D}_k$;

(T3) a kinetic generator $\widehat{K}_k := \widehat{H}_{\mathrm{c}} + \widehat{H}_{\mathrm{grav}} + \widehat{H}^{(k)}_{\mathrm{m}}[\widehat{\mathrm{g}}] + \alpha\,\widehat{\Sigma}_{\delta t}$, essentially self-adjoint on $\mathcal{D}_k$;

(T4) a Kato–Rellich domination constant $a \in [0, 1)$ with relative bound $\left\|\kappa^{-1}\widehat{Q}^{(k)}[f]\Psi\right\| \le a\left\|\widehat{K}_k\Psi\right\| + b\,\|\Psi\|$ for all $\Psi \in \mathcal{D}_k$ and some $b \ge 0$ (so that $\kappa^{-1}\widehat{Q}^{(k)}[f]$ is the correctly dimensionalised perturbation; $\kappa = 8\pi G/c^4$ has units of inverse energy density times length, and a is dimensionless), where $\widehat{Q}^{(k)}[f]$ is the symmetrically ordered gravity–matter coupling

$$\widehat{Q}^{(k)}[f] \;:=\; \tfrac{1}{2}\big(\widehat{G}^{(Q)\,\mu\nu}[\widehat{\mathrm{g}}; f]\,\widehat{T}^{\mathrm{m}}_{\mu\nu}[\widehat{\mathrm{g}}; f] \;+\; \widehat{T}^{\mathrm{m}}_{\mu\nu}[\widehat{\mathrm{g}}; f]\,\widehat{G}^{(Q)\,\mu\nu}[\widehat{\mathrm{g}}; f]\big); \tag{62}$$

(T5) **Spectral absolute continuity at 0.** Let $E_{\mathfrak{C}_k}(\cdot)$ denote the projection-valued measure of the self-adjoint closure of $\widehat{\mathfrak{C}}_k$. There exists $\epsilon_0 > 0$ and a nuclear core $\Phi_k \subset \mathcal{D}_k$ such that for every $\psi, \varphi \in \Phi_k$ the spectral measure $\lambda \mapsto \langle\psi|E_{\mathfrak{C}_k}(d\lambda)|\varphi\rangle$ is absolutely continuous with respect to Lebesgue measure on $(-\epsilon_0, \epsilon_0)$ with Radon–Nikodym density continuous and uniformly bounded at $\lambda = 0$ (so that the rigged $\delta(\widehat{\mathfrak{C}}_k)$ of (29) is a well-defined sesquilinear form on $\Phi_k \times \Phi_k$);

(T6) **Slow-branch (analyticity / band-limited) selection.** The physical state space is the maximal $\widehat{\mathfrak{C}}_k$- invariant subspace $\Phi^{\mathrm{bl}}_k \subset \Phi_k$ on which the joint spectral projection of $\widehat{H}_{\mathrm{c}}$ satisfies $E_{\widehat{H}_{\mathrm{c}}}\big([-\hbar/\delta_\star, \hbar/\delta_\star]\big)\Psi_k = \Psi_k$ for some fixed $\delta_\star > \delta t$ (a strict band-limit on the clock generator), and on which the induced family of low-energy solutions is real-analytic in the deformation parameter $\mu := \alpha\delta t^2/\hbar^2$ at $\mu = 0$. The band-limit is a strictly stronger condition than mere real-analyticity of $\delta \mapsto e^{-i\delta\widehat{H}_{\mathrm{c}}/\hbar}\Psi_k$ at $\delta = 0$ (analytic vectors with Gaussian spectral tails exist; (T6) excludes them). The band-limited formulation is exactly what the Taylor-remainder bound of Prop. 10.27 requires, and it implies μ-analyticity of the low-energy expansion (the two notions of analyticity are no longer conflated; see Rem. 11.57 below). Henceforth Φ^{bl}_k replaces the earlier notation Φ^{an}_k throughout. *Scope of the invariance claim.* For a generic constraint $\widehat{\mathfrak{C}}_k = \widehat{K}_k - \kappa^{-1}\widehat{Q}^{(k)}$ the clock band-limit $E_{\widehat{H}_{\mathrm{c}}}([-\hbar/\delta_\star, \hbar/\delta_\star])$ is *not* preserved by $\widehat{\mathfrak{C}}_k$: the gravity– matter coupling $\widehat{Q}^{(k)}$ and the matter generator inside $\widehat{K}_k$ act on tensor legs distinct from the clock factor and do not commute with the clock projector, so the largest $\widehat{\mathfrak{C}}_k$-invariant subspace contained in a fixed band can in principle be $\{0\}$. The $\widehat{\mathfrak{C}}_k$- invariance asserted here is established only under Pos. 11.30 (Prop. 11.33 (ii)), where every factor is finite-dimensional and $\Phi^{\mathrm{bl}}_k = \Phi_k$. Outside that scope Φ^{bl}_k is read as the band-limited test core on which the group-averaging map η_k is evaluated (cf. the definition $\mathcal{H}^{(k),\mathrm{bl}}_{\mathrm{phys}} := \eta_k(\Phi^{\mathrm{bl}}_k)$ in Thm. 11.64 (iii)), not as a constraint-invariant subspace.

(T7) **Boundary data lie in the test space.** Both rigged boundary vectors used in (T0) and in (67)–(68), namely

$$\Psi^{(k)\,\mathrm{rigg}}_{\mathrm{i}} \;:=\; \iota_i\big(\sqrt{w(k)}\,|\Psi^{(k)}_{\mathrm{i}}\rangle\big), \qquad \Psi^{(k)\,\mathrm{rigg}}_{\mathrm{f}} \;:=\; \iota_f\big((\mathbb{1}_{\mathrm{geom}}\otimes J_k\Lambda_k^{1/2}J_k)\,|\Psi^{(k)}_{\mathrm{f}}\rangle\big),$$

obtained from the clock-time-smeared embeddings $\iota_{i,f} : \mathcal{H}_{\mathrm{geom}}\otimes\mathcal{H}^{(k)}_{\mathrm{matter}} \to \Phi_k$ of Rem. 11.58 below (Gaussian time-window of width $\sigma_t \ll \delta t$ around $\tau_{i,f}$, replacing distributional sharp clock states $|\tau_{i,f}\rangle$), lie in $\Phi^{\mathrm{bl}}_k \subset \Phi_k$. Equivalently, the boundary-value problem (67)–(68) is interpreted as a smeared-time boundary problem in the rigged setup of (T5) on the test space, not as a sharp-time eigenvalue constraint in $\mathcal{H}_{\mathrm{clock}}$. (This makes both the rigged pairing of (T0) and the realisability claim of Thm. 11.64 (iv) well-defined on the same pair of objects.)

(T8) **Symmetric domain of the gravity–matter coupling.** The smeared operators of (T2) satisfy the joint domain inclusions

$$\widehat{T}^{\mu\nu}_{\mathrm{m}}[\hat{\mathfrak{g}}; f]\,(\mathcal{D}_k) \;\subset\; \mathcal{D}(\widehat{G}^{(Q)\,\mu\nu}[\hat{\mathfrak{g}}; f]), \quad \widehat{G}^{(Q)\,\mu\nu}[\hat{\mathfrak{g}}; f]\,(\mathcal{D}_k) \;\subset\; \mathcal{D}(\widehat{T}^{\mu\nu}_{\mathrm{m}}[\hat{\mathfrak{g}}; f]),$$

so that the symmetrically ordered product $\widehat{Q}^{(k)}[f]$ of (62) is a well-defined symmetric operator on $\mathcal{D}_k$ (i.e. both compositions in (62) map $\mathcal{D}_k$ into $\mathcal{H}$ and the resulting operator equals its formal adjoint on $\mathcal{D}_k$). Without this domain inclusion the formal Hermitian ordering of (62) is not enough to guarantee symmetry of the unbounded product, so the Kato–Rellich step in Thm. 11.64 (i) requires (T7) explicitly.

(T9) **Strengthened spectral compatibility.** For every $k \in \mathcal{K}$ with $w(k) = p_0(k) > 0$ and the boundary vectors of (T6), the limit

$$\rho^{(k)}_{\mathrm{rigg}}(0) \;:=\; \lim_{\epsilon\downarrow 0}\,(2\epsilon)^{-1}\left\langle \Psi^{(k)\,\mathrm{rigg}}_{\mathrm{f}} \;\middle|\; E_{\mathfrak{C}_k}([-\epsilon,\epsilon]) \;\middle|\; \Psi^{(k)\,\mathrm{rigg}}_{\mathrm{i}}\right\rangle \tag{63}$$

exists and is non-zero, *and* the zero-spectral slice of $\overline{\widehat{\mathfrak{C}}_k}$ on the band-limited test space Φ^{bl}_k is a *one-dimensional subspace* of $\mathcal{H}^{(k),\mathrm{bl}}_{\mathrm{phys}}$ — concretely, the diagonal positive semi-definite rigged density form $\rho^{(k)}_{\mathrm{rigg}}(0)[\Psi;\Psi] := \lim_{\epsilon\downarrow 0}(2\epsilon)^{-1}\langle\Psi|E_{\mathfrak{C}_k}([-\epsilon,\epsilon])|\Psi\rangle$ exists on Φ^{bl}_k as a non-zero positive semi-definite sesquilinear form whose kernel has codimension one in Φ^{bl}_k (rank-one zero-spectral slice; the earlier phrasing "the functional $\Psi \mapsto \rho^{(k)}_{\mathrm{rigg}}(0)[\Psi^{(k)\,\mathrm{rigg}}_{\mathrm{f}};\Psi]$ has one-dimensional range" was vacuous because any non-zero $\mathbb{C}$-valued linear functional trivially has range a one-dimensional subspace of $\mathbb{C}$; the substantive content is the codimension-one kernel of the diagonal form, which is exactly the rank-one density hypothesis of Prop. 11.66). This codimension-one kernel is the structural condition used in part (iv) of Thm. 11.64 to deduce uniqueness up to scale of the boundary-realising universal state. (T8) is strictly stronger than (T0): it fixes both that the spectral pairing has a non-trivial δ- function singularity at $\lambda = 0$ *on the rigged boundary vectors of (T6) inclusive of the Tomita conjugation factor* $J_k\Lambda_k^{1/2}J_k$, that the renormalised limit is finite and non-zero, not merely that the pre-limit numerator is non-zero, *and* that the zero-mode slice is one-dimensional so the functional $\rho^{(k)}_{\mathrm{rigg}}(0;\,\cdot\,)$ determines the boundary-matching universal state up to scale. This subsumes both (T0) and the rank-one density condition of Prop. 11.66: under (T6) and (T5), Prop. 11.66 provides a sufficient structural condition for (T8); see Rem. 11.59.

(T10) **Endpoint-preserving gauge.** The gauge group of physical equivalence on the kernel of $\widehat{\mathfrak{C}}_k$ is restricted to those constraint deformations $\Psi_k \mapsto \Psi_k + \widehat{\mathfrak{C}}_k\chi_k$ with $\chi_k \in \Phi^{\mathrm{bl}}_k$ satisfying the endpoint-vanishing conditions

$$\iota_i^*\big(\widehat{\mathfrak{C}}_k\chi_k\big) \;=\; 0, \qquad \iota_f^*\big(\widehat{\mathfrak{C}}_k\chi_k\big) \;=\; 0, \tag{64}$$

where $\iota_{i,f}^* : \Phi_k' \to (\mathcal{H}_{\rm geom} \otimes \mathcal{H}_{\rm matter}^{(k)})'$ are the adjoint smeared evaluations of (T6). Denote this restricted gauge orbit by $\widehat{\mathfrak{C}}_k\, \Phi_k^{{\rm bl},\,\partial}$. The unrestricted constraint range $\widehat{\mathfrak{C}}_k \Phi_k^{\rm bl}$ generically contains deformations that violate (67)– (68); quotienting by it would identify states with different boundary data. (T9) restricts the gauge to the largest subgroup that preserves the two-boundary problem.

(T11) **Matter junction across the Israel shell.** The matter Heisenberg propagator across the terminal slice Σ_{τ_f} is shell-blind:

$$U_{\rm m}[g_k](\tau_f^+, \tau_f^-) \;=\; \mathbb{1}_{\mathcal{H}_{\rm matter}^{(k)}} \qquad \text{for every } k \in \mathcal{K}_+, \tag{65}$$

equivalently the ϵ-normalized Israel–Darmois surface stress-energy $S_{ab}^{(k)}$ of (21) couples only to the gravitational sector and induces no matter-side boundary phase or matching unitary on $\mathcal{H}_{\rm matter}^{(k)}$. Combined with the adaptedness restriction (23), (T10) gives, in the zero-record sector, $U[g_k](\tau_f, \tau_i) = U[g_0](\tau_f, \tau_i)$ on the matter algebra at τ_f, hence the seed-prior identity (60) extends across the shell (with nontrivial records the identity holds at zeroth order in κ and the exact weights are the Born-map fixed point (x-c) of Conj. 10.2). Shell-blindness is *not* assumed at the interior record cones $\mathcal{N}_m$, whose null shells (20) couple to matter only through the (x-b)-bounded junction data. Without (T10) a generic Israel shell could induce a finite matter boundary unitary, breaking the seed-prior self-consistency.

(T12) **Centralizer-preserving Heisenberg propagation.** The matter Hamiltonian $\widehat{H}_{\rm m}^{(k)}[\widehat{\mathfrak{g}}]$ of (T3) commutes with the dual modular flow of the *seed* background state ω on the matter algebra $\widetilde{\mathcal{N}}$ in the form $[\widehat{H}_{\rm m}^{(k)}[\widehat{\mathfrak{g}}], \Delta_{\widehat{\omega}}^{it}] = 0$ for all $t \in \mathbb{R}$ (this is the (CP) hypothesis of Conj. 11.1 restated at the matter-Hamiltonian level; the seed dual flow $\sigma^{\widehat{\omega}}$, not the conditioned σ^{ω_k}, is the operationally relevant one because $\{E_k\} \subset \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ by (39) and the whole Tomita construction lives in $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$). Consequently the Heisenberg propagate $\Lambda_k(\tau)$ of $E_k \in \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ (39) remains in the centralizer $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ for every $\tau \in [\tau_i, \tau_f]$, so the entire-analyticity of $\Lambda_k^{1/2}$ used in Conj. 11.1 is preserved under matter dynamics. Without (T11) the propagated effect can leave the centralizer, invalidating the simplification $\sigma_z^{\widehat{\omega}}(\Lambda_k^{1/2}) = \Lambda_k^{1/2}$.

(T13) **Universal sample-space full coverage.** Under Pos. 11.30, the matter sector is finite-dimensional and $p_* := \min\{p_0(k) : k \in \mathcal{K}_+\} > 0$ (where $\mathcal{K}_+ := \{k \in \mathcal{K} : p_0(k) > 0\}$). The Banach fixed-point threshold $\varepsilon_{\rm BFP}$ of Conj. 10.2 is taken strictly below the seed gap, $\varepsilon_{\rm BFP} \in (0, p_*)$ (canonically $\varepsilon_{\rm BFP} = p_*/2$ per Prop. 10.3), so that $\{k \in \mathcal{K} : 0 < p_0(k) \leq \varepsilon_{\rm BFP}\} = \emptyset$ and the fixed-point statement of Conj. 10.2 holds uniformly for every $k \in \mathcal{K}_+$ without low-probability fallback. This makes $\sum_{k \in \mathcal{K}_+} p_0(k) = 1$, so the universal sample space (24) is a probability measure (no sub-probability defect, no kinematic extension). Prior formulations in which $\varepsilon_{\rm BFP} \geq p_*$ required an *out-of-scope fallback* $g_k := g_0$ on low- probability sectors that did not in fact solve (19); (T12) excludes that regime by construction, absorbing the low-probability-sector defect into the scope restriction provided by Pos. 11.30.

(T14) **Sample-space stability of the Part I observer.** The Part I worldline/filtration data $(\gamma, \tau, \mathcal{A}(\Sigma_\tau))$ used to construct $\rho_2[g](\tau \mid k)$, $\mathbb{P}_\gamma^{[g]}$, and $\Omega_\gamma^{[g]}$ are assumed to persist for every $g \in \mathcal{B}_{g_0} \cup \{g_k : k \in \mathcal{K}_+\}$: τ remains a g-Cauchy-temporal function, γ remains g-timelike with $\tau(\gamma(\tau)) = \tau$, and the local-algebra hypotheses (M1)–(M3) of Ax. 2.2 hold for every such g on the same algebra $\mathcal{A}$. The weighted Sobolev ball $\mathcal{B}_{g_0}$ is restricted to the open subset on which (T13) holds; this is automatic on a sufficiently small ball by structural stability of Cauchy temporal functions [9].

Under (T0)–(T13), the constraint operator

$$\boxed{\widehat{\mathfrak{C}}_k \;:=\; \widehat{K}_k - \kappa^{-1}\, \widehat{Q}^{(k)}[f], \qquad \widehat{\mathfrak{C}}_k\, \Psi_k \;=\; 0 \quad \forall k \in \mathcal{K},} \tag{66}$$

is essentially self-adjoint on $\mathcal{D}_k$ by Kato–Rellich applied to (T4) and (T7) with relative bound $a < 1$ (Thm. 11.64 (i)). The domain inclusion (T7) upgrades the formal Hermitian ordering of (62) into a bona fide symmetric operator on $\mathcal{D}_k$, removing the operator-domain gap that the relative-bound (T4) alone does not close. The symmetric ordering in (62) also replaces the earlier formal expression "$\widehat{T}^{1/2}\widehat{G}\widehat{T}^{1/2}$", which would have required $\widehat{T}^{\mu\nu} \succeq 0$ (false in general for matter stress-energy components). Tensor factors and summation over $\mu\nu$ are implicit; $\kappa = 8\pi G/c^4$ is the Einstein coupling, consistently with Conj. 10.2 (iv). Self-adjointness, well-definedness of the rigged kernel, and slow-branch uniqueness are the content of Thm. 11.64 below. As in Conj. 10.13, $\ker \widehat{\mathfrak{C}}_k$ is taken in the rigged-Hilbert sense (28)–(29): the physical state space is the distributional kernel $\mathcal{H}^{(k)}_{\mathrm{phys}} := \{\Psi_k \in \Phi'_k : \widehat{\mathfrak{C}}_k \Psi_k = 0 \text{ in } \Phi'_k\}$ on a Gel'fand triple $\Phi_k \subset \mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}} \subset \Phi'_k$ with $\Phi_k \subset \mathcal{D}_k$ a nuclear core for $\widehat{\mathfrak{C}}_k$, equipped with the group-averaging inner product $\langle \eta_k(\psi)|\eta_k(\varphi)\rangle_{\mathrm{phys}} := \langle \psi|\delta(\widehat{\mathfrak{C}}_k)|\varphi\rangle$. Hilbert-space orthogonal projection onto $\ker \widehat{\mathfrak{C}}_k$ is generically the zero operator (clock and geometry sectors carry continuous spectrum) and is not used.

The constraint is subject to the *two-boundary conditions*

$$\langle \tau_i|\Psi_k \;=\; \sqrt{w(k)}\,|\Psi^{(k)}_{\mathrm{i}}\rangle \in \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}}, \tag{67}$$

$$\langle \tau_f|\Psi_k \;=\; \big(\mathbb{1}_{\mathrm{geom}} \otimes J_k \Lambda_k^{1/2} J_k\big)|\Psi^{(k)}_{\mathrm{f}}\rangle \in \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}}, \tag{68}$$

where J_k is the modular conjugation of ω_k on its standard form (so the right-action $J_k\Lambda_k^{1/2}J_k$ realises the $\sqrt{E_k}$-effect on the right factor of the GNS purification), $|\Psi^{(k)}_{\mathrm{f}}\rangle \in \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}^{(k)}_{\mathrm{matter}}$ is the corresponding terminal state, and the sector weights $w(k) = p_0(k)$ are fixed by the seed prior (59) of Conj. 10.2. By Lem. 2.5 applied to the seed background g_0,

$$w(k) \;=\; p_0(k) \;=\; \mathrm{Tr}\big(\rho[g_0](\tau_f)\, E_k\big), \qquad \sum_k w(k) = 1, \tag{69}$$

so the law on $\mathcal{K}$ is an input from the two-boundary data in the zero-record sector, where (60) holds exactly under (T10); with a nontrivial record protocol the exact weights are the Born-map fixed point p of (x-c) of Conj. 10.2, with $w = p_0$ recovered at $O(\kappa^0)$, and the back-reaction otherwise modifies only the per-sector wavefunctions Ψ_k via the metrics g_k.

Compatibility condition. The simultaneous boundary problem (66)–(68) has a non-trivial solution in sector k iff the *strengthened* rigged boundary-to-boundary pairing (63) of (T8) is non-zero, equivalently

$$\rho^{(k)}_{\mathrm{rigg}}(0) \;:=\; \big\langle \Psi^{(k)\,\mathrm{rigg}}_{\mathrm{f}} \;\big|\; \delta(\widehat{\mathfrak{C}}_k) \;\big|\; \Psi^{(k)\,\mathrm{rigg}}_{\mathrm{i}} \big\rangle \;\neq\; 0, \tag{70}$$

where $\delta(\widehat{\mathfrak{C}}_k)$ is the spectral density of $\widehat{\mathfrak{C}}_k$ at 0 in the rigged-Hilbert sense of (29), and the test elements $\Psi^{(k)\,\mathrm{rigg}}_{\mathrm{i}}, \Psi^{(k)\,\mathrm{rigg}}_{\mathrm{f}} \in \Phi^{\mathrm{bl}}_k$ are the clock-time-smeared rigged boundary vectors of (T6) (these include the Tomita conjugation factor $J_k\Lambda_k^{1/2}J_k$ on the terminal side, so the pairing is computed on exactly the data prescribed by (67)–(68), not on the bare $\Psi^{(k)}_{\mathrm{f}}$). Under (T8) every sector with $p_0(k) > 0$ satisfies (70), so $\mathcal{K}_+ = \{k : p_0(k) > 0\}$ and the seed prior $\{p_0(k)\}_{k\in\mathcal{K}_+}$ is already a probability law on $\mathcal{K}_+$ with $w(k) = p_0(k)$ on $\mathcal{K}_+$; no rescaling of weights is required or applied. (If (T8) fails for some $k \in \mathcal{K}$ with $p_0(k) > 0$, the standing scope of Conj. 11.56 excludes that data; the conjecture is not extended to a sector-pruning fall-back regime, since such pruning would re-introduce the branch-dependent vs. branch-independent weight tension that (T12) is designed to eliminate.) On the solution space $\bigoplus_{k\in\mathcal{K}_+} \ker \widehat{\mathfrak{C}}_k$ subject to (67)–(68) the following *reduction theorem* holds.

Remark 11.57 ((T6) band-limit vs. analyticity). The earlier formulation of (T6) asserted real-analyticity of $\delta \mapsto e^{-i\delta \widehat{H}_{\mathrm{c}}/\hbar}\Psi_k$ at $\delta = 0$ as "equivalent" to a band limit on $\widehat{H}_{\mathrm{c}}$. This equivalence is false: bounded spectral support implies entire analyticity, but analytic vectors with Gaussian (or any rapidly decaying) spectral tails can be analytic without being band-limited. (T6) now imposes the band limit directly, which is the strictly stronger condition actually used in the proof of Thm. 11.64 (iii) to bound the Taylor remainder of Prop. 10.27 on $\Phi_k^{\rm bl}$ uniformly. Furthermore, δ-analyticity of the clock-translation orbit and μ-analyticity of the low-energy expansion are distinct: (T6) now states both, with the band limit guaranteeing the second via the spectral-mapping theorem applied to the polynomial expansion of (26) on $E_{\widehat{H}_{\rm c}}([-\hbar/\delta_\star, \hbar/\delta_\star])\Phi_k$. Because the exact (zero-tail) band limit is an idealization no physical state satisfies (thermal/Hadamard tails $\sim e^{-\beta E_\star}$), the physical form of this input is the ε-tail robustness hypothesis Hyp. 11.35 (Rem. 11.34, Rem. 11.36).

Remark 11.58 (Smearing of sharp-time boundary data (T6)). In Conj. 10.13 the boundary embeddings $\langle\tau_i|, \langle\tau_f| : \mathcal{H}_{\rm sys} \to \Phi$ formally use sharp clock states $|\tau_{i,f}\rangle$, which are generalised eigenvectors in Φ' and not in Φ. To make the rigged pairing (63) act on test elements rather than on distributions, (T6) replaces the sharp embeddings by Gaussian time- windowed versions

$$\iota_i(\psi) \;:=\; \int_{\mathbb{R}} g_{\sigma_t}(t-\tau_i)\,|t\rangle \otimes \psi\, dt, \qquad \iota_f(\psi) \;:=\; \int_{\mathbb{R}} g_{\sigma_t}(t-\tau_f)\,|t\rangle \otimes \psi\, dt,$$

with $g_{\sigma_t}(t) \;:=\; (2\pi\sigma_t^2)^{-1/2}e^{-t^2/(2\sigma_t^2)}$ and $0 < \sigma_t \ll \delta t$. For any $\psi \in \mathcal{H}_{\rm geom} \otimes \mathcal{H}_{\rm matter}^{(k)}$ the smeared image $\iota_{i,f}(\psi)$ is in $\Phi_k^{\rm bl}$ provided ψ has band-limited spectral content with respect to the matter+geometry generators (which is automatic on the finite-dimensional matter sector of (S3), and stable for any $\sigma_t < \hbar\delta_\star/(2\hbar) = \delta_\star/2$ in the band- limited clock sector). This is the precise sense in which the boundary-value problem (67)–(68) is well-posed in the rigged setup, removing the type mismatch between sharp-time data and nuclear test space. As $\sigma_t \to 0$ the smeared boundary recovers the formal sharp-time limit in Φ_k'.

Remark 11.59 ((T8) versus (T0)). (T8) is logically equivalent to a strengthened (T0) on the rigged boundary vectors of (T6) instead of the bare boundary vectors. The earlier hypothesis (T0) asserted a non-zero limit on the bare $(\Psi_{\rm i}^{(k)}, \Psi_{\rm f}^{(k)})$, which omitted both the clock-time smearing of (T6) and the Tomita conjugation factor $J_k\Lambda_k^{1/2}J_k$ of (68). (T8) computes the pairing on exactly the data that the boundary-value problem imposes, closing the gap between the compatibility number and the actual boundary problem; under (T6) the strengthened pairing is the operationally meaningful one. Prop. 11.66 below gives a rank-one density condition on the smeared rigged boundary vectors that is sufficient for both (T0) and (T8); under (T6), (T8) is therefore generic.

Remark 11.60 (State-relative reading of the master equation; thermal-time identification). The kinetic generator $\widehat{K}_k$ of (T3) contains the clock Hamiltonian $\widehat{H}_{\rm c}$. Under Pos. 11.11, $\widehat{H}_{\rm c}$ is not a manifold-coordinate-derived quantity but the *modular Hamiltonian of the seed state* ω:

$$\widehat{H}_{\rm c} \;=\; -\log\Delta_\omega, \qquad \sigma_t^\omega(A) \;=\; e^{-it\widehat{H}_{\rm c}/\hbar}\,A\,e^{it\widehat{H}_{\rm c}/\hbar} \quad (A \in \widetilde{\mathcal{N}}), \tag{71}$$

on the CLPW-dressed crossed product $\widetilde{\mathcal{N}} = \mathcal{N} \rtimes_{\sigma^\omega} \mathbb{R}$. The master equation (66) is therefore labelled not by a manifold coordinate but by a seed state ω; different seed states give different master equations, all on the same kinematical rigged space $\Phi_k \subset \mathcal{H}_{\rm clock} \otimes \mathcal{H}_{\rm geom} \otimes \mathcal{H}_{\rm matter}^{(k)} \subset \Phi_k'$. "Time" in (66) is the state-derived Connes–Rovelli thermal-time parameter, not a preferred coordinate; the diffeomorphism invariance of the underlying gravitational theory is preserved because (66) specifies ω, not a coordinate. In this sense the master equation is the operator-algebraic, state-relative TBRME associated with the choice ω of seed equilibrium, and the label k indexes terminal two-boundary data on top of that choice.

Remark 11.61 ("Master equation" means constraint, not Lindblad generator). The term *master equation* is used here in the Wheeler–DeWitt/constraint sense (a self-adjoint constraint $\widehat{\mathfrak{C}}_k\Psi_k = 0$ on the universal state), *not* in the Gorini–Kossakowski–Sudarshan–Lindblad (GKLS) sense of a time-local dissipative generator $\partial_t\rho = \mathcal{L}\rho$ of a completely positive trace-preserving (CPTP) semigroup [382, 381]. Equation (66) carries no dissipator, generates no one-parameter dynamical semigroup, and imposes no time-local positivity-preservation condition; the complete-positivity content of the framework resides in the *static* two-boundary conditioning channel $x \mapsto L_k^* x L_k$ of (40), rather than in (66). Per-slice non-unitarity of the projected dynamics (Prop. 10.14) is therefore not a positivity defect: the conserved unitary object lives on the history Hilbert space, and the conditioned matter state $\rho_2(\tau \mid k)$ is positive and normalised at every τ by Lem. 3.6 and Thm. 3.7.

Proposition 11.62 (Connes-cocycle covariance of the master equation)**.** *Let ω, ω' be two cyclic-separating normal states on $\mathcal{N}$ with modular operators $\Delta_\omega, \Delta_{\omega'}$ and modular Hamiltonians $\widehat{H}_{\mathrm{c}}^{(\omega)} := -\log \Delta_\omega$, $\widehat{H}_{\mathrm{c}}^{(\omega')} := -\log \Delta_{\omega'}$. Denote the Connes cocycle by*

$$u_t \;:=\; (D\omega' : D\omega)_t \;\in\; \mathcal{U}(\mathcal{N}), \qquad t \in \mathbb{R}, \tag{72}$$

with the defining intertwining property $\sigma_t^{\omega'}(A) = u_t\, \sigma_t^{\omega}(A)\, u_t^*$ for $A \in \mathcal{N}$ [39]. Let $\widehat{\mathfrak{C}}_k^{(\omega)}$ and $\widehat{\mathfrak{C}}_k^{(\omega')}$ denote the constraint operators (66) *with clock generator $\widehat{H}_{\mathrm{c}}^{(\omega)}$ and $\widehat{H}_{\mathrm{c}}^{(\omega')}$ respectively, built from the same matter sector, geometry sector, and coupling kernel $f^{\mu\nu}$, and from the same terminal effects $\{E_k\} \subset \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} \cap \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}'}}$ (jointly dressed-centralized, the two crossed products being identified by the canonical Connes-cocycle isomorphism; a non-empty condition in the centralizer-rich scope of Conj. 11.1). Assume (T0)–(T13) hold for* both *choices of seed state (so the rigged structure, band-limit, and centralizer conditions are satisfied simultaneously). Then the two constraint operators are related by the unitarily implemented Connes-cocycle map: there exists a family of unitaries $\{\widehat{U}_t^{(\omega'\leftarrow\omega)}\}_{t\in\mathbb{R}}$ on $\mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{geom}} \otimes \mathcal{H}_{\mathrm{matter}}^{(k)}$ extending $u_t \in \mathcal{U}(\mathcal{N})$ to the kinematical rigged product, satisfying the 1-cocycle identity $\widehat{U}_{s+t}^{(\omega'\leftarrow\omega)} = \widehat{U}_s^{(\omega'\leftarrow\omega)}\, \sigma_s^{\omega}(\widehat{U}_t^{(\omega'\leftarrow\omega)})$, and obeying the intertwining relation*

$$\widehat{\mathfrak{C}}_k^{(\omega')} \;=\; \widehat{U}_t^{(\omega'\leftarrow\omega)}\, \widehat{\mathfrak{C}}_k^{(\omega)}\, \big(\widehat{U}_t^{(\omega'\leftarrow\omega)}\big)^* \;+\; \Delta\widehat{H}_{\mathrm{c}}^{(\omega'\leftarrow\omega)}, \tag{73}$$

on $\mathcal{D}_k$, where $\Delta\widehat{H}_{\mathrm{c}}^{(\omega'\leftarrow\omega)} := -\log \Delta_{\omega'} + \log \Delta_\omega$ is the relative modular Hamiltonian (a self-adjoint, generically unbounded, σ^ω-affiliated operator on $\mathcal{N}$). In particular:

- (i) *The rigged physical kernel is ω-covariant: $\widehat{U}_t^{(\omega'\leftarrow\omega)}(\mathcal{H}_{\mathrm{phys}}^{(k),\omega}) = \mathcal{H}_{\mathrm{phys}}^{(k),\omega'}$ as rigged subspaces of Φ_k', so the set of master-equation solutions transforms covariantly under change of seed state.*
- (ii) *The reduction limits (R1)–(R5) of Thm. 11.68 hold with either seed-state labelling, with the semiclassical limit of Cor. 5.4 labelled by the common terminal data $(\rho_{\mathrm{i}}, \{E_k\})$ not by the modular-time label.*
- (iii) *In the thermal-equilibrium limit $\omega' = \omega$, the cocycle reduces to the identity $u_t = \mathbb{1}$, the intertwining* (73) *is trivial, and the master equation is ω-stationary.*

Thus the master equation specifies a seed state, not a coordinate; its dependence on the choice of ω is purely the modular covariance (73)*. Observers in different thermodynamic states experience different times, consistent with the Connes–Rovelli thermal-time hypothesis [33], and the master equation is compatible with all such choices simultaneously on the rigged kinematical product.*

Proof. The existence of $u_t \in \mathcal{U}(\mathcal{N})$ and its 1-cocycle property are Tomita–Takesaki theory [39, Ch. VIII]. Extension to $\widehat{U}_t^{(\omega'\leftarrow\omega)}$ on the rigged kinematical product is by standard lift of a unitary automorphism of $\mathcal{N}$ through the GNS representation and tensoring with $\mathbb{1}_{\mathrm{clock}} \otimes \mathbb{1}_{\mathrm{geom}}$: since $u_t \in \mathcal{N}$ and (T11) gives $[\widehat{H}_{\mathrm{m}}, \Delta_{\widehat{\omega}}^{it}] = 0 = [\widehat{H}_{\mathrm{m}}, \Delta_{\widehat{\omega}'}^{it}]$ (the latter by the common dressed-centralizer hypothesis $\{E_k\} \subset \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} \cap \widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}'}}$ combined with Hyp. 11.13 stated for both ω and ω'), the lift acts as $\widehat{U}_t^{(\omega'\leftarrow\omega)}|_{\mathcal{H}_{\mathrm{matter}}^{(k)}} = \pi_\omega(u_t)$ on the GNS factor and as the identity on $\mathcal{H}_{\mathrm{clock}} \otimes \mathcal{H}_{\mathrm{geom}}$. The intertwining (73) then follows by direct computation: conjugation by $\widehat{U}_t^{(\omega'\leftarrow\omega)}$ sends $\sigma_t^\omega \to \sigma_t^{\omega'}$ on the matter factor, hence $-\log \Delta_\omega \to -\log \Delta_{\omega'}$ up to the additive shift

$\Delta\widehat{H}_{\mathrm{c}}^{(\omega'\leftarrow\omega)}$ which is the content of the Connes–Radon–Nikodym relation $\sigma_t^{\omega'}(A) = u_t\sigma_t^{\omega}(A)u_t^*$ applied to unbounded $A = -\log\Delta_\omega$ via the functional-calculus extension. The other terms of $\widehat{K}_k$ ($\widehat{H}_{\mathrm{grav}}, \widehat{H}_{\mathrm{m}}^{(k)}, \alpha\widehat{\Sigma}_{\delta t}$) commute with u_t since $\widehat{H}_{\mathrm{m}}^{(k)}$ is affiliated with $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}}$ by Hyp. 11.13 and the clock-frame operators are on the clock factor; $\widehat{Q}^{(k)}[f]$ intertwines under conjugation by $\widehat{U}_t^{(\omega'\leftarrow\omega)}$ because its gravity–matter constituents $\widehat{G}^{(Q)\,\mu\nu}, \widehat{T}^{\mathrm{m}}_{\mu\nu}$ transform as tensor observables under $\sigma^\omega \to \sigma^{\omega'}$ in the dressed crossed-product algebra. (i) is immediate from the intertwining applied to $\ker\widehat{\mathfrak{C}}_k$. (ii) follows because the reduction limits (R1)–(R5) are formulated in terms of the two-boundary data $(\rho_{\mathrm{i}}, \{E_k\})$ and the rigged kernel, both of which transform covariantly under (73). (iii) is the triviality of the cocycle at fixed seed, $(D\omega : D\omega)_t = \mathbb{1}$, which is the KMS-stationarity of ω. □

Remark 11.63 (Operational meaning of state-relative time). Prop. 11.62 makes explicit that the "non-canonicity" of time in the master equation is a feature, not a defect. What used to be a protocol-relative ambiguity (Conj. 11.10 admissibility condition (ii) depending on a chosen clock-coupled protocol) is absorbed into a manifest covariance: change of seed state $\omega \to \omega'$ produces a covariant transformation of the master equation via the Connes cocycle, and any observable whose definition involves ω (modular flow, thermal-time evolution, Araki relative entropy of Rem. 11.12 (ii)) transforms accordingly. The set of solutions $\{\mathcal{H}_{\mathrm{phys}}^{(k),\omega}\}_\omega$ is a single unitary-equivalence class modulo Connes cocycles, and the physical content of Thm. 11.73 is invariant under this change of state. The Connes-cocycle covariance therefore replaces the old "canonicity" claim of Conj. 11.10 with a sharper statement: the master equation is canonical up to, and only up to, the modular cocycle of the underlying matter algebra, which is the correct operator-algebraic notion of equivalence for thermal equilibrium observers.

Theorem 11.64 (Conditional well-posedness of the master equation)**.** *Assume the standing hypotheses (T0)–(T13) of Conj. 11.56 (i.e. the operator-algebraic data $\widehat{T}_{\mathrm{m}}^{\mu\nu}[\widehat{\mathfrak{g}}; f]$, $\widehat{G}^{(Q)\,\mu\nu}[\widehat{\mathfrak{g}}; f]$, $\widehat{K}_k$ are given on a common form-core $\mathcal{D}_k$ with the stated self-adjointness, Kato–Rellich relative bound $a < 1$, symmetric domain inclusion (T7), absolute continuity of the spectral measure of $\widehat{\mathfrak{C}}_k$ at 0, the band-limited slow- branch test space $\Phi_k^{\mathrm{bl}} \subset \Phi_k \subset \mathcal{D}_k$, rigged boundary data in Φ_k^{bl}, strengthened spectral compatibility (T8), endpoint-preserving gauge (T9), shell-blind matter junction (T10), centralizer-preserving Heisenberg propagation (T11), full-coverage sample space (T12), and observer-stable metric ball (T13)). Then:*

(i) (Self-adjointness.) *The constraint operator $\widehat{\mathfrak{C}}_k = \widehat{K}_k - \kappa^{-1}\widehat{Q}^{(k)}[f]$ is essentially self-adjoint on $\mathcal{D}_k$, with unique self-adjoint extension $\overline{\widehat{\mathfrak{C}}_k}$.*

(ii) (Rigged kernel.) *The distributional kernel $\mathcal{H}_{\mathrm{phys}}^{(k)} = \{\Psi \in \Phi_k' : \widehat{\mathfrak{C}}_k\Psi = 0$ in $\Phi_k'\}$ is non-trivial; the group-averaging map $\eta_k : \Phi_k \to \mathcal{H}_{\mathrm{phys}}^{(k)}$ of* (29) *(with $\widehat{C}_{\alpha,\delta t}$ replaced by $\widehat{\mathfrak{C}}_k$) is well-defined, and $\langle\eta_k(\psi)|\eta_k(\varphi)\rangle_{\mathrm{phys}}$ is a finite, sesquilinear, positive semi-definite form on $\Phi_k \times \Phi_k$, descending to a Hilbert inner product on $\Phi_k/\ker\eta_k$. Non-triviality is witnessed by the strict non-vanishing $\rho_{\mathrm{rigg}}^{(k)}(0) \neq 0$ of (T8) for at least one (in fact every) $k \in \mathcal{K}_+$.*

(iii) (Slow-branch uniqueness.) *The restriction $\mathcal{H}_{\mathrm{phys}}^{(k),\mathrm{bl}} := \eta_k(\Phi_k^{\mathrm{bl}})$ selected by (T6) is a closed subspace of $\mathcal{H}_{\mathrm{phys}}^{(k)}$ on which the Taylor expansion of Prop. 10.27 converges uniformly (using the band-limit of (T6) in the form of Rem. 11.57); the fast branch of Rem. 10.29 lies in $\mathcal{H}_{\mathrm{phys}}^{(k)} \setminus \mathcal{H}_{\mathrm{phys}}^{(k),\mathrm{bl}}$ and is excluded by (T6). The induced family of solutions is real-analytic in $\mu = \alpha\delta t^2/\hbar^2$ at $\mu = 0$ on $\mathcal{H}_{\mathrm{phys}}^{(k),\mathrm{bl}}$ by the spectral-mapping argument of Rem. 11.57.*

(iv) (Two-boundary realisability.) *Under (T8), every $k \in \mathcal{K}_+ := \{k : p_0(k) > 0\}$ admits a non-zero $\Psi_k \in \mathcal{H}_{\mathrm{phys}}^{(k),\mathrm{bl}}$ satisfying* (67)–(68) *(in the smeared sense of (T6) and Rem. 11.58), unique up to overall scale. The sector weight is $w(k) = p_0(k)$ by* (59)*; consistency with the back-reacted dynamics g_k is automatic by* (60) *under (T10) (shell-blind matter junction)* in the zero-record sector*; with a nontrivial record protocol the identity holds at $O(\kappa^0)$ and the exact weight is the Born-map fixed point p of (x-c) of Conj. 10.2, $\|p - p_0\| \leq O(\kappa T L_U M)$.*

(v) (Sector uniqueness modulo endpoint-preserving gauge.) *On each $k \in \mathcal{K}_+$ the solution of* (66)–(68) *is unique modulo the endpoint-preserving gauge transformations of (T9), $\Psi_k \mapsto \Psi_k + \widehat{\mathfrak{C}}_k \chi_k$ with $\chi_k \in \Phi_k^{\mathrm{bl},\partial}$ satisfying* (64). *The unrestricted constraint range $\widehat{\mathfrak{C}}_k \Phi_k^{\mathrm{bl}}$ generically violates the boundary conditions and is not the relevant gauge orbit.*

(vi) (Universal sample space is a probability measure.) *Under (T12), the universal sample space $(\Omega_\gamma^{\mathrm{univ}}, \mathbb{P}_\gamma^{\mathrm{univ}})$ of* (24) *(extended to all $k \in \mathcal{K}$ with $p_0(k) > 0$, with $g_k := g_0$ on $0 < p_0(k) \le \varepsilon_{\mathrm{BFP}}$) satisfies $\mathbb{P}_\gamma^{\mathrm{univ}}(\Omega_\gamma^{\mathrm{univ}}) = 1$, and Prop. 10.7 extends in its record-relative and structural form (i)–(iv) to the full support, with the zero-record clause (iii) covering the $\varepsilon_{\mathrm{BFP}}$ sectors.*

Proof. (i) By (T2)–(T3), $\widehat{K}_k$ is essentially self-adjoint on $\mathcal{D}_k$ (Nelson + Kato–Rellich applied to the bounded clock-shift $\alpha\widehat{\Sigma}_{\delta t}$, exactly as in the Status paragraph of Conj. 10.13). By (T7), the formal product $\widehat{Q}^{(k)}[f]$ of (62) is a well-defined symmetric operator on $\mathcal{D}_k$: the domain inclusions of (T7) ensure both compositions in the symmetrization map $\mathcal{D}_k$ into $\mathcal{H}$, and the real-tensor smearing of (T2) guarantees Hermiticity of the ordering. By (T4), the resulting symmetric $\kappa^{-1}\widehat{Q}^{(k)}[f]$ is relatively bounded with respect to $\widehat{K}_k$ with relative bound $a < 1$. The Kato–Rellich theorem then gives essential self-adjointness of $\widehat{\mathfrak{C}}_k$ on $\mathcal{D}_k$ and uniqueness of the self-adjoint extension. (ii) By (T5), the spectral projection-valued measure of $\overline{\widehat{\mathfrak{C}}_k}$ admits a continuous Radon–Nikodym density at $\lambda = 0$ in the $\Phi_k \times \Phi_k$ sesquilinear sense, so the limit $\langle\psi|\delta(\widehat{\mathfrak{C}}_k)|\varphi\rangle := \lim_{\epsilon\downarrow 0}(2\epsilon)^{-1}\langle\psi|E_{\mathfrak{C}_k}([-\epsilon,\epsilon])|\varphi\rangle$ exists, is finite, and is sesquilinear and positive semi-definite by the spectral theorem (positivity uses that $E_{\mathfrak{C}_k}([-\epsilon,\epsilon])$ is a positive operator; this positivity is available precisely because $\widehat{\mathfrak{C}}_k$ is genuinely self-adjoint, which in turn requires the *symmetric* (anticommutator) ordering of the gravity–matter coupling in (62). The discarded ordering "$\widehat{T}^{1/2}\widehat{G}\widehat{T}^{1/2}$" would not even be well-defined without $\widehat{T}_{\mathrm{m}}^{\mu\nu} \succeq 0$ (false for generic matter stress-energy components), and absent self-adjointness the spectral measure of $\widehat{\mathfrak{C}}_k$ need not be positive — breaking positive-semidefiniteness of the group-averaging inner product. Positivity of the physical inner product is therefore downstream of the symmetric ordering choice). Cauchy–Schwarz on the resulting form gives the inner-product structure on $\Phi_k/\ker\eta_k$. Non-triviality follows from (T8): the strengthened pairing $\rho_{\mathrm{rigg}}^{(k)}(0) \neq 0$ on the rigged boundary vectors of (T6) is by definition a non-vanishing matrix element of $\delta(\widehat{\mathfrak{C}}_k)$ between elements of Φ_k^{bl}, hence the rigged kernel is non-trivial as a sesquilinear form on $\Phi_k^{\mathrm{bl}} \times \Phi_k^{\mathrm{bl}}$. (Note that (T5) by itself does not imply non-triviality: a continuous spectral density may still vanish identically near $\lambda = 0$, or vanish at $\lambda = 0$ on every test pair; (T8) is the additional zero-density non-vanishing condition.) (iii) The band-limit of (T6) implies $E_{\widehat{H}_{\mathrm{c}}}([-\hbar/\delta_\star, \hbar/\delta_\star])\Psi_k = \Psi_k$ on Φ_k^{bl}, hence the spectral calculus of $\widehat{\Sigma}_{\delta t} = \cos(\delta t\widehat{H}_{\mathrm{c}}/\hbar) - \mathbb{1}$ on Φ_k^{bl} has spectral support in $[-2, 0]$ with uniformly bounded Taylor remainder $\left\|(\cos(\delta t\widehat{H}_{\mathrm{c}}/\hbar) - 1 + \frac{1}{2}(\delta t\widehat{H}_{\mathrm{c}}/\hbar)^2)\Psi_k\right\| \le C\delta t^4(\hbar/\delta_\star)^4\,\|\Psi_k\|$ uniformly. This converts δ-analyticity of the clock translation orbit into μ-analyticity of the low-energy expansion of Prop. 10.27 on $\mathcal{H}_{\mathrm{phys}}^{(k),\mathrm{bl}}$ (the two notions of analyticity being related by the spectral-mapping theorem on the band-limited subspace, see Rem. 11.57). The fast branch of Rem. 10.29 has spectral support of $\widehat{H}_{\mathrm{c}}$ at $|\xi| = \sqrt{2/\mu}/\hbar \gg \hbar/\delta_\star$, hence violates the band-limit. Closedness of $\eta_k(\Phi_k^{\mathrm{bl}})$ in $\mathcal{H}_{\mathrm{phys}}^{(k)}$ follows from the closedness of the spectral projection in the rigged-Hilbert topology (group-averaging commutes with bounded spectral functions of $\widehat{H}_{\mathrm{c}}$). (iv) For each $k \in \mathcal{K}_+$, (T8) asserts (63) is non-zero on the rigged boundary vectors of (T6). The rigged pairing $\rho_{\mathrm{rigg}}^{(k)}(0) \neq 0$ defines, up to scale, the unique element of $\mathcal{H}_{\mathrm{phys}}^{(k),\mathrm{bl}}$ obtained by group-averaging $\Psi_{\mathrm{i}}^{(k)\,\mathrm{rigg}} = \iota_i(\sqrt{w(k)}\Psi_{\mathrm{i}}^{(k)})$ onto its $\widehat{\mathfrak{C}}_k$-zero-mode component: any other $\Psi_k' \in \mathcal{H}_{\mathrm{phys}}^{(k),\mathrm{bl}}$ matching the boundary data must have the same image under the linear functional $\langle\Psi_{\mathrm{f}}^{(k)\,\mathrm{rigg}}|\delta(\widehat{\mathfrak{C}}_k)|\cdot\rangle$, which on the one-dimensional zero-mode component is fixed by the compatibility number (63) up to scale. The scale is then set by $w(k) = p_0(k)$ via (59). The conservation- and Born-consistency of p_0 extended across the terminal slice is (60), valid under

(T10) in the zero-record sector: the shell-blind matter junction makes the matter Heisenberg propagator continuous across Σ_{τ_f}, so the Born trace at τ_f is unchanged by the gravitational shell. (T10) does not touch the interior record cones; there the exact Born-consistent weight is the (x-c) fixed point p, equal to p_0 at $O(\kappa^0)$, which is the caveat recorded in the statement of (iv). (v) Standard constraint-quantisation gauge invariance: any two solutions of $\widehat{\mathfrak{C}}_k\Psi = 0$ in Φ_k' that project to the same equivalence class in $\mathcal{H}^{(k),\mathrm{bl}}_{\mathrm{phys}}$ and satisfy the same boundary data (67)–(68) differ by an element of the endpoint-preserving constraint range $\widehat{\mathfrak{C}}_k\Phi_k^{\mathrm{bl},\partial}$ of (T9), since the difference must lie in $\widehat{\mathfrak{C}}_k\Phi_k^{\mathrm{bl}}$ *and* have vanishing rigged endpoint pairings, which is exactly (64). (vi) By (T12), $\mathcal{K}_+ = \{k : p_0(k) > 0\}$ has total seed-prior mass $\sum_{k\in\mathcal{K}_+} p_0(k) = 1$ on the original seed background (Lem. 2.5 on g_0). Sectors with $p_0(k) \in (0, \varepsilon_{\mathrm{BFP}}]$ are assigned $g_k := g_0$ by (T12), so the disjoint union of fibres $\Omega_\gamma^{[g_k]}|_{K=k}$ is defined for every $k \in \mathcal{K}_+$ and the weighted sum $\mathbb{P}^{\mathrm{univ}}_\gamma$ of (24) has total mass $\sum_{k:p_0(k)>0} p_0(k) = 1$. Prop. 10.7 on the adaptedness restriction (23) is unchanged on the $\varepsilon_{\mathrm{BFP}}$ sectors (where $g_k = g_0$ trivially, the zero-record clause (iii)) and holds on the back-reacted sectors in the record-relative form (i)–(ii), with the structural clause (iv) metric-independent; the conclusion (25) therefore holds on the full sample space. □

Remark 11.65 (Status of Thm. 11.64). The theorem is *conditional*: hypotheses (T1)–(T2) (existence of $\widehat{T}^{\mu\nu}_{\mathrm{m}}$, $\widehat{G}^{(Q)\,\mu\nu}$ as essentially self-adjoint smeared operators on a common core in QFT-on-curved-spacetime), (T5) (absolute continuity of the spectrum of the coupled constraint at 0), (T7) (joint domain inclusion of the gravity–matter ordering), and (T11) (centralizer- preserving matter dynamics) are open analytic problems of canonical quantum gravity, not solved here. (T8), (T10), (T12), (T13) are structural conditions on the two-boundary data, the Israel shell, the sample space, and the metric ball, respectively, that close previously identified gaps; each is genuinely independent of (T0)–(T6) and is needed for the conclusion. What is rigorous is the conditional implication: *given* all of (T0)–(T13), the master equation (66) is well-posed in the rigged-Hilbert sense, its physical state space is non-trivial on the boundary data, the slow branch is uniquely selected, the weight law is uniquely conservation- and Born-consistent and terminal-shell-stable — the seed prior exactly in the zero-record sector, its (x-c) Born-map fixed point p (equal to p_0 at $O(\kappa^0)$) under nontrivial record protocols — the gauge orbit preserves the boundary problem, and the universal sample space is a probability measure. This is the precise sense in which Conj. 11.56 is now an *airtight conditional theorem* rather than a free-standing conjecture: every previously identified inferential gap (whether about boundary-vector domains, operator-domain symmetry of the gravity–matter ordering, distinction between δ- and μ-analyticity, gauge-orbit compatibility with endpoint data, terminal-shell back-reaction on matter, centralizer escape under Heisenberg propagation, sample- space normalisation, or Part I observer stability across the metric ball) is now an explicitly named hypothesis (T6)–(T13), and the proof is a Kato–Rellich + rigged-Hilbert + spectral-calculus calculation with no remaining inferential gaps.

Proposition 11.66 (Sufficient condition for generic compatibility (T0) and (T8))**.** *In the setting of Thm. 11.64 (i)–(iii), assume in addition the* rank-one rigged density hypothesis*: for every $k \in \mathcal{K}$ with $p_0(k) > 0$, the limit*

$$\rho^{(k)}_{\mathrm{rigg}}(0) \;:=\; \lim_{\epsilon\downarrow 0}(2\epsilon)^{-1}\big\langle \Psi_{\mathrm{f}}^{(k)\,\mathrm{rigg}} \;\big|\; E_{\mathfrak{C}_k}([-\epsilon,\epsilon]) \;\big|\; \Psi_{\mathrm{i}}^{(k)\,\mathrm{rigg}}\big\rangle \text{ exists in } \mathbb{C} \text{ and is non-zero,}$$

and the diagonal densities $\rho^{(k)}_{\mathrm{rigg}}(0)[\Psi;\Psi] := \lim_{\epsilon\downarrow 0}(2\epsilon)^{-1}\langle\Psi|E_{\mathfrak{C}_k}([-\epsilon,\epsilon])|\Psi\rangle \geq 0$ exist and define a non-zero positive semi-definite sesquilinear form on Φ_k^{bl} whose kernel has codimension one in Φ_k^{bl} (rank-one zero-spectral slice on the band-limited test space; this strengthens the prior formulation, which only stipulated pre-limit non-vanishing for every ϵ but did not by itself imply that the rigged density at $\lambda = 0$ is non-zero — pre-limit non-vanishing is necessary but not sufficient, as the pre-limit may decay slower than ϵ giving a vanishing limit, or faster giving a finite but 0 *limit). Here*

$\Psi_{\mathrm{i}}^{(k)\,\mathrm{rigg}}, \Psi_{\mathrm{f}}^{(k)\,\mathrm{rigg}} \in \Phi_k^{\mathrm{bl}}$ *are the clock-time-smeared rigged boundary vectors of (T6)/Rem. 11.58 (with the Tomita conjugation factor* $J_k\Lambda_k^{1/2}J_k$ *on the terminal side). Then both (T0) and (T8) hold (including the one-dimensional zero-spectral-slice clause of (T8)): every* $k \in \mathcal{K}_+$ *satisfies the rigged boundary-to-boundary compatibility* (70)/(63)*, and* $\mathcal{K}_+ = \{k : p_0(k) > 0\}$ *without sector pruning.*

Proof. By hypothesis $\rho^{(k)}_{\mathrm{rigg}}(0)$ exists and is non-zero in $\mathbb{C}$, which is exactly (63) of (T8). The rank-one (codimension-one kernel) condition on the diagonal positive semi-definite form gives the one-dimensional zero-spectral slice clause of (T8) by Cauchy–Schwarz: the off-diagonal pairing $\rho^{(k)}_{\mathrm{rigg}}(0)[\Psi;\Phi]$ factors through the one-dimensional quotient $\Phi_k^{\mathrm{bl}}/\ker$, hence the functional $\Psi \mapsto \rho^{(k)}_{\mathrm{rigg}}(0)[\Psi_{\mathrm{f}}^{(k)\,\mathrm{rigg}};\Psi]$ has range a one-dimensional subspace of $\mathbb{C}$. The smeared pairing of (T0) is a fortiori non-zero, since it tests the same spectral density on the rigged boundary pair. ☐

Remark 11.67 (Genericity of (T0) and (T8)). The hypothesis of Prop. 11.66 is generic in a precise measure-theoretic sense: in finite-dimensional truncations $\mathcal{H}^{(N)}_{\mathrm{geom}}$ of the geometry sector, the set of two-boundary data $(\Psi_{\mathrm{i}}^{(k)\,\mathrm{rigg}}, \Psi_{\mathrm{f}}^{(k)\,\mathrm{rigg}})$ violating it has Lebesgue measure zero on the joint unit sphere of $\Phi_k^{\mathrm{bl}} \cap \mathcal{H}^{(N)}_{\mathrm{geom}}$ (the violating set is contained in a finite union of proper algebraic subvarieties cut out by orthogonality to the discrete part of $\widehat{\mathfrak{C}}_k$'s spectrum near 0, which is empty under (T5)). The same statement extends to the infinite-dimensional rigged setting modulo the standard caveats of (T5) and the band-limited condition of (T6). Hence (T0) and (T8) both hold for generic two-boundary rigged data, and the sector-pruning fall-back regime is non-generic. (As Rem. 11.59 explains, (T8) is the operationally meaningful version since it tests the actual rigged boundary problem; under (T6), Prop. 11.66 gives (T8) directly.)

Theorem 11.68 (Five-fold reduction of the master equation)**.** *Assume Conjectures 11.1, 11.4, 11.5, 11.8, 11.10 and 11.56 (including all of (T0)–(T13) of Conj. 11.56, so that Thm. 11.64 supplies rigged-Hilbert well-posedness of the master equation* (66)*). Assume the joint-fixed-point Born condition* (69) *(with the operative prior given by the (x-c) fixed point p of Conj. 10.2 under nontrivial record protocols,* $p = p_0$ *in the zero-record sector and at* $O(\kappa^0)$*), and (for (R1) only) the additional* external *choice of an admissible worldline* γ *and measurement protocol* $\mathcal{P}_\gamma$ *in the sense of Sec. 5 (the master equation by itself has no observer structure; the worldline/protocol pair is what extracts a posterior). The reductions below are* conditional regimes*: each requires the explicit auxiliary input listed in its body, and is not advertised as an unconditional consequence of* (66) *alone. Specifically: (R1) requires the external observer* $(\gamma, \mathcal{P}_\gamma)$ *and the limit* $\delta t \to 0$*; (R2) requires collapse of* $\mathcal{H}_{\mathrm{geom}}$ *to a one-dimensional sub-Hilbert space and* $\widetilde{\mathcal{N}} = \mathcal{B}(\mathcal{H}_{\mathrm{sys}})$*; (R3) requires* $\widehat{\mathsf{g}} = g_0$ *held classical; (R4) requires the chosen WKB/Born–Oppenheimer family* $\Psi_k = e^{iS_k/\hbar}\chi_k$ *and the small-entropy regime of Conj. 11.4; (R5) requires* $\widehat{H}_{\mathrm{m}}[\widehat{\mathsf{g}}] = 0$*, matter decoupling* $\widehat{T}^{\mu\nu}_{\mathrm{m}}[\widehat{\mathsf{g}}; f] = 0$*, the trivial terminal POVM* $\{E_k\} = \{\mathbb{1}\}$*, and the Hilbert-space identification* $\mathcal{H}_{\mathrm{grav}} \cong \mathcal{H}_{\mathrm{s}}$ *of Conj. 10.13. None of these auxiliary inputs is derived from* (66)*; they are formal regimes that select a sector of the universal state space. For each* $k \in \mathcal{K}_+$*, let* $\Psi_k(t, \mathsf{g}, \phi)$ *denote the k-th direct-sum component of the universal state* $\Psi = (\Psi_{k'})_{k' \in \mathcal{K}_+} \in \mathcal{U}$ *of* (58) *(no inner-product projection* $\langle k|\cdot$ *across the classical superselection sectors of* (61) *is implied or used;* Ψ_k *is simply the k-component of the tuple, in agreement with the direct-sum structure of* $\mathcal{U}$*). Then:*

(R1) ***Standard QM limit.*** *In the regime* $\delta t \to 0$ *at fixed* α *(so* $\mu = \alpha\delta t^2/\hbar^2 \to 0$*) and with* $\widehat{\mathsf{g}}$ *held at a classical background* g_0*, projecting* (66) *onto* $\langle t| \otimes \langle g_0|$ *yields* $i\hbar\partial_t\psi_k = \widehat{H}_{\mathrm{m}}[g_0]\psi_k$*, i.e. the ordinary TDSE on* (g_0, t)*. Given* $(\gamma, \mathcal{P}_\gamma)$ *the posterior process* π^γ_τ *and the retrocausal state* $\rho^\gamma_{\mathrm{RC}}$ *of Sec. 5 are constructed externally from the per-sector wavefunctions* ψ_k*; averaging* $\rho_2(\tau \mid k)$ *over k with weights* $\pi^\gamma_\tau(k)$ *recovers* $\rho^\gamma_{\mathrm{RC}}$ *and, after taking the prior expectation under* $\mathbb{P}_\gamma$*, the forward state* $\rho(\tau)$ *of Def. 2.4 (Cor. 5.4). The posterior* π^γ_τ *is* not *an output of the master equation alone.*

(R2) ***Part I.*** *In the regime where* $\mathcal{H}_{\mathrm{geom}}$ *collapses to a one-dimensional subspace spanned by a fixed classical background geometry* $|g_0\rangle$, *with* $\widehat{G}^{(Q)}_{\mu\nu}[\widehat{\mathrm{g}}]|g_0\rangle = 0$ *(interpreted as "no back-reaction" in this limit) and* $\widetilde{\mathcal{N}} = \mathcal{B}(\mathcal{H}_{\mathrm{sys}})$ *a finite- dimensional type I factor, the standard-form vector* $\Psi_k \in \mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}^*_{\mathrm{sys}} \cong \mathcal{H}^{(k)}_{\mathrm{matter}}$ *is the GNS purification of* $\rho_2(\tau \mid k)$ *on the doubled Hilbert space, and represents the density operator* $\rho_2(\tau \mid k)$ *of Def. 3.1 via the modular-conjugate partial trace* $\rho_2(\tau \mid k) = \mathrm{Tr}_{\mathrm{aux}}|\Psi_k\rangle\langle\Psi_k|$, *where* $\mathrm{Tr}_{\mathrm{aux}}$ *traces over the auxiliary GNS Hilbert-space factor* $\mathcal{H}^*_{\mathrm{sys}}$ *(the modular conjugate* J*-image of* $\mathcal{H}_{\mathrm{sys}}$, *on which the commutant* $\mathcal{B}(\mathcal{H}_{\mathrm{sys}})'$ *acts; the geometry trace is trivial here since* $\mathcal{H}_{\mathrm{geom}}$ *is one-dimensional, so the purification structure must come from the GNS doubling, not from geometry). The reduced density operator on* $\mathcal{H}_{\mathrm{sys}}$ *is generally* mixed *(it is exactly* $\rho_2(\tau \mid k)$, *which is mixed whenever* ρ_{i} *is). Given additionally the external observer data* $(\gamma, \mathcal{P}_\gamma)$ *of the conjecture's preamble (inherited from (R1); (R2) by itself produces only the per-sector conditioned state* $\rho_2(\tau \mid k)$, *not a posterior process), every result of Secs. 2–9 is recovered verbatim.*

(R3) ***Modified TDSE.*** *In the regime* $\widehat{\mathrm{g}} = g_0$ *fixed and with* $\alpha, \delta t$ *not taken to zero, projecting* (66) *onto* $\langle t|$ *yields exactly the symmetric finite-difference TDSE* (31) *(Prop. 10.14), whose low-energy effective generator is Hermitian and Planck-suppressed by Prop. 10.27.*

(R4) ***Semiclassical Einstein equation.*** *In the regime of small retrodictive entropy* $\mathcal{S}(\rho_{\mathrm{i}}, \{E_k\}) < C$ *(Conj. 11.4)* and *a chosen WKB family* $\Psi_k = e^{iS_k[\mathrm{g}]/\hbar}\chi_k[\mathrm{g}]$ *concentrated on a classical geometry* g_ω *(the Born–Oppenheimer ansatz), projecting* (66) *onto* $\mathcal{H}_{\mathrm{geom}}$*-expectation against* $|\chi_k[\mathrm{g}]|^2$ *yields* (22) *to leading order in* $\hbar$. *The Born–Oppenheimer remainder is bounded by the standard semiclassical residual* $O(\hbar\,\|\nabla S_k\|^{-1})$ *from gradient-expansion of* χ_k*; Pinsker's inequality enters separately, controlling only the trace-norm fluctuation* $\|\rho_2[g](\tau \mid k) - \rho_{\mathrm{i}}\|_1 \le \sqrt{2\,\mathcal{S}}$ *that feeds into the* k*-sector source. Combined with hypotheses (S1)–(S8) of Conj. 11.4 (the full package, not merely the original (S1)–(S4); in particular (S5) operator-Lipschitz, (S6-a)–(S6-b) junction-regularity inheritance with the explicit Barrabès–Israel null-junction operator-norm bound* $C_{\mathrm{J}}(g_0)$ *of Conj. 10.2 (vi-b), (S7) Lipschitz extension to the full interval, and (S8) seed-background contraction, all of which are required for the contraction estimate to close uniformly across the metric ball), this gives a contraction of the joint prefix/sector iteration whose fixed point is the unique record-adapted* Barrabès–Israel weak *solution of* (22)*: the assembled random metric* g_ω *of Conj. 10.2, agreeing near each* x *with the prefix metric* $g_{r(\omega,x)}$, *equal to the branch metric* $g_{K(\omega)}$ *on* $J^+(\Sigma_{\tau_f})$ *(and, in the zero-record sector, throughout), smooth off the record null cones* $\mathcal{N}_m = \partial J^+(\gamma(s_m))$ *with the prescribed null-shell data* (20), *and — only when Ax. 7.1 fails — carrying the* ϵ*-normalized terminal Lanczos jump* (21) *on* Σ_{τ_f}*; under Ax. 7.1 the terminal shell vanishes a.s. and* g_ω *is continuous across* Σ_{τ_f}. *All of this holds on the universal sample space* $\Omega^{\mathrm{univ}}_\gamma$ *of* (24). *(This is a mean-field reduction, not a variational saddle point:* (66) *is a constraint, not a stationarity condition of an action functional, so "saddle point" would be a category error. "Unique classical solution" is moreover a category error in the presence of the shells, since the metric is not smooth across the record cones (nor, absent Ax. 7.1, across* Σ_{τ_f}*); the correct notion is the Barrabès–Israel weak solution as in Conj. 10.2.)*

(R5) ***Wheeler–DeWitt sector.*** *In the regime* $\widehat{H}_{\mathrm{m}}[\widehat{\mathrm{g}}] = 0$ *together with matter decoupling* $\widehat{T}^{\mu\nu}_{\mathrm{m}}[\widehat{\mathrm{g}}; f] = 0$ *(so that the matter source of (T1) vanishes and the coupled constraint* $\widehat{\mathfrak{C}}_k = \widehat{K}_k - \kappa^{-1}\widehat{Q}^{(k)}[f]$ *reduces to the pure geometry constraint* $\widehat{K}_k = \widehat{H}_{\mathrm{c}} \otimes \mathbb{1} + \alpha\widehat{\Sigma}_{\delta t} \otimes \mathbb{1} + \mathbb{1} \otimes \widehat{H}_{\mathrm{grav}}$ *on the identification* $\mathcal{H}_{\mathrm{grav}} \cong \mathcal{H}_{\mathrm{s}}$ *of Conj. 10.13) and the sector label* k *rendered trivial by the matter-free terminal POVM* $\{E_k\} = \{\mathbb{1}\}$, (66) *reduces to the modified Wheeler–DeWitt constraint* (27) *with parameters* $\alpha, \delta t$, *while the composition-law argument of Conj. 11.5 (axioms (A)–(E)) selects the self-adjoint shift* $\widehat{\Sigma}_{\delta t}$ *uniquely. The low-energy phenomenology is governed by Rem. 10.30.*

Proof. Each reduction is a direct spectral-calculus/projection computation on the universal constraint $\widehat{\mathfrak{C}}_k\Psi_k = 0$ of (66), whose well-posedness is supplied by Thm. 11.64.

(R1). Under (T6) $\Psi_k \in \Phi_k^{\rm bl}$, so by the uniform band-limit on $\widehat{H}_{\rm c}$ the spectral calculus of $\widehat{\Sigma}_{\delta t} = \cos(\delta t\widehat{H}_{\rm c}/\hbar) - \mathbb{1}$ has Taylor remainder bound $\left\|\widehat{\Sigma}_{\delta t} + \frac{1}{2}(\delta t\widehat{H}_{\rm c}/\hbar)^2\right\| \le C\delta t^4(\hbar/\delta_\star)^4$ uniformly on $\Phi_k^{\rm bl}$ (Prop. 10.27). Holding $\widehat{\mathsf{g}} = g_0$ fixed, the kernel of $\widehat{\mathfrak{C}}_k$ on $|t\rangle \otimes |g_0\rangle \otimes \mathcal{H}_{\rm matter}^{(k)}$ projects onto $i\hbar\partial_t\psi_k = \widehat{H}_{\rm m}[g_0]\psi_k + O(\delta t^2)$; taking $\delta t \to 0$ kills the remainder and recovers the TDSE. The external posterior process π_τ^γ is constructed as in Sec. 5 from $(\gamma, \mathcal{P}_\gamma)$ and the per-sector wavefunctions ψ_k, and Cor. 5.4 gives $\mathbb{E}_{\mathbb{P}_\gamma}[\rho_{\rm RC}^\gamma(\tau)] = \rho(\tau)$.

(R2). With $\mathcal{H}_{\rm geom}$ collapsed to a one-dimensional sub-Hilbert space spanned by $|g_0\rangle$ and $\widehat{G}_{\mu\nu}^{(Q)}[\widehat{\mathsf{g}}]|g_0\rangle = 0$, the coupling $\widehat{Q}^{(k)}[f]$ of (62) vanishes on the retained sector, so $\widehat{\mathfrak{C}}_k = \widehat{K}_k$ acts only on $\mathcal{H}_{\rm clock} \otimes \mathcal{H}_{\rm matter}^{(k)}$. With $\widetilde{\mathcal{N}} = \mathcal{B}(\mathcal{H}_{\rm sys})$ a type I factor, Conj. 11.1(ii) identifies the GNS standard-form vector $\Psi_k \in \mathcal{H}_{\rm matter}^{(k)} \cong \mathcal{H}_{\rm sys} \otimes \mathcal{H}_{\rm sys}^*$ as the purification of the density operator $\rho_2(\tau \mid k)$ of Def. 3.1. The partial trace $\mathrm{Tr}_{\rm aux}|\Psi_k\rangle\langle\Psi_k|$ over the modular-conjugate factor $\mathcal{H}_{\rm sys}^*$ recovers $\rho_2(\tau \mid k)$, and all statements of Secs. 2–9 follow by the Part I proofs applied to this density operator, given the external observer data $(\gamma, \mathcal{P}_\gamma)$.

(R3). With $\widehat{\mathsf{g}} = g_0$ held classical, the gravity sector drops and $\widehat{\mathfrak{C}}_k = \widehat{H}_{\rm c} + \widehat{H}_{\rm m}[g_0] + \alpha\widehat{\Sigma}_{\delta t}$. Projection onto $\langle t|$ yields exactly the symmetric finite-difference TDSE (31) of Prop. 10.14; the low-energy effective Hamiltonian is Planck-suppressed and Hermitian by Prop. 10.27.

(R4). Fix a WKB/Born–Oppenheimer family $\Psi_k = e^{iS_k[\mathsf{g}]/\hbar}\chi_k[\mathsf{g}]$ concentrated on g_ω. Taking the $\mathcal{H}_{\rm geom}$-expectation of $\widehat{\mathfrak{C}}_k\Psi_k = 0$ against $|\chi_k[\mathsf{g}]|^2$ yields the Einstein-tensor expectation $\langle\widehat{G}_{\mu\nu}^{(Q)}\rangle = \kappa\langle\widehat{T}_{\mu\nu}^{\rm m}\rangle$ at leading order in $\hbar$, with Born–Oppenheimer remainder $O(\hbar\,\|\nabla S_k\|^{-1})$. Under small retrodictive entropy $\mathcal{S} < C$ (Conj. 11.4) Pinsker's inequality controls $\|\rho_2[g](\tau \mid k) - \rho_{\rm i}\|_1 \le \sqrt{2\,\mathcal{S}}$; combined with (S1)–(S8) of Conj. 11.4 the contraction (vii)/(x-a) of Conj. 10.2 closes uniformly on $\mathcal{B}_{g_0}$, prefix by prefix along the record tree, and supplies the unique record-adapted Barrabès–Israel weak solution $g_\omega \in \mathcal{B}_{g_0}$ of (22) — the causal-wedge assembly of the prefix metrics $g_{r(\omega,x)}$, equal to $g_{K(\omega)}$ on $J^+(\Sigma_{\tau_f})$ and continuous across Σ_{τ_f} under Ax. 7.1 — on the universal sample space (24).

(R5). Setting $\widehat{H}_{\rm m} = 0$ and $\widehat{T}_{\rm m}^{\mu\nu} = 0$ eliminates the matter sector from (66), leaving the pure-geometry constraint $\widehat{H}_{\rm c} \otimes \mathbb{1} + \alpha\widehat{\Sigma}_{\delta t} \otimes \mathbb{1} + \mathbb{1} \otimes \widehat{H}_{\rm grav}$. Under the trivial terminal POVM $\{E_k\} = \{\mathbb{1}\}$ the sector label k is trivial, and Conj. 10.13 identifies $\mathcal{H}_{\rm grav} \cong \mathcal{H}_{\rm s}$ to give the modified Wheeler–DeWitt constraint (27). Axioms (A)–(E) of Conj. 11.5, applied as in the witness computation at the end of Sec. 11.1, select $\widehat{\Sigma}_{\delta t}$ uniquely up to the trivial rescaling. □

Remark 11.69 (What is new and what is not). None of (R1)–(R5) is new on its own: (R1), (R3), (R5) have been discussed in some form in the canonical quantum gravity literature since [8]; (R4) is the content of the semiclassical Einstein programme. What the master equation proposes is that a *single* constraint (66) with *single* pair of two-boundary conditions (67)–(68) has all five reductions as limits, and that Part I of this paper is already the rigorous finite-dimensional content of (R2). The novel ingredients are (a) the presence of the terminal-label factor $\ell^2(\mathcal{K})$ and the two-boundary condition (68), which encode retrocausality at the level of the constraint rather than as a post-hoc conditioning step; and (b) the self-adjoint symmetric shift $\alpha\widehat{\Sigma}_{\delta t}$, whose operator form is selected by the composition-law argument of Conj. 11.5 and whose Planck-suppressed low-energy effective Hamiltonian is established in Prop. 10.27 (Rem. 10.30).

Remark 11.70 (Why this is not a "God equation"). Equation (66) is not a theory of everything. It *presumes*: the existence of a constructed $(\mathcal{H}_{\rm geom}, \widehat{H}_{\rm grav})$, a renormalised stress-energy operator on curved spacetime, a well-defined operator ordering for $\widehat{G}_{\mu\nu}^{(Q)}$, and the five conjectures of Sec. 11.1. Each of these is a substantial open problem. The master equation is the concise target of a programme, not a derivation of physics from nothing. We record it because, given the structure of Part I, it is the *unique* form consistent with every reduction (R1)–(R5), and because writing it down makes precise what a full theory in this framework would have to be.

Remark 11.71 (Falsifiability). Even as a conjecture, (66) makes a quantitative low-energy prediction through the dimensionful combination $\mu := \alpha\delta t^2/\hbar^2$ (with units of inverse energy). Projecting to the matter sector gives the symmetric finite-difference TDSE (31), and Prop. 10.27 shows that the leading correction to the matter Hamiltonian is the Hermitian operator $-\frac{\mu}{2}\widehat{H}_{\mathrm{s}}^2$ with relative size $\mu\langle\widehat{H}_{\mathrm{s}}\rangle/2$. The natural Planck identification $\alpha = E_{\mathrm{Pl}}$, $\delta t = t_{\mathrm{Pl}}$ gives $\mu = 1/E_{\mathrm{Pl}}$, hence a relative correction $\langle\widehat{H}_{\mathrm{s}}\rangle/(2E_{\mathrm{Pl}}) \sim 10^{-28}$ for laboratory energies—compatible with current precision-spectroscopy and optical-clock bounds on unitary deviations [19]. The asymmetric forward-shift form (Rem. 10.16), with dimensionless $\lambda := \alpha\delta t/\hbar$, would produce an order-one non-Hermitian factor $(1+i\lambda)^{-1}$ multiplying $\widehat{H}_{\mathrm{s}}$ and is excluded for $|\lambda| \sim 1$ by the same data; it is not the form predicted by (66). Genuinely testable signatures of (66) therefore live either in quadratic-in-energy spectroscopic shifts at the $\langle\widehat{H}_{\mathrm{s}}\rangle/E_{\mathrm{Pl}}$ level, or in strong-gravity regimes where $\mu\langle\widehat{H}_{\mathrm{s}}\rangle$ is no longer Planck-suppressed.

The Master Unification Theorem

We now consolidate Thm. 11.64 (well-posedness) and Thm. 11.68 (five-fold reduction) into a single statement that is the central conditional result of Part II. Let H denote the following *hypothesis bundle*, written out explicitly so that no unlisted assumption enters:

(H0) **EFT scope.** Pos. 11.30 of Sec. 11.4 holds: all physical matter states, terminal effects $\{E_k\}$, initial state ρ_{i}, and Heisenberg propagation are restricted to the range of $P_{\le E_\star}$ for a fixed cutoff $E_\star \lesssim E_{\mathrm{P}}$. The matter Hilbert space on any compact spatial region is finite-dimensional, the matter Hamiltonian has discrete spectrum in $[0, E_\star]$, and the modular Hamiltonian K_ω of Pos. 11.11 is a bounded self-adjoint operator on the GNS space of $\omega\restriction_{P_{\le E_\star}\mathcal{N}P_{\le E_\star}}$. *(Co-scoping.)* This finite-dimensional band-limited compression applies to the *matter* sector only; the clock and gravity/horizon sectors retain their infinite-dimensional (type III_1 / crossed-product) structure. The type-III hypotheses below are accordingly asserted each on the algebra it is a predicate about: (H3)'s Tomita conditioning (Conj. 11.1) and (H4)'s horizon-centraliser structure (Conj. 11.8) live on the *uncompressed* algebra $\mathcal{N}$ and its crossed product, whereas the finite-dim reductions of (H4) discharged by Prop. 11.33 and the record-completeness of (H1) operate on the compressed $\mathrm{Ran}(P_{\le E_\star})$; the two are related by the observer compression Π, not identified. On the compressed matter sector the type-III subtleties of (H3) reduce to finite-dim linear algebra and are correspondingly weaker there; they do real work only on the uncompressed clock/gravity algebras. The witness (H6) is finite-dim in matter and infinite-dim in clock/gravity by construction.

(H1) **Part I axioms.** Ax. 2.1 (two-boundary data), Ax. 2.2 (local algebras, (M1)–(M3)), and Ax. 7.1 (terminal record-completeness) hold on the fixed globally hyperbolic slab $\mathcal{M}_{[\tau_i,\tau_f]}$, restricted to $\mathrm{Ran}(P_{\le E_\star})$ under (H0).

(H2) **Gravity bridges.** Conj. 10.2 (final-state semiclassical Einstein equation, including the full hypothesis package (i)–(x) — in particular the null-junction solvability (vi-b), the observer stability (ix), and the record-tree closure (x-a)–(x-d) — the branch postulate (B), the record postulate (R) with its record-permanence admissibility clause (RP), and the causal adaptedness restriction (23)) and Conj. 10.13 (modified Wheeler–DeWitt constraint with its stated canonical-quantisation hypotheses) hold.

(H3) **Speculative closures (reduced bundle).** Under the two structural postulates — Pos. 11.11 (thermal-time) of Sec. 11.2 and Pos. 11.18 (entropic-gravity) of Sec. 11.3, which we adopt as primitives of (H3) — the following shrinks the original five-conjecture closure: Conj. 11.1 (Tomita two-boundary conditioning, now *with (CP) replaced by the consolidated Hyp. 11.13*), Conj. 11.4 (relative-entropy smallness, with (S3) gap now a theorem by Prop. 11.33(iii) under Pos. 11.30, the remainder (S1)–(S2), (S4)–(S8) restricted to the conditioning-perturbation contraction; coupling-existence is absorbed by Pos. 11.18), Conj. 11.5 (composition-law selection of the clock shift via (A)–(E)), and Conj. 11.8 (horizon subalgebra as modular centralizer, with (KE)). Conj. 11.10 is *not* listed: under Pos. 11.11 it is a theorem (Prop. 11.15). Conj. 10.2 is *not* listed as an independent hypothesis either: its coupling-existence content is absorbed into Pos. 11.18 (Prop. 11.26), and the residual contraction-estimate content is inherited via Conj. 11.4. All listed items are taken on the common scope of Conj. 11.55.

(H4) **Master-equation structural hypotheses (reduced).** The residual hypotheses of Conj. 11.56 hold for every $k \in \mathcal{K}_+$: (T0), (T3), (T4), (T8)–(T10), (T12)–(T13). (T1), (T2) are promoted to theorems by Prop. 11.26 under Pos. 11.18, and (T7) by the same postulate (Rem. 11.27; independently Prop. 12.3 under Pos. 11.30). (T11) is consolidated with (CP) into the single commutation Hyp. 11.13 under Pos. 11.11. (T5) spectral absolute continuity, (T6) band-limited selection, and the nuclearity portion of (T8) are promoted to theorems by Prop. 11.33 under Pos. 11.30 of (H0). Under this cumulative reduction, Thm. 11.64 applies and Thm. 11.68 has its hypotheses met.

(H5) **Regime data.** The auxiliary regime inputs of Thm. 11.68 are available as external formal data: for (R1) a worldline γ with measurement protocol $\mathcal{P}_\gamma$ and the limit $\delta t \to 0$; for (R2) collapse of $\mathcal{H}_{\rm geom}$ to a one-dimensional subspace and $\widetilde{\mathcal{N}} = \mathcal{B}(\mathcal{H}_{\rm sys})$; for (R3) $\widehat{\mathsf{g}} = g_0$ held classical; for (R4) a chosen WKB/Born–Oppenheimer family $\Psi_k = e^{iS_k/\hbar}\chi_k$ together with Conj. 11.4; for (R5) $\widehat{H}_{\rm m} = 0$, $\widehat{T}^{\mu\nu}_{\rm m} = 0$, trivial terminal POVM, and the identification $\mathcal{H}_{\rm grav} \cong \mathcal{H}_{\rm s}$ of Conj. 10.13.

(H6) **Non-vacuous common scope.** The common-scope witness model exhibited in the final bullet of Conj. 11.55 (de Sitter static patch with spectrally-cut-off thermal matter, Page–Wootters clock generated by the boost, de Sitter horizon, and the spectral-theorem verification of (A)–(E)) is admitted, so that (H0)–(H5) are jointly realisable on at least one concrete background. The de Sitter witness satisfies Pos. 11.30 canonically via its spectral cutoff $P_{\le E_\star}$ on the wedge thermal state, which is the origin of the witness's finite-dimensional matter sector.

Remark 11.72 (Named residual: the H0 co-scoping / bare–dressed compatibility). The bundle uses one seed-modular structure in two scopes that H0 treats differently, and their compatibility is a load-bearing assumption we now name explicitly rather than absorb silently. On the *matter* leg, H0 compresses to $\mathrm{Ran}(P_{\le E_\star})$, where Tomita–Takesaki reduces to finite-dim linear algebra and the dressed clock generator is bounded with discrete spectrum — this is what Prop. 11.33 (T5),(T6),(S3) and Prop. 11.26 operate on. On the *clock/gravity/horizon* legs, the construction retains the uncompressed type III_1 modular flow of $-\log\Delta_\omega$ — *unbounded*, full-line spectrum — which the thermal-time clock, the Pauli time-of-arrival evasion (Thm. Y.1), the de Sitter boost witness, the KMS/half-sided-modular conservation law (Pos. 11.18, EG1–EG2), and the ANEC route all require. These two readings of the same symbol $\widehat{H}_{\rm c} = -\log\Delta_\omega$ are *not* identified; they are related by the observer compression Π. **The assumption (H0-co-scoping):** *the compression* Π *preserves, on the clock/gravity legs, the modular/KMS data those legs require, so that the bounded-generator finite-dim rigour (matter) and the unbounded-generator thermal time (clock/gravity) hold simultaneously without collision.* This is stated, not proved. Its sharpest stress point is EG1, where the type III modular density $-\log\Delta_\omega$ is smeared against a *matter* stress tensor that H0 wants finite-dim — exactly the compressed/uncompressed straddle. We flag this as a residual on the same footing as the other named items of Rem. 12.16: the master implication is stated *modulo* the co-scoping assumption, and a proof that Π is KMS-data-preserving (or a counterexample) is an open problem the framework does not settle.

Theorem 11.73 (Master Unification Theorem)**.** *Assume* H=(H0)–(H6)*. Then the Two-Boundary Retrocausal Master Equation* (66) *on the universal state space* $\mathcal{U}$ *of* (58) *with two-boundary rigged data* (67)–(68) *is a single well-defined mathematical object with the following simultaneous properties:*

(U1) ***Well-posedness.*** *The constraint $\widehat{\mathfrak{C}}_k\Psi_k = 0$ is essentially self-adjoint on the common core $\mathcal{D}_k$ of (T1); the rigged-Hilbert kernel on $\Phi_k^{\mathrm{bl}} \subset \Phi_k$ is a non-trivial subspace of $\mathcal{H}_{\mathrm{phys}}^{(k),\mathrm{bl}}$; the slow branch is uniquely selected by (T6); every $k \in \mathcal{K}_+$ admits a non-zero solution satisfying the two-boundary data, unique up to scale modulo the endpoint-preserving gauge of (T9); the universal sample space* (24) *has total mass one under the operative prior (the seed prior p_0 in the zero-record sector, its (x-c) Born fixed point p — with $\sum_k p(k) = 1$ and $p = p_0$ at $O(\kappa^0)$ — under nontrivial record protocols; see the gloss after* (24)*). (Thm. 11.64.)*

(U2) ***Quantum-mechanical reduction.*** *Projecting* (66) *onto a fixed classical background $\langle g_0|$ and taking the $\delta t \to 0$ limit yields the ordinary Schrödinger equation $i\hbar\partial_t\psi_k = \widehat{H}_{\mathrm{m}}[g_0]\psi_k$; given the worldline/protocol data of (H5), the Part I retrocausal-state process $\rho_{\mathrm{RC}}^{\gamma}$ and its posterior martingale π_τ^{γ} are constructed externally, and Cor. 5.4 gives $\mathbb{E}_{\mathbb{P}_\gamma}[\rho_{\mathrm{RC}}^{\gamma}(\tau)] = \rho(\tau)$. (Thm. 11.68, (R1)–(R2).)*

(U3) ***Semiclassical general-relativistic reduction.*** *In the small-entropy WKB regime $\mathcal{S}(\rho_{\mathrm{i}}, \{E_k\}) < C$ of Conj. 11.4, the $\mathcal{H}_{\mathrm{geom}}$- expectation of* (66) *against $|\chi_k[\mathrm{g}]|^2$ reduces to the final-state semiclassical Einstein equation* (22) *with unique record-adapted Barrabès–Israel weak solution $g_\omega \in \mathcal{B}_{g_0}$ on the universal sample space of* (24)*: near each x, g_ω agrees with the prefix metric $g_{r(\omega,x)}$, equals the branch metric $g_{K(\omega)}$ on $J^+(\Sigma_{\tau_f})$ (and, in the zero-record sector, throughout), and is smooth off the record null cones $\mathcal{N}_m$, whose shells carry the null data* (20)*; the terminal ϵ-normalized Lanczos jump* (21) *arises only when Ax. 7.1 fails, and under Ax. 7.1 (assumed in (H1)) g_ω is a.s. continuous across Σ_{τ_f}. (Thm. 11.68, (R4).)*

(U4) ***Quantum-geometry reduction.*** *In the matter-decoupling limit $\widehat{H}_{\mathrm{m}} = 0$, $\widehat{T}_{\mathrm{m}}^{\mu\nu} = 0$ with trivial terminal POVM,* (66) *reduces to the modified Wheeler–DeWitt constraint* (27) *on $\mathcal{H}_{\mathrm{grav}} \cong \mathcal{H}_{\mathrm{s}}$, and the composition-law axioms (A)–(E) of Conj. 11.5 select the self-adjoint clock-shift $\alpha\widehat{\Sigma}_{\delta t}$ uniquely up to the trivial rescaling. (Thm. 11.68, (R5).)*

(U5) ***Modified-TDSE reduction.*** *With $\widehat{\mathrm{g}} = g_0$ held classical and $\alpha, \delta t$ retained, projection of* (66) *onto $\langle t|$ yields exactly the retrocausal symmetric finite-difference TDSE* (31) *(Prop. 10.14; derivation and effective form in App. C), with Planck-suppressed Hermitian low-energy effective Hamiltonian (Prop. 10.27). (Thm. 11.68, (R3).)*

(U6) ***Non-vacuity (of scope).*** *The joint scope of (U1)–(U5) is non-empty: the de Sitter static-patch witness model of (H6) realises all of (H0)–(H5), in particular supplies a concrete $(\mathcal{H}_{\mathrm{clock}}, \mathcal{H}_{\mathrm{geom}}, \mathcal{H}_{\mathrm{matter}}^{(k)})$ on which* (66) *is meaningful and the five reductions are computable. This is* scope *non-emptiness, a property of the de Sitter construction; it is* distinct *from non-vacuity* of the two-boundary phenomenon *($f_k > 0$), which is the separate finite-dimensional theorem Prop. 11.53 and is* not *here shown to be realised inside the* same *de Sitter model that meets the five scopes — their coexistence is the open (no known obstruction) matter-supported $\wedge$ flow-invariant conjunction of Rem. L.7.*

In particular, ordinary nonrelativistic/relativistic quantum mechanics (U2) and general relativity in its semiclassical Einstein form (U3) are simultaneous *reductions of the single equation* (66) *on the single universal state space $\mathcal{U}$ (in the precise sense delineated in Rem. 11.74 below). This is the conditional unification claimed in the title and abstract.*

Proof. (U1) is Thm. 11.64 applied under (H4); that theorem's hypotheses are exactly (T0)–(T13) of Conj. 11.56, which is (H4). (U2)–(U5) are (R1)–(R2), (R4), (R5), (R3) respectively of Thm. 11.68; that theorem's hypotheses are Conj. 11.1, 11.4, 11.5, 11.8, 11.10, 11.56, the joint-fixed-point Born condition (69), and the regime data of its statement, all of which are enumerated in (H2), (H3), (H4), (H5). The Planck-cutoff scope of (H0) supplies Prop. 11.33, which discharges (T5), (T6), (S3), and the nuclearity portion of (T8) as theorems within the scope of H — *modulo the H0 co-scoping residual* (Rem. 11.72): the bound on the dressed clock generator underlying (T6)/(T5) holds on the compressed corner provided the observer compression Π preserves the clock-sector modular data, which is the standing co-scoping assumption — so (H4) is satisfied as stated. (U6) is the witness model of Conj. 11.55's *scope* non-vacuity bullet, admitted as (H6), canonically satisfying (H0) via the wedge spectral cutoff (it establishes scope non-emptiness, not the two-boundary phenomenon $f_k > 0$, which is the separate Prop. 11.53). No additional hypothesis *beyond* H *and the co-scoping residual* is used in the deduction of any (Un); the implication $\mathsf{H} \Rightarrow$ (U1) & ... & (U6) is therefore unconditional *as an implication* (modulo co-scoping; see Rem. 11.75 and Rem. 11.72 for the separate question of whether H is satisfiable at cosmological scale). □

Remark 11.74 (In what sense "simultaneous"). "Simultaneous" means *co-derivable from the one constraint* $\widehat{\mathfrak{C}}_k$ *on the one state space* $\mathcal{U}$ — not that both exact faces are non-trivially present in a single regime. The coupling operator $\widehat{Q}^{(k)}[f] = \frac{1}{2}\{\widehat{G}^{(Q)\mu\nu}, \widehat{T}^m_{\mu\nu}\}$ is a genuine, non-factorised matter–geometry interaction (it does not commute with the clock projector and does not block-diagonalise), so this is *not* the trivial "$A{=}0 \wedge B{=}0$" conjunction. The structure is a correspondence-principle one: (U2) exact QM is the frozen-geometry limit ($\widehat{Q} \to 0$, $\mathcal{H}_{\rm geom}$ collapsed), while (U3) semiclassical Einstein is the WKB dynamical-geometry limit (where $\widehat{Q}$'s expectation *is* the source), and these two *exact* faces are complementary limits, not co-present — just as Newtonian gravity and special relativity are complementary limits of general relativity. Where they *do* genuinely couple is the semiclassical middle (U3/U5 together): there the metric g_ω is sourced by the branch-conditioned $\langle \widehat{T} \rangle$ *while* matter evolves by the modified TDSE on g_ω — a single coupled system in which a Schrödinger evolution and the Einstein equation hold at once. The framework's genuinely *new* content (the retrocausal two-boundary terminal condition and the composition-law-selected clock shift) is load-bearing for the QM/TDSE arm and *not* for the GR arm (R4 survives $\alpha{=}0$, $\{E_k\}{=}\{\not\Bbbk\}$): the GR arm is competently-rendered but standard semiclassical gravity. The unification's content is therefore "one constraint whose limits are QM and GR, genuinely coupled in the semiclassical regime," a *sectored/correspondence* unification — stronger than a bookkeeping conjunction, weaker than "both exact faces in one regime."

Remark 11.75 (Precise meaning of "conditional unification"). Thm. 11.73 is the conditional implication $\mathsf{H} \Rightarrow$ unification. The conditional implication itself is an unconditional theorem: its proof uses no hypothesis beyond H. Every step is either a direct invocation of Thm. 11.64 or Thm. 11.68, or a bookkeeping identification of hypotheses. What is conjectural is H itself — more precisely, the Planck-cutoff postulate (H0), items (H2) (semiclassical Einstein coupling and modified Wheeler–DeWitt constraint), (H3) (the four residual speculative closures), and the residual structural items of (H4) not discharged by Prop. 11.33 or Prop. 11.26. Notably, (T5), (T6), (T1), (T2), (T7), and the centralizer conditions flagged earlier in Rem. 11.65 (existence of essentially self-adjoint smeared $\widehat{T}^{\mu\nu}_{\rm m}, \widehat{G}^{(Q)\,\mu\nu}$, spectral absolute continuity of $\widehat{\mathfrak{C}}_k$ at 0, joint domain inclusion of the gravity–matter ordering, centralizer-preserving matter Heisenberg propagation) are no longer in the residual open list under the three postulates (H0), Pos. 11.11, Pos. 11.18. What remains open is the contraction estimate of Conj. 11.4, the composition-law selection of Conj. 11.5, the horizon-algebra structure of Conj. 11.8, the non-(CP) parts of Conj. 11.1, and the rigged-Hilbert structural conditions (T0), (T3), (T4), (T8)–(T10), (T12), (T13) of Conj. 11.56. A resolution of these open problems *jointly* upgrades Thm. 11.73 from conditional to unconditional, and then QM and GR are simultaneously reductions of (66). We do not claim such a joint resolution; we claim, and prove as a theorem, the conditional implication. This is what "conditional unification of QM and GR" means in this paper: the five reductions (U2)–(U5) hold *if* H does, and the "if" is made arithmetically precise.

"Unconditional" means unconditional as an implication, not bundle-satisfiability. The theorem asserts $\mathsf{H} \Rightarrow$ (U1)–(U6) using no hypothesis beyond H; it does *not* assert that H is jointly satisfiable at every scale, and that satisfiability is not uniform across scales. At finite/laboratory scale the bundle is satisfiable unconditionally: the finite-dim witness Prop. 11.53 exhibits a two-boundary datum with $f_k > 0$ realising the operative content, independent of the type III / Cartan machinery. At *cosmological* scale, joint satisfiability of (H1) and (H2) rides on the CHT terminal-masa selection (clause (x-d) of Conj. 10.2, tied to Conj. L.1) and hence on the flow-

invariance vise of Rem. L.7: this is currently *genuinely open, with no known obstruction but no positive evidence for the given flow* (the horizon/boost flow is inner, so the question is the tractable conjugacy problem of whether the dressed-energy group normalises a matter-crossing Cartan; normaliser-flow constructions show the target set of good flows is non-empty but not that the *given* flow is in it, and the central-abelian structure of the flow gives a mild adverse hint). Were the vise ever proven obstructed, Ax. 7.1 would be unsatisfiable at cosmological scale (Rem. L.3, outcome (i)), rendering H vacuous *there* — but the implication would remain a valid theorem, and the finite-dim/laboratory arm (including the falsifiable modified-TDSE predictions) would be untouched. The unification claim is therefore unconditional as an implication and non-vacuous unconditionally at laboratory scale; its cosmological non-vacuity is the one genuinely open frontier.

Sec. 12 (Part III) sharpens this remark: all of (T0), (T3), (T4), (T7)–(T10), (T12), (T13) are theorems of the present paper under the three postulates and Hyp. 12.9 (Props. 12.1, 12.3, 12.5, 12.7, 12.11, 12.13) Conj. 11.5 is Thm. B.5 (App. B); and the remaining four conjectures of Part II — Conj. 11.4 (QFT-regime), Conj. 11.8 (dynamical), Conj. 11.1 (non-(CP) portion), and the Planck-cutoff Pos. 11.30 (CLPW derivation) — are reduced, of the original external set (E1)–(E4) of Sec. 12.5, to the two surviving residuals (E2a),(E3a): (E1) is retired into the Planck-cutoff postulate H0, (E4) is closed unconditionally, and (E2),(E3) shrink to (E2a),(E3a). The sharpened master theorem Thm. 12.15 is conditional modulo (E2a),(E3a) plus the single primitive postulate Pos. 11.18 (item (E0)) and the H0 co-scoping residual (Rem. 11.72); the residual hypothesis bundle H' of the sharpened theorem contains no bespoke conjecture of the present paper beyond the named structural hypotheses of the record-adapted layer ((vi-b), (x-a), (x-c), (x-d), (RP)) and the shell-summability residual (E5), all of which are enumerated in Rem. 12.16.

Remark 11.76 (Why this is the unique form). Given (H1)–(H3), every structural choice entering the master equation is forced: the kinetic generator $\widehat{K}_k$ is the clock–geometry–matter sum forced by the Page–Wootters clock and the Wheeler–DeWitt constraint; the symmetric shift $\alpha\widehat{\Sigma}_{\delta t}$ is forced (up to rescaling) by (A)–(E) of Conj. 11.5 via Prop. 11.6; the symmetric ordering of the gravity–matter coupling (62) is forced by self-adjointness together with (T7); the universal state space $\mathcal{U}$ with the classical superselection structure (61) is forced by the fact that $\mathcal{K}$ is a classical record; the two-boundary rigged data are forced by the Tomita lift of Def. 3.1 under Conj. 11.1(ii). (66) is therefore not *a* conjectural unification but the unique one consistent with (H1)–(H3). Consequently, if any alternative unified equation exists, it must violate at least one of the five speculative conjectures of Sec. 11.1; every such alternative is then either out of scope here or is a direct falsifier of one of (H2)–(H3).

Revised summary. Sections 2–7 constitute a rigorous probabilistic/operator-algebraic framework. Within its stated scope — finite-dimensional matter on a fixed globally hyperbolic background with two-boundary data — every theorem is proved from the explicitly stated axioms, with no unlisted hypotheses. The principal load-bearing axioms are the causal-propagation axiom (M3) of Axiom 2.2 (a postulate in the present setting; a theorem in fully local QFT) and the terminal record-completeness Axiom 7.1. Sections 10–11.6 are the conjectural Part II, whose central output is Thm. 11.73 (Master Unification Theorem): *conditionally on the explicitly enumerated hypothesis bundle* H, *the Two-Boundary Retrocausal Master Equation* (66) *on the universal state space* $\mathcal{U}$ *is a single well-defined constraint whose simultaneous reductions are ordinary quantum mechanics, the rigorous Part I conditioned quantum mechanics, the semiclassical Einstein equation, the modified Wheeler–DeWitt constraint, and the retrocausal modified TDSE*. We claim this conditional unification as a theorem of the present paper, in the precise sense of Rem. 11.75: the implication $\mathsf{H} \Rightarrow$ unification is proved unconditionally, and the unconditional truth of the unification reduces to the explicit open analytic and structural problems packaged in (H2)–(H4). Every ingredient of H is stated with the precise analytic conditions that would elevate it, each of the five reductions is derived from a direct spectral-calculus or projection argument, and a concrete witness (de Sitter static patch) shows the common scope is non-empty. Conditional unification is what Part II delivers; unconditional unification modulo the residual external programme is what Part III (Sec. 12) delivers via the sharpened master theorem Thm. 12.15: after the structural discharge of every (Tn) of Conj. 11.56 to a theorem of this paper, the only remaining internal postulate is the matter–geometry coupling (E0) — in the scope-split form of App. I, the nonlinear axiom Ax. I.10 (E0-lin being a theorem, Thm. I.1) (plus the H0 co-scoping residual, Rem. 11.72), and the only remaining external items are the two surviving residuals (E2a),(E3a) of Sec. 12.5 (the original external set (E1)–(E4) having been discharged: (E1)→H0, (E4) closed, (E2),(E3)→(E2a),(E3a)).

12 Sharpening the hypothesis bundle: residual structural discharge and the external programme

The Master Unification Theorem 11.73 is conditional on the bundle H =(H0)–(H6), of which the speculative closures (H3) and the structural items of (H4) not already discharged by Prop. 11.33, Prop. 11.26, and Hyp. 11.13 remain open. In this section we *tighten* H by (a) promoting each structural (T)-condition of Conj. 11.56 that is tractable under the three postulates Pos. 11.11, 11.18, 11.30 to a theorem of this paper, and (b) reducing each of the remaining speculative conjectures of (H3) to a single named open problem of the ambient operator-algebraic/QFT-on-curved-spacetime literature. The outcome is a *sharpened* master unification theorem (Thm. 12.15) whose residual hypothesis bundle is dominated by the primitive postulates Pos. 11.18 (the Jacobson–Bianchi–Myers Clausius identification, later scope-split in App. I) and Pos. 11.11, together with the named external open problems of current quantum gravity research and the named structural hypotheses of the record-adapted layer; the complete reconciled list is Rem. 12.16. This is the strongest honest form the unification claim of the present paper can take today: every conjectural ingredient that is internal to the framework is either discharged or carried as an explicitly named commitment; what remains is either an operationally clean axiom about matter–geometry coupling, a named structural hypothesis of the record-adapted gravitational layer, or an explicit reduction to a named problem outside the framework's scope.

We do not claim the residual external problems of Sec. 12.5 below are solved; we claim, and prove as a theorem in this section, that *if* each of them is resolved within its own ambient literature in the form stated, then Thm. 12.15 becomes unconditional modulo Pos. 11.18. The distinction from Thm. 11.73 is that the reduction of H is now *external*: each residual item is a specified open problem of the ambient literature, not a bespoke conjecture of the present framework.

12.1 Structural (T)-conditions discharged as theorems under (H0)

Under Pos. 11.30 (the (H0) postulate), six of the thirteen structural conditions (T0)–(T13) of Conj. 11.56 are theorems of finite-dimensional linear algebra and classical operator theory rather than open analytic hypotheses. Prop. 11.33 already discharges (T5), (T6), (S3) of Conj. 11.4, and the nuclearity clause of (T8). In this subsection we promote (T0), (T7), (T10), and (T13) to theorems. The residual items among (T3), (T4), (T8), (T9), (T12) are treated in Sec. 12.2 and Sec. 12.3 below.

Proposition 12.1 ((T13) Sample-space stability is automatic on a sufficiently small metric ball)**.** *Let $(\mathcal{M}, g_0)$ be a smooth globally hyperbolic Lorentzian 4-manifold with smooth Cauchy temporal function $\tau : \mathcal{M} \to \mathbb{R}$, let $\gamma : \mathbb{R} \to \mathcal{M}$ be a smooth g_0-timelike worldline with $\tau(\gamma(\tau)) = \tau$, and let $\{\mathcal{A}(R)\}_{R\subset\mathcal{M}}$ be a local net satisfying Ax. 2.2 on the slab $\mathcal{M}_{[\tau_i,\tau_f]}$. Let $\mathcal{B}^{(s)}_{g_0}$ denote the weighted Sobolev ball of radius ϵ around g_0 in the pointwise norm $\|g - g_0\|_{H^s(\mathcal{M}_{[\tau_i,\tau_f]},g_0)}$ on compactly-supported Lorentzian metrics, with $s \geq 3$ (so $H^s \hookrightarrow C^1$ by Sobolev embedding on the compact* *slab $\mathcal{M}_{[\tau_i,\tau_f]}$). Then there exists $\epsilon = \epsilon(g_0, \gamma, \tau) > 0$ such that for every $g \in \mathcal{B}^{(s)}_{g_0}$ with radius ϵ:*

(i) *g is globally hyperbolic on $\mathcal{M}_{[\tau_i,\tau_f]}$ and τ is a smooth g-Cauchy temporal function;*

(ii) *γ is g-timelike, with $g(\dot\gamma, \dot\gamma) < 0$ on $[\tau_i, \tau_f]$;*

(iii) *The local-algebra assignment $\{\mathcal{A}(R)\}$ satisfies Ax. 2.2(M1)–(M3) with respect to the g-causal structure, for every $g \in \mathcal{B}^{(s)}_{g_0}$.*

In particular, (T13) of Conj. 11.56 holds with $\mathcal{B}_{g_0} := \mathcal{B}^{(s)}_{g_0}$ for any sufficiently small ϵ.

Proof. (i)–(ii) are the structural-stability theorems for globally hyperbolic Lorentzian metrics. The set of globally hyperbolic Lorentzian metrics on a fixed manifold is open in the C^0 topology on the space of Lorentz metrics, and a fixed smooth Cauchy temporal function of g_0 remains a smooth Cauchy temporal function of g provided $\|g - g_0\|_{C^0}$ is smaller than an explicit constant controlled by the minimum of $-g_0(\nabla_{g_0}\tau, \nabla_{g_0}\tau) > 0$ on the compact slab (Bernal–Sánchez [9]; see also the stability statements for globally hyperbolic manifolds in the same reference and in [10] Ch. 8). The Sobolev embedding $H^s(\mathcal{M}_{[\tau_i,\tau_f]}) \hookrightarrow C^1$ for $s \geq 3$ on the compact 4-slab converts the required C^0 (indeed C^1) openness into Sobolev-ball openness; choose ϵ smaller than the structural constant times the embedding constant. The timelike condition on γ is $g(\dot\gamma, \dot\gamma) < 0$, which is an open condition on g in C^0 and hence in H^s for $s \geq 3$; the same ϵ (up to further reduction) makes (ii) hold.

(iii) The local-algebra assignment $\{\mathcal{A}(R)\}$ is defined as a subset of $\mathcal{L}(\mathcal{H})$ for each open $R \subset \mathcal{M}_{[\tau_i,\tau_f]}$ (Ax. 2.2); only the partial order on R under inclusion and the causal-separation relation depend on the metric. Isotony (M1) depends only on set inclusion, hence is metric-independent. Microcausality (M2) depends on the causal-separation relation, which is preserved by (i); explicitly, if R_1, R_2 are g_0-spacelike-separated and ϵ is chosen so that the g-lightcones deviate by less than the g_0-spacelike margin of (R_1, R_2), then R_1, R_2 remain g-spacelike-separated, and (M2) for g follows from (M2) for g_0. Causal propagation (M3) is the Heisenberg propagation bound under $U(\tau', \tau)$; the propagator is specified by the abstract Hamiltonian $H(\tau)$ (Def. 2.4), independent of g, and the causal domain $J_g(R)$ varies upper-semicontinuously in g by (i), so $J_g(R) \subset J_{g_0}(R_\epsilon)$ for an ϵ-thickened region R_ϵ, and (M3) is preserved under outer-regular extension. Outer regularity (1) is metric-free. □

Remark 12.2 (Effect on (T13)). Prop. 12.1 makes the phrase "restricted to the open subset on which (T13) holds" of (T13) of Conj. 11.56 redundant: the weighted Sobolev ball is automatically contained in that subset for any sufficiently small radius ϵ. Hence (T13) is no longer a structural hypothesis to be imposed; it is a corollary of the ambient choice of $\mathcal{B}_{g_0}$ via [9]. (T13) is therefore struck from the residual list in the hypothesis bundle of the sharpened theorem.

Proposition 12.3 ((T7) Symmetric domain of the gravity–matter coupling under (H0))**.** *Under Pos. 11.30 (all matter states restricted to* $\mathrm{Ran}(P_{\leq E_\star})$*, so the matter Hilbert space on any compact spatial region is finite-dimensional) and Pos. 11.18 (matter stress-energy is the local density of the bounded modular Hamiltonian K_ω on the cut-off GNS space), the joint domain inclusions of (T7) hold:*

$$\widehat{T}^{\mu\nu}_{\mathrm{m}}[\hat{\mathfrak{g}}; f]\,(\mathcal{D}_k) \;\subset\; \mathcal{D}(\widehat{G}^{(Q)\,\mu\nu}[\hat{\mathfrak{g}}; f]), \quad \widehat{G}^{(Q)\,\mu\nu}[\hat{\mathfrak{g}}; f]\,(\mathcal{D}_k) \;\subset\; \mathcal{D}(\widehat{T}^{\mu\nu}_{\mathrm{m}}[\hat{\mathfrak{g}}; f]).$$

The symmetrised product $\widehat{Q}^{(k)}[f]$ of (62) *is a bounded symmetric operator on $\mathcal{D}_k$, hence a theorem of finite-dimensional linear algebra rather than an open operator-domain hypothesis.*

Proof. Under Pos. 11.30, the matter Hilbert space is finite-dimensional on the (compact) spatial support of f, so $\widehat{T}^{\mu\nu}_{\mathrm{m}}[\hat{\mathfrak{g}}; f]$ is a bounded self-adjoint operator: it is a finite linear combination, with test-function smearing weights, of bounded matter observables on a finite-dimensional Hilbert space. By Prop. 11.26, under Pos. 11.18, $\widehat{G}^{(Q)\,\mu\nu}[\hat{\mathfrak{g}}; f]$ is the area-response operator of the bounded modular Hamiltonian K_ω against f; its smeared form is a bounded self-adjoint operator on the GNS space of $\omega\restriction_{P_{\leq E_\star}}$, which is itself finite-dimensional on any compact spatial region by the same cutoff. Every bounded operator on a finite-dimensional space has domain equal to the whole space, so both inclusions of (T7) hold trivially. The symmetrised product $\widehat{Q}^{(k)}[f] = \frac{1}{2}(\widehat{G}^{(Q)}\widehat{T}_{\mathrm{m}} + \widehat{T}_{\mathrm{m}}\widehat{G}^{(Q)})$ is a sum and product of bounded self-adjoint operators, hence bounded symmetric. □

Remark 12.4 (Effect on (T7)). Under Pos. 11.30 and Pos. 11.18, (T7) is a theorem and the Kato–Rellich relative bound $a < 1$ of (T4) becomes a bound on bounded operators, hence is trivially satisfied with $a = 0$ on the bounded part. The non-trivial relative-bound content of (T4) is then concentrated on the clock–geometry sector $\widehat{H}_{\mathrm{c}} + \widehat{H}_{\mathrm{grav}}$, addressed in Sec. 12.2.

Proposition 12.5 ((T10) Matter shell-blindness under the terminal matter collar)**.** *Assume Pos. 11.30 and the* matter-collar condition *accompanying the adaptedness restriction* (23) *of Conj. 10.2: the matter field operators have support bounded away from the terminal slice* Σ_{τ_f} *within a collar* $\{\tau \in [\tau_f - \delta_{\mathrm{mat}}, \tau_f]\}$ *of uniformly positive width. Then the matter Heisenberg propagator across the Israel–Darmois shell at* Σ_{τ_f} *satisfies*

$$U_{\mathrm{m}}[g_k](\tau_f^+, \tau_f^-) \;=\; \mathbb{1}_{\mathcal{H}^{(k)}_{\mathrm{matter}}} \qquad \textit{for every } k \in \mathcal{K}_+, \tag{74}$$

i.e. (T10) of Conj. 11.56 is a theorem.

Proof. By the Israel–Darmois junction theorem [20], the thin-shell stress-energy $S^{(k)}_{ab}$ of (21) on Σ_{τ_f} couples only to the extrinsic-curvature jump $[K^{(k)}_{ab}]$ of the gravitational sector; the bulk Einstein equation holds separately on $\mathcal{M}_{[\tau_i,\tau_f]}$ and on $\mathcal{M}_{[\tau_f,\infty)}$, and the matter fields extend continuously across Σ_{τ_f} without a distributional jump. Under the matter-collar condition, the matter Hamiltonian $\widehat{H}_{\mathrm{m}}[g_k](\tau)$ vanishes on $[\tau_f - \delta_{\mathrm{mat}}, \tau_f]$, so the matter Heisenberg evolution on a shrinking neighborhood $[\tau_f - \varepsilon, \tau_f + \varepsilon]$ of the shell is generated by a Hamiltonian of uniformly bounded operator norm $\left\|\widehat{H}_{\mathrm{m}}[g_k](\tau)\right\| \le E_\star$ (Pos. 11.30, so in particular locally bounded), multiplied by the temporal interval 2ε.

Hence in the limit $\varepsilon \downarrow 0$ the matter Heisenberg propagator between $\tau_f - \varepsilon$ and $\tau_f + \varepsilon$ is

$$U_{\mathrm{m}}[g_k](\tau_f + \varepsilon, \tau_f - \varepsilon) \;=\; \mathcal{T}\mathrm{exp}\Big(-\tfrac{i}{\hbar}\int_{\tau_f-\varepsilon}^{\tau_f+\varepsilon} \widehat{H}_{\mathrm{m}}[g_k](\tau)\, d\tau\Big) \;=\; \mathbb{1} + O(\varepsilon E_\star/\hbar),$$

using the matter-collar condition $\widehat{H}_{\mathrm{m}}[g_k](\tau)\restriction_{[\tau_f-\delta_{\mathrm{mat}},\tau_f]} = 0$ on the left endpoint side and boundedness on the right. Taking $\varepsilon \downarrow 0$, the limit exists and equals $\mathbb{1}_{\mathcal{H}^{(k)}_{\mathrm{matter}}}$, as claimed. The extrinsic distributional source $[K^{(k)}_{ab}]$ appears in the gravitational Einstein equation with a δ-function support on Σ_{τ_f}, but does not enter $\widehat{H}_{\mathrm{m}}[g_k](\tau)$, which depends on g_k via the induced metric on the spacelike slice Σ_τ and not on its extrinsic curvature discontinuity. The matter-side boundary phase or matching unitary advertised as a logical possibility in (T10) is therefore zero by construction. □

Remark 12.6 (Scope of Prop. 12.5). The terminal matter-collar condition is itself a postulate accompanying Conj. 10.2 (it travels with the adaptedness restriction (23) but is logically separate from it), and it is a physical one: matter is not co-localised with the terminal measurement slice, i.e. the terminal POVM records a gravitational final-boundary rather than a matter field profile. Under this condition, Prop. 12.5 makes (T10) a corollary of the Israel–Darmois junction and the Pos. 11.30 boundedness. Without the collar, a finite matter boundary phase could arise from the distributional shell; (T10) would then be a genuine hypothesis rather than a theorem. No analogous collar is assumed at the interior record cones $\mathcal{N}_m$ of (19): their null shells couple to matter through the (x-b)-bounded junction data, and the exact seed-prior identity is correspondingly replaced by the Born-map fixed point (x-c) outside the zero-record sector. The framework's scope coincides with (23) plus the collar, so the theorem holds throughout that scope.

Proposition 12.7 ((T0) Automatic under (H0) for generic boundary data)**.** *Under Pos. 11.30 the following hold:*

(i) *the constraint operator* $\widehat{\mathfrak{C}}_k$ *restricted to the finite-dimensional physical subspace* $\mathcal{H}^{(k),\mathrm{bl}}_{\mathrm{phys}} = \Phi^{\mathrm{bl}}_k$ *(Prop. 11.33(ii),(iv)) has pure-point spectrum, and zero is a simple eigenvalue of* $\widehat{\mathfrak{C}}_k \restriction_{\Phi^{\mathrm{bl}}_k}$ *on an open dense subset of the boundary-data parameter space;*

(ii) *for boundary data in this open dense subset, the rank-one rigged density hypothesis of Prop. 11.66 holds automatically, hence (T0) and (T8) are theorems;*

(iii) *the complementary subset (where zero is either absent from the spectrum or of higher multiplicity) is Lebesgue null in the joint Hilbert-space norm topology on* $(\rho_{\mathrm{i}}, \{E_k\})$.

Proof. (i) Under Pos. 11.30, $\Phi^{\mathrm{bl}}_k = \Phi_k$ is finite-dimensional (Prop. 11.33(iv)), and every self-adjoint operator on a finite-dimensional space has pure-point spectrum. The map from boundary data $(\rho_{\mathrm{i}}, \{E_k\}) \in \mathcal{S}(\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}) \times (\mathcal{E}(\mathcal{H}^{\mathrm{phys}}_{\mathrm{matter}}))^{\mathcal{K}}$ (density operators and effects on the cut-off matter space) to the finite matrix representation of $\widehat{\mathfrak{C}}_k \restriction_{\Phi^{\mathrm{bl}}_k}$ is a real-*analytic* map of the boundary data on the open set $p_0(k) = \mathrm{Tr}(\rho[g_0](\tau_f)E_k) > 0$ supplied by Ax. 2.1 (its entries carry $\rho^{1/2}$, $\Lambda_k^{1/2}$ and $1/p_0(k)$, all analytic there). The subset of boundary data for which zero has multiplicity ≥ 2 is the common zero locus of the $(\dim \Phi^{\mathrm{bl}}_k - 1) \times (\dim \Phi^{\mathrm{bl}}_k - 1)$ minors of the matrix — a proper real-analytic subvariety, Lebesgue null and closed nowhere-dense. The subset where zero is absent entirely is open, as is the subset where zero is simple.

(ii) When zero is simple and present, let $\Psi_0^{(k)}$ be the unique (up to scale) zero-eigenvector of $\widehat{\mathfrak{C}}_k \restriction_{\Phi^{\mathrm{bl}}_k}$. The rigged density at $\lambda = 0$ is $\rho^{(k)}_{\mathrm{rigg}}(0)[\Psi; \Psi'] = \langle \Psi | \Psi_0^{(k)} \rangle \overline{\langle \Psi' | \Psi_0^{(k)} \rangle}$, a rank-one positive semi-definite sesquilinear form on Φ^{bl}_k with kernel of codimension one — the rank-one rigged density hypothesis of Prop. 11.66. The rigged pairing on the boundary vectors is $\rho^{(k)}_{\mathrm{rigg}}(0) = \langle \Psi_{\mathrm{f}}^{(k)\,\mathrm{rigg}} | \Psi_0^{(k)} \rangle \overline{\langle \Psi_{\mathrm{i}}^{(k)\,\mathrm{rigg}} | \Psi_0^{(k)} \rangle}$, which is non-zero iff neither boundary vector is orthogonal to $\Psi_0^{(k)}$. The orthogonality subspaces are each of codimension one in Φ^{bl}_k; on the joint boundary-data subset avoiding both, the rigged pairing is non-zero. Under the two-boundary positivity axiom Ax. 2.1 with $p_0(k) = \mathrm{Tr}(\rho[g_0](\tau_f)E_k) > 0$, the Born trace itself is non-zero, which by explicit computation ((59)–(69)) factors through the same pairing. Hence $p_0(k) > 0 \Rightarrow \rho^{(k)}_{\mathrm{rigg}}(0) \neq 0$ on the generic open subset.

(iii) The complementary set (zero absent or non-simple) is contained in a finite union of proper real-analytic subvarieties, hence Lebesgue null in the joint unit-ball topology of the data space $(\rho_{\mathrm{i}}, \{E_k\}) \in \mathcal{S} \times \mathcal{E}^{\mathcal{K}}$. □

Remark 12.8 (Comparison with Rem. 11.67). Prop. 12.7 strengthens the genericity statement of Rem. 11.67 to a quantitative open-dense theorem under Pos. 11.30: the complement is a proper real-analytic subvariety and Lebesgue null, and the rank-one density hypothesis of Prop. 11.66 is automatic on the open dense subset. (T0) and (T8) are therefore theorems of finite-dimensional linear algebra under (H0), modulo the (measure-zero) real-analytic exception set.

12.2 Nelson-core discharge of (T3) and (T4)

The residual non-trivial content of (T3) (essential self-adjointness of the kinetic generator $\widehat{K}_k$) and (T4) (Kato–Rellich relative-bound) under Pos. 11.30 and Pos. 11.18 lives on the clock–geometry sector $\widehat{H}_{\mathrm{c}} + \widehat{H}_{\mathrm{grav}}$, since the matter sector is finite-dimensional and the gravity–matter coupling is bounded (Prop. 12.3). Both conditions are standard consequences of the Nelson analytic-vector theorem[1] once the gravity sector is realised on a Hilbert space with a dense core

[1]Throughout, 'Nelson' refers to E. Nelson's *analytic-vector theorem* on essential self-adjointness ([32], Thm. X.39), an operator-theoretic result. It is unrelated to Nelson *stochastic mechanics* (the diffusion-process derivation of the Schrödinger equation): the present framework involves no diffusion, no osmotic velocity, and no Madelung decomposition, and the master equation is a deterministic operator identity (cf. the proof of Prop. 13.5,

of joint analytic vectors for $\widehat{H}_{\text{grav}}$.

Hypothesis 12.9 (Gravity-sector Hilbert realisation)**.** The gravity-sector Hilbert space $\mathcal{H}_{\text{grav}}$ is a separable Hilbert space, the gravity Hamiltonian $\widehat{H}_{\text{grav}}$ is a densely defined symmetric operator with a *dense core* $\mathcal{D}_{\text{grav}} \subset \mathcal{H}_{\text{grav}}$ *of analytic vectors*: for every $\psi \in \mathcal{D}_{\text{grav}}$ there exists $t > 0$ with $\sum_{n\geq 0} \left\| \widehat{H}^n_{\text{grav}}\psi \right\| t^n/n! < \infty$.

Remark 12.10 (Standard realisations). Hyp. 12.9 is satisfied by the standard canonical-quantum-gravity Hilbert-space realisations considered in the literature: the Ashtekar–Lewandowski space of cylindrical functions on spin networks (see [186] and references therein), on which polynomial operators in volume and area have cylindrical-function cores of analytic vectors; the reduced-phase Hilbert space of loop quantum cosmology, where $\widehat{H}_{\text{grav}}$ has discrete spectrum and every algebraic vector is analytic; and finite-dimensional truncations of canonical geometrodynamics such as minisuperspace models, on which every polynomial operator is analytic trivially. In the $\mathcal{H}_{\text{grav}} \cong \mathcal{H}_{\text{sys}}$ regime of Thm. 11.68 (R5) and Conj. 10.13, the gravity sector reduces to the system Hilbert space on which $\widehat{H}_{\text{s}}$ is bounded under Pos. 11.30; boundedness trivially implies Hyp. 12.9 with entire analytic vectors.

Proposition 12.11 ((T3) and (T4) essential self-adjointness via Nelson)**.** *Assume Pos. 11.30, Pos. 11.18, Pos. 11.11, and Hyp. 12.9. Let* $\mathcal{D}^{\text{an}}_k := \mathcal{H}^{\text{bl}}_{\text{c}} \otimes_{\text{alg}} \mathcal{D}_{\text{grav}} \otimes_{\text{alg}} \mathcal{H}^{(k)}_{\text{matter}} \subset \mathcal{D}_k$*, where* $\mathcal{H}^{\text{bl}}_{\text{c}} = E_{\widehat{H}_{\text{c}}}([-\hbar/\delta_\star, \hbar/\delta_\star])\mathcal{H}_{\text{clock}}$ *is the band-limited clock sector (under Pos. 11.30, finite-dim for* $\delta_\star \leq \hbar/E_\star$*) and* $\mathcal{D}_{\text{grav}}$ *is the analytic-vector core of Hyp. 12.9. Then:*

(i) *the kinetic generator* $\widehat{K}_k = \widehat{H}_{\text{c}} + \widehat{H}_{\text{grav}} + \widehat{H}^{(k)}_{\text{m}}[\widehat{\mathfrak{g}}] + \alpha\widehat{\Sigma}_{\delta t}$ *is essentially self-adjoint on* $\mathcal{D}^{\text{an}}_k$ (this is (T3))*;*

(ii) *the perturbation* $\kappa^{-1}\widehat{Q}^{(k)}[f]$ *is bounded on* $\mathcal{D}_k$ *(Prop. 12.3), so the Kato–Rellich relative-bound of (T4) is trivially* $a = 0$*,* $b = \left\|\kappa^{-1}\widehat{Q}^{(k)}[f]\right\| < \infty$ (this is (T4))*;*

(iii) *consequently the constraint* $\widehat{\mathfrak{C}}_k = \widehat{K}_k - \kappa^{-1}\widehat{Q}^{(k)}[f]$ *is essentially self-adjoint on* $\mathcal{D}^{\text{an}}_k$ *by Kato–Rellich.*

Proof. (i) The relevant object here is the *band-limited compressed corner* $\mathcal{H}^{\text{bl}}_{\text{c}} = P_{\leq E_\star}\mathcal{H}_{\text{c}}$ of the clock factor — a finite-dimensional subspace of the (infinite-dim, $L^2(\mathbb{R})$) clock sector, carrying the *dressed* band-limited generator (not the full clock sector, which retains its infinite-dimensional structure per the H0 co-scoping, Rem. 11.72). On this corner $\widehat{H}_{\text{c}}\restriction_{\mathcal{H}^{\text{bl}}_{\text{c}}}$ is bounded self-adjoint with discrete spectrum. The matter factor $\mathcal{H}^{(k)}_{\text{matter}}$ is finite-dimensional under Pos. 11.30, so $\widehat{H}^{(k)}_{\text{m}}[\widehat{\mathfrak{g}}]$ is bounded self-adjoint for every (fixed) value of $\widehat{\mathfrak{g}}$ in the semiclassical sector, and bounded as an operator-valued function on $\mathcal{H}_{\text{grav}}$ by Pos. 11.18's identification with a local density of the bounded modular Hamiltonian. The clock shift $\alpha\widehat{\Sigma}_{\delta t} = \alpha(\cos(\delta t\widehat{H}_{\text{c}}/\hbar) - \mathbb{1})$ is a bounded self-adjoint function of the bounded clock Hamiltonian. Only $\widehat{H}_{\text{grav}}$ on $\mathcal{D}_{\text{grav}}$ is unbounded; by Hyp. 12.9, $\mathcal{D}_{\text{grav}}$ is a dense analytic-vector core, so $\widehat{H}_{\text{grav}}$ is essentially self-adjoint there by Nelson's analytic-vector theorem ([32], Thm. X.39). The remaining terms of $\widehat{K}_k$ — the bounded modular-Hamiltonian density and the bounded clock shift $\alpha\widehat{\Sigma}_{\delta t}$ — are bounded symmetric operators, so $\widehat{K}_k = \widehat{H}_{\text{grav}} + (\text{bounded symmetric})$ is essentially self-adjoint on $\mathcal{D}^{\text{an}}_k$ by Kato–Rellich ([32], Thm. X.12) with relative bound 0 (a bounded symmetric perturbation preserves essential self-adjointness on any core of $\widehat{H}_{\text{grav}}$).

(ii) By Prop. 12.3, $\widehat{Q}^{(k)}[f]$ is bounded on $\mathcal{D}_k$ with $\left\|\widehat{Q}^{(k)}[f]\right\| \leq \left\|\widehat{T}^{\mu\nu}_{\text{m}}[f]\right\| \left\|\widehat{G}^{(Q)\,\mu\nu}[f]\right\|$, each factor finite under (H0). Hence $\left\|\kappa^{-1}\widehat{Q}^{(k)}[f]\Psi\right\| \leq 0 \cdot \left\|\widehat{K}_k\Psi\right\| + \kappa^{-1}\left\|\widehat{Q}^{(k)}[f]\right\| \|\Psi\|$, i.e. $a = 0$ and $b = \kappa^{-1}\left\|\widehat{Q}^{(k)}[f]\right\|$.

'no random-walk master equation term'). In particular the Wallstrom single-valuedness / quantization-of-circulation obstruction to stochastic mechanics does not arise here.

(iii) Kato–Rellich ([32], Thm. X.12) applied to (i) self-adjoint $\widehat{K}_k$ plus (ii) relative bound $a = 0 < 1$ gives essential self-adjointness of the sum $\widehat{\mathfrak{C}}_k$ on $\mathcal{D}_k^{\mathrm{an}}$. □

Remark 12.12 (Effect on (T3)(T4)). Under Hyp. 12.9 (satisfied by every standard canonical-gravity realisation considered in the literature) and the three postulates, (T3) and (T4) are theorems of the present paper. They are discharged to the specific structural hypothesis Hyp. 12.9 on the gravity-sector realisation. On every background where the Chandrasekaran–Longo–Penington–Witten construction is established — de Sitter, stationary Killing-horizon, and cosmological FLRW — that hypothesis is itself *discharged to a theorem* (Thm. H.1, Rem. H.4), so (T3),(T4) are unconditional there; on a generic globally hyperbolic $(\mathcal{M}, g)$ it reverts to a hypothesis of the ambient canonical-gravity literature. See Sec. 12.5 for the precise external reduction.

12.3 Consolidation of (T8), (T9), (T12) via the band-limited rigged boundary

Under Pos. 11.30 the rigged-Hilbert structure on Φ_k^{bl} collapses to ordinary finite-dimensional linear algebra (Prop. 11.33(iv)): the test space $\Phi_k = \Phi_k^{\mathrm{bl}}$ is already finite-dimensional, the rigged dual Φ_k' is the same finite-dimensional space (via the inner product), and the spectral delta $\delta(\widehat{\mathfrak{C}}_k)$ collapses to the orthogonal projector $P_0^{(k)}$ onto the zero-eigenspace. In this setting (T8), (T9), (T12) reduce to bounded-operator statements on Φ_k^{bl}.

Proposition 12.13 (Rigged-boundary collapse under (H0))**.** *Under Pos. 11.30 the following hold:*

(i) *The rigged spectral density $\delta(\widehat{\mathfrak{C}}_k)$ on the finite-dim Φ_k^{bl} equals the orthogonal projector $P_0^{(k)}$ onto* $\ker \widehat{\mathfrak{C}}_k \restriction_{\Phi_k^{\mathrm{bl}}}$*;*

(ii) ***(T8 strengthened-compatibility).*** *For every $k \in \mathcal{K}_+$ on the open dense subset of boundary data of Prop. 12.7, the rigged pairing $\rho_{\mathrm{rigg}}^{(k)}(0) = \langle \Psi_{\mathrm{f}}^{(k)\,\mathrm{rigg}} | P_0^{(k)} | \Psi_{\mathrm{i}}^{(k)\,\mathrm{rigg}} \rangle$ exists, is finite and non-zero, and the zero-spectral slice* $\mathrm{Ran}(P_0^{(k)})$ *is a one-dimensional subspace of Φ_k^{bl};*

(iii) ***(T9 endpoint-preserving gauge).*** *The restricted constraint-range $\widehat{\mathfrak{C}}_k \Phi_k^{\mathrm{bl},\partial}$ equals the zero subspace on $\Phi_k^{\mathrm{bl}} \cap \ker \iota_i^* \cap \ker \iota_f^*$, i.e. quotienting by the endpoint-preserving gauge is trivial on the boundary data satisfying* (64)*;*

(iv) ***(T12 BFP threshold below seed gap).*** *The Banach fixed-point threshold $\varepsilon_{\mathrm{BFP}}$ of Conj. 10.2 satisfies $\varepsilon_{\mathrm{BFP}} < p_* := \min\{p_0(k) : k \in \mathcal{K}_+\}$ automatically: under Pos. 11.30, $\mathcal{K}$ is finite, the seed-prior probability $p_* > 0$ is strictly positive on $\mathcal{K}_+$, and the canonical choice $\varepsilon_{\mathrm{BFP}} = p_*/2$ (Prop. 10.3) is admissible.*

Proof. (i) On a finite-dimensional space the spectral measure of a self-adjoint operator is the discrete sum $E_{\mathfrak{C}_k} = \sum_j \delta_{\lambda_j} P_{\lambda_j}$ of projectors onto eigenspaces; the rigged delta at $\lambda = 0$ is the projector $P_0^{(k)}$ onto the zero-eigenspace, by the spectral theorem.

(ii) is Prop. 12.7(i)–(ii) in projector form.

(iii) The constraint range $\widehat{\mathfrak{C}}_k \Phi_k^{\mathrm{bl}}$ is the orthogonal complement of $P_0^{(k)}$ in Φ_k^{bl} (finite-dim spectral theorem). The endpoint evaluation maps ι_i^*, ι_f^* are linear functionals on $\Phi_k^{\mathrm{bl},*}$; their simultaneous kernel is a codimension-two subspace on the generic open subset. The intersection with the constraint range is a subspace of dimension $\dim \Phi_k^{\mathrm{bl}} - 3$ generically, and quotienting the constraint range by the endpoint-preserving part is a well-defined finite-dimensional linear-algebra operation with no new content beyond linear algebra on Φ_k^{bl}.

(iv) The minimum $p_* = \min\{p_0(k) : k \in \mathcal{K}_+\} > 0$ because $\mathcal{K}$ is finite (under Pos. 11.30 the terminal POVM is on a finite-dimensional space with finitely many effects). Hence any $\varepsilon_{\mathrm{BFP}} \in (0, p_*)$ is an admissible choice; Prop. 10.3's canonical $\varepsilon_{\mathrm{BFP}} = p_*/2$ is in this range. □

Remark 12.14 (Effect on (T8), (T9), (T12)). Under Pos. 11.30, the rigged-Hilbert structural conditions (T8), (T9), (T12) collapse to finite-dimensional linear-algebra statements (Prop. 12.13), and are theorems of the present paper once the generic dense open condition of Prop. 12.7 is assumed on boundary data. The reduction of rigged-Hilbert content to finite-dimensional projector content under the band-limited hypothesis is analogous to the reduction of Bohm–Gadella rigged-Hilbert analysis of resonances to finite-matrix spectral calculations on a low-energy effective subspace (see, e.g., the review in [31] Ch. XII for the Gel'fand-triple formalism and its specialisation to discrete-spectrum subspaces).

12.4 Sharpened Master Unification Theorem

We now consolidate Prop. 12.1, Prop. 12.3, Prop. 12.5, Prop. 12.7, Prop. 12.11, and Prop. 12.13 into a sharpened form of the Master Unification Theorem. Let H' denote the following reduced hypothesis bundle:

(H'0) **Three structural postulates and the consolidated commutation hypothesis.** Pos. 11.30 (Planck-scale matter cutoff; its ε-tail robustness is the separately flagged Hyp. 11.35), Pos. 11.11 (thermal-time identification of clock Hamiltonian with modular Hamiltonian), and Pos. 11.18 (entropic-gravity identification of matter stress-energy with modular Hamiltonian density); together with Hyp. 11.13 (the consolidation of (CP) and (T11) under Pos. 11.11), which after the two-branch restructuring of Conj. 11.1 is needed *only* for the branch-(II) (entire-analytic centralizer) special case: branch (I), the Araki-conditioning branch carrying the physical two-boundary content, bypasses it entirely (App. G).

(H'1) **Part I axioms.** Ax. 2.1, Ax. 2.2, Ax. 7.1 hold on the fixed globally hyperbolic slab $\mathcal{M}_{[\tau_i,\tau_f]}$ under the support restriction (H0) of $\mathrm{Ran}(P_{\leq E_\star})$.

(H'2) **Gravity-sector realisation.** Hyp. 12.9 on $\mathcal{H}_{\mathrm{grav}}$ (dense analytic-vector core of $\widehat{H}_{\mathrm{grav}}$) — discharged to a theorem (Thm. H.1, merged with Hyp. D.1) on every CLPW-established background, a genuine hypothesis only in its generic-$(\mathcal{M}, g)$ scope (Rem. H.4).

(H'3) **Adaptedness restriction, terminal matter collar, and record-layer structural hypotheses.** The causal adaptedness restriction (23) of Conj. 10.2 (record-adapted metric sourcing, Def. 10.1, with the record postulate (R)) together with the matter-collar condition of Prop. 12.5 keeping the matter sector away from the terminal slice Σ_{τ_f}; and the named structural hypotheses of the record-adapted gravitational layer: null-junction solvability Conj. 10.2 (vi-b), uniform record-tree contraction (x-a), the self-consistent Born/record-law fixed point (x-c), and terminal-POVM metric-ball stability (x-d) (the latter tied to the Cartan-masa selection of Conj. L.1, which travels with it).

(H'4) **Regime data and witness.** The regime data of (H5) and the non-vacuity witness of (H6) of Thm. 11.73.

(H'5) **Residual external inputs (sharpened).** After the discharge appendices App. D (Prop. D.2), App. E (Thm. E.7, Cor. E.8), App. F (Thm. F.2), and App. G (Thms. G.1, G.2), the residuals (E3)–(E4) are *theorems of the present paper* modulo one *named* ambient-literature input (Hyp. F.1) and zero inputs for (E4), (E2) is a *conditional theorem* modulo Hyp. E.1 with the scope residuals (E2b)–(E2d) of Rem. E.10, while (E1) is *restated as the explicit Wilsonian scope postulate* Pos. 11.30 (H0), motivated but not derived by the Type II trace (Prop. D.2, Rem. D.3). We relabel the surviving inputs as (E2a), (E3a) of Sec. 12.5; (E4) has no residual. Concretely: (E2a-ω) Hyp. E.2 holds on generic compact-Cauchy globally hyperbolic backgrounds admitting a real-analytic representative g_0 (uniform analytic-collar Sobolev Hadamard parametrix with mixed C^2 coincidence jets, microlocal programme; the open-ball form Hyp. E.1 is the unconditional fallback; the analytic slice register this input's Option A booking needs is proved unconditionally over U_ω in Appendix E2a (Thm. E2a.83, Part (A)), and the fused first law holds modulo clauses (ii)–(iii) (ibid., Part (B); the clause (i) difference form is ported to a theorem, App. E2a)); (E3a) Hyp. F.1 holds beyond the perturbative-around-isolated-horizon regime [227, 228]. To these we add the record-layer residual **(E5)** (*a.s. shell summability for uniformly informative protocols*): the exponential-selection constants $(C_{\mathrm{UI}}, \delta')$ of Conj. 10.2 (x-b) are uniform over the metric ball $\mathcal{B}_{g_0}$, so that the cumulative null-shell data are a.s. summable on the protocol class $\mathfrak{P}_{\mathrm{UI}}$ (Sec. 12.5.6); and the admissibility clause (RP) (record permanence) of Def. 10.1, which fixes the protocol class on which (E5) and the record-adapted layer are asserted.

Theorem 12.15 (Sharpened Master Unification Theorem)**.** *Assume* H' *=(H'0)–(H'5). Then all of (T0)–(T13) of Conj. 11.56* except (T11) *are theorems; (T11), consolidated with (CP) into Hyp. 11.13 under (H'0), remains a hypothesis, and is needed only for the branch-(II) special case of Conj. 11.1 (branch (I) is discharged by App. G without it):*

- *(T5), (T6), nuclearity of* Φ_k*, Hadamard-free well-posedness of Prop. 11.26: discharged by Prop. 11.33 under (H'0);*
- *(T1), (T2): discharged by Prop. 11.26 under (H'0);*
- *(T11) and (CP) of Conj. 11.1: consolidated into Hyp. 11.13 under (H'0) and carried as a hypothesis there (branch-(II) only), not discharged;*
- *(T13): discharged by Prop. 12.1 under (H'0)–(H'1);*
- *(T7): discharged by Prop. 12.3 under (H'0);*
- *(T10): discharged by Prop. 12.5 under (H'0) & (H'3);*
- *(T0), (T8) (strengthened-compatibility), (T9), (T12): discharged by Prop. 12.7 and Prop. 12.13 under (H'0) on the open dense subset of generic boundary data;*
- *(T3), (T4): discharged by Prop. 12.11 under (H'0) & (H'2).*

Conditional on the two remaining postulates Pos. 11.18 of (H'0) as a primitive structural postulate (not *an external open problem — see Sec. 12.5, (E0)) and the Planck-cutoff postulate H0 (Pos. 11.30, motivated but not derived by App. D), and conditional on the two named ambient-literature inputs (E2a), (E3a) of (H'5) (each attached to a specific external research programme, as discharged — conditionally, with the (E2) scope of Rem. E.10 — in Apps. E, F), the Two-Boundary Retrocausal Master Equation* (66) *is a single well-defined mathematical object whose five simultaneous reductions (U2)–(U5) of Thm. 11.73 (five reductions distributed over four items, (U2) carrying both (R1) and (R2); (U1) is well-posedness) coincide with ordinary quantum mechanics, the rigorous Part I two-boundary conditioned quantum mechanics on curved spacetime, the semiclassical Einstein field equation of general relativity, the modified Wheeler–DeWitt constraint on pure geometry, and the retrocausal modified TDSE of App. C (Prop. C.2). Item (E4) is discharged by App. G (Thms. G.1, G.2), using Araki's 1973 relative-Hamiltonian perturbation theory and the Haagerup dual-weight theorem (the generic unbounded-generator regime handled by the standard KMS-perturbation extension in the proof). The non-vacuity witness (U6) is unchanged from Thm. 11.73.*

In other words: every bespoke conjecture of Part II is either a theorem of the present paper, discharged to a specific named ambient-literature open problem, or carried as an explicitly named structural hypothesis of the bundle H'. *The complete residual commitment set — the postulates of (H'0), the realisation and record-layer hypotheses of (H'2)–(H'3), and the external inputs of (H'5) — is reconciled item by item in Rem. 12.16; item (E4) is already unconditionally closed.*

Proof. By Thm. 11.73, under the full bundle H =(H0)–(H6) the conclusions (U1)–(U6) hold. It suffices to show that H' implies H modulo (E1)–(E5).

(H0) of Thm. 11.73 is Pos. 11.30 of (H'0). (H1) is Part I axioms, (H'1). (H2) (the gravity bridges) consists of Conj. 10.2 and Conj. 10.13. The former's coupling-existence content is Pos. 11.18 of (H'0) via Prop. 11.26; the Banach fixed-point (contraction) content is the QFT-regime statement of (E2); the adaptedness restriction (with (R)/(RP)) and matter collar are (H'3), as are the named record-layer hypotheses (vi-b), (x-a), (x-c), (x-d); clause (ix) is (T13), discharged by Prop. 12.1 under (H'0)–(H'1); the shell-summability clause (x-b) is the residual (E5) of (H'5) on the (RP)-admissible protocol class $\mathfrak{P}_{\mathrm{UI}}$; the remaining (i)–(vi-a) and (viii) clauses are structural and discharged by Prop. 11.33 and Prop. 12.13(iv) under (H'0). The latter Conj. 10.13 is discharged by Thm. B.5 (operator-cosine uniqueness of App. B) under the band-limit of (H'0).

(H3) (speculative closures) consists of Conj. 11.1, Conj. 11.4, Conj. 11.5, Conj. 11.8. Conj. 11.5 is Thm. B.5 under (H'0) (App. B). The (CP) portion of Conj. 11.1 is Hyp. 11.13 under Pos. 11.11 of (H'0); the non-(CP) portion is (E4). Conj. 11.4's finite-dim contraction bound is Prop. 11.49 under (H'0); the QFT-regime extension is (E2). Conj. 11.8 on stationary horizons is closed by [87, 88, 89, 92]; the dynamical-horizon extension is (E3).

(H4) (master-equation structural hypotheses) consists of (T0)–(T13). By the enumerated propositions above, each (Tn) is either a theorem under (H'0)–(H'3) or is consolidated into Hyp. 11.13 under Pos. 11.11 (the latter carried in (H'0) as a hypothesis, branch-(II) only).

(H5) (regime data) and (H6) (non-vacuity witness) are (H'4).

Hence $\mathsf{H}' \Rightarrow \mathsf{H}$ modulo (E1)–(E5), and Thm. 11.73 applies conditional on the resolution of (E1)–(E5) in their ambient literatures. □

Remark 12.16 (Master residual commitment list). The following list reconciles, in one place, *every* commitment that remains open after Parts I–III and the discharge appendices. The shorter summaries elsewhere (the abstract; Rem. 11.75; the pre-predictions consolidation before Rem. 12.18) point here and assert nothing beyond this list.

(i) *EFT scope.* The Planck-cutoff postulate H0 (Pos. 11.30), together with its ε-tail error budget Hyp. 11.35 (stated, not proven; the flagged highest-value UV strengthening). (Within the H0 scope, the Unruh–Schützhold adiabatic-vacuum condition (Rem. 11.32, the Hawking-robustness item) is a sub-condition consumed only by a dynamical-flux extension not made here; the equilibrium KMS recovery of Thm. M.1 is insensitive to it.)

(ii) *Thermal time.* Pos. 11.11, together with the *H0 co-scoping / bare–dressed compatibility residual* (Rem. 11.72): the assumption that the observer compression Π preserves the clock/gravity modular–KMS data, so the finite-dim matter-sector rigour and the unbounded-generator thermal-time sector coexist without collision. Stated, not proven; sharpest stress point EG1.

(iii) *Matter–geometry coupling*, in the scope-split form of App. I superseding the primitive Pos. 11.18: the nonlinear Clausius closure Ax. I.10 — whose sharpest generic-diamond residual is the modular-second-variation saturation Hyp. I.5 (the $n = 2$ buck-stop) together with the trace/conformal inversion seam of Rem. I.6; the QRF-dressed Type II realisation Hyp. D.1 (through which the linearised AdS-anchored sector, Thm. I.1, and the Type II trace structure are theorems); and the de Sitter-sector identification Hyp. I.3 for every cosmological-facing use of the Clausius relation.

(iv) *Consolidated commutation.* Hyp. 11.13 ((CP)+(T11)), needed only for branch (II) of Conj. 11.1; branch (I) is closed unconditionally by App. G.

(v) *Named external inputs.* (E2a) Hyp. E.1 (with the scope residuals (E2b)–(E2d) of Rem. E.10) and (E3a) Hyp. F.1.

(vi) *Record-adapted gravitational layer.* The record postulate (R) with its record-permanence admissibility (RP) (Def. 10.1); null-junction solvability Conj. 10.2 (vi-b) — which now includes the record no-gravitational-radiation postulate $[C_{AB}]^{\mathrm{TT}} = 0$ and, on the regular record cone $\mathcal{N}_m^{\mathrm{reg}}$, the codimension-$\geq 2$ caustic residual (vi-b$'$) and non-crossing scope (vi-b$''$); uniform record-tree contraction (x-a); the self-consistent Born/record-law fixed point (x-c); terminal-POVM metric-ball stability (x-d); and the shell-summability residual (E5) = (x-b) on the uniformly informative protocol class $\mathfrak{P}_{\mathrm{UI}}$ (Sec. 12.5.6).

(vii) *Terminal-POVM selection.* Conj. L.1 in its Cartan-masa form (App. L).

(viii) *Part I load-bearing axioms.* Ax. 7.1 (terminal record-completeness; it is what makes the terminal shell vanish a.s.), Ax. 2.1, Ax. 2.2, and the self-consistent achronal-ANEC admissibility condition on the two-boundary data $(\rho_{\mathrm{i}}, \{E_k\}, g_0)$ of Rem. 10.12.

(ix) *Regime data and witness.* The regime data of (H5)/(H'4) and the de Sitter static-patch witness of (H6).

(x) *Framework-level items outside the master-theorem chain* (consumed by their appendices, not by Thm. 12.15 itself): Hyp. Q.1 (orbit stratification for the physical-Hilbert-space normalisation, App. Q) and Conj. K.1 (Type II trace-stable inductive limit, App. K).

(xi) *Emergent-G inputs* (outside the master-theorem chain, consumed only by Prop. 11.20): ($\mathrm{H}_\star 1$) $E_\star$ as the sole scale anchoring the induced-gravity coefficient, and ($\mathrm{H}_\star 2$) the imported induced-gravity / Sakharov heat-kernel coefficient ($\mathrm{str}[k_1] \approx -14.83$, whence $g \approx 0.42$), of the correct sign and $O(1)$. These bear on the emergent status of G (Prop. 11.20) only, not on Thm. 12.15.

The sharpened theorem's claim is one of *concentration, not concealment*: every item is named, individually falsifiable or dischargeable, and attached to a specific section of this paper or a specific ambient-literature programme. The finest-grained count is of order eighteen *named* items, but that number reflects itemisation rather than independent conjectural burden — items (viii)–(ix) are scope declarations rather than conjectures, item (iv) is bypassed on the physical branch, and the genuine future-discharge load sits on the few structural items (i), (iii), (v), and (vi)/(E5). Reduced to conjectural essentials the residual is the coupling axiom (E0) plus the two ambient inputs (E2a),(E3a) together with the co-scoping compatibility residual, which is the short list the abstract quotes.

Remark 12.17 (Comparison with Thm. 11.73). Thm. 12.15 strictly strengthens Thm. 11.73: every bespoke conjecture internal to the framework is discharged to either a theorem of the paper (Props. 12.1, 12.3, 12.5, 12.7, 12.11, 12.13) or an explicit external reduction to a named ambient-literature problem ((E1)–(E4) of Sec. 12.5). With one named exception, the residual hypothesis bundle H' contains no conjecture of the form "there exists an operator with such-and-such properties" whose existence is not either established here or reduced to a specific external theorem; the exception is Hyp. 12.9 (existence of a dense analytic-vector core for the gravity generator). On every background where the Chandrasekaran–Longo–Penington–Witten construction is established — de Sitter, stationary Killing-horizon, and cosmological FLRW — this hypothesis is in fact *discharged to a theorem* of the present paper (Thm. H.1, which also merges it with Hyp. D.1; Rem. H.4); what remains genuinely open is only its *scope* on a generic globally hyperbolic $(\mathcal{M}, g)$, where no distinguished boost generator is available and the realisation reverts to the ambient canonical-gravity programme. (T3),(T4) are correspondingly conditional only in that generic-scope residual (Rem. 12.12).

This is as strong as the unification claim can honestly be made today: Pos. 11.18 is a standalone physical postulate (discussed as (E0) in Sec. 12.5), and each (E1)–(E4) is a single external open problem with a specific literature address, not a free-floating conjecture.

12.5 Residual external programme

We give the precise external reduction of each residual item (E0)–(E5) of (H'5) to a named open problem of the ambient operator-algebraic / QFT-on-curved-spacetime literature (the record-layer residual (E5) is bespoke in origin but reduces, Sec. 12.5.6, to a stability question for ambient repeated-measurement limit theorems). For each, we identify (i) the ambient literature's statement of the problem, (ii) what partial progress is already known, and (iii) what the reduction supplies if resolved.

Status update (post-discharge). Apps. E– G upgrade items (E3)–(E4) from external open problems to *theorems of the present paper*, (E3) modulo the single named ambient-literature input Hyp. F.1 and (E4) unconditionally; item (E2) is discharged *conditionally* (Rem. E.10): a conditional theorem modulo Hyp. E.1 (E2a), scoped to quasi-free seeds of the free massive scalar, with the further scope residuals (E2b)–(E2d). Item (E1) is *not* upgraded: App. D restates it as the explicit Wilsonian scope postulate Pos. 11.30 (H0), motivated but not derived by the Type II trace (Prop. D.2). The surviving residuals are relabelled (E2a), (E3a); (E1a) is retired and (E4a) does not exist. The narrative of this section below records the pre-discharge ambient status of each item; the *post-discharge* residual of each (En) is the corresponding (Ena) input identified in Remarks D.3, E.10, F.3, and G.3.

12.5.1 (E0) Pos. 11.18 as a primitive postulate (not an external open problem)

Pos. 11.18 is the Jacobson–Bianchi–Myers identification of the semiclassical Einstein equation as an equation of state derived from the first law of entanglement:

$$\widehat{T}^{\mathrm{m}}_{\mu\nu} \;=\; \text{local density of } K_\omega, \qquad \widehat{G}^{(Q)}_{\mu\nu} \;=\; \text{area-response of } K_\omega, \tag{75}$$

with the Clausius identity $\widehat{G}^{(Q)}_{\mu\nu} = 8\pi G\,\widehat{T}^{\mathrm{m}}_{\mu\nu}/c^4$ automatic as an equality of modular-flow-derived objects.

Ambient status. Unlike (E1)–(E4), (E0) is not framed as an open problem reducible to a single named programme; the derivation of the Einstein equation from entanglement thermodynamics is an active but unresolved programme ([56, 116, 59, 58, 106, 117]). The present framework takes (E0) as a primitive postulate on the matter–geometry coupling, analogous in status to the assumption of the first law of thermodynamics in classical statistical mechanics: the postulate is operationally clean and has been verified to linear order on a range of backgrounds (holographic CFTs via [58, 106, 115]), but a fully general constructive derivation beyond linear order and for non-holographic matter is open.

Sharpest known limitation. The most direct challenge to the generality of the first-law assumption underlying (E0) is Wang–Braunstein [60], who prove that the first-law analogue invoked in entropic-gravity derivations fails for generic surfaces away from horizons — it holds on horizons, remains an excellent approximation on stretched horizons, but on general surfaces only in the spherically symmetric case. The present framework evades this objection by construction: its modular Hamiltonians K_ω are attached to causal-diamond / Rindler-type horizons (the Casini–Huerta–Myers and Bisognano–Wichmann sector, cf. App. I), precisely the loci where the first law is on its firmest footing. The restriction to such horizon loci is, however, a genuine scope feature of (E0)/(E0-nl) and not a free choice; a fully general constructive non-horizon derivation remains open.

Content of (E0) as an axiom. What Thm. 12.15 delivers under the axiomatic reading of (E0) is: *if one accepts the Jacobson Clausius identification* (75) *as a matter–geometry coupling axiom*, then QM and GR are simultaneous reductions of (66). The axiomatic reading is exactly the "equation of state" reading of the Einstein equation, and it is the weakest standalone matter–geometry coupling consistent with the modular-flow structure of the framework.

Constructive upgrade programme. A constructive derivation of (E0) from entanglement thermodynamics on a generic non-holographic matter sector (beyond the linearised regime of [58]) would promote (E0) from postulate to theorem and remove the last primitive axiom of Thm. 12.15. This programme is external to the present paper and out of scope; it is identified as the deepest residual of the sharpened master theorem.

12.5.2 (E1) Planck-cutoff postulate derivation via CLPW Type II crossed product

Ambient statement. On a von Neumann algebra $\mathcal{N}$ of type III associated to a bounded spacetime region in quantum field theory, the crossed product $\mathcal{N} \rtimes_\sigma \mathbb{R}$ by the modular automorphism group $\sigma^{\widehat{\omega}}$ of an equilibrium state is a type II algebra admitting a semifinite trace τ_{CLPW}; under Chandrasekaran–Longo–Penington–Witten [18] the trace is densely defined on the gravity-dressed algebra on the de Sitter static patch, and its spectrum corresponds to the renormalised generalised entropy of the subregion. The natural spectral cutoff $P_{\le E_\star}$ on the matter sector then corresponds to the high-frequency cutoff of τ_{CLPW}.

Partial progress. CLPW construct the Type II crossed product explicitly for the de Sitter static patch and show the resulting trace is semi-finite; extensions to Killing horizons of stationary black holes are in [87, 88, 92]; cosmological extensions in [89, 96, 97]; thermal-time interpretation via unsharp observables in [101]; relation to gravitational edge modes via quantum reference frames in [102, 103].

Reduction. A CLPW-style construction of the Type II crossed product on a non-stationary background with a spectral cutoff of the form $P_{\leq E_\star}$ on the matter sector would *derive* Pos. 11.30 from the Type II structure of the gravitationally-dressed algebra, rather than impose it as a Wilsonian EFT scope postulate. Under such a derivation, (H'0)'s Planck-cutoff component becomes a theorem of operator-algebraic quantum gravity on the ambient spacetime, and the framework of the present paper becomes fully sub-Planckian without an external cutoff scale.

Caveat on what the Type II structure delivers. The crossed-product trace renders the renormalised generalised entropy S_{gen} UV-finite *without* any energy cutoff: the type III_1 matter algebra and its unbounded modular/matter Hamiltonian are retained, and only the trace/entropy is regularised [18, 92, 88]. This is strictly weaker than Pos. 11.30, which band-limits the matter Hamiltonian to a finite-dimensional spectral window on compact regions. Deriving the latter requires an additional spectral-truncation mechanism beyond the CLPW trace, which the present paper does not supply; the "*derive*" and "fully sub-Planckian" language above should accordingly be read as the aspirational target of the former (E1) derivation programme — now retired (Prop. D.2, Rem. D.3) — rather than an established equivalence, and Pos. 11.30 is kept as an independent EFT postulate.

External programme. The CLPW programme is active research ([18, 87, 88, 89, 92, 97, 96]); the extension to non-stationary backgrounds with explicit matter-sector spectral cutoffs is the specific open problem.

12.5.3 (E2) QFT-regime entropy-smallness via spacetime-smeared QEIs and modular-energy estimates

Ambient statement. The smeared quantum stress-energy of Hadamard states of the free massive scalar on a globally hyperbolic spacetime admits a *spacetime-smeared*, state-independent bound (Sanders [207]): $0 \leq \omega(\phi(f)^*\phi(f)) \leq C_S(\omega(\widehat{T}^{\text{ren}}_{\mu\nu}[t^\mu t^\nu F^2]) + c_S)$ for every Hadamard ω, every smooth timelike t^μ, and $F \equiv 1$ on $\operatorname{supp} f$; in particular $\langle \widehat{T}^{\text{ren}}[ttF^2]\rangle_\omega \geq -c_S$. Along null directions, state-independent worldline bounds provably fail for free fields in four dimensions [210]; semi-local null-integrated bounds exist for all 2d QFTs satisfying the QNEC and for interacting CFTs in $d > 2$ on nearly-null strips (Fliss–Rolph [209]), and the null leg of the present framework is routed through relative-entropy monotonicity (Rem. 11.19 (v); [234, 236, 214, 237]). For Gaussian states, differences of bounded functions of modular Hamiltonians are quantitatively controlled by energy differences (Chialastri–Sanders [208], ultrastatic, $m > 0$).

Partial progress. Prop. 11.49 gives the finite-dim toy bound with explicit constants; Cor. 11.51 extends it to every finite matter dimension under (H'0). App. E proves, *conditionally on* (E2a) (Hyp. E.1), the Sobolev–Lipschitz bound

$$\left|\langle \widehat{T}^{\text{m}}_{\mu\nu}[f]\rangle_{\omega_g} - \langle \widehat{T}^{\text{m}}_{\mu\nu}[f]\rangle_{\omega_{g'}}\right| \;\leq\; L(g_0, K)\, \|f\|_{H^s}\, \|g - g'\|_{H^s}$$

with the finite constant (109) (Thm. E.7), discharging the (S4)/(S5) constants of Conj. 11.4 on the H0 cutoff sector and yielding the finite contraction constant (110) together with the conditioned-source QEI floor (111) (Cor. E.8).

Reduction. What remains for a full discharge of (E2) is the named input (E2a) — uniform Sobolev regularity of the Hadamard parametrix with mixed C^2 coincidence jets on generic compact-Cauchy backgrounds — together with the scope extensions (E2b)–(E2d) of Rem. E.10 (Dirac/Maxwell; modular-energy estimates beyond ultrastatic seeds; interacting fields via Much–Passegger–Verch [219]).

External programme. (E2a) is a specific finite-regularity microlocal problem (Hintz–Vasy type, [221]); (E2c) is a direct extension of [208]; both are single-paper-sized and have active partial progress. No claim of unconditional discharge is made.

12.5.4 (E3) Dynamical-horizon algebra via Ashtekar–Krishnan + quasi-local modular flow

Ambient statement. For a trapping or apparent horizon (Ashtekar–Krishnan dynamical horizon) $\mathcal{H}$ in a generic globally hyperbolic spacetime, the horizon subalgebra $\widetilde{\mathcal{A}}(\mathcal{H})$ carries a quasi-local modular automorphism group generated by the locally-defined surface gravity, whose fixed-point subalgebra is the centraliser $\widetilde{\mathcal{A}}(\mathcal{H})^{\sigma^{\widehat{\omega}_{\mathcal{H}}}}$.

Partial progress. The stationary (Killing) horizon case is closed by [87, 88, 92] for bifurcate Killing horizons on Schwarzschild, Kerr, and de Sitter static patch, and by [89] for cosmological stationary cases. Quantum reference frame techniques of [97, 96] supply the observer- dependent modular structure needed for the crossed product. The Longo–Morsella programme for local modular flow on subregions without a global Killing structure supplies the quasi-local modular Hamiltonian.

Reduction. Conj. 11.8 on a dynamical horizon reduces to (a) defining $\widehat{\omega}_{\mathcal{H}}$ as a quasi-local KMS state for the Longo–Morsella quasi-local modular flow on $\widetilde{\mathcal{A}}(\mathcal{H})$; (b) verifying that the centraliser-resolving POVM protocol of (KE) of Conj. 11.8 admits an equivariant extension under Ashtekar–Krishnan dynamical-horizon diffeomorphisms via the QRF-intertwining of [96]; (c) establishing measure-theoretic isomorphism of the record σ-algebra $\mathcal{G}_\gamma$ across MASA choices in the dynamical setting, which generalises the stationary case closure.

External programme. The specific open problem is the Longo–Morsella quasi-local modular flow construction plus the QRF-intertwining equivariance on Ashtekar–Krishnan horizons; the ingredients are individually available in the cited ambient literature, and the combination is identified as the external reduction.

Related contemporaneous construction. A closely related contemporaneous construction is that of Cao, Faulkner and Wang [247], who build Type II gravitational algebras on spacetimes with *two* extremal surfaces: there the crossed product yields a weight that restricts to a *trace* on the intermediate region, and area-*difference* operators whose relative entropies reproduce differences of generalised entropy controlling entanglement-wedge phase transitions. Their setting is holographic (a two-boundary AdS wormhole) and addresses entanglement-wedge transitions rather than two-boundary retrocausal conditioning; modulo verifying the area-difference trace lives on the middle region in both cases, it is naturally read as independent operator-algebraic evidence that a two-area Type II structure with an area-difference trace is well-defined, complementary to the present non-holographic, Ashtekar–Krishnan dynamical-horizon realisation.

12.5.5 (E4) Non-(CP) Tomita two-boundary conditioning via Araki perturbation of KMS states

Ambient statement. On a type III von Neumann algebra $\mathcal{N}$ with faithful normal state ω and modular automorphism group σ^ω, the perturbed state $\omega_k(x) := p(k)^{-1}\omega(L_k^* x L_k)$ for $L_k = \Lambda_k^{1/2} \in \mathcal{N}_+^{1/2}$ admits an Araki perturbation theory: if L_k is a bounded positive element of $\mathcal{N}$ (not necessarily in the centraliser $\mathcal{N}^{\sigma^\omega}$), the perturbed state ω_k is KMS for the perturbed modular flow σ^{ω_k}, whose Connes cocycle $(D\omega_k : D\omega)_t$ is the Araki perturbation cocycle [52, 39].

What is and is not at stake for complete positivity. We stress that the map $x \mapsto p(k)^{-1}\omega(L_k^* x L_k)$ defining ω_k is completely positive for every bounded positive L_k; the obstruction overcome in App. G is the loss of entire-analyticity/centraliser-invariance, not any failure of complete positivity. The genuine non-complete-positivity question — whether the associated Petz *recovery* channel is fully completely positive or only 2-positive on a type III factor — is addressed by Faulkner–Hollands [198], whose Haagerup-reduction estimates control the relative-entropy change of the conditioned state (for the underlying quantum relative entropy and Petz recovery formalism see [240]).

Partial progress. Conj. 11.1 under the (CP) hypothesis (centraliser-preservation: $[\widehat{H}_{\mathrm{m}}, \Delta_\omega^{it}] = 0$) gives normality, finite-dim reduction, and prior-average identity (items (i)–(iii) of Conj. 11.1) via the entire-analytic simplification $\sigma_z^\omega(\Lambda_k^{1/2}) = \Lambda_k^{1/2}$. Under Pos. 11.11 the centraliser condition consolidates to the single commutation Hyp. 11.13, closing the (CP) portion. In finite dimensions the per-branch content of the prior-average item (iii) is now sharp: the t_i-leg marginal of the conditioned state-over-time is the symmetrised (Margenau–Hill) operator $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$, which by Thm. 3.10 is positive iff $[\rho(\tau), \Lambda_k(\tau)] = 0$ (projective-effect case; the POVM converse fails), and whose prior-average $\sum_k p(k)\, M_k = \frac{1}{2}\{\rho, \sum_k \Lambda_k\} = \rho \succeq 0$ recovers the matter-side no-signalling identity Thm. 6.1 verbatim — so any per-branch non-classicality of the (non-(CP)) conditioning is irreducibly per-branch and washes out under (iii).

Reduction. The non-(CP) portion of Conj. 11.1 reduces to (a) extending the Petz transpose $\omega_k(x) = p(k)^{-1}\omega(L_k^* x L_k)$ to the full type III factor without the centraliser simplification — direct by Araki's perturbation theory of KMS states; (b) establishing normality of ω_k on the crossed product $\widetilde{\mathcal{N}} = \mathcal{N} \rtimes \mathbb{R}$ via the standard crossed-product lift of Araki's perturbed KMS state; (c) the finite-dim reduction (ii) and prior-average (iii) of Conj. 11.1 both follow from the crossed-product lift by finite-dim restriction, verbatim.

External programme. The Araki perturbation theory is classical [52, 39]; the specific open problem is its combination with the rigged-Hilbert two-boundary data of the present framework to give a clean (non-(CP)) realisation of ω_k. This is a technical application of the ambient theory, not a new structural problem, and is the most tractable of (E1)–(E4).

12.5.6 (E5) A.s. record-shell summability on uniformly informative protocols

Statement. Hypothesis (x-b) of Conj. 10.2: for (RP)-admissible protocols of QND-repeated type whose per-step outcome likelihoods separate every pair of live terminal sectors with a step-uniform gap $\delta > 0$ (the class $\mathfrak{P}_{\mathrm{UI}}$), the exponential-selection constants $(C_{\mathrm{UI}}, \delta')$ of the posterior concentration $1 - \pi^\gamma_{s_n}(K) \leq C_{\mathrm{UI}} e^{-\delta' n}$ are uniform over the metric ball $\mathcal{B}_{g_0}$, so that the cumulative Barrabès–Israel null-shell data $\sum_m \left\|(\mu, j, p)^{(\omega,m)}\right\|_{L^\infty(\mathcal{N}_m^{\mathrm{reg}})}$ are a.s. finite and, for κT small, keep $g_\omega \in \mathcal{B}_{g_0}$. On the certified quadrupole-free source class (where $[C_{AB}]^{\mathrm{TT}} = 0$, so the shell data (μ, j^A, p) *is* the full transverse-curvature jump $[C_{AB}] = \frac{1}{2}[\mathcal{L}_N g]$), the metric increment across each cone is the retarded integral of that derivative-jump over the bounded ball extent $\Lambda_{\max}$, giving the *one-sided* bound $\left\|g_{s_m} - g_{s_m^-}\right\|_{C^0/\mathrm{BV}} \leq \Lambda_{\max} \left\|(\mu, j^A, p)^{(m)}\right\|_{L^\infty}$. Hence shell-mass summability *implies* metric-increment total-variation summability — which is exactly what the C^0/BV terminal-limit requires. We do *not* claim the reverse bound (it fails: an oscillatory derivative-jump can integrate to an arbitrarily small net increment), but the upper direction is the only one the existence argument uses. This summability is not merely a ball-containment bookkeeping bound: it is the *existence condition for the terminal-limit metric.* Under it, the pre-terminal assembly $g_\omega\restriction_{J^-(\Sigma_{\tau_f})}$ is a Cauchy sequence in the C^0/BV completion of $\mathcal{B}_{g_0}$ whose increments are the shell layers, hence has a C^0/BV limit as $\tau \uparrow \tau_f$; without (E5) infinitely many cones would accumulate unbounded total variation and no C^0/BV terminal-limit metric would exist. Under (E5) that limit exists; and it *coincides* with the post-terminal branch extension $\widetilde{g}_K$ on Σ_{τ_f} not automatically but by the (x-c) fixed-point matching (Prop. 10.10(ii)), which pins the two one-sided limits to the same terminal data. The pre-terminal accumulation limit and the post-terminal g_K extension are thus a single continuous (a.s. shell-free, Ax. 7.1) junction at Σ_{τ_f} precisely because (x-c) matches them.

Ambient status. The fixed-background ingredients are theorems of the repeated-measurement literature: exponential sector selection and purification for uniformly informative discrete protocols [177, 175, 176]. What is genuinely open is the *uniformity* of those constants over prefix-adapted propagator families $U[g_r]$ on a Sobolev ball — a perturbation-stability question for repeated-measurement limit theorems that the ambient literature has not addressed, and the reason (x-b) is carried as a residual rather than cited as a theorem. Outside $\mathfrak{P}_{\mathrm{UI}}$ the claim fails in general (diffusive-limit posterior paths have a.s. infinite total variation; the thin-shell class is then the wrong regularity class, cf. [130]), so (E5) is a genuine scope restriction, not a technicality.

Reduction. A proof of parametric stability of the Benoist et al. exponential-selection estimates under norm-Lipschitz perturbations of the Kraus family (constants uniform on a compact parameter ball) would discharge (E5) verbatim, with the Lipschitz input supplied by (S5) of Conj. 11.4.

12.6 Revised summary under the sharpened theorem

Sharpened summary (post-discharge). Under the sharpened hypothesis bundle H' of Sec. 12.4 and the discharge appendices Apps. D–G, the dominant internal postulate of the unification framework that remains an axiom is the Jacobson–Bianchi–Myers identification Pos. 11.18 (item (E0) of Sec. 12.5; scope-split in App. I into Thm. I.1, Hyp. D.1, Hyp. I.3, and Ax. I.10); the complete reconciled commitment list is Rem. 12.16. The external residuals (E3)–(E4) are each upgraded to a theorem of the present paper and (E2) to a conditional theorem with explicit scope, while (E1) reverts to an explicit postulate:

- (E1) $\Rightarrow$ Pos. 11.30 (H0), an explicit Wilsonian scope postulate; the Type II trace structure of Hyp. D.1 yields UV-finite trace and entropy (Prop. D.2), which motivates but does not derive the cutoff (Rem. D.3).
- (E2) $\Rightarrow$ Thm. E.7 (Sobolev-Lipschitz bound with finite constant (109)) and Cor. E.8 (QFT-regime contraction (110) and conditioned-source QEI floor), *conditionally on* Hyp. E.1 (uniform Sobolev Hadamard parametrix with mixed C^2 coincidence jets; residual (E2a)), and scoped to quasi-free seeds of the free massive scalar (residuals (E2b)–(E2d), Rem. E.10).
- (E3) $\Rightarrow$ Thm. F.2 (dynamical-horizon Type II crossed product), modulo Hyp. F.1 (quasi-local modular flow with τ-dependent generator; residual (E3a)).
- (E4) $\Rightarrow$ Thms. G.1 and G.2 (Araki non-(CP) realisation), *unconditionally* — no residual.

The sharpened master unification theorem (Thm. 12.15) now states that under the axiomatic reading of (E0), the Planck-cutoff postulate H0 (Pos. 11.30), the named ambient- literature inputs (E2a), (E3a), and the named record-layer hypotheses of (H'3) with the shell-summability residual (E5), quantum mechanics and general relativity are simultaneous reductions of a single equation. (E4) is closed unconditionally. This is the strongest honest form the unification claim of the present paper can take today: every bespoke conjecture internal to the framework is discharged to a theorem of the paper, restated as an explicit postulate, or carried as a named structural hypothesis of the bundle, and each remaining ambient-literature input is a single, specific, addressable problem of operator-algebraic or QFT-on-curved-spacetime research with active partial progress.

What remains external is: (E0-nl) a single clean Fréchet- differentiability axiom on the area functional (Ax. I.10, with an explicit buck-stop remark in App. I); the realisation hypotheses Hyp. D.1 and (for the cosmological sector) Hyp. I.3; the named ambient-literature inputs (E2a) Hyp. E.1 (uniform Sobolev Hadamard parametrix with mixed C^2 coincidence jets) and (E3a) Hyp. F.1 (Longo–Morsella quasi-local modular flow on Ashtekar– Krishnan dynamical horizons beyond the perturbative-isolated- horizon regime); and the record-layer residual (E5) = (x-b) of Sec. 12.5.6, with the remaining record-layer hypotheses ((vi-b), (x-a), (x-c), (x-d), (RP)) carried in (H'3) per Rem. 12.16. Everything else is a theorem of the present paper: (E0-lin) by Thm. I.1 (FGHMVR + Oh–Park + CHM), conditional on Hyp. D.1; Hyp. 12.9 by Thm. H.1 (CLPW/DEHK gravity-sector realisation); (E1) reverts to the explicit postulate Pos. 11.30 (H0), motivated by Prop. D.2; (E2) by Thm. E.7 (conditional on (E2a), scoped per Rem. E.10); (E3) by Thm. F.2; (E4) by Thms. G.1, G.2 (unconditional). Standard Model matter is accommodated without new axiom by Thm. J.1 and Cor. J.3 (App. J); UV behaviour is organised by the Type II trace-stable inductive-limit conjecture Conj. K.1 (App. K), with the big-bang singularity explicitly flagged as out of scope; terminal POVM selection at cosmological scale is Conj. L.1 (App. L); Hawking spectrum and AdS/CFT are recovered as consistency-check theorems Thm. M.1, Thm. M.2 (App. M). A finite-dimensional numerical toy model demonstrating the five reductions is specified in App. N (Prop. N.1). Empirical content is in Sec. 13 (Props. 13.2, 13.5, 13.8, 13.12, 13.14). Five further foundational items have explicit closure: Wood–Spekkens fine-tuning evasion via the modular-theoretic character of the terminal POVM (Thm. P.1, App. P); Kuchař's four classical objections to Page–Wootters discharged via thermal time + Höhn–Smith–Lock trinity + PW=CLPW (Thm. O.1, App. O); Lorentz covariance preserved as a QRF-gauge equivalence (Rem. H.3); wave-function normalisation via the Fewster–Janssen–Loveridge–Rejzner–Waldron rigging-map / CLPW trace equivalence and the renormalised group-averaging of [299] (Thm. Q.2, App. Q); and the Bianchi–Myers chiral-anomaly invocation of Thm. J.1 is honestly restricted to the parity-even sector with explicit $\mathcal{O}(v^2/E_{\mathrm{P}}^2)$ Higgs-VEV scope and three explicitly-flagged ambient open items (Rem. J.2).

Four further foundational concerns are addressed: the Reeh–Schlieder property under ABL conditioning is preserved verbatim in the strictly-positive POVM case and controlled in the projective case (Thm. R.1, App. R); conformal compatibility under the Planck cutoff is honestly stated as EFT-precise, not operator-exact, with the cutoff sector reproducing CHM/BW modular flow up to $O((E/E_\star)^\alpha)$ (all orders only for a smoothed band edge) (Prop. S.1, App. S); the cluster of standard Bell-style no-gos (PBR, Frauchiger–Renner, Colbeck–Renner, Bong et al. Local Friendliness, Brogioli 2024 complexity-theoretic) is consolidated and addressed in a single combined evasion theorem (Thm. T.1, App. T); and the cosmological-constant prediction Prop. 13.12 is refined with an explicit variance formula $\delta\Lambda/\Lambda \sim H_0/E_{\mathrm{P}} \sim 10^{-61}$ (Prop. U.1, App. U) plus a Boltzmann-brain truncation lemma (Lem. U.2) showing the two-boundary structure structurally precludes Albrecht–Sorbo recurrence-dominated tails. The numerical companion notebook `tbrme_numerical_demo.py` promised by App. N is delivered as supplementary material and illustrates the reduction structure of (U1), (U2), (U5) on decoupled finite-dimensional subsystems; (U3), (U4) schematic at low resolution. Full FLRW-coupled verification on the $\sim 5 \times 10^5$-dimensional $\widehat{H}_{\mathrm{tot}}$ is deferred to the companion numerical paper.

Six final foundational concerns are addressed in Apps. V–Z: cluster decomposition is preserved on the bulk principal-series sector and confined to the algebra centre (Thm. V.1); causal-set discreteness is compatible as a UV substrate via [338] pAQFT-on-causets continuum limit (Rem. W.1); empirical Lorentz-dispersion bounds are sector-split: the thermal-time sector predicts $\xi = \eta = 0$ exactly (Prop. X.1), while the Σ_δ matter sector induces a subluminal linear photon dispersion whose GRB timing bounds are the binding constraint on μ, excluding the Planck-natural convention (Prop. X.2, Rem. X.4); the time-of-arrival / Pauli no-go is structurally evaded by the unbounded modular spectrum and a self-adjoint conjugate operator $\widehat{T}$ is explicitly constructed (Thm. Y.1); the observer problem is addressed by adoption of the perspective-neutral / subsystem- relativity stance [102, 96, 364] with predictions observer-invariant and algebra-type-data frame-relative gauge (Prop. Z.1); and Thm. R.1 is refined with an explicit commutant clause via Bratteli–Robinson cyclic- separating duality. Uniqueness of the TBRME is scoped to a separate companion paper, outlined in Rem. 12.18 below. The complete residual commitment set is reconciled in Rem. 12.16; in its terms: if the matter–geometry items ((E0-nl) Ax. I.10, Hyp. D.1, Hyp. I.3), the ambient inputs (E2a), (E3a), the record-layer items ((vi-b), (x-a), (x-c), (x-d), (RP), and (E5) of Sec. 12.5.6), Hyp. 11.35 (ε-tail robustness of H0), Hyp. Q.1 (orbit stratification), Conj. L.1 (Cartan-masa terminal-POVM uniqueness), and Conj. K.1 (Type II trace-stable inductive limit) are resolved, the unification is unconditional within sub-Planckian EFT scope (H0 and Pos. 11.11 retained as scope postulates, Hyp. 11.13 needed only on the branch-(II) clock sector).

Remark 12.18 (Uniqueness companion paper). Thm. 11.73 and Thm. 12.15 establish that the TBRME *has* the five simultaneous reductions (U2)–(U5) (together with the well-posedness

(U1)); they do not establish that the TBRME is the *unique* rigged-Hilbert constraint equation on $\mathcal{U}$ with these reductions. A uniqueness-modulo-centraliser statement (analogous to App. B's Kisyński closure for the clock-shift cocycle) is natural and expected. The intended form is: any densely-defined essentially self-adjoint constraint $\widehat{C}$ on $\mathcal{U}$ satisfying seven axioms (clock-covariance + d'Alembert cocycle; modular covariance on the CLPW sector; direct-sum locality in k; TDSE flat-space limit; semiclassical-Einstein WKB limit with Pinamonti–Drago–Hack rigidity; modified-Wheeler–DeWitt pure- geometry limit; Part I finite-dim consistency) coincides with $\lambda\,\widehat{\mathfrak{C}}_k$ for some $\lambda > 0$, up to (a) overall scale, (b) a centraliser-trivial automorphism of $\mathcal{M}_{\mathrm{geom}}$, and (c) Giulini–Marolf self-adjoint-extension freedom where the deficiency indices are non-trivial [280, 281]. The natural structural template is Giulini–Marolf constraint quantization uniqueness (outer frame) plus Kisyński (clock sector, imported from App. B) plus Takesaki–Connes modular uniqueness (gravity sector) plus Sanders-type semiclassical Einstein rigidity [194]. All ingredients exist in the ambient literature; the work is assembly and verification of axiom compatibility. This is scoped to a standalone ∼25–40 page companion paper ("Uniqueness of the Two-Boundary Retrocausal Master Equation: a Giulini–Marolf–Kisyński theorem for the TBRME"), outlined in the accompanying supplementary document.

13 Predictions and empirical differentiators

This section records five concrete predictions of the framework, each derivable from the machinery of Parts I–III and the discharge appendices. The purpose is to convert the paper's status from "conditional unification master theorem" to "framework with explicit testable content". Each prediction is stated with the sharpest form the present machinery supports; each is distinguished from the nearest competitor (standard semiclassical QM+GR, string-theoretic AdS/CFT, LQG, asymptotic safety, or a classical–quantum hybrid) by at least one signature.

Prop. 11.39 (Page–Geilker evasion) is the existing empirical differentiator, now sharpened to a real-time switching prediction: each apparatus tracks the realized record history, switching at the recorded decay time (Prop. 11.39); Props. 13.2, 13.5, 13.8, 13.12, 13.14 below sharpen the framework's empirical commitments on five independent fronts: black-hole unitarity (Page curve from two-boundary martingale convergence), tabletop quantum gravity (Bose–Marletto–Vedral coherent-entanglement commitment), cosmology (parity-even primordial-spectrum corrections from the clock-shift cocycle $\widehat{\Sigma}_\delta$; the earlier parity-odd claim is retracted in Rem. 13.10), the cosmological-constant problem (Type II renormalisation of the vacuum energy), and laboratory metrology (universal quadratic phonon-cavity softening, Sec. 13.5 — the designated near-term *laboratory* falsifier).

13.1 Page curve from two-boundary martingale convergence

Let $\mathcal{H} \subset \mathcal{M}$ be an evaporating black hole with Ashtekar–Krishnan dynamical horizon foliated by MOTSs $\{S_\tau\}$, carrying the Type II_∞ crossed-product algebra $\widetilde{\mathcal{A}}(\mathcal{H}) \rtimes_\sigma \mathbb{R}$ of Thm. F.2 (the evaporating-black-hole algebra is a II_∞ factor [90]), and let $\{E_k\}_{k\in\mathcal{K}}$ be a centraliser-resolving terminal POVM satisfying Ax. 7.1. Let $\pi_\tau(k)$ be the ABL posterior of Sec. 5.

Remark 13.1 (Two-sector type reconciliation). The type assignment is an attribute of the pair (algebra, observer), not of the spacetime [97, 96] (cf. Prop. Z.1(B)). A laboratory observer on a relatively compact diamond carries the type II_1 algebra of Thm. H.1(i), whose entropy is bounded above by that of its maximum-entropy (tracial) state. The evaporating Ashtekar–Krishnan horizon carries the type II_∞ algebra of Thm. F.2: its quasi-local charge is unbounded along the MOTS foliation and no positive-energy projection is imposed, which is the same mechanism that makes the evaporating-black-hole algebra a II_∞ factor in [90] and removes the maximum-entropy state in [89]. The two assignments are simultaneously consistent: the lab algebra embeds in the horizon-era algebra as a compression by a $\tau_{\mathcal{H}}$-finite projection (a II_1 corner of a II_∞ factor), so

$S_{\text{gen}}(\tau)$ computed with $\tau_{\mathcal{H}}$ can grow without bound while every compactly-dressed observer's entropy stays finite. The unbounded growth of S_{gen} before the Page time *requires* $\tau_{\mathcal{H}}(\mathbb{1}) = \infty$ and is therefore a genuinely II_∞ statement; it would be inconsistent on the II_1 laboratory sector, and the framework does not assert it there.

Proposition 13.2 (Page curve from martingale convergence)**.** *Under Pos. 11.30 (H0), Thm. F.2 and Pos. 11.11 (and, for a non-stationary horizon, conditional on the quasi-local modular-flow Hyp. F.1, item (E3a), which underlies the Type II_∞ structure of Thm. F.2):*

(i) *$\pi_\tau(k)$ is a bounded martingale with respect to the filtration $\{\mathcal{G}_\tau\}$ generated by the horizon subalgebra foliation, a.s. convergent to $\mathbf{1}_{\{K=k\}}$ along the protocol sequence.*

(ii) *The Type II modular entropy $S_{\text{gen}}(\tau) := -\log \tau_{\mathcal{H}}(\rho_2(\tau \mid K)\, P_{\leq E_\star})$ is a continuous function of τ whose τ-derivative changes sign at the* Page time *τ_P defined by the QES/island variational principle [389, 390] in the Type II setting of [246, 244].*

(iii) *(*Subleading drift.*) The clock-shift cocycle $\widehat{\Sigma}_\delta = \cos(\delta \widehat{H}_{\text{c}}/\hbar) - \mathbb{1}$ of Thm. B.5 induces an $\mathcal{O}(\delta^2/t_{\text{P}}^2)$ drift in $\pi_\tau(k)$ and shifts the Page time by $\tau_P \mapsto \tau_P\big(1 + \mathcal{O}(\delta^2/t_{\text{P}}^2)\big)$.*

Proof sketch. (i) is Thm. 7.3 applied branch-by-branch on the Type II_∞ crossed product, with the quasi-local modular flow of Hyp. F.1 as the filtration-generating dynamics. This identifies the martingale limit with the sharp indicator $\mathbf{1}_{\{K=k\}}$ only in the record-complete regime (Ax. 7.1); for an evaporating horizon, where information about K may be hidden behind the horizon and Ax. 7.1 can fail, the limit is instead the centraliser conditional expectation $\mathbb{E}_{\mathbb{P}_\gamma}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}^{\gamma}_{\tau_f-}]$ of Prop. 9.1, whose residual horizon-hidden entropy (Cor. 9.2) governs the late-time floor. (ii) follows from coincidence of the modular trace $\tau_{\mathcal{H}}$ with the generalised entropy functional [18, 92, §3] and the QES/island variational principle in the Type II setting [246, 244]. (iii) is Taylor expansion of $\widehat{\Sigma}_\delta = -\frac{1}{2}(\delta \widehat{H}_{\text{c}}/\hbar)^2 + \mathcal{O}(\delta^4)$ and modular-flow propagation through the martingale structure of (i). □

Remark 13.3 (Relation to algebraic Page-curve programmes). Three recent purely algebraic derivations of the Page curve in the Type II setting—the relative continuous-dimension construction of Gomez [246], the complementary-recovery / approximate-QEC description of Zhong [248], and the two-area gravitational algebras of Cao–Faulkner–Wang [247]—obtain the curve from, respectively, Murray–von Neumann dimension flow, algebraic relative entropy, and operator-valued weights between two boundary areas. None employs a filtration or a Bayesian posterior. The distinctive content of the present route is that the falling branch is a *martingale-convergence statement* for the ABL record (Prop. 13.2(i)), and that the two-boundary structure is realised as ABL conditioning rather than as two independent area operators; the Cao–Faulkner–Wang two-area construction is the natural benchmark for this comparison.

Remark 13.4 (Differentiator). The Page curve is reproduced by any unitary evaporation model; the distinctive signatures of the framework are (a) derivation of the *falling branch*—the late-time purification of the conditioned radiation state $\rho_2(\tau \mid K)$—from two-boundary record completeness as a martingale-convergence statement rather than as an island postulate, the Page *time* τ_P itself being the Type II QES crossover of Thm. F.2, and (b) the $\mathcal{O}(\delta^2/t_{\text{P}}^2)$ shift of τ_P from the clock-shift cocycle, absent from standard semiclassical QM+GR, AdS/CFT (no δ parameter), and LQG.

13.2 Coherent gravity-mediated entanglement at Bose–Marletto–Vedral scales

Proposition 13.5 (Coherent entanglement from Clausius coupling)**.** *Under Pos. 11.18 (Jacobson–Bianchi–Myers Clausius coupling) and its linearised discharge Thm. I.1, the gravitational interaction between two mesoscopic mass superpositions m_1, m_2 at separation d, coupling time t, in the regime $G m_1 m_2 t/(\hbar d) \sim 1$ of Bose–Marletto–Vedral [121, 124, 118], induces a coherent* *reduced-state entanglement negativity*

$$\mathcal{N} \;=\; \frac{Gm_1 m_2\, t}{\hbar d} \;-\; \Gamma_{\rm DSW}(\kappa)\, t \;+\; \mathcal{O}(G^2), \tag{76}$$

where $\Gamma_{\rm DSW}(\kappa)$ is the Danielson–Satishchandran –Wald horizon-decoherence rate [119], proportional to the effective surface gravity κ of any background Killing horizon (including acceleration horizons for non- inertial observers), and $\mathcal{O}(G^2)$ collects nonlinear corrections outside the linearised Clausius regime governed by Ax. I.10.

For terrestrial BMV parameters ($m \sim 10^{-14}\,$kg, $d \sim 10^2\,\mu$m, $t \sim 1\,$s) the Danielson–Satishchandran–Wald factor is negligible, $\Gamma_{\rm DSW} t \ll \phi$, so $e^{-\Gamma_{\rm DSW} t} \approx 1$: the soft-graviton which-path flux requires coherence times of order years or near-horizon separations to compete [119, 120]. Systematic scans of the (m, d, t) parameter space for QGEM-type protocols are given in [140]. The floor becomes the dominant ceiling only near a genuine Killing horizon, including the acceleration-horizon variant of Rem. 13.6.

Proof sketch. The Clausius identification $\widehat{G}^{(Q)}_{\mu\nu} = (8\pi G/c^4)\widehat{T}^{\rm m}_{\mu\nu}$ of Pos. 11.18 (discharged in the linearised regime by Thm. I.1) is an operator identity on the matter-plus-gravity crossed product, implementing gravity as a local unitary coherent interaction of stress-energy with the modular Hamiltonian's area-response. Structurally this is the Christodoulou et al. [118] linearised-QG locally-mediated-entanglement derivation, yielding the BMV leading-order negativity $Gm_1m_2t/(\hbar d)$. The mass-independence of this leading negativity in the appropriate parameter window is analysed in [122]. The DSW floor is the universal decoherence rate from soft-graviton flux across any background Killing horizon, derived in [119]. Absence of stochastic diffusion (no random-walk master equation term) distinguishes the framework both from the Diósi–Penrose collapse class — which nonetheless predicts gravitationally induced entanglement below a length scale [127] — and from the Oppenheim post-quantum classical-gravity class, whose decoherence–diffusion trade-off mandates a stochastic diffusion term [128]: Pos. 11.18 is a deterministic operator identity, not a stochastic coupling. This is the discriminator sharpened by Carney, Karydas, Scharnhorst, Singh and Taylor [141]: explicit microscopic models in which the Newtonian potential arises from free-energy extremization of an information reservoir generically replace coherent two-body entanglement by position-diagonal Lindblad dissipators, predicting anomalous decoherence and the *absence* of gravitationally-induced entanglement. The present identification (47)–(48) is *not* of that dissipative class — it is with the self-adjoint, unitarily-implemented modular Hamiltonian $K_\omega = -\log \Delta_\omega$, which generates a $*$-automorphism group with no Lindblad part — so the coherent Newtonian phase, and hence the BMV negativity, is preserved by construction, and consistency with GIE is a constraint the framework satisfies rather than a prediction it violates. □

Remark 13.6 (Sharp commitment). The framework *commits* to the quantum-gravity side of the BMV dichotomy: entanglement generation at BMV scales will be coherent (not stochastic) and bounded above by the DSW horizon-decoherence floor. We stress, however, that following Aziz–Howl [142] and Di Biagio [143] (with rebuttals [144]), entanglement generation *per se* no longer cleanly discriminates a quantum from a suitably field-theoretic classical mediator — a point already flagged here via the Diósi–Penrose prediction of gravitationally induced entanglement [127]. The framework's genuinely distinguishing commitment is therefore not the presence of entanglement but its *deterministic, non-stochastic* character (no random-walk master-equation term of the kind mandated by the Oppenheim decoherence–diffusion trade-off [128], and no length-scale threshold of the kind in the Diósi–Penrose prediction [127]), which a measured coherence-versus-stochasticity bound can test. A secondary distinctive prediction is that an acceleration-horizon variant of the BMV protocol (entanglement witness in a uniformly accelerated frame) will show surface-gravity-proportional coherence loss $\Gamma_{\rm DSW}(\kappa) \propto \kappa$, unique to the Clausius + two-boundary framework and absent from ordinary linearised QG. This κ-proportionality is consistent with the warm-horizon analysis of Wilson-Gerow, Dugad and Chen [123], who show—taking the Rindler horizon as the worked example—that the DSW decoherence is intrinsically tied to the horizon (an inertial finite-temperature laboratory does *not* decohere at the DSW rate), with the rate fixed by the surface gravity. The framework's prediction is thus that the acceleration-horizon coherence loss tracks κ (equivalently the Unruh temperature $T_U = \hbar\kappa/2\pi k_B c$), distinguishing it from any inertial-frame decoherence model. Quantitatively, however, $\Gamma_{\rm DSW}(\kappa)$ is unobservably small for any terrestrial protocol: Danielson, Satishchandran and Wald [119, 120] show the horizon-decoherence rate is negligible unless the superposition is held near a black-hole-scale κ with large masses and separations. For a laboratory acceleration horizon the floor lies many orders of magnitude below both the leading BMV negativity $Gm_1m_2t/(\hbar d)$ and ambient decoherence, so the acceleration-horizon variant is a matter of principle rather than a 2026–2035 target. The near-term observable is the leading coherent negativity together with the absence of a stochastic term, not the $\Gamma_{\rm DSW}$ floor itself.

13.3 Primordial-spectrum signatures from $\widehat{\Sigma}_\delta$: the parity-even correction, and a retraction

Lemma 13.7 (Parity-evenness of the $\widehat{\Sigma}_\delta$-modified dynamics)**.** *Let $U_{\rm P}$ be the unitary implementing spatial parity on the Born–Oppenheimer perturbation sector, $\zeta_{\mathbf{k}} \mapsto \zeta_{-\mathbf{k}}$, $h^{(\pm 2)}_{\mathbf{k}} \mapsto h^{(\mp 2)}_{-\mathbf{k}}$, extended by* $\mathbb{1}$ *on the clock factor. Assume:*

(P1) *the seed state ω of Pos. 11.11 is parity-invariant, $\omega \circ \operatorname{Ad} U_{\rm P} = \omega$ with $U_{\rm P}\mathcal{N}U_{\rm P}^* = \mathcal{N}$. This holds for FRW (parity centred on any comoving worldline is an isometry of the flat slicing, and the Bunch–Davies two-point function depends only on geodesic separation), for the de Sitter static patch (the observer-centred parity — r fixed, antipodal map on the S^2 fibres, i.e. $\mathbf{x} \mapsto -\mathbf{x}$ about the worldline — is an improper isometry mapping the patch to itself and fixing the horizon; the* global *antipodal parity, which maps the patch to its antipode, is not used), and for Schwarzschild-type wedges via the centre-fixing antipodal map on the sphere fibres. It* fails *for the flat Rindler wedge: the* edge-centred *spatial parity (reflection about the codimension-2 edge $\partial W_{\rm R}$) maps $W_{\rm R}$ cleanly to its causal complement, so $U_{\rm P}\mathcal{N}U_{\rm P}^* = \mathcal{N}'$; an interior-point-centred parity instead maps $W_{\rm R}$ to a boosted half-space, not to the complement, and so does not even implement the algebra flip. Either way no point-centred parity preserves $\mathcal{N}$, and the flat wedge is excluded from the scope of this lemma (a wedge-preserving transverse reflection exists but is a different unitary and yields strictly weaker conclusions);*

(P2) $[U_{\rm P}, \widehat{H}_{\rm s}] = 0$ *(true for Einstein–Hilbert gravity minimally coupled to scalar matter at every order in perturbations: all vertices are built from δ_{ij} and ∂_i; no ϵ_{ijk} occurs).*

Then:

(i) $[U_{\rm P}, \widehat{H}_{\rm c}] = 0$, *hence* $[U_{\rm P}, \widehat{\Sigma}_\delta] = 0$ *and* $[U_{\rm P}, \widehat{C}_{\alpha,\delta t}] = 0$.

(ii) *The effective generator $h(\widehat{H}_{\rm s})$ of Prop. C.2 commutes with $U_{\rm P}$, and the modified adiabatic (Bunch–Davies) vacuum coincides with the $U_{\rm P}$-invariant vacuum of $\widehat{H}_{\rm s}$ on the band (strict monotonicity of h: no level folding). Hence every in–in correlator obeys $\langle \mathcal{O} \rangle = \langle U_{\rm P}\mathcal{O}U_{\rm P}^* \rangle$, and every parity-odd correlator — scalar, tensor, or mixed — vanishes identically, at all orders in μ and at all orders of in–in perturbation theory.*

(iii) *The dispersion correction is helicity-blind: h acts by functional calculus on $\widehat{H}_{\rm s}$, whose tensor-sector restriction is degenerate in helicity, so $\omega_+(k) = \omega_-(k) = \omega(k)\big[1 - \frac{\mu\hbar\omega(k)}{2} + \dots\big]$: no graviton-helicity splitting, no birefringence, no chiral gravitational-wave production.*

Proof. (i) In the GNS representation of ω, (P1) yields $U_{\rm P}\Omega = \Omega$. For $A \in \mathcal{N}$, $U_{\rm P}S_\omega U_{\rm P}^* A\Omega = U_{\rm P}S_\omega(U_{\rm P}^* A U_{\rm P})\Omega = U_{\rm P}(U_{\rm P}^* A^* U_{\rm P})\Omega = A^*\Omega$, so $U_{\rm P}S_\omega U_{\rm P}^* = S_\omega$ on the core $\mathcal{N}\Omega$, hence as closed operators; uniqueness of the polar decomposition $S_\omega = J\Delta_\omega^{1/2}$ gives $[U_{\rm P}, \Delta_\omega] = [U_{\rm P}, J] = 0$, hence $[U_{\rm P}, \widehat{H}_{\rm c}] = 0$ by the spectral theorem. The crossed-product lift and CLPW dressing are functorial in $(\mathcal{N}, \omega)$-preserving unitaries, and $\widehat{\Sigma}_\delta = \cos(\delta \widehat{H}_{\rm c}/\hbar) - \mathbb{1}$ is a bounded real Borel function of $\widehat{H}_{\rm c}$. (ii) h is a real Borel function (Prop. C.2), so $[U_{\rm P}, h(\widehat{H}_{\rm s})] = 0$ by (P2) and functional calculus; the ground states of $h(\widehat{H}_{\rm s})$ and $\widehat{H}_{\rm s}$ coincide on the band by strict monotonicity. In–in expectations of fields evolved by $e^{-ih(\widehat{H}_{\rm s})t/\hbar}$ in a $U_{\rm P}$-invariant state are $U_{\rm P}$-covariant term by term; for $U_{\rm P}\mathcal{O}U_{\rm P}^* = -\mathcal{O}$ this gives $\langle\mathcal{O}\rangle = -\langle\mathcal{O}\rangle = 0$. (iii) is immediate from helicity degeneracy and functional calculus. □

Proposition 13.8 (CMB correction from the clock-shift cocycle; parity-even)**.** *Assume Thm. B.5, the Born–Oppenheimer expansion of Conj. 10.13 at inflationary scale* $H_{\rm inf} \ll E_{\rm P}$*, and the matching* (99)*. Then:*

(i) *the scalar power spectrum is*

$$\mathcal{P}(k) \;=\; \mathcal{P}_0(k)\Big[1 + c_P\,\kappa H_{\rm inf}^2 \big(k_\star/k\big)^3\Big] + \mathcal{O}\big(\kappa^2 H_{\rm inf}^4\big), \qquad c_P \;=\; 4 - 2\gamma_E - 2\log(-2k\eta), \quad (77)$$

in the convention $\kappa = m_{\rm P}^{-2}$ *of [251], with* $c_P \approx 1.5$ *at horizon crossing (the non-unitary de Sitter treatment gives* $c_P \approx 0.988$ *[250]); the dimensionless spectral coefficient, previously denoted "*α*", is renamed* c_P *to avoid collision with the constraint coupling* α *of Conj. 10.13, and is identical to the matching coefficient* c_0 *of* (99)*. The tensor spectrum receives a correction of the same* $(k_\star/k)^3$ *form with an* $\mathcal{O}(1)$*-different coefficient [250], identical for both graviton helicities (Lem. 13.7(iii));*

(ii) *every parity-odd correlator of the scalar and tensor perturbations vanishes identically (Lem. 13.7(ii)); in particular* $f_{\rm NL}^{\rm odd} = 0$ *and* $\tau_{\rm NL}^{\rm odd} = 0$.

Proof sketch. (i) On the perturbation sector the constraint reduces fibrewise (Prop. C.2) to $i\hbar\partial_t\psi = h(\widehat{H}_{\rm s})\psi$ with $h(E) = E - \frac{\mu}{2}E^2 + \mathcal{O}(\alpha\,\delta t^4 E^4/\hbar^4)$ (98); the matching (99) fixes $\mu = 2c_0 H_{\rm inf}/m_{\rm P}^2$ precisely by equating this quadratic-in-energy correction to the unitary Born–Oppenheimer correction of [251, Eqs. (158)–(160)], whose spectral form is (77): the $(k_\star/k)^3$ envelope and the $\log(-2k\eta)$ running (the e-fold count to horizon exit) are those of the Kiefer–Krämer family [249, 250, 251]. Because μ is fixed by this very comparison, the $\mathcal{O}(\kappa H_{\rm inf}^2)$ term is a *reproduction* — a consistency of the cosine cocycle with the established quantum-gravitational correction — not an independent prediction; the first framework-distinctive residue is the quartic term of (98), of relative order $\kappa^2 H_{\rm inf}^4$ and dependent on the $(\alpha, \delta t)$ split convention (Rem. C.4). (ii) is Lem. 13.7. □

Remark 13.9 (Amplitude honesty). At the observational ceiling $H_{\rm inf}/m_{\rm P} \lesssim 2.5 \times 10^{-5}$ set by the tensor-to-scalar bound, the fractional correction (77) is $\lesssim 10^{-9}$ at $k = k_\star$ and decays as $(k_\star/k)^3$: it is largest precisely on the cosmic-variance-dominated largest scales, and is unobservable by any currently projected CMB or LSS program. It is recorded as a consistency statement, not as a falsifier; the designated near-term *laboratory* falsifier is the phonon-cavity channel of Sec. 13.5 (the framework's leading falsifier overall being photon time-of-flight, Rem. X.4).

Remark 13.10 (Retraction of the parity-odd bispectrum claim). Earlier versions of this manuscript advertised, as the flagship consequence of Prop. 13.8, a parity-odd scalar bispectrum "$f_{\rm NL}^{\rm odd} = \gamma\,(H_{\rm inf}/E_{\rm P})\,\mathcal{S}_\delta[k_1, k_2, k_3]$". That claim is *retracted*, for three independent reasons, in decreasing order of severity. *(a) Kinematic.* A parity-odd *scalar bispectrum* is forbidden by rotational and translational invariance alone: with $\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 = 0$ the momenta are coplanar, every invariant of the configuration is a function of the parity-even side lengths (k_1, k_2, k_3), and the unique pseudoscalar $\mathbf{k}_1 \cdot (\mathbf{k}_2 \times \mathbf{k}_3)$ vanishes identically. The displayed object was ill-posed as stated; a parity-odd scalar signal can first appear in the connected *trispectrum*, through the imaginary part of the correlator, odd under $\operatorname{sgn} \mathbf{k}_1 \cdot (\mathbf{k}_2 \times \mathbf{k}_3)$ [255, 256]. *(b) Dynamical.* No parity-odd correlator — bispectrum or trispectrum, scalar or tensor — is derivable from $\widehat{\Sigma}_\delta$ at all: Lem. 13.7

shows the modified dynamics and state are exactly parity-even. The earlier proof sketch asserted a “parity-breaking kernel” from “coupling to the tensor-mode polarisation basis”; no such coupling is derived anywhere in this paper, no ϵ-tensor, gravitational Chern–Simons, axial-chemical-potential, or helicity-dependent structure occurs in the framework (indeed Thm. J.1 is deliberately restricted to the parity-even anomaly sector), and the tensor-sector correction is helicity-blind (Lem. 13.7(iii)). Sourcing parity-odd primordial correlators is known to require precisely such structures — a pseudoscalar coupling ($f(\phi)R\widetilde{R}$, $\phi F\widetilde{F}$), a helicity-split dispersion, a chiral non-Bunch–Davies state, or parity-violating exchange of massive spinning fields — and even given one, the parity-odd scalar trispectrum vanishes at tree level under scale invariance and Bunch–Davies conditions [255]; the yes-go routes (scale-invariance breaking, ghost-condensate-type dispersion, massive spinning exchange [256, 260]) all presuppose a parity-odd vertex that $\widehat{\Sigma}_\delta$ — a real, even function of the parity-commuting clock generator — does not supply; its $\log(-2k\eta)$ scale-breaking is inert for parity purposes. The only way to reintroduce a parity-odd correlator in the two-boundary framework would be to *posit* a parity-non-invariant terminal effect in Conj. L.1: an unmotivated new axiom, not a consequence of $\widehat{\Sigma}_\delta$, and we do not adopt it. *(c) Observational.* The anchor has evaporated: the claimed BOSS 4PCF parity signal does not survive reanalysis (0–2.5σ depending on analysis choices, with an unmodelled $\sim 6\sigma$ bias term in the original covariance treatment [254]); the DESI DR1 LRG 4PCF is consistent with zero parity violation ([257]; the auto-correlation excess of [258] is in significant tension with its own null cross-patch test and is attributed to covariance mis-estimation), compressed kurto-spectrum searches find no signal in either BOSS or DESI [259], and the Planck PR4 parity-odd tensor-bispectrum analysis is likewise null [253].

Remark 13.11 (Distinguishability from LQC, in principle). The dressed-metric LQC corrections of Agullo et al. [186] give a low-k *suppression* of $\mathcal{P}(k)$ only below the bounce scale k_I (or for non-local-in-time initial data), with *enhancement* at observable large scales for standard initial data; (77) is a low-k *enhancement* ($c_P > 0$ near horizon crossing) with a $(k_\star/k)^3$ envelope and $\log(-2k\eta)$ modulation. The distinguishing feature is therefore the *shape* — a monotonic $(k_\star/k)^3$ envelope with logarithmic modulation, versus LQC's non-monotonic oscillatory large-scale structure — rather than a clean opposite sign, and by Rem. 13.9 the comparison is a matter of principle, not a foreseeable measurement.

13.4 Thermal-time cosmological constant

Proposition 13.12 (Absence of the Planckian vacuum-energy catastrophe)**.** *Under Pos. 11.11 (Connes–Rovelli thermal time), the CLPW Type II trace structure (Hyp. D.1, Prop. D.2), and the de Sitter-facing scope of the Clausius identification (Hyp. I.3, or the postulate form of Pos. 11.18 on static-patch diamonds; Rem. I.4), the matter-sector vacuum-energy contribution to the effective cosmological constant Λ_{eff} is independent of the UV Planck scale and scales as*

$$\Lambda_{\text{eff}} \;=\; \Lambda_0 \;+\; c_\Lambda\, H_0^2 \;+\; \mathcal{O}\big(H_0^2 \cdot (H_0/E_{\text{P}})^2\big), \tag{78}$$

where Λ_0 is the modular-RG fixed point of the entropic-gravity Clausius coupling on the tracial state of the static-patch crossed product $\widetilde{\mathcal{N}}_{\text{dS}}$ of [18], H_0 the Hubble scale, and $c_\Lambda \in \mathbb{R}$ an $\mathcal{O}(1)$ constant set by the de Sitter surface-gravity normalisation. In particular, the naive quartic vacuum energy $\sim E_{\text{P}}^4$ is absorbed — exactly as the cosmological-constant counterterm of ordinary renormalised QFT absorbs it — by the induced-gravity (Sakharov) cosmological counterterm, not *by the Type II trace; the single state-independent additive constant of the Type II trace instead fixes the generalised-entropy zero-point, an entropy shift on the $A/4G$ term that cancels in every entropy difference (and so cannot absorb an energy density). The non-trivial content is not the removal of this leading divergence — which standard renormalisation already achieves — but that the Type II trace fixes the generalised-entropy zero-point* canonically and state-independently *while the Planckian vacuum energy is absorbed by the induced-gravity (Sakharov) cosmological* *counterterm; conditional on Type II trace-finiteness (see the proof), the framework then claims the radiative* stability *of* $\Lambda_{\text{eff}} - \Lambda_0$ *at* $\mathcal{O}(H_0^2)$*, not merely the cancellation of one divergence.*

Proof sketch. The CLPW Type II trace τ is defined modulo a single finite, state-independent additive constant in $\log \tau$ — the generalised-entropy zero-point of the tracial (maximum-entropy) state, which the Type II$_1$ structure fixes canonically ([18, §3]; Witten eq. (3.47), [238]). This is an *entropy* zero-point, not an energy density. The Planckian matter vacuum energy $\sim E_{\text{P}}^4$ is a distinct object, absorbed — exactly as the Λ counterterm of renormalised QFT absorbs it — by the induced-gravity (Sakharov) cosmological counterterm, the E_{P}^4 partner of the E_{P}^2 term that induces $1/G$ ([403, 404, 405]; the single heat-kernel str-sum with two allocations). With the entropy zero-point canonically fixed, Pos. 11.11's identification of the clock with the modular Hamiltonian makes the cosmological constant the modular-RG fixed point of the Clausius coupling: $\partial_\tau S_{\text{gen}} = 0$ on the tracial state gives Λ_{eff} as an eigenvalue of the thermal-time flow, expressible in Hubble units up to surface-gravity normalisation [89, 263]. The claimed sub-leading stability $\mathcal{O}(H_0^2 \cdot (H_0/E_{\text{P}})^2)$ is *conditional* on Type II trace-finiteness (Prop. D.2), which delivers finiteness of the trace and generalised entropy but *not* the loop-order vacuum-energy scaling; that scaling is here posited, not computed. □

Remark 13.13 (Buck-stop and relation to competitors). Prop. 13.12 does not *derive* the numerical value of Λ_{eff}: the constants Λ_0, c_Λ remain free parameters set by the static-patch surface gravity. What it predicts is the *absence of the* E_{P}^4 *catastrophe*: the matter-sector vacuum energy does not contribute a Planckian term to Λ_{eff}, because the induced-gravity (Sakharov) cosmological counterterm absorbs it while the Type II trace fixes the generalised-entropy zero-point canonically. This is a sharper *structural* statement than standard QFT (which suffers the 10^{120}-catastrophe), comparable in philosophy to asymptotic safety's fixed-point control of the ultraviolet Λ-flow — the infrared Λ remaining a relevant coupling matched to observation, not predicted [187] — and distinct from the string landscape's anthropic tuning. The $c_\Lambda H_0^2$ running coincides term-for-term with the running-vacuum model $\rho_{\text{vac}}(H) = c_0 + \frac{3\nu}{8\pi} E_{\text{P}}^2 H^2 + \mathcal{O}(H^4)$ derived from QFT in curved spacetime [330] and with holographic dark energy under a Hubble infrared cutoff [332]; here that same H^2 term arises instead as the thermal-time RG fixed point of the Type II Clausius coupling. The holographic dark-energy cutoff invoked above itself traces to the black-hole effective-field-theory bound of Cohen, Kaplan, and Nelson [326], who first argued that forbidding states of Schwarzschild radius larger than the infrared box size ties the ultraviolet and infrared cutoffs by $\Lambda_{\text{UV}}^4 \lesssim E_{\text{P}}^2/L^2$ and hence yields a vacuum energy $\rho_{\text{vac}} \sim E_{\text{P}}^2 H^2$ — the same H^2 term, reached by a second, independent black-hole argument. Two cautions fix its relation to the present work. *Which cutoff is which:* the CKN infrared-locked cutoff is a *vacuum-energy* scale, $\sim (E_{\text{P}}^2 H_0^2)^{1/4} \sim 10^{-3}\,\text{eV}$, and must *not* be identified with the anchoring scale $E_\star$ of Rem. 11.31, which is a Planckian degrees-of-freedom cutoff; conflating them would drive $g = G E_\star^2/\hbar c^5$ down by some sixty orders of magnitude and destroy the induced-gravity accounting. They are distinct cutoffs on distinct observables that surface in the same str-sum at different allocations (Rem. D.4). *The* $w = 0$ *evasion:* a Hubble-scale infrared cutoff of the naive holographic form yields an equation of state $w = 0$ for the resulting dark energy [327], a well-known defect of the simplest holographic models. The present route does not inherit it: the H^2 term here is generated as a thermal-time RG fixed point of the Type II Clausius coupling, not by saturating an infrared-cutoff energy budget, so its equation of state is set by the running-vacuum structure rather than pinned to $w = 0$. The genuinely new content is the *mechanism*, not the H^2 scaling: the Planckian vacuum energy is absorbed by the induced-gravity counterterm with the generalised-entropy accounting fixed canonically by the state-independent Type II trace ambiguity, rather than tuned order-by-order, so that running-vacuum and holographic dark energy supply an *independent*, observationally constrained QFT route to the same term and thereby render the prediction falsifiable through their existing phenomenology.

A caveat forecloses a tempting over-reading of this falsifiability, which an adversarial scoping pass confirmed is a dead end. One might hope the coincidence with running vacuum promotes the coefficient ν to a *content-fixed prediction*, since the renormalisation-group running of the vacuum energy density $\rho_{\rm vac}$ (the running-vacuum coefficient $\nu_{\rm eff}$) is governed by the *same* a_1 heat-kernel supertrace that fixes g in Rem. 11.31 ([331], Eq. (7.3): $\nu_{\rm eff} = \frac{1}{2\pi}(\frac{1}{6}-\xi)\, M_X^2/E_{\rm P}^2\,[1+\mathcal{O}(m^2/M_X^2)]$, with M_X the mass of the heaviest *contributing* field). It does not. The coefficient is sourced only by fields at the matching scale M_X; the framework's native scale is the anchor $E_\star \approx 0.65\,E_{\rm P}$, which would give $\nu_{\rm eff} \sim 10^{-2}$ — already excluded, by more than an order of magnitude, by $\dot{G}/G$ and running-vacuum bounds [330, 333]. Lifting ν into an observable-but-allowed window would require a sub-Planckian intermediate scale $M_X \sim 10^{-1}E_{\rm P}$ that the framework does not possess and that, being a new field in the same weighted supertrace, would shift $\mathrm{str}[k_1]$ and de-tune $g \approx 0.42$ — the very No-Fundamental-G result it is meant to extend. The framework therefore does *not* predict an observable ν: in its Type II G-running branch it predicts ν unobservably small, a *consistency requirement* passed by the current null fits rather than a live signal. The falsifiable content of Prop. 13.12 is accordingly the *structural* statement — an H^2 running with the $E_{\rm P}^4$ term absent, shared with running vacuum and holographic dark energy — not a numerical prediction for ν.

13.5 Laboratory falsifier: universal quadratic softening in macroscopic phonon cavities

The retraction of Rem. 13.10 removes the channel previously advertised as the framework's distinctive near-term observable. Its replacement was already implicit in Rem. C.4: the universal quadratic-in-energy softening $E \mapsto E - \frac{\mu}{2}E^2$ of (98) is directly — and *presently* — constrained by precision phonon-cavity metrology. This is the leading *laboratory* channel; the binding constraint on μ is astrophysical — the induced subluminal photon time-of-flight dispersion of Prop. X.2, bounded by GRB timing at $\mu < 1.0 \times 10^{-29}\,\mathrm{eV}^{-1}$ (Rem. X.4), 5.3 decades below the phonon-cavity ceiling (80).

Proposition 13.14 (Amplitude–frequency softening; current bound)**.** *Let a mechanical mode of angular frequency ω and excitation energy E (above the equilibrium state) lie in the band-limited sector of Prop. C.2. Then:*

(i) *(*Universal softening.*) The modified generator shifts the levels by $\delta E_n = -\frac{\mu}{2}E_n^2 + \mathcal{O}(\alpha\,\delta t^4 E_n^4/\hbar^4)$ and the mode frequency by*

$$\frac{\tilde{\omega}(E) - \omega}{\omega} \;=\; -\,\mu E \;+\; \mathcal{O}\big(\mu\hbar\omega,\; \mu^2 E^2\big), \tag{79}$$

with the same *μ — negative shift for $\alpha > 0$ — for every mode, every material, and every effective mass: the correction is a function of the mode energy alone.*

(ii) *(*Current bound, under a no-cancellation assumption on the intrinsic crystal anharmonicity; see the proof sketch.*) The cryogenic quartz BAW amplitude–frequency measurement of Campbell–Tobar–Galliou–Goryachev [252] bounds the identical observable, with the published fit performed for the* hardening *sign and the entire measured net nonlinearity conservatively attributed to the new-physics term. Their strongest fitted exclusion, $\beta_0 < 62.4$ (3σ; mode $B_{3,0,0}$ at $\omega_r/2\pi = 5.505$ MHz, effective mass $m_{\rm eff} = 8.5$ mg), transcribes to*

$$|\mu| \;\lesssim\; \beta_0^{\max}\,\frac{m_{\rm eff}c^2}{E_{\rm P}^2} \;\approx\; 2.0\times10^{-24}\ \mathrm{eV}^{-1}, \qquad \textit{i.e.} \qquad E_{\rm QG,1} = 1/\mu \;\gtrsim\; 5.0\times10^{14}\ \mathrm{GeV}. \tag{80}$$

Proof sketch. (i) With $E_n = \hbar\omega(n+\frac{1}{2})$, the shifted spacing is $E'_{n+1} - E'_n = \hbar\omega\big[1 - \mu E_n - \frac{\mu\hbar\omega}{2}\big]$; classically, in action-angle variables, $H' = H - \frac{\mu}{2}H^2$ gives $\omega' = \partial H'/\partial I = \omega(1-\mu E)$ exactly. Universality is functional calculus: h depends on $\widehat{H}_{\rm s}$ only. (ii) Ref. [252] fits $\tilde{\omega}_r = \omega_r[1 + \frac{\beta_0}{2}(m_{\rm eff}\omega_r x_0\, c/E_{\rm P})^2]$ to the measured frequency perturbation versus drive amplitude

x_0. With mode energy $E = \frac{1}{2}m_{\rm eff}\omega_r^2 x_0^2$, i.e. $(m_{\rm eff}\omega_r x_0\, c)^2 = 2m_{\rm eff}c^2 E$, their observable is $\Delta\omega/\omega = \beta_0\, m_{\rm eff}c^2 E/E_{\rm P}^2$, matching (79) in modulus with $|\mu| = \beta_0\, m_{\rm eff}c^2/E_{\rm P}^2$. Numerically $m_{\rm eff}c^2 = 4.77\times10^{30}\,\mathrm{eV}$ and $E_{\rm P}^2 = 1.49\times10^{56}\,\mathrm{eV}^2$, so $\beta_0^{\max} = 62.4$ gives $|\mu| \lesssim 2.0\times10^{-24}\,\mathrm{eV}^{-1}$. Two caveats, flagged honestly: the published fit is for the hardening sign ($\beta_0 > 0$) whereas (79) is a softening for $\alpha > 0$, and [252] attributes the entire measured (net-hardening) nonlinearity to the new-physics term without independently subtracting the intrinsic Tiersten–Stevens anharmonicity — so a softening $-\mu E$ masked by intrinsic hardening of comparable size is not excluded by the published analysis alone. The transcription therefore holds under a *no-cancellation assumption* (supported, but not proven, by the consistency of the three net-positive fitted modes); a dedicated re-fit with the softening sign would fix the exact factor. Second, the mode-energy calibration inherits the effective-mass and circuit-model systematics of [252]. The transcription is otherwise exact within the single-mode harmonic model. □

Remark 13.15 (Universality discriminator; freedom from the soccer-ball ambiguity). GUP/minimum-length phenomenology suffers the composite-body ("soccer-ball") ambiguity: it must decide which mass — constituent, centre-of-mass, effective — enters the deformed commutator, and the inferred β_0 acquires an unphysical platform dependence (as [252] itself cautions). The present channel is free of this ambiguity: $\delta E_n = -\frac{\mu}{2}E_n^2$ depends on the *mode excitation energy alone.* Two sharp discriminators follow. (1) *Universality:* the framework predicts one global μ — equivalently, an apparent GUP parameter scaling as $\beta_0^{\rm app} \propto 1/m_{\rm eff}$ across modes and platforms — with fixed (negative) sign for $\alpha > 0$; crystal anharmonicity, the dominant mundane background, is mode-, material-, and temperature-dependent with either sign. The clean protocol is a multi-mode, two-material consistency test (e.g. quartz and sapphire cavities, ≥ 3 overtones each): a single fitted softening coefficient per unit energy across all modes and both materials is the framework's signature; scatter falsifies universality and bounds μ at the smallest fitted coefficient. (2) *Zero-point insensitivity:* writing the total energy as $E_{\rm ref} + E_{\rm mode}$, all $E_{\rm ref}$-dependent contributions shift ω by amplitude-*independent* amounts absorbed in the frequency calibration; the amplitude-dependent signal isolates $-\mu E_{\rm mode}$ exactly, so the test is well posed independently of the zero of $\widehat{H}_{\rm s}$ (which the Born–Oppenheimer split fixes as an excitation Hamiltonian in any case).

Remark 13.16 (MICROSCOPE forces the excitation reading of $\widehat{H}_{\rm s}$). The excitation split invoked in Rem. 13.15(2) is *forced* by equivalence-principle data, not a convention. If the softening $-\frac{\mu}{2}\widehat{H}_{\rm s}^2$ acted on the rest-mass-inclusive Hamiltonian of a composite body, the correction would be nonlinear in total energy and hence composition-dependent in free fall: two macroscopic test bodies of rest energies $E_{1,2} = M_{1,2}c^2$ would acquire an Eötvös parameter $\eta \sim \frac{\mu}{2}|E_1 - E_2|$, which for the kg-scale MICROSCOPE test masses ($Mc^2 \sim 5.6\times10^{35}\,\mathrm{eV}$) exceeds the final MICROSCOPE bound $\eta(\mathrm{Ti},\mathrm{Pt}) < 4\times10^{-15}$ [347] by at least *sixteen orders of magnitude* for every candidate value of μ considered in this paper (matched, Planck-natural, or the BAW ceiling). Even the weaker per-atom reading (softening acting on single-atom rest energies; $\Delta(Mc^2)_{\rm Pt-Ti} \approx 137\,\mathrm{GeV}$ per atom) would force $\mu < 2\eta/\Delta(Mc^2) \approx 5.8\times10^{-26}\,\mathrm{eV}^{-1}$, $35\times$ tighter than (80). MICROSCOPE thereby independently confirms the Born–Oppenheimer excitation reading: $\widehat{H}_{\rm s}$ enters the softening as an excitation Hamiltonian, and rest-mass contributions belong to the reference sector absorbed in calibration.

Remark 13.17 (Validity ceiling of the universality claim; constraint on the $(\alpha,\delta t)$ split). The universal softening (79) is the quadratic approximation of the exact cosine correction $\alpha(\cos(\delta t\, E/\hbar) - 1)$, valid only for modes with $E \ll \hbar/\delta t$; for $E \gtrsim \hbar/\delta t$ the exact correction stops growing and oscillates, with fractional frequency-shift amplitude saturating at $\sim \mu\hbar/\delta t = \lambda$, the subcriticality parameter of Prop. C.2 (D4). The universality claim of Rem. 13.15 therefore carries an explicit validity ceiling $E < \hbar/\delta_\star < \hbar/\delta t$. The ceiling has teeth for macroscopic *coherent* modes: the Earth's rotational energy ($\sim 2\times10^{29}\,\mathrm{J} \approx 10^{48}\,\mathrm{eV}$) exceeds any sub-Planckian band edge, and under the (now excluded) Planck-natural split $(\alpha,\delta t) = (E_{\rm P}, \hbar/E_{\rm P})$, i.e. $\lambda = 1$, such modes would suffer $O(1)$ frequency distortions — against length-of-day stability at the $\sim 10^{-9}$ level — excised only by the slow-branch axiom (Rem. 10.28), which removes above-band states from the sector where the reduction holds. Made explicit, the resulting constraint on the split (beyond the physical combination μ) is: $\hbar/\delta t$ must exceed the highest coherent mode energies to which the softening is applied, while the saturated amplitude $\lambda = \mu\hbar/\delta t$ stays below celestial-mechanics precision ($\sim 10^{-9}$). On the matched branch ($\mu \sim 10^{-33}\,\mathrm{eV}^{-1}$) this window is comfortably nonempty — e.g. $\hbar/\delta t \in [10^{15}, 10^{24}]\,\mathrm{eV}$, i.e. $\delta t \in [10^{4}, 10^{13}]\,t_{\mathrm{Pl}}$ — but it is a genuine restriction: the mode-level analogue of the GUP soccer-ball ambiguity, stated here as an explicit scope clause rather than left implicit in the band-limit hypothesis.

Remark 13.18 (Reach, and what would falsify what). Eq. (80) is the leading *laboratory* exclusion — modulo the sign caveat of Prop. 13.14 (ii): the published fit bounds the net hardening coefficient, so the softening branch is excluded only under the stated no-cancellation assumption. Subject to that assumption, $E_{\mathrm{QG},1} < 5 \times 10^{14}\,\mathrm{GeV}$ is ruled out in the laboratory. The previously advertised "4.4-decade window" between this bound and the Planck-natural point has, however, collapsed: the *binding* ceiling is the astrophysical photon-timing bound $\mu < 1.0 \times 10^{-29}\,\mathrm{eV}^{-1}$ (140), 5.3 decades below (80), and the Planck-natural normalization of Rem. 11.37 ($\mu = 1/E_{\mathrm{P}} \approx 8.2 \times 10^{-29}\,\mathrm{eV}^{-1}$) is already *dead*, excluded by GRB timing by a factor 7.6–8.2 (Rem. X.4); the Born–Oppenheimer-matched value (99), $\mu = 2c_0 H_{\mathrm{inf}}/m_{\mathrm{P}}^2 \sim 10^{-33}\,\mathrm{eV}^{-1}$, sits 3–4 decades below the photon ceiling and nine decades below the BAW bound. What closes which window is therefore restated: (i) BAW improvements chase the *laboratory* channel toward the photon ceiling — a β_0-class sensitivity gain of 5.3 decades (not 4.4) is needed before phonon cavities constrain territory that GRB timing has not already excluded, and a laboratory detection corresponding to $\mu > 1.0 \times 10^{-29}\,\mathrm{eV}^{-1}$ is pre-excluded by the photon bound; (ii) a detected softening that passes the universality test of Rem. 13.15 *below* the photon ceiling would be discovery-level and would fix $(\alpha, \delta t)$ empirically within the split window of Rem. 13.17; (iii) a mode- or material-dependent signal is ordinary anharmonicity and constrains nothing. No quantitative next-generation sensitivity projection is published; the platform developments identified in [252] — single-phonon-regime operation (attainable for $\omega/2\pi > 200\,\mathrm{MHz}$ at $10\,\mathrm{mK}$), higher-Q overtone families, quantum-limited readout — target exactly the systematics that limit the current fit, and gram-scale sapphire resonators supply the second material. Two further laboratory channels probe the same universal μ on platforms independent of the phonon cavity and independently improving: the hydrogen–deuterium $1S$–$2S$ isotope-shift comparison [356, 357] and Yb^+ King-plot nonlinearity searches [354, 355]. These experiments publish limits on interparticle Yukawa couplings and King-plot nonlinearities rather than on μ directly; we do not here carry out the nontrivial, energy-lever-dependent transcription of those limits onto the universal softening coefficient μ, and so cite them only as complementary precision-spectroscopy channels probing the same observable — both reaching sensitivities well short of the phonon-cavity ceiling (80) and of the binding photon bound of Prop. X.2. We state the windows and the protocol, not a timeline.

Testability triage. Of the five predictions, exactly one — the universal quadratic softening — is both framework-specific and constrained by existing data, and it is constrained through two channels. The ranked falsifiers are: (1) *the two-channel quadratic-softening falsifier*: (1a) *photon time-of-flight* (Prop. X.2): binding — GRB timing already gives $\mu < 1.0 \times 10^{-29}\,\mathrm{eV}^{-1}$, excluding the Planck-natural convention outright and turning the matched branch into a sharp sign-and-linearity commitment (subluminal, strictly linear, $|\xi_{\mathrm{eff}}| = \mu E_{\mathrm{P}} \sim 10^{-5}$–$10^{-4}$; Rem. X.4); (1b) *BAW amplitude–frequency softening* (Prop. 13.14): the leading *laboratory* channel, quantitative and already biting — current data exclude $E_{\mathrm{QG},1}$ below $5 \times 10^{14}\,\mathrm{GeV}$ (under the no-cancellation sign caveat of Prop. 13.14 (ii)), with any laboratory detection above the photon ceiling pre-excluded (Rem. 13.18); (2) *coherent, non-stochastic BMV coupling* (Prop. 13.5): long-term — entanglement generation alone no longer cleanly discriminates the mediator class

(Rem. 13.6), the framework-specific Γ_{DSW} correction is terrestrially unobservable, and the surviving commitment is the deterministic, non-stochastic character of the coupling; (3) *thermal-time cosmological-constant handle* (Prop. 13.12): conditional and structural (no numerical Λ), falsifiable indirectly through the running-vacuum/holographic phenomenology whose H^2 term it reproduces (Rem. 13.13). The Page-time shift of Prop. 13.2 and the parity-even spectrum correction of Prop. 13.8 are Planck-suppressed consistency statements beyond foreseeable reach (Rem. 13.9). The parity-odd channel formerly ranked first is *retracted* (Rem. 13.10): kinematically ill-posed at the bispectrum level, not derivable from $\widehat{\Sigma}_\delta$ at any level (Lem. 13.7), blocked at tree level on the only correlator where it could live [255], and observationally unanchored after the BOSS reanalysis and the DESI DR1 nulls [254, 257, 258, 259].

A The finite-dimensional two-boundary core

This appendix makes the paper self-contained by inlining, at full rigor, the finite-dimensional two-boundary core of which Secs. 3–7 are the curved-spacetime, worldline development. Time is here an abstract parameter $t \in [t_i, t_f]$ (the body of the paper uses the Cauchy time function τ along a worldline γ); $\mathcal{H}$ is a finite-dimensional Hilbert space of dimension d except in Secs. A.6–A.7, which lift the dimension restriction. Relative to the previously circulated standalone note, three repairs are incorporated silently: the completion convention of Rem. A.7 is now explicit, the Lévy-upward citation in Rem. A.19 is corrected, and the trace-norm computation in Cor. A.31 is stated a.s. rather than pointwise. Secs. A.5–A.7 then strengthen the core along three axes: concrete instrument filtrations (with the exact non-demolition interface for Born consistency), infinite dimensions and the type III boundary, and quantitative collapse without the record-completeness axiom.

A.1 Setup and Born invariance

The system is prepared at time t_i in a density operator $\rho_{\mathrm{i}} \succeq 0$ with $\operatorname{Tr}\rho_{\mathrm{i}} = 1$ and evolves via a unitary propagator $U(t, t_i)$.

Remark A.1 (Propagator identities). We assume, for all $t_1 \le t_2 \le t_3$,

$$U(t_3, t_1) = U(t_3, t_2)U(t_2, t_1), \qquad U(t, t) = \mathbb{1}, \qquad U(t_2, t_1)^\dagger = U(t_1, t_2).$$

At a final time $t_f > t_i$ a detector implements a POVM $\{E_k\}_{k\in\mathcal{K}}$ with $E_k \succeq 0$, $\sum_{k\in\mathcal{K}} E_k = \mathbb{1}$, $\mathcal{K}$ finite.

Definition A.2 (Forward state)**.** For $t \in [t_i, t_f]$, $\rho(t) := U(t, t_i)\, \rho_{\mathrm{i}}\, U^\dagger(t, t_i)$.

Definition A.3 (Backward effect operators)**.** For $k \in \mathcal{K}$ and $t \in [t_i, t_f]$, $\Lambda_k(t) := U^\dagger(t_f, t)\, E_k\, U(t_f, t)$. Each $\Lambda_k(t) \succeq 0$ and $\sum_{k\in\mathcal{K}} \Lambda_k(t) = \mathbb{1}$.

Definition A.4 (Born distribution)**.** For $t \in [t_i, t_f]$, $p_t(k) := \operatorname{Tr}\big(\rho(t)\Lambda_k(t)\big)$.

Lemma A.5 (Time invariance)**.** *For all $t \in [t_i, t_f]$ and $k \in \mathcal{K}$,*

$$p_t(k) = \operatorname{Tr}\big(\rho_{\mathrm{i}}\, U^\dagger(t_f, t_i) E_k U(t_f, t_i)\big), \qquad \sum_{k\in\mathcal{K}} p_t(k) = 1.$$

Proof. By the propagator identities and cyclicity of the trace,

$$p_t(k) = \operatorname{Tr}\big(U(t, t_i)\rho_{\mathrm{i}} U^\dagger(t, t_i)\, U^\dagger(t_f, t) E_k U(t_f, t)\big) = \operatorname{Tr}\big(\rho_{\mathrm{i}}\, U^\dagger(t_f, t_i) E_k U(t_f, t_i)\big),$$

independent of t. Moreover $\sum_k p_t(k) = \operatorname{Tr}\big(\rho(t) \sum_k \Lambda_k(t)\big) = \operatorname{Tr}\rho(t) = 1$. □

Definition A.6 (Time-invariant Born law)**.** $p(k) := p_t(k)$ for any $t \in [t_i, t_f]$.

A.2 Measurement records and the posterior martingale

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space; without loss of generality replace it by its $\mathbb{P}$-completion (same notation).

Remark A.7 (Completion convention). Let $\mathscr{N} := \{A \subseteq \Omega : \exists N \in \mathcal{F},\ \mathbb{P}(N) = 0,\ A \subseteq N\}$; completeness of $(\Omega, \mathcal{F}, \mathbb{P})$ means $\mathscr{N} \subseteq \mathcal{F}$. For a sub-$\sigma$-algebra $\mathcal{G} \subseteq \mathcal{F}$ we *define* its $\mathbb{P}$-completion by augmentation with the ambient null class,

$$\overline{\mathcal{G}} := \{A \subseteq \Omega : \exists B \in \mathcal{G},\ A \triangle B \in \mathscr{N}\}.$$

This is a σ-algebra (complements: $A^c \triangle B^c = A \triangle B$; countable unions: $(\bigcup_i A_i) \triangle (\bigcup_i B_i) \subseteq \bigcup_i (A_i \triangle B_i)$), it contains $\mathcal{G} \cup \mathscr{N}$, and it equals $\sigma(\mathcal{G} \cup \mathscr{N})$, since any σ-algebra containing $\mathcal{G}$ and $\mathscr{N}$ contains each $A = B \triangle (A \triangle B)$.

Lemma A.8 (Sub-σ-algebra completion stays inside a complete ambient space). *If $(\Omega, \mathcal{F}, \mathbb{P})$ is complete and $\mathcal{G} \subseteq \mathcal{F}$, then $\overline{\mathcal{G}} \subseteq \mathcal{F}$.*

Proof. If $A \in \overline{\mathcal{G}}$, there is $B \in \mathcal{G}$ with $A \triangle B \subseteq N$ for some $N \in \mathcal{F}$, $\mathbb{P}(N) = 0$. Completeness gives $A \triangle B \in \mathcal{F}$, hence $A = B \triangle (A \triangle B) \in \mathcal{F}$. □

Let $\{\mathcal{F}_t\}_{t \in \mathbb{Q},\, t < t_f}$ be an increasing filtration with $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for rationals $s \le t < t_f$, and let $\overline{\mathcal{F}}_t$ denote the $\mathbb{P}$-completion of $\mathcal{F}_t$; by Lemma A.8, $\overline{\mathcal{F}}_t \subseteq \mathcal{F}$.

Lemma A.9 (Completion preserves filtration monotonicity). *If $\mathcal{F}_s \subseteq \mathcal{F}_t$ then $\overline{\mathcal{F}}_s \subseteq \overline{\mathcal{F}}_t$; in particular $(\overline{\mathcal{F}}_t)_{t \in \mathbb{Q},\, t < t_f}$ is an increasing filtration.*

Proof. If $A \in \overline{\mathcal{F}}_s$, choose $B \in \mathcal{F}_s$ with $\mathbb{P}(A \triangle B) = 0$; then $B \in \mathcal{F}_t$, hence $A \in \overline{\mathcal{F}}_t$. □

Definition A.10 (Left-limit σ-algebra). $\overline{\mathcal{F}}_{t_f-} := \bigvee_{t < t_f,\, t \in \mathbb{Q}} \overline{\mathcal{F}}_t$.

Lemma A.11 (Completion invariance of conditional expectation). *If $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G} \subseteq \mathcal{F}$, then $\mathbb{E}[X \mid \overline{\mathcal{G}}] = \mathbb{E}[X \mid \mathcal{G}]$ a.s.*

Proof. Let $Y := \mathbb{E}[X \mid \mathcal{G}]$; Y is $\overline{\mathcal{G}}$-measurable. For $A \in \overline{\mathcal{G}}$ choose $B \in \mathcal{G}$ with $\mathbb{P}(A \triangle B) = 0$; then $\int_A Y\, d\mathbb{P} = \int_B Y\, d\mathbb{P} = \int_B X\, d\mathbb{P} = \int_A X\, d\mathbb{P}$, so Y satisfies the defining property of $\mathbb{E}[X \mid \overline{\mathcal{G}}]$. □

Definition A.12 (Terminal outcome random variable). $K : (\Omega, \mathcal{F}) \to (\mathcal{K}, 2^{\mathcal{K}})$ is an $\mathcal{F}$-measurable random variable.

Axiom A.13 (Born-consistent outcome). $\mathbb{P}(K = k) = p(k)$ for all $k \in \mathcal{K}$.

Definition A.14 (Posterior). For rational $t < t_f$, $\pi_t(k) := \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_t]$, with a fixed version chosen for each of the countably many pairs (t, k).

Remark A.15 (Zero-probability outcomes). If $p(k) = \mathbb{P}(K = k) = 0$ then $\mathbf{1}_{\{K=k\}} = 0$ a.s., hence $\pi_t(k) = 0$ a.s. for all rational $t < t_f$.

Definition A.16 (Rational cofinal sequence). A sequence (t_n) is *rational cofinal below t_f* if $t_n \in \mathbb{Q}$, $t_n < t_f$, $t_n \uparrow t_f$. (Cofinality — for every rational $t < t_f$ there is n with $t \le t_n$ — is then automatic, since $\sup_n t_n = t_f > t$. Such sequences exist: choose rationals $t_f - \frac{1}{n} < t_n < t_f$.)

Lemma A.17 (Left-limit join along cofinal sequences). *If (t_n) is rational cofinal below t_f, then $\bigvee_{n \ge 1} \overline{\mathcal{F}}_{t_n} = \bigvee_{t < t_f,\, t \in \mathbb{Q}} \overline{\mathcal{F}}_t = \overline{\mathcal{F}}_{t_f-}$.*

Proof. Each t_n is rational, so $\bigvee_n \overline{\mathcal{F}}_{t_n} \subseteq \overline{\mathcal{F}}_{t_f-}$. Conversely, for rational $t < t_f$ choose n with $t \le t_n$; by Lemma A.9, $\overline{\mathcal{F}}_t \subseteq \overline{\mathcal{F}}_{t_n} \subseteq \bigvee_n \overline{\mathcal{F}}_{t_n}$. Since the right-hand side is a σ-algebra containing every generator $\overline{\mathcal{F}}_t$, it contains the σ-algebra they generate. □

Lemma A.18 (Lévy upward theorem, discrete-time form)**.** *Let $(\mathcal{G}_n)_{n\geq 1}$ be an increasing sequence of sub-σ-algebras of $\mathcal{F}$ and $\mathcal{G} := \bigvee_{n\geq 1} \mathcal{G}_n$. If $X \in L^1$, then $\mathbb{E}[X \mid \mathcal{G}_n] \to \mathbb{E}[X \mid \mathcal{G}]$ a.s. and in L^1.*

Remark A.19 (Citation). Lemma A.18 is Lévy's upward theorem: Durrett [183, Thm. 4.6.8] (4th ed.: Thm. 5.5.7); see also Williams [184, §14.2]. Lévy's 0–1 law is [183, Thm. 4.6.9].

Lemma A.20 (Posterior normalization and martingale property)**.** *For all rational $t < t_f$: $0 \leq \pi_t(k) \leq 1$ a.s., $\sum_{k\in\mathcal{K}} \pi_t(k) = 1$ a.s., and for each k the family $\{\pi_t(k)\}_{t\in\mathbb{Q},\, t<t_f}$ is a martingale: for rationals $s \leq t < t_f$, $\mathbb{E}[\pi_t(k) \mid \overline{\mathcal{F}}_s] = \pi_s(k)$ a.s.*

Proof. Monotonicity of conditional expectation gives $0 \leq \pi_t(k) \leq 1$ a.s.; linearity gives $\sum_k \pi_t(k) = \mathbb{E}[\sum_k \mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_t] = 1$ a.s. For $s \leq t$ the tower property (Lemma A.9) gives $\mathbb{E}[\pi_t(k) \mid \overline{\mathcal{F}}_s] = \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_s] = \pi_s(k)$ a.s. □

Theorem A.21 (Martingale limit)**.** *Let (t_n) be rational cofinal below t_f. Then for each $k \in \mathcal{K}$,*

$$\pi_{t_n}(k) \longrightarrow \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_{t_f-}] \quad \text{a.s. and in } L^1.$$

Proof. Apply Lemma A.18 with $\mathcal{G}_n := \overline{\mathcal{F}}_{t_n}$ and $X := \mathbf{1}_{\{K=k\}} \in L^1$; by Lemma A.17, $\bigvee_n \overline{\mathcal{F}}_{t_n} = \overline{\mathcal{F}}_{t_f-}$. □

A.3 Posterior collapse

Axiom A.22 (Terminal identifiability / record-completeness)**.** K is $\overline{\mathcal{F}}_{t_f-}$-measurable.

Corollary A.23 (Posterior collapse)**.** *For any rational cofinal (t_n), $\pi_{t_n}(k) \to \mathbf{1}_{\{K=k\}}$ a.s. for all $k \in \mathcal{K}$ simultaneously.*

Proof. Fix k. Since $\mathbf{1}_{\{K=k\}}$ is $\overline{\mathcal{F}}_{t_f-}$-measurable and integrable, $\mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_{t_f-}] = \mathbf{1}_{\{K=k\}}$ a.s.; combine with Theorem A.21. As $\mathcal{K}$ is finite, the union of the $|\mathcal{K}|$ exceptional null sets is null, giving the simultaneous statement. □

Remark A.24 ($\overline{\mathcal{F}}_{t_f-}$ versus the terminal σ-algebra). Define $\overline{\mathcal{F}}_{t_f} := \overline{\mathcal{F}}_{t_f-} \vee \sigma(K)$. Axiom A.22 is precisely the statement $\overline{\mathcal{F}}_{t_f} = \overline{\mathcal{F}}_{t_f-}$ modulo $\mathbb{P}$-null sets: the terminal measurement adds no information not already contained in the pre-terminal record. The limit in Theorem A.21 always lives at t_f-; the possible strict jump $\overline{\mathcal{F}}_{t_f-} \subsetneq \overline{\mathcal{F}}_{t_f}$ is quantified exactly by the residual conditional entropy $H(K \mid \overline{\mathcal{F}}_{t_f-})$ of Thm. A.40 below. Note also that under the convention of Rem. A.7 the join $\overline{\mathcal{F}}_{t_f-}$ contains $\mathscr{N}$ and is closed under null modification (by the identity $A = B\triangle(A\triangle B)$), so no re-completion is needed.

A.4 Retrocausal present state

Definition A.25 (Conditioned states)**.** For $t \in [t_i, t_f]$,

$$\rho_2(t \mid k) := \begin{cases} \dfrac{\rho^{1/2}(t)\, \Lambda_k(t)\, \rho^{1/2}(t)}{p(k)}, & p(k) > 0, \\ \mathbb{1}/d, & p(k) = 0. \end{cases}$$

Definition A.26 (Random conditioned state)**.** $\rho_2(t \mid K) := \sum_{k\in\mathcal{K}} \mathbf{1}_{\{K=k\}}\, \rho_2(t \mid k)$.

Lemma A.27. *For each t and k, $\rho_2(t \mid k) \succeq 0$ and $\operatorname{Tr} \rho_2(t \mid k) = 1$.*

Proof. For $p(k) = 0$ the claim is clear. For $p(k) > 0$, $\rho^{1/2}(t)\Lambda_k(t)\rho^{1/2}(t) = \big(\Lambda_k^{1/2}(t)\rho^{1/2}(t)\big)^* \big(\Lambda_k^{1/2}(t)\rho^{1/2}(t)\big) \succeq 0$, and by cyclicity $\operatorname{Tr}\big(\rho^{1/2}(t)\Lambda_k(t)\rho^{1/2}(t)\big) = \operatorname{Tr}\big(\rho(t)\Lambda_k(t)\big) = p(k)$. □

Definition A.28 (Retrocausal state)**.** For rational $t < t_f$, $\rho_{\mathrm{RC}}(t) := \sum_{k\in\mathcal{K}} \pi_t(k)\, \rho_2(t \mid k)$.

Remark A.29 (Measurability). With the versions of Definition A.14 fixed, $\rho_{\mathrm{RC}}(t)$ is an $\overline{\mathcal{F}}_t$-measurable random matrix (entrywise, equivalently Borel for $\mathbb{C}^{d\times d} \cong \mathbb{R}^{2d^2}$). In fact, for fixed t it takes values in the *finite-dimensional* subspace $\mathrm{span}\{\rho_2(t \mid k)\}_{k\in\mathcal{K}}$, so measurability is automatic in any ambient topology — an observation that survives infinite dimensions verbatim (Thm. A.38). Since the index set $\mathbb{Q} \cap [t_i, t_f) \times \mathcal{K}$ is countable, all a.s. statements below hold simultaneously outside a single null set.

Lemma A.30 (Retrocausal state is a density operator)**.** *For each rational $t < t_f$, $\rho_{\mathrm{RC}}(t) \succeq 0$ and $\mathrm{Tr}\, \rho_{\mathrm{RC}}(t) = 1$ a.s.*

Proof. By Lemma A.27 and Lemma A.20: a.s. a convex combination of density operators. □

Corollary A.31 (Trace-norm collapse)**.** *Under Axiom A.22, for any rational cofinal (t_n),*

$$\|\rho_{\mathrm{RC}}(t_n) - \rho_2(t_n \mid K)\|_1 \longrightarrow 0 \quad a.s.$$

Proof. Write $\rho_{\mathrm{RC}}(t_n) - \rho_2(t_n \mid K) = \sum_k (\pi_{t_n}(k) - \mathbf{1}_{\{K=k\}})\rho_2(t_n \mid k)$. Since $\|\rho_2(t_n \mid k)\|_1 = \mathrm{Tr}\, \rho_2(t_n \mid k) = 1$ (for $A \succeq 0$, $\|A\|_1 = \mathrm{Tr}\, A$), the triangle inequality gives

$$\|\rho_{\mathrm{RC}}(t_n) - \rho_2(t_n \mid K)\|_1 \le \sum_{k\in\mathcal{K}} \left|\pi_{t_n}(k) - \mathbf{1}_{\{K=k\}}\right|.$$

Set $\pi_{t_n}(K) := \sum_k \pi_{t_n}(k)\mathbf{1}_{\{K=k\}}$. On $\{K = j\}$, and a.s. on the full-measure set where $\sum_k \pi_{t_n}(k) = 1$ for all n (Lemma A.20, Rem. A.29), the right-hand side equals $(1 - \pi_{t_n}(j)) + \sum_{k\neq j} \pi_{t_n}(k) = 2(1 - \pi_{t_n}(j))$, i.e. a.s. it equals $2(1 - \pi_{t_n}(K))$. By Corollary A.23, $\pi_{t_n}(K) \to 1$ a.s. □

Remark A.32 (Positioning within quantum filtering theory; scope of the axioms). Two structural observations. *First*, Axiom A.13 is not used anywhere in Lemma A.20, Theorem A.21, or Corollaries A.23 and A.31: the collapse chain is pure measure theory, and the quantum content enters only through Axiom A.13 (which fixes the marginal of K and yields the prior-average identity $\mathbb{E}[\rho_{\mathrm{RC}}(t)] = \sum_k p(k)\rho_2(t \mid k) = \rho(t)$, cf. Lem. 5.3) and through Definition A.25. This is simultaneously the source of the chain's robustness (Thm. A.38) and its known ceiling — nothing yet ties the abstract filtration to quantum instruments; Sec. A.5 closes this. *Second*, in filtering language Theorem A.21 is the closed-martingale property of the exact filter of the terminal pointer with respect to the output filtration — a structure going back to Belavkin's nondemolition filtering [180] (modern exposition: [181]), proved with rates for repeated QND chains by Bauer–Bernard and Bauer–Benoist–Bernard [178, 177] and Benoist–Pellegrini [179], and complemented by the Maassen–Kümmerer purification theorem [182]: the trajectory of repeated perfect measurement on a finite quantum system a.s. either asymptotically purifies or hits a family of *dark subspaces* on which the evolution is unitary — dark subspaces being precisely the obstruction to the instrument realization of Axiom A.22. The novelty here is (a) the two-boundary object filtered (the ABL terminal outcome rather than a static pointer), (b) the abstract-record formulation, with Thm. A.33 supplying its concrete instrument realization and identifying the QND fixed-point condition (81) as a sufficient condition for undisturbed Born consistency (the exact boundary being $\widetilde{\Lambda}_k^{(0)} = \Lambda_k(t_i)$), and (c) the type-free predual collapse of Thm. A.38, which decouples the collapse mechanism from the existence of density operators.

A.5 Strengthening I: the concrete instrument filtration

The abstract filtration of Sec. A.2 is now realized by physical intermediate measurements. The subtlety is that intermediate instruments *disturb*: the posterior must be defined with respect to the physically realized record law, and Axiom A.13 does not survive unmodified.

Theorem A.33 (Posterior martingale on the physically realized record filtration)**.** *Let $\mathcal{H}$ be a Hilbert space, ρ_{i} a density operator, and fix times $t_i = s_0 < s_1 < \cdots < s_N < s_{N+1} = t_f$, $N < \infty$. For $0 \le m \le N$ let $\mathcal{U}_m := \operatorname{Ad} U(s_{m+1}, s_m)$, and at each s_m $(1 \le m \le N)$ let $\{\mathcal{I}_j^{(m)}\}_{j \in J_m}$ be an instrument: each $\mathcal{I}_j^{(m)}$ completely positive, $\mathcal{L}_m := \sum_j \mathcal{I}_j^{(m)}$ trace-preserving, J_m countable. Let $\{E_k\}_{k \in \mathcal{K}}$ be the terminal POVM, $\mathcal{K}$ finite. On $\Omega := J_1 \times \cdots \times J_N \times \mathcal{K}$ with the product σ-algebra, define*

$$\mathbb{P}(y_1, \dots, y_N, k) := \operatorname{Tr}\big[E_k\, (\mathcal{U}_N \circ \mathcal{I}_{y_N}^{(N)} \circ \mathcal{U}_{N-1} \circ \cdots \circ \mathcal{I}_{y_1}^{(1)} \circ \mathcal{U}_0)(\rho_{\mathrm{i}})\big].$$

Let Y_m, K be the coordinate variables, $\mathcal{F}_m := \sigma(Y_1, \dots, Y_m)$, and $\pi_m(k) := \mathbb{P}(K = k \mid \mathcal{F}_m)$. Then:

(i) *$\mathbb{P}$ is a probability measure. (Positivity: CP maps and $E_k \succeq 0$; normalization: $\sum_k E_k = \mathbb{1}$ and trace preservation.)*

(ii) *(*The martingale property survives disturbance.*) For each k, $(\pi_m(k))_{0 \le m \le N}$ is a bounded $(\mathcal{F}_m)$-martingale closed by $\mathbf{1}_{\{K=k\}}$, with $\sum_k \pi_m(k) = 1$ a.s.; i.e. Lemma A.20 holds verbatim on the concrete filtration.*

(iii) *(*Quantum representation; disturbed backward effects.*) Let $\Phi_{>m} := \mathcal{U}_N \circ \mathcal{L}_N \circ \cdots \circ \mathcal{L}_{m+1} \circ \mathcal{U}_m$ and $\widetilde{\Lambda}_k^{(m)} := \Phi_{>m}^\dagger(E_k)$, where $\Phi^\dagger$ is the Heisenberg (unital, CP) dual. Then $\{\widetilde{\Lambda}_k^{(m)}\}_k$ is a POVM and, on every record cylinder of positive probability,*

$$\pi_m(k) = \operatorname{Tr}\big[\widetilde{\Lambda}_k^{(m)}\, \hat{\rho}_{y_{1:m}}\big], \qquad \hat{\rho}_{y_{1:m}} := \frac{(\mathcal{I}_{y_m}^{(m)} \circ \mathcal{U}_{m-1} \circ \cdots \circ \mathcal{I}_{y_1}^{(1)} \circ \mathcal{U}_0)(\rho_{\mathrm{i}})}{\operatorname{Tr}[\,\cdot\,]}.$$

(iv) *(*Born consistency must be modified.*) $\mathbb{P}(K = k) = \widetilde{p}(k) := \operatorname{Tr}[\rho_{\mathrm{i}}\, \widetilde{\Lambda}_k^{(0)}]$ (*disturbed *Born law), and the time-invariance Lemma A.5 holds for the pair (unconditional disturbed state, disturbed backward effect). The original Axiom A.13, $\mathbb{P}(K = k) = p(k)$, holds for* all *ρ_{i} iff $\widetilde{\Lambda}_k^{(0)} = \Lambda_k(t_i)$ for all k, for which the following non-demolition fixed-point condition is sufficient:*

$$\mathcal{L}_m^\dagger\big(\Lambda_k(s_m)\big) = \Lambda_k(s_m) \qquad \textit{for all } m, k, \tag{81}$$

with $\Lambda_k(t)$ the undisturbed backward effects of Definition A.3.

Proof. (i) Immediate from the stated properties. (ii) Any process of the form $\mathbb{P}(K = k \mid \mathcal{F}_m)$ is a martingale by the tower property; boundedness and normalization as in Lemma A.20. *The point is that no dynamical input enters*: disturbance by intermediate instruments changes the measure $\mathbb{P}$, not the Bayesian structure of conditional expectations with respect to the realized record process. (iii) By definition of conditional probability on the discrete cylinder, $\mathbb{P}(K = k \mid Y_{1:m} = y_{1:m}) = \operatorname{Tr}[E_k \Phi_{>m}(\rho_{y_{1:m}})] / \operatorname{Tr}[\rho_{y_{1:m}}] = \operatorname{Tr}[\Phi_{>m}^\dagger(E_k)\hat{\rho}_{y_{1:m}}]$ with $\rho_{y_{1:m}}$ the unnormalized conditioned state; positivity of $\widetilde{\Lambda}_k^{(m)}$ because duals of CP maps preserve positivity, and $\sum_k \widetilde{\Lambda}_k^{(m)} = \Phi_{>m}^\dagger(\mathbb{1}) = \mathbb{1}$ because duals of trace-preserving maps are unital. (iv) The first claim is (iii) at $m = 0$. For the second: $\mathcal{U}_N^\dagger(E_k) = \Lambda_k(s_N)$; by (81), $\mathcal{L}_N^\dagger(\Lambda_k(s_N)) = \Lambda_k(s_N)$; by the propagator identities, $\mathcal{U}_{N-1}^\dagger(\Lambda_k(s_N)) = \Lambda_k(s_{N-1})$; descending induction gives $\widetilde{\Lambda}_k^{(0)} = \Lambda_k(t_i)$. The "for all ρ_{i}" converse is the standard state-independence argument: $\operatorname{Tr}[\rho(\widetilde{\Lambda} - \Lambda)] = 0$ for all states iff $\widetilde{\Lambda} = \Lambda$. $\square$

Theorem A.34 (Infinite QND record chain: the abstract axioms are realized)**.** *Let $s_m \uparrow t_f$ be rational, with instruments as in Thm. A.33 satisfying the fixed-point condition* (81) *for all $m \ge 1$. Then the family*

$$\mu_m(y_{1:m}, k) := \operatorname{Tr}\big[\Lambda_k(s_m)\, (\mathcal{I}_{y_m}^{(m)} \circ \mathcal{U}_{m-1} \circ \cdots \circ \mathcal{I}_{y_1}^{(1)} \circ \mathcal{U}_0)(\rho_{\mathrm{i}})\big]$$

is Kolmogorov-consistent, hence extends to a unique law $\mathbb{P}$ of $(Y_1, Y_2, \dots, K)$ on $\big(\prod_m J_m\big) \times \mathcal{K}$. With $\mathcal{F}_t := \sigma(Y_m : s_m \le t)$, Axiom A.13 holds in its original *Born form, and Lemmas A.9,* *A.17, A.20, Theorem A.21, and Corollaries A.23, A.31 hold verbatim on this concrete filtration; Axiom A.22 becomes the statement that the record sequence identifies K, i.e. $K \in \bigvee_m \sigma(Y_{1:m})$ mod null.*

Proof. Consistency in k: $\sum_k \Lambda_k(t) = \mathbb{1}$ and trace preservation. Consistency in m: summing μ_{m+1} over y_{m+1} replaces $\mathcal{I}^{(m+1)}_{y_{m+1}}$ by $\mathcal{L}_{m+1}$; then, with ρ_y the unnormalized conditioned state at s_m,

$$\mathrm{Tr}[\Lambda_k(s_{m+1})\,\mathcal{L}_{m+1}(\mathcal{U}_m(\rho_y))] = \mathrm{Tr}[\mathcal{L}^\dagger_{m+1}(\Lambda_k(s_{m+1}))\,\mathcal{U}_m(\rho_y)] = \mathrm{Tr}[\Lambda_k(s_{m+1})\,\mathcal{U}_m(\rho_y)] = \mathrm{Tr}[\Lambda_k(s_m)\,\rho_y] = \mu_m,$$

using (81) and $\mathcal{U}^\dagger_m(\Lambda_k(s_{m+1})) = \Lambda_k(s_m)$. Kolmogorov extension applies since all factors are countable discrete. Axiom A.13: the $m = 0$ marginal. The measure-theoretic lemmas apply because they use nothing about $\mathbb{P}$ beyond its existence. □

Remark A.35 (Relation to the passive law of Sec. 5). The worldline construction (9) is Born-consistent *by construction* (passive/retrodictive weighting); the active sequential-Born law of Sec. 6 need not be (remark after Thm. 6.1). Thm. A.33(iv) sharpens that "need not" to an exact operator criterion: the active law satisfies $\mathbb{P}(K = k) = p(k)$ for all initial states iff the disturbed backward effects coincide with the undisturbed ones, for which (81) is the natural sufficient (QND) condition.

Conjecture A.36 (Non-QND accumulation)**.** Without (81) and with infinitely many instruments accumulating at t_f, even the joint law of $(Y_{1:\infty}, K)$ requires the weak* limits $\widetilde{\Lambda}^{(m)}_k := \lim_{r\to\infty}(\mathcal{U}^\dagger_m \circ \mathcal{L}^\dagger_{m+1} \circ \cdots \circ \mathcal{L}^\dagger_r \circ \mathcal{U}^\dagger_r)(E_k)$ to exist. *Conjecture:* if these dual compositions converge in the weak* topology for each k (e.g. under ergodicity/relaxation hypotheses on the average channels), the consistent family exists and Thm. A.33(ii)–(iii) persist with $\widetilde{p}(k)$ in place of $p(k)$. This is genuinely open in general: intermediate disturbance can destroy Born consistency and, in the accumulating limit, even the existence of the two-boundary joint law; it can never destroy the martingale property.

A.6 Strengthening II: infinite dimensions and the type III interface

Theorem A.37 (Separable-Hilbert-space transcription)**.** *Let $\mathcal{H}$ be separable, ρ_i a density operator (positive, trace-class,* $\mathrm{Tr}\,\rho_\mathrm{i} = 1$*), $U(t,s)$ unitaries satisfying the propagator identities, $\{E_k\}_{k\in\mathcal{K}}$ a POVM with $\mathcal{K}$ finite. Then:*

- (i) *Lemmas A.8–A.20, Theorem A.21, and Corollary A.23 hold verbatim: they are pure measure theory, dimension-free.*
- (ii) *Lemma A.5 holds verbatim: $\rho(t)\Lambda_k(t)$ is trace-class (trace-class ideal) and* $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ *for A trace-class, B bounded.*
- (iii) *(*Conditioned state exists.*) For $p(k) > 0$, $\rho^{1/2}(t)\Lambda_k(t)\rho^{1/2}(t)$ is positive and trace-class with trace $p(k)$; hence $\rho_2(t \mid k)$ of Definition A.25 is a density operator, with the $p(k) = 0$ branch $\mathbb{1}/d$ replaced by an arbitrary fixed density operator σ_0.*
- (iv) *Lemma A.30 and Corollary A.31 hold verbatim in $(\mathcal{T}(\mathcal{H}), \|\cdot\|_1)$, and $\rho_\mathrm{RC}(t)$ is strongly measurable because it takes values in the finite-dimensional subspace* $\mathrm{span}\{\rho_2(t \mid k)\}_{k\in\mathcal{K}}$ *(Rem. A.29).*

Proof. Only (iii) needs an argument. $\rho^{1/2}(t)$ is Hilbert–Schmidt: $\|\rho^{1/2}\|^2_{\mathrm{HS}} = \mathrm{Tr}\,\rho = 1$. Then $\Lambda_k\rho^{1/2}$ is Hilbert–Schmidt (two-sided ideal), and the product of two Hilbert–Schmidt operators is trace-class, so $\rho^{1/2}\Lambda_k\rho^{1/2} \in \mathcal{T}(\mathcal{H})$. Positivity: $\rho^{1/2}\Lambda_k\rho^{1/2} = (\Lambda_k^{1/2}\rho^{1/2})^*(\Lambda_k^{1/2}\rho^{1/2}) \succeq 0$. Trace: for Hilbert–Schmidt A, B, $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$, hence $\mathrm{Tr}(\rho^{1/2} \cdot \Lambda_k\rho^{1/2}) = \mathrm{Tr}(\Lambda_k\rho^{1/2} \cdot \rho^{1/2}) = \mathrm{Tr}(\Lambda_k\rho) = p(k)$. □

Theorem A.38 (Algebra-free collapse; valid for type III). *Let $\mathcal{N}$ be any von Neumann algebra with predual $\mathcal{N}_*$. Keep the probabilistic data $(\Omega, \mathcal{F}, \mathbb{P})$, filtration, K, Axiom A.22, and $\pi_t(k)$ of Secs. A.2–A.3, and for each rational $t < t_f$, $k \in \mathcal{K}$ let $\omega_k(t) \in \mathcal{N}_*$ be a normal state (any assignment). Define $\omega_{\mathrm{RC}}(t) := \sum_k \pi_t(k)\, \omega_k(t)$ and $\omega_2(t \mid K) := \sum_k \mathbf{1}_{\{K=k\}}\, \omega_k(t)$. Then $\omega_{\mathrm{RC}}(t)$ is a.s. a normal state, and for any rational cofinal $t_n \uparrow t_f$,*

$$\|\omega_{\mathrm{RC}}(t_n) - \omega_2(t_n \mid K)\|_{\mathcal{N}_*} \;\le\; 2\big(1 - \pi_{t_n}(K)\big) \xrightarrow[n\to\infty]{} 0 \quad a.s.$$

No trace, no density operator, no $\rho^{1/2}$, and no complete positivity are used.

Proof. A normal state has $\|\omega\|_{\mathcal{N}_*} = \omega(\mathbb{1}) = 1$, so $\omega_{\mathrm{RC}}(t)$ is a.s. a convex combination of normal states, hence a normal state. The triangle inequality gives $\|\omega_{\mathrm{RC}}(t_n) - \omega_2(t_n \mid K)\| \le \sum_k \big|\pi_{t_n}(k) - \mathbf{1}_{\{K=k\}}\big|$, and the $2(1 - \pi_{t_n}(K))$ identity and its a.s. convergence to 0 under Axiom A.22 are exactly the computations of Corollary A.31 and Corollary A.23, which are algebra-independent. □

Remark A.39 (The precise type III interface). Thm. A.38 shows the collapse *mechanism* is type-free; what does *not* survive in type III is the construction of the conditioned family $\omega_k(t)$ by the symmetric recipe $\rho^{1/2}\Lambda_k\rho^{1/2}/p(k)$, since $\rho^{1/2}$ does not exist. The paper's replacement is the KMS-perturbation state $\omega_k(x) = p(k)^{-1}\omega(L_k^* x L_k)$, $L_k = \sigma^{\omega}_{-i/2}(\Lambda_k^{1/2})$ the modular half-twist (App. G Setup), of App. G (Thms. G.1, G.2; ambient discussion Sec. 12.5.5). The residual interface, stated honestly: (1) *ordering ambiguity* — in finite dimensions ω_k has density $L_k\rho L_k/p(k)$, whereas Definition A.25 uses $\rho^{1/2}L_k^2\rho^{1/2}/p(k)$; they agree only after the modular half-twist $\sigma^{\omega}_{-i/2}$, which in type III exists only for L_k entire-analytic for σ^{ω} (Tomita algebra); for general L_k only the KMS-perturbation ordering is available; (2) the scalar side needs only *some* t-consistent family $\omega_k(t)$ with $\mathbb{P}(K = k) = \omega(\Lambda_k)$, both supplied by App. G; what remains open is its combination with the rigged-Hilbert two-boundary data (the scoping of Sec. 12.5.5); (3) localizing the instrument records of Sec. A.5 in local algebras interacts with Reeh–Schlieder under ABL conditioning (App. R) — genuine work remains there.

A.7 Strengthening III: quantitative collapse without record-completeness

The following theorem quantifies the failure mode of Axiom A.22; it is the quantitative companion of Prop. 9.1 and of the accessible-record residual entropy Cor. 9.2, and it closes, in entropy form, the interpolation gap noted after Prop. 9.1.

Theorem A.40 (Quantitative residual collapse). *Assume the setting of Secs. A.1–A.4* without *Axiom A.22. Let $\pi_\infty(k) := \mathbb{E}[\mathbf{1}_{\{K=k\}} \mid \overline{\mathcal{F}}_{t_f-}]$, $\pi_\infty(K) := \sum_k \mathbf{1}_{\{K=k\}}\pi_\infty(k)$, and define the residual conditional entropy (natural logarithm;* $0 \log 0 := 0$*)*

$$H := H\big(K \mid \overline{\mathcal{F}}_{t_f-}\big) = \mathbb{E}\Big[-\sum_k \pi_\infty(k)\log\pi_\infty(k)\Big] = \mathbb{E}\big[-\log\pi_\infty(K)\big] \in [0, \log|\mathcal{K}|].$$

Then for any rational cofinal $t_n \uparrow t_f$:

(i) *(*Exact posterior residual.*)* $\sum_k \big|\pi_{t_n}(k) - \mathbf{1}_{\{K=k\}}\big| \to 2(1 - \pi_\infty(K))$ *a.s. and in L^1, and*

$$\limsup_n \|\rho_{\mathrm{RC}}(t_n) - \rho_2(t_n \mid K)\|_1 \le 2\big(1 - \pi_\infty(K)\big) \quad a.s.$$

(ii) *(*Entropy bounds, Pinsker-type.*)* *Almost surely* $2(1-\pi_\infty(K)) \le \min\big\{-2\log\pi_\infty(K),\ \sqrt{-2\log\pi_\infty(K)}\big\}$, *and in expectation*

$$\mathbb{E}\Big[\limsup_n \|\rho_{\mathrm{RC}}(t_n) - \rho_2(t_n \mid K)\|_1\Big] \;\le\; 2\,\mathbb{E}[1 - \pi_\infty(K)] \;\le\; 2\min\big\{H,\ \sqrt{H/2}\big\}.$$

(iii) *(Converse: the failure mode is exactly $H > 0$.) With $P_e := \mathbb{E}[1 - \max_k \pi_\infty(k)] \le \mathbb{E}[1 - \pi_\infty(K)]$, Fano's inequality gives $H \le h_b(P_e) + P_e \log(|\mathcal{K}| - 1)$, where h_b is the binary entropy. Consequently*

$$H = 0 \iff \mathbb{E}[1 - \pi_\infty(K)] = 0 \iff K \textit{ is } \overline{\mathcal{F}}_{t_f-}\textit{-measurable} \iff \textit{Axiom A.22},$$

and each is equivalent to $\pi_\infty(k) \in \{0, 1\}$ a.s. for all k.

Proof. (i) By Theorem A.21, $\pi_{t_n}(k) \to \pi_\infty(k)$ a.s. for each of the finitely many k, and $\sum_k \pi_\infty(k) = 1$ a.s. (limit of normalized posteriors). The algebra of Corollary A.31 applied to π_∞ gives $\sum_k \left|\pi_\infty(k) - \mathbf{1}_{\{K=k\}}\right| = 2(1-\pi_\infty(K))$ a.s.; L^1 convergence by boundedness (≤ 2) and dominated convergence. The trace-norm bound is the estimate of Corollary A.31 with $\|\rho_2(t_n \mid k)\|_1 = 1$, followed by the a.s. limit of the scalar bound. (Only a lim sup: the states $\rho_2(t_n \mid k)$ need not converge.)

(ii) Pointwise, the total variation and relative entropy between the point mass δ_K and the posterior π_∞ are $\mathrm{TV}(\delta_K, \pi_\infty) = 1-\pi_\infty(K)$ and $\mathrm{KL}(\delta_K \| \pi_\infty) = -\log \pi_\infty(K)$; Pinsker's inequality $\mathrm{TV}^2 \le \frac{1}{2}\mathrm{KL}$ gives $2(1-\pi_\infty(K)) \le \sqrt{-2\log \pi_\infty(K)}$, while $1-x \le -\log x$ gives the linear bound (sharper when $-\log \pi_\infty(K) < \frac{1}{2}$). In expectation: conditioning on $\overline{\mathcal{F}}_{t_f-}$,

$$\mathbb{E}[1 - \pi_\infty(K) \mid \overline{\mathcal{F}}_{t_f-}] = \sum_k \pi_\infty(k)(1 - \pi_\infty(k)) \le \sum_k -\pi_\infty(k) \log \pi_\infty(k)$$

(termwise $q(1-q) \le -q \log q$ for $q \in [0, 1]$); taking expectations, $\mathbb{E}[1-\pi_\infty(K)] \le H$. The $\sqrt{H/2}$ bound follows from the pointwise Pinsker bound, Jensen, and the identity $\mathbb{E}[-\log \pi_\infty(K)] = H$; the latter holds because $\mathbb{E}[\mathbf{1}_{\{K=k\}} f(\pi_\infty(k))] = \mathbb{E}[\pi_\infty(k) f(\pi_\infty(k))]$ for $\overline{\mathcal{F}}_{t_f-}$-measurable $f(\pi_\infty(k))$, and $\mathbb{P}(K = k,\ \pi_\infty(k) = 0) = \mathbb{E}[\mathbf{1}_{\{\pi_\infty(k)=0\}} \pi_\infty(k)] = 0$.

(iii) Fano's inequality applied to the MAP estimator $\widehat{K} := \arg\max_k \pi_\infty(k)$ (measurable selection is trivial for finite $\mathcal{K}$), which is $\overline{\mathcal{F}}_{t_f-}$-measurable with error probability P_e. The chain of equivalences: $H = 0$ iff $\pi_\infty(k) \in \{0, 1\}$ a.s. (strict positivity of $-q \log q$ on $(0, 1)$) iff each $\mathbf{1}_{\{K=k\}}$ has an $\overline{\mathcal{F}}_{t_f-}$-measurable version iff K does (finite $\mathcal{K}$), which is Axiom A.22; and $\mathbb{E}[1-\pi_\infty(K)] = 1 - \sum_k \mathbb{E}[\pi_\infty(k)^2] = 0$ iff $\pi_\infty(k) \in \{0, 1\}$ a.s. □

Remark A.41 (Sharpness; state-level residual can vanish without Axiom A.22). The upper bound in (i) is saturated when the conditioned states have mutually orthogonal supports: then, on $\{K = j\}$, the trace norm of $\sum_{k \ne j} \pi_\infty(k) \rho_2(\cdot \mid k) - (1 - \pi_\infty(j))\rho_2(\cdot \mid j)$ is additive over the orthogonal blocks and equals $\sum_{k \ne j} \pi_\infty(k) + (1 - \pi_\infty(j)) = 2(1 - \pi_\infty(K))$. Conversely, no lower bound in terms of H is possible: if $\rho_2(t \mid j) = \rho_2(t \mid k)$ for all j, k, then $\rho_{\mathrm{RC}}(t) = \rho_2(t \mid K)$ identically while H is arbitrary. Residual terminal entropy is state-visible only to the extent that the conditioned states are distinguishable; in the horizon setting of Cor. 9.2 this means horizon-hidden entropy can be invisible in the exterior retrocausal state.

B Operator-cosine uniqueness for the clock shift: symmetrized d'Alembert reduction in the Kisyński–Sova framework

This appendix upgrades Conj. 11.5 from conjecture to theorem on the band-limited subspace $\mathcal{H}_{[E_\star]} := E_{\widehat{H}_c}([-E_\star, E_\star])\mathcal{H}$ of (T6). Two structural points are made explicit at the outset, because an earlier version of this appendix got the first one wrong. *First*, the clock-shift cocycle (axiom (D) below) is the *inhomogeneous* (Wilson-type) companion of d'Alembert's functional equation, and the shifted operator $C(\delta) := B(\delta) + \mathbb{1}$ does *not* satisfy the homogeneous operator-cosine equation (82) for a generic admissible B (within the admissible family exactly the members $B = P\widehat{\Sigma}_\delta$ with P an orthogonal projection in $\{\widehat{H}_c\}'$ do — including $B \equiv 0$ — so cosine-class membership cannot select the normalized member); consequently the Kisyński–Sova classification of operator cosine functions (Thm. B.1) cannot be applied to C, and it is nowhere invoked below in the uniqueness direction. *Second*, the actual uniqueness mechanism is elementary: symmetrizing the cocycle in (δ_1, δ_2) (Lem. B.2) collapses it, on each spectral fibre of $\widehat{H}_{\mathrm{c}}$, to a one-line linear identity that fixes the cos-*shape* with no regularity input at all; strong continuity in δ is used only to classify the $\ker \widehat{H}_{\mathrm{c}}$ block and to pass from rational to real δ; and the TDSE normalization (E) fixes the fibre *scale*. The classical scalar d'Alembert classification of Aczél [29] (§2.4.1: the continuous solutions of $g(s+t) + g(s-t) = 2g(s)g(t)$ on $\mathbb{R}$ are $g = 0$, $g = 1$, $g = \cos \lambda s$, $g = \cosh \lambda s$; see Stetkær [78] for the modern group-theoretic treatment) is the ambient background, but even it is not needed: the cosine kernel on the right-hand side of (D) is *known*, so the cos-vs-cosh trichotomy never has to be adjudicated, and no positivity argument ($\widehat{\Sigma}_\delta \preceq 0$) is required to exclude the cosh branch. The operator-cosine theory of Kisyński [72, 73, 74], Sova [75], and Fattorini [76, 77] enters only in the *existence* direction: it certifies that the selected solution is consistent, $\cos(\delta \widehat{H}_{\mathrm{c}}/\hbar)$ being the genuine cosine operator function with generator $-(\widehat{H}_{\mathrm{c}}/\hbar)^2$.

Cosine operator functions: the homogeneous classification (context)

Theorem B.1 (Kisyński–Sova cosine operator functions)**.** (Sova [75]; Kisyński [72, 73, 74]; Fattorini [77, 76].) *Let X be a Banach space and let $C : \mathbb{R} \to \mathcal{L}(X)$ be strongly continuous with $C(0) = \mathbb{1}$, satisfying d'Alembert's homogeneous operator equation*

$$C(s+t) + C(s-t) \;=\; 2\, C(s)\, C(t) \qquad \textit{for all } s, t \in \mathbb{R}, \tag{82}$$

i.e. a cosine operator function. *Then its generator A, defined by $A\psi := \lim_{t\to 0} 2t^{-2}(C(t)\psi - \psi)$ on the subspace of X where the limit exists, is densely defined and closed, and C is uniquely determined by A: two strongly continuous cosine operator functions with the same generator coincide, and $u(t) := C(t)u_0$ solves the abstract wave problem $u''(t) = Au(t)$, $u(0) = u_0$, $u'(0) = 0$ for $u_0 \in \mathcal{D}(A)$, uniquely among mild solutions. If $X = \mathcal{H}$ is a Hilbert space and T is self-adjoint on $\mathcal{H}$, then $C(t) := \cos(tT)$ (spectral calculus) is a cosine operator function with generator $A = -T^2 \preceq 0$ and $\|C(t)\| \le 1$; conversely, every uniformly bounded strongly continuous cosine operator function on a Hilbert space is, after passage to an equivalent inner product, of the form $\cos(tA_+^{1/2})$ with $A_+ \succeq 0$ self-adjoint [77]. In our application the relevant member is $C(\delta) = \cos(\delta \widehat{H}_{\mathrm{c}}/\hbar)$, with generator $-(\widehat{H}_{\mathrm{c}}/\hbar)^2 \preceq 0$ (note the sign: in the convention $u'' = Au$ the oscillatory, i.e. cosine rather than hyperbolic-cosine, branch corresponds to a* nonpositive *self-adjoint generator).*

Symmetrization of the inhomogeneous cocycle

Lemma B.2 (Swap identity and fibrewise rigidity)**.** *Let $\mathcal{V}$ be a complex Hilbert space, $\kappa \in \mathbb{R}$, and let $D \subseteq \mathbb{R}$ be a subgroup. Suppose $b : D \to \mathcal{L}(\mathcal{V})$ satisfies $b(-\delta) = b(\delta)$ and the shifted (inhomogeneous) d'Alembert identity*

$$b(\delta_1 + \delta_2) + b(\delta_1 - \delta_2) \;=\; 2\cos(\kappa\delta_1)\, b(\delta_2) + 2\, b(\delta_1) \qquad (\delta_1, \delta_2 \in D). \tag{83}$$

(i) (Swap identity.) $\big(\cos(\kappa\delta_1) - 1\big) b(\delta_2) = \big(\cos(\kappa\delta_2) - 1\big) b(\delta_1)$ *for all $\delta_1, \delta_2 \in D$.*

(ii) *If $\kappa \neq 0$ and D is dense in $\mathbb{R}$, there is a unique $G \in \mathcal{L}(\mathcal{V})$ with $b(\delta) = (\cos(\kappa\delta) - 1)\, G$ for all $\delta \in D$; conversely, every b of this form (arbitrary G) satisfies* (83)*. No continuity or measurability in δ is assumed.*

(iii) *If $\kappa = 0$, $D = \mathbb{R}$, and b is weakly continuous, then* (83) *is the quadratic (parallelogram) equation $b(\delta_1 + \delta_2) + b(\delta_1 - \delta_2) = 2b(\delta_1) + 2b(\delta_2)$, and its solutions are exactly $b(\delta) = \delta^2 Q$ with $Q := b(1) \in \mathcal{L}(\mathcal{V})$.*

Proof. (i) Interchange $\delta_1 \leftrightarrow \delta_2$ in (83), use $b(\delta_2 - \delta_1) = b(\delta_1 - \delta_2)$ (evenness), and subtract the two identities.

(ii) The zero set $\{\delta \in \mathbb{R} : \cos(\kappa\delta) = 1\} = (2\pi/|\kappa|)\mathbb{Z}$ is discrete, so density of D supplies $\delta_0 \in D$ with $\cos(\kappa\delta_0) \neq 1$. Put $G := (\cos(\kappa\delta_0) - 1)^{-1}b(\delta_0)$; then (i) with $\delta_1 = \delta_0$ gives $b(\delta) = (\cos(\kappa\delta) - 1)G$ for every $\delta \in D$, including those δ with $\cos(\kappa\delta) = 1$, where it forces $b(\delta) = 0$. Uniqueness follows by evaluating at δ_0. The converse is the product-to-sum identity $\cos x \cos y = \frac{1}{2}[\cos(x+y)+\cos(x-y)]$: for $b = (\cos(\kappa\,\cdot)-1)G$ both sides of (83) equal $(2\cos(\kappa\delta_1)\cos(\kappa\delta_2)-2)G$.

(iii) Setting $\delta_1 = \delta_2 = 0$ gives $b(0) = 0$. For $\psi \in \mathcal{V}$ put $q(\delta) := \langle\psi, b(\delta)\psi\rangle$; q is continuous, $q(0) = 0$, and satisfies the scalar parallelogram equation. Induction on n ($\delta_1 = n\delta$, $\delta_2 = \delta$) gives $q(n\delta) = n^2 q(\delta)$ for $n \in \mathbb{N}$, hence $q(r\delta) = r^2 q(\delta)$ for $r \in \mathbb{Q}$ (evenness of q is the case $\delta_1 = 0$), and continuity yields $q(\delta) = \delta^2 q(1)$. Since a bounded operator on a *complex* Hilbert space is determined by its diagonal quadratic form (polarization), $b(\delta) = \delta^2 b(1)$. The converse is $(\delta_1 + \delta_2)^2 + (\delta_1 - \delta_2)^2 = 2\delta_1^2 + 2\delta_2^2$. □

Remark B.3 (Relation to the scalar Aczél–d'Alembert classification). Equation (83) is the inhomogeneous, Wilson-type companion of d'Alembert's equation with *known* cosine kernel; this is why Lem. B.2(ii) needs no regularity, whereas the classification of the homogeneous scalar equation $g(\delta_1 + \delta_2) + g(\delta_1 - \delta_2) = 2g(\delta_1)g(\delta_2)$ — continuous nonzero solutions on $\mathbb{R}$: $g = 1$, $\cos(\lambda\,\cdot)$, $\cosh(\lambda\,\cdot)$ (Aczél [29], §2.4.1; Stetkær [78]) — requires at least measurability. Had axiom (D) been posed with an unknown kernel $g(\delta_1)$ in place of $\cos(\delta_1\widehat{H}_c/\hbar)$, the Aczél trichotomy would enter and the cosh branch would have to be excluded by boundedness of B; with the kernel fixed, neither step arises.

Reduction of the inhomogeneous clock-shift cocycle

Proposition B.4 (Reduction of Conj. 11.5: what (A)–(D) do and do not fix)**.** *Let the band-limited subspace $\mathcal{H}_{[E_\star]}$ of (T6) be separable, let $\widehat{H}_c$ be self-adjoint on $\mathcal{H}_{[E_\star]}$ with $\left\|\widehat{H}_c\restriction_{\mathcal{H}_{[E_\star]}}\right\| \leq E_\star < \infty$, and let $B : \mathbb{R} \to \mathcal{L}(\mathcal{H}_{[E_\star]})$ be a strongly continuous map with $B(0) = 0$, satisfying axioms (A)–(D) of Conj. 11.5 on $\mathcal{H}_{[E_\star]}$:*

(a) *(A) $B(\delta) = B(\delta)^*$ for all $\delta \in \mathbb{R}$.*

(b) *(B) $\widehat{T}_s B(\delta)\widehat{T}_{-s} = B(\delta)$ for all $s \in \mathbb{R}$, where $\widehat{T}_s = e^{-is\widehat{H}_c/\hbar}$. Only the commutant property $B(\delta) \in \{\widehat{H}_c\}'$ implied by this is used below; the stronger reading $B(\delta) \in \{\widehat{H}_c\}''$ in the parenthetical of Conj. 11.5(B), which is equivalent to commutation only for multiplicity-free clock spectra, is* not *needed.*

(c) *(C) $B(-\delta) = B(\delta)$.*

(d) *(D) The inhomogeneous cocycle*

$$B(\delta_1 + \delta_2) + B(\delta_1 - \delta_2) \;=\; 2\cos(\delta_1\widehat{H}_c/\hbar)B(\delta_2) \;+\; 2B(\delta_1) \tag{84}$$

holds as an operator identity on $\mathcal{H}_{[E_\star]}$ for all $\delta_1, \delta_2 \in \mathbb{R}$.

Write $\widehat{\Sigma}_\delta := \cos(\delta\widehat{H}_c/\hbar) - \mathbb{1}$ and $E_0 := E_{\widehat{H}_c}(\{0\})$ (the projection onto $\ker\widehat{H}_c$*). Then:*

(i) (Operator swap identity.) *From (C) and (D) alone, with no continuity input,*

$$\widehat{\Sigma}_{\delta_1} B(\delta_2) \;=\; \widehat{\Sigma}_{\delta_2} B(\delta_1) \qquad \text{for all } \delta_1, \delta_2 \in \mathbb{R}. \tag{85}$$

(ii) (Kernel block.) *$B(\delta)E_0 = \delta^2 Q$ for a unique self-adjoint $Q \in \mathcal{L}(E_0\mathcal{H}_{[E_\star]})$.*

(iii) (Fibre rigidity on the rational grid.) *Fix a direct-integral diagonalisation of $\widehat{H}_c$ on $(\mathbb{1} - E_0)\mathcal{H}_{[E_\star]} \cong \int_{\sigma^*}^{\oplus} \mathcal{H}_\lambda\, d\mu(\lambda)$, where $\sigma^* := \sigma(\widehat{H}_c) \setminus \{0\}$, $\widehat{H}_c$ acts as multiplication by λ, and $\{\widehat{H}_c\}'$ acts by decomposable operators (spectral multiplicity theory; [30], Part II). Then there are a μ-null set N and bounded operators $g_\lambda \in \mathcal{L}(\mathcal{H}_\lambda)$ for $\lambda \notin N$ such that*

$$B(\delta)_\lambda \;=\; \big(\cos(\delta\lambda/\hbar) - 1\big)\, g_\lambda \qquad \text{for all } \delta \in \mathbb{Q},\ \mu\text{-a.e. } \lambda. \tag{86}$$

The cos-shape *is thus forced on every fibre with no regularity assumption; the fibre* scale *g_λ is left free by (A)–(D).*

(iv) (No selection without (E).) *Conversely, (A)–(D) admit the strictly larger solution family*

$$B(\delta) \;=\; G\,\widehat{\Sigma}_\delta \;+\; \delta^2\, Q\, E_0, \qquad G = G^* \in \{\widehat{H}_{\rm c}\}', \quad Q = Q^* \in \mathcal{L}(E_0\mathcal{H}_{[E_\star]}),$$

and, more generally, norm-bounded families whose fibre scale is not even essentially bounded, e.g. $B(\delta) = f_\delta(\widehat{H}_{\rm c})$ *with* $f_\delta(\lambda) = (\cos(\delta\lambda/\hbar) - 1)/(\lambda/\hbar)^2$*,* $f_\delta(0) := -\delta^2/2$*, for which* $\|f_\delta\|_\infty \le \delta^2/2$*. In particular axiom (D) fixes neither the fibre scale nor a common normalization, and the shifted operator* $C(\delta) := B(\delta) + \mathbb{1}$ *satisfies the homogeneous equation* (82) *exactly for the projection-scaled members* $B = P\widehat{\Sigma}_\delta$*,* P *an orthogonal projection in* $\{\widehat{H}_{\rm c}\}'$ *(including* $B \equiv 0$*,* $P = 0$*): by (D),*

$$C(\delta_1+\delta_2) + C(\delta_1-\delta_2) = 2\cos(\delta_1\widehat{H}_{\rm c}/\hbar)\, C(\delta_2) + 2\big[B(\delta_1) + \mathbb{1} - \cos(\delta_1\widehat{H}_{\rm c}/\hbar)\big],$$

where the bracket vanishes iff $B = \widehat{\Sigma}_\delta$ *— a* sufficient *but not necessary route into the cosine class, whose full intersection with the admissible family is the projection-scaled set above. Cosine-class membership therefore cannot single out the normalized member* $P = \mathbb{1}$*. (Setting* $\delta_2 = 0$ *in* (84) *is the tautology* $2B(\delta_1) = 2B(\delta_1)$ *and carries no information.) All selection power beyond the* cos*-shape therefore resides in the TDSE normalization (E).*

Proof. (i) Interchange $\delta_1 \leftrightarrow \delta_2$ in (84), use $B(\delta_2 - \delta_1) = B(\delta_1 - \delta_2)$ from (C), and subtract; the terms $B(\delta_1+\delta_2)$ and $B(\delta_1-\delta_2)$ cancel, leaving (85).

(ii) By (B) and Stone's theorem, $B(\delta)$ commutes with every bounded Borel function of $\widehat{H}_{\rm c}$, in particular with E_0; hence $B(\delta)$ preserves $E_0\mathcal{H}_{[E_\star]}$. On that block $\cos(\delta_1\widehat{H}_{\rm c}/\hbar)E_0 = E_0$, so (84) restricts to the parallelogram equation, and Lem. B.2(iii) (weak continuity follows from strong continuity) gives $B(\delta)E_0 = \delta^2 Q$ with $Q = B(1)E_0$, self-adjoint by (A).

(iii) The diagonalisation exists because $\mathcal{H}_{[E_\star]}$ is separable and $\widehat{H}_{\rm c}$ is self-adjoint ([30], Part II: the abelian von Neumann algebra $\{\widehat{H}_{\rm c}\}''$ is the diagonalisable algebra of a direct integral over a scalar spectral measure μ, and its commutant $\{\widehat{H}_{\rm c}\}'$ is the algebra of decomposable operators; adjoints, products, and equalities of decomposable operators hold fibrewise, the latter μ-a.e.). For each ordered pair $(\delta_1, \delta_2) \in \mathbb{Q}^2$, (85) yields a μ-null set $N_{\delta_1\delta_2}$ off which

$$\big(\cos(\delta_1\lambda/\hbar) - 1\big)B(\delta_2)_\lambda = \big(\cos(\delta_2\lambda/\hbar) - 1\big)B(\delta_1)_\lambda.$$

Let N be the union over the countably many rational pairs. For $\lambda \notin N$, $\lambda \neq 0$, the set $\{\delta : \cos(\delta\lambda/\hbar) = 1\} = (2\pi\hbar/\lambda)\mathbb{Z}$ is discrete, so there is $\delta_0 = \delta_0(\lambda) \in \mathbb{Q}$ with $\cos(\delta_0\lambda/\hbar) \neq 1$; Lem. B.2(ii) applied on the fibre with $D = \mathbb{Q}$ and $\kappa = \lambda/\hbar$ gives (86) with $g_\lambda := (\cos(\delta_0\lambda/\hbar) - 1)^{-1}B(\delta_0)_\lambda$. (If desired, $\lambda \mapsto g_\lambda$ is measurable on each Borel set $\Lambda_j := \{\lambda : \; j(\lambda) = j\}$, $j(\lambda)$ the least index in a fixed enumeration $\{t_j\}$ of $\mathbb{Q}$ with $\cos(t_j\lambda/\hbar) \neq 1$; measurability is not used below.)

(iv) That $G\widehat{\Sigma}_\delta + \delta^2 Q E_0$ satisfies (A)–(D) is a direct computation: (A) holds since G, Q are self-adjoint and commute with the functional calculus of $\widehat{H}_{\rm c}$; (B) since $G \in \{\widehat{H}_{\rm c}\}'$ and $E_0, \widehat{\Sigma}_\delta \in \{\widehat{H}_{\rm c}\}''$; (C) since cosine and δ^2 are even; (D) reduces, using $[G, \cos(\delta_1\widehat{H}_{\rm c}/\hbar)] = 0$ and $\cos(\delta_1\widehat{H}_{\rm c}/\hbar)E_0 = E_0$, to the product-to-sum identity and the parallelogram identity as in Lem. B.2. For $f_\delta(\widehat{H}_{\rm c})$ the same checks apply fibrewise with $g(\lambda) = \hbar^2/\lambda^2$; the bound $\|f_\delta\|_\infty \le \delta^2/2$ is $|\cos x - 1| \le x^2/2$. The displayed identity for $C = B + \mathbb{1}$ is (D) rewritten; the bracket vanishes for all δ_1 iff $B(\delta_1) = \cos(\delta_1\widehat{H}_{\rm c}/\hbar) - \mathbb{1}$, while the full solution set of the homogeneous equation within the admissible family ($B = G\widehat{\Sigma}_\delta + \delta^2 Q E_0$) is obtained by direct substitution: the equation holds iff $Q = 0$ and $G^2 = G = G^*$, i.e. $B = P\widehat{\Sigma}_\delta$ with P an orthogonal projection in $\{\widehat{H}_{\rm c}\}'$. □

Theorem B.5 (Uniqueness of the symmetric clock shift on the band-limited subspace)**.** *Let* $\mathcal{H}_{[E_\star]}$*,* $\widehat{H}_{\rm c}$*, and* B *satisfy the hypotheses of Prop. B.4 (separability, self-adjointness, strong continuity, axioms (A)–(D)), together with the TDSE normalization*

$$(\mathrm{E}') \qquad \lim_{\delta\to 0} \frac{2}{\delta^2}\, B(\delta)\psi \;=\; -\,\alpha\, \frac{\widehat{H}_{\rm c}^2}{\hbar^2}\,\psi \qquad \textit{for all } \psi \in \mathcal{D}(\widehat{H}_{\rm c}^2) = \mathcal{H}_{[E_\star]}, \tag{87}$$

for some fixed $\alpha \in \mathbb{R}$. (E′) is the second-order Peano form of axiom (E) of Conj. 11.5 and is implied by the twice-strong-differentiability form stated there, via the vector-valued Taylor formula with $B(0) = 0$ and $B'(0) = 0$. Then

$$B(\delta) \;=\; \alpha\,\widehat{\Sigma}_\delta \;=\; \alpha\big(\cos(\delta\widehat{H}_\mathrm{c}/\hbar) - \mathbb{1}\big) \qquad \text{on } \mathcal{H}_{[E_\star]},\ \text{for all } \delta \in \mathbb{R}. \tag{88}$$

If $\widehat{H}_\mathrm{c}\restriction_{\mathcal{H}_{[E_\star]}} \neq 0$, the constant α is uniquely determined by the family B; since $\widehat{\Sigma}_{-\delta} = \widehat{\Sigma}_\delta$, the only reparametrization freedom is $\delta \mapsto -\delta$ at fixed α. (The residual freedom "$(\alpha,\delta) \mapsto (-\alpha,-\delta)$" claimed in an earlier version of this appendix is spurious.) Conversely, $\alpha\widehat{\Sigma}_\delta$ satisfies (A)–(D) and (E′), so the axiom package is consistent and (88) is exactly its solution set.

Proof. Step 1 (the kernel block dies). By Prop. B.4(ii), $B(\delta)E_0 = \delta^2 Q$. For $\psi \in E_0\mathcal{H}_{[E_\star]}$ we have $\widehat{H}_c^2\psi = 0$, so (E′) gives $2Q\psi = \lim_{\delta\to 0} 2\delta^{-2}B(\delta)\psi = 0$; hence $Q = 0$ and $B(\delta)E_0 = 0 = \alpha\widehat{\Sigma}_\delta E_0$.

Step 2 (the fibre scale is α). Work in the direct integral of Prop. B.4(iii) and fix a countable dense set $\{\psi_m\} \subset (\mathbb{1} - E_0)\mathcal{H}_{[E_\star]}$. Apply (E′) along the rational sequence $\delta_n = 1/n$: for each m,

$$v_n^{(m)} := 2n^2 B(\tfrac{1}{n})\psi_m \;\longrightarrow\; -\alpha\,(\widehat{H}_\mathrm{c}/\hbar)^2\psi_m \quad \text{in norm.}$$

Norm convergence in $\int^\oplus \mathcal{H}_\lambda\, d\mu$ is $L^2(\mu)$-convergence of the fibre fields, so a subsequence converges fibrewise μ-a.e.; a diagonal extraction yields one subsequence (n_i) and one μ-null set $N' \supseteq N$ that work for every m simultaneously. Off N', (86) gives

$$(v_{n_i}^{(m)})_\lambda = 2n_i^2\big(\cos(\lambda/(n_i\hbar)) - 1\big)\, g_\lambda(\psi_m)_\lambda \;\longrightarrow\; -(\lambda/\hbar)^2\, g_\lambda(\psi_m)_\lambda,$$

since the scalar prefactor converges to $-(\lambda/\hbar)^2$ and multiplies a fixed vector. Comparing the two limits and cancelling $(\lambda/\hbar)^2 \neq 0$ on σ^*: $g_\lambda(\psi_m)_\lambda = \alpha\,(\psi_m)_\lambda$ for all m and μ-a.e. λ. Since $\{(\psi_m)_\lambda\}_m$ is total in $\mathcal{H}_\lambda$ for μ-a.e. λ (the decomposable projection onto the fibrewise closed span of $\{(\psi_m)_\lambda\}$ fixes every ψ_m, hence equals the identity by density, hence its fibres equal $\mathbb{1}_{\mathcal{H}_\lambda}$ a.e.), we conclude $g_\lambda = \alpha\mathbb{1}_{\mathcal{H}_\lambda}$ for μ-a.e. λ.

Step 3 (rational reassembly). For $\delta \in \mathbb{Q}$, the fibres of $B(\delta)$ and of $\alpha\widehat{\Sigma}_\delta$ agree μ-a.e. on σ^* (Step 2 with (86)) and on the kernel block (Step 1); equality of decomposable operators is fibrewise a.e.-equality, so $B(\delta) = \alpha\widehat{\Sigma}_\delta$ for every rational δ.

Step 4 (real δ). Both $\delta \mapsto B(\delta)\psi$ (hypothesis) and $\delta \mapsto \alpha\widehat{\Sigma}_\delta\psi$ (dominated convergence in the spectral calculus) are continuous on $\mathbb{R}$; they agree on the dense subgroup $\mathbb{Q}$, hence everywhere. This proves (88).

Step 5 (uniqueness of α; existence). If $\widehat{H}_\mathrm{c} \neq 0$ then $\widehat{H}_\mathrm{c}^2 \neq 0$ and (E′) determines α from the family B; equivalently, $\widehat{\Sigma}_\delta \neq 0$ for suitable δ and (88) fixes α. Conversely, for $B = \alpha\widehat{\Sigma}_\delta$: (A) and (C) are immediate (real bounded even function of a self-adjoint operator), (B) holds by functional calculus, (D) is the product-to-sum identity $\cos x \cos y = \frac{1}{2}[\cos(x+y) + \cos(x-y)]$ under the spectral calculus, and (E′) follows by dominated convergence using the pointwise limit $2\delta^{-2}(\cos(\delta\lambda/\hbar) - 1) \to -(\lambda/\hbar)^2$ and the δ-uniform domination $|2\delta^{-2}(\cos(\delta\lambda/\hbar) - 1)| \leq (\lambda/\hbar)^2$, integrable against $d\mu_\psi$ for $\psi \in \mathcal{D}(\widehat{H}_\mathrm{c}^2)$. At no point was the homogeneous operator-cosine hypothesis of Thm. B.1 imposed on $C = B + \mathbb{1}$; Thm. B.1 is consistent with the result in the existence direction, $\cos(\delta\widehat{H}_\mathrm{c}/\hbar)$ being the cosine operator function generated by $-(\widehat{H}_\mathrm{c}/\hbar)^2 \preceq 0$. □

Remark B.6 (Band limit and boundedness of $\widehat{H}_\mathrm{c}$ are inessential). The proof of Thm. B.5 uses only: separability of the Hilbert space (spectral multiplicity theory), self-adjointness of $\widehat{H}_\mathrm{c}$, boundedness of each $B(\delta)$, strong continuity of $\delta \mapsto B(\delta)$, and (E′) on the dense domain $\mathcal{D}(\widehat{H}_\mathrm{c}^2)$ (with $\{\psi_m\}$ in Step 2 chosen dense in $\mathcal{H}$ and contained in $\mathcal{D}(\widehat{H}_\mathrm{c}^2)$). It therefore holds verbatim for an arbitrary, possibly unbounded, self-adjoint clock Hamiltonian on a separable Hilbert space; the band limit of (T6) serves only to make $\mathcal{D}(\widehat{H}_\mathrm{c}^2) = \mathcal{H}_{[E_\star]}$ and to match the scope of Pos. 11.30. This discharges residual item (i) of Rem. B.8.

Remark B.7 (Relation to covariant relational clocks). The covariance axiom (B) is morally the operator-cosine counterpart of the covariance-with-respect-to-the-clock-Hamiltonian condition under which the Page–Wootters, relational-Dirac-observable, and deparametrization formulations of relational quantum dynamics coincide — the "trinity" of Höhn–Smith–Lock [102]. We state this only as a structural analogy, not an established equivalence. The band-limit of (T6) and Pos. 11.30 confine $\widehat{H}_c$ to a bounded spectral window; if that window is additionally taken discrete (as a regularized clock), the pertinent comparison is with the periodic-clock relational dynamics of Chataignier–Höhn–Lock–Mele [262], where relational observables are only transiently invariant per clock cycle.

Remark B.8 (Effect on Conj. 11.5's status). Thm. B.5 upgrades Conj. 11.5 from conjecture to theorem on the band-limited subspace $\mathcal{H}_{[E_\star]}$, which under Pos. 11.30 is the entire physical matter+clock Hilbert space. The residual open content of Conj. 11.5 beyond this appendix is: (i) extension to unbounded $\widehat{H}_c$ on the full Hilbert space without the band-limit — now closed by Rem. B.6, since the proof of Thm. B.5 uses only self-adjointness and separability; (ii) extension to non-self-adjoint or non-covariant candidate shifts, which are excluded by axioms (A)–(C) and not relevant on the physical subspace. Under Pos. 11.30 and App. B, the Master Unification Theorem's hypothesis bundle H no longer contains Conj. 11.5 as an independent conjectural item; it is replaced by Thm. B.5, proved via the symmetrized d'Alembert reduction of Lem. B.2 and Prop. B.4, with the Kisyński/Sova/Fattorini cosine-operator-function theory as consistency context.

C Derivation of the retrocausal modified TDSE: motivation, reduction, and consistency

This appendix absorbs and supersedes an earlier companion note (January 2026) that presented a six-step "derivation" of the modified TDSE from particle compression to a singularity; its salvageable content is integrated here under an explicit claim taxonomy: the collapse material below is *motivation and carries no logical weight*, the actual derivation chain is Conj. 10.13 (constraint and cocycle, Eq. (26)) + Thm. B.5 (uniqueness of the symmetric clock shift, App. B) + Sec. 10.3 (the two-time link state, Definition 10.17–Prop. 10.23, replacing the note's heuristic "temporal bifurcation"), and the consistency checks (fiberwise reduction to the modified TDSE, first-law compatibility, structural invariances) occupy the remaining subsections of this appendix.

C.1 Physical motivation: Oppenheimer–Snyder collapse and the minimal clock increment

Nothing in this subsection is load-bearing. The modified constraint of Conj. 10.13 and the form $\widehat{\Sigma}_\delta = \cos(\delta\widehat{H}_c/\hbar) - \mathbb{1}$ of Eq. (26) are fixed by Thm. B.5, with the band-limit supplied by Pos. 11.30 via Prop. 11.33(T6). What follows motivates only the *existence* of a minimal operational clock increment $\delta \gtrsim \hbar/E_\star$, by exhibiting the generic situation in which general relativity drives matter beyond the scope of Pos. 11.30 in finite proper time. Metric signature in this subsection is $(-,+,+,+)$.

Classical collapse reaches Planck density in finite proper time. The Oppenheimer–Snyder interior [1] is the closed FLRW dust solution

$$ds^2 = -c^2\,d\tau^2 + a(\tau)^2\big(d\chi^2 + \sin^2\!\chi\,d\Omega^2\big), \qquad 0 \le \chi \le \chi_0, \tag{89}$$

matched to a Schwarzschild exterior across the comoving surface $\chi = \chi_0$; here τ is proper time along the dust, $a(\tau)$ has dimension of length, and χ is dimensionless. Covariant conservation of the dust stress tensor $T^{\mu\nu} = \rho\, c^2 u^\mu u^\nu$ (ρ the rest-mass density, $\mathrm{kg\,m^{-3}}$) gives

$$\rho(\tau) \;=\; \rho_0\big(a_0/a(\tau)\big)^3, \qquad \frac{8\pi G\rho_0}{3} = \frac{c^2}{a_0^2} \;\text{(collapse from rest at } a_0\text{)}, \tag{90}$$

consistent with the Friedmann equation $(\dot a/a)^2 = \frac{8\pi G}{3}\rho - c^2/a^2$ (both sides $\mathrm{s^{-2}}$). Each dust element is a *geodesic at fixed comoving coordinates* (${\Gamma^\mu}_{\tau\tau} = 0$ in comoving gauge); no worldline "moves toward the center". The compression is the shrinkage of proper volume $\propto a^3$, and the invariant signal of classical breakdown is the divergence of curvature scalars: in the near-singularity regime (where the spatial-curvature term c^2/a^2 is subdominant to $\frac{8\pi G}{3}\rho \propto a^{-3}$),

$$\rho(\tau) = \frac{1}{6\pi G\,(\tau_s-\tau)^2}, \qquad R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} = \frac{80}{27\,c^4(\tau_s-\tau)^4} = \frac{320\pi^2}{3}\Big(\frac{G\rho}{c^2}\Big)^2, \tag{91}$$

where τ_s is the comoving time of the classical singularity [dimensions: $(6\pi G\rho)^{-1/2}$ is a time; the Kretschmann scalar is $\mathrm{m^{-4}}$, matching $(G\rho/c^2)^2$]. Inverting (91): at density ρ the remaining proper time to the classical singularity is $\tau_s - \tau = (6\pi G\rho)^{-1/2}$, so the Planck density $\rho_P = c^5/(\hbar G^2)$ is crossed at

$$\tau_s - \tau \;=\; (6\pi)^{-1/2}\,t_\mathrm{P} \;\approx\; 0.23\,t_\mathrm{P}, \qquad t_\mathrm{P} = \sqrt{\hbar G/c^5}, \tag{92}$$

one Planck time before the classical singularity. Classical GR thus *generically* exits the validity scope of Pos. 11.30 in finite proper time.

Semiclassical fields blueshift and lose adiabaticity at the same crossing. A test scalar on (89) obeys $(\Box - m^2c^2/\hbar^2)\psi = 0$ with the exact d'Alembertian

$$\Box\psi \;=\; -\frac{1}{c^2}\,\frac{1}{a^3}\,\partial_\tau\big(a^3\partial_\tau\psi\big) + \frac{1}{a^2}\,\Delta_{S^3}\psi, \tag{93}$$

where Δ_{S^3} is the (dimensionless) Laplace–Beltrami operator on the unit 3-sphere with eigenvalues $-k(k+2)$, $k \in \mathbb{N}_0$. A comoving mode therefore satisfies $\ddot\chi_k + 3H\dot\chi_k + \omega_k^2\chi_k = 0$ with physical frequency

$$\omega_k(\tau) \;=\; \sqrt{\frac{c^2\,k(k+2)}{a(\tau)^2} + \frac{m^2c^4}{\hbar^2}} \qquad [\mathrm{s^{-1}}], \tag{94}$$

and WKB phase S obeying the eikonal equation $g^{\mu\nu}\partial_\mu S\,\partial_\nu S + m^2c^2 = 0$, i.e. $E^2 = m^2c^4 + p^2c^2$ with $E = -\partial_\tau S$ and $p = \partial_\chi S/a$ [S in J s; E energy, p momentum]. As $a \to 0$ the *spatial-gradient* term diverges ($g^{\tau\tau} = -1/c^2$ identically in comoving gauge, so no divergence originates in the time sector): modes blueshift, physical wavelengths $2\pi a/\sqrt{k(k+2)} \to 0$, and WKB energies $E = \hbar\omega_k$ grow without bound — *sharper* localization, not spreading. Simultaneously the WKB validity parameter [4] for effectively massless modes,

$$\frac{|\dot\omega_k|}{\omega_k^2} \;=\; \frac{|H|}{\omega_k} \;=\; \frac{|H|\,a}{c\sqrt{k(k+2)}} \;\propto\; (\tau_s-\tau)^{-1/3} \;\longrightarrow\; \infty \tag{95}$$

(dimensionless: $|H|$ in $\mathrm{s^{-1}}$, a/c in s; $|H| \propto (\tau_s-\tau)^{-1}$, $a \propto (\tau_s-\tau)^{2/3}$ in the dust regime), so every comoving mode exits the adiabatic regime before the singularity, while $|H|$ itself reaches t_P^{-1} at the crossing (92). Within $O(1)$ Planck times of the classical singularity, *every* semiclassical validity condition ($\rho \ll \rho_P$, curvature $\ll \ell_\mathrm{P}^{-2}$, $|\dot\omega_k|/\omega_k^2 \ll 1$) fails together.

The operational moral: a minimal clock increment. Under Pos. 11.30, physical matter states — including any physical clock — are supported below $E_\star \lesssim E_{\rm P}$. A clock so band-limited cannot resolve proper-time increments below $\delta \approx \hbar/E_\star \gtrsim t_{\rm P}$: the frequencies that the collapse drives past $E_\star/\hbar$ carry no operational meaning inside the framework's scope, and "evolution by less than δ" is not a physical statement. The natural clock-evolution primitive is therefore not the generator $\widehat{H}_{\rm c}$ but the finite shift $\widehat{S}_\delta = e^{-i\delta\widehat{H}_{\rm c}/\hbar}$, whose self-adjoint symmetrisation is exactly the cocycle $\widehat{\Sigma}_\delta$ of Eq. (26); that this is the *unique* admissible modification is the content of Thm. B.5, and the collapse story plays no role in that proof. Independently, effective loop-quantum-cosmology treatments of the Oppenheimer–Snyder model [2] find that the classical singularity is replaced by new temporal structure (a bounce) precisely at Planck-order critical density (with [3] bounding any such bounce by the horizon-formation threshold while remaining agnostic about the resolution mechanism), corroborating — again as motivation, not derivation — that the Planck-density crossing is where the classical time parameter must be superseded.

Remark C.1 (Retraction of the heuristic "temporal delocalization"). The earlier companion note argued that collapse forces the wave function to "delocalize temporally" because "$g^{tt} \to \infty$". Both claims are incorrect: $g^{tt} = -1/c^2$ in comoving gauge, the divergence is in the spatial sector $g^{ij} \propto a^{-2}$, and the corrected WKB analysis yields diverging energies and *sharper* localization until adiabaticity fails, after which no WKB statement survives. The motivation retained above is purely operational (finite clock resolution under the cutoff); the rigorous replacement for the note's "temporal bifurcation" step is the two-time link state of Sec. 10.3, and the derivation of $\widehat{\Sigma}_\delta$ rests entirely on Thm. B.5.

C.2 From constraint to modified TDSE: exact fiberwise reduction and α-matching

The delay–advance equation (31) of Prop. 10.14 is not a first-order ODE and does not by itself generate a one-parameter group on $\mathcal{H}_{\rm sys}$. This subsection closes that gap on the slow branch: the constraint of Conj. 10.13 reduces, *exactly and fibrewise*, to a genuine single-time Schrödinger equation with the bounded self-adjoint generator correction $\alpha\widehat{\Sigma}_{\delta t}$, and the Born–Oppenheimer matching then fixes the single physical combination $\mu = \alpha\,\delta t^2/\hbar^2$ against the quantum-gravitational power-spectrum corrections of Kiefer–Krämer [249] and Chataignier–Krämer [251].

Proposition C.2 (Effective single-branch modified TDSE; exact fiberwise reduction)**.** *Assume Conj. 10.13 with the slow-branch axiom (Rem. 10.28). Precisely, assume:*

(D1) $\widehat{H}_{\rm s}$ *self-adjoint on* $\mathcal{D}(\widehat{H}_{\rm s}) \subset \mathcal{H}_{\rm sys}$*, bounded below, with* $E_0 := \inf\sigma(\widehat{H}_{\rm s})$*;*

(D2) $\widehat{H}_{\rm c}$ *self-adjoint with absolutely continuous spectrum and rigged eigenbasis* $\{|t\rangle\}$*,* $\langle t|\widehat{H}_{\rm c} = -i\hbar\partial_t\langle t|$*, on the Gel'fand triple* $\Phi \subset \mathcal{H}_{\rm clock} \otimes \mathcal{H}_{\rm sys} \subset \Phi'$ *of Conj. 10.13;*

(D3) *(*slow branch / band limit*) the joint clock-spectral support of* Ψ *lies in* $[-\hbar/\delta_\star, \hbar/\delta_\star]$ *with* $\delta_\star > \delta t$ *(Rem. 10.28);*

(D4) *(*subcriticality*)* $\lambda := |\alpha|\,\delta t/\hbar < 1$.

Let $\Psi \in \mathcal{H}^{\rm slow}_{\rm phys}$ *solve* $\widehat{C}_{\alpha,\delta t}\Psi = (\widehat{H}_{\rm c} \otimes \mathbb{1} + \mathbb{1} \otimes \widehat{H}_{\rm s} + \alpha\widehat{\Sigma}_{\delta t} \otimes \mathbb{1})\Psi = 0$ *of* (27)*, and set* $\psi(t) := \langle t|\Psi$*. Then:*

(i) *(*Fiberwise reduction.*) On each joint spectral fibre* $(\xi, E) \in \sigma(\widehat{H}_{\rm c}) \times \sigma(\widehat{H}_{\rm s})$ *the constraint reads* $\xi + E + \alpha(\cos(\delta t\,\xi/\hbar) - 1) = 0$*. Since* $\partial_\xi[\xi + \alpha(\cos(\delta t\,\xi/\hbar) - 1)] \geq 1 - \lambda > 0$ *by* (D4)*, this has a* unique *real solution* $\xi(E) =: -h(E)$*; using evenness of the cosine,* h *is the unique fixed point of*

$$h(E) \;=\; E + \alpha\big(\cos(\delta t\, h(E)/\hbar) - 1\big), \tag{96}$$

obtained by Banach iteration with contraction rate λ*. Implicit differentiation of* (96) *gives* $h'(E) = (1 + \alpha(\delta t/\hbar)\sin(\delta t\, h/\hbar))^{-1} \geq (1+\lambda)^{-1} > 0$*, so* $E \mapsto h(E)$ *is continuous and* *strictly increasing, and* $\widehat{H}^{\rm ex}_{\rm mod} := h(\widehat{H}_{\rm s})$ *is self-adjoint on* $\mathcal{D}(\widehat{H}_{\rm s})$ *by spectral calculus (the difference* $h(\widehat{H}_{\rm s}) - \widehat{H}_{\rm s}$ *is bounded by* (ii) *below).*

(ii) *(*Effective single-branch TDSE.*) The conditional state satisfies* exactly $i\hbar\,\partial_t\psi(t) = h(\widehat{H}_{\rm s})\,\psi(t)$, *and with*

$$\widehat{H}_{\rm mod} \;:=\; \widehat{H}_{\rm s} + \alpha\,\widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm s}), \qquad \widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm s}) = \cos(\delta t\,\widehat{H}_{\rm s}/\hbar) - \mathbb{1}, \tag{97}$$

one has the fibrewise bound

$$\big|h(E) - E - \alpha(\cos(\delta t\,E/\hbar) - 1)\big| \;\le\; \lambda\,|\alpha|\,\min\Big\{2, \tfrac{1}{2}\big(\delta t\,(|E| + 2|\alpha|)/\hbar\big)^2\Big\},$$

i.e. $\widehat{H}^{\rm ex}_{\rm mod} = \widehat{H}_{\rm mod} + O(\lambda) \times$ *(leading correction); to first order in* λ *the effective single-branch evolution is* $i\hbar\,\partial_t\psi = (\widehat{H}_{\rm s} + \alpha\widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm s}))\psi$*. On the constraint surface* $\widehat{H}_{\rm c} = -\widehat{H}_{\rm s} + O(\alpha)$ *and* $\widehat{\Sigma}_{\delta t}$ *is even, so writing the clock-side shift* $\widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm c})$ *as the system-side* $\widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm s})$ *is the on-shell form, exact up to a further* $O(\lambda)$*.*

(iii) *(*Self-adjointness, semiboundedness, unitarity.*)* $\widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm s})$ *is bounded self-adjoint with spectrum in* $[-2, 0]$*; by Kato–Rellich* $\widehat{H}_{\rm mod}$ *is self-adjoint on* $\mathcal{D}(\widehat{H}_{\rm s})$*, bounded below by* $E_0 - 2|\alpha|$*, and generates a strongly continuous unitary group* $U_{\rm mod}(t) = e^{-i\widehat{H}_{\rm mod}t/\hbar}$ *by Stone's theorem. The same statements hold for* $\widehat{H}^{\rm ex}_{\rm mod} = h(\widehat{H}_{\rm s})$*.*

(iv) *(*$\delta t \to 0$ expansion.*) Strongly on* $\mathcal{D}(\widehat{H}_{\rm s}^2)$*,*

$$\alpha\,\widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm s}) \;=\; -\frac{\mu}{2}\,\widehat{H}_{\rm s}^2 + O\Big(\frac{\alpha\,\delta t^4}{\hbar^4}\,\widehat{H}_{\rm s}^4\Big), \qquad \mu := \frac{\alpha\,\delta t^2}{\hbar^2}, \tag{98}$$

with $\Big\|(\widehat{\Sigma}_{\delta t} + \frac{1}{2}(\delta t\widehat{H}_{\rm s}/\hbar)^2)\psi\Big\| \le \frac{\delta t^4}{24\hbar^4}\Big\|\widehat{H}_{\rm s}^4\psi\Big\|$ *on* $\mathcal{D}(\widehat{H}_{\rm s}^4)$ *(fibrewise Lagrange remainder* $|\cos x - 1 + x^2/2| \le x^4/24$*), hence operator-norm bound* $\delta t^4 E_\star^4/(24\hbar^4)$ *on an* $E_\star$*-band-limited subspace. This reproduces* $\widehat{H}_{\rm eff} = \widehat{H}_{\rm s} - \frac{\mu}{2}\widehat{H}_{\rm s}^2$ *of Prop. 10.27 identically: the fibrewise route and the Taylor route agree. The correction is a GUP-type quadratic-in-energy redshift* $E \mapsto E - \mu E^2/2$*; strict monotonicity of* h *(no level folding) on the band requires* $\mu E_\star < 1$*, which* (D3) *together with* (D4) *enforces (*$\mu E_\star = \lambda \cdot \delta t\,E_\star/\hbar < 1$*).*

(v) *(*Born time-invariance: knitting condition.*) Let* $\rho(\tau) = U_{\rm mod}(\tau - \tau_i)\,\rho_{\rm i}\,U_{\rm mod}(\tau - \tau_i)^\dagger$ *and* $\Lambda_k(\tau) = U_{\rm mod}(\tau_f - \tau)^\dagger E_k U_{\rm mod}(\tau_f - \tau)$*, i.e. forward state* and *backward effects transported with the* same $\widehat{H}_{\rm mod}$*. Then* $p(k) = \mathrm{Tr}(\rho(\tau)\Lambda_k(\tau))$ *is* τ*-independent, normalized and non-negative: the proof of Lem. A.5 (App. A) uses only the propagator group law, unitarity, and trace cyclicity, all supplied by* (iii)*. If instead the effects are transported with* $\widehat{H}_{\rm s}$ *while the state moves with* $\widehat{H}_{\rm mod}$*, the weight acquires the drift* $\partial_\tau p = \frac{i}{\hbar}\,\mathrm{Tr}\big(\rho(\tau)\,[\alpha\widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm s}), \Lambda_k(\tau)]\big) \ne 0$ *for generic non-commuting* $(\rho_{\rm i}, E_k)$*; the invariance is a property of the common-generator structure, not an identity.*

Proof. (i) Monotonicity of $\xi \mapsto \xi + \alpha(\cos(\delta t\,\xi/\hbar) - 1)$ is immediate from (D4); existence of the root follows since the function tends to $\pm\infty$ (the cosine term is bounded). The substitution $h = -\xi$ and evenness of the cosine give (96), whose right-hand side is a contraction with rate $\sup_h |\alpha(\delta t/\hbar)\sin(\delta t\,h/\hbar)| \le \lambda < 1$. (ii) On the fibre, $|h - E| = |\alpha||\cos(\delta t\,h/\hbar) - 1| \le |\alpha|\min\{2, (\delta t\,h/\hbar)^2/2\}$ and $|h| \le |E| + 2|\alpha|$; then $|h(E) - E - \alpha(\cos(\delta t E/\hbar) - 1)| = |\alpha||\cos(\delta t\,h/\hbar) - \cos(\delta t\,E/\hbar)| \le |\alpha|(\delta t/\hbar)|h - E|$, giving the displayed bound. The conditional-state statement follows from the joint spectral representation: on the solution fibre, $\langle t|$ picks up the phase $e^{i\xi t/\hbar} = e^{-ih(E)t/\hbar}$, so $i\hbar\partial_t\psi = h(\widehat{H}_{\rm s})\psi$ (this is Prop. 10.14 resolved fibrewise rather than left as a delay–advance equation). (iii) Spectral calculus and Kato–Rellich for the bounded self-adjoint perturbation $\alpha\widehat{\Sigma}_{\delta t}$ ($\Big\|\alpha\widehat{\Sigma}_{\delta t}\Big\| \le 2|\alpha|$); Stone's theorem. (iv) Fibrewise Lagrange remainder, dominated convergence on the spectral measure. (v) The invariance computation is verbatim Lem. A.5 with $U(t', t) = e^{-i\widehat{H}_{\rm mod}(t'-t)/\hbar}$; the drift formula follows from $\dot\rho = -\frac{i}{\hbar}[\widehat{H}_{\rm mod}, \rho]$, $\dot\Lambda_k = -\frac{i}{\hbar}[\widehat{H}_{\rm s}, \Lambda_k]$ and trace cyclicity. □

For the dispersion-relation consequence of (98) for freely propagating massless modes — an effective subluminal linear time-of-flight dispersion, whose GRB bounds are the binding empirical constraint on μ — see Prop. X.2 and Rem. X.4 (App. X).

Remark C.3 (Dimensional autopsy of the superseded asymmetric form). The companion note's "$\alpha\Delta t\,\psi$" term (recovered on the retarded sector in Rem. 10.16) admits only two dimensional readings, and both fail. If Δt is a c-number time interval, then with the note's $\alpha \sim 1/4G \sim E_{\mathrm{Pl}}^2$ (natural units) the term is a scalar multiple of ψ: an unobservable global phase, not a temporal link. If instead $\Delta t\psi$ denotes the forward difference $\psi(t+\delta t) - \psi(t)$, then $[\alpha] =$ energy is forced (as in Conj. 10.13), so $\alpha \sim 1/4G$ is wrong by one power of energy; moreover the forward shift is not self-adjoint and the resulting non-unitary evolution is excluded by the spectroscopy bounds of Rem. 10.30. The symmetric cocycle $\widehat{\Sigma}_{\delta t}$ with $[\alpha] =$ energy, $[\delta t] =$ time, $[\mu] = \text{energy}^{-1}$, λ dimensionless repairs both readings simultaneously.

Born–Oppenheimer matching of α

The leading correction $-\frac{\mu}{2}\widehat{H}_{\mathrm{s}}^2$ of (98) has exactly the operator structure of the Born–Oppenheimer quantum-gravitational correction to the functional Schrödinger equation (Kiefer–Singh; Kiefer–Krämer [249]): quadratic in the matter Hamiltonian, weighted by an inverse background energy. Matching the two on the perturbation sector at inflationary scale therefore fixes the single combination μ. In the convention $\kappa := m_{\mathrm{P}}^{-2}$ (rescaled Planck mass m_{P}, as adopted in [251]), the corrected scalar and tensor spectra are $\mathcal{P}(k) = \mathcal{P}_0(k)\,[1 + c_0\,\kappa H_{\mathrm{inf}}^2(k_\star/k)^3]$ with $c_0 \approx 0.988$ in the de Sitter limit of the Brizuela–Kiefer–Krämer treatment [250] (the original Kiefer–Krämer analysis [249], as published, yields a *suppression* with a different coefficient and Planck-mass convention; the sign and coefficient were settled in the subsequent literature), while the unitary Born–Oppenheimer treatment of Chataignier–Krämer [251, Eqs. (158)–(160)] gives $c_0 = 4 - 2\gamma_E - 2\log(-2k\eta) \approx 1.5$ at horizon crossing, with secular logarithmic running. Since $\widehat{H}_{\mathrm{mod}}$ is unitary by construction, the Chataignier–Krämer value is the appropriate comparator. The matching equation is

$$\mu \;=\; \frac{\alpha\,\delta t^2}{\hbar^2} \;=\; \frac{2\,c_0\,H_{\mathrm{inf}}}{m_{\mathrm{P}}^2}, \qquad c_0 \approx 1.5 \text{ (CK)}, \quad c_0 \approx 0.988 \text{ (BKK, de Sitter)}. \tag{99}$$

Because $\widehat{\Sigma}_{\delta t}$ is even in δt, the constraint enters the low-energy sector *only* through μ: the matching fixes the product $\alpha\,\delta t^2$ and nothing else; the split into α and δt is convention. The minimal (Planck-time) convention $\delta t := \hbar/E_{\mathrm{P}}$ gives $\alpha = \mu\,E_{\mathrm{P}}^2 = 2c_0 H_{\mathrm{inf}}(E_{\mathrm{P}}/m_{\mathrm{P}})^2 \approx 3H_{\mathrm{inf}}$: under this convention α is inflation-scale, not Planck-scale. The subcriticality parameter is then $\lambda = \mu E_{\mathrm{P}} = 2c_0 H_{\mathrm{inf}} E_{\mathrm{P}}/m_{\mathrm{P}}^2 \sim 10^{-5} \ll 1$, so hypothesis (D4) of Prop. C.2 holds with an enormous margin, and the $O(\lambda)$ discrepancy between $\widehat{H}_{\mathrm{mod}}$ and $h(\widehat{H}_{\mathrm{s}})$ is negligible. The $O(1)$ kinematic factor relating the generator-level matching to $\Delta\mathcal{P}/\mathcal{P}$ is imported from [251, Eqs. (117)–(136)], not re-derived here; an independent converged coefficient is the scope of the companion numerical paper (App. N).

Remark C.4 (Tension between the cutoff scaling and the Born–Oppenheimer matching). Rem. 11.37 adopts the scaling $\delta t \sim \hbar/E_\star$, $\alpha \sim E_\star$, which would give $\mu \sim 1/E_\star = 1/E_{\mathrm{P}}$ at the Planck-natural point. The matching (99) instead gives $\mu = 2c_0 H_{\mathrm{inf}}/m_{\mathrm{P}}^2$, smaller by a factor $\sim H_{\mathrm{inf}}/m_{\mathrm{P}} \sim 10^{-5}$. Both cannot hold simultaneously: the scaling of Rem. 11.37 is a normalization convention on the $(\alpha, \delta t)$ split, whereas (99) is the matched prediction for the physical combination μ. In the laboratory, both sit far below current sensitivity: the phonon-cavity bound of Campbell et al. [252] constrains quadratic-in-energy Hamiltonian corrections at the level $\mu \lesssim 2\times 10^{-24}\,\mathrm{eV}^{-1}$, i.e. $E_{\mathrm{QG},1} = 1/\mu \gtrsim 5\times 10^{14}\,\mathrm{GeV}$, which neither value approaches (see Sec. 13.5).

The tension is, however, now resolved *empirically*, by astrophysical rather than laboratory data. By Prop. X.2 (App. X), the same μ controls photon time-of-flight through the induced subluminal linear dispersion $v_g = c(1 - \mu E)$, and the GRB timing bounds — Fermi-LAT GRB 090510, $E_{\mathrm{QG},1} > 7.6\, E_{\mathrm{P}}$ [346]; LHAASO GRB 221009A, $E_{\mathrm{QG},1} > 8.2\, E_{\mathrm{P}}$ (subluminal, Table I of [345]) — give $\mu < 1.0 \times 10^{-29}\,\mathrm{eV}^{-1}$ (140). The Planck-natural point $\mu = 1/E_{\mathrm{P}} = 8.2 \times 10^{-29}\,\mathrm{eV}^{-1}$ is thereby excluded by a factor 7.6–8.2, while the matched value (99), $\sim 10^{-33}\,\mathrm{eV}^{-1}$, survives by 3–4 decades. The convention question is settled by data in favour of the Born–Oppenheimer matching; the dimensional tension above is retained as documentation of why the Planck-natural normalization was entertained at all.

C.3 Thermodynamic consistency: Araki-perturbed equilibrium and the modified first law

The superseded companion note closed its derivation with a "first-law check" of the form $\delta S = (\delta A + \alpha\,\delta\Delta t)/4G = \delta E/T$. That check fails twice over. First, it is dimensionally incoherent: the note's modified TDSE requires $[\alpha\Delta t] =$ energy, hence $[\alpha] = E^2$ in natural units, while its entropy variation requires $[\alpha\,\delta\Delta t] = [\delta A] = E^{-2}$, hence $[\alpha] = E^{-1}$ — no assignment of $[\alpha]$ satisfies both (Rem. C.3 performs the units autopsy of the underlying $\alpha \sim 1/4G$ identification). Second, it is vacuous: Δt ($= \delta t$) is a fixed structural parameter of the dynamics, selected once by the constraint of Conj. 10.13, not a state function varied in a first-law process, so $\delta(\Delta t) \equiv 0$ and the "check" collapses to the unmodified Bekenstein first law without ever probing the temporal link. This subsection replaces it with a genuine thermodynamic consistency check of the $\widehat{\Sigma}_{\delta t}$-modified dynamics, formulated with Araki's perturbation theory of KMS states. The thermodynamic dimensional assignments (those of the dynamical parameters $\alpha, \delta t, \mu, \lambda$ are fixed in Rem. C.3) are:

Symbol	Meaning	SI dimension	$\hbar = c = 1$
$K_\omega = -\log \Delta_\omega$	modular Hamiltonian	1	1
$\beta_{\mathrm{H}} = 2\pi/\hbar\kappa$	Hartle–Hawking inverse temperature	J^{-1}	E^{-1}
$\beta_{\mathrm{H}}\,\alpha\widehat{\Sigma}_{\delta t}$	modular perturbation	1	1
δS	entropy variation ($k_B = 1$)	1	1
$\delta\langle K\rangle$	modular-energy variation	1	1

Both sides of the first law below are dimensionless and every term of the modified Clausius relation (100) is an energy; this is the corrected form of the superseded note's Eq. (12).

Proposition C.5 (First-law consistency of the $\widehat{\Sigma}_{\delta t}$-modified dynamics)**.** *Let $(\mathcal{M}, g)$ carry a bifurcate Killing horizon of surface gravity κ, let $\mathcal{A}_R$ be the CLPW Type II_∞ crossed product of Thm. M.1, and let $\omega = \omega_{\mathrm{HH}}$ be the* equilibrium Hartle–Hawking state*, which is $(\sigma, \beta_{\mathrm{H}})$-KMS at $\beta_{\mathrm{H}} = 2\pi/\hbar\kappa$ for the horizon-boost flow σ_t generated, via Pos. 11.11, by the dressed clock generator $\widehat{H}_{\mathrm{c}}$ [275, 276, 18]. (We use the physical calibration $\widehat{H}_{\mathrm{c}} = K_\omega/\beta_{\mathrm{H}}$, so that $\widehat{H}_{\mathrm{c}}$ carries energy units; the modular-units convention of Pos. 11.11 is recovered at $\beta_{\mathrm{H}} = 1$.) Fix $\delta t > 0$, $\alpha \in \mathbb{R}$, and set*

$$V \;:=\; \alpha\widehat{\Sigma}_{\delta t} \;=\; \alpha\big(\cos(\delta t\,\widehat{H}_{\mathrm{c}}/\hbar) - \mathbb{1}\big), \qquad \widehat{H}_{\mathrm{mod}} \;:=\; \widehat{H}_{\mathrm{c}} + V,$$

the clock-side perturbation of Conj. 10.13 (its on-shell system-side form differs by $O(\lambda)$, Prop. C.2 (ii)). Let $(L, J, \mathcal{P}^\natural)$ be the standard-form data of $(\mathcal{A}_R, \Omega)$ with Ω the vector representative of ω, and let $\mathcal{H}_{[E_\star]} := E_{\widehat{H}_{\mathrm{c}}}([-E_\star, E_\star])\mathcal{H}$ be the band-limited sector of Pos. 11.30. Then:

(i) ***(Existence of a modified equilibrium at the same temperature.)*** *$V = V^* \in \mathcal{A}_R$ with $\|V\| \le 2|\alpha|$ (for $\alpha \ge 0$, $-2\alpha\mathbb{1} \preceq V \preceq 0$) and $\left\|V\restriction_{\mathcal{H}_{[E_\star]}}\right\| \le |\alpha|\min\{2,\, \delta t^2 E_\star^2/2\hbar^2\}$. The Araki vector $\Omega_V := e^{-\beta_{\mathrm{H}}(L+V)/2}\Omega$ is well defined, lies in $\mathcal{P}^\natural$, and is cyclic and separating; the perturbed state $\omega_V := \langle\Omega_V, \cdot\,\Omega_V\rangle/\|\Omega_V\|^2$ is faithful, normal, and $(\sigma^V, \beta_{\mathrm{H}})$-KMS* *for the modified flow σ^V generated by $\widehat{H}_{\rm mod}$ [50, 195, 239]. In particular the modified dynamics possesses an exact equilibrium at the* unrenormalised *Hawking temperature $T_{\rm H} = \hbar\kappa/2\pi$: bounded Araki perturbations shift the state and the modular Hamiltonian, never β. Quantitatively, Peierls–Bogoliubov and Golden–Thompson [50] give $e^{-\beta_{\rm H}\alpha\,\omega(\widehat{\Sigma}_{\delta t})} \le \|\Omega_V\|^2 \le \omega\big(e^{-\beta_{\rm H}\alpha\widehat{\Sigma}_{\delta t}}\big)$, and the relative-entropy identities of Araki perturbation theory yield the two-sided bound*

$$S(\omega\|\omega_V)+S(\omega_V\|\omega) \;=\; \beta_{\rm H}\big(\omega(V)-\omega_V(V)\big) \;\le\; 2\beta_{\rm H}\,\|V\| \;\le\; \beta_{\rm H}\,|\alpha|\,\delta t^2E_\star^2/\hbar^2 \quad \text{on } \mathcal{H}_{[E_\star]},$$

so the modified equilibrium is entropically $O(\beta_{\rm H}|\alpha|\delta t^2E_\star^2/\hbar^2)$-close to $\omega_{\rm HH}$.

(ii) ***(Modified first law.)*** *The relative modular operator satisfies* $\log\Delta_{\Omega_V,\Omega} = \log\Delta_\Omega - \beta_{\rm H}V$, *i.e. on the algebra side $K_{\omega_V} = \beta_{\rm H}(\widehat{H}_{\rm c}+\alpha\widehat{\Sigma}_{\delta t})$. For any differentiable family ψ_s, $s\in(-1,1)$, of algebra-side (inner) perturbations of ω_V with $\psi_0=\omega_V$, the first law of entanglement [54, 58] (Sec. 11.3) holds verbatim, $\delta S = \delta\langle K_{\omega_V}\rangle$ (first variations in s at $s=0$), and splits as*

$$T_{\rm H}\,\delta S \;=\; \delta\langle\widehat{H}_{\rm c}\rangle+\alpha\,\delta\langle\widehat{\Sigma}_{\delta t}\rangle \;=\; \delta\langle\widehat{H}_{\rm c}\rangle-\frac{\mu}{2}\,\delta\langle\widehat{H}_{\rm c}^2\rangle+O\Big(\frac{|\alpha|\,\delta t^4}{\hbar^4}\,\delta\langle\widehat{H}_{\rm c}^4\rangle\Big), \qquad \mu=\frac{\alpha\,\delta t^2}{\hbar^2}, \tag{100}$$

every term an energy. Equivalently, with the Bekenstein calibration of Pos. 11.18, $\delta A/4G\hbar = \beta_{\rm H}\,\delta\langle\widehat{H}_{\rm mod}\rangle$: the Clausius relation holds exactly at $T_{\rm H}$ provided the energy entering it is the modified energy $\langle\widehat{H}_{\rm mod}\rangle$, *and the temporal-link correction to the modular energy is second order in the band energy, vanishing at first order in the equilibrium (soft) limit.*

(iii) ***(Hawking benchmark, KMS form.)*** *Since $[\widehat{\Sigma}_{\delta t},\widehat{H}_{\rm c}]=0$, the modified flow acts diagonally on the boost spectrum. Set $\varepsilon(E) := E+\alpha\Sigma_{\delta t}(E)$, $\Sigma_{\delta t}(E) = \cos(\delta t\,E/\hbar)-1$, and let $a_E\in\mathcal{A}_R$ be an eigenoperator of σ^V connecting the kernel fibre to the boost fibre $E=\hbar\omega\in(0,E_\star]$, so $\sigma_t^V(a_E) = e^{-it\varepsilon(E)/\hbar}a_E$ (using $\Sigma_{\delta t}(0)=0$), with $[a_E,a_E^\dagger] = \Bbb{1}$. If $\varepsilon(E)>0$, the KMS condition for ω_V — a property of the* state, *not of the trace τ — gives $\omega_V(a_Ea_E^\dagger) = e^{\beta_{\rm H}\varepsilon(E)}\omega_V(a_E^\dagger a_E)$, hence*

$$\omega_V(\widehat{N}_E) = \big(e^{\beta_{\rm H}\varepsilon(E)}-1\big)^{-1} = \Big(e^{E/T_{\rm eff}(E)}-1\Big)^{-1}, \qquad \frac{T_{\rm eff}(E)}{T_{\rm H}} = 1+\frac{\mu E}{2}+O\big(\mu^2E^2,\ |\alpha|\,\delta t^4E^3/\hbar^4\big). \tag{101}$$

As $E\to 0$, $\omega_V(\widehat{N}_{\hbar\omega})\to(e^{2\pi\omega/\kappa}-1)^{-1}$: the benchmark of Thm. M.1 is recovered exactly *in the equilibrium limit. This sharpens the scoping of Thm. M.1: the Bose–Einstein occupation is obtained here as a KMS-state property of the equilibrium Hartle–Hawking state under the (perturbed) boost flow — the equilibrium-only reading under which that theorem is to be understood — not as a property of the tracial weight, and no statement about a dynamical Hawking* flux *is made. The frequency-dependent tilt $\Delta T/T_{\rm H} = \mu E/2$ is controlled by the same physical combination μ fixed by the Born–Oppenheimer matching* (99)*; at the matched value $\mu = 2c_0H_{\rm inf}/m_{\rm P}^2$ it is $\sim c_0H_{\rm inf}E/m_{\rm P}^2$, Planck-suppressed.*

(iv) ***(Consistency constraint on α.)*** *No constraint on α arises at first order in the equilibrium limit, because the correction in* (100) *is $O(E^2)$. The genuine constraint is positivity of the perturbed frequencies on the band, $\varepsilon(E)>0$ for all $E\in(0,E_\star]$, without which* (101) *turns negative (level inversion; loss of the particle interpretation underlying the benchmark). Since $0\ge\Sigma_{\delta t}(E)\ge -\delta t^2E^2/2\hbar^2$, this holds for all $\alpha\le 0$, and for $\alpha>0$ whenever*

$$\alpha \;<\; \frac{2\hbar^2}{\delta t^2E_\star} \qquad\Longleftrightarrow\qquad \mu E_\star < 2, \tag{102}$$

the condition being asymptotically sharp as $\delta t\,E_\star/\hbar\to 0$; moreover $\varepsilon'(\xi) = 1-(\alpha\,\delta t/\hbar)\sin(\delta t\,\xi/\hbar)\ge 1-\lambda>0$ under the subcriticality hypothesis (D4) *of Prop. C.2, so no level crossing occurs* *on the band. Under the cutoff-natural normalization convention $\delta t = \hbar/E_\star$ of Rem. 11.37,* (102) *reads $\alpha < 2E_\star$ and is equivalent to $\lambda < 2$; it is therefore strictly weaker than both the subcriticality hypothesis* (D4) *($\lambda < 1$) and the no-folding condition $\mu E_\star < 1$ of Prop. C.2 (iv), which coincide at this convention ($\mu E_\star = \lambda$ when $\delta t\, E_\star/\hbar = 1$). Whenever the fibrewise reduction of Sec. C.2 is valid, thermodynamic stability thus holds automatically: the cutoff-natural convention passes the check with factor-2 headroom, the dynamical conditions bind first, and couplings with $\mu E_\star \geq 2$ (e.g. $\alpha \geq 2E_\star$ at this convention) are* excluded *— the corrected Step 6 is a real check, not a tautology.*

Proof. (o) $V \in \mathcal{A}_R$. In the CLPW crossed product the dressed clock generator $\widehat{H}_{\mathrm{c}}$ is self-adjoint and affiliated to $\mathcal{A}_R$ (Pos. 11.11); hence the bounded Borel function $\widehat{\Sigma}_{\delta t} = (\cos(\delta t \cdot /\hbar) - 1)(\widehat{H}_{\mathrm{c}})$ lies in $\mathcal{A}_R$ with $-2\mathbb{1} \preceq \widehat{\Sigma}_{\delta t} \preceq 0$, and $|\cos x - 1| \leq \min\{2, x^2/2\}$ gives the band-limited norm bound. (This step fails in the undressed Type III wedge algebra, where $-\log \Delta_\omega$ is *not* affiliated to $\mathcal{N}$; the crossed product [18] is what renders the perturbation algebraic.)

(i) is Araki's bounded-perturbation theory of KMS states [50], in the standard-form/Liouvillian formulation of [239, Thm. 5.1] (textbook form [195, Ch. 5.4]): for $V = V^* \in \mathcal{A}_R$ and a $(\sigma, \beta_{\mathrm{H}})$-KMS ω with standard vector Ω, one has $\Omega \in \operatorname{dom} e^{-\beta_{\mathrm{H}}(L+V)/2}$, $\Omega_V \in \mathcal{P}^\natural$ cyclic and separating, ω_V is $(\sigma^V, \beta_{\mathrm{H}})$-KMS, $\log \Delta_{\Omega_V} = -\beta_{\mathrm{H}} L_V$ with $L_V = L + V - JVJ$, and $S(\omega \| \omega_V) = \log \|\Omega_V\|^2 + \beta_{\mathrm{H}}\omega(V)$, $S(\omega_V \| \omega) = -\log \|\Omega_V\|^2 - \beta_{\mathrm{H}}\omega_V(V)$. Adding the last two identities and using $|\omega(V) - \omega_V(V)| \leq 2\|V\|$ gives the entropy bound; positivity of relative entropy applied to the first identity gives Peierls–Bogoliubov, and Golden–Thompson $\|\Omega_V\| \leq \left\| e^{-\beta_{\mathrm{H}} V/2}\Omega \right\|$ is Araki's von Neumann-algebraic Golden–Thompson inequality [51].

(ii) The relative-modular identity $\log \Delta_{\Omega_V, \Omega} = \log \Delta_\Omega - \beta_{\mathrm{H}} V$ is part of the same theorem. For an inner perturbation $\psi_s = \omega_V(u_s^* \cdot u_s)$ with $u_s \in \mathcal{U}(\mathcal{A}_R)$, the commutant contribution drops identically: $JVJ \in \mathcal{A}_R'$, so $\psi_s(JVJ) = \langle \Omega_V, u_s^* JVJ u_s \Omega_V \rangle = \omega_V(JVJ)$ for all s, whence $\delta\langle L_V \rangle = \delta\langle \widehat{H}_{\mathrm{c}} + V \rangle$ (the identification of the algebra-side part of L with the boost energy is the standard Bisognano–Wichmann splitting used throughout Sec. 11.3). The first-order identity $\delta S = \delta\langle K \rangle$ for perturbations of a faithful normal state is the entanglement first law [54, 58], applied to the pair (ω_V, K_{ω_V}) — it is state-generic, hence form-invariant under the perturbation; this is precisely why the Clausius leg of Pos. 11.18 survives the $\widehat{\Sigma}_{\delta t}$ modification. The expansion is the strong Taylor expansion of $\widehat{\Sigma}_{\delta t}$ on analytic vectors (Eq. (26)), with fibrewise Lagrange remainder $|\cos x - 1 + x^2/2| \leq x^4/24$ as in Prop. C.2 (iv).

(iii) Since $\widehat{\Sigma}_{\delta t}$ is a function of $\widehat{H}_{\mathrm{c}}$, σ^V preserves the boost spectral fibres and a_E is a σ^V-eigenoperator of frequency $\varepsilon(E)/\hbar$. The β_{H}-KMS condition for ω_V under σ^V, applied to $F(t) = \omega_V(a_E^\dagger \sigma_t^V(a_E))$, gives the detailed-balance relation $\omega_V(a_E a_E^\dagger) = e^{\beta_{\mathrm{H}}\varepsilon(E)}\omega_V(a_E^\dagger a_E)$, and the CCR normalisation yields (101); no tracial weight is invoked at any step. The expansion of T_{eff} follows from $\varepsilon(E) = E(1 - \mu E/2) + O(|\alpha| \delta t^4 E^4/\hbar^4)$.

(iv) $\varepsilon(E) \geq E(1 - \mu E/2)$ from $1 - \cos x \leq x^2/2$; monotonicity of $E \mapsto \mu E/2$ makes $E = E_\star$ the binding case, giving (102); sharpness follows since $1 - \cos x = x^2/2 + O(x^4)$. The derivative bound uses $|\sin x| \leq 1$ and (D4); the comparison with λ and $\mu E_\star$ is arithmetic: $\mu E_\star = \lambda \cdot (\delta t\, E_\star/\hbar)$. □

Remark C.6 (Scope: equilibrium only, and circularity disclosure). Everything above is scoped to the *equilibrium Hartle–Hawking state* at $T_{\mathrm{H}} = \kappa/2\pi$: Eq. (101) is a KMS-state property of ω_V (detailed balance under the modified boost flow), not a property of the Type II trace τ, and no statement about a dynamical Hawking flux from a collapsing background is made or implied. As throughout, the surface gravity κ enters only through Pos. 11.11's identification of the modular flow with the horizon boost, so T_{H} is calibrated by, not derived independently of, κ: the check certifies internal consistency, not an independent prediction of $\kappa/2\pi$.

Remark C.7 (The cosine resummation is thermodynamically essential). Araki's theory applies because $\widehat{\Sigma}_{\delta t}$ is *bounded.* The quadratic truncation $V_2 = -\alpha\, \delta t^2 \widehat{H}_{\mathrm{c}}^2/2\hbar^2$ is unbounded below for $\alpha > 0$ on an unbounded clock spectrum, where neither Araki's construction nor its lower-bounded (Sakai) or L^p-controlled extensions [239, 197] guarantee a perturbed KMS state: the truncated dynamics has no certified equilibrium. Thermodynamic consistency thus *selects* the resummed cocycle over its polynomial truncations; on the band-limited sector $\mathcal{H}_{[E_\star]}$ the two agree to $O(\delta t^4 E_\star^4/\hbar^4)$, so the check is insensitive to the truncation exactly where the framework is defined.

Remark C.8 (Rigor ledger for Prop. C.5). (a) The textbook citation [195, Ch. 5.4] is section-level; the theorem content used is [50] and [239, Thm. 5.1]. (b) Only *existence* of the modified equilibrium and the structural first law are proved; *dynamical* stability (return to equilibrium of perturbed states under σ^V) would require L^1-asymptotic-abelianness-type hypotheses [195, Ch. 5.4] not established for the horizon crossed product, and is not claimed. (c) The identification of the algebra-side part of L with the boost energy uses the formal Bisognano–Wichmann splitting $L = H_R - JH_RJ$, at the same level of rigor as its use in Sec. 11.3. (d) Item (iii) presumes a CCR eigen-mode $a_E \in \mathcal{A}_R$ for the modified flow — the same "asymptotic number operator" hypothesis as Thm. M.1 itself. (e) The necessity direction of (102) is asymptotic (sharp as $\delta t\, E_\star/\hbar \to 0$); sufficiency is unconditional.

C.4 Structural compatibility: Born invariance, no-signalling, conservation, unitarity

This subsection verifies that the modified generator $\widehat{H}_{\rm mod}$ of (97) (equivalently the clock-side $\widehat{H}_{\rm c} + \alpha\widehat{\Sigma}_{\delta t}$ of Rem. 10.24, or the exact fibrewise $h(\widehat{H}_{\rm s})$ of Prop. C.2(i)) is compatible with every structural theorem of the two-boundary core. The organizing observation is that the proofs of those theorems use exactly four ingredients: (1) the propagator identities of Rem. A.1; (2) kinematic POVM completeness ($\sum_k E_k = \mathbb{1}$, $\sum_j M_j^{(1)*} M_j^{(1)} = \mathbb{1}$); (3) microcausality (M2) of Ax. 2.2; and (4) causal propagation (M3b), which enters only in Case 1 of Thm. 6.1. The perturbation $\alpha\widehat{\Sigma}_{\delta t}$ touches only ingredient (4), and harmlessly, because $\widehat{\Sigma}_{\delta t} = \cos(\delta t\, \widehat{H}_{\rm c}/\hbar) - \mathbb{1}$ is a bounded real Borel function of $\widehat{H}_{\rm c}$ alone, hence lies in the abelian clock von Neumann algebra $\{\widehat{H}_{\rm c}\}''$, whose tensor embedding $\{\widehat{H}_{\rm c}\}'' \otimes \mathbb{C}\mathbb{1}$ is contained in the commutant of every local matter algebra $\mathbb{1} \otimes \mathcal{A}(O)$. The five propositions below are corollaries of this affiliation.

Proposition C.9 (K1: Born time-invariance under the modified dynamics). *Let $\widehat{H}_{\rm mod}$ be any self-adjoint operator and $U_{\rm mod}(t_2, t_1) := e^{-i\widehat{H}_{\rm mod}(t_2 - t_1)/\hbar}$. Then, with $\rho(t) := U_{\rm mod}(t, t_i)\rho_{\rm i} U_{\rm mod}(t, t_i)^\dagger$ and $\Lambda_k(t) := U_{\rm mod}(t_f, t)^\dagger E_k U_{\rm mod}(t_f, t)$ built from the* same *propagator,*

$$p_t(k) = \mathrm{Tr}\big(\rho(t)\Lambda_k(t)\big) = \mathrm{Tr}\big(\rho_{\rm i}\, U_{\rm mod}(t_f, t_i)^\dagger E_k\, U_{\rm mod}(t_f, t_i)\big)$$

is independent of $t \in [t_i, t_f]$, with $\sum_k p_t(k) = 1$. Moreover every *statement of App. A — Lem. A.5, the conditioned-state results Defs. A.25–A.28 and Lems. A.27, A.30, the martingale/collapse chain Lem. A.20, Thm. A.21, Cors. A.23, A.31, and Axioms A.13, A.22 — holds verbatim with $U \to U_{\rm mod}$: no statement of that appendix refers to a generator, only to the propagator identities of Rem. A.1 and to the measure-theoretic layer, which does not see the dynamics at all. Prop. C.2(v) is the special case $\widehat{H}_{\rm mod} = \widehat{H}_{\rm s} + \alpha\widehat{\Sigma}_{\delta t}(\widehat{H}_{\rm s})$ of* (97) *(numerically verified in App. N), and Rem. 10.24(a) is the clock-side case $\widehat{H}_{\rm c} + \alpha\widehat{\Sigma}_{\delta t}$; both are absorbed here by reference. One caveat of* reading, *already imposed by Prop. 10.14: the per-slice delay–advance equation* (31) *does not define a propagator from single-slice data, so K1 applies to the generators that do define one — the clock/constraint generator, the history-space operator of Prop. 10.14, or the effective generators of Prop. C.2 and Prop. 10.27 on $\mathcal{H}^{\rm slow}_{\rm phys}$ (Rem. 10.28) — not to a fictitious per-slice propagator for* (31).

Proof. Stone's theorem gives a strongly continuous unitary group, and the two-parameter family $U_{\rm mod}(t_2, t_1)$ satisfies the composition, identity and reversal identities of Rem. A.1. Time invariance is then word-for-word the proof of Lem. A.5: cyclicity of the trace and the composition law collapse $U_{\rm mod}(t,t_i)^\dagger U_{\rm mod}(t_f,t)^\dagger = U_{\rm mod}(t_f,t_i)^\dagger$; normalization follows from $\sum_k \Lambda_k(t) = U_{\rm mod}(t_f,t)^\dagger(\sum_k E_k)U_{\rm mod}(t_f,t) = \mathbb{1}$. For the second claim, inspection of App. A shows the dynamics enters only through $U(\cdot,\cdot)$ and Rem. A.1; the filtration results are purely measure-theoretic. □

Proposition C.10 (K2: preservation of bulk no-signalling)**.** *The addition of $\alpha\widehat{\Sigma}_{\delta t}$ preserves Thm. 6.1, at three levels of resolution.*

(a) Constraint level (exact). *On $\mathcal{H}_{\rm clock}\otimes\mathcal{H}_{\rm sys}$ the modified constraint splits as*

$$\widehat{C}_{\alpha,\delta t} = \big(\widehat{H}_{\rm c} + \alpha\widehat{\Sigma}_{\delta t}\big)\otimes\mathbb{1} + \mathbb{1}\otimes\widehat{H}_{\rm s},$$

with the two summands strongly commuting, so $e^{-is\widehat{C}_{\alpha,\delta t}/\hbar} = e^{-is(\widehat{H}_{\rm c}+\alpha\widehat{\Sigma}_{\delta t})/\hbar}\otimes e^{-is\widehat{H}_{\rm s}/\hbar}$. Since $\alpha\widehat{\Sigma}_{\delta t}\otimes\mathbb{1}\in\{\widehat{H}_{\rm c}\}''\otimes\mathbb{C}\mathbb{1}$ lies in the commutant of every local matter algebra $\mathbb{1}\otimes\mathcal{A}(O)$, conjugation of $\mathbb{1}\otimes\mathcal{A}(O)$ by $e^{-is\widehat{C}_{\alpha,\delta t}/\hbar}$ coincides with conjugation by $\mathbb{1}\otimes e^{-is\widehat{H}_{\rm s}/\hbar}$: microcausality (M2) and causal propagation (M3b) are inherited verbatim *from the unmodified axioms, both cases of the proof of Thm. 6.1 go through unchanged, and $\sum_j P_M(j,l) = P_\varnothing(l)$ holds exactly.*

(b) Retro-direction (exact, generator-free). *Case 2 of the proof of Thm. 6.1 (intervention to the* future *of the distant region) uses only trace preservation, never (M3b); hence no-signalling into the past holds for the modified dynamics without any locality hypothesis on the generator whatsoever. The time-nonlocality of $\widehat{\Sigma}_{\delta t}$ cannot create backwards-in-time signalling.*

(c) Effective level (scoped). *On the slow branch the per-slice generator is $\widehat{H}_{\rm eff} = \widehat{H}_{\rm s} - \frac{\mu}{2}\widehat{H}_{\rm s}^2$ (Prop. 10.27, Prop. C.2(iv)); the correction $-\frac{\mu}{2}\widehat{H}_{\rm s}^2$ is not a local integral of densities, so (M3b) for $e^{-i\widehat{H}_{\rm eff}t/\hbar}$ can fail by cone-leakage of order $O(\mu E_\star^2/\hbar)$ on the $E_\star$-band-limited subspace. This is not a physical signal: the effective description is itself accurate only to the same order in the truncation (Prop. C.2(ii)), and the exact operational statistics are those of item (a), which are exactly no-signalling. Any apparent acausal tail of $\widehat{H}_{\rm eff}$ is an artifact of truncating the δt-expansion.*

Proof. (a) The summands are functions of commuting self-adjoint operators on the joint analytic core Φ of Conj. 10.13, so the group factorizes by the spectral theorem; membership $\widehat{\Sigma}_{\delta t}\in\{\widehat{H}_{\rm c}\}''$ is bounded Borel functional calculus, and $\{\widehat{H}_{\rm c}\}''\otimes\mathbb{C}\mathbb{1}\subset(\mathbb{C}\mathbb{1}\otimes B(\mathcal{H}_{\rm sys}))'\subset(\mathbb{1}\otimes\mathcal{A}(O))'$ for every O. Hence $e^{-is\widehat{C}_{\alpha,\delta t}/\hbar}(\mathbb{1}\otimes A)e^{is\widehat{C}_{\alpha,\delta t}/\hbar} = \mathbb{1}\otimes e^{-is\widehat{H}_{\rm s}/\hbar}Ae^{is\widehat{H}_{\rm s}/\hbar}$ for all $A\in\mathcal{A}(O)$: the displayed steps of Thm. 6.1 — the (M3b) inclusion $U_{21}^\dagger\mathcal{A}(O_2)U_{21}\subset\mathcal{A}(J(O_2)\cap\Sigma_{\tau_1})$, the normal-separation argument, and the (M2) commutation $[F_l^{(1)},M_j^{(1)}]=0$ — are literally the same computation. (b) is read off from Case 2 of the proof of Thm. 6.1, which eliminates the later intervention by $\sum_j M_j^{(1)*}M_j^{(1)} = \mathbb{1}$ and cyclicity alone. (c) On $E_{\widehat{H}_{\rm s}}([-E_\star,E_\star])\mathcal{H}_{\rm sys}$ one has $\left\|\frac{\mu}{2}\widehat{H}_{\rm s}^2\psi\right\|\le\frac{\mu}{2}E_\star^2\,\|\psi\|$, and a Duhamel estimate bounds

$$\big\|e^{-i\widehat{H}_{\rm eff}t/\hbar}A\,e^{i\widehat{H}_{\rm eff}t/\hbar} - e^{-i\widehat{H}_{\rm s}t/\hbar}A\,e^{i\widehat{H}_{\rm s}t/\hbar}\big\| \;\le\; \frac{\mu E_\star^2\,|t|}{\hbar}\,\|A\|$$

for A in the band-limited compression, which is the stated $O(\mu)$ leakage; exactness is restored at level (a). □

Corollary C.11 (K3: conservation of charges)**.** *Let Q be self-adjoint and commute with the spectral measure of $\widehat{H}_{\rm c}$. Then $[Q,\widehat{\Sigma}_{\delta t}]=0$, since $\widehat{\Sigma}_{\delta t}$ is a bounded Borel function of $\widehat{H}_{\rm c}$; hence $[Q,\widehat{H}_{\rm c}+\alpha\widehat{\Sigma}_{\delta t}]=0$. Conversely, under the subcriticality condition* (D4) *of Prop. C.2, i.e. $\lambda=|\alpha|\ \delta t/\hbar<1$, the function $g(x):=x+\alpha(\cos(\delta t\,x/\hbar)-1)$ has $g'(x)=1-\frac{\alpha\,\delta t}{\hbar}\sin(\delta t\,x/\hbar)\ge 1-\lambda>0$, so g is strictly increasing with continuous inverse on its range, $\{g(\widehat{H}_{\rm c})\}''=\{\widehat{H}_{\rm c}\}''$, and*

$$[Q,\widehat{H}_{\rm c}+\alpha\widehat{\Sigma}_{\delta t}]=0 \iff [Q,\widehat{H}_{\rm c}]=0:$$

the modified dynamics conserves exactly *the charges of the unmodified clock dynamics, no more and no fewer. The same statement holds on the system side for* $\widehat{H}_{\rm mod} = g(\widehat{H}_{\rm s})$ *of* (97) *and for the exact* $h(\widehat{H}_{\rm s})$*, whose strict monotonicity* $h'(E) \geq (1+\lambda)^{-1} > 0$ *is Prop. C.2(i). Matter-sector charges* $\mathbb{1} \otimes Q_{\rm m}$ *commute with* $\alpha\widehat{\Sigma}_{\delta t} \otimes \mathbb{1}$ *unconditionally; consequently Thm. 4.1 (outcome-wise conservation of conditioned expectations), whose proof uses only unitary covariance (Thm. 3.7) and the Heisenberg relation* (6)*, holds for the modified propagator under the* same *hypothesis* $\nabla^\mu \widehat{O}_{\mu\nu\cdots} = 0$*; and Prop. 11.24 (KMS stationarity) is untouched, since under Pos. 11.11 the clock generator is* $-\log \Delta_\omega$ *and* $\widehat{\Sigma}_{\delta t} \in \{\widehat{H}_{\rm c}\}''$ *preserves the modular centraliser on which that proposition lives.*

Proof. Commutation with the spectral measure passes to every bounded Borel function (functional calculus). For the converse: g strictly increasing and continuous is a Borel isomorphism onto its range, so $\widehat{H}_{\rm c} = g^{-1}(g(\widehat{H}_{\rm c}))$ is a Borel function of $g(\widehat{H}_{\rm c})$, giving the mutual inclusion of the generated abelian von Neumann algebras and hence the displayed equivalence. The matter-sector statements are the tensor-commutant observation of Prop. C.10(a) together with an inspection of the proof of Thm. 4.1, which never differentiates the generator, only conjugates by the propagator. □

Proposition C.12 (K4: self-adjointness, unitarity, trace preservation)**.** (i) *Self-adjointness of* $\widehat{H}_{\rm mod}$ *on* $\mathcal{D}(\widehat{H}_{\rm s})$*, semiboundedness, and unitarity of* $U_{\rm mod}(t) = e^{-i\widehat{H}_{\rm mod}t/\hbar}$ *are Prop. C.2(iii); we record only that for the one-sector generators* ($g(\widehat{H}_{\rm s})$, $h(\widehat{H}_{\rm s})$, $\widehat{H}_{\rm c} + \alpha\widehat{\Sigma}_{\delta t} = g(\widehat{H}_{\rm c})$) *spectral calculus alone already suffices, since each is a real Borel function* g *of a self-adjoint operator with* $g(x) - x$ *bounded, so Kato–Rellich is sufficient but not necessary there. Kato–Rellich ([32, Thm. X.12], relative bound* $a = 0$, $b = 2\,|\alpha|$*) is genuinely needed only for the history-space operator* $\widehat{H}_{\rm s} \otimes \mathbb{1}_t + \alpha\widehat{\Sigma}^{\rm h}_{\delta t}$ *of Prop. 10.14, where* $\widehat{\Sigma}^{\rm h}_{\delta t}$ *is* not *a function of* $\widehat{H}_{\rm s}$. (ii) *Conjugation by* $U_{\rm mod}$ *preserves positivity, trace, and trace norm. Consequently Lems. A.27 and A.30 (conditioned and retrocausal states are density operators) and Cor. A.31 (trace-norm collapse) hold under the modified dynamics without modification.*

Proof. (i) is Prop. C.2(iii) and the spectral theorem. (ii) $\mathrm{Tr}(U\rho U^\dagger X) = \mathrm{Tr}(\rho\, U^\dagger X U)$ and unitary invariance of the trace norm; the cited finite-dimensional proofs use only positivity of $\Lambda_k(t)$, Tr-normalization (both supplied by K1), and the measure-theoretic layer. □

Proposition C.13 (K5: compatibility of the conditioning (CP-)structure with the modified evolution)**.** *Let* $U := U_{\rm mod}(\tau, \tau')$ *as in K1, and define* $\rho_2(\tau \mid k)$ *by Def. A.25 with* $\rho(\tau), \Lambda_k(\tau)$ *built from* $U_{\rm mod}$*. Then:*

- (i) *(*Intertwining.*)* $\rho_2(\tau \mid k) = U\,\rho_2(\tau' \mid k)\,U^\dagger$ *for all* τ, τ', k*: conditioning on the terminal outcome commutes with the modified evolution exactly as in Thm. 3.7, and the von Neumann equation of Cor. 3.8 holds with* $H \to \widehat{H}_{\rm mod}$ *on* $\mathcal{D}(\widehat{H}_{\rm mod})$*.*
- (ii) *(*Where linearity/CP lives.*) For fixed* k *the sandwich* $A \mapsto \rho^{1/2} A\, \rho^{1/2}$ *is completely positive; the branch map* $\rho \mapsto \rho_2(\tau \mid k)$ *is nonlinear only through the scalar normalization* $p(k)^{-1}$ *(measurement update, not dynamics), and its prior average is the* identity channel*:* $\sum_k p(k)\,\rho_2(\tau \mid k) = \rho^{1/2}(\sum_k \Lambda_k)\rho^{1/2} = \rho(\tau)$*, the no-signalling identity of Thm. 6.1. This is why the Gisin–Polchinski obstruction (nonlinear* dynamics $\Rightarrow$ *superluminal signalling [284, 285]) does not apply: the dynamics* $e^{-i\widehat{H}_{\rm mod}t/\hbar}$ *is linear and unitary, and all nonlinearity is confined to Bayesian conditioning, which averages away.*
- (iii) *(*Probability side untouched.*) Defs. A.26, A.28 and the collapse chain (Lem. A.20, Thm. A.21, Cors. A.23, A.31) involve the dynamics only through the constant* $p(k)$ *of K1 and the trace-one positivity of K4(ii); hence* $\rho_{\rm RC}(\tau)$ *remains a measurable density-valued martingale mixture and* $\|\rho_{\rm RC}(t_n) - \rho_2(t_n \mid K)\|_1 \to 0$ *a.s. under the modified dynamics.*

Proof. (i) is the proof of Thm. 3.7 with $U(\tau, \tau') \to U_{\rm mod}(\tau, \tau')$: it uses only (1) covariance of the forward state, $\rho(\tau) = U\rho(\tau')U^\dagger$; (2) uniqueness of the positive square root, giving $\rho(\tau)^{1/2} =$

$U\rho(\tau')^{1/2}U^\dagger$; (3) the composition law $U_{\mathrm{mod}}(\tau_f,\tau') = U_{\mathrm{mod}}(\tau_f,\tau)U$, giving $\Lambda_k(\tau) = U\Lambda_k(\tau')U^\dagger$; and (4) τ-independence of $p(k)$ (K1). All four hold for U_{mod}; the $p(k) = 0$ branch is trivial since $U(\mathbb{1}/d)U^\dagger = \mathbb{1}/d$. Cor. 3.8 follows by differentiating on $\mathcal{D}(\widehat{H}_{\mathrm{mod}})$ (K4). (ii) Complete positivity of the sandwich is Kraus form with the single Kraus operator $\rho^{1/2}$; the average identity is $\sum_k \Lambda_k(\tau) = \mathbb{1}$ (K1). (iii) is a re-run of Lems. A.27, A.30 and Cor. A.31, which need only $\rho_2(\tau \mid k) \succeq 0$, $\operatorname{Tr}\rho_2(\tau \mid k) = 1$, and the measure-theoretic layer, none of which sees the generator. □

Remark C.14 (The temporal link is algebraic, not geometric: no wormhole is used or permitted). The superseded companion note (absorbed into this appendix; Rem. C.1) phrased its Steps 4–5 as the replacement of the classical singularity by a "wormhole network" mediating the temporal link. That language is retired, for three reasons that together constitute a consistency requirement rather than a loss. (i) *What the temporal link actually is.* The mathematical content of the note's "bifurcated wave function" is, in the corrected framework, the *two-boundary conditioning structure*: an initial datum ρ_{i} on Σ_{τ_i}, a terminal POVM $\{E_k\}$ on Σ_{τ_f}, and the bounded self-adjoint clock-shift cocycle $\widehat{\Sigma}_{\delta t}$ of (26) entering the constraint of Conj. 10.13. The link is carried by operator algebra — the backward effects $\Lambda_k(\tau)$, the conditioned states of Def. A.25, and the two-time link state of Def. 10.17, whose non-classical content is exactly the Margenau–Hill negativity of Thm. 3.10 — not by any bulk geometry connecting the two boundaries; the item-by-item translation of the bifurcation vocabulary is Rem. 10.25. (ii) *Why it must not be geometrized.* The framework *forbids* its own geometrization as a traversable bridge: the self-consistent achronal averaged null energy condition (Rem. 10.12; Graham–Olum [391], Wall [55]), imposed as the admissibility condition on the two-boundary data $(\rho_{\mathrm{i}}, \{E_k\}, g_0)$, is precisely the condition excluding traversable wormholes, closed timelike curves and causality violation in every admissible per-sector solution g_k (cf. Rem. 10.24(b)). A traversable temporal wormhole would therefore be inconsistent with the framework's own energy conditions, and would moreover threaten the bulk no-signalling theorem that the conditioning structure provably satisfies (Prop. C.10). (iii) *No step needs one.* In the corrected chain, Steps 1–3 (Sec. C.1) are wormhole-free; Step 4's role is played by Conj. 10.13, the d'Alembert uniqueness selection of $\widehat{\Sigma}_{\delta t}$ (Prop. 11.6, Thm. B.5) and the derived link structure of Def. 10.17; Step 5 is Prop. 10.14 / Prop. C.2, with the note's asymmetric form recovered on the retarded sector (Rem. 10.16) and dimensionally repaired in Rem. C.3; and Step 6's first-law check uses horizon area (Prop. C.5), not traversability. Entanglement is treated algebraically throughout; the only remaining wormhole references elsewhere in this paper are the achronal-ANEC *prohibition* itself and a contrastive citation of the holographic two-boundary construction of [247], whose AdS-wormhole setting is explicitly not relied upon. ER=EPR is not assumed.

Remark C.15 (Position relative to published retrocausal and modified-Schrödinger proposals). The no-signalling structure established above is not automatic for modified-TDSE proposals, and the $\widehat{\Sigma}_{\delta t}$ form avoids the known failure modes by construction. (i) *Nonlinear-dynamics obstruction.* Gisin [284] and Polchinski [285] showed that deterministic nonlinear modifications of the Schrödinger equation generically permit superluminal signalling; this is the standard no-go against modified dynamics, with only narrow non-superluminal exceptions [286]. The present modification is categorically outside its scope: $\alpha\widehat{\Sigma}_{\delta t}$ is a *linear, bounded, self-adjoint* addition to the generator, and all nonlinearity is confined to per-branch conditioning whose prior average is the identity channel (Prop. C.13(ii)) — the same structural reason the two-state-vector/ABL formalism [5] is signalling-free. The known ABL-adjacent failure mode — operational access to the post-selection before τ_f — is exactly what the active law of Thm. 6.1 and the remark following it quarantine: influence on the non-local terminal quantity $\Lambda_k(\tau_1)$, never on any spacelike-separated marginal (cf. the NBTS analysis of [85]). (ii) *Hidden-variable fine-tuning.* Sutherland's causally symmetric Bohm model [287] achieves no-signalling only statistically, by averaging over final boundary conditions, and sits inside the Wood–Spekkens fine-tuning critique [282], with the tuning argued to be symmetry-protected in [283]. The present framework's answer is stronger in kind: there is no hidden-variable layer to be fine-tuned — no-signalling is an operator-algebraic theorem, exact at the constraint level (Prop. C.10(a)) and generator-free in the retro-direction (Prop. C.10(b)) — though the operational (causal-structure-independent) reading of fine-tuning and the Local Friendliness theorems remain open engagements (Sec. 11). (*iii*) *Transactional interpretation.* Cramer's transactional picture [288] shares the two-boundary intuition; its known difficulties (contingent-absorber consistency) concern the transaction narrative, not a proven signalling violation, and nothing here borrows from it.

D CLPW Type II trace finiteness: status of (E1) as the scope postulate (H0)

This appendix records the honest operator-algebraic status of item (E1) of Sec. 12.5. The Chandrasekaran–Longo–Penington–Witten Type II crossed-product construction [18] and its extensions [87, 88, 92, 89, 90, 96] render the gravitational trace and generalised entropy UV-finite *without* any energy cutoff; they do *not* produce a spectral band-limit of the matter Hamiltonian. Accordingly, Pos. 11.30 is retained as an explicit Wilsonian effective-field-theory scope postulate (H0), and the Type II structure below is offered as *motivation* for its consistency, not as a derivation.

QRF-dressed Type II crossed product

Let $(\mathcal{M}, g)$ be globally hyperbolic, $\mathcal{O} \subset \mathcal{M}$ a relatively compact causal diamond, and $\mathcal{N} = \mathcal{N}(\mathcal{O})$ the type III_1 local matter algebra in the sense of Haag–Kastler / locally covariant QFT [10, 16, 220]. Fix a Hadamard state ω with modular operator Δ_ω and modular automorphism group $\sigma_t^\omega = \mathrm{Ad}(\Delta_\omega^{it})$. Let R be a covariant-observer quantum reference frame [96, 97, 102, 103] dressed to the boost/ADM generator of the wedge associated with $\mathcal{O}$, and write $\hat{H}_R$ for the constraint-reduced dressed generator.

Hypothesis D.1 (QRF-dressed Type II realisation)**.** The QRF-dressed crossed product

$$\widetilde{\mathcal{N}} \;:=\; \mathcal{N} \rtimes_\sigma \mathbb{R}_{\hat{H}_R}$$

is a type II factor with a faithful normal semifinite trace τ_R, unique up to scale, and $\hat{H}_R$ is self-adjoint on a dense core of $\widetilde{\mathcal{N}}$. Its subtype is fixed by the spectrum of the dressed generator: if $\hat{H}_R$ is bounded below (a positive-energy observer on a relatively compact diamond or the de Sitter static patch), the observer-projected algebra $\Pi\widetilde{\mathcal{N}}\Pi$ is type II_1 with $\tau_R(\mathbb{1}) < \infty$ [18, 96, 87]; if the dressed charge is unbounded (asymptotic boundary, evaporating horizon, or a cosmology not future-eternal de Sitter), the algebra is type II_∞ with $\tau_R(\mathbb{1}) = \infty$ and no maximum-entropy state [18, 87, 90, 89]. We take Hyp. D.1 as the ambient-literature input for the present appendix; on de Sitter, stationary, and cosmological backgrounds it is a theorem of the cited literature, and its extension to generic globally hyperbolic $(\mathcal{M}, g)$ remains an ambient-literature programme.

Type II trace finiteness

Proposition D.2 (Type II trace-finiteness: UV-finite entropy without a band-limit)**.** *Assume Hyp. D.1. Then:*

(i) *(*II_1 sector.*) Normalising $\tau_R(\mathbb{1}) = 1$, every normal state on $\Pi\widetilde{\mathcal{N}}\Pi$ admits a density operator $\rho \in L^1(\Pi\widetilde{\mathcal{N}}\Pi, \tau_R)$ with finite von Neumann entropy relative to the tracial state, and the tracial state itself is the maximum-entropy state [18].*

(ii) *(*II_∞ sector.*) Every spectral projection of the dressed generator bounded above is trace-finite: $\tau_R\big(\mathbf{1}_{(-\infty,E]}(\hat{H}_R)\big) < \infty$ for every $E < \infty$, while $\tau_R(\mathbb{1}) = \infty$. (In the canonical* *q-representation of the crossed product the dual trace is* $\tau_R(a) = \int_{\mathbb{R}} dq\, e^{\beta q} \langle q|a|q\rangle$ *[238, 18], whose weight decays towards* $-\infty$*; hence every half-line band bounded above is finite and the divergence sits entirely at* $q \to +\infty$*.)*

(iii) *(*No threshold.*) In both sectors the set* $\{E \in \mathbb{R} : \tau_R(\mathbf{1}_{(-\infty,E]}(\hat{H}_R)) < \infty\}$ *is all of* $\mathbb{R}$*, so its supremum is* $+\infty$*:* no finite trace-finiteness threshold exists. *The Type II structure renders trace and generalised entropy UV-finite while the underlying type III*$_1$ *matter algebra and its unbounded matter Hamiltonian are retained unchanged. In particular the Planck-scale spectral cutoff of Pos. 11.30 is* not *derivable from Hyp. D.1; the modular Hamiltonian* $-\log \Delta_\omega$ *(a state property) and the matter Hamiltonian* $\widehat{H}_{\mathrm{m}}$ *(an algebra generator) are distinct operators, and trace-finiteness of spectral bands of the former implies no band-limit on the latter.*

Proof. (i) is the CLPW positivity compression: with the observer energy bounded below, the projected crossed product carries a finite trace, hence is type II$_1$ after normalisation [18][96, §3.2]; finiteness of the trace makes L^1 density operators and the maximum-entropy (tracial) state standard consequences of finite von Neumann algebra theory. (ii) is the canonical dual trace formula on the crossed product [238, 18]: the weight $e^{\beta q}$ integrates finitely over every half-line $(-\infty, E]$ and divergently over $\mathbb{R}$. (iii) is immediate from (i)–(ii): in the II$_1$ sector every projection is trace-finite, and in the II$_\infty$ sector (ii) holds for every finite E; in neither case does the supremum saturate at a finite value. The final statement is the modular-vs-matter-Hamiltonian distinction: the crossed-product trace weights states, not the spectrum of $\widehat{H}_{\mathrm{m}}$, which retains its full unbounded range on the type III$_1$ seed algebra. □

Remark D.3 (Status of (E1): postulate, motivated but not derived). Pos. 11.30 is an explicit Wilsonian EFT scope postulate (H0), exactly as stated in Sec. 11.4. What Prop. D.2 establishes is strictly weaker and strictly true: Type II gravitational algebras give UV-finiteness of trace and generalised entropy with no cutoff input, which makes the *consistency* of a sub-Planckian scope natural, and shows that no structure of the framework resists the cutoff. Deriving $E_\star \lesssim E_{\mathrm{P}}$ as a theorem would require a spectral-truncation mechanism beyond the crossed-product trace; no such mechanism exists in the present literature, and an earlier version of this appendix that claimed one (via a "modular trace-class cutoff" and a Faulkner–Speranza double-scaling representation) was erroneous on three counts: the positive-energy static-patch algebra is II$_1$ rather than II$_\infty$ [18, 96], the trace-finiteness supremum never saturates (Prop. D.2(iii)), and the cited double-scaling representation does not appear in [92]. The external item (E1a) is retired together with the claimed derivation: what remains of (E1) is the postulate H0 itself, motivated by Prop. D.2.

Remark D.4 (The framework is bounded on both ends by one physics; scope of the claim). Prop. D.2(i) supplies more than ultraviolet finiteness. In the II$_1$ sector the tracial (maximum-entropy) state caps the generalised entropy of the observer's algebra at a finite value, $S \le S_{\max}$. For the de Sitter static patch that carries the framework's global cosmological witness, this ceiling is the Gibbons–Hawking horizon entropy [324],

$$S \;\le\; S_{\mathrm{dS}} \;=\; \frac{A_{\mathrm{dS}}}{4G} \;=\; \frac{3\pi}{\Lambda G}\,, \tag{103}$$

the finite trace of the type-II$_1$ algebra being the operator-algebraic incarnation of Bousso's N-bound on the number of accessible degrees of freedom under a positive cosmological constant [325]. Both ends descend from the same induced-gravity heat-kernel expansion of Prop. 11.20: its quadratic supertrace $\mathrm{str}[k_1]$ anchors the ultraviolet band edge through $E_\star^2 \to 1/G$ (Pos. 11.30, Rem. 11.31), while its quartic supertrace $\mathrm{str}[k_0]$ is the divergence the Sakharov cosmological counterterm must absorb at the infrared ceiling (103). The dimensionless span between the two is the pure number

$$S_{\mathrm{dS}} \;=\; \frac{3\pi}{\Lambda G} \;\sim\; \Big(\frac{r_{\mathrm{dS}}}{\ell_{\mathrm{Pl}}}\Big)^2 \;\approx\; 10^{122}, \qquad r_{\mathrm{dS}} = \sqrt{3/\Lambda}\,. \tag{104}$$

Two cautions are essential, and we state them plainly because a first draft of this remark overstated the point. *First, this is not a derivation of the cosmological-constant hierarchy.* The supertraces str$[k_0]$ and str$[k_1]$ are algebraically distinct and feed *logically independent* bare counterterms Λ_0 and G_0 [404, 405]; fixing G does not fix Λ, and the observed smallness of Λ remains an independent subtraction put in by hand in the standard EFT sense. Correspondingly the single state-independent additive constant of the Type II trace (Prop. D.2(i), [18]) lands entirely on the Newton/area term $A/4G$ and cancels in every entropy *difference*, so it cannot fix Λ dynamically. An adversarial verification pass confirmed that the two poles therefore do *not* cross-check one another: (104) is a *relabeling* of the hierarchy, not an explanation of it. *Second, the relabeling is not empty.* Solving (104) for Newton's constant,

$$G \;=\; \frac{3\pi}{\Lambda\, N}, \qquad N := S_{\mathrm{dS}} \approx 10^{122}, \tag{105}$$

recasts the dimensionful G as a dimensionless *address* N — how many Planck cells deep the accessible slab sits between the de Sitter ceiling and the band-edge floor — so that "why is G so small" is exchanged for "why is $N \approx 10^{122}$." The framework does not answer the latter: N enters as a free boundary condition, not a derived quantity. What the two-ended structure *does* establish is the weaker, true statement that both bounds are governed by one holographic/heat-kernel physics and that our position between them is a *single* large number rather than two independent hierarchies — the Dirac large-number question sharpened, not resolved.

We are deliberate about what this does and does not establish.

(a) **Different bound-types, one physics.** The lower pole is a *degrees-of-freedom floor* (a spectral band edge on the matter Hamiltonian, Pos. 11.30); the upper pole is an *entropy ceiling* (a finite trace on the observer algebra, (103)). These are the same physics — the induced-gravity str-sum — allocated to two different observables, not two instances of one inequality. Reading them as a single bracketing statement is a physical synthesis, not a derived equivalence.

(b) **Existence still posited.** What is derived is the Planckian *order* of $E_\star$ (Rem. 11.31) and the finite *value* of the entropy ceiling (Prop. D.2(i)); what remains posited is the *existence* of a single sharp band edge $E_\star$ (H0) and, for the ceiling, a positive Λ with a static-patch observer whose energy is bounded below (Hyp. D.1). No finite trace-finiteness *threshold* on the matter spectrum is thereby derived (Prop. D.2(iii)): the ceiling is a property of the state (the tracial one), not a truncation of the spectrum.

(c) **Beyond the bounds is unreachable, not merely unmodelled.** States exceeding S_{dS} are not states the framework declines to describe; they are actively excluded by the same tracial maximum-entropy structure that supplies the ceiling, consistently with the Boltzmann-brain suppression of Lem. U.2. Whether anything "lives" above the ceiling or below the floor is, by construction, outside the operational content of the theory — a limit of what is *sayable* here, not a claim that nothing is there.

Remark D.5 (The count N is the framework's one irreducible input: the cosmological-constant problem stated exactly). We state, as precisely as the framework permits, the single dimensionful quantity it does *not* fix, and why — because naming a theory's one load-bearing gap sharply is itself a result. Prop. 11.20 *content-fixes* the dimensionless response coefficient $g = (E_\star/E_{\mathrm{Pl}})^2 \approx 0.42$ from the Standard-Model heat-kernel supertrace; what remains unfixed is the *absolute count*

$$N \;=\; S_{\mathrm{dS}} \;=\; \frac{A_{\mathrm{dS}}}{4G} \;=\; \frac{3\pi}{\Lambda G} \;\sim\; \Big(\frac{r_{\mathrm{dS}}}{\ell_{\mathrm{Pl}}}\Big)^2 \;\approx\; 10^{122},$$

equivalently the de Sitter radius $r_{\mathrm{dS}} = \sqrt{3/\Lambda}$, the cosmological constant Λ, or the dimensionful part of G via (105). Its freedom is *structural*, not incidental: (i) the finite trace of the Type-II$_1$ factor is unique only up to its overall normalisation $\tau_R(\Bbbk)$, a free positive scalar; every equilibrium, KMS, tracial, or maximum-entropy condition is homogeneous under the rescaling

$\tau_R \mapsto \lambda\tau_R$ (i.e. $N \mapsto N + \log\lambda$: the entropy zero-point, and with it $N = S_{\mathrm{dS}}$, shifts *additively* — the multiplicative rescaling acts on the trace face, the additive shift on the entropy face of the same freedom), and so fixes only dimensionless *shape*, never the scale; (ii) the two-boundary condition provably selects a *state*, not the operator *spectrum* (Rem. I.8), whereas N is the absolute *scale* of that state's trace — a normalisation, not a matter-spectrum quantity — which no reweighting of outcomes on the fixed algebra can set; and (iii) no finite trace-finiteness threshold on the matter spectrum exists to be quantised (Prop. D.2(iii)). Consequently *no* modular, thermal-time, two-boundary, or entropy-flux construction internal to the framework can output N: it would require a state-independent, spectrum-level, *absolute* (non-difference) input from outside the modular structure, which the framework by design does not contain.

An invariant-theoretic sharpening of clause (i). The scalar freedom of clause (i) — uniqueness of the trace up to $\tau_R \mapsto \lambda\tau_R$ — holds for *every* II_1 factor and is not, by itself, what makes N physically irrecoverable; it is removed by the canonical choice $\tau_R(\mathbb{1}) = 1$, under which the physical count e^N ($N = S_{\mathrm{dS}} \approx 10^{122}$ being the corresponding entropy) reappears as the *amplification index* relating the observer algebra to a unit-normalised cell. Whether that index is an isomorphism *invariant* or a gauge label is governed by a sharper object: the fundamental group $\mathcal{F}(\mathcal{A}_{\mathrm{cosm}}) := \{t > 0 : \mathcal{A}^t_{\mathrm{cosm}} \cong \mathcal{A}_{\mathrm{cosm}}\}$. Because $\mathcal{A}_{\mathrm{cosm}}$ is the hyperfinite — equivalently (Connes [165]) injective, equivalently amenable — II_1 factor (App. L, via Connes–Feldman–Weiss [41]), one has $\mathcal{F}(\mathcal{A}_{\mathrm{cosm}}) = \mathbb{R}_+$ (Murray–von Neumann [164]): every amplification is isomorphic, so the index carries *no* algebra-internal information and N is pure gauge — clause (i) in its strongest form. The contrapositive is exact and is the only structural escape within von Neumann carriers: a carrier on which N is even a *well-defined* invariant must have $\mathcal{F} \neq \mathbb{R}_+$, hence (Connes) must be *non-amenable* — a rigid factor of the Popa class [166, 167], outside the hyperfinite class this framework and all three carriers of Rem. D.6 inhabit. Two honest limits attach. *(a) Well-defined is not computed*: even on a rigid carrier the index is a meaningful isomorphism *label*, not a *value* — outputting $\approx 10^{122}$ would require a matter-tied dimensionless invariant of that magnitude, and the Standard-Model heat-kernel sector supplies only $\mathcal{O}(1)$– $\mathcal{O}(10)$ data ($\mathrm{str}[k_1] = -14.83$, $g = 0.42$), with no finite trace-finiteness threshold to host it (Prop. D.2(iii)). *(b)* A rigid seed is a Hagedorn/nuclearity-type object at the Planckian scale that Pos. 11.30 places out of scope, so this is a statement about *where* an N-fixing completion could live, not a construction of one.

This is not a defect peculiar to the present construction; it is the *cosmological-constant problem*, in the framework's own language: *fix the de Sitter scale from the spectral (Dirac/heat-kernel) data rather than presuppose it as the background on which that data is computed.* Two observations sharpen, without closing, the residual. *First, the correct target-shape is a curvature, not a density.* The observed value sits naturally when $\Lambda = 3/r^2_{\mathrm{dS}}$ is read off a fixed geometric *length* as a curvature ($\sim (\ell_{\mathrm{Pl}}/r_{\mathrm{dS}})^2 \approx 10^{-122}$ in Planck units); the notorious E^4_{Pl} catastrophe is the mismatch of a vacuum-energy *density*, which Prop. 13.12's Sakharov counterterm already absorbs. The un-derived quantity is therefore the geometric *scale*, and a derivation must target the length and read Λ as its curvature, not compute an energy density. *Second*, a mechanism that makes the vacuum energy *non-gravitating* (thereby evading the additive-constant obstruction) generically does so by rendering it *metric-independent* — exactly the property that removes any restoring gradient which could fix the scale, so that within that class, evading the additive catastrophe and fixing the scale are mutually exclusive. The honest status is thus sharp: this is a conditional unification whose *sole* un-fixed dimensionful input is N; its freedom is structural — two of the three obstructions above, state-selection (Rem. I.8) and the absent trace-finiteness threshold (Prop. D.2(iii)), are theorems; and its resolution is the cosmological-constant problem itself — genuinely new (spectral, scale-fixing) physics, not a gap in the present derivation.

Remark D.6 (External carriers of the absolute count: the interfaces and their uniform limit). The preceding remark proves the framework cannot generate the absolute count N; we record here that the *input type* it demands — a state-independent, spectrum-level, absolute normalisation

— is now *realised* by three existing external structures, and that each supplies the type while none supplies the value, for reasons that are uniform across all three. *(i) Microscopic Type-II_1 models.* The double-scaled SYK chord algebra is a Type-II_1 factor whose tracial no-chord state is cyclic and separating [150, 151] — the algebraic skeleton of Hyp. D.1 realised in a UV-complete quantum mechanics, with conjectural de Sitter dictionaries [152]. But the double-scaled construction *normalises the trace at the outset* ($\mathrm{Tr}\,\mathbb{1} = 1$; [153, footnote 3]), so the absolute count is stripped before physics begins — an external instantiation, verbatim, of clause (i) above; and in the sharpest independent microscopic dS_3 model the absolute normalisation *cancels* from the entropy-matching identity [154, §4.4]. *(ii) Gravitational path integrals.* The path integral computes precisely the trace normalisation the framework leaves free — but as a *coupling transducer*: the JT normalisation carries the free additive constant S_0 [155], the de Sitter one-loop normalisation "leaves undetermined a state-independent $\mathcal{O}(1/G)$ contribution" [156], i.e. these output N *from* G, the inverse of what (105) needs; and the baby-universe completion pins the ensemble to its input coupling [157]. We note also a *two-boundary neutrality* observation (at minisuperspace–WKB level, and subject to the open derivation of Conj. 10.13): the two-boundary amplitudes of this framework, taken between realised (classically allowed) slices, carry no wavefunction-of-the-universe weight $e^{\pm N}$ — that weight arises solely from one-boundary capping — so the Hartle–Hawking-versus-tunneling contour dispute is *out-of-formalism* here, and the residual is exactly the external prior on N, the same trace seam posited in Prop. 13.12. *(iii) Spectral integrality.* Volume quantisation in noncommutative geometry [158] breaks the trace-scaling gauge from $\mathbb{R}_+$ to $\mathbb{Z}$ — the unique known external axiom with the demanded *form* — but quantises without selecting: every sufficiently large integer is admissible by its own classification, so "why this integer" is clause-for-clause the problem above. Two further external mechanisms sharpen the boundary negatively: vacuum-energy sequestering fixes its residual only on finite (crunching) spacetime volumes — the Type-II_∞ branch of Hyp. D.1, where no finite N exists — while its manifestly local, future-eternal version leaves the residual "a priori completely arbitrary" [159, 160]; the causal-set everpresent-Λ mechanism [168, 169] is the same Type-II_∞-branch instance in fluctuation language — it derives the fluctuation *amplitude* $\delta\Lambda \sim V^{-1/2}$ about a mean its own authors leave undetermined (the mean of Λ being the unfixed integration constant of unimodular gravity), the amplitude coupling-pinned to a posited sprinkling density; it re-parametrises the freedom $\tau_R(\mathbb{1})$, it does not close it; and in asymptotically safe gravity Λ is a *relevant* direction, free by definition, so the content-fixed $g \approx 0.42$ of Prop. 11.20 intersects every renormalisation-group trajectory once and selects none. The uniform mechanism across all of these is the second observation of Rem. D.5, externalised: whatever renders the input absolute and radiatively stable (normalised trace, coupling transduction, topological protection, global constraint, stochastic substrate, relevance) is exactly what renders it value-blind. Environmental selection over an ensemble of Type-II_1 sectors remains available as a *consistency account* — within a single model no terminal effect can select or shift N (a corollary of clauses (i)–(ii) above), so selection requires an ensemble in Weinberg's original sense [161, 162, 163], with the prior over sectors provably not internal — an account, not a derivation. The conclusion of Rem. D.5 therefore stands verbatim; what this remark changes is the *boundary* of the open problem: the missing ingredient is one additional independent equation on the one-dimensional freedom $\tau_R(\mathbb{1})$, and every currently known external structure supplies only re-parametrisations of the first.

E Spacetime-smeared QEI and modular-energy estimates: conditional discharge of (E2)

This appendix records what the published quantum-energy-inequality and modular-theory literature, as of this writing, actually delivers for item (E2) of Sec. 12.5, and proves a Sobolev–Lipschitz bound on the smeared expected stress-energy that is *conditional* on one named input of the microlocal / Hadamard-parametrix literature (Hyp. E.1, labelled (E2a) below). Under that input, the two QFT-regime constants that Conj. 11.4 previously had to assume — the stress-energy Lipschitz constant $L_T^{(0)}$ of (S5) and the operator-norm bound M of (S4) — become theorems on the Planck-cutoff sector of Pos. 11.30, and the contraction constant C of Prop. 11.49 is finite and strictly positive with all inputs published or named. An earlier version of this appendix asserted an unconditional discharge via a worldline QEI scaling ansatz and a bound involving $\left\|\log \Delta_{\omega_0} \restriction_{\mathcal{A}(\mathcal{O})}\right\|$; both steps were defective (the modular Hamiltonian of a local type III_1 algebra is unbounded, so that norm is generically infinite, and by Fewster–Roman [210] no worldline QEI controls general smeared tensor components) and have been removed. The present version claims strictly less and proves what it claims.

Setup: quasi-free seed on a Sobolev ball of Hadamard-compatible metrics

Fix $(\mathcal{M}, g_0)$ globally hyperbolic with compact Cauchy slice Σ and a *quasi-free (Gaussian)* Hadamard state ω_0 for the free minimally coupled scalar field of mass $m > 0$ [10, 16]. The restriction to quasi-free seeds is not a loss for the regimes used in this paper: KMS (Hartle–Hawking-type) states of the linear field and Sorkin–Johnston-type states are quasi-free by construction. Let $n = \dim \mathcal{M}$ ($n = 4$ in the physical regime) and fix $s > n/2 + 6$ (so $H^s \hookrightarrow C^6$, $H^{s-4} \hookrightarrow C^2$, and $H^{s-6} \hookrightarrow C^0$; the index is calibrated to *two distinct* consumers, which earlier versions of this appendix conflated: the $\Box R$- and $\nabla\nabla R$-type terms of the renormalisation polynomial $\Theta[g]$ are *four*-jets of g (coincidence *values*, C^0 at $s > n/2 + 4$), while the diagonal C^2 regularity of the Hadamard *finite part* W_g is governed by the first-order parametrix remainder $\Box_g v_1$, a *six*-jet of g (a remainder *function*, C^0 only at $s > n/2 + 6$); the standing index is set by the stronger requirement). For a neighbourhood $\mathcal{U} \subset H^s(\mathcal{M}, \mathrm{Sym}^2 T^*\mathcal{M})$ of g_0 small enough that every $g \in \mathcal{U}$ is globally hyperbolic (Chruściel–Grant stability of global hyperbolicity under C^0-small perturbations [222], which H^s-smallness implies for $s > n/2$), let ω_g be the Hadamard state obtained from ω_0 by the Brunetti–Fredenhagen–Verch relative Cauchy evolution [220]; since the field equation is linear, relative Cauchy evolution acts by pullback on two-point functions and ω_g is again quasi-free. Write $\omega_g^{(2)} = H_g + W_g$ for the Hadamard split into singular parametrix and smooth finite part, and

$$\langle \widehat{T}^{\mathrm{m}}_{\mu\nu} \rangle_{\omega_g}(x) \;=\; \big[\mathcal{D}^g_{\mu\nu} W_g\big](x,x) \;+\; \Theta_{\mu\nu}[g](x)$$

for the point-split Hadamard-renormalised stress-energy, with $\mathcal{D}^g_{\mu\nu}$ the standard second-order point-splitting bidifferential operator and $\Theta[g]$ the local curvature polynomial fixed by the renormalisation prescription [10, 16].

Hypothesis E.1 (Uniform Sobolev Hadamard parametrix)**.** There exist a neighbourhood $\mathcal{U} \subset H^s(\mathcal{M}, \mathrm{Sym}^2 T^*\mathcal{M})$ of g_0, $s > n/2+6$, and for every compact $K \subset \mathcal{M}$ constants $C_H(K), C_W(K), W_*(K) < \infty$ such that for all $g, g' \in \mathcal{U}$:

- (i) (singular part, dual Lipschitz) $\left|(H_g - H_{g'})(u \otimes v)\right| \le C_H(K)\, \|g - g'\|_{H^s}\, \|u\|_{H^{s-2}} \|v\|_{H^{s-2}}$ for u, v supported in K;
- (ii) (finite part, *mixed* second-order *coincidence jets*, Lipschitz) for the multi-indices the point-splitting construction consumes — $|a|+|b| \le 1$ together with the mixed pair $|a| = |b| = 1$ — $\sup_{x\in K}\left|[\partial_x^a \partial_y^b (W_g - W_{g'})](x,x)\right| \le C_W(K)\, \|g - g'\|_{H^s}$, and $\sup_{g\in\mathcal{U}} \sup_{x\in K}\left|[\partial_x^a \partial_y^b W_g](x,x)\right| \le W_*(K)$. (The mixed-jet form is exactly what the minimally-coupled point-split $\langle \widehat{T}_{\mu\nu} \rangle$ consumes — each of its terms distributes one derivative per slot, so the pure same-slot jets $(2,0)$, $(0,2)$ are never formed; this matters, because at finite metric regularity the Hadamard split is necessarily truncated, the truncation residual is C^1 but not C^2 *across* the light cone, the full $C^2(K \times K)$ norm of an earlier formulation is unavailable, and the pure same-slot *diagonal* jets of the residual diverge as well — only the *mixed* diagonal jets remain finite.);

(iii) (QEI-constant uniformity) for fixed smearing data (t, f, F) as in (106) below, the constants $c_S(g, t, F), C_S(g, t, f, F)$ of Sanders' stress-tensor bound are locally uniform on $\mathcal{U}$: bounded, and continuous in g in the H^s topology.

The primary form adopted below is the *analytic-collar* restatement of this hypothesis (Option A of the fence paragraph following it); the open-ball form above is retained verbatim as the unconditional fallback (Option B), and every existing consumer continues to cite it. The analytic restatement narrows only the metric-ball class, not the physical scope: the standard backgrounds (Minkowski, Schwarzschild, Kerr, de Sitter, FRW, ultrastatic) are real-analytic.

Hypothesis E.2 (Uniform Sobolev Hadamard parametrix over an analytic ball)**.** Let g_0 be a *real-analytic* globally hyperbolic metric with compact Cauchy slice. Fix a complex-collar radius $\rho_a > 0$ and a sup-bound $\varepsilon_a > 0$, and set

$$U_\omega := \big\{\, g \in H^s(\mathcal{M}, \mathrm{Sym}^2 T^*\mathcal{M}) : g \text{ real-analytic, } g \text{ extends holomorphically to the collar } \{|\mathrm{Im}\, z| < \rho_a\} \text{ of a fixe}$$

$s > n/2 + 6$, with (ρ_a, ε_a) chosen small enough that the C^0-image of U_ω has radius $< \varepsilon_{\mathrm{GH}}$, so every $g \in U_\omega$ is globally hyperbolic (Chruściel–Grant stability of global hyperbolicity under C^0-small perturbations [222], *exactly* as in the smooth case and by the *same* C^0 bound — global hyperbolicity is a C^0 property, unchanged by the analytic restriction; U_ω is H^s-dense but has empty H^s-interior, the disclosed cost). For every compact $K \subset \mathcal{M}$ there exist constants $C_H(K), C_W(K), W_*(K) < \infty$ such that for all $g, g' \in U_\omega$:

(i) (singular part, dual Lipschitz) $\big|(H_g - H_{g'})(u \otimes v)\big| \le C_H(K)\, \|g - g'\|_{H^s}\, \|u\|_{H^{s-2}} \|v\|_{H^{s-2}}$ for u, v supported in K;

(ii) (finite part, *mixed* second-order *coincidence jets*, Lipschitz) for the multi-indices the point-splitting construction consumes — $|a|+|b| \le 1$ together with the mixed pair $|a| = |b| = 1$ — $\sup_{x\in K} \big|[\partial_x^a \partial_y^b (W_g - W_{g'})](x, x)\big| \le C_W(K)\, \|g - g'\|_{H^s}$, and $\sup_{g\in U_\omega} \sup_{x\in K} \big|[\partial_x^a \partial_y^b W_g](x, x)\big| \le W_*(K)$. (The mixed-jet form is exactly what the minimally-coupled point-split $\langle \widehat{T}_{\mu\nu}\rangle$ consumes — each of its terms distributes one derivative per slot, so the pure same-slot jets $(2, 0)$, $(0, 2)$ are never formed; this matters, because at finite metric regularity the Hadamard split is necessarily truncated, the truncation residual is C^1 but not C^2 *across* the light cone, the full $C^2(K \times K)$ norm of an earlier formulation is unavailable, and the pure same-slot *diagonal* jets of the residual diverge as well — only the *mixed* diagonal jets remain finite.);

(iii) (QEI-constant uniformity) for fixed smearing data (t, f, F) as in (106) below, the constants $c_S(g, t, F), C_S(g, t, f, F)$ of Sanders' stress-tensor bound are locally uniform on U_ω: bounded, and continuous in g on U_ω *in its subspace H^s topology.* (*Fence* F-iii: U_ω has no H^s-interior, so this is the subspace-topology continuity, *not* the open-set continuity of an H^s neighbourhood; only the boundedness $\sup_{U_\omega} c_S =: c_S^*$ and this subspace-continuity are consumed downstream — Cor. E.8 never differentiates c_S in g nor takes an H^s-open modulus of it.)

The constants depend only on $(g_0, \rho_a, \varepsilon_a, s, K)$: no Sobolev index beyond s, no C^k norm, enters. *Fence* F-diff: because g' ranges over U_ω it is itself real-analytic in every difference estimate (i),(ii); this is used only through $\|g - g'\|_{H^s}$, the uniform sup W_*, and Banach-algebra Lipschitz of the D-coefficients (polynomial in $(g', g'^{-1}, dg', d^2 g')$, needing only $s - 2 > n/2$), none of which requires g' analytic, so the restriction is costless for the paper's uses; a merely-smooth (non-analytic) comparison g' is *not* served, which must be flagged wherever a consumer feeds one.

Remark E.3 (Guard F-shock: the boundary the analytic route must never cross). The back-reacted Barrabès–Israel null-shell metrics of the physical back-reaction ball $\mathcal{B}_{g_0}$ (Conj. 10.2) carry a curvature δ-shock across a null hypersurface; they are at best C^0 (metric) / distributional (curvature), hence *categorically not real-analytic* and *not* admitted to U_ω. They are *not* consumers of Hyp. E.2 today (they enter the separate record-adapted fixed-point layer, which does not cite this hypothesis). Thm. E.7 and every (E2a) consumer *must never* be wired to range g' over $\mathcal{B}_{g_0}$'s shock metrics: that wiring would be a *false discharge* (the analytic hypothesis cannot serve a non-analytic g'). This is the single fence the route must hold.

Status (companion appendix). The analytic slice register that the primary (Option A) booking of this hypothesis requires is now proved in print: Appendix E2a (Thm. E2a.83, Part (A)) establishes, *unconditionally over* U_ω, the ball-uniform fold count, the linear device bound on the fold stratum with the near-flat sliver closed at the binding cut, the resulting $(D3)/(D1)$ and the $\partial_y : (c, p) \mapsto (c + \frac{1}{2}, p - 1)$ booking, and hence the promoted slice register $s > 11$ (T2) / $s > \frac{23}{2}$ (T3). The three clauses (i)–(iii) of this hypothesis are *not* thereby proved: clause (ii) is reduced to $s > 11$ but remains plausible (the $N = 2$ Hadamard construction and full constant-tracking are not in print), and clause (iii) remains a named input, while the difference form of clause (i) is now a *theorem* (ported by full re-derivation as App. E2a, item (M3)). Accordingly the hypothesis is *retained*, and the fused free-field first law (Prop. E.4) is a theorem *modulo the finite named list* {clause (ii), clause (iii)} (Thm. E2a.83, Part (B); Appendix E2a.24). What the companion chain flips is the metric-ball *class* (open H^s ball $\to$ analytic collar U_ω) and the disclosed *status* of the slice register (now a theorem over U_ω), not the hypothesis itself.

Two admissible bookings of the ∂_y slice (the elected primary is Option A; Option B is the unconditional fallback). The half-derivative at the transport-averaged ∂_y leg admits two honest bookings, differing only in the regularity hypothesis on the metric class and in the slice register they buy. They are *not* mutually exclusive: Option B is a strict weakening of Option A's target, so B may ship now and A be pursued to upgrade the slice later.

OPTION A — the analytic slice $s > 11$ over U_ω Book $\partial_y : (c, p) \mapsto (c+\frac{1}{2}, p-1)$ over the *analytic* class U_ω (Hyp. E.2). The slice register promotes to

$$\boxed{s > 11} \quad \text{(slice, T2)}, \qquad s > \tfrac{23}{2} \quad \text{(T3)},$$

the unique zero-slack inequality of the chain (the commutator branch $L^\infty_t H^{\min(s-19/2, 7/2^-)}(\Sigma_t)$ is $C^0(\Sigma)$ iff $s - \frac{19}{2} > \frac{3}{2}$ iff $s > 11 = n/2+9$), in place of the naive slice $s > 11.5$ / locked-4d $s > 12$; the physical anchors R0/HB1 revive at the promoted row *with a real-analyticity hypothesis* on g_0. *Status:* the supporting chain (the uniform analytic fold count, the linear sublevel device on the fold stratum, and the near-flat–sliver weigh-in at the binding cut $\mu_* = (\Lambda\rho)^{-1} < \kappa_V$) is now written up and proved in print as Thm. E2a.83, Part (A) (Appendix E2a), a theorem *unconditional over* U_ω; it is established end-to-end over U_ω at $s > 11$. The hypothesis form of Hyp. E.2 is nonetheless *retained*, because clauses (ii)–(iii) are named inputs, not consequences of that chain (clause (ii) plausible, clause (iii) named; the difference form of clause (i) is now a ported theorem, App. E2a); the fused first law is a theorem modulo that finite list (Thm. E2a.83, Part (B)). *Price:* an analyticity hypothesis on g (dense in H^s but H^s-interior-empty), tolerated by every consumer (the consumer audit is a THEOREM: all sites analytic-safe) with fences F-iii / F-diff / F-shock. The promoted bookings (Thm. E.7's slice feeders via T1$'$, the T2/T3 ladders) are supplied by Thm. E2a.83, Part (A) as a theorem over U_ω.

OPTION B — the naive close-out over the full H^s ball Book $\partial_y : (c, p) \mapsto (c+1, p-1)$ over the full H^s ball $\mathcal{U}$ (Hyp. E.1 unchanged, *no* analyticity). The slice register is

$$s > 11.5 \quad \text{(slice)}, \qquad s > 12 \quad \text{(locked-4d)},$$

anchors R0/HB1 9/8.5, HB3 8.5/8, HB2/HB4 (T1)-free $s > 8$. *Status:* consumer-safe, the booking of record; T1$'$ PLAUSIBLE, R1 closed-as-undischarged, R2–R5 retired on the

T1′-conditional path. *Price:* none — the unconditional fallback with no regularity cost; the only open items are the two author-gated envelope-locus edits, which may be left gated indefinitely or applied.

The trade. Option A buys a half-derivative ($s > 11$ vs 11.5 slice; 23/2 vs 12 in locked-4d) at the cost of the analyticity hypothesis; Option B is unconditional and, being a strict weakening of A's target, holds regardless. The elected configuration is **A primary, B fallback**. (E-env) and the R3/R4 conditionality are untouched under either choice.

Proposition E.4 (Free-field first-law extraction, uniform over the analytic ball)**.** *Let g_0 be real-analytic (Hyp. E.2), and let the finite-part / QEI-constant family $\{C_W, W_*, D_*, C_D, C_\Theta\}$ be uniform over the analytic ball U_ω in its subspace H^s topology (this is precisely Hyp. E.2, clauses (ii)–(iii), with the F-iii fence). Fix the free massive ($m > 0$) minimally-coupled scalar and a finite-relative-entropy quasi-free state class on a compact Cauchy region K. Then the per-point first-law extraction — the pointwise null–null equation of state $\delta G_{\mu\nu}k^\mu k^\nu = (8\pi G/c^4)\,\delta\langle\widehat{T}_{\mu\nu}\rangle_\omega k^\mu k^\nu$ along every null k through every point of K — holds* uniformly in the point and in the null direction*, with the uniformity constant depending only on $(g_0, \rho_a, \varepsilon_a, s, K, m)$.*

Remark E.5 (what Prop. E.4 does and does not buy). (i) The direction/point uniformity is automatic from the compact sphere plus parametrix smoothness once the ball-uniform finite-part family is granted; that family *is* (E2a-ω), so the free-field proposition and (E2a-ω) have *fused* — one hypothesis gates both. (ii) It does *not* upgrade the input (N1) of Cor. J.5: (N1) is the *interacting*-SM tensor completion, and an unconditional free-field statement leaves the interacting seam (B1 wedge modular / B2 curved rate) untouched. (iii) The state class is pinned to finite-relative-entropy quasi-free perturbations; arbitrary locally-squeezed states have divergent relative entropy and are excluded (the inherited state-class boundary). (iv) The analytic-collar anchor g_0 is native to every physical background (de Sitter, Schwarzschild, Kerr, Minkowski, FRW, ultrastatic are real-analytic), so the proposition applies to the physical g_0 with no extra hypothesis; it does *not* extend to the non-analytic shock metrics of $\mathcal{B}_{g_0}$ (Guard F-shock).

Clause (ii) strengthens the C^0 clause of earlier versions of this hypothesis to second-order coincidence jets; this is what the point-splitting construction of $\langle\widehat{T}_{\mu\nu}\rangle$ actually consumes. The honest status of the ambient literature is as follows. The finite-regularity microlocal programme establishes the *singularity structure*: [221] proves Sobolev wavefront-set inclusions for the (anti-symmetric) causal propagator at a *fixed* metric of finite Hölder regularity; it does not construct the Hadamard parametrix subtraction or bound the symmetric finite part W_g in any C^k norm, so the C^2-jet comparison of clause (ii) is not in print even at fixed g, and the uniformity in g over the ball $\mathcal{U}$ is a further strengthening — both are packaged here as (E2a). Clause (iii) is of a different character: the constants of [207] are built by reference-Hadamard subtraction (no parametrix enters that construction) through explicit chart and Fourier-integral expressions, so their H^s-continuity in g at fixed metric is a matter of constant-tracking rather than new analysis; no such derivation is in print, so we retain it as part of the named input rather than assert it as a consequence. Hyp. E.1 is a finite-regularity strengthening of the locally-covariant smoothness of $\omega_g^{(2)}$ in g established in [220, 16]; it is the named external input (E2a), on the same footing as Hyp. D.1.

Published QEI inputs and component scoping

Timelike-timelike, spacetime-smeared. Sanders' stress-tensor bound [207] states: for the free scalar of mass $m > 0$ on *any* globally hyperbolic $(\mathcal{M}, g)$, any smooth timelike vector field t^μ, and $f, F \in C_c^\infty(\mathcal{M})$ with $F \equiv 1$ on $\operatorname{supp} f$, there are state-independent constants $c_S, C_S > 0$ (depending on g, t, f, F only) with

$$0 \;\le\; \omega\big(\phi(f)^*\phi(f)\big) \;\le\; C_S\Big(\omega\big(\widehat{T}^{\mathrm{ren}}_{\mu\nu}[t^\mu t^\nu F^2]\big) + c_S\Big) \qquad \text{for every Hadamard state } \omega, \tag{106}$$

whence in particular the *spacetime-smeared absolute QEI* $\langle \widehat{T}^{\rm ren}_{\mu\nu}[t^\mu t^\nu F^2]\rangle_\omega \geq -c_S(g,t,F)$. By Hyp. E.1 (iii) the constants c_S, C_S are locally uniform in $g \in \mathcal{U}$. This replaces both the worldline QEI [199] and the absolute QEI of Fewster–Smith [218] as the quantitative input of this appendix; the cosmological budget for the energy-condition violation such bounds permit is quantified in [212].

Null components. For the free massless scalar in $n = 4$ there is provably *no* state-independent lower bound on null-geodesic-averaged $\langle \widehat{T}_{\mu\nu}k^\mu k^\nu\rangle$ [210]; the same reference proves that null-contracted stress-energy averaged along *timelike* worldlines does satisfy a QEI on any globally hyperbolic spacetime. State-independent *semi-local* lower bounds on null-integrated $\langle T_{vv}\rangle$ exist in all 2d QFTs satisfying the QNEC and, on nearly-null strips with uniform transverse smearing, in interacting CFTs in $d > 2$ (Fliss–Rolph [209]); free fields in $d > 2$ are explicitly outside their scope. Accordingly, the present appendix claims QEI control only for the timelike-contracted smeared components — which are the components sourcing the Clausius relation (47)–(48) in the interior of the local diamond — while the null-horizon leg of Pos. 11.18 is routed, as in Rem. 11.19 (v), through monotonicity of Araki relative entropy (Ceyhan–Faulkner [234], Hollands–Longo [236]), not through a QEI.

Modular-energy input (replacement for the deleted modular-norm term)

For quasi-free states, quasi-equivalence of the ω_g family with the seed folium on the compact slice Σ is classical (Verch [211]). What was missing until recently is *quantitative* control of the modular data. For the free scalar of mass $m > 0$ on an *ultrastatic* spacetime, Chialastri–Sanders [208] prove, at the one-particle level and for Gaussian perturbations of the ground state with positive trace-class relative energy density and total relative energy $E_\delta < \infty$,

$$\mathrm{tr}\Big|\, i\coth\tfrac{K_\delta}{2} - i\coth\tfrac{K_0}{2}\Big| \;\leq\; \frac{8E_\delta}{m}, \qquad \Big\|\cosh^{-1}\tfrac{K_\delta}{2}\Big\|^2_{\rm HS} \;\leq\; \frac{8E_\delta}{m}, \tag{107}$$

with the analogous $\sinh^{-1}$ estimate; here K_δ, K_0 are the one-particle modular Hamiltonians of the perturbed and unperturbed states. Differences of *bounded functions* of modular Hamiltonians are thus controlled by *energy differences*, with constants explicit in the mass gap. This is the finite, published replacement for the term $\big\|\log\Delta_{\omega_0}\restriction_{\mathcal{A}(\mathcal{O})}\big\|$ used in earlier versions of this appendix, which is generically infinite for type III$_1$ local algebras and is deleted.

Lemma E.6 (Energy-Lipschitz continuity of the RCE family)**.** *Assume Hyp. E.1 (ii). Then the relative energy on the compact slice Σ satisfies*

$$E_\delta(\omega_g,\omega_{g'}) \;\leq\; C_E(g_0)\,\|g-g'\|_{H^s}, \qquad g,g' \in \mathcal{U},$$

with $C_E(g_0)$ built from $C_W(K_\Sigma), W_*(K_\Sigma)$ and $\mathrm{vol}_{g_0}(\Sigma)$, K_Σ a compact neighbourhood of Σ. Consequently, in the ultrastatic benchmark regime — and provided the pairs (ω_g,ω_0) satisfy the standing hypotheses of [208], i.e. ω_0 is the ultrastatic ground state and the relative energy densities are positive and trace-class (neither is automatic for the RCE family — positivity indeed fails generically *for record-conditioned perturbations, which are Bogoliubov-squeezed, so the estimate is retained under its own hypotheses as a cited refinement; no displayed consumer uses it as an operator statement — see the residual (E2c) below) — the modular differences* (107) *are Lipschitz in g: $\mathrm{tr}\,|i\coth(K_g/2) - i\coth(K_{g'}/2)| \leq 8\,C_E(g_0)\big(\|g-g_0\|_{H^s} + \|g'-g_0\|_{H^s}\big)/m$ by the triangle inequality through the seed (* (107) *requires the ultrastatic ground state in one slot, so the modulus is* seed-anchored*; a genuinely pairwise $\|g-g'\|$ form does not follow from this route).*

Proof. The relative energy density of two quasi-free states is the 00-component of the point-split difference of their two-point functions at coincidence, i.e. second-order coincidence jets of $W_g - W_{g'}$ plus curvature-polynomial terms; clause (ii) bounds the former by $C_W(K_\Sigma)\|g-g'\|_{H^s}$ in

$C^0(\Sigma)$. The curvature-polynomial terms contain, besides the quadratic-in-curvature pieces (two-jets of g, handled by the Banach algebra $H^{s-2} \hookrightarrow C^0$), the $\Box R$- and $\nabla\nabla R$-type renormalisation terms, which are *four*-jets of g; with $s > n/2+4$ these are Lipschitz $H^s \to H^{s-4} \hookrightarrow C^0(K_\Sigma)$, so their restriction to Σ is controlled by continuity alone (no trace estimate is needed once $s-4 > n/2$; for the borderline range $n/2+2 < s \leq n/2+4$ one would instead need the trace theorem onto Σ, which is why the standing index was raised). Integration over the compact Σ gives the claim. The second statement combines (107) for (ω_g, ω_0) and $(\omega_{g'}, \omega_0)$. $\square$

Operator-Lipschitz theorem

Theorem E.7 (Sobolev-Lipschitz bound on the smeared expected stress-energy, conditional on (E2a))**.** *Assume Hyp. E.1 and fix $s > n/2+6$. For every compact $K \subset \mathcal{M}$ there is a finite constant $L(g_0, K)$, depending only on g_0, K, the constants of Hyp. E.1, and the renormalisation constants of Θ, such that for every test tensor $f \in H^s_c(\mathrm{Sym}^2 T^*\mathcal{M})$ with $\operatorname{supp} f \subset K$ and every $g, g' \in \mathcal{U}$,*

$$\big|\langle \widehat{T}^{\mathrm{m}}_{\mu\nu}[f]\rangle_{\omega_g} - \langle \widehat{T}^{\mathrm{m}}_{\mu\nu}[f]\rangle_{\omega_{g'}}\big| \;\leq\; L(g_0,K)\,\|f\|_{H^s}\,\|g-g'\|_{H^s}, \tag{108}$$

with

$$L(g_0,K) \;=\; c_{n,s}(K)\Big[\,D_*(g_0,K)\,C_W(K) \;+\; W_*(K)\,C_{\mathcal{D}}(g_0,K) \;+\; C_\Theta(g_0,K)\,\Big], \tag{109}$$

where $D_*(g_0,K) := \sup_{g\in\mathcal{U}} \|\text{coefficients of } \mathcal{D}^g\|_{C^0(K)}$ is the uniform coefficient bound of the point-splitting operator on the ball, $C_{\mathcal{D}}$ is the Lipschitz constant of the coefficients of $(\mathcal{D}^g_{\mu\nu}, \mathrm{dvol}_g)$ in g and C_Θ that of the curvature polynomial $\Theta[g]$. In particular $L(g_0,K) < \infty$: no QEI constant and no modular norm enters the Lipschitz modulus.

Proof. Write $\langle \widehat{T}^{\mathrm{m}}_{\mu\nu}[f]\rangle_{\omega_g} = \int_K f^{\mu\nu}\big[\mathcal{D}^g_{\mu\nu} W_g\big](x,x)\,\mathrm{dvol}_g + \int_K f^{\mu\nu}\Theta_{\mu\nu}[g]\,\mathrm{dvol}_g$ and estimate the difference in four parts. (1) $[\mathcal{D}^g_{\mu\nu}(W_g - W_{g'})](x,x)$: the coefficients of $\mathcal{D}^g$ are bounded on $\mathcal{U}$ by $D_*(g_0,K)$ (finite by $H^{s-1} \hookrightarrow C^1$ and Sobolev multiplication, after shrinking $\mathcal{U}$ if necessary), and Hyp. E.1 (ii) bounds the jet by $C_W(K)\|g-g'\|_{H^s}$ uniformly on K; this part contributes $D_* C_W$. (2) $[(\mathcal{D}^g_{\mu\nu} - \mathcal{D}^{g'}_{\mu\nu})W_{g'}](x,x)$: the coefficients (inverse metric, Christoffel, curvature-coupling terms) are polynomial in $(g, g^{-1}, \partial g, \partial^2 g)$, hence Lipschitz $H^s \to C^0(K)$ with constant $C'_{\mathcal{D}}(g_0,K)$ since $s-2 > n/2$ makes H^{s-2} a Banach algebra embedded in C^0; this is paid against the uniform jet bound $W_*(K)$ of (ii). (3) $\Theta[g] - \Theta[g']$: the renormalisation polynomial contains quadratic-in-curvature terms (two-jets of g) but also the $\Box R$- and $\nabla\nabla R$-type terms of the $[v_1]$ anomaly and the $I_{\mu\nu}, J_{\mu\nu}$ ambiguities — *four*-jets of g. The quadratic pieces are Lipschitz $H^s \to H^{s-2}$ by the Banach-algebra property; the four-jet pieces are linear in $\partial^4 g$ with H^{s-2}-algebra coefficients, hence Lipschitz $H^s \to H^{s-4}$ by the multiplication estimate $H^{s-2} \times H^{s-4} \to H^{s-4}$ ($s-2 > n/2$). Since $s > n/2+4$ (a fortiori under the standing index $s > n/2+6$), $H^{s-4} \hookrightarrow C^0(K)$, giving the constant $C_\Theta(g_0,K)$; the Θ-terms consume $s > n/2+4$, while the standing index $s > n/2+6$ is pinned by the six-jet parametrix-remainder requirement recorded in the Setup. (4) $\mathrm{dvol}_g - \mathrm{dvol}_{g'}$: Lipschitz with a constant absorbable into $C_{\mathcal{D}}$. Smearing against f costs $\|f\|_{L^1(K)} \leq c_{n,s}(K)\|f\|_{H^s}$ by Hölder and compactness of K. Summing the four parts gives (108) with (109), where $C_{\mathcal{D}} := C'_{\mathcal{D}} +$ the volume-form constant. Note that clauses (i) and (iii) of Hyp. E.1 are not consumed here: (iii) enters only through the QEI (106) (in Cor. E.8 (iv)), and (i) only as the microlocal control that makes (iii) plausible; the modular leg uses (ii) via Lem. E.6. $\square$

Corollary E.8 (QFT-regime entropy-smallness bound, conditional on (E2a))**.** *Assume Pos. 11.30 (H0), the scoping hypotheses (S1)–(S3), (S6) of Conj. 11.4 on the cutoff sector* $\operatorname{Ran} P_{\leq E_\star}$*, and Hyp. E.1. Fix the smearing data (f,t,F) of* (106) *with* $\operatorname{supp} f \subset K$*. Then on the Sobolev ball $\mathcal{U}$:*

(i) ((S5) stress leg discharged) *the zero-entropy Lipschitz constant of the source map satisfies $L^{(0)}_T \leq L^{\mathrm{QFT}}_T := L(g_0,K)\,\|f\|_{H^s} < \infty$ by Thm. E.7;*

(ii) ((S4) discharged) *the uniform stress-energy norm bound holds with* $M \le M_{\mathcal{U}} := c_T(g_0, f)\,(1+E_\star)^2 < \infty$*, where* c_T *is the H-bound constant of the free scalar [199, 213], locally uniform on* $\mathcal{U}$ *by Thm. E.7;*

(iii) *provided the seed-background contraction (S8) holds in the form* $\mu\, L_T^{\mathrm{QFT}} < 1$ *with* $\mu = \kappa T\, \|\mathcal{G}_{g_0}\|_{\mathrm{op}}$*, the contraction constant of Conj. 11.4 satisfies the finite, strictly positive bound*

$$C \ \ge\ \frac{2\,\big(1-\mu L_T^{\mathrm{QFT}}\big)^2\, p_*^2\, \eta}{\mu^2\, M_{\mathcal{U}}^2} \ >\ 0 \qquad \textit{nats}; \tag{110}$$

(iv) (conditioned-source QEI floor) *if the prior* ρ_{i} *is a Hadamard state of the seed folium (as in the standing setup, where it is induced by the quasi-free seed* ω_0*), then for every* k *with* $p_0(k) \ge p_*$*,*

$$\big\langle \widehat{T}^{\mathrm{ren}}_{\mu\nu}[t^\mu t^\nu F^2]\big\rangle_{\rho_2[g_0](\tau_i|k)} \ \ge\ -c_S(g_0, t, F)\ -\ M_{\mathcal{U}}\,\sqrt{2\,\mathcal{S}(\rho_{\mathrm{i}}, \{E_k\})}, \tag{111}$$

with transport to $g \in \mathcal{U}$ *and* $\tau \in [\tau_i, \tau_f]$ *via the (S5)/(S7) constants; this quantifies the anomalous negativity of the retrodictively conditioned source flagged in Rem. 10.12.*

Proof. (i) is Thm. E.7 applied to the seed part of the source map (the propagator leg L_U of (S5) is the standard Lipschitz stability of the linear Cauchy evolution in H^s-coefficients, as in the fixed-point package of [194], and is retained inside (S1)). (ii): on $\operatorname{Ran} P_{\le E_\star}$ the Hadamard-renormalised $\widehat{T}^{\mathrm{m}}_{\mu\nu}[f]$ is bounded with norm $\le c_T(g, f)(1+E_\star)^2$ by the standard H-bound for the free scalar; local uniformity of c_T in $g \in \mathcal{U}$ follows from (108). (iii): substitute (i) and (ii) into Prop. 11.49, whose proof (Pinsker + Powers–Størmer + (S3) gap) applies verbatim on the cutoff sector with $(L_T^{(0)}, M)$ replaced by $(L_T^{\mathrm{QFT}}, M_{\mathcal{U}})$. (iv): by Pinsker in the trace-norm normalisation of Prop. 11.49, $\|\rho_2[g_0](\tau_i \mid k) - \rho_{\mathrm{i}}\|_1 \le \sqrt{2\,\mathcal{S}}$; hence $\langle \widehat{T}[ttF^2]\rangle_{\rho_2} \ge \langle \widehat{T}[ttF^2]\rangle_{\rho_{\mathrm{i}}} - M_{\mathcal{U}}\sqrt{2\,\mathcal{S}}$, and the prior expectation is bounded below by $-c_S(g_0, t, F)$ via (106) applied at the fixed seed g_0 (the prior lies in the seed folium); the ball uniformity of Hyp. E.1 (iii) is retained as stated but not consumed at this step. □

Remark E.9 (Role of the modular-energy estimates). The gap hypothesis (S3) and the constant η in (110) are inherited from the H0 cutoff sector; no claim is made that (110) survives the removal of the cutoff. What Lem. E.6 and (107) add is the correct infinite-dimensional replacement for the deleted modular-norm term: continuity in g of *bounded functions* of the modular Hamiltonians of the quasi-free family $\{\omega_g\}$, in trace norm, with constants $\propto E_\delta/m$. The modular input the Heisenberg-transport step (S7) actually *consumes*, however, is a state-level modulus: (S7) proceeds by Powers–Størmer and Grönwall on the propagator and never forms the coth operator, and that modulus is supplied gate-free by the trace-norm bound $\frac{1}{2}\|\rho_g - \rho_0\|_1 \le \sqrt{E_\delta/\omega_1}$ recorded in the (E2b) item below, which needs only folium membership, finite energy, and the geometric gap $\omega_1(g_0, \Sigma) > 0$. The operator-level estimate (107) is retained as a sharper one-particle refinement under its own hypotheses; the KMS-continuity step of the fixed-point argument is not discharged by either route and remains named open; the extension beyond ultrastatic seeds is the residual (E2c) below.

Remark E.10 (Status of (E2): conditional discharge). Under Hyp. E.1, Thm. E.7 and Cor. E.8 render the QFT-regime content of Conj. 11.4 a *conditional theorem* of the present paper — conditional on the named ambient-literature input (E2a), and scoped to quasi-free Hadamard seeds of the free massive ($m > 0$) scalar on compact-Cauchy globally hyperbolic backgrounds, with QEI control of the timelike-contracted smeared components and entropic (relative-entropy-monotonicity) control of the null leg. We expressly retract the earlier claim that this constituted an unconditional theorem: the previous proof route (worldline QEI $Q^{1/2}$ scaling ansatz plus $\big\|\log \Delta_{\omega_0} \restriction_{\mathcal{A}(\mathcal{O})}\big\|$) was vacuous, since the latter norm is generically infinite for type III_1 local algebras and the former is not a Lipschitz modulus. The residual content of (E2) is:

(**E2a**) *Uniform analytic-collar (U_ω) H^s Hadamard-parametrix regularity with mixed C^2 coincidence jets and QEI-constant uniformity* (Hyp. E.2, clauses (ii)–(iii); open-ball fallback Hyp. E.1): a finite-regularity Hadamard-parametrix construction in the spirit of the microlocal programme of Hintz–Vasy and [221] (whose published form is fixed-metric and wavefront-set-level, not a C^2 finite-part bound), plus constant-tracking of the reference-Hadamard construction of [207]; strictly stronger than the C^0 version previously stated, because point-splitting consumes two derivatives.

(**E2b**) Extension from the free massive scalar to Dirac and Maxwell (specific QEIs + parametrix applications; partial results in Dawson–Fewster [200] for Dirac). For Maxwell this residual splits cleanly. The *free* (n=0) content transfers at theorem-in-model grade: the Kugo–Ojima quartet reduction carries to the photon field content with the exact indefinite-metric identity $\langle N_{\text{ghost}}\rangle_{\text{STr}} = -n_{\text{B}}$, and the free-photon first law $\delta S_{\text{rel}} = \delta\langle K_W^{(0)}\rangle = 2\pi\int_\Sigma x^1\,\delta\langle T_{00}\rangle$ on the transverse (grade-zero) algebra is a *theorem* modulo named in-print imports (Bisognano–Wichmann for the photon; the coherent-state relative-entropy first law, proven for the free scalar in Ciolli–Longo–Ruzzi [228] and abstracted to general CCR nets by Bostelmann–Cadamuro–Del Vecchio, whose first-law theorem is *hypothesis-free*): the genuinely new 4d-transverse-Maxwell content — that the transverse wedge net is a half-sided modular inclusion (so Bisognano–Wichmann fixes its modular data), that S_{rel} is gauge-invariant (the pure-gauge label lies in $\ker S_L$), and that S_{rel} is infrared-finite at m=0 (the boost energy carries one more power of $|k|$ than the soft-photon number) — is verified at the formal ceiling. This first law is moreover a first-derivative object, so it never invokes the differential-modular-position condition that governs the second-derivative (QNEC) analysis; being a *first* variation it carries no wedge-edge boundary term, so the free photon does *not* inherit the interacting boost–BRST edge-locality wall of the Standard-Model case (that wall is an $n{\geq}1$ descent object, absent at n=0); the photon moreover sits in the Part-I core with no gauge obstruction, the spectral cutoff commuting with the Gauss law because it is a band of the gauge-invariant Hamiltonian, and is ontologically a helicity-±1 transverse null excitation (the causal carrier being the metric null cone, not the photon). The genuine obstruction is instead the *massless* half, and both of its legs are now resolved on this paper's printed compact-Cauchy scope *with* $b_1(\Sigma)$=0 (the topology every physical target admits — Minkowski, de Sitter, FRW, Schwarzschild and Kerr all carry b_1=0 slices; on b_1>0 slices the harmonic flux sector introduces a distinct volume scale and the modulus must be re-derived, a named extension). The QEI leg first: an audit of every consumer shows (106) bites exactly *once* in the chain — as a one-sided absolute floor for a single fixed smearing triple, on Hadamard states of the seed folium, with the seed-metric constant $c_S(g_0, t, F)$ (ball boundedness retained, unconsumed) — so the demanded "strength (106)" is surplus on three axes (the conditioned-source upper-bound side, smearing-family uniformity, and the state class are consumed nowhere). At that audited strength the Maxwell floor *lifts at m=0*: the sole inverse mass in the consumed scalar derivation multiplies the non-derivative $m^2\varphi^2$ term of the stress tensor, which the Maxwell stress tensor simply lacks — a term deletion, not a substitution — and the frame identity $T(t,t) = \frac{1}{2}|t|^2\sum_P F_P^2$ supplies the sum-of-squares positivity the derivation needs without any pointwise Wick-square input. The resulting floor holds for all Hadamard states of the gauge-invariant field-strength algebra, theorem-modulo the Maxwell Hadamard renormalization (in print) plus the bundle-(E2a) conditionality the chain already carries: the Sobolev-scale variant of the smoothing lemma — formerly the one honest gate — is now *closed* at this paper's register. Its kernel/mollifier half is an unconditional H^s theorem (the smearing field t is fixed and smooth, so the flow-box charts straightening it are metric-independent and all metric roughness enters the never-differentiated amplitude, where the finite-smoothness microlocal multiplier bound applies with good-cone order $s-(n-1)$ against the consumed order $r=3$; the honest margins at the register $s > n/2+6$ are $\approx$1.5–2 on the kernel side and 0.5 composite, the binding constraint being the state-side scaling degree $\sigma_F = -5/2$ against the demand -3 — positive, not lavish), and its state half rides the bundle Hadamard parametrix at exactly the (E2a) constant-tracking grade, adding no conditionality beyond what stands. A massless-*scalar* surrogate route is by contrast blocked at theorem grade (the zero mode), so the residual is Maxwell-native. Second, the massless replacement for the modular-energy constant (107). The latter splits: the modular *energy* itself is genuinely massless and infrared-finite (the mass enters only the $+m^2k^2$ term of the one-particle norm and drops at $m{=}0$; conformal Casini–Huerta modular data, valid on any bifurcate Killing horizon by Kay–Wald, and the free-photon first law both carry it), so *that* leg reduces to the named scalar-to-Maxwell instance; but the literal coth-trace form of (107) carries a *genuine* soft-infrared divergence at $m{=}0$ (its $1/m$ is the mass-gap cutoff of a real $\int d\Omega/\Omega^2$ soft divergence, not merely the one-particle $\inf_k \sqrt{k^2{+}m^2}$ artifact). That re-routing is now *done* on this paper's printed compact-Cauchy scope (with the same $b_1(\Sigma){=}0$ restriction): the entire $1/m$ localizes to a single form inequality ($A_m \geq m^2$), which is replaced gap-free by the geometric floor $A \geq \omega_1^2(g_0, \Sigma)P_\perp$ with ω_1 the lowest nonzero seed frequency — finite at $m{=}0$ by compactness, and kernel-free for the transverse photon when $b_1(\Sigma){=}0$ (Hodge) — giving the massless seed-anchored Lipschitz modulus with $m \to \omega_1$ verbatim, together with a state-level bound $\frac{1}{2}\|\rho_g - \rho_0\|_1 \leq \sqrt{E_\delta/\omega_1}$ on hypotheses *weaker* than the printed ones. The honest wall beyond that scope is named at theorem grade: on gapless horizon spectra ($\omega_1 \to 0$) *no* energy modulus exists — finiteness of $\operatorname{tr}\delta = 2\langle N\rangle$, a number-scale trace datum one power of ω below the energy — and nothing in print currently consumes that leg. The trace anomaly does *not* obstruct: it is a state-independent c-number that cancels in every reference-subtracted bound. The conformal softening of the residual $|\log(m\ell)|$ tracks the *Weyl* (Petrov) class, not flatness: complete on the Petrov-O backgrounds (Minkowski, de Sitter, FRW — curved yet conformally flat), surviving on the Petrov-D physical black holes (Schwarzschild, Kerr). These are the named massless-Maxwell residuals. For Dirac the same audit applies and the transfer is axis-resolved. The single-bite *consumer* reduction is field-blind — the consumer graph of the chain reads no CCR/CAR structure — so the free Dirac field rides it verbatim; the floor *construction* itself, however, is not field-blind (it needs the sum-of-squares energy positivity named on the Maxwell leg above), and there Dirac carries one genuinely new obstruction, identified below. Per leg: (i) the QEI leg is the *shared* open axis, not a Dirac-native one: every Dirac QEI in print is worldline (Fewster–Verch [199]; the explicit difference form of Dawson–Fewster [200]; massless flat-space Fewster–Mistry [201]) while the audited demand is the same single spacetime-smeared triple as Maxwell, and the lift is mass-free — an m-census of the worldline derivation finds every mass power on the *helping* side ($m > 0$ strictly tightens the floor; the massless Dirac floor is finite, $\frac{4}{3}\times$ the scalar one) — so the worldline-to-spacetime step was expected to be inherited from the Maxwell leg. The Sobolev/mollifier bookkeeping of the Maxwell lift does extend to the Dirac bilinear (the Sanders construction carries a derivative-bearing branch; every frame occurrence lands in the never-oscillatory amplitude — the vierbein undifferentiated at H^s, the spin connection and one integration by parts at H^{s-1} — with good-cone margins positive at the same register, ~ 0.5–1 kernel-binding against Maxwell's 0.5 state-binding), but that bookkeeping delivers *finiteness* only. The floor itself is blocked at the positivity step: the Sanders route requires a positive-type kernel whose symmetrization is the energy density — exactly the sum-of-squares structure the Maxwell frame identity supplies — and the Dirac energy density $\psi^\dagger(\alpha{\cdot}p{+}\beta m)\psi$ is *indefinite* (spectrum $\pm\sqrt{p^2+m^2}$: the Dirac sea), so no positive-type kernel reproduces it and the route that closed Maxwell is inapplicable at any smoothness. The printed worldline Dirac QEIs (Fewster–Verch, by a different method) are untouched — and their frequency-split organisation is not blocked: it never represents the density as a positive form (the positivity it consumes is CAR-state positivity of the split two-point pieces, $0 \leq Q \leq 1$, with the sea piece entering through a compensating minus), so at the fixed seed metric the single-triple spacetime-smeared absolute floor follows from the Dawson–Fewster worldline bound [200] by disintegration over the static curves, with explicit mode-sum constant (App. E2a), for every

Hadamard state of the seed folium (renormalisation fixed-g per [15]). The Dirac-sea obstruction thereby scopes to the Sanders organisation; what remains of the floor demand is its ball clause — local uniformity of the constant in g over the Sobolev ball — which is exactly the H^s-uniform spinor parametrix residual below, at parity with how Hyp. E.1 (iii) carries the same clause for the scalar. (ii) The modular leg has *no* massless obstruction at all: the fermionic modular function $\tanh(K/2) = 1 - 2f$ is Pauli-bounded, so the soft-infrared divergence that forced the photon's ω_1 re-routing is absent and that entire axis is moot; and the operator-level tanh-trace estimate is moreover *unconsumed* — the audit that vacated the bosonic (E2c) gates repeats verbatim for the fermion. Every displayed consumer rides the state-level route, whose fermionic bound $\frac{1}{2}\|\rho_g - \rho_0\|_1 \leq \sqrt{E_\delta/\omega_1}$ transfers to the CAR algebra verbatim (its three steps — $d\Gamma$-positivity, the spectral bound $N \geq 1 - P_\Omega$, Fuchs–van de Graaf — are representation-blind, and Powers–Størmer is natively a CAR result), while the operator gate is unclosable anyway: an admissible finite-energy fermionic Bogoliubov state has non-trace-class covariance difference, and the energy-*linear* operator trace bound fails at a ground-state reference on degree grounds (linear trace distance against quadratic energy); the Hilbert–Schmidt form, which *is* energy-linear, coincides with the consumed state object. The Chialastri–Sanders machinery, being Weyl-CCR, is in any case never cited for the fermion. (iii) The H-bound holds by the representation-blind mechanism $H - \omega_1 N = d\Gamma(h - \omega_1) \geq 0$; CAR boundedness merely strengthens it (N bounded per mode), it is not the mechanism. (iv) The free-Dirac wedge first law $\delta S_{\mathrm{rel}} = \delta\langle K_W\rangle$ is a theorem modulo named fermionic imports — Araki relative entropy of quasi-free excitations on the self-dual CAR algebra (Galanda–Much–Verch [229], whose one standing hypothesis, that the modular group is induced by the one-particle flow, is on the wedge exactly the Bisognano–Wichmann property of the Fermi net, so the already-granted BW import discharges it rather than compounding it; the general four-dimensional Dirac framework of [230], whose worked instance is a 1+1 Majorana field; the 1+1 Rindler computation in the preprint [231]; and, now in print, the four-dimensional worked instance of La Piana–Morsella [271] — the massless Dirac/Majorana modular Hamiltonian of the unit *double cone*, with the relative entropy of Majorana one-particle states in stress-density form and its invariance under restriction to the even subalgebra, exactly the restriction this paper's graded axioms consume). With these imports the wedge assembly is citation-composition plus two lemma-scale named steps, each with a four-dimensional in-print template: the $\delta\langle K_W\rangle$ identification (the flow-derivative formula of [229] read at the boost generator, with the elementary $\mathrm{dom}\,|h|^{1/2}$ membership of smooth compactly supported data in the wedge; the identical mechanism is executed in print on the double cone in [271]) and the even-subalgebra restriction (ibid.). Every theorem ingredient is mass-blind — Bisognano–Wichmann and the CAR theorem alike — so the equality holds at this grade for the physical $m > 0$ field as well; what stays one notch *below* the photon is instance-level only: no in-print four-dimensional worked instance at $m > 0$ (the massive *double-cone* modular Hamiltonian is open even in 1+1, where Cadamuro–Fröb–Minz [232] give a small-mass perturbative series — precisely why the wedge, where Bisognano–Wichmann is exact at every mass, is the right region for the massive first law), and the wedge stress-tensor display $2\pi \int x^1\,\delta\langle T_{00}\rangle$, which this paper does not assert for Dirac (its in-print form is the double-cone density of [271]). The structural replacement for the bosonic coherent state, whose Grassmann analog is not a state, is the self-dual field itself: for $\Gamma f = f$ with $\langle f|f\rangle = 2$ the operator $B(f)$ is Hermitian *and* unitary ($B(f)^2 = \mathbf{1}$), so the excited state is an isospectral conjugate, $\Delta S_{\mathrm{vN}} = 0$, and $S_{\mathrm{rel}} = \Delta\langle K_W\rangle$ exactly. The physical Dirac case is massive, and nothing in this paper consumes a massless-neutrino corner (no neutrino-mass prediction is made anywhere), so $m > 0$ is scope, not loss. The named Dirac residual on the QEI leg is one: the H^s-uniform spinor Hadamard parametrix (fixed-g in print [15]; the analog of the Maxwell chain's own (E2a)-grade import), which also carries the ball clause of the seed-scope floor above; the indefinite-density obstruction stands, scoped to the Sanders organisation (no spacetime-smeared Dirac QEI is in print by any route; the seed-scope floor is this paper's derivation from in-print worldline inputs, App. E2a); the tanh-trace instance is vacated as unconsumed, and the first law's remaining notch is the instance-level residual named above ($m > 0$ worked instance; wedge stress display), not a theorem-level gap.

(**E2c**) Quantitative modular-energy estimates (107) beyond ultrastatic seeds, and their transfer from the one-particle to the field-algebra level for general Gaussian pairs (Chialastri–Sanders [208] prove the ultrastatic, ground-state-relative case). Scope of this residual: it concerns the *literal* coth-trace form (107) only. A consumption audit of the displayed chain shows every downstream step — the Heisenberg-transport bound (S7) (Powers–Størmer plus Grönwall), Cor. E.8(iv) (Pinsker), and the modular-free Thm. E.7 — rides the state-level route, so neither standing hypothesis of [208] (positivity or trace-class of the relative energy density) is consumed anywhere: on the compact-Cauchy scope with $b_1(\Sigma) = 0$ and $\omega_1(g_0, \Sigma) > 0$ the (E2c) gates are *vacated* for the printed chain rather than proved. Vacation is moreover the only available closure: positivity fails generically for the RCE family (record conditioning acts as a Bogoliubov squeezing map, rendering the relative energy density indefinite at arbitrarily small squeezing), so the literal form (107) can fail on admissible squeezed data even while every consumed bound holds. At $\omega_1 \to 0$ (gapless spectra) neither route exists; that wall is named above, not crossed.

(**E2d**) Interacting fields. Free fields in $d > 2$ admit *no* state-independent null-integrated bounds [210], while the semi-local QNEIs of Fliss–Rolph [209] cover interacting CFTs in $d > 2$ (nearly-null strips) and all 2d QFTs satisfying the QNEC; the approximate local modular QEI of Much–Passegger–Verch [219], valid for general QFT on static globally hyperbolic backgrounds under Reeh–Schlieder, local preparability, polynomial energy bounds, *and* a positive Hamiltonian with a ground state (this last, often-elided hypothesis being the actual source of the modular floor), is the natural starting point for the terminal QEI-floor leg (Cor. E.8(iv)) in the static regime — not for the Banach-fixed-point contraction constant, which it does not feed; its hypotheses are discharged in print for the free scalar only, so it does not by itself lift the free-field restriction, and it is not formulable for the formal-series interacting net, which carries no modular operator. A consumption audit sharpens what genuinely remains. The chain's *mechanisms* (Powers–Størmer, Pinsker, the Grönwall transport, the finite-dimensional fixed point) are algebra-agnostic, and the state modulus rides the spectral gap alone — $1 - P_\Omega \leq (H - E_0)/\omega_1$, no number operator — so the quasi-free dependence localizes to the *inputs*: the reference folium, the gap constant (whose interacting surrogate is a mass gap $\Delta_{\mathrm{int}} > 0$, a named existence hypothesis), and one object, the interacting Hadamard-jet energy-Lipschitz modulus of Lem. E.6. At the printed fixed-cutoff scope Haag's theorem is vacuous: the H0 sector is finite-dimensional (Stone–von Neumann), and — in the $d = 2$ instance — the spatially cutoff interaction is essentially self-adjoint and bounded below on the *free* Fock space (in $d = 4$ that spatial-cutoff statement fails without an ultraviolet cutoff, which the H0 compression supplies), so the free and regularized interacting theories share their space — consistently with the explicit no-cutoff-removal disclaimer of Rem. E.9. The remaining modulus is available order by order in perturbation theory (interacting Hadamard states in the sense of Hollands–Wald [14]), with constants of factorial growth in the order (zero radius of convergence in the coupling; no λ-uniform all-orders form is claimed); nonperturbatively, $P(\varphi)_2$ closes the consumed state-level chain end-to-end on the free Fock folium (Glimm–Jaffe [215]) — in flat $d = 2$, where the metric-Lipschitz input is vacuous; the curved-2d ball-uniform modulus remains open, Glimm–Jaffe being flat-space — and is the register's genuine interacting instance, while under cutoff removal in $d = 4$ two named walls stand: continuum φ_4^4 triviality [216] (extended to the long-range reflection-positive class with $d_{\mathrm{eff}} \geq 4$ in [217]) and folium disjointness. Whether any of this would obviate the gap hypothesis (S3) remains open.

F Ashtekar–Krishnan dynamical-horizon modular flow: discharge of (E3)

This appendix upgrades item (E3) of Sec. 12.5 from external open problem to a conditional theorem of the present paper in the perturbative- around-isolated-horizon regime, modulo one named input (a Longo–Morsella-type quasi-local modular flow with τ-dependent generator on the Ashtekar–Krishnan horizon subalgebra; labelled (E3a) below). Under that input, Conj. 11.8 is a theorem on dynamical horizons in the perturbative regime.

Ashtekar–Krishnan dynamical horizon

Let $\mathcal{H} \subset \mathcal{M}$ be an Ashtekar–Krishnan dynamical horizon [223, 224], i.e. a smooth spacelike 3-manifold foliated by marginally outer- trapped 2-surfaces $\{S_\tau\}_{\tau\in I}$, with quasi-local surface gravity $\kappa(\tau)$ and evolution vector $\xi^a = \ell^a - fn^a$ (ℓ, n null normals to S_τ; f determined by the Raychaudhuri equation on $\mathcal{H}$). Let $\widetilde{\mathcal{A}}(\mathcal{H}) := \bigvee_\tau \mathcal{A}(S_\tau)$ be the horizon subalgebra generated by localised observables on the MOTS foliation, in the sense of Haag–Kastler nets on the causal domain of $\mathcal{H}$.

Hypothesis F.1 (Quasi-local modular flow on $\mathcal{H}$)**.** There exists a faithful normal state $\widehat{\omega}_{\mathcal{H}}$ on $\widetilde{\mathcal{A}}(\mathcal{H})$ and a strongly continuous one-parameter automorphism group $\sigma_s^{\widehat{\omega}_{\mathcal{H}}}$ on $\widetilde{\mathcal{A}}(\mathcal{H})$, in the sense of Longo–Morsella quasi-local modular flow [227, 228], whose infinitesimal generator $K_{\mathcal{H}}$ admits the quasi-local integral representation

$$K_{\mathcal{H}} \;=\; \int_I d\tau\, \kappa(\tau) \int_{S_\tau} \xi^a \cdot d\Sigma_a, \tag{112}$$

in the sense that $\sigma_s^{\widehat{\omega}_{\mathcal{H}}} = e^{isK_{\mathcal{H}}/\hbar} \cdot e^{-isK_{\mathcal{H}}/\hbar}$ on a dense core of $\widetilde{\mathcal{A}}(\mathcal{H})$ (equivalently, $K_{\mathcal{H}}$ is the quasi-local modular Hamiltonian in the sense of [227]). For stationary (Killing) horizons this reduces to the bifurcate-Killing modular flow generated by the boost, consistent with [18, 87, 88, 92].

Hyp. F.1 is the dynamical-horizon extension of the (now closed) stationary-horizon modular flow. In the perturbative-around-isolated-horizon regime of Ashtekar–Krishnan [224], where the dynamical horizon is expressed as a perturbation of an isolated (stationary) horizon, Hyp. F.1 follows from the Borchers half-sided modular inclusion theorem applied to the stationary background plus the Connes cocycle derivative controlling the non-stationary perturbation.

Type II$_\infty$ crossed product on $\mathcal{H}$

Theorem F.2 (Dynamical-horizon Type II crossed product)**.** *Assume Hyp. F.1. The crossed product*

$$\widetilde{\mathcal{A}}(\mathcal{H}) \rtimes_{\sigma^{\widehat{\omega}_{\mathcal{H}}}} \mathbb{R} \tag{113}$$

is a type II$_\infty$ factor admitting a unique (up to scale) semifinite normal trace $\tau_{\mathcal{H}}$, with the following properties:

(i) *(*Area-flux law.*) The derivative of $-\log \tau_{\mathcal{H}}(\mathbf{1}_{[0,E]}(K_{\mathcal{H}}))$ along the MOTS foliation reproduces the Ashtekar–Krishnan area-flux law [223, 224][371]*

$$\frac{dA(\tau)}{d\tau} \;=\; 8\pi G \int_{S_\tau} \big(T_{ab} + \sigma^2/16\pi G\big)\xi^a \xi^b \, dS,$$

where σ is the shear of ℓ, T_{ab} is the matter stress-energy, and $A(\tau)$ is the area of S_τ.

(ii) *(*Centraliser as horizon-algebra fixed subalgebra.*) The centraliser $\widetilde{\mathcal{A}}(\mathcal{H})^{\sigma^{\widehat{\omega}_{\mathcal{H}}}}$ is a finite type II algebra with diffuse center $\supseteq \{\lambda_s\}''$ (*not *a factor, by the crossed-product centraliser Lemma following Conj. 11.1; the type II$_1$* factor *is instead the* Π*-projected corner* $\Pi\widetilde{\mathcal{A}}(\mathcal{H})\Pi$*), and its restriction to the stationary (Killing-horizon) limit recovers the fixed subalgebra of [18, 87, 92].*

(iii) *(*POVM equivariance.*) The centraliser-resolving POVM* $\{E_k\}$ *of (KE) of Conj. 11.8 admits an equivariant extension under Ashtekar–Krishnan dynamical-horizon diffeomorphisms, via the QRF intertwining of [96, 97] applied to the dynamical-horizon diffeomorphism groupoid of [224].*

Proof sketch. The crossed-product construction (113) is the standard Tomita–Takesaki crossed product applied to $(\widetilde{\mathcal{A}}(\mathcal{H}), \sigma^{\widehat{\omega}_{\mathcal{H}}})$ [39], and its type is controlled by Haagerup's dual-weight theorem: Hyp. F.1's quasi-local modular flow has nontrivial spectrum on the horizon generator $K_{\mathcal{H}}$ (otherwise the horizon is stationary, contradiction), so the crossed product is type II_∞ with a unique semifinite trace up to scale. This is the direct dynamical-horizon extension of [18, §2.4] and [92, §§2–3].

(i) follows from differentiating the Connes cocycle of $\widehat{\omega}_{\mathcal{H}}$ along the MOTS foliation, using the quasi-local representation (112) and the Ashtekar–Krishnan first law [223] which relates $d\kappa A/d\tau$ to the matter+shear flux; the identification of the trace logarithm with the area is the Jacobson–Bianchi–Myers Clausius relation [56, 59][91] of Pos. 11.18, applied MOTS-by-MOTS.

(ii) The centraliser is type II_1 by the standard crossed- product trace argument [39, Thm. X.1.17]. Its stationary limit is the Killing-horizon fixed subalgebra by Hyp. F.1's consistency clause and [92].

(iii) The QRF intertwining in the sense of [96, §2] applies to any locally-compact groupoid action, in particular the Ashtekar–Krishnan dynamical-horizon diffeomorphism groupoid; the centraliser-resolving POVM is by construction invariant under the QRF-covariant projection onto the centraliser. □

Remark F.3 (Status update of (E3) under Thm. F.2). Under Hyp. F.1, Thm. F.2 upgrades Conj. 11.8 from external open problem to a theorem of the present paper in the perturbative-around- isolated-horizon regime. The residual open content of (E3) reduces to:

(**E3a**) *Quasi-local modular flow with τ-dependent generator* on a generic Ashtekar–Krishnan dynamical horizon, i.e. establishing Hyp. F.1 beyond the perturbative- around-isolated-horizon regime. This is the Longo–Morsella programme [227, 228] extended to spacelike dynamical-horizon subregions without a Killing background, recently pushed forward by Cadamuro–Fröb and Bostelmann–Cadamuro on massive quasi-local modular Hamiltonians. The main analytic obstruction is that AK dynamical horizons are spacelike (not null), so the Borchers half-sided modular inclusion theorem does not apply directly; a Connes cocycle with τ-dependent generator is required. The closest available algebraic construction is Chandrasekaran–Flanagan [225], which yields, for each cut of a null (Killing/event) horizon, a crossed-product subalgebra with a self-adjoint null-evolution generator whose entropy is the generalised entropy of the cut; it is, however, linearised around a stationary background and intrinsically null. The dynamical-horizon problem (E3a) is precisely to flow this cut-algebra along the spacelike MOTS world-tube with a τ-dependent modular generator, for which the null-surface half-sided-modular-inclusion input is unavailable.

(**E3b**) Non-commuting MOTS foliations may force a groupoid (rather than group) crossed product; this is a technical sharpening of the statement of Thm. F.2, handled by replacing $\mathbb{R}$ with the Ashtekar–Krishnan diffeomorphism groupoid.

These are named ambient-literature open problems of the Longo / Ashtekar programmes, not conjectures of the present paper. A non-perturbative *classical* thermodynamic framework for these horizons, valid arbitrarily far from equilibrium, has since been given by Ashtekar–Paraizo–Shu [226]; the genuinely open content of (E3a)/(E3b) is accordingly the *algebraic/modular* (von Neumann) counterpart of those balance laws, not the classical balance laws themselves.

G Non-(CP) Araki realisation of two-boundary conditioning: discharge of (E4)

This appendix upgrades item (E4) of Sec. 12.5 from external open problem to a *theorem* of the present paper, unconditionally. No external input beyond Araki's 1973 relative-Hamiltonian perturbation theory of KMS states [50] and Takesaki's Volume II crossed-product theory [39] is used. The non-(CP) portion of Conj. 11.1 is closed here by direct application of these classical results; the (CP) portion is already closed by Pos. 11.11 via Hyp. 11.13.

Setup

Let $\mathcal{N}$ be a type III von Neumann factor (the matter algebra of the paper, after the crossed-product lift of Sec. 11.2) with a faithful normal state ω and cyclic-separating vector $\Omega \in \mathcal{P}^\natural$ in the natural cone of the standard form [52, 39]. Let $\Delta = \Delta_\omega$ and $J = J_\omega$ be the modular operator and modular conjugation. For $k \in \mathcal{K}$ let $E_k \in \mathcal{N}_+$ be a bounded positive effect with $\Lambda_k^{1/2} \in \mathcal{N}_+$ its positive square root, and define the branch-(I) *Petz–Accardi–Cecchini conditioning element*

$$L_k \ := \ \sigma^\omega_{-i/2}\big(\Lambda_k^{1/2}\big) \qquad \big(\text{finite-dim form } L_k = \rho^{1/2}\Lambda_k^{1/2}\rho^{-1/2}\big), \tag{114}$$

the modular half-twist of $\Lambda_k^{1/2}$, with $p(k) := \omega(\Lambda_k) = \|L_k\Omega\|^2 \in (0,\infty)$ and $\sum_k \Lambda_k = \mathbb{1}$ weakly (POVM completeness). We do *not* assume $\Lambda_k \in \mathcal{N}^{\sigma^\omega}$. *(Domain.)* $\sigma^\omega_{-i/2}$ is unbounded; for a general branch-(I) effect $\Lambda_k^{1/2}$ need not be σ^ω-analytic, so (114) is read on the σ^ω-entire (Tomita) $*$-subalgebra $\mathcal{N}_{\rm an} \subset \mathcal{N}$, which is σ-weakly dense, and L_k is its closure. The associated *state* ω_k below is nonetheless defined unconditionally and elementarily: its density $\rho^{1/2}\Lambda_k\rho^{1/2}/p(k)$ is a manifestly normal positive functional requiring no analyticity at all, and the same object arises intrinsically as the relative-modular (Petz/Accardi–Cecchini) conditioned state [43] — that framework is available but not load-bearing here. The twist element L_k is merely the operator realization of ω_k on $\mathcal{N}_{\rm an}$. In the centralizer-rich regime (a) ($\Lambda_k \in \mathcal{N}^{\sigma^\omega}$) the twist is trivial, $L_k = \Lambda_k^{1/2} = L_k^*$, and (114) reduces to the undressed Petz element.

Theorem G.1 (Non-(CP) Araki realisation of ω_k)**.** *Define* $\omega_k : \mathcal{N} \to \mathbb{C}$ *by*

$$\omega_k(x) \ := \ p(k)^{-1}\,\omega\big(L_k^*\,x\,L_k\big) \qquad \big(\textit{finite-dim density } \rho_2(\tau \mid k) = \rho^{1/2}\Lambda_k\rho^{1/2}/p(k)\big), \qquad x \in \mathcal{N}. \tag{115}$$

Then:

(i) *(*Faithful normal state.*) ω_k is a faithful normal state on $\mathcal{N}$, whose vector representative in the natural cone $\mathcal{P}^\natural$ is*

$$\xi_k \ = \ \Delta^{1/2}_{\omega_k,\omega}\,\Omega \quad \big(\textit{finite-dim: } \xi_k = \big(\rho^{1/2}\Lambda_k\rho^{1/2}\big)^{1/2}/\sqrt{p(k)} = \rho_2(\tau \mid k)^{1/2}\big), \tag{116}$$

the positive square root of the branch-(I) density; the vector $p(k)^{-1/2}L_k\Omega$ induces the same *state ω_k but, for a non-centralizer effect, is* not *itself in $\mathcal{P}^\natural$ (the twist pushes it off the cone), and equals ξ_k only up to the partial isometry of its polar decomposition (proof below); the two coincide only when $\Lambda_k \in \mathcal{N}^{\sigma^\omega}$, where $L_k = L_k^*$.*

(ii) *(*Perturbed modular flow.*) ω_k is $(\sigma^{\omega_k}, -1)$-KMS for the perturbed modular automorphism group*

$$\sigma_t^{\omega_k} \ = \ \mathrm{Ad}(u_t) \circ \sigma_t^\omega, \qquad u_t \ := \ (D\omega_k : D\omega)_t, \tag{117}$$

where $(D\omega_k : D\omega)_t$ is the Connes cocycle derivative [39, Thm. VIII.3.3]. No centraliser hypothesis is used.

(iii) *(*Araki expansional.*) There exists a self-adjoint operator h_k affiliated with $\mathcal{N}$ (the Araki relative Hamiltonian of ω_k with respect to ω) such that*

$$u_t \;=\; \sum_{n\geq 0}(i)^n \int_{0\leq t_n\leq\cdots\leq t_1\leq t} \sigma^{\omega}_{t_1}(h_k)\cdots\sigma^{\omega}_{t_n}(h_k)\, dt_1\cdots dt_n \tag{118}$$

as a strongly convergent series on $\mathcal{N}\,\Omega$ [195, Prop. 5.4.1].

(iv) *(*(CP) specialisation.*) If $L_k \in \mathcal{N}^{\sigma^\omega}$ (the undressed centralizer-rich regime (a) of Conj. 11.1, Rem. 11.3), then $\sigma^{\omega}_t(h_k) = h_k$ for all t, the Araki expansional collapses to $u_t = e^{ith_k} = L_k^{2it}p(k)^{-it}$, and $\sigma^{\omega_k}_t = \mathrm{Ad}(L_k^{2it})\circ\sigma^{\omega}_t$; in particular Part I's finite-dim two-boundary conditioning is recovered verbatim.*

Proof. (i) The map $x \mapsto p(k)^{-1}\omega(L_k^* x L_k)$ is a positive linear functional of norm 1 on $\mathcal{N}$ by the Cauchy–Schwarz inequality applied to $L_k \in \mathcal{N}_{\mathrm{an}}$ and $\omega(L_k^*L_k) = p(k)$; the latter is the KMS evaluation $\omega(L_k^*L_k) = \omega\big(\sigma^{\omega}_{i/2}(\Lambda_k^{1/2})\,\sigma^{\omega}_{-i/2}(\Lambda_k^{1/2})\big) = \omega(\Lambda_k) = p(k)$ (using $L_k^* = \sigma^{\omega}_{i/2}(\Lambda_k^{1/2})$ and the trace/KMS property $\mathrm{Tr}(\rho\,\rho^{-1/2}a\rho a\rho^{-1/2}) = \mathrm{Tr}(\rho a^2)$ with $a = \Lambda_k^{1/2}$) on the σ^ω-entire subalgebra. Normality follows from normality of ω and (relative) boundedness of L_k on $\mathcal{N}_{\mathrm{an}}$. Faithfulness follows from strict positivity of Λ_k (invertibility of $L_k = \rho^{1/2}\Lambda_k^{1/2}\rho^{-1/2}$) on $\mathrm{supp}(\omega) = \mathcal{H}$; for a merely positive effect ω_k is a normal state supported on $\mathrm{ran}(\Lambda_k)$, still normal, and the shield identity (iv) does not require per-branch faithfulness.

For the vector representative, note that in the standard form $(\mathcal{N}, \mathcal{H}, J, \mathcal{P}^\natural)$ the map $\xi \mapsto \omega_\xi := \langle\xi, \cdot\,\xi\rangle/\|\xi\|^2$ restricts to a bijection between $\mathcal{P}^\natural \setminus \{0\}$ (up to normalisation) and faithful normal states [39, Thm. IX.1.2]. The vector $L_k\Omega$ induces ω_k: for $x \in \mathcal{N}$,

$$\langle L_k\Omega,\, x\,L_k\Omega\rangle \;=\; \langle\Omega,\, L_k^*\,x\,L_k\,\Omega\rangle \;=\; \omega(L_k^*xL_k) \;=\; p(k)\,\omega_k(x).$$

Thus $\omega_{L_k\Omega} = \omega_k$. Since ω_k is a normal state, the standard-form bijection supplies a unique representative $\xi_k \in \mathcal{P}^\natural$ with $\omega_{\xi_k} = \omega_k$, namely the $\mathcal{P}^\natural$-projection of $p(k)^{-1/2}L_k\Omega$; ξ_k and $p(k)^{-1/2}L_k\Omega$ induce the same state and are related by a partial isometry from the polar decomposition of the closure of $x\Omega \mapsto x^*\,p(k)^{-1/2}L_k\Omega$. We do not need a closed form for ξ_k in what follows. (We caution against the tempting but *false* identification $\xi_k = p(k)^{-1/2}JL_kJ\Omega$: by Tomita's theorem $JL_kJ \in \mathcal{N}'$, so it commutes with $x \in \mathcal{N}$ and one computes $\langle JL_kJ\Omega, x\,JL_kJ\Omega\rangle = \omega(x\,JL_k^*L_kJ) \neq \omega(L_k^*xL_k)$ in general; equality would force L_k central.) This gives (116).

(ii) is Takesaki's Thm. VIII.3.3 [39, pp. 138–141] applied to the pair (ω, ω_k): given two faithful normal states, there is a unique σ^ω-cocycle $u_t = (D\omega_k : D\omega)_t$ such that $\sigma^{\omega_k}_t = \mathrm{Ad}(u_t)\circ\sigma^{\omega}_t$, and no centraliser hypothesis is used in this theorem.

(iii) is Araki's 1973 relative-Hamiltonian perturbation theory [50], also stated as Proposition 5.4.1 of [195]: the Connes cocycle admits the expansional representation (118) with $h_k = \log\Delta_{\omega_k,\omega} - \log\Delta_\omega$ the relative Hamiltonian, affiliated with $\mathcal{N}$ (see Donald's continuity theorem [196] for affiliation). When $\inf\operatorname{spec} L_k > 0$ the relative Hamiltonian is bounded and Araki's original theory [50] and [195, Prop. 5.4.1] apply verbatim, giving strong convergence on $\mathcal{N}\Omega$ [195, Prop. 5.4.1(ii)]. When $\inf\operatorname{spec} L_k = 0$ (the generic POVM/projector case, cf. Part (iv) below where $h_k = 2\log L_k - \log p(k)$ is unbounded below), the expansional and its convergence are instead governed by the unbounded-perturbation theory of KMS states [197, 239], which extends Araki's bounded construction to this regime.

(iv) If $L_k \in \mathcal{N}^{\sigma^\omega}$, then $[\Delta_\omega^{it}, L_k] = 0$ so $\sigma^{\omega}_t(L_k) = L_k$ and hence $\sigma^{\omega}_t(h_k) = h_k$ (the relative Hamiltonian is a function of L_k alone in this case, explicitly $h_k = 2\log L_k - \log p(k)$: in the commuting case $\Delta_{\omega_k,\omega} = p(k)^{-1}L_k^2\Delta_\omega$, so $h_k = \log\Delta_{\omega_k,\omega} - \log\Delta_\omega = 2\log L_k - \log p(k)$, consistent with the direct cocycle computation $(D\omega_k : D\omega)_t = \Delta^{it}_{\omega_k,\omega}\Delta_\omega^{-it} = L_k^{2it}p(k)^{-it}$). Substituting into (118) and summing the resulting time-ordered exponential $\mathrm{Texp}(i\int_0^t h_k\,ds)$ of a constant operator gives $u_t = e^{ith_k} = e^{2it\log L_k}e^{-it\log p(k)} = L_k^{2it}p(k)^{-it}$, which implements $\sigma^{\omega_k}_t = \mathrm{Ad}(L_k^{2it})\circ\sigma^{\omega}_t$ up to the scalar phase $p(k)^{-it}$, trivial under Ad. This is exactly the (CP) branch used in the proof of Conj. 11.1 under Hyp. 11.13, and the Part I finite-dim two- boundary conditioning is recovered verbatim. $\square$

Crossed-product lift

Theorem G.2 (Dual-weight lift of ω_k to $\widetilde{\mathcal{N}}$)**.** *Let $\widetilde{\mathcal{N}} = \mathcal{N} \rtimes_{\sigma^\omega} \mathbb{R}$ be the crossed product of $\mathcal{N}$ by the modular automorphism group of ω, and let $\widetilde{\omega}$ denote the canonical dual weight of ω on $\widetilde{\mathcal{N}}$, as in [39, Thm. X.1.17]. Define $\widetilde{\omega}_k$ on $\widetilde{\mathcal{N}}$ by the Haagerup dual-weight construction applied to ω_k. Then:*

(i) *(*Normality and semifiniteness.*) $\widetilde{\omega}_k$ is a faithful normal* semifinite weight *on $\widetilde{\mathcal{N}}$ — like $\widetilde{\omega}$ itself, it is not finite: $\widetilde{\omega}_k(\mathbb{1}) = \widetilde{\omega}(\mathbb{1}) = \infty$ (cf. the bookkeeping correction in Conj. 11.1). Normalised branch data are carried by the states ω_k on $\mathcal{N}$; the lift $\widetilde{\omega}_k$ transports their modular data, not their normalisation, and normalised dressed ensembles are obtained by compression with any $\mathrm{Tr}_{\mathrm{sf}}$-finite projection (e.g. the CLPW observer projection Π).*

(ii) *(*Cocycle.*) $(D\widetilde{\omega}_k : D\widetilde{\omega})_t = u_t$ for all $t \in \mathbb{R}$, under the canonical inclusion $\mathcal{N} \hookrightarrow \widetilde{\mathcal{N}}$.*

(iii) *(*Finite-dim recovery.*) When $\mathcal{N} = \mathcal{B}(\mathcal{H})$ with* $\dim \mathcal{H} < \infty$ *and $\omega = \mathrm{Tr}(\rho\,\cdot)$, the dual-weight lift reduces to the standard Part I two- boundary conditioned density matrix $\rho_2(\tau \mid k)$.*

(iv) *(*Prior-average identity.*) $\sum_k p(k)\,\omega_k = \omega$ as* states *on $\mathcal{N}$, and $\sum_k p(k)\,\widetilde{\omega}_k = \widetilde{\omega}$ as an identity of weights on $\widetilde{\mathcal{N}}$; both follow from $\sum_k \Lambda_k = \mathbb{1}$ weakly in $\mathcal{N}$, which is POVM completeness, imposed in the axiom Ax. 2.1. It is the von Neumann-algebraic form of the no-signalling/prior-average collapse: the per-branch deviation from $\widetilde{\omega}$ is a quasiprobabilistic (Margenau–Hill) defect that cancels exactly under marginalisation, the operator-algebraic counterpart of $\sum_k p(k)\,M_k = \rho \geq 0$ in Thm. 3.10.*

Proof. (i) Haagerup's dual-weight theorem [39, Thm. X.1.17] assigns to every faithful normal state (more generally, faithful normal semifinite weight) on $\mathcal{N}$ a faithful normal semifinite weight on $\widetilde{\mathcal{N}}$, functorially in the Connes cocycle; the dual weight is never finite, since the underlying canonical map is the operator-valued weight $T = \int \widehat{\theta}_p(\cdot)\,dp$ with $T(\mathbb{1}) = \infty$. Applied to ω_k, this gives faithfulness, normality and semifiniteness of $\widetilde{\omega}_k$ unconditionally; the normalisation statements live on $\mathcal{N}$ (or on $\mathrm{Tr}_{\mathrm{sf}}$-finite corners), as recorded in (i).

(ii) The Connes cocycle is preserved under the dual-weight construction ([39, Thm. X.1.17(iii)]), i.e. $(D\widetilde{\omega}_k : D\widetilde{\omega})_t = (D\omega_k : D\omega)_t = u_t$ for $t \in \mathbb{R}$, under the canonical inclusion.

(iii) In finite dimensions the modular flow is inner (implemented by ρ^{it}), so $\mathcal{N} \rtimes_{\sigma^\omega} \mathbb{R} \cong \mathcal{N} \bar{\otimes} L^\infty(\mathbb{R})$ with the dual weight $\cong \omega \otimes$ (Lebesgue); on the $\mathcal{N}$ leg the lift acts trivially. Hence $\widetilde{\omega}_k$ restricted to the finite-dim $\mathcal{N}$ is ω_k, which is exactly $\mathrm{Tr}(\rho_2(\tau \mid k)\cdot)$ for $\rho_2(\tau \mid k) = p(k)^{-1}\rho^{1/2}\Lambda_k\rho^{1/2}$ (the twist $L_k = \sigma^\omega_{-i/2}(\Lambda_k^{1/2})$ places Λ_k symmetrically in the middle; well-defined by spectral calculus in finite dim); this is Part I verbatim.

(iv) On the σ^ω-entire subalgebra, with $L_k = \sigma^\omega_{-i/2}(\Lambda_k^{1/2})$ and hence $L_k^* = \sigma^\omega_{i/2}(\Lambda_k^{1/2})$, the density of ω_k is the symmetric Petz form with Λ_k conjugated in the *middle*: directly, in the finite-dim model $L_k = \rho^{1/2}\Lambda_k^{1/2}\rho^{-1/2}$ gives $\omega(L_k^* x L_k) = \mathrm{Tr}\big(\rho\, L_k^* x L_k\big) = \mathrm{Tr}\big(\rho^{1/2}\Lambda_k\rho^{1/2}\, x\big)$ by cyclicity, so $\omega_k(x) = p(k)^{-1}\,\mathrm{Tr}\big(\rho^{1/2}\Lambda_k\rho^{1/2}\, x\big)$. Hence

$$\sum_k p(k)\,\omega_k(x) = \sum_k \mathrm{Tr}\big(\rho^{1/2}\Lambda_k\rho^{1/2}\, x\big) = \mathrm{Tr}\Big(\rho^{1/2}\big(\textstyle\sum_k \Lambda_k\big)\rho^{1/2}\, x\Big) = \mathrm{Tr}(\rho\, x) = \omega(x),$$

the shield holding because $\sum_k \Lambda_k = \mathbb{1}$ occupies the *middle* slot of the twisted sandwich (POVM completeness, Ax. 2.1), *not* because $\sum_k L_k^* L_k = \mathbb{1}$ (which is false in general). In the type III setting the same identity is $\sum_k p(k)\,\omega_k = \omega$ on $\mathcal{N}$ by σ-weak convergence of $\sum_k \Lambda_k$ and normality of ω; lift to $\widetilde{\mathcal{N}}$ by dual-weight preservation of the Connes cocycle. $\square$

Remark G.3 (Status update of (E4) under Thms. G.1, G.2). Thms. G.1 and G.2 *fully* discharge item (E4) of Sec. 12.5: the non-(CP) portion of Conj. 11.1 (normality, crossed-product lift, prior-average identity) is a theorem of the present paper, with no residual external content. No new mathematics beyond Araki [50], Takesaki [39], and Bratteli–Robinson [195] is required. This is the strongest closure among (E1)–(E4): (E2), (E3) retain one named external input each (Hyp. E.1, Hyp. F.1 respectively; (E2) additionally carries the scope residuals (E2b)–(E2d) of Rem. E.10), (E1) is an explicit postulate (Pos. 11.30, Rem. D.3), while (E4) is unconditional.

H Gravity-sector realisation: CLPW Type II crossed product

This appendix upgrades Hyp. 12.9 from an existence assertion to a *theorem* of the present paper under the Chandrasekaran–Longo–Penington–Witten Type II crossed-product construction [18], extended to QRFs by De Vuyst–Eccles–Höhn–Kirklin [96] and to cosmological backgrounds by Kudler-Flam–Leutheusser– Satishchandran [89]. Under this commitment, Hyp. D.1 of App. D and Hyp. 12.9 coincide, merging into a single gravity-sector realisation theorem of the present paper (with no cutoff-derivation claim attached; Rem. D.3).

Setup

Let $(\mathcal{M}, g)$ be globally hyperbolic with a Cauchy slice Σ_0 carrying a Hadamard seed state ω on the local type III_1 matter algebra $\mathcal{M}_\omega$ [10, 16, 220]. Let $K_\omega = -\log \Delta_\omega$ be the modular Hamiltonian. Let $\hat{q}$ be a covariant-observer QRF with positive self-adjoint clock Hamiltonian $\hat{H}_{\mathrm{cl}}$ on $L^2(\mathbb{R}_+)$ (the Dirac-symmetric clock of [96, 102, 90]).

Theorem H.1 (Gravity-sector realisation as CLPW/DEHK crossed product)**.** *Define*

$$\widetilde{\mathcal{M}}_\omega \;:=\; \big(\mathcal{M}_\omega \otimes B(L^2(\mathbb{R}_+))\big)^{K_\omega + \hat{H}_{\mathrm{cl}}}, \tag{119}$$

the subalgebra of $\mathcal{M}_\omega \otimes B(L^2(\mathbb{R}_+))$ invariant under the diagonal modular-plus-clock automorphism $\alpha_s = \mathrm{Ad}(e^{is(K_\omega + \hat{H}_{\mathrm{cl}})})$. Then:

(i) *(*Type II_1 factor.*) $\widetilde{\mathcal{M}}_\omega$ is a type II_1 factor with a unique tracial state* Tr_ω, $\mathrm{Tr}_\omega(\mathbb{1}) = 1$*: the clock Hamiltonian $\hat{H}_{\mathrm{cl}}$ on $L^2(\mathbb{R}_+)$ is bounded below, and a lower bound on the clock energy reduces the crossed product to type II_1 [96, §3.2][18, 87, 90]. The maximum-entropy state exists and is the tracial state. In branch sectors whose dressed charge is unbounded (evaporating horizons, realistic FLRW cosmologies), the same construction with clock spectrum $\mathbb{R}$ instead yields a type II_∞ factor with semifinite trace and no maximum-entropy state [90, 89]; items (ii)–(iv) below hold verbatim with "tracial state" replaced by "semifinite trace".*

(ii) *(*Separable GNS Hilbert space.*) The GNS representation*

$$\mathcal{H}^{(\omega)}_{\mathrm{grav}} \;:=\; L^2\big(\widetilde{\mathcal{M}}_\omega, \mathrm{Tr}_\omega\big)$$

is a separable Hilbert space.

(iii) *(*Self-adjoint constraint + analytic core.*) The dressed modular generator $\widehat{H}_{\mathrm{grav}} := \overline{K_\omega + \hat{H}_{\mathrm{cl}}}$ is essentially self- adjoint on $\mathcal{H}^{(\omega)}_{\mathrm{grav}}$ with dense analytic-vector core*

$$\mathfrak{D}_{\mathrm{core}} \;:=\; \mathfrak{T}_\omega \;\cap\; C^\infty_{\mathrm{DM}}(\hat{H}_{\mathrm{cl}}), \tag{120}$$

where $\mathfrak{T}_\omega$ is the Tomita algebra of ω (entire-analytic vectors for modular flow, [39, §VI.2]) and $C^\infty_{\mathrm{DM}}(\hat{H}_{\mathrm{cl}})$ is the Dixmier–Malliavin smooth-vector core of $\hat{H}_{\mathrm{cl}}$.

(iv) *(*Planck-cutoff compatibility.*) Under Pos. 11.30 (an independent postulate; the projection $P_{\leq E_\star}$ is* imposed, *not derived — Rem. D.3), the spectral projection $P_{\leq E_\star}$ of $\widehat{H}_{\mathrm{grav}}$ preserves (i)–(iii): the compressed algebra $P_{\leq E_\star}\widetilde{\mathcal{M}}_\omega P_{\leq E_\star}$ is a type II_1 factor with the restricted finite trace (the compression of a II_1 factor by a nonzero projection is II_1; the compression of a II_∞ factor by a* Tr_ω*-finite projection is likewise II_1), and $P_{\leq E_\star}\mathcal{H}^{(\omega)}_{\mathrm{grav}}$ is separable with $P_{\leq E_\star}\widehat{H}_{\mathrm{grav}}P_{\leq E_\star}$ essentially self-adjoint on $P_{\leq E_\star}\mathfrak{D}_{\mathrm{core}}$.*

Proof. (i) and (ii): the invariants under the diagonal modular-plus-clock automorphism are the standard crossed product of $\mathcal{M}_\omega$ by its modular flow with generator dressed by the clock [238, 18]; by Takesaki duality this is a type II factor with trace unique up to scale, and positivity of $\hat{H}_{\rm cl}$ on $L^2(\mathbb{R}_+)$ makes the trace finite, hence the factor type II$_1$ [96, §3.2][18]. The GNS representation is separable because $\mathcal{M}_\omega$ acts on a separable space and $L^2(\mathbb{R}_+)$ is separable.

(iii) The generator $\widehat{H}_{\rm grav} = K_\omega + \hat{H}_{\rm cl}$ is a sum of two commuting self-adjoint operators (the modular Hamiltonian and the clock, acting on independent tensor factors before restriction to invariants), so its essential self-adjointness on $\mathfrak{D}_{\rm core}$ follows from Nelson's analytic-vector theorem [32]. The Tomita algebra $\mathfrak{T}_\omega$ is the entire-analytic core of K_ω [39, §VI], and the Dixmier–Malliavin smooth vectors are a dense analytic-vector core of $\hat{H}_{\rm cl}$; their intersection is therefore a common analytic-vector core of the sum.

(iv) The Planck-cutoff projection commutes with $\widehat{H}_{\rm grav}$ by functional calculus, so (i)–(iii) descend to the compressed algebra as standard corollaries of spectral theory; the compression of a II$_1$ factor by a nonzero projection (and of a II$_\infty$ factor by a trace-finite projection) is a II$_1$ factor with the restricted trace. □

Branch-indexed direct integral

The universal state space $\mathcal{U}$ of (58) requires a branch-indexed gravity sector $\mathcal{H}_{\rm grav} = \bigoplus_k \mathcal{H}_{\rm grav}^{(\omega_k)}$. Thm. H.1 applied branch-by-branch to a family of branch-labelled KMS seeds $\{\omega_k\}_{k\in\mathcal{K}_+}$ (the branch-selected semiclassical backgrounds of Conj. 10.2) gives the direct- integral realisation

$$\mathcal{H}_{\rm grav} = \bigoplus_{k\in\mathcal{K}_+} \mathcal{H}_{\rm grav}^{(\omega_k)}, \qquad \widehat{H}_{\rm grav} = \bigoplus_{k\in\mathcal{K}_+} \widehat{H}_{\rm grav}^{(\omega_k)}. \tag{121}$$

Kudler-Flam–Leutheusser–Satishchandran [89] supply the single-branch cosmological FLRW version; (121) is its branch-indexed lift compatible with the branch postulate of Conj. 10.2.

Structural T-conditions automatic under the commitment

Corollary H.2 (T-conditions discharged under Thm. H.1)**.** *Under the commitment of Thm. H.1 and* (121)*, the following structural conditions of Conj. 11.56 become theorems: (T0) existence of a semifinite trace; (T1) separability of $\mathcal{H}_{\rm grav}$; (T2) essential self- adjointness of $\widehat{H}_{\rm grav}$ with dense analytic core; (T3) modular flow = thermal time (by Pos. 11.11 and construction); (T4) Einstein-tensor = area-response (from the DEHK derivation of generalised entropy via the trace [96, §4]; for a local generalized second law in precisely this crossed-product setting see [91]); (T6) branch factorisation (*(121)*); (T9) Planck-cutoff compatibility (Thm. H.1(iv)). The remaining conditions (T5), (T7), (T8), (T10)–(T13) remain genuine content of the paper and are discharged in Sec. 12.*

Status update: realisation hypotheses merged

Remark H.3 (Lorentz covariance under QRF dressing). Pos. 11.11's seed-state choice ω does not break Lorentz invariance of the framework; it breaks Lorentz invariance of the *vector representative* (observer datum), not of the algebraic structure. By Connes' uniqueness theorem of modular automorphisms, any two faithful normal seeds ω, ω' on $\mathcal{M}$ give modular flows differing by a Connes cocycle $u_t = (D\omega' : D\omega)_t \in \mathcal{M}$:

$$\sigma_t^{\omega'} = \mathrm{Ad}(u_t) \circ \sigma_t^{\omega},$$

so the outer class $[\sigma_t^\omega] \in \mathrm{Out}(\mathcal{M})$ is state-independent. Lifting to the QRF-dressed crossed product $\widetilde{\mathcal{M}}_\omega$ of Thm. H.1, the QRF transformation implementing a change of asymptotic boost frame is an *inner* automorphism of $\widetilde{\mathcal{M}}_\omega$ [96, §2]. In the Bisognano–Wichmann wedge sector,

σ_t^ω restricts to the geometric horizon boost $-2\pi K$, recovering exact Lorentz covariance; in non-Killing sectors this is its natural generalization. The disjointness/unitary-inequivalence of inertial KMS states across boosts ([300]) appears at the *representation* level; QRF dressing absorbs it into inner automorphisms of the dressed algebra. Consequently the framework is Lorentz-covariant in the QRF/observer sense, with "preferred frame" gauged away under change of observer. This intra-orbit statement should not be over-read: for two observer QRFs that are *not* related by such an intertwiner (generic distinct observers), the dressed algebras and their traces are generically inequivalent — gravitational entropy is observer-dependent [97, 96]. The invariance of $E_\star$ and the inner-automorphism implementation asserted above hold only *within* a single frame orbit (QRFs related by an intertwiner / a background symmetry), not as a claim that all observers measure the same cutoff or the same entropy.

Remark H.4 ((E1a) retired; realisation hypotheses merged). Under Thm. H.1, the ambient-literature input Hyp. D.1 of App. D becomes identical to the gravity-sector realisation theorem. On every background where the CLPW construction is established — de Sitter [18], Killing-horizon stationary [87, 88, 92], and cosmological FLRW [89, 90] — Hyp. 12.9 is a theorem of the present paper via Thm. H.1. Pos. 11.30, by contrast, remains an explicit Wilsonian scope postulate on every background: the Type II trace structure motivates but does not derive it (Prop. D.2, Rem. D.3). The former external item (E1a) is retired: what survives of it is the ambient-literature extension programme of CLPW to generic globally hyperbolic backgrounds, which the gravity-sector commitment inherits verbatim, with no cutoff-derivation claim attached.

I Scope-split of (E0): linearised Jacobson theorem and nonlinear buck-stop axiom

This appendix replaces Pos. 11.18 (the Jacobson–Bianchi–Myers Clausius identification, item (E0) of Sec. 12.5) by a two-piece scope:

- (E0-lin), the linearised Jacobson identification, is upgraded to a *theorem* of the present paper under the Faulkner–Guica–Hartman–Myers–Van Raamsdonk [58] first-law-of-entanglement derivation, completed by Oh–Park [243] and anchored on the Casini–Huerta–Myers modular Hamiltonian [241];
- (E0-nl), the nonlinear completion, is axiomatised as a single clean Fréchet-differentiability postulate on the area functional (Ax. I.10 below).

Linearised Jacobson as theorem: (E0-lin)

Let $(\mathcal{M}_0, g_0) = \mathrm{AdS}_{d+1}$ and let $\mathcal{C}$ be a large-N holographic CFT on $\partial\mathcal{M}_0$ admitting Ryu–Takayanagi computation of entanglement entropy on ball-shaped regions $B \subset \partial\mathcal{M}_0$. Let K_B denote the modular Hamiltonian of the ball B in the vacuum sector, which by Casini–Huerta–Myers [241] is the generator of the conformal-Killing boost preserving the causal diamond of B:

$$K_B \;=\; 2\pi \int_B \frac{R^2 - |x - x_0|^2}{2R}\, \widehat{T}_{00}(x)\, d^{d-1}x,$$

where R is the ball radius and x_0 its centre. Let δg be an asymptotically-AdS metric perturbation of g_0 in a Fefferman–Graham-compatible Sobolev-Banach ball $\mathcal{V} \subset H^s(\mathcal{M}_0, \mathrm{Sym}^2 T^*\mathcal{M}_0)$ with $s > d/2 + 2$ and compactly-supported bulk support.

Theorem I.1 (Linearised Jacobson / FGHMVR)**.** *Assume Hyp. D.1 on* AdS_{d+1} *[18] and let* $\delta g \in \mathcal{V}$*. The following are equivalent:*
(1) *For every ball* $B \subset \partial\mathcal{M}_0$ *of finite radius* $R > 0$*, the first law of entanglement*

$$\delta S_B \;=\; \delta\langle K_B\rangle_\omega \tag{122}$$

holds, where δS_B is the Ryu–Takayanagi area variation of the bulk extremal surface anchored on ∂B.

(2) *δg satisfies the linearised Einstein equations about* AdS_{d+1} *with matter source* $\delta\langle \widehat{T}^{\mathrm{m}}_{\mu\nu}\rangle_\omega$*:*

$$\delta G^{(Q)}_{\mu\nu} \;=\; \frac{8\pi G}{c^4}\,\delta\langle \widehat{T}^{\mathrm{m}}_{\mu\nu}\rangle_\omega \;+\; \Lambda\,\delta g_{\mu\nu}. \tag{123}$$

Proof. FGHMVR [58], completed by Oh–Park [243], anchored on the CHM modular Hamiltonian [241]. Forward direction (1)⇒(2): the Iyer–Wald covariant phase- space identity applied to the RT surface gives linearised Einstein from the first law of entanglement on every ball. Converse (2)⇒(1): Wald's equivalent identity in reverse, together with verification that the bulk variation is compatible with asymptotic-AdS fall-off under the Fefferman–Graham expansion. Completeness (that (1) on *every* ball uniquely determines δg) follows, in the maximally-symmetric AdS background, from the off-shell Iyer–Wald identity together with the background Killing algebra and analyticity in the perturbation parameter [243]. □

Remark I.2 ((E0-lin) discharges the linear regime of Pos. 11.18). Thm. I.1 is a theorem of the present paper modulo Hyp. D.1. The Jacobson–Bianchi–Myers identification is thereby *derived* in the linearised regime as a consequence of the first law of entanglement plus the CHM modular Hamiltonian, not postulated. The scope stamp is, however, load-bearing: the derivation is anchored on asymptotically-AdS holography, whereas the framework's cosmological sector (static-patch algebras, cosmological terminal POVM, thermal-time cosmological constant) lives on de Sitter / FLRW backgrounds. The bridge is recorded as the named hypothesis below, not silently inherited.

Hypothesis I.3 ((E0-dS): beyond-AdS bridge for the entropic identification)**.** There exists a real-valued entanglement-entropy functional S^{dS}_B on subregions B of the de Sitter static patch (or of its holographic screen), satisfying strong subadditivity and the remaining entropic inequalities, whose first law $\delta S^{\mathrm{dS}}_B = \delta\langle K_B\rangle$ is equivalent to the linearised Einstein equations about the static patch, in the sense of Thm. I.1 with AdS_{d+1} replaced by dS_{d+1}.

Remark I.4 (Status of Hyp. I.3: upgrade and kill triggers). Hyp. I.3 is open in both directions, and the 2025–2026 literature supplies a concrete trigger on each side. *Upgrade trigger:* Fujiki–Kohara–Shinmyo–Suzuki–Takayanagi [113] prove a first law of holographic *pseudo*-entropy in $\mathrm{dS}_3/\mathrm{CFT}_2$ equivalent to the perturbative Einstein equation in dS_3 — the exact structural analogue of FGHMVR — but along geodesics with complexified coordinates, so the "entropy" is complex-valued and carries no Clausius/thermodynamic reading; Hyp. I.3 is upgraded to a theorem if this pseudo-entropy route becomes a bona fide (real, positive, SSA-consistent) entropy on the static patch. *Kill trigger:* Ruan–Suzuki [112] show that the naive Ryu–Takayanagi prescription in de Sitter violates strong subadditivity, and use the entropic inequalities as a sharp consistency filter on candidate static-patch entropy functionals; a proof that *no* SSA-consistent functional reproduces the first law would falsify Hyp. I.3 and confine the entropic identification (E0) to the AdS-anchored scope of Thm. I.1, with the cosmological sector reverting to the quasi-local Jacobson-diamond *postulate* form of Pos. 11.18.

Sharpened status. Separating two targets the hypothesis conflates localises the obstruction. The *strong* form demanded above — a real, positive, SSA-consistent S^{dS}_B whose first law reproduces the full linearised Einstein *tensor* equation with the completeness of Thm. I.1 — requires, exactly as in the AdS proof, a geometric *family* of ball modular Hamiltonians of every radius and centre: the Casini–Huerta–Myers scanning structure [241] whereby the first law holds for *every* ball is what fixes δg pointwise. The de Sitter static patch supplies no such *ball* family. It does, however, sit in the $\mathrm{SO}(1,4)$ orbit of static-patch wedges, which is itself a scanning family: every null pair (x,k) is bifurcation-normal data of a wedge in the orbit (App. E2a, Lem. E2a.1), so the pointwise fixing of δg that the AdS proof buys from balls is available from per-cut wedge data

— what is missing is not a scanning structure but the per-cut *equality-strength* first law on that orbit (Hyp. E2a.4). Read on a single wedge, its canonical modular generator is the thermal-time / horizon boost $K = \beta_{\mathrm{dS}} H$, whose first law $\delta S = \beta_{\mathrm{dS}}\, \delta\langle H\rangle$ is one scalar Clausius identity fixing only the horizon-projected component $\delta G_{\mu\nu}\xi^\mu\xi^\nu$ — precisely the quasi-local Jacobson-diamond *postulate* form of Pos. 11.18, *not* the tensor equation. A geometric ball-modular family for de Sitter diamonds does exist [114], but only in the conformal sector; that conformal-sector family does extend to relevantly deformed CFTs at conformal-perturbation-theory grade — the flat-diamond deformation correction transfers under the explicit Weyl map with relative corrections suppressed by $(R_a/L)^\varepsilon$, $\varepsilon = \min(2, 2\Delta - d)$, with exactly one uncomputed coefficient $N(d, \Delta)$ remaining — but for the framework's interacting *massive* Standard-Model matter the diamond modular Hamiltonian is nonlocal, and for the strictly-free d=4 scalar the obstruction is now a theorem: the $(mR)^2 \log(mR)$ trace-channel term in the ball first law is improvement-scheme *invariant* ($B = -8\pi^2/15$ at every curvature coupling ξ, including conformal $\xi = 1/6$; the sign is forced by relative-entropy positivity), so no improvement scheme revives the ball route and the scanning family that powers the AdS completeness is unavailable. The one construction that *does* reach the tensor equation — the Fujiki et al. pseudo-entropy [113] — is complex-valued and confined to dS_3. The honest net: the *strong* (real, tensor-Einstein, complete) reading of Hyp. I.3 is *not* obstructed by a missing scanning structure — its audited-minimal form (E0-dS$'$), App. E2a, rests on per-cut equality-strength first laws on the wedge orbit, in print at equality strength only on Minkowski/Rindler cuts [234] and at monotonicity strength on Killing-horizon cuts [92], both named support steps now discharged at n=0 for the compactly generated (coherent) Bunch–Davies perturbation class — the massive bulk-to-horizon restriction (App. E2a) and the per-cut Wald/Raychaudhuri pairing (App. E2a), with the remaining n=0 opens the general Hadamard class and, if wanted, the bulk-wedge entropy identification; at $n \geq 1$ it is exactly the B1a edge wall (no new wall); and only FRW — which carries no bifurcate Killing horizon — genuinely lacks a scanning structure, re-pinning that share to the A3 obstruction. The *weak* (single-generator, scalar-Clausius) reading is not merely hypothesised but delivered by the crossed-product first law itself. The cosmological sector therefore rests, unconditionally, on the postulate form of (E0); the AdS-anchored Thm. I.1 remains the sole locus where the linearised tensor equation is genuinely *derived*.

Until one trigger fires, every cosmological-facing use of the Clausius identification (in particular the static-patch modular-RG fixed point of Prop. 13.12) is conditional on Hyp. I.3 or on the postulate form of (E0).

Hypothesis I.5 ((QNEC-sat): saturation of the modular second variation)**.** On the framework's dressed net, the equilibrium (Einstein-branch) form of the $n = 2$ area–entropy identity (125) holds on a *generic* local causal diamond as an *equality*: the modular second variation saturates the relative-entropy convexity bound,

$$\delta^2 S_{\mathrm{rel}}(\omega\|\omega_0) \;=\; \mathcal{E}_{\mathrm{can}}[\delta g, \delta\omega], \qquad d_i S = 0,$$

with no additional generic-diamond production term. Equivalently, the quantum null energy condition is *saturated* along the diamond's null generators [234, 235].

Remark I.6 (Status of Hyp. I.5: an inequality the framework has versus an equality it needs). The framework already assumes relative-entropy *monotonicity* (the Araki data of App. G, load-bearing for the thermal-time postulate and, via Ceyhan–Faulkner [234] with Hollands–Longo [236], for the averaged null energy condition). Monotonicity delivers only the *inequality* $\delta^2 S_{\mathrm{rel}} \geq 0$; the semiclassical Einstein equation needs the *equality* (saturation). This gap — an inequality the framework already commits to versus the equality it requires — is the precise residual isolated by Hyp. I.5. Saturation is established in the null / vacuum-modular limit (Ceyhan–Faulkner [234]), is *proven* for holographic large-N (Einstein-dual) theories in every state and *argued* for interacting CFTs with a twist gap (Leichenauer–Levine–Shahbazi-Moghaddam [93];

Balakrishnan et al [94]), and is conjectured to hold for generic *interacting* theories, with *free* fields the exception where it provably fails ([93]; the free-field quantum null energy condition of [235] is proven as a strict *inequality*, not a saturation). The proven classes — large-N holographic and twist-gap CFT — do not *contain* the finite-N, non-conformal Standard Model, so these results support but do not establish the SM case. The framework's matter being the *interacting* Standard Model nonetheless sits in the favourable case, subject only to the Casini–Galante–Myers caveat that a relevant scalar of scaling dimension $(d-2)/2 < \Delta < d/2$ can spoil the local first-law form of the modular second variation ([107]; cf. the low-Δ obstruction of Rem. J.2) — a window the physical (interacting, $d=4$ weakly-coupled) Higgs escapes: the mass operator $|H|^2$ acquires a *positive* one-loop anomalous dimension for $\lambda_H > 0$ (of order $\lambda_H/4\pi^2$), so $\Delta = 2+\gamma > d/2 = 2$. (The escape is specific to the weakly-coupled $d=4$ regime; at a Wilson–Fisher fixed point $\Delta_{|H|^2}$ can instead fall below $d/2$, back inside the window.) *Effect on the primitive count.* Granting Hyp. I.5, the matter-stress identification (EG1) and the null–null Einstein source are theorems downstream of primitives the framework *already* assumes — relative-entropy monotonicity, the half-sided modular inclusion of the Type III_1 seed net (which remains a structure of that seed; what descends to the dressed Type II crossed product is not the inclusion itself but the relative-entropy monotonicity, and the associated ANEC/QNEC inequality, which survives the boost crossing — a theorem of the ambient literature on Killing horizons [92, 95] — so the geometry-consuming inclusion itself is never invoked on the physical non-Killing background, only its monotonicity shadow), and finite averaged null energy of ω (Ceyhan–Faulkner); what remains postulated is then Hyp. I.5 together with the inversion from the null–null component to the full tensor equation. That inversion is often packaged as a holographic theorem in maximally-symmetric AdS (Oh–Park–Sin [243]), but on a *generic* diamond it decomposes cleanly and only its last step is irreducible. *(i) Trace-free part — a local theorem.* Requiring the null-null equation of state for *every* null direction through a point reconstructs the trace-free Einstein equation exactly, by the pointwise linear-algebra fact that a symmetric tensor annihilated by all null-null contractions is proportional to $g_{\mu\nu}$ (Jacobson's local-Rindler-horizon move [56]); this needs no holography, no AdS and no maximal symmetry, so the Oh–Park–Sin route is one option, not a necessity. *(ii) Promotion to a constant.* The residual scalar $c(x)$ is forced to a constant by the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ (geometric, free) together with $\nabla^\mu \langle T_{\mu\nu}\rangle = 0$ (a mild input, itself a theorem of covariant Hadamard renormalisation on a fixed background), plus the all-u^a isotropy of step (i) and a standard Hadamard/analyticity regularity condition. *(iii) Coefficient normalisation.* The fixed-*volume* equilibrium constraint fixes the correct *coefficient* of the trace part (the fixed-*radius* variation provably gives it wrong, off by $(d+1)/3$ [116]). *(iv) The irreducible seam — the trace* value. What remains is the *value* of the additive cosmological/trace constant Λ, and this is provably unreachable from null-modular data: null-null contraction annihilates $c\, g_{\mu\nu}$, so full general relativity and unimodular gravity share *identical* null-null content and differ only here, and even the correct fixed-volume derivation leaves Λ an undetermined integration constant [116]. This single choice — the dynamical status of the trace sector, general relativity versus unimodular — is a genuine extra input the null-modular substrate cannot supply, and the framework posits it separately (the free fixed-point value Λ_0 — equivalently the free trace normalisation of Rem. D.5; the single Type-II additive constant of Prop. 13.12 is the *entropy* zero-point, a different face of that same freedom, and cannot absorb an energy density — whose scaling is *posited, not computed*). The fixed-volume constraint therefore supplies the trace *coefficient*, not the trace *value*. The saturation postulate is thus strictly sharper than, and does not enlarge, the buck-stop of Ax. I.10: it *names* the $n=2$ generic-diamond residual as the QNEC-saturation equality. Moreover this equality has a natural equilibrium reading: QNEC saturation is exactly the vanishing of the *second* null-shape derivative $S''_{\mathrm{rel}} = 0$ (Leichenauer–Levine–Shahbazi-Moghaddam [93]), a second-order shape-stationarity of the monotone S_{rel} under null deformations of the entangling cut. This is, however, *strictly stronger* than the first-order modular-time (KMS) stationarity the thermal-time seed supplies: the modular equilibrium delivers only the first law ($\partial_a S_{\rm rel} = 0$ at first order) and monotonicity only the inequality ($\partial_a^2 S_{\rm rel} \geq 0$), while saturation is the second-order *equality* $\partial_a^2 S_{\rm rel} = 0$ — not implied by either, as witnessed by its generic *failure* for free-field states (a strict inequality; only the free vacuum trivially saturates) versus its holding in interacting theories — an interaction- and state-dependent property, hence not the (state-independent) KMS/thermal-time equilibrium of the seed. It therefore remains a genuine, if natural, residual, not a consequence of the framework's modular-boost equilibrium. *Band-limit status.* The finite-dimensional band-limit of Pos. 11.30 does not *evade* the Casini–Galante–Myers obstruction: it regulates the small-radius limit at $R \gtrsim 1/E_\star \gtrsim \ell_{\rm P}$, which coincides with the $R \gg \ell_{\rm P}$ floor the local Clausius derivation already requires, so the framework only ever operates where the offending $R^{2\Delta}$ term is parametrically suppressed: with the two-scale hierarchy $\mu_{\mathcal{O}} \ll \mu_{0,T} \sim E_\star$ ($\mu_{\mathcal{O}}$ the matter IR scale, $\mu_{0,T}$ the stress-tensor scale) the Casini–Galante–Myers safe condition $(\mu_{\mathcal{O}} R)^{2\Delta} \ll (\mu_{0,T} R)^d \ll 1$ holds, so at the floor $R \sim 1/E_\star$ the dangerous term is suppressed by $(\mu_{\mathcal{O}}/E_\star)^{2\Delta}$ (positive exponent). Separately, Prop. S.1 establishes that the CHM modular flow and the *first law* $\delta S = \delta\langle K\rangle$ are reproduced on the cutoff sector up to $O((E/E_\star)^\alpha)$ — i.e. at the accuracy to which the semiclassical Einstein equation is itself stated. What the band-limit does *not* separately bound is the second-variation saturation equality $\delta^2 S_{\rm rel} = \mathcal{E}_{\rm can}$ itself: that remains the named residual of Hyp. I.5, parametrically controlled at the framework's operating scales but not thereby turned into an exact identity. *Upgrade trigger:* a proof of exact QNEC saturation for the interacting SM sector promotes the $n = 2$ generic-diamond clause of Ax. I.10 from postulate to theorem. An active ambient programme (Cao–Faulkner–Wang [247]) makes the area operator an *intrinsic observable* of the gravitational algebra, with a genuine trace-entropy $S_{\rm gen} = A/4G + S_{\rm bulk}$, rather than imported from the Ryu–Takayanagi dictionary; but this settles only that the area is a well-defined observable, *not* the dynamical closure — in these constructions the area operator is still *defined* via the boost/kink generator of a fixed Einstein solution at an extremal surface, with the Einstein equations used as input. The generic-diamond dynamical closure upgrades to a theorem only if that defining area bracket is derived *from* the crossed-product structure on a generic, non-extremal, not-assumed-Einstein diamond — which no current result does; the trigger is thus intrinsic-*area*, not intrinsic-*dynamics*, and the closure remains open.

Proposition I.7 (No-go: the crossed-product/thermal-time structure cannot upgrade QNEC monotonicity to saturation)**.** *Let $\mathcal{M}$ be the Type-III$_1$ seed factor of the dressed net, with modular flow σ_t, and let $\mathcal{M} \rtimes_\sigma \mathbb{R}$ be the Type-II crossed product by which the observer clock (thermal time, Pos. 11.11) gauges σ, carrying a faithful semifinite trace and dual weight. Write the generalised entropy in the CLPW additive form*

$$S_{\rm gen} \;=\; \langle \beta\hat{q}\rangle \;+\; S_{\rm rel}(\psi\|\Omega) \;+\; c_0,$$

with $\langle\beta\hat{q}\rangle$ the observer/clock (area) sector and c_0 the state-independent additive constant ([18]; cf. Ax. I.10). Then the QNEC-saturation equality of Hyp. I.5 — the vanishing of the nonnegative saturation defect $\mathcal{D} := \delta^2 S_{\rm rel} - \mathcal{E}_{\rm can} \geq 0$ — is not implied *by any property of the trace, the dual weight $e^{\beta q}$, the tracial (maximum-entropy) state, the thermal-time identification σ_t = physical time, or the two-boundary seed*-state *choice.*

Proof. Two steps. *(i) Orthogonality.* The crossed-product data enters $S_{\rm gen}$ additively through the independent clock sector $\langle\beta\hat{q}\rangle$ and leaves the matter relative entropy $S_{\rm rel}$ — hence its second null-shape variation $\delta^2 S_{\rm rel}$ and the defect $\mathcal{D}$ — equal to its bare Type-III value, computed by the seed modular data alone. No trace, weight, or tracial-state property can therefore constrain $\mathcal{D}$. *(ii) Free-field witness.* The entire construction is available to *free*-field matter: the free wedge algebra is Type III$_1$ (Bisognano–Wichmann/Araki) and CLPW build the Type II$_1$ trace and maximum-entropy state on the free graviton-plus-matter system [18]. For free fields the defect is *provably nonzero*, $\mathcal{D} \sim \langle\partial_v\phi\rangle^2 \neq 0$ — the square of the *coherent field amplitude* (the one-point function), not a fluctuation — [93]. Hence any implication "crossed-product data $\Rightarrow \mathcal{D} = 0$" would force free-field saturation, contradicting [93]; by modus tollens no such implication exists. *Thermal time* is an interpretive relabelling of σ (free fields carry it); the half-sided modular inclusion supplies the monotonicity *sign* $\mathcal{D} \geq 0$, not its vanishing (the free null-cut net has the inclusion yet $\mathcal{D} \neq 0$); and the two-boundary seed selects a *state*, injecting into $\langle \beta\hat{q} \rangle$, not into the spectral data controlling $\mathcal{D}$. □

Remark I.8 (What the no-go localises, and the residual it leaves). Prop. I.7 converts the saturation residual from a bare admission into a sharp localisation. The framework robustly *owns* (a) the monotonicity inequality $\mathcal{D} \geq 0$ (half-sided modular inclusion positivity / the generalised second law) and (b) the clock/area sector $\langle \beta\hat{q} \rangle$; the *single* missing ingredient for the $n = 2$ generic-diamond closure of Ax. I.10 is a *twist gap* on the seed's correlation data — the absence of non-identity light-ray (defect) operators of twist $\tau \leq d - 2$ besides the stress tensor, equivalently the anomalous-dimension washout of the minimal-twist higher-spin tower whose replica contact term is the $\langle \partial_v \phi \rangle^2$ obstruction ([94]; the interacting-CFT saturation of Rem. I.6). Two consequences. *First*, this input is interaction-dependent and *provably not derivable* from the crossed-product / thermal-time / two-boundary structure (Prop. I.7): adopting it *is* the twist-gap theorem of [94], so the "derive saturation from the framework's own algebra" route collapses into the interacting/holographic routes, and the algebraic apparatus contributes only the inequality and the area sector, never the equality. *Second*, the framework-internal alternative — imposing $\langle \partial_v \phi \rangle^2 = 0$ on the seed via the *final* boundary condition — is both circular (it handpicks a saturating configuration) and insufficient (the equation of state requires $\mathcal{D} = 0$ in *every* state [93], whereas state-selection reaches at most a measure-zero set). The honest residual is therefore precise: saturation for the framework's matter is neither a structural theorem nor a state-selection artefact but a *spectral (twist-gap) property of the two-boundary seed's correlation functions*, whose independent justification within the present axioms is the genuine open problem. This localisation rested on one structural fact — that the two-boundary condition injects only into $\langle \beta\hat{q} \rangle$ because it selects a *state* and not the operator *spectrum* — which we now *discharge* rather than assume. The two-boundary axioms (Ax. 2.1, Ax. 2.2, Conj. 10.13) are *matter-agnostic*: they specify an initial state ρ_i and terminal effects $\{E_k\} \subset \mathcal{N}$ and reference no twist, OPE, or operator-dimension data, and a terminal effect $E_k \in \mathcal{N}$ reweights outcomes on the *fixed* algebra $\mathcal{N}$ — it cannot delete operators from it. The free-field parity test upgrades this from presumption to theorem: a free-field TBRME carries the *identical* two-boundary data, yet $\mathcal{D} \neq 0$ is proven there [93]; were the two-boundary condition to *supply* the twist gap it would equally — and falsely — supply it for free matter, where no gap exists to be found. Hence the condition provably selects a state, and the one framework-internal route to $\mathcal{D} = 0$ that remains, imposing $\langle \partial_v \phi \rangle^2 = 0$ on the future cut via a terminal effect, is a state-selection reaching at most the measure-zero mean-field-free configuration ($\langle \partial_v \phi \rangle = 0$ on the cut, whereas the generic slowly-varying macroscopic states the framework operates in have $\langle \partial_v \phi \rangle \neq 0$), not the every-state equality the equation of state requires. The no-go's single assumption is thereby closed: no property of the crossed product, thermal time, *or the retrocausal two-boundary condition* supplies the saturation equality; the twist gap remains an un-derived spectral input, and that — not any residual ambiguity about what the two-boundary condition can do — is the genuine open problem.

Remark I.9 (Regime relaxation: the linearised equation of state closes in-regime; the nonlinear residual sharpens but does not close). The framework uses the Einstein equation only as a *macroscopic* equation of state, on diamonds of radius $R \gtrsim 1/E_\star$ (Pos. 11.30) in slowly-varying near-vacuum states, so one might hope it needs saturation only *effectively* — $\mathcal{D} \to 0$ to the order the equation is stated — rather than as the exact identity $\mathcal{D} = 0$. An adversarial analysis of this relaxation splits it cleanly across the two variational orders. *Linearised branch* (n=1)*: closes in-regime, and needs no saturation.* The linearised semiclassical Einstein equation follows from *first-order* relative-entropy stationarity $\delta S_{\rm rel} = 0$ — the first law, a theorem (Thm. I.1, Cor. J.3) — into which saturation never enters. Its only obstruction, the Casini–Galante–Myers first-law danger term, is parametrically suppressed by $(\mu_{\mathcal{O}}/E_\star)^{2\Delta}$ (the Higgs escaping the window with $\Delta = 2 + \gamma > d/2$, Rem. I.6), and the modular flow and first law are reproduced on the cutoff sector to $O((E/E_\star)^\alpha)$ (Prop. S.1), the accuracy the semiclassical equation is itself stated at. This branch is *genuinely discharged in-regime. Nonlinear branch* (n=2)*: sharpens, does not close.* The framework's actual route (Hyp. I.5) is the *second*-order equality $\delta^2 S_{\rm rel} = \mathcal{E}_{\rm can}$, and here the relaxation does *not* help, for a structural reason: the saturation defect enters the smeared second variation at the *same* power of the smearing scale ℓ^{d-2} as the Einstein source $2\pi\langle T_{vv}\rangle$, so $\mathcal{D}/\langle T_{vv}\rangle = O(1)$ at *every* wavelength — it carries no relative power of R and is not suppressed by operating at large R [93]. Indeed the free-field defect $\mathcal{D} \sim \langle \partial_v \phi\rangle^2$ is the square of the *coherent field amplitude*, which is *largest* in exactly the slowly-varying macroscopic states the framework uses and *survives* coarse-graining (which suppresses short-distance fluctuations, not the classical mean field). What the relaxation does achieve is to weaken the requirement from the exact identity to the parametric bound

$$\frac{\mathcal{D}}{\langle T_{vv}\rangle} \lesssim (\mu_{\mathcal{O}} R)^{2\gamma} \qquad \text{at } R \sim 1/E_\star,$$

whose conformal proxy is a strictly positive minimal-twist anomalous dimension $\gamma > 0$ — the twist gap of [94] that Prop. I.7 shows the framework cannot supply. An adversarial analysis of this reduced condition against the *physical* Standard Model, however, forbids reading it as a closure, on three levels we record honestly. *(i) The sign is proven, but only in a CFT.* In a genuine interacting conformal theory the sole exactly-conserved twist-$(d-2)$ operators are the stress tensor and conserved spin-1 currents; every broken higher-spin current is lifted to $\gamma_s > 0$ (Coleman–Mandula; Maldacena–Zhiboedov; the explicit weak-coupling $\gamma_s \sim g^2 \log s$ of Gross–Wilczek). *(ii) The condition is ill-posed for the actual SM.* The twist gap, the minimal-twist γ, and the very "light-ray spectrum" are conformal constructs (defined through $SO(d,2)$ representation theory and proved to control saturation only for CFTs, [94]); the physical Standard Model is *not* conformal — mass thresholds, a confining infrared with a mass gap, running couplings, no fixed point — so there is no global dilatation operator whose spectrum γ would label, and below $\Lambda_{\rm QCD}$ the classification does not exist at all. The residual is therefore not a clean "$\gamma > 0$" condition but the *existence of a non-conformal generalisation of the saturation-versus-twist-gap mechanism* for a massive, confining, running theory — which the literature does not currently supply. *(iii) Even granting the conformal proxy, the suppression is inadequate in-regime.* The factor $(\mu_{\mathcal{O}} R)^{2\gamma}$ depends on the enormous $\mu_{\mathcal{O}}/E_\star$ hierarchy only *logarithmically*, so a realistic weak-coupling $\gamma \sim 10^{-2}$ merely halves the $O(1)$ defect rather than suppressing it; and at the operating floor $R \sim 1/E_\star$ asymptotic freedom drives the minimal-twist $\gamma \to 0$, so the suppression $\to 1$ exactly where the framework needs it — approaching the free-field limit [93] that provably fails to saturate. The honest net is a clean split with a sharply-drawn residual: the *linear* gravitational response is a theorem in-regime; the *nonlinear* response is postulated, and its residual is not one positive-exponent condition away from closure but a genuinely open problem — the formulation of a non-conformal saturation mechanism for the Standard Model, of which the CFT twist gap is only the (in-regime inadequate) conformal-limit shadow.

Nonlinear completion as single clean axiom: (E0-nl)

Axiom I.10 (Nonlinear Clausius closure of the area functional, Einstein branch)**.** Fix a reference pair (g_0, ω_0) with ω_0 Hadamard. For every local causal diamond $\mathcal{O} \subset (\mathcal{M}, g)$ with Hadamard state ω in a neighbourhood of (g_0, ω_0), the area functional

$$A \,:\, (g, \omega) \longmapsto \mathrm{Area}(\partial\mathcal{O}; g, \omega)$$

is Fréchet-differentiable to all orders under simultaneous variation of (g,ω), and satisfies the *area–entropy identity*

$$\frac{\delta^n A[\partial\mathcal{O};\,(g,\omega)]}{4G\hbar} \;=\; \delta^n\langle K_{\omega_0}\rangle_\omega \;-\; \delta^n S_{\mathrm{rel}}(\omega\,\|\,\omega_0) \qquad \text{for every } n\geq 1, \tag{124}$$

with a single state-independent additive constant $c_0(\omega_0)\in\mathbb{R}$ at order zero (the species-independent UV renormalisation of the entanglement area law into G [59, 56]), where S_{rel} is the Araki relative entropy on the diamond algebra. The right-hand side is organised by the *exact* identity $\Delta S = \Delta\langle K_{\omega_0}\rangle - S_{\mathrm{rel}}$ (sign convention: $S_{\mathrm{rel}}\geq 0$, so the term $-\delta^n S_{\mathrm{rel}}\leq 0$ at $n=2$ enters as an area *deficit* against the free modular energy $\delta^n\langle K_{\omega_0}\rangle$), so (124) is the statement that the Bekenstein area functional computes the *finite variations* of the diamond entanglement entropy at every Fréchet order. (The all-orders reading "computes the entanglement entropy itself" would double-count against $S_{\mathrm{gen}} = A/4G + S_{\mathrm{out}}$; it is the order-zero split carrying the single state-independent constant $c_0(\omega_0)$ that separates the fixed UV area-law piece from the state-dependent *variations* tracked by (124), so only the variations are asserted.) Its low orders are *not* free:

(i) at $n=1$, $\delta S_{\mathrm{rel}}=0$ (first-order stationarity of relative entropy) and (124) reduces to the first law $\delta A/4G\hbar = \delta\langle K_{\omega_0}\rangle$, the content discharged at linear order by Thm. I.1;

(ii) at $n=2$, the deficit term is the manifestly *state-dependent*, quadratic Hollands–Wald canonical energy of the perturbation,

$$\delta^2 S_{\mathrm{rel}}(\omega\,\|\,\omega_0) \;=\; \mathcal{E}_{\mathrm{can}}[\delta g,\delta\omega] \;\geq\; 0 \qquad [17,\,106], \tag{125}$$

strictly positive on perturbations carrying graviton or bulk-matter flux; no state-independent constant can stand in for it. *Scope of the $n=2$ identity.* As an equality of $\delta^2 S_{\mathrm{rel}}$ with the bulk canonical energy, (125) is a *theorem* only in the holographic / Ryu–Takayanagi setting, where the diamond is an entanglement wedge and S_{rel} is computed by the RT/HRT prescription: it is the second-order relative-entropy = canonical-energy identity of [106], with $\mathcal{E}_{\mathrm{can}}$ the Hollands–Wald bulk canonical energy [17]. On a *generic* (non-holographic) local causal diamond $\mathcal{O}\subset(\mathcal{M},g)$ the same equality is imposed as a *postulate* — the $n=2$ instance of the buck-stop of Rem. I.12, named as Hyp. I.5 — namely that the modular second variation is exhausted by the canonical energy with no additional generic-diamond term; the framework asserts it, it is not derived at generic scope;

(iii) for $n\geq 3$ the identity is the conjectural Einstein-branch completion (the buck-stop below), likewise a theorem only on the holographic/RT scope and a postulate on the generic diamond.

Remark I.11 (Corrective note on the previous formulation of Ax. I.10). An earlier formulation of this axiom asserted $\delta^n A = \delta^n\langle K_\omega\rangle + c_n(\omega_0)$ for $n\geq 2$ with *state-independent* constants $c_n(\omega_0)$. That statement is retracted: at $n=2$ it contradicts the canonical-energy structure (125) established by Hollands–Wald [17] and, holographically, by the axiom's own cited evidence [106] ($\delta^2\langle K\rangle - \delta^2 S = \mathcal{E}_{\mathrm{can}}$ is quadratic in the perturbation, hence state-dependent, and positive precisely on the flux-carrying perturbations a constant cannot track). The corrected form (124) carries the relative-entropy deficit explicitly at every order; the state-independent freedom is confined to the single order-zero constant $c_0(\omega_0)$, where it is true (UV area-law renormalisation).

Remark I.12 (Content and buck-stop). Ax. I.10 packages the following into one Fréchet-analytic condition on one functional: (a) nonlinear Jacobson (Einstein on dynamical backgrounds as an equation of state at all orders); (b) the nonlinear extension of FGHMVR: [106] establishes the $n=2$ case (125) holographically (second-order relative entropy = bulk canonical energy in the entanglement wedge), and Lewkowycz–Parrikar [383] give a *conditional* consistency argument — *if* the Ryu–Takayanagi formula holds for all subregions and bulk and boundary modular flows coincide (JLMS), *then* shape deformations of the modular Hamiltonian reproduce the nonlinear bulk Einstein equations in the Hollands–Iyer–Wald formalism. We stress this is a consistency statement conditional on RT-for-all-subregions + bulk=boundary modular flow, *not* an independent $n \geq 3$ derivation of Einstein from entanglement; it strengthens the holographic *evidence* for Ax. I.10 but does not close it. A clean all-orders Fréchet identity remains beyond what is rigorously established, and the buck-stop survives for the generic-diamond $n \geq 2$ and all non-holographic / non-RT sectors; (c) positivity and monotonicity of the deficit term: relative entropy positivity gives the *inequality* $\delta^2 S_{\rm rel} \geq 0$ (the stability/quantum-Fisher content of (125), the convexity statement adjacent to the quantum focussing conjecture at $n = 2$) — but the Einstein branch needs the *equality* (saturation) of Hyp. I.5, which monotonicity alone does not supply: the inequality is what the framework already has via the Ceyhan–Faulkner averaged null energy route, the equality is the named residual. This positivity is, moreover, *confined to* n=2: the deficit coefficients obey the exact parity identity $\mathrm{sign}(\delta^n S_{\rm rel}) = (n{-}1)(-1)^n\,\mathrm{sign}(\kappa_n)$, with κ_n the connected Kubo–Mori cumulants of the modular generator, so for the physically-stable interacting scalar ($\kappa_n > 0$, robust across the massive window) the odd orders are *negative* ($\delta^3 S_{\rm rel} < 0$) — the double zero $\delta^0 S_{\rm rel} = \delta^1 S_{\rm rel} = 0$ of the nonnegative $S_{\rm rel}(\lambda)$ protects exactly the one curvature inequality $\delta^2 S_{\rm rel} \geq 0$ (the Kubo–Mori metric) and no more. The all-orders dissipative correction of (d) is therefore a *signed*, order-alternating quantity beyond second order, not a monotone one; (d) the operator-ordered Clausius identity of Pos. 11.18 at all orders, with the dissipative correction now carried explicitly by $\delta^n S_{\rm rel}$ rather than absorbed into a constant.

Buck-stop. Ax. I.10 is not derivable from the present paper's other postulates, nor from currently-known ambient-literature results, beyond the linearised case closed by Thm. I.1. Its full derivation from entanglement thermodynamics on a generic non-holographic matter sector is co-extensive with the unresolved programme of deriving quantum gravity from entanglement [56, 116, 59, 106, 117, 108]; the present paper takes this axiom as the single remaining primitive matter–geometry coupling postulate of the framework (absent Hyp. I.5; under it the residual shrinks further to the saturation equality, Rem. I.13) and makes no claim to derive it internally.

Two scope restrictions are explicit in the Einstein-branch form of (124). First, it presupposes the *area-entropy (equilibrium) branch* of the Clausius hierarchy: by Eling–Guedens–Jacobson [109] and Chirco–Liberati [110], a nonzero internal entropy-production term $d_i S$ is generically required for consistency with the contracted Bianchi identity once the horizon entropy depends on curvature invariants, so Ax. I.10 commits the framework to the Einstein (area-entropy) branch and excludes $f(R)$/higher-curvature equations of state; in the corrected form the reversible deficit is the relative-entropy term itself, and what the Einstein branch asserts is that *no additional* production term beyond $\delta^n S_{\rm rel}$ appears. Second, this branch commitment is *formulation-dependent* in the ambient thermodynamic-gravity literature: Pai, Namboothiri and Mathew [111] show that the Rindler-horizon and apparent-horizon formulations of horizon thermodynamics assign the entropy-production term different roles (Bianchi-consistency bookkeeping in one, dynamical source in the other), so "equilibrium branch" is a choice of thermodynamic frame rather than a derived fact. Promoting Ax. I.10 to a theorem would require either proving $d_i S = 0$ in the present modular setting — in a specified frame — or carrying the production term explicitly.

Remark I.13 (Replacement of (E0) by (E0-lin) theorem + (E0-nl) axiom). Under App. I, Pos. 11.18 is replaced by the two-piece scope (Thm. I.1, Ax. I.10). The paper's residual internal content is then *one* clean axiom Ax. I.10 rather than the full Jacobson–Bianchi–Myers identification. This is the strongest honest form (E0) takes today: its linearised content is a holographic theorem of the ambient literature; its nonlinear completion is a single Fréchet area–entropy axiom with an explicit buck-stop remark; and its beyond-AdS (cosmological) extension is the named open hypothesis Hyp. I.3 with explicit upgrade and kill triggers (Rem. I.4). A further reduction is available *within* this scope: under the saturation hypothesis Hyp. I.5, the first-law content (EG1) and the null–null Einstein source are themselves theorems downstream of the framework's already-assumed relative-entropy monotonicity (the half-sided modular inclusion remaining a structure of the Type III$_1$ seed, Rem. I.6), so the residual matter–geometry primitive shrinks to the single QNEC-saturation *equality* on the generic diamond (an inequality-versus-equality gap, Rem. I.6), with the standing upgrade trigger of Cao–Faulkner–Wang [247]. What (E0) postulates, in its sharpest form, is therefore not "the Einstein equation" but this one saturation equality: the statement that the modular second variation is exhausted by the gravitational canonical energy with no generic-diamond production term.

J Standard Model accommodation: Clausius–SM compatibility and spectral-triple consistency

This appendix shows that the Standard Model matter content (SU(3)×SU(2)×U(1) with three fermion generations, chirality, anomaly cancellation, and the Higgs mechanism) is accommodated by the framework of Parts I–III *without a new axiom*: the three load-bearing postulates (Pos. 11.11, Pos. 11.18, Pos. 11.30) are stated at a level of generality (type III$_1$ local net, Hadamard state, globally hyperbolic $(\mathcal{M}, g)$) that the SM content drops into verbatim, once the ambient locally-covariant pAQFT machinery is invoked.

Locally covariant pAQFT for the SM

The ingredients are by now classical:

- Dirac/Weyl fields on globally hyperbolic Lorentzian manifolds: the locally covariant quantization of [220], extended to Fermi fields by Sanders, Verch, and Brunetti–Dütsch– Fredenhagen–Rejzner [265].
- Renormalized Yang–Mills on curved spacetime: Hollands [264], in the locally covariant BV framework.
- BV/BRST quantization of Higgs-type scalar sectors: Fredenhagen–Rejzner [266].

Together these deliver, on any globally hyperbolic $(\mathcal{M}, g)$ and for any Hadamard seed state ω, a type III$_1$ local net $\mathcal{A}_{\mathrm{SM}}(\mathcal{O})$ carrying the full SM gauge and matter content. Each ingredient is a theorem of ambient pAQFT (see the survey [272]); the paper imports them.

Anomaly structure and the Clausius correction

Theorem J.1 (Clausius–SM compatibility, parity-even sector with explicit Higgs scope)**.** *Let $\mathcal{A}_{\mathrm{SM}}(\mathcal{O})$ be the SM local net in pAQFT on globally hyperbolic $(\mathcal{M}, g)$ with Hadamard state ω and modular Hamiltonian $K_\omega = -\log \Delta_\omega$. Assume:*

(SM1) *The 't Hooft anomaly cocycle $[\mathcal{A}_{\mathrm{SM}}] \in H^4(\mathrm{B}G, \mathbb{Z})$ for $G = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ vanishes on the full three-generation content (standard SM hypercharge assignment; gauge anomaly cancellation per generation is exact).*

(SM2) *The matter-sector chiral anomalies are taken in the* parity-even *sector only: the trace anomaly is the Duff $a\, E_4 + c\, W^2$ combination assembled from [304]-renormalised $\langle T^\mu_\mu \rangle$ in interacting QFT on curved spacetime, with coefficients summed over SM field content. The parity-odd Pontryagin contribution to the Weyl-fermion trace anomaly is* set to zero*, following Bastianelli–Broccoli and Fröb–Zahn [301] (Hadamard / path-integral regularisation); the surviving controversy with the Bonora et al. result [302] is recorded as an open ambient-literature item below.*

(SM3) *In the perturbative-Higgs-VEV expansion (BFR pAQFT BV-BRST framework [266]) the Higgs VEV v back-reacts on the modular generator as $K_\omega \to K_\omega + (v^2/E_{\mathrm{P}}^2)\, \delta K + \mathcal{O}(v^4/E_{\mathrm{P}}^4)$, with δK a state-dependent local boost-density shift.*

Then there exists a renormalisation scheme in which the heat flux across a Rindler wedge W_p with bifurcation surface $B \subset \partial W_p$ obeys the modified Clausius identity

$$\delta Q \;:=\; \int_B \langle \widehat{T}^{\mathrm{SM}}_{ab} \rangle_\omega \, \chi^a \, d\Sigma^b \;=\; T \, dK_\omega \;+\; \hbar \, \mathcal{A}^{\mathrm{even}}_{\mathrm{grav}}[g] \;+\; \mathcal{O}(v^2/E_{\mathrm{P}}^2), \tag{126}$$

where $T = (2\pi)^{-1}$ is the Unruh temperature, and $\mathcal{A}^{\mathrm{even}}_{\mathrm{grav}}[g]$ is the parity-even, state-independent *pure-gravity trace/ conformal anomaly density (local in curvature invariants). The $\mathcal{O}(v^2/E_{\mathrm{P}}^2)$ Higgs back-reaction shift on both sides of* (126) *is matched, so the Clausius identification holds at leading order in v/E_{P}.*

Proof sketch. The Fröb–Zahn Weyl-anomaly theorem [304] establishes the rigorous state-independent trace anomaly in interacting QFT on curved spacetime; its parity-even sector decomposes as $a\,E_4 + c\,W^2$ with coefficients depending on field content and is state-independent local in curvature. The Hollands–Wald free- field result [16] is recovered as the free limit. SM gauge anomaly cancellation per generation (SM1) is standard Bilal/Tong material; with parity-odd Pontryagin set to zero per (SM2), the matter side of the Clausius identity carries only the a- and c-anomaly. The Bianchi–Myers extension [59] of Jacobson's equation of state, applied here only to the *higher-curvature gravity entropy* of Wald type (its proper domain), gives (126)'s $\mathcal{A}^{\mathrm{even}}_{\mathrm{grav}}[g]$ as the corresponding gravity-side response. The Higgs-VEV back-reaction (SM3) is the BFR pAQFT computation of σ^{ω}_t on the Higgs-broken vacuum to leading order in v; both δQ and $T\,dK_\omega$ receive matched $\mathcal{O}(v^2/E_{\mathrm{P}}^2)$ corrections, so the identification survives at leading order. □

Remark J.2 (Explicit residual scope of Thm. J.1). The refined Thm. J.1 explicitly does *not* cover three items of the ambient literature:

- (i) The parity-odd Pontryagin contribution to the Weyl-fermion trace anomaly: contested between [302, 303] (non-vanishing) and [301] (zero by Hadamard/ path-integral). This is a live ambient open problem; the paper takes the (SM2) regularisation choice and flags the alternative.
- (ii) Higgs-VEV back-reaction beyond leading order $\mathcal{O}(v^4/E_{\mathrm{P}}^4)$ in the BFR pAQFT framework. No published result; would be a short pAQFT companion paper.
- (iii) Survival of the CLPW Type II crossed product when $\mathcal{N}$ is the *interacting* SM type III_1 factor rather than free scalar. Witten's background-independent algebra programme [305] and 2025 crossed-product extensions are suggestive but not a proof; remains explicitly open.
- (iv) The Casini–Galante–Myers low-dimension obstruction to the underlying first law. [107] showed that for relevant operators of low conformal dimension the ball-by-ball identification $\delta S_B = \delta\langle K_B \rangle$ used in the Jacobson-type derivation (and hence in Thm. I.1 and its SM corollary Cor. J.3) can fail; they also give conditions to circumvent it. Since the SM matter sector is non-conformal (massive/Higgs deformations, App. J), the linearised (E0-lin) discharge for SM matter holds only modulo their circumvention conditions. This obstruction is folded into the buck-stop of Ax. I.10.

Corollary J.3 ((E0-lin) with SM matter: scalar-Clausius form)**.** *Under Thm. J.1, the horizon-projected scalar component of Thm. I.1 (linearised Jacobson) holds for SM matter, augmented by the state-independent anomaly shift of* (126). *In particular: the horizon-projected component $\delta G_{\mu\nu}\chi^\mu\chi^\nu$ of the linearised Einstein equation is derived as a scalar Clausius identity from the first law of entanglement for SM matter, with chirality and gauge content contributing only through an additive c-number renormalization. The full tensor equation is NOT derived: it is pending the cut-localised first law (E0-dS′) on the de Sitter wedge orbit (Hyp. E2a.4, App. E2a; at $n \geq 1$ this is the B1a wall, and FRW lacks the wedge orbit altogether — Rem. I.4), leaving the SM cosmological sector on the postulate form of (E0).*

Corollary J.4 (Exact n=0 pointwise null–null equation of state for SM matter, with mass-conserving interacting tail — the leading-order closure of the (N1) residual)**.** *Work in the setting of*

Thm. J.1: the SM local net $\mathcal{A}_{\rm SM}(\mathcal{O})$ in pAQFT (BFR BV–BRST) on globally hyperbolic $(\mathcal{M}, g)$, Hadamard state ω of finite relative entropy, hypotheses (SM1)–(SM3) *in force, on the weak-field region of Cor. J.5 ($GM/Rc^2 \ll 1$), with the SM held in the Casini–Galante–Myers-safe window (weakly coupled $d = 4$; the physical Higgs at $\Delta = 2 + \gamma > d/2$, Rem. I.6). Expand every interacting object as a formal power series in the SM couplings. Then the* exact *n=0 pointwise null–null equation of state*

$$\delta G_{\mu\nu}\, k^\mu k^\nu \;=\; \frac{8\pi G}{c^4}\, \delta\langle \widehat{T}^{\rm SM,(0)}_{\mu\nu}\rangle_\omega\, k^\mu k^\nu \;+\; \mathcal{R}_{\geq 1}(x;k), \qquad \int_{S_r} \mathcal{R}_{\geq 1}\, dS = 0 \;\;(r \geq R) \tag{127}$$

holds along every *null direction k through* every *point x of the weak-field region,* uniformly *in (x, k), at first order in δ, where:*

(i) *the equality $\delta G\, kk = (8\pi G/c^4)\, \delta\langle \widehat{T}^{\rm SM,(0)}\rangle_\omega\, kk$ is* exact *at n=0 — Bisognano–Wichmann gives the wedge modular Hamiltonian as the boost charge and Longo/Ciolli–Longo–Ruzzi give the free first law, the free two-state relative modular operator genuinely existing (PIECE 2 (1)); the interacting tail is carried by $\mathcal{R}_{\geq 1}$,* not *placed on the load-bearing RHS;*

(ii) *the remainder $\mathcal{R}_{\geq 1}$ is the entire interacting tail $n \geq 1$. It is* mass-conserving*: it integrates to zero enclosed-mass change through any sphere S_r, $r \geq R$ (App. J, Step 4), so it leaves $g_{\rm N} = GM/R^2$ exactly invariant.* (*Its purely gravitational-nonlinear and non-Killing parts are additionally* $(GM/Rc^2)^2$*-suppressed with an $\mathcal{O}(1)$ constant $C_{\rm EFT}$, App. J, Step 3, Earth: $\approx 5 \times 10^{-19}$; the interacting-matter part is controlled by mass conservation, not that power counting*)*;*

(iii) *the uniformity constant of the n=0 equality depends only on $(g, \rho, \varepsilon, K, \{m_{\rm SM}\})$ and the CGM-window data, not on (x, k) (direction-uniformity is kinematic, App. J; point-uniformity holds by the free-field template of Prop. E.4 at a single order-independent Sobolev threshold).*

Consequently the linearised tensor equation (123) *with the n=0 SM source follows on the weak-field region* at leading order*, through the matter-agnostic inversion of Rem. I.6 (steps (i)–(iii)), the trace* value Λ *remaining the irreducible measured seam (step (iv)), and $g_{\rm N} = GM/R^2$ follows by the shell-theorem mass-conservation step (App. J, Step 4), which routes the whole interacting tail into the measured M. The n=1 interacting-matter first-law* identity *— that the interacting boost charge equals $2\pi \int x^1 \delta\langle T^{\rm SM,(1)}_{00}\rangle$ — is* not *used here; it is the interacting boost–Noether identity* (R-i)*, exact at n=0 but open at $n \geq 1$ (theorem-modulo-*(R-i)*; the leading slice of the named companion problem* B1a*). Thus the leading-order part of premise* (N1) *of Cor. J.5 needed by the deliverable is a theorem; the residual* (N1) *is the all-orders pointwise* ($\star$)*, which reduces to* B1a *(Rem. J.6(b)) and is inert at planetary field strength.*

Proof sketch. Four ingredients, each in print or kinematic. The load-bearing pointwise RHS carries only the exact n=0 term; the whole interacting tail is routed into the mass-conserving remainder $\mathcal{R}_{\geq 1}$, so the $g_{\rm N} = GM/R^2$ deliverable needs neither an interacting all-orders object nor the n=1 interacting first-law identity (R-i).

(1) The n=0 leg is exact. At zeroth interacting order the SM matter reduces to its free (Gaussian) content, for which the wedge modular Hamiltonian is *exactly* the boost generator $2\pi \int_B x^1 T_{00}$ by Bisognano–Wichmann, mass-independent to all orders, and the first-law identity $\delta S_{\rm rel} = \delta\langle K\rangle = 2\pi \int_B x^1\, \delta\langle T_{00}\rangle$ is the Araki/Ciolli–Longo–Ruzzi relative-entropy statement [228], exact at n=0. No interacting relative modular operator is invoked at this order, so the type-obstruction of Rem. J.2(iii) is not yet active.

(2) The n=1 correction: the gravitational dressing *leg is closed on Killing loci; the interacting-matter first-law* identity *is theorem-modulo-(R-i) and is kept off the load-bearing RHS.* The Faulkner–Speranza finite-cut audit [92] shows the finite-cut modular Hamiltonian approaches the flat boost form with an $\mathcal{O}(\varepsilon^1)$ correction that *vanishes by explicit integration* (their eq. 5.25,

$\varepsilon \sim g'/g$ the crossed-product dressing parameter), so on Killing horizons the n=1 *dressing* correction vanishes. This is a distinct, orthogonal channel from the interacting-matter first law: the n=1 interacting-matter stress tensor $\langle \widehat{T}^{\mathrm{SM},(1)} \rangle$ is a well-defined Drago–Pinamonti–Rejzner / Fröb–Zahn renormalised object [304] (*it exists*), but the identity that would place it on the pointwise RHS as boost flux — $\delta K_\omega^{(1)} = 2\pi \int x^1 \delta \langle T_{00}^{(1)} \rangle$, the n=1 slice of the interacting boost–Noether identity (R-i) — is *open* (exact at n=0/Longo; at $n \geq 1$ the leading slice of B1a). Thm. J.1 delivers only the abstract-K_ω scalar Clausius identity, not this identification. Accordingly the n=1 interacting-matter term is *not* on the load-bearing RHS of (127); it sits in the mass-conserving remainder $\mathcal{R}_{\geq 1}$ (routed into the measured M by ingredient (4)), and where recorded as a first-law clause it is graded theorem-modulo-(R-i). On generic non-Killing weak-field patches the residual n=1 dressing correction is $\mathcal{O}(GM/Rc^2)$ and likewise enters $\mathcal{R}_{\geq 1}$.

(3) Direction- and point-uniformity at the truncated order. Direction-uniformity is kinematic and matter-independent: the transverse projector spectrum is $\{0, 0, 1, 1\}$ on the celestial sphere, and both the 2π-Unruh temperature and $8\pi G/c^4$ are SO(d−1)-scalars (App. J). Point-uniformity holds at leading order by the free-field extraction template of Prop. E.4: at each fixed order the UV singularity structure is the order-independent free Hadamard parametrix (interacting vertices sit at distinct interior points), so "ball-uniform at each order" reduces to a single order-independent finite-regularity threshold, reused at n=0, 1 — no new input per order.

(4) The interacting tail is inert on g_N by mass conservation, and its gravitational part is EFT-suppressed. The remainder $\mathcal{R}_{\geq 1}$ is the whole interacting tail $n \geq 1$. Its control is twofold. *(4a)* By Newton's shell theorem the entire tail — interacting-matter n=1 term included — integrates to zero enclosed-mass change for $r \geq R$: the $n \geq 1$ corrections to $\langle \widehat{T}_{00}^{\mathrm{SM}} \rangle$ are covariant functionals whose spatial integral is the correction to the total gravitating mass, fixed to the *measured* M (already inclusive of all interacting binding and self-energy), so $g_\mathrm{N} = GM/R^2$ is exactly untouched (App. J, Step 4). *This is why the deliverable never needs (R-i). (4b)* The purely gravitational-nonlinear and non-Killing parts are, in addition, $(GM/Rc^2)^2$-suppressed with an $\mathcal{O}(1)$ prefactor C_EFT (App. J, Steps 2–3). The $\mathcal{O}(v^2/E_\mathrm{P}^2)$ Higgs back-reaction of (SM3) is matched on both sides of (127) exactly as in (126) and does not enter the leading balance.

Combining (1)–(4): the per-wedge scalar identity of Cor. J.3 is promoted, *at n=0*, to the pointwise, every-direction, uniform equality (127), and $g_\mathrm{N} = GM/R^2$ follows because the interacting tail is mass-conserving. The n=1 interacting-matter first-law identity is not used (it is (R-i), theorem-modulo-(R-i)). The full bookkeeping — the exact form of C_EFT, the shell-theorem mass conservation covering the interacting tail, and the order-independent uniformity threshold — is App. J. □

The scalar-Clausius form above is exactly one conditional link short of the classical-recovery statement a unification owes the reader. We record that statement with its single lever named, rather than leave it implicit.

Corollary J.5 (Conditional Newtonian limit: classical recovery for terminally decohered bodies, modulo the linearised E0 tensor residual)**.** *Grant the two declared inputs:*

(N1) (Linearised tensor completion for SM matter — the named residual.) *On the weak-field region of interest, the per-wedge scalar-Clausius identity* (126) *(Thm. J.1, Cor. J.3) upgrades to the* pointwise *null–null equation of state $\delta G_{\mu\nu} k^\mu k^\nu = (8\pi G/c^4)\, \delta \langle \widehat{T}_{\mu\nu}^{\mathrm{SM}} \rangle_\omega k^\mu k^\nu$ along* every *null direction k through* every *point — precisely the content Cor. J.3 records as underived for interacting massive SM matter (pending the cut-localised first law (E0-dS′) on the wedge orbit, Hyp. E2a.4; Rem. I.4). By the inversion steps of Rem. I.6 — (i) pointwise trace-free reconstruction [56], (ii) promotion of the residual scalar to a constant via contracted Bianchi plus covariant conservation (with the isotropy and regularity inputs stated there), (iii) fixed-volume normalisation of the trace coefficient [116] — this yields the linearised tensor equation in the form* (123) *with source $\delta \langle \widehat{T}_{\mu\nu}^{\mathrm{SM}} \rangle_\omega$ on that region, the trace* value Λ *remaining undetermined (the irreducible seam, step (iv)). At linearised* *order this input is strictly weaker than Hyp. I.5, which implies it on generic diamonds but whose $n=2$ saturation content the first-law branch does not consume (Rem. I.9); the low-dimension first-law caveat is inherited unchanged from Rem. J.2(iv).*

(N2) (One-time dimensionful calibration.) *The single dimensionful input of Rem. D.5 is taken at its measured value: G calibrated once against any one gravitating system, and Λ (equivalently the count N) at its observed magnitude. The framework fixes only the dimensionless response coefficient (Prop. 11.20); nothing below re-derives G or Λ.*

Let B *be a body of total rest mass M and mean radius R that is*

(a) terminally decohered*: it carries (RP)-admissible, terminally readable configuration records (Def. 10.1), so that under Ax. 7.1 the realized branch source is its* definite *classical rest-mass density $\rho(\mathbf{x})\,c^2$ with $\int \rho\, d^3x = M$ (Thm. 7.3; per-branch sourcing as in the proof of Prop. 11.43);*

(b) quasi-static and weak-field*: $\partial_t \rho \approx 0$, source velocities $\ll c$, $GM/Rc^2 \ll 1$, and ρ spherically symmetric to leading multipole order.*

Then, writing g_{N} for the surface acceleration (the symbol g being reserved for the anchor ratio (49)*):*

(i) (Poisson.) *The* 00*-component of the (N1) equation reduces, exactly as in Prop. 11.43(i), to $\nabla^2\Phi = 4\pi G\rho$ with $\Phi = -\frac{1}{2}c^2 h_{00}$; for the measured Λ the trace-seam term contributes an acceleration $\Lambda c^2 r/3 \sim 10^{-29}\,\mathrm{m\,s^{-2}}$ at planetary radii and is dropped.*

(ii) (Exterior field.) *Outside the body $\Phi(r) = -GM/r$ for $r \geq R$, so a slow test body falls with $g_{\mathrm{N}}(r) = GM/r^2$; at the surface*

$$g_{\mathrm{N}} = \frac{GM}{R^2}.$$

(iii) (Earth.) *With the* measured *inputs $GM_{\oplus} = 3.986 \times 10^{14}\,\mathrm{m^3\,s^{-2}}$ and $R_{\oplus} = 6.371 \times 10^6\,\mathrm{m}$,*

$$g_{\mathrm{N},\oplus} = \frac{GM_{\oplus}}{R_{\oplus}^2} \approx 9.82\,\mathrm{m\,s^{-2}} \approx 9.8\,\mathrm{m\,s^{-2}},$$

the residual per-mille offsets from locally measured values (rotation, oblateness $J_2 \sim 10^{-3}$) lying outside the static spherical idealisation.

Modulo the named residual (N1) *and the one-time calibration* (N2)*, the framework therefore* entails *the Newtonian limit for decohered matter: given a terminally decohered body of measured (M, R), the exterior surface acceleration is GM/R^2 — classical recovery for planets, with Earth's $9.8\,\mathrm{m\,s^{-2}}$ the worked instance.*

Proof. (N1) supplies the linearised tensor equation with SM source on the region; premise (a) supplies the definite branch source (the Page–Geilker step: what gravitates is the realized decohered configuration — Prop. 11.39, Thm. 7.3, and the sourcing argument opening the proof of Prop. 11.43). Steps 1–3 of that proof are textbook GR arithmetic (Rem. 11.44) once the linearised equation and the source are granted; they use neither the origin of the equation nor the point-mass form of ρ, so they apply verbatim with ρ the body's continuum rest-mass density, giving (i). For (ii), the decaying solution of $\nabla^2\Phi = 4\pi G\rho$ for a (leading-multipole) spherical ρ is $\Phi = -GM/r$ outside the support, by the Green's-function superposition of Step 4 (Newton's shell theorem: only the enclosed total M enters for $r \geq R$), and the geodesic reading of Step 3 gives $\ddot{\mathbf{r}} = -\nabla\Phi$, i.e. $g_{\mathrm{N}}(r) = GM/r^2$. (iii) is arithmetic on measured inputs, with $GM_{\oplus}/R_{\oplus}c^2 \approx 7 \times 10^{-10}$ certifying the weak-field premise (b). □

Remark J.6 (Honest scope of Cor. J.5: theorem links, the single conditional link, and what is not exercised)*.* The corollary's chain has exactly one conditional link; naming it is the point of stating the corollary here.

(a) Theorems (in-paper). The static weak-field reduction Einstein → Poisson → GM/R^2 is the arithmetic of Prop. 11.43 (Steps 1–4), and record-classicality — that a terminally decohered body gravitates as its definite realized configuration, never as a superposition — is

Thm. 7.3 under Ax. 7.1 with the per-branch sourcing of Def. 10.1 (the Page–Geilker resolution, Prop. 11.39). Likewise the per-wedge scalar Clausius identity for SM matter is a theorem (Thm. J.1: null-surface modular theory is universal, not conformal-only), as is the trace-free tensor reconstruction *given* the pointwise null–null equation of state in every direction (Rem. I.6(i), pointwise linear algebra [56]).

(b) Conditional — the one named lever, its exact weight, and what is now closed. Input (N1) is the linearised tensor completion for interacting massive SM matter. For conformal matter the linearised tensor equation is *derived* outright, on the AdS anchor of Rem. I.4 (Thm. I.1, via the Casini–Huerta–Myers scanning family [241, 58]); for rock and iron the corresponding derivation is pending the per-cut first law (E0-dS′) (Rem. I.4, App. E2a), and Cor. J.3 delivers one scalar identity per wedge — integrated over the horizon, not pointwise. (N1) is exactly the gap between the two: the uniform pointwise, every-null-direction upgrade of the per-wedge first-law identity, for interacting SM matter, *to all orders* in the SM couplings.

This gap now splits into two clearly separated tiers. **Tier 1 (the physics deliverable, closed).** The Newtonian recovery $g_{\mathrm{N}} = GM/R^2$ is a *theorem*: Cor. J.4 establishes the *exact* $n{=}0$ pointwise null–null equation of state (127) uniformly over the weak-field region (Bisognano–Wichmann + Longo, the free two-state relative modular operator genuinely existing), and the shell-theorem mass-conservation step routes the entire interacting tail into the measured M, so the exterior monopole — and hence g_{N} — is exactly untouched (App. J, Step 4). The leading-order Newtonian link is therefore *closed*, and classical recovery for planets is *derived at leading interacting order* — resting on the $n{=}0$ equation of state and mass conservation, *not* on any interacting first-law identity (the deliverable never references the interacting boost–Noether identity (R-i)). The gravitational-nonlinear tail is additionally $(GM/Rc^2)^2$-suppressed, so no more than leading order is ever exercised at planetary strength, part (c).

Tier 2 (the all-orders pointwise ($\star$), open — reduced to B1a, now mapped). What remains open is the *all-orders* link: the same equality (127) with the remainder $\mathcal{R}_{\geq 1} \to 0$, i.e. the pointwise ($\star$) as a formal power series to all orders at the rigor grade of Thm. J.1 (all-orders-formal-pAQFT; the honest ceiling, since a nonperturbative statement would outrun that theorem's own formal-series premise). This is *already open at $n{=}1$*, not merely all-orders: the $n{=}1$ interacting-matter first-law identity is the interacting boost–Noether identity (R-i) $w_{2\pi}(K) = 2\pi \int x^1 \langle T_{00}^{\mathrm{int}} \rangle$ for the Bogoliubov map R_V, exact at $n{=}0$/Longo but open at $n \geq 1$. The link is not a new opaque residual — it is the single named companion problem B1a, whose structure is now mapped into three pieces:

- *Existence of the surrogate:* construct the interacting relative-free-energy surrogate $\widehat{S}_{\mathrm{rel}}$ for the SM pAQFT net on the local Rindler wedge (the two-state relative modular *operator* does not exist for the formal-series interacting net, so $\widehat{S}_{\mathrm{rel}}$ must be the surrogate cocycle functional, not an operator). This is now a *theorem outright*: the sole named gate (a boost-KMS strip-integrability lemma) has been discharged (formal-series, per-order, massive scalar window; the interacting kernels enter as fixed test data and the strip integral converges by microcausal wavefront control), so the surrogate *exists*; what remains open of B1a is exclusively the first-law question below.
- *The naive first law is negative:* the literal $\delta\widehat{S}_{\mathrm{rel}} = 2\pi \int x^1\, \delta\langle T_{00}^{\mathrm{SM}} \rangle$ at $n{=}1$ is a *negative result* — the surrogate is a relative free energy, hence stationary at the reference boost-KMS state (its first variation vanishes identically), while the boost flux has generically nonzero first variation; the order mismatch makes the literal equality false.
- *The surviving open core is (R-i):* the true $n{=}1$ content is not $\delta\widehat{S}_{\mathrm{rel}}$ but (R-i), the interacting boost–Noether identity, standard-shaped and strictly weaker than ($\star$). Its *scalar avatar is now a theorem at $n{=}1$*: for the massive $\lambda\varphi^4$ interaction on the wedge the obstructing term vanishes identically, $\mathfrak{O}^{(1)} = 0$, by an exact thermal-Wick/KMS Bose-orthogonality mechanism — the interaction's thermal-Wick diagonal is quadratic in the occupation fluctuation $d = n - \bar{n}$ while the Bisognano–Wichmann modular generator is *linear* in it, and the exact Bose moment ratio

$\langle d^3\rangle/\langle d^2\rangle = 1+2\bar{n}$ makes them orthogonal at every boost frequency (the vanishing is specific to the modular generator; non-modular charges violate it). The scalar n=1 statement now has independent first-order company in the literature: for interacting-coherent excitations $W(f)\Omega$ of the massive $\lambda\varphi^4$ wedge theory, the $O(\lambda)$ Araki–Uhlmann relative entropy with respect to the wedge vacuum (the boost-KMS reference) organizes into the boost Noether charge of the classical causal interacting solution, $S_{\rm rel} = 2\pi\int_{x^0=0} x^1\, T_{00}(\varphi_f^{\rm int})\, d^3x + O(\lambda^2)$ [233] (arXiv preprint, July 2026, unrefereed) — the two-state, coherent-class counterpart of the same first-order first law, obtained in a conservation-fixed Hadamard scheme with one free improvement-type $:\!\varphi^2\!:$ constant (the vacuum-/thermal-Wick ordering seam of the present proof is an ambiguity in the same channel). That computation *consumes* the interacting Bisognano–Wichmann identification rather than proving the Ward identity, so it corroborates but does not reprove $\mathfrak{O}^{(1)} = 0$; it is first-order-only (its all-orders form is there an explicit conjecture) and wedge-interior, touching neither the $n \geq 2$ tail nor the edge-locality wall below. The SM lift of this identity at n=1 is now resolved *per sector*: the Higgs and both gauge self-couplings are theorems by the same mechanism (the quartic by the Bose ratio; the cubic vanishing identically by odd ladder count), and the fermionic sector is non-load-bearing at this order (Pauli n^2=n forbids a single-fermion quadratic; the Yukawa vertex is linear per field — the Fermi-Dirac moment ratio $1-2f$, computed exactly, never enters). The ghost sector does *not* self-orthogonalize (its raw contribution is provably nonzero); the state-level resolution is the *thermal Kugo–Ojima cancellation*, now exhibited to machine precision in the quartet-KMS model: the indefinite Krein/ghost-number metric maps the fermionic ghost occupation onto *minus* the Bose occupation exactly, so ghost+antighost cancel longitudinal+scalar by the *metric* — the occupations are separately nonzero at the Unruh temperature and no Bose=Fermi moment match is involved — and the super-trace of any BRST-exact observable vanishes at finite temperature (the genuine Araki–KMS transplant, needing only $[Q,H]$=0 and the odd grading). This state-level cancellation is *equivalent* (via a KMS Ward identity) to the single remaining gate: the horizon locality of the boost–BRST descent $[Q_{\rm int}, K_W^{\rm int}] = 0$ on the non-compact Rindler wedge. The descent equation is now written explicitly at n=1 ($[Q^{(1)}, K_W^{(0)}] = -[Q_0, K_W^{(1)}]$), and its obstruction is provably *not* cohomological ($H^{+1}(Q) = 0$, Koszul-exact) nor the $1/\omega$ horizon-bulk mechanism (the physical boost cocycle is boost-invariant, so the source carries zero boost weight): the entire residual is the *edge locality* of the descent solution at $x^1 = 0$ — the boundary term $[x^1 Y^1]_{x^1=0}$ of the modular-moment integration by parts. The edge two-point structure is now computed: the composite edge exponents are exhibited (Fulling-mode exponent 0; $\langle:\!\varphi^2\!:\rangle_{\rm ren} \sim \rho^{-2}$, the local-temperature square; connected $\langle:\!\varphi^n\!:\,:\!\varphi^n\!:\rangle_c \sim \rho^{-2n}$, subtraction-independent; the UV-before-edge limit order forced), the *one-point* edge term vanishes by ghost number (necessary, not sufficient), the $O(g)$ Araki dressing is edge-benign, and the genuine gate is the ghost-number-neutral *two-point* norm, whose *leading* edge power survives by a *dimensional floor* (the descent current has dimension three, so its base edge norm scales as ρ^{-4}; the graded norm itself is *indefinite* — an earlier positivity framing of this coefficient was falsified — but no torsor-reachable ordering shift escapes the floor). The gate now resolves to a per-ordering × per-test-function-class table: under the KMS-positive and Krein-graded orderings the boundary term *diverges* for the generic smooth smearing this paper's first-law consumer uses ($\alpha_f = 0$, against the finiteness threshold $\alpha_f > p - \frac{3}{2}$). Both final doors are now adjudicated: BRST-closure and the pAQFT/Hadamard construction do select the Minkowski-covariant ordering, and it is BRST-compatible on the wedge — but its apparent vanishing is a *one-point* statement; the gate object is the two-point *norm*, which is subtraction-independent and unavoidably boost-KMS-referenced, so the covariant cell sits on the same ρ^{-4} dimensional floor and *no ordering closes the gate*. Likewise the consumer audit: all wedge first-law loci collapse to one genuine edge-sampling object (the boost-moment first law itself, whose weight x^1 is the booked moment, not an independent taper), the CHM ball avoids the gate structurally rather than by taper, and an imposed edge-vanishing smearing is internally consistent but *changes the theorem* (the fenced charge misses a collar deficit and the full-charge limit reopens the gate). The sole surviving datum is the retarded-product edge-locality of the boost-stress charge at the bifurcation surface (in print for the gauge current only) — a continuum, type-III_1 question beyond any finite model. The SM n=1 first law is accordingly *not* a theorem at the present ceiling: the scalar theorem stands, and the wall has its name. It now also has a necessary-and-sufficient form (the necessity direction (R-i) $\Rightarrow$ (W1) is promoted from sketch to a form-grade theorem on the fenced massive-scalar window: its reduction (R-i) $\Rightarrow \mathrm{diag}(K_\omega - 2\pi K_{\mathrm{boost}}) = 0$ is Bisognano–Wichmann-free and non-circular through the intrinsic coherent-entropy identity, the polarization real part $\mathrm{Re}\langle\xi|K_\omega - 2\pi K_{\mathrm{boost}}|\eta\rangle = 0$ on the standard subspace is unconditional, and the remaining imaginary/Kubo–Martin–Schwinger half — pinned only by the same truncated-Reeh–Schlieder and formal-Takesaki density–polarization scaffolding that decomposes (W1) — is now *discharged* together with that scaffolding at quadratic-form grade, modulo the same imports as (W1) above; the Standard-Model instance still rests on (W1)-for-the-SM, i.e. the open m=0 infrared gate): the clause holds if and only if (W1) the interacting Bisognano–Wichmann identification for the BRST-physical SM wedge net at n=1 *and* (W2) collar control at fence removal, the latter equivalent to the edge datum itself through the $\delta^{-1/2}$ collar rate at the ρ^{-4} floor (discharge requires the physical descent current's connected KMS edge exponent $p < 3/2$ on the collar reading, $p < 1$ for the boundary value; the dimensional floor is $p = 2$ — a half-power gap closable only by BRST-exactness of the physical current's edge limit, which no local ghost-number-zero counterterm supplies). Two corollaries of the sharpening: the semicontinuity route class (lower-semicontinuous, martingale, and nested-wedge fence-removal arguments) is closed as a class — such arguments deliver the inequality, never the rate equality — and the ghost gate is route-borne rather than law-intrinsic: on the fenced class the n=1 first law is edge-free, theorem-modulo-(W1) plus assembly, since the edge term is a boundary artifact of the one-sided moment presentation of the charge, which the Araki–Bisognano–Wichmann presentation never forms. The input (W1) itself is now converted from an open dynamical postulate into a finite named checklist: at n=1 on the fenced class, and given the printed free Bisognano–Wichmann, Lorentz master-Ward, and vacuum-stability anchors, (W1) is *equivalent* to the two-sided boost-KMS boundary family — computed on the massive scalar window by the exact Euclidean angular-slicing identity (the one-vertex Gell-Mann–Low integral equals the angular integral of rotated half-plane boost charges, the 2π period being the KMS period), of which the printed $\mathfrak{O}^{(1)} = 0$ is precisely the one-point member. The four checklist items that were named-open — the per-order angular (Klein–Landau-class) Osterwalder–Schrader step, the formal-series Takesaki transfer, truncated Reeh–Schlieder, and the collar/tail bound — are now *written and discharged* at quadratic-form grade (the Takesaki transfer by reduction to the free modular uniqueness at zeroth order through the homogeneity of the linearised Kubo–Martin–Schwinger equation, the naive interacting-Takesaki route being dead since the formal-series net carries no modular operator; truncated Reeh–Schlieder by a unipotent triangular cascade over the free Reeh–Schlieder property, which does not transport verbatim; the angular step by free Klein–Landau positivity plus linearity in the fixed vertex, the interacting Euclidean truncation being a signed measure to which Klein–Landau does not apply). So on the fenced *massive-scalar* window (W1) at n=1 is an unconditional form-grade theorem modulo the strip lemma — whose m=0 case is not an independent open but is precisely the infrared gate named below (the honest Standard-Model wall), entering through its k=2 scheme (:φ^2: improvement, b_0 zero-mode) channel alone, the quartic interaction channel's leading tail being m=0-safe — and the Epstein–Glaser bounded-region existence (in print): the form-grade transport of free Takesaki is no longer a carried input but an explicit quadratic-form lemma, the free modular commutation pushed order by order through the unipotent triangular Reeh–Schlieder cascade, every row preserving the fenced common form domain on which the perturbed generator is definable as a form (the boost-charge form being bounded relative to neither the free modular generator nor its square), consuming only free Takesaki at n=0 (in print) and evaluating no interacting modular operator anywhere — discharged, not carried, and no open-ended dynamical lemma remains. All at quadratic-form grade, with the cocycle generator taken in the compact-support device register (in the sharp-time register the coefficient is form-grade only, and *provably* so: no self-adjoint operator affiliated to the fenced wedge algebra, with the vacuum in its domain, generates the $O(\lambda)$ increment there — a theorem-grade *non-affiliation*, not a limitation of technique, driven by the fence-uniform boost-spectral growth $d\mu \gtrsim \nu^4$ (bulk coincidence, not edge; equivalently the Λ^3 time-integrated vector norm), of which only the positive lower bound is consumed, supplied by positivity of the leading Wick term. Operator grade is carried by, and *only* by, the compact-support device register: the necessary-and-sufficient condition is the *conjunction* of compact/decaying spatial support (an infrared/volume condition) and an H^2 boost-time smearing (the ultraviolet condition), a single fixed H^2 profile sufficing — nothing weaker, no spatial fence and no boost-covariant dressing, reaches it. The quartic vertex is the first degree to fail sharp; the quadratic c'/improvement channel is operator-grade even sharp. The screening that makes $[K_1, B]\Omega$ a genuine vector is vector-goodness of the derivation, strictly weaker than affiliation of the generator; the sole ν^4-immune operator-grade object in the algebra is the bounded Connes cocycle, which is not a one-parameter group and which exists as an operator only if the perturbed state ω_P is normal — a constructive-field-theory question that is *not* in fact orthogonal to the boost ultraviolet: both standard perturbative routes to ω_P normality are obstructed by the same ν^4 bulk coincidence, the finite-relative-entropy (Bogoliubov–Kubo–Mori) criterion diverging as Λ^4 and the Araki–Dereziński–Jakšić–Pillet natural-cone expansional as Λ^3 (two Λ-powers milder — one modular integration — than the generator's Λ^5, since K_1 is form-bounded relative to neither K_0 nor K_0^2). This is a route-closure, not a non-affiliation of the cocycle: whether a non-perturbative representative renders ω_P normal — the divergence lives purely at the Kubo–Martin–Schwinger strip boundary, the modular midpoint $\Delta_0^{1/4} K_1 \Omega$ being finite — remains open, so the cocycle escape hatch is neither opened nor provably closed at $O(\lambda)$. This is a milder, register-curable ultraviolet non-affiliation, categorically distinct from the type-III$_1$ one-sided-charge non-affiliation below (trace-theoretic, crossed-product-cured, impervious to smearing)).

The ceiling is moreover *dimensional, not structural*: the obstruction measure scales as $d\mu \sim \nu^{2\Delta-d}$ in the short-distance dimension Δ of the interaction density, so in d=2, where constructive field theory supplies the non-perturbative vacuum, every $P(\varphi)_2$ density ($\Delta = 0^+$, logarithmic) yields an *integrable* boost-spectral tail ν^{-2}: the same sharp-time fenced moment charge that is provably non-affiliated here is an operator with the vacuum in its domain for every polynomial coupling, the ultraviolet half of the register conjunction evaporates (the infrared/volume half is dimension-blind), and on the weakly coupled window the first law $\delta S_{\mathrm{rel}} = \delta\langle K\rangle$ assembles at operator grade from in-print inputs (Glimm–Jaffe–Spencer Wightman axioms; the self-adjoint locally correct boost generator; the Bisognano–Wichmann pair applied to the essentially self-adjoint $P(\varphi)_2$ field — an assembly of in-print theorems, the conjunction not itself in print; Araki relative entropy) — while Yukawa$_2$ stays marginally obstructed ($d\mu \sim \nu^0 \log\nu$). The two-dimensional case also locates the cocycle escape hatch honestly: the interacting vacuum is locally Fock but not expected wedge-normal relative to the free algebra even there, so the hatch is bypassed by the native interacting representation, whose absence in d=4 — non-perturbative ω_P — *is* the wall.

No step of the checklist consumes edge control, so (W1) and (W2) are independent at n=1, dependence re-entering exactly if the fence is dropped or the one-sided presentation is chosen; and the honest Standard-Model wall is the infrared gate alone — which includes the scheme channel, since the c'/improvement Kubo integrals themselves log-diverge at m=0. The framework's own geometric floor ω_1, which removes the massless coth soft-infrared divergence, does *not* reach this channel: that cure is a spectral floor on the co-exact one-form

($b_1(\Sigma)$=0, Hodge) sector, whereas the c'/improvement $:\varphi^2:$ channel is spin-0 and carries the 0-form constant zero mode (b_0=1, ungapped by b_1=0), whose m=0 boost moment admits no ω_1 modulus — so ω_1 is *necessary* (it fixes the transverse modular data and closes the coth and transverse-photon legs) but *not sufficient* — now *unconditionally*: the former "one open lever" (whether thermal Kugo–Ojima empties the b_0 site, removing the would-be-longitudinal seam and closing the gate by the ω_1 route) is closed in the negative. The site is live at m=0 for a structural, coupling-independent reason — the improvement-type $c'{:}\varphi^2{:}$ ambiguity is already present for the free *ungauged* scalar ($O(g^0)$), where the Kugo–Ojima ghost sector is empty, and no Ward identity fixes c' (it multiplies an identically-conserved total-derivative improvement). The quartet mechanism cannot reach it: at zero *or* finite temperature it empties a scheme site only where that site's operator is BRST-*exact* (the super-trace and the KMS Ward identity annihilate exact insertions *only*), and BRST-exactness (membership in $\operatorname{im} Q$) is temperature-independent, so no thermal upgrade — KMS state, thermal-field doubling, or a Tolman-redshifted thermal mass — can enlarge what is emptied; an $O(g)$ ghost-sector cancellation cannot remove an $O(g^0)$ seam, and the super-trace in any case presupposes a trace the type-III$_1$ algebra lacks. The site's finite home is therefore the type-II crossed product named next, not the ω_1 route. The wall also has an invariant characterization: in the fixed-background type-III$_1$ wedge algebra the one-sided boost charge is not merely divergent but *not affiliated to the algebra at all* (no trace, no density matrix — the divergence is the quantitative shadow of non-affiliation), while the corner charge becomes affiliated, self-adjoint, and finite precisely in the type-II crossed product that is this framework's own central structure — i.e. the gravitational (clock-adjoined) algebra is the *unique finite home* of the boost–BRST corner charge among the classes examined. The identity the paper's n=1 clause consumes lives in the fixed-background algebra, so this does not close the gate; it locates it invariantly — and the located wall is now identity-level, not merely object-level. On the clock-adjoined algebra an m=0 first law (the register-native one), $\delta S_{\text{gen}} = \delta\langle \widehat{H}_{\text{grav}}\rangle$, is a theorem for first-order normal-state variations about the dressed reference in the CLPW class — the charge is the affiliated corner generator $\widehat{H}_{\text{grav}}$ of Thm. H.1, the entropy the Type-II trace entropy in the CLPW additive form (up to the state-independent c_0); the interacting n=1 strong form is conditional only on the cocycle escape hatch above (normality of ω_P), the state-variation form unconditional — and the b_0 improvement ambiguity provably enters no register datum: c' multiplies a total derivative, so it touches neither the modular data nor the trace, whose sole normalisation freedom is the state-independent trace-scaling gauge already spent, canonically and orthogonally to c', by the cosmological-constant accounting. The wall is equally rigid in the other direction: the clause is equivalent to the interacting Bisognano–Wichmann identification (W1) plus collar control (W2), one relatum of which (the geometric boost witness) is not affiliated to the clock-adjoined algebra, while the one-sided flux form and the per-wedge scan are not transportable to it, and the improvement torsor is orthogonal to the trace gauge (a state-dependent first-order channel that no state-independent normalisation can absorb). The crossed product thus hosts the charge and the first law kinematically — with the area and charge sectors fused by construction, the dressed identity constrains nothing about g — while the identity the clause consumes, the boost–Noether presentation (R-i) that alone carries the c' torsor whose Kubo integrals log-diverge at m=0, remains open precisely in the fixed-background comparison register: one more instance of the type-II structure doing work the type-III$_1$ setting cannot, together with a precise statement of the work it cannot do. At $n \geq 2$ the orthogonality is provably first-order-only ($\mathfrak{O}^{(2)} > 0$, sign-definite, off-diagonal-dominated); structurally $\mathfrak{O}^{(n)}$ decomposes as a modular first law whose second-order term is the (non-negative) Bogoliubov–Fisher entropy production of the dressed state. The precise formulation is the Clausius–σ identity $\delta Q^{(n)} = T\,\delta S^{(n)} - T\sigma^{(n)}$ with $\sigma^{(n)} \geq 0$ *saturated* at $n \leq 1$ (a theorem-grade rearrangement); Thm. J.1 consumes a pure-equality chain at leading order and Cor. J.3 imports only the saturated n=1 first law, so the linearised-$E0$

derivation *survives* the dissipative reading untouched. Whether $\sigma^{(2)}$ is *physical* production remains open with the failure mode named: it is switching-independent (a state function of the two equilibria, not a quench artifact), but $\sim 99.97\%$ of it is reversible modular work; the genuine-production remainder is *robust* (closed-form-certified, not a numerical floor) and carries a *two-quantum* $\bar{n}^2$ signature that identifies the *collisional* process class; the multi-mode analysis has now narrowed it N-exactly to the ν=0 *diagonal* (elastic/forward, Kubo-static) sector — the off-diagonal candidates (thermal decay width, pair decoherence, inelastic $2 \to 2$ collision) carry *zero* production weight at every N — with the surviving member the Kubo-static/elastic-forward response (inseparable within the diagonal at finite mode count). The δt assignment is genuinely absent at finite mode number (full quantum recurrence, no Ruelle time; only a continuum spectral density relaxes), so a unique relaxation time and any absolute dissipative coefficient require the field-theoretic continuum; the dissipative term is not yet consumable, and no measurable prediction is claimed (for massive matter at realistic accelerations the effect is suppressed beyond observability).

This is the problem designated B1a; its answer is not guaranteed positive, and it is strictly weaker than (N1) as originally stated (it is the surrogate first-law / boost–Noether problem, not the pointwise EoS restated).

Its weight should still be read precisely. It sits at first order in the interacting couplings: the surrogate first law in its *literal* $\delta\widehat{S}_{\rm rel}$ = flux form is there definitely negative (as above), and the live residual is the boost–Noether identity (R-i), which no $n \geq 2$ saturation enters (Rem. I.9); the heavier Hyp. I.5 would imply the all-orders ($\star$) but is not what the corollary consumes. Conversely, nothing here upgrades any part of the saturation residual: the $n \geq 2$ closure is provably *not* derivable from the framework's own crossed-product/thermal-time/two-boundary structure (Prop. I.7), and its non-conformal mechanism for the physical SM remains a genuinely open problem (Rem. I.9). Absent even the all-orders (N1), the identical Newtonian conclusion still holds in the framework at leading order via Cor. J.4, and (beyond leading order) downstream of the *postulated* record-adapted equation (Conj. 10.2) — the route of Prop. 11.43 and Rem. 11.45. The content of Cor. J.5 is thus sharpened: closing the leading-interacting-order residual (done, Cor. J.4) already converts classical recovery for planets from postulate-based to *derived at leading interacting order*; closing the all-orders residual — the single named companion problem B1a — would convert it to *derived to all orders* at the grade of Thm. J.1.

(c) Not exercised at planetary field strength. The nonlinear completion Ax. I.10 (its $n \geq 2$ clauses, and with them the entire saturation question) is not load-bearing here: the expansion parameter is $GM_\oplus/R_\oplus c^2 \approx 7 \times 10^{-10}$, so nonlinear corrections enter at relative order $(GM/Rc^2)^2 \sim 5 \times 10^{-19}$ and the linearised clause of (N1) suffices. The trace-value seam is likewise inert: with the measured Λ the induced acceleration at $r \sim R_\oplus$ is $\sim 10^{-29}\,\mathrm{m\,s^{-2}}$, i.e. $\sim 10^{-30}$ of $g_{\mathrm{N},\oplus}$; the corollary consumes Λ (equivalently N, Rem. D.5) only as a measured input, exactly as it consumes G, M, and R. The framework's Planck-suppressed dynamical sector leaves the static potential unmodified (Rem. 11.44(b)), so it does not touch the law. Nothing here predicts $M_\oplus$, $R_\oplus$, G, or Λ: what the corollary delivers, modulo (N1)–(N2), is the *law* $g_{\mathrm{N}} = GM/R^2$, not the planet. The analytic-ball restriction of Hyp. E.2 (the input (E2a-ω)) does *not* touch this corollary: its input (N1) is routed through the scalar-Clausius / linearised-$E0$ branch (Thm. J.1, Cor. J.3, and the inversion of Rem. I.6), not the $E2$ Hadamard-parametrix chain, so the Newtonian limit and the recovered $g_N = GM/R^2$ are independent of the metric-ball regularity class — whether that class is the open H^s ball or the analytic collar U_ω.

The EFT power-counting budget for Cor. J.4

We exhibit the facts the corollary's remainder control rests on: the $(GM/Rc^2)^2$ suppression of the discarded *gravitational* tail, the $\mathcal{O}(1)$ geometric prefactor C_{EFT}, and — the control that actually protects the deliverable — the shell-theorem mass conservation that leaves $g_{\mathrm{N}} = GM/R^2$ untouched *for the entire interacting tail* $n \geq 1$*, the interacting-matter* n=1 *term included.* Nothing here consumes an all-orders fact, and nothing here consumes the n=1 interacting first-law identity (R-i): every estimate is either a within-a-finite-order statement about objects that exist, or a conserved-charge statement about the total mass.

Step 1 (the two independent small parameters). Two expansions run in parallel and must not be conflated:

- the *gravitational* weak-field parameter $\epsilon_g := GM/Rc^2$ (Earth: 7×10^{-10}), controlling the metric perturbation $h_{\mu\nu} = \mathcal{O}(\epsilon_g)$ and hence the linearisation of $G_{\mu\nu}$;
- the *matter* interacting-order parameter, the SM couplings $\{g, g', g_s, y_t, \lambda\}$ entering the formal pAQFT series for $\langle \widehat{T}^{\mathrm{SM}} \rangle$. The load-bearing RHS of (127) carries the n=0 term in *this* parameter; the whole $n \geq 1$ tail is the remainder $\mathcal{R}_{\geq 1}$.

The corollary's remainder $\mathcal{R}_{\geq 1}$ has two parts with two different controls. Its *gravitational-nonlinear and non-Killing* part, re-expressed in the gravitational counting, obeys $|\mathcal{R}^{\mathrm{grav}}_{\geq 1}| \leq C_{\mathrm{EFT}}\, \epsilon_g^2 \, \|\delta \langle \widehat{T}^{\mathrm{SM}} \rangle\|$ (Steps 2–3); the fence is that no bound on the matter series' n-dependence (radius of convergence, $\{C_n\}$ growth) is used. Its *interacting-matter* part (the n=1 term dropped from the RHS, and the $O(\lambda^2)$ tail) carries no such pointwise smallness — it is controlled by mass-conservation alone (Step 4), which is what protects g_{N}.

Step 2 (why the dropped gravitational *tail is ϵ_g^2, not ϵ_g, on the null cut).* This step bounds the gravitational-nonlinear part of $\mathcal{R}_{\geq 1}$; the interacting-matter part is handled by Step 4, not here. The leading balance equates two $\mathcal{O}(\epsilon_g)$ quantities: $\delta G\, kk = \mathcal{O}(\epsilon_g)$ (one power of h) and $(8\pi G/c^4)\, \delta \langle \widehat{T}^{\mathrm{SM},(0)} \rangle\, kk$, calibrated to it by G. The first *gravitational* correction to *this equality* that the truncation discards is the back-reaction of the higher matter tail onto the geometry, which is one further factor of the source relative to the leading source, i.e. $\mathcal{O}(\epsilon_g)$ smaller — but on the *null-null* projection the linear-in-mass channel is annihilated. This is the spectral face of the antilocality clearance: the scalar-bilinear = trace channel, the only channel carrying an $\mathcal{O}(\epsilon_g)$-linear term, couples through $g_{ab}k^ak^b = k^2 = 0$ on the null cut and is annihilated there. The surviving spin-2 channel's threshold behaviour is $\propto (s - 4m^2)$, converting the would-be m-linear $(ml)\log(ml)$ "diamond disease" into the m-quadratic wedge rate (symbolically exact: spin-2 amplitude $a_2 = 2m^2/3 - s/6$ vanishes at threshold, edge exponent 5/2). Hence the first surviving correction to (127) is $\mathcal{O}(\epsilon_g^2)$, not $\mathcal{O}(\epsilon_g)$. FENCE: this uses the free-field spectral computation as the leading spectral density; the interacting spin-2 Källén–Lehmann density ρ_2^{int} is twist-gap-controlled (plausible) but its exact coefficient is the companion item B2-nonKilling and is NOT claimed here — for the leading-order truncation only the ϵ_g^2 POWER (not the coefficient) is load-bearing. *Status of the companion item.* A proof-level Killing-consumption audit narrows B2-nonKilling drastically: of the chain's apparent Bisognano–Wichmann/Kay–Wald consumers, the load-bearing majority consume no Killing structure at all (Thm. 4.1 is unitary covariance plus trace cyclicity; Lem. 11.25 is Hadamard and state-independent; the E2 transport is modular-free), while the geometric identification $K_\omega = 2\pi \int \xi\, T$ is consumed only in its null–null projection at free-field order, discharged in print at first-law-*equality* strength on Minkowski/Rindler cuts (Ceyhan–Faulkner), at monotonicity/GSL strength on cuts of Killing horizons (Faulkner–Speranza), with the invariant-Hadamard state support in place on the de Sitter static patch and Schwarzschild; Kerr is *excluded* — no isometry-invariant Hadamard state exists there (Kay–Wald), so the identification is not available in print on Kerr. B2's own remaining share is the two seed-side pins already recorded above (the H0 co-scoping straddle and finite averaged null energy of ω on complete cuts) together with the promotion of Hyp. F.1 beyond the perturbative-around-isolated-horizon regime; the interacting rate, the FRW tensor completion, and the finite-part-family uniformity belong to B1a, Hyp. I.3, and (E2a-ω) respectively, and are not double-counted here. On FRW — which carries no bifurcate Killing horizon — the curved rate remains a genuinely open named wall: the exact null-cut route is scope-empty there, and the approximate-boost route fails the contraction step (its curvature defect $\sim \ell^2|\mathrm{Riem}|$ costs two derivatives, which the plain Banach step cannot absorb; and its leading power is $(\ell/L)^2$, not the optimistic $(\ell/L)^4$).

Step 3 (the prefactor $C_{\rm EFT}$, exhibited). Collecting the geometric factors of Step 2: $C_{\rm EFT} = \sup_{x\in K}\|\Pi^{\perp}(x)\| \cdot c_{\rm BR}$, where $\Pi^{\perp}$ is the transverse ($\{0,0,1,1\}$-spectrum) projector onto the null cut and $c_{\rm BR}$ is the back-reaction kernel norm of the linearised Einstein operator on the weak-field region. Both are $\mathcal{O}(1)$, dimensionless, and direction-independent (the projector spectrum and $8\pi G/c^4$ are SO($d-1$)-scalars, PIECE 2 (3)); on the static spherical weak-field background $\|\Pi^{\perp}\| = 1$ and $c_{\rm BR} = \mathcal{O}(1)$ by the explicit Green's function of ∇^2 (Cor. J.5 proof, Step 4). So $C_{\rm EFT} = \mathcal{O}(1)$; for Earth $C_{\rm EFT}\,\epsilon_g^2 \approx C_{\rm EFT} \times 5 \times 10^{-19}$, i.e. $\sim 10^{-18}$ fractional, $\sim 10^{-30}$ of $g_{\rm N,\oplus}$ after the inversion (matching the $\sim 10^{-19}$ nonlinear estimate already recorded in Rem. J.6(c), l.19589).

Step 4 (mass conservation: the entire interacting tail does not move $g_{\rm N}$ — the control the deliverable rests on). Independently of its magnitude, the discarded tail cannot shift the Newtonian law, because it is mass-conserving — and this holds for the *whole* $n \geq 1$ tail, the interacting-matter n=1 term included, which is why the deliverable never references the n=1 first-law identity (R-i). The $n \geq 1$ corrections to $\langle \widehat{T}_{00}^{\rm SM}\rangle$ are built from local composite operators of the SM fields (renormalised, DPR/Fröb–Zahn); their integral over a spatial slice is the correction to the *total* enclosed rest-energy, fixed to Mc^2 by the definition of M as the realized decohered configuration's rest mass ($\int \rho\, d^3x = M$, Cor. J.5 premise (a); the measured GM is the full gravitating mass, already inclusive of all interacting binding and self-energy). By Newton's shell theorem (Cor. J.5 proof, Step 4: only the enclosed total M enters for $r \geq R$), any redistribution of the source at fixed enclosed M leaves the exterior potential $\Phi = -GM/r$ — and hence $g_{\rm N} = GM/R^2$ — exactly invariant. The gravitational tail additionally perturbs the *interior* multipole structure at relative order ϵ_g^2; the interacting-matter tail redistributes the interior source with no such smallness, but neither shifts the exterior monopole. This is why the n=0-anchored corollary already delivers the full planetary deliverable: the interacting matter physics is, for the exterior field, exactly inert by the shell theorem — *not* because it is pointwise small, but because it conserves the enclosed mass.

Step 5 (the order-independent uniformity threshold). The point-uniformity constant of Cor. J.4(iii) is order-independent because the UV singularity controlling the finite-part extraction at each order is the *same* free Hadamard parametrix (interacting vertices are integrated over distinct interior points and do not collide with the null cut). Thus the ball-uniform finite-part family $\{C_W, W_*, D_*, C_D, C_\Theta\}$ of Prop. E.4 controls every order at one Sobolev threshold s_0; "uniform at each order" costs no new input beyond the n=0, 1 instances actually used. FENCE: this is a per-order statement; it does NOT assert the family is uniform *in* n (that would be a nonperturbative-in-disguise claim). Only the $n \leq 1$ orders are consumed.

Chamseddine–Connes spectral triple as natural entry point

The Connes–Rovelli thermal-time postulate Pos. 11.11 is a modular-flow statement on an abstract operator algebra; the Chamseddine–Connes spectral-action principle [267, 270, 269] derives the SM Lagrangian from a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ with $\mathcal{A}$ almost-commutative [268] (tensor product of a continuous gravity algebra with a finite generation-sector algebra $\mathcal{A}_F$) and D a generalized Dirac operator. The Dixmier trace of $|D|^{-d}$ reproduces the Einstein–Hilbert plus SM action.

Remark J.7 (Generation-count blindness under thermal time). The modular flow of a faithful normal state on an almost- commutative spectral triple is unique up to inner automorphism (Connes' uniqueness theorem of modular automorphisms). The finite generation-sector algebra $\mathcal{A}_F$ enters only through inner automorphism classes and is *invisible to the thermal-time flow.* Concretely: the paper's three structural postulates are insensitive to the choice of three vs. any other number of generations, and to the choice of gauge group within an anomaly-consistent class. This is a previously unnoticed consistency result: the framework is matter-content agnostic at the level of structural postulates, and SM-specific input enters only through the anomaly-cocycle vanishing condition of Thm. J.1.

Remark J.8 (Residual technical items). Thm. J.1 is an application of existing pAQFT theorems plus Bianchi–Myers; it does not address two ambient-literature-residual items: (i) survival of the CLPW Type II crossed-product construction (App. H) when $\mathcal{N}$ is the interacting-SM type III_1 factor rather than a free scalar net (plausible but unproven; the crossed product uses only modular theory, which exists for any type III_1 factor); (ii) the Higgs VEV's back-reaction on the modular generator. Both are standalone follow-up items. The paper takes (i) as a reasonable ambient-literature expectation and does not rely on (ii) for any statement in the main body.

K UV completion via Type II trace-stable inductive limit

This appendix organises the status of the Wilsonian EFT scope postulate Pos. 11.30 (which remains a postulate; Rem. D.3) as conjecturally the finite-Λ restriction of a *single universal σ-finite Type II_∞ object* via a trace-stable inductive-limit construction, and records the honest status of singularity resolution under the CLPW machinery.

Setup

For each cutoff $\Lambda \leq E_\star$ let $\mathcal{A}_\Lambda$ be the cutoff-compressed CLPW crossed- product algebra constructed on the spectral subspace $P_\Lambda \mathcal{H}_\mathrm{m} \subset \mathcal{H}_\mathrm{m}$ of App. H: a type II_1 factor with normalised finite trace τ_Λ (Thm. H.1(iv)).

Conjecture K.1 (Type II trace-stable UV completion)**.** There exists a directed system of trace-preserving conditional expectations

$$E_{\Lambda' \leftarrow \Lambda} \; : \; \mathcal{A}_{\Lambda'} \;\longrightarrow\; \mathcal{A}_\Lambda \qquad (\Lambda \leq \Lambda'), \qquad \tau_\Lambda \circ E_{\Lambda' \leftarrow \Lambda} \;=\; \tau_{\Lambda'},$$

satisfying the Markov/compatibility conditions $E_{\Lambda' \leftarrow \Lambda} \circ E_{\Lambda'' \leftarrow \Lambda'} = E_{\Lambda'' \leftarrow \Lambda}$ for $\Lambda \leq \Lambda' \leq \Lambda''$, such that the inductive limit

$$(\mathcal{A}_\infty, \tau_\infty) \;:=\; \varinjlim_{\Lambda \to \infty} (\mathcal{A}_\Lambda, \tau_\Lambda) \tag{128}$$

exists as a σ-finite semifinite von Neumann algebra with faithful normal semifinite trace τ_∞. Thermal-time equilibrium states on $\mathcal{A}_\infty$ restrict on each $\mathcal{A}_\Lambda$ to the two-boundary retrocausal state of the paper.

Under Conj. K.1, Pos. 11.30 becomes the finite-Λ slice of a universal Type II_∞ object, and the paper's sub-Planckian scope is the EFT restriction of a trace-stable UV-complete construction. The finite-Λ members of the directed system are type II_1; semifiniteness with $\tau_\infty(\mathbb{1}) = \infty$ is a property of the limit only, arising because the compatible traces are rescaled along the system (cf. the hyperfinite II_∞ inductive limit of [37]).

Remark K.2 (The inductive limit enforces the trace freedom). For *any* well-posed AFD-preserving directed system — one built, as in (128), from trace-preserving conditional expectations or inclusions of the hyperfinite (approximately finite-dimensional) II_1 members $\mathcal{A}_\Lambda$ — the limit $\mathcal{A}_\infty$ is again approximately finite-dimensional, hence amenable, hence has fundamental group $\mathbb{R}_+$ on its II_1 core (Rem. D.5). Such a UV completion therefore *enforces* the one-dimensional trace freedom of the absolute count rather than breaking it: it cannot, by its own construction, supply the state-independent absolute input that Rem. D.5 shows N requires. By the contrapositive there recorded, any von Neumann completion that fixes N must be *non-amenable* — a Hagedorn/nuclearity-type object at the Planckian seed that Pos. 11.30 places out of scope. This is not an argument against Conj. K.1; it is the observation that the trace-stable inductive-limit programme and the count-fixing problem are, by construction, orthogonal — the former is a statement about dimensionless UV structure, the latter about the absolute scale the former leaves free.

Status of singularity resolution

Remark K.3 (Singularity status under the Type II machinery). The honest assessment of classical-singularity resolution under the gravity-sector commitment of App. H and Conj. K.1 is:

(a) (*BH exterior + near-horizon.*) **Automatic** in the Type II framework. Leutheusser–Liu [273] and Jensen–Sorce–Speranza [87] show the crossed product extends smoothly across the horizon; modular flow remains well-defined.

(b) (*BH interior up to the classical singularity.*) **Partially automatic.** Half-sided modular translations reach the singularity as a well-defined algebra homomorphism, but operator norms of curvature invariants diverge — the algebra persists, geometric interpretation does not. A type-transition / trace- renormalization prescription at $r = 0$ is an additional structural postulate the present paper *does not supply.*

(c) (*de Sitter / inflationary past.*) **Automatic** at the level of the horizon algebra [18, 89], with zero-mode of the inflaton preventing a maximum-entropy state but consistent with the paper's sub-Planckian scope.

(d) (*Big Bang itself.*) **Not automatic.** No published operator-algebraic construction extends τ_∞ to the $t = 0$ boundary; [89] is explicitly asymptotic-to-dS in the past. This is the genuine programme-level residual of the UV-completion programme and is explicitly flagged as out of scope of the present paper.

(e) (*Firewall / Page-time interior.*) **Automatic modulo island prescription**, and consistent with Prop. 13.2.

Remark K.4 (Scope declaration). The present paper claims Conj. K.1 as a *conjecture*, not a theorem; its appendix-scale status is the statement that the paper's Planck-cutoff postulate is the finite-Λ restriction of a conjectural universal Type II_∞ object. Proving existence of the inductive limit (128) even in a specific toy model (e.g. JT + matter, or $\mathrm{AdS}_3/\mathrm{CFT}_2$) is a standalone paper; extending τ_∞ across the big-bang boundary is currently open research. The paper explicitly does not claim UV completeness.

Remark K.5 (Relation to existing inductive/renormalization constructions). Conj. K.1 is the gravitational-sector analogue of structures already realised in the holographic and lattice operator-algebra literature, and is best read against them. Gesteau [36] constructs a nested family of von Neumann factors connected by faithful normal conditional expectations obeying the same compatibility relation as here, interprets the cutoff parameter as a renormalization-group flow, and proves the Susskind–Uglum compensation between the renormalizations of the area term and the bulk-entropy term. The operator-algebraic renormalization programme of Morinelli–Morsella–Stottmeister–Tanimoto [38] is the rigorous inductive-limit-of-von-Neumann-algebras machinery, building the continuum free field as a scaling limit of lattice algebras via wavelet scaling maps. Chemissany–Gesteau–Jahn–Murphy–Shaposhnik [37] exhibit the hyperfinite type II_∞ factor as an inductive limit of finite type-I tensor-network algebras — a worked instance of the target object of the present conjecture. Relative to these, the content specific to Conj. K.1 is twofold: the directed system is to be built for the CLPW/DEHK *gravitational* crossed product of App. H (not a holographic code or a lattice field algebra), and the limiting equilibrium states are required to restrict on each $\mathcal{A}_\Lambda$ to the two-boundary retrocausal state of the paper. This last requirement has no analogue in the cited constructions and is the genuine open content; it is also why the appendix records the conjecture as open rather than as a transcription of an existing theorem.

L Cosmological-Horizon Terminal POVM: selection principle for $\{E_k\}$

This appendix supplies a selection principle for the terminal POVM $\{E_k\}_{k\in\mathcal{K}}$ of Ax. 2.1 at cosmological scale, derived from the CLPW Type II_1 algebra at the future cosmological horizon plus the centraliser-resolving KE condition of Conj. 11.8. Without such a principle, the terminal POVM is an input; under this appendix it is the terminal POVM *with conjugacy class uniquely determined up to conjugation by an automorphism of the horizon algebra — Connes–Feldman–Weiss uniqueness of Cartan masas in the hyperfinite II_1 factor [41] — of which only the flow-commuting, QRF-implemented subgroup is Tier-(B) gauge (Rem. L.8); representative canonicity within the class is open* by three natural consistency conditions (what remains genuinely open is recorded in Rem. L.7 and the kill-test Rem. L.3).

Setup

Let $\mathcal{I}^+_{\mathrm{cosm}}$ be the future cosmological horizon of an asymptotic de Sitter observer (a covariant-observer QRF in the sense of [96, 97]). Let

$$\mathcal{A}_{\mathrm{cosm}} \;:=\; \Pi\left(\mathcal{A}_{\mathrm{QFT}} \rtimes_{\sigma^\phi} \mathbb{R}\right)\Pi$$

be the observer-projected CLPW Type II_1 crossed-product algebra associated with the observer's static patch, where Π is the observer positive-energy projection (the unprojected crossed product is type II_∞ [18]), with unique tracial state τ and centraliser $\mathcal{A}^\tau_{\mathrm{cosm}} = \{a \in \mathcal{A}_{\mathrm{cosm}} : \tau(ax) = \tau(xa)\ \forall x\}$.

We adopt throughout the future-eternal-de Sitter idealisation, in which bounding the observer energy from below renders $\mathcal{A}_{\mathrm{cosm}}$ type II_1 [18]. For a comoving observer in a realistic FLRW universe asymptotically de Sitter only in the past, the accessible algebra is instead type II_∞ with no maximum-entropy state [89, 90]; the type II_1 statements below (in particular the maximally-mixed terminal state and clause (C)) then require the semifinite-trace reformulation of Prop. D.2(ii) / Thm. F.2.

Conjecture L.1 (Cosmological-Horizon Terminal POVM)**.** The terminal POVM $\{E_k\}_{k\in\mathcal{K}}$ of Ax. 2.1 at cosmological scale is the projection-valued measure (PVM) on $\mathcal{A}_{\mathrm{cosm}}$, whose *conjugacy class* under $\mathrm{Aut}(\mathcal{A}_{\mathrm{cosm}})$ is unique by Connes–Feldman–Weiss (clause (A)); the choice of representative within the class is *not* in general Tier-(B) QRF gauge (see Rem. L.8), and is fixed only up to the flow-commuting subgroup by clause (B), satisfying:

(A) *Cartan masa condition:* $\{E_k\}$ generates a *Cartan* maximal abelian subalgebra $\mathcal{M} \subset \mathcal{A}_{\mathrm{cosm}}$: a masa whose normaliser $\mathcal{N}_{\mathcal{A}_{\mathrm{cosm}}}(\mathcal{M}) = \{u \text{ unitary} : u\mathcal{M}u^* = \mathcal{M}\}$ generates $\mathcal{A}_{\mathrm{cosm}}$ as a von Neumann algebra (equivalently, admitting a normal faithful τ-preserving conditional expectation onto it and a unitary normaliser acting ergodically). Cartan-ness is the operator-algebraic form of *record objectivity*: a normaliser-rich masa is one whose "pointer" projections admit an algebra-generating family of symmetries that copy and permute records, the redundancy/objectivity structure of quantum-Darwinist records in the algebraic reading of Girard–Cheng–Cao [149] (objectivity as algebraic local recoverability); a *singular* masa, by contrast, admits no non-trivial record-copying symmetries and cannot support redundant, observer-shared records.

(B) *Equivariance with respect to the horizon/QRF flow (KE at cosmological scale):* $\mathcal{M}$ is globally invariant under the dressed horizon evolution (the lift of the boost/QRF-reorientation flow to $\mathcal{A}_{\mathrm{cosm}}$), and the τ-preserving normal conditional expectation $E_{\mathcal{M}} : \mathcal{A}_{\mathrm{cosm}} \to \mathcal{M}$ (which exists and is unique for a Cartan masa) commutes with the branch-conditioning channel Φ_K of Conj. 11.1, i.e. the (KE) condition of Prop. 11.9 holds with $\mathcal{Z} = \mathcal{M}$. (*Correction:* an earlier version stated this clause against the *trace* — "every normal τ-preserving conditional expectation onto the centraliser of τ" — which is vacuous in the II_1 idealisation, since the τ-centraliser is all of $\mathcal{A}_{\mathrm{cosm}}$ and the only such expectation is the identity. The non-vacuous invariance requirement is against the horizon/QRF flow, as stated here.)

(C) *Extremal-trace decomposition:* $\{E_k\}$ resolves τ into its extremal tracial components on the center $Z(\mathcal{A}^{\tau}_{\mathrm{cosm}})$:

$$\tau \;=\; \sum_{k\in\mathcal{K}} \tau(E_k)\,\tau_k, \qquad \tau_k \text{ extremal tracial state on } E_k\mathcal{A}_{\mathrm{cosm}}E_k.$$

Uniqueness is up to measurable relabelling of $\mathcal{K}$ and the Tier-(B) automorphism gauge above.

Why Cartan + hyperfiniteness closes the conjugacy question. The local QFT algebra of the static patch is the (unique, [40]) injective type III_1 factor; injectivity is preserved by the crossed product with the (amenable) modular $\mathbb{R}$-action and by compression to the corner $\Pi\widetilde{\mathcal{N}}\Pi$, so $\mathcal{A}_{\mathrm{cosm}}$ is the *hyperfinite* II_1 factor $\mathcal{R}$. By the Connes–Feldman–Weiss theorem [41] (amenable measured equivalence relations are hyperfinite), all Cartan masas of $\mathcal{R}$ are conjugate by an automorphism of $\mathcal{R}$: clause (A) therefore determines the *conjugacy class* of $\mathcal{M}$ *uniquely up to* $\mathrm{Aut}(\mathcal{A}_{\mathrm{cosm}})$. The forward inclusion is *expected* (but not proved here — see Rem. L.8): different QRF orientations of Prop. Z.1(B) are expected to act by flow-commuting automorphisms and hence select conjugate Cartan masas, with the *marginal* record law invariant under any masa change by Prop. 11.9(1); the flow-commutativity of that action is not established. What *is* unconditional is that the converse fails: $\mathrm{Aut}(\mathcal{R})$ is far larger than the finite-parameter QRF gauge, so the residual freedom is *not* in general frame-relative gauge (Rem. L.8); representative selection is clause (B), open. Without the Cartan requirement no such statement is available: general masas of $\mathcal{R}$ are wildly non-classifiable (non-conjugate singular families, Pukánszky-invariant moduli), which is precisely why clause (A) is stated for Cartan masas.

Clause (C) requires care in the factor case. For the eternal-de Sitter static patch the centraliser $\mathcal{A}^{\tau}_{\mathrm{cosm}}$ of the tracial state coincides with the whole algebra $\mathcal{A}_{\mathrm{cosm}}$, which is a type II_1 *factor* [18]; its center $Z(\mathcal{A}^{\tau}_{\mathrm{cosm}}) = \mathbb{C}\mathbb{1}$ is then trivial, the trace is unique and already extremal, and the extremal-tracial decomposition of (C) is itself trivial — it does *not* by itself close the uniqueness. Uniqueness in this idealisation therefore rests on (A)+(B): (A) fixes the conjugacy class (Cartan/CFW, above), and (B) would rigidify the representative within the class up to the flow-commuting subgroup of $\mathrm{Aut}(\mathcal{A}_{\mathrm{cosm}})$, whose *QRF-implemented* part is Tier-(B) gauge — the identification of that subgroup with QRF gauge being itself expected but *unproven* (Rem. L.8), and the representative selection correspondingly open. At the level of record *laws* this reproduces, and sharpens, the measure-theoretic record-process invariance of Prop. 11.9. Clause (C) can acquire nontrivial content only in a regime where the centraliser carries a nontrivial center — plausibly the type II_∞ comoving-observer setting of [89], where it would select the center-labelled components — but whether that center is nontrivial there is left open (Rem. L.7).

Remark L.2 (The selected PVM is not a family of clock windows). The CHT-selected effects have genuine matter support and feed *branch (I)* of Conj. 11.1. Indeed the spectral masa of the dressed observer-energy generator $\widehat{P}$ alone is contained in the image of $\{\lambda_s\}''$ in the corner, which lies in (the corner of) the dressed centralizer $\widetilde{\mathcal{N}}^{\sigma^{\widehat{\omega}}} \cong L^\infty(\mathbb{R})$ in the geometric-ergodic regime — an *abelian* algebra whose conditioning is provably trivial on the matter algebra (Rem. 11.2); records drawn from it carry zero matter information and cannot satisfy the record-completeness axiom Ax. 7.1 for any matter observable. A Cartan masa of the II_1 factor $\mathcal{A}_{\mathrm{cosm}}$, by contrast, cannot lie in any proper sub-von Neumann algebra that fails to generate $\mathcal{A}_{\mathrm{cosm}}$ (its normaliser generates), so $\mathcal{M}$ necessarily contains effects with non-trivial matter components; these are exactly the terminal data used by the witness model of Conj. 11.55, whose per-branch Margenau–Hill negativity $f_k > 0$ certifies a non-classical two-boundary link. Whether the $\widehat{P}$-spectral masa is even maximal abelian in $\mathcal{A}_{\mathrm{cosm}}$ — and if so, Cartan or singular — is the decidable kill-test of Rem. L.3.

Remark L.3 (Decidable kill-test for the Cartan/CFW route). The following is a sharp, decidable open question on which clause (A) can be falsified: *in the CLPW II_1 factor $\mathcal{A}_{\mathrm{cosm}} = \Pi(\mathcal{A}_{\mathrm{QFT}} \rtimes_{\sigma^\phi} \mathbb{R})\Pi$, is the spectral masa generated by the dressed observer-energy generator $\widehat{P}$ (compressed to the corner) maximal abelian, and if so is it Cartan or singular?* Three outcomes: (i) if it is Cartan, the physically distinguished energy records are already of the CFW class and clause (A) is realised by the most natural candidate — but then, being energy windows, they sit in the matter-trivial sector of Rem. 11.2, and clause (B) must select a *different* Cartan representative with matter support (the two are still Aut-conjugate by CFW, and the conjugating automorphism necessarily moves matter content into the masa) — but the conjugating automorphism that injects matter content is generically *not* flow-commuting, so it can break clause (B) invariance; whether a matter-supported Cartan representative that is *also* flow-invariant exists is the open conjunction of Rem. L.7(i); (ii) if it is singular, no record-copying normaliser exists for pure energy records — consistent with their matter-triviality — and clause (A) forces the terminal masa to be genuinely different from the energy masa, as the witness model assumes; (iii) if it is not even maximal abelian, its extensions to masas carry matter components automatically. Outcomes (ii) and (iii) are consistent with the present formulation; outcome (i) would demand that the selection mechanism in clause (B) do real work, and if additionally the horizon/QRF flow were shown to fix the energy masa uniquely among flow-invariant Cartan masas, the CFW route as a *selection* principle would die (the selected records would be matter-trivial and Ax. 7.1 unsatisfiable at cosmological scale). We record this as the sharpest currently decidable test of Conj. L.1.

Remark L.4 (Relation to Thm. F.2). Conj. L.1 is the cosmological-scale analogue of Thm. F.2: Thm. F.2 resolves the terminal POVM on an Ashtekar–Krishnan black-hole dynamical horizon via the Type II_∞ crossed-product trace; Conj. L.1 promotes the construction to the future cosmological horizon via the Type II_1 trace of [18]. The trace types track the *idealisation* rather than the horizon kind: the eternal-de Sitter static patch is II_1, whereas a comoving observer in a realistic FLRW cosmology asymptotically de Sitter only in the past is II_∞ [89, 90], like the black-hole case. The two are nonetheless versions of a single programme: terminal POVM = KE-resolving PVM on the centraliser of the tracial state of a Type II horizon crossed product, subject in the II_1 factor case to the record-process caveat above.

Remark L.5 (External inputs required). Conj. L.1 requires three ambient-literature inputs, each available and active:

- (i) A Naimark/Stinespring-type measurement-scheme treatment of the CLPW algebra via QRF measurement schemes [298];
- (ii) A uniqueness theorem for the terminal masa. For the *Cartan* class this is no longer external: it is Connes–Feldman–Weiss [41] (uniqueness up to automorphism in the hyperfinite II_1 factor). What remains external is the QRF-fixing of the equivariant representative within the conjugacy class (clause (B)), which [96] almost delivers;
- (iii) A gluing result showing that the *observer-invariant* predictions extracted from the per-patch CHT-POVM (the conditional probabilities of Prop. Z.1, not the masa itself, which is frame-dependent by [97, 96]) descend to a single object on Σ_{τ_f}. A fully covariant (frame-superposed) realisation is provided by the $\mathrm{SO}(1,d)$-averaged covariant-observer crossed product of [98]; a canonical observer-independent global PVM does not follow automatically and remains the open content of this input.

Remark L.6 (Status vs. prior proposals). Horowitz–Maldacena final-state projection [65, 66] specifies a maximally-entangled final state at BH singularities, not a POVM on a cosmological Cauchy slice, and is non-unitary under environment interactions. Kent's Lorentzian final-boundary programme [65] postulates a late-time mass-energy distribution with Born probability as a phenomenological input. Conj. L.1 is distinctive in deriving $\{E_k\}$ from the algebraic structure of the CLPW Type II_1 crossed product, not postulating it.

Remark L.7 (Open problem: canonical masa selection). Under the Cartan restatement of clause (A), Connes–Feldman–Weiss [41] supplies the uniqueness-up-to-conjugacy theorem that earlier versions of this remark identified as missing: the conjugacy class of $\mathcal{M}$ is unique. The choice of representative is fixed only up to the flow-commuting automorphism subgroup (clause (B)) and is *not* in general absorbed into Tier-(B) QRF gauge (Rem. L.8). What remains genuinely open is: (i) *canonicity within the class* — exhibiting a horizon/QRF-flow-*equivariant* Cartan representative (clause (B)) that *also* carries matter support (clause (A) non-trivially), and showing the flow-commuting automorphism subgroup acts transitively on such representatives so that the gauge absorption is complete. This is the framework's sharpest structural risk (the "flow-invariance vise"): clause (B) demands the terminal masa be invariant under the horizon/boost flow, while a matter-supported (hence non-flow-commuting) terminal measurement is what makes the record non-trivial. Whether a single Cartan masa can be simultaneously flow-invariant *and* matter-supported is **open** — but it is a *more tractable* question than the "ergodic flow" language suggests, and the structural evidence does not lean against it. The key point, easy to miss, is that on the crossed product $\mathcal{A}_{\mathrm{cosm}}$ the dressed horizon/boost flow is *inner*: by the crossed-product centraliser Lemma following Conj. 11.1, $\sigma_t^{\widehat{\omega}} = \mathrm{Ad}(\lambda_t)$ is implemented by the clock/translation unitary group $\{\lambda_t\} \subset \mathcal{A}_{\mathrm{cosm}}$. So clause (B) is *not* the exotic demand that an exogenous weakly-mixing ergodic $\mathbb{R}$-flow fix a Cartan setwise; it is the concrete demand that the *one fixed unitary group* $\{\lambda_t\}$ *normalise* a Cartan masa $\mathcal{M}$ while acting on it non-trivially — matter support being exactly the normaliser elements $N(\mathcal{M}) \setminus \mathcal{M}$, which generate all of $\mathcal{R}$ (Feldman–Moore [42]). The weak-mixing / full-Connes-spectrum $\Gamma(\alpha) = \mathbb{R}$ language pertains to the boost on the *underlying* type III_1 algebra $\mathcal{N}$ (Borchers–Buchholz), *not* to the fixed-point structure of the inner flow on the II_1 factor $\mathcal{A}_{\mathrm{cosm}}$, whose relative commutant $\{\lambda_t\}' \cap \mathcal{A}_{\mathrm{cosm}}$ is large. The honest open question is therefore a *conjugacy* question: does the given dressed-energy group $\{\lambda_t\}$ (spectrum of $\widehat{P}$) conjugate, by an automorphism of $\mathcal{R}$, into the normaliser of some Cartan on which it acts non-trivially? Two facts frame it. *(No obstruction.)* The invariant-subalgebra-rigidity results (Amrutam–Jiang [44], Kalantar–Panagopoulos, Ioana–Spaas) concern Γ-invariant subalgebras of *group* von Neumann algebras under discrete conjugation and do *not* transfer to this continuous inner flow; the outer-flow classification of [45, 46] classifies flows up to cocycle conjugacy but says *nothing* about Cartan preservation. No obstruction theorem is known, and the inner-flow structure gives no *a priori* reason to expect one. *(Construction direction, and its limit.)* By Feldman–Moore/CFW the normaliser of a Cartan generates $\mathcal{R}$, and measure-preserving $\mathbb{R}$-flows whose orbits lie in the generating equivalence relation lift to normaliser flows that keep $\mathcal{M}$ invariant while acting non-trivially — so flow-invariant *and* matter-supported Cartans exist in abundance *for suitable flows.* But this construction is free to *choose* the flow to suit a given Cartan, whereas clause (B) does not have that freedom: the flow $\{\lambda_t\} = \{e^{it\widehat{P}}\}$ is *given.* So the construction proves only that the target set of good flows is non-empty, *not* that the fixed $\{\lambda_t\}$ lies in it. The naïve candidate, the energy masa containing $\widehat{P}$, is invariant but *centralised* (the flow acts trivially on it), hence matter-trivial (kill-test outcome (i)); and $\{\lambda_t\}''$ is abelian and central in the centralizer ($\{\lambda_s\}'' \subseteq Z(\widetilde{\mathcal{N}}^{\sigma})$), while the Kawahigashi classification of one-parameter flows fixing a Cartan concerns flows fixing it *elementwise* (acting trivially), so a single inner flow generated by one $\widehat{P}$ tends to *centralise* rather than *permute* a masa. We therefore record the status as **genuinely open: no known obstruction, but no positive evidence for the specific $\{\lambda_t\}$, and a faint adverse structural tilt** — neither the "leaning obstructed" of the retracted exogenous-ergodic-flow framing nor "plausibly achievable." (Note that "matter-supported" in the strong sense — $\mathcal{M}$ not inside the state centralizer — is a priori stronger than "the flow acts non-trivially on $\mathcal{M}$"; the two co-vary for the energy masa but need not in general.) What is unestablished is the *conjunction* with matter support for that boost flow; the paper's cited intrinsic-observer construction [365] fixes the observer only up to QRF freedom and does not deliver this masa. The cosmological realizability of Ax. 7.1 rides on this open conjunction; the finite-dimensional non-vacuity (Prop. 11.53) is independent of it. Additionally open: (ii) the *kill-test* of Rem. L.3 (Cartan-or-singular status of the dressed-energy spectral masa); and (iii) the type II_∞ comoving-observer setting of [89], where hyperfiniteness persists but the trace is only semifinite and the CFW argument needs the semifinite Cartan theory. The classification obstruction for *general* masas (Dixmier Cartan-vs-singular [30]; wild singular moduli) is real but no longer bites: it is exactly what the Cartan restriction excludes.

Remark L.8 (The residual freedom is $\mathrm{Aut}(\mathcal{R})$, not QRF gauge). Connes–Feldman–Weiss determines the terminal Cartan masa $\mathcal{M}$ only up to conjugation by $\mathrm{Aut}(\mathcal{A}_{\mathrm{cosm}}) = \mathrm{Aut}(\mathcal{R})$, the automorphism group of the hyperfinite II_1 factor. This group is *enormous*: it contains the outer group $\mathrm{Out}(\mathcal{R})$ and is far larger than the finite-parameter QRF frame gauge $\{C_\alpha\}$ of Prop. Z.1(B). We correct an overclaim of earlier versions: it is *not* true that $\mathrm{Aut}(\mathcal{R})$ is "absorbed into" the Tier-(B) QRF gauge. The intended *forward* direction is the following, and we are careful to mark which part is proved and which is expected: distinct QRF orientations C_α are *expected* to act by (a subgroup of) flow-commuting automorphisms and hence to select *conjugate* Cartan masas — the flow-commutativity of the C_α-action on $\mathcal{A}_{\mathrm{cosm}}$ is *not* proved here (Prop. Z.1(B),(D) establish QRF-covariance of the Type-II data and fix the observer only up to QRF freedom, not flow-commutativity), so this forward inclusion is itself conjectural. What *is* unconditional is the converse's falsity: not every $\mathrm{Aut}(\mathcal{R})$-conjugate masa is QRF-related, since a generic element of $\mathrm{Aut}(\mathcal{R})$ is implemented by no QRF reorientation — which suffices to refute "$\mathrm{Aut}(\mathcal{R})$ is gauge." Consequently: (i) the *marginal* record process is invariant under *any* change of terminal masa (a fortiori under the CFW conjugacy class), as established — for the horizon II_∞ setting, by the generic separable-abelian argument of Prop. 11.9(1), which transfers to the cosmological II_1 factor unchanged; the stronger invariance of the *joint* law with the terminal label K is Prop. 11.9(2) and holds only under (KE), which is *not* established for a matter-supported cosmological masa — so it is the marginal, not the full record law, that survives unconditionally; (ii) fixing the physical record *content* frame-independently requires the flow-commuting automorphism subgroup to act *transitively* on the physically admissible representatives (flow-invariant *and* matter-supported), which is exactly the open transitivity of Rem. L.7(i) and is *not* delivered by CFW. The genuine content of clause (A) is therefore conjugacy-class uniqueness, not representative uniqueness; representative selection is clause (B), open.

M Hawking recovery and AdS/CFT reduction: canonical benchmarks

This appendix records two recovery theorems that the framework is expected to satisfy as consistency checks with known physics: the Hawking radiation spectrum and the AdS/CFT reduction. Both are theorems of the 2022–2026 ambient literature ([18, 92, 87, 88, 244, 273]); the paper cites and restates.

(B5a) Hawking recovery

Theorem M.1 (Hawking recovery from CLPW Type II modular flow)**.** *Let $(\mathcal{M}, g)$ be globally hyperbolic with a bifurcate Killing horizon $\mathcal{H}$ of surface gravity κ, and let $\mathcal{A}_R$ be the CLPW Type II_∞ crossed-product algebra of Thm. F.2 specialised to the Killing case, with faithful normal semifinite trace τ (a II_∞ factor admits no tracial* state*) and equilibrium Hartle–Hawking KMS state ω_{HH}. Under Pos. 11.11 identifying the clock Hamiltonian with the modular Hamiltonian of the horizon boost, for any asymptotic number operator $\widehat{N}_\omega \in \mathcal{A}_R$ with horizon-boost frequency ω,*

$$\omega_{\mathrm{HH}}(\widehat{N}_\omega) \;=\; \left(e^{2\pi\omega/\kappa} - 1\right)^{-1}. \tag{129}$$

The expectation is taken in the equilibrium KMS state ω_{HH} *at $T_{\mathrm{H}} = \kappa/(2\pi)$, not in the trace τ (whose weight is state-independent and carries no Bose–Einstein distribution); the claim is scoped to the equilibrium (Hartle–Hawking) temperature, not to a dynamical Hawking flux. In particular, the Hawking temperature $T_{\mathrm{H}} = \kappa/(2\pi)$ is a corollary of Tomita–Takesaki, not an independent input — with the caveat that the surface gravity κ enters through the same geometric boost data that calibrates thermal time, so the identification is a consistency recovery rather than an independent derivation.*

Proof sketch. Bisognano–Wichmann plus Sewell [275] plus Kay–Wald [276] give the KMS condition for ω_{HH} on the wedge algebra at inverse temperature $\beta_{\text{H}} = 2\pi/\kappa$ under the horizon-boost flow; Summers–Verch modular-inclusion [277] extends to bifurcate horizons. [88] and [92] promote this to the CLPW Type II crossed product. On number-type observables commuting with the dressed clock, the KMS condition for ω_{HH} under the boost flow gives the Bose–Einstein distribution (129); the semifinite trace τ enters only as the reference weight defining density operators on $\mathcal{A}_R$, not as the expectation functional. $\square$

(B5b) AdS/CFT reduction

Theorem M.2 (AdS/CFT reduction of the framework)**.** *Let $\mathcal{M} = \text{AdS}_{d+1}$ with conformal boundary $\partial\mathcal{M} = \mathbb{R} \times S^{d-1}$ carrying a large-N holographic CFT$_d$ with vacuum modular algebra $(\mathcal{M}_{\partial,\text{CHM}}, \Delta_{\text{CHM}})$ in the sense of [241]. Let ω_{TFD} be the thermofield-double state of the CFT at inverse temperature β. If the two-boundary retrocausal boundary states $\omega_\pm$ of Sec. 5 satisfy the identification $\omega_- \otimes \omega_+ = \omega_{\text{TFD}}$, then the CLPW Type II$_\infty$ crossed-product algebra $\mathcal{A}_{\text{bulk}}$ of App. H is unitarily equivalent, in the $G_{\text{N}} \to 0$ limit, to the CFT's single-trace modular algebra $\mathcal{M}^{\text{crossed}}_{\partial,\text{CHM}}$ via the bulk/boundary extrapolate dictionary. The linearised Einstein equation (Thm. I.1) then follows from the boundary first law of entanglement.*

Proof sketch. Algebra-level reduction is [18, 244, 273, 274]: subregion-subalgebra duality in the Type II setting identifies the bulk crossed product with the boundary modular algebra via the CHM modular Hamiltonian and the JLMS equality of bulk and boundary modular flow [245] in the entanglement wedge. The only new step is the $\omega_- \otimes \omega_+ = \omega_{\text{TFD}}$ identification, which is a short lemma: the TFD is the unique pure state on the doubled boundary Hilbert space whose modular flow is the bulk-extended boost and whose single-boundary restriction is the KMS state at β. Linearised Einstein then follows from Thm. I.1 plus the FGHMVR argument [58, 243]. $\square$

Remark M.3 (Status)*.* Thms. M.1, M.2 are consistency checks: they establish that the framework recovers (not merely contains) two canonical benchmarks of quantum gravity. Both are direct applications of the 2022–2026 operator-algebraic QG literature and require no new mathematics beyond the two-boundary $\to \omega_{\text{TFD}}$ identification lemma of (B5b). The (B5b) identification is an $N \to \infty$ / code-subspace statement: at finite N subregion bulk reconstruction degrades above a logarithmic UV scale $\Lambda_{\text{crit}} \sim \ln N$ [242], so it reduces the leading large-N data rather than furnishing an all-orders equivalence.

N Numerical toy model: finite-dimensional TBRME on FLRW

This appendix specifies a minimal computable toy model in which the Two-Boundary Retrocausal Master Equation (TBRME) can be numerically integrated and its five computable reductions explicitly tested. The purpose is to convert the paper from "mathematical framework" to "framework with working model". *Labelling convention.* The items (U1)–(U5) of this appendix renumber the five *computable* reductions independently of the six properties (U1)–(U6) of Thm. 11.73: Appendix-L (U1) TDSE corresponds to Theorem (U2) (quantum-mechanical reduction), Appendix-L (U2) Part I two-boundary to the Part I posterior of Theorem (U2), and Appendix-L (U3),(U4),(U5) to Theorem (U3),(U4),(U5) respectively; the Theorem's well-posedness item (U1) is exercised implicitly by constructibility of $\widehat{H}_{\text{tot}}$.

Toy-model specification

The universal Hilbert space $\mathcal{U} = \bigoplus_{k\in\mathcal{K}_+} \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{geom}} \otimes \mathcal{H}^{(k)}_{\text{matter}}$ is truncated as:

- *Clock sector.* $\mathcal{H}_{\text{clock}} = \mathbb{C}^{N_c}$, $N_c = 64$: zero-mode of a free massive scalar ϕ_0 on a compact interval with Dirichlet cutoff, $\widehat{H}_{\text{c}} = \widehat{p}^2_{\phi_0}/2 + m^2\phi_0^2/2$.

- *Geometry sector.* $\mathcal{H}_{\text{geom}} = \mathbb{C}^{N_a}$, $N_a = 128$: log-scale-factor $b = \log a$ on a finite grid, $\widehat{H}_{\text{geom}}$ generating the WDW kinetic term $\widehat{\pi}_b^2/(24\,a)$.
- *Matter sector.* $\mathcal{H}^{(k)}_{\text{matter}} = \mathbb{C}^{N_m}$, $N_m = 20$: truncation of a single inhomogeneous conformally-coupled mode χ_k per branch, with back-reaction coupling $\chi_k^2 R(a)$ to the scale factor.
- *Branch index.* $\mathcal{K}_+ = \{k_1, k_2, k_3\}$: three representative comoving wavenumbers.

Total dimension $\dim \mathcal{U} = 3 \cdot N_c \cdot N_a \cdot N_m \approx 5 \times 10^5$ (sparse). The total Hamiltonian is

$$\widehat{H}_{\text{tot}} \;=\; \widehat{H}_{\text{c}} \otimes \mathbb{1} \otimes \mathbb{1} \;+\; \mathbb{1} \otimes \widehat{H}_{\text{WDW}} \otimes \mathbb{1} \;+\; \sum_{k \in \mathcal{K}_+} \widehat{H}^{(k)}_{\text{matter}} \;+\; \widehat{H}_{\text{backreact}}.$$

Two-boundary data: ρ_{i} Gaussian centred at WKB slow-roll inflationary point; terminal POVM $\{E_k\}$ a rank-1 coherent-state projector in (a, π_a) per branch.

Numerical verification of the five reductions

Proposition N.1 (Numerical verification of (U1)–(U5))**.** *Under the toy-model specification above, the five reductions of Thm. 11.73 admit the following numerical tests:*

(U1) *(*TDSE.*) Freezing a and tracing out clock and geometry sectors reduces the TBRME to the standard Schrödinger equation on $\mathcal{H}^{(k)}_{\text{matter}}$; comparison to direct TDSE integration is exact up to truncation order.*

(U2) *(*Part I two-boundary.*) Restricting to $\mathcal{H}_{\text{clock}} \otimes \mathcal{H}^{(k)}_{\text{matter}}$ and truncating a to a single mode reproduces the ABL posterior $\pi_\tau(k)$ and the martingale convergence of Thm. 7.3. The per-branch two-time quasiprobability attached to this posterior is the Margenau–Hill operator $M_k = \frac{1}{2p}\{\rho, \Lambda\}$ of Thm. 3.10; numerically, $M_k \not\succeq 0$ (temporal non-classicality / Leggett–Garg negativity) exactly on branches where the forward state and backward effect fail to commute, while the prior average $\sum_k p(k) M_k = \rho \succeq 0$ recovers classicality by the no-signalling identity (Thm. 6.1). Verifying both the per-branch negativity and its vanishing under marginalization is a direct finite-dimensional test.*

(U3) *(*Semiclassical Einstein.*) Ehrenfest evolution on a with back-reaction $\langle T^\mu_\mu \rangle$ from the matter sector is structurally compatible with the Pinamonti–Siemssen higher-derivative cosmological semiclassical-Einstein equation of [194]; a quantitative ($\lesssim$ few-percent) comparison at converged WDW grid resolution is the scope of the companion numerical paper.*

(U4) *(*Modified Wheeler–DeWitt.*) Born–Oppenheimer expansion to $\mathcal{O}(1/E_{\text{P}}^2)$ is expected to reproduce the Kiefer–Krämer power-spectrum correction coefficients of [249, 251]; the converged coefficient match under a refined WDW grid is the scope of the companion numerical paper.*

(U5) *(*Modified TDSE.*) On a single branch with $\mathcal{H}_{\text{clock}} \otimes \mathcal{H}^{(k)}_{\text{matter}}$ only, the TBRME admits a closed-form analytic benchmark agreeing with the exact fiberwise reduction of Prop. C.2 (Sec. C.2) verbatim.*

Implementation. The toy model is numerically tractable on a laptop (compute budget $\approx$ one week). Recommended implementation stack: Python with QuTiP 5.3 (sparse tensor-product Hamiltonians, Page–Wootters-compatible `mesolve` / propagator, built-in partial-trace and POVM primitives); Julia/QuantumOptics.jl as a performance alternative. Reference benchmarks: [278] for hybrid LQC numerics (U3, U4); [279] for Page–Wootters clock simulation primitives (U2). (U1), (U2), (U5) are end-to-end tractable on a laptop; (U3), (U4) at full convergence are the scope of a companion numerical paper. The companion notebook is released alongside the paper as supplementary material. □

Remark N.2 (Status)*.* Prop. N.1 converts the TBRME from a formal expression into a finite-dimensional object with explicit numerical content. No claim in the main body of the paper relies on Prop. N.1; it is provided as empirical support for the structural theorems of the paper. The companion numerical paper extends (U3), (U4) to converged precision.

O Kuchar-evasion: thermal time, Trinity, and PW=CLPW

This appendix discharges Kuchař's four classical objections to the Page–Wootters formulation ([297]: (K1) frozen formalism, (K2) inner-product problem, (K3) clock-reparametrisation / multiple-choice gauge, (K4) absence of an actual Schrödinger evolution) for the TBRME machinery, under the paper's three structural postulates plus the CLPW Type II crossed-product commitment of App. H.

Setup

Take the Page–Wootters universal Hilbert space $\mathcal{U} = \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{rest}}$ with constraint $\widehat{C}\Psi = 0$. Under Pos. 11.11, $\widehat{H}_{\text{c}} = -\log \Delta_\omega$. Under App. H, $\widetilde{\mathcal{N}} = \mathcal{N} \rtimes_{\sigma^\omega} \mathbb{R}$ is a Type II factor with faithful normal semifinite trace Tr_ω (II_∞ as unprojected crossed product; the observer-projected CLPW algebra with dressed generator bounded below is II_1 with finite trace, Thm. H.1(i)).

Theorem O.1 (Kuchar-evasion for the TBRME)**.** *Under Pos. 11.11 and the CLPW Type II crossed- product realisation $\widetilde{\mathcal{N}}$ of Thm. H.1, the Page–Wootters framework of the TBRME satisfies (K1)–(K4) as follows:*

(K1) *(*Frozen formalism resolved.*) The constraint $\widehat{C}\Psi = 0$ on $\mathcal{U}$ is the kernel of the modified Wheeler–DeWitt operator $\widehat{\Sigma}_\delta = \cos(\delta \widehat{H}_{\text{c}}/\hbar) - \not\Vdash$ (Conj. 10.13, Thm. B.5). Conditioning on the covariant clock POVM associated with $\widehat{H}_{\text{c}} = -\log \Delta_\omega$ yields the modular-time TDSE Prop. 10.14; the flow is* not *parameter time but the modular automorphism σ_t^ω, which is genuine dynamics on $\widetilde{\mathcal{N}}$.*

(K2) *(*Physical inner product.*) By Haagerup duality (Thm. G.2), the dual-weight trace* Tr_ω *on $\widetilde{\mathcal{N}}$ supplies the physical inner product on the constraint surface; finite on band-limited states by Pos. 11.30 (H0) together with the trace-finiteness of Prop. D.2. Equivalently this is the Marolf group-averaging inner product [280] across the modular orbit.*

(K3) *(*Clock-reparametrisation gauge.*) Change of clock subsystem $\omega \to \psi$ acts by the Connes cocycle $u_t = (D\psi : D\omega)_t \in \mathcal{U}(\mathcal{N})$; conditional probabilities transform covariantly, and observables on the rest of $\mathcal{U}$ are gauge-invariant up to the cocycle. This is the operator-algebraic form of covariance under clock reparametrisation.*

(K4) *(*Schrödinger dynamics for physical observables.*) The Höhn–Smith–Lock trinity [102, 103], lifted to the CLPW setting by the De Vuyst–Eccles–Höhn–Kirklin PW=CLPW result [96], identifies Page–Wootters conditional probabilities with the Heisenberg dynamics of relational Dirac observables on each frequency superselection sector. Hence expectations of physical observables genuinely evolve in modular time; Page–Wootters is not merely a correlation calculus. Indeed, the per-branch content that distinguishes genuine relational dynamics from a static joint-correlation table is made precise by the Margenau–Hill witness of Thm. 3.10: the t_i-leg marginal $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$ of the conditioned state-over-time is non-positive exactly when the forward state and the backward effect fail to commute, $[\rho(\tau), \Lambda_k(\tau)] \neq 0$, certifying temporal non-classicality ("entanglement across time") on each branch while marginalizing to the unconditional $\rho(\tau) \succeq 0$ by the no-signalling identity. (Macrorealist exclusion from this negativity is conditional on non-invasive measurability, and the clean projective iff is for sharp effects.)*

Proof sketch. (K1), (K4) reduce to the Höhn–Smith–Lock trinity [102, Prop. 4.5], [103, Thm. 3.2] applied to the modular-time clock $\widehat{H}_{\text{c}} = -\log \Delta_\omega$ via the PW=CLPW identification of [96]. As in (K4) and the structural-backbone remark of Sec. 11.2, the trinity and PW=CLPW equivalences are established *per frequency superselection sector* of $\widehat{C}$ [103, 96]; the discharge of (K1),(K4) is therefore understood sector-wise, the relativistic clock $\widehat{H}_{\text{c}} = -\log \Delta_\omega$ entering through a covariant POVM rather than a sharp self-adjoint time operator across sectors. (K2) is Haagerup dual-weight Thm. G.2 plus the Type II trace-finiteness of Prop. D.2 on the band-limited sectors of Pos. 11.30. (K3) is the Connes cocycle derivative [39, Thm. VIII.3.3]: change of clock state by

$\omega \to \psi$ gives $\sigma_t^\psi = \mathrm{Ad}(u_t) \circ \sigma_t^\omega$ with $u_t \in \mathcal{U}(\mathcal{N})$, and conditional-probability formulae transform by the same cocycle. □

Remark O.2 (Status). Thm. O.1 discharges Kuchař's four objections to a corollary of theorems already in the paper (Thm. B.5, Prop. D.2, Thm. G.2, Thm. H.1), the postulate Pos. 11.30, plus the Höhn–Smith– Lock trinity and the PW=CLPW result. No new functional analysis is required.

The thermal-time-specific objections (Swanson, Chua)

The objections (K1)–(K4) discharged above are Kuchař's objections to the *Page–Wootters / Dirac-constraint* programme (frozen formalism, inner product, multiple-choice, absence of Schrödinger evolution), not the distinct, thermal-time-*specific* objections that a referee may have in mind. The canonical catalogue of the latter is Swanson's [34], on which Chua [359] builds.

Remark O.3 (Swanson's three conceptual problems). Distinct from (K1)–(K4), Swanson [34] isolates three conceptual objections to the thermal-time hypothesis itself: (S1) the flow of time in genuinely non-equilibrium seed states; (S2) background-independence; and (S3) gauge-invariance. We do not claim to *discharge* these; the framework offers the following resources. Against (S2)–(S3): the Connes-cocycle covariance of the master equation (Prop. 11.62) means a change of seed $\omega \to \omega'$ acts by the cocycle $(D\omega' : D\omega)_t$, and all *physical* predictions are QRF- and seed-invariant (Prop. Z.1). This bears on, but does not by itself settle, (S2)–(S3): Swanson's objections concern whether the *choice* of coarse-grained seed is itself background/gauge data, a question the cocycle covariance reframes rather than dissolves. Against (S1) we concede a genuine scope limitation: the entropy-monotonicity of Prop. 11.15 presupposes a faithful seed ω for which σ_t^ω is the relaxation flow, and for a generic far-from-equilibrium seed the modular parameter need not coincide with any geometric or proper time. The de Sitter witness of Conj. 11.55 sidesteps (S1) by sitting in the Bisognano–Wichmann case, where $-\log \Delta_\omega$ *is* the geometric boost; outside such geometric cases (S1) is flagged as a limitation rather than a discharged objection.

P Wood–Spekkens evasion: terminal data as modular- theoretic invariant

This appendix records the framework's evasion of the Wood–Spekkens fine-tuning no-go theorem [282] for retrocausal hidden-variable models, via Conj. L.1's dynamical selection of the terminal POVM as a modular-theoretic invariant.

Theorem P.1 (Wood–Spekkens evasion)**.** *In the framework of Sec. 5 with terminal POVM* $\{E_k\}$ *determined by Conj. L.1 (App. L), the model satisfies the operational predictions of standard QM and* avoids *the Wood–Spekkens 2015 fine-tuning condition [282], because:*

(a) *The terminal data* $\{E_k\}$ *are not free parameters but the unique centraliser-PVM of the CLPW Type* II_1 *tracial state on the cosmological-horizon algebra (Conj. L.1); they cannot be adjusted to suppress signalling.*

(b) *The retrocausal posterior* $\rho_{\mathrm{RC}}^\gamma$ *enters as an ABL-martingale conditional expectation on observables localised in the past of* τ_f*, so by the causal-propagation axiom (M3) of Ax. 2.2 every retrocausal "arrow" terminates in a region operationally indistinguishable from a forward-evolved Heisenberg observable; the retrocausal degrees of freedom act as a gauge in the sense of Pearl–Spekkens [282, 289]. We stress that this is a statement of no-superluminal-*signalling *(NSS); the centraliser-side retrocausal capacity (the Ji–Lloyd–Wilde-style influence of Rem. 6.4 and Conj. L.1) is a genuine no-superluminal-*causation *(NSC) violation on* *postselected sub-ensembles. The operator-algebraic certificate of this postselected influence is the per-branch Margenau–Hill witness of Thm. 3.10: on each retrocausally conditioned branch k the symmetric two-boundary marginal $M_k = \frac{1}{2p}\{\rho, \Lambda_k\}$ fails positivity exactly when the forward state and backward effect do not commute, $[\rho(\tau), \Lambda_k(\tau)] \neq 0$ (projective case), yet $\sum_k p(k) M_k = \rho \succeq 0$, so the negativity is irreducibly per-branch and is annihilated by the prior average — the same marginalisation that yields bulk NSS (Thm. 6.1). By Vilasini–Colbeck [295, 296] NSS neither implies nor is implied by NSC, and embedding an NSC-violating influence in a frame-covariant spacetime generically requires causal fine-tuning; here that fine-tuning is supplied lawfully by modular invariance (Conj. L.1) rather than by hand, but it is not eliminated (cf. Rem. P.3).*

(c) *The internal-cancellation mechanism of Almada et al. [283] and the lawhood-as-constraint framework of Adlam [289] are here* derived consequences *of modular invariance rather than imposed symmetries, putting the framework in the Price–Wharton "W-as-edge-of-wedge" equivalence class [294] where the centraliser-PVM plays the role of W (cf. the critical reading of the constrained-collider mechanism as co-causal/acausal rather than a genuine mechanism in Stuckey–Silberstein [293]).*

Faithfulness is violated only on a measure-zero set of two-boundary data outside the centraliser class of Conj. L.1, and the violation is removable by a unique gauge fixing.

Remark P.2 (Conditional status). Thm. P.1 is conditional on Conj. L.1 (centraliser-PVM uniqueness) holding; if uniqueness fails, the framework reverts to the Almada-style "symmetry-natural fine-tuning" defence [283], which is weaker but still defensible. This is the only place in the paper where evasion of an external no-go theorem is conditional on a paper conjecture rather than an established theorem.

Remark P.3 (Scope of the evasion). Thm. P.1 evades the *causal* fine-tuning of Wood–Spekkens [282] (unfaithfulness imposed as an adjustable parameter to suppress signalling); it does *not* claim to evade the causal-structure-independent notion of *operational* fine-tuning of Catani–Leifer [290], under which any property holding operationally (here bulk no-signalling, Thm. 6.1) while breaking ontologically (the centraliser-side retrocausal influence of Conj. L.1) is fine-tuned regardless of whether its value is fixed by a symmetry. The precise operator-algebraic content of this "operational-while-ontological" split is Thm. 3.10: the operationally invariant quantity is the prior-averaged state $\sum_k p(k) M_k = \rho \succeq 0$ (no-signalling, Thm. 6.1), while the ontological non-classicality is the per-branch quasiprobability negativity $f_k > 0$, present iff $[\rho(\tau), \Lambda_k(\tau)] \neq 0$. Our claim is the weaker and more precise one that the requisite unfaithfulness is a *derived, lawlike constraint* (modular invariance) in the sense of Adlam [289], not a tunable coincidence — i.e. lawful fine-tuning, not its absence.

Q Wave-function normalisation on $\mathcal{U}$: refined algebraic quantisation via the CLPW trace

This appendix establishes a normalisable physical Hilbert space $\mathcal{H}^{(k)}_{\text{phys}}$ for solutions of the TBRME constraint $\widehat{\mathfrak{C}}_k \Psi_k = 0$ on the kinematical space $\mathcal{H}^{(k)}_{\text{kin}} = \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{geom}} \otimes \mathcal{H}^{(k)}_{\text{matter}}$, via the Marolf rigging map / CLPW Type II trace equivalence of Fewster–Janssen–Loveridge–Rejzner–Waldron [298] and the renormalised group-averaging construction of [299].

Setup

Constraint solutions of $\widehat{\mathfrak{C}}_k \Psi_k = 0$ generically have 0 in the continuous spectrum of $\widehat{\mathfrak{C}}_k$, so no normalisable $\mathcal{H}^{(k)}_{\text{kin}}$-eigenvectors exist; rigged-Hilbert machinery is required.

Hypothesis Q.1 (Stabiliser-orbit hypotheses). Assume:

(H1) The modular automorphism group σ_t^{Φ} acts with compact (or amenable, finite-volume) stabilisers on a dense domain $\mathcal{D}_k \subset \mathcal{H}_{\mathrm{kin}}^{(k)}$ (orbit-type stratification).
(H2) The Page–Wootters clock translations are unitarily implemented and amenable.
(H3) The matter sector is finite-dimensional under Pos. 11.30.

(H1) is the obstruction identified by [299] for cosmological constraints; it is non-trivial for closed FLRW but is preserved on each orbit-type stratum. (H2) is a standard Page–Wootters clock condition. (H3) is Pos. 11.30. The status of (H1) is sharpened by the linearization-(in)stability analysis of De Vuyst–Eccles–Höhn–Kirklin [99]: whether linearized constraint solutions integrate to exact ones (and hence whether the residual modular/boost constraints carry an essentially unambiguous crossed-product justification) is topology-dependent, holding on closed (compact-Cauchy) spacetimes but becoming contingent in the presence of boundaries or absent isometries. (H2) is realised rigorously for non-perturbative quantum gravity, including the many-independent-symmetry ("multi-fingered time") generalisation that matches our sector-indexed family $\{k\}$, by the group-field-theory Page–Wootters construction of Calcinari–Gielen [100].

Theorem Q.2 (TBRME physical Hilbert space)**.** *Assume Hyp. Q.1. The renormalised group-averaging rigging map*

$$\eta_k \;:\; \mathcal{D}_k \;\longrightarrow\; \mathcal{D}_k^*, \qquad \eta_k(\psi)[\chi] \;=\; \int_{(\mathbb{R}\times\mathbb{R})/\mathrm{Stab}_x} d\dot{g}\;\langle \psi, U(g)\chi\rangle, \tag{130}$$

converges absolutely on each orbit-type stratum, and its image

$$\mathcal{H}_{\mathrm{phys}}^{(k)} \;:=\; \overline{\eta_k(\mathcal{D}_k)} \tag{131}$$

is a separable Hilbert space carrying a non-degenerate inner product, isomorphic via GNS to the standard form of the CLPW Type II trace Tr_ω *of Thm. H.1. Every* $\Psi_k \in \mathcal{H}_{\mathrm{phys}}^{(k)}$ *satisfies* $\widehat{\mathfrak{C}}_k \Psi_k = 0$ *in the rigged sense and admits a finite physical norm*

$$\|\Psi_k\|_{\mathrm{phys}}^2 \;=\; \mathrm{Tr}_\omega(|\Psi_k\rangle\langle\Psi_k|), \tag{132}$$

unique up to the central ambiguity (rescaling of the clock zero).

Proof sketch. By Fewster–Janssen–Loveridge–Rejzner–Waldron [298], on a kinematical Hilbert space carrying a modular flow with compact (amenable) stabilisers, the crossed-product/conditional-expectation construction is *equivalent* to Marolf's rigging-map construction; both agree with BRST and with the path integral. Apply this equivalence to the TBRME constraint with the modular flow of Pos. 11.11 and the clock translations of Hyp. Q.1(H2): the rigging map on each orbit-type stratum produces the same physical Hilbert space as the Type II GNS construction of Thm. H.1.

For non-compact stabilisers (closed FLRW; JT gravity [299]) the naive group-averaging integral diverges; the renormalised rigging map of [299, §3] replaces $\int dg$ by the quotient $\int_{G/\mathrm{Stab}_x} d\dot{g}$, restoring absolute convergence. The result is positive-definite and equals the CLPW Type II trace on the central-orbit sector.

The clock-zero rescaling ambiguity is the standard central ambiguity of any group-averaging construction; it does not affect Born-rule predictions, which depend on $\Psi_k/\|\Psi_k\|$. □

Remark Q.3 (Status). Thm. Q.2 closes the wave-function normalisation problem for the TBRME under Hyp. Q.1. The orbit-type hypothesis (H1) is non-trivial: [299] (Dec. 2025) shows naive group averaging fails for cosmological constraints without it, but their renormalised rigging map (130) restores convergence on each stratum. Under Pos. 11.30 the orbit-type stratification of the TBRME on $\mathcal{U}$ is finite, so (H1) holds; this should be made explicit in any computational specialisation.

Remark Q.4 (Type II_1 strengthening). Thm. Q.2 is insensitive to the II_1-vs-II_∞ subtype of Thm. H.1: the FJLRW equivalence and the renormalised rigging map require only a faithful normal semifinite trace. In the type II_1 regime of Thm. H.1(i) the trace is a normal faithful *state*, so the trace vector $\Omega_\tau \in L^2(\widetilde{\mathcal{M}}_\omega, \mathrm{Tr}_\omega)$ is a unit vector and the physical norm (132) obeys $\|\Psi_k\|^2_{\mathrm{phys}} \leq \|\Psi_k\|^2 \,\mathrm{Tr}_\omega(\mathbb{1}) < \infty$ with no semifiniteness domain restriction: normalisability of the physical wave function is automatic rather than conditional. Only the II_∞ branch sectors (evaporating/dynamical horizons, realistic FLRW; Thm. F.2, [90, 89]) need the semifinite-domain version.

R Reeh–Schlieder for ABL conditioning

This appendix shows the framework is not exposed to a new Reeh–Schlieder paradox under ABL two-boundary conditioning, and identifies precisely when the cyclic-separating property is preserved versus lost (in a controlled, expected way).

Theorem R.1 (Reeh–Schlieder for ABL conditioning)**.** *Let $\mathcal{A}(\mathcal{O})$ be a Haag–Kastler net of type III_1 on a globally hyperbolic spacetime, ω a faithful normal Hadamard state with cyclic-separating GNS vector $\Omega \in \mathcal{P}^\natural$ for which $(\mathcal{A}(\mathcal{O}), \Omega)$ has the Reeh–Schlieder property for every relatively compact $\mathcal{O}$ ([306], [307]). Let $\{E_k\} \subset \mathcal{A}(\Sigma_{\tau_f})$ be a POVM with $L_k := \Lambda_k(\tau)^{1/2}$ bounded. Then, via the Araki-perturbation realisation of App. G — which requires* no *centralizer hypothesis, so this theorem covers the undressed (non-(CP)) branch of Rem. 11.3 with $E_k \in \mathcal{N}$:*

(i) *The conditioned ABL state $\rho_2(\tau \mid k)$ defines a faithful normal state ω_k on $\mathcal{A}(\mathcal{O})$ (Thm. G.1), with natural-cone representative $\xi_k = \Delta^{1/2}_{\omega_k,\omega}\Omega$ (116), which reduces to $\xi_k = p(k)^{-1/2} L_k \Omega$ in the special case $L_k \in \mathcal{A}^{\sigma^\omega}$ (the centralizer-rich regime (a) of Conj. 11.1).*

(ii) *ξ_k is cyclic-separating for $\mathcal{A}(\mathcal{O})$ for every relatively compact $\mathcal{O}$* iff *L_k* has dense range in *$\mathcal{H}$.*

(iii) *In the POVM/strictly-positive case (*$\ker L_k = 0$*), Reeh–Schlieder is preserved verbatim.*

(iv) *In the projective/non-faithful case (E_k a projector with non-trivial kernel), Reeh–Schlieder is lost on* $\ker L_k$ *and preserved on* $\overline{\mathrm{ran}}\, L_k$*.*

(v) *In all cases, microcausality of the net is intact, and ω_k-correlators between spacelike-separated regions satisfy the same no-signalling identity as ω (Thm. 6.1); no new Reeh–Schlieder paradox is generated by conditioning. The very identity invoked here, $\sum_k p(k)\, M_k = \frac{1}{2}\{\rho, \sum_k \Lambda_k\} = \rho \succeq 0$, is the prior-averaging collapse of the per-branch Margenau–Hill witness (Thm. 3.10): any temporal-quasiprobability negativity $f_k > 0$ encoding non-commutation of the forward state with the backward effect Λ_k lives irreducibly within a single conditioning branch and is annihilated by marginalisation over k, which is why ABL conditioning generates no signalling and no new Reeh–Schlieder pathology.*

(vi) *(*Commutant case.*) By the elementary cyclic/separating duality (Bratteli–Robinson [195] Prop. 2.5.3: ξ cyclic for $\mathcal{M}$ iff separating for $\mathcal{M}'$; ξ separating for $\mathcal{M}$ iff cyclic for $\mathcal{M}'$), conclusion (ii) is equivalent to: ξ_k is cyclic and separating for $\mathcal{A}(\mathcal{O})'$. Under Haag duality $\mathcal{A}(\mathcal{O})' = \mathcal{A}(\mathcal{O}^c)$, this gives the Reeh–Schlieder property symmetrically across causal complements, with modular conjugation $J_k = J$ unchanged (by uniqueness of the standard form [195, §2.5]: every faithful normal state on $\mathcal{N}$ has the same modular conjugation J, that of the standard form $(\mathcal{N}, \mathcal{H}, J, \mathcal{P}^\natural)$; only the modular* operator *changes, $\Delta_{\omega_k} \neq \Delta_\omega$, governed by the Connes cocycle $u_t = (D\omega_k : D\omega)_t$ of Thm. G.1(ii)). The location of L_k relative to $\mathcal{O}$ affects physical interpretation (which net element implements the post-selection) but not the algebraic cyclic-separating conclusion.*

Proof. (i): faithfulness of ω_k on $\mathcal{A}(\mathcal{O})$ holds iff L_k is injective, i.e. $\ker L_k = 0$ (Thm. G.1(i)); for $L_k = L_k^*$ bounded this is equivalent to dense range, $\overline{\mathrm{ran}}\, L_k = (\ker L_k)^\perp$. (ii): when $\ker L_k = 0$, ω_k is faithful normal, so its natural-cone representative ξ_k is cyclic and separating for $\mathcal{A}(\mathcal{O})$ automatically by the standard form [39, Thm. IX.1.2], and Reeh–Schlieder for $(\mathcal{A}(\mathcal{O}), \Omega)$ transfers to $(\mathcal{A}(\mathcal{O}), \xi_k)$ verbatim; no dense-range computation on a closed-form vector is needed. (iii): when $\ker L_k \neq 0$, put $p = \mathrm{s}(L_k)$; ω_k is faithful normal on $p\mathcal{A}(\mathcal{O})p$, the standard form of which lives on $p\mathcal{H} = \overline{\mathrm{ran}}\, L_k$, so Reeh–Schlieder holds there and fails on $\ker L_k$. (iv) is the dense-range criterion read off from (ii)–(iii). (v) is Thm. 6.1 applied at slice τ and slice τ_f, plus the fact that ABL conditioning is a Bayesian update on the classical index k, not an operator insertion; the conditional expectation $E_{\mathcal{A}(\mathcal{O})}$ to a spacelike-separated region commutes with the Bayes update. $\square$

Remark R.2 (Status). Thm. R.1 settles the Reeh–Schlieder issue: ABL conditioning preserves cyclic-separating in the generic POVM case (the operationally relevant one for Conj. L.1); in the projective/non-faithful case the loss of Reeh–Schlieder on $\ker L_k$ is exactly the post-selection collapse of the state space onto outcomes compatible with k, which is a *feature* (the operational content of conditioning), not a bug. No new functional analysis is required beyond Thm. G.1 and the standard Sanders/Strohmaier–Verch–Wollenberg curved-spacetime Reeh–Schlieder transfer. The existential RS transfer used here admits an explicit, error-controlled constructive form for coherent/Hadamard states via a Bisognano–Wichmann analytic-continuation scheme. We note for completeness that RS-preservation is a property of the relativistic type-III$_1$ matter net and is not automatic in the presence of charge superselection that can force would-be cyclic generators out of the local algebra (cf. the Galilean Bargmann-mass obstruction of [393], which has no relativistic analogue once Hermitian field combinations are used).

S Conformal compatibility of the Planck cutoff: EFT-precise, not operator-exact

This appendix records the honest status of CFT structures (CHM modular Hamiltonian, Bisognano–Wichmann boost, AdS/CFT dictionary) under Pos. 11.30's spectral cutoff $P_{\le E_\star}$. Specifically, it disclaims a naive operator-exact intertwining that would be false, and states the EFT-precise result that is true and is what the framework actually uses.

Disclaimer: the boost generator does not commute with $P_{\le E_\star}$

The CHM modular generator K_B for a ball region B ([241]) and the Bisognano–Wichmann wedge boost K_W are unbounded self-adjoint operators on the full Fock space; they *do not commute* with $P_{\le E_\star}$ as projection operators on the matter Fock space, because Lorentz/conformal boosts mix energy shells (an energy-E momentum eigenstate is mapped under boost parameter s to an eigenstate at energy $Ee^{|s|}$ in the worst case). The naive identity $[K, P_{\le E_\star}] = 0$ is therefore *false.*

EFT-precise compatibility

Proposition S.1 (Cutoff-sector CHM/BW compatibility)**.** *Let $\mathcal{A}(D)$ be the local algebra of a causal diamond $D \subset \mathcal{M}$ of radius R, and let K_D be the CHM modular generator ([241]). Let $P_{\le E_\star}$ be the matter spectral projection of Pos. 11.30 with $E_\star R \gg 1$. Then for every Hadamard state ω with $\omega(\widehat{H}_{\mathrm{m}}^2) \le E^2 \ll E_\star^2$, every local observable $A \in \mathcal{A}(D)$ with $\|[A, \mathbb{1} - P_{\le E_\star}]\| \le \epsilon$, and every $s \in \mathbb{R}$:*

$$\left|\omega(e^{isK_D} A e^{-isK_D}) - \omega(P_{\le E_\star} e^{isK_D} A e^{-isK_D} P_{\le E_\star})\right| \;\le\; C(s)\big(E/E_\star\big)^\alpha + \mathcal{O}(\epsilon), \tag{133}$$

for some $\alpha > 0$ and an s-locally bounded $C(s)$ (boost-broadening factor $e^{|s|}$ in the worst case; gentler on the Faulkner–Speranza half-sided modular translation construction [92] where the null generator preserves the null-energy spectrum exactly). In the $E_\star \to \infty$ limit the cutoff drops out and the standard CHM theorem is recovered. The cutoff sector $P_{\le E_\star}\mathcal{H}$ therefore reproduces CHM/BW modular flow up to $O((E/E_\star)^\alpha)$ — all orders only for a smoothed band edge or the null-translation $C(s) = 0$ case of (133) *— in the EFT regime relevant to Thm. I.1 and Thm. M.2.*

Proof sketch. Standard EFT/Wilsonian argument: a state of energy $E \ll E_\star$ acted on by a boost of parameter s has its energy support broadened to $[Ee^{-|s|}, Ee^{|s|}] \ll E_\star$ for any finite s, so the cutoff projection is trivial on the relevant subspace. The leakage (133) is suppressed by powers of $E/E_\star$, with exponent α depending on the smearing of A (Pye–Donnelly–Kempf bandlimited QFT [309] gives $\alpha \to \infty$ for A smeared on the cutoff scale, i.e. for a smoothed band edge; the sharp projection $P_{\leq E_\star}$ alone gives only power-law suppression from the hard edge). The Faulkner–Speranza null- translation case is sharper: null translations preserve any covariant null-energy band exactly, so (133) holds with $C(s) = 0$ on the null component used in Hyp. D.1. $\square$

Remark S.2 (Status of AdS/CFT under the cutoff). Thm. M.2's AdS/CFT reduction uses Prop. S.1 in its EFT form: the CLPW/CFT modular structure is recovered up to $O((E/E_\star)^\alpha)$ on the cutoff sector; the framework is an EFT in the bulk, in line with the standard low-energy reading of AdS/CFT [308]. Thm. I.1's linearised Jacobson is similarly EFT-precise: the CHM modular Hamiltonian on cutoff matter satisfies the first law of entanglement up to corrections suppressed by $(E/E_\star)^\alpha$. No new philosophical commitment is incurred beyond Pos. 11.30's declared EFT scope.

Why not a covariant cutoff. One might avoid the boost-broadening factor $e^{|s|}$ altogether by replacing the rest-frame projection $P_{\leq E_\star} = E_{\widehat{H}_\mathrm{m}}([0, E_\star])$ with a Lorentz-covariant spectral cutoff on the d'Alembertian, $|k_\mu k^\mu| < E_\star^2$, with which a single boost generator does commute. This trade is not free, and it fails on both horns. First, Funai and Menicucci [311] show that a covariant bandlimit induces genuine acausal signalling between spacelike-separated probes and blurs temporal ordering on the scale $1/E_\star$, even while spacelike commutators are formally preserved. Second, Pye [310] shows that the covariant bandlimit is largely *vacuous where it is needed*: a cutoff on the d'Alembertian spectrum fails to regulate precisely the quantities the present framework's cutoff must control — e.g. the entanglement entropy between spatial regions — because the on-shell modes of arbitrarily high frequency survive inside any covariant band. The two results together close both covariant escape routes and strengthen the deliberate rest-frame choice made here: the framework keeps the non-covariant $P_{\leq E_\star}$, which preserves exact microcausality and conformal/causal structure at the cost of the EFT-suppressed boost-broadening (133); the covariant null-energy band of the Faulkner–Speranza construction is the one place where boost-exactness and causality coexist, which is why Hyp. D.1's Killing-horizon anchor sits there.

T Bell-style no-go evasion beyond Wood–Spekkens

This appendix consolidates the framework's evasion of the standard family of Bell-style no-go theorems (PBR preparation-independence, Frauchiger–Renner, Colbeck–Renner, Bong et al. Local Friendliness, Brogioli 2024 complexity-theoretic) into a single combined theorem, extending Thm. P.1.

Theorem T.1 (Combined Bell-style no-go evasion). *Assume Conj. L.1, Pos. 11.11, Ax. 7.1, and Thm. 6.1. Then the two-boundary framework of the present paper satisfies:*

(1) Bell–LR. *Outcome correlations factorise conditional on the terminal record $\sigma_T \in \sigma(\{E_k\})$, with bulk no-signalling preserved by Thm. 6.1; the framework falls in the Wharton–Argaman locally-mediated class [320, 317] (an explicit all-at-once, locally-mediated realisation with the same weak-value/boundary-constraint structure is given by [317]). In the taxonomy of Adlam–Hance–Hossenfelder–Palmer [316] the framework is* non-dynamical (all-at-once), future-input-dependent retrocausal without operational signalling*: by Thm. 6.1 the active law carries no marginal signal, while the non-invariance of $p(k)$ (Remark following Thm. 6.1) is the all-at-once, non-dynamical retrocausal influence confined to postselected sub-ensembles.*

In particular the framework is not *superdeterministic: measurement independence at the initial slice is retained (item (2)). The cosmic Bell tests sharpen this separation empirically: with measurement settings drawn from quasar light emitted at redshift $z \approx 3.9$, Rauch et al. [27] constrain any past-common-cause (superdeterministic) mechanism to have acted more than* 7.8 *Gyr ago, while such setting-history constraints are structurally blind to future-boundary conditioning — the terminal-record mechanism operating here is untouched by them, which is exactly the experimental signature of the taxonomy placement above.*

(2) PBR preparation-independence. *Preparation independence holds for the initial slice; terminal-side PIP-violation is confined to the centraliser-PVM and is invariant under modular flow (no fine-tuning, by Thm. P.1, in line with the Leifer–Pusey [312] characterisation of retrocausal psi-epistemic models).*

(3) Frauchiger–Renner. *Agents' reasoning chains are well-defined only relative to the unique CHT-POVM terminal record (Conj. L.1), blocking the cross-Heisenberg-cut combination of (Q),(C),(S) that drives the Frauchiger–Renner contradiction; cf. Adlam [289] and Sutherland's zigzag [321].*

(4) Colbeck–Renner. *The CHT-POVM is the maximal abelian centraliser of the CLPW Type II_1 tracial state, so no predictive extension beyond $\{E_k\}$ exists; the Colbeck–Renner extension hypothesis fails trivially.*

(5) Bong et al. Local Friendliness. *The framework reproduces LF-inequality violations only by accepting a* lawful *(modular-invariant) setting-dependence of past hidden variables fixed by Conj. L.1. By Ying et al. [291] no nonclassical or cyclic causal model evades LF without some fine-tuning; the framework does not contradict this — its evasion is precisely of the* lawful-fine-tuning *type of Thm. P.1 (cf. Rem. P.3), not a fine-tuning-free causal model, and so it lies* within, *not outside, the Ying et al. characterisation while remaining outside the scope of the Bong et al. [313]* classical-*LR assumptions.*

(6) Brogioli 2024 complexity-theoretic no-go. *The CHT-POVM terminal record is a horizon-geometric PVM fixed by modular invariance, not an efficiently-specifiable post-selection projector; the complexity-theoretic obstruction of [314] therefore does not apply.*

(7) Timelike absoluteness of observed events. *Mukherjee–Hance [315] prove a time-ordered analogue of the Local-Friendliness no-go for sequential (timelike-separated) measurements, deriving a causal inequality from Absoluteness of Observed Events (AOE) together with No-Retrocausality (NRC), Screening-via- Pseudo-Events and Axiological Time Symmetry. This is the timelike no-go directly relevant to the present worldline-conditioned construction. The two-boundary framework abandons* NRC *— not AOE — in a principled, modular-invariant manner (the terminal record $\{E_k\}$ is the centraliser-PVM of Conj. L.1, fixed by the future boundary), so the Mukherjee–Hance inequality is evaded by exactly the mechanism of item (5): a future-setting dependence of past hidden variables that is not fine-tunable. Bulk no-signalling (Thm. 6.1) is preserved throughout.*

(8) Shrapnel–Costa. *The framework does* not *attempt a noncontextual ontology: by Shrapnel–Costa [292] no model — including one whose ontology is a global constraint such as our centraliser-PVM terminal record — reproduces quantum statistics from noncontextual ontic properties. The terminal record $\sigma_T \in \sigma(\{E_k\})$ is explicitly a* contextual *datum (its conditional structure depends on the measured POVM context), so the framework is consistent with, rather than excluded by, [292]; the classical-ontology motivation that this no-go removes is one we do not claim.*

Remark T.2 (Status). Thm. T.1 is a corollary of Thm. P.1 plus the centraliser- maximality of CHT-POVM and Thm. 6.1; no genuinely new theorem is required, but the consolidated statement closes the cluster of standard Bell-style no-gos that any retrocausal QM+GR framework must explicitly address in the post-Wood–Spekkens era. Items (1) and (4) follow in one line; (2), (3), (5), (7) require the modular-invariance input of Thm. P.1 (items (5) and (7) being the spacelike and timelike faces of the same future-setting-dependence mechanism); (6) needs only the non-algorithmic character of the CHT-POVM. We note the Local-Friendliness family is itself under active revision: Walleghem–Catani [318] strengthen it to a generalized-contextuality-based "Noncontextual Friendliness" no-go, while Okon [319] argues such extended-Wigner's-friend no-gos impose weaker constraints on reality than is usually claimed; either reading leaves the present evasion — by abandonment of no-retrocausality rather than of absoluteness of observed events — intact.

U Cosmological-constant fluctuations: refinement of Prop. 13.12

This appendix refines Prop. 13.12 by addressing the Albrecht–Sorbo [322] / Banks–Fischler [323] / Carroll [328] cosmological-constant fluctuation problem and the Verlinde–Zurek modular-fluctuation enhancement [329]. Prop. 13.12 as stated is a *one-point* prediction (central value of $\Lambda_{\rm eff}$); Type II trace renormalisation absorbs the state-independent additive divergence but does *not* automatically suppress $\langle(\Delta T_{\mu\nu})^2\rangle$. The Kudler-Flam–Leutheusser–Satishchandran (KLS) construction [89] for inflationary FLRW shows the inflaton zero-mode is central in the algebra and $\mathbb{R}$-valued, so it generates physical (finite, non-zero) fluctuations of $\Lambda_{\rm eff}$.

Proposition U.1 (CC variance under thermal time + Type II_∞)**.** *Under Pos. 11.11, Pos. 11.30 (H0), and the KLS type II_∞ comoving-observer algebra [89] (Prop. D.2(ii)), the matter- sector contribution to $\Lambda_{\rm eff}$ obeys both:*

(i) Central value *(Prop. 13.12):*

$$\Lambda_{\rm eff} \;=\; \Lambda_0 + c_\Lambda H_0^2 + \mathcal{O}\big(H_0^2\,(H_0/E_{\rm P})^2\big). \tag{134}$$

(ii) Variance *(*new*; KLS-derived):*

$$\langle\Lambda_{\rm eff}^2\rangle - \langle\Lambda_{\rm eff}\rangle^2 \;=\; c_{\rm var}\,(H_0/E_{\rm P})^2\,H_0^4, \tag{135}$$

with $c_{\rm var} \in \mathbb{R}$ a state-dependent $O(1)$ functional of the two-boundary thermal state on the central inflaton zero-mode of [89]. Equivalently $\delta\Lambda/\Lambda \sim H_0/E_{\rm P} \sim 10^{-61}$, far below DESI/Planck precision.

Proof sketch. (i) is Prop. 13.12 unchanged. (ii) uses the KLS construction: the inflaton zero-mode $\tilde{q}$ is central in $\widetilde{\mathcal{N}}_{\rm dS}$ and $\mathbb{R}$-valued; its canonical-commutator scaling $G_N\,H_0^2$ from the CLPW first law $\delta H_{\rm mod} = 2\pi\delta A/(8\pi G_N)$ gives (135) via direct two-point computation of $\langle\delta\Lambda\,\delta\Lambda\rangle_{\rm thermal}$ on the two-boundary state. The Aalsma–Bak analysis [329] gives a modular-Hamiltonian variance $\langle\Delta K^2\rangle \sim \varepsilon\,S_{\rm dS}$ (with ε the slow-roll parameter and $S_{\rm dS}$ the de Sitter entropy), equivalent to a horizon-*size* fluctuation $\delta L \sim \sqrt{\ell_{\rm Pl}L}$ of Verlinde–Zurek type — that is, $\sqrt{\ell_{\rm Pl}L}$ is the size fluctuation, not the modular-Hamiltonian fluctuation. We argue (Rem. U.3) that this horizon-mode channel is sub-leading to the central-zero-mode channel (135) for the cosmological-constant observable in semiclassical two-boundary states; because the two scale oppositely in $\ell_{\rm Pl}/L$ (an enhancement versus a suppression) this requires comparison rather than assertion. □

Lemma U.2 (Boltzmann-brain truncation by future boundary)**.** *Under Ax. 7.1 (terminal record-completeness), Pos. 11.11, and the CLPW Type II_∞ modular cutoff, thermal time on the two-boundary state* never reaches the Poincaré-recurrence regime *within the algebra's modular flow. The terminal POVM $\{E_k\}$ truncates modular time at τ_f, before the recurrence- dominated tail of [322, 323, 328] can dominate. Boltzmann-brain contributions to the cosmological state are therefore explicitly absent from the present framework's predictions.*

Proof sketch. The two-boundary structure of Sec. 5 restricts modular time to the finite slab $\mathcal{M}_{[\tau_i,\tau_f]}$; modular flow is a finite-time Bayesian-conditioned process, not an asymptotic dynamical system. The Poincaré-recurrence regime requires modular time greatly exceeding the de Sitter Hubble time, which is incompatible with terminal-conditioning at any finite τ_f. The KLS centrality of the inflaton zero-mode [89] prevents a maximum-entropy state from forming in the bulk-modular sector independently. Boltzmann- brain dominance is therefore structurally precluded in the framework, not merely empirically suppressed. □

Remark U.3 (Status). Prop. U.1 and Lem. U.2 together complete the cosmological-constant prediction of the framework: central value via Prop. 13.12 (no E_{P}^4 catastrophe via Type II trace renormalisation), variance via (135) (suppressed by $(H_0/E_{\mathrm{P}})^2$, empirically negligible), and Boltzmann-brain truncation via the two-boundary structure. The KLS inflationary FLRW construction [89] and the Aalsma–Bak modular-fluctuation analysis [329] are the ambient inputs.

Remark U.4 (Observational status: DESI DR2). DESI DR2 baryon-acoustic-oscillation data combined with the CMB and Type Ia supernovae favour an evolving dark-energy equation of state (w_0, w_a) over a pure cosmological constant at up to 4.2σ [333] (the high end is dataset-combination-dependent). Eq. (134) as written (static $\Lambda_0 + c_\Lambda H_0^2$) reproduces a cosmological constant at the present epoch. We do not derive a redshift dependence here; we note only the conjectural handle that *if* the present-epoch term $c_\Lambda H_0^2$ is the static limit of a running $c_\Lambda H(z)^2$ contribution — as it is in the running-vacuum models [330] whose H^2 term it shares (Rem. 13.13) — then $\Lambda_{\mathrm{eff}}(z) = \Lambda_0 + c_\Lambda H(z)^2$ would yield an effective $w(z) \neq -1$ of running-vacuum type, providing a falsifiable handle on c_Λ against DESI DR2. The running reading is, however, *BBN-capped*: inserting $\Lambda_{\mathrm{eff}}(z) = \Lambda_0 + c_\Lambda H(z)^2$ into the Friedmann constraint gives $3(1 - c_\Lambda/3)H^2 = 8\pi G\rho + \Lambda_0$, i.e. a fractional rescaling $\delta H/H \approx c_\Lambda/6$ of the radiation-era expansion rate (where ρ dominates and Λ_0 is negligible), and the N_{eff}/deuterium sensitivity of big-bang nucleosynthesis, $\delta H/H \lesssim 1.3$–2%, then forces $|c_\Lambda| \lesssim 0.08$–$0.12$ — one order of magnitude below the $\mathcal{O}(1)$ surface-gravity normalisation of Prop. 13.12. The honest consequence: if the DESI $w_0 w_a$ preference consolidates at $\geq 5\sigma$, the framework's running escape cannot respond at $\mathcal{O}(1)$ amplitude; only the static reading (present-epoch $c_\Lambda H_0^2$ with free Λ_0, which BBN does not constrain) survives unmodified, and it predicts no $w(z)$ evolution at all. The clash is in fact sharper than an amplitude worry, on two counts. *Shape.* DESI DR2's central preference is *quintom-B* — $w_0 > -1$ today, $w_a < 0$, the dark-energy equation of state *crossing* $w = -1$ in the past (near $z \sim 0.3$–0.5). A single-component running-vacuum $\Lambda(H)$ term of the form $\Lambda_0 + c_\Lambda H(z)^2$ is structurally *non-crossing*: its effective $w(z)$ touches -1 only at $z = 0$ and stays one-sided (quintessence-like for one sign of c_Λ, phantom-like for the other), so it cannot reproduce a phantom crossing for *any* value or sign of c_Λ — crossing requires a genuinely composite sector (running vacuum *plus* a phantom-matter component, *wXCDM* [334]) that the present construction does not contain. *Amplitude.* Mapping the running-vacuum contribution onto the CPL parametrisation gives $w_a = \mathcal{O}(c_\Lambda)$ with a prefactor $\lesssim \Omega_{\mathrm{m}}/\Omega_\Lambda \approx 0.45$, so reproducing DESI's $|w_a| \sim 0.8$–1.1 would require $c_\Lambda \sim 2$ — some 15–40× above the BBN cap and past the perturbative regime in which the H^2 expansion is defined; saturating the cap ($c_\Lambda = 0.12$) yields only $|w_a| \lesssim 0.05$, roughly 20× too small. Consistently, a direct fit of the $\Lambda(t)$CDM running-vacuum model to DESI DR2 drives the running parameter toward zero, favouring ΛCDM [335]. The honest reading is therefore that the running *reading* is already disfavoured, on shape before amplitude, and only the static reading survives. One caveat cuts the other way: if DESI's apparent crossing is an artefact of the $w_0 w_a$ parametrisation rather than a physical feature — as it is for thawing quintessence with $w \geq -1$, which fits the data comparably [336] — then a one-sided, non-crossing $w(z)$ is not excluded, and the framework's shape-class survives as a live (shape-only, amplitude-free) prediction. Either way the *scale* is untouched: a time-dependent N would be a free *function* rather than a free constant, and the trace-rescaling homogeneity of Rem. D.5 rescales $N(\tau) \mapsto \lambda N(\tau)$ globally, leaving the absolute scale free at every epoch. Establishing that the thermal-time construction (which fixes c_Λ from present-epoch surface gravity) actually generates such a running is left open; the present framework is at minimum compatible with, and testable by, the DESI dynamical-dark-energy programme.

One piece of that question is now settled at the mechanism level: the framework's own second-order dissipative sector (the Clausius-σ term) *does* run at the de Sitter horizon — the static patch shares the wedge's modular structure, and the de Sitter quasinormal tower supplies the continuum relaxation time $\delta t = 1/(3H)$ the finite-mode analysis provably lacked — but its amplitude is dimensionally pinned at $\mathcal{O}\big((H/M_{\rm Pl})^2\big)$ per Hubble time (microscopic at every epoch; no fit is possible or attempted), and because $\sigma \geq 0$ the induced equation of state is one-signed: $w(z) \geq -1$, *non-crossing*, quintessence-side. This is a falsifiable sign: a $\geq 5\sigma$ parametrization-independent phantom crossing — the behaviour DESI DR2 centrally prefers — would rule out the framework's dissipative sector as a dark-energy contributor. The dissipative route therefore does *not* resolve the tension recorded above; it takes a definite, testable side in it.

A distinct, stochastic alternative — the causal-set everpresent-Λ mechanism [168, 169], in which Λ sign-fluctuates with amplitude $\delta\Lambda \sim V^{-1/2}$ tracking the ambient four-volume — would also mimic a departure from a static constant. It is, however, *excluded within this framework*, not merely disfavoured: Prop. U.1 gives a *static*, radiatively negligible variance $\delta\Lambda/\Lambda \sim (H_0/E_{\rm P}) \sim 10^{-61}$, some sixty orders below the $\mathcal{O}(1)$-per-Hubble amplitude that everpresent-Λ requires, and Lem. U.2 truncates the modular-time tail before the fluctuation-dominated regime that mechanism lives in can form. The framework therefore predicts *against* everpresent-Λ. A positivity-floored everpresent-Λ fit to Planck+DESI DR2 remains, as of this writing, an un-run test of that alternative — of it, not of the present construction.

V Sector-resolved cluster decomposition at cosmological scales

This appendix records the honest status of cluster decomposition under Pos. 11.11 + Thm. H.1 + Conj. L.1. Cluster fails on the algebra centre (inflaton zero-mode + discrete-series logarithmic mode + CHT-POVM-induced central charges) but is preserved on the bulk principal-series sector where it is physically required.

Theorem V.1 (Sector-resolved cluster decomposition)**.** *Under Pos. 11.11, Thm. H.1, and Conj. L.1, the dressed observable algebra factorises as*

$$\mathcal{A}_{\rm dress} \;=\; \mathcal{A}_{\rm bulk}^{\rm princ} \otimes \mathcal{A}_{\rm centre}, \qquad \mathcal{A}_{\rm centre} \;=\; Z(\mathcal{A}_{\rm dress}), \tag{136}$$

where $\mathcal{A}_{\rm centre}$ contains the inflaton zero-mode $\tilde{q}$ of [89], any discrete-series logarithmic modes from massless / gauge sectors of [337], and the CHT-POVM-induced central charge. For local observables $A, B \in \mathcal{A}_{\rm bulk}^{\rm princ}$ in causally-disjoint regions $\mathcal{O}_1, \mathcal{O}_2$ with comoving separation r,

$$\big|\omega_{(\rho_{\rm i},E_k)}(A\,\tau_y B) - \omega(A)\omega(B)\big| \;\leq\; C\,r^{-3}\big(1 + \mathcal{O}(H_0/E_{\rm P})\big), \qquad r \to \infty, \tag{137}$$

with the r^{-3} decay rate of the principal series [337]. On the central sector, cluster fails by an $\mathcal{O}(H_0^2/E_{\rm P}^2)$ contribution from $\tilde{q}$ and an additive $\log r$ piece from any discrete- series modes.

Proof sketch. Factorisation (136) is the standard direct-integral decomposition of a type II_∞ algebra over its centre, with the centre identified with the inflaton zero-mode + discrete-series logarithmic sector following KLS and Allameh–Goodhew–Premkumar [89, 337]. Bulk principal-series cluster (137) follows from the late-time dS principal-series two-point function $r^{-3\pm 2i\rho}$ result of [337]. ABL conditioning by E_k shifts the bulk correlator by $\mathcal{O}(\|E_k - \mathbb{1}/d\|)$, which vanishes in the maximally- mixed cosmological-horizon limit of Conj. L.1; hence post-selection does not propagate cluster violation into the bulk principal sector. Type II_∞ commutant disjointness prevents the CHT-POVM central component from acting on the bulk sector. □

Remark V.2 (Distinction from Horowitz–Maldacena)*.* Thm. V.1's ABL conditioning differs from the Horowitz–Maldacena final-state proposal [65] in that the CHT-POVM is *informationally complete* (a maximal abelian centraliser- PVM rather than a single pure final state), so the Horowitz–Maldacena super-Tsirelson correlations do not arise. The bulk cluster bound (137) is preserved at the principal-series level; cluster failure is strictly confined to the algebra centre.

W Causal-set compatibility: continuum-limit status

Remark W.1 (Causal-set discreteness compatibility). The framework's continuum machinery (Haag–Kastler nets, modular flow, CHM modular Hamiltonian, CLPW Type II crossed product) is compatible with a causal-set substrate at sprinkling density $\rho \sim \ell_P^{-4}$ in the following sense: by Dable-Heath–Fewster–Rejzner–Woods [338], perturbative AQFT admits a direct construction on a fixed causal set (for suitable discretised wave operators, including the preferred-past d'Alembertian), with an explicit notion of continuum limit via embedded causets $(C_n, \preceq_n, \ell_n) \to (\mathcal{M}, g)$. The Sorkin–Johnston (SJ) state on a causet [339, 340] is covariant, distinguished, quasi-free, and recovers the Minkowski/conformal vacuum in the centre of the diamond in appropriate continuum limits. The bare SJ state is, however, *not* Hadamard at the boundary of a causal diamond [341], and by Fewster–Verch [342] no covariant prescription yields a Hadamard state on every globally hyperbolic spacetime; we therefore invoke the *softened* SJ state of [341], a boundary-regularised modification engineered to satisfy the Hadamard condition. Its GNS triple supplies a modular operator Δ_Ω whose continuum limit is *conjectured* to be the Bisognano–Wichmann/CHM modular operator of the Hadamard vacuum invoked in Pos. 11.11; establishing this convergence at the level of the boundary-supported modular generator is recorded as open caveat (i) below. Lorentz invariance is preserved by Poisson sprinkling at Planck density (Bombelli–Henson–Sorkin 2009), so the smooth Killing/conformal-Killing structures invoked in Thm. I.1 (CHM modular Hamiltonian) and Thm. M.2 (boundary-CFT modular flow) survive in the coarse-grained continuum.

The Planck-scale matter cutoff Pos. 11.30 at $E_\star \lesssim E_P$ is the natural energy associated with the sprinkling density. The framework is the *continuum effective description* of an underlying causet structure; it does not require continuous spacetime as a fundamental ontological assumption. Open caveats: (i) the *bare* SJ modular operator is provably non-Hadamard at the diamond boundary [341], so its naive continuum limit is *not* the CHM modular operator; the conjectured agreement via the softened SJ state holds only for that boundary-regularised state and remains unproven at the level of the boundary-supported modular generator (and, moreover, the softened construction of [341] is established only in $1+1$D, so its extension to the 3+1D CHM ball is not yet shown); (ii) conformal-Killing symmetries are emergent, not exact, on any finite causet — and in de Sitter spacetime, the cosmological background of the CLPW generalised-observer construction used here, the SJ state is least controlled: its Hadamard status is unclear and it need not coincide with a de Sitter-invariant α-vacuum [340, 343], so the substrate compatibility of the de Sitter modular flow (Pos. 11.11 in the cosmological sector) is correspondingly weaker than in the flat causal-diamond case; (iii) the II_1-vs-II_∞ subtype of the dressed algebra, which tracks boundedness of the dressed observer charge (Thm. H.1(i)) rather than any Planck-scale trace renormalisation, is conjecturally compatible with the causet substrate but unproven; (iv) the present discussion is kinematic (sprinkling into a fixed $(\mathcal{M}, g)$); the *dynamical* emergence of a manifold-like continuum from causal-set growth/path-integral dynamics remains an open problem of causal-set theory itself, since non-manifoldlike Kleitman–Rothschild orders dominate the unweighted sum and, while their suppression by the Benincasa–Dowker–Glaser action is a significant step, it does not yet establish a continuum limit [344]. These are recorded as out-of-scope items rather than residuals of the framework.

X Lorentz-violating dispersion bounds: thermal-time sector exact ($\xi = \eta = 0$); Σ_δ matter sector subluminal linear

This appendix is sector-split. The thermal-time/geometric sector generates *no* Lorentz-violating dispersion (Prop. X.1); the Σ_δ matter-sector correction of Prop. C.2 *does* induce an effective subluminal linear time-of-flight dispersion for freely propagating massless modes (Prop. X.2), and GRB photon timing is thereby the *binding* empirical constraint on the physical combination

μ (Rem. X.4). An earlier version of this appendix claimed $\xi = \eta = 0$ for the framework as a whole; that claim was over-scoped, since its proof covers only the modular-flow sector.

Proposition X.1 (Absence of derived Lorentz-violating dispersion in the thermal-time/geometric sector)**.** *Under Pos. 11.11 on the SM matter algebra plus Pos. 11.30 at $E_\star \lesssim E_{\mathrm{P}}$, the thermal-time/geometric sector of the framework (modular flow, QRF dressing, and the spectral cutoff) generates* zero *Lorentz-violating dispersion-relation coefficients at all orders accessible to current experiments:*

$$E^2 \;=\; p^2 + m^2 + \xi\,\frac{p^3}{E_{\mathrm{P}}} + \eta\,\frac{p^4}{E_{\mathrm{P}}^2} + \cdots, \qquad \xi \;=\; \eta \;=\; 0 \text{ at all orders in } E/E_{\mathrm{P}}. \tag{138}$$

Proof sketch. The Connes–Rovelli thermal-time flow σ_t^ω is intrinsic to the algebraic structure and does not modify the on-shell dispersion of any field operator: a thermal state selects a preferred *frame* (the rest frame of the modular bath), not a deformation of $E^2 = p^2 + m^2$. Bisognano–Wichmann gives modular flow = boost on every Killing wedge *exactly*, with no E^3/E_{P} or E^4/E_{P}^2 corrections. Away-from-Killing deformations enter the Connes cocycle $u_t \in \mathcal{U}(\mathcal{N})$ (an inner automorphism of the dressed algebra) and are absorbed by QRF-gauge equivalence (Rem. H.3); they do not produce single-particle dispersion-relation deviations. This scope is deliberate: the proof covers every dispersion effect generated by modular flow and the cutoff, and does *not* cover the Σ_δ matter-sector generator correction, treated next. □

Proposition X.2 (Induced subluminal linear dispersion from the Σ_δ softening)**.** *Assume Conj. 10.13 with the slow-branch axiom (Rem. 10.28), so that band-limited modes evolve with the softened generator $h(\widehat{H}_{\mathrm{s}}) = \widehat{H}_{\mathrm{s}} - \frac{\mu}{2}\widehat{H}_{\mathrm{s}}^2 + \mathcal{O}(\mu^2\widehat{H}_{\mathrm{s}}^3 + \mu\,\delta t^2\widehat{H}_{\mathrm{s}}^4/\hbar^2)$ (Prop. C.2). A freely propagating massless mode of frequency $\omega(k) = c|k|$ then acquires the phase evolution $e^{-i\,h(\hbar\omega(k))\,t/\hbar}$, so its group velocity is*

$$v_g(E) \;=\; \frac{1}{\hbar}\frac{d\,h(\hbar\omega)}{dk} \;=\; c\,h'(E) \;=\; c\big(1 - \mu E\big) + \mathcal{O}\big(\mu^2E^2 + \mu\,\delta t^2 E^3/\hbar^2\big), \qquad E = \hbar\omega: \tag{139}$$

an energy-dependent, subluminal *(for $\alpha > 0$), strictly* linear *time-of-flight dispersion with quantum-gravity scale $E_{\mathrm{QG},1} = 1/\mu$; in the convention of* (138) *this is an effective linear coefficient of magnitude $|\xi_{\mathrm{eff}}| = \mu\,E_{\mathrm{P}}$ with the subluminal sign. The universality argument of Prop. 13.14 (functional calculus in $\widehat{H}_{\mathrm{s}}$ alone: "every mode, every material") provides no exemption for photon modes; the* same *μ therefore governs phonon-cavity softening and photon time-of-flight. This is stated as a prediction of the framework, not hidden as a scope restriction.*

Remark X.3 (Two null structures: softened propagating modes vs. the exact-c metric front). Prop. X.2 softens *propagating field modes* — the photon ($v_g = c(1-\mu E)$) and, by Lem. 13.7(iii), the graviton ($\omega_\pm = \omega[1 - \frac{\mu\hbar\omega}{2}]$, helicity-blind). It does *not* soften the null structure of the metric g_ω itself. The record null shells of Sec. 12.5.6 (the surfaces $\mathcal{N}_m = \partial J^+(\gamma(s_m))$ carrying the Barrabès–Israel surface layers) are a classical distributional feature of g_ω, not a band-limited mode evolved by the softened generator $h(\widehat{H}_{\mathrm{s}})$: they lie on the *exact* null cones of g_ω and propagate at the unsoftened metric speed c. Three points follow.

(i) No contradiction between the two null speeds. The softened modes travel at $v_g < c$ (subluminal, for $\mu > 0$); the shell front travels at exactly c on the metric cone. Both are on or inside the same exact null cone of g_ω, so both are causal; the mode dispersion is a within-cone retardation of the field, not a superluminal excursion. In particular the shell front is the *fastest* carrier of the record it broadcasts: it outruns the electromagnetic signal of the same event, which is itself subluminal by (139).

(ii) Why Prop. 10.7(iv) is stated on the exact cone. The causal-safety clause Prop. 10.7(iv) quantifies over regions spacelike to the record events with respect to $J^+(\gamma(s_m))$ — the *exact* metric cone $\mathcal{N}_m$, not the (strictly narrower) photon cone $v_g < c$. This is forced: the metric

deviation itself is carried by the shell front on the exact cone, so a region that is spacelike to the photon cone but timelike to the exact cone can already see the branch geometry (the impulsive tidal kick of the shell, cf. the detectability discussion around Prop. 10.7). Stating the no-signalling guarantee on the exact cone is therefore the conservative (correct) choice; a photon-cone statement would be an under-count of the causal reach and would fail on the shell-straddling regions.

(iii) GW170817 is untouched. The $|c_{\mathrm{gw}} - c|/c \lesssim 10^{-15}$ constraint of Rem. X.4 concerns *propagating* gravitational-wave and gamma-ray modes, both of which receive the *same* $-\mu E$ shift at equal energy (139) (photon and graviton alike, by the universality of $h(\widehat{H}_{\mathrm{s}})$ and the helicity-blindness of Lem. 13.7(iii)); the common shift cancels in the difference $c_{\mathrm{gw}} - c_\gamma$ to the measured precision, and at GW/gamma-ray energies the residual is $\mu E \lesssim 10^{-28}$ on the matched branch. The exact-c shell front is a distinct object and does not enter this comparison: the GW170817 bound tests the mode sector, the shell front lives in the metric sector, and neither is in tension with the other.

Remark X.4 (Empirical status: GRB photon timing is the binding constraint on μ). By Prop. X.2, the published linear time-of-flight bounds apply directly to $1/\mu$:

- $E_{\mathrm{QG},1} > 7.6\,E_{\mathrm{P}}$ (subluminal, 95% CL) from Fermi-LAT observations of GRB 090510 [346];
- $\xi < 0.1$ (subluminal $E_{\mathrm{QG},1} > 1.0 \times 10^{20}\,\mathrm{GeV} \approx 8.2\,E_{\mathrm{P}}$) from LHAASO observations of GRB 221009A [345].

The binding constraint is therefore

$$\mu \;=\; \frac{1}{E_{\mathrm{QG},1}} \;<\; \frac{1}{8.2\,E_{\mathrm{P}}} \;\approx\; 1.0 \times 10^{-29}\ \mathrm{eV}^{-1}, \tag{140}$$

5.3 decades below the laboratory (BAW) ceiling (80). Three consequences. (i) The Planck-natural convention $\mu = 1/E_{\mathrm{P}} \approx 8.2 \times 10^{-29}\,\mathrm{eV}^{-1}$ (Rem. 11.37) is *excluded*, by a factor 7.6–8.2; this resolves the convention tension of Rem. C.4 empirically, in favour of the Born–Oppenheimer-matched branch (99), $\mu = 2c_0 H_{\mathrm{inf}}/m_{\mathrm{P}}^2 \sim 10^{-33}\,\mathrm{eV}^{-1}$, which survives (140) by 3–4 decades. (ii) Photon time-of-flight, not the phonon cavity, is the framework's leading falsifier; the BAW channel of Sec. 13.5 remains the leading *laboratory* falsifier. (iii) On the matched branch the framework makes a genuine falsifiable prediction: a subluminal, strictly linear dispersion with $|\xi_{\mathrm{eff}}| = \mu E_{\mathrm{P}} = \lambda \sim 10^{-5}$–$10^{-4}$ (the subcriticality parameter; H_{inf}- and convention-dependent), i.e. $E_{\mathrm{QG},1} = 1/\mu \sim 10^4$–$10^5\,E_{\mathrm{P}}$ — four decades below the current bound $\xi < 0.1$, beyond foreseeable timing reach in magnitude, but with the *sign* (subluminal), *linearity* (no $\xi = 0, \eta \neq 0$ escape), and *universality* (one global μ across every mode and material) as sharp commitments: a confirmed superluminal, dominantly quadratic, or mode/material-dependent dispersion falsifies the Σ_δ sector outright. (One order deeper, implicit differentiation of the exact fibrewise generator (96) fixes the whole correlated tower, $v_g/c = 1 - \mu E + \frac{3}{2}(\mu E)^2 + \frac{1}{6}(\mu\,\delta t^2/\hbar^2)E^3 + \cdots$: the induced quadratic coefficient is *slaved* to the linear one, $\eta_{\mathrm{eff}} = +\xi_{\mathrm{eff}}^2$ — recovery sign, the net v_g staying subluminal, independent of the $(\alpha, \delta t)$ split — while split sensitivity first enters at the cubic term, bounded in-band by $(\delta t\,E/\hbar)^2/6 < 1/6$ of the linear term and $\lesssim 10^{-4}$ of it for the most energetic GRB photons anywhere in the split window of Rem. 13.17. Every coefficient beyond the linear one is thus determined by $(\mu, \delta t)$ alone; there is no independent quadratic coefficient to tune, which is what "strict linearity" commits to.) We emphasise, to avoid overselling near-term testability, that on the matched branch it is these *form* commitments (sign, linearity, universality), *not* the magnitude, that are the live empirical content; and that the laboratory (BAW) channel bites only modulo the no-cancellation assumption of Prop. 13.14(ii), so the "two-channel" framing is a statement about *form*-falsifiability, not about imminent magnitude detection. The remaining Lorentz tests are consistency checks of the thermal-time sector (Prop. X.1), which the framework passes identically: $|c_{\mathrm{gw}} - c|/c \lesssim 10^{-15}$ from GW170817/GRB 170817A [348] (a comparison of *propagating* GW and γ modes, both of which receive the same $-\mu E$ shift at equal energy so that it cancels in the difference — the exact-c null-shell front of Rem. X.3 is a distinct, metric-sector object and does not enter this bound; at GW/gamma-ray energies $\mu E \lesssim 10^{-28}$ on the matched branch); $\eta \lesssim 2.8 \times 10^{14}$ from $E_{\mathrm{QG},2} > 6 \times 10^{-8}\, E_{\mathrm{P}}$ (LHAASO), against which the induced quadratic coefficient $\mathcal{O}(\mu^2 E^2)$ is utterly negligible; $\sim 10^{-21}$ on electron-sector SME c-coefficients from Yb^+ optical-clock comparison [352]; and $\sim 10^{-17}$ on photon-sector SME parameters from rotating cavities [353] (Kostelecký–Russell data tables [351]). Tentative reports of an energy-dependent light-speed variation from gamma-ray-burst timing at $E_{\mathrm{LV}} \sim 3 \times 10^{17}\,\mathrm{GeV}$ [349, 350] are already in tension with the bounds adopted here and, if confirmed, would falsify the matched-branch magnitude prediction as well.

Y Time-of-arrival in modular time: Pauli evasion via unbounded modular spectrum

This appendix shows that the Pauli no-go on time operators [358] is structurally evaded under Pos. 11.11: the modular Hamiltonian $\widehat{H}_{\mathrm{c}} = -\log \Delta_\omega$ has spectrum on all of $\mathbb{R}$, so a self-adjoint time operator canonically conjugate to it exists by Stone–von Neumann.

Theorem Y.1 (Modular time-of-arrival operator)**.** *Let ω be a faithful normal seed state on the matter algebra, and let $\widehat{H}_{\mathrm{c}} = -\log \Delta_\omega$ be the modular Hamiltonian under Pos. 11.11. Then:*

(i) *(*Pauli evasion.*) The spectrum of $\widehat{H}_{\mathrm{c}}$ is all of $\mathbb{R}$. The Tomita relation $J\Delta_\omega^{1/2} = \Delta_\omega^{-1/2} J$ forces* $\mathrm{spec}(\widehat{H}_{\mathrm{c}})$ *to be only* symmetric about 0 *(it conjugates Δ_ω to Δ_ω^{-1});* fullness $\mathrm{spec}(\widehat{H}_{\mathrm{c}}) = \mathbb{R}$ *with absolutely continuous spectrum is the Type III$_1$ property of the seed — Connes invariant $S(\mathcal{N}) = [0, \infty)$ [273] — which is already assumed of $\mathcal{N}$ in Conj. 11.1 and holds for the Bisognano–Wichmann wedge vacuum of part (iii). (A generic $\beta = 1$ KMS state on a Type III$_\lambda$ factor, $0 < \lambda < 1$, instead has* $\mathrm{spec}(\widehat{H}_{\mathrm{c}}) \subset (\log \lambda)\,\mathbb{Z}$*, symmetric but not full, and would* not *support the construction.) The Pauli theorem [358] requires $\widehat{H} \geq 0$ and therefore does not apply.*

(ii) *(*Spectral construction of $\widehat{T}$.*) On the spectral representation $\widehat{H}_{\mathrm{c}} = \int \lambda\, dE(\lambda) \cong L^2(\mathbb{R}, dE)$, define*

$$\widehat{T} \ :=\ -i\hbar\, \partial_\lambda \quad \text{on } L^2(\mathbb{R}, dE) \ \cong\ L^2(\mathbb{R}) \otimes \mathcal{K}, \tag{141}$$

where $\mathcal{K}$ carries the (uniform, countably infinite) spectral multiplicity of the Type III$_1$ modular Hamiltonian $\widehat{H}_{\mathrm{c}}$. Then $\widehat{T}$ is essentially self-adjoint on the dense Schwartz domain, $[\widehat{T}, \widehat{H}_{\mathrm{c}}] = i\hbar$, and the pair $(\widehat{T}, \widehat{H}_{\mathrm{c}})$ is unitarily equivalent to $(\widehat{Q}, \widehat{P}) \otimes \mathbb{1}_{\mathcal{K}}$ on $L^2(\mathbb{R}) \otimes \mathcal{K}$ — a uniform direct integral of Schrödinger pairs, irreducible on each fibre. (The load-bearing conclusions $[\widehat{T}, \widehat{H}_{\mathrm{c}}] = i\hbar$ and essential self-adjointness of $\widehat{T}$ are insensitive to the multiplicity.)

(iii) *(*Operational meaning.*) $\widehat{T}$ is the* modular-time *observable: it measures the parameter of the modular automorphism group σ_t^ω, coinciding with proper time in the Bisognano–Wichmann/Unruh wedge case (where modular flow is geometric boost). Outside this geometric (Bisognano–Wichmann / CHM) class no proper-time interpretation is claimed: $\widehat{T}$ is then the state-relative modular parameter only, and the identification of modular time with a geometric or proper time is known to hold only when $-\log \Delta_\omega$ coincides with a spacetime generator [34, 359]; the de Sitter witness of Conj. 11.55 lies in this geometric class by construction. For non-faithful seeds it generalises to the van Neerven–Portal POVM construction [101].*

Remark Y.2 (Status). Thm. Y.1 is a clean structural consequence of Pos. 11.11: the modular-Hamiltonian framework automatically solves a 90-year-old QM obstruction that any positive-spectrum Hamiltonian cannot. Chua's circularity worry [359] (thermal time presupposes dynamics it claims to ground) is broken by the two-boundary closure of the framework: the seed ω is fixed by the two-boundary condition $(\rho_{\mathrm{i}}, \{E_k\})$ together with Conj. L.1, not by prior dynamics. We flag that this breaking of the circularity is itself conditional on the terminal-POVM selection

Conj. L.1; what is unconditional is that, *given* a fixed seed ω, no prior dynamical time is presupposed — the seed is fixed boundary-theoretically, not evolved into being. No new analytic input beyond Stone–von Neumann and the Tomita spectral structure is required.

We stress that $\widehat{H}_c$ in Thm. Y.1 is the *bare* modular generator $-\log\Delta_\omega$ of the Type III$_1$ matter algebra (continuous spectrum on $\mathbb{R}$), *not* the gravitationally dressed, bounded-below observer Hamiltonian $\hat{q} \geq 0$ of the CLPW Type II crossed product (App. H), whose boost generator has discrete spectrum bounded below after the dressing; as already noted following Pos. 11.11 and in App. H, these are distinct operators and only the former carries the global modular time. The Pauli obstruction is thus evaded at the level of the bare modular flow. For the dressed, semibounded, discrete-spectrum generator a self-adjoint conjugate also exists, but as the Galapon/Hiroshima–Teranishi *unbounded* time operator [360, 362] rather than the Schrödinger pair of Thm. Y.1, so the evasion is robust in either regime. The same role-inversion of energy and time underlies the recent self-adjoint constructions of [361, 363]; Thm. Y.1 is the modular-theoretic instance of this now-established family of evasions, so Pauli's original argument — which tacitly assumes the time operator preserves $\mathrm{dom}(\widehat{H})$ — is non-binding here.

Remark Y.3 (Chua's second prong). Chua's second prong — that reading σ_t^ω as *dynamics* rather than as a passive geometric/spatial redescription presupposes an antecedent dynamical notion — is not dissolved by the time operator alone. The framework's operational reading of modular time via $\widehat{T}$ (Thm. Y.1), the covariant clock POVM of [101], and the dynamical content carried by the master equation (66) (Heisenberg evolution of relational Dirac observables, Thm. O.1 (K4)) *bear on* this prong by supplying a measurable parameter and an evolution law fixed by the two-boundary data; but whether this licenses calling the modular flow genuine dynamics, as opposed to a state-relative redescription, remains a conceptual commitment we record rather than a result we prove.

Z Observer invariance via the perspective-neutral framework

This appendix records the framework's stance on the "problem of the observer" under the Höhn–Smith–Lock perspective-neutral approach [102, 103], the De Vuyst–Höhn–Tsobanjan QRF equivalence theorem [364], and the De Vuyst–Eccles– Höhn–Kirklin observer-dependence results [96, 97].

Proposition Z.1 (Observer-invariance and frame-relative gauge data)**.** *Under Pos. 11.11, Thm. H.1, and the perspective-neutral QRF framework [102, 103, 364], the framework distinguishes two tiers:*

(A) Observer-invariant predictions. *Conditional probabilities $P(k \mid \omega, C_\alpha)$ for the terminal POVM $\{E_k\}$ realised as a relational observable on Σ_{τ_f} are* independent of the QRF choice C_α for ideal QRFs *(perspective-neutral / algebraic / effective QRF equivalence theorem [364]). For non-ideal QRFs (degenerate spectra, non-sharp orientations, finite resolution) the guarantee weakens to invariance of physical* expectation values*; conditional probabilities and entropic quantities may then carry residual frame-dependence [368], controlled by inter-frame coherence/correlation terms. All physical predictions of the framework (Sec. 13) are observer-invariant.*

(B) Frame-relative gauge data. *The Type II$_\infty$ vs. Type II$_1$ classification of $\mathcal{A}_{C_\alpha}$ and the generalised entropy $S_{\mathrm{gen}}(\omega; C_\alpha)$ are* QRF-relative *[96, 97]; they transform covariantly but are not invariants. They are not directly measurable by any single observer's protocol (entropy* differences *under fixed C_α are operational). Beyond differences, specific QRF*-invariant *information-theoretic combinations are known: the sum of entanglement and subsystem coherence is conserved under QRF transformations [366], and a family of diagonal Rényi invariants is observer-independent for arbitrary subsystems [367]. These furnish frame-independent entropic observables complementary to the frame-relative $S_{\mathrm{gen}}(\omega; C_\alpha)$ (they are* *not yet shown to coincide with the Type II generalised entropy).*

(C) Definition of "observer". *An observer is any quantum subsystem satisfying the DEHK ideal-QRF axioms [96] (covariant clock POVM, sufficient pointer coherence). No appeal to consciousness, agency, or Wigner-friend observation is required; the measurement-paradox face of the observer problem is handled by Thm. T.1's Frauchiger–Renner clause.*

(D) Cosmological-horizon ambiguity. *Different cosmological observers may see different future horizons; the Speranza intrinsic-observer construction [365] obtains the observer (and hence the horizon) intrinsically from modular flow, fixing it* up to QRF transformation*; the residual QRF freedom is precisely the Tier-(B) gauge above. Selecting a canonical Σ_{τ_f} POVM within this family is the content of Conj. L.1, not an output of [365].*

Remark Z.2 (Status). Prop. Z.1 adopts the consensus 2024–2026 perspective-neutral / subsystem-relativity stance [102, 96, 364]: predictions are observer-invariant, certain algebraic-data quantities are frame-relative gauge. No new principle is required; the framework cites and adopts. This addresses the non-Wignerian face of the observer problem; the Wignerian face is handled by Thm. T.1.

E2a The analytic platform: U_ω, the geometric package, and the ball-uniform fold count

This appendix converts the analytic-collar restatement of (E2a), Hyp. E.2, from a hypothesis carried by the slice machinery into a *theorem about the class U_ω and the slice register it supports.* Part 1 (this section) assembles the analytic platform on which that theorem rests: the class U_ω (cited from the paper), the fixed-atlas geometric package ($R1$)–($R5$) that every phase estimate consumes, and the single object that finite C^k regularity could not supply — a *ball-uniform* bound N_0 on the number of fold branches of the geodesic phase. Parts 2–3 consume this platform to close the device cascade and state the (honestly modulo'd) first-law corollary.

What Part 1 proves, exactly. The count $N_0(\rho_a, \varepsilon_a, g_0)$ is finite and ball-uniform over U_ω (Theorem E2a.17); the geometric exports it is built from are theorems relative to their stated interfaces (Table 1). *What Part 1 does not claim.* It does not close (D3), book the transport average, or touch the physical clauses (i)–(iii) of Hyp. E.2; those are Parts 2–3 and remain, respectively, a theorem over U_ω (the slice register) and a theorem *modulo* the named clauses (the fused statement).

E2a.1 The class U_ω (cited) and its H^s-embedding

The analytic ball is the object U_ω of Hyp. E.2: for a real-analytic globally hyperbolic anchor g_0 with compact Cauchy slice, a complex-collar radius $\rho_a > 0$, and a collar sup-bound $\varepsilon_a > 0$ small enough that the C^0-image of U_ω has radius $< \varepsilon_{\mathrm{GH}}$ (so every $g \in U_\omega$ is globally hyperbolic by Chruściel–Grant [222]), U_ω collects the H^s metrics ($s > n/2+6$) that extend holomorphically to the collar of a fixed analytic atlas and lie within ε_a of g_0 there. We do not restate the definition; we record only the two quantitative facts Part 1 uses, in the atlas notation of the geometric package below.

Fix the atlas of $\mathbb{R}_t \times \Sigma^3$ ($n = 4$) and the compacts $K_\Sigma \subset K' \subset K'' := \{z : \mathrm{dist}(z, K') \le 4\rho_A\}$ (ρ_A the atlas margin); $|\cdot|$ is the chart-Euclidean norm. For $\rho_a > 0$ write $\mathcal{K}_{\rho_a} := \{z \in \mathbb{C}^n : \mathrm{dist}(z, K'') < \rho_a\}$ for the open complex ρ_a-collar, and

$$\|F\|_{\mathcal{K}_{\rho_a}} := \sup_{z \in \mathcal{K}_{\rho_a}} \max_{a,b} |F_{ab}(z)| \tag{142}$$

for the *collar sup-norm* (holomorphic extensions of continuous real fields are unique, so the membership constraint of U_ω is well posed). We take ε_a small enough that $\inf_{g \in U_\omega} \min_{\mathcal{K}_{\rho_a}} |\det g| =:$

$d_\omega > 0$ (possible since $\det g_0$ is bounded below on the compact $\overline{\mathcal{K}_{\rho_a}}$ and $g \mapsto \det g$ is collar-Lipschitz).

Lemma E2a.1 ($U_\omega \hookrightarrow U$, embedding constant exhibited)**.** *There is a ball-uniform embedding constant, $C_{\mathrm{emb}} := C_{\mathrm{cw}}(K'', n, s)\,(1+\rho_a^{-s})\,|K''|^{1/2}$, with*

$$\|g-g_0\|_{H^s(K'')} \;\le\; C_{\mathrm{emb}}\,\|g-g_0\|_{\mathcal{K}_{\rho_a}} \qquad (g \in U_\omega). \tag{143}$$

Consequently, if $\varepsilon_a \le \varepsilon_0/C_{\mathrm{emb}}$ then $U_\omega \subseteq U := \{g : \|g-g_0\|_{H^s(K'')} < \varepsilon_0\}$, and every export of the geometric package below (all of $(R1)$–$(R5)$, Fact F2, Convention S2-0) holds verbatim on U_ω with its U-constants.

Proof. For F holomorphic on $\mathcal{K}_{\rho_a}$ and $|\alpha| \le s$, the Cauchy estimate on the polydisc of radius ρ_a about each $z_0 \in K''$ gives $\sup_{K''}|\partial^\alpha F| \le s!\,\rho_a^{-s}\|F\|_{\mathcal{K}_{\rho_a}}$; summing the $\le C(n,s)$ derivatives of order $\le s$ and integrating over K'' gives (143) after absorbing constants into C_{cw} and summing the n^2 components. Apply to $F = g_{ab} - (g_0)_{ab}$. The embedding, and hence the verbatim transfer of every U-export (each a sup over $U \supseteq U_\omega$), is immediate. ☐

Remark E2a.2 (what pins U_ω, and the one disclosed cost). U_ω is fixed by exactly three data: the anchor g_0 (real-analytic), the collar radius ρ_a, and the collar sup-bound ε_a; the count N_0 below is a function of these three alone — *no Sobolev index and no C^k norm* enters it, which is the entire gain over U. The disclosed cost, recorded here and respected throughout Parts 2–3: U_ω is H^s-dense but has empty H^s-interior, so every uniformity claimed over it is a supremum over the ball or a pair-Lipschitz modulus, never an H^s-open property. This is exactly the scope in which the paper carries Hyp. E.2 (Option A, the analytic-collar restatement).

E2a.2 The fixed-atlas geometric package $(R1)$–$(R5)$

Every phase estimate in Parts 1–3 — the fold count N_0, the device sublevel bound, the coarea measure — reads the geometry through a fixed package of exports, established once over the H^s ball U and inherited on U_ω by Lemma E2a.1. We state the package as one exports table (Table 1) and prove the load-bearing entries; the routine composition-stability and difference legs $(R2)$, $(R6)$ are recorded but not reproved (they are classical for flows of H^{s-1} fields, $s-1 > n/2+1$, at the S1a primitives). All constants are named functions of the ball data $(g_0, \varepsilon_0, s, K_\Sigma, \text{atlas})$ alone — ball-uniform by construction, never read off examples.

Setup. Write $v = v(x,y) := y - x$ for the chord and, for $g \in U$, $\Gamma = \Gamma[g]$ for the fixed-atlas Levi-Civita symbols. The setup fixes ε_0 so small that $d_0 := \inf_{g\in U}\min_{K''}|\det g| > 0$ (uniform non-degeneracy); with $G_k := \|g_0\|_{C^k} + C_S^{(k)}\varepsilon_0$ and $G_s := \|g_0\|_{H^s} + \varepsilon_0$ one has $\sup_U \|g\|_{C^k} \le G_k$, so

$$K_\Gamma \;:=\; \sup_{g\in U}\|\Gamma[g]\|_{C^1(K'')} \;\le\; c_n\,(1+G_3)^{2n}(1+d_0^{-2}) \;<\; \infty \tag{144}$$

is finite and ball-uniform (rational map of $(g, \partial g)$ with denominators $\ge d_0$), bounding both $|\Gamma|$ and $|\partial\Gamma|$. No smallness of K_Γ is available or used: bent charts of flat metrics have $\Gamma \ne 0$.

Uniform flow. For $g \in U$, $z \in K'$, $|p| \le \rho_1$, the geodesic IVP $\gamma'' = -\Gamma(\gamma)(\gamma',\gamma')$, $\gamma(0) = z$, $\gamma'(0) = p$ has a unique solution on $[0,1]$ with image in $B(z, \tfrac{6}{5}\rho_1) \subset B(z, 4\rho_A)$ and, writing $E_z(p) := \gamma(1; z, p)$ and $W(t) := \partial_p\gamma$,

$$|\gamma'| \le \tfrac{16}{15}|p|, \quad |\gamma''| \le \tfrac{3}{2}K_\Gamma|p|^2, \quad \|W(t) - t\,\mathrm{Id}\| \le 3K_\Gamma|p|t^2, \quad \|d(E_z)_p - \mathrm{Id}\| \le 3K_\Gamma|p| \le \tfrac{3}{16}, \tag{145}$$

so E_z is a bi-Lipschitz C^2 diffeomorphism on $B(0,\rho_1)$ with $\frac{13}{16} \le \|d(E_z)_p\|^{\pm 1} \le \frac{19}{16}$ and $E_z(B(0,\rho_1)) \supseteq B(z, \rho_1/2)$ (comparison $\dot y = K_\Gamma y^2$ for the speed; Duhamel for W; Neumann for the inverse). This is the exponential-map primitive $(R2)$ consumes.

export	statement (ball-uniform)	grade / provenance
$(R1)$	$\gamma_{x,y}(t) = x + tv + q$, $\|q\|_{C_t^2} \le \kappa_2 \|v\|^2$, $q(0) = q(1) = 0$, $\partial_y q(0) = 0$; $\kappa_2 = 14K_\Gamma$, $c_{\mathrm{geo}} = 1 + \kappa_2\rho_1/4 \le \frac{39}{32}$, $\|\partial_y \gamma\| \le c_{\mathrm{geo}} t$	THEOREM (§E2a.2)
$(R2)$	H^{s-1}-uniform E_y; $\|F \circ E_y\|_{H^\kappa} \le c_E \|F\|_{H^\kappa}$, $0 \le \kappa \le s-1$	THEOREM rel. $(P1)$–$(P2)$
$(R3)$	$\|\phi' - e\cdot v\|$, $\|\phi''\| \le \kappa_2\|v\|^2$; $\|\phi'''\|_{C^0} \le \kappa_3\|v\|^3$; $\phi'' = -e\cdot\Gamma(\dot\gamma,\dot\gamma)$; $\kappa_3 = \frac{27}{8}K_\Gamma(1+2K_\Gamma)$	THEOREM rel. S1a, (NC)
$(R4)$	$c_{\mathrm{ch}}^{-1}\rho \le \|v\| \le c_{\mathrm{ch}}\rho$ ($c_{\mathrm{ch}} = \frac{3}{2}\lambda_g^{1/2}$); velocity chart G_t a C^1-diffeo, $\|(dG_t)^{-1}\| \le 2$	THEOREM rel. S1a, (NC)
$(R5)$, F2	$d\theta(\{\|e\cdot v\| \le s\rho\}) \le c_{\mathrm{fan}} s$, $c_{\mathrm{fan}} = c_{\mathrm{dens}}\, c_n\, c_{\mathrm{ch}}$	THEOREM rel. S1a/S1b
$(R6)$	2-plane worst-section reduction; d_yC paraproduct split; soft legs book $\ge s-3$	THEOREM rel. $(R1)$, $(R2)$
S2-0	$\bar\kappa_2 := c_{\mathrm{ch}}^2\kappa_2$; cone $\{\|e\cdot v\| < 2\bar\kappa_2\rho^2\}$ stationary-free	convention (A-1)

Table 1: The T1′-rung geometric exports, condensed from `t1p_S1a`/`S1b`/`S1c` (the provisional radius $\rho_1 = \min(\rho_A, 1, \frac{1}{16(1+K_\Gamma)})$ is the setup constant they share). Each is a theorem relative to its named interface; the loci carry no oscillatory content, no per-θ object, no pointwise envelope. $\lambda_g \ge 1$ is the ball-uniform ellipticity constant, $\lambda_g^{-1}|\xi|^2 \le g_{ij}\xi^i\xi^j \le \lambda_g|\xi|^2$; (NC) is the Gauss-lemma distance-realization fact.

Proposition E2a.3 (($R1$): chord/velocity expansion)**.** *On $N_1 := \{(x,y) : |v| \le \rho_1/4\}$ let $\gamma_{x,y}(t) := \gamma(t; x, E_x^{-1}(y))$ and write $\gamma_{x,y}(t) = x + tv + q(t;x,y)$. Then, with $\kappa_2 := 14K_\Gamma$ and $c_{\mathrm{geo}} := 1 + \kappa_2\rho_1/4 \le \frac{39}{32}$:* (a) *$q(0) = q(1) = 0$ and $\|q\|_{C_t^2} \le \kappa_2|v|^2$;* (b) *$\partial_y q(0) = 0$, $\|\partial_y q\|_{C_t^1} \le \kappa_2|v|$, whence $|\partial_y\gamma_{x,y}(t)| \le c_{\mathrm{geo}}\, t$ (the y-derivative vanishes at the x-end);* (c) *q is jointly C^1 in (x,y). No smallness of κ_2 is claimed.*

Proof. Set $p := E_x^{-1}(y)$, so $|p| \le \frac{6}{5}|v|$ and $|p - v| \le \frac{3}{4}K_\Gamma|p|^2 \le \frac{27}{25}K_\Gamma|v|^2$ by (145) at $t = 1$. *(a)* $q(0) = 0$, $q(1) = y - x - v = 0$; $q'' = \gamma''$ gives $\|q''\|_\infty \le \frac{3}{2}K_\Gamma|p|^2 \le \frac{11}{5}K_\Gamma|v|^2$; $q' = (\gamma' - p) + (p - v)$ gives $\|q'\|_\infty \le \frac{10}{3}K_\Gamma|v|^2$; the Dirichlet Green function of d^2/dt^2 ($\max_t \int |G| = \frac{1}{8}$) gives $\|q\|_\infty \le \frac{1}{8}\|q''\|_\infty \le \frac{11}{40}K_\Gamma|v|^2$; the maximum of the three coefficients is ≤ 14, so $\|q\|_{C_t^2} \le \kappa_2|v|^2$. *(b)* With $d(E_x)_p = W(1)$ the chain rule gives $\partial_y\gamma(t) = W(t)W(1)^{-1}$, so $\partial_y q(t) = (W(t) - t\,\mathrm{Id})W(1)^{-1} + t(W(1)^{-1} - \mathrm{Id})$; the brackets (145) give $|\partial_y q(t)| \le 10K_\Gamma|v|\,t \le \kappa_2|v|\,t$ (so $\partial_y q(0) = 0$) and, differentiating, $\|\partial_t\partial_y q\| \le 14K_\Gamma|v|$. Then $|\partial_y\gamma| \le t + |\partial_y q| \le (1 + \kappa_2|v|)t \le c_{\mathrm{geo}} t$ since $\kappa_2|v| \le 14K_\Gamma\rho_1/4 \le \frac{7}{32}$. *(c)* The flow is jointly C^1 in (t,z,p) and $(x,y) \mapsto p$ is C^1 by the inverse function theorem for $(x,p) \mapsto (x, E_x(p))$. □

The remaining exports fix the calibrated radius and the phase derivatives. Let $\lambda_g \ge 1$ be the ball-uniform ellipticity constant of the setup and (NC) the Gauss-lemma fact of Table 1. Set $c_{\mathrm{ch}} := \frac{3}{2}\lambda_g^{1/2} \ge 1$ and choose the calibrated radius $\rho_0 \le \rho_1$ so that (i) $c_{\mathrm{ch}}\rho_0 \le \rho_1/4$ (E_y-domain / N_1 membership), (ii) $\kappa_2 c_{\mathrm{ch}}\rho_0 \le \frac{1}{2}$ (velocity normalization), (iii) $c_{E2}\lambda_g^{1/2}\rho_0 \le \frac{1}{4}$ (velocity-chart margin, $c_{E2} := \sup_U \|E_y\|_{C^2} < \infty$ by $H^{s-1} \subset C^2$), and (iv) $\rho_0 \le \rho_{\mathrm{nc}}$ (distance realization). Every entry is ball-uniform, so $\rho_0 > 0$ is.

Lemma E2a.4 (($R4$): chord bi-Lipschitz and the velocity chart)**.** *Let $g \in U$, $y \in K'$, $0 < \rho \le \rho_0$.* (i) *No point of $\{d_g(\cdot,y) \le \rho_0\}$ is conjugate to y, and for $d_g(x,y) = \rho$ the chord obeys*

$$c_{\mathrm{ch}}^{-1}\rho \;\le\; |v(x,y)| \;\le\; c_{\mathrm{ch}}\rho. \tag{146}$$

(ii) *On the frozen unit sphere $\Sigma_y := \{\theta : |\theta|_{g(y)} = 1\}$, setting $x(\theta) := E_y(\rho\theta)$, the velocity field $G_t(\theta) := -\gamma'_{x(\theta),y}(t)/\rho = d(E_y)_{(1-t)\rho\theta}[\theta]$ (so $G_1 = \mathrm{Id}$) extends to a C^1 map with $\|dG_t - \mathrm{Id}\| \le \frac{1}{2}$, hence a C^1-diffeomorphism onto its image with $\|(dG_t)^{-1}\| \le 2$, for every $t \in [0,1]$.*

Proof. *(i)* By (NC) and ellipticity, $w := E_y^{-1}(x)$ has $|w| \le \lambda_g^{1/2}\rho \le \rho_1$, so (145) applies at $p = w$ and $d(E_y)$ is invertible on the whole normal ball (no conjugate point); $\gamma_{x,y}(t) = E_y((1-t)w)$. Then $v = E_y(0) - E_y(w)$ gives $\frac{13}{16}|w| \le |v| \le \frac{19}{16}|w|$, and $\lambda_g^{-1/2}\rho \le |w| \le \lambda_g^{1/2}\rho$ yields (146) ($\frac{19}{16}\lambda_g^{1/2} \le \frac{3}{2}\lambda_g^{1/2}$ above, $\frac{13}{16}\lambda_g^{-1/2} \ge \frac{2}{3}\lambda_g^{-1/2}$ below). *(ii)* From $\gamma_{x(\theta),y}(t) = E_y((1-t)\rho\theta)$, $dG_t[h] = d(E_y)_u[h] + d^2(E_y)_u[(1-t)\rho h, \theta]$ with $u := (1-t)\rho\theta$, $|u| \le \lambda_g^{1/2}\rho$; the two terms are $\le 3K_\Gamma\lambda_g^{1/2}\rho_0 \le \frac{1}{32}$ (by (i)-margin and $\rho_1 \le \frac{1}{16K_\Gamma}$) and $\le c_{E2}\lambda_g^{1/2}\rho_0 \le \frac{1}{4}$, so $\|dG_t - \mathrm{Id}\| \le \frac{1}{2}$; Neumann gives $\|(dG_t)^{-1}\| \le 2$. □

Proposition E2a.5 ($(R3)$: phase derivative bounds; Convention S2-0)**.** *Let $g \in U$, $d_g(x,y) \le \rho_0$, $|e| = 1$, and $\phi(t) := e \cdot \gamma_{x,y}(t)$. With $\kappa_3 := \frac{27}{8}K_\Gamma(1+2K_\Gamma)$:* (i) *$|\phi'(t) - e\cdot v| \le \kappa_2|v|^2$ and $|\phi''(t)| \le \kappa_2|v|^2$, and the registry identity $\phi''(t) = -e\cdot\Gamma(\gamma(t))(\gamma'(t),\gamma'(t))$ holds;* (ii) *$\phi \in C^3([0,1])$ with $\|\phi'''\|_{C^0} \le \kappa_3|v|^3$;* (iii) *a stationary point $\phi'(t_*) = 0$ forces $|e\cdot v| \le \kappa_2|v|^2$. Setting Convention S2-0, $\bar\kappa_2 := c_{\mathrm{ch}}^2\kappa_2$, the cone $\{|e\cdot v| < 2\bar\kappa_2\rho^2\}$ contains all stationary directions while its complement Θ_{osc} carries the phase floor $|\phi'| \ge \frac{1}{2}|e\cdot v|$ — no middle band exists.*

Proof. By $(R1)$, $\gamma' = v + q'$ and $\phi'' = e\cdot q''$ with $\|q'\|, \|q''\| \le \kappa_2|v|^2$, giving (i); the identity is the geodesic equation contracted with e. For (ii), differentiating the identity and using $|\gamma''| \le K_\Gamma|\gamma'|^2$, $|\gamma'| \le \frac{3}{2}|v|$ gives $|\phi'''| \le K_\Gamma(1+2K_\Gamma)(\frac{3}{2})^3|v|^3 = \kappa_3|v|^3$. (iii) is immediate from the ϕ' bound. Inserting (146) ($|v| \le c_{\mathrm{ch}}\rho$) calibrates each power to $c_{\mathrm{ch}}^k\rho^k$; $c_{\mathrm{ch}}^2\kappa_2 = \bar\kappa_2$ gives the cone/floor dichotomy. □

The fan measure $(R5)$ *and Fact F2.* Fix $g \in U$, $y \in K_\Sigma$, $0 < \rho \le \rho_0$. Let $\mathrm{d}S_y$ be the round surface measure that the frozen unit sphere $\Sigma_y = \{|\theta|_{g(y)} = 1\}$ inherits from $g(y)$, and $d\theta := \omega_y^{-1}\mathrm{d}S_y$ ($\omega_y := \mathrm{d}S_y(\Sigma_y)$) the *normalized fan measure*, a probability measure on Σ_y independent of ρ, Λ; the bending of the fan lives in the chart $x(\theta)$, carried by the Jacobian bounds, so $d\theta$ itself needs no curvature hypothesis. This is the measure Parts 2–3 integrate.

Proposition E2a.6 (Fact F2: the shell-measure bound)**.** *Write $\sigma(\theta) := |e\cdot v(x(\theta),y)|/\rho$. There is a ball-uniform $c_{\mathrm{fan}} = c_{\mathrm{dens}}\, c_n\, c_{\mathrm{ch}}$, with $c_{\mathrm{dens}} = (\frac{5}{3})^{n-1}\lambda_g^{n-1}$ and $c_n = 2|S^{n-2}|/|S^{n-1}|$ ($c_4 = 4/\pi$), such that for all $g \in U$, $y \in K_\Sigma$, $0 < \rho \le \rho_0$, $|e| = 1$, $s > 0$,*

$$d\theta\big(\{\theta : |e\cdot v(x(\theta),y)| \le s\rho\}\big) \ \le\ c_{\mathrm{fan}}\, s. \tag{147}$$

Proof. Let $V(\theta) := v(x(\theta),y)/\rho = -(E_y(\rho\theta) - E_y(0))/\rho$; by (145), $dV_\theta = -d(E_y)_{\rho\theta}$ is invertible with $\|(dV_\theta)^{-1}\| \le \frac{4}{3}$ and $\frac{3}{4} \le |V| \le \frac{5}{4}$. The radial projection $\Phi := R\circ V$, $R(w) = w/|w|$, is then a C^1-diffeomorphism of Σ_y onto an open subset of S^{n-1}, and every singular value of $d\Phi_\theta$ (from the $g(y)$-domain to the round target) is $\ge \frac{3}{4}\cdot\frac{4}{5}\cdot\lambda_g^{-1/2}$: the V-factor contributes $\ge \frac{3}{4}$ (min–max on $\|(dV)^{-1}\| \le \frac{4}{3}$), the R-factor $\ge (1/|w|) \ge \frac{4}{5}$ on directions transverse to the radius, and the metric bridge $\ge \lambda_g^{-1/2}$. Hence $J_\Phi \ge (\frac{3}{4}\cdot\frac{4}{5}\cdot\lambda_g^{-1/2})^{n-1}$, and the area formula with $\omega_y \ge \lambda_g^{-(n-1)/2}|S^{n-1}|$ gives $\Phi_* d\theta \le c_{\mathrm{dens}}\,\omega$ on S^{n-1}. Since $|v| \le c_{\mathrm{ch}}\rho$ (eq. (146)) gives $\sigma \le c_{\mathrm{ch}}|e\cdot u|$ with $u = \Phi(\theta)$, the target set is contained in $\Phi^{-1}(\{|e\cdot u| \le c_{\mathrm{ch}}s\})$, and the equatorial-band estimate $\omega(\{|e\cdot u| \le r\}) \le c_n r$ yields (147). The bending is absorbed entirely into the one-sided Jacobian $\|(dV)^{-1}\| \le \frac{4}{3}$ — a transversality floor, not a curvature bound. □

Remark E2a.7 (the worst-section reduction $(R6)$, recorded)*.* The scalar model $I_\Lambda(x,y;e) = \int_0^1 e^{i\Lambda\phi(t)}w\,dt$ depends on the geodesic only through $\phi = e\cdot\gamma_{x,y}$, hence only through the projection onto $\mathrm{span}(e,v)$. So any pointwise-in-θ phase bound proved on the planar ($n=2$) section transfers to the $n=4$ fan unchanged (the $\mathrm{span}(e,v)$-section is worst), while $d\theta$ fibers over the section with the transverse directions contributing only measure (Prop. E2a.6). No pointwise-in-θ object crosses an interface: the sectioning transports the planar model, then $d\theta$ is integrated before export. The soft (d_yC paraproduct) and difference legs book register $\ge s-3$ by composition stability $(R2)$ and Moser, with margin $\frac{11}{2}$ over the consumers' worst demand $s - \frac{17}{2}$; these are recorded from `t1p_S1c` and not reproved here.

E2a.3 The ball-uniform fold count N_0

The geometric package supplies, on the H^s ball U, the device sublevel bound only in the form $d\theta(S_\mu) \le C_{\rm sub}\max(\mu, \rho\mu^{1/2})$; the *linear* form $d\theta(S_\mu) \le c_{\rm sub}\mu$ that Part 2 needs to close (D3) requires a ball-uniform count N_0 of the e-relevant fold branches of the geodesic phase, i.e. of the zeros of ϕ'' along admissible geodesics. Over U this count is $+\infty$: the ripple $g^{(N)} = \mathrm{diag}(1, 1 + a_N \sin(Nz_1))$ with $a_N = \frac{1}{2}\varepsilon_0 C_s^{-1} N^{-s}$ stays in U (the H^s-amplitude $a_N N^s \equiv \frac{1}{4}\varepsilon_0$ is pinned) while $\#\{t : \phi''(t) = 0\} \sim N\rho/\pi \to \infty$; and finite C^k regularity cannot supply the count, because the Rolle recursion needs a ϕ'''-lower floor no (R)-item provides. The gain of U_ω is that the count *is* ball-uniform there. The mechanism is one fact: the ripple's zero-forcing power lives in its mode N, and a uniform collar sup-bound caps the mode exponentially.

The analytic-ODE package and the phase on a t-strip

The tool is the holomorphic Cauchy–Kovalevskaya / Cauchy–Lipschitz theorem for the geodesic ODE with a holomorphic vector field, quoted at the strength used with explicit constants. Its sole non-textbook input is a *ball-uniform* field bound; everything downstream is uniform because that bound is.

Remark E2a.8 (analytic-ODE package; classical, quoted). For $g \in U_\omega$ (so each g_{ab} is holomorphic on $\mathcal{K}_{\rho_a}$ and $|\det g| \ge d_\omega > 0$ there): *(i)* the Christoffel symbols $\Gamma[g]$ are holomorphic on the shrunk collar $\mathcal{K}_{\rho_a/2}$ with

$$M_\Gamma \;:=\; \sup_{g\in U_\omega} \|\Gamma[g]\|_{\mathcal{K}_{\rho_a/2}} \;\le\; c_n\, \rho_a^{-1}\big(\|g_0\|_{\mathcal{K}_{\rho_a}} + \varepsilon_a\big)\big(1 + d_\omega^{-1}\big)^n \;<\; \infty \tag{148}$$

(one Cauchy derivative costs ρ_a^{-1}; ∂g and $g^{-1} = (\det g)^{-1}\,\mathrm{adj}\, g$ holomorphic by Lemma E2a.1); *(ii)* for $z \in K'$ and $|p| \le \rho_a/(8M_\Gamma)$ the complex-time geodesic IVP has a unique holomorphic solution on the disc $\{|\tau| \le \frac{1}{4}\}$ with $\gamma \in \mathcal{K}_{\rho_a/4}$ and $|\dot\gamma| \le 2|p|$ (Picard iteration, contraction $\le \frac{1}{2}$). References: Hille, *ODE in the Complex Domain*, Thm 2.2.2; Ilyashenko–Yakovenko §1.4. The literature is used strictly inside the analytic class it is stated for, never promoted to C^k.

Proposition E2a.9 (the phase lives on a definite t-strip; ball-uniform sup)**.** *There are ball-uniform $r_t > 0$ and $M''_\phi < \infty$, depending on $(\rho_a, \varepsilon_a, g_0)$ only through M_Γ,*

$$r_t \;:=\; \min\Big(\tfrac{1}{4},\ \tfrac{\rho_a}{16\, M_\Gamma\, c_{\rm ch}\rho_0}\Big), \tag{149}$$

such that for every $g \in U_\omega$, $y \in K_\Sigma$, $0 < \rho \le \rho_0$, $x \in \mathrm{fan}_\rho(y)$, $|e| = 1$, the phase $\phi(t) = e\cdot\gamma_{x,y}(t)$ extends holomorphically to the strip $\Sigma_{r_t} := \{\tau \in \mathbb{C} : |\operatorname{Im}\tau| < r_t\}$ with

$$\sup_{\tau\in\Sigma_{r_t}} |\phi''(\tau)| \;\le\; M''_\phi \;:=\; 4\, M_\Gamma\, c_{\rm ch}^2\, \rho^2. \tag{150}$$

Proof. On $\mathrm{fan}_\rho(y)$ the geodesic solves $\ddot\gamma = -\Gamma(\gamma)(\dot\gamma,\dot\gamma)$ with $|\dot\gamma(0)| \le c_{\rm ch}\rho$; the field $(\zeta,\eta) \mapsto (\eta, -\Gamma(\zeta)(\eta,\eta))$ is holomorphic on $\mathcal{K}_{\rho_a/2}\times\mathbb{C}^n$ with $\|\Gamma\| \le M_\Gamma$ (Remark E2a.8). Applying (ii) at each base point $\gamma(t_0)$, $t_0 \in [0,1]$, with momentum bounded by $2c_{\rm ch}\rho_0$ extends the flow to a complex-time disc of t_0-uniform radius $\tau'_0 = \rho_a/(16M_\Gamma c_{\rm ch}\rho_0)$; the discs glue by uniqueness of continuation to the strip Σ_{r_t}, $r_t = \min(\frac{1}{4}, \tau'_0)$. On the strip $|\dot\gamma(\tau)| \le 2c_{\rm ch}\rho$ (Picard, on the whole disc), so *directly from the ODE* (not a Cauchy shrinkage of ϕ) $|\phi''(\tau)| = |e\cdot\Gamma(\gamma)(\dot\gamma,\dot\gamma)| \le M_\Gamma(2c_{\rm ch}\rho)^2 = M''_\phi$, which is $O(\rho^2)$ — the same quadratic scale as the real $(R3)$ bound. $\square$

Remark E2a.10 (why the Jensen ratio is ρ-bounded — the crux). The ρ^2 in (150) is load-bearing. The anchor floor m_0 (below) is *also* $O(\rho^2)$: for a ϕ''-nondegenerate curved anchor, $(\phi^0)'' = -e\cdot\Gamma[g_0](\dot\gamma^0,\dot\gamma^0) \sim (\text{curvature})\cdot\rho^2$. Hence the Jensen numerator and denominator share the ρ-scale and the ratio $M''_\phi/m_0 =: \mathcal{R}_0$ is ρ-bounded ($\sim \rho_0^2/\rho_{\min}^2$), not ρ-free. Had one used the lossy $O(1)$ Cauchy bound $\sim r_t^{-2}M_\phi$ for the numerator against the $O(\rho^2)$ floor, the ratio would blow up as $\rho \to 0$ — a spurious divergence of N_0. The ρ-honest ODE bound removes it.

Lemma E2a.11 (ripple exclusion)**.** *Fix any $\rho_a, \varepsilon_a > 0$. For the refuting ripple $g^{(N)} = \mathrm{diag}(1, 1+a_N \sin(Nz_1))$, $a_N = \frac{1}{2}\varepsilon_0 C_s^{-1} N^{-s}$ (in U for all N), the collar sup-norm blows up:*

$$\|g^{(N)} - g_0^{\mathrm{flat}}\|_{\mathcal{K}_{\rho_a}} = a_N \cosh(N\rho_a) = \tfrac{1}{2}\varepsilon_0 C_s^{-1} N^{-s} \cosh(N\rho_a) \xrightarrow[N\to\infty]{} +\infty, \tag{151}$$

because $e^{N\rho_a}$ dominates N^{-s} for every fixed $\rho_a > 0$. Hence there is a finite threshold $N_*(\rho_a, \varepsilon_a)$ with $g^{(N)} \notin U_\omega$ for all $N \geq N_*$: the ripples in U_ω have $N < N_*$, a finite set.

Proof. $\sup_{|\operatorname{Im} z_1| \leq \rho_a} |\sin(Nz_1)| = \cosh(N\rho_a)$ (attained at $\operatorname{Im} z_1 = \rho_a$, $\sin^2(N \operatorname{Re} z_1) = 1$); multiply by a_N. The logarithm is $N\rho_a - s\log N + O(1) \to +\infty$, so the sublevel $\{N : \text{collar-norm} \leq \varepsilon_a\}$ is bounded. □

Remark E2a.12 (the exclusion is structural, not amplitude-tuned). Lemma E2a.11 is a statement about the *mode* N, not the amplitude: for any amplitude a, $\|a\sin(N\cdot)\|_{\mathcal{K}_{\rho_a}} = a\cosh(N\rho_a)$, so keeping the collar norm $\leq \varepsilon_a$ forces $a \leq 2\varepsilon_a e^{-N\rho_a}$ — on U_ω a mode-N oscillation has exponentially small amplitude and can bend ϕ'' by at most $\sim aN^2\rho^2 \to 0$. This is the analytic replacement for the missing ϕ'''-floor: not a floor on ϕ''', but a collar-supplied ceiling on the number of sign changes. A C^k ball caps aN^k (polynomial, lets $N \to \infty$ at fixed cost); the collar caps $ae^{N\rho_a}$ (exponential).

Stratification, anchor floor, and the per-tile Jensen count

Jensen counts zeros of a holomorphic function that is not identically zero; along admissible geodesics ϕ'' can vanish identically, exactly on the flat / geodesically-straight configurations. We split these off explicitly, so the count is applied only where it is licensed and the identically-vanishing stratum is carried by the device measure, never by a division by zero. Fix once the tiling of $[0,1]$ into $P := \lceil 4/r_t \rceil$ subintervals $I_1, \ldots, I_P$ of length $\leq r_t/4$, and write the tile-j working strip $\Sigma^{(j)}_{r_t/4} := \{\tau \in \Sigma_{r_t/4} : \operatorname{Re}\tau \in I_j\}$.

Definition E2a.13 (the two strata; a per-tile floor)**.** For a threshold $m_0 > 0$ (fixed below, an anchor datum), requiring the working-strip sup $\geq m_0$ *on every tile*, set

$$\mathcal{N}_{m_0} := \Big\{(x,y,e) : \min_{1\leq j\leq P} \sup_{\tau \in \Sigma^{(j)}_{r_t/4}} |\phi''_{x,y,e}(\tau)| \geq m_0\Big\}, \qquad \mathcal{V}_{m_0} := \text{(complement)}. \tag{152}$$

On $\mathcal{N}_{m_0}$ every tile carries a valid Jensen floor; on $\mathcal{V}_{m_0}$ some tile has strip-sup $< m_0$ (the phase is near-flat there) and the count is not invoked. The identically-vanishing set $\{\phi'' \equiv 0\} \subseteq \mathcal{V}_{m_0}$ for every $m_0 > 0$. A single *global* floor would bound zeros only in one $O(r_t)$-segment; the per-tile condition is the one the count actually requires.

The threshold is chosen from the anchor g_0, not the perturbed g. Because the collar transfer $\sup_{\Sigma_{r_t/4}} |\phi''_g - \phi''_{g_0}| \leq C'_{\mathrm{tr}} \|g - g_0\|_{\mathcal{K}_{\rho_a}} \leq C'_{\mathrm{tr}}\varepsilon_a$ holds (holomorphic-Lipschitz dependence of the geodesic flow on the field, via a Grönwall estimate on the definite complex-time disc), setting $m_0 := \frac{1}{2} m_0^{\mathrm{anch}}$ with

$$m_0^{\mathrm{anch}} := \inf_{(x,y,e,\rho)\in\mathcal{T}} \min_{1\leq j\leq P} \sup_{\tau\in\Sigma^{(j)}_{r_t/4}} \big|(\phi^0_{x,y,e})''(\tau)\big| \tag{153}$$

and imposing $\varepsilon_a \leq m_0^{\mathrm{anch}}/(4C'_{\mathrm{tr}})$ makes the perturbed strip-sup $\geq \frac{3}{4} m_0^{\mathrm{anch}} > m_0$ for every anchor-nondegenerate datum: on U_ω the nonvanishing stratum *contains* every anchor-nondegenerate configuration, and $\mathcal{V}_{m_0}$ is confined to a collar-neighbourhood of the anchor's own degenerate set. This is the precise sense in which the analytic anchor supplies the floor.

Lemma E2a.14 (anchor floor is positive; finite-dimensional Euclidean compactness only)**.** *Let g_0 be real-analytic and not everywhere ϕ''-flat. Then $m_0^{\text{anch}} > 0$, where the infimum* (153) *is over the* compact *datum set*

$$\mathcal{T} \;:=\; \{(x, y, e, \rho) : y \in K_\Sigma,\;\rho_{\min} \le \rho \le \rho_0,\; x \in \mathrm{fan}_\rho(y),\; |e| = 1\} \setminus \mathcal{D}_0^{\varepsilon'}, \tag{154}$$

$\mathcal{D}_0^{\varepsilon'}$ an open neighbourhood of the anchor-degenerate set $\mathcal{D}_0 = \{(x, y, e, \rho) : (\phi^0)'' \equiv 0\}$, and $\rho_{\min} > 0$ a fixed floor.

Proof. $\mathcal{T} \subset \mathbb{R}^n \times S^{n-1} \times [\rho_{\min}, \rho_0]$ is closed and bounded, hence compact in the *Euclidean* topology; the metric is the single fixed g_0, so *no function-space compactness is used.* The map $(x, y, e, \rho) \mapsto \min_j \sup_{\Sigma^{(j)}_{r_t/4}} |(\phi^0)''|$ is continuous (joint holomorphy of the anchor flow in (τ, data), Remark E2a.8(ii) at $g = g_0$). Pointwise it is > 0: off $\mathcal{D}_0$, $(\phi^0)''$ is real-analytic and $\not\equiv 0$, so it has isolated zeros and its sup over each length-positive tile is > 0. A positive continuous function on a compact set attains a positive minimum. □

Remark E2a.15 (the topology each constant lives in; why the ball need not be compact). Every compactness step here is finite-dimensional-Euclidean or a one-point (fixed-metric) statement — never a function-space compactness of U_ω. Ball-uniformity across U_ω is obtained *not* by compactness but by the collar margin $\|g - g_0\|_{\mathcal{K}_{\rho_a}} < \varepsilon_a$ propagating the anchor floor by an explicit Lipschitz constant. This is essential: bounded holomorphic functions on a *fixed* collar are Montel-compact only in the local-uniform topology of a strictly smaller collar, so the sup-norm ball is not sup-norm-compact. The collar shrinkage a Montel argument would force is exactly the $\rho_a \to \rho_a/2 \to r_t$ shrinkage already tracked as explicit radius loss in $r_t, M''_\phi, C'_{\text{tr}}$.

Lemma E2a.16 (per-tile Jensen count on $\mathcal{N}_{m_0}$)**.** *Let $(x, y, e) \in \mathcal{N}_{m_0}$ and $g \in U_\omega$. Then $\#\{t \in [0, 1] : \phi''_{x,y,e}(t) = 0\} \le N_0$, where*

$$\boxed{N_0 \;=\; N_0(\rho_a, \varepsilon_a, g_0) \;:=\; \left\lceil \tfrac{4}{r_t} \right\rceil \frac{\log(M''_\phi/m_0)}{\log(3/\sqrt{2})} \;+\; \left\lceil \tfrac{4}{r_t} \right\rceil \;<\; \infty} \tag{155}$$

with r_t of (149)*, M''_ϕ of* (150)*, $m_0 = \frac{1}{2} m_0^{\text{anch}}$; equivalently $[0, 1]$ splits into $\le 1 + N_0$ maximal subintervals on each of which ϕ'' is single-signed.*

Proof. The strip has half-width $r_t \le \frac{1}{4}$, thinner than $[0, 1]$, so we tile at the strip scale. Fix tile I_j and a centre $\tau_*^{(j)} \in \overline{\Sigma^{(j)}_{r_t/4}}$ where the tile-j working-strip sup is attained; on $\mathcal{N}_{m_0}$, $|\phi''(\tau_*^{(j)})| \ge m_0 > 0$ for *every* j (the per-tile floor). Set $R := \frac{3}{4} r_t$, $r := \frac{\sqrt{2}}{4} r_t$; then *(g1)* $D(\tau_*^{(j)}, R) \subseteq \Sigma_{r_t}$ (as $|\operatorname{Im} \tau_*^{(j)}| + R \le r_t$), so $|\phi''| \le M''_\phi$ on its circle by (150); *(g2)* every real zero $t \in I_j$ lies in $D(\tau_*^{(j)}, r)$ (both t and $\operatorname{Re} \tau_*^{(j)}$ in the length-$(r_t/4)$ tile, $|\operatorname{Im} \tau_*^{(j)}| \le r_t/4$); *(g3)* the radius ratio $R/r = 3/\sqrt{2}$ is absolute. Jensen's counting formula on $D(\tau_*^{(j)}, R)$ (Levin, *Lectures on Entire Functions*, Lect. 1; Titchmarsh §3.61) for $\phi'' \not\equiv 0$ gives, for the inner-disc zero count,

$$n(r) \log \tfrac{R}{r} \;\le\; \tfrac{1}{2\pi} \int_0^{2\pi} \log |\phi''(\tau_*^{(j)} + R e^{i\theta})|\, d\theta - \log |\phi''(\tau_*^{(j)})| \;\le\; \log \tfrac{M''_\phi}{m_0}, \tag{156}$$

so by (g2)–(g3) the zeros of ϕ'' in I_j number $n_j \le \log(M''_\phi/m_0)/\log(3/\sqrt{2})$. Summing over $j = 1, \ldots, P$ and adding one boundary zero per tile-junction gives (155). Every quantity is ball-uniform, uniform over $g \in U_\omega$ and over $\mathcal{N}_{m_0}$. □

Theorem E2a.17 (N_0-analytic: the ball-uniform fold count)**.** *Fix a real-analytic anchor g_0 (not everywhere ϕ''-flat), a collar radius $\rho_a \in (0, \rho_a^0]$, and $\varepsilon_a > 0$ with $\varepsilon_a \le \min\big(\varepsilon_0/C_{\text{emb}},\; m_0^{\text{anch}}/(4C'_{\text{tr}})\big)$ (so $U_\omega \subseteq U$ and the anchor floor propagates). Then there is a ball-uniform constant*

$$N_0(\rho_a, \varepsilon_a, g_0) \;=\; \left\lceil \tfrac{4}{r_t} \right\rceil \Big(1 + \frac{\log \mathcal{R}_0}{\log(3/\sqrt{2})}\Big), \qquad \mathcal{R}_0 = \frac{M''_\phi}{m_0} = \frac{8 M_\Gamma c_{\text{ch}}^2}{c_0} \cdot \frac{\rho_0^2}{\rho_{\min}^2} \;\; (\rho\text{-}bounded), \tag{157}$$

with ingredients exhibited as functions of $(\rho_a, \varepsilon_a, g_0)$*:* r_t *of* (149), M_Γ *of* (148), $M''_\phi = 4M_\Gamma c_{\rm ch}^2 \rho^2$, *and* $m_0 = \frac{1}{2} c_0 \rho_{\min}^2$ *with* $c_0 := \rho_{\min}^{-2} \inf_{\mathcal{T}} \min_j \sup_{\Sigma^{(j)}_{r_t/4}} |(\phi^0)''| > 0$*, such that for every* $g \in U_\omega$*, every* $(x, y, e) \in \mathcal{N}_{m_0}$*, every* $0 < \rho \leq \rho_0$*,*

$$\#\{t \in [0,1] : \phi''_{x,y,e}(t) = 0\} \ \leq \ N_0.$$

Grade: THEOREM*, relative to* $(R1), (R3), (R4), (R5)$ *and Hyp. E.2.*

Proof. Assemble Lemmas E2a.11, E2a.14, E2a.16 and Proposition E2a.9: r_t, M''_ϕ, m_0 are ball-uniform (Prop. E2a.9, Lem. E2a.14); the count is (155); the ratio $\mathcal{R}_0$ is ρ-bounded (Remark E2a.10). Positivity of m_0 needs only the finite-dimensional Euclidean compactness of Lemma E2a.14, and ball-uniformity across U_ω the collar margin (Remark E2a.15). □

Remark E2a.18 (what N_0 settles, and what it defers)*.* Theorem E2a.17 supplies exactly the object the S3c device decision named and finite C^k could not deliver: the ball-uniform branch bound $Z_0 := N_0$, from which the device constant $D = 1 + N_0$ is ball-uniform over U_ω. The count itself is *unconditional* over U_ω. What is *not* proved here, and is deferred to Part 2: the pure-linear device bound $d\theta(S_\mu) \leq c_{\rm sub}\mu$ ($c_{\rm sub} = c_{\rm vel}(1 + N_0)$) holds on the fold stratum $\mathcal{N}_{m_0}$ and for $\mu \geq \kappa_V$ (a fixed ball-uniform threshold), with a count-free $\mu^{1/2}$ remainder confined to the near-flat regime $\mu < \kappa_V$ where ϕ'' is uniformly small and no genuine fold occurs; whether (D3) then closes over U_ω is Part 2's cascade, not a claim of Part 1. On the near-vanishing stratum $\mathcal{V}_{m_0}$ no zero count of an identically-vanishing ϕ'' is ever taken — only the device measure is used there.

E2a.4 The fan model integral and the oscillatory branch

Recall the near-diagonal fan geometry of Part 1. For $g \in U_\omega$ (Def. E2a.1), $y \in K_\Sigma$, $0 < \rho \leq \rho_0$, a unit covector e ($|e| = 1$), and frequency $\Lambda > 0$, the chord is $v = v(x, y) := y - x$, the phase is $\phi(t) = \phi(t; x, y, e) := e \cdot \gamma_{x,y}(t)$ with $\gamma_{x,y}(t) = x + tv + q(t; x, y)$ the geodesic ((R1), Lemma E2a.3), and $\sigma(\theta) := |e \cdot v(\theta)|/\rho$ is the normalized direction-cosine of the fan point $x(\theta) \in \mathrm{fan}_\rho(y) := \{x : d_g(x, y) = \rho\}$ carrying the normalized measure $d\theta$. The transport weight class is

$$\mathcal{W}(b) \ := \ \big\{\tilde{w} \in C^1([0,1]) : \ \tilde{w}(0) = 0, \ \|\tilde{w}'\|_{C^0([0,1])} \leq b\big\} \qquad (b > 0), \tag{158}$$

so that $|\tilde{w}(t)| \leq bt$, $\int_0^1 |\tilde{w}| \, dt \leq b/2$, and $\int_0^1 |\tilde{w}'| \, dt \leq b$; the physical weight $w(\cdot; x, y)$ lies in $\mathcal{W}(c_w)$ ((W), Part 1). The scalar *model integral* is

$$I_\Lambda[\tilde{w}](x, y; e) \ := \ \int_0^1 e^{i\Lambda\phi(t)} \, \tilde{w}(t) \, dt, \qquad I_\Lambda(x, y; e) := I_\Lambda[w(\cdot; x, y)](x, y; e). \tag{159}$$

Write $\bar{\kappa}_2 := c_{\rm ch}^2 \kappa_2$ for the fan-calibrated curvature constant of record (Convention S2-0, Part 1), so that (R3) (Lemma E2a.5) reads, for $x \in \mathrm{fan}_\rho(y)$ and all $t \in [0, 1]$,

$$\sup_t \big|\phi'(t) - e \cdot v\big| \leq \bar{\kappa}_2 \rho^2, \qquad \sup_t \big|\phi''(t)\big| \leq \bar{\kappa}_2 \rho^2, \qquad \sup_t \big|\phi'''(t)\big| \leq \bar{\kappa}_3 \rho^3, \tag{160}$$

$\bar{\kappa}_3 := c_{\rm ch}^3 \kappa_3$, all ball-uniform (a supremum over $U_\omega \times K_\Sigma$, uniform also in y, e, ρ, Λ; no smallness of κ_2 is claimed or used). Fact F2 (Lemma E2a.6) supplies the fan-shell measure bound

$$d\theta\big(\{\theta : \ \sigma(\theta) \leq s\}\big) \ \leq \ c_{\rm fan} \, s \qquad (\text{all } s > 0), \qquad c_{\rm fan} = c_{\rm dens} c_n c_{\rm ch}. \tag{161}$$

The fan splits, with no middle band, into the *oscillatory region* on which the phase derivative never vanishes and its stationary complement, the *cone*:

$$\Theta_{\rm osc}(y, \rho; e) := \{\sigma \geq 2\bar{\kappa}_2 \rho\}, \qquad \mathcal{C}(y, \rho; e) := \{\sigma < 2\bar{\kappa}_2 \rho\}, \qquad \Theta_{\rm osc} \sqcup \mathcal{C} = \mathrm{fan}_\rho(y). \tag{162}$$

The threshold $2\bar{\kappa}_2 \rho^2$ places the stationary set $S(x, e) := \{t : \phi'(t) = 0\}$ strictly inside $\mathcal{C}$ with margin factor 2: by (160), $S \neq \emptyset$ only if $|e \cdot v| \leq \bar{\kappa}_2 \rho^2$, i.e. $\sigma \leq \bar{\kappa}_2 \rho$.

Lemma E2a.19 (oscillatory branch; one integration by parts, multiplicity-free). *Let $g \in U_\omega$, $y \in K_\Sigma$, $0 < \rho \le \rho_0$, $|e| = 1$, $\Lambda > 0$, and $\theta \in \Theta_{\rm osc}$. Then for every $\tilde w \in \mathcal{W}(b)$:*

(i) (phase floor) $|\phi'(t)| \ge |e \cdot v|/2 > 0$ *on* $[0, 1]$*, and* $\bar\kappa_2\rho^2 \le |e \cdot v|/2$*;*

(ii) (exact decomposition) $I_\Lambda[\tilde w] = B_1[\tilde w] + R[\tilde w]$ *with*

$$B_1[\tilde w] = \frac{\tilde w(1)\, e^{i\Lambda\phi(1)}}{i\Lambda\, \phi'(1)}, \qquad R[\tilde w] = -\frac{1}{i\Lambda}\int_0^1 e^{i\Lambda\phi}\Big[\frac{\tilde w'}{\phi'} - \frac{\tilde w\, \phi''}{\phi'^2}\Big]dt; \tag{163}$$

(iii) (bounds) $|B_1[\tilde w]| \le \dfrac{2b}{\Lambda|e \cdot v|}$, $|R[\tilde w]| \le \dfrac{4b}{\Lambda|e \cdot v|}$*, hence* $|I_\Lambda[\tilde w]| \le \dfrac{6b}{\Lambda|e\cdot v|}$*;*

(iv) (trivial cap, all θ) $|I_\Lambda[\tilde w]| \le \int_0^1 |\tilde w|\, dt \le b/2$.

In particular for the transport weight, $|I_\Lambda| \le C_{\rm osc}(\Lambda|e \cdot v|)^{-1}$ on $\Theta_{\rm osc}$ with $C_{\rm osc} := 6c_w$ (ball-uniform, multiplicity-free).

Proof. On $\Theta_{\rm osc}$, $|e\cdot v| \ge 2\bar\kappa_2\rho^2$, so (160) gives $|\phi'| \ge |e\cdot v| - \bar\kappa_2\rho^2 \ge |e\cdot v|/2 > 0$: (i), and $S(x, e) = \emptyset$. As ϕ'' is continuous, $u := \tilde w/(i\Lambda\phi') \in C^1$; writing $e^{i\Lambda\phi} = (i\Lambda\phi')^{-1}\frac{d}{dt}e^{i\Lambda\phi}$ and integrating by parts, the $t = 0$ boundary term vanishes ($\tilde w(0) = 0$, the load-bearing weight hypothesis) and the $t = 1$ term is B_1, giving (ii). For (iii), $|B_1| \le |\tilde w(1)|/(\Lambda|\phi'(1)|) \le 2b/(\Lambda|e \cdot v|)$, and the two R-integrands are bounded pointwise using $|\phi'| \ge |e \cdot v|/2$ and $\sup|\phi''| \le \bar\kappa_2\rho^2 \le |e \cdot v|/2$ (from (i)): $\int |\tilde w'|/|\phi'| \le 2b/|e \cdot v|$ and $\int |\tilde w||\phi''|/\phi'^2 \le b\,\bar\kappa_2\rho^2 \cdot 4/|e \cdot v|^2 \le 2b/|e \cdot v|$, whence $|R| \le 4b/(\Lambda|e \cdot v|)$. This is the step that makes the estimate multiplicity-free: on $\Theta_{\rm osc}$ the second derivative is dominated by the first-derivative floor, so no sign-cell decomposition of ϕ'' is needed. (iv) is $|e^{i\Lambda\phi}| = 1$. □

Proposition E2a.20 (the fan-integrated oscillatory bound). *For all $\Lambda > 0$, $|e| = 1$, $0 < \rho \le \rho_0$, $y \in K_\Sigma$, $g \in U_\omega$,*

$$\int_{\Theta_{\rm osc}(y,\rho;e)} \big|I_\Lambda(x(\theta), y; e)\big|^2\, d\theta \;\le\; C'_{\rm osc}\,\Lambda^{-1}\rho^{-1}, \qquad C'_{\rm osc} := 15\, c_{\rm fan}\, c_w^2, \tag{164}$$

$C'_{\rm osc}$ ball-uniform, explicit, and carrying no *multiplicity input (no bound on the number of sign changes of ϕ'' is assumed). For a general weight $\tilde w \in \mathcal{W}(b)$ replace c_w by b.*

Proof. On $\Theta_{\rm osc}$, Lemma E2a.19(iii),(iv) give $|I_\Lambda| \le \min(c_w/2,\, 6c_w(\Lambda\rho\sigma)^{-1})$ since $|e \cdot v| = \rho\sigma$. Let $\sigma_* := 12(\Lambda\rho)^{-1}$ be the crossover (weight-independent: $6b/(b/2) = 12$), and split $\Theta_{\rm osc} = L \sqcup \bigcup_{k\ge0} H_k$ with $L = \{\sigma \le \sigma_*\}$, $H_k = \{2^k\sigma_* < \sigma \le 2^{k+1}\sigma_*\}$. By F2, $d\theta(L) \le c_{\rm fan}\sigma_*$ and $\int_L |I_\Lambda|^2 \le (c_w^2/4) \cdot 12c_{\rm fan}(\Lambda\rho)^{-1} = 3c_{\rm fan}c_w^2(\Lambda\rho)^{-1}$; on H_k, $|I_\Lambda|^2 \le 36c_w^2(\Lambda\rho)^{-2}(2^k\sigma_*)^{-2}$ and $d\theta(H_k) \le c_{\rm fan}2^{k+1}\sigma_*$, so $\int_{H_k} |I_\Lambda|^2 \le 72c_{\rm fan}c_w^2(\Lambda\rho)^{-2}\sigma_*^{-1}2^{-k}$ and $\sum_k \le 144c_{\rm fan}c_w^2(\Lambda\rho)^{-2}\sigma_*^{-1} = 12c_{\rm fan}c_w^2(\Lambda\rho)^{-1}$. Adding, $\int_{\Theta_{\rm osc}} |I_\Lambda|^2 \le 15c_{\rm fan}c_w^2\Lambda^{-1}\rho^{-1}$, uniform in (y, e, g, Λ, ρ). □

Remark E2a.21 (the boundary ledger B_1; killed twin). The $t = 1$ term B_1 of (163) is exported, not absorbed. By (R1) $q(1) = 0$, so $\gamma_{x,y}(1) = y$ and $e^{i\Lambda\phi(1)} = e^{i\Lambda e\cdot y}$ is *independent of* x: B_1 carries no Λ-scale x-oscillation and its θ-shell mass is (via the same F2 dyadic sum, restricted to the B_1-part of (164)) $\int_{\Theta_{\rm osc}} |B_1|^2\, d\theta \le 8c_{\rm fan}c_w^2\Lambda^{-1}\rho^{-1}$. The $t = 0$ boundary term vanishes identically by $\tilde w(0) = 0$; had it not, its phase $e^{i\Lambda e\cdot x}$ would oscillate at x-frequency Λ and destroy the decay — this is why $w(0) = 0$ is load-bearing. *Grade of §E2a.4:* THEOREM relative to the $S1$ interface $(R1), (R2), (R3), (R5)$ imported in Part 1; every constant is exhibited and $\bar\kappa_2$ enters $C'_{\rm osc}$ only through the *region* $\Theta_{\rm osc}$ (which it can only shrink), never through the constant, so $C'_{\rm osc} = 15c_{\rm fan}c_w^2$ is $\bar\kappa_2$-free.

E2a.5 The multiplicity device: a count-free sublevel measure

The oscillatory branch was multiplicity-free. The stationary cone $\mathcal{C}$ is not: there I_Λ can carry a caustic, and controlling its *fan-measure* (never a per-θ supremum, which is refuted by folds — §E2a.6) requires a device. We supply it here as a sublevel-measure bound of the phase, amplitude-free. The functional is

$$F(\theta) \;:=\; \inf_{t\in[0,1]} \max\Big(\frac{|\phi'(t)|}{\rho},\ \frac{|\phi''(t)|}{\bar\kappa_2\rho^2}\Big), \qquad S_\mu := \{\theta\in\Sigma_y : F(\theta)<\mu\} \quad (\mu>0), \tag{165}$$

where $\Sigma_y = \{|\theta|_{g(y)}=1\}$ is the fan sphere. Thus $\theta\in S_\mu$ iff some witness time t_0 has $|\phi'(t_0)|<\mu\rho$ *and* $|\phi''(t_0)|<\mu\bar\kappa_2\rho^2$ simultaneously — a near-degenerate (fold) stationary point. The max (both branches small at the *same* t) is irreducible: the $|\phi'|$-branch alone fails on curved charts (a fold gives $\inf_t|\phi'|=0$ on an $O(\rho)$-wide band), the $|\phi''|$-branch alone fails on flat charts ($\phi''\equiv 0$, branch set = whole sphere). F is continuous, so each S_μ is open and $d\theta$-measurable.

Two engine lemmas come from (R3),(R4) alone. Write the normalized velocity component $\Xi(t,\theta) := \phi'(t;x(\theta),y)/\rho = -e\cdot G_t(\theta)$, where $G_t:\Sigma_y\to\mathbb{R}^n$ is the velocity chart (R4, Lemma E2a.4), a ball-uniform C^1 diffeomorphism onto its image with $\|dG_t-\mathrm{Id}\|\le\frac12$, $\|(dG_t)^{-1}\|\le 2$, uniformly in t; then $\partial_t\Xi = \phi''/\rho$ and $|\partial_t\Xi|\le\bar\kappa_2\rho$.

Lemma E2a.22 (equatorial confinement)**.** *$S_\mu\subseteq\{\sigma\le\mu+\bar\kappa_2\rho\}\subseteq\{\sigma\le\mu+\kappa_b\}$, where $\kappa_b := \bar\kappa_2\rho_0\le\frac{7}{32}c_{\mathrm{ch}}$ is ball-uniform.*

Proof. If $\theta\in S_\mu$ with witness t_0 then $|\phi'(t_0)|<\mu\rho$, and (160) gives $|e\cdot v|\le|\phi'(t_0)|+\bar\kappa_2\rho^2<(\mu+\bar\kappa_2\rho)\rho$; divide by ρ and use $\rho\le\rho_0$. The bound $\bar\kappa_2\rho_0\le\frac{7}{32}c_{\mathrm{ch}}$ is the Part 1 radius budget. □

Lemma E2a.23 (per-time velocity sublevel)**.** *There is a ball-uniform $c_{\mathrm{vel}} = 2c_{\mathrm{dens}}c_n$ with: for every fixed $t\in[0,1]$ and every $s>0$, $d\theta(\{\theta : |\Xi(t,\theta)|\le s\})\le c_{\mathrm{vel}}\,s$, uniformly in t.*

Proof. Fix t. Since $\Xi(t,\cdot) = -e\cdot G_t$, the sublevel set is $G_t^{-1}(G_t(\Sigma_y)\cap\{w : |e\cdot w|\le s\})$: the F2 estimate with the chord chart replaced by G_t, whose tangential Jacobian is $\ge\|(dG_t)^{-1}\|^{-1}\ge\frac12$ (pushforward density $\le 2^{n-1}$, absorbed into c_{dens}) and whose round slab band has measure $\le c_n s$. The bound uses only $\|(dG_t)^{-1}\|\le 2$ of (R4) — the exact bent-chart-robust ingredient of F2 — and is *insensitive* to whether the t-slice contains a fold (G_t is a diffeomorphism for every t). This is why the device survives the refutation fold. □

Theorem E2a.24 (count-free sublevel bound; ball-uniform)**.** *With $c_{\mathrm{vel}} = 2c_{\mathrm{dens}}c_n$, $\kappa_3' := c_{\mathrm{ch}}^3\kappa_3$ (from (160), $\sup|\phi'''|\le\kappa_3'\rho^3$), and $\kappa_b = \frac{7}{32}c_{\mathrm{ch}}$, set the ball-uniform constant*

$$C_{\mathrm{sub}} \;:=\; (2+\kappa_b)\,c_{\mathrm{vel}}\,\max\big(1,\ \tfrac{1}{\sqrt2}(\kappa_3')^{1/2}\big). \tag{166}$$

Then for every $g\in U_\omega$, $y\in K_\Sigma$, $0<\rho\le\rho_0$, $|e|=1$, and all $\mu>0$,

$$d\theta(S_\mu) \;\le\; C_{\mathrm{sub}}\max\big(\mu,\ \rho\,\mu^{1/2}\big) \;=\; \begin{cases}(2+\kappa_b)c_{\mathrm{vel}}\,\mu, & \mu\ge\kappa_3'\rho^2/2,\\ \frac{2+\kappa_b}{\sqrt2}c_{\mathrm{vel}}(\kappa_3')^{1/2}\rho\,\mu^{1/2}, & \mu<\kappa_3'\rho^2/2.\end{cases} \tag{167}$$

This is a genuine $C^{1,1}$-sublevel bound with an exhibited ball-uniform constant, proved with no *bound on the number of folds and* no *ϕ'''-lower-floor.*

Proof. Fix (g,y,ρ,e); $t\mapsto\Xi(t,\theta)$ is C^2 with $\|\partial_t^2\Xi\|_{C^0}\le\kappa_3'\rho^2$ (160). Put $\delta := \min(1,(2\mu/(\kappa_3'\rho^2))^{1/2})\in(0,1]$. *Step 1: each $\theta\in S_\mu$ carries a t-window of length $\ge\delta$ on which $|\Xi|<(2+\kappa_b)\mu$.* For a witness t_0 ($|\Xi(t_0)|<\mu$, $|\partial_t\Xi(t_0)| = |\phi''(t_0)|/\rho<\bar\kappa_2\rho\,\mu$), Taylor gives for $|t-t_0|\le\delta$

$$|\Xi(t)-\Xi(t_0)|\le|\partial_t\Xi(t_0)|\delta+\tfrac12\kappa_3'\rho^2\delta^2\le\bar\kappa_2\rho\mu\cdot\delta+\tfrac12\kappa_3'\rho^2\delta^2\le\kappa_b\mu+\mu,$$

using $\bar\kappa_2\rho \le \kappa_b$ and $\delta \le 1$ for the first term and $\frac{1}{2}\kappa_3'\rho^2\delta^2 \le \mu$ (definition of δ) for the second. Hence $|\Xi(t)| < (2+\kappa_b)\mu$ on $J_\theta := [t_0-\delta, t_0+\delta] \cap [0,1]$, $|J_\theta| \ge \delta$. *Step 2: Fubini against Lemma E2a.23.* With $A := \{(t,\theta) : |\Xi(t,\theta)| < (2+\kappa_b)\mu\}$, Step 1 gives $\int_0^1 \mathbf{1}_A\,dt \ge \delta$ for $\theta \in S_\mu$, so by Tonelli

$$\delta\, d\theta(S_\mu) \le \int_{S_\mu}\int_0^1 \mathbf{1}_A\,dt\,d\theta \le \int_0^1 d\theta(\{|\Xi(t,\cdot)| < (2{+}\kappa_b)\mu\})\,dt \le (2+\kappa_b)c_{\mathrm{vel}}\mu,$$

whence $d\theta(S_\mu) \le (2+\kappa_b)c_{\mathrm{vel}}\mu/\delta$. *Step 3.* If $\mu \ge \kappa_3'\rho^2/2$ then $\delta = 1$; else $\delta = (2\mu/(\kappa_3'\rho^2))^{1/2}$ and $d\theta(S_\mu) \le \frac{2+\kappa_b}{\sqrt{2}}c_{\mathrm{vel}}(\kappa_3')^{1/2}\rho\,\mu^{1/2}$. Both are dominated by $C_{\mathrm{sub}}\max(\mu, \rho\mu^{1/2})$. Every constant is $c_{\mathrm{vel}}, \kappa_3', \rho_0$ — ball-uniform, independent of (y,e,Λ) and of the fold structure. $\square$

Remark E2a.25 (Claim (the LINEAR device bound) — PLAUSIBLE over the naive ball, with the exact obstruction). *Claim:* $d\theta(S_\mu) \le c_{\mathrm{sub}}\,\mu$ for a ball-uniform c_{sub} and all $\mu > 0$. **This is graded PLAUSIBLE, not a theorem, over the full H^s ball** (it is true and numerically robust, but the ball-uniform proof has the honest obstruction stated next); it is upgraded to a THEOREM over U_ω in Theorem E2a.41.

The obstruction, stated exactly. The linear bound needs, beyond the engine (R4)+(R3) of Theorem E2a.24, a ball-uniform bound N_0 on the number of *e-relevant fold branches* — the connected components of $\{(t,\theta) : \phi''(t;\theta) = 0,\ |\phi'(t;\theta)| \le 2\bar\kappa_2\rho^2\}$, equivalently the number of distinct near-zero critical values of $\phi'(\cdot;\theta)$ as θ ranges over the confinement band $\mathcal{B}$. *Given* N_0, each branch is θ-transversal (its critical value has $|\partial_\theta\Xi|$ bounded below by the diffeomorphism floor of (R4) on $\mathcal{B}$), so its $\{$crit. value $\le \mu\rho\}$-slice is $O(\mu)$ by Lemma E2a.23, and $d\theta(S_\mu) \le c_{\mathrm{vel}}N_0\mu$ with $c_{\mathrm{sub}} = c_{\mathrm{vel}}N_0$. *But* N_0 is not ball-uniformly bounded at finite C^k regularity: $\#\{\phi'' = 0\}$ obeys only the Rolle recursion $\le 1 + \#\{\phi''' = 0\}$, which needs a ϕ'''*-lower* floor to terminate, and no (R)-item supplies one — (R3) gives only the *upper* $|\phi'''| \le \kappa_3'\rho^3$. Consequently over the full H^s ball c_{sub} ball-uniform is **PLAUSIBLE**, not proved; only C_{sub} of the $\mu^{1/2}$ form (Theorem E2a.24) is unconditional. *The entire content of the analytic narrowing to U_ω is that* (N0-an) *(Lemma E2a.17, Part 1) supplies precisely this ball-uniform* N_0; this is executed in §E2a.8 (Theorem E2a.41).

Remark E2a.26 (what the device does and does not do). The device gates *only* the cone bound (D3): the oscillatory branch (§E2a.4) is multiplicity-free, so no per-θ multiplicity and no per-θ supremum on the cone is produced or used here. S_μ never approaches the e-poles $\{\sigma = 1\}$ (on the cone $\sigma < 2\bar\kappa_2\rho_0$); the pole margin is intrinsic to the diffeomorphism floor of Lemma E2a.23, which does not degrade with σ. *Grade: $\mu^{1/2}$ bound* THEOREM *relative to* $(R1), (R3), (R4), (R5)$; *linear bound* PLAUSIBLE *over the full ball with the exact N_0 obstruction above.*

E2a.6 The stationary cone: the fold-aware cell ledger and the $\bar\kappa_2$-cancellation

We now own the cone $\mathcal{C} = \{\sigma < 2\bar\kappa_2\rho\}$ (162) and prove the decisive fan-integrated bound (D3). The banned object is a per-θ supremum on the cone: it is *refuted by folds*. The (R1),(R3)-admissible fold $q = \bar\kappa_2\rho^2[(t-\frac{1}{2})^3/3 - (t-\frac{1}{2})/12]$ gives ϕ' an interior double zero, forcing $|I_\Lambda| \sim (\Lambda\bar\kappa_2\rho^2)^{-1/3} \gg (\Lambda\bar\kappa_2\rho^2)^{-1/2}$ (van der Corput III), with van der Corput II inapplicable ($\phi'' = 0$ at the stationary point). We therefore never bound I_Λ pointwise in θ: on a fold cell the caustic amplitude is admitted *only* inside the $d\theta$-integral, against its F2-small width. Write $K := \bar\kappa_2\rho^2$, $s_\flat := (\Lambda\rho)^{-1}$; the regime $X := \Lambda K \ge 1$ is where a caustic can form (for $X < 1$ the whole cone is deep core and the trivial cap already gives (D3), (177)).

We use the van der Corput lemma with C^1 amplitude (classical; e.g. Stein, *Harmonic Analysis*, VIII.1): for $\psi \in \mathcal{W}(b)$ and $\Phi = \Lambda\phi$, $|\int_J e^{i\Phi}\psi| \le A_k\Lambda_0^{-1/k}(2b)$ when $|\Phi^{(k)}| \ge \Lambda_0$ ($k \ge 2$, or $k = 1$ with Φ' monotone), with $A_2 \le 8$, $A_3 \le 16$; hence (II) $|\int_J e^{i\Lambda\phi}\psi| \le 16b(\Lambda\nu)^{-1/2}$ if $|\phi''| \ge \nu$ single-signed on J, and (III) $\le 32b(\Lambda\tau)^{-1/3}$ if $|\phi'''| \ge \tau$ — the latter valid *through* an interior double zero of ϕ'.

Definition E2a.27 (the three-piece partition of the cone; the device depth)**.** The classifying invariant is the device depth $m(\theta) := F(\theta)$ of (165) (on the cone $m \in [0, 3\bar{\kappa}_2\rho]$). Partition $\mathcal{C} = \mathrm{Cone}_{\mathrm{rim}} \sqcup \mathrm{Cone}_{\mathrm{core}} \sqcup \mathrm{Cone}_{\mathrm{bad}}$ with

$$\mathrm{Cone}_{\mathrm{rim}} = \{\tfrac{3}{2}\bar{\kappa}_2\rho \le \sigma < 2\bar{\kappa}_2\rho\}, \quad \mathrm{Cone}_{\mathrm{bad}} = \{\sigma \le s_\flat\}, \quad \mathrm{Cone}_{\mathrm{core}} = \{s_\flat < \sigma < \tfrac{3}{2}\bar{\kappa}_2\rho\},$$

pairwise-disjoint and $d\theta$-measurable. The core is split *inside the ledger* by m: the fold width $\mathrm{Cone}_{\mathrm{fold}} = \mathrm{Cone}_{\mathrm{core}} \cap \{m < \mu_\sharp\}$, $\mu_\sharp := (\Lambda K)^{-1/3}$, and the vdC-II shells $\mathrm{Cone}_{\mathrm{II},l} = \mathrm{Cone}_{\mathrm{core}} \cap \{2^{-l-1} < m/(3\bar{\kappa}_2\rho) \le 2^{-l}\}$, $0 \le l \le L := \lceil \frac{1}{3}\log_2(\Lambda K)\rceil$.

The mechanism linking m to the vdC-II floor is the *depth–floor identity*: at a stationary point t_j ($\phi'(t_j) = 0$), $\max(|\phi'|/\rho, |\phi''|/K) = |\phi''(t_j)|/K \ge m(\theta)$, so $|\phi''(t_j)| \ge m(\theta)K$. Pure geometry (R3) gives only upper ϕ'', ϕ''' bounds; the *measure* of each depth shell is supplied by the device (Theorem E2a.24) in the cone-scaled form

$$d\theta(\{m < \mu\}) \;\le\; c^\star_{\mathrm{sub}}\,(\bar{\kappa}_2\rho)\,\mu, \qquad c^\star_{\mathrm{sub}} := c_{\mathrm{sub}}/(\bar{\kappa}_2\rho) \text{ (or } C_{\mathrm{sub}}/(\bar{\kappa}_2\rho) \text{ count-free)}, \tag{168}$$

dimensionless and ball-uniform; this cone scale is forced (at $\mu \ge 1$ the sublevel set is the whole cone, of measure $O(c_{\mathrm{fan}}\bar{\kappa}_2\rho)$), not silently assumed.

Lemma E2a.28 (the per-cell L^2(fan) ledger)**.** *Assume* $(R1), (R3), (R4), (R5)$ + *F2 and the device sublevel bound* (168) *with scalar output*

$$D \;:=\; \begin{cases} 1 + N_0, & \textit{count route (over } U_\omega\textit{)}, \\ c_{\mathrm{sub}}^{1/2}, & \textit{sublevel route}, \end{cases} \qquad D \ge 1 \textit{ ball-uniform.} \tag{169}$$

Fix (y, ρ, e, g), $\Lambda \ge 1$, $X = \Lambda K \ge 1$, $w \in \mathcal{W}(b)$. *Then each cell of Definition E2a.27 obeys*

$$\int_{\mathrm{Cone}_{\mathrm{rim}}} |I_\Lambda|^2 d\theta \le 288\, c_{\mathrm{fan}} b^2\, (\Lambda K)^{-1}\Lambda^{-1}\rho^{-1} \le 288\, c_{\mathrm{fan}} b^2\, \Lambda^{-1}\rho^{-1}, \tag{170}$$

$$\sum_{l=0}^{L} \int_{\mathrm{Cone}_{\mathrm{II},l}} |I_\Lambda|^2 d\theta \le 2^{10}\, c^\star_{\mathrm{sub}} b^2 D^2\, (L{+}1)\, \Lambda^{-1}\rho^{-1}, \tag{171}$$

$$\int_{\mathrm{Cone}_{\mathrm{fold}}} |I_\Lambda|^2 d\theta \le 2^{11}\, c_{\mathrm{fold}}\, b^2\, \Lambda^{-1}\rho^{-1}, \tag{172}$$

$$\int_{\mathrm{Cone}_{\mathrm{bad}}} |I_\Lambda|^2 d\theta \le \tfrac{1}{4}\, c_{\mathrm{fan}} b^2\, \Lambda^{-1}\rho^{-1}, \tag{173}$$

with $c_{\mathrm{fold}} = c^\star_{\mathrm{sub}}$ *(sublevel route) or* $(1+N_0)^2\eta^{-2/3}$ *(count route,* η *the* ϕ'''*-floor fraction). Summing, with the shell log* $\ell_\Lambda := L + 1 \le 1 + \frac{1}{3}\log_2(\Lambda K)$ *absorbed by the open profile caps* p^-*,*

$$\int_{\mathcal{C}} |I_\Lambda|^2 d\theta \;\le\; C_{\mathrm{part}}\, \ell_\Lambda\, D^2\, \Lambda^{-1}\rho^{-1}, \qquad C_{\mathrm{part}} := 2^{12}\, c_w^2\, (c_{\mathrm{fan}} + c_{\mathrm{fold}}); \tag{174}$$

$\bar{\kappa}_2$ *cancels in every line.*

Proof. Throughout $|e \cdot v| = \rho\sigma$ and F2 gives $d\theta\{\sigma \le s\} \le c_{\mathrm{fan}} s$. *Rim.* $\sigma \ge \frac{3}{2}\bar{\kappa}_2\rho$ gives $|\phi'| \ge |e{\cdot}v| - K \ge \frac{1}{3}\rho\sigma > 0$, so one integration by parts (Lemma E2a.19) gives $|I_\Lambda| \le 18b/(\Lambda\rho\sigma) \le 12b/(\Lambda K)$; the rim band has measure $\le 2c_{\mathrm{fan}}\bar{\kappa}_2\rho$, so $\int_{\mathrm{rim}} |I_\Lambda|^2 \le (12b/(\Lambda K))^2 \cdot 2c_{\mathrm{fan}}\bar{\kappa}_2\rho = 288 c_{\mathrm{fan}} b^2 (\Lambda K)^{-1}\Lambda^{-1}\rho^{-1}$ (using $K^{-1}\bar{\kappa}_2\rho = \rho^{-1}$: $\bar{\kappa}_2$ cancels; $(\Lambda K)^{-1} \le 1$). *vdC-II shells.* On $\mathrm{Cone}_{\mathrm{II},l}$, $m_l/2 \le m \le m_l$, $m_l = 3\bar{\kappa}_2\rho\, 2^{-l}$; the depth–floor identity gives every stationary point $|\phi''(t_j)| \ge \frac{1}{2}m_l K =: \nu_l$, so vdC-II summed over the device-controlled cells gives $|I_\Lambda|^2 \le 2^9 D^2 b^2 (\Lambda m_l K)^{-1}$; (168) gives $d\theta(\mathrm{Cone}_{\mathrm{II},l}) \le c^\star_{\mathrm{sub}}\bar{\kappa}_2\rho\, m_l$, so the shell integral is l-independent, $= 2^9 c^\star_{\mathrm{sub}} D^2 b^2\, \bar{\kappa}_2\rho/(\Lambda K) = 2^9 c^\star_{\mathrm{sub}} D^2 b^2 \Lambda^{-1}\rho^{-1}$ (m_l and $\bar{\kappa}_2$ both cancel); summing $l = 0, \dots, L$ gives (171). *Fold.* On $\mathrm{Cone}_{\mathrm{fold}} = \{m < \mu_\sharp\}$: the sublevel route continues the m-layer-cake below $\mu_\sharp$ and prices the tail by the trivial cap on its sublevel-small measure $\le c^\star_{\rm sub}\bar\kappa_2\rho\,\mu_\sharp$, giving $\le (b/2)^2$ times it, i.e. $c^\star_{\rm sub}b^2\Lambda^{-1}\rho^{-1}$ (no vdC-III, no ϕ'''-floor); the count route uses vdC-III ($|\phi'''|\ge \eta K$ on each of $\le 1{+}N_0$ cells, amplitude $(\Lambda K)^{-1/3}$) admitted only under $\int d\theta$ against the width $\mu_\sharp$, trading $(\Lambda K)^{-1/3}(\Lambda K)^{-2/3}=(\Lambda K)^{-1}$, giving (172). *Bad.* The trivial cap $|I_\Lambda|\le b/2$ and $d\theta\{\sigma\le s_\flat\}\le c_{\rm fan}s_\flat$ give (173), no device input. Each line is $\le$ (const)$D^2\Lambda^{-1}\rho^{-1}$ with $\bar\kappa_2$ absent (structurally: measure $\propto \bar\kappa_2\rho\,(\Lambda K)^{-\alpha}$ times amplitude^2 $\propto (\Lambda K)^{-(1-\alpha)}=\bar\kappa_2\rho(\Lambda K)^{-1}=\Lambda^{-1}\rho^{-1}$), giving (174). □

Theorem E2a.29 (the fixed-fibre cone bound (D3); the $\bar\kappa_2$-cancellation)**.** *Let $g\in U_\omega$, $y\in K_\Sigma$, $0<\rho\le\rho_0$, $|e|=1$, $\Lambda\ge 1$. Under the cell ledger of Lemma E2a.28, Fact F2, and the device scalar D* (169)*,*

$$\boxed{\int_{\mathcal{C}(y,\rho;e)}\big|I_\Lambda(x(\theta),y;e)\big|^2\,d\theta\ \le\ C_2\,D^2\,\Lambda^{-1}\rho^{-1}}\tag{175}$$

with the ball-uniform, $\bar\kappa_2$-free constant of record

$$C_2\ :=\ \underbrace{16\,A_{\rm II}^2c_{\rm fan}c_w^2}_{\textit{vdC-II}}+\underbrace{6\,A_{\rm fold}^2c_{\rm fan}c_w^2c_\tau^{-2/3}}_{\textit{fold}}+\underbrace{\tfrac14 A_{\rm bad}c_w^2}_{\textit{bad}}+\underbrace{16\,c_{\rm fan}c_w^2}_{\textit{endpoint}},\tag{176}$$

depending only on $(U_\omega,K_\Sigma,n,c_w)$ through $c_{\rm fan},c_w,A_{\rm II},A_{\rm fold},A_{\rm bad},c_\tau$ (the per-cell amplitude/measure constants, all $\bar\kappa_2$-free), and independent of y,e,ρ,Λ and of $\bar\kappa_2$. The device factor D^2 is carried only *by the bad cell; the vdC-II, fold, and endpoint masses are D-free.*

Proof. The partition is $d\theta$-measurable and a.e. disjoint, so $\int_{\mathcal C}|I_\Lambda|^2=\sum_{\rm cells}$; bound the four cells by Lemma E2a.28, and the endpoint B_1-mass on the cone by the same F2 dyadic sum as Remark E2a.21 (restricted to the cone side), $\int_{\mathcal C}|B_1|^2\le 8c_{\rm fan}c_w^2\Lambda^{-1}\rho^{-1}$. Adding, with $D\ge 1$ so that the D-free terms are $\le(\cdot)D^2$, collects the four constituents into C_2. Every constituent cancelled $\bar\kappa_2$ internally and was uniform in (y,e,ρ,Λ), so C_2 carries no power of $\bar\kappa_2$ and is ball-uniform. □

Remark E2a.30 (the cancellation is an equality of scalings; the fold is subdominant). The dominant vdC-II cancellation is exact: the cone is *as small* ($d\theta(\mathcal C)\asymp\bar\kappa_2\rho$) as the stationary amplitude squared is large ($(\bar\kappa_2\rho^2)^{-1}$), both governed by the single scale $\bar\kappa_2$, so the product is $\bar\kappa_2^0\rho^{-1}$ — the flat-layer count $\Lambda^{-1}\rho^{-1}$. The caustic is real (it kills the per-θ supremum), but it occupies an Airy sliver of σ-width $\sim\Lambda^{-2/3}\bar\kappa_2^{1/3}$: its amplitude$^2\times$width $=(\Lambda\bar\kappa_2\rho^3)^{-1/3}\Lambda^{-1}\rho^{-1}$ comes in *below* the flat count by $(\Lambda\bar\kappa_2\rho^3)^{-1/3}\le 1$. Pricing the fold on the whole cone would over-count by a factor $(\Lambda\bar\kappa_2\rho^2)^{1/3}$; the honest accounting prices it on its thin sliver, where the mass survives with room to spare while the supremum does not survive at all. Consistent with the measured curved fan slope $-\frac12$ ($\|I_\Lambda\|_{L^2(\mathcal C)}\sim(\Lambda\rho)^{-1/2}$), which is chart-independent precisely because $\bar\kappa_2$ has cancelled on both sides of the partition.

Remark E2a.31 (the trivial small-X regime). For $X=\Lambda\bar\kappa_2\rho^2<1$ the whole-cone trivial cap already gives (D3): $\int_{\mathcal C}|I_\Lambda|^2\le(c_w/2)^2\,d\theta(\mathcal C)\le\frac12 c_{\rm fan}c_w^2\bar\kappa_2\rho<\frac12 c_{\rm fan}c_w^2\Lambda^{-1}\rho^{-1}$, so (175) holds for all $\Lambda\ge 1$ without an $X\ge 1$ hypothesis; the ledger machinery is needed only for $X\ge 1$, where the caustic can form.

$$X<1:\qquad \int_{\mathcal C}|I_\Lambda|^2\,d\theta\ \le\ \tfrac12\,c_{\rm fan}c_w^2\,\Lambda^{-1}\rho^{-1}.\tag{177}$$

Grade of §E2a.6: THEOREM relative to $(R1),(R3),(R4),(R5)+F2$ and the device export D — the identical relative grade the oscillatory branch carries against the $S1$ interface. The cancellation bookkeeping (Lemma E2a.28, Theorem E2a.29) is unconditional; the conditionality is exactly the device scalar D, whose linear value $D=(1+N_0)$ is PLAUSIBLE over the full ball and THEOREM over U_ω (§E2a.8). No per-θ supremum on the cone is stated, used, or implied.

E2a.7 The near-flat sliver: (D3) closes device-free at the binding cut

The device scalar D of (169) is bounded, over U_ω, by the count N_0 of (N0-an) — but only where a genuine fold is present. The count-free split (Cor. an-split, Lemma E2a.41, Part 1) proves the pure linear device bound $d\theta(S_\mu) \le c_{\rm sub}\mu$ for $\mu \ge \kappa_V := m_0/(\bar\kappa_2\rho^2) = O(1)$; the (D3) bad cell, however, is fed at the binding cut $\mu_\star := s_\flat = (\Lambda\rho)^{-1} \to 0$, so for $\Lambda > \Lambda_0(\rho) := 1/(\rho\kappa_V)$ one has $\mu_\star < \kappa_V$, and on the near-flat stratum $\mathcal{V}_{m_0}$ (Def. an-strata, Lemma E2a.13, Part 1: per-tile strip-sup $\sup_\tau |\phi''(\tau)| < m_0$) only the count-free $\mu^{1/2}$ remainder is available. This is the last load-bearing gap of the naive cascade. *A1 is the theorem that the gap is harmless*: over any $\mathcal{V}_{m_0}$ fibre, the (D3) cone mass closes at the target $\Lambda^{-1}\rho^{-1}$ with device output $D = 1$ — device-free, floor-free, count-free.

Three fences constrain the mechanism, each a booked overclaim home. *(No floor.)* $\mathcal{V}_{m_0}$ is *defined* by ϕ''-smallness; a lower ϕ''-floor on it would be circular and is never used — only the *upper* whole-cone bound and the first-derivative confinement enter. *(No device / no per-θ object.)* D is set to 1 and never consumed; the sole sublevel input is the count-free measure (Theorem E2a.24, itself device-free), used only to bound a fan-measure. *(The naive-cap trap.)* The seductive reading "cap the whole device-sublevel bad set by the trivial cap" is *false* on $\mathcal{V}_{m_0}$ and is exhibited and rejected below.

Lemma E2a.32 (whole-cone quadratic flatness; an UPPER bound, no floor)**.** *For every $g \in U_\omega$ and every fibre θ in the cone,*

$$\bar s := \sup_{t\in[0,1]} |\phi''(t;\theta)| \;\le\; M_{\phi''} = 4M_\Gamma c_{\rm ch}^2\rho^2 \;\le\; \mathcal{R}_0\, m_0, \qquad \mathcal{R}_0 := \frac{8M_\Gamma c_{\rm ch}^2}{c_0}\,\frac{\rho_0^2}{\rho_{\min}^2} = O(1), \tag{178}$$

and consequently $|\phi'(t;\theta) - e\cdot v| = |\int_0^t \phi''| \le M_{\phi''}$, so $\phi'(t) \in [\rho\sigma - M_{\phi''}, \rho\sigma + M_{\phi''}]$ for all t. This is an over-estimate, never a floor, and holds on the WHOLE cone. (Grade: THEOREM; it is Prop. an-strip's whole-cone bound, Lemma E2a.9 of Part 1, imported verbatim.)

Proof. $\sup_{[0,1]}|\phi''| \le \sup_{\Sigma_{r_t}} |\phi''| \le M_{\phi''}$ is Prop. an-strip restricted to the real axis; $M_{\phi''} \le \mathcal{R}_0 m_0$ is the Part 1 ratio bound. The ϕ'-window is the fundamental theorem of calculus, $\phi'(t) = \phi'(0) + \int_0^t \phi''$, absorbing the $O(\bar\kappa_2\rho^2) \le M_{\phi''}$ endpoint drift of (R3) into the stated window. □

Two regimes of a fibre follow, comparing $\rho\sigma$ to the flatness window $M_{\phi''} = O(\rho^2)$ (Λ-independent). If $\sigma > \sigma_\dagger := 2M_{\phi''}/\rho = 8M_\Gamma c_{\rm ch}^2\rho = O(\rho)$, then ϕ' is single-signed on $[0,1]$ (it cannot reach 0), the fibre is non-stationary, and IBP applies. If $\sigma \le \sigma_\dagger$, ϕ' may vanish; this is a thin equatorial band of fan-measure $\le c_{\rm fan}\sigma_\dagger = O(\rho)$, where the device count would have been spent.

Lemma E2a.33 (the whole device-sublevel bad set is NOT cap-priceable on $\mathcal{V}_{m_0}$)**.** *Let $\Theta_\star := \{\theta : \inf_t |\phi'(t;\theta)|/\rho < \mu_\star\}$ at the binding cut $\mu_\star = s_\flat$. On a $\mathcal{V}_{m_0}$ fibre family whose ϕ' has span $\Delta := \sup_\theta(\sup_t \phi' - \inf_t \phi')/\rho > 0$ (a fixed $O(m_0/\rho)$ constant, present whenever $\phi'' \not\equiv 0$),*

$$d\theta(\Theta_\star) \;\asymp\; c_{\rm fan}(\Delta + 2\mu_\star) \;\xrightarrow[\Lambda\to\infty]{}\; c_{\rm fan}\Delta = O(m_0/\rho) > 0, \tag{179}$$

so the trivial-cap pricing $(b/2)^2 d\theta(\Theta_\star)$ is a NON-DECAYING $O(m_0/\rho)$ constant and does not close (D3). The device's $\mu_\star$-measure bound $A_{\rm bad}D^2\mu_\star$ relies on D^2 being LARGE (it counts the $1 + N_0$ interior dips), which is exactly what is unavailable device-free.

Proof. For a fixed fibre shape, $\Xi(t,\theta) = \phi'(t;\theta)/\rho$ sweeps an interval of length $\ge \Delta$ in t; by the equatorial confinement (Lemma E2a.22), up to the transversal chart G_t (R4), $\theta \mapsto \Xi$ is a shift of that sweep-interval by $-\sigma$. Hence $\inf_t |\Xi(t,\theta)| < \mu_\star$ holds precisely on a σ-band of width $\Delta + 2\mu_\star$, of fan-measure $\asymp c_{\rm fan}(\Delta + 2\mu_\star)$ (F2, two-sided); as $\Lambda \to \infty$, $\mu_\star \to 0$ and the measure tends to $c_{\rm fan}\Delta > 0$. □

The cap $|I_\Lambda| \le b/2$ is an upper bound everywhere, but produces a *decaying* mass only where the set is $O(\mu_\star)$-thin. On the deep core $\{\sigma \le s_\flat\}$ the fan-measure IS $O(\mu_\star)$; on the annulus $\{s_\flat < \sigma \le \sigma_\dagger\}$ it is $O(\rho)$, and there one MUST use that I_Λ is genuinely small (the phase oscillates: $\sup_t |\phi'| \ge \rho\sigma - M_{\phi''} \gtrsim \rho\sigma > \Lambda^{-1}$), cashed not by a per-$\theta$ bound (which re-needs the count) but by the $\int d\theta$ fan-kernel of Lemma E2a.35.

Lemma E2a.34 (deep core: trivial cap $\times$ F2 measure)**.** *On* $\mathrm{Cone}_{\mathrm{bad}} = \{\sigma \le s_\flat\}$,

$$\int_{\mathrm{Cone}_{\mathrm{bad}}} |I_\Lambda|^2 \, d\theta \;\le\; \Big(\tfrac{b}{2}\Big)^2 c_{\mathrm{fan}} s_\flat \;=\; \tfrac{1}{4} c_{\mathrm{fan}} b^2 \, \Lambda^{-1} \rho^{-1}, \tag{180}$$

with no device, no floor, no count — exactly the bad cell of Lemma E2a.28 with $D = 1$.

Proof. $|I_\Lambda| \le \int_0^1 |w| \le b/2$ unconditionally (Lemma E2a.19(iv)); $d\theta\{\sigma \le s_\flat\} \le c_{\mathrm{fan}} s_\flat$ (F2). Multiply. $\square$

The annulus $\mathcal{A} = \{s_\flat < \sigma \le \sigma_\dagger\}$ is where ϕ' may vanish at interior times and the trivial cap is loose; its mass is $\Theta(\Lambda^{-1}\rho^{-1})$, not subdominant. A per-θ pointwise bound there would re-import the fold count (IBP re-imports $\mathrm{Var}(1/\phi') \sim 1 + N_0$); we instead bound its mass at the $\int d\theta$ level by the fan-kernel, using only θ-transversality (R4), which is fold-blind.

Lemma E2a.35 (fan-kernel off-diagonal decay; θ-non-stationary phase)**.** *For $t, s \in [0,1]$ set, over the near-equatorial band $\mathcal{E} := \{\sigma \le \sigma_\dagger\} \supseteq \mathcal{A}$,*

$$\mathcal{K}(t,s) \;:=\; \int_{\mathcal{E}} w(t)\, \overline{w(s)}\, e^{i\Lambda(\phi(t;\theta) - \phi(s;\theta))} \, d\theta. \tag{181}$$

There is a ball-uniform C_K with, for all t, s,

$$|\mathcal{K}(t,s)| \;\le\; b^2 \min\!\Big(c_{\mathrm{fan}}\sigma_\dagger,\; \frac{C_K}{\Lambda\rho\,|t-s|}\Big). \tag{182}$$

Grade: THEOREM *relative to* $(R1), (R3), (R4)$*; no fold count, no floor.*

Proof. The diagonal-cap branch is $|\mathcal{K}| \le b^2 d\theta(\mathcal{E}) \le b^2 c_{\mathrm{fan}} \sigma_\dagger$ (F2). For the decaying branch, since $\phi(t;\theta) - \phi(s;\theta) = \rho \int_s^t \Xi(u,\theta)\, du$ with $\Xi = -e \cdot G_u$, the θ-phase is $\Psi = \Lambda\rho\, \Phi_{ts}$, $\Phi_{ts}(\theta) = -e \cdot \int_s^t G_u(\theta)\, du$. Its directional derivative along $\hat{n} := e/|e|$ is

$$\partial_{\hat{n}} \Phi_{ts} = -\int_s^t e \cdot dG_u(\theta)[\hat{n}]\, du, \qquad e \cdot dG_u[\hat{n}] = |e| + e \cdot (dG_u - \mathrm{Id})[\hat{n}] \ge |e| - \tfrac{1}{2}|e| = \tfrac{1}{2}$$

for every u (using $\|dG_u - \mathrm{Id}\| \le \frac{1}{2}$, R4, $|e| = 1$; the integrand is single-signed and $\ge \frac{1}{2}$). Hence

$$|\partial_{\hat{n}} \Phi_{ts}(\theta)| \;\ge\; \tfrac{1}{2}\, |t-s| \qquad (\theta \in \mathcal{E}), \tag{183}$$

with NO dependence on the u-fold structure of ϕ'' (G_u is a diffeomorphism for every u; the bound uses only $\|dG_u - \mathrm{Id}\| \le \frac{1}{2}$), so $|\partial_{\hat{n}} \Psi| \ge \frac{1}{2}\Lambda\rho|t-s|$ on $\mathcal{E}$.

Single-signedness of $\partial_{\hat{n}}\Phi_{ts}$ on $\mathcal{E}$ (the ρ-smallness clause). To run van der Corput I as a genuine non-stationary (single-signed, no interior critical point) phase, we verify $\partial_{\hat{n}}\Phi_{ts}$ does not swing back through 0 across $\mathcal{E}$. The second derivative is controlled by the same diffeomorphism data: $\|\partial_\theta^2 \Phi_{ts}\| \le |t-s| \sup_u \|d^2 G_u\| \le C_2''|t-s|$ ($d^2 G_u$ ball-uniform, R4/S1b), so the total variation of $\partial_{\hat{n}}\Phi_{ts}$ across $\mathcal{E}$ — a set of θ-diameter $\le \sigma_\dagger = O(\rho)$ — is $\le C_2''\sigma_\dagger|t-s|$. Hence, provided the Λ-independent ρ-smallness condition

$$C_2''\,\sigma_\dagger \;<\; \tfrac{1}{2} \qquad \Big(\text{true for } \rho \le \rho_0 \text{ small, since } \sigma_\dagger = 8 M_\Gamma c_{\mathrm{ch}}^2 \rho = O(\rho),\; C_2'' \;\Lambda\text{-free ball-uniform}\Big) \tag{184}$$

holds, the variation never overcomes the floor $\frac{1}{2}|t-s|$ of (183): $\partial_{\hat n}\Phi_{ts}$ keeps its sign, i.e. Φ_{ts} is strictly monotone along $\hat n$ on $\mathcal{E}$, with no interior critical point — the honest hypothesis of vdC I. (The count is set by the diffeomorphism curvature C_2'', independent of the t-folds of ϕ''; the condition tightens the standing regime by at most a fixed shrink of ρ_0, and being Λ-independent it never competes with the $\Lambda\to\infty$ decay.)

The $(n-2)$-dimensional transverse fan integration ($n>2$). The band $\mathcal{E}\subseteq\Sigma_y$ is $(n-1)$-dimensional; the vdC I above is a 1-dimensional IBP along the $\hat n$-flow $\{\theta+\tau\hat n\}$. Foliate $\mathcal{E}=\bigsqcup_{\theta'\in\mathcal{E}^\perp}\ell_{\hat n}(\theta')$ by its $\hat n$-integral segments, $\mathcal{E}^\perp$ the $(n-2)$-dimensional transverse slice. On EACH fibre $\ell_{\hat n}(\theta')$ the floor (183) and the single-signedness (184) hold uniformly (both pointwise in θ), so vdC I gives $\le b^2C_K'/(\Lambda\rho|t-s|)$ ON EACH FIBRE, with C_K' uniform in θ'. Integrating over $\mathcal{E}^\perp$ contributes ONLY its bounded transverse fan-measure $|\mathcal{E}^\perp|\le c_{\mathrm{fan}}^\perp\sigma_\dagger^{n-2}=O(\rho^{n-2})$ (F2, θ-diameter $O(\rho)$ in every direction), NO extra power of Λ (the transverse directions carry no Λ-oscillation). Absorbing this bounded $O(\rho^{n-2})$ factor into C_K' (equivalently into c_{fan}, which already carries the n-dependent fan volume c_n) keeps a single ball-uniform C_K. For $n=2$ ($\mathcal{E}^\perp$ a point) it is the vdC I bound itself. Both give $|\mathcal{K}|\le b^2C_K'/(\Lambda\rho|t-s|)$; combining with the diagonal cap and setting C_K so the two branches match at the crossover yields (182). The only inputs are the $\Xi=-e\cdot G_u$ transversality (R4, fold-blind) and the (R1)/(R4) curvature data. ☐

Lemma E2a.36 (near-equatorial band mass; device-free, count-free)**.**

$$\int_{\mathcal{E}}|I_\Lambda(\theta)|^2\,d\theta\;=\;\int_0^1\!\!\int_0^1\mathcal{K}(t,s)\,dt\,ds\;\le\;C_{\mathcal{E}}\,b^2\,\Lambda^{-1}\rho^{-1}\,\ell_\Lambda,\qquad \ell_\Lambda:=1+\log_+(\Lambda\rho^2),\tag{185}$$

$C_{\mathcal{E}}:=2C_K+2c_{\mathrm{fan}}$ ball-uniform, the log absorbed by the open profile caps p^-. No device, no floor, no count.

Proof. $\int_{\mathcal{E}}|I_\Lambda|^2\,d\theta=\int_0^1\int_0^1\mathcal{K}(t,s)\,dt\,ds$ (Fubini, bounded integrand). Split the square at the crossover $\ell_0:=C_K/(\Lambda\rho c_{\mathrm{fan}}\sigma_\dagger)$: on $|t-s|\le\ell_0$ the cap $c_{\mathrm{fan}}\sigma_\dagger$ integrates to $\le 2C_K/(\Lambda\rho)$, on $|t-s|>\ell_0$ the tail $\int_{\ell_0}^1(C_K/\Lambda\rho)u^{-1}du=(C_K/\Lambda\rho)\log(1/\ell_0)$. Since $\sigma_\dagger=O(\rho)$, $\ell_0^{-1}=O(\Lambda\rho^2)$ and $\log(1/\ell_0)\le\ell_\Lambda$; summing gives (185). ☐

Theorem E2a.37 (A1; the near-flat sliver closes (D3) at the binding cut, $D=1$)**.** *Let $g\in U_\omega$, $y\in K_\Sigma$, $0<\rho\le\rho_0$, $|e|=1$, $\Lambda\ge1$, and let the cone fibre lie in the near-flat stratum $\mathcal{V}_{m_0}$. Then*

$$\int_{\mathcal{C}(y,\rho;e)}|I_\Lambda(\theta)|^2\,d\theta\;\le\;C_{A1}\,b^2\,\ell_\Lambda\,\Lambda^{-1}\rho^{-1},\qquad C_{A1}:=\tfrac14c_{\mathrm{fan}}+C_{\mathcal{E}}+C_{\mathrm{rim}},\tag{186}$$

$\ell_\Lambda=1+\log_+(\Lambda\rho^2)$, C_{A1} ball-uniform, exhibited, and independent of the device output D (here $D=1$), of any ϕ''-floor, and of the fold count N_0. *In the (D3) reading C_2D^2 of Theorem E2a.29 the sliver contributes $C_2^{\mathcal{V}}=C_{A1}b^2$ with $D=1$: the device factor is ABSENT on $\mathcal{V}_{m_0}$.*

Proof. Partition the cone $\{\sigma<2\bar\kappa_2\rho\}$ into $\{\sigma\le s_\flat\}$ (deep core) $\sqcup$ $\mathcal{A}=\{s_\flat<\sigma\le\sigma_\dagger\}\subseteq\mathcal{E}$ $\sqcup$ $\{\sigma_\dagger<\sigma<2\bar\kappa_2\rho\}$ (rim); this is well-defined since $\sigma_\dagger=2M_{\phi''}/\rho\le2\bar\kappa_2\rho$ (both $O(\rho^2)$; if $\sigma_\dagger\ge2\bar\kappa_2\rho$ the rim is empty and $\mathcal{E}$ is the whole cone, only improving the estimate). *Deep core:* Lemma E2a.34, $\le\frac14c_{\mathrm{fan}}b^2\Lambda^{-1}\rho^{-1}$. *Band $\mathcal{E}\supseteq\mathcal{A}$:* Lemma E2a.36, $\le C_{\mathcal{E}}b^2\Lambda^{-1}\rho^{-1}\ell_\Lambda$ (which already dominates the deep core; the explicit deep-core line is kept only to exhibit the cap term). *Rim:* on $\{\sigma>\sigma_\dagger\}$, Lemma E2a.32 gives $\inf_t|\phi'|\ge\rho\sigma-M_{\phi''}>\frac12\rho\sigma>0$, so ϕ' is monotone and one IBP gives $|I_\Lambda|\le18b/(\Lambda\rho\sigma)$; layer-caking against F2,

$$\int_{\mathrm{rim}}|I_\Lambda|^2d\theta\le\int_{\sigma_\dagger}^{2\bar\kappa_2\rho}\Big(\tfrac{18b}{\Lambda\rho\sigma}\Big)^2c_{\mathrm{fan}}\,d\sigma\le\frac{324b^2c_{\mathrm{fan}}}{\Lambda^2\rho^2\sigma_\dagger}=\frac{162b^2c_{\mathrm{fan}}}{\Lambda^2\rho\,M_{\phi''}}=:C_{\mathrm{rim}}'\Lambda^{-2}\rho^{-3},$$

with $C_{\mathrm{rim}}'=162c_{\mathrm{fan}}/(4M_\Gamma c_{\mathrm{ch}}^2)$ (via $\sigma_\dagger=2M_{\phi''}/\rho$, $M_{\phi''}=4M_\Gamma c_{\mathrm{ch}}^2\rho^2$), which is $\le C_{\mathrm{rim}}\Lambda^{-1}\rho^{-1}$ in the regime $\Lambda\rho^2\ge1$ (a factor $(\Lambda\rho^2)^{-1}\le1$ smaller — the rim is subdominant). Summing the three device-free lines gives (186); $D=1$ throughout, no floor, no count, no per-θ supremum crossed any interface (the fold amplitude was never invoked — the annulus by the fan-kernel, the rim by IBP, the deep core by the cap). $\square$

Remark E2a.38 (scope honesty: A1 does NOT dissolve the device on $\mathcal{N}_{m_0}$). Since Theorem E2a.37 uses only the whole-cone UPPER bound $\sup|\phi''| \le M_{\phi''}$ and R4, its bound (186) is in fact numerically valid on the whole cone, $\mathcal{N}_{m_0}$ included. This is CONSISTENT with, and does NOT overturn, the device machinery of §E2a.6: the object the device/count $D = 1+N_0$ bounds is the *priced ceiling* (per-cell amplitude$^2\times$measure, a legitimate immediately-squared intermediate that overshoots at $\Lambda^{+1/2}$ on the deep vdC-II shell), NOT the *honest* $\int |I_\Lambda|^2 d\theta$ (already bounded). A1's fan-kernel bounds the honest integral directly (a TT^* route that never passes through the priced ceiling), so it lands at target on any fibre. *We therefore do NOT claim A1 removes the need for the device on* $\mathcal{N}_{m_0}$: the auditable per-cell ledger (Theorem E2a.29) remains the reference closure there, and N_0 remains load-bearing for its bad-cell measure. A1's genuine and ONLY claim is the $\mathcal{V}_{m_0}$ gap, where the device measure $A_{\rm bad}D^2\mu_\star$ is unavailable below κ_V (Cor. an-split gives only $\mu^{1/2}$) and A1 supplies the missing closure device-free. Over-reading (186) as "the device is dispensable" would be exactly the kind of overclaim the honesty discipline polices.

Corollary E2a.39 (the sliver in the (D3) ledger; $\bar\kappa_2$-free closure over U_ω). *Write $\Sigma_y = \mathcal{B}_\mathcal{N} \sqcup \mathcal{B}_\mathcal{V}$ (Cor. an-split fibre stratum split, Lemma E2a.41), with $\mathcal{B}_\mathcal{V}$ the near-flat sliver $\{$fibre$\in \mathcal{V}_{m_0}\}$ and $\mathcal{B}_\mathcal{N}$ the fold stratum $\{$fibre$\in \mathcal{N}_{m_0}\}$. Summing the fixed-fibre (D3) over the two strata,*

$$\int_\mathcal{C} |I_\Lambda|^2 d\theta \;\le\; \underbrace{C_2(1+N_0)^2\Lambda^{-1}\rho^{-1}}_{\mathcal{B}_\mathcal{N},\ \textit{Thm E2a.29},\ D=1+N_0} \;+\; \underbrace{C_{A1}b^2\ell_\Lambda\Lambda^{-1}\rho^{-1}}_{\mathcal{B}_\mathcal{V},\ \textit{Thm E2a.37},\ D=1}\;, \tag{187}$$

so (D3) closes on the WHOLE cone at THEOREM grade over U_ω,

$$\boxed{\int_{\mathcal{C}(y,\rho;e)} |I_\Lambda(\theta)|^2\, d\theta \;\le\; C_2^{\rm tot}\,\ell_\Lambda\,\Lambda^{-1}\rho^{-1}, \qquad C_2^{\rm tot} := C_2(1+N_0)^2 + C_{A1}b^2} \tag{188}$$

$C_2^{\rm tot}$ ball-uniform and $\bar\kappa_2$-free (each summand cancels $\bar\kappa_2$: the $\mathcal{B}_\mathcal{N}$ term by Theorem E2a.29, the $\mathcal{B}_\mathcal{V}$ term because C_{A1} carries only $c_{\rm fan}, C_K, C_{\rm rim}$, all $\bar\kappa_2$-free). The device factor $(1+N_0)^2$ multiplies ONLY the fold-stratum term; on the sliver the device is ABSENT ($D=1$).

Proof. The $\mathcal{B}_\mathcal{N}$ line is Theorem E2a.29 restricted to the fold stratum, where N_0 is a THEOREM (Lemma E2a.17) so $D = 1+N_0$ is ball-uniform. The $\mathcal{B}_\mathcal{V}$ line is Theorem E2a.37. Adding gives (188); the common ℓ_Λ log (the shell-log on $\mathcal{B}_\mathcal{N}$, the kernel-log on $\mathcal{B}_\mathcal{V}$) is absorbed by p^- identically. $\square$

Remark E2a.40 (what A1 discharges; grade). With Corollary E2a.39, the fixed-fibre (D3), hence the cascade's (D3)/(D1)/T1$'$/fence, upgrade from "THEOREM on the fold stratum $\mathcal{B}_\mathcal{N}$" to "THEOREM over U_ω": the last load-bearing gap of the naive cascade (the $\mathcal{V}_{m_0}$ weigh-in at $\mu_\star < \kappa_V$) is closed. The binding cut $\mu_\star = (\Lambda\rho)^{-1} < \kappa_V$ is HARMLESS: on $\mathcal{V}_{m_0}$ the (D3) mass never sees the device, closing device-free by the flat-layer/fan-kernel route, which is Λ^{-1} (not the $\Lambda^{-1/2}$ of the count-free $\mu^{1/2}$ remainder — so that remainder is dominated, not binding). *STATUS of §E2a.7:* THEOREM relative to $(R1), (R3), (R4), (R5) + F2$ and Part 1's Prop. an-strip (the whole-cone $\sup|\phi''| \le M_{\phi''} = O(\rho^2)$ bound), Def. an-strata, and the degenerate $\{\phi'' \equiv 0\}$ split. Every constant is ball-uniform over U_ω with $(\rho_a, \varepsilon_a, g_0)$ explicit. Fences honored: no device / no per-θ object / no zero count on $\mathcal{V}_{m_0}$; no ϕ''-floor (only the upper $M_{\phi''}$); the naive whole-bad-set cap exhibited and rejected (Lemma E2a.33). The single-signedness clause (184) and the $(n-2)$-transverse integration of Lemma E2a.35 are the run-completion of the fan-kernel; both are Λ-independent geometric facts, so they never compete with the decay.

E2a.8 The linear device over U_ω, and (D3) end-to-end

We now discharge Claim E2a.25. Its proof structure named exactly where the count enters (§E2a.5): *given* a ball-uniform N_0, each fold branch is θ-transversal and its sublevel slice is $O(\mu)$, so $d\theta(S_\mu) \le c_{\mathrm{vel}} N_0 \mu$; the sole missing input is N_0, and (N0-an) (Lemma E2a.17) supplies it over U_ω. Nothing else in the device proof changes.

Theorem E2a.41 (device sublevel bound over U_ω; scoped to Cor. an-split)**.** *Assume* (N0-an) *with constant N_0. Set $c_{\mathrm{sub}} := c_{\mathrm{vel}}(1+N_0)$ ($c_{\mathrm{vel}} = 2c_{\mathrm{dens}}c_n$), let C_{sub} be the count-free constant of Theorem E2a.24, and $\kappa_V := m_0/(\bar\kappa_2\rho^2) \le c_0/(2\bar\kappa_2) = O(1)$. Then for every $g \in U_\omega$, $y \in K_\Sigma$, $0 < \rho \le \rho_0$, $|e| = 1$,*

$$d\theta(S_\mu) \le c_{\mathrm{sub}}\,\mu \quad (\mu \ge \kappa_V), \qquad d\theta(S_\mu) \le c_{\mathrm{sub}}\,\mu + C_{\mathrm{sub}}\,\rho\,\mu^{1/2} \quad (0 < \mu < \kappa_V), \tag{189}$$

c_{sub} ball-uniform over U_ω, carrying the (ρ_a, ε_a) dependence solely through $N_0 = N_0(g_0, \rho_a, \varepsilon_a, \dots)$. The pure linear bound $d\theta(S_\mu) \le c_{\mathrm{sub}}\mu$ holds for $\mu \ge \kappa_V$ on all of Σ_y and for all *$\mu > 0$ on the fold stratum $\mathcal{B}_{\mathcal{N}}$; the sole degradation is the count-free $\mu^{1/2}$ remainder confined to $\mu < \kappa_V$ on the near-flat sliver $\mathcal{B}_{\mathcal{V}}$ (where no zero count is taken).* ***No unqualified all-μ linear claim is made:*** *the pure linear form over all of Σ_y for $\mu < \kappa_V$ is the sliver weigh-in supplied by A1 (Theorem E2a.37), not by this theorem.* Grade: THEOREM (both lines) relative to $(R1), (R3), (R4), (R5)$ and (N0-an).

Proof. This is Cor. an-split (Lemma E2a.41) with the same constants; we reproduce the fold-stratum mechanism, which is what the pure linear line rests on. Partition by the FIBRE stratum, $\mathcal{B} = \mathcal{B}_{\mathcal{N}} \sqcup \mathcal{B}_{\mathcal{V}}$ (per-tile strip-sup $\ge m_0$ vs. $< m_0$); the degenerate $\{\phi'' \equiv 0\}$ lies in $\mathcal{B}_{\mathcal{V}}$ and is disposed of count-free (Part 1). *Part A ($\mathcal{B}_{\mathcal{N}}$).* Here $\phi'' \not\equiv 0$ and the per-tile floor is met, so (N0-an) applies: $S_\mu \cap \mathcal{B}_{\mathcal{N}}$ is covered by $\le 1 + N_0$ e-relevant fold branches Γ_k, on each of which the critical value $c_k(\theta) = \Xi(t_k(\theta), \theta)$ is a single-valued C^1 function (implicit function theorem, off the measure-zero merge locus), θ-transversal with $|\partial_\theta c_k| = |\partial_\theta \Xi|$ bounded below by the (R4) diffeomorphism floor. Hence $\{\theta \in \Gamma_k : |c_k| < \mu\}$ is $O(\mu)$-thin, of measure $\le c_{\mathrm{vel}}\mu$ (Lemma E2a.23); any $\theta \in S_\mu \cap \mathcal{B}_{\mathcal{N}}$ has its witness within $O(\mu)$ of a critical time (Taylor, as in Theorem E2a.24 Step 1), hence on one branch with $|c_k| < (1+\kappa_b)\mu$, so summing over $\le 1 + N_0$ branches (absorbing $1 + \kappa_b$ into c_{vel}) gives $d\theta(S_\mu \cap \mathcal{B}_{\mathcal{N}}) \le (1+N_0)c_{\mathrm{vel}}\mu = c_{\mathrm{sub}}\mu$ for all $\mu > 0$. *Part B ($\mathcal{B}_{\mathcal{V}}$).* Here $\sup_{[0,1]} |\phi''| < m_0$: no genuine fold, and the count is not invoked. Two facts (Cor. an-split(b1)–(b2)), not their unqualified union: (b1) the count-free $d\theta(S_\mu \cap \mathcal{B}_{\mathcal{V}}) \le C_{\mathrm{sub}} \max(\mu, \rho\mu^{1/2})$ for all $\mu > 0$ (Theorem E2a.24); (b2) for $\mu \ge \kappa_V$ the ϕ''-branch is inactive so $S_\mu \cap \mathcal{B}_{\mathcal{V}}$ is the pure velocity sublevel, measure $\le c_{\mathrm{vel}}\mu \le c_{\mathrm{sub}}\mu$. For $\mu < \kappa_V$ only (b1)'s $C_{\mathrm{sub}}\rho\mu^{1/2}$ is proved; the pure linear form there is *not* claimed. *Assembly.* For $\mu \ge \kappa_V$, Part A + Part B(b2) give $d\theta(S_\mu) \le c_{\mathrm{sub}}\mu$; for $0 < \mu < \kappa_V$, Part A gives $c_{\mathrm{sub}}\mu$ on $\mathcal{B}_{\mathcal{N}}$ and Part B(b1) gives $C_{\mathrm{sub}}\rho\mu^{1/2}$ on $\mathcal{B}_{\mathcal{V}}$ — exactly (189). Every constant is ball-uniform over U_ω. The residual pure-linear closure over $\mathcal{B}_{\mathcal{V}}$ for $\mu < \kappa_V$ is supplied device-free by A1, not by this argument. □

Remark E2a.42 (the only new input is (N0-an); the topology it needs). Part A is verbatim the device proof of §E2a.5 with N_0 now finite and ball-uniform, applied ONLY on the fold stratum. The count N_0 is the sole object drawn from U_ω rather than U; every other ingredient (the velocity floor $\|(dG_t)^{-1}\| \le 2$, F2, the Taylor drift) is an H^s-ball fact holding a fortiori on $U_\omega \subset U$ (Prop. an-subset, Part 1). (N0-an) delivers N_0 with its own collar shrinkage; this theorem inherits that shrinkage and needs NO compactness of a fixed-collar sup-ball. The sliver $\mathcal{B}_{\mathcal{V}}$ is handled count-free in Part B; the count is never spent there.

Proposition E2a.43 ((D3) over U_ω: THEOREM end-to-end)**.** *Feed the linear device measure of Theorem E2a.41 into the cell ledger in the cone-scaled form* (168)*, $d\theta\{m < \mu\} \le c^\star_{\mathrm{sub}}(\bar\kappa_2\rho)\mu$*

with $c^\star_{\rm sub} = c_{\rm sub}/(\bar\kappa_2\rho)$ ball-uniform, on the fold stratum $\mathcal{B}_\mathcal{N}$; combine with the device-free sliver closure (Theorem E2a.37) on $\mathcal{B}_\mathcal{V}$. Then, for every $g \in U_\omega$, $y \in K_\Sigma$, $0 < \rho \le \rho_0$, $|e| = 1$, $\Lambda \ge 1$,

$$\int_{\mathcal{C}(y,\rho;e)} |I_\Lambda(x(\theta), y; e)|^2 \, d\theta \;\le\; C_2^{\rm tot}\, \ell_\Lambda\, \Lambda^{-1}\rho^{-1} \tag{190}$$

with $C_2^{\rm tot} = C_2(1 + N_0)^2 + C_{A1}b^2$ ball-uniform over U_ω, $\bar\kappa_2$ ABSENT, and $\ell_\Lambda = 1 + \log_+(\Lambda\rho^2)$ absorbed by the open profile caps p^-. ***Grade: THEOREM end-to-end over U_ω.*** *The "given (A1)" conditional of the fold-stratum-only closure is discharged because A1 (Theorem E2a.37) IS a theorem; the linear device that the fold stratum needs is a THEOREM over U_ω (Theorem E2a.41) — the one node that is only PLAUSIBLE over the full H^s ball (Claim E2a.25).*

Proof. On $\mathcal{B}_\mathcal{N}$ this is Theorem E2a.29 with the device scalar $D = (c_{\rm vel}(1 + N_0))^{1/2}$ instantiated from Theorem E2a.41 (S-route); every device cell of the ledger lands at Λ-exponent 0 with $\bar\kappa_2$ cancelled (the deep vdC-II shell and the route-S fold tail cancel $\bar\kappa_2$ identically; the bad-set cap carries $\bar\kappa_2\rho \le \kappa_b = \frac{7}{32}c_{\rm ch} = O(1)$, hence bounded on the target column), so the ledger sums to $C_2D^2\Lambda^{-1}\rho^{-1} = C_2(1 + N_0)^2\Lambda^{-1}\rho^{-1}$ up to the absorbed constant $c_{\rm vel}$. On $\mathcal{B}_\mathcal{V}$ it is Theorem E2a.37, $\le C_{A1}b^2\ell_\Lambda\Lambda^{-1}\rho^{-1}$ with $D = 1$. Adding is Corollary E2a.39, giving (190). The only change from the naive terminus is that $c_{\rm sub}$ is now a THEOREM over U_ω rather than PLAUSIBLE; the cancellation arithmetic is unchanged. □

Remark E2a.44 (this is the closing, not a re-pricing; grade summary of Part 2). The naive terminus was: the $\mu^{1/2}$ THEOREM form overshoots the cut by $\Lambda^{+1/2}$, and only the linear ($\beta = 1$) form closes, which needs N_0. Theorem E2a.41 delivers the $\beta = 1$ form at THEOREM grade over U_ω on the fold stratum; A1 closes the near-flat sliver device-free. This is NOT a new device and NOT a re-pricing of the count-free device: it is the SAME linear device the terminus named as the unique closer, with its one missing input (a ball-uniform N_0) supplied by (N0-an) over the analytic class, plus the device-free A1 sliver. *Grade ledger of this part.* The oscillatory branch (§E2a.4), the fixed-fibre cone bound (§E2a.6), the count-free device (§E2a.5), and A1 (§E2a.7) are each THEOREM at their stated relative grades. The one PLAUSIBLE node — the linear device bound over the full ball (Claim E2a.25) — is upgraded to THEOREM over U_ω by (N0-an) (Theorem E2a.41); no grade is promoted, and the analytic narrowing is the sole content that lifts it. Consequently (D3) is a THEOREM end-to-end over U_ω (Proposition E2a.43) — the export consumed by the $S4/\mathrm{T1}'$-Main assembly of Part 3.

E2a.9 The analytic slice register: from the Schur assembly to $T1'$-Main over U_ω

This part discharges the booking line elected as Option A in the fence paragraph following Hyp. E.2 — the promotion $\partial_y : (c, p) \mapsto (c + \frac{1}{2}, p - 1)$ over the analytic ball U_ω, and with it the slice register $s > 11$ (T2) / $s > \frac{23}{2}$ (T3). Three things are established, at the grades the companion dependency tree supports and no higher:

- **A.6** the **Schur assembly** (S4a's per-Λ fan bound (D1) and frequency-transfer matrix (D4); S4b's discrete Schur summation to the extraction (D5) and the Leibniz profile booking) delivering *$T1'$-Main over U_ω at the consumed register* $(c+\frac{1}{2}, p-1)$, the *exact* statement the T2/T3 ladders read (Theorem E2a.51) — transcribed from the corrected cascade post-A2, with A1 a theorem so the composite is **THEOREM end-to-end over** U_ω;
- **A.6′** the **$S1$-transfer / collar bookkeeping** (§E2a.11): which $S1$ exports restrict to U_ω verbatim, which the analytic collar *improves*, and the one exponential-map object that must be *re-proved* with an explicit collar shrinkage $\rho_a \to \rho'_a$;

A.6″ the **calibration record** (Remark E2a.60): a numerics-witness note, honest that the reproducibility scripts *confirm* the exhibited ball-uniform constants and *do not* prove the bounds.

Remark E2a.45 (what this part proves, and what it does not). The object proved here is the *analytic slice register over* U_ω: the booking $(c+\frac{1}{2}, p-1)$ and the fence $s > 11$, unconditional over U_ω once the Part-1/Part-2 imports (the ball-uniform fold count N_0, Lemma E2a.17; the near-flat sliver Theorem E2a.37; (D3), Proposition E2a.43) are in hand. It is *not* a proof of Hyp. E.2: clauses (i)–(iii) (the singular-part dual-Lipschitz bound, the mixed-jet finite-part family $\{C_W, W_*\}$, and the Sanders QEI-constant uniformity) are the paper's *named input* (E2a-ω), consumed by Thm. E.7 and Prop. E.4; nothing in this chain derives the Hadamard parametrix or the Sanders constants over a metric family. Those clauses are collected, at their honest grades, in the closing modulo-list of this appendix; this part cites them by name only, never at theorem grade. All scope is the free massive ($m > 0$) minimally-coupled scalar of (106); the jets are *mixed* ($|a|+|b| \le 1$ with the mixed pair $|a| = |b| = 1$ only); the class U_ω is H^s-dense with empty H^s-interior, so every uniformity below is a supremum over the ball or a pair-Lipschitz modulus.

E2a.10 A.6 The Schur assembly: (D1), the frequency-transfer matrix (D4), and the extraction (D5)

The $T1'$ rung (assembled over the H^s ball U and, by the restriction of §E2a.11, over U_ω with the same constants) reduces the ∂_y half-derivative booking to a single anisotropic estimate on the oscillatory integral $I_\Lambda(x(\theta), y; e)$ over the geodesic fan $\mathrm{fan}_\rho(y)$. The assembly consumes two fan-integrated square bounds and produces the register map. We recall the two inputs at their consumed strength, then state the assembly.

Standing data and the two consumed square bounds. Throughout: $g \in U_\omega$, $y \in K_\Sigma$, $0 < \rho \le \rho_0$, $|e| = 1$, $\Lambda \ge 1$; $x = x(\theta) \in \mathrm{fan}_\rho(y)$, chord $v = y - x$, and $\sigma(\theta) := |e \cdot v|/\rho$. "Ball-uniform" means a supremum over $U_\omega \times K_\Sigma$, independent of (y, e, ρ, Λ, h). The fan splits disjointly, $\mathrm{fan}_\rho = \Theta_{\mathrm{osc}} \sqcup \mathcal{C}$ with $\Theta_{\mathrm{osc}} = \{\sigma \ge 2\bar{\kappa}_2\rho\}$ and $\mathcal{C}$ its complement, where $\bar{\kappa}_2 = c_{\mathrm{ch}}^2\kappa_2$. Two bounds enter (each a Λ-oscillatory / stationary mass of order $(\Lambda\rho)^{-1}$):

$$\text{(D2)} \quad \int_{\Theta_{\mathrm{osc}}} |I_\Lambda|^2\, d\theta \;\le\; 15\, c_{\mathrm{fan}}\, c_w^2\, (\Lambda\rho)^{-1} \qquad (S2;\ Z\text{-free}), \tag{191}$$

$$\text{(D3)} \quad \int_{\mathcal{C}} |I_\Lambda|^2\, d\theta \;\le\; C_2\, D^2\, \Lambda^{-1}\rho^{-1}, \quad D = c_{\mathrm{sub}}^{1/2} = \big(c_{\mathrm{vel}}(1+N_0)\big)^{1/2} \quad \text{(Prop. E2a.43)}. \tag{192}$$

Here (192) is the sole place the analytic gain enters the assembly: the device scalar D carries the ball-uniform fold count N_0 (Lemma E2a.17) through $c_{\mathrm{sub}} = c_{\mathrm{vel}}(1+N_0)$, and C_2 is ball-uniform with $\bar{\kappa}_2$ *absent*. Over the naive H^s ball D is not ball-uniform (the ripple family drives $N_0 \to \infty$); over U_ω it is (the entire raison d'être of the analytic class). Everything else in this subsection is the $S4$ bookkeeping, unchanged by the class.

Proposition E2a.46 ((D1): the per-Λ fan bound over U_ω; $S4a$)**.** *For all $\Lambda \ge 1$, $|e| = 1$, $0 < \rho \le \rho_0$, $y \in K_\Sigma$ and $g \in U_\omega$,*

$$\big\|I_\Lambda(\cdot, y; e)\big\|_{L^2(\mathrm{fan}_\rho, d\theta)} \;\le\; C_{\mathrm{fan}}\, (\Lambda\rho)^{-1/2}, \qquad C_{\mathrm{fan}} = \sqrt{15\, c_{\mathrm{fan}}}\, c_w + \sqrt{C_2}\, D, \tag{193}$$

C_{fan} ball-uniform over U_ω, independent of (y, e, ρ, Λ, h). This is an L^2-over-the-fan statement; it is not *upgraded, here or downstream, to a pointwise-in-x envelope or a pointwise-in-θ supremum.*

Proof. Minkowski on the disjoint union $\mathrm{fan}_\rho = \Theta_{\mathrm{osc}} \sqcup \mathcal{C}$: $\|I_\Lambda\|_{L^2(\mathrm{fan}_\rho)} \le (\int_{\Theta_{\mathrm{osc}}} |I_\Lambda|^2)^{1/2} + (\int_{\mathcal{C}} |I_\Lambda|^2)^{1/2}$; insert (191), (192), both $(\Lambda\rho)^{-1}$-masses. The flat sliver $\mathcal{F} := \{\sigma \le (\Lambda\rho)^{-1}\}$ is *not* a third region: for $X := \Lambda\bar{\kappa}_2\rho^2 \geq \frac{1}{2}$ one has $\mathcal{F} \subseteq \mathcal{C}$ and $\mathcal{F}$ lies inside the $S3b$ bad-cell ledger at the same threshold $\mu_* = (\Lambda\rho)^{-1}$, so its trivial-cap mass is already counted inside (192); nothing is double-counted. The boundary term B_1 enters (191) jointly with the interior remainder (it is $I_\Lambda = B_1 + R$ that (D2) bounds), so no B_1-only mass is added at this step. Collecting the two square roots gives (193). $\square$

The extraction to the half-derivative register needs one more object: the frequency-transfer matrix $T_{M,\Lambda}$, measuring how much x-frequency-M mass the hard leg $\partial_y C_\Lambda$ (an Λ-oscillatory object) deposits after the output projection P^x_M. The point — the $S4a$ STEP-2 fence — is that a pointwise-in-x statement must *not* be smuggled back: the transfer exponent β is *proved* from the fan geometry, at its honest worst-case value, not asserted at the heuristic's strength.

Theorem E2a.47 ((D4): the frequency-transfer matrix, $\beta = \frac{1}{2}$ proved; $S4a$)**.** *Let $g \in U_\omega$, $y \in K_\Sigma$, $0 < \rho \leq \rho_0$, $|e| = 1$, $\Lambda \geq 1$, and $T_{M,\Lambda} := \|P^x_M\, \partial_y C_\Lambda(\cdot, y)\|_{L^2_x(\text{near-diag})}$. Then*

$$T_{M,\Lambda} \;\leq\; C_{\text{Schur}}\, \Lambda^{1/2-a}\, \varepsilon_\Lambda\, \omega(M/\Lambda), \qquad \omega(u) = \begin{cases} u^{1/2}, & 0 < u \leq c_{\text{geo}}, \\ 0, & u > c_{\text{geo}}, \end{cases} \tag{194}$$

with C_{Schur} ball-uniform over U_ω (independent of M, Λ, y, e, h), the transfer exponent $\beta = \frac{1}{2}$, and $\varepsilon_\Lambda = \Lambda^a\|h_\Lambda\|_{L^2}$ the H^a frequency envelope ($\sum_\Lambda \varepsilon_\Lambda^2 \leq \|h\|^2_{H^a}$).

Proof (the affine-synthesis exponent count; $\beta = \frac{1}{2}$ is the boundary order of vanishing, not a heuristic). Work on the planar worst section (the phase sees γ only through its $\operatorname{span}(e, v)$ projection), coordinatize by the chord cosine $X := e \cdot v$, and read $\partial_y C_\Lambda$ as a function of X; P^x_M is frequency-M projection in the variable conjugate to X. Because $\partial_x\gamma = (1-t)\mathrm{Id} + O(\kappa_2|v|)$, the t-slice feeds output x-frequency $\approx \Lambda(1-t)$, so P^x_M selects $1 - t \approx M/\Lambda$. Writing the hard leg $\partial_y C_\Lambda(X) = i\Lambda A_\Lambda e^{i\Lambda(e\cdot y)} \int_0^1 (e^\top \partial_y\gamma)\, w\, e^{-i\Lambda[(1-t)X-b]}\, dt$ with $A_\Lambda = \Lambda^{-a}\varepsilon_\Lambda\|\chi\|^{-1}_{L^2}$ and bend $b = e \cdot q = O(\kappa_2\rho^2)$, its magnitude is the *unit-amplitude affine synthesis* of the profile $F(t) := J(t)\, w(t)$, $J(t) := |e^\top \partial_y\gamma(t)| = t + O(\kappa_2\rho)$: in the flat-bend limit the X-transform is $|\widehat{\partial_y C_\Lambda}(k)| = 2\pi A_\Lambda\, F(1 - k/\Lambda)\mathbf{1}_{[0,\Lambda]}(k)$ (the amplitude Λ of ∇h_Λ cancels the Λ^{-1} of the change of variables $t \mapsto k = \Lambda(1-t)$). By Plancherel the dyadic-M shell mass is then $A_\Lambda\Lambda^{1/2}(\int_{u_0}^{2u_0} |F(1-u)|^2 du)^{1/2}$ with $u_0 = M/\Lambda$; if F vanishes to order $\beta - \frac{1}{2}$ at $t = 1$ this is $\asymp A_\Lambda\Lambda^{1/2}u_0^\beta$, so $\beta =$ (order of vanishing of $J \cdot w$ at $t = 1$) $+ \frac{1}{2}$. For the class-worst weight $w(1) \neq 0$ (only $w(0) = 0$ is imposed) and the geometric factor $J(1) = 1$ *exactly* (differentiate $\gamma_{x,y}(1) = y$ in y: $\partial_y\gamma(1) = \mathrm{Id}$, whence $\partial_y q(1) = 0$ and $J(1) = |e^\top| = 1$), the order is 0, so $\beta = \frac{1}{2}$ — the honest worst case (a weight forced to $w(1) = 0$ would only *improve* β to $\frac{3}{2}$). The exponent survives (i) the finite fan $|X| \leq X_{\max} = c_{\text{ch}}\rho$ (a Λ-independent Dirichlet smearing, factor $1 + O((MX_{\max})^{-1})$) and (ii) the nonzero bend $b = O(\kappa_2\rho^2)$: crucially $b(1, X) = 0$ ($q(1) = 0$), so the bend does not shift the $t = 1$ endpoint value $F(1)$ that *sets* β — the exponent is a boundary datum, invariant under the interior bend. The support cut $\omega(u) = 0$ for $u > c_{\text{geo}}$ is the killed $t \sim 0$ pile-up: at $t = 0$ both $J(t) \to 0$ and $w(t) \to w(0) = 0$, so $F = O(t^2)$ kills the shell mass as $u \to 1$ ($c_{\text{geo}} \geq 1$ the safe ball-uniform cutoff, the $O(\kappa_2\rho)$ tail of J pushing it just past 1). Assembling with $A_\Lambda = \Lambda^{-a}\varepsilon_\Lambda\|\chi\|^{-1}_{L^2}$ gives (194) with $C_{\text{Schur}} = C_0\, c_{\text{geo}} c_w\, c_{\text{dens}}^{1/2} \|\chi\|^{-1}_{L^2}$ ball-uniform; the ξ-direction integral is moved outside L^2_x by Minkowski using (D1)'s e-uniformity. No per-θ quantity is used — only the fan-integrated (D1) and the L^2_X synthesis mass. $\square$

Remark E2a.48 (the fence held: $\beta = \frac{1}{2}$ is proved, not assumed). The transfer exponent is the single place the naive rung could smuggle a pointwise envelope back in. It does not: $\beta = \frac{1}{2}$ is the order of vanishing of $J \cdot w$ at the $t = 1$ boundary plus $\frac{1}{2}$, an exponent-counting identity on the affine synthesis (Theorem E2a.47), machine-confirmed and curvature-insensitive (Remark E2a.60); the refuted pointwise-in-x envelope and per-θ cone supremum appear nowhere and are not implied. The load-bearing weight condition is $w(0) = 0$: it kills the $t \sim 0$ pile-up (the $\omega = 0$ cutoff) and the $t = 0$ boundary twin. The value $w(1)$ is left *free*, which is exactly why the honest worst-case β is $\frac{1}{2}$ and not larger.

S4b: **the discrete Schur summation and the extraction (D5).** The hard leg $\partial_y C = \sum_\Lambda \partial_y C_\Lambda$ has its $H_x^{a-1/2}$ norm controlled by the frequency-transfer matrix through $\|\partial_y C\|^2_{H_x^{a-1/2}} \le \sum_M M^{2(a-1/2)}(\sum_\Lambda T_{M,\Lambda})^2$. The exponent identity $M^{a-1/2}\,\omega(M/\Lambda)\,\Lambda^{1/2-a} = (M/\Lambda)^{a-1/2+\beta}$, which at $\beta = \frac{1}{2}$ is exactly $(M/\Lambda)^a$, turns the summand into a dyadic Schur kernel $K(M,\Lambda) = (M/\Lambda)^\gamma \mathbf{1}[M/\Lambda \le c_{\text{geo}}]$, $\gamma := a - \frac{1}{2} + \beta$, one-sided in the octave difference. The hypothesis floor is $a > 2$ (every consuming locus has $a > 3$), so $\gamma > \frac{3}{2} > 0$ with room.

Theorem E2a.49 ((D5): the extraction, with a finite Schur constant c_a; *S4b*)**.** *With* $\gamma = a - \frac{1}{2} + \beta > 0$,

$$\|\partial_y C(\cdot, y)\|^2_{H_x^{a-1/2}} \;\le\; C^2_{\text{Schur}}\, c_a\, \|h\|^2_{H^a}, \qquad c_a := \Big(\frac{c_{\text{geo}}^\gamma}{1 - 2^{-\gamma}}\Big)^2, \tag{195}$$

ball-uniform over U_ω, independent of y and (linearly) of h; at the target $\beta = \frac{1}{2}$, $c_a = (c^a_{\text{geo}}/(1 - 2^{-a}))^2$.

Proof. Cauchy–Schwarz in the ε-weight gives $(\sum_\Lambda K\varepsilon_\Lambda)^2 \le (\sum_\Lambda K)(\sum_\Lambda K\varepsilon^2_\Lambda)$. The row sum $R_M = \sum_\Lambda K(M,\Lambda)$ is uniformly bounded — on dyadic $\Lambda \ge M/c_{\text{geo}}$, $K = (M/\Lambda)^\gamma \le c^\gamma_{\text{geo}}$ and $\sum_{i\ge 0} 2^{-\gamma i} = (1 - 2^{-\gamma})^{-1}$, so $R_M \le c^\gamma_{\text{geo}}(1 - 2^{-\gamma})^{-1} = c_a^{1/2} =: R$; the column sum $S_\Lambda = \sum_M K(M,\Lambda)$ is bounded by the same R (the kernel is one-sided, $M \le c_{\text{geo}}\Lambda$). Summing over M and exchanging (Tonelli, all terms ≥ 0) yields the double sum $\le C^2_{\text{Schur}} R^2 \sum_\Lambda \varepsilon^2_\Lambda \le C^2_{\text{Schur}} c_a \|h\|^2_{H^a}$ by $\sum_\Lambda \varepsilon^2_\Lambda \le \|h\|^2_{H^a}$. The geometric tails converge precisely because $\gamma > 0$; K is *summable*, not merely bounded, so no $(\Lambda\rho)$-log is generated (the $\omega = 0$ support cut of Theorem E2a.47 makes K one-sided). $\square$

The extraction is a bound in the slab-transverse x-norm at fixed ρ-shell content; it reconciles with the slab register through the near-diagonal disintegration $\|F\|^2_{L^2_x(\text{near-diag})} = \int_0^{\rho_0} \rho^{n-1}\|F\|^2_{L^2(\text{fan}_\rho)}\, d\rho$ ($n = 4$: $\rho^3\, d\rho$). The fan currency's ρ^{-1} (from (D1)) is integrable against $\rho^3\, d\rho$, and the profile Leibniz rule books the two legs at the binding column, giving the consumed register:

Proposition E2a.50 (the profile column: the Leibniz booking $(c + \frac{1}{2}, p - 1)$; *S4b*)**.** *Let* Π *be a profile factor of* 4*d height* $p \ge 2$*. Then* $\partial_y[C\,\Pi] = (\partial_y C)\Pi + C(\partial_y\Pi)$ *books at the binding column* $(c + \frac{1}{2}, p - 1)$ *in the slab near-diagonal, uniformly in* y*: the leg* $(\partial_y C)\Pi$ *books* $(c + \frac{1}{2}, p)$ *(Theorem E2a.49 gives* $\partial_y C \in H_x^{a-1/2}$*, i.e. column* $c \mapsto c + \frac{1}{2}$*; multiplication by* Π *preserves the profile height* p *by the pair calculus), and* $C(\partial_y\Pi)$ *books* $(c, p-1)$ *(the direct profile computation:* $\partial_y\Pi$ *lowers the height by one, leaving the coefficient column* c *untouched). The Leibniz sum sits at the smaller profile column* $(p-1)$ *and the shared coefficient column* $c+\frac{1}{2}$*, with* $y \mapsto \partial_y[C\Pi](\cdot, y)$ *continuous into* $H^{\min(s-c-1/2,(p-1)^-)}$.

The boundary branch B_1 of the t-integration by parts carries an *x-independent* phase $e^{i\Lambda(e\cdot y)}$ (frozen because $\gamma_{x,y}(1) = y$ identically, $q(1) = 0$), so it deposits only in the low-M rows and, after the ρ-integral, contributes a finite ball-uniform additive amount, absorbed into c_a; the $t = 0$ boundary twin vanished under $w(0) = 0$. Collecting the interior hard leg, the boundary branch, and the soft legs:

Theorem E2a.51 ($T1'$-Main over U_ω at $(c + \frac{1}{2}, p - 1)$ — the statement the ladders consume)**.** *Let* $g \in U_\omega(g_0, \rho_a, \varepsilon_a)$ (Def. E2a.1)*, and let* h *be a polynomial in* $(g, \ldots, \partial^j g)$ *with* $h \in H^a$*,* $a = s - j > 2$*, and transport weights* $w \in \mathcal{W}(c_w)$ $(\tilde w(0) = 0)$*. Then the* ∂_y *booking of the transport-averaged coefficient obeys, in the slab near-diagonal, uniformly in* y *and linearly in* h*,*

$$\big\|\, \partial_y\, C(\cdot, y)\, \big\|_{H^{a-1/2}(x\text{-}slab,\ near\text{-}diag)} \;\le\; K_{T1}\, \|h\|_{H^a}, \tag{196}$$

i.e. the register books

$$\boxed{\partial_y : (c,p) \longmapsto \big(c+\tfrac{1}{2},\ p-1\big)} \qquad \textit{on transport-averaged terms} \tag{197}$$

(T1′-b), with the anisotropic LP envelope (T1′-LP) supplied by (D1), Proposition E2a.46. The constant

$$K_{T1} \;=\; C_{\mathrm{Schur}}\, c_a^{1/2} \;+\; (S1c \textit{ soft-leg constants}), \qquad c_a \textit{ built from } C_{\mathrm{fan}}^2, \tag{198}$$

is ball-uniform over U_ω and independent of y and of h (linear). For the Lipschitz-in-g form (T1′-c), both g and the comparison g' range over U_ω (both real-analytic; fence F-diff*): $\|\partial_y[(C_g - C_{g'})\Pi]\|_{(c+1/2,p-1)} \le K'_{T1}\, C_{\mathrm{poly}}\, \|g-g'\|_{H^s}$, with K'_{T1} ball-uniform over $U_\omega \times U_\omega$.*

Proof. The interior hard leg gives $C_{\mathrm{Schur}} c_a^{1/2} \|h\|_{H^a}$ (Theorem E2a.49); the soft legs add the $S1c$ soft-leg constant at register $\ge s-3 \ge a-\frac{1}{2}$ (no oscillatory input); the boundary rows add a ball-uniform amount folded into c_a; the profile Leibniz booking is Proposition E2a.50. Sum by the triangle inequality; y-uniformity and y-continuity hold because every constant is y-independent and the same estimates apply to the y-increment (the geometric data $(\gamma_{x,y}, \partial_y\gamma, w)$ are jointly C^1 in y). Ball-uniformity over U_ω is inherited from Theorem E2a.49 and (D3), Proposition E2a.43. The difference form is the same bound applied to $h_g - h_{g'}$ (linearity, $\|h_g - h_{g'}\|_{H^a} \le C_{\mathrm{poly}} \|g-g'\|_{H^s}$) with the γ/w-legs from $S1c$ at rate $\|g-g'\|_{H^s}$; the quantifier ranges over analytic g' only (Remark E2a.53). □

Remark E2a.52 (the grade of Theorem E2a.51, read from the files)*.* The two feeder families carry *relative* grades, and the composite grade is their honest conjunction:

- The $S4$ assembly (Propositions E2a.46, E2a.50, Theorems E2a.47, E2a.49) is **THEOREM relative to the $S1/S2/S3b$ interface** — the identical relative grade the $S2$ carries against $S1$; its Schur/extraction bookkeeping is unconditional and machine-verified, and $\beta = \frac{1}{2}$ is *proved* (Theorem E2a.47), not asserted.

- The interface it consumes, (D3) of Proposition E2a.43, is **THEOREM on the fold stratum $\mathcal{B}_{\mathcal{N}}$, and THEOREM end-to-end over U_ω GIVEN (A1)** (the near-flat $\mathcal{B}_{\mathcal{V}}$ sliver at the binding cut $\mu_* = (\Lambda\rho)^{-1} < \kappa_V$). Since (A1) is itself a theorem here — the sliver closes (D3) device-free, Theorem E2a.37 — the "GIVEN (A1)" conditional is *discharged.*

Consequently Theorem E2a.51 is **THEOREM end-to-end over** U_ω, with all constants ball-uniform in $(g_0, \rho_a, \varepsilon_a)$: the booking $(c+\frac{1}{2}, p-1)$ holds unconditionally over the analytic ball. (Absent a proof of the sliver it would be only PLAUSIBLE end-to-end; the appendix carries it at theorem grade precisely because Part 2 proves the sliver.) This is the established end-to-end status — the analytic chain walked N_0-analytic → cascade (post-A2) → A1 theorem → (D3) → the $(c+\frac{1}{2}, p-1)$ booking → slice $s > 11$, THEOREM end-to-end over U_ω.

The constant chain (where (ρ_a, ε_a) enters). The dependence on the analytic parameters enters *once*, at the head of the chain, through the fold count, and propagates monotonically:

$$\underbrace{N_0(g_0,\rho_a,\varepsilon_a)}_{\text{Lem. E2a.17}} \longrightarrow c_{\mathrm{sub}} = c_{\mathrm{vel}}(1{+}N_0) \longrightarrow D = c_{\mathrm{sub}}^{1/2} \longrightarrow \underbrace{C_2 D^2}_{\text{(D3)}} \longrightarrow C_{\mathrm{fan}} \longrightarrow c_a \sim C_{\mathrm{fan}}^2 \longrightarrow \underbrace{K_{T1} = C_{\mathrm{Schur}} c_a^{1/2}}_{\text{Thm. E2a.5}} \tag{199}$$

Every arrow is an established $S3/S4$ step. No other constant in the chain sees the collar: they are all H^s-ball facts holding over $U_\omega \subseteq U$ with the U-constants (Lemma E2a.1). Shrinking ρ_a or tightening ε_a can change K_{T1} only through N_0's own dependence, so $K_{T1} = K_{T1}(g_0, \rho_a, \varepsilon_a, K_\Sigma, j, c_w)$.

Remark E2a.53 (the difference form: both metrics analytic (fence F-diff)). The Lipschitz-in-g form of Theorem E2a.51 requires *both* g *and* the comparison g' to lie in U_ω, i.e. both real-analytic. The right-hand norm is still $\|g - g'\|_{H^s}$ (the difference of two analytic metrics measured in H^s; legitimate since $U_\omega \subset H^s$), but the *quantifier* ranges over analytic g' only. A merely-smooth (non-analytic) comparison g' is *not* served — the disclosed honest cost of the analytic route, flagged at every consumer that feeds a difference estimate (this is fence F-diff of Hyp. E.2, and Guard F-shock forbids ever wiring g' to the non-analytic Barrabès–Israel shock metrics).

Corollary E2a.54 (the revived slice fence over U_ω)**.** *For $g, g' \in U_\omega$, the T2 slice ladder fires at the promoted threshold*

$$\boxed{s > 11} \quad \textit{(slice register, T2)}; \qquad s > \tfrac{23}{2} \quad \textit{(T3)}, \tag{200}$$

in place of the naive-fallback slice $s > 11.5$ / locked-4d $s > 12$. The value $s > 11$ is the unique zero-slack inequality of the chain: the commutator branch at $L^\infty_t H^{\min(s-19/2,\,7/2^-)}(\Sigma_t)$ is $C^0(\Sigma)$ iff $s - \frac{19}{2} > \frac{3}{2}$ iff $s > 11 = n/2 + 9$; it is an upper *bound for the fence (method-relative), not sharp-from-below. The $\frac{1}{2}$-order gain over the naive row is exactly the promoted booking $(c + \frac{1}{2})$ vs $(c + 1)$ of* (197)*.*

Proof. Inherits Theorem E2a.51 (THEOREM end-to-end over U_ω, Remark E2a.52): the fence value is the arithmetic of the promoted $(c + \frac{1}{2})$ booking. The purely arithmetic Sobolev embedding/multiplication/trace uses that fix the value hold for $g, g' \in U_\omega \subseteq U$ verbatim; only the hypothesis *class* narrows. □

Remark E2a.55 (where the analytic hypothesis must appear in the ladders). Because Theorem E2a.51 holds only over U_ω and only for analytic comparison g', every T2/T3 hypothesis line that quantified over the H^s ball $\mathcal{U}$ must be read over U_ω: the $(T1)$ import line, the $(N\text{-}a)$ normal-form difference estimate (whose comparison g' must be declared analytic), the calculus line $\partial_y : (c, p) \mapsto (c + \frac{1}{2}, p - 1)$ "[$(T1)$, only this]" (citation swap only, booking value unchanged), the $(T1'\text{-LP})$ envelope locus, and the T3 standing index. The register arithmetic, the zero-slack site, and the values $s > 11$ / $s > \frac{23}{2}$ are identical under the narrowing; only which metrics are admitted changes. **(E-env), (D0), and the R3/R4 conditionality are untouched** — they are U-facts inherited on $U_\omega \subset U$, orthogonal to this slice chain (they gate the data-side ladder, not the register here).

E2a.11 A.6′ The $S1$ transfer and the collar bookkeeping

The Schur assembly of A.6 was stated over U_ω "with the same constants" as the $S1$ rung. This subsection justifies that phrase: it inventories the $S1$ corpus (proved over the H^s ball U) against the analytic ball U_ω, and books the one object that does *not* transfer by restriction. The inventory is trichotomous.

(T-R) Restriction: the $S1$ bulk transfers verbatim. Every $S1$ statement has the form $\forall g \in U : P(g)$ with P a closed predicate, and each named constant is a supremum $\sup_{g \in U} Q(g)$ of a g-continuous $Q \geq 0$. Because $U_\omega \subseteq U$ (the embedding Lemma E2a.1: a holomorphic extension bounded by ε_a on the collar has, by Cauchy estimates, $\|g - g_0\|_{H^s(K)} \leq C_{\mathrm{em}} \varepsilon_a$, so $U_\omega \subseteq U$ for ε_a small), restriction to the subset is pure quantifier logic: $\forall g \in U \Rightarrow \forall g \in U_\omega$, and $\sup_{U_\omega} Q \leq \sup_U Q$.

Proposition E2a.56 ($(T\text{-}R)$: the $S1$ exports restrict to U_ω with the same constants)**.** *Assume $\varepsilon_a \leq \varepsilon_0 / C_{\mathrm{em}}$ so $U_\omega \subseteq U$ (Lem. E2a.1). Then every lemma, proposition, and exported constant of the $S1a/S1b/S1c$ rung holds verbatim with U replaced by U_ω, the constants unchanged (a fortiori bounded by their U-values). In particular the exports the Schur assembly consumes — $(R1)$ chord/velocity expansion ($\kappa_2, c_{\mathrm{geo}} \leq \frac{39}{32}$), $(R2)$ exp-map comparability c_E, $(R3)$ phase* *bounds* (κ_3, $|\phi''| \le \kappa_2|v|^2$), *(R4) velocity chart, (R5)/F2 fan measure* $c_{\rm fan} = c_{\rm dens}c_n c_{\rm ch}$, *(R6) hard-leg paraproduct and S1c-soft — transfer with* no re-proof. (*Caveat: these transfer as* real-domain *statements; the holomorphy of the exp map is the separate (T-X) re-proof below, and S1c-diff's comparison g' must also lie in U_ω, Remark E2a.53.*)

Proof. Each cited statement is $\forall g \in U : P(g)$ with P closed and each constant $\sup_{g\in U} Q(g)$, $Q \ge 0$; the base points, radii, directions, and weights are quantified over $K_\Sigma, (0,\rho_0], S^{n-1}, \mathcal{W}(c_w)$, none of which changes. Quantifier restriction to $U_\omega \subseteq U$ gives the verbatim transfer with $\sup_{U_\omega} Q \le \sup_U Q$. No proof is re-run. □

(T-C) Improvement: one collar datum controls every C^k. Restriction gives the *same* constants, hence the *same* verdicts — it does not by itself close (D3). The analytic gain is a genuinely new object: Cauchy's inequality turns the single collar sup-bound $M_a := \|g_0\|_{\sup,\rho_a} + \varepsilon_a$ into a control of *every* derivative, $\sup_K |D^\alpha g| \le |\alpha|!\,\rho_a^{-|\alpha|} M_a$, valid for all α with *no* Sobolev window. Consequently every $S1$ constant becomes a closed-form function of the *two* analytic parameters (ρ_a, ε_a) (through M_a) and the setup floor $d_0 = \inf_U \min_K |\det g| > 0$: no Sobolev index s, no per-order embedding constant, appears. Two things genuinely improve — the top-Sobolev composition caveat of $(R2)$ becomes an *unconditional* theorem on U_ω (Cauchy bounds all indices at once), and $S1c$-soft's booked register extends from $\ge s-3$ to *any* finite order. One thing does *not*: the numeric values of the load-bearing device constants κ_2, κ_3 are *not* asserted smaller (bent analytic charts have $\Gamma \ne 0$; the improvement is qualitative — all orders, no window, unconditional — not a re-pricing of the count-free device, which is forbidden).

Lemma E2a.57 ((*T-C*): the all-order phase majorant — the object H^s cannot supply)**.** *For $g \in U_\omega(\rho_a, \varepsilon_a)$ there are ball-uniform constants $\tau_0 = \tau_0(\rho_a, M_a, d_0) > 0$ and $C_\phi = C_\phi(\rho_a, M_a, d_0)$, independent of (g, x, y, e), such that the phase $\phi(t) = e \cdot \gamma_{x,y}(t)$ extends holomorphically to the strip $S_{\tau_0} = \{\Re t \in [0,1],\ |\Im t| \le \tau_0\}$ with $|\phi^{(k)}(t)| \le C_\phi\, k!\, \tau_0^{-k}$ for every $k \ge 0$ (a geometric majorant, no Sobolev cap). On the non-degenerate stratum $\phi'' \not\equiv 0$, ϕ'' is the restriction of a nonzero holomorphic function, and a per-tile Jensen count over $\lceil 1/\tau_0 \rceil$-many concentric-disc tiles caps $\#\{t \in [0,1] : \phi''(t) = 0\}$ by a* finite ball-uniform *bound — exactly the input Lemma E2a.17 promotes to the fold count N_0.*

The majorant is the exact contrapositive of ripple exclusion: membership $g \in U_\omega$ *is* the finiteness of the strip sup $\sup_{S_{\tau_0}} |\phi''| < \infty$, and the refuting ripple $b_N = a_N \sin(N z_1)$ (which is H^s-pinned, $a_N N^s = \frac{1}{2}\varepsilon_0$, but collar-infinite, $\|b_N\|_{\sup,\rho_a} \ge a_N \frac{1}{2} e^{N\rho_a} \to \infty$) is excluded precisely because for it this sup is $+\infty$. The same collar bound $M_a < \infty$ both removes the ripple and bounds the count; this is the entire mechanism by which analyticity supplies N_0, and everything else in the transfer is platform.

(T-X) Re-proof: complexifying the exp map, with collar shrinkage. The one $S1a$ object that does not transfer by restriction is the geodesic flow / exponential map *as a holomorphic object*: the phase strip of Lemma E2a.57(i) needs $\gamma_{x,y}(t)$ holomorphic in t, whereas restriction gives only the real ($C^{\ge 5}$) map. Holomorphy is a new construction — the complex-domain IVP — and it costs a *collar shrinkage*, because the complex-time flow must keep the (now complex) geodesic image inside the collar on which the Christoffel field $\Gamma[g]$ is holomorphic.

Theorem E2a.58 ((*T-X*): the complexified exp map with explicit shrinkage $\rho_a \to \rho_a'$)**.** *Let $g \in U_\omega(\rho_a, \varepsilon_a)$. Define the shrunken analytic radius*

$$\rho_a' := \frac{\rho_a}{16\,(1 + K_\Gamma^{\mathbb{C}})}, \qquad \tau_0 := \rho_a', \tag{201}$$

$K_\Gamma^{\mathbb{C}} = \sup_{U_\omega} \sup_{K_{\mathbb{C}}''(\rho_a/2)} (|\Gamma| + |\partial\Gamma|) < \infty$ *ball-uniform* (Γ *is rational in the holomorphic* g_{ij} *with denominator* $\det g \ge d_0/2$ *on the half-collar, Cauchy-bounded*)*. Then for complex data* (z, p) *with*

$\Re z \in K'$, $|\Im z| \le \rho'_a$, $|p| \le \rho'_a$, the complex-time geodesic IVP has a unique holomorphic solution $\gamma(\cdot; z, p)$ on the strip S_{τ_0} with image in $K''_{\mathbb{C}}(\rho_a/2)$; it restricts on the reals to the $S1a$ objects (identity theorem), all $S1a$-2 brackets hold on the complex domain with the same constants (up to a $\le \frac{1}{15}$ enlargement from the shrinkage margin), and the phase $\phi = e \cdot \gamma$ is holomorphic on S_{τ_0} with $\sup_{S_{\tau_0}} |q| \le C_\phi |v|^2$ — which is Lemma E2a.57(i).

Remark E2a.59 (the topology bookkeeping — two collars, never interchanged). Two topologies are in play, kept apart throughout. (i) The collar sup-norm on a *fixed* ρ_a: balls in it are *not* compact (bounded sets in a sup-norm on a fixed collar are not precompact). (ii) The local-uniform (Montel) topology after the shrinkage $\rho_a \to \rho'_a$: on the smaller collar the ball is relatively compact (Montel: uniformly bounded holomorphic families are normal). The fold count Lemma E2a.17 is stated with its own shrinkage built in (Theorem E2a.58), so *no* constant in the slice chain ever invokes compactness of a fixed-collar sup-ball — the discipline against citing Montel where it does not apply. Every uniformity in A.6 is a plain supremum over U_ω or a pair-Lipschitz modulus, and U_ω is H^s-*dense* with *empty* H^s-interior; downstream coercivity floors therefore use the collar-margin attainment note (the infimum over the *open* U_ω is controlled by the collar margin, *not* by attainment on a closed H^s-ball, which fails).

E2a.12 A.6″ Calibration record (a numerics witness)

Remark E2a.60 (what the reproducibility scripts do, and what they do not). The exhibited ball-uniform constants and exponents of A.6 are accompanied by an independent numerical calibration, recorded in the companion reproducibility scripts `t1p_calibrate.py` (with the `ref12` integrator), `s4a_beta_final.py` / `s4a_beta_finitefan.py`, and `s4b_arith_check.py` / `s4b_schur_check.py`. The record is a *witness*, not a proof; it is stated here at that grade and no higher. *What the numerics confirm.*

(i) The (D1) exponent $-\frac{1}{2}$: on both curved-fan charts the measured $L^2(\mathrm{fan}_\rho)$ slope is -0.496 (against the theoretical $-\frac{1}{2}$), and the proved-bound witness $\sup_\Lambda (\Lambda\rho)^{1/2} \|I_\Lambda\|_{L^2}$ *plateaus* at ≈ 1.44 (non-increasing over the top three octaves within 1%), confirming Proposition E2a.46 is a bound the data *approach from below* but do not exceed.

(ii) The (D4) transfer exponent $\beta = \frac{1}{2}$: on the exact affine synthesis the small-u shell-mass slope converges to $\beta \in \{0.484, 0.492, 0.494\}$ for the three admissible non-vanishing-at-t=1 profiles ($\to \frac{1}{2}$), rises to ≈ 1.49 for a weight that vanishes at $t = 1$ (the $\beta = \frac{3}{2}$ improvement) and to ≈ 2.49 for a doubly-vanishing weight, and the finite curved fan reproduces $\beta \approx \frac{1}{2}$ *bend-insensitively* (identical to three significant figures across bend strengths $\kappa_2\rho \in \{0, 0.15, 0.35\}$ and across $\Lambda \in \{4000, \ldots, 32000\}$) — confirming that the exponent of Theorem E2a.47 is a *boundary datum*, invariant under the interior curvature.

(iii) The Schur/extraction arithmetic: the exponent identity $M^{a-1/2}\omega(M/\Lambda)\Lambda^{1/2-a} = (M/\Lambda)^{a-1/2+\beta}$ and the finiteness of the row/column Schur sums are machine-checked with identically-zero symbolic residual, and the empirical Schur ratio is non-increasing in the octave count and dominated by c_a — confirming Theorem E2a.49 generates no hidden $(\Lambda\rho)$-log.

What the numerics do NOT do. They do not prove any bound and they do not saturate any bound. The proofs are Propositions E2a.46, E2a.50 and Theorems E2a.47, E2a.49, E2a.51; the figures neither establish the estimates nor exhibit an extremizer — (D1), for instance, is a bound the numerics can only *confirm*, not *saturate* (the Θ_{osc} part alone over-covers the empirical plateau). No calibration figure enters the constant chain (199): it certifies that the *exhibited* ball-uniform constants are of the claimed order and sign, and that the derived slopes match the proved exponents, on the finite grid tested. The complete gate suite runs 17/17 PASS with quadrature stability to $\sim 10^{-11}$ ("doubling the quadrature resolution changes no reported digit at 10^{-8}"); this is a reproducibility check on the *integrator and the exponent bookkeeping*, not a certificate of the theorem. In particular the numerics say *nothing* about the analytic gain itself — the ball-uniformity of N_0 over U_ω (Lemma E2a.17) is a Jensen/majorant *theorem* (§E2a.11), not a numerical finding; the calibration only witnesses the $S4$ exponents that feed on $D = c_{\rm sub}^{1/2}$ once N_0 is in hand.

Remark E2a.61 (the scope of what A.6–A.6$''$ establish)*.* Sections A.6–A.6$''$ establish, at **THEOREM** grade over U_ω, the analytic slice register: the Schur assembly books $\partial_y : (c,p) \mapsto (c+\frac{1}{2}, p-1)$ (Theorem E2a.51, end-to-end over U_ω because the near-flat sliver A1 is proved, Remark E2a.52) and the slice fence promotes to $s > 11$ (T2) / $s > \frac{23}{2}$ (T3) (Corollary E2a.54), all constants ball-uniform in $(g_0, \rho_a, \varepsilon_a)$ via the $S1$ transfer and collar bookkeeping (§E2a.11), and independently calibrated (Remark E2a.60). This is the support the paper's Option A booking line (the boxed $s > 11$ at the fence paragraph following Hyp. E.2) rests on. It is *not* a proof of Hyp. E.2: clauses (i)–(iii) — the singular-part dual-Lipschitz bound (a named input; the single-metric $(1,1)$ jet is C^0 at $s > 8$ but the family-uniform difference bound is part of the named E2a black box), the mixed-jet finite-part family $\{C_W, W_*\}$ (PLAUSIBLE: reduced to $s > 11$ modulo the not-in-print N=2 family-uniform Hadamard construction and the unexecuted multi-index constant-tracking), and the Sanders QEI-constant uniformity (a named input, un-derived in print, F-iii-fenced) — remain the paper's external input, consumed by Prop. E.4 and Thm. E.7, and are collected with their exact grades in the appendix's closing modulo-list. The fused free-field first law (Prop. E.4) is therefore a theorem *modulo* that finite named list; what A.6–A.6$''$ contribute is that the *slice register* it invokes is now a theorem over U_ω rather than a naive-fallback input.

E2a.13 The trace audit, the dimensional refund, and the transverse data legs

The two ladders below put objects on the codimension-one Cauchy slice $\Sigma^3 \subset \mathcal{M}^4$ in three distinct ways, and the fence value $s > 11$ (as opposed to $s > 11.5$) turns on accounting for those three ways exactly. This subsection isolates the accounting as three THEOREM-grade lemmas, all established in the trace audit and here restated over the analytic class U_ω of Part 1 (Def. E2a.1); nothing in it is new mathematics, and none of it depends on the analytic restriction beyond the uniform spacelikeness that $U_\omega \subset \mathcal{U}$ already supplies.

Lemma E2a.62 (Uniform transversality of a spacelike slice to null cones; empty glancing set)**.** *Let Σ be the compact Cauchy slice of $(\mathcal{M}, g_0)$ and let U_ω be the analytic ball of Def. E2a.1, chosen (by the ε_a-cap of Part 1) inside a C^0-neighbourhood $\mathcal{U}$ small enough that Σ is* uniformly spacelike over the ball*: there is $\delta_0 = \delta_0(g_0, \Sigma, \varepsilon_a) > 0$ with $g(v,v) \geq \delta_0 |v|_e^2$ for all $v \in T\Sigma$ and all $g \in U_\omega$. Then:*

- (i) *For every $g \in U_\omega$ and every g-null hypersurface $\mathcal{C}$ (in particular every light cone of a spectator point, away from its vertex), Σ and $\mathcal{C}$ intersect* transversally at every common point*: at $p \in \Sigma \cap \mathcal{C}$, $T_p\mathcal{C}$ contains its null generator $\ell \neq 0$ with $g(\ell,\ell) = 0$, while every nonzero $v \in T_p\Sigma$ has $g(v,v) > 0$; hence $T_p\Sigma \neq T_p\mathcal{C}$ and the two distinct hyperplanes span $T_p\mathcal{M}$.* The glancing set is empty — *not merely measure-zero. The transversality angle is bounded below by $c_0(\delta_0, \|g\|_{C^0(\text{collar})}) > 0$, uniformly over $\Sigma \times U_\omega$.*
- (ii) (Near-diagonal comparability.) *There are $d_0, c_\Sigma, C_\Sigma > 0$, ball-uniform, with $c_\Sigma\, d_\Sigma(x,y)^2 \leq \sigma_g(x,y) \leq C_\Sigma\, d_\Sigma(x,y)^2$ for $x, y \in \Sigma$, $d_\Sigma(x,y) \leq d_0$ (Synge world function $\sigma_g(x,y) = \frac{1}{2} g_{\mu\nu}(x)(y-x)^\mu (y-x)^\nu + O(|y-x|^3)$ with C^1-uniform constants; the form restricted to the uniformly spacelike $T\Sigma$ is positive-definite with constant δ_0). Hence every near-diagonal radial profile $r^p \log r$ ($r^2 \sim 2\sigma_g$) restricts on Σ to the model $\rho^p \log \rho$, $\rho = d_\Sigma$, with ball-uniform constants and no glancing degeneration anywhere.*

Proof. The uniformly spacelike bound $g(v,v) \geq \delta_0 |v|_e^2$ over U_ω is the Chruściel–Grant C^0-stability of the causal structure [222] applied to the ε_a-collar cap; every step is a fixed pointwise linear-algebra fact about a Lorentzian form on a spacelike hyperplane, uniform in the bounded C^0 data. (i): a null generator ℓ and a spacelike v cannot be parallel, so $T_p\Sigma \neq T_p\mathcal{C}$, and the intersection angle is a continuous positive function of the bounded C^0 metric data on the compact $\Sigma \times U_\omega$, hence bounded below. (ii): Taylor expansion of σ_g with C^1-uniform remainder over the ball; positive-definiteness on $T\Sigma$ is the δ_0-bound. □

Remark E2a.63 (Slice-trace accounting at $s > 11$: the refund identity). The three ways the $N{=}2$ chain lands objects on $\Sigma^3 \subset \mathcal{M}^4$, with their exact costs, are:

(a) *The final $C^0(\Sigma)$ consumer.* Either $H^{s-9}(\text{slab}) \hookrightarrow C^0(\text{slab})$ (four-dimensional Morrey, $s - 9 > 2$) followed by free restriction of a continuous function, or the trace $H^{s-9}(\text{slab}) \to H^{s-19/2}(\Sigma)$ (costs $\frac{1}{2}$; needs $s-9 > \frac{1}{2}$) followed by the three-dimensional Morrey embedding $H^t(\Sigma^3) \hookrightarrow C^0(\Sigma)$, $t > \frac{3}{2}$: the demand $s - \frac{19}{2} > \frac{3}{2}$ is again exactly $s > 11$, because $\frac{1}{2} + \frac{3}{2} = 2$. *The standard $\frac{1}{2}$ trace cost is exactly refunded by the drop of the Morrey threshold from $n/2 = 2$ to $3/2$.* The reading "$s - \frac{19}{2} > 2$, so $s > \frac{23}{2}$" *double-charges*: it pays the trace and then demands the four-dimensional Morrey index of a three-dimensional consumer. In the energy register actually used below, the Tice estimate outputs the slice norms $L^\infty_t H^k(\Sigma_t)$ natively (both tangential and ∂_0 components, via the first-order reduction), so no trace is taken at all and the Morrey call $t > \frac{3}{2}$ on $H^t(\Sigma_t)$ carries a built-in unspent $\frac{1}{2}$ over the true three-dimensional threshold.

(b) *Sources.* The estimate consumes $\|F\|_{L^1_t H^k(\Sigma_t)}$ or $\|F\|_{L^2_t H^k(\Sigma_t)}$, *time-integrated* slice norms: by Fubini ($(1 + |\xi|^2)^k \le (1 + \tau^2 + |\xi|^2)^k$) and Cauchy–Schwarz in t, $\|F\|_{L^1_t H^k(\Sigma_t)} \le \sqrt{T}\,\|F\|_{H^k(\text{slab})}$ — *no trace theorem, no $\frac{1}{2}$, on any source term.*

(c) *Single-time data.* Only the Cauchy-data legs at Σ_0 (and any slab-defined coefficient restricted at fixed time) genuinely pay the $\frac{1}{2}$; each such site is carried below with a named margin $\ge \frac{1}{2}$ against its demand, and the profile data legs restrict by Lem. E2a.62(ii).

Consequently the slice fence stays $s > 11$; no step of the chain forces $s > \frac{23}{2}$ in the slice register. The refund identity $\text{trace}(\frac{1}{2}) + \text{Morrey}_{3d}(\frac{3}{2}) = 2 = \text{Morrey}_{4d}$ is the trace-audit content, THEOREM.

Lemma E2a.64 (Transverse restriction of cone profiles to a spacelike slice; honest sharp height $\frac{\alpha}{2} + \frac{1}{2}$ at $\alpha > 0$ (the transverse correction; the printed height $\alpha + \frac{1}{2}$ stands only at $\alpha = 0$))**.** *Under the hypotheses of Lem. E2a.62, let $y \in K_\Sigma$, let $r(\cdot) = r_g(\cdot, y)$ be the near-cone profile coordinate ($r^2 \sim 2|\sigma_g|$ near the light cone of y, away from the vertex), and let $\alpha \ge 0$. Then the restriction to Σ_0 of $r^\alpha \log r$ (times any factor smooth-modulo-H^{s-2}) lies in $H^t(\Sigma_0)$ for every $t < \frac{\alpha}{2} + \frac{1}{2}$ when $\alpha > 0$ (in the σ-grading: $\sigma^k \log \sigma$, $k = \alpha/2 \ge 1$, restricts at $H^{k+\frac{1}{2}^-}$), and for every $t < \frac{1}{2}$ when $\alpha = 0$, with constants uniform in $y \in K_\Sigma$ and $g \in U_\omega$; both exponents are* sharp *for the codimension-one intersection geometry. In the vertex-on-slice regime the radial model of Lem. E2a.62(ii) gives the better height $\alpha + \frac{3}{2}$ (correct as printed), and the y-uniform level over the collar is* $\min(\frac{\alpha}{2} + \frac{1}{2}, \alpha + \frac{3}{2})^- = (\frac{\alpha}{2} + \frac{1}{2})^-$. *The previously printed transverse height $\alpha + \frac{1}{2}$ is correct at $\alpha = 0$,* under*books (safe side) at the $\alpha < 0$ singular legs, and* over*books by exactly $\alpha/2$ at every $\alpha > 0$: the two printed errors are named in Rem. E2a.65, the sharp counterexample and the seed record in Rem. E2a.66.* Grade: THEOREM as corrected (sharp, y-uniform); the $\alpha = 0$ instance of the old statement is unchanged THEOREM; the $\alpha < 0$ (singular-leg) applications of the old rule *under*book — the safe side, untouched.

Proof. By Lem. E2a.62(i) the cone of y meets Σ_0 transversally at angle $\ge c_0 > 0$ in a compact codimension-one submanifold $\mathcal{S}_y \subset \Sigma_0$ (a topological 2-sphere, or the empty set, in which case there is nothing to prove). The same transversality forces $\sigma_g(\cdot, y)|_{\Sigma_0}$ to vanish *simply* on $\mathcal{S}_y$: at $p \in \mathcal{S}_y$ the gradient of σ_g is the nonzero null generator, and a nonzero null covector cannot annihilate the uniformly spacelike $T_p\Sigma_0$ (it would then be proportional to its timelike conormal), so $d(\sigma_g|_{\Sigma_0})(p) \neq 0$, with slope bounded below by the c_0, δ_0-data. Hence in c_0-uniform tubular coordinates $(u, \omega) \in (-u_0, u_0) \times \mathcal{S}_y$ one has $\sigma_g|_{\Sigma_0} = e(u, \omega)\, u$ with $e \neq 0$, so $r|_{\Sigma_0} \sim (2|e|\,|u|)^{1/2}$ is

Hölder-$\frac{1}{2}$ in u — *not* C^1-comparable to u — and the profile restricts as $|u|^{\alpha/2}\log|u|$ times a non-vanishing bounded factor. The one-dimensional model $\widehat{|u|^{\alpha/2}\log|u|}\,\chi(\xi) = O(|\xi|^{-\alpha/2-1}\log|\xi|)$ gives $\int(1+\xi^2)^t|\cdot|^2\,d\xi < \infty$ iff $2t-\alpha-2 < -1$ iff $t < \frac{\alpha}{2}+\frac{1}{2}$ — strict, open, no rounding (at $\alpha = 6$ exactly: $(d/du)^3[u^3\log|u|] = 6\log|u| + 11$, so $|\widehat{u^3\log|u|}| = 6\pi|\xi|^{-4}$ modulo a delta term, and H^t iff $t < \frac{7}{2}$); tangential smoothness contributes no loss (finite atlas on $\mathcal{S}_y$). At $\alpha = 0$ the profile is $\log r = \frac{1}{2}\log|\sigma_g| + O(1) \sim \log|u|$ and the printed height $\frac{1}{2}^-$ is unchanged. Near the degenerate limit (the vertex approaching Σ_0, $\mathcal{S}_y$ shrinking) the profile tends to the radial model $\rho^\alpha\log\rho$ of Lem. E2a.62(ii), which lies in H^t for $t < \alpha+\frac{3}{2}$ — *more* regular; two-zone patching across the collar gives the y-uniform bound at the worse exponent $\frac{\alpha}{2}+\frac{1}{2}$ (the transverse slope factor shrinks the norm, not the level, as the vertex approaches the slice). □

Remark E2a.65 (The transverse correction note: two printed errors, both named; what is gated and what is not). The earlier form of this note took the transverse height $\alpha+\frac{1}{2}$ of Lem. E2a.64 to be the honest availability of a single-time cone-profile restriction (the naïve radial $\alpha+\frac{3}{2}$ overstates it by one half-order — that direction was, and remains, correct) and called the correction fence-neutral. The present correction sharpened the count downward: the $\alpha > 0$ claim was itself false as previously printed, and the honest transverse height is $\frac{\alpha}{2}+\frac{1}{2}$, as now stated. TWO distinct printed errors are corrected, and both are named:

(a) *The restriction rule at r-graded $\alpha > 0$.* The old proof step "in tubular coordinates the profile is $u^\alpha\log|u|$ times a C^1-bounded factor" requires r/u bounded, which is false: $r|_{\Sigma_0} \sim (2\tau|u|)^{1/2}$ at spectator height τ is Hölder-$\frac{1}{2}$ in the tubular coordinate ($r/|u|^{1/2}$ is what is bounded), and no C^1 tubular coordinate makes $\sqrt{\text{linear}}$ linear. The no-glancing lemma itself forces $\sigma_g|_{\Sigma_0}$ to vanish *simply* on the crossing sphere, so one power of σ is one power of the slice coordinate but *two* powers of the profile coordinate r: the old rule double-counts the transverse vanishing by $\alpha/2$.

(b) *The per-derivative eikonal grading.* The grading "one r-power per derivative" on profile columns (geometric root: the eikonal identity $|\nabla\sigma|_g = r$) confuses the Lorentzian g-norm with the coordinate gradient: $|\nabla\sigma|_g \to 0$ at the cone while the *coordinate* gradient is $O(1)$ there, so near the crossing each derivative on $\sigma^k\log\sigma$ costs one σ-order, i.e. *two* r-powers (honest r-grading of the N=2 tail: $\partial T \sim r^4\log r$, $\partial^2 T \sim r^2\log r$ — not $r^5\log r$, $r^4\log r$).

Both errors push the same way (overbooking of $\alpha > 0$ tail legs); the sharp in-scope counterexample and the honest seed column are recorded in Rem. E2a.66. *Consequently the following printed sites are GATED — each stands at its printed value MODULO the transverse re-derivation; the arithmetic is deliberately kept, not deleted:* the A1/A2 source-rung slab profile bookings and their (206) consumption, the A3 availability column, the A4 ∂^2-caps, the B2 commutator consumption, the B3/Lem. E2a.75 slice caps, the clause (i)/(ii) profile caps of Lem. E2a.67, the raised caps of Lem. E2a.69, the N=1 calibration numerology (Rem. E2a.68; its DEATH verdict stands *a fortiori*), and the (D0) port display (App. E2a). *Not* gated (untouched by the correction): the coefficient columns everywhere, including the $s > 11$ commutator pin; Thm. E2a.51; Lem. E2a.73, Lem. E2a.74, Lem. E2a.62; the refund identity Rem. E2a.63; every $\alpha \le 0$ (singular-leg) booking (the old rule is exact at $\alpha = 0$ and *under*books the $\alpha < 0$ legs — the safe side); the diagonal coincidence values as objects; and the (D0) port's Hadamard-1932 lever and w_0-smoothness lemmas.

Remark E2a.66 (The (D0) record at the seed: existence THEOREM, the honest column, the sharp counterexample, and the open repair). Established by the seed re-derivation, at the fixed real-analytic seed (g_0, ω_0) on the seed-adjacent slice, collar-localized throughout (DoD cutoff of Lem. E2a.74; no global-in-x slice norm):

(i) *Existence of the slice restriction (THEOREM, unconditional).* For every spectator y in the short slab (the vertex-on-slice endpoint included) and every jet leg $L \in \{\mathrm{id}, \partial_t, \partial_y, \partial_t\partial_y\}$, the restriction $(L\,W_{g_0,2})(\cdot, y)|_{\Sigma_0}$ exists, classically and as a distributional trace, uniformly and continuously in y: the wave front set of the tail is null-conormal to the cone (null covectors over the vertex), $N^*\Sigma_0$ is timelike, and the glancing set is empty (Lem. E2a.62(i)), so Hörmander's restriction criterion (*The Analysis of Linear Partial Differential Operators I*, Thm. 8.2.4) applies. Existence survives everything below — the obstruction is purely at the level count.

(ii) *The honest seed column (THEOREM given the paper's standing seed-Hadamard premise — (P-a) of the (D0) port, App. E2a; sharp).* With the finite-order split $W_{g_0,2} = w_{N'} + T_{N'}$, $N' = 9$, the levels are decided by the $k=3$ tail $v_3\sigma^3 \log \sigma$:

$$W_{g_0,2}|_{\Sigma_0} \in H^{7/2^-}, \quad \partial_t W_{g_0,2}|_{\Sigma_0} \in H^{5/2^-}, \quad \partial_y W_{g_0,2}|_{\Sigma_0} \in H^{5/2^-}, \quad \partial_t\partial_y W_{g_0,2}|_{\Sigma_0} \in H^{3/2^-},$$

each sharp — versus the printed A3/port column $(13/2, 11/2, 11/2, 9/2)^-$.

(iii) *The sharp counterexample (in scope: free massive scalar).* Flat massive vacuum: $\sigma|_{\Sigma_0}$ vanishes simply on the crossing sphere ($\sigma|_{t=0} = \tau u + u^2/2$ at spectator height τ), and the log coefficient has $v_3 = m^8/(18432\pi^2) \neq 0$ (all $v_k > 0$), so $W_{g_0,2}|_{\Sigma_0} = v_3\tau^3u^3 \log|u| \times$ (nonvanishing analytic) + (smoother) $\in H^t$ iff $t < \frac{7}{2}$. This refutes the old $\alpha + \frac{1}{2}$ rule at $\alpha = 6, 5, 4$ and the printed column by three full orders at the top leg. Compactness scope: the content is collar-local, and the compact-slice setting is restored at zero cost on the flat cylinder $\mathbb{R} \times \mathbb{T}^3$ (real-analytic, globally hyperbolic, massive vacuum Hadamard; on a short collar the image sums are smooth and the $v_3\sigma^3 \log \sigma$ term is unchanged).

(iv) *Consequence for (D0) (NEGATIVE-RESULT, sharp).* The seed supplies the $b=1$ field slot at $H^{5/2^-}$ sharp; the printed single-metric ladder consumes it strictly above that under every reading ($\sigma_1 > \frac{3}{2}$ strict at every $s > 11$, so demand $> \frac{5}{2}$: missed by 0^+ at the bare slot, by one order at the ∂^2-consumer, by three versus the printed cap; the $b=0$ chain clears its bare slots for $s < 11.5$ and dies 0^+ at its ∂^2-consumer). (D0) is OBSTRUCTED on the printed route — a real obstruction at the cone crossing, not a write-out gap.

(v) *Repair: OPEN.* Raising the truncation N buys one transverse order per unit N on every data leg; the k-counts suggest $b=0$ heals at $N=3$ and $b=1$ needs $N \geq 4$, but the see-saw reused printed source columns that are themselves gated (the slab register $p = \alpha + 2$ is the vertex-zone height; near the crossing the honest slab profile of the $\sigma^2 \log \sigma$ factor is $\frac{5}{2}^-$, under which the $b=0$ source rung survives only for $s < 11.5$ — spot-checked at $b=0$ only). The $N \geq 4$ see-saw is PROVISIONAL; no repaired fence value is derived or quoted here, and the source columns must be re-derived before any is. The alternative repair direction is a consumer rewiring that stops feeding slice-Sobolev norms of the tail across crossing spheres. Neither is executed here.

E2a.14 The slice-register ladder at $s > 11$ over U_ω

The device of Parts 1–3 promotes the transport-averaged ∂_y booking to $(c, p) \mapsto (c + \frac{1}{2}, p - 1)$ over the analytic ball U_ω (Thm. E2a.51). With that booking in hand the $N=2$ difference-system energy estimate closes in the *slice-native* energy norms $L_t^\infty H^k(\Sigma_t)$ (Tice-native, including the ∂_0 components as first-order-reduction unknowns) at the promoted slice register $s > 11$, at zero slack. The lemma below is the slice-register close-out; its $b=0$ chain uses no ∂_y booking and is unconditional, its $|b| = 1$ chain rides Thm. E2a.51.

Lemma E2a.67 ($N=2$ two-slot difference-system energy estimate, slice register; the mixed $(1,1)$ coincidence value at the Lipschitz rate for $s > n/2 + 9 = 11$, zero slack)**.** *Let $n = 4$, $m > 0$, minimal coupling $\xi = 0$; fix the compact Cauchy slice Σ of the real-analytic globally hyperbolic $(\mathcal{M}, g_0)$, a compact causal slab $K_\Sigma \subset \mathcal{M}$ foliated by $\{\Sigma_t\}_{t\in[0,T]}$ in a fixed finite atlas, uniformly spacelike over the ball with constant δ_0 (Lem. E2a.62), a slab $\widetilde{K} \supset\supset K_\Sigma$, and the analytic ball $U_\omega = U_\omega(g_0, \rho_a, \varepsilon_a)$ of Def. E2a.1, with the standing index*

$$s > \tfrac{n}{2} + 9 = 11, \qquad \delta := s - 11 > 0.$$

For $g, g' \in U_\omega$ let $W_{g,2}$ be the $N{=}2$ truncated Hadamard finite part of the parametrix package (N-a) *of §E2a.17, set $D_2 := W_{g,2} - W_{g',2}$, and let D_2 solve, on $K_\Sigma \times K_\Sigma$ near the diagonal,*

$$(\Box_g + m^2)\, D_2(\cdot, y) = f_2(\cdot, y), \qquad f_2 = -\big(S_{g,2} - S_{g',2}\big) - (\Box_g - \Box_{g'})\, W_{g',2}, \tag{202}$$

$S_{g,2} := (\Box_g + m^2) W_{g,2}$ the parametrix residual. Then, with every constant finite and ball-uniform at each fixed $s > 11$ (NOT uniform as $s \downarrow 11$: the top three-dimensional Morrey constant diverges at the open fence — the honest zero-slack behaviour):

(i) (field rungs, slice-native) *For $|b| \le 1$, with $\sigma_0 := \min(s-9, \frac{9}{2}^-)$ and $\sigma_1 := \min(s - \frac{19}{2}, \frac{7}{2}^-)$,*

$$\|\partial_y^b D_2(\cdot, y)\|_{L_t^\infty H^{\sigma_{|b|}+1}(\Sigma_t)} + \|\partial_0 \partial_y^b D_2(\cdot, y)\|_{L_t^\infty H^{\sigma_{|b|}}(\Sigma_t)} \le C_E^{(b)}\, \|g - g'\|_{H^s},$$

uniformly in $y \in K_\Sigma$; the ∂_0-components are Tice unknowns of the first-order reduction, not *traces.*

(ii) (the mixed $(1,1)$ package) *For $|a| \le 1$, $|b| \le 1$,*

$$\sup_y \big\|\partial_x^a \partial_y^b D_2(\cdot, y)\big\|_{L_t^\infty H^{\min(s-\frac{19}{2}, \frac{7}{2}^-)}(\Sigma_t)} \le C_E^{(1,1)}\, \|g - g'\|_{H^s},$$

with y-continuity; consequently $\partial_x^a \partial_y^b D_2 \in C^0$ near the diagonal of the closed slab (joint (t, x, y)-continuity; the slab-C^0 consumers read the same output, the refund spent once).

(iii) (mixed coincidence values at the rate; consumer interface) *The diagonal values exist, are continuous on Σ, and*

$$\max_{|a|+|b| \le 1 \,\cup\, |a|=|b|=1} \ \sup_{x \in \Sigma} \big|[\partial_x^a \partial_y^b D_2](x, x)\big| \le C_W^{(2),\mathrm{sl}}(g_0, K_\Sigma; s)\, \|g - g'\|_{H^s}, \tag{203}$$

with the explicit chain (208)*. In particular the A-contracted consumer scalar*

$$\rho_{\mathrm{kin}}^{(2)} := A_{00}^{ab}\, [\partial_x^a \partial_y^b D_2]_{\mathrm{diag}}, \qquad A_{\mu\nu}^{ab} = \delta_{(\mu}^a \delta_{\nu)}^b - \tfrac{1}{2} g_{\mu\nu} g^{ab}, \quad A_{00}^{00} = \tfrac{1}{2},$$

is $C^0(\Sigma)$ at the rate: the hypothesis of Lem. E.6 of the paper is supplied directly on the slice, with no flux and no codimension-one trace step on the solution side; the single-metric sups $W_*^{(2)}$ hold at the same registers with the commutator branch absent (pin $s > 10$, margin $1 + \delta$).

The fence is pinned by the commutator *branch at $L_t^\infty H^{\min(s-19/2, 7/2^-)}(\Sigma_t)$: $C^0(\Sigma)$ iff $s - \frac{19}{2} > \frac{3}{2}$ iff $s > 11$ — the unique zero-slack inequality of the chain.*

Grade: THEOREM end-to-end over U_ω, at $s > 11$, GIVEN the analytic slice chain of Parts 1–3 (Thm. E2a.51), and consuming the parametrix package (N-a) and difference data (N-c) of §E2a.17 at their audited grades. [Correction gate: the profile caps $\frac{9}{2}^-, \frac{7}{2}^-$ inside σ_0, σ_1 of clause (i), the clause-(ii) cap, the clause-(iii) C^0 extraction, and the single-metric sups $W_*^{(2)}$ consume the gated data-cap columns of Ladders A/B below; they stand at the printed values MODULO the transverse re-derivation (Rem. E2a.65). Under the honest crossing pricing the $b{=}1$ field slot is supplied at $H^{5/2^-}$ sharp versus demand H^{σ_1+1} with $\sigma_1 > \frac{3}{2}$ strict — missed at every $s > 11$ — and the $b{=}0$ chain dies 0^+ at its ∂^2-consumer (Rem. E2a.66); the coefficient columns and the $s > 11$ commutator pin are untouched.] *The three scope pins are enforced verbatim: minimally-coupled* free massive *($m > 0$) scalar; the* mixed *jet set $|a| + |b| \le 1$ together with $|a| = |b| = 1$ (the pure same-slot $(2,0), (0,2)$ jets are never formed and in fact diverge for the truncated residual — Rem. E2a.77); and the class U_ω, on which every uniformity is a supremum or a pair-Lipschitz modulus (fences F-iii, F-diff of Hyp. E.2), never an H^s-open property.*

Proof. Register conventions. A *slab pair* (c,p) abbreviates membership in $H^{\min(s-c,p^-)}$ near the diagonal of the slab, c the coefficient ceiling, p the four-dimensional profile height ($r^\alpha \log r$ has $p=\alpha+2$ on $\mathbb{R}^4$ *in the vertex/diagonal zone* — correction gate: off the diagonal, near the cone crossing, σ vanishes simply in the slab too, so the honest slab height of $\sigma^k \log\sigma$ is $(k+\frac{1}{2})^-$, not $2k+2$; its single-time restriction has slice height $\frac{\alpha}{2}+\frac{1}{2}$ at $\alpha>0$ and $\frac{1}{2}$ at $\alpha=0$ by the corrected Lem. E2a.64; the profile columns of this proof predate the correction and are carried GATED, Rem. E2a.65). A *slice register* is $L_t^\infty H^{\min(s-c,p^-)}(\Sigma_t)$, the same pair tagged sl. The calculus is

$$\partial_x : (c,p) \mapsto (c{+}1, p{-}1); \qquad \partial_y \text{ on transport-averaged coefficients} : (c,p) \mapsto (c{+}\tfrac{1}{2}, p{-}1) \text{ [Thm. E2a.51, only th}$$

∂_y on explicit profile factors books $(c, p{-}1)$ (direct computation, not an averaging claim); ∂_y on non-averaged bitensor/weight factors books $(c{+}1, p)$ at register $\geq s-3$ (worst demand $s-\frac{17}{2}$: margin $\frac{11}{2}$, never binding). The wave solve is $+1$ on both columns over the consumed source register, capped by the data legs (Rung 2; the cap column is carried at every solve at the transverse heights). No $(c, p{-}1)$ booking on averaged coefficients occurs anywhere; no $+2$ gain anywhere. [Correction gate on the profile decrement: $p \mapsto p{-}1$ per derivative on profile factors is the vertex-zone/eikonal grading — error (b) of Rem. E2a.65; near the cone crossing the honest cost is one σ-order, i.e. two r-powers, per derivative. The profile columns derived below stand at their printed values MODULO the transverse re-derivation.]

Rung 1 (first-order reduction; integer slice-energy estimate). Write $\Box_g = -a^{00}\partial_t^2 + 2a^{0j}\partial_t\partial_j + a^{jk}\partial_j\partial_k + b^\mu\partial_\mu$ in the fixed foliation and set $U=(u,\partial_t u,\nabla_x u)$, a symmetric-hyperbolic system with $A^0 \geq \theta I$,

$$\theta = \min\Big(1, \inf_{K_\Sigma\times U_\omega} \lambda_{\min}(a^{jk}/a^{00})\Big) \geq \theta_* := \min\big(1, \tfrac{1}{2}\theta_0^{\mathrm{anch}}\big) > 0$$

ball-uniform *by the collar margin, not by attainment* (U_ω is H^s-dense with empty H^s-interior, hence not H^s-closed; the old closed-ball attainment clause is void): the anchor floor $\theta_0^{\mathrm{anch}} = \min_{K_\Sigma}\lambda_{\min}(a_{g_0}) > 0$ holds over the compact slab at the analytic g_0 (Chruściel–Grant nondegeneracy [222]); a^{jk}/a^{00} is algebraic in g with $a^{00} \geq d_{00} > 0$ (the scalar collar-margin floor of Lem. E2a.72; the determinant floor d_ω alone does not floor a^{00}), hence C_a-Lipschitz in the collar C^0 norm (Weyl 1-Lipschitz for $\lambda_{\min}$); the caps $\varepsilon_a \leq d_{00}/C_{\mathrm{inv}}$ and $\varepsilon_a \leq \theta_0^{\mathrm{anch}}/(2C_a)$, imposed with Part 1's, give $\lambda_{\min}(a_g) \geq \frac{1}{2}\theta_0^{\mathrm{anch}}$ for all $g\in U_\omega$ (proved in Lem. E2a.72), and only $1/\theta \leq 1/\theta_*$ is consumed downstream. Tice Thm. 1.2.1 [206] applies at every integer $0\leq k\leq 6$ (the demand $\|\nabla A^\mu\|_{L_t^\infty H^{k-1}(\Sigma_t)}$ needs $s \geq k+\frac{1}{2}$; at $k=6$, $s\geq \frac{13}{2}$, margin $\frac{9}{2}+\delta$), giving for data on Σ_0 and forcing F

$$\|u\|_{L_t^\infty H^{k+1}(\Sigma_t)} + \|\partial_t u\|_{L_t^\infty H^k(\Sigma_t)} \leq \sqrt{Q_* e^{P_* T}}\Big(\|(u,\partial_t u)|_{\Sigma_0}\|_{H^{k+1}\times H^k} + \|F\|_{L_t^1 H^k(\Sigma_t)}\Big), \quad (204)$$

slice-native on both components; *no trace theorem is applied to the solution anywhere in this proof.* (The $k=0$ base case is the Friedrichs L^2 estimate, the linear estimate of Kato 1970, Prop. 3.4, that HKM '77 cite; provenance verified.)

Rung 2 (fractional rung, $+1$ exactly). Real interpolation of the per-t solution maps between the integer rungs $k\in\{0,\dots,6\}$: for real $\sigma\in[0,6]$,

$$\|u\|_{L_t^\infty H^{\sigma+1}(\Sigma_t)} + \|\partial_t u\|_{L_t^\infty H^\sigma(\Sigma_t)} \leq C_E(\sigma)\Big(\|(u,\partial_t u)|_{\Sigma_0}\|_{H^{\sigma+1}\times H^\sigma} + \sqrt{T}\|F\|_{L_t^2 H^\sigma(\Sigma_t)}\Big), \quad (205)$$

with t-continuity into $H^{\sigma'}$ for $\sigma' < \sigma+1$ (Tice Thm. 1.2.1(2) + density). *The gain is exactly* $+1$, never $+2$.

Rung 3 (source consumption: Fubini, no trace charge). For $k\geq 0$, in the fixed atlas,

$$\|F\|_{L_t^2 H^k(\Sigma_t)} \leq c_{\mathrm{fol}}\|F\|_{H^k(\mathrm{slab})} \qquad [(1+|\xi|^2)^k \leq (1+\tau^2+|\xi|^2)^k], \quad (206)$$

the refund audit's mechanism (b) (Rem. E2a.63), THEOREM: every slab-booked source is consumed at its slab register with no $\frac{1}{2}$; slice-booked sources are consumed via $L^2_t \le \sqrt{T}\,L^\infty_t$. Legality ($k \ge 0$) per source: worst is $\sigma_1 \ge \frac{3}{2} + \delta > 0$.

Rung 4 (slice products; coefficient traces). Per t, $H^{s-1/2}(\Sigma_t) \times H^\sigma(\Sigma_t) \to H^\sigma(\Sigma_t)$ for $0 \le \sigma \le s - \frac{1}{2}$ (Kato–Ponce/Moser on Σ^3; algebra demand $s - \frac{1}{2} > \frac{3}{2}$, margin $9 + \delta$), constants t-uniform. Metric factors enter by slab-to-slice *coefficient-leg* trace, $\|(g - g')_*|_{\Sigma_t}\|_{H^{s-1/2}(\Sigma_t)} \le c_{\mathrm{alg}} c_{\mathrm{tr}} \|g - g'\|_{H^s}$ (the refund audit's mechanism (c), $s > \frac{1}{2}$, margin $\frac{21}{2} + \delta$); each pays its honest $\frac{1}{2}$ at its reduced register — these are coefficient traces, not solution traces.

Ladder A (single-metric spectator solves at g'; slice-native outputs; data caps carried). By the normal form (N-a-NF) of (N-a), S'_2 books $(8, 6)$: coefficient $\Box_{g'} v'_2 \sim \partial^8 g' \in H^{s-8}$, profile $r^4 \log r$ ($p = 6$; $\partial^5 \sim r^{-1}$ is the last locally-L^2 derivative, $\partial^6 \sim r^{-2}$ never consumed).

A1. *Source, $b = 0$:* $\|S'_2\|_{L^2_t H^{\min(s-8,6^-)}} \le c_{\mathrm{fol}} L_S^{(0)}$ by (206); legality margin $3 + \delta$.

A2. *Source, $b = 1$ [Thm. E2a.51]:* $\partial_y S'_2 = (\partial_y[\Box_{g'} v'_2])\Pi$ books $(8+\frac{1}{2}, 5)$ by the promoted booking (tail hypothesis $a = s - 8 > \frac{3}{2}$: margin $\frac{3}{2} + \delta$), plus $(\Box_{g'} v'_2)(\partial_y \Pi)$ at $(8, 5)$, plus bitensor/weight legs at register $\ge s - 3$ and R_{sm}-legs at $(8.5, \ge 6)$. Sum: $(8.5, 5)$; legality margin $\frac{5}{2} + \delta$. [Correction gate on A1/A2 and their (206) consumption: the slab profile columns 6 resp. 5 are vertex-zone heights; near the cone crossing the honest slab profile of the $\sigma^2 \log \sigma$ factor is $\frac{5}{2}^-$, under which the b=0 source rung survives only for $s < 11.5$ and the b=1 rung tightens correspondingly (spot-checked at b=0; the dedicated re-derivation must sweep every slab booking meeting the cone off-diagonal). The printed bookings and legality margins stand MODULO the transverse re-derivation (Rem. E2a.65); the Fubini inequality (206) itself is untouched.]

A3. *Solves with the data-cap column.* Availabilities on Σ_0 (Lem. E2a.64; on the BFV surface the legs are g_0-seed data, profile caps only; the conservative H^s-background coefficient ceilings clear each demand by exactly $\frac{1}{2}$): the four legs W'_2, $\partial_t W'_2$, $\partial_y W'_2$, $\partial_t \partial_y W'_2$ restrict at transverse heights $\rho^6 \log \rho \in H^{13/2^-}$, $\rho^5 \log \rho \in H^{11/2^-}$, $\rho^5 \log \rho \in H^{11/2^-}$, $\rho^4 \log \rho \in H^{9/2^-}$ respectively [Correction gate: this column was computed with the old $\alpha + \frac{1}{2}$ tail rule at the r-powers $\alpha = 6, 5, 5, 4$; the honest crossing values under the corrected Lem. E2a.64 are $(\frac{7}{2}, \frac{5}{2}, \frac{5}{2}, \frac{3}{2})^-$, each sharp at the seed (Rem. E2a.66); the printed column stands MODULO the transverse re-derivation — gated, not deleted] (the ∂_y-slotted coefficient legs book $+\frac{1}{2}$ in the slab by Thm. E2a.51 *before* the $\frac{1}{2}$ trace). Rung 2 gives, uniformly and continuously in y,

$$W'_2 \in (7, \tfrac{13}{2})_{\mathrm{sl}}, \quad \partial_t W'_2 \in (8, \tfrac{11}{2})_{\mathrm{sl}}, \quad \partial_y W'_2 \in (\tfrac{15}{2}, \tfrac{11}{2})_{\mathrm{sl}}, \quad \partial_t \partial_y W'_2 \in (\tfrac{17}{2}, \tfrac{9}{2})_{\mathrm{sl}},$$

constants $\mathcal{W}_b^{(2)} = C_E(\cdot)[(\text{N-a) data} + \sqrt{T} c_{\mathrm{fol}} L_S^{(b)}]$ (the $s - 7$ finite-part register is *derived*: $+1$ from Rung 2, not $+2$).

A4. *The ∂^2-package.* Spatial–spatial: two derivatives off A3; time–spatial: one off the ∂_t-components — both land $(\frac{19}{2}, \frac{7}{2})_{\mathrm{sl}}$ (b=1), $(9, \frac{9}{2})_{\mathrm{sl}}$ (b=0). Time–time via the solved equation, its new ingredient the single-time restriction of the source (a coefficient-type trace, mechanism (c)): $\partial_y S'_2|_{\Sigma_t} \in H^{\min(s-9,\, \frac{9}{2}^-)}(\Sigma_t)$ t-uniformly (legality margin $2+\delta$) vs demand $(\frac{19}{2}, \frac{7}{2})$: margins $(\frac{1}{2}, 1)$; $b = 0$: $S'_2|_{\Sigma_t} \in H^{\min(s-\frac{17}{2}, \frac{11}{2}^-)}$ vs $(9, \frac{9}{2})$: margins $(\frac{1}{2}, 1)$. Net

$$\partial_x^\alpha \partial_y^b W'_2 \in (\tfrac{19}{2}, \tfrac{7}{2})_{\mathrm{sl}}\ (|b|{=}1), \qquad (9, \tfrac{9}{2})_{\mathrm{sl}}\ (b{=}0), \qquad |\alpha| \le 2, \tag{207}$$

uniformly/continuously in y, constants $\mathcal{W}^{(2)}_{\partial^2, b}$. [Correction gate: the profile caps $\frac{7}{2}$ ($|b|$=1), $\frac{9}{2}$ (b=0) inherit the gated A3 column and the gated per-derivative grading; the honest crossing profiles are $\frac{1}{2}^-$ (b=1) and $\frac{3}{2}^-$ (b=0). The display stands MODULO the transverse re-derivation (Rem. E2a.65).]

Ladder B (difference solves at g). ∂_y commutes with $\Box_{g,x}$: $(\Box_g + m^2)\partial_y^b D_2 = \partial_y^b f_2$.

B1. *Branch 1 (transport difference), at the rate:* $\partial_y^b(S_{g,2} - S_{g',2})$ books $(8+\frac{b}{2}, 6-b)$: at $b=1$, Leibniz gives $(\partial_y[\Box_g v_2 - \Box_{g'} v_2'])\Pi$ [Thm. E2a.51, $(8.5, 5)$, rate C_{v_2}] $+(\Box_g v_2 - \Box_{g'} v_2')(\partial_y \Pi)$ [$(8,5)$] + profile-difference legs (coefficient column ≤ 2.5, margin 6) $+R_{\mathrm{sm}}$-differences ($(8.5, \ge 6)$, profile margin ≥ 1). Consumed by (206): legality margins $3+\delta$, $\frac{5}{2}+\delta$. *Alone this branch pins only $s > 10$ (slice); not the pin.*

B2. *Branch 2 (the commutator — the pin), slice-native, at the rate:*

$$(\Box_g - \Box_{g'})\partial_y^b W_2' = (g_*^{\mu\nu} - g_*'^{\mu\nu})\partial_\mu\partial_\nu\partial_y^b W_2' + (b_g^\mu - b_{g'}^\mu)\partial_\mu\partial_y^b W_2'.$$

The two ∂_x are irreducibly transverse: they act on the *solved* spectator (207), not the raw coefficient depth. Rung-4 products per t:

$$\|(g-g')_*\partial^2\partial_y^b W_2'\|_{L_t^\infty H^{\kappa_b}(\Sigma_t)} \le c_{\mathrm{alg}} c_{\mathrm{tr}} c_{\mathrm{mult}} \mathcal{W}^{(2)}_{\partial^2,b} \|g-g'\|_{H^s}, \quad \kappa_1 = \min(s-\tfrac{19}{2}, \tfrac{7}{2}^-),\ \kappa_0 = \min(s-9, \tfrac{9}{2}^-),$$

the Christoffel leg softer by $(1,1)$. Branch 2 exceeds Branch 1 by $(1, \frac{3}{2})$ at $b=1$: *the commutator pins under the slice calculus too.* Consumed as $L_t^2 \le \sqrt{T} L_t^\infty$: the source never leaves slice norms. [Correction gate: the profile branches $\frac{7}{2}^-, \frac{9}{2}^-$ of κ_1, κ_0 consume the gated (207); the honest slice profile of $\partial^2\partial_y^b W_2'$ near the crossing is $\frac{1}{2}^-$ ($|b|$=1), $\frac{3}{2}^-$ (b=0). The coefficient branches $s - \frac{19}{2}$, $s-9$ — the $s > 11$ pin — are untouched; the commutator consumption stands MODULO the transverse re-derivation (Rem. E2a.65).]

B3. *Data* [(N-c) corrected, ∂_y-slotted, consumed transverse-weakened]. By (N-c) of §E2a.17 (Lem. E2a.75, grade PLAUSIBLE) the difference data $(\partial_y^b D_2, \partial_0 \partial_y^b D_2)|_{\Sigma_0}$ are Lipschitz at the rate in $H^{\min(s-7, \frac{11}{2}^-)} \times H^{\min(s-8, \frac{9}{2}^-)}$, against the demands $H^{\sigma_{|b|}+1} \times H^{\sigma_{|b|}}$; at $|b|=1$ the interface is consumed with margin ≥ 1 per slot (over-delivered under the corrected booking even after the transverse weakening). On the BFV splitting surface $C_{\Delta_2} = 0$. [Correction gate: the consumed caps and the quoted margins ride the same $\alpha > 0$ tail pricing (Lem. E2a.75) and stand MODULO the transverse re-derivation (Rem. E2a.65).]

B4. *Solves.* Rung 2 at $\sigma_0 = \min(s-9, \frac{9}{2}^-)$, $\sigma_1 = \min(s - \frac{19}{2}, \frac{7}{2}^-)$ (legality margins $2+\delta$, $\frac{3}{2}+\delta$) gives (i), with $C_E^{(b)} = C_E(\sigma_b)[C_{\Delta_2} + \sqrt{T} c_{\mathrm{fol}} C_{v_2} c_\Pi + \sqrt{T} c_{\mathrm{alg}} c_{\mathrm{tr}} c_{\mathrm{mult}} \mathcal{W}^{(2)}_{\partial^2,b}]$.

The mixed $(1,1)$ package and the two-slot step. With a spatial: one derivative off B4; a temporal: the ∂_t-Tice components (no equation conversion at $|a| \le 1$, no trace). All in $(\frac{19}{2}, \frac{7}{2})_{\mathrm{sl}}$ at the rate: $C_E^{(1,1)} = \max_b C_E^{(b)}$. y-uniformity/continuity ride A1–B4 on y-increments; slot symmetry ($W_{g,2}, W_{g',2}$ symmetric biscalars) plus the two-slot continuity lemma (Arendt–Kreuter vector-valued continuity, fence-free) give joint continuity near the diagonal. Lower jets, exhibited: $(0,0)$ at $(8, \frac{11}{2})$ (coefficient margin $\frac{3}{2}+\delta$, profile 4); $(1,0)$ at $(9, \frac{9}{2})$ (margins $\frac{1}{2}+\delta$, 3); $(0,1)$ at $(\frac{17}{2}, \frac{9}{2})$ (margins $1+\delta$, 3). The mixed set $|a|+|b| \le 1 \cup |a| = |b| = 1$ is exactly covered; the pure same-slot $(2,0), (0,2)$ jets are never formed (Rem. E2a.77).

Top rung (three-dimensional Morrey, zero slack, no rounding). Set $\varepsilon_* := \frac{1}{2}\min(\delta, 1) > 0$, $\kappa_* := \min(s - \frac{19}{2}, \frac{7}{2} - \varepsilon_*)$, $\sigma' := \frac{1}{2}(\frac{3}{2} + \kappa_*)$. The package lies in $L_t^\infty H^{\min(s-19/2,\, 7/2-\varepsilon)}(\Sigma_t)$ for *every* $\varepsilon > 0$, in particular at ε_*. (a) $\kappa_* > \frac{3}{2}$: coefficient branch $s - \frac{19}{2} > \frac{3}{2} \iff \delta > 0$ — *the standing index verbatim, margin identically δ: the unique zero-slack inequality*; profile branch $\frac{7}{2} - \varepsilon_* \ge 3 > \frac{3}{2}$, absolute margin $\frac{3}{2}$. (b) $\frac{3}{2} < \sigma' < \kappa_*$, both strict iff $\delta > 0$ (the same fence, not a new demand). (c) t-continuity into $H^{\sigma'}$ (Rung 2) plus Adams–Fournier on the compact three-slice, $H^{\sigma'}(\Sigma_t) \hookrightarrow C^0(\Sigma_t)$, $\sigma' > \frac{3}{2}$ strict, t-uniform norm $c_{\sigma'}^{\mathrm{Mor},3d}(\Sigma)$, plus joint (t,x)-continuity, hence C^0 on the closed slab. Nowhere is $\kappa_* \ge 2$ used, nowhere a four-dimensional Morrey index, nowhere a trace of the solution. As $s \downarrow 11$: $\sigma' \downarrow \frac{3}{2}$ and $c^{\mathrm{Mor},3d}_{\sigma'} \uparrow \infty$ — finite at each $s > 11$, degenerate at the open fence.

Refund audit (single spend). The three-vs-four-dimensional Morrey drop ($2 \to \frac{3}{2}$; refund identity of Rem. E2a.63) is consumed exactly once, at the top-rung extraction on the *solution* side. Sources: Fubini (206), no trace. Data and coefficient restrictions: pay their honest $\frac{1}{2}$ against exhibited margins (A3/A4, B3) — healing, not refund. The slab-C^0 consumers are served by the same single extraction; no second trace exists to re-spend on.

Coincidence extraction and the constant chain. The diagonal values exist, are continuous, and obey (iii) with

$$C^{(2),\mathrm{sl}}_{W}(g_0, K_\Sigma; s) = c^{\mathrm{Mor},3d}_{\sigma'(s)}(\Sigma)\sqrt{Q_* e^{P_* T}}\Big[\sqrt{T} c_{\mathrm{alg}} c_{\mathrm{tr}} c_{\mathrm{mult}} \mathcal{W}^{(2)}_{\partial^2} + \sqrt{T} c_{\mathrm{fol}} C_{v_2} c_\Pi + C_{\Delta_2}\Big], \qquad (208)$$

$\mathcal{W}^{(2)}_{\partial^2} = \max_b \mathcal{W}^{(2)}_{\partial^2,b}$ the Ladder-A chain times the ∂^2-conversion constants. Provenance: $\theta, \Lambda_{U_\omega}$ (Rung 1); Q_*, P_*, T (Tice); $C_E(\sigma)$ (Rung 2); c_{fol} (Rung 3); $c_{\mathrm{mult}}, c_{\mathrm{tr}}, c_{\mathrm{alg}}$ (Rung 4/A4); c_Π (explicit radial integrals); $C_{v_2}, L^{(b)}_S$ and the data legs ((N-a) + Lem. E2a.64; data-leg heights GATED per Rem. E2a.65); C_{Δ_2} ((N-c), $= 0$ on BFV); $c^{\mathrm{Mor},3d}_{\sigma'}$ (Adams–Fournier, $\sigma' > \frac{3}{2}$); the Thm. E2a.51 envelope constant inside $L^{(1)}_S$ and the C_{v_2}-legs. Every factor is finite and ball-uniform at each fixed $s > 11$; $C^0(\Sigma)$ iff $s > 11$, zero slack, the commutator the pin. □

Remark E2a.68 (Calibration: the transverse data-cap column is load-bearing). The same machinery at $N{=}1$ dies, as it must: seed residual $r^4 \log r$; the ∂_y-data legs $\rho^3 \log \rho \in H^{7/2^-}$ give $\partial_y W \in (\frac{11}{2}, \frac{7}{2})_{\mathrm{sl}}$, hence $\partial^2 \partial_y W \in (\frac{15}{2}, \frac{3}{2})_{\mathrm{sl}}$ and the mixed $(1,1)$ value at $(\frac{15}{2}, \frac{3}{2})_{\mathrm{sl}}$: profile $\frac{3}{2}^-$ vs three-dimensional Morrey $> \frac{3}{2}$ — an open-vs-open mismatch, dead at *every* s; in four dimensions the same chain caps at H^{2^-} vs Morrey > 2, reproducing the data wall exactly from the energy side. At $N{=}2$ the healed profile $r^6 \log r$ lifts every cap two orders; the caps clear all demands and the *coefficient* column pins $s > 11$. A ladder written with the radial data heights $\alpha + \frac{3}{2}$ (Rem. E2a.65) would overstate the $N{=}1$ profile columns by one and misplace the wall; the transverse height $\alpha + \frac{1}{2}$ previously printed at Lem. E2a.64 is what made the calibration read as exact. [Correction gate: this remark's profile numbers are the old $\alpha > 0$ pricing — the honest $N{=}1$ ∂_y-data leg is $H^{3/2^-}$, not $H^{7/2^-}$, so the "reproduce the data wall exactly" numerology dissolves, while the $N{=}1$ DEATH verdict stands *a fortiori* (lower availability dies harder). The $N{=}2$ "caps clear all demands" clause is what Rem. E2a.66 refutes at the cone crossing; it stands only MODULO the transverse re-derivation (Rem. E2a.65).]

E2a.15 The locked-4d variant at $s > \frac{23}{2}$

The slice ladder of §E2a.14 spends the dimensional refund of Rem. E2a.63 once, on the solution side, to reach the three-dimensional Morrey threshold $\frac{3}{2}$ and hence $s > 11$. The companion *locked-4d* variant does *not* spend that refund: it is derived and audited entirely in the four-dimensional slab register (all products, traces and constants slab-priced), quotes the fence at the four-dimensional Morrey demand $s - \frac{19}{2} > 2$, and therefore reads $s > \frac{23}{2}$. It is the register in which the physical anchors are locked (runs 5–10), and it is stated here as the honest fallback quote of the same estimate; the two calculi are never mixed inside a single derivation.

Lemma E2a.69 ($N{=}2$ two-slot difference-system energy estimate, locked-4d register; the mixed $(1,1)$ coincidence value at the rate for $s > n/2 + \frac{19}{2} = \frac{23}{2}$). *Under the hypotheses of Lem. E2a.67 but with the standing index raised to*

$$s > \tfrac{n}{2} + \tfrac{19}{2} = \tfrac{23}{2},$$

$D_2 = W_{g,2} - W_{g',2}$ *solving* (202), *the conclusions (i)–(iii) of Lem. E2a.67 hold verbatim with the slice registers* $(\cdot)_{\mathrm{sl}}$ *replaced throughout by the four-dimensional slab registers and the top* *profile caps raised by one half-order — explicitly, with the solve levels* $\sigma_0 = \min(s-9, 5^-)$, $\sigma_1 = \min(s-\frac{19}{2}, 4^-)$*, the mixed* $(1,1)$ *package lands in* $L_t^\infty H^{\min(s-19/2,4^-)}(\Sigma_t)$*, and the fence is pinned by the commutator branch at* $H^{\min(s-19/2,4^-)}$*:* $C^0(\Sigma)$ *iff* $s-\frac{19}{2} > 2$ *iff* $s > \frac{23}{2}$*, the profile column clearing by* 2^-*.* [Correction gate: the raised profile caps $5^-, 4^-$ and the "clearing by 2^-" clause ride the same gated data-cap columns — the slab-transverse count across the cone is the same $u^k \log|u|$ in one codimension — and stand MODULO the transverse re-derivation (Rem. E2a.65); the coefficient pin $s > \frac{23}{2}$ is untouched.]

Grade: THEOREM end-to-end over U_ω, at $s > \frac{23}{2}$, GIVEN the analytic slice chain of Parts 1–3 (Thm. E2a.51), and consuming (N-a)/(N-c) of §E2a.17 at their audited grades. *Same three scope pins as Lem. E2a.67.*

Proof. Identical to the proof of Lem. E2a.67 through Rung 4 and both ladders, with two changes of register discipline. First, the integer Tice rung (204) is run to $k = 7$ (demand $s \geq \frac{15}{2}$, margin 4 below $s > \frac{23}{2}$) so that the fractional interpolation window of Rung 2 covers the Ladder-A ceiling 6^- for every $s > \frac{23}{2}$, closing a latent range mismatch of the earlier ladder at $s \geq 13$. Second — and this is the only substantive difference — the extraction is performed in the locked four-dimensional slab bookkeeping: the mixed $(1,1)$ package sits in $H^{\min(s-19/2,4^-)}(\text{slab})$, and the coincidence value is read by the four-dimensional Morrey embedding $H^t(\text{slab}) \hookrightarrow C^0(\text{slab})$, $t > n/2 = 2$, followed by free restriction of a continuous function. Mechanically the codimension-one slice embedding needs only $\frac{3}{2}$; that unspent $\frac{1}{2}$ is exactly the dimensional refund of Rem. E2a.63, which is *not* spent here and is what Lem. E2a.67 spends to reach $s > 11$. No step mixes the two calculi. The constant chain is (208) with $c_{\sigma'}^{\text{Mor},3d}$ replaced by the four-dimensional $c_{s-19/2}^{\text{Mor},4d}(K_\Sigma)$ and every slab-priced factor unchanged; Branch 2 (the commutator: two irreducibly transverse ∂_x on the solved $N=2$ finite part at register $s-7$, one ∂_y at $+\frac{1}{2}$ by Thm. E2a.51) is the pin, Branch 1 ($\Box_g v_2 \sim \partial^8 g$) pinning only $s > \frac{21}{2}$. $C^0(\Sigma)$ iff $s > \frac{23}{2}$. □

Remark E2a.70 (Why both registers are carried). The two ladders are the two admissible bookings of the paper's fence paragraph (Options A and B there), both over the *same* analytic class U_ω: Lem. E2a.67 buys the sharper slice register $s > 11$ by spending the dimensional refund; Lem. E2a.69 is the locked-4d quote $s > \frac{23}{2}$ that carries no refund and in which the physical anchors are locked. Both are method-relative upper bounds for the fence, not sharp-from-below. The analytic device of Parts 1–3 licenses the $+\frac{1}{2}$ ∂_y booking in *both*; the choice between $s > 11$ and $s > \frac{23}{2}$ is a choice of extraction register, not of hypothesis.

E2a.16 Anchor restatements and the collar-margin attainment notes

Promoting the slice register from the paper's standing $s > n/2+6 = 8$ to the ladder value $s > 11$ requires restating the physical thermodynamic anchors at the promoted row. This subsection records their restated grades, distinguishing the *unconditional* core (the $(T1)$-free anchors, THEOREM at $s > 8$) from the anchors that revive at $s > 11$ only *with* the analyticity hypothesis (hence a class-narrowed THEOREM, not unconditional). It also records the coercivity-floor attainment note the two ladders' Rung 1 depends on, now proved at THEOREM grade by the third run of the Part 1 collar-margin engine (Lem. E2a.72).

Remark E2a.71 (Anchor rows, restated). (a) *The* $(T1)$*-free core — unconditional THEOREM.* The anchors HB2 and HB4 Part-A are established by $b = 0$ chains that use no spectator ∂_y booking and hence no averaging rung: they hold at $s > 8$ *verbatim*, independent of Option A/B and of the analytic restriction. *Grade: THEOREM ((*$T1$*)-free,* $s > 8$*)* — the unconditional core, independently re-verified.

(b) *The anchor rows R0/HB1/HB3 — THEOREM at their rows,* $(T1')$*-modulo under Option A.* R0 (the single-metric $N=1$ finite-part regularity) and the physical anchors HB1, HB3 hold at their stated rows (R0/HB1 slice 8 modulo $T1'$; HB3 8.5/8) via the verified restatement of the anchor appendix. Under the analytic route their revival at the *promoted* row $s >$

11 rides Thm. E2a.51 (the analytic $T1'$ over U_ω): given the analyticity hypothesis they discharge at $s > 11$, but this rides the same U_ω restriction and is therefore a *class-narrowed* THEOREM, not an unconditional one. *Grade: THEOREM at the anchor row; $s > 11$ revival is ($T1'$, analytic)-modulo.* These carry the flag (modulo $T1'$, analytic) wherever quoted at the promoted row.

The historical "theorem-modulo" blanket on the anchor imports of the bulk clause was localized to R0/HB1/HB3; HB2/HB4 Part-A were established ($T1$)-free at that time and are not modulo anything. No anchor is cited at an uncorrected strength anywhere in either ladder.

Lemma E2a.72 (Collar-margin coercivity floor: the attainment note, proved)**.** *Fix the standing gauge data of the two ladders: the compact causal slab $K_\Sigma \subset K''$ foliated by $\{\Sigma_t\}_{t\in[0,T]}$ in the fixed finite atlas (Lem. E2a.67), with the anchor nondegeneracy that dt and ∂_t are both g_0-timelike on K_Σ (Chruściel–Grant nondegeneracy of the anchor [222]; equivalently $\theta(g_0) > 0$, a one-point statement at the fixed g_0 — the exact analogue of Thm. E2a.17's "not everywhere ϕ''-flat"). Let $d_\omega > 0$ be the standing determinant floor of Part 1, normalize $\varepsilon_a \le 1$, and write $a_g := a_g^{jk}/a_g^{00}$ for the Rung-1 quotient form, a real symmetric $(n{-}1)\times(n{-}1)$ matrix at each real slab point. Then, with the existence constants (fixed by $(g_0, \rho_a, \mathrm{atlas}, n)$ alone)*

$$2d_{00} := \min_{\text{charts}} \min_{K_\Sigma} a_{g_0}^{00} > 0, \qquad \theta_0^{\text{anch}} := \min_{\text{charts}} \min_{K_\Sigma} \lambda_{\min}(a_{g_0}) > 0,$$

$$B_{\text{inv}} := \frac{n!\,\big(\|g_0\|_{\mathcal{K}_{\rho_a}} + 1\big)^{n-1}}{d_\omega}, \qquad C_{\text{inv}} := n\,B_{\text{inv}}^2, \qquad R_a := \max_{K_\Sigma} \|a_{g_0}^{jk}\|_{\text{op}}, \qquad C_a := \frac{C_{\text{inv}}}{d_{00}} + \frac{R_a\,C_{\text{inv}}}{2d_{00}^2},$$

and under the two further caps $\varepsilon_a \le d_{00}/C_{\text{inv}}$ and $\varepsilon_a \le \theta_0^{\text{anch}}/(2C_a)$ (imposed with Part 1's caps; harmless, since every export is a $\forall$-statement over U_ω and survives shrinking ε_a, and $g_0 \in U_\omega$ keeps the class nonempty), every $g \in U_\omega$ satisfies, at every point of the slab,

$$a_g^{00} \;\ge\; d_{00} \;>\; 0, \qquad \lambda_{\min}(a_g) \;\ge\; \theta_0^{\text{anch}} - C_a\varepsilon_a \;\ge\; \tfrac{1}{2}\,\theta_0^{\text{anch}}.$$

Consequently $\theta = \min\big(1, \inf_{K_\Sigma\times U_\omega} \lambda_{\min}(a_g)\big) \ge \theta_* := \min\big(1, \tfrac12\theta_0^{\text{anch}}\big) > 0$: the Rung-1 floor is ball-uniform by the collar margin, not by attainment on the ball. *No infimum over the H^s-dense, H^s-interior-empty (hence non-H^s-closed) U_ω is attained or needed — a lower bound on an infimum requires no minimiser; this is the honest replacement of the false "H^s-closed ball, infimum attained" clause. Only $1/\theta \le 1/\theta_*$ is consumed downstream; $d_{00}, \theta_0^{\text{anch}}, C_{\text{inv}}, C_a$ are existence constants, not numerically certified, and are* not quoted downstream. ***Grade: THEOREM*** *over U_ω, relative to the standing gauge data above and the Part 1 determinant floor — the third run of the Part 1 engine (anchor floor by finite-dimensional compactness at the fixed g_0, collar-margin propagation), after m_0^{anch} (Lem. E2a.14) and δ_0 (Lem. E2a.62).*

Proof. Step 0 (algebraic coefficients; the cap controls the slab). Chartwise the principal part of $\Box_g$ is $g^{\mu\nu}\partial_\mu\partial_\nu$, so $a^{00} = -g^{00}$ and $a^{jk} = g^{jk}$ are entries of $g^{-1} = \mathrm{adj}(g)/\det g$ — algebraic in the entries of g with denominator floored by d_ω. The slab lies in the collar, $K_\Sigma \subset K'' \subset \mathcal{K}_{\rho_a}$, and g is real at real points, so $\sup_{K_\Sigma} \max_{a,b} |g_{ab} - (g_0)_{ab}| \le \|g - g_0\|_{\mathcal{K}_{\rho_a}} < \varepsilon_a$: the ε_a-cap controls exactly the slab C^0 norm the coefficient map consumes.

Step 1 (anchor floors at the fixed g_0; finite-dimensional Euclidean compactness only, as in Lem. E2a.14). $a_{g_0}^{00}$ is continuous and pointwise positive on the slab (dt g_0-timelike), so $2d_{00} > 0$ on the compact K_Σ per the finite atlas. In lapse/shift form $g_0 = -N^2dt^2 + h_{jk}(dx^j + \beta^j dt)(dx^k + \beta^k dt)$ the inverse identities give $a^{00} = N^{-2}$ and $a_{g_0} = N^2h^{jk} - \beta^j\beta^k$; for a spatial covector $\xi \ne 0$, Cauchy–Schwarz in the h^{-1} inner product gives $\xi{\cdot}a_{g_0}\xi \ge (N^2 - |\beta|_h^2)|\xi|_{h^{-1}}^2 > 0$ pointwise, since $N^2 - |\beta|_h^2 = -g_0(\partial_t, \partial_t) > 0$ (∂_t g_0-timelike). The continuous positive function $(t, x, e) \mapsto e{\cdot}a_{g_0}(t, x)\,e$ on the compact $K_\Sigma \times S^{n-2}$ (slab $\times$ unit covector sphere, per chart, g_0 *fixed*) attains a positive minimum: $\theta_0^{\text{anch}} > 0$. No function-space compactness is used, and — unlike Lem. E2a.14

— no degenerate stratum need be excised: the anchor nondegeneracy makes it empty on the slab.

Step 2 (collar-Lipschitz propagation, non-circular). For $g \in U_\omega$: $\|g\|_{C^0(K_\Sigma)} \le \|g_0\|_{\mathcal{K}_{\rho_a}} + 1$ and $|\det g| \ge d_\omega$, so the adjugate bound gives $\|g^{-1}\|_{\mathrm{op}} \le B_{\mathrm{inv}}$, and the resolvent identity $g^{-1} - g_0^{-1} = -g^{-1}(g - g_0)\, g_0^{-1}$ gives $\|g^{-1} - g_0^{-1}\|_{\mathrm{op}} \le C_{\mathrm{inv}}\varepsilon_a$ on the slab. Hence first $a_g^{00} \ge 2d_{00} - C_{\mathrm{inv}}\varepsilon_a \ge d_{00}$ under the first cap (a matrix entry is bounded by the operator norm); and second, since a principal-submatrix compression does not increase the operator norm,

$$a_g - a_{g_0} = \frac{a_g^{jk} - a_{g_0}^{jk}}{a_g^{00}} + a_{g_0}^{jk}\,\frac{a_{g_0}^{00} - a_g^{00}}{a_g^{00}\,a_{g_0}^{00}}, \qquad \|a_g - a_{g_0}\|_{\mathrm{op}} \le C_a\,\varepsilon_a.$$

The constant chain $d_\omega \to B_{\mathrm{inv}} \to C_{\mathrm{inv}} \to d_{00} \to C_a$ is well founded in the fixed data; no cap's right-hand side depends on ε_a (the normalization $\varepsilon_a \le 1$ breaks that loop).

Step 3 (Weyl). a_g, a_{g_0} are real symmetric on the real slab, so Weyl's perturbation theorem ($\lambda_{\min}$ is 1-Lipschitz in the operator norm; Bhatia, *Matrix Analysis*, Cor. III.2.6) gives $\lambda_{\min}(a_g) \ge \theta_0^{\mathrm{anch}} - C_a\varepsilon_a \ge \frac{1}{2}\theta_0^{\mathrm{anch}}$ under the second cap, for every $g \in U_\omega$ at every slab point.

Step 4 (the infimum, without attainment). Every element of $\{\lambda_{\min}(a_g(t,x)) : g \in U_\omega,\ (t,x) \in K_\Sigma\}$ is $\ge \frac{1}{2}\theta_0^{\mathrm{anch}}$, so its infimum is too; hence $\theta \ge \theta_*$. The only minimum ever *attained* is over the finite-dimensional compact $K_\Sigma \times S^{n-2}$ at the single fixed g_0 — a one-point statement in function space. That U_ω has empty H^s-interior is irrelevant to a lower bound on an infimum; ball-uniformity is by the collar margin, never by ball compactness (Rem. E2a.15). □

E2a.17 The data legs: (N-a), (N-c), and the named residues

The two ladders consume two data inputs at the diagonal collar: the $N{=}2$ parametrix package (N-a) (the source and single-metric data of Ladder A) and the difference data (N-c) (the Σ_0-data of Ladder B). This subsection states each at its audited grade. The gluing and short-slab truncation are THEOREM; the difference-data interface is PLAUSIBLE (theorem-structure with verified margins); and the state-leg envelope and the long-slab traversal are *named residues* that this appendix does not close. None of the PLAUSIBLE/named items is ever cited at THEOREM grade by either ladder.

The parametrix package (N-a). For $g \in U_\omega$ the $N{=}2$ truncated Hadamard finite part $W_{g,2} = \omega_g^{(2)} - H_g^{(2)}$ is built from the local geometric biscalars $\sigma_g, \Delta_g^{1/2}$ and the Hadamard coefficients v_0, v_1, v_2, which are universal curvature polynomials — jets of g in $H^{s-2}, H^{s-4}, H^{s-6}$ — by the transport recursion of [13, Thm. 2.1] (equivalently the Decanini–Folacci transport representation; the recursion is the same, and we cite the Hadamard-coefficient key already in the paper). The consumed registers are $v_2 \sim \partial^6 g \in H^{s-6}$ and $\Box_g v_2 \sim \partial^8 g \in H^{s-8}$, and the residual has the normal form

$$\text{(N-a-NF)} \qquad S_2' = (\Box_{g'} v_2')\,\sigma_{\mathrm{geo}}^2 \log\sigma_{\mathrm{geo}} + R_{\mathrm{sm}},$$

$\Box_{g'} v_2'$ transport-averaged of register H^{s-8}, profile $r^4 \log r$ (pair $(8,6)$; correction gate: the profile column 6 is the vertex-zone height, Rem. E2a.65), R_{sm} strictly softer. The transport average is *linear* in its curvature-polynomial argument, so every estimate carries the Lipschitz rate $\|h_g - h_{g'}\|_{H^a} \le C_{\mathrm{poly}}\|g - g'\|_{H^s}$ with $C_{v_2} = \frac{1}{12}\kappa_\Delta C_{\mathrm{poly}}$; the single-metric data of $W_{g,2}$ on Σ_0 are finite at the transverse leg heights of the corrected Lem. E2a.64 (honest seed column: Rem. E2a.66; the printed A3 column is gated). *Grade: the transport core and its registers are the* (N-a) *(twice-verified); the family-uniform finite-regularity* construction *of v_2 over the ball is folklore-writable but not in print as a ball-uniform statement — the same status the paper already grants R0's construction — so* (N-a) *as a family-uniform package is a NAMED INPUT, not reproved here.*

Lemma E2a.73 (Collar gluing of the N=2 parametrix; defect bookkeeping; honest multiplicity). *Let $\{U_a\}_{a\le N(K)}$ be the ball-uniform convex-normal cover of $\widetilde{K}$ (Chen–LeFloch / Cheeger–Gromov–Taylor injectivity bound; convexity radius $r_{\text{conv}}(n, K_0, V_0) > 0$ from g_0-data and $\sup_{g\in U_\omega} \|g\|_{H^s}$), $\{\chi_a\}$ a subordinate partition of unity with $\|\chi_a\|_{C^2} \le L_\chi$, and on the collar $V_\delta = \{d_{g_0}(x, y) < \delta_c\}$ set $\theta_a = \chi_a(x)\chi_a(y)/\sum_b \chi_b(x)\chi_b(y)$, $H^{(2)}_g = \sum_a \theta_a Z^{(2),a}_g$, $W_{g,2} = \omega^{(2)}_g - H^{(2)}_g$. Then: (i) θ_a is smooth with $\sum_a \theta_a \equiv 1$ and $\|\theta_a\|_{C^2} \le c(N(K), L_\chi)$, once $\delta_c \le (2m(K)^2L_\chi)^{-1}$; (ii) on overlaps $Z^{(2),a}_g - Z^{(2),b}_g$ is a* smooth *curvature biscalar (the $1/\sigma$ pole and $\sigma^k \log\sigma$ branch are patch-independent geometric biscalars and cancel; with one global length scale the* log *term vanishes), worst content $v_2 \sim \partial^6 g \in H^{s-6}$, no profile constraint; (iii) the defect source $S^{\text{glu}}_{g,2} = \sum_a \theta_a S^{(a)}_{g,2}$ + (defect legs) books $(8, \infty)$ in the defect legs and rides the main rung $(8, 6)$, Lipschitz at the rate in the difference; (iv) every constant scales* linearly *in the overlap multiplicity $m(K) = \operatorname{ess\,sup}_x \#\{a : x \in \operatorname{supp}\chi_a\} \le N(K) < \infty$ (*no Besicovitch sharpening is claimed *— only finiteness and ball-uniformity are consumed); (v) the fixed-g residual source is collar-supported by construction.* Grade: THEOREM *(parts (i)–(v); no Besicovitch).*

Lemma E2a.74 (Short-slab domain-of-dependence truncation). *Let $c_* = c_*(\Lambda_{U_\omega}, \theta)$ be the ball-uniform coordinate light speed of the first-order reduction and $T_c := \delta_c/(4c_*)$. Fix y, a short slab $[t_0, t_0 + T_c]$, and let $u(\cdot, y)$ solve $(\Box_g + m^2)u = F(\cdot, y)$ there (F arbitrary). Truncating source and data by a cutoff ψ ($= 1$ on $d_{g_0}(\cdot, y) \le \delta_c/2$, $= 0$ off $d \le \frac{3}{4}\delta_c$) gives $u = \tilde u$ on $\{d \le \delta_c/4\} \times [t_0, t_0 + T_c]$ (finite speed + Kato/Tice uniqueness), so the Tice estimate controls the collar norms of u while consuming only the collar norms of F and of the data — no $[\Box_g, \psi]$-commutator on u is ever formed. Constants: the Tice factor $\sqrt{Q_* e^{P_* T_c}}$, ball-uniform.* Grade: THEOREM.

Lemma E2a.75 (N=2 difference data at the corrected ∂_y-slotted levels). *Under the hypotheses of Lem. E2a.73, with Σ_0 the initial slice and K_y a compact spectator neighbourhood, the ∂_y-slotted Cauchy data of $D_2 = W_{g,2} - W_{g',2}$ are Lipschitz at the rate in exactly the norms the two ladders consume: for $j \in \{0, 1\}$,*

$$\sup_{y\in K_y} \Big[\big\| \partial^j_y D_2(\cdot, y)|_{\Sigma_0} \big\|_{H^{\min(s-7,\,6^-)}(\Sigma_0)} + \big\| \partial_0 \partial^j_y D_2(\cdot, y)|_{\Sigma_0} \big\|_{H^{\min(s-8,\,5^-)}(\Sigma_0)} \Big] \le C_{\Delta_2}(g_0, \Sigma_0, K_y)\, \|g - g'\|_{H^s},$$

together with the single-metric uniform analogue for $W_{g',2}$. The coefficient-difference legs (P1) and profile-difference legs (P2) are PROVED, with margins verified against both ladders' demands (traced ∂_y-slotted coefficient legs at exactly $\frac{1}{2}$, profile legs $\ge \frac{1}{2}$) [Correction gate: the slice caps $6^-, 5^-$ and the profile-leg margins were priced with the old $\alpha > 0$ tail rule and stand MODULO the transverse re-derivation, Rem. E2a.65]*; on the BFV splitting surface $C_{\Delta_2} = 0$ identically (RCE pullback for the states, local-geometry agreement for the glued parametrices; N-independent). The state-leg residue is REDUCED to the envelope* (E-env) *below.* Grade: PLAUSIBLE *(theorem-structure with verified margins; the end-to-end write-out is the data step, and the state legs rest on the un-proved* (E-env)*). It is consumed by both ladders' rung B3 only at the transverse-weakened caps, where every demand clears by ≥ 1 per slot; it is NEVER cited at THEOREM grade.*

Remark E2a.76 (The named residues: (E-env), #A, (D0), #B' — printed, not closed). The following data-side items are *not* theorems and are *not* closed in this appendix; they are the residues of the N=2 difference-data write-out and gate the data-side (interacting / N-c) programme, *not* the analytic slice chain of Parts 1–3 (which cites only the R-interface and the analytic ODE package, never these). They are stated here so that any consumer that feeds one is on notice.

- (E-env) — *the state-leg kernel envelope.* $|\partial^j K_2[g, g']| \le C_K r^{3-j} \|g - g'\|_{H^s}$, $j \le 6$ (the N=2-healed ceiling; at N=1 the ceiling is $j < 3$, the data wall). *Given* (E-env) the state legs of Lem. E2a.75 follow by the convolution machinery (total kernel-jet demand $\le 6 < 7$); (E-env) itself is *unproven at finite regularity* — the data-side twin of the analytic curved rung. *Grade: PLAUSIBLE* (its arithmetic shadow — kernel powers, ceiling, jet counts — is verified; nothing downstream of it is missing).

- #A — *the long-slab traversal.* By Lem. E2a.74 a short slab suffices *provided* its data are supplied at the demand levels; the zero-data route across the perturbation region P does not supply them, because over a long slab the domain of dependence widens to $O(1)$ and consumes the difference field at collar-exterior points where D_2 carries the displaced-cone singular difference. The short-slab Lem. E2a.74 is THEOREM, but #A itself is *discharged if and only if* (E-env) holds; it rides (E-env). *Grade: named gate, (E-env)-conditional.*
- (D0) — *the seed-data input.* The Cauchy-data levels the single-metric Ladder A consumes on a seed-adjacent slice; open at the $b = 1$ field slot. It gates the single-metric write-out $W_*^{(2)}$. *Grade: named residue — upgraded by the seed re-derivation to a confirmed OBSTRUCTION of the printed route (NEGATIVE-RESULT, sharp): the seed supplies $(\frac{7}{2}, \frac{5}{2}, \frac{5}{2}, \frac{3}{2})^-$, the b=1 field slot $H^{5/2^-}$ sharp versus consumed demand $> \frac{5}{2}$ at every $s > 11$ (Rem. E2a.66). The gate does NOT lift; repair is OPEN ($N \geq 4$ truncation see-saw, PROVISIONAL, source columns to be re-derived — no repaired fence value is quoted; or a consumer rewiring off slice-Sobolev tails). The (D0) port's closure display (App. E2a) is gated accordingly; its lever and w_0 lemmas stand.*
- #B' — *the off-collar single-metric variant.* The kernel variant needed to run the single-metric write-out *across patches P*. *Grade: named gate.*

These are the Group-II items of the audit: verified ORTHOGONAL to the slice THEOREM subtree, on the modulo-list because they are non-THEOREM, and named here so no consumer promotes them silently.

E2a.18 Clause (ii) assembly: the mixed jets and the consumer scalar $\rho_{\mathrm{kin}}^{(2)}$

The two ladders deliver the mixed $(1,1)$ coincidence value of D_2 at the Lipschitz rate (Lem. E2a.67(iii), Lem. E2a.69(iii)) and the single-metric sups $W_*^{(2)}$. Assembling these into clause (ii) of Hyp. E.2 (the finite-part family $C_W(K), W_*(K)$) is where the honest boundary of this appendix lies: the fence value $s > 11$ is a THEOREM-grade result, but clause (ii) *as a whole* is PLAUSIBLE (REDUCED, not proved), because the family-uniform finite-regularity Hadamard construction is not in print and the full multi-index constant-tracking is not executed. Clause (ii) is therefore carried, here and in the paper, as the NAMED INPUT (E2a) it is; it is *never* cited at THEOREM.

Remark E2a.77 (The mixed jets, and why the pure same-slot jets are excluded). The consumer object is the minimally-coupled point-split $\langle \widehat{T_{\mu\nu}} \rangle$, whose every term distributes *one* derivative per slot; the jet set it consumes is therefore $|a|+|b| \leq 1$ together with the mixed pair $|a| = |b| = 1$, exactly as in clause (ii) of Hyp. E.2. The pure same-slot jets $(2,0)$ and $(0,2)$ are *never formed*, and this is not a convenience: at finite metric regularity the Hadamard split is necessarily truncated, the truncation residual is C^1 but not C^2 *across* the light cone, so the full $C^2(K \times K)$ norm is unavailable and the pure same-slot diagonal jets of the residual *diverge*. Only the mixed diagonal jets remain finite. The A-contraction $A_{\mu\nu}^{ab} = \delta_{(\mu}^{a} \delta_{\nu)}^{b} - \frac{1}{2} g_{\mu\nu} g^{ab}$ (so $A_{00}^{00} = \frac{1}{2}$) projects onto exactly this mixed set: the consumer scalar $\rho_{\mathrm{kin}}^{(2)} = A_{00}^{ab} [\partial_x^a \partial_y^b D_2]_{\mathrm{diag}}$ reads only mixed jets, and the ladder supplies it in $C^0(\Sigma)$ at the rate. No step of this appendix ever forms or claims a pure same-slot second jet, and no statement is made about "all second jets" or the full $C^2(K \times K)$ norm.

Remark E2a.78 (The assembled constant C_W, and its honest grade). On a single convex-normal chart the mixed $(1,1)$ difference jet closes at $s > 11$ via the N=2 truncated finite part, and the clause-(ii) constant assembles as

$$C_W(K) = N_{\mathrm{geo}}(K)\, c^{\mathrm{Mor}}(K) \sqrt{Q_* e^{P_* T(K)}} \big[c_{\mathrm{mult}}(s)\, W_*^{(2)}(K) + C_{\mathrm{trans}}(K) + C_{\mathrm{data}}(K) \big],$$

each factor named, finite and ball-uniform: the covering/injectivity factor $N_{\text{geo}}(K) \le c_n$ globalizing the per-chart estimate over compact K (overlap multiplicity, no Besicovitch sharpening, Lem. E2a.73); the Morrey constant $c^{\text{Mor}}(K)$ (Adams–Fournier, three- or four-dimensional per register); the Tice factor $\sqrt{Q_* e^{P_* T(K)}}$ (Lem. E2a.67 Rung 1, [206]); the spectator sup $W_*^{(2)}(K)$ (the single-metric N=2 jets, C^0 at $s > 10$); the difference-data constants $C_{\text{trans}}, C_{\text{data}}$ (both $= 0$ on the BFV surface; Lem. E2a.75). The fence $s > 11$ is verified by two independent derivative-budget methods, each calibrated against two established ground truths (R0's single-metric C^0 at $s > 8$; the N=1 difference jet failing at every s), and is $+3$ over the paper's standing $s > n/2 + 6 = 8$ — a fence trade the author must pay to fire the flip.

<u>*Grade*</u>*: PLAUSIBLE (clause (ii) is REDUCED, not proved).* The fence value $s > 11$ is THEOREM-grade; the constant structure is written and internally consistent; but two honest gaps are *not closed* here:

(a) the N=2 finite-regularity Hadamard-parametrix *construction* with quantitative v_2, family-uniform in g over the ball — folklore-writable, NOT in print as a ball-uniform statement (the same status the paper already grants R0's construction);
(b) the exact multi-index constant-tracking of every product through the N=2 commutator source and the Tice estimate — single-paper-sized, not executed here.

Therefore clause (ii) stays a NAMED INPUT: it is C_W, W_* of Hyp. E.2, consumed by Lem. E.6, Thm. E.7 and Prop. E.4. This appendix's contribution to clause (ii) is the ladder that closes the fence at $s > 11$ over U_ω and the assembled constant structure — *not* a proof of the clause. **Clause (ii) is never cited at THEOREM grade.**

E2a.19 What Part 4 establishes, and its place in the appendix

Part 4 publishes, at THEOREM grade over U_ω: the trace-audit trio (Lem. E2a.62, the refund identity Rem. E2a.63, the Fubini source consumption (206)); the sharp transverse restriction Lem. E2a.64 *as corrected* (honest $\alpha > 0$ tail height $\frac{\alpha}{2} + \frac{1}{2}$; both printed errors named in Rem. E2a.65; the seed record — existence THEOREM, honest column, sharp counterexample — Rem. E2a.66); the slice-register ladder Lem. E2a.67 at $s > 11$; and the locked-4d variant Lem. E2a.69 at $s > \frac{23}{2}$ — each a THEOREM end-to-end over U_ω *given* the analytic slice chain of Parts 1–3 (Thm. E2a.51), whose "given" is discharged because that chain is itself a THEOREM, and with the two ladders' profile/data-cap columns carried GATED per Rem. E2a.65 (coefficient columns and both fence pins untouched). It publishes, at their audited sub-theorem grades and never promoted: the gluing and short-slab truncation (Lem. E2a.73, Lem. E2a.74, THEOREM); the difference-data interface (Lem. E2a.75, PLAUSIBLE); the named residues (E-env)/#A/(D0)/#B' (Rem. E2a.76); the collar-margin coercivity floor (Lem. E2a.72, THEOREM — the former attainment note, proved); and clause (ii) itself (Rem. E2a.78, PLAUSIBLE — the NAMED INPUT (E2a), clause (ii)).

The one honest boundary this part draws, repeated so it cannot be missed: the fence value $s > 11$ is a THEOREM, but clause (ii) — and hence the fused first-law statement Prop. E.4 of the paper — is *not* a theorem; it is a theorem MODULO the finite named list {clause (ii) C_W/W_* PLAUSIBLE (Rem. E2a.78); clause (iii) Sanders constant-tracking, named-un-derived} (the clause (i) difference form, formerly on this list, is now a ported theorem, App. E2a). The appendix therefore titles itself as the analytic slice chain over U_ω (the THEOREM of Parts 1–3 plus the ladders of Part 4), and states the first law as a corollary MODULO those three clauses — never as an "*E2a-ω is a theorem*" claim.

E2a.20 Clause (i): the singular part (single-metric theorem; difference form now a theorem, ported — App. E2a)

Clause (i) of Hyp. E.2 asserts the dual-Lipschitz bound on the singular Hadamard part,

$$\big|(H_g - H_{g'})(u \otimes v)\big| \;\le\; C_H(K)\,\|g-g'\|_{H^s}\,\|u\|_{H^{s-2}}\,\|v\|_{H^{s-2}}, \qquad u, v \text{ supported in } K, \quad (209)$$

ball-uniform for $g, g' \in U_\omega$. Its status splits into a *single-metric* half that is a theorem in the corpus, and a *difference* half, now a ported theorem (App. E2a, item (M3)).

Lemma E2a.79 (Singular jet, single-metric, $s > n/2 + 6$)**.** *Let $g \in U_\omega$ and $s > n/2 + 6$. The* $(1, 1)$ *coincidence jet of the $N = 1$ Hadamard finite part W_g is C^0 on the compact K; more generally the singular part H_g acts by* (209) *at a* fixed *metric with a finite $C_H(K, g)$, on the standard Hadamard-parametrix footing [13, 16, 276]. The threshold is $s > n/2 + 6$ (equivalently $s > 8$ at $n = 4$): the coincidence jet inherits the Sobolev order of the residual wave source through the $O(1)$ elliptic symbol $\xi^2/(\xi^2 + m^2)$, and the jet-*function *fails to be C^0 at $s \le n/2 + 6$.*

Provenance and grade. This is the established ground truth **GT1** of the difference-fence analysis: the single-metric $(1, 1)$ jet of W_g is C^0 at $s > 8$ and fails at $s \le 8$ (two independent symbol-counting methods, calibrated to GT1/GT2, agree). It is a **theorem** at the single-metric level (grade established in the earlier record; the difference-fence recomputation reproduces it as GT1). The construction of the parametrix itself is the standard local Hadamard/Riesz construction inside one convex normal chart, cited – not reproved – from [13, 276], exactly as the paper's Setup already does. *Grade: THEOREM (single metric, $s > n/2 + 6$).* □

Remark E2a.80 (The difference form of clause (i): ported to a theorem (App. E2a)). The bound (209) that clause (i) actually asserts is the *difference* $H_g - H_{g'}$, uniform over the *family* $U_\omega \times U_\omega$. What Lemma E2a.79 establishes is the single-metric jet; the family-uniform difference bound is *not* separately derived in the companion corpus. It is packaged as part of the named input (E2a), alongside clause (ii), on the same [220, 16] microlocal smoothness footing. Two mitigating facts, both verified in the dependency audit, keep this the lowest-risk item of the modulo-list: (a) no quantitative slice-register consumer of the T2/T3 ladders exercises the singular difference bound – it feeds only the pivot Thm. E.7, so it never enters the analytic slice-chain theorem of §E2a.22 below; and (b) at a fixed metric the estimate is a theorem (Lemma E2a.79). It nonetheless remains a named input for the *difference* statement, and is carried on the modulo-list as item (M3) (§E2a.24). *Grade: THEOREM (single metric here; the family-uniform difference form ported by full re-derivation as App. E2a, item* (M3)*).* This companion appendix establishes the single-metric jet; the family-uniform difference form is promoted to a theorem on import of the port (App. E2a), and clause (i) is retained in the hypothesis as a package for its microlocal role.

E2a.21 Clause (iii): the QEI-constant uniformity (a named input, at its audited grade)

Clause (iii) asserts that, for fixed smearing data (t, f, F), the constants $c_S(g, t, F), C_S(g, t, f, F)$ of Sanders' quantum-energy-inequality stress-tensor bound [207, 208] (entering (106)) are locally uniform on U_ω: *bounded*, $\sup_{U_\omega} c_S =: c_S^* < \infty$, and *continuous in g on U_ω in its subspace H^s topology* (Fence F-iii). We record its grade exactly, and do *not* upgrade it.

Remark E2a.81 (Clause (iii) is retained as a named input; its status is un-derived in print). The Sanders constants are constructed by reference-Hadamard subtraction through explicit chart / Fourier-integral expressions, so their H^s-continuity in g at fixed smearing is, as the paper states, "a matter of constant-tracking rather than new analysis" – but *no such derivation is in print*, and the corpus does *not* supply one. The companion chain's only contact with clause (iii) is the scoping repair **Fence F-iii**: the modulus is read in the *subspace H^s* topology of U_ω

(which has empty H^s-interior), not the open-set continuity of an H^s neighbourhood. This is a scope narrowing, *not* a proof: it is costless precisely because the sole downstream consumer, Cor. E.8, never differentiates c_S in g and never takes an H^s-open modulus of it – it consumes only the boundedness c_S^* and the subspace-continuity. Two orientation points bound the risk: (a) the underlying QEI is the *free massive* ($m > 0$) scalar bound; the massless null case has provably *no* state-independent bound [210], so clause (iii) cannot be extended off the free massive scope; (b) nothing in the analytic slice-chain theorem (§E2a.22) touches c_S – clause (iii) enters only the fused first-law corollary, as an explicit named hypothesis. *Grade: NAMED INPUT (PLAUSIBLE): the constant-tracking is un-derived in print; only the F-iii subspace-topology scoping is established.* Carried on the modulo-list as item (M2) (§E2a.24).

Remark E2a.82 (Clause (ii) is likewise a named input, reduced but not proved – forward pointer). For completeness of the modulo-list bookkeeping we recall the grade of clause (ii), the load-bearing finite-part clause, established in §E2a.18 (part 4): its finite part W_g mixed-jet Lipschitz bound closes at the fence $s > 11 = n/2 + 9$ via the $N = 2$ truncated Hadamard finite part, with $C_W(K)$ assembled from named ball-uniform factors; but it is *reduced, not proved*, modulo two honest gaps – the $N = 2$ finite-regularity Hadamard-parametrix construction with quantitative v_2, family-uniform in g, is not in print as a ball-uniform statement (the same status the paper already grants the $N = 1$ construction), and the exact multi-index constant-tracking through the $N = 2$ commutator source and the energy estimate is not executed. *Grade: PLAUSIBLE.* It is the classic overclaim home: clause (ii) is a named input (E2a) and *must never be cited at THEOREM.* Carried on the modulo-list as item (M1) (§E2a.24).

E2a.22 The master theorem

We now state, at exactly the scope the dependency tree supports, what this appendix proves. The tree does *not* prove Hyp. E.2 (its three clauses are the named inputs (E2a-ω) consumed by Thm. E.7 and Prop. E.4, not derived by any companion file: clause (ii) is PLAUSIBLE, clauses (i) difference-form and (iii) are named inputs). What it *does* prove is the analytic slice register over the class U_ω; the physical first law then follows *modulo the finite list of the three clauses.* The master theorem therefore has two parts – an **unconditional** part over U_ω (the slice chain) and a **conditional** part (the fused first law), with the modulo-list carried as *explicit named hypotheses* (M1)–(M3).

Theorem E2a.83 (Analytic slice register over U_ω, and the first-law extraction modulo the Hadamard/QEI clauses)**.** *Let g_0 be a real-analytic globally hyperbolic metric with compact Cauchy slice; fix a complex-collar radius $\rho_a > 0$ and a sup-bound $\varepsilon_a > 0$ with (ρ_a, ε_a) small enough that the C^0-image of $U_\omega := U_\omega(g_0, \rho_a, \varepsilon_a)$ (Def. E2a.1) has radius $< \varepsilon_{\mathrm{GH}}$, so every $g \in U_\omega$ is globally hyperbolic ([222], a C^0 property inherited by the analytic subset); let $s > n/2 + 6$. Then:*

(A) [Unconditional over U_ω – the analytic slice chain]. *Over U_ω:*

(A1) *the fold-branch count $N_0(\rho_a, \varepsilon_a, g_0)$ is finite and ball-uniform (Lem. E2a.17);*

(A2) *the linear device measure bound $d_\theta(S_\mu) \le c_{\mathrm{sub}}\,\mu$, with $c_{\mathrm{sub}} = c_{\mathrm{vel}}(1 + N_0)$, holds on the whole cone – on the fold stratum B_N and for $\mu \ge \kappa_V$ (Cor. E2a.41), and the near-flat sliver B_V is closed device-free at the binding cut $\mu_* = (\Lambda\rho)^{-1} < \kappa_V$ (Thm. E2a.37);*

(A3) *consequently* (D3)*,* (D1)*, and the transport-averaged booking $\partial_y : (c, p) \mapsto (c + \frac{1}{2}, p - 1)$ (Thm. E2a.51) hold end-to-end over U_ω, with all constants ball-uniform in $(\rho_a, \varepsilon_a, g_0)$; all difference estimates range* both *g and g' over U_ω (both real-analytic; Fence F-diff);*

(A4) *the slice register therefore promotes to $s > 11$ (slice, T2) and $s > \frac{23}{2}$ (locked-4d, T3) (Cor. E2a.54), the unique zero-slack inequality of the chain ($s - \frac{19}{2} > \frac{3}{2} \iff s > 11 = n/2 + 9$).*

(B) [Conditional – the fused first law, modulo the three named clauses]. *Assume additionally the finite list of named inputs:*

(M1) (clause (ii), finite part). *Hyp. E.2(ii): the finite part W_g admits the mixed-jet Lipschitz bound $C_W(K)$ and the uniform bound $W_*(K)$, ball-uniform over U_ω;*

(M2) (clause (iii), QEI constants). *Hyp. E.2(iii): the Sanders QEI constants are bounded ($\sup_{U_\omega} c_S =: c_S^* < \infty$) and subspace-$H^s$-continuous on U_ω (Fence F-iii);*

(M3) (clause (i), difference form). *Hyp. E.2(i): the singular difference $H_g - H_{g'}$ is dual-Lipschitz* (209)*, ball-uniform over U_ω.*

Fix the free massive ($m > 0$) minimally-coupled scalar and a finite-relative-entropy quasi-free state class on a compact Cauchy region K. Then the per-point first-law extraction – the pointwise null–null equation of state

$$\delta G_{\mu\nu}\, k^\mu k^\nu \;=\; \frac{8\pi G}{c^4}\, \delta\langle \widehat{T}_{\mu\nu}\rangle_\omega\, k^\mu k^\nu$$

along every null k through every point of K – holds uniformly in the point and in the null direction*, with uniformity constant depending only on $(g_0, \rho_a, \varepsilon_a, s, K, m)$.*

Grades, and where each part is proved. Part (A) is a ***theorem****, unconditional over U_ω.* Each cited node is established at THEOREM grade in §§E2a.20–E2a.13 (parts 1–4 and the present part): (A1) is Lem. E2a.17 (N0-analytic, graded THEOREM over U_ω); (A2) is Cor. E2a.41 together with Thm. E2a.37 (A1, with the kernel-lemma completion); (A3) is the cascade Prop. E2a.43, Prop. E2a.46, Thm. E2a.51, each of which the corpus grades “THEOREM on B_N; THEOREM end-to-end over U_ω *given* (A1)” – and the “given (A1)” condition is discharged precisely because (A2)'s Thm. E2a.37 *is* a theorem, so the composite is THEOREM end-to-end over U_ω; (A4) is Cor. E2a.54, inheriting (A3), with the fence held at $s > 11$ by the trace-audit trio and anchor restatements (§E2a.13). This is the end-to-end verdict. *Part (B) is a* ***theorem modulo*** *{*(M1),(M2),(M3)*}*: granting those three named inputs, (B) is exactly Prop. E.4 of the main text, whose slice-register feeders (the $s > 11$ register and the $(c + \frac{1}{2})$ booking) are now supplied by Part (A) as a theorem over U_ω rather than as a naive-fallback input; the point/direction uniformity is automatic from the compact sphere and parametrix smoothness once the ball-uniform finite-part family is granted. The three hypotheses (M1)–(M3) are *not* discharged by this appendix: (M1) is graded PLAUSIBLE (§E2a.18, Rem. E2a.82), (M2) a named input (Rem. E2a.81), while (M3), the clause (i) difference form, is now a *ported theorem* (Rem. E2a.80; App. E2a); (M1) and (M2) are stated as explicit hypotheses and printed in the modulo-list (§E2a.24). □

E2a.23 Scope of the master theorem

Remark E2a.84 (The four scope pins Thm. E2a.83 is stated at – and never beyond). The theorem claims exactly the following scope; each pin is a place a looser statement would overclaim, and is held tight.

(1) *Mixed jets, never all jets.* Where the physical consumer enters (Part (B)), the coincidence-jet scope is $|a|+|b| \le 1$ together with the mixed pair $|a| = |b| = 1$ *only.* The pure same-slot jets $(2, 0)$, $(0, 2)$ are never formed – the minimally-coupled point-split distributes one derivative per slot – and in fact the pure same-slot *diagonal* jets of the truncated residual *diverge.* The statement therefore never reads “all second jets” or “the full $C^2(K \times K)$ norm”: that formulation is unavailable at finite metric regularity (the truncation residual is C^1 but not C^2 across the light cone).

(2) *Free massive $m > 0$ minimally-coupled scalar.* Not interacting, not massless. The QEI input (106) [207] is the free massive scalar; the massless null case has provably *no* state-independent bound [210], so the scope cannot be relaxed to massless.

(3) *The class is U_ω, not an open H^s ball.* U_ω is H^s-dense with *empty* H^s-interior (the disclosed cost). Every uniformity in Thm. E2a.83 is a *supremum* over the ball or a *pair-Lipschitz*

modulus; no H^s-open property is claimed (Fences F-iii, F-diff). Coercivity floors use the collar-margin *attainment* note (part E2a.16), not closed-ball attainment – the false "H^s-closed ball attained" clause was struck. Both members g, g' of every difference estimate are analytic; a merely-smooth comparison g' is *not* served (Fence F-diff), and the non-analytic C^0/shock metrics of the back-reaction ball are categorically excluded (Guard F-shock, Rem. E.3).

(4) $s > 11$ *slice* / $s > \frac{23}{2}$ *locked-4d.* Here $s > 11 = n/2 + 9$ at $n = 4$; the locked-4d register is $s > \frac{23}{2}$. These are method-relative *upper* bounds for the fence (the honest $+\frac{1}{2}$-derivative analytic booking), not sharp-from-below. Do *not* quote the naive-fallback $s > 11.5$ / $s > 12$ for the analytic route – those belong to Option B (Hyp. E.1).

Remark E2a.85 (What Part (A) proves that the paper needed, and what it does not touch). The single downstream purpose of Part (A) is to discharge the one PLAUSIBLE cap of the naive T1′ rung. In the full H^s ball the *linear* device bound that $(D3)$ needs to book $(c + \frac{1}{2})$ is only PLAUSIBLE: it requires a ball-uniform fold count N_0, and a C^k norm cannot supply one without a ϕ'''-floor (the Z_0 negative-result; Route A refuted). N0-analytic (Lem. E2a.17) supplies exactly this N_0 over the narrowed class U_ω. Accordingly, Part (A) narrows the *class*, not the clauses; it does *not* promote the naive rung as a whole (that aggregate stays PLAUSIBLE over the full ball – Option B's story). Part (A) cites only $(R1)$, $(R3)$, $(R4)$, $(R5)$, $F2$ and the classical holomorphic-ODE package (each a theorem relative to its interface); it does *not* cite the data-side items (E-env), $(D0)$, or #B′, which gate the separate N-c data ladder and Option B and are orthogonal to the slice chain (§E2a.24).

Remark E2a.86 (What remains open – including the interacting seam). Even with (M1)–(M3) granted, Thm. E2a.83 is a *free-field* statement, and several boundaries are untouched.

(a) *The three named clauses themselves.* (M1) clause (ii) is PLAUSIBLE (reduced to $s > 11$ modulo the $N = 2$ Hadamard construction and the constant-tracking); (M2) clause (iii) is a named input, un-derived in print for the family; (M3) clause (i) difference-form is now a ported theorem (App. E2a). The literal flip of Hyp. E.2 to a theorem is therefore *not* achieved; what is achieved is the class-and-register upgrade of Part (A) plus the first law modulo this finite list.

(b) *The interacting seam.* Part (B) does *not* upgrade the input (N1) of the Newtonian-limit corollary (Cor. J.5): (N1) is the *interacting* Standard-Model tensor completion, and an unconditional free-field statement leaves the interacting seam (the B1 wedge-modular / B2 curved-rate items) entirely untouched. The analytic-ball route of this appendix is about the free massive scalar; the interacting completion is a separate programme.

(c) *The state class.* The extraction is pinned to finite-relative-entropy quasi-free perturbations; arbitrary locally-squeezed states have divergent relative entropy and are excluded (the inherited state-class boundary).

(d) *The physical scope is nonetheless native.* The analytic-collar anchor g_0 is real-analytic for every standard background (Minkowski, Schwarzschild, Kerr, de Sitter, FRW, ultrastatic), so Thm. E2a.83 applies to the physical g_0 with no extra hypothesis – but it does *not* extend to the non-analytic shock metrics of the back-reaction ball (Guard F-shock).

E2a.24 The modulo-list, in print

The corollary of record (Part (B) of Thm. E2a.83) is a theorem *modulo the finite named list below.* We print it, so no reader mistakes the fused first-law statement for an unconditional theorem, and so no consumer cites any of these at THEOREM grade. Exactly three items are on the critical path; each is listed as **exact open content** – its **consumer** – its **verified grade**.

(M1) **Clause (ii) – finite part W_g mixed-jet Lipschitz + uniform bound (C_W, W_*).** *Open content:* the mixed-jet Lipschitz bound $\sup_{x\in K}|[\partial_x^a\partial_y^b(W_g - W_{g'})](x,x)| \le C_W(K)\|g - g'\|_{H^s}$ ($|a|+|b|\le 1$ and $|a|=|b|=1$) and the uniform $\sup_{U_\omega}\sup_{x\in K}|[\partial_x^a\partial_y^b W_g](x,x)| \le W_*(K)$, ball-uniform. Reduced to $s>11$ via the $N=2$ truncated Hadamard finite part with $C_W(K)$ assembled from named ball-uniform factors, *modulo two honest gaps not closed*: (a) the $N=2$ finite-regularity Hadamard-parametrix construction with quantitative v_2, family-uniform in g, is not in print as a ball-uniform statement; (b) the exact multi-index constant-tracking through the $N=2$ commutator source and the energy estimate is not executed. Over U_ω both gaps are discharged *as family-uniformity questions* by the Cauchy-estimate package now written out in App. E2a (one polydisc estimate on the fixed collar replaces the multi-index tape; the transport recursion runs as a linear holomorphic ODE; the seed triangle controls W_*), and the seed-data residue (D0) is closed at the seed by the classical convergent-Hadamard-series lever, verified at the primary source in its native non-parabolic category (App. E2a; given the paper's standing seed-Hadamard premise). [Correction gate: the "(D0) is closed at the seed" clause just stated does not stand as printed — the convergent-Hadamard-series lever supplies the seed slice-restrictions at existence grade only (Rem. E2a.66, item (i): THEOREM, unconditional), while the seed availability column is obstructed at the honest heights $(\frac{7}{2},\frac{5}{2},\frac{5}{2},\frac{3}{2})^-$ (the two printed errors named at Rem. E2a.65; sharp flat-massive counterexample $v_3 = m^8/(18432\pi^2) \neq 0$ at Rem. E2a.66), so (D0) remains gated as a confirmed obstruction of the printed route. The lever and the w_0-smoothness lemmas stand, and the discharge of gaps (a),(b) for the finite-part family rests on the Cauchy-estimate package, not on this (D0) closure.] The honest modulo list is therefore: the state-leg envelope (E-env), to which the carrier Lem. E2a.75 reduces the difference data and which drives its PLAUSIBLE grade (the BFV-surface vanishing $C_{\Delta_2}=0$ itself holds identically, relocated to an anchoring role by #A; revival condition attached), and #B′ (the single-metric off-collar variant of (E-env)) for patch-traversal consumers; no collar$\to H^s$ shortcut exists (the collar and H^s topologies are transverse; the collar interpolation modulus is Hölder-$\frac{1}{2}$, sharp). *Consumer:* Lem. E.6, Thm. E.7, Prop. E.4 (the finite-part family $\{C_W, W_*, \dots\}$); the load-bearing clause of the first-law extraction. *Grade: PLAUSIBLE* (the fence value $s>11$ inside it is a theorem-grade derivative-budget result; the clause as a whole is reduced, not proved).

(M2) **Clause (iii) – Sanders QEI-constant uniformity (c_S^*, subspace-continuity).** *Open content:* for fixed smearing, the Sanders constants $c_S(g,t,F), C_S(g,t,f,F)$ are bounded on U_ω ($\sup_{U_\omega} c_S =: c_S^*$) and continuous in g in the subspace H^s topology (Fence F-iii); the constant-tracking that would establish this is not in print. *Consumer:* Cor. E.8 (the QEI budget), which consumes only c_S^* and subspace-continuity – never differentiates c_S in g, never takes an H^s-open modulus. The QEI bites exactly once, at the *seed* — the displayed floor constant is $c_S(g_0,t,F)$ and the transport legs (S5)/(S7) are QEI-free — so the strength consumed at proof level is the *single-metric* Sanders theorem at g_0; the ball boundedness c_S^* and the subspace continuity are stated but unconsumed. Revival condition: any future consumer applying (106) at a g-native state revives the ball clause. *Grade: NAMED INPUT* (its only companion touch is the F-iii scoping repair; the tracking itself is un-derived in print).

(M3) **Clause (i) difference form – singular part H_g dual-Lipschitz.** *Open content:* the difference bound (209), ball-uniform over the *family*; the single-metric jet is a theorem (Lem. E2a.79) but the family-uniform difference bound is not separately in print. *Consumer:* the singular leg of Thm. E.7; it is exercised by *no* quantitative slice-ladder consumer (lowest risk of the three). At *proof* level the consumer set is empty — the pivot's own proof note (Thm. E.7) declares clause (i) unconsumed, C_H is formed by no proof in the paper (grep-exhaustive), and Prop. E.4 premises clauses (ii)–(iii) only; the clause is retained in the hypothesis for its microlocal role. The written-with-constants difference theorem over the full H^s ball ($C_H = C_{\rm sob}^3\,\kappa$) is now ported by full re-derivation as App. E2a, closing the content outright. *Grade: THEOREM* (ported, App. E2a; formerly NAMED INPUT — THEOREM at single-metric level with the family-uniform difference bound in the named black box).

Remark E2a.87 (The remaining ledger residues are Option-B / data-side and orthogonal to the slice chain). For the reader tracking the full no-go ledger: the further open items (E-env) (the data-side kernel envelope, PLAUSIBLE), ($D0$) and #B′ (the N-a data residues gating lem : W2star across patches; ($D0$) is OBSTRUCTED on the printed route, Rem. E2a.66), the #A collar architecture (the long-slab restatement, a negative result, discharged iff (E-env)), the transverse correction with its gated profile columns (Rem. E2a.65, over the corrected lem : transverse-data — no longer fence-neutral bookkeeping at the data-cap sites), and the HB1/HB3 imports of the bulk clause – are *not* on the critical path for Thm. E2a.83. Part (A) cites none of them (it cites only $(R1), (R3), (R4), (R5), F2$ and the holomorphic-ODE package); they gate the separate N-c data-side ladder and the Option B close-out. The companion cascade's own disclaimer records this: these items are U-facts, inherited on U_ω, and left untouched by the analytic sub-programme. Only the three critical items (M1)–(M3) enter the fused first law, and only they are listed above.

E2a The seed-fixed spacetime-smeared Dirac floor

This appendix records the derivation, from in-print worldline inputs, of the one-sided absolute lower bound on the smeared renormalised Dirac energy density at the audited-minimal strength consumed by the (E2b) discussion: a *single* fixed smearing triple, Hadamard states of the seed folium, fixed seed metric. The organising fact is that the Fewster–Verch route [199] never represents the energy density as a positive quadratic form — the positivity it consumes is CAR-*state* positivity of the frequency-split two-point pieces (the Pauli window $0 \le Q \le 1$), with the negative-frequency (sea) piece entering through a compensating minus sign — so the indefinite-density obstruction that blocks the Sanders positive-type-kernel organisation for Dirac does not touch it.

Standing data. Seed $(M, g_0) = (\mathbb{R} \times \Sigma,\, -dt^2 \oplus h_\Sigma)$ with Σ a compact spin Cauchy slice, $\dim M = 4$; Dirac mass $m > 0$ (for $m = 0$ see Rem. E2a.3); ω_0 the ultrastatic ground state with its canonical positive-frequency split P_0 (the positive spectral projection of the time flow; the ground state is Hadamard by Sahlmann–Verch [202]); h the one-particle Dirac Hamiltonian on $L^2(\Sigma; \mathbb{C}^4)$: self-adjoint, discrete spectrum $\{E_k\}$ with $|E_k| \ge m$, smooth eigensections u_k; $\gamma_x(t) = (t, x)$ the unit-speed static worldlines. $\widehat{T}^{\rm ren}_{\mu\nu}$ is the fixed-g_0 Hadamard-renormalised Dirac stress tensor [15].

Theorem E2a.1 (seed-fixed spacetime-smeared absolute Dirac floor). *Let $t^\mu = (\partial_t)^\mu$ and $F \in C_c^\infty(M; \mathbb{R})$. Then for every Hadamard state ω of the Dirac field on (M, g_0),*

$$\big\langle \widehat{T}^{\rm ren}_{\mu\nu}[t^\mu t^\nu F^2] \big\rangle_\omega \;\ge\; -c_{\rm fl}(g_0, \partial_t, F), \qquad c_{\rm fl} := \int_\Sigma d{\rm vol}(x)\, B_x + \Big| \int_M F^2 \,\big\langle \widehat{T}^{\rm ren}(\partial_t, \partial_t) \big\rangle_{\omega_0} \Big| \;<\; \infty,$$

where, with $g_x := F(\cdot, x) \in C_0^\infty(\mathbb{R}; \mathbb{R})$,

$$B_x = \frac{1}{2\pi} \int_0^\infty d\mu\, \mu\, \big(S_\mu(x) + S_\mu^\Gamma(x)\big), \qquad S_\mu(x) = \sum_{k:\, E_k > 0} w_k(x)\, |\widehat{g_x}(\mu + E_k)|^2,$$

and $S_\mu^\Gamma(x) = \sum_{k:\, E_k < 0} w_k^\Gamma(x)\, |\widehat{g_x}(\mu + |E_k|)|^2$ (the mirrored argument, from the $(\mu, -\mu)$ evaluation of the Dawson–Fewster bound [200]), with $w_k(x), w_k^\Gamma(x) \ge 0$ the spin-frame contractions of the *mode densities of u_k at x. In particular the floor holds for the audited single triple, one-sided, on Hadamard states of the seed folium.*

Proof. (1) *Disintegration.* For Hadamard ω the function $x \mapsto \langle \widehat{T}^{\mathrm{ren}}_{\mu\nu} \rangle_\omega(x)$ is smooth (fixed-g_0 renormalisation, [15]); $d\mathrm{vol}_{g_0} = dt\, d\mathrm{vol}_\Sigma$ and F is compactly supported, so by Fubini $\langle \widehat{T}^{\mathrm{ren}}[ttF^2] \rangle_\omega = \int_\Sigma d\mathrm{vol}(x) \int_\mathbb{R} dt\, g_x(t)^2 \, \langle \widehat{T}^{\mathrm{ren}}(\partial_t, \partial_t) \rangle_\omega(t, x)$.

(2) *Reference conversion.* Pointwise on M, $\langle \widehat{T}^{\mathrm{ren}} \rangle_\omega = \langle {:}T{:} \rangle_\omega + \langle \widehat{T}^{\mathrm{ren}} \rangle_{\omega_0}$, with ${:}T{:}$ the ω_0-normal-ordered density: both subtractions are Hadamard subtractions of the same classical density, so they differ by the smooth ω-independent c-number $\langle \widehat{T}^{\mathrm{ren}} \rangle_{\omega_0}$ (the renormalised, reference-free reading of the Fewster–Verch bound run in the absolute direction). The c-number term integrates to a finite constant on the compact support of F^2.

(3) *Per-curve bound.* Fix $x \in \Sigma$. The static curve γ_x is smooth, timelike, unit-speed, and the weight g_x^2 with g_x real C_0^∞ is exactly the Dawson–Fewster class; their explicit worldline quantum weak energy inequality [200] with reference ω_0 gives $\int dt\, g_x(t)^2 \, \langle {:}T{:}(\partial_t, \partial_t) \rangle_\omega(t, x) \geq -B_x$ for every Hadamard ω, with B_x as displayed.

(4) *Mode evaluation.* On the seed, the ground-state two-point function restricted to positive frequency along γ_x is the mode sum $\sum_{E_k>0} w_k(x)\, e^{-iE_k(t-t')}$ (convergence in $\mathscr{D}'(\mathbb{R}^2)$, locally uniformly in x; the standard ground-state expansion from P_0), and $[\bar{g}_x \otimes g_x\, e^{-iE(t-t')}]^\wedge(-\mu, \mu) = |\widehat{g_x}(\mu + E)|^2 \geq 0$; termwise summation is justified by step (5). The Γ-piece runs identically on the negative spectral subspace, the $(\mu, -\mu)$ evaluation producing the mirrored argument $|\widehat{g_x}(\mu + |E_k|)|^2$.

(5) *Envelope.* Elliptic sup-norm bound on compact Σ: $\sup_x w_k(x) \leq C(1 + |E_k|)^4$ (Sobolev embedding $H^2(\Sigma) \hookrightarrow C^0$ applied to u_k, with $\|u_k\|_{H^2} \leq C(1 + E_k^2)\|u_k\|_2$); Weyl asymptotics for the first-order elliptic h: $\#\{k : |E_k| \leq E\} \leq C(1 + E)^3$; Paley–Wiener decay $\sup_x |\widehat{g_x}(\xi)| \leq C_N(1 + |\xi|)^{-N}$ for all N. Hence

$$\int_0^\infty d\mu\, \mu\, S_\mu(x) \leq C_N \sum_k (1{+}|E_k|)^4 \int_0^\infty \frac{\mu\, d\mu}{(1 + \mu + |E_k|)^N} \leq C \sum_k (1{+}|E_k|)^{6-N} < \infty \qquad (N > 9),$$

uniformly in x; dominated convergence gives x-continuity of B_x, hence $\int_\Sigma B_x\, d\mathrm{vol} < \infty$ (Σ compact). All constants depend on (g_0, F) only.

(6) *Summation.* Integrating (3) over $(\Sigma, d\mathrm{vol})$ and combining with (1)–(2) gives $\langle \widehat{T}^{\mathrm{ren}}[ttF^2] \rangle_\omega \geq -\int_\Sigma B_x\, d\mathrm{vol} - \left| \int F^2 \langle \widehat{T}^{\mathrm{ren}} \rangle_{\omega_0} \right| = -c_{\mathrm{fl}}$. □

Remark E2a.2 (imports, each at printed strength). (a) The Dawson–Fewster worldline bound [200] (published), whose proof is the Fewster–Verch frequency-split mechanism [199]; (b) the seed ground state, its canonical split, and its Hadamard property (Araki–Weinless via the Fewster–Verch construction; Sahlmann–Verch passivity [202], published); (c) the fixed-g Dirac Hadamard renormalisation [15] (published). Steps (1), (2), (4), (5), (6) are derived here and are elementary given (a)–(c); no step consumes an unpublished estimate. The theorem is therefore closed at the fixed seed metric. The remaining *ball clause* of the audited strength — local uniformity of c_{fl} in g — is exactly the H^s-uniform spinor Hadamard parametrix residual of the (E2b) discussion, at parity with how Hyp. E.1 (iii) carries the same clause on the scalar leg.

Remark E2a.3 (massless case). For $m = 0$ the Fewster–Verch reference exists via their variable-mass deformation; it is Hadamard but non-canonical, and zero modes of h (harmonic spinors) can obstruct a gapped split on special Σ. A reference change shifts c_{fl} by a finite smooth c-number only. The physical case of the (E2b) discussion is $m > 0$; $m = 0$ costs a non-canonical reference, not the floor.

E2a The de Sitter tensor completion: the audited-minimal form (E0-dS′)

A proof-level audit of every displayed consumer of Hyp. I.3 shows that the ball-modular family of every radius and centre — the Casini–Huerta–Myers scanning structure [241] — is the shape of the AdS-anchored *route* (Thm. I.1), not of the framework's need: what the printed inversion steps consume is the pointwise null–null equation of state at every (x, k), Hadamard conservation, the linearised Bianchi identity, and the posited Λ seam. The following three lemmas show that this consumed strength follows from per-cut first laws on the $\mathrm{SO}(1,4)$ orbit of static-patch wedges of dS_4; the audited-minimal hypothesis (E0-dS′) then replaces the ball family. Throughout, $\mathrm{dS}_4 \subset \mathbb{R}^{1,4}$ is the unit hyperboloid.

Lemma E2a.1 (null coverage by the wedge orbit)**.** *Let $x \in \mathrm{dS}_4$ and let $k \in T_x\mathrm{dS}_4$ be null. Then there is a wedge W in the* $\mathrm{SO}(1,4)$ *orbit of the static patch such that x lies on the bifurcation surface ∂W of its bifurcate Killing horizon and k is a null normal of ∂W at x (the generator direction of one horizon sheet). The wedges with this property form a 2-parameter family.*

Proof. The orthogonal complement $k^\perp \subset T_x\mathrm{dS}_4$ is a degenerate 3-plane containing k; its lifts to spacelike 2-planes $P \subset k^\perp$ transverse to k form a 2-parameter family. For any such P, the totally geodesic surface $\Sigma = \mathrm{dS}_4 \cap \Pi$, with $\Pi \subset \mathbb{R}^{1,4}$ the linear 3-plane with $T_x\Sigma = P$, is a totally geodesic round 2-sphere through x whose normal bundle at x is the Lorentzian 2-plane $P^\perp \ni k$, containing exactly two null rays, one of them $\mathbb{R}k$. The subgroup of $\mathrm{SO}(1,4)$ fixing Σ up to rotation contains a boost whose bifurcate Killing horizon has bifurcation surface Σ and generators along the null normals; its wedge is an $\mathrm{SO}(1,4)$-image of the static patch, since $\mathrm{SO}(1,4)$ acts transitively on (totally geodesic bifurcation sphere, null normal) data. □

Lemma E2a.2 (null–null determination plus identities; trace closure without trace-channel data)**.** *Let $E_{\mu\nu}$ be a symmetric tensor field on a region of an* on-shell *semiclassical background ($G^{(0)}_{\mu\nu} + \Lambda g^{(0)}_{\mu\nu} = \kappa\langle T^{(0)}_{\mu\nu}\rangle_{\mathrm{BD}}$, equivalently $\langle T\rangle$ taken Bunch–Davies-relative so that the background combination $E^{(0)} = 0$; the background field equations are consumed here — they are what make $E_{\mu\nu}$ gauge-consistent across the per-wedge adapted frames, by Stewart–Walker invariance of the linearised combination, which requires $E^{(0)} = 0$), of the form $E = \delta G + \Lambda\,\delta g - \kappa\,\delta\langle T\rangle$ with $\nabla^\mu \delta\langle T_{\mu\nu}\rangle = 0$ (Hadamard conservation). If $E_{\mu\nu}k^\mu k^\nu = 0$ for every null k at every point, then $E_{\mu\nu} = c\, g^{(0)}_{\mu\nu}$ with c* constant*: the linearised Einstein equation holds up to a shift of Λ.*

Proof. Pointwise, a symmetric tensor annihilated by all null–null contractions is proportional to the metric (polarisation over the null cone; the local-Rindler move of step (i) of the printed inversion remark, [56]). So $E = c(x)\, g^{(0)}$. The linearised contracted Bianchi identity about an Einstein background gives $\nabla^\mu(\delta G_{\mu\nu} + \Lambda\,\delta g_{\mu\nu}) = 0$ identically; with matter conservation, $0 = \nabla^\mu E_{\mu\nu} = \partial_\nu c$. The constant c is the trace-value seam of step (iv) of the printed inversion: it renormalises Λ and is posited separately, exactly as in print. The trace part of the *equation* thus closes by identities alone, consuming no modular data at any scale. □

Lemma E2a.3 (second cut-variation: the smeared-to-pointwise step, with its cost stated)**.** *Fix a wedge W as in Lem. E2a.1, horizon coordinates (v, Ω) with affine v along the generators. Suppose: (a) for every cut $V(\cdot)$ in a C^2 neighbourhood of the bifurcation cut the* equality-strength *first law holds at first order in the perturbation, with the geometric pairing*

$$\frac{\delta\big[A_\infty - A(V)\big]}{4G} \;=\; \delta S(V) \;=\; \delta\langle K_V\rangle \;=\; 2\pi \int d\Omega\,\sqrt{h} \int_{V(\Omega)}^{\infty} dv\,\big(v - V(\Omega)\big)\,\delta\langle T_{vv}(v,\Omega)\rangle,$$

both sides twice differentiable in V (the geometric side is the future-normalised area deficit *— the literal $\delta A(V)$ reading returns the anti-Einstein sign under the second cut-variation; A_∞ drops out* *of both variations); (b) the* first moment $\int dv\,(v{-}V)\,|\delta\langle T_{vv}\rangle|$ *is finite on every complete generator (an existence premise for the display, strictly stronger than the averaged-null-energy pin the paper elsewhere assumes); and (c) for cut vector fields with nonvanishing* ξ *the geometric side is read as the first-order* Wald *entropy variation — for the improved stress tensor the improvement term in* T_{vv} *equals the first-order Wald surface term exactly, so the pairing is with Wald entropy and no improvement ambiguity leaks into the display, the effective coupling being* $1/G_{\rm eff} = 1/G - 8\pi\xi\langle\varphi^2\rangle_{\rm BD}$ *(spacetime-constant on* dS, *so the* Λ *seam and the Bianchi identity are intact; the correction rescales the coefficient, not the form). Then* $\delta^2/\delta V(\Omega)^2$ *yields the pointwise null–null equation of state* $\delta G_{kk}(x) = 8\pi G_{\rm eff}\,\delta\langle T_{kk}(x)\rangle$ *at every point of every generator (matter side:* $\delta^2\langle K_V\rangle/\delta V^2 = 2\pi\,\delta\langle T_{vv}(V,\Omega)\rangle\sqrt{h}$*; geometric side: linearised Raychaudhuri,* $\partial_v\delta\theta = -\delta R_{vv}$*, integrated against the same weight). Combined with Lem. E2a.1 and Lem. E2a.2 this gives the linearised tensor equation about* dS_4 *up to the* Λ *seam.*

Hypothesis E2a.4 ((E0-dS$'$): cut-localised first law on the de Sitter wedge orbit — the audited-minimal form)**.** On the $\mathrm{SO}(1,4)$ orbit of static-patch wedges of dS_4, in a Bunch–Davies-class Hadamard state of the massive (SM) matter net, the equality-strength first law of Lem. E2a.3 — with its premises (a)–(c), including the finite first moment per generator — holds for every cut in a C^2 neighbourhood of the bifurcation cut of every wedge, at the order in the couplings at which the tensor equation is asserted.

Remark E2a.5 (audit content and status). (1) Hyp. E2a.4 implies the full strength every displayed consumer of Hyp. I.3 uses; the printed ball-family demand (every radius and centre, the SSA functional, the two-sided equivalence) is consumed by *no* downstream consumer. (2) What remains open inside Hyp. E2a.4: at $n{=}0$ (free massive matter) the per-cut *equality* is in print only on Minkowski/Rindler cuts (Ceyhan–Faulkner [234]), with monotonicity/GSL strength on cuts of Killing horizons [92] and the de Sitter-wedge KMS structure in print [49]; both named support steps are now discharged at $n{=}0$ for the compactly generated (coherent) Bunch–Davies perturbation class — the massive bulk-to-horizon restriction (App. E2a: mass-inertness from the exact Bunch–Davies closed form, the DMP injection at printed strength, the first-moment premise *proved* for the class) and the per-cut pairing (App. E2a: exact horizon chart, complete boundary-term list, the improvement/Wald lemma with its $G_{\rm eff}$ bookkeeping); the remaining $n{=}0$ opens are the general Bunch–Davies-class Hadamard perturbations (the first moment stays premise (b) there) and, if wanted, the identification of horizon cut-algebra entropy with bulk wedge-algebra entropy. At $n \geq 1$ it *is* the B1a interacting first-law wall — no new wall. (3) Scope fence: the wedge orbit exists on exact dS only; on FRW there is no bifurcate Killing horizon, so the FRW share of the completion re-pins to the A3 obstruction and stays walled — and that is the only habitat of the marginal-Δ diamond disease, which lives in the small-ball trace-channel extraction that this route never consumes: $k^2 = 0$ annihilates the trace channel on the null–null projection, there is no small-radius dial on the wedge orbit (every bifurcation surface is a great sphere pinned at the Hubble scale), and the trace part of the equation closes by the identities of Lem. E2a.2.

E2a Ports for the analytic-collar appendix: the clause-(i) difference theorem, the family-uniformity package, and the seed-data closure

This appendix executes, at full re-derivation grade, the three ports left open by App. E2a: (i) the clause-(i) difference theorem ($C_H = C_{\rm sob}^3\,\kappa$, every sub-constant displayed), stated in exactly the form item (M3) names; (ii) the Cauchy-estimate family-uniformity package for the clause-(ii) gaps of item (M1), with the four required corrections folded in at their marked loci; (iii) the seed-data residue (D0), closed at the seed via the classical convergence of the Hadamard series for analytic coefficients — verified at the primary source in its native non-parabolic category — given the paper's standing seed-Hadamard premise (correction gate: the (D0) closure does *not* stand as printed; the port's Prop. display and T-leg calculus are gated at the honest column $(\frac{7}{2}, \frac{5}{2}, \frac{5}{2}, \frac{3}{2})^-$ and (D0) remains gated — see Rem. E2a.65 and Rem. E2a.66; the port's lever and w_0 lemmas stand). Grades are carried per item; nothing here upgrades the clause-(ii) grade (the BFV conditionality survives, as recorded in item (M1)).

E2a.1 The clause-(i) difference theorem, ported

P-M3. Clause (i), difference form: the dual-Lipschitz singular-part theorem over the full H^s ball, with constants

Setting (the paper's own). $n = 4$, $\mathcal{M} \cong \mathbb{R}_t \times \Sigma^3$ with the fixed g_0-atlas and compacts $K \subset K' \subset K''$ of the appendix Setup; $|\cdot|$ and ∂ are chart-Euclidean; $r := |x-y|$; t is the fixed g_0-time function and $t_\Delta := t(x) - t(y)$. Fix $s > n/2 + 6 = 8$ and the ball

$$U := \{g \in H^s : \ \|g - g_0\|_{H^s(K'')} \le \varepsilon_0\},$$

with ε_0 small enough that (u1) $\inf_U \min_{K''} |\det g| =: d_U > 0$; (u2) the C^0-image of U has radius $< \varepsilon_{\mathrm{GH}}$ (global hyperbolicity, [222] — the hypothesis' own smallness); (u3) the cone-transversality constants c_*, c_t of Lemma E2a.3(iii) below are positive. This is exactly the class $\mathcal{U}$ of Hyp. hyp:hadamard-uniform (Option B); by an17:lem-embed, $U_\omega \subseteq U$ once $\varepsilon_a \le \varepsilon_0/C_{\mathrm{emb}}$, so everything below holds verbatim over U_ω with the same constants.

H_g is the appendix's own singular part: the N=2 truncated glued Hadamard parametrix of Lem. an17:G,

$$H_g = \sum_{a \le N(K)} \theta_a \, Z_g^{(2),a}, \qquad Z_g^{(2),a}(x,y) = \frac{1}{8\pi^2}\Big[\frac{\Delta_g^{1/2}(x,y)}{\sigma_{g,\epsilon}(x,y)} + v_g^{(2)}(x,y)\,\log\frac{\sigma_{g,\epsilon}(x,y)}{\ell^2}\Big], \tag{210}$$

$v_g^{(2)} = \sum_{k=0}^{2} v_k[g]\,\sigma_g^k$, on the bi-collar $V_{\delta_c} = \{x \in K, \ r \le \delta_c\}$ of the ball-uniform convex-normal cover (radius δ_c: Chen–LeFloch / Cheeger–Gromov–Taylor, cited at its printed site an17:G, THEOREM; the cutoffs θ_a are *g-independent*, built from the g_0-atlas partition of unity, $\|\theta_a\|_{C^2} \le c_\theta$, overlap multiplicity $m(K)$). Here σ_g is the world function, $\Delta_g^{1/2}$ the van Vleck determinant, $v_k[g]$ the Hadamard coefficients of the transport recursion ([13, Thm. 2.1]), ℓ the single global length scale (an17:G(ii)), and $\sigma_{g,\epsilon} = \sigma_g + 2i\epsilon t_\Delta + \epsilon^2$ the prescription of [276], $\epsilon \downarrow 0$. The pairing $H_g(u \otimes v)$ is the $\epsilon \downarrow 0$ limit, which exists per metric on the cited footing ([13, 276]; this is the single-metric half, established as Lem. an17:clausei-single); every bound below is uniform in $\epsilon \in (0,1]$, so it passes to the limit.

Theorem E2a.1 (Clause (i), difference form, over the full H^s ball; every constant displayed)**.** *Let $n = 4$, $s > n/2 + 6$, U as above, and set $\alpha := \min\big(\frac{1}{2}, \frac{s-8}{2}\big) \in (0, \frac{1}{2}]$. There is a constant*

$$C_H(K) \ = \ C_{\mathrm{sob}}(K)^3\,\kappa(g_0, K),$$

with $C_{\mathrm{sob}}(K)$ and $\kappa(g_0,K)$ the displayed constants (212) *and* (219) *below, such that for all $g, g' \in U$ and all u, v supported in K,*

$$\big|(H_g - H_{g'})(u \otimes v)\big| \ \le \ C_H(K)\,\|g - g'\|_{H^s}\,\|u\|_{H^{s-2}}\,\|v\|_{H^{s-2}}. \tag{211}$$

The bound is ball-uniform (it depends on U only through $(g_0, \varepsilon_0, s, K, \delta_c, \ell)$), state-independent (no state enters (210)*), and holds a fortiori over $U_\omega \times U_\omega$. Sharper form proved en route: one test slot needs only* sup*-norm, $|(H_g - H_{g'})(u \otimes v)| \le C_{\mathrm{sob}}\,\kappa\,\|g - g'\|_{H^s}\|u\|_{C^0}\|v\|_{C^{1,\alpha}}$ (one embedding, on the metric difference alone).*

The constant inventory. All constants are finite and ball-uniform; each is defined at its first use and collected here.

$$C_{\rm sob}(K) := \max\bigl\{C^{\rm Mor}_{s\to(6,\alpha)}(K''),\ C^{\rm Mor}_{(s-2)\to(1,\alpha)}(K)\bigr\}, \tag{212}$$

the two Morrey–Sobolev embedding constants $\|\cdot\|_{C^{6,\alpha}(K'')} \le C^{\rm Mor}_{s\to(6,\alpha)}\|\cdot\|_{H^s(K'')}$ (legal: $s \ge 8+2\alpha > 8$, $n = 4$) and $\|\cdot\|_{C^{1,\alpha}(K)} \le C^{\rm Mor}_{(s-2)\to(1,\alpha)}\|\cdot\|_{H^{s-2}(K)}$ (legal: $s-2 > 6 > 3+\alpha$); Adams–Fournier. The three factors of $C^3_{\rm sob}$ are the three norm conversions of (211): one on the metric difference ($\|h\|_{C^6} \le C_{\rm sob}\|h\|_{H^s}$, $h := g-g'$), one on each test slot. Ball-uniform geometry constants: $\Lambda_U := 1+\sup_{g\in U}\|g\|_{C^{6,\alpha}(K'')} \le 1 + C_{\rm sob}(\|g_0\|_{H^s(K'')}+\varepsilon_0)$; d_U and δ_c as above; $|K|$, $|K''|$ chart volumes; $\Lambda_{\rm vol} := \sup_U \sup_{K''}\sqrt{|\det g|} \le c\,\Lambda_U^2$. The masses of the two singular profiles ($n = 4$; the r^{-2} kernel is where $L^1_{\rm loc}$ at $n = 4$ enters):

$$M_2 := \sup_{x\in K}\int_{r\le\delta_c} r^{-2}\,dy = \tfrac{|S^3|}{2}\delta_c^2 = \pi^2\delta_c^2, \qquad M_{\log} := \sup_{x\in K}\int_{r\le\delta_c}\Bigl(1+\bigl|\log\tfrac{\Lambda_\sigma r^2}{\ell^2}\bigr|\Bigr)dy < \infty. \tag{213}$$

Lemma E2a.2 (L1: Christoffel difference — exact identity, then the constant)**.** *For $g, g' \in U$, $h := g - g'$, with ∇' the g'-connection,*

$$\delta\Gamma^\lambda{}_{\mu\nu} := \Gamma[g]^\lambda{}_{\mu\nu} - \Gamma[g']^\lambda{}_{\mu\nu} = \tfrac12\, g^{\lambda\rho}\bigl(\nabla'_\mu h_{\rho\nu} + \nabla'_\nu h_{\rho\mu} - \nabla'_\rho h_{\mu\nu}\bigr), \tag{214}$$

an exact tensor identity (no remainder). Consequently, for $0 \le k \le 5$,

$$\|\delta\Gamma\|_{C^k(K'')} \le \kappa_\Gamma(k)\,\|h\|_{C^{k+1}(K'')}, \qquad \kappa_\Gamma(k) := c_k(n)\,\Lambda_U^{2k+7}\,d_U^{-2(k+1)}. \tag{215}$$

Proof. The difference of two torsion-free metric connections is a tensor; evaluating $\nabla' g = \nabla'(g'+h) = \nabla' h$ in the Koszul formula for $\Gamma[g]$ written in ∇'-form gives (214) directly (or: verify in g'-normal coordinates at a point, where $\Gamma[g'] = 0$ and (214) is the Koszul formula for h; both sides are tensors, so the identity is global). For (215): $g^{-1} = \mathrm{cof}(g)^{\mathsf T}/\det g$ with cof polynomial of degree $n-1 = 3$ in the entries of g, so $\|g^{-1}\|_{C^k(K'')} \le c_k\Lambda_U^{3+k} d_U^{-(k+1)}$ (Leibniz on the quotient, $|\det g| \ge d_U$); $\nabla' h = \partial h + \Gamma'\cdot h$ with $\|\Gamma'\|_{C^k} \le c\Lambda_U^{k+4} d_U^{-(k+1)}$ by the same count at the single metric g'. Multiply out and absorb the numerical factors into $c_k(n)$. $\square$

Lemma E2a.3 (L2: world-function difference at the Gronwall rate; the rescaled normal form; uniform cone transversality)**.** *There are ball-uniform constants $\Lambda_\sigma, \kappa_\sigma, \Lambda_S, \kappa_S, c_*, c_t > 0$, displayed in the proof, such that on V_{δ_c} (per convex-normal chart a, all statements uniform in a):*

- (i) *(rescaled normal form) $\sigma_g(x, x+r\omega) = r^2 S_g(x,\omega,r)$, $|\omega| = 1$, with $S_g \in C^3$, $\|S_g\|_{C^3(K\times S^3\times[0,\delta_c])} \le \Lambda_S$, and $S_g(x,\omega,0) = \frac12\, q_{g,x}(\omega,\omega)$ where $q_{g,x}$ is the metric quadratic form at x; in particular $|\partial^j_{(x,y)}\sigma_g| \le \Lambda_\sigma r^{(2-j)_+}$, $j \le 2$;*
- (ii) *(difference legs, Lipschitz, real side) for $j = 0,1,2$,*

$$\bigl|\partial^j_{(x,y)}(\sigma_g - \sigma_{g'})\bigr| \le \kappa_\sigma\,\|h\|_{C^{j+1}(K'')}\,r^{2-j}, \qquad \textit{equivalently} \quad \|S_g - S_{g'}\|_{C^j} \le \kappa_S\,\|h\|_{C^{j+1}(K'')};$$

- (iii) *(uniform transversality) for every $\tau \in [0,1]$ the interpolant $S_\tau := (1-\tau)S_{g'} + \tau S_g$ satisfies, on the cone collar $\{|S_\tau| \le \theta_*\}$, $|\nabla_\omega S_\tau| \ge c_* > 0$ and $|t_\Delta| \ge c_t\, r$ with $\operatorname{sign} t_\Delta$ constant on each (future/past) sheet; $\theta_* = \theta_*(d_U, \Lambda_U) > 0$.*

Proof. (i) In the chart, the g-exponential flow $\Phi^t_g(x,\xi)$ solves $\ddot\gamma = -\Gamma[g](\dot\gamma,\dot\gamma)$; on the compact shell $\{x\in K, |\xi| \le 2\delta_c\}$ its C^3-norm in $(x,\xi,t)_{t\in[0,1]}$ is bounded by the Picard/Gronwall constant $\kappa_{\rm fl} := c\,\exp\bigl(c\,\delta_c\,\|\Gamma\|_{C^3(K'')}\bigr) \le c\exp\bigl(c\,\delta_c\,\Lambda_U^7 d_U^{-4}\bigr)$ (coefficients $\Gamma[g] \in C^{4,\alpha}$ since $g \in C^{6,\alpha}$; three derivatives of the flow consume $\|\Gamma\|_{C^3}$). With $\xi = \exp_x^{-1}(y)$ (well defined and C^3, Jacobian $\ge \frac12$

on V_{δ_c} after the one ball-uniform shrink of δ_c built into an17:G) and $\sigma_g(x,y) = \frac{1}{2}q_{g,x}(\xi,\xi)$, homogeneity of the geodesic flow in (ξ,t) gives $\xi = r\,\Xi_g(x,\omega,r)$ with $\Xi_g \in C^3$, $\Xi_g(x,\omega,0) = \omega$; set $S_g := \frac{1}{2}q_{g,x}(\Xi_g,\Xi_g)$. Then $\Lambda_S \le c\,\Lambda_U\kappa_{\mathrm{fl}}^2$ and $\Lambda_\sigma \le c\,\Lambda_S$. (ii) The flows of g,g' differ by a linear-inhomogeneous (variational) ODE with source $\delta\Gamma(\dot\gamma,\dot\gamma)$: Gronwall gives $\|\Phi_g - \Phi_{g'}\|_{C^j(\mathrm{shell})} \le \kappa_{\mathrm{fl}}^2\,\|\delta\Gamma\|_{C^j}$, $j \le 2$, hence $\|\Xi_g - \Xi_{g'}\|_{C^j} \le c\,\kappa_{\mathrm{fl}}^3\|\delta\Gamma\|_{C^j}$ (inverse-function difference at Jacobian $\ge \frac{1}{2}$), and with (215), $\kappa_S := c\,\Lambda_U\kappa_{\mathrm{fl}}^3\big(1+\kappa_\Gamma(3)\big)$, $\kappa_\sigma := c\,\kappa_S$. The r-grading in the unscaled form is the chain rule applied to $r^2S(x,\omega,r)$, $\omega = (y-x)/r$: each (x,y)-derivative costs one factor r at worst. (iii) At $r=0$: $\nabla_\omega S_\tau = q_{\tau,x}(\omega,\cdot)$ with $q_\tau := (1-\tau)q_{g'} + \tau q_g$ — again a Lorentzian metric in the convex ball U, $|\det q_\tau| \ge d'_U > 0$ for ε_0 small (continuity of det on the C^0-ball). On the set $\{S_\tau = 0, |\omega| = 1\}$, ω is q_τ-null, and for a Lorentzian form with $|\det| \ge d'_U$, $\|q\| \le \Lambda_U$ one has $|q(\omega,\cdot)| \ge c(d'_U,\Lambda_U) > 0$ on unit null vectors (else ω would be a null eigenvector of q with vanishing eigenvalue, contradicting $|\det q| \ge d'_U$); perturbing in $r \le \delta_c$ and in $|S_\tau| \le \theta_*$ costs a factor $\frac{1}{2}$ for θ_*,δ_c small ball-uniformly: $c_* := \frac{1}{2}c(d'_U,\Lambda_U)$. For c_t: on the q_τ-null cone, $|\omega^0| \ge c(d'_U,\Lambda_U) > 0$ (time component of unit null vectors is bounded below when the cones are pinched between two g_0-cones, which is (u2)-smallness), and $t_\Delta = -r\,\omega^0\,\partial_0 t + O(\Lambda_S r^2)$, so $c_t := \frac{1}{2}c(d'_U,\Lambda_U)\inf_{K''}|\partial_0 t|$ after one more shrink of δ_c; the sign is constant per sheet because ω^0 has a fixed sign on each component of the null cone. The time leg extends from the collar to the full timelike side $\{S_\tau \le \theta_*\}$ by the same linear algebra: on unit vectors with $q_\tau(\omega,\omega) \le \theta_*$, a time component $|\omega^0| < c'$ would force $q_\tau(\omega,\omega) \ge c(d'_U,\Lambda_U) - O(c') > \theta_*$ (the spatial block is positive definite with determinant bounded below), a contradiction; so $|\omega^0| \ge c'(d'_U,\Lambda_U) > 0$, hence $|t_\Delta| \ge c_t r$ with per-sheet fixed sign, on all of $\{S_\tau \le \theta_*\}$, $r \le \delta_c$. □

Lemma E2a.4 (L3: van Vleck and Hadamard-coefficient legs — transport/Gronwall, sup and difference)**.** *With Λ_S,κ_S as in L2, there are displayed ball-uniform $\Lambda_\Delta,\kappa_\Delta,\Lambda_v,\kappa_v$ with, on V_{δ_c}:*

(i) $\|\Delta_g^{\pm1/2}\|_{C^{1,\alpha}(V)} \le \Lambda_\Delta$ *and* $\|\Delta_g^{1/2} - \Delta_{g'}^{1/2}\|_{C^{0,\alpha}(V)} \le \kappa_\Delta\|h\|_{C^4(K'')}$*;*

(ii) *for $k \le 2$:* $\|v_k[g]\|_{C^{0,\alpha}(V)} \le \Lambda_v$ *and* $\|v_k[g]-v_k[g']\|_{C^0(V)} \le \kappa_v\|h\|_{C^{2k+2}(K'')} \le \kappa_v\|h\|_{C^6(K'')}$*.*

Proof. (i) $\Delta_g(x,y) = -\det\big(-\partial_x\partial_y\sigma_g\big)\big(|\det g(x)||\det g(y)|\big)^{-1/2}$; by L2(i)–(ii) the entries $\partial_x\partial_y\sigma_g$ are $C^{1,\alpha}$ with norm $\le c\Lambda_S$ and difference $\le c\kappa_S\|h\|_{C^4}$ in $C^{0,\alpha}$ (one derivative above the $j=2$ leg, obtained by running L2(ii) at $j=2$ on the $C^{1,\alpha}$ flow bounds — the flow is C^3, one order to spare); the determinant is a cubic polynomial, and on V_{δ_c} the matrix is invertible with determinant $\ge \frac{1}{2}d_U$ (coincidence value $\det(-\partial^2\sigma) = |\det g|$, plus the ball-uniform shrink of δ_c), so the square roots are Lipschitz functions of the entries: $\Lambda_\Delta := c\,\Lambda_S^4 d_U^{-3/2}$, $\kappa_\Delta := c\,\Lambda_S^3 d_U^{-3/2}\kappa_S$. (ii) The recursion of [13, Thm. 2.1], $2\nabla\sigma\cdot\nabla v_k + (\Box\sigma - 2n + 4k)v_k = -\Box v_{k-1}$ (v_{-1}-convention anchoring v_0 at $\Delta^{1/2}$-data), is for each k a linear transport ODE along the L2 geodesics: in the rescaled variable it reads $\big(2r\partial_r + A_k(x,\omega,r)\big)v_k = f_k$ with $A_k = \Box\sigma - 2n + 4k + O(r)$, $|A_k| \le c\Lambda_S$, and source f_k a universal polynomial in the jets $(\partial^{\le 2}\sigma, \Delta^{\pm1/2}, \partial^{\le 2k+2}g, g^{-1})$. Duhamel along $[0,r]$ with the regular-singular weight $r^{-A_k/2}$-type factor (the indicial root is positive for $k \ge 0$, $n=4$: $\Box\sigma \to 2n$ at coincidence, so $A_k \to 4k > 0$ for $k \ge 1$ and the $k=0$ equation fixes $v_0 = \Delta^{1/2}$-transported data) gives the sup bound by Gronwall, $\Lambda_v := c\,e^{c\Lambda_S\delta_c}P\big(\Lambda_S,\Lambda_\Delta,\Lambda_U,d_U^{-1}\big)$ (P a displayed universal polynomial; the $C^{0,\alpha}$ grade rides the same bound one order down, legal since $g \in C^{6,\alpha}$ and f_2 consumes exactly $\partial^6 g \in C^{0,\alpha}$). The difference leg is the same Duhamel applied to $(v_k[g] - v_k[g'])$, whose ODE has coefficient difference $\le c(\kappa_S + \kappa_\Gamma(1))\|h\|_{C^2}$ and source difference $\le cP\cdot\|h\|_{C^{2k+2}}$ (linearity of f_k-assembly in its top jet; lower-order factors estimated by the sup bounds already obtained — this is the linear-in-top-jet structure the appendix's (N-a) block records at l.3219–3222): $\kappa_v := c\,e^{c\Lambda_S\delta_c}P\cdot\big(1 + \kappa_S + \kappa_\Gamma(1)\big)$. No step differentiates in g; every difference is pair-Lipschitz on the real side. □

Lemma E2a.5 (L4: uniform cone pairing — the analytic core)**.** *Fix $x \in K$ and a phase $\rho = r^2S$ with $S \in \{S_\tau\}_{\tau\in[0,1]}$ as in L2 (so $\|S\|_{C^3} \le \Lambda_S$ — L2(i) proves the C^3 flow bound — transversality (c_*,θ_*), time leg $(c_t, \Lambda_T := \|t_\Delta/r\|_{C^1})$, and $\rho_\epsilon := \rho + 2i\epsilon t_\Delta + \epsilon^2$. Let $F(y) = r^a\Phi(\omega,r)$ in $y = x + r\omega$, $\operatorname{supp} F \subset \{r \le \delta_c\}$. Then, uniformly in $\epsilon \in (0,1]$ and in τ:*

(P1) *if $a \geq 0$ and $\Phi \in C^{0,\alpha}$:* $\Big|\int \frac{F}{\rho_\epsilon}\,dy\Big| \leq \kappa_P\, \|\Phi\|_{C^{0,\alpha}}\, \frac{\delta_c^{a+2}}{a+2}$;

(P2) *if $a \geq 2$ and $\Phi \in C^{1,\alpha}$:* $\Big|\int \frac{F}{\rho_\epsilon^2}\,dy\Big| \leq \kappa_P\, \|\Phi\|_{C^{1,\alpha}}\, \frac{\delta_c^{a}}{a}$;

(P3) *if $a \geq 0$ and $\Phi \in C^{0}$:* $\int |F|\,\big|\log(\rho_\epsilon/\ell^2)\big|\,dy \leq \kappa_P\, \|\Phi\|_{C^0}\, \delta_c^{a+2}\big(1+|\log\delta_c^2/\ell^2|\big)$;

with the single displayed constant

$$\kappa_P := c_n\big(1+\Lambda_S+c_*^{-1}\big)^4\Big[\theta_*^{-2}\varkappa^{-2}+(\pi+4)+\tfrac{2}{\alpha}\theta_*^\alpha+\big(1+c_t^{-2}\big)\big(1+\Lambda_T\big)\Big], \qquad \varkappa := \min\big(\tfrac12,\ \sqrt{2}\,c_t/\sqrt{\Lambda_S}\big). \tag{216}$$

Proof. Rescale: $dy = r^3\,dr\,d\omega$, so $\int F\rho_\epsilon^{-2}dy = \int_0^{\delta_c} r^{a-1}\big[\int_{S^3}\Phi\,(S+i\eta)^{-2}d\omega\big]dr$ with $\eta(\omega) := (2\epsilon t_\Delta+\epsilon^2)/r^2 = \eta_0 T(\omega,r)+\epsilon^2/r^2$, $\eta_0 := 2\epsilon/r$, $T := t_\Delta/r$ ($|T| \geq c_t$ with fixed sign per sheet on the cone collar, L2(iii)); similarly $\int F\rho_\epsilon^{-1}dy = \int_0^{\delta_c} r^{a+1}[\,\cdot\,]dr$ with $(S+i\eta)^{-1}$. The real shift ϵ^2/r^2: where $\epsilon^2 \geq \theta_* r^2/2$, i.e. $r \leq \epsilon\sqrt{2/\theta_*}$, split by the size of $r^2|S|$. If $r^2|S| \leq \epsilon^2/2$ then $|\rho_\epsilon| \geq \operatorname{Re}\rho_\epsilon = r^2S+\epsilon^2 \geq \epsilon^2/2$; otherwise $r^2 \geq \epsilon^2/(2\Lambda_S)$ (since $|S| \leq \Lambda_S$) and the time leg — valid on the whole set $\{S_\tau \leq \theta_*\}$, collar plus timelike interior, by L2(iii)'s extension — gives $|\rho_\epsilon| \geq |\operatorname{Im}\rho_\epsilon| = 2\epsilon|t_\Delta| \geq 2\epsilon c_t r \geq \sqrt{2}\,c_t\,\epsilon^2/\sqrt{\Lambda_S}$. Hence $|\rho_\epsilon| \geq \epsilon^2\varkappa$ with $\varkappa := \min\big(\tfrac12,\ \sqrt{2}\,c_t/\sqrt{\Lambda_S}\big)$, and $|\rho_\epsilon| \geq \varkappa\,\theta_* r^2/2$ on this range, so the slice integrand is $\leq \|\Phi\|_\infty(2/\theta_*)^2\varkappa^{-2}|S^3|$ pointwise and this radial range contributes $\leq c\|\Phi\|_\infty\theta_*^{-2}\varkappa^{-2}\delta_c^a/a$; elsewhere absorb $\epsilon^2/r^2 \leq \theta_*/2$ into S (transversality survives with $c_*/2$, $\theta_*/2$). It remains to bound the slice integrals $J_p := \int_{S^3}\Phi(S+i\eta_0T)^{-p}d\omega$, $p=1,2$, uniformly in (r,η_0). Split S^3: on $\{|S|>\theta_*\}$, $|J_p^{\rm far}| \leq \|\Phi\|_\infty|S^3|\theta_*^{-p}$. On the cone collar, the level flow of $S/|\nabla S|^2$ gives coordinates (z,ζ), $z \in Z = \{S=0\}$, $S=\zeta$, $|\zeta| \leq \theta_*$, with Jacobian J_z, $\|J_z\|_{C^{1,\alpha}} \leq c(1+\Lambda_S+c_*^{-1})^4$ (one further implicit-function differentiation, licensed by $\|S\|_{C^3} \leq \Lambda_S$ at $|\nabla S| \geq c_*$ — the $p=2$ fiber Taylor split consumes $J_z \in C^{1,\alpha}$; the null sphere Z has at most two sheets, the future and past cones of the Lorentzian normal form, L2(iii)). Per fiber, with $G := \Phi J_z$ and T frozen at its foot value T_0 ($|T_0| \geq c_t$, sign fixed), $a := \eta_0 T_0 \neq 0$: the exact complex antiderivatives give the four moment bounds, each uniform in $a \neq 0$ ($L := \theta_*$),

$$\Big|\int_{-L}^{L}\frac{d\zeta}{(\zeta+ia)^2}\Big| = \Big|\frac{-2L}{(L+ia)(-L+ia)}\Big| \leq \frac{2}{L}, \qquad \Big|\int_{-L}^{L}\frac{\zeta\,d\zeta}{(\zeta+ia)^2}\Big| = \Big|\log\frac{L+ia}{-L+ia}+\frac{ia}{\zeta+ia}\Big|_{-L}^{L}\Big| \leq \pi+2, \tag{217}$$

$$\Big|\int_{-L}^{L}\frac{d\zeta}{\zeta+ia}\Big| = \Big|\log\frac{L+ia}{-L+ia}\Big| \leq \pi, \qquad \int_{-L}^{L}\frac{|\zeta|^\beta\,d\zeta}{|\zeta+ia|^2} \leq \int_{-L}^{L}|\zeta|^{\beta-2}d\zeta = \tfrac{2}{\beta-1}L^{\beta-1} \quad (\beta>1), \tag{218}$$

all uniform in $a \neq 0$ (the middle identity uses $|L+ia| = |-L+ia|$, so the log has modulus $\leq \pi$, its imaginary part). For $p=2$ Taylor-split $G = G_0+\zeta G_0'+R$, $|R| \leq [G']_\alpha|\zeta|^{1+\alpha}$, and apply (217)–(218) with $\beta = 1+\alpha$: the fiber integral is $\leq \big(\frac{2}{\theta_*}+\pi+2+\frac{2}{\alpha}\theta_*^\alpha\big)\|G\|_{C^{1,\alpha}}$. For $p=1$ split $G = G_0+(G-G_0)$, $|G-G_0| \leq [G]_\alpha|\zeta|^\alpha$: fiber $\leq (\pi+\frac{2}{\alpha}\theta_*^\alpha)\|G\|_{C^{0,\alpha}}$. Unfreezing T: $|\eta(\zeta)-\eta_0T_0| \leq \eta_0\Lambda_T|\zeta|/c_*$ and $|\zeta+i\eta| \geq \max(|\zeta|,\eta_0c_t)/2$ on the fiber, so the freezing error for $p=2$ is $\leq c\|G\|_\infty\Lambda_Tc_*^{-1}\int\eta_0|\zeta|\max(|\zeta|,\eta_0c_t)^{-3}d\zeta \leq c\|G\|_\infty\Lambda_Tc_*^{-1}(1+c_t^{-2})$, and for $p=1$ $\leq c\|G\|_\infty\Lambda_Tc_*^{-1}(1+c_t^{-1})$ — uniform in $\eta_0>0$. Integrate over $z \in Z$ ($|Z| \leq c_n$) and the two sheets, then radially: $\int_0^{\delta_c}r^{a-1}dr = \delta_c^a/a$ for (P2) ($a \geq 2$; in unrescaled terms the integrand is dominated by $r^ar^{-4}\cdot r^3 = r^{a-1}$, the $n=4$ $L^1_{\rm loc}$ of the r^{-2} kernel at $a=2$, mass M_2 of (213)), and $\int_0^{\delta_c}r^{a+1}dr = \delta_c^{a+2}/(a+2)$ for (P1). For (P3): $|\log(\rho_\epsilon/\ell^2)| \leq 2|\log r|+|\log|S+i\eta||+|\log\ell^2|+2\pi$ and $\int_{S^3}|\log|S+i\eta||\,d\omega \leq c(1+|\log\theta_*|)$ uniformly in η (on fibers $|\log|\zeta+ia|| \leq |\log|\zeta||+c$ for $|\zeta| \leq \theta_* \leq c$), so the radial integral converges to the stated mass. Collect (216). $\square$

Proof of Theorem E2a.1 (assembly; four legs). Fix $\epsilon \in (0,1]$ and a chart a; drop the chart index. With $w := \sigma_g-\sigma_{g'}$ and the linear interpolant $\sigma_{\tau,\epsilon} := (1-\tau)\sigma_{g',\epsilon}+\tau\sigma_{g,\epsilon} = r^2S_\tau+2i\epsilon t_\Delta+\epsilon^2$ (same t_Δ: the prescription is g-independent), the exact calculus identities $\sigma_{g,\epsilon}^{-1}-\sigma_{g',\epsilon}^{-1} = -\int_0^1 w\,\sigma_{\tau,\epsilon}^{-2}\,d\tau$

and $\log\sigma_{g,\epsilon} - \log\sigma_{g',\epsilon} = \int_0^1 w\,\sigma_{\tau,\epsilon}^{-1}\,d\tau$ give

$$8\pi^2\big(Z_g^{(2)} - Z_{g'}^{(2)}\big) = \underbrace{\frac{\Delta_g^{1/2} - \Delta_{g'}^{1/2}}{\sigma_{g,\epsilon}}}_{\text{(A)}} - \underbrace{\Delta_{g'}^{1/2}\int_0^1 \frac{w\,d\tau}{\sigma_{\tau,\epsilon}^2}}_{\text{(B)}} + \underbrace{\big(v_g^{(2)} - v_{g'}^{(2)}\big)\log\frac{\sigma_{g,\epsilon}}{\ell^2}}_{\text{(C)}} + \underbrace{v_{g'}^{(2)}\int_0^1 \frac{w\,d\tau}{\sigma_{\tau,\epsilon}}}_{\text{(D)}}.$$

Pair each leg with $\theta_a\,u(x)v(y)$, integrate in y at fixed x by L4, then in x. The graded weights, from L2–L3 (all differences real-side, none through any collar): (A) $F = (\Delta_g^{1/2} - \Delta_{g'}^{1/2})\theta_a u(x)v$: $a = 0$, $\|\Phi\|_{C^{0,\alpha}} \le c_\theta\kappa_\Delta\|h\|_{C^4}\,|u(x)|\,\|v\|_{C^{0,\alpha}}$; apply (P1). (B) $F = \Delta_{g'}^{1/2} w\,\theta_a u(x)v$: $w = r^2W$ with $\|W\|_{C^{1,\alpha}} \le c\,\kappa_S\|h\|_{C^3}$ (L2(ii), $j \le 2$, plus interpolation to the α-grade), so $a = 2$, $\|\Phi\|_{C^{1,\alpha}} \le c\,c_\theta\Lambda_\Delta\kappa_S\|h\|_{C^3}\,|u(x)|\,\|v\|_{C^{1,\alpha}}$; apply (P2) for each τ (the phase family $\{S_\tau\}$ is admissible by L2(iii)) and integrate $d\tau$. (C) $v_g^{(2)} - v_{g'}^{(2)} = \sum_{k\le 2}\big[(v_k[g] - v_k[g'])\sigma_g^k + v_k[g']\,(\sigma_g^k - \sigma_{g'}^k)\big]$ and $\sigma_g^k - \sigma_{g'}^k = w\sum_{j<k}\sigma_g^j\sigma_{g'}^{k-1-j}$, so $\|v_g^{(2)} - v_{g'}^{(2)}\|_{C^0(V)} \le \kappa'_C\|h\|_{C^6}$ with $\kappa'_C := c\big[\kappa_v(1+\Lambda_\sigma\delta_c^2)^2 + \Lambda_v\kappa_\sigma\delta_c^2(1+\Lambda_\sigma\delta_c^2)\big]$; apply (P3), $a = 0$, weight $|u(x)|\|v\|_{C^0}$. (D) $F = v_{g'}^{(2)} w\,\theta_a u(x)v$: $a = 2$, $\|\Phi\|_{C^{0,\alpha}} \le c\,c_\theta\Lambda_v(1+\Lambda_\sigma\delta_c^2)^2\kappa_S\|h\|_{C^2}\,|u(x)|\,\|v\|_{C^{0,\alpha}}$; apply (P1) with $a = 2$, per τ. Summing the four legs, the charts ($\le m(K)$ overlapping terms), and integrating $|u(x)|$ over K:

$$\big|(H_g - H_{g'})(u\otimes v)\big| \le \kappa(g_0,K)\,\|h\|_{C^6(K'')}\|u\|_{C^0(K)}\|v\|_{C^{1,\alpha}(K)}, \tag{219}$$

$$\kappa(g_0,K) := \frac{|K|\,m(K)\,c_\theta\,\kappa_P}{8\pi^2}\Big[\underbrace{\kappa_\Delta\tfrac{\delta_c^2}{2}}_{\text{(A)}} + \underbrace{c\,\Lambda_\Delta\kappa_S\tfrac{\delta_c^2}{2}}_{\text{(B)}} + \underbrace{\kappa'_C\,\delta_c^2\big(1+|\log\tfrac{\delta_c^2}{\ell^2}|\big)}_{\text{(C)}} + \underbrace{c\,\Lambda_v(1+\Lambda_\sigma\delta_c^2)^2\kappa_S\tfrac{\delta_c^4}{4}}_{\text{(D)}}\Big].$$

This is the sharper (asymmetric) form. Since $\|h\|_{C^6} \le \|h\|_{C^{6,\alpha}} \le C_{\rm sob}\|h\|_{H^s}$, $\|u\|_{C^0} \le C_{\rm sob}\|u\|_{H^{s-2}}$ and $\|v\|_{C^{1,\alpha}} \le C_{\rm sob}\|v\|_{H^{s-2}}$ (212), the symmetric form (211) follows with $C_H = C_{\rm sob}^3\kappa$. All bounds are uniform in $\epsilon \in (0,1]$; letting $\epsilon \downarrow 0$ against the per-metric limits (cited footing) completes the proof. No state was used at any step. $\square$

Remark E2a.6 (Three-way fidelity: what this port states, against the appendix demand and the established theorem). (a) *Appendix demand.* Display (211) is verbatim eq:an17-clausei = clause (i) of Hyp. hyp:hadamard-uniform (Option B) and of Hyp. hyp:hadamard-uniform-omega: same pairing, same three norms H^s, H^{s-2}, H^{s-2}, same support condition, and the quantifier is over the *pair* (g,g') ranging over the full ball U — strictly containing the $U_\omega\times U_\omega$ range item (M3) asks for (an17:lem-embed). H_g is the appendix's own object (210) (the N=2 glued parametrix, one global ℓ). No hypothesis is strengthened; the only weakening is in the *conclusion's* favor (full ball, asymmetric sharper form). (b) *The established clause-(i) theorem.* The constant reproduces its architecture exactly — $C_H = C_{\rm sob}^3\kappa$ with the cube accounting for the three norm conversions (212); the Christoffel difference is the exact identity (214); all transport legs are Gronwall along geodesics (L2, L3); the r^{-2} kernel enters as the a=2, p=2 leg of L4, $L^1_{\rm loc}$ at $n = 4$ with mass M_2 (213); the result is state-independent. The suppressed arguments of the compact notation are displayed here: $\kappa = \kappa(g_0, K; s, \varepsilon_0, \delta_c, \ell, \text{atlas})$. (c) *Single-metric consistency.* Applying L4 legs (P1)/(P3) to H_g alone (weights $\Lambda_\Delta, \Lambda_v$ instead of the κ-differences) re-proves the single-metric Lem. an17:clausei-single bound with $C_H(K,g) \le C_{\rm sob}^2\,\kappa'(g_0,K)$, consistent with GT1.

Remark E2a.7 (Sharpening the test-space threshold: $t > 3$, not $t > 2$). A prior side note records that the H^{s-2} test space is over-strong and "any H^t, $t > 2$ suffices". The re-derivation *confirms the over-strength but corrects the threshold*: the legs (A), (C), (D) need only $C^{0,\alpha}$ ($t > 2+\alpha$) in the v-slot and C^0 in the u-slot, but the leading leg (B) — the r^{-2} kernel — consumes one full transversal derivative of the test pair: $\mathrm{Fp}(\sigma_\tau^{-2})$ pairs at the Lipschitz rate only against $C^{1,\alpha}$, i.e. H^t, $t > 3+\alpha$, in one slot. At $2 < t \le 3$ the pairing rate degrades to Hölder $\|g - g'\|^{t-2}$: a shifted first-order pole tested against $C^{0,\beta}$ moves at rate d^β, not d (numerically confirmed at exponents 0.50 vs 1.00; the mechanism is the same Hölder-vs-Lipschitz conversion loss as the

EP2 fence, now on the real side). Consequence for consumers: none — the clause consumes H^{s-2}, $s-2>6$, with room to spare on every leg; but the side note should read "any H^t, $t>3$, in one slot ($t>2$ in the other)". This corrects a side *remark*, not any printed statement of the paper or appendix.

Remark E2a.8 (Grade and consumers; what this port does NOT do). This port closes item (M3) of sec:an17-modulo at THEOREM grade: the difference form (211) is now *derived*, ball-uniform over the family, at the hypothesis' own standing fence $s > n/2+6$ (no re-fence: the proof consumes $\|h\|_{C^6}$, $\|g\|_{C^{6,\alpha}}$ with $\alpha = \min(\frac{1}{2}, \frac{s-8}{2})$, both inside H^s, $s>8$). Consumer effect (per the printed audit, appendix l.3588–3591): clause (i)-diff feeds only the singular leg of thm:E2-Lipschitz and no quantitative slice-register consumer, so no downstream constant changes; the pivot's conditionality on (M1) and (M2) is untouched. This port says *nothing* about the finite part W_g (clause (ii)/M1: still PLAUSIBLE at its printed modulo-list) or the QEI constants (clause (iii)/M2: reduced-at-consumed-strength, named). The paper's hypothesis statements are unchanged; this port supplies the clause-(i) derivation only.

E2a.2 The family-uniformity package (Props A/C and Lemma B), written out

P-AC. The family-uniformity package for clause (ii): Propositions A and C and Lemma B, written out

Scope and wiring (read first). Target: the two printed gaps of Rem. an17:CW-grade (appendix l.3394–3402): (a) the N=2 finite-regularity Hadamard-parametrix *construction* with quantitative v_2, family-uniform over the ball, not in print; (b) the exact multi-index constant-tracking through the N=2 commutator source and the Tice estimate, not executed. On the analytic class U_ω both are discharged *as family-uniformity questions* below; what survives is the modulo-list, restated in Theorem E2a.14. **Wiring order [W1]:** Proposition E2a.10 → Proposition E2a.12 → Theorem E2a.14 → Lemma E2a.15. Lemma E2a.15 (the seed triangle for W_*) is *consumer-facing only*: it is stated *after* the C_W chain closes and is cited by *no* proposition of this package. The in-ladder spectator sups of Proposition E2a.12 are wired to the single-metric route (an17:Nc-corrected's own uniform analogue + an17:DoD transport, ($D0$)-gated), *never* to Lemma E2a.15 — citing Lemma B there would be circular (Lemma B derives W_* *from* C_W), so that wiring is avoided here.

The four conditions folded into the package.

W1 *Circular wiring* (a naive Prop. C would cite Lemma B for in-ladder spectator sups; Lemma B derives W_* from C_W). Fix: the wiring order above; spectator sups re-routed at Prop. E2a.12(C3). Marked there.

W2 *#B′ omitted* (the appendix books lem:W2star gated by ($D0$) *and* #B′, l.3325–3327 and l.3951–3952; listing only ($D0$) would be incomplete). Fix: #B′ restored to the modulo-list with its conditional discharge sentence, Rem. E2a.13. Marked there.

W3 *Grade inheritance* ("$C_{\Delta_2}=0$ on BFV" lives inside an17:Nc-corrected, printed grade "PLAUSIBLE …NEVER cited at THEOREM grade", l.3289–3293; citing it bare would drop that grade). Fix: every BFV-vanishing citation below carries the carrier grade and the revival condition. Marked at each use.

W4 *Seed tag* (a bare Lemma B would label W_*^{seed} "THEOREM"; the verdict's own residue (res1) makes it ($D0$)-gated at the b=1 slot). Fix: Lemma E2a.15 tags it "THEOREM modulo ($D0$)". Marked there.

Lemma E2a.9 (EP1′: spacetime polydisc Cauchy estimates, with the $\sqrt{n}$ polyradius made explicit)**.** *Let $g \in U_\omega$, $F := g - g_0$ (components F_{ab}, holomorphic on $\mathcal{K}_{\rho_a} = \{z \in \mathbb{C}^n : \mathrm{dist}(z, K'') <$*

$\rho_a\}$, $\|F\|_{\mathcal{K}_{\rho_a}} \le \varepsilon_a$). *Then for every multi-index* α *and every* z_0 *with* $\operatorname{dist}(z_0, K'') \le \rho_a/2$,

$$|\partial^\alpha F(z_0)| \;\le\; \alpha!\,\Big(\frac{2\sqrt{n}}{\rho_a}\Big)^{|\alpha|}\varepsilon_a, \qquad \textit{hence} \quad \sup_{U_\omega}\sup_{\mathcal{K}_{\rho_a/2}} |\partial^\alpha(g-g_0)| \le \alpha!\,\Big(\frac{2\sqrt{n}}{\rho_a}\Big)^{|\alpha|}\varepsilon_a. \tag{220}$$

(The owed $\sqrt{n}$ parenthetical: the collar is a chart-Euclidean tube in $\mathbb{C}^n$, so the largest polydisc centred at z_0 and contained in $\mathcal{K}_{\rho_a}$ has polyradius $(\rho_a/2)/\sqrt{n}$ per coordinate — a corner of a polydisc of polyradius r lies at Euclidean distance $r\sqrt{n}$. A naive EP1 displays $(2/\rho_a)^{|\alpha|}$, and the appendix's own an17:lem-embed proof (l.166–175) says "polydisc of radius ρ_a"; both are constant-level slips with no structural effect — the appendix absorbs its version into the unspecified $C_{\mathrm{cw}}(K'', n, s)$, so nothing printed is false — but every EP1-derived constant below carries the honest $\sqrt{n}$.)

Proof. The polydisc $P(z_0; r, \dots, r)$, $r = \rho_a/(2\sqrt{n})$, satisfies $\sup_{w\in P}|w - z_0| \le r\sqrt{n} = \rho_a/2$, so $P \subset \mathcal{K}_{\rho_a}$ when $\operatorname{dist}(z_0, K'') \le \rho_a/2$. The iterated one-variable Cauchy inequality on P gives $|\partial^\alpha F(z_0)| \le \alpha!\, r^{-|\alpha|}\sup_P |F| \le \alpha!\,(2\sqrt{n}/\rho_a)^{|\alpha|}\varepsilon_a$. This is the appendix's own method (an17:lem-embed, printed site l.166–175), iterated to all orders: ball-uniform C^k bounds for *every* k with geometric constants in $(\rho_a, \varepsilon_a, n)$ alone — no Sobolev index, no C^k norm of a sample metric. It is the licensed analytic tool of the package (spacetime variable only; the EP0/EP2 fences remain in force: nothing below differentiates in g, and no difference estimate routes through the collar). □

Proposition E2a.10 (A: the (N-a) parametrix package is family-uniform over U_ω — gap (a) discharged on the analytic class)**.** *Let* $U_\omega = U_\omega(g_0, \rho_a, \varepsilon_a)$, $n = 4$, $s > 11$, $K \subset \mathcal{M}$ *compact. There exist* $\delta_c^{\mathbb{C}} > 0$ *and constants* $\Lambda_0, \Lambda_1, \Lambda_2, C_{v_2}^{\mathrm{fam}}$, *all functions of* $(g_0, \rho_a, \varepsilon_a, K, \mathrm{atlas})$ *alone, such that for all* $g, g' \in U_\omega$*:*

(A1) *(uniform normal geometry, holomorphic) every* $x \in K$ *has a* g*-convex normal chart of radius* $\ge \delta_c$ *(the an17:G radius), and* $\sigma_g, \Delta_g^{\pm 1/2}$ *exist on the bi-collar* V_{δ_c}*, are real-analytic, and extend holomorphically to the uniform complex bi-collar* $\mathcal{V}^{\mathbb{C}} := \{(z,w) : z \in \mathcal{K}_{\rho_a/4},\ |w - z| < \delta_c^{\mathbb{C}}\}$ *with* $\sup_{U_\omega}\sup_{\mathcal{V}^{\mathbb{C}}}\big(|\sigma_g| + |\Delta_g^{\pm 1/2}|\big) \le \Lambda_0$*;*

(A2) *(quantitative* v_k*, all orders from one sup) the Hadamard coefficients* v_0, v_1, v_2 *of the transport recursion ([13, Thm. 2.1], cited in print at appendix l.3208–3210) exist on* V_{δ_c}*, extend holomorphically to* $\mathcal{V}^{\mathbb{C}}$*, and satisfy, for every bi-multi-index* β*,*

$$\sup_{U_\omega}\sup_{\mathcal{V}^{\mathbb{C}}_{1/2}} |\partial^\beta v_k[g]| \;\le\; \beta!\,\Big(\frac{4\sqrt{2n}}{\min(\rho_a, \delta_c^{\mathbb{C}})}\Big)^{|\beta|}\Lambda_k \qquad (k \le 2),$$

($\mathcal{V}^{\mathbb{C}}_{1/2}$ *the half bi-collar; the* $\sqrt{2n}$ *is EP1′ in the* $2n$ *complex bi-variables); in particular the consumed registers* $v_2 \in H^{s-6}$, $\Box_g v_2 \in H^{s-8}$ *of (N-a-NF) hold with ball-uniform norms* $\le C_{v_2}^{\mathrm{reg}} := c(s)\,|V_{\delta_c}|^{1/2}\max_{|\beta|\le s}\beta!\,(4\sqrt{2n}/\min(\rho_a, \delta_c^{\mathbb{C}}))^{|\beta|}\Lambda_2$*;*

(A3) *(difference legs at the Lipschitz rate, real side only) for the same registers,*

$$\|v_k[g] - v_k[g']\|_{H^{s-2k-2}(V_{\delta_c})} + \|\Box_g v_2[g] - \Box_{g'} v_2[g']\|_{H^{s-8}(V_{\delta_c})} \;\le\; C_{v_2}^{\mathrm{fam}}\,\|g - g'\|_{H^s};$$

(A4) *(normal form) the glued* $N{=}2$ *residual has the printed normal form (N-a-NF) (appendix l.3213–3218):* $S_2' = (\Box_{g'} v_2')\,\sigma_{\mathrm{geo}}^2 \log\sigma_{\mathrm{geo}} + R_{\mathrm{sm}}$*, pair* $(8,6)$*,* R_{sm} *strictly softer, all constants ball-uniform.*

Proof. (A1). The ball-uniform *real* convex-normal radius δ_c is in print at THEOREM grade: an17:G, l.3233–3236 (Chen–LeFloch / Cheeger–Gromov–Taylor from g_0-data and $\sup_{U_\omega}\|g\|_{H^s}$). For the holomorphic extension: on $\mathcal{K}_{\rho_a/2}$ the Christoffels $\Gamma[g]$ are holomorphic with $M_\Gamma := \sup_{U_\omega}\|\Gamma[g]\|_{\mathcal{K}_{\rho_a/2}} < \infty$ — this is the appendix's own analytic-ODE package, an17:rem-CK (printed site l.465–482, display an17:eq-MGamma; the denominators carry $|\det g| \ge d_\omega$, l.142–150), which Part (A) of the appendix already consumes (rem:an17-orthogonal: Part (A) "cites only $(R1), (R3), (R4), (R5), F2$ and the holomorphic-ODE package"). The complex geodesic flow $\ddot\gamma = -\Gamma[g](\dot\gamma, \dot\gamma)$ with data (z, ξ), $z \in \mathcal{K}_{\rho_a/4}$, $|\xi| \le 2\delta_c^{\mathbb{C}}$, has a holomorphic solution on $t \in [0,1]$ by the uniform-strip Picard iteration an17:prop-strip (printed site l.484–500), provided $\delta_c^{\mathbb{C}} \le c\,(1 + M_\Gamma)^{-1} \min(\rho_a/4, \delta_c)$ — one displayed shrink, ball-uniform. Holomorphy of $(z, \xi) \mapsto \gamma$ is Picard's uniform limit of holomorphic iterates. The exponential Jacobian $\partial\xi/\partial w$ at $\xi = \exp_z^{-1}(w)$ equals $\mathbb{1} + O(\delta_c^{\mathbb{C}} M_\Gamma)$, invertible after one more displayed shrink; then $\sigma_g(z, w) = \frac{1}{2} q_{g,z}(\xi, \xi)$ and $\Delta_g^{1/2} = |\det g(z)|^{-1/4} |\det g(w)|^{-1/4} |\det(-\partial_z \partial_w \sigma_g)|^{1/2}$ are holomorphic with $\Lambda_0 := c\,(1 + \varepsilon_a + \|g_0\|_{\mathcal{K}_{\rho_a}})(1 + M_\Gamma)^2 (\delta_c^{\mathbb{C}})^2 + c\, d_\omega^{-1/2} (1 + M_\Gamma)^2$; non-vanishing of $\Delta^{1/2}$: $\Delta^{1/2} = 1 + O(\sigma)$ with the O Cauchy-bounded via EP1$'$ on its bi-collar Hessian, so a final displayed shrink of $\delta_c^{\mathbb{C}}$ gives $|\Delta^{1/2}| \ge \frac{1}{2}$ on $\mathcal{V}^{\mathbb{C}}$, ball-uniformly. *(A2).* The recursion $2\nabla\sigma\cdot\nabla v_k + (\Box\sigma - 2n + 4k) v_k = -\Box v_{k-1}$ is, along the (A1) holomorphic geodesics in the rescaled radial variable, a *linear* regular-singular ODE $(2r\partial_r + A_k) v_k = f_k$ with holomorphic coefficients bounded by $c\Lambda_0$ on $\mathcal{V}^{\mathbb{C}}$ and indicial root $4k > 0$ ($k \ge 1$; $k = 0$ transports $\Delta^{1/2}$-data), and source f_k a universal polynomial in $(\partial^{\le 2}\sigma, \Delta^{\pm 1/2}, \partial^{\le 2k+2} g, g^{-1})$ — every factor holomorphic and EP1$'$-bounded on $\mathcal{V}^{\mathbb{C}}$. Gronwall along the radial integral gives the *single* sup bound $\sup_{\mathcal{V}^{\mathbb{C}}} |v_k| \le \Lambda_k := c\, e^{c\Lambda_0 \delta_c^{\mathbb{C}}} P_k\big(\Lambda_0, M_\Gamma, d_\omega^{-1}, (2\sqrt{n}/\rho_a)^{2k+2} (2k+2)!\, \varepsilon_a + \|g_0\|_{C^{2k+2}}\big)$, P_k a displayed universal polynomial (finite recursion, $k \le 2$). *One* sup bound then generates every derivative bound: the iterated Cauchy estimate of EP1$'$ in the $2n$ bi-collar variables on $\mathcal{V}^{\mathbb{C}}$ gives the displayed all-order form — this is where analyticity replaces the entire multi-index tape of gap (b) at the construction level. Restricting to the real points and integrating over V_{δ_c} yields the H^{s-6}/H^{s-8} register bounds with $C_{v_2}^{\mathrm{reg}}$ as displayed ($\Box_g$ costs two more Cauchy derivatives and one g^{-1}-factor, absorbed). *(A3).* Real side. The difference legs never touch the collar (EP2): they are the Gronwall/Christoffel-difference transport route, now *ported in full* as P-M3 Lemmas L1–L3 (port:lem-gamma / port:lem-sigma / port:lem-vk: the exact identity $\delta\Gamma = \frac{1}{2} g^{-1}(\nabla' h + \nabla' h - \nabla' h)$, flow Gronwall, transport Duhamel — a standard derivation, here re-derived with explicit constants), upgraded from C^0- to H^{s-2k-2}-difference norms by the linearity of the transport average in its top curvature-polynomial argument, which is in print at the (N-a) block (appendix l.3219–3222: $\|h_g - h_{g'}\|_{H^a} \le C_{\mathrm{poly}} \|g - g'\|_{H^s}$, with $C_{v_2} = \frac{1}{12} \kappa_\Delta C_{\mathrm{poly}}$): lower-order factors are estimated by the (A2) sup bounds, the top jet enters linearly, so $C_{v_2}^{\mathrm{fam}} := c\, C_{\mathrm{poly}}\, e^{c\Lambda_0 \delta_c} \big(1 + \Lambda_2\big)^2 \big(1 + \kappa_\Gamma(1) + \kappa_S\big)$ with κ_Γ, κ_S the displayed P-M3 constants (in the corrected difference-count form $\kappa_\Gamma(k) = c_k(n) \Lambda_U^{2k+7} d_U^{-2(k+1)}$). All placements are Morrey/Banach-algebra at $s > 11$ on the real side. *(A4).* Gluing: an17:G(ii)–(v) is THEOREM in print (l.3233–3254) and consumes exactly (A2)–(A3): defect legs book $(8, \infty)$, the main rung $(8, 6)$ (N-a-NF), the log-branch cancellation on overlaps is patch-independent geometry with the one global scale ℓ, and every constant scales linearly in the multiplicity $m(K)$ (an17:G(iv), no Besicovitch). Nothing in this proof estimates anything not already named. $\square$

Remark E2a.11 (What (A) does not do). It never touches the state ($\omega_g^{(2)}$ enters only through the data/spectator routes of Prop. E2a.12(C3) and the verdict's residues); it makes no H^s-open statement (F-iii respected: sups and pair-moduli over U_ω only); it never continues across a shock (every object lives on the fixed collar of real-analytic members; Barrabès–Israel metrics are outside U_ω; Guard F-shock, unification.tex l.17172–17185); and it asserts *no* analyticity in the metric parameter (EP0): every g-difference is real-side Lipschitz, never a collar conversion (EP2's Hölder-only counterexample stands as the fence).

Proposition E2a.12 (C: gap (b) vacates over U_ω to the printed register arithmetic; the spectator sups correctly routed)**.** *Grant Proposition E2a.10. Consider the ladder source $f_2 = -(S_{g,2} - S_{g',2}) - (\Box_g - \Box_{g'}) W_{g',2}$ and the Rung-4 slice products of the two ladders (an17:slice-ladder, an17:slice-ladder-4d — THEOREM over U_ω at their printed sites). Then:*

(C1) *every constant appearing in the ladders' estimates of f_2 and of the Rung-4 products is a polynomial in the finite named list*

$$\Big\{\alpha!\big(\tfrac{2\sqrt{n}}{\rho_a}\big)^{|\alpha|}\varepsilon_a\ (|\alpha|\le s{+}1);\ \Lambda_0,\Lambda_1,\Lambda_2;\ C^{\mathrm{fam}}_{v_2};\ d_\omega^{-1};\ Q_*,P_*,T_c;\ c^{\mathrm{Mor}},c_{\mathrm{mult}},c_{\mathrm{tr}},c_{\mathrm{alg}},c_\Pi,c_{\mathrm{fol}};\ m(K),L_\chi\Big\},$$

each entry ball-uniform and named in print or in Prop. E2a.10 (EP1′; an17:DoD l.3256–3268 for Q_*,P_*,T_c; the ladder constant display and its provenance note, appendix l.3010–3035, for $c_{\mathrm{mult}},c_{\mathrm{tr}},c_{\mathrm{alg}},c_\Pi,c_{\mathrm{fol}},c^{\mathrm{Mor}}$; an17:G(i),(iv) for $m(K),L_\chi$);

(C2) *consequently the* content *of gap (b) over U_ω is not "track every multi-index constant" (each is automatically ball-uniform by (C1)) but only the finite* term inventory*: that f_2 and the Rung-4 products contain no term outside the booked registers — and that inventory is in print: the normal form (N-a-NF) (Prop. E2a.10(A4)), the gluing defect legs (an17:G(iii), THEOREM), and the commutator leg $(\Box_g-\Box_{g'})W_{g',2}$, whose coefficients are $(g-g')$-linear with Banach-algebra placements at $s>11$;*

(C3) ***[W1 — the spectator sups]*** *the $W_{g',2}$-factor of the commutator leg consumes the* single-metric *uniform bounds at the ∂_y-slotted registers. These are supplied by the* single-metric uniform analogue *recorded inside Lem. an17:Nc-corrected's own statement (l.3280–3281: "together with the single-metric uniform analogue for $W_{g',2}$"), fed through the an17:DoD short-slab data route (THEOREM, l.3256–3268) from the seed-adjacent Cauchy data — and those data are exactly the $(D0)$ residue at the $b{=}1$ field slot (an17:named-residues, l.3321–3324). They are therefore cited here as* $(D0)$-gated inputs*, NOT as Lemma E2a.15 consequences: Lemma E2a.15 presupposes C_W and may not feed this proposition. (Lemma B is* not *cited at this locus — that would be circular; the gate is carried into Theorem E2a.14(res1) instead.)*

Proof. (C1) Fix any product term $T=\prod_j \partial^{\gamma_j}c_j\cdot\partial^\beta u$ appearing in the first-order reduction of the source or in a Rung-4 slice product ($c_j\in\{a^{\mu\nu},b^\mu,\theta_a,\chi_a,\sigma,\Delta^{\pm1/2},v_k\}$, $u\in\{D_2,W_{g',2}\}$). The ladders (THEOREM) book T's *register*; only the constant was open. Every coefficient factor is a g-jet (EP1′: ball-uniform at all orders, with the $\sqrt{n}$), an atlas/cutoff datum (L_χ, $m(K)$: fixed by an17:G), or a package biscalar (Prop. E2a.10(A1)–(A2): ball-uniform at all orders). Moser/tame product constants at fixed s are polynomials in finitely many C^k norms of the coefficients — all just listed — times the universal $c_{\mathrm{mult}}(s),c_{\mathrm{alg}}$; the energy step contributes the Tice factor $\sqrt{Q_*e^{P_*T_c}}$ (an17:DoD, THEOREM). The list is finite and each entry named, so the composite is the claimed polynomial. (C2) What this does NOT prove: completeness of the inventory is *cited*, not re-derived — (N-a-NF) + an17:G(iii) + the displayed commutator leg, at their printed grades. (C3) The re-routing is definitional (which in-print statement is cited); the mathematical content — that the single-metric ∂_y-slotted sups exist at the consumed registers given seed data at the $b{=}1$ slot — is precisely the $(D0)$-gated single-metric write-out $W_*^{(2)}$ named at an17:named-residues; no circular step remains: C_W-chain $\not\ni$ Lemma E2a.15. $\square$

Remark E2a.13 (#B′ and its conditional discharge sentence)*.* The appendix books the single-metric write-out lem:W2star as gated by $(D0)$ *and* #B′ (the off-collar/across-patches kernel variant; printed at l.3325–3327 and in the M1 modulo item l.3951–3952). Accordingly #B′ is listed here as a residue. *The owed discharge sentence, conditional-tagged:* at the *seed* instance the perturbation region P is empty, so no patch traversal occurs and #B′ is not exercised; for general $g'\in U_\omega$ the spectator route runs entirely from BFV/RCE surface data through the an17:DoD short slab (no patch traversal), so #B′ is bypassed *if and only if* the BFV/RCE anchoring identity holds — i.e. this discharge **[W3]** rides "$C_{\Delta_2}=0$ on the BFV surface", whose carrier is Lem. an17:Nc-corrected at printed grade *PLAUSIBLE* (l.3289–3293: "NEVER cited at THEOREM grade"), and is therefore itself only PLAUSIBLE-conditional. *Revival condition:* any consumer that un-anchors the state family from BFV/RCE re-opens both (E-env) *and* #B′ at full strength.

Theorem E2a.14 (M1 verdict: the family-uniformity content of clause (ii) closes over U_ω; the honest modulo-list)**.** *Over U_ω, at $s > 11$, grant the printed ladder theorems (an17:slice-ladder, an17:slice-ladder-4d, an17:G, an17:DoD — all THEOREM at their printed sites) and Propositions E2a.10 and E2a.12. Then clause (ii) of Hyp. hyp:hadamard-uniform-omega holds for the paper's BFV/RCE-anchored state family, with $C_W(K)$ exactly as assembled at an17:CW-grade (l.3374–3384) and $W_*(K)$ from Lemma E2a.15,* modulo precisely the following modulo-list*:*

(res1) *(D0) — the seed-adjacent Cauchy-data write-out of the single-metric $W_{g_0,2}$ at the b=1 field slot (an17:named-residues l.3321–3324): fixed-metric,* not *a uniformity statement; it gates the spectator registers $W_*^{(2)}$ of Prop. E2a.12(C3) through the data-transport route, whose transport constants are ball-uniform (EP1′/an17:DoD);*

(res2) ***[W3]*** *the BFV-vanishing "$C_{\Delta_2} = 0$ on the BFV surface", cited* only *at its carrier's printed grade: Lem. an17:Nc-corrected, PLAUSIBLE, never at THEOREM (l.3289–3293). Where it enters: the $C_{\text{trans}} = C_{\text{data}} = 0$ simplification of the an17:CW-grade assembly and the $\#B'$ bypass of Rem. E2a.13. Revival: un-anchoring from BFV/RCE re-opens (E-env) and $\#B'$;*

(res3) ***[W2]*** *$\#B'$ — listed, with its conditional discharge sentence in Rem. E2a.13 (BFV-conditional, hence riding* (res2)*);*

(res4) *(E-env) — exercised only* off *the BFV anchoring, hence not on the Part-B consumer path for the paper's own state class; same carrier grade discipline as* (res2).

The two printed gaps of an17:CW-grade are thereby discharged as family-uniformity questions*: (a) by Prop. E2a.10 (the construction is classical-Hadamard on analytic metrics, quantitative by EP1′), (b) by Prop. E2a.12 (every constant automatically ball-uniform; the term inventory is in print). The full-length Props A/C write-out is itself provided here. What blocks the word THEOREM for clause (ii) as a whole is* (res1)–(res2)*: a long-standing fixed-metric data residue, plus one PLAUSIBLE-grade carrier.* Grade: clause (ii)/M1 remains PLAUSIBLE. *No grade line moves with this write-out alone; and a* (D0) *closure alone would* not *lift* (res2) *(the BFV conditionality survives it).*

Lemma E2a.15 (B: seed-anchored triangle — consumer-facing only, stated after the C_W chain)**.** *Suppose the mixed-jet Lipschitz bound of clause (ii) holds over U_ω with constant $C_W(K)$ (Theorem E2a.14, modulo its list). Let $W_*^{\text{seed}}(K) := \max_{|a|+|b|\le 1 \,\cup\, |a|=|b|=1} \sup_{x\in K} \big|[\partial_x^a \partial_y^b W_{g_0,2}](x,x)\big|$ — the seed mixed jets, finite at $s > 10$ (the established single-metric ground truth, appendix l.2752–2753 and l.3383).* ***[W4]*** *$W_*^{\text{seed}}(K)$ carries the grade* THEOREM *modulo* (D0)*: its data feed at the* b=1 *slot is exactly* (res1) *(a bare THEOREM tag would be inconsistent with it). Then for every $g \in U_\omega$,*

$$W_*(K) \;\le\; W_*^{\text{seed}}(K) \;+\; C_W(K)\, C_{\text{emb}}\, \varepsilon_a,$$

since $\|g - g_0\|_{H^s(K'')} \le C_{\text{emb}}\varepsilon_a$ (an17:lem-embed, l.157–175). Hence the two-constant demand $\{C_W, W_*\}$ of M1 collapses to the single demand C_W — the F2 seed-anchored pattern of the paper's own modulus (lem:E2-energy-lipschitz's triangle, unification.tex l.17438–17443). Wiring note [W1]: this lemma is consumed only downstream (by the clause-(ii) consumers of hyp:hadamard-uniform-omega); no proposition of this package cites it.

Remark E2a.16 (Grade discipline; fences; what remains)*.* (i) This write-out moves *no* printed grade: clause (ii)/M1 is PLAUSIBLE before and after; the contribution is that its open content is now the *written* finite list of Theorem E2a.14 (all four conditions folded in), not the two unbounded-looking gaps (a),(b). (ii) Fences: F-diff (both metrics in U_ω throughout); F-iii (sups and pair-moduli only, never an H^s-open modulus); F-shock (fixed collar, no continuation); EP2 (no collar-to-H^s conversion anywhere — all Lipschitz legs real-side); EP4 (no Montel step anywhere in this package). (iii) The remaining mathematics is (res1) ((*D*0), now closed at the seed by the classical analytic-convergence lever — App. E2a, the (D0) port; correction gate: that closure is obstructed as printed — the port display is gated at the honest column and

($D0$) remains gated, Rem. E2a.66) and the (res2) carrier grade, which no port can lift (it is a data-side PLAUSIBLE of the appendix itself). (iv) With this write-out and the clause-(i) port in hand, an17:CW-grade's gaps (a),(b) may be re-pointed to the discharge-modulo-list of Theorem E2a.14.

E2a.3 The seed-data closure (D0) via the convergent Hadamard series

Correction gate: the closure below is OBSTRUCTED as printed (its display and T-leg calculus are gated at the honest seed column; the Hadamard-1932 lever and the w_0-smoothness lemma stand untouched); see Rem. E2a.65 and Rem. E2a.66.

Port (D0): the single-seed-metric Cauchy-data write-out at the $b = 1$ slot, via the classical analytic-convergence lever

Scope and objects (EP0-d0). Throughout, g_0 is the U_ω anchor: real-analytic on K'', holomorphic on the fixed complex collar $\mathcal{K}_{\rho_a}$ with $\|g_0\|_{\mathcal{K}_{\rho_a}} < \infty$ and $\inf_{\mathcal{K}_{\rho_a}} |\det g_0| \geq d_\omega > 0$ (an17:sub-Uomega, l.126–150). The operator is $P_0 = \Box_{g_0} + m^2$. The state is the *seed* ω_0, the paper's standing Hadamard state at g_0 (unification.tex l.17037–17042: the family ω_g is *defined* as the Brunetti–Fredenhagen–Verch relative Cauchy evolution of ω_0; the seed slot is $g = g_0$, no transport). $H^{(N)}_{g_0}$ is the glued N-truncated Hadamard parametrix of Lem. an17:G (l.3231–3254): $H^{(N)}_{g_0} = \sum_a \theta_a Z^{(N),a}_{g_0}$ with $Z^{(N),a}_{g_0} = \frac{1}{8\pi^2}\big[\Delta^{1/2}/\sigma + \big(\sum_{k\leq N} v_k\sigma^k\big)\log(\sigma/\ell^2)\big]$ chartwise on the convex-normal cover, one *global* length scale ℓ (an17:G(ii)), and $W_{g_0,2} = \omega_0^{(2)} - H^{(2)}_{g_0}$ (the (N-a) package, l.3203–3229). The v_k are the transport-recursion coefficients cited in print at (N-a) [Moretti 2003, Thm. 2.1]; in the source-verbatim normalisation of [Fewster–Smith, gr-qc/0702056, eqs. (97)–(98)] ($n = 4$, $\mu^2 = m^2\ell^2$ rescaled):

$$0 = \ell^2(\Box_g + m^2)\Delta^{1/2} + 2\nabla v_0 \cdot \nabla\sigma + 4v_0 + v_0\,\Box_g\sigma, \tag{D0-T0}$$
$$0 = \ell^2(\Box_g + m^2)v_j + 2(j{+}1)\nabla v_{j+1} \cdot \nabla\sigma - 4j(j{+}1)v_{j+1} + (j{+}1)v_{j+1}\,\Box_g\sigma, \tag{D0-Tj}$$

$j \geq 0$: each v_{j+1} is determined from v_j by a linear transport ODE along the geodesics from the vertex, regular at the vertex. *What this port consumes and what it does not:* of the three classically distinct statements — (S1) convergence of the coefficient series $v = \sum_{k\geq 0} v_k\sigma^k$ with analytic sum; (S2) the summed object being an exact elementary solution ($P_0E = \delta$, partie-finie bookkeeping); (S3) the finite-part data regularity consumed by Ladder A — the port consumes *only* (S1), plus the two printed premises:

- (P-a) *(seed-Hadamard premise; paper-level, standing.)* ω_0 is Hadamard on the seed spacetime; equivalently (Radzikowski's theorem; the fixed-neighbourhood form as in [Kay–Wald 1991; Radzikowski 1996; Sanders 2010]) there is a causal normal neighbourhood $\mathcal{N} \supset \Sigma_0$ and a diagonal collar $V_H \subset \mathcal{N}\times\mathcal{N}$, both N-independent, with $\omega_0^{(2)} - H^{(N)}_{g_0} \in C^N(V_H)$ for every N. This is the paper's own anchor hypothesis, restated — *not* a new conditionality; it is printed here so nothing is promoted silently.
- (P-b) *(collar-shrink licence; fence-neutral.)* The gluing collar δ_c of Lem. an17:G may be shrunk: an17:G(i) demands only the upper bound $\delta_c \leq (2m(K)^2L_\chi)^{-1}$ (l.3242); the short-slab length $T_c = \delta_c/(4c_*)$ of Lem. an17:DoD shrinks with it (l.3259) and the DoD truncation is stated for every such slab; Lem. an17:transverse-data is local in the collar. No Sobolev fence moves ($s > 11$ untouched).

(S2) is *never* used below; no partie-finie identity enters. The ball-uniform version of (S1) over all of U_ω (same majorants, by the uniform collar Cauchy estimates of an17:lem-embed/EP1) is *available but not consumed*: (D0) is the single-seed-metric residue (res1), and no parameter-analyticity in g is asserted anywhere (EP0).

Lemma E2a.17 (The lever, (S1) at the seed: the Hadamard coefficient series converges on a uniform diagonal collar of the analytic anchor)**.** *There exist $\delta_{\mathrm{conv}} = \delta_{\mathrm{conv}}(g_0, \rho_a, d_\omega, m^2, K'') > 0$ and $M_{\mathrm{tail}} < \infty$ such that, on*

$$V_{\mathrm{conv}} := \{(x, y) \in K' \times K' : \ d_{g_0}(x, y) < \delta_{\mathrm{conv}}\},$$

the series $v(x, y) = \sum_{k \geq 0} v_k(x, y)\, \sigma_{g_0}(x, y)^k$ of (D0-T0)–(D0-Tj) *at $g = g_0$ converges absolutely and uniformly, together with all (x, y)-derivatives; its sum v is real-analytic on V_{conv} (holomorphic on a complex neighbourhood thereof), uniformly in the vertex. In particular the $N{=}2$ tail factorises,*

$$v - v^{(2)} \ = \ \sigma^3\, \varrho, \qquad v^{(2)} := \textstyle\sum_{k \leq 2} v_k \sigma^k, \qquad \varrho := \textstyle\sum_{k \geq 3} v_k \sigma^{k-3},$$

with ϱ analytic on V_{conv} and $\|\varrho\|_{C^6(\overline{V'_{\mathrm{conv}}})} \leq M_{\mathrm{tail}}$ on the closed half-collar V'_{conv} ($\delta_{\mathrm{conv}}/2$-collar). [Classical: Hadamard 1932, nos. 52, 63–66; verified at source — see the citation record below. The U_ω-scope transfer and the jet upgrade are re-derived in the proof.]

Proof. Step 1 (reduction to the classical hypotheses; re-derived). Hadamard's construction of the elementary solution requires: a linear second-order operator with coefficients holomorphic in a neighbourhood of the vertex, and a non-degenerate (non-parabolic) principal part there [Hadamard 1932, nos. 52, 63, 65]. At the seed both hold *with quantitative collar data*: g_0 and $g_0^{-1} = (\det g_0)^{-1} \operatorname{adj} g_0$ are holomorphic on $\mathcal{K}_{\rho_a}$ with the sup bound $\|g_0\|_{\mathcal{K}_{\rho_a}}$ and the determinant floor d_ω (an17:sub-Uomega l.148–150), m^2 is constant, and the normal (Lipschitz) variables he works in exist holomorphically at a uniform radius: the complex-time geodesic IVP is solved by the analytic-ODE package an17:rem-CK (l.465–482; Picard contraction on $\mathcal{K}_{\rho_a/4}$, field bound M_Γ of eq. an17:eq-MGamma), so the exponential map, its inverse on the uniform convex-normal cover of Lem. an17:G, and hence σ_{g_0} and $\Delta_{g_0}^{1/2}$ (Jacobian determinant of a holomorphic map; nonvanishing near the diagonal since $\Delta \to 1$) are holomorphic on a complex diagonal collar whose radius depends only on $(\rho_a, \|g_0\|_{\mathcal{K}_{\rho_a}}, d_\omega)$ through M_Γ. This is the same reduction the appendix already exercises for the phase (an17:prop-strip, l.484–500); nothing new is assumed.

Step 2 (identification of the series; re-derived). For $n = 4$ Hadamard's exponent is $p = (m_{\dim} - 2)/2 = 1$: his elementary solution [1932, eq. (51)/(53), no. 65] is $U_0\Gamma^{-1} - \mathcal{V} \log \Gamma + w$ with the algebraic part the *single* term $U_0\Gamma^{-1}$ and $\Gamma = 2\sigma$ (his squared geodesic distance; the constant rebooking $\sigma^k \mapsto \Gamma^k$ multiplies v_k by 2^{-k} and leaves every radius statement unchanged). His transport hierarchy for the log-coefficient expansion $\mathcal{V} = \sum_h \mathcal{V}_h \Gamma^h$ [1932, eqs. (44′), (52), (49′)] is, term by term in Γ^h, the hierarchy (D0-T0)–(D0-Tj): both are the unique regular-at-the-vertex solutions of the same singular linear transport ODEs along the geodesics from y (the appendix's printed recursion site: (N-a), l.3208, [Moretti 2003, Thm. 2.1]; the Fewster–Smith display above is that recursion verbatim at $n = 4$). Uniqueness of regular solutions of the transport hierarchy identifies $\mathcal{V}_h = 2^{-h}\, c_{\mathrm{nrm}}\, v_h$ with the single overall normalisation c_{nrm} fixed by v_0-vs-U_0 matching at the vertex ($\Delta^{1/2}(y, y) = 1$).

Step 3 (the classical convergence; cited at verified source). Under Step 1's hypotheses, Hadamard proves: the series (43)-type expansions converge for $|\Gamma|$ small, by direct majorants [1932, no. 63], with radius *uniform* as the vertex ranges over any compact strictly interior region [no. 64: "la convergence … sera uniforme"]; the even-case log coefficient $\mathcal{V}$ is built by "exactement les mêmes opérations" from conoid data [no. 65], hence its series converges with the same uniformity, and the whole elementary solution exists "pour toute équation non parabolique … sous l'hypothèse que les coefficients sont analytiques" [no. 65, closing display; no. 66 extends verbatim to both elliptic and hyperbolic signatures]. Because his data are holomorphic (no. 52) the convergence is convergence of holomorphic functions on a complex collar, and the sum depends analytically on the vertex when the free element is chosen by an analytic law [no. 65, footnote]; the transport hierarchy makes no free choice below the w-level, and $\mathcal{V}$ involves none. Fix $\delta_{\mathrm{conv}} > 0$ as (half) the resulting uniform radius over the compact K'.

Step 4 (jets and the factorisation; re-derived). A locally uniformly convergent series of holomorphic functions converges with all derivatives (Weierstrass), locally uniformly; restriction to the real collar gives the all-jet convergence and the analyticity of v on V_{conv}. Subtracting the polynomial-in-σ part $v^{(2)}$ (analytic, finitely many terms of the same series) leaves $v - v^{(2)} = \sum_{k\geq 3} v_k\sigma^k = \sigma^3\varrho$ with $\varrho = \sum_{k\geq 3} v_k\sigma^{k-3}$ again a locally uniformly convergent series of holomorphic functions, hence analytic; $M_{\mathrm{tail}} := \|\varrho\|_{C^6}$ on the closed half-collar is finite by compactness. *Constants emitted:* δ_{conv}, M_{tail} — both functions of $(g_0, \rho_a, d_\omega, m^2, K'')$ only, consumed downstream only as finite named constants (the same consumption mode as W_*-type constants in print). □

Citation record for Lemma E2a.17 (existence premise verified at source). Primary, read verbatim (public-domain scan): J. Hadamard, *Le problème de Cauchy et les équations aux dérivées partielles linéaires hyperboliques* (Hermann, Paris, 1932; the definitive French edition of the 1923 Yale *Lectures*), Livre II, ch. III: no. 52 (holomorphic-data hypothesis), no. 63 (majorant convergence proof), no. 64 (vertex-uniformity; conoid-data variant), no. 65 (even m: log term; "On a ainsi réussi à calculer la solution élémentaire (pour toute équation non parabolique) sous l'hypothèse que les coefficients sont analytiques"), no. 66 (elliptic and hyperbolic in common). Classical corroboration, named: M. Riesz, Acta Math. 81 (1949) 1–223, ch. IV (analytic Lorentzian case). Modern witnesses, verbatim: Fewster–Smith [gr-qc/0702056, after eq. (96)]: "The series does not actually converge except in analytic spacetimes"; Khavkine–Moretti [arXiv:1412.5945, §3.3]: "The limit exists in the analytic case, but in the smooth general case the sequence diverges", with the smooth-case cutoff-sum construction pinned at [Friedlander, *The Wave Equation on a Curved Space-Time* (CUP 1975), §4.3]. The Lorentzian-category check is Hadamard's own nos. 65–66 (non-parabolic; both signatures); no elliptic-to-Lorentzian transfer is needed.

Lemma E2a.18 (the seed exact finite part is C^∞ on the shrunken collar)**.** *Let $\delta'_c := \min(\delta_c, \delta_{\mathrm{conv}}/2, \delta_H)$, where δ_H is a collar radius with $V_{\delta_H} \subset V_H \cap V_{\mathrm{conv}}$ (premise (P-a) and Lemma E2a.17), and let $H^{(\infty)}_{g_0}$ denote the glued parametrix of Lem. an17:G with the chartwise truncations $Z^{(N),a}_{g_0}$ replaced by the convergent sums $Z^{(\infty),a}_{g_0} = \frac{1}{8\pi^2}[\Delta^{1/2}/\sigma + v\log(\sigma/\ell^2)]$. Then on the collar $V_{\delta'_c} = \{d_{g_0}(x,y) < \delta'_c\}$ (intersected with $\mathcal{N}\times\mathcal{N}$):*

- (i) *$Z^{(\infty),a}_{g_0}$ is patch-independent, so $H^{(\infty)}_{g_0} = \frac{1}{8\pi^2}[\Delta^{1/2}/\sigma + v\log(\sigma/\ell^2)]$ exactly ($\sum_a \theta_a \equiv 1$, an17:G(i); the members $\sigma, \Delta^{1/2}, v_k$ are geometric biscalars and the length scale ℓ is global, an17:G(ii));*
- (ii) *the $N{=}2$ tail is exact and analytic-profiled: $T := H^{(\infty)}_{g_0} - H^{(2)}_{g_0} = \frac{1}{8\pi^2}\sigma^3\varrho\log(\sigma/\ell^2)$ with ϱ as in Lemma E2a.17 (chartwise, $Z^{(\infty),a} - Z^{(2),a}$ is the* same *series tail on every patch, so the gluing weights drop out);*
- (iii) *$w_0 := \omega^{(2)}_0 - H^{(\infty)}_{g_0} \in C^\infty(V_{\delta'_c})$, with all jet sup-norms $w^{(j)}_* := \|w_0\|_{C^j(\overline{V_{\delta''_c}})} < \infty$ on any strictly smaller closed collar $\delta''_c < \delta'_c$.*

Proof. (i), (ii) are display arithmetic given an17:G(i)–(ii) and Lemma E2a.17: on each patch overlap the chartwise members $\sigma, \Delta^{1/2}, v_k$ agree (geometric biscalars; one global ℓ kills the branch mismatch, an17:G(ii)), hence so do the convergent sums; $\sum_a \theta_a \equiv 1$ collapses the gluing. For (iii): for every N,

$$w_0 = \big(\omega^{(2)}_0 - H^{(N)}_{g_0}\big) - \big(H^{(\infty)}_{g_0} - H^{(N)}_{g_0}\big).$$

The first bracket is $C^N(V_{\delta'_c})$ by premise (P-a) — the N-independence of the neighbourhood is exactly the fixed-neighbourhood form of the Hadamard condition [Kay–Wald 1991; Radzikowski 1996; Sanders 2010]. The second bracket is $\frac{1}{8\pi^2}\sigma^{N+1}\varrho_N\log(\sigma/\ell^2)$ with ϱ_N analytic (Lemma E2a.17, same factorisation at order N): a function with an $(N{+}1)$-fold zero on the cone times log, hence C^{N,α^-} across the cone (transverse model $u^{N+1}\log|u|$; radial model $r^{2N+2}\log r$ is better, exactly the two model heights of Lem. an17:transverse-data's proof), and analytic off it. So $w_0 \in C^N$ on the *fixed* collar $V_{\delta_c'}$ for every N, i.e. $w_0 \in C^\infty$; the jet sups are finite on compact closures. *No partie-finie identity, no fundamental-solution property of $H^{(\infty)}$, is used* — only (S1) and (P-a). $\square$

Proposition E2a.19 ((D0) executed: the Ladder-A seed Cauchy data at all four slots, $b = 1$ included)**.** *Premises: (P-a), (P-b), Lemmas E2a.17–E2a.18. Work on the shrunken collar δ_c'' and the correspondingly shortened DoD slab (Lem. an17:DoD at $T_c'' = \delta_c''/(4c_*)$). Then the four Ladder-A data legs of $W_{g_0,2}(\cdot, y) = w_0(\cdot, y) + T(\cdot, y)$ on Σ_0 exist and are finite at exactly the availability heights printed at A3 (l.2882–2900), uniformly and continuously in the spectator $y \in K_y$:*

$$W_{g_0,2}|_{\Sigma_0} \in H^{13/2^-}(\Sigma_0), \quad \partial_t W_{g_0,2}|_{\Sigma_0} \in H^{11/2^-}, \quad \partial_y W_{g_0,2}|_{\Sigma_0} \in H^{11/2^-}, \quad \partial_t\partial_y W_{g_0,2}|_{\Sigma_0} \in H^{9/2^-},$$

[Correction gate on this display: it inherits the two printed errors corrected at Lem. an17:transverse-data (corrected form; both named at Rem. an17:transverse-note) — (a) the $\alpha + \frac{1}{2}$ restriction rule at r-graded $\alpha > 0$, (b) the one-r-power-per-derivative eikonal grading — and is FALSE as printed: the honest, sharp seed column is $H^{7/2^-}, H^{5/2^-}, H^{5/2^-}, H^{3/2^-}$ (flat massive vacuum counterexample, $v_3 = m^8/(18432\pi^2) \neq 0$; appendix record Rem. an17:d0-record). The display and the T-leg calculus below stand only MODULO the transverse re-derivation — gated, not deleted; the w_0-legs and Lemmas E2a.17–E2a.18 are untouched.] *with norms bounded by $c_{\rm tr}(K)\big[w_*^{(9)} + M_{\rm tail}\, c_{\rm prof}(K)\big]$ ($w_*^{(9)}$: two slot derivatives plus seven slice derivatives; any $w_*^{(j)}$, $j \geq 9$, works), $c_{\rm prof}$ the transverse-restriction constant of Lem. an17:transverse-data at the four profile exponents. In particular the $b = 1$* field *slots (∂_y, $\partial_t\partial_y$) are supplied, and the (D0) residue of an17:named-residues (l.3321–3324) is CLOSED at the seed: the single-metric write-out $W_*^{(2)}$ is no longer (D0)-gated.* [Correction gate: this closure sentence does not stand as printed — at the honest heights the b=1 field slot supplies $H^{5/2^-}$ sharp versus every consumed demand $> \frac{5}{2}$; (D0) remains gated, now as a confirmed obstruction of the printed route (Rem. an17:d0-record).] (For general $g' \in U_\omega$ the spectator data reduce to seed data by the BFV/RCE surface anchoring at its carrier's grade; that reduction is *not* part of (D0) and is not claimed here — see Rem. E2a.20.)

Proof. Split each leg as w_0-leg + T-leg on the DoD-localised collar piece of Σ_0 (the cutoff ψ of Lem. an17:DoD confines every consumed norm to $\{d_{g_0}(\cdot, y) \leq \frac{3}{4}\delta_c''\}$, compact).

w_0-legs. By Lemma E2a.18(iii) each of the four jets (id, $\partial_t, \partial_y, \partial_t\partial_y$; at most two slot derivatives) is C^∞ on the closed collar; a C^k function on a compact three-dimensional slice piece lies in $H^t(\Sigma_0)$ for every $t \leq k$, so all four demands ($t < \frac{13}{2}$) are met with norm $\leq c(K)\, w_*^{(9)}$ (two slot derivatives + seven slice derivatives; any $j \geq 9$ works). Continuity and uniformity in $y \in K_y$: the jets are continuous in (x, y) on a compact closure.

T-legs (the tail is re-derived; its consumption is booked by the printed profile calculus). $T = \frac{1}{8\pi^2}\sigma^3\varrho \log(\sigma/\ell^2)$ with $\|\varrho\|_{C^6} \leq M_{\rm tail}$ (Lemma E2a.17). In the profile coordinate of Lem. an17:transverse-data ($r^2 \sim 2|\sigma|$ near the cone, away from the vertex) the appendix's (N-a) profile calculus — $\sigma^k \log\sigma$ carries profile $r^{2k}\log r$, one r-power per derivative (the printed convention: S_2''s $\sigma^2\log\sigma$ is booked $r^4 \log r$ with "$\partial^5 \sim r^{-1}$", l.2870–2871; its geometric root is the eikonal grading $|\nabla\sigma|_g = r$) — books

$$T \sim r^6 \log r, \qquad \partial T = \big[3\sigma^2\varrho\,\partial\sigma + \sigma^3\partial\varrho\big]\log(\sigma/\ell^2) + \sigma^2\varrho\,\partial\sigma \;\sim\; r^5 \log r, \qquad \partial^2 T \;\sim\; r^4 \log r,$$

[Correction gate: this is error (b) of Rem. an17:transverse-note — $|\nabla\sigma|_g$ is the Lorentzian g-norm, which vanishes at the cone, while the coordinate gradient is $O(1)$ there; near the crossing each derivative on $\sigma^k \log\sigma$ costs *two* r-powers, so the honest r-grading is $\partial T \sim r^4 \log r$, $\partial^2 T \sim r^2 \log r$. The bookings of this paragraph stand MODULO the transverse re-derivation.] each $\sim$ meaning: a finite sum of terms at profile exponent $\geq$ the displayed one, coefficients bounded by

M_{tail} on the closed collar; the ∂_y and ∂_t slots grade identically (σ a symmetric analytic biscalar; the vertex-adjacent radial regime is *better*, the $\rho^\alpha \log\rho \in H^{\alpha+3/2^-}$ model height inside Lem. an17:transverse-data's proof). These are *verbatim* the profile assignments the printed A3 column gives the same four legs; no new grading convention is introduced, and the established trace-audit that underwrites the printed calculus is consumed, not re-derived. Lem. an17:transverse-data (THEOREM, sharp, y-uniform; the "times any factor smooth-modulo-H^{s-2}" hypothesis is satisfied by the *analytic* factors here, under the standing no-glancing hypotheses of Lem. an17:no-glancing at the seed) applied at $\alpha = 6, 5, 5, 4$ gives restrictions in $H^{13/2^-}, H^{11/2^-}, H^{11/2^-}, H^{9/2^-}$ respectively, norms $\leq M_{\text{tail}}\, c_{\text{prof}}(K)$, uniformly and continuously in $y \in K_y$.

Summing the two legs gives the display, which is verbatim the A3 availability column (l.2886–2890: transverse heights $\rho^6 \log\rho \in H^{13/2^-}$, $\rho^5 \log\rho \in H^{11/2^-}$, $\rho^5 \log\rho \in H^{11/2^-}$, $\rho^4 \log\rho \in H^{9/2^-}$) — now with the b=1 *field* slots carried by $w_0 \in C^\infty$ instead of a finite-regularity cap. The margins against Rung 2's demands are therefore $\geq$ the printed ones (at the analytic seed the "conservative H^s-background coefficient ceilings" of A3 are strict surplus), and Rung 2 of Lem. an17:slice-ladder consumes the column as printed; no step of the ladder is re-derived here. [Correction gate on this closing paragraph: error (a) of Rem. an17:transverse-note — the restriction rule applied at the r-powers $\alpha = 6, 5, 5, 4$; the corrected Lem. an17:transverse-data gives $\frac{\alpha}{2} + \frac{1}{2}$, so the honest column is $(\frac{7}{2}, \frac{5}{2}, \frac{5}{2}, \frac{3}{2})^-$, sharp at the seed. The w_0-leg paragraph above is untouched (the sound half); the T-leg margins and the "consumes the column as printed" clause stand only MODULO the transverse re-derivation.] ☐

Remark E2a.20 (post-(D0) modulo-list, grade arithmetic, and what does NOT move). *(a) What this port closes.* [Correction gate: as printed, nothing — the discharge claim below is gated. Existence of the four legs is established (THEOREM, unconditional), but at the honest heights $(\frac{7}{2}, \frac{5}{2}, \frac{5}{2}, \frac{3}{2})^-$, not the printed ones; (D0) remains gated (Rem. an17:d0-record).] The (D0) entry of an17:named-residues (l.3321–3324) is discharged at the seed: the four A3 data legs exist at the printed heights, b=1 included, given (P-a) (the paper's standing seed-Hadamard premise) and (P-b) (fence-neutral collar shrink). The single-metric seed write-out $W_*^{(2)}$ loses its (D0) gate; condition W4's internal contradiction resolves in res1's favour with the gate now *closed*: Lemma B's W_*^{seed} tag reads "THEOREM given Prop. E2a.19" (not "THEOREM modulo (D0)"). *(b) What survives — the post-(D0) M1 modulo-list.* Updating the modulo-list {(D0); BFV-vanishing @ PLAUSIBLE; #B′ discharge sentence; Props A/C write-out}:

- **BFV-vanishing @ PLAUSIBLE — SURVIVES, expected.** (D0)'s closure is BFV-free (the seed write-out consumes no difference data), but clause (ii)'s assembly still consumes the (N-c) difference legs *on the BFV surface* at the carrier grade of Lem. an17:Nc-corrected (l.3289–3293: "PLAUSIBLE ... NEVER cited at THEOREM grade"). A (D0) closure does NOT lift it; the resisting step for M1 remains conditional on it, with the Props A/C state-side difference estimate the unconditional fallback.
- **#B′ — unchanged** (the across-patches variant; the W2 discharge sentence is owed in the Props A/C port, conditional-tagged).
- **Props A/C + Lemma B full write-out** — given in the family-uniformity package of this appendix; not this port's claim.

(c) Grade arithmetic. M1's printed grade (an17:CW-grade, PLAUSIBLE) does *not* move on this port alone. With all three ports applied and the corresponding re-pointing of an17:CW-grade in place, the honest M1 label is "theorem MODULO {BFV-vanishing @ PLAUSIBLE; #B′ for patch-traversal consumers}" — still *not* THEOREM, and the printed grade line stays PLAUSIBLE. (E-env) and #A are data-side residues orthogonal to this port (an17:named-residues l.3329–3331) and are not touched. *(d) Revival conditions.* (D0) re-opens if: the seed anchor is taken non-analytic (the lever's hypothesis fails); or a consumer feeds seed data beyond the collar (then #B′, not (D0)); or the seed state's Hadamard premise (P-a) is weakened below all-orders.

E2a The massive bulk-to-horizon restriction on the de Sitter wedge horizon: the n=0 support step (s1-ii)

This appendix discharges, at $n = 0$ (free massive matter, arbitrary curvature coupling), the first named support step of Hyp. E2a.4: the restriction of the massive Bunch–Davies bulk data to the horizon of a de Sitter wedge lands on the *universal massless chiral horizon net*, where the Ceyhan–Faulkner-class per-cut equality [234] lives natively; composing, the equality-strength per-cut first law of Lem. E2a.3(a), together with its finite-first-moment premise (b), holds on every wedge of the $\mathrm{SO}(1,4)$ orbit for the compactly generated perturbation class defined below. The scope fences are collected in Rem. E2a.7; the geometric side (premise (c) and the Wald/Raychaudhuri pairing) is the companion step (s1-iii) and is not consumed here.

Throughout, $\mathrm{dS}_4 = \{X \in \mathbb{R}^{1,4} : X \cdot X = H^{-2}\}$ with $X \cdot X = -(X^0)^2 + \sum_{i=1}^{4}(X^i)^2$ (the paper elsewhere sets $H = 1$); $R = 12H^2$; the field is the real scalar with $(\Box - m^2 - \xi R)\phi = 0$ and effective mass $m^2_{\rm eff} := m^2 + \xi R > 0$, and

$$\nu := \left(\tfrac{9}{4} - \tfrac{m^2_{\rm eff}}{H^2}\right)^{1/2}, \qquad h_\pm := \tfrac{3}{2} \pm \nu,$$

so that $\nu \in i\,\mathbb{R}$ (principal series; all SM masses, $m \gg H$) or $\nu \in (0, \frac{3}{2})$ (complementary series); in all cases $\mathrm{Re}\, h_- > 0$. $\omega_{\rm BD}$ denotes the Bunch–Davies state, whose two-point function is the boundary value of

$$F(Z) \;=\; \frac{H^2\,\Gamma(h_+)\Gamma(h_-)}{16\pi^2}\;{}_2F_1\!\Big(h_+, h_-; 2; \frac{1+Z}{2}\Big), \qquad Z(x,x') := H^2\, X(x)\cdot X(x'), \qquad (221)$$

with the Wightman $i\epsilon$ prescription $Z \mapsto Z - i\epsilon\,(X^0 - X'^0)$.

Lemma E2a.1 (wedge-horizon geometry and the boundary dictionary)**.** *Let $W = \{X^1 > |X^0|\}$ be the reference wedge and $\mathcal{H} := \{X^0 = X^1\} \cap \mathrm{dS}_4$ the null hypersurface carrying its future horizon. Then:*

- *(i) $\mathcal{H} = \{X(v,\hat n) = (v, v, \hat n/H) : v \in \mathbb{R},\ \hat n \in S^2\} \cong \mathbb{R} \times S^2$; each generator $v \mapsto X(v,\hat n)$ is a complete null geodesic of* dS_4 *with* affine *parameter v; the cross-metric is the round sphere of radius $1/H$ (area measure $\sqrt{h}\,d^2\Omega = H^{-2}dS^2$), every cross-section a great sphere. The bifurcation surface of W is the cut $v = 0$; cuts $v = V(\hat n)$ of Lem. E2a.3 are cross-sections of this one surface.*
- *(ii) For any two points of $\mathcal{H}$, $Z = \hat n \cdot \hat n'$* exactly*: Z is independent of v, v', and $1 - Z = \frac{H^2}{2}\, d_{\rm ch}(\hat n, \hat n')^2$ with $d_{\rm ch}$ the chordal distance on the radius-$1/H$ sphere.*
- *(iii) The wedge boost (the (X^0, X^1) boost of* $\mathrm{SO}(1,4)$*, the modular/KMS flow of $\omega_{\rm BD}$ on $\mathcal{A}(W)$ [49]) acts on $\mathcal{H}$ as the dilations $v \mapsto e^s v$, fixing the bifurcation cut.*
- *(iv)* $\mathrm{SO}(1,4)$ *acts transitively on the wedge orbit (Lem. E2a.1) and carries the data (i)–(iii) of any wedge to any other; every statement below is therefore orbit-uniform once proved on W.*
- *(v) $\mathcal{H}$ is at the same time the past cosmological horizon $\Im^-$ of the expanding Poincaré-type patch $P^+ = \{X^0 > X^1\}$, i.e. exactly the surface of the Dappiaggi–Moretti–Pinamonti bulk-to-boundary construction [203]; their Bondi coordinate ℓ on $\Im^-$ is an* affine *function of v, with the same slope on every generator.*

Proof. (i) $X(v,\hat n)\cdot X(v,\hat n) = -v^2 + v^2 + |\hat n|^2/H^2 = H^{-2}$; the curve is a straight null line of $\mathbb{R}^{1,4}$ contained in the hyperboloid, hence a complete null geodesic of the induced metric with v affine. (ii) $H^2 X \cdot X' = -vv' + vv' + \hat n \cdot \hat n' = \hat n \cdot \hat n'$; $d^2_{\rm ch} = |\hat n - \hat n'|^2/H^2 = 2(1-Z)/H^2$.

(iii) The boost $(X^0, X^1) \mapsto (X^0 \cosh s + X^1 \sinh s,\ X^0 \sinh s + X^1 \cosh s)$ maps $(v, v, \hat{n}/H) \mapsto (e^s v, e^s v, \hat{n}/H)$. (iv) is Lem. E2a.1 plus the transitivity used in its proof. (v) In the planar chart of P^+ ($a(\tau) = -1/(H\tau)$, $\tau \in (-\infty, 0)$, conformal to Minkowski) one has, along $\tau \to -\infty$, $r \to \infty$ with $\tau + r \to v_*$: $X^0 \to v_*$, $X^1 \to v_*$ and $|\vec{X}_\perp| \to 1/H$, so the boundary point is $X(v_*, \hat{n})$ — the null coordinate $\tau + r$ of the chart converges on $\Im^-$ to the ambient affine parameter v of (i). The Bondi coordinate of [203] is $\ell(U) = -\gamma^{-1} \tan U + k_-$ with $U = \arctan(\tau + r)$, i.e. $\ell = -\gamma^{-1}(\tau + r) + k_-$: an affine function of $\tau + r$, hence of v, with generator-independent slope $-\gamma^{-1} > 0$. All five statements are verified symbolically/numerically in `ds2_check.py` (C1). □

Lemma E2a.2 (mass-inert restriction: the horizon state is the universal chiral vacuum)**.** *Let $J(f) := \int_{\mathcal{H}} \partial_v \phi\, f\, dv \sqrt{h}\, d^2\Omega$, $f \in C_c^\infty(\mathbb{R} \times S^2; \mathbb{R})$, be the horizon current smeared on $\mathcal{H}$, and let $\mathcal{A}_{\mathcal{H}}$ be the CCR algebra it generates, with cut subalgebras $\mathcal{A}_{\mathcal{H}}(V) := \{J(f) : \operatorname{supp} f \subset \{v > V(\hat{n})\}\}''$ for any continuous cut V. Then the pullback of $\partial_v \partial_{v'} \langle \phi\phi \rangle_{\omega_{\mathrm{BD}}}$ to $\mathcal{H} \times \mathcal{H}$ exists as a bidistribution and equals*

$$\omega_{\mathcal{H}}\big(\partial_v \phi(v, \hat{n})\, \partial_{v'} \phi(v', \hat{n}')\big) = -\frac{1}{4\pi} \frac{\delta^2_{\sqrt{h}}(\hat{n}, \hat{n}')}{(v - v' - i0)^2}, \tag{222}$$

independently of (m, ξ)*: the induced quasi-free state $\omega_{\mathcal{H}}$ on $\mathcal{A}_{\mathcal{H}}$ is the vacuum of the ultralocal chiral $U(1)$-current net fibred over $(S^2, \sqrt{h})$, i.e. the same state for every $m^2_{\text{eff}} > 0$ and the same as for the massless field on a Minkowski null plane. In particular the per-cut data $(\mathcal{A}_{\mathcal{H}}(V), \omega_{\mathcal{H}})$ carries no m^2-dependence at* any *cut, bifurcation or not: the cut variation of Lem. E2a.3 moves the subalgebra only, never the surface or the state.*

Proof. Near $Z = 1$ the hypergeometric connection formula for parameters $(h_+, h_-; 2)$, $2 - h_+ - h_- = -1$, gives

$$F(Z) = \frac{U_0}{1 - Z} + Q(Z) \log(1 - Z) + R(Z), \qquad U_0 = \frac{H^2 \Gamma(h_+)\Gamma(h_-)}{16\pi^2} \cdot \frac{\Gamma(2)\Gamma(1)}{\Gamma(h_+)\Gamma(h_-)} \cdot 2 = \frac{H^2}{8\pi^2},$$

with Q, R analytic at $Z = 1$; the pole coefficient U_0 is *mass-independent* (the $\Gamma(h_\pm)$ cancel exactly; numerically confirmed for $m^2_{\text{eff}}/H^2 = 0.5$, 4, 100, `ds2_check.py` C2), while all (m, ξ)-dependence sits in Q and R. Restrict to $\mathcal{H} \times \mathcal{H}$ using Lem. E2a.1(ii): $1 - Z = \frac{H^2}{2} d^2_{\text{ch}}$ is *v, v'-free*; the entire v-dependence of the boundary value enters through the prescription $1 - Z \mapsto \frac{H^2}{2} d^2_{\text{ch}} + i\epsilon(v - v')$ ($X^0 - X'^0 = v - v'$ on $\mathcal{H}$; the sign is fixed by positivity of the limit state, below). Writing $\rho := d_{\text{ch}}$ and absorbing $2/H^2$ into ϵ:

$$F\big|_{\mathcal{H} \times \mathcal{H}} = \frac{1}{4\pi^2 (\rho^2 + i\epsilon\, \Delta v)} + Q \log\big(\tfrac{H^2}{2} \rho^2 + i\epsilon \Delta v\big) + R.$$

Apply $\partial_v \partial_{v'} = -\partial^2_{\Delta v}$ and smear against C_c^∞ test functions: (pole) $-\partial^2_{\Delta v}(\rho^2 + i\epsilon \Delta v)^{-1} = 2\epsilon^2 (\rho^2 + i\epsilon \Delta v)^{-3}$, whose transverse integral against the measure $\sqrt{h}\, d^2\Omega$ (flat to leading order at coincidence, $\int 2\pi\rho\, d\rho$) converges dominatedly to $-1/(4\pi(\Delta v - i0)^2)$ times the transverse evaluation at $\hat{n} = \hat{n}'$ — i.e. to (222); (log) $-\partial^2_{\Delta v} \log(\cdots) = -\epsilon^2 (\rho^2 + i\epsilon \Delta v)^{-2}$, whose transverse integral is $O(\epsilon)$ and vanishes; (analytic) R and the Q-prefactor variations are v-free after $\epsilon \to 0$ and are annihilated identically. (Numerics: C3a reproduces $-1/(4\pi \Delta v^2)$ to 10^{-4}; C3b the $O(\epsilon)$ rate.) The limit kernel (222) has ℓ-Fourier transform supported on positive frequency ($\propto k\, \Theta(k)\, \delta^2_{\sqrt{h}}$), i.e. it is precisely the boundary state λ of [203], Eq. (20) there — the unique $\Im^-$-symmetric positive-energy pure quasi-free state — read in the affine parameter via Lem. E2a.1(v); positivity fixes the $-i0$ orientation and confirms $\omega_{\mathcal{H}}$ is a state. Since U_0 is universal, $\omega_{\mathcal{H}}$ is (m, ξ)-independent; it coincides with the Minkowski null-plane restriction of the massless (or any massive) Minkowski vacuum, for which the same computation with $\sigma = |y - y'|^2$ gives the same kernel. The final clause holds because (222) is a statement about the state on the *whole* net $\mathcal{A}_{\mathcal{H}}$ of the fixed complete surface $\mathcal{H}$; cuts index subalgebras of it. □

Remark E2a.3 (consistency with the printed massive bulk-to-boundary corpus). For $m_{\text{eff}}^2 > \frac{5}{48}R$ (equivalently $|\operatorname{Re}\nu| < 1$; all principal-series and hence all SM masses) Theorem 4.4 of [203] proves, on the expanding patch P^+: (a) every bulk wavefunction restricts to their boundary space $S(\Im^-)$ (an L^2-type space: $\psi, \partial_\ell\psi \in L^2$, their Eq. (17)) symplectically, giving the isometric injective $*$-homomorphism $\imath : \mathcal{W}(P^+) \to \mathcal{W}(\Im^-)$ of their Theorem 4.2; and (b) $\lambda \circ \imath$ *is* the Bunch–Davies state. Lemma E2a.2 is the two-sided, wedge-adapted form of the state content of (b) and was derived independently above; the net-level injection (a) is quoted here at exactly this printed strength and scope (expanding-patch Weyl algebra; no wedge- or cut-localised net correspondence is claimed in [203], and none is consumed below). For $0 < m_{\text{eff}}^2 \le \frac{5}{48}R$ their Remark 4.4 states the extension on an enlarged boundary space; we do not consume it.

Lemma E2a.4 (restriction of the perturbation class; the first moment is finite, not assumed). *Let the* compactly generated class $\mathcal{P}$ *consist of the coherent perturbations* $\omega_f := \omega_{\text{BD}}(W(f)^* \cdot W(f))$, $f \in C_c^\infty(\mathrm{dS}_4;\mathbb{R})$ *real test functions (in particular those supported in a wedge), and finite convex/normal combinations thereof. Write* $\psi := Ef$ *for the causally propagated classical solution. Then, on every generator of every orbit-wedge horizon:*

(i) $\psi|_{\mathcal{H}}$ *is smooth with* $|\psi(v,\hat{n})| \le C(1+|v|)^{-\min(1,\operatorname{Re}h_-)}$ *and* $|\partial_v\psi(v,\hat{n})| \le C(1+|v|)^{-\min(1,\operatorname{Re}h_-)-1}$*: the rate is governed by the cone-delta mechanism — the universal residue of the commutator function on the past cone of the perturbation's support caps the decay at* v^{-1} *for plane-straddling supports, while wedge-supported perturbations reach* $v^{-\operatorname{Re}h_-}$*; for principal-series (SM) masses* $\min(1,\operatorname{Re}h_-) = 1$.

(ii) $\delta\langle T_{vv}\rangle := \langle T_{vv}\rangle_{\omega_f} - \langle T_{vv}\rangle_{\omega_{\text{BD}}} = (\partial_v\psi)^2$ *exactly (free field, minimal coupling; for* $\xi \neq 0$ *the improvement term is handled by the companion Wald-pairing lemma, the improvement/Wald identity), and the* first moment *of Lem. E2a.3(b),* $\int_V^\infty dv\,(v-V)\,|\delta\langle T_{vv}\rangle| < \infty$, converges on every complete generator, for every $m_{\text{eff}}^2 > 0$*: the integrand is* $O(v^{-1-2\min(1,\operatorname{Re}h_-)})$ *—* $O(v^{-3})$ *for plane-straddling and* $O(v^{-4})$ *for wedge-supported perturbations at SM masses — integrable in either case.*

(iii) the horizon one-particle datum obeys $\psi,\ \partial_v\psi \in L^2(\mathbb{R}\times S^2, dv\,\sqrt{h}\,d^2\Omega)$ *whenever* $\operatorname{Re}h_- > \frac{1}{2}$ *(all principal-series masses), consistently with the boundary space* $S(\Im^-)$ *of [203], Thm. 4.4(a).*

Proof. (i) $\psi(x) = \int E(x,x')f(x')\,d\mu(x')$ with $E = 2\operatorname{Im}F$ the commutator function, a function of Z (and time orientation). Along a generator, $Z(X(v,\hat{n}),x') = v\,H^2(x'^1 - x'^0) + H\,\hat{n}\cdot\vec{x}'_\perp$ is *linear* in v with slope bounded away from 0 uniformly on the compact $\operatorname{supp} f$ (for f supported off the null plane $x'^0 = x'^1$, e.g. in any wedge or its complement). A compact support *straddling* the plane is not detachable (a second orbit wedge would bound ψ on a *different* horizon), and there the slope degenerates on the plane's trace; the decay is then set by the cone-delta mechanism: the leading singularity of E at $Z = 1$ is the universal $2\pi U_0\,\delta(Z-1)$ (residue U_0 mass-independent, Lem. E2a.2), and along a generator $Z - 1 = v\,H^2(x'^1 - x'^0) + \cdots$, so $\int f\,2\pi U_0\,\delta(Z-1)\,d\mu = \frac{2\pi U_0}{v}\int_{\text{plane}} f$, giving $\psi = O(v^{-1})$ for plane-straddling supports; the smooth interior contributes the faster $O(v^{-\operatorname{Re}h_-})$, so the envelope is $O(v^{-\min(1,\operatorname{Re}h_-)})$ as stated. For f supported off the plane the slope is bounded away from 0 and the connection formula for (221) at $Z \to \infty$ gives $|F|,|E| \le C|Z|^{-\operatorname{Re}h_-}$ and $|F'|,|E'| \le C|Z|^{-\operatorname{Re}h_- -1}$ (envelope verified numerically, `ds2_check.py` C4: complementary exponent to 2%; principal envelope $|Z|^{-3/2}$ bounded); differentiation in v multiplies by the bounded slope. (ii) The coherent shift is classical: $\phi \mapsto \phi + \psi$ under $\operatorname{Ad}W(f)$, so $\delta\langle(\partial_v\phi)^2\rangle = (\partial_v\psi)^2$ with the normal-ordering constant cancelling in the difference; the moment integrand is $(v-V)\,O((1+v)^{-2\operatorname{Re}h_- -2})$, integrable at $v = \infty$ iff $\operatorname{Re}h_- > 0$, which holds for every $m_{\text{eff}}^2 > 0$. (iii) $2\operatorname{Re}h_- > 1$. This *proves* premise (b) of Lem. E2a.3 for $\mathcal{P}$ on every generator; the witness state ($T = (1+v^2)^{-1}$: finite ANEC, divergent first moment) shows the premise is *not* implied by the paper's ANEC pin and genuinely needed the falloff proof above for the class. $\square$

Lemma E2a.5 (transfer: the dS horizon net is the Minkowski null-plane net, so the per-cut machinery applies verbatim)**.** *The pair* $(\mathcal{A}_{\mathcal{H}}, \omega_{\mathcal{H}})$ *of Lem. E2a.2, with its cut ordering* $V \le V'$ *and dilation action of Lem. E2a.1(iii), is isomorphic — as a cut-indexed net with state — to the free-field Minkowski null-plane net with its vacuum, by any measure isomorphism* $(S^2, \sqrt{h}\, d^2\Omega) \to (\mathbb{R}^2, d^2y)$ *(both are standard non-atomic* σ*-finite measure spaces; the kernel* (222) *is ultralocal transversally, and no transverse derivative or transverse geometry enters any cut object). In particular, at* $n = 0$*:*

(i) the vacuum modular Hamiltonian of the cut algebra $\mathcal{A}_{\mathcal{H}}(V)$ *is* $K_V = 2\pi \int_{S^2} \sqrt{h}\, d^2\Omega \int_{V(\hat{n})}^{\infty} dv\, (v - V(\hat{n}))\, T_{vv}(v, \hat{n})$*, with* $T_{vv} = :(\partial_v \phi)^2:_{\omega_{\mathcal{H}}}$ *[[204], null-plane Markov/factorisation; [55], per-generator half-sided structure — free-field facts whose proofs consume only the transferred data];*

(ii) translations $v \mapsto v + a$ *and cut flows are unitarily implemented in the GNS representation of* $\omega_{\mathcal{H}}$ *with positive generators (half-sided modular inclusion), and the Ceyhan–Faulkner equality machinery [234], whose hypotheses are exactly (HSMI) + finite* $\langle P \rangle$ *+ finite* S_{rel} *[their Eqs. (3.1)–(3.2)], applies to* $(\mathcal{A}_{\mathcal{H}}, \omega_{\mathcal{H}})$ *with no Minkowski-specific residue:* no bulk isometry generating v-translations is needed — *on* dS *none exists (the result of Lem. E2a.1(v)) — because the translations act on the* horizon *net and preserve the kernel* (222) *manifestly.*

Proof. Ultralocality: (222) factorises as $\delta^2_{\sqrt{h}} \otimes$ (chiral kernel); the GNS Hilbert space is the Bose Fock space over $L^2(S^2, \sqrt{h}) \otimes \mathfrak{h}_{\mathrm{chiral}}$, and every cut algebra is generated fibrewise; a measure isomorphism therefore carries nets, states, cut ordering, dilations and translations onto their null-plane counterparts and back. The v-translation invariance of (222) is manifest (Z is v-free; the kernel depends on $v - v'$ only), so translations are $\omega_{\mathcal{H}}$-preserving automorphisms, unitarily implementable in GNS; positivity of the generator and the HSMI property then follow from the transferred null-plane statements (or intrinsically: the fibred chiral vacuum satisfies Bisognano–Wichmann-type dilation KMS at the $V = 0$ cut, which is also the restriction of the bulk boost KMS [49] by Lem. E2a.1(iii) — the two structures agree). Points (i)–(ii) are then the quoted printed results read through the isomorphism. □

Theorem E2a.6 ((s1-ii) discharged at $n = 0$ for the compactly generated class)**.** *Let* W *be any wedge of the* $\mathrm{SO}(1,4)$ *orbit,* $\mathcal{H}$ *its extended horizon,* $m^2_{\mathrm{eff}} > 0$*, and* $\omega_f \in \mathcal{P}$ *a compactly generated perturbation of* ω_{BD}*. Then for every cut* V *in a* C^2 *neighbourhood of the bifurcation cut:*

$$\delta S(V) \;=\; \delta\langle K_V \rangle \;=\; 2\pi \int_{S^2} \sqrt{h}\, d^2\Omega \int_{V(\hat{n})}^{\infty} dv\, \big(v - V(\hat{n})\big)\, \delta\langle T_{vv}(v, \hat{n})\rangle, \tag{223}$$

where $S(V)$ *is the (relative-entropy-renormalised) entropy of the horizon cut algebra* $\mathcal{A}_{\mathcal{H}}(V)$*,* K_V *its vacuum modular Hamiltonian, and* $\delta\langle T_{vv}\rangle$ *the horizon restriction of the bulk stress expectation (Lem. E2a.4(ii)); both sides are finite (Lem. E2a.4), twice differentiable in* V *along the* C^2 *cut family for* ψ *as in Lem. E2a.4(i) (dominated by the* $v^{-2\,\mathrm{Re}\, h_- - 2}$ *envelope), and the matter side of the second cut-variation of Lem. E2a.3 follows:* $\delta^2\langle K_V\rangle / \delta V(\hat{n})^2 = 2\pi\, \delta\langle T_{vv}(V, \hat{n})\rangle \sqrt{h}$*. Consequently premises (*a_{matter}*) and (b) of Lem. E2a.3 hold at* $n = 0$ *for* $\mathcal{P}$ *on* every *wedge of the orbit: what remains of Hyp. E2a.4 at* $n = 0$ *is exactly the geometric pairing (premise (c) and* $\delta A/4G$*, the companion step (s1-iii)); at* $n \ge 1$ *it remains the B1a wall.*

Proof. Chain: (1) $\omega_{\mathrm{BD}} \mapsto \omega_{\mathcal{H}}$ = universal chiral vacuum (Lem. E2a.2); $\omega_f \mapsto$ coherent excitation of $\omega_{\mathcal{H}}$ with one-particle datum $\partial_v \psi$ (the shift is local-classical, so restriction commutes with the coherent conjugation). (2) K_V as displayed (Lem. E2a.5(i)); $\delta\langle K_V\rangle$ finite by Lem. E2a.4(ii) — the first moment *is* the modular energy. (3) First order in the perturbation: $\delta S = \delta\langle K_V\rangle$ is the entanglement first law (Araki: $S_{\mathrm{rel}} \ge 0$ vanishing to first order); at coherent-class strength the stronger per-cut statement holds *exactly*: $S_{\mathrm{rel}}(\omega_f \| \omega_{\mathcal{H}}; \mathcal{A}_{\mathcal{H}}(V)) = \langle K_V\rangle_{\mathrm{cl}}[\psi] = 2\pi \int (v -$

$V)(\partial_v\psi)^2$, the coherent-state relative entropy of the chiral net [205] read through Lem. E2a.5, whose premises (finite $\langle P\rangle$, finite $S_{\rm rel}$ at a reference cut) are discharged by Lem. E2a.4; the Ceyhan–Faulkner sum rule [234] supplies the same equality abstractly on the transferred HSMI, so the display is doubly anchored. (4) Second cut-variation of the right side of (223) in $V(\hat n)$: the boundary term from the lower limit vanishes linearly ($(v-V)|_{v=V}=0$) and the second variation evaluates the integrand at the cut, giving $2\pi\,\delta\langle T_{vv}(V,\hat n)\rangle\sqrt{h}$; the differentiations are justified by the envelope of Lem. E2a.4(i). (5) Orbit uniformity: Lem. E2a.1(iv). □

Remark E2a.7 (scope fences — what is and is not claimed). (F1) *Entropy reading.* $\delta S(V)$ in (223) is the entropy of the *horizon* cut algebra — the object the displayed consumers of Hyp. E2a.4 differentiate. The identification of $S(\mathcal{A}_{\mathcal{H}}(V{=}0))$-variations with bulk wedge-algebra entropy variations ("restriction faithfulness", $\mathcal{A}(W)$ vs $\mathcal{A}_{\mathcal{H}}(0)$ duality for the massive field) is *not claimed and not consumed*: no printed massive-dS statement supplies it ([203] inject the expanding-patch algebra, not the wedge algebra; Rem. E2a.3), and the per-cut display never uses it. (F2) *Perturbation class.* The first-moment premise is *proved* here for $\mathcal{P}$ (Lem. E2a.4); for general Bunch–Davies-class Hadamard perturbations it remains the stated premise (b) of Lem. E2a.3 — the witness state shows it does not follow from ANEC finiteness. (F3) *Coupling.* For $\xi\neq 0$ the display pairs with *Wald* entropy via the improvement identity, consumed in the companion step (s1-iii), not here. (F4) *Imports at printed strength.* [203]: expanding-patch Weyl injection + boundary state = Bunch–Davies, $m^2_{\rm eff}>\frac{5}{48}R$ for the L^2-type boundary space ($\psi,\ \partial_\ell\psi\in L^2$; DMP Eq. (17)) (all SM masses); [204, 55]: free-field null-plane cut modular data; [234]: HSMI equality machinery at stated premises; [205]: coherent-state entropy on the chiral net. Nothing is cited beyond these scopes; the dS-specific content (mass-inertness, dictionary, falloff) is derived, not imported. (F5) $n\geq 1$ stays the B1a wall; FRW stays re-pinned to A3 (no bifurcate Killing horizon), exactly as in Rem. E2a.5.

E2a The per-cut Wald/Raychaudhuri pairing on the de Sitter wedge horizon: the geometric side of Lem. E2a.3

Premise (a) of Lem. E2a.3 equates a quantum object (the per-cut modular energy) to a geometric one (the cut-localised area/Wald variation). The quantum side is the support step (s1-ii). This appendix proves the *geometric* side outright: an exact per-cut identity on the de Sitter wedge horizon between the future-normalised area deficit and the linearised Raychaudhuri flux integrated against the modular weight $(v-V(\Omega))$, with every boundary term explicit (Thm. E2a.5); the finite-first-moment premise (b) of Lem. E2a.3 is consumed at the $v\to\infty$ boundary and *only* there (Lem. E2a.3, Rem. E2a.4); the $\xi\neq 0$ improvement/Wald identity (premise (c)) is proved as Lem. E2a.7, including the de Sitter condensate term $\xi\langle\phi^2\rangle_{\rm BD}\,\delta G_{vv}$ that renormalises $G\to G_{\rm eff}$ coherently on both sides; and the second cut-variation step that Lem. E2a.3 consumes is Prop. E2a.8, with the twice-differentiability of premise (a) *derived* rather than assumed. Everything here is classical Lorentzian geometry plus measure theory; no field content beyond a Hadamard-smooth linear perturbation enters.

Conventions. Signature $(-,+,+,+)$; $R^\rho{}_{\sigma\mu\nu}=\partial_\mu\Gamma^\rho_{\nu\sigma}-\cdots$, $R_{\mu\nu}=R^\rho{}_{\mu\rho\nu}$; $\mathrm{dS}_4\subset\mathbb{R}^{1,4}$ is the hyperboloid $X\cdot X=\ell^2$, so $R_{\mu\nu}=3\ell^{-2}g_{\mu\nu}$, $R=12\ell^{-2}$, $\Lambda=3\ell^{-2}$ ($\ell=1$ reproduces the unit-hyperboloid normalisation of App. E2a). W denotes the wedge $\{X^1>|X^0|\}$, $\xi_W=X^1\partial_0+X^0\partial_1$ its boost. Capital indices A,B run over the S^2 cross-sections, γ_{AB} is the unit round metric, $\hat n(\omega)\in S^2\subset\mathbb{R}^3$ the unit embedding.

Lemma E2a.1 (the wedge horizon in exact null-adapted coordinates)**.** *Let* $N:=\{X^0=X^1\}\cap\mathrm{dS}_4$*. Define*

$$X(u,v,\omega)\;=\;\big(v,\ v,\ \ell\,\hat n(\omega)\big)\;+\;u\left(\tfrac{v^2}{\ell^2}+1,\ \tfrac{v^2}{\ell^2}-1,\ \tfrac{2v}{\ell}\,\hat n(\omega)\right).$$

Then:

(i) $X(u,v,\omega) \in \mathrm{dS}_4$ for all (u,v,ω), and in these coordinates the metric is exactly

$$ds^2 \;=\; -4\,du\,dv \;+\; \frac{4u^2}{\ell^2}\,dv^2 \;+\; \Big(\ell + \frac{2uv}{\ell}\Big)^2 \gamma_{AB}\,d\omega^A d\omega^B,$$

an exact dS_4 chart ($R_{\mu\nu} = 3\ell^{-2} g_{\mu\nu}$ identically) covering a neighbourhood of $N = \{u = 0\}$.

(ii) $N \cong \mathbb{R}_v \times S^2$; its generators $v \mapsto X(0,v,\omega)$ are straight null lines of $\mathbb{R}^{1,4}$ lying in dS_4, hence null geodesics of dS_4 with v an affine *parameter, complete in both directions ($v \in \mathbb{R}$). Equivalently: $\Gamma^{\mu}_{vv}|_{u=0} = 0$ for all μ. The bifurcation sphere of W is $B = \{u = 0, v = 0\}$; the $v < 0$ half of N is a horizon sheet of the antipodal wedge W', exactly as the $v < 0$ half of the Rindler null plane; the $v \to +\infty$ endpoint lies on $\mathscr{I}^+$ at* infinite *affine parameter — no far bifurcation surface or pole is met at finite v, and the second horizon sheet $\{X^0 = -X^1\}$ meets N only at B.*

(iii) On N the induced (degenerate) metric is $h_{AB} = \ell^2\gamma_{AB}$, independent of v: every constant-v cross-section S_v is a round sphere of radius ℓ, and the background expansion, shear and twist of the generator congruence vanish identically on N, $\theta = \sigma_{AB} = \omega_{AB} \equiv 0$; consequently $R_{vv}|_N = 0 = G_{vv}|_N$, and $g_{vv} = 4u^2/\ell^2$ vanishes quadratically *at N.*

(iv) (Exact cut-area formula.) For any cut, i.e. any continuous $V : S^2 \to \mathbb{R}$ with graph $\mathcal{C}_V = \{(0, V(\omega), \omega)\} \subset N$, the induced area element of $\mathcal{C}_V$ at ω equals $\sqrt{h}\,(V(\omega),\omega)\,d^2\omega$ — the cross-section area element evaluated along the graph, with no *$\partial_A V$ contribution at any order. This holds for the exact perturbed metric as long as N is null with generator ∂_v.*

(v) For $v \neq 0$ the cut S_v is extremal along the generator direction ($\theta_k = 0$) but not *totally geodesic: its expansion along the second null normal l (normalised $k \cdot l = -2$) is $\theta_l = 4v/\ell^2 \neq 0$. Hence $S_{v\neq 0}$ is the bifurcation surface of* no *wedge in the* $\mathrm{SO}(1,4)$ *orbit, and the cut-translated wedge W_V is* not *an isometric image of W — unlike Minkowski, where null translations are isometries. The cut-variation of Lem. E2a.3 genuinely leaves the orbit of Lem. E2a.1 for every $V \neq 0$.*

Proof. Write $p(v,\omega) = (v, v, \ell\hat{n})$, $k = \partial_v p = (1,1,0)$, $l(v,\omega) = (\frac{v^2}{\ell^2}+1, \frac{v^2}{\ell^2}-1, \frac{2v}{\ell}\hat{n})$. One checks $p\cdot p = \ell^2$, $k \cdot k = 0$, $l \cdot l = 0$, $p \cdot l = 0$, $k \cdot l = -2$; hence $X \cdot X = \ell^2 + 2u\,p\cdot l + u^2\,l\cdot l = \ell^2$ exactly, giving (i) up to the metric components, which are the pullbacks $g_{uu} = l \cdot l = 0$, $g_{uv} = l \cdot (k + u\partial_v l) = -2 + \frac{u}{2}\partial_v(l\cdot l) = -2$, $g_{uA} = \frac{u}{2}\partial_A(l \cdot l) = 0$, $g_{vv} = (k + u\partial_v l)^2 = 4u^2/\ell^2$, $g_{vA} = 0$, $h_{AB} = (\ell + 2uv/\ell)^2\gamma_{AB}$, each by the dot products of the displayed frame vectors ($k \cdot \partial_v l = 0$, $(\partial_v l)^2 = 4/\ell^2$, etc.); $R_{\mu\nu} = 3\ell^{-2}g_{\mu\nu}$ is verified symbolically from this line element (ds2_pairing.py, P1). For (ii): a straight line of $\mathbb{R}^{1,4}$ contained in the hyperboloid has vanishing ambient acceleration, hence vanishing tangential acceleration, and is an affinely parametrised geodesic of the induced metric; completeness is $v \in \mathbb{R}$ in the exact chart (h_{AB} degenerates only at $u = -\ell^2/2v$, never on N); $\Gamma^{\mu}_{vv}|_{u=0} = 0$ is checked symbolically (P2). The global statements are the elementary identifications $\{X^0 = X^1\} \cap \{X^0 = -X^1\} \cap \mathrm{dS}_4 = B$ and $\partial W' \supset \{v < 0\}$-half, from the definition of W, W'. (iii) is read off the line element at $u = 0$; $R_{vv}|_N = 0$ follows from $R_{\mu\nu} \propto g_{\mu\nu}$ and $g_{vv}|_N = 0$. For (iv): the graph tangents are $e_A = \partial_A + \partial_A V\,\partial_v$, and since N is null with normal-and-tangent generator ∂_v, the ambient metric restricted to TN annihilates ∂_v: $g(\partial_v, \cdot)|_{TN} = 0$. Hence $g(e_A, e_B) = h_{AB}(V(\omega),\omega)$ exactly, for the *perturbed* metric as well, which is (iv). For (v): the l-second fundamental form of S_v is $\mathrm{II}^{(l)}_{AB} = -\partial_A\partial_B X \cdot l$ with $X = p(v,\cdot)$; using $\partial_A\partial_B(\ell\hat{n}) = -\ell^{-1}h_{AB}\hat{n}$ one gets $\theta_l = h^{AB}\mathrm{II}^{(l)}_{AB} = 4v/\ell^2$, verified as $\partial_u \ln\sqrt{h}\,|_{u=0} = 4v/\ell^2$ in the chart (P2). A totally geodesic 2-sphere would have $\theta_l = \theta_k = 0$; bifurcation surfaces of orbit wedges are totally geodesic (Lem. E2a.1), so $S_{v\neq 0}$ is none such, and an isometry carrying W to W_V would carry B to a totally geodesic S_V, a contradiction. □

Remark E2a.2 (gauge, and what the display is a functional of). Fix the linear perturbation $\delta g_{\mu\nu}$ (Hadamard-smooth; at n=0 it is the metric response sourced by the state perturbation) in *horizon gauge*: N remains a null hypersurface and $\delta g_{v\mu}|_N = 0$, so ∂_v remains an affinely parametrised null generator field on N. Such a gauge exists: put the perturbed metric in Gaussian null form about N (the standard GNC construction applies to any smooth null hypersurface) and pull back; the residual freedom on N is the per-generator affine group $v \mapsto a(\omega)\, v + b(\omega)$. Every object in the pairing below — the cut, the weight $(v - V(\Omega))$, $\delta\theta$, δR_{vv}, the area elements — transforms covariantly under this residual group, and the identity of Thm. E2a.5 is verified to be form-invariant under it (the weight is degree 1, dv degree 1, δR_{vv} degree -2). The pairing is thus a statement about the triple (perturbed horizon, affine structure, cut), not about a coordinate choice. In this gauge $g_{vv}|_N = 0 = \delta g_{vv}|_N$, so on N

$$\delta G_{vv} = \delta R_{vv} - \tfrac{1}{2}\,\delta(g_{vv}R) = \delta R_{vv}, \qquad \Lambda\,\delta g_{vv} = 0:$$

the null–null linearised Einstein–Λ combination on N is $\delta R_{vv} - 8\pi G\,\delta\langle T_{vv}\rangle$, which is the E_{kk} of Lem. E2a.2. Extension off N does not enter: $\delta R_{vv}|_N$ is fixed by the intrinsic data $\delta h_{AB}(v,\omega)$ through the exact congruence identity $R_{vv} = -\partial_v\theta - \frac{1}{2}\theta^2 - \sigma_{AB}\sigma^{AB}$ (affine, twist-free), whose linearisation about $\theta = \sigma = 0$ (Lem. E2a.1(iii)) is

$$\partial_v\,\delta\theta = -\,\delta R_{vv} \text{ on } N, \qquad \delta\theta = \partial_v\Big(\frac{\delta\sqrt{h}}{\sqrt{h}}\Big), \quad \delta\sqrt{h} := \tfrac{1}{2}\sqrt{h}\,h^{AB}\delta h_{AB},$$

with no shear contribution at linear order ($\sigma^{AB}_{\rm bg}\,\delta\sigma_{AB} = 0$). Both statements, and their independence of affine-preserving extensions of δg off N, are verified symbolically (P3): a trace perturbation $\delta h_{AB} = f(v)h_{AB}$ gives $\delta R_{vv}|_N = -f''(v)$; a transverse-traceless δh_{AB} gives $\delta R_{vv}|_N = 0$; adding $\delta g_{vv} = u^2 q(v)$ changes neither.

Lemma E2a.3 (weighted tail calculus: the entire consumption of the finite-first-moment premise)**.** *Fix a generator ω and $V \in \mathbb{R}$, and let $F \in C^0([V,\infty))$ satisfy the* finite first moment

$$\text{(FM)} \qquad M_V(\omega) := \int_V^\infty (v - V)\,|F(v)|\,dv \;<\; \infty.$$

Let $g \in C^1$ solve $g' = -F$ on $[V,\infty)$ and let $G \in C^2$ solve $G' = g$. Then:

(i) $g(\infty) := \lim_{v\to\infty} g(v)$ exists (the tail $\int_v^\infty |F| \le (v-V)^{-1}M_V < \infty$ for $v > V$), and

$$(v - V)\,\big|g(v) - g(\infty)\big| \;\le\; \int_v^\infty (s - V)\,|F(s)|\,ds \;\xrightarrow[v\to\infty]{}\; 0,$$

the tail of the convergent moment integral.

(ii) (Tonelli.) $\displaystyle\int_V^\infty \big|g(v) - g(\infty)\big|\,dv \le \int_V^\infty\int_v^\infty |F(s)|\,ds\,dv = M_V(\omega).$

(iii) (Exact weighted identity; both boundary terms explicit.)

$$\int_V^\infty (v - V)\,F(v)\,dv \;=\; \underbrace{\Big[-(v - V)\big(g(v) - g(\infty)\big)\Big]_{v=V}^{v\to\infty}}_{=\,0\,-\,0} \;+\; \int_V^\infty \big(g(v) - g(\infty)\big)\,dv,$$

where the term at $v = V$ vanishes because the weight *vanishes at the cut (this is the entire boundary contribution at the bifurcation cut, for every cut including $V = 0$ and cuts crossing B), and the term at $v \to \infty$ vanishes by (i), i.e. by (FM) — and by nothing else.*

(iv) If moreover $g(\infty) = 0$, then $G(\infty) := \lim G(v)$ exists (absolutely, by (ii)) and

$$\int_V^\infty (v - V)\, F(v)\, dv \;=\; G(\infty) - G(V).$$

If instead $g(\infty) \neq 0$, then $G(v) = g(\infty)\, v + O(1)$ diverges linearly and no finite normalisation of G at infinity exists.

(v) Sharpness: (FM) is not implied by $\int_V^\infty |F| < \infty$. Witness $F(v) = (1+v^2)^{-1}$ (the divergent-moment witness): the ANEC-type integral is finite (π over the full line) while $M_V = \infty$ and G diverges logarithmically — the display in (iv) has no finite left-hand side.

Proof. (i): $g(v) - g(w) = \int_v^w F$, and $\int_v^\infty |F| \le (v-V)^{-1} \int_v^\infty (s-V)|F| \to 0$, so g is Cauchy at ∞; the displayed bound is $(v - V)|\int_v^\infty F| \le \int_v^\infty (v - V)|F(s)|\, ds \le \int_v^\infty (s - V)|F(s)|\, ds$, whose $v \to \infty$ limit is 0 by dominated convergence applied to the tails of the finite measure $(s-V)|F(s)|\, ds$. (ii) is Tonelli for $g(v) - g(\infty) = \int_v^\infty F(s)\, ds$. (iii): integrate by parts on $[V, R]$, $\int_V^R (v - V)F = -\int_V^R (v - V)\, d\big(g - g(\infty)\big)$, and send $R \to \infty$ using (i) at the upper end; at $v = V$ the weight $(v - V)$ vanishes and $g(V) - g(\infty)$ is finite, so the lower boundary term is exactly 0. (iv): integrate (iii) once more, absolutely convergent by (ii); the divergence statement is immediate. (v): direct computation, reproduced numerically in P5. □

Remark E2a.4 (the located consumption). Premise (b) of Lem. E2a.3 (finite first moment per complete generator) is consumed *exactly twice*, both times at the $v \to \infty$ end of Lem. E2a.3 with $F = \delta R_{vv}(\cdot, \omega)$, $g = \delta\theta$, $G = \delta\sqrt{h}/\sqrt{h}$: once in (iii) to kill the far boundary term $(v - V)\,\delta\theta$, once in (iv)/(ii) to make the late-time area limit exist. It is *not* consumed at the bifurcation cut (the weight vanishes there identically), and it is *not* consumed in the second cut-variation (Prop. E2a.8), which is local at the cut. Under the linearised field equation the two consumptions transfer verbatim to $F = 8\pi G\, \delta\langle T_{vv}\rangle$, which is the form stated in print. The witness (v) shows the premise is genuinely stronger than the ANEC-type pin: it must be *verified* for Bunch–Davies-class perturbations (falloff of $\delta\langle T_{vv}\rangle$ along the generators, affine parametrisation, weight linear in v); that verification is the (s1-ii)/H4 obligation, not part of this geometric arm.

Theorem E2a.5 (per-cut Wald/Raychaudhuri pairing on the de Sitter wedge horizon)**.** *Let N be the wedge horizon of Lem. E2a.1, δg a Hadamard-smooth linear perturbation in horizon gauge (Rem. E2a.2), and assume, per generator $\omega \in S^2$ and uniformly on S^2:*

(FM) $\sup_\omega \int_{V_0}^\infty (1 + |v|)\, \big|\delta R_{vv}(v, \omega)\big|\, dv < \infty$ *for some $V_0 \le \inf V$ (the finite first moment; under the linearised field equation this is premise (b) of Lem. E2a.3);*

(FN) $\lim_{v\to\infty} \delta\theta(v, \omega) \;=\; 0$ *for every ω (*future normalisation*: the perturbed horizon settles; by Lem. E2a.3(i) the limit exists under (FM), and by (iv) it vanishes if and only if the late-time area variation is finite; (FN) is thus tied to the finiteness of the display and is carried as an explicit hypothesis rather than a separate dynamical assumption).*

Then the late-time area variation $\delta A_\infty := \lim_{w\to\infty} \int_{S^2} d^2\Omega\, \delta\sqrt{h}\,(w, \Omega)$ exists, and for every *continuous cut $V : S^2 \to \mathbb{R}$,*

$$\boxed{\frac{1}{4G}\Big[\delta A_\infty - \delta A[V]\Big] \;=\; \frac{1}{4G}\int_{S^2} d^2\Omega\, \sqrt{h} \int_{V(\Omega)}^\infty dv\, \big(v - V(\Omega)\big)\, \delta R_{vv}(v, \Omega)}$$

with $\delta A[V] = \int d^2\Omega\, \delta\sqrt{h}(V(\Omega), \Omega)$ (exact, Lem. E2a.1(iv)), and with the complete list of boundary terms: at $v = V(\Omega)$ the integration by parts produces $(v - V)\,\delta\theta|_{v=V} = 0$ (weight vanishes at the cut; no contribution at or near the bifurcation cut for any cut); at $v \to \infty$ it produces

$-\lim_v(v-V)\,\delta\theta(v,\Omega)=0$ *by (FM)+(FN) via Lem. E2a.3(i,iii). If in addition the linearised semiclassical field equation holds on* N, $\delta R_{vv}=8\pi G\,\delta\langle T_{vv}\rangle$ *(Rem. E2a.2: on* N *this* is *the null–null Einstein–Λ equation), the right-hand side equals*

$$2\pi\int_{S^2}d^2\Omega\,\sqrt{h}\int_{V(\Omega)}^{\infty}dv\,(v-V(\Omega))\,\delta\langle T_{vv}(v,\Omega)\rangle \;=\; \delta\langle K_V\rangle_{\text{flux}},$$

the geometric flux form of the modular energy in premise (a) of Lem. E2a.3. Conversely — the direction the framework consumes — premise (a) as an identity in V *plus Prop. E2a.8 returns the pointwise equation.*

Proof. Fix ω. In horizon gauge the congruence identity linearises to $\partial_v\delta\theta=-\delta R_{vv}$ on N (Rem. E2a.2, verified P3), so Lem. E2a.3 applies with $F=\delta R_{vv}$, $g=\delta\theta$, $G=\delta\sqrt{h}/\sqrt{h}$ (background $\sqrt{h}$ is v-independent, Lem. E2a.1(iii), so $G'=\delta\theta$ integrates $\delta\theta=\partial_v(\delta\sqrt{h}/\sqrt{h})$ exactly). Parts (iii)+(iv) with $g(\infty)=0$ (FN) give, per generator,

$$\int_V^{\infty}(v-V)\,\delta R_{vv}\,dv=\frac{\delta\sqrt{h}(\infty,\omega)-\delta\sqrt{h}(V,\omega)}{\sqrt{h}}\,,$$

absolutely convergent with both boundary terms as listed. Multiply by $\sqrt{h}/4G$ and integrate over S^2: the uniform bound in (FM) dominates the ω-integral (compact S^2, continuous data), so Fubini applies, δA_∞ exists, and the boxed display follows, using Lem. E2a.1(iv) to identify $\int d^2\Omega\,\delta\sqrt{h}(V(\Omega),\Omega)$ with the area variation of the graph cut with no $\partial_A V$ correction. The field-equation substitution is pointwise on N. Numerical verification of the identity, of the boundary-term decay rates, and of the failure mode when (FM) is dropped: P4, P5. □

Remark E2a.6 (reading of the printed display: the deficit normalisation, and a sign trap). Premise (a) of Lem. E2a.3 writes the *deficit* pairing $\delta[A_\infty-A(V)]/4G=\delta\langle K_V\rangle$ with $\delta\langle K_V\rangle$ the *positive*-weight flux displayed above. Thm. E2a.5 shows the geometric object equal to that flux is the *future-normalised area deficit* $\delta[A_\infty-A(V)]/4G$: for a positive-energy perturbation crossing the horizon after the cut, $\delta\langle K_V\rangle>0$ and the area *grows* from the cut to $\mathscr{I}^+$, so $\delta A(V)<\delta A_\infty$ and the deficit is positive. The literal reading "$\delta A(V)$, normalised so that $\delta A_\infty=0$" has the opposite sign, and matters: under the second cut-variation the literal reading returns $\delta R_{vv}=-8\pi G\,\delta\langle T_{vv}\rangle$ (anti-Einstein), while the deficit reading returns the Einstein sign (Prop. E2a.8; the V-independent constant δA_∞ drops out of first *and* second variations, so the consumption of Lem. E2a.3 is unaffected once the reading is fixed). The consistent display convention is to write $\delta A(V)$ as $\delta\big[A_\infty-A(V)\big]$ (equivalently, state the teleological normalisation together with the orientation "$\delta S(V)$ decreases toward the future equilibrium"), so that premise (a), the Ceyhan–Faulkner-normalised first law it imports, and the Raychaudhuri pairing carry one consistent sign.

Lemma E2a.7 (the $\xi\neq 0$ improvement/Wald identity, with the de Sitter condensate tracked)**.** *Let the matter sector contain a free scalar with nonminimal coupling,* $L=-\frac{1}{2}[(\nabla\phi)^2+m^2\phi^2+\xi R\phi^2]$, *so that*

$$T^{(\xi)}_{\mu\nu}=T^{\min}_{\mu\nu}+\xi\big[G_{\mu\nu}-\nabla_\mu\nabla_\nu+g_{\mu\nu}\Box\big]\phi^2,$$

and let $\delta\langle\cdot\rangle$ *denote the semiclassical linear response about* $(\mathrm{dS}_4,\omega_{\mathrm{BD}})$, *with* $\langle\phi^2\rangle_{\mathrm{BD}}$ *the (renormalised, dS-invariant, hence constant) Bunch–Davies condensate. Assume, per generator and uniformly on* S^2, (FM_ϕ) $\int(1+|v|)\,|\partial_v^2\delta\langle\phi^2\rangle|\,dv<\infty$ *and* (FN_ϕ) $\partial_v\delta\langle\phi^2\rangle\to 0$ *as* $v\to\infty$ *(again forced by finiteness, Lem. E2a.3(i,iv), which also give existence of* $\delta\langle\phi^2\rangle(\infty,\omega)$*). Define*

$$\frac{1}{G_{\text{eff}}}:=\frac{1}{G}-8\pi\xi\langle\phi^2\rangle_{\mathrm{BD}},\qquad \delta S_{\mathrm{W}}(V):=\frac{\delta A[V]}{4G_{\text{eff}}}-2\pi\xi\int_{S^2}d^2\Omega\,\sqrt{h}\,\delta\langle\phi^2\rangle(V(\Omega),\Omega),$$

the first-order variation of the Wald entropy $S_{\mathrm{W}}=\oint\sqrt{h}\,[1/4G-2\pi\xi\langle\phi^2\rangle]$ *of the* $R/16\pi G-\frac{1}{2}\xi R\phi^2$ *sector. Then:*

(I) *(Exact structure of the null–null response on N.) In horizon gauge, using $\Gamma^{\mu}_{vv}|_N = 0$, $\nabla_\mu\langle\phi^2\rangle_{\rm BD} = 0$, $g_{vv}|_N = \delta g_{vv}|_N = 0$ and $G_{vv}|_N = 0$ (Lem. E2a.1),*

$$\delta\langle T^{(\xi)}_{vv}\rangle\big|_N = \delta\langle T^{\rm min}_{vv}\rangle\big|_N - \xi\,\partial_v^2\,\delta\langle\phi^2\rangle + \xi\,\langle\phi^2\rangle_{\rm BD}\,\delta G_{vv}\big|_N,$$

and the last (condensate) term is of the same *(first) order as the others; on N, $\delta G_{vv} = \delta R_{vv}$.*

(II) *(The telescoping identity; boundary terms explicit.) Per generator, for every cut value V,*

$$\int_V^\infty (v-V)\left(-\partial_v^2\delta\langle\phi^2\rangle\right)dv = \delta\langle\phi^2\rangle(\infty,\omega) - \delta\langle\phi^2\rangle(V,\omega),$$

by Lem. E2a.3 with $F = -\partial_v^2\delta\langle\phi^2\rangle$, $g = \partial_v\delta\langle\phi^2\rangle$, $G = \delta\langle\phi^2\rangle$: the cut boundary term carries the vanishing weight, the far term is killed by $(\mathrm{FM}_\phi) + (\mathrm{FN}_\phi)$. *The improvement contribution to the weighted flux, $2\pi\xi\oint\sqrt{h}\,[\delta\langle\phi^2\rangle(\infty) - \delta\langle\phi^2\rangle(V)]$, is therefore cut-localised — nothing is left on the horizon bulk — and cancels* exactly *against the ξ-piece of the Wald-entropy deficit, $\delta S^{(\xi)}(\infty) - \delta S^{(\xi)}(V) = 2\pi\xi\oint\sqrt{h}\,[\delta\langle\phi^2\rangle(V) - \delta\langle\phi^2\rangle(\infty)]$ with $\delta S^{(\xi)}(V) := -2\pi\xi\oint\sqrt{h}\,\delta\langle\phi^2\rangle(V)$, when moved across the first law: this is the improvement-identity surface-term statement, made sign-explicit.*

(III) *(Equivalence of pairings; no leakage.) Assume the linearised semiclassical equation on N with the full nonminimal source, $\delta R_{vv} = 8\pi G\,\delta\langle T^{(\xi)}_{vv}\rangle|_N$. By (I) this is equivalent to*

$$\delta R_{vv} = 8\pi G_{\rm eff}\left[\delta\langle T^{\rm min}_{vv}\rangle - \xi\,\partial_v^2\delta\langle\phi^2\rangle\right] \quad \text{on } N,$$

and under (FM)+(FN)+(FM_ϕ)+(FN_ϕ) *— which also give existence of $\delta S^\infty_{\rm W} := \lim_{w\to\infty}\delta S_{\rm W}(V \equiv w) = \delta A_\infty/4G_{\rm eff} - 2\pi\xi\oint\sqrt{h}\,\delta\langle\phi^2\rangle(\infty)$ — the two per-cut first laws*

$$(a_{\rm W})\quad \delta S^\infty_{\rm W} - \delta S_{\rm W}(V) = 2\pi\int d^2\Omega\,\sqrt{h}\int_V^\infty dv\,(v-V)\,\delta\langle T^{\rm min}_{vv}\rangle,$$

$$(a_A)\quad \frac{\delta A_\infty - \delta A[V]}{4G_{\rm eff}} = 2\pi\int d^2\Omega\,\sqrt{h}\int_V^\infty dv\,(v-V)\,\left[\delta\langle T^{\rm min}_{vv}\rangle - \xi\partial_v^2\delta\langle\phi^2\rangle\right]$$

are equivalent *— their difference is the identity (II) — and both hold, by Thm. E2a.5 applied with $G \to G_{\rm eff}$. A constant renormalisation-scheme shift $\langle\phi^2\rangle_{\rm BD} \mapsto \langle\phi^2\rangle_{\rm BD} + c$ moves $1/G_{\rm eff}$ and the condensate source term by the same combination, so the equivalence is scheme-covariant.*

(IV) *(Pointwise output is pairing-independent.) The second cut-variation (Prop. E2a.8) of $(a_{\rm W})$ and of (a_A) returns the* same *pointwise equation, the display in (III): no improvement ambiguity reaches the equation of state. Premise (c) of Lem. E2a.3 is thereby* derived, *in the sharpened form: the pairing is with Wald entropy and the* minimal *(chiral) flux — the flux the horizon net actually carries — or equivalently with the area and the full improved flux, with $G_{\rm eff}$ carrying the condensate in either reading.*

Proof. (I): expand $\delta\langle\xi[G_{vv} - \nabla_v\nabla_v + g_{vv}\Box]\phi^2\rangle$; the $\delta(g_{vv}\Box\phi^2)$ term dies on N by $g_{vv} = \delta g_{vv} = 0$; $\delta(\nabla_v\nabla_v\phi^2) = \partial_v^2\delta\langle\phi^2\rangle - \delta\Gamma^\mu_{vv}\partial_\mu\langle\phi^2\rangle_{\rm BD} = \partial_v^2\delta\langle\phi^2\rangle$ (constant condensate; background $\Gamma^\mu_{vv}|_N = 0$ removes the $\Gamma\cdot\partial\delta\phi^2$ cross term as well); $\delta(G_{vv}\phi^2) = \delta G_{vv}\,\langle\phi^2\rangle_{\rm BD} + G_{vv}\delta\langle\phi^2\rangle$ and $G_{vv}|_N = 0$. The Wald density $1/4G - 2\pi\xi\langle\phi^2\rangle$ is the standard $-2\pi\,\partial L/\partial R_{\mu\nu\rho\sigma}\,\epsilon_{\mu\nu}\epsilon_{\rho\sigma}$ evaluation for the two R-linear sectors, normalised so the Einstein sector returns $A/4G$; its first variation at the cut, using Lem. E2a.1(iv) and the constancy of $\langle\phi^2\rangle_{\rm BD}$ (which pairs with $\delta\sqrt{h}$ to give the $G \to G_{\rm eff}$ shift of the area term), is the displayed $\delta S_{\rm W}(V)$. (II) is Lem. E2a.3(iii,iv) under the stated mapping. (III): substitute (I) into the field equation and solve for δR_{vv} (the condensate term is proportional to δR_{vv} on N); apply Thm. E2a.5 verbatim with the constant $1/4G_{\rm eff}$ in place of

$1/4G$ and the equivalent source; then add/subtract (II) weighted by $2\pi\xi\sqrt{h}$ and integrate over S^2 to pass between ($\mathrm{a_W}$) and (a_A). (IV) is Prop. E2a.8 applied to both forms; the difference of the two second variations is the second cut-variation of the exact identity (II), which vanishes identically. Numerical check of (II), of the equivalence, and of the G_{eff} algebra: P6. □

Proposition E2a.8 (second cut-variation: the smeared-to-pointwise step, with differentiability derived)**.** *Under the premises of Thm. E2a.5 (and, if $\xi \neq 0$, of Lem. E2a.7), let $V \in C^0(S^2)$, $f \in C^0(S^2)$, and $V_s := V + sf$. Then both sides of the per-cut first law — in either reading ($\mathrm{a_W}$) or (a_A) — are twice continuously differentiable in s, with*

$$\frac{d^2}{ds^2}\Big|_0 \frac{\delta A_\infty - \delta A[V_s]}{4G_{\mathrm{eff}}} = \frac{1}{4G_{\mathrm{eff}}}\int_{S^2} d^2\Omega\, f(\Omega)^2\,\sqrt{h}\,\delta R_{vv}(V(\Omega),\Omega),$$

$$\frac{d^2}{ds^2}\Big|_0 2\pi\int d^2\Omega\sqrt{h}\int_{V_s}^{\infty} dv\,(v - V_s)\,\Phi(v,\Omega) = 2\pi\int_{S^2} d^2\Omega\, f(\Omega)^2\,\sqrt{h}\,\Phi(V(\Omega),\Omega)$$

for Φ the relevant flux density ($\Phi = \delta\langle T^{\min}_{vv}\rangle - \xi\partial_v^2\delta\langle\phi^2\rangle$ in (a_A); $\Phi = \delta\langle T^{\min}_{vv}\rangle$ in ($\mathrm{a_W}$), whose extra Wald term contributes $+2\pi\xi\int f^2\sqrt{h}\,\partial_v^2\delta\langle\phi^2\rangle(V,\Omega)$ to the geometric side — the same reshuffle). Consequently, if the per-cut first law holds as an identity for all cuts in a C^0 (a fortiori C^2) neighbourhood of the bifurcation cut, then for every f the two displays agree, and since both densities are continuous, polarisation in f gives the pointwise *null–null equation on N:*

$$\delta R_{vv}(x) \;=\; 8\pi G_{\mathrm{eff}}\left[\delta\langle T^{\min}_{vv}\rangle - \xi\,\partial_v^2\delta\langle\phi^2\rangle\right](x) \;=\; 8\pi G\,\delta\langle T^{(\xi)}_{vv}(x)\rangle, \qquad x \in N,$$

i.e. $E_{kk}(x) = 0$ in the sense of Lem. E2a.2 (Rem. E2a.2: $\delta G_{vv} = \delta R_{vv}$ and $\Lambda\delta g_{vv} = 0$ on N). Ranging over the wedge orbit (Lem. E2a.1) yields $E_{kk} = 0$ at every null pair (x,k), the input consumed by Lem. E2a.2. The finite first moment is not *consumed in this step: the second variation is local at the cut; (FM) enters only through the integrated display it varies (Rem. E2a.4).*

Proof. Geometric side: by the exact cut-area formula (Lem. E2a.1(iv)), $\delta A[V_s] = \int d^2\Omega\,\delta\sqrt{h}(V_s(\Omega),\Omega)$, and $\delta\sqrt{h}(\cdot,\Omega) \in C^\infty$ (Hadamard smoothness) with $\partial_v\delta\sqrt{h} = \sqrt{h}\,\delta\theta$, $\partial_v^2\delta\sqrt{h} = \sqrt{h}\,\partial_v\delta\theta = -\sqrt{h}\,\delta R_{vv}$ on N (Rem. E2a.2). Differentiation under $\int d^2\Omega$ twice is dominated on the compact S^2 by continuity of these derivatives on the compact range of V_s, $|s| \le 1$. Note the second variation is purely diagonal in Ω — the exact area formula has no $\partial_A V$ terms to produce off-diagonal or $(\partial f)^2$ contributions, at any order. Flux side: with $I(w,\Omega) := \int_w^\infty \Phi\,dv$ and $J(w,\Omega) := \int_w^\infty (v-w)\Phi\,dv$, (FM) gives absolute convergence of both and $\partial_w J = -I$, $\partial_w I = -\Phi(w,\Omega)$ (fundamental theorem of calculus; the boundary term of $\partial_w J$ carries the vanishing weight). Both are continuous in (w,Ω), and the uniform (FM) bound dominates the Ω-integral, so $\frac{d}{ds}\int d^2\Omega\sqrt{h}\,J(V_s,\Omega) = -\int d^2\Omega\sqrt{h}fI(V_s,\Omega)$ and $\frac{d^2}{ds^2} = \int d^2\Omega\sqrt{h}f^2\Phi(V_s,\Omega)$, continuous in s: premise (a)'s twice-differentiability is thereby derived from (FM) plus Hadamard smoothness. Pointwise step: the equality $\int f^2[\rho_1 - \rho_2]d^2\Omega = 0$ for all $f \in C^0(S^2)$ with ρ_i continuous forces $\rho_1 = \rho_2$ (evaluate on peaked f). Applying it at every admissible V delivers the pointwise equation at every point $(V(\Omega),\Omega)$ swept by the cut family: for the family of Lem. E2a.3 (all cuts in a C^2 neighbourhood of the bifurcation cut — already the constant cuts $V \equiv c$, $|c| < \varepsilon$, suffice) this is a neighbourhood of B in N, *per wedge.* Globality is then supplied by Lem. E2a.1, not by sliding along a single horizon: every (x,k) is bifurcation data of a 2-parameter family of orbit wedges, and the near-B output of each wedge covers it. (One cannot instead transport the equation along one N with the boost of W: the boost is a background isometry but the perturbation carries no invariance.) The $\xi \neq 0$ bookkeeping is Lem. E2a.7(IV). Numerical check of both second variations against the pointwise density, including the s-family: P7. □

Remark E2a.9 (status: what is proved, what is consumed, what is not delivered here). (1) The geometric side of Lem. E2a.3 *closes at theorem grade*: Thm. E2a.5 (pairing, explicit boundary terms), Lem. E2a.7 (premise (c) derived, condensate tracked), Prop. E2a.8 (the $\delta^2/\delta V^2$ consumption, differentiability derived). No resisting step. (2) Premise (b) (finite first moment) is consumed exactly at the two $v \to \infty$ loci of Lem. E2a.3, transferred verbatim to $\delta\langle T_{vv}\rangle$ by the field equation; the divergent-moment witness shows it does not follow from the ANEC pin, and its verification for Bunch–Davies-class perturbations along the affinely complete generators (weight linear in affine v, far end on $\mathscr{I}^+$) is an obligation of the quantum arm (H4), not of this one. (3) Two items to note: the deficit reading of the printed display (Rem. E2a.6; sign-critical for the second variation), and the condensate term in Lem. E2a.7(I) ($G \to G_{\text{eff}}$ on de Sitter, where $\langle\phi^2\rangle_{\text{BD}} \neq 0$; premise (c) as printed is correct for the ∂_v^2 piece and silently absorbs the condensate iff G is read as G_{eff}). (4) Not delivered here: premise (a)'s quantum content — that the deformed-cut modular energy *is* the displayed flux and that the per-cut *equality* $\delta S = \delta\langle K_V\rangle$ holds on the dS wedge orbit (support step (s1-ii)); Lem. E2a.1(v) is a structural caution for that arm: unlike Minkowski, the cut-translated wedges are not isometric images of W (no null-translation isometry on N), so the Ceyhan–Faulkner-class argument must be run on the half-sided-inclusion/chiral structure of the horizon net itself [234, 92], as the printed status remark already frames it. (5) Relation to print: the boost weight and KMS structure of the dS wedge are as in [49]; the local-Rindler polarisation step consuming the output is Lem. E2a.2 [56]; horizon-slice entropy bookkeeping at cuts follows the Killing-horizon conventions of [55].

References

[1] J. R. Oppenheimer, H. Snyder, *On Continued Gravitational Contraction*, Phys. Rev. **56**, 455 (1939).

[2] J. Lewandowski, Y. Ma, J. Yang, C. Zhang, *Quantum Oppenheimer–Snyder and Swiss Cheese Models*, Phys. Rev. Lett. **130**, 101501 (2023), arXiv:2210.02253.

[3] J. Ben Achour, S. Brahma, J.-P. Uzan, *Bouncing compact objects I: Quantum extension of the Oppenheimer–Snyder collapse*, JCAP **03** (2020) 041, arXiv:2001.06148.

[4] S. Winitzki, *Cosmological particle production and the precision of the WKB approximation*, Phys. Rev. D **72**, 104011 (2005), arXiv:gr-qc/0510001.

[5] Y. Aharonov, P. G. Bergmann, J. L. Lebowitz, *Time Symmetry in the Quantum Process of Measurement*, Phys. Rev. **134**, B1410 (1964).

[6] R. Silva, Y. Guryanova, N. Brunner, N. Linden, A. J. Short, S. Popescu, *Pre- and post-selected quantum states: density matrices, tomography, and Kraus operators*, Phys. Rev. A **89**, 012121 (2014), arXiv:1308.2089.

[7] J. Fullwood, *Quantum dynamics as a pseudo-density matrix*, Quantum **9**, 1719 (2025), arXiv:2304.03954.

[8] D. N. Page, W. K. Wootters, *Evolution without evolution*, Phys. Rev. D **27**, 2885 (1983).

[9] A. N. Bernal, M. Sánchez, *Smoothness of time functions and the metric splitting of globally hyperbolic spacetimes*, Commun. Math. Phys. **257**, 43 (2005).

[10] R. M. Wald, *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics*, University of Chicago Press (1994).

[11] R. M. Wald, *The back reaction effect in particle creation in curved spacetime*, Commun. Math. Phys. **54**, 1 (1977).

[12] R. M. Wald, *Trace anomaly of a conformally invariant quantum field in curved spacetime*, Phys. Rev. D **17**, 1477 (1978).

[13] V. Moretti, *Comments on the stress-energy tensor operator in curved spacetime*, Commun. Math. Phys. **232**, 189 (2003).

[14] S. Hollands, R. M. Wald, *Conservation of the stress tensor in interacting quantum field theory in curved spacetimes*, Rev. Math. Phys. **17**, 227 (2005).

[15] C. Dappiaggi, T.-P. Hack, N. Pinamonti, *The extended algebra of observables for Dirac fields and the trace anomaly of their stress-energy tensor*, Rev. Math. Phys. **21**, 1241 (2009).

[16] S. Hollands, R. M. Wald, *Quantum fields in curved spacetime*, Phys. Rep. **574**, 1 (2015).

[17] S. Hollands, R. M. Wald, *Stability of black holes and black branes*, Commun. Math. Phys. **321**, 629 (2013), arXiv:1201.0463.

[18] V. Chandrasekaran, R. Longo, G. Penington, E. Witten, *An algebra of observables for de Sitter space*, JHEP **02** (2023) 082, arXiv:2206.10780.

[19] A. D. Ludlow, M. M. Boyd, J. Ye, E. Peik, P. O. Schmidt, *Optical atomic clocks*, Rev. Mod. Phys. **87**, 637 (2015).

[20] W. Israel, *Singular hypersurfaces and thin shells in general relativity*, Nuovo Cim. B **44**, 1 (1966); erratum ibid. **48**, 463 (1967).

[21] C. Barrabès, W. Israel, *Thin shells in general relativity and cosmology: The lightlike limit*, Phys. Rev. D **43**, 1129 (1991).

[22] P. T. Chruściel, E. Delay, G. J. Galloway, R. Howard, *Regularity of horizons and the area theorem*, Ann. Henri Poincaré **2**, 109 (2001), arXiv:gr-qc/0001003.

[23] P. T. Chruściel, J. H. G. Fu, G. J. Galloway, R. Howard, *On fine differentiability properties of horizons and applications to Riemannian geometry*, J. Geom. Phys. **41**, 1 (2002), arXiv:gr-qc/0011067.

[24] E. Poisson, *A reformulation of the Barrabès–Israel null-shell formalism*, arXiv:gr-qc/0207101 (2002).

[25] T. Dray, G. 't Hooft, *The effect of spherical shells of matter on the Schwarzschild black hole*, Commun. Math. Phys. **99**, 613 (1985).

[26] J. Jezierski, *Geometry of crossing null shells*, J. Math. Phys. **44**, 641 (2003), arXiv:gr-qc/0406084.

[27] D. Rauch, J. Handsteiner, A. Hochrainer, J. Gallicchio, A. S. Friedman, et al., *Cosmic Bell test using random measurement settings from high-redshift quasars*, Phys. Rev. Lett. **121**, 080403 (2018), arXiv:1808.05966.

[28] B. Bertotti, L. Iess, P. Tortora, *A test of general relativity using radio links with the Cassini spacecraft*, Nature **425**, 374 (2003).

[29] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, New York (1966).

[30] J. Dixmier, *Von Neumann Algebras*, North-Holland, Amsterdam (1981).

[31] M. Reed, B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, Academic Press, New York (1980).

[32] M. Reed, B. Simon, *Methods of Modern Mathematical Physics, Vol. II: Fourier Analysis, Self-Adjointness*, Academic Press, New York (1975).

[33] A. Connes, C. Rovelli, *Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories*, Class. Quant. Grav. **11**, 2899 (1994), arXiv:gr-qc/9406019.

[34] N. Swanson, *Can quantum thermodynamics save time?*, Philosophy of Science **88**(2) (2021) 281–302, doi:10.1086/711569.

[35] H. Liu, *Lectures on entanglement, von Neumann algebras, and emergence of spacetime*, arXiv:2510.07017 (2025).

[36] E. Gesteau, *Large N von Neumann algebras and the renormalization of Newton's constant*, Commun. Math. Phys. **406**, 40 (2025), arXiv:2302.01938.

[37] W. Chemissany, E. Gesteau, A. Jahn, D. Murphy, L. Shaposhnik, *On infinite tensor networks, complementary recovery and type II factors*, J. Phys. A **58**, 435301 (2025), arXiv:2504.00096.

[38] V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Commun. Math. Phys. **387**, 299–360 (2021), arXiv:2010.11121.

[39] M. Takesaki, *Theory of Operator Algebras II*, Encyclopaedia of Mathematical Sciences **125**, Springer, Berlin (2003). See Ch. VIII (Tomita–Takesaki theory and the Connes cocycle $(D\omega' : D\omega)_t$).

[40] U. Haagerup, *Connes' bicentralizer problem and uniqueness of the injective factor of type III_1*, Acta Math. **158** (1987) 95–148.

[41] A. Connes, J. Feldman, B. Weiss, *An amenable equivalence relation is generated by a single transformation*, Ergodic Theory Dynam. Systems **1** (1981) 431–450.

[42] J. Feldman, C. C. Moore, *Ergodic equivalence relations, cohomology, and von Neumann algebras. II*, Trans. Amer. Math. Soc. **234** (1977) 325–359.

[43] L. Accardi, C. Cecchini, *Conditional expectations in von Neumann algebras and a theorem of Takesaki*, J. Funct. Anal. **45** (1982) 245–273.

[44] T. Amrutam, Y. Jiang, *On invariant von Neumann subalgebras rigidity property*, arXiv:2205.10700.

[45] C. Houdayer, A. Marrakchi, *Uniqueness of almost periodic outer flows on the hyperfinite type II_1 factor*, arXiv:2605.02781.

[46] T. Masuda, R. Tomatsu, *Rohlin flows on von Neumann algebras*, Mem. Amer. Math. Soc. **244** (2016), no. 1153, arXiv:1206.0955.

[47] A. Connes, *Une classification des facteurs de type III*, Ann. Sci. Éc. Norm. Supér. (4) **6** (1973) 133–252.

[48] A. Connes, *Almost periodic states and factors of type III_1*, J. Funct. Anal. **16** (1974) 415–445.

[49] H.-J. Borchers, D. Buchholz, *Global properties of vacuum states in de Sitter space*, Ann. Inst. H. Poincaré Phys. Théor. **70** (1999) 23–40, arXiv:gr-qc/9803036.

[50] H. Araki, *Relative Hamiltonian for faithful normal states of a von Neumann algebra*, Publ. RIMS Kyoto **9**, 165–209 (1973).

[51] H. Araki, *Golden–Thompson and Peierls–Bogolubov inequalities for a general von Neumann algebra*, Commun. Math. Phys. **34**, 167–178 (1973).

[52] H. Araki, *Relative entropy of states of von Neumann algebras*, Publ. RIMS Kyoto **11**, 809 (1976).

[53] J. D. Bekenstein, *Black holes and entropy*, Phys. Rev. D **7**, 2333 (1973).

[54] J. Bhattacharya, M. Nozaki, T. Takayanagi, T. Ugajin, *Thermodynamical property of entanglement entropy for excited states*, Phys. Rev. Lett. **110**, 091602 (2013), arXiv:1212.1164.

[55] A. C. Wall, *A proof of the generalized second law for rapidly changing fields and arbitrary horizon slices*, Phys. Rev. D **85**, 104049 (2012).

[56] T. Jacobson, *Thermodynamics of spacetime: the Einstein equation of state*, Phys. Rev. Lett. **75**, 1260 (1995).

[57] S. Weinberg, E. Witten, *Limits on massless particles*, Phys. Lett. B **96**, 59 (1980).

[58] T. Faulkner, M. Guica, T. Hartman, R. C. Myers, M. Van Raamsdonk, *Gravitation from entanglement in holographic CFTs*, JHEP **03** (2014) 051, arXiv:1312.7856.

[59] E. Bianchi, R. C. Myers, *On the architecture of spacetime geometry*, Class. Quant. Grav. **31**, 214002 (2014).

[60] Z.-W. Wang, S. L. Braunstein, *Surfaces away from horizons are not thermodynamic*, Nat. Commun. **9**, 2977 (2018), arXiv:2207.04390.

[61] E. Verlinde, *On the origin of gravity and the laws of Newton*, JHEP **04** (2011) 029.

[62] T. Padmanabhan, *Thermodynamical aspects of gravity: new insights*, Rep. Prog. Phys. **73**, 046901 (2010).

[63] D. N. Page, C. D. Geilker, *Indirect evidence for quantum gravity*, Phys. Rev. Lett. **47**, 979 (1981).

[64] R. T. Powers, E. Størmer, *Free states of the canonical anticommutation relations*, Commun. Math. Phys. **16**, 1 (1970).

[65] A. Kent, *Solution to the Lorentzian quantum reality problem*, Phys. Rev. A **90**, 012107 (2014), arXiv:1311.0249.

[66] A. Kent, *Tests of quantum gravity near measurement events*, Phys. Rev. D **103**, 064038 (2021), arXiv:2010.11811.

[67] A. Kent, *Real-world interpretations of quantum theory*, Found. Phys. **42**, 421 (2012), arXiv:0708.3710.

[68] S. Fedida, A. Kent, *The thermodynamics of readout devices and semiclassical gravity*, Phys. Rev. A **113**, 012214 (2026), arXiv:2509.07096.

[69] J. J. Halliwell, *Decoherence in quantum cosmology*, Phys. Rev. D **39**, 2912 (1989).

[70] J. J. Halliwell, *Decoherent histories and hydrodynamic equations*, Phys. Rev. D **58**, 105015 (1998), arXiv:quant-ph/9805062.

[71] J. B. Hartle, *Spacetime quantum mechanics and the quantum mechanics of spacetime*, in *Gravitation and Quantizations, Les Houches LVII*, North-Holland (1995), arXiv:gr-qc/9304006.

[72] J. Kisyński, *On operator-valued solutions of d'Alembert's functional equation, I*, Colloq. Math. **23**, 107–114 (1971).

[73] J. Kisyński, *On operator-valued solutions of d'Alembert's functional equation, II*, Studia Math. **42**, 43–66 (1972).

[74] J. Kisyński, *On cosine operator functions and one-parameter groups of operators*, Studia Math. **44**, 93 (1972).

[75] M. Sova, *Cosine operator functions*, Rozprawy Mat. **49**, 1 (1966).

[76] H. O. Fattorini, *Second Order Linear Differential Equations in Banach Spaces*, North-Holland Math. Studies **108**, Amsterdam (1985).

[77] H. O. Fattorini, *Uniformly bounded cosine functions in Hilbert space*, Indiana Univ. Math. J. **20**, 411–425 (1970).

[78] H. Stetkær, *Functional Equations on Groups*, World Scientific, Singapore (2013).

[79] A. Soulas, *A proof that no-signalling implies microcausality in QFT*, Found. Phys. **55**, 22 (2025), arXiv:2309.07715.

[80] T. Eggeling, D. Schlingemann, R. F. Werner, *Semicausal operations are semilocalizable*, Europhys. Lett. **57**, 782 (2002), arXiv:quant-ph/0104027.

[81] F. Calderón, *The causal axioms of algebraic quantum field theory: A diagnostic*, Stud. Hist. Phil. Sci. **104**, 98 (2024), arXiv:2401.06504.

[82] H. Bostelmann, C. J. Fewster, M. H. Ruep, *Impossible measurements require impossible apparatus*, Phys. Rev. D **103**, 025017 (2021), arXiv:2003.04660.

[83] R. Simmons, M. Papageorgiou, M. Christodoulou, Č. Brukner, *Factorisation conditions and causality for local measurements in QFT*, arXiv:2511.21644 (2025).

[84] J. Mandrysch, M. Navascués, *Quantum field measurements in the Fewster–Verch framework*, Lett. Math. Phys. **115**, 115 (2025), arXiv:2411.13605.

[85] Y. Guryanova, R. Silva, A. J. Short, P. Skrzypczyk, N. Brunner, S. Popescu, *Exploring the limits of no backwards in time signalling*, Quantum **3**, 211 (2019), arXiv:1708.00669.

[86] D. Beckman, D. Gottesman, M. A. Nielsen, J. Preskill, *Causal and localizable quantum operations*, Phys. Rev. A **64**, 052309 (2001), arXiv:quant-ph/0102043.

[87] K. Jensen, J. Sorce, A. J. Speranza, *Generalized entropy for general subregions in quantum gravity*, JHEP **12** (2023) 020, arXiv:2306.01837.

[88] J. Kudler-Flam, S. Leutheusser, G. Satishchandran, *Generalized black hole entropy is von Neumann entropy*, arXiv:2309.15897.

[89] J. Kudler-Flam, S. Leutheusser, G. Satishchandran, *Algebraic observational cosmology*, arXiv:2406.01669.

[90] C.-H. Chen, G. Penington, *A clock is just a way to tell the time: gravitational algebras in cosmological spacetimes*, arXiv:2406.02116 (2024).

[91] M. Ali, V. Suneeta, *A local generalized second law in crossed product constructions*, Phys. Rev. D **111** (2025) 024015, arXiv:2404.00718.

[92] T. Faulkner, A. J. Speranza, *Gravitational algebras and the generalized second law*, arXiv:2405.00847.

[93] S. Leichenauer, A. Levine, A. Shahbazi-Moghaddam, *Energy is entanglement*, Phys. Rev. D **98**, 086013 (2018), arXiv:1802.02584.

[94] S. Balakrishnan, V. Chandrasekaran, T. Faulkner, A. Levine, A. Shahbazi-Moghaddam, *Entropy variations and light ray operators from replica defects*, arXiv:1906.08274 (2019).

[95] J. Kudler-Flam, S. Leutheusser, A. A. Rahman, G. Satishchandran, A. J. Speranza, *A covariant regulator for entanglement entropy: proofs of the Bekenstein bound and QNEC*, Phys. Rev. D **111**, 105001 (2025), arXiv:2312.07646.

[96] J. De Vuyst, S. Eccles, P. A. Höhn, J. Kirklin, *Crossed products and quantum reference frames: on the observer-dependence of gravitational entropy*, JHEP **07** (2025) 063 [Erratum *ibid.* **10** (2025) 234], arXiv:2412.15502.

[97] J. De Vuyst, S. Eccles, P. A. Höhn, J. Kirklin, *Gravitational entropy is observer-dependent*, JHEP **07** (2025) 146, arXiv:2405.00114.

[98] B. Chen, J. Xu, *An algebra for covariant observers in de Sitter space*, arXiv:2511.00622.

[99] J. De Vuyst, S. Eccles, P. A. Höhn, J. Kirklin, *Linearization (in)stabilities and crossed products*, JHEP **05** (2025) 211, arXiv:2411.19931.

[100] A. Calcinari, S. Gielen, *Relational dynamics and Page–Wootters formalism in group field theory*, Quantum **9**, 1610 (2025), arXiv:2407.03432.

[101] J. van Neerven, P. Portal, *Thermal time as an unsharp observable*, J. Math. Phys. **65**, 032105 (2024), arXiv:2306.13774.

[102] P. A. Höhn, A. R. H. Smith, M. P. E. Lock, *Trinity of relational quantum dynamics*, Phys. Rev. D **104**, 066001 (2021), arXiv:1912.00033.

[103] P. A. Höhn, A. R. H. Smith, M. P. E. Lock, *Equivalence of approaches to relational quantum dynamics in relativistic settings*, Front. Phys. **9**, 587083 (2021), arXiv:2007.00580.

[104] S. Gielen, L. Menéndez-Pidal, *Unitarity, clock dependence and quantum recollapse in quantum cosmology*, Class. Quantum Grav. **39**, 075011 (2022), arXiv:2109.02660.

[105] J. Weberszpil, O. Sotolongo-Costa, *Entropy as a clock: foundations and parametrizations of emergent time*, Int. J. Theor. Phys. **65** (2026) 15, doi:10.1007/s10773-025-06212-1.

[106] T. Faulkner, F. M. Haehl, E. Hijano, O. Parrikar, C. Rabideau, M. Van Raamsdonk, *Nonlinear gravity from entanglement in conformal field theories*, JHEP **08** (2017) 057, arXiv:1705.03026.

[107] H. Casini, D. A. Galante, R. C. Myers, *Comments on Jacobson's "Entanglement equilibrium and the Einstein equation"*, JHEP **03** (2016) 194, arXiv:1601.00528.

[108] A. Svesko, *Equilibrium to Einstein: entanglement, thermodynamics, and gravity*, Phys. Rev. D **99**, 086006 (2019), arXiv:1810.12236.

[109] C. Eling, R. Guedens, T. Jacobson, *Nonequilibrium thermodynamics of spacetime*, Phys. Rev. Lett. **96**, 121301 (2006), arXiv:gr-qc/0602001.

[110] G. Chirco, S. Liberati, *Nonequilibrium thermodynamics of spacetime: the role of gravitational dissipation*, Phys. Rev. D **81**, 024016 (2010), arXiv:0909.4194.

[111] V. A. Pai, V. S. Namboothiri, T. K. Mathew, *Subtleties in non-equilibrium horizon thermodynamics of modified gravity theories*, arXiv:2604.04731 (2026).

[112] S.-M. Ruan, Y. Suzuki, *Toward a consistent definition of holographic entanglement entropy in de Sitter space*, arXiv:2508.14478 (2025).

[113] K. Fujiki, M. Kohara, K. Shinmyo, Y. Suzuki, T. Takayanagi, *Entropic interpretation of Einstein equation in dS/CFT*, arXiv:2511.07915 (2025).

[114] M. B. Fröb, *Modular Hamiltonian for de Sitter diamonds*, arXiv:2308.14797 (2023).

[115] P. Paul, P. Roy, *Linearized Einstein's equation around pure BTZ from entanglement thermodynamics*, Gen. Rel. Grav. **51** (2019) 155, arXiv:1803.06484.

[116] T. Jacobson, *Entanglement equilibrium and the Einstein equation*, Phys. Rev. Lett. **116**, 201101 (2016), arXiv:1505.04753.

[117] M. Van Raamsdonk, *Building up spacetime with quantum entanglement*, Gen. Rel. Grav. **42**, 2323 (2010), arXiv:1005.3035.

[118] M. Christodoulou, A. Di Biagio, M. Aspelmeyer, C. Brukner, C. Rovelli, R. Howl, *Locally mediated entanglement in linearised quantum gravity*, Phys. Rev. Lett. **130**, 100202 (2023), arXiv:2202.03368.

[119] D. L. Danielson, G. Satishchandran, R. M. Wald, *Black holes, soft hair, and decoherence*, arXiv:2205.06279; and *Killing horizons decohere quantum superpositions*, Int. J. Mod. Phys. D **31**, 2241003 (2022), arXiv:2301.00026.

[120] D. L. Danielson, G. Satishchandran, R. M. Wald, *Local description of decoherence of quantum superpositions by black holes and other bodies*, Phys. Rev. D **111**, 025014 (2025), arXiv:2407.02567.

[121] S. Bose, A. Mazumdar, G. W. Morley, H. Ulbricht, M. Toroš, M. Paternostro, A. A. Geraci, P. F. Barker, M. S. Kim, G. Milburn, *Spin entanglement witness for quantum gravity*, Phys. Rev. Lett. **119**, 240401 (2017), arXiv:1707.06050.

[122] L. Braccini, A. Serafini, S. Bose, *Mass-independent gravitationally induced entanglement*, arXiv:2602.19306 (2026).

[123] J. Wilson-Gerow, A. Dugad, Y. Chen, *Decoherence by warm horizons*, Phys. Rev. D **110**, 045002 (2024), arXiv:2405.00804.

[124] C. Marletto, V. Vedral, *Gravitationally induced entanglement between two massive particles is sufficient evidence of quantum effects in gravity*, Phys. Rev. Lett. **119**, 240402 (2017), arXiv:1707.06036.

[125] J. Barandes, *The ABL rule and the perils of post-selection*, arXiv:2602.07402 (2026).

[126] J. Barandes, *The Trouble with Weak Values*, arXiv:2602.09380 (2026).

[127] D. Trillo, M. Navascués, *Diósi–Penrose model of classical gravity predicts gravitationally induced entanglement*, Phys. Rev. D **111**, L121101 (2025), arXiv:2411.02287.

[128] J. Oppenheim, C. Sparaciari, B. Šoda, Z. Weller-Davies, *Gravitationally induced decoherence vs space-time diffusion: testing the quantum nature of gravity*, Nat. Commun. **14**, 7910 (2023), arXiv:2203.01982.

[129] J. Oppenheim, *A postquantum theory of classical gravity?*, Phys. Rev. X **13**, 041040 (2023), arXiv:1811.03116.

[130] A. Tilloy, L. Diósi, *Sourcing semiclassical gravity from spontaneously localized quantum matter*, Phys. Rev. D **93**, 024026 (2016), arXiv:1509.08705.

[131] A. Tilloy, L. Diósi, *Principle of least decoherence for Newtonian semiclassical gravity*, Phys. Rev. D **96**, 104045 (2017), arXiv:1706.01856.

[132] A. Tilloy, *General quantum-classical dynamics as measurement based feedback*, SciPost Phys. **17**, 083 (2024), arXiv:2403.19748.

[133] D. Kafri, J. M. Taylor, G. J. Milburn, *A classical channel model for gravitational decoherence*, New J. Phys. **16**, 065020 (2014), arXiv:1401.0946.

[134] B. Helou, B. J. J. Slagmolen, D. E. McClelland, Y. Chen, *Measurable signatures of quantum mechanics in a classical spacetime*, Phys. Rev. D **96**, 044008 (2017), arXiv:1612.06310.

[135] Y. Liu, H. Miao, Y. Chen, Y. Ma, *Semiclassical gravity phenomenology under the causal-conditional quantum measurement prescription*, Phys. Rev. D **107**, 024004 (2023), arXiv:2207.05966.

[136] L. Diósi, *A healthier stochastic semiclassical gravity: world without Schrödinger cats*, Gen. Relativ. Gravit. **57**, 62 (2025), arXiv:2501.18486.

[137] G. T. Horowitz, J. Maldacena, *The black hole final state*, JHEP **02** (2004) 008, arXiv:hep-th/0310281.

[138] U. Yurtsever, G. Hockney, *Signaling and the black hole final state*, arXiv:hep-th/0402060 (2004).

[139] N. Tibau Vidal, C. Marletto, V. Vedral, G. Chiribella, *Bose–Marletto–Vedral experiment without observable spacetime superpositions*, arXiv:2506.21122 (2025).

[140] M. Schut, A. Mazumdar, *Parameter scanning in a quantum-gravity-induced entanglement of masses (QGEM) experiment with electromagnetic screening*, arXiv:2502.12474 (2025).

[141] D. Carney, M. Karydas, T. Scharnhorst, R. Singh, J. M. Taylor, *On the quantum mechanics of entropic forces*, Phys. Rev. X **15**, 031038 (2025), arXiv:2502.17575.

[142] J. Aziz, R. Howl, *Classical theories of gravity produce entanglement*, Nature **646**, 813–817 (2025), doi:10.1038/s41586-025-09595-7, arXiv:2510.19714.

[143] A. Di Biagio, *The simple reason why classical gravity can entangle*, arXiv:2511.02683 (2025).

[144] C. Marletto, J. Oppenheim, V. Vedral, E. Wilson, *Classical gravity cannot mediate entanglement*, arXiv:2511.07348 (2025).

[145] M. Bojowald, *Singularities in loop quantum cosmology*, arXiv:2507.08116 (2025).

[146] C. J. Riedel, W. H. Zurek, M. Zwolak, *Objective past of a quantum universe: redundant records of consistent histories*, Phys. Rev. A **93**, 032126 (2016), arXiv:1312.0331.

[147] F. G. S. L. Brandão, M. Piani, P. Horodecki, *Generic emergence of classical features in quantum Darwinism*, Nat. Commun. **6**, 7908 (2015), arXiv:1310.8640.

[148] J. K. Korbicz, *Roads to objectivity: quantum Darwinism, spectrum broadcast structures, strong quantum Darwinism — a review*, Quantum **5**, 571 (2021), arXiv:2007.04276.

[149] M. Girard, G. Cheng, C. Cao, *Demystifying objectivity with operator algebra quantum error correction*, arXiv:2606.06588 (2026).

[150] H. W. Lin, *The bulk Hilbert space of double scaled SYK*, JHEP **11** (2022) 060, arXiv:2208.07032.

[151] J. Xu, *Von Neumann algebras in double-scaled SYK*, JHEP **03** (2026) 016, arXiv:2403.09021.

[152] V. Narovlansky, H. Verlinde, *Double-scaled SYK and de Sitter holography*, JHEP **05** (2025) 032, arXiv:2310.16994.

[153] M. Berkooz, O. Mamroud, *A cordial introduction to double scaled SYK*, Rep. Prog. Phys. **88**, 036001 (2025), arXiv:2407.09396.

[154] S. Collier, L. Eberhardt, B. Mühlmann, *A microscopic realization of dS_3*, SciPost Phys. **18**, 131 (2025), arXiv:2501.01486.

[155] G. Penington, E. Witten, *Algebras and states in JT gravity*, arXiv:2301.07257.

[156] A. Blommaert, J. Kudler-Flam, E. Y. Urbach, *Absolute entropy and the observer's no-boundary state*, arXiv:2505.14771.

[157] D. Marolf, H. Maxfield, *Transcending the ensemble: baby universes, spacetime wormholes, and the order and disorder of black hole information*, JHEP **08** (2020) 044, arXiv:2002.08950.

[158] A. H. Chamseddine, A. Connes, V. Mukhanov, *Geometry and the quantum: basics*, JHEP **12** (2014) 098, arXiv:1411.0977.

[159] N. Kaloper, A. Padilla, *Sequestering the Standard Model vacuum energy*, Phys. Rev. Lett. **112**, 091304 (2014), arXiv:1309.6562.

[160] N. Kaloper, A. Padilla, D. Stefanyszyn, G. Zahariade, *Manifestly local theory of vacuum energy sequestering*, Phys. Rev. Lett. **116**, 051302 (2016), arXiv:1505.01492.

[161] S. Weinberg, *Anthropic bound on the cosmological constant*, Phys. Rev. Lett. **59**, 2607 (1987).

[162] H. Martel, P. R. Shapiro, S. Weinberg, *Likely values of the cosmological constant*, Astrophys. J. **492**, 29 (1998), arXiv:astro-ph/9701099.

[163] R. Bousso, J. Polchinski, *Quantization of four-form fluxes and dynamical neutralization of the cosmological constant*, JHEP **06** (2000) 006, arXiv:hep-th/0004134.

[164] F. J. Murray, J. von Neumann, *On rings of operators. IV*, Ann. of Math. **44** (1943) 716–808.

[165] A. Connes, *Classification of injective factors. Cases* II_1, II_∞, III_λ, $\lambda \neq 1$, Ann. of Math. **104** (1976) 73–115.

[166] S. Popa, *On a class of type* II_1 *factors with Betti numbers invariants*, Ann. of Math. **163** (2006) 809–899, arXiv:math/0209310.

[167] S. Popa, S. Vaes, *Strong rigidity of generalized Bernoulli actions and computations of their symmetry groups*, Adv. Math. **217** (2008) 833–872, arXiv:math/0605456.

[168] M. Ahmed, S. Dodelson, P. B. Greene, R. Sorkin, *Everpresent* Λ, Phys. Rev. D **69** (2004) 103523, arXiv:astro-ph/0209274.

[169] S. Das, A. Nasiri, Y. K. Yazdi, *Aspects of everpresent Λ (I): a fluctuating cosmological constant from spacetime discreteness*, JCAP **10** (2023) 047, arXiv:2304.03819.

[170] J. Torvinen, E. Keski-Vakkuri, N. Pranzini, *Quantum Darwinism and the quality of Petz recovery*, arXiv:2605.06848 (2026).

[171] S. Bhattacharya, *Universal recoverability of quantum states in tracial von-Neumann algebras*, arXiv:2512.08418 (2025).

[172] C. J. Riedel, W. H. Zurek, M. Zwolak, *The rise and fall of redundancy in decoherence and quantum Darwinism*, New J. Phys. **14**, 083010 (2012), arXiv:1205.3197.

[173] X. Cao, Z. Nussinov, *Redundancy from subsystem thermalization*, arXiv:2603.15743 (2026).

[174] A. B. Tank, *Functional information in quantum Darwinism: an operational measure of objectivity*, arXiv:2509.17775 (2025).

[175] T. Benoist, J.-L. Fatras, C. Pellegrini, *Limit theorems for quantum trajectories*, Stochastic Process. Appl. **164**, 288 (2023), arXiv:2302.06191.

[176] T. Benoist, L. Greggio, C. Pellegrini, *Exponentially fast selection of sectors for quantum trajectories beyond non-demolition measurements*, Ann. Henri Poincaré (2025), arXiv:2407.18864.

[177] M. Bauer, T. Benoist, D. Bernard, *Repeated quantum non-demolition measurements: convergence and continuous-time limit*, Ann. Henri Poincaré **14**, 639 (2013), arXiv:1206.6045.

[178] M. Bauer, D. Bernard, *Convergence of repeated quantum nondemolition measurements and wave-function collapse*, Phys. Rev. A **84**, 044103 (2011), arXiv:1106.4953.

[179] T. Benoist, C. Pellegrini, *Large time behavior and convergence rate for quantum filters under standard non demolition conditions*, Commun. Math. Phys. **331**, 703–723 (2014), arXiv:1306.1430.

[180] V. P. Belavkin, *Quantum stochastic calculus and quantum nonlinear filtering*, J. Multivariate Anal. **42**, 171–201 (1992).

[181] L. Bouten, R. van Handel, M. R. James, *An introduction to quantum filtering*, SIAM J. Control Optim. **46**, 2199–2241 (2007), arXiv:math/0601741.

[182] H. Maassen, B. Kümmerer, *Purification of quantum trajectories*, arXiv:quant-ph/0505084 (2005); published in *Dynamics & Stochastics: Festschrift in honor of M. S. Keane*, IMS Lecture Notes–Monogr. Ser. **48** (2006).

[183] R. Durrett, *Probability: Theory and Examples*, 5th ed., Cambridge Series in Statistical and Probabilistic Mathematics **49**, Cambridge University Press (2019).

[184] D. Williams, *Probability with Martingales*, Cambridge University Press (1991).

[185] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019), arXiv:1903.11544.

[186] I. Agullo, A. Wang, E. Wilson-Ewing, *Loop quantum cosmology: relation between theory and observations*, arXiv:2301.10215 (2023).

[187] A. Bonanno, A. Eichhorn, H. Gies, J. M. Pawlowski, R. Percacci, M. Reuter, F. Saueressig, G. P. Vacca, *Critical reflections on asymptotically safe gravity*, Front. Phys. **8**, 269 (2020), arXiv:2004.06810.

[188] A. Eichhorn, M. Schiffer, *Asymptotic safety of gravity with matter*, arXiv:2212.07456 (2023).

[189] S. Caron-Huot, Y.-Z. Li, *Gravity and a universal cutoff for field theory*, JHEP **02** (2025) 115, arXiv:2408.06440.

[190] S. de Alwis, A. Eichhorn, A. Held, J. M. Pawlowski, M. Schiffer, F. Versteegen, *Asymptotic safety, string theory and the weak gravity conjecture*, Phys. Lett. B **798**, 134991 (2019), arXiv:1907.07894.

[191] R. Percacci, G. P. Vacca, *Asymptotic safety, emergence and minimal length*, Class. Quantum Grav. **27**, 245026 (2010), arXiv:1008.3621.

[192] W. G. Unruh, R. Schützhold, *On the universality of the Hawking effect*, Phys. Rev. D **71**, 024028 (2005), arXiv:gr-qc/0408009.

[193] B. A. Juárez-Aubry, *Recent developments in semiclassical gravity*, arXiv:2509.02051 (2025).

[194] P. Meda, N. Pinamonti, D. Siemssen, *Existence and uniqueness of solutions of the semiclassical Einstein equation in cosmological models*, Ann. Henri Poincaré **22**, 3965 (2021), arXiv:2007.14665.

[195] O. Bratteli, D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics*, Vol. II, 2nd ed., Springer (1997), Ch. 5.4.

[196] M. J. Donald, *Continuity of relative Hamiltonians*, Commun. Math. Phys. **136**, 625 (1991).

[197] R. Correa da Silva, *Perturbations of KMS states and noncommutative L^p-spaces*, arXiv:1811.04514.

[198] T. Faulkner, S. Hollands, *Approximate recoverability and relative entropy II: 2-positive channels of general von Neumann algebras*, Lett. Math. Phys. **112**, 26 (2022), arXiv:2010.05513.

[199] C. J. Fewster, R. Verch, *A quantum weak energy inequality for Dirac fields in curved spacetime*, Commun. Math. Phys. **225**, 331 (2002), arXiv:math-ph/0105027; and C. J. Fewster, *A general worldline quantum inequality*, Class. Quantum Grav. **17**, 1897 (2000), arXiv:gr-qc/9910060.

[200] S. P. Dawson, C. J. Fewster, *An explicit quantum weak energy inequality for Dirac fields in curved spacetimes*, Class. Quantum Grav. **23**, 6659 (2006), arXiv:gr-qc/0604106.

[201] C. J. Fewster, B. Mistry, *Quantum weak energy inequalities for the Dirac field in flat spacetime*, Phys. Rev. D **68**, 105010 (2003), arXiv:gr-qc/0307098.

[202] H. Sahlmann, R. Verch, *Passivity and microlocal spectrum condition*, Commun. Math. Phys. **214**, 705 (2000), arXiv:math-ph/0002021.

[203] C. Dappiaggi, V. Moretti, N. Pinamonti, *Cosmological horizons and reconstruction of quantum field theories*, Commun. Math. Phys. **285**, 1129 (2009), arXiv:0712.1770.

[204] H. Casini, E. Testé, G. Torroba, *Modular Hamiltonians on the null plane and the Markov property of the vacuum state*, J. Phys. A **50**, 364001 (2017).

[205] R. Longo, *Entropy of coherent excitations*, Lett. Math. Phys. **109**, 2587 (2019).

[206] I. Tice, *Quasilinear symmetric hyperbolic systems*, lecture notes, Department of Mathematical Sciences, Carnegie Mellon University (2017), Thm. 1.2.1 (higher-order spatial regularity); the linear estimate develops the Friedrichs symmetric-hyperbolic method and is the estimate cited (via Kato, J. Fac. Sci. Univ. Tokyo **17** (1970) 241, Prop. 3.4) by Hughes–Kato–Marsden 1977.

[207] K. Sanders, *Stress tensor bounds on quantum fields*, Commun. Math. Phys. **405**, 132 (2024), arXiv:2308.15218.

[208] A. Chialastri, K. Sanders, *Quasi-equivalence of Gaussian states and energy estimates for functions of modular Hamiltonians*, Commun. Math. Phys. **406**, 285 (2025), arXiv:2506.03907.

[209] J. R. Fliss, A. Rolph, *Curious QNEIs from QNEC: new bounds on null energy in quantum field theory*, arXiv:2510.26247 (2025).

[210] C. J. Fewster, T. A. Roman, *Null energy conditions in quantum field theory*, Phys. Rev. D **67**, 044003 (2003), arXiv:gr-qc/0209036.

[211] R. Verch, *Local definiteness, primarity and quasiequivalence of quasifree Hadamard quantum states in curved spacetime*, Commun. Math. Phys. **160**, 507 (1994).

[212] E. Moghtaderi, B. R. Hull, J. Quintin, G. Geshnizjani, *How much null-energy-condition breaking can the Universe endure?*, Phys. Rev. D **111**, 123552 (2025), arXiv:2503.19955.

[213] E.-A. Kontou, K. Sanders, *Energy conditions in general relativity and quantum field theory*, Class. Quantum Grav. **37**, 193001 (2020), arXiv:2003.01815.

[214] I. Ben-Dayan, *The quantum focusing conjecture and the improved energy condition*, JHEP **02** (2024) 132, arXiv:2310.14396.

[215] J. Glimm, A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed., Springer, New York (1987).

[216] M. Aizenman, H. Duminil-Copin, *Marginal triviality of the scaling limits of critical 4D Ising and* φ_4^4 *models*, Ann. Math. **194**, 163 (2021), arXiv:1912.07973.

[217] R. Panis, *Triviality of the scaling limits of critical Ising and* φ^4 *models with effective dimension at least four*, Ann. Probab. **54**, 892 (2026), DOI 10.1214/25-AOP1782, arXiv:2309.05797.

[218] C. J. Fewster, C. J. Smith, *Absolute quantum energy inequalities in curved spacetime*, Ann. Henri Poincaré **9**, 425 (2008), arXiv:gr-qc/0702056.

[219] A. Much, A. G. Passegger, R. Verch, *An approximate local modular quantum energy inequality in general quantum field theory*, arXiv:2210.01145 (2022).

[220] R. Brunetti, K. Fredenhagen, R. Verch, *The generally covariant locality principle — a new paradigm for local quantum field theory*, Commun. Math. Phys. **237**, 31 (2003), arXiv:math-ph/0112041.

[221] Y. E. Sanchez Sanchez, E. Schrohe, *The Sobolev wavefront set of the causal propagator in finite regularity*, Ann. Henri Poincaré **26** (2025) 1375–1406, arXiv:2203.04362. (Fixed-metric finite-regularity result; cf. also the Hintz–Vasy microlocal finite-regularity propagator theory.)

[222] P. T. Chruściel, J. D. E. Grant, *On Lorentzian causality with continuous metrics*, Class. Quantum Grav. **29** (2012) 145001, arXiv:1111.0400.

[223] A. Ashtekar, B. Krishnan, *Dynamical horizons and their properties*, Phys. Rev. D **68**, 104030 (2003), arXiv:gr-qc/0308033; and *Dynamical horizons: energy, angular momentum, fluxes, and balance laws*, Phys. Rev. Lett. **89**, 261101 (2002), arXiv:gr-qc/0207080.

[224] A. Ashtekar, B. Krishnan, *Quasi-local black hole horizons: recent advances*, Living Rev. Relativ. (2025), arXiv:2502.11825; and *Dynamical horizon segments and spacetime isometries*, arXiv:2506.10190 (2025).

[225] V. Chandrasekaran, É. É. Flanagan, *Subregion algebras in classical and quantum gravity*, arXiv:2601.07915 (2026).

[226] A. Ashtekar, D. E. Paraizo, J. Shu, *Thermodynamics of dynamical black holes beyond perturbation theory*, arXiv:2604.00170 (2026).

[227] R. Longo, G. Morsella, *The massless modular Hamiltonian*, Commun. Math. Phys. **400**, 1181 (2023), arXiv:2012.00565; and R. Longo, *Entropy distribution of localised states*, Commun. Math. Phys. **373**, 473 (2020).

[228] F. Ciolli, R. Longo, G. Ruzzi, *The information in a wave*, Commun. Math. Phys. **379**, 979 (2020); and *Relative entropy and curved spacetimes*, J. Geom. Phys. **172**, 104416 (2022), arXiv:2107.06787.

[229] S. Galanda, A. Much, R. Verch, *Relative entropy of fermion excitation states on the CAR algebra*, Math. Phys. Anal. Geom. **26**, 21 (2023), arXiv:2305.02788.

[230] S. Galanda, *Relative entropy for fermionic quantum field theory*, M.Sc. thesis, Universität Leipzig (2022), arXiv:2210.10746.

[231] F. Finster, A. Much, *The relative fermionic entropy in two-dimensional Rindler spacetime*, arXiv:2505.14076 (2025).

[232] D. Cadamuro, M. B. Fröb, C. Minz, *Modular Hamiltonian for fermions of small mass*, Ann. Henri Poincaré **26**, 4071 (2025), doi:10.1007/s00023-024-01508-0, arXiv:2312.04629.

[233] M. B. Fröb, A. Much, K. Papadopoulos, *Relative entropy for $\lambda\varphi^4$ in the Rindler wedge*, arXiv:2607.07810 [hep-th] (2026). Preprint, unrefereed.

[234] F. Ceyhan, T. Faulkner, *Recovering the QNEC from the ANEC*, Commun. Math. Phys. **377**, 999 (2020), arXiv:1812.04683.

[235] R. Bousso, Z. Fisher, S. Leichenauer, A. C. Wall, *A quantum focusing conjecture*, Phys. Rev. D **93**, 064044 (2016), arXiv:1506.02669.

[236] S. Hollands, R. Longo, *A new proof of the QNEC*, Commun. Math. Phys. **406**, 269 (2025), arXiv:2503.04651.

[237] I. Ben-Dayan, A. Srivastava, *Towards a Proof of the Improved Quantum Null Energy Condition*, arXiv:2601.18860 (2026).

[238] E. Witten, *Gravity and the crossed product*, JHEP **10** (2022) 008, arXiv:2112.12828.

[239] J. Dereziński, V. Jakšić, C.-A. Pillet, *Perturbation theory of W*-dynamics, Liouvilleans and KMS-states*, Rev. Math. Phys. **15**, 447 (2003).

[240] M. Ohya, D. Petz, *Quantum Entropy and Its Use*, Springer (1993; corr. 2004), Part IV (perturbation theory).

[241] H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, arXiv:1102.0440.

[242] S. Terashima, *Holography at finite N: breakdown of bulk reconstruction for subregions*, arXiv:2508.11592 (2025).

[243] E. Oh, I. Y. Park, S.-J. Sin, *Complete Einstein equation from the generalized First Law of Entanglement*, Phys. Rev. D **98**, 026020 (2018), arXiv:1709.05752.

[244] V. Chandrasekaran, G. Penington, E. Witten, *Large N algebras and generalized entropy*, JHEP **04** (2023) 009, arXiv:2209.10454.

[245] D. L. Jafferis, A. Lewkowycz, J. Maldacena, S. J. Suh, *Relative entropy equals bulk relative entropy*, JHEP **06** (2016) 004, arXiv:1512.06431.

[246] C. Gomez, *The algebraic Page curve*, arXiv:2403.09165 (2024); see also the follow-up Type II / QRF treatments of [244, 92].

[247] X. Cao, T. Faulkner, Z. Wang, *Gravitational Algebras with Two Areas*, arXiv:2512.04435 (2025).

[248] H. Zhong, *An algebraic description of the Page transition*, JHEP **04** (2026) 160, arXiv:2601.11363.

[249] C. Kiefer, M. Krämer, *Quantum gravitational contributions to the cosmic microwave background anisotropy spectrum*, Phys. Rev. Lett. **108**, 021301 (2012), arXiv:1103.4967.

[250] D. Brizuela, C. Kiefer, M. Krämer, *Quantum-gravitational effects on gauge-invariant scalar and tensor perturbations during inflation: the de Sitter case*, Phys. Rev. D **93**, 104035 (2016), arXiv:1511.05545.

[251] L. Chataignier, M. Krämer, *Unitarity of quantum-gravitational corrections to primordial fluctuations in the Born–Oppenheimer approach*, Phys. Rev. D **103**, 066005 (2021), arXiv:2011.06426.

[252] W. M. Campbell, M. E. Tobar, S. Galliou, M. Goryachev, *Improved constraints on minimum length models with a macroscopic low loss phonon cavity*, Phys. Rev. D **108**, 102006 (2023), arXiv:2304.00688.

[253] O. H. E. Philcox, M. Shiraishi, *Testing graviton parity and Gaussianity with Planck T-, E- and B-mode bispectra*, Phys. Rev. D **109**, 063522 (2024), arXiv:2312.12498.

[254] A. Krolewski, S. May, K. Smith, H. Hopkins, *No evidence for parity violation in BOSS*, JCAP **08** (2024) 044, arXiv:2407.03397.

[255] G. Cabass, S. Jazayeri, E. Pajer, D. Stefanyszyn, *Parity violation in the scalar trispectrum: no-go theorems and yes-go examples*, JHEP **02** (2023) 021, arXiv:2210.02907.

[256] G. Cabass, M. M. Ivanov, O. H. E. Philcox, *Colliding ghosts: constraining inflation with the parity-odd galaxy four-point function*, Phys. Rev. D **107**, 023523 (2023), arXiv:2210.16320.

[257] J. Hou et al., *Parity-odd four-point correlation function from DESI Data Release 1 luminous red galaxy sample*, arXiv:2512.20132.

[258] Z. Slepian et al., *Measurement of parity-violating modes of the DESI Year 1 luminous red galaxies' 4-point correlation function*, arXiv:2508.09133.

[259] Z. Gao, M. S. Cagliari, A. Moradinezhad Dizgah, Z. Vlah, *Testing parity with composite-field spectra of BOSS and DESI luminous red galaxies*, arXiv:2604.06021.

[260] Y. Bao, L.-T. Wang, Z.-Z. Xianyu, Y.-M. Zhong, *Anatomy of parity-violating trispectra in galaxy surveys*, Phys. Rev. D **112**, 103536 (2025), arXiv:2504.02931.

[261] M. Rotondo, *A Wheeler–DeWitt equation with time*, Universe **8**, 580 (2022), arXiv:2201.00809.

[262] L. Chataignier, P. A. Höhn, M. P. E. Lock, F. M. Mele, *Relational dynamics with periodic clocks*, New J. Phys. **28**, 034504 (2026), arXiv:2409.06479.

[263] C. Rovelli, M. Smerlak, *Thermal time and the Tolman–Ehrenfest effect: temperature as the speed of time*, Class. Quantum Grav. **28**, 075007 (2011), arXiv:1005.2985.

[264] S. Hollands, *Renormalized quantum Yang–Mills fields in curved spacetime*, Rev. Math. Phys. **20**, 1033 (2008), arXiv:0705.3340.

[265] R. Brunetti, M. Dütsch, K. Fredenhagen, K. Rejzner, *The unitary master Ward identity: time slice axiom, Noether's theorem and anomalies*, Lett. Math. Phys. **112**, 101 (2022), arXiv:2103.05740.

[266] K. Fredenhagen, K. Rejzner, *Batalin–Vilkovisky formalism in the functional approach to classical and quantum field theory*, Commun. Math. Phys. **317**, 697 (2013), arXiv:1101.5112; and *Perturbative algebraic quantum field theory*, Commun. Math. Phys. **345**, 741 (2016), arXiv:1503.08802.

[267] A. H. Chamseddine, A. Connes, *The spectral action principle*, Commun. Math. Phys. **186**, 731 (1997), arXiv:hep-th/9606001.

[268] W. D. van Suijlekom, *Noncommutative Geometry and Particle Physics*, 2nd ed., Mathematical Physics Studies (Springer, 2024), ISBN 978-3-031-59120-4.

[269] A. H. Chamseddine, A. Connes, W. D. van Suijlekom, *Beyond the spectral standard model: emergence of Pati–Salam unification*, JHEP **11** (2013) 132, arXiv:1304.8050.

[270] A. H. Chamseddine, A. Connes, M. Marcolli, *Gravity and the standard model with neutrino mixing*, Adv. Theor. Math. Phys. **11**, 991 (2007), arXiv:hep-th/0610241.

[271] F. La Piana, G. Morsella, *The fermionic massless modular Hamiltonian*, Commun. Math. Phys. **406**, 81 (2025), arXiv:2405.07888.

[272] *Perturbative algebraic quantum field theory and beyond* (2025 review), arXiv:2512.14227.

[273] S. Leutheusser, H. Liu, *Causal connectability between quantum systems and the black hole interior in holographic duality*, Phys. Rev. D **108**, 086019 (2023), arXiv:2110.05497; and *Subalgebra–subregion duality*, Phys. Rev. D **111**, 066021 (2025), arXiv:2212.13266.

[274] S. Ali Ahmad, R. Jefferson, *Crossed product algebras and generalized entropy for subregions*, SciPost Phys. Core **7**, 020 (2024), arXiv:2306.07323.

[275] G. L. Sewell, *Quantum fields on manifolds: PCT and gravitationally induced thermal states*, Ann. Phys. **141**, 201 (1982).

[276] B. S. Kay, R. M. Wald, *Theorems on the uniqueness and thermal properties of stationary, nonsingular, quasifree states on spacetimes with a bifurcate Killing horizon*, Phys. Rep. **207**, 49 (1991).

[277] S. J. Summers, R. Verch, *Modular inclusion, the Hawking temperature, and quantum field theory in curved spacetime*, Lett. Math. Phys. **37**, 145 (1996).

[278] P. Diener, B. Gupt, P. Singh, *Chimera: a hybrid approach to numerical loop quantum cosmology*, Class. Quantum Grav. **31**, 025013 (2014), arXiv:1310.4795.

[279] N. L. Diaz, P. Braccia, M. Larocca, J. M. Matera, R. Rossignoli, M. Cerezo, *Parallel-in-time quantum simulation via Page and Wootters quantum time*, Phys. Rev. Research **7**, 033294 (2025), arXiv:2308.12944.

[280] D. Giulini, D. Marolf, *A uniqueness theorem for constraint quantization*, Class. Quantum Grav. **16**, 2479 (1999), arXiv:gr-qc/9902045; and D. Marolf, *Refined algebraic quantization: systems with a single constraint*, arXiv:gr-qc/9508015.

[281] L. Chataignier et al., *A formalism for the ambiguities of the Wheeler–DeWitt equation*, arXiv:2407.03077 (2024).

[282] C. J. Wood, R. W. Spekkens, *The lesson of causal discovery algorithms for quantum correlations: causal explanations of Bell-inequality violations require fine-tuning*, New J. Phys. **17**, 033002 (2015), arXiv:1208.4119.

[283] D. Almada, K. Ch'ng, S. Kintner, B. Morrison, K. B. Wharton, *Are retrocausal accounts of entanglement unnaturally fine-tuned?*, arXiv:1510.03706 (2016).

[284] N. Gisin, *Weinberg's non-linear quantum mechanics and supraluminal communications*, Phys. Lett. A **143**, 1 (1990).

[285] J. Polchinski, *Weinberg's nonlinear quantum mechanics and the Einstein–Podolsky–Rosen paradox*, Phys. Rev. Lett. **66**, 397 (1991).

[286] A. Kent, *Nonlinearity without superluminality*, Phys. Rev. A **72**, 012108 (2005), arXiv:quant-ph/0204106.

[287] R. I. Sutherland, *Causally symmetric Bohm model*, Stud. Hist. Phil. Mod. Phys. **39**, 782 (2008), arXiv:quant-ph/0601095.

[288] J. G. Cramer, *The transactional interpretation of quantum mechanics*, Rev. Mod. Phys. **58**, 647 (1986).

[289] E. Adlam, *Two roads to retrocausality*, Synthese (2022), arXiv:2201.12934; and *Laws of nature as constraints*, Found. Phys. (2022), arXiv:2109.13836.

[290] L. Catani, M. S. Leifer, *A mathematical framework for operational fine tunings*, Quantum **7**, 948 (2023), arXiv:2003.10050.

[291] Y. Yīng, M. M. Ansanelli, A. Di Biagio, E. Wolfe, D. Schmid, E. G. Cavalcanti, *Relating Wigner's friend scenarios to nonclassical causal compatibility, monogamy relations, and fine tuning*, Quantum **8**, 1485 (2024), arXiv:2309.12987.

[292] S. Shrapnel, F. Costa, *Causation does not explain contextuality*, Quantum **2**, 63 (2018), arXiv:1708.00137.

[293] W. M. Stuckey, M. Silberstein, *Price–Wharton constrained colliders: co-causation or no causation?*, arXiv:2406.19419 (2024).

[294] H. Price, K. B. Wharton, *Bell correlations as selection artefacts*, arXiv:2309.10969 (2023); and *W as the edge of a wedge*, arXiv:2404.13928 (2024).

[295] V. Vilasini, R. Colbeck, *General framework for cyclic and fine-tuned causal models and their compatibility with space-time*, Phys. Rev. A **106**, 032204 (2022), arXiv:2109.12128.

[296] V. Vilasini, R. Colbeck, *The standard no-signalling constraints in Bell scenarios are neither sufficient nor necessary for preventing superluminal signalling with general interventions*, arXiv:2311.18465 (2023); see also *Information-processing in theories constrained by no superluminal causation vs no superluminal signalling*, Phys. Rev. Research **8**, 013070 (2026), arXiv:2402.12446.

[297] K. V. Kuchař, *Time and interpretations of quantum gravity*, Int. J. Mod. Phys. D **20**, 3 (2011) [reprint of 1991 lectures].

[298] C. J. Fewster, M. Janssen, L. Loveridge, K. Rejzner, J. Waldron, *Quantum reference frames, measurement schemes and the type of local algebras in quantum field theory*, Commun. Math. Phys. (2024), arXiv:2403.11973; and *Crossed products, conditional expectations and constraint quantization*, arXiv:2312.16678 (Nucl. Phys. B 2024).

[299] E. Alonso-Monsalve, *The Hilbert space of gauge theories: group averaging and the quantization of Jackiw–Teitelboim gravity*, arXiv:2512.03030 (2025).

[300] D. Buchholz, K. Solveen, *Disjointness of inertial KMS states and the role of Lorentz symmetry in thermalization*, arXiv:2402.14794 (2024).

[301] M. B. Fröb, J. Zahn, *Trace anomaly for chiral fermions via Hadamard subtraction*, JHEP **10** (2019) 223, arXiv:1904.10982.

[302] L. Bonora, S. Giaccari, B. Lima de Souza, *Trace anomalies in chiral theories revisited*, JHEP **07** (2014) 117, and Eur. Phys. J. C **77**, 511 (2017); see also JHEP **12** (2023) 064.

[303] C.-Y. Liu, *Investigation of Pontryagin trace anomaly using Pauli–Villars regularization*, arXiv:2202.13893 (v3, 2025).

[304] M. B. Fröb, J. Zahn, *The Weyl anomaly in interacting quantum field theory on curved spacetimes*, Ann. Henri Poincaré (2025), doi:10.1007/s00023-025-01635-2, arXiv:2504.17854.

[305] E. Witten, *A background-independent algebra in quantum gravity*, JHEP **03** (2024) 077, arXiv:2308.03663.

[306] K. Sanders, *On the Reeh–Schlieder property in curved spacetime*, Commun. Math. Phys. **288**, 271 (2009).

[307] A. Strohmaier, R. Verch, M. Wollenberg, *Microlocal analysis of quantum fields on curved spacetimes: analytic wavefront sets and Reeh–Schlieder theorems*, J. Math. Phys. **43**, 5514 (2002).

[308] A. Belin, J. de Boer et al., *Approximate CFTs and random tensor models*, arXiv:2308.03829 (2023).

[309] J. Pye, W. Donnelly, A. Kempf, *Locality and entanglement in bandlimited quantum field theory*, Phys. Rev. D **92**, 105022 (2015), arXiv:1508.05953.

[310] J. Pye, *Covariant bandlimitation from generalized uncertainty principles*, J. Phys. Conf. Ser. **1275**, 012025 (2019), arXiv:1903.12160.

[311] N. Funai, N. C. Menicucci, *Locality and entanglement harvesting in covariantly bandlimited scalar fields*, Phys. Rev. D **110**, 105001 (2024), arXiv:2408.01043.

[312] M. S. Leifer, M. F. Pusey, *Is a time-symmetric interpretation of quantum theory possible without retrocausality?*, Proc. R. Soc. A **473**, 20160607 (2017).

[313] K.-W. Bong et al., *A strong no-go theorem on the Wigner's friend paradox*, Nat. Phys. **16**, 1199 (2020).

[314] D. Brogioli, *A no-go theorem for sequential and retrocausal hidden-variable theories based on computational complexity*, arXiv:2409.11792 (2024).

[315] S. Mukherjee, J. R. Hance, *Limits of absoluteness of observed events in timelike scenarios: a no-go theorem*, Phys. Rev. Research (2026), doi:10.1103/1c6t-45g8, arXiv:2510.26562.

[316] E. Adlam, J. R. Hance, S. Hossenfelder, T. N. Palmer, *Taxonomy for physics beyond quantum mechanics*, Proc. R. Soc. A **480**, 20230779 (2024), arXiv:2309.12293.

[317] K. Wharton, R. Sutherland, T. Amza, R. Liu, J. Saslow, *A localized reality appears to underpin quantum circuits*, arXiv:2412.05456 (2024).

[318] L. Walleghem, L. Catani, *An extended Wigner's friend no-go theorem inspired by generalized contextuality*, arXiv:2502.02461 (2025).

[319] E. Okon, *Reassessing the strength of a class of Wigner's friend no-go theorems*, Entropy **27**, 563 (2025), arXiv:2204.12015.

[320] K. B. Wharton, N. Argaman, *Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics*, Rev. Mod. Phys. **92**, 021002 (2020).

[321] R. I. Sutherland, *How retrocausality helps*, Found. Phys. **47**, 174 (2017).

[322] A. Albrecht, L. Sorbo, *Can the universe afford inflation?*, Phys. Rev. D **70**, 063528 (2004), arXiv:hep-th/0405270.

[323] T. Banks, W. Fischler, *Sinks in the landscape, Boltzmann brains, and the cosmological constant problem*, arXiv:hep-th/0611043 (2006).

[324] G. W. Gibbons, S. W. Hawking, *Cosmological event horizons, thermodynamics, and particle creation*, Phys. Rev. D **15**, 2738 (1977).

[325] R. Bousso, *Positive vacuum energy and the N-bound*, JHEP **0011**, 038 (2000), arXiv:hep-th/0010252.

[326] A. G. Cohen, D. B. Kaplan, A. E. Nelson, *Effective field theory, black holes, and the cosmological constant*, Phys. Rev. Lett. **82**, 4971 (1999), arXiv:hep-th/9803132.

[327] S. D. H. Hsu, *Entropy bounds and dark energy*, Phys. Lett. B **594**, 13 (2004), arXiv:hep-th/0403052.

[328] S. M. Carroll, *Why Boltzmann brains don't fluctuate into existence from the de Sitter vacuum*, arXiv:1505.02780 (2015).

[329] L. Aalsma, S.-E. Bak, *Modular fluctuations in cosmology*, Phys. Rev. D **112**, 026017 (2025), arXiv:2503.04886.

[330] J. Solà Peracaula, *Running Vacuum in the expanding Universe: a unified QFT paradigm for Inflation and Dark Energy*, arXiv:2606.05352 (2026).

[331] C. Moreno-Pulido, J. Solà Peracaula, *Running vacuum in quantum field theory in curved spacetime: renormalizing $\rho_{\rm vac}$ without $\sim m^4$ terms*, Eur. Phys. J. C **80**, 692 (2020), arXiv:2005.03164.

[332] T.-N. Li, Y.-H. Li, G.-H. Du, P.-J. Wu, L. Feng, J.-F. Zhang, X. Zhang, *Revisiting holographic dark energy after DESI 2024*, Eur. Phys. J. C **85**, 608 (2025), arXiv:2411.08639.

[333] DESI Collaboration, *DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints*, Phys. Rev. D **112**, 083515 (2025), arXiv:2503.14738.

[334] J. Solà Peracaula, *Composite running vacuum in the Universe: implications on the cosmological tensions*, arXiv:2410.20382 (2024).

[335] O. Avsajanishvili, *Background dynamics and observational constraints of flat and non-flat $\Lambda(t)CDM$ models from $H(z)$ and DESI DR2 BAO measurements*, arXiv:2603.03468 (2026).

[336] B. R. Dinda, R. Maartens, *Physical vs phantom dark energy after DESI: thawing quintessence in a curved background*, arXiv:2504.15190 (2025).

[337] N. Allameh, S. Goodhew, S. Premkumar, *Revisiting zero modes and cluster decomposition at the late-time boundary of de Sitter*, arXiv:2510.24859 (2025).

[338] E. Dable-Heath, C. J. Fewster, K. Rejzner, N. Woods, *Algebraic classical and quantum field theory on causal sets*, Phys. Rev. D **101**, 065013 (2020), arXiv:1908.01973.

[339] A. Mathur, S. Surya, *Sorkin–Johnston state in a 2D causal diamond*, arXiv:1906.07952 (2019).

[340] Y. K. Yazdi et al., *Principles for a distinguished global vacuum on causal sets*, arXiv:2412.07832 (2024).

[341] Y. R. Zhu, Y. K. Yazdi, *On the (Non)Hadamard property of the SJ state in a $1+1D$ causal diamond*, Class. Quantum Grav. **41**, 045007 (2024), arXiv:2212.10592.

[342] C. J. Fewster, R. Verch, *On a recent construction of 'vacuum-like' quantum field states in curved spacetime*, Class. Quantum Grav. **29**, 205017 (2012), arXiv:1206.1562.

[343] S. Surya, X. Nomaan, Y. K. Yazdi, *Studies on the SJ vacuum in de Sitter spacetime*, JHEP **07**, 009 (2019), arXiv:1812.10228.

[344] S. Carlip, *Causal sets and an emerging continuum*, Gen. Rel. Grav. **56**, 95 (2024), arXiv:2405.14059.

[345] Z. Cao et al. (LHAASO Collaboration), *Stringent tests of Lorentz invariance violation from LHAASO observations of GRB 221009A*, Phys. Rev. Lett. **133**, 071501 (2024), arXiv:2402.06009.

[346] V. Vasileiou et al., *Constraints on Lorentz invariance violation from Fermi-Large Area Telescope observations of gamma-ray bursts*, Phys. Rev. D **87**, 122001 (2013), arXiv:1305.3463.

[347] P. Touboul et al. (MICROSCOPE Collaboration), *MICROSCOPE mission: final results of the test of the equivalence principle*, Phys. Rev. Lett. **129**, 121102 (2022), arXiv:2209.15487.

[348] B. P. Abbott et al. (LIGO Scientific, Virgo, Fermi-GBM, INTEGRAL), *Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A*, Astrophys. J. Lett. **848**, L13 (2017), arXiv:1710.05834.

[349] H. Xu, B.-Q. Ma, *Light speed variation from gamma-ray bursts*, Astropart. Phys. **82**, 72 (2016), arXiv:1607.03203.

[350] H. Song, B.-Q. Ma, *Lorentz invariance violation from gamma-ray bursts*, Astrophys. J. **983**, 9 (2025), arXiv:2504.00918.

[351] V. A. Kostelecký, N. Russell, *Data tables for Lorentz and CPT violation*, v19, Feb. 2026 edition, arXiv:0801.0287.

[352] C. Sanner et al., *Optical clock comparison for Lorentz symmetry testing*, Nature **567**, 204 (2019).

[353] M. Nagel et al., *Direct terrestrial test of Lorentz symmetry in electrodynamics to* 10^{-18}, Nat. Commun. **6**, 8174 (2015), arXiv:1412.6954.

[354] I. Counts, J. Hur, D. P. L. Aude Craik, H. Jeon, C. Leung, J. C. Berengut, A. Geddes, A. Kawasaki, W. Jhe, V. Vuletić, *Evidence for nonlinear isotope shift in* Yb^+ *search for new boson*, Phys. Rev. Lett. **125**, 123002 (2020), arXiv:2004.11383.

[355] M. Door, C.-H. Yeh, M. Heinz, F. Kirk, C. Lyu, T. Miyagi, J. C. Berengut, *et al.*, *Probing new bosons and nuclear structure with ytterbium isotope shifts*, Phys. Rev. Lett. **134**, 063002 (2025), arXiv:2403.07792.

[356] C. G. Parthey, A. Matveev, J. Alnis, R. Pohl, T. Udem, U. D. Jentschura, N. Kolachevsky, T. W. Hänsch, *Precision measurement of the hydrogen–deuterium* $1S$–$2S$ *isotope shift*, Phys. Rev. Lett. **104**, 233001 (2010).

[357] R. M. Potvliege, A. Nicolson, M. P. A. Jones, M. Spannowsky, *Deuterium spectroscopy for enhanced bounds on physics beyond the Standard Model*, Phys. Rev. A **108**, 052825 (2023), arXiv:2309.03732.

[358] W. Pauli, *Die allgemeinen Prinzipien der Wellenmechanik*, in *Handbuch der Physik*, Bd. 24, Springer (1933); English translation in W. Pauli, *General Principles of Quantum Mechanics*, Springer (1980).

[359] E. Y. S. Chua, *The time in thermal time*, J. Gen. Philos. Sci. (2025), doi:10.1007/s10838-024-09691-8, arXiv:2407.18948.

[360] E. A. Galapon, *Pauli's theorem and quantum canonical pairs: the consistency of a bounded, self-adjoint time operator canonically conjugate to a Hamiltonian with non-empty point spectrum*, Proc. R. Soc. Lond. A **458**, 451 (2002), arXiv:quant-ph/9908033.

[361] E. A. Galapon, R. F. Caballar, R. T. Bahague Jr., *Confined quantum time of arrivals*, Phys. Rev. Lett. **93**, 180406 (2004), arXiv:quant-ph/0302036.

[362] F. Hiroshima, N. Teranishi, *Self-adjointness of unbounded time operators*, Lett. Math. Phys. **115**, 95 (2025), arXiv:2405.02851.

[363] T. Shushi, *On quantum systems with a self-adjoint time operator without violating Pauli's theorem*, EPL **151**, 38001 (2025), doi:10.1209/0295-5075/adf3e0.

[364] J. De Vuyst, P. A. Höhn, A. Tsobanjan, *On the relation between perspective-neutral, algebraic, and effective quantum reference frames*, arXiv:2507.14131 (2025).

[365] A. J. Speranza, *An intrinsic cosmological observer*, Class. Quantum Grav. **42**, 21 (2025), arXiv:2504.07630.

[366] C. Cepollaro, A. Akil, P. Cieśliński, A.-C. de la Hamette, Č. Brukner, *Sum of Entanglement and Subsystem Coherence Is Invariant under Quantum Reference Frame Transformations*, Phys. Rev. Lett. **135**, 010201 (2025), arXiv:2406.19448.

[367] A.-C. de la Hamette, *Observer-Dependent Entropy and Diagonal Rényi Invariants in Quantum Reference Frames*, arXiv:2603.23598 (2026).

[368] S. C. Garmier, L. Hausmann, E. Castro-Ruiz, *The Perspectives of Non-Ideal Quantum Reference Frames*, arXiv:2512.19343 (2025).

[369] K. Ji, S. Lloyd, M. M. Wilde, *Retrocausal capacity of a quantum channel*, accepted Phys. Rev. Lett. (2026), 10.1103/znyd-npk5, arXiv:2509.08965.

[370] S. Lloyd, L. Maccone, R. Garcia-Patron, V. Giovannetti, Y. Shikano, S. Pirandola, L. A. Rozema, A. Darabi, Y. Soudagar, L. K. Shalm, A. M. Steinberg, *Closed timelike curves via post-selection: theory and experimental demonstration*, Phys. Rev. Lett. **106**, 040403 (2011), arXiv:1005.2219.

[371] A. Mondal, K. Prabhu, *Perturbative semiclassical entropy of dynamical black holes*, arXiv:2603.02320 (2026).

[372] J. Cotler, C.-M. Jian, X.-L. Qi, F. Wilczek, *Superdensity operators for spacetime quantum mechanics*, JHEP **09** (2018) 093, arXiv:1711.03119.

[373] N. L. Díaz, R. Rossignoli, *Spacetime quantum mechanics for bosonic and fermionic systems*, Phys. Rev. Research **8**, 013132 (2026), arXiv:2506.10250.

[374] H. Bostelmann, D. Cadamuro, *Fermionic integrable models and graded Borchers triples*, Lett. Math. Phys. **114**, 130 (2024), arXiv:2112.14686.

[375] S. Doplicher, R. Haag, J. E. Roberts, *Fields, observables and gauge transformations. I*, Comm. Math. Phys. **13**, 1–23 (1969).

[376] L. E. Ballentine, Y. Yang, J. P. Zibin, *Inadequacy of Ehrenfest's theorem to characterize the classical regime*, Phys. Rev. A **50**, 2854 (1994).

[377] W. H. Zurek, *Decoherence, einselection, and the quantum origins of the classical*, Rev. Mod. Phys. **75**, 715 (2003), arXiv:quant-ph/0105127.

[378] F. Hernández, D. Ranard, C. J. Riedel, *Ehrenfest's theorem beyond the Ehrenfest time*, arXiv:2306.13717 (2023).

[379] F. Hernández, D. Ranard, C. J. Riedel, *Classical correspondence beyond the Ehrenfest time for open quantum systems with general Lindbladians*, Commun. Math. Phys. **406**, 4 (2025), arXiv:2307.05326.

[380] F. Hernández, D. Ranard, C. J. Riedel, *The threshold for quantum-classical correspondence is $D \sim \hbar^{4/3}$*, arXiv:2512.17623 (2025).

[381] V. Gorini, A. Kossakowski, E. C. G. Sudarshan, *Completely positive dynamical semigroups of N-level systems*, J. Math. Phys. **17**, 821 (1976).

[382] G. Lindblad, *On the generators of quantum dynamical semigroups*, Commun. Math. Phys. **48**, 119 (1976).

[383] A. Lewkowycz, O. Parrikar, *The holographic shape of entanglement and Einstein's equations*, JHEP **05** (2018) 147, arXiv:1802.10103.

[384] A. Neumaier, *The Born rule — 100 years ago and today*, Entropy **27**, 415 (2025), arXiv:2502.08545.

[385] M. Ridley, *Quantum probability from temporal structure*, Quantum Rep. **5**, 496 (2023), arXiv:2112.10929.

[386] M. Ridley, *Many retrocausal worlds: a foundation for quantum probability*, Eur. J. Philos. Sci. **15** (2025), DOI:10.1007/s13194-025-00696-8, arXiv:2510.02505.

[387] M. Ridley, E. Adlam, *Time and event symmetry in quantum mechanics*, Quantum Stud. Math. Found. (2024), arXiv:2312.13524.

[388] J. Zhang, *Summing to Uncertainty: On the Necessity of Additivity in Deriving the Born Rule*, arXiv:2603.06211 (2026).

[389] N. Engelhardt, A. C. Wall, *Quantum extremal surfaces: holographic entanglement entropy beyond the classical regime*, JHEP **01** (2015) 073, arXiv:1408.3203.

[390] T. Faulkner, A. Lewkowycz, J. Maldacena, *Quantum corrections to holographic entanglement entropy*, JHEP **11** (2013) 074, arXiv:1307.2892.

[391] N. Graham, K. D. Olum, *Achronal averaged null energy condition*, Phys. Rev. D **76**, 064001 (2007), arXiv:0705.3193.

[392] J. Kirklin, *Generalised second law beyond the semiclassical regime*, arXiv:2412.01903 (2025).

[393] L. A. Pachón, *Galilean Reeh–Schlieder obstruction*, arXiv:2604.26271 (2026).

[394] H. Yan et al., *First result for testing semiclassical gravity effect with a torsion balance*, Phys. Rev. D **111**, 082007 (2025), arXiv:2411.17817.

[395] J. J. Halliwell, *Leggett-Garg inequalities and no-signaling in time: A quasiprobability approach*, Phys. Rev. A **93**, 022123 (2016); arXiv:1508.02271.

[396] J. J. Halliwell, *Comparing conditions for macrorealism: Leggett-Garg inequalities versus no-signaling in time*, Phys. Rev. A **96**, 012121 (2017); arXiv:1704.01485.

[397] J. Fullwood and A. J. Parzygnat, *On quantum states over time*, Proc. R. Soc. A **478**, 20220104 (2022).

[398] S. H. Lie and N. H. Y. Ng, *Uniqueness of quantum state over time function*, Phys. Rev. Research **6**, 033144 (2024); arXiv:2308.12752.

[399] S. H. Lie and J. Fullwood, *Multipartite quantum states over time from two fundamental assumptions*, Phys. Rev. Lett. **135**, 230204 (2025); arXiv:2410.22630.

[400] Z. Jia, K. Modi, and D. Kaszlikowski, *Temporal Kirkwood–Dirac quasiprobability distribution and unification of temporal state formalisms through temporal Bloch tomography*, arXiv:2601.05294.

[401] M. M. Wilde and A. Mizel, *Addressing the clumsiness loophole in a Leggett-Garg test of macrorealism*, Found. Phys. **42**, 256 (2012); arXiv:1001.1777.

[402] M. J. Duff, L. B. Okun, G. Veneziano, *Trialogue on the number of fundamental constants*, JHEP **03** (2002) 023, arXiv:physics/0110060.

[403] A. D. Sakharov, *Vacuum quantum fluctuations in curved space and the theory of gravitation*, Dokl. Akad. Nauk SSSR **177** (1967) 70 [Sov. Phys. Dokl. **12** (1968) 1040]; reprinted Gen. Rel. Grav. **32** (2000) 365.

[404] M. Visser, *Sakharov's induced gravity: a modern perspective*, Mod. Phys. Lett. A **17** (2002) 977, arXiv:gr-qc/0204062.

[405] S. N. Solodukhin, *Entanglement entropy of black holes*, Living Rev. Relativity **14** (2011) 8, arXiv:1104.3712.

[406] T. Jacobson, *Black hole entropy and induced gravity*, arXiv:gr-qc/9404039.